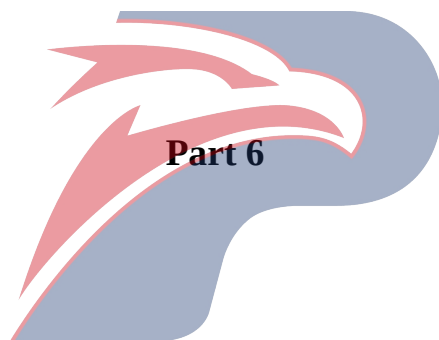


Maintenance Manual

MY F Series

Engine

(6HK1 type)



POWERSTAR

ISUZU



Isuzu Motors Co., Ltd.
Service Marketing Department

Engine

Engine Mechanical Parts

(6HK1)

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6HK1-TC engine

Precautions for maintenance

When performing engine maintenance, the following points should be carefully observed to avoid damage to the engine and to maintain the reliability of engine performance.

- When lifting or supporting the engine, do not lift the jack on the oil pan.

When placing the engine on the ground, engine parts or wooden frames should be used to support the engine feet and the flywheel housing.

- When removing the air intake system, cover the air intake to prevent debris from entering the cylinder. If debris is allowed to enter, the engine may cause severe damage to cylinders and other components.

- When repairing the engine body, be sure to remove the battery ground wire. If the battery ground cable is not removed during maintenance, the harness or electrical components may be damaged. If you need to supply power when performing checks or other operations, be sure not to cause a short circuit.

- When running the engine for the first time, a sufficient amount of engine oil should be painted on the rotating surface to lubricate and protect these parts.

- When disassembling valve-related parts, pistons, piston rings, connecting rods, connecting rod bearings, and crankshaft journal bearings, be sure to place the parts neatly.

- When reassembling a component, it should be assembled in its original position before disassembly.

- After the gasket, oil seal, O-ring, etc. have been removed, new parts should be replaced.

- Thoroughly remove the left liquid filler and remove the oil, water or dirt from these parts. Use the specified liquid filler and reassemble the parts.

- Apply components within 7 minutes of applying fluid filling.

If they are not installed well after more than 7 minutes, the liquid filler should be removed and repainted.

- When assembling or installing the components, tighten them with the specified torque strength and ensure that

the components are installed correctly.

This precautions for engine operation

All holes and voids in the fuel line of the fuel system injector are manufactured to exact specifications. These parts are very sensitive to dust or debris, and dust or debris entering these parts can damage the fuel lines. Therefore, as much as possible, dust or debris should be prevented from entering these parts.

Technical service information

How to use plastic gauges

Types	Measuring range (mm/inch)
PG-1(Green)	0.025(0.001)— 0.076(0.003)
PR-1(Red)	0.051(0.002)— 0.152(0.006)
PB-1(blue)	0.102(0.004)— 0.229(0.009)

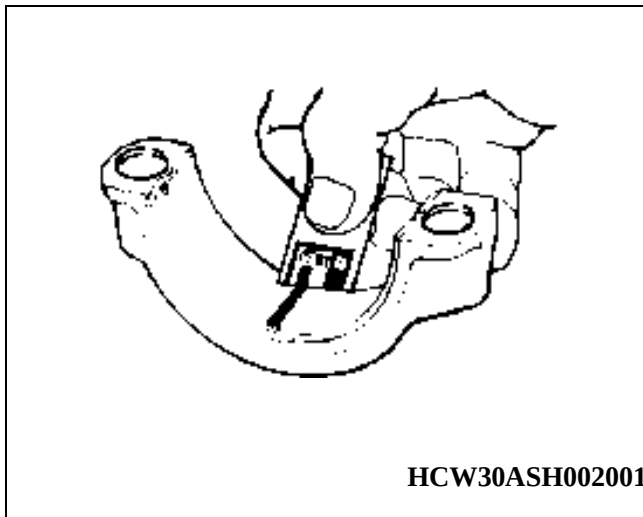
<Example> How to measure the clearance between the connecting rod bearing and the crank pin

- Clean the connecting rod and bearing, then assemble the connecting rod.
- Cut the plastic gauge's crank pin width and place it in parallel, while avoiding the crank pin hole.
- Align the connecting rod mark and cap label with each other, then assemble the crank pin and apply molybdenum disulfide lubricant to the threaded part of the fastening bolt and the bearing surface, and alternately tighten the cap to the specified torque.

Note:

Never use a plastic pressure gauge to move the connecting rod.

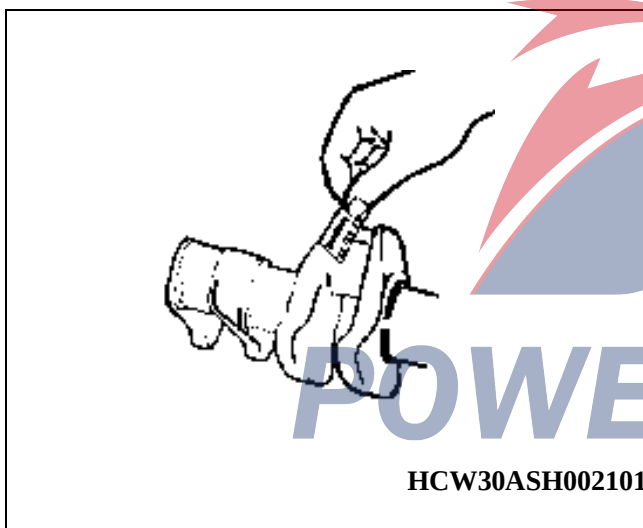
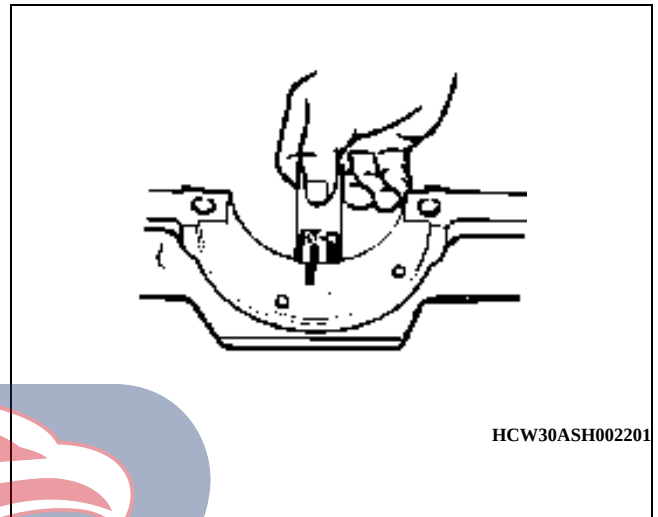
- Gently remove the gap and connecting rod and use the scale on the bag to measure the plastic gauge's extrusion width (gap).



Note:

Do not rotate the crankshaft when using plastic line clearance rules.

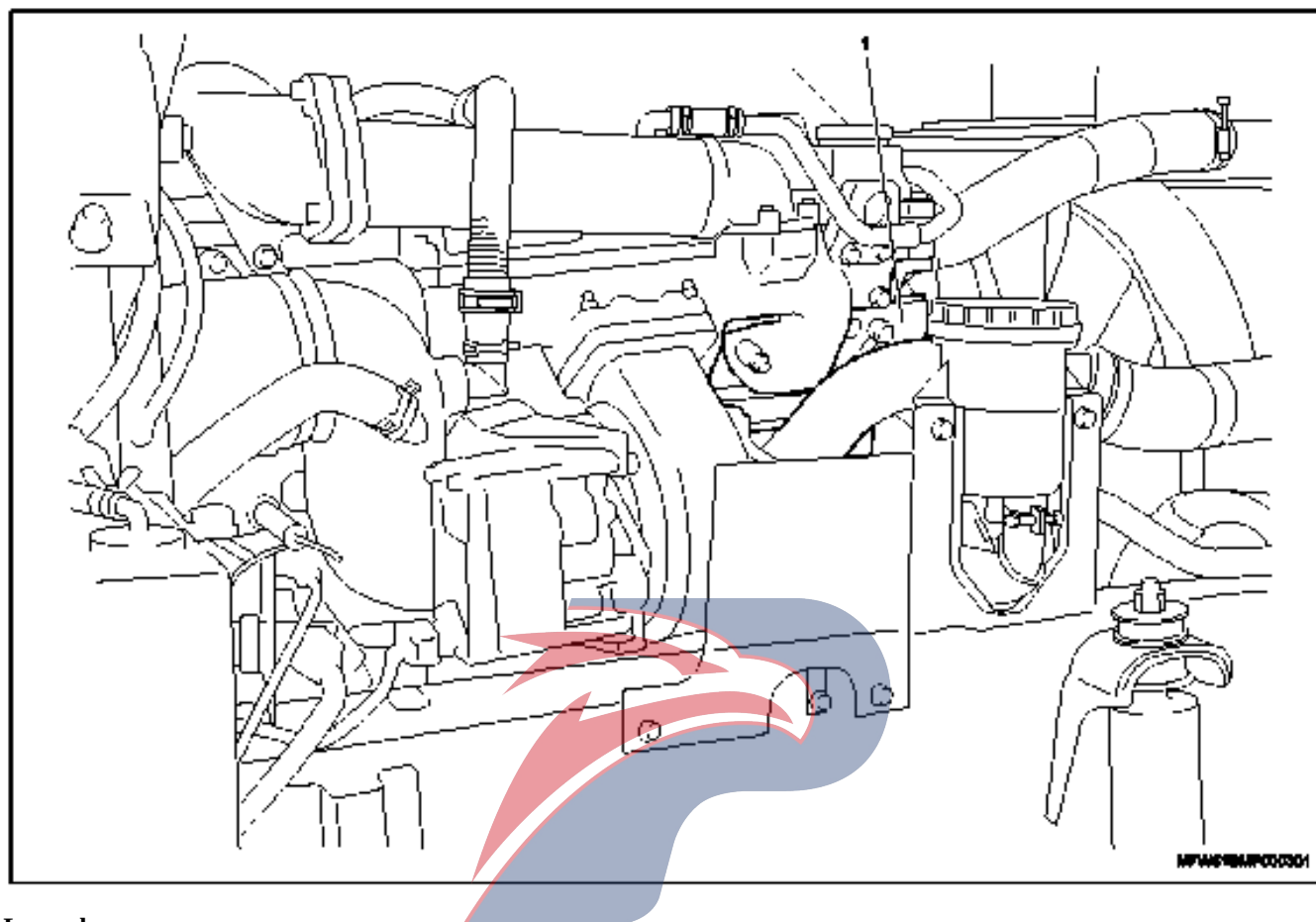
- Gently remove the crankcase and use the scale on the bag to measure the plastic gauge's extrusion width (gap).



<Example> Measuring the clearance between the crankshaft bearing and the crankshaft journal

- Clean the cylinder block, the crankshaft bearing mounting surface and the bearing, then install the cylinder block on the crankcase.
- Gently place the crankshaft on the cylinder block and turn for approximately 30 turns until it is fixed in place.
- The plastic line gap gauge is cut according to the crankshaft journal width and placed in parallel, while avoiding the journal oil hole.
- Gently place the crankcase on the cylinder block and apply molybdenum disulfide to the threaded portion of the fastening bolt and the bearing surface and tighten with the specified torque.

Type description



Legend

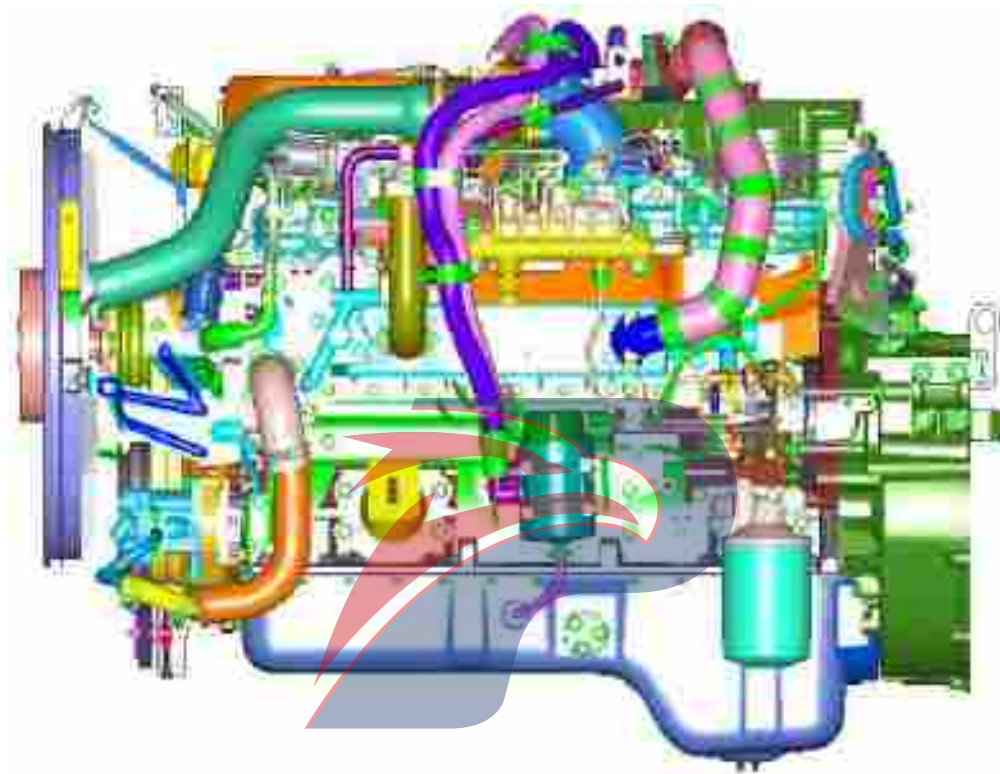
1. Engine number imprinting position (6HK1 -...)

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Description of function and operation

Engine structure

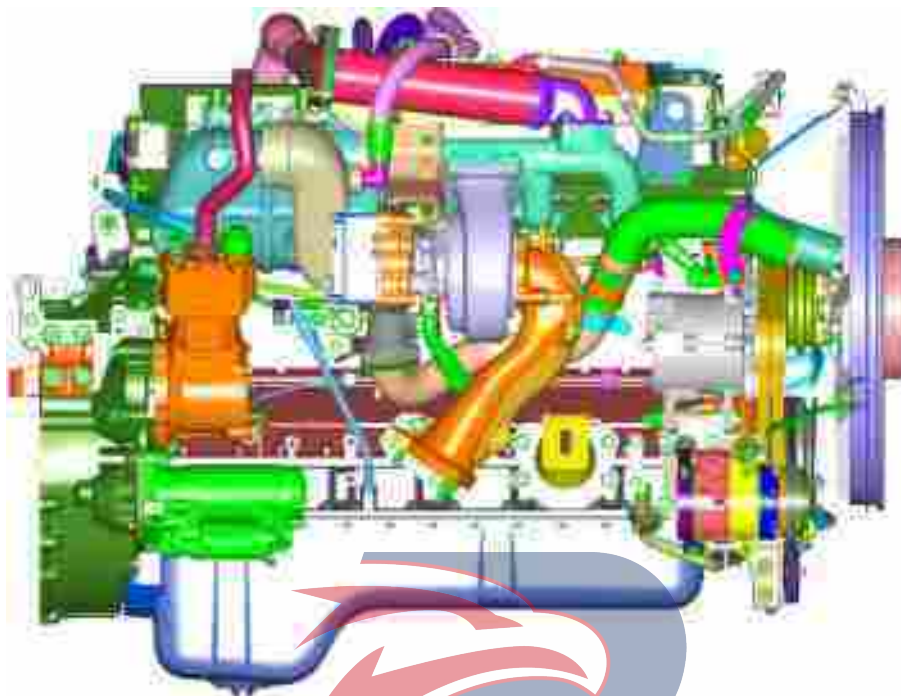
Left



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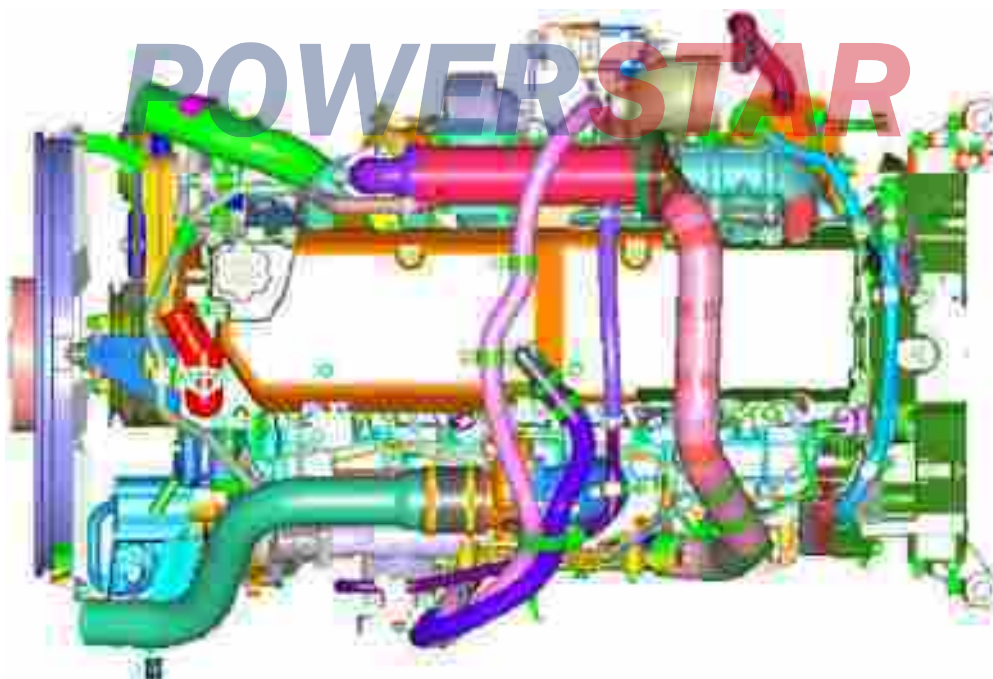
MFW61BMF000501

Right



MFW61BMF000601

Upper surface



MFW61BMF000401

Electronic control engine control

The control device can control the fuel injection volume, fuel injection timing, EGR, idling speed and injection intake/exhaust.

Cylinder block

The cylinder block is made of cast iron and has a rigid structure that is symmetrical about the center of the crankshaft, which is equal to the center-to-center distance of each hole. The bearing cover adopts a trapezoidal frame structure and is installed by an angle method.

Cylinder liner

The cylinder liner is aligned with the inside diameter of the cylinder block, and the left side of the cylinder liner is labeled with the rating for the optional cylinder liner.

Piston

The piston is made of aluminum alloy and the combustion chamber is round and reentrant.

Cylinder head

The cylinder heads are made of cast iron and each valve has four valves. The cylinder head bolts are mounted by the angle method.

Crankshaft

Using a soft nitriding process, the journal diameter is marked on the No. 1 counterweight.

EGR device

ECM executes control function according to the coolant temperature, engine speed and engine load and purges with EGR.

The main components include EGR valves, EGR coolers, exhaust brake valves and various sensors.

Connecting rod cover bolt

Connecting rod cover bolts are mounted by angle method.

Common rail device electronic fuel injection system

The common rail device electronic fuel injection system consists of an oil supply pump that sets a target pressure and supplies high pressure fuel, a common rail device that measures high pressure fuel, and an injector that reduces the amount of high pressure fuel and injects fuel. The controls are based on the signal. ECM controls the injection timing and injection volume under various

driving conditions.

Injector

The injector uses 7 orifice nozzles. The opening and closing of the injector head solenoid valve is used to adjust the injection timing and injection quantity.

Fuel filter with sediment separator

The filter is equipped with one precipitation separator to eliminate the water through the difference of diesel oil and water specific weight. It is also provided with an indicator to indicate when there is water accumulation.

Preheating system

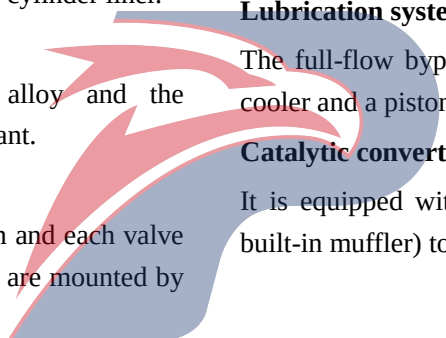
The preheat system uses the QOS II system. In addition, the controller has been integrated in the engine control module (ECM).

Lubrication system

The full-flow bypass unit filter uses a water-cooled oil cooler and a piston cooler injector.

Catalytic converter

It is equipped with oxidation catalysis converter (with built-in muffler) to reduce the exhaust gas emission.



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Functional Checklist

Valve clearance inspection and adjustment
Measurement of compression force
Judgment of the overhaul time



Functional check:**Valve clearance inspection and adjustment**

1. Remove the top cover.
2. Check the valve play.
 - Rotate the crankshaft forward and align the top dead center of the first cylinder.
 - Insert a 0.4 mm (0.0157 in) gauge into the gap between the rocker arm and the bridge to check the gap size and adjust the gap if necessary.

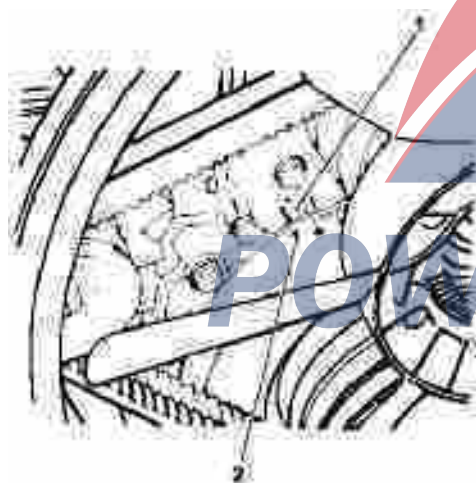
Valve play

Intake valve 0.4 mm (0.0157 inch)

Exhaust valve 0.4 mm (0.0157 inch)

Note:

Adjust in the cooling state.



MFW41BSH024901

Legend

1. Gear box cover mark
2. Crankshaft damper 0 degree mark
3. Valve clearance adjustment.

Note:

Adjust the height of the bridge correctly (in contact with both ends of the two valve shafts).

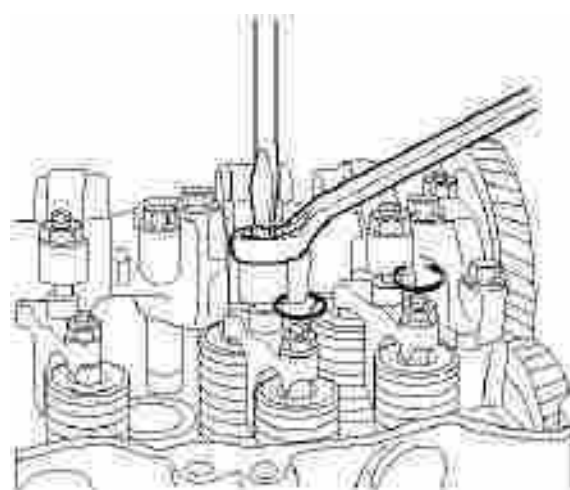
- Completely loosen the bridge and rocker adjustment screws.



BLW51BSH005501

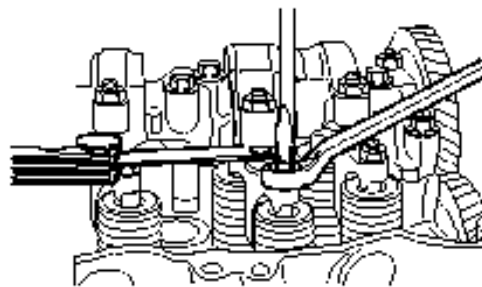
- Insert a 0.4 mm (0.0157 in) thickness gauge between the rocker tip (valve side) and the cover (attached to the top of the bridge), tighten the adjustment screw on the rocker arm, and secure arm adjustment nut on the rocker where it is difficult to insert and remove the gauge.

Tightening torque N m (kgf m/lb ft)
22. (2,2/16)



BLW51BSH005601

- After inserting the gauge, make sure that the tip of the adjusting screw and the edge of the valve shaft are in contact with each other. After gently tightening the bridge adjusting screw, the gauge is difficult to move.
- Then, make sure that the other side of the valve shaft does not move or tilt. If any movement or tilt occurs, slightly loosen the bridge adjustment screw and adjust the end of the valve shaft for proper contact. Valve bridge clearance: less than 0.1 mm (0.0040 inches).
- Once both sides of the valve shaft are properly adjusted and contacted, use a flat head (slotted screwdriver) to fix the bridge adjustment screw, and then tighten the bridge adjustment nut.



BLW51BSH005701

Tightening torque N m (kgf m/lb ft)
22 (2.2 / 16)

Note:

Must be precisely adjusted. If the bridge adjustment screw is not adjusted, there may be problems such as tilting of the bridge, depression, burning, cracking of the guide rod, and the like.

Tightening torque		N m (kgf m/lb ft)
Rocker arm adjusting screw/nut:		22 (2.2 / 16)
Bridge adjustment screw/nut:		22 (2.2 / 16)

Valve clearance adjustment table

Cylinder number		1		2		3		4		5		6	
Status	Valve arrangement	Intake	Exhaust	Intake	Exhaust	Intake	Exhaust	Intake	Exhaust	Intake	Exhaust	Intake	Exhaust
Need to adjust the valve	No. 1 cylinder is set at the top dead center of compression	○	○	○			○	○			○		
	No. 6 cylinder is set at the top dead center of compression				○	○			○	○		○	○

Injection order (1-5-3-6-2-4)

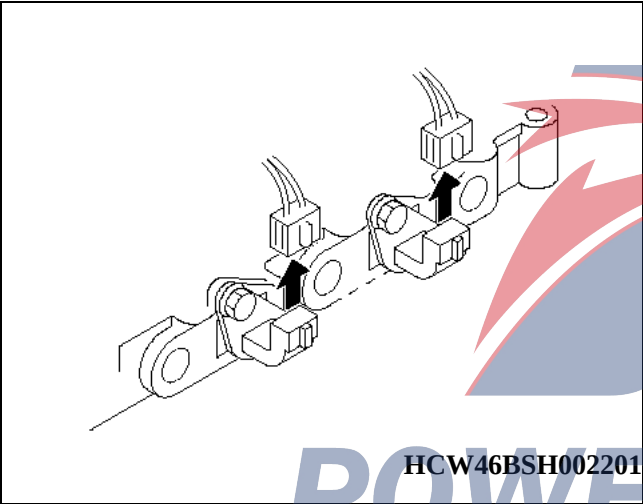
4. Install the top cover.

Measurement of compression force

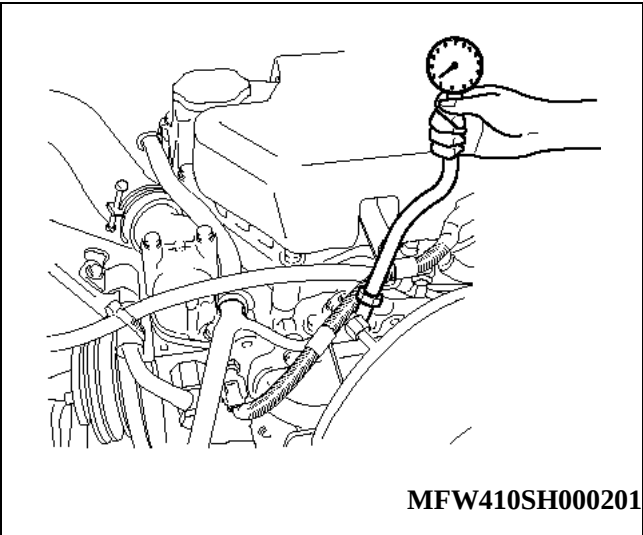
- Preheat the engine.
- Disconnect the negative battery terminal and remove all electric glow plugs.
- Remove the indicator harness connector attached to the lower top cover. (No fuel injection)

Note:

After removing the harness connector, the ECM will determine if there is a fault and record the DTC. Be sure to clear the ECM memory after completing the measurement. (For ECM memory clearing procedure, please refer to "Engine Control System")



- Connect the negative battery terminal.
- Rotate the starter to remove debris from the cylinder.
- Install special tools for pressure gauge fittings and pressure gauges.



Compression connector: 5-8840-2815-0

Compressor: 5-8840-2675-0

- Turn the starter and check the compression force.

Compression	MPa(kg / cm2 / lb / in2) / 200rpm
Standard value	3.2 (33 / 469)
Limit value	2.3 (23 / 327)
Pressure difference between cylinders	294 kPa (3.0)

- Each cylinder is measured in turn.

Note:

Use a fully charged battery to keep the engine above 200 rpm.

- Remove the special tool compressor.
- Disconnect the negative battery terminal.
- Connect the indicator harness connector to the lower top cover.
- Put on all glow plugs.

Tightening torque N • m (kg force • m/lb • feet)
20 (2.0 / 14)

- Connect the negative battery terminal.

Judgment of the overhaul time

Engine fuel consumption growth rate

Assume that the consumption rate of a new vehicle (the number of kilometers a litre of oil can travel) is 100%. When the consumption rate is less than 50%, the vehicle needs to be overhauled.

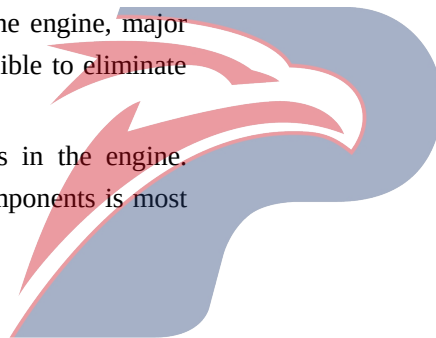
Fuel consumption rate growth rate

Assume that the consumption rate of the new car (km / l) is 100%. When the consumption rate is lower than 60%, the vehicle needs to be overhauled.

Abnormal noise in the engine

If abnormal noise is generated inside the engine, major repairs should be made as soon as possible to eliminate the malfunction.

There are many unusual noise sources in the engine. However, excessive wear of engine components is most likely to cause engine fires or debris.



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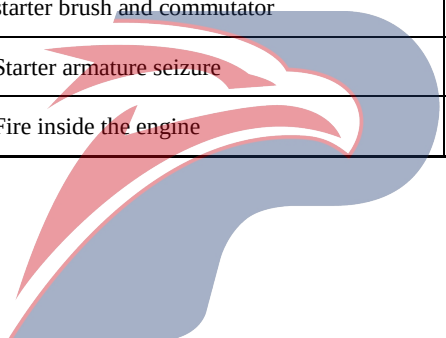
Failure symptoms list

- No engine revolution
- No increase in engine speed
- Produce a lot of black exhaust gas
- Produce a lot of white exhaust gas
- Engine knock
- Engine speed failure
- Poor performance of Booster



Phenomenon: There is no engine revolution

Phenomenon	Probable causes	Correction method
Starter without rotating speed	Bad battery relay	Replace
	Wiring failure	Connection or repair
	Insufficient battery capacity	Recharge or replace
	Starter stuck or worn/damaged	Repair or replace
	Internal function fault of starter	Repair after dismantling
The starter can rotate and cannot mesh the flywheel	Flywheel starter gear wear	Repair or replace
	Improper start switch adjustment	Readjust or repair
Starter pinion meshes with connecting rod gear but does not rotate	Battery capacity is too small	Charge or repair
	Insufficient connection pressure between the starter brush and commutator	Repair after dismantling
	Starter armature seizure	Repair after dismantling
	Fire inside the engine	Repair after dismantling



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Phenomenon: No increase in engine speed

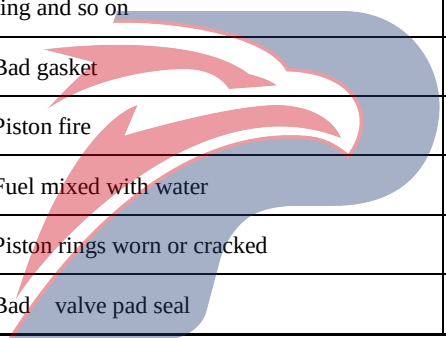
Phenomenon	Probable causes	Correction method
Fuel cannot reach the feed pump	Mix air in a fuel system	Vent
	Air enters the fuel pipe	Degas or replace
	Exhausted fuel	Oil charge
	Filter plugging at the fuel suction section	Clean or replace
	Fuel pipe blockage	Clean or replace
	Feed pump failure	Repair after dismantling
	Cold season = Fuel is not suitable for use in cold areas	Put in the right fuel
	Fuel filter plugged	Replace
	Engine control system failure	Diagnosis of engine control systems
Fuel reaches the feed pump	Jet pipe connection loose	Tighten each part
	Overflow valve is not sealed	Replace
	Provide internal pump failure	Replace
	Wiring failure or disconnection	Repair or replace
	Rotary sensor failure	Replace
	Engine control system failure	Diagnosis of engine control systems
Insufficient or unstable fuel injection	Mix air in a fuel system	Vent
	Fuel pipe blockage	Clean or replace
	Feed pump failure	Replace
	The injector nozzle stuck	Replace
	Wiring failure or disconnection	Repair or replace
	Fuel filter plugged	Replace
	Engine control system failure	Diagnosis of engine control systems

Phenomenon: Produce a lot of white exhaust gas

Phenomenon	Probable causes	Correction method
Injection timing malfunction	Engine control system failure	Diagnosis of engine control systems
Injector injection abnormal	Nozzle stuck	Replace
	Engine control system failure	Diagnosis of engine control systems
Bad compression	Valve clearance too big/too large	Adjust
	Stem stuck (valve open)	Repair after dismantling
	Bad valve spring	Replace
	Seat wear	Repair after dismantling
	Air leakage due to bad performance of piston ring and so on	Repair after dismantling
	Bad gasket	Repair after dismantling
	Piston fire	Repair after dismantling
Poor fuel	Fuel mixed with water	Replace
	Use fuel of poor quality	Replace
Air intake malfunction	Pipe plug	Repair or replace
	Air filter element is blocked	Clean or replace
Engine control system failed	Various sensors and other faults	Repair or replace
	Engine control system failure	Diagnosis of engine control systems
EGR failure	EGR valve stuck	Repair or replace
	Exhaust brake valve seizure	Repair or replace
	Engine control system failure	Diagnosis of engine control systems

Phenomenon: Produce a lot of white exhaust gas

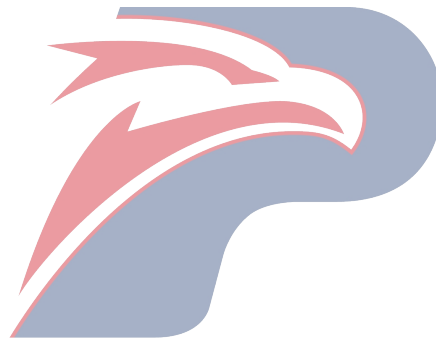
Phenomenon	Probable causes	Correction method
Injection timing malfunction	Engine control system failure	Diagnosis of engine control systems
Engine control system failed	Various sensors and other faults	Replace
	Control equipment failure	Replace
	Engine control system failure	Diagnosis of engine control systems
Bad compression	Valve clearance too big/too small	Adjust
	Stem stuck (valve open)	Repair after dismantling
	Bad valve spring	Replace
	Seat wear	Repair after dismantling
	Air leakage due to bad performance of piston ring and so on	Repair after dismantling
	Bad gasket	Repair after dismantling
	Piston fire	Repair after dismantling
Poor fuel	Fuel mixed with water	Replace
Excessive fuel consumption	Piston rings worn or cracked	Repair after dismantling
	Bad valve pad seal	Repair after dismantling



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Phenomenon: Engine knock

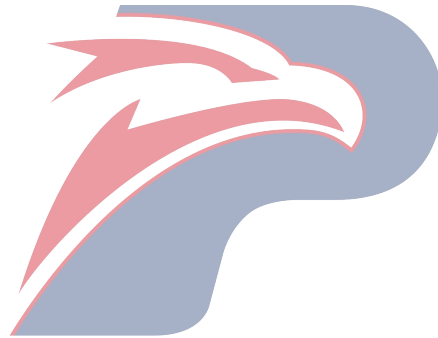
Phenomenon	Probable causes	Correction method
Injection timing malfunction	Engine control system failure	Diagnosis of engine control systems
Engine control system failed	Various sensors and other faults	Replace
	Control equipment failure	Replace
	Engine control system failure	Diagnosis of engine control systems
Fuel	Use fuel of poor quality	Replace
Air intake malfunction	Pipe plug	Repair or replace
	Engine control system failure	Diagnosis of engine control systems
Engine failure	Fragments in a cylinder	Repair or replace
	Pistons, bearings waiting for fire	Repair or replace



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Phenomenon: Engine speed difference

Phenomenon	Probable causes	Correction method
The number of revolutions cannot increase	Control equipment failure	Replace
	Engine control system failure	Diagnosis of engine control systems
Unstable revolution	Control equipment failure	Replace
	Engine control system failure	Diagnosis of engine control systems
	Fuel filter	Replace filter element
	Injector failure	Replace
	Fuel mixed with water	Replace
	Mix air in a fuel system	Vent
	Exhaust brake valve seizure	Repair or replace



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Phenomenon: bad charging

Phenomenon	Probable causes	Correction method
Can't charge	Broken wire, short circuit, poor connection	Repair or replace
	Generator internal fault	Repair after dismantling
Insufficient charging	Line short circuit, poor connection	Connection or repair
	Generator internal fault	Repair after dismantling
	Generator drive belt loose	Readjust or repair
	The battery is broken	Replace
Overcharged	Line short circuit	Connection or repair
	Generator internal fault	Repair after dismantling

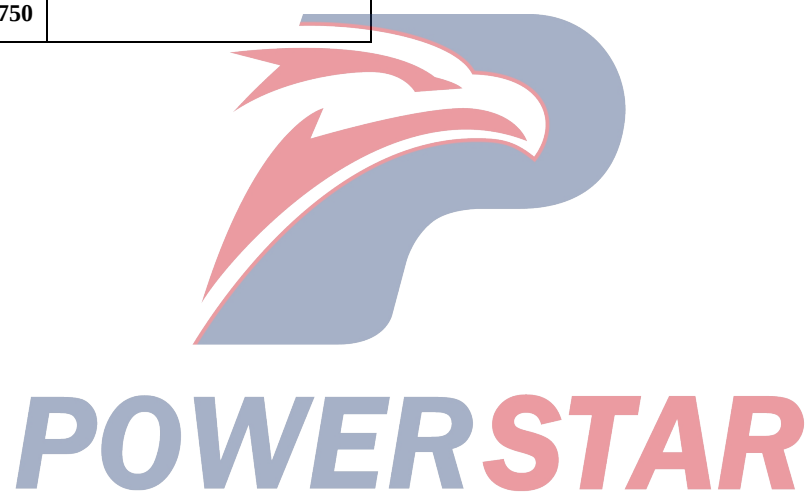
Specifications

Model		6HK1-TCN	6HK1-TCC	6HK1-TCS
Item				
Engine type		Water-cooled 4 cycles inline 6 cylinder 24 valve OHC IDI diesel engine		
Cylinder arrangement and number - inside diameter - stroke	mm(inch)	L6—Φ115(4.53)—125(4.92)		
Total displacement	L (in ³)	7.79(475.35)		
Compression ratio		17.5		
Compression	kPa (kg/cm ² /lb/in ²)/rpm	3240(33/469)/200		
Size (length / width / height)	mm(inch)	1281(50.43)/854(33.62)/946(37.24)	1322(52.05)/854(33.62) /1010(39.76)	
Dry weight (excluding air filter)	Kg (lb)	630(1389)		
Fuel injection timing	BTDC	0°		
Fuel injection sequence		1—5—3—6—2—4		
Jet pump type		EFI system (common rail type)		
Governor type		Electronic		
Timer type		Electronic		
Nozzle model		DLL—P		
Inlet and exhaust valve clearance (cold)	mm(inch)	0.4(0.016)		
Intake valve opening and closing timing	Open	17°(BTDC)		
Intake valve opening and closing timing	Close	42°(ABDC)		
Exhaust valve opening and closing timing	Open	48°(BBDC)		

Model		6HK1-TCN	6HK1-TCC	6HK1-TCS
Item				
Exhaust valve opening and closing timing	Close	14°(ATDC)		
Fuel filter type		Filter paper type		
Oil filter type		Total flow filter paper type		
Oil pump type		Gear type		
Oil cooler type (including drive axle EGR cooler)		Water-cooled type		
Engine oil quantity	Liter (British gallon)	Total capacity: 13.5 (2.97) and 19.5 (4.29); Oil trays: 11.0 (2.42) and 16.0 (3.52) With combination oil filter		
Cooling system		Water-cooled forced circulation		
Total cooling water capacity	Liter (British gallon)	29(6.38)	30(6.60)	
Pump type		Centrifugal gear type		
Incubator type		Wax pill type (with or without shaking valve (one for each))		
Air filter type		Cyclone filter paper		
type of battery	Type - Quantity	75D23R(MF)or 95E41 R(MF)or 115E41 R(MF)—2		
Generator capacity	V/A	24—50, 24—60, 24—90		
Starter	Volts·kW	24—5.0		
Idle speed	Rotate per minute	500—550		
Turbocharger type		HG9006J(Honeywell)		
Intercooler type		Aluminum tube and end plate type		
Incubator valve opening temperature	℃(°F)	82(180)		
Incubator fully open temperature/head	℃/mm (°F /inch)	95/8(203/0.315)		
EGR		equipment		

Special tools

Graphic representation	Tool number/description
 5884028150	5-8840-2815-0 Pressure Gauge Connector
 5884026750	5-8840-2675-0 compressed table



Engine assembly

Disassembly

Note:

Precautions for removing and replacing the engine:

- Ensure to stop the wheel rotating.
- Always use an engine belt that can withstand the weight of the engine.
- Do not stand under the engine when the engine is suspended by a strap.
- Do not place your hand where it may get stuck.

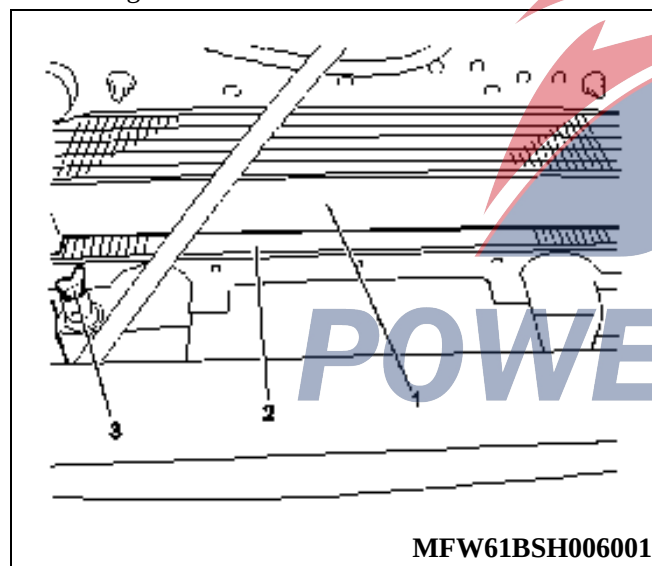
1. Remove the negative terminal of the battery cable.

2. Remove the engine and transmission cover.

3. Remove the transmission.

- Refer to "disassembly and replacement of transmission".

4. Discharge coolant.



Legend

1. Intercooler

2. Radiator

3. Release plug

5. Disconnect the EGR valve air duct.

6. Remove the harness connector.

- Disconnect switches, sensors and terminals for each device.

7. Disconnect the solenoid valve tubing (the connection of the cylinder to the solenoid valve).

8. Disconnect the exhaust brake air line and air hose connection.

9. Remove the cab rear member.

- Remove the relay and solenoid valve connector attached to the cab rear section and remove the cab rear section.

10. Disconnect the flexible joints of the air compressor governor tube and air duct.

11. Disconnect starter connector and terminals. Remove the harness bracket

12. Remove the motor developing circuit.

13. Remove the air-conditioning compressor connecting bolts, remove the engine-side piping (do not damage it), and secure it to the side of the chassis with a belt or other tool.

14. Remove the power steering pump.

- Remove the power steering pump bracket.

- Remove the power steering pump and use a belt or other tool to hang it on the side of the vehicle.

15. Remove the intake pipe (from air filter to turbocharger).

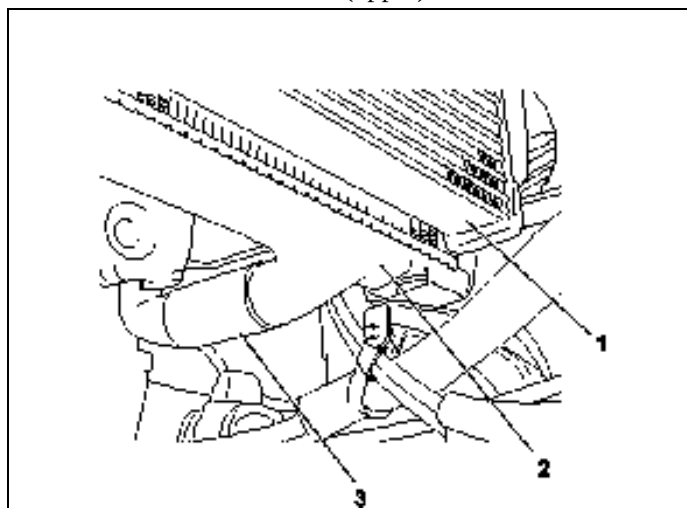
16. Remove the oil level meter pipe.

17. Remove the thermal protector from the turbocharger rubber hose and remove the intake pipe (from turbocharger to intercooler).

18. When lifting the engine, remove the power steering tank connection bolt and loosen it to avoid obstacles.

19. Remove the radiator hose (upper side).

20. Remove the radiator hose (upper).



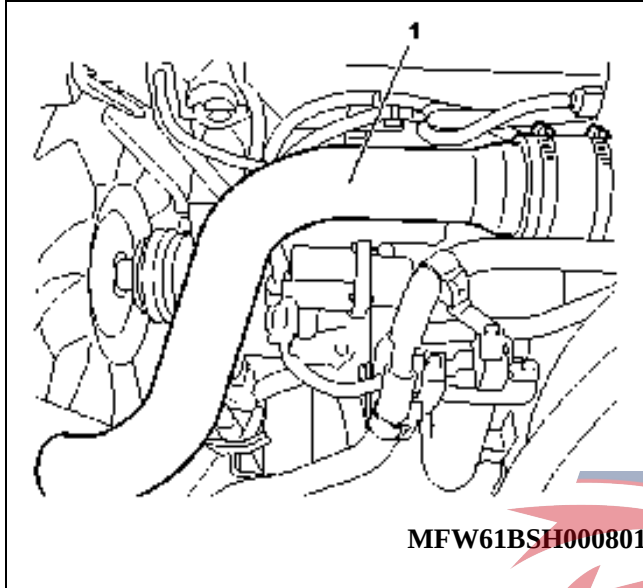
Legend

1. Intercooler

2. Radiator

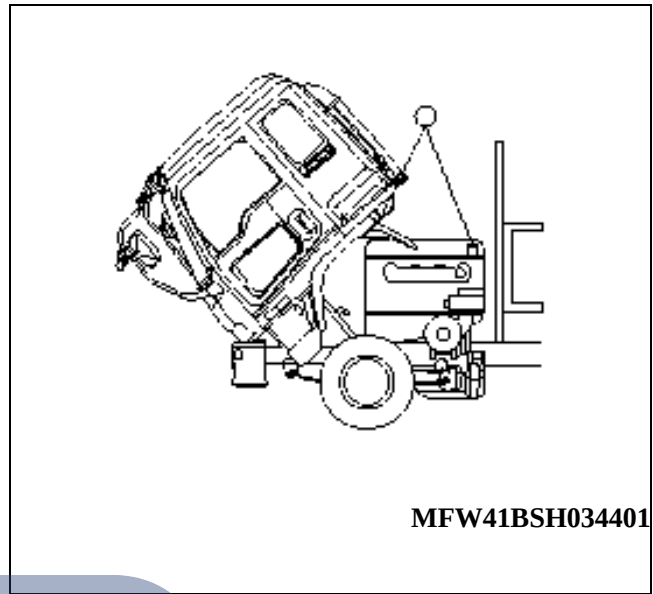
3. Radiator hose (lower side)

21. Remove the bell holder.
22. Remove the cooling fan.
23. Remove the intake pipe (1) (intake cooler to intake throttle).

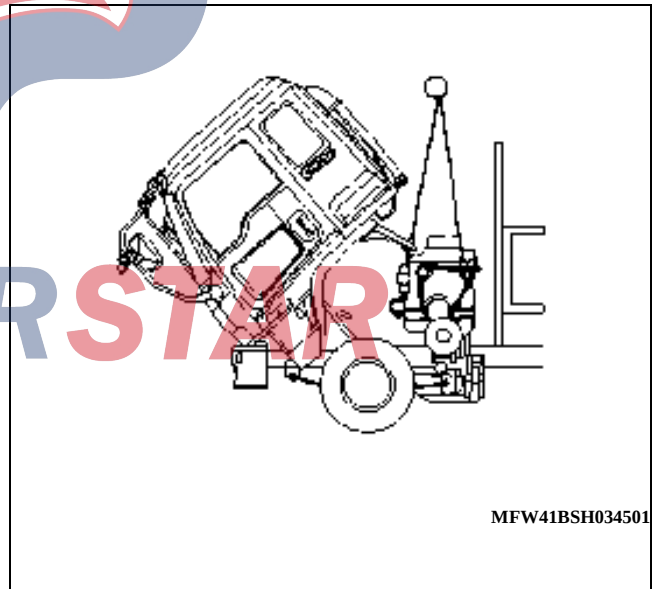


24. Remove the bracket, harness connector and grounding wire on the left side of the engine (on the left side of the vehicle).
25. Remove the heater hose.
26. Remove the fuel filter.
 - Disconnect the sediment separator connector.
 - Remove the fuel hose.
 - Remove the fuel filter assembly.
27. Remove the engine frame.
 - Remove the engine before lifting the engine frame.
 - Remove the mounting bolts from the engine frame.
28. Remove engine components.

- Slowly operate the crane to raise the engine so that it crosses the top of the chassis frame.



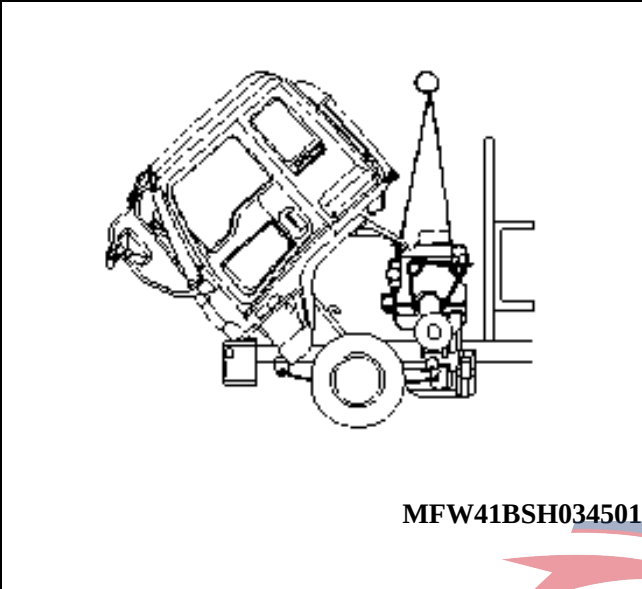
- Rotate the engine 90° from the top of the chassis frame and remove the engine components.



Reassemble

1. Install engine components.
 - Install slings on engine hooks and cranes, then install engines.

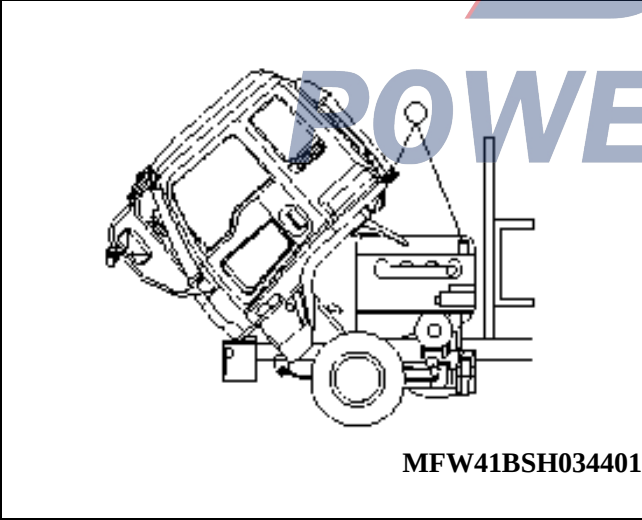
- Operate the crane to move the engine to the installation position.



- Rotate the engine 90° and slowly lower it to the engine mounting position.

Caution:

Do not stand under the engine when the engine is suspended by a strap.



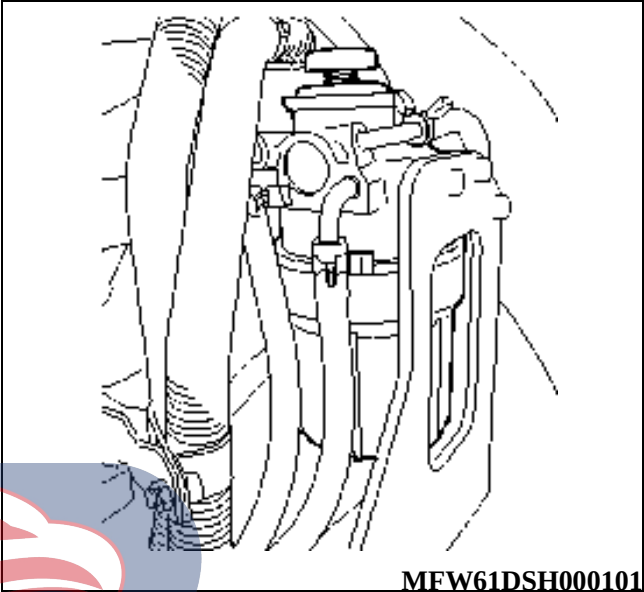
2. Install engine bracket.

- After the connecting nut of engine bracket is fully tightened, release the crane.
- Tighten the front and rear engine brackets according to the specified torque.

Tightening torque		N m (kgf m/lb ft)
Front side:		83 (8.5 / 61)
Rear side:		216 (22 / 159)

3. Install the fuel filter.

- Install the fuel filter assembly.
- Install fuel hose.
- Reconnect the precipitation separator connector.

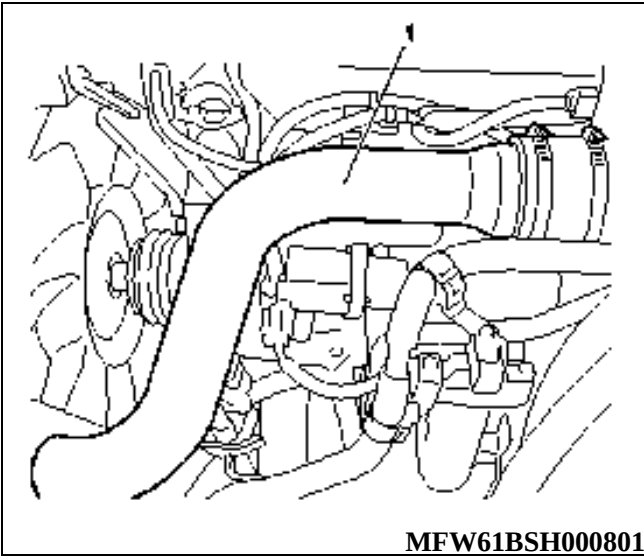


4. Install heater hoses.

- 5. Install the harness on the left side of the engine (on the left side of the vehicle) and connect the connector.
- 6. Install intake pipe (1) (from intake cooler to intake throttle).

- Tighten the clamps.

Tightening torque	N m (kgf m/lb ft)
6 (0.6 / 52)	

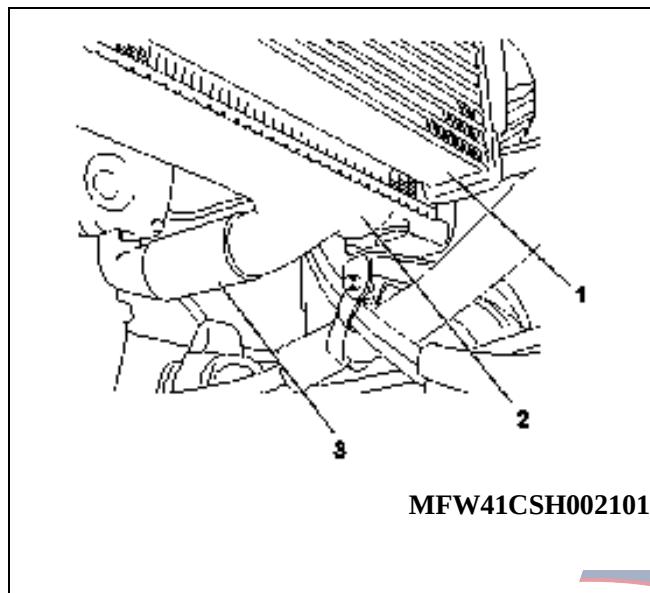


7. Put on a cooling fan.

8. Install the horn bracket.

9. Install the radiator hose (upper side).

10. Install the radiator hose (upper).



Legend

1. Intercooler

2. Radiator

3. Radiator hose (lower side)

11. Install the power steering tank and tighten it to the specified torque.

12. Install intake manifold (from turbocharger to intercooler) and install turbocharger hose thermal protector.

13. Put on the oil level gauge conduit.

14. Install the air intake pipe (from air filter to turbocharger).

- Tighten the clamps.

Tightening torque	N m (kgf m/lb ft)
6 (0.6 / 52)	

15. Install power steering pump.

- Install the power steering pump and tighten the bolts according to the specified torque.
- Install the power steering pump bracket.

16. Install the air conditioning compressor.

- Adjust the adjusting bolt with a force of approximately 98 N (10 kgf / 72 lb ft). The deviation between the belt between crankshaft pulley and idler is approximately 10 mm (0.3937 in). After adjustment, tighten the idler

center nut.

17. Connect the alternator wiring again.

18. Connect the starter circuit and install the C terminal bracket.

19. Connect the air compressor governor tube and the flexible joint section of the air duct.

20. Install the cab rear member and tighten to the specified torque.

- Install the cab rear member and connect the relay and solenoid connector to the cab rear member.

Tightening torque	N m (kgf m/lb ft)
103 (10.5 / 76)	

21. Connect the exhaust brake air duct to the air hose connection.

22. Connect the solenoid valve duct connection (gas cylinder and solenoid valve connection).

23. Install the harness connector.

- Connect terminals to switches, sensors, and devices.

24. Connect the EGR valve air pipe.

25. Pour into the cooling water.

26. Install the transmission assembly.

- Refer to "disassembly and replacement of transmission".

27. Install engine and transmission cover.

28. Install the negative terminal of the battery cable.

Note:

When the engine is loaded, please check the following:

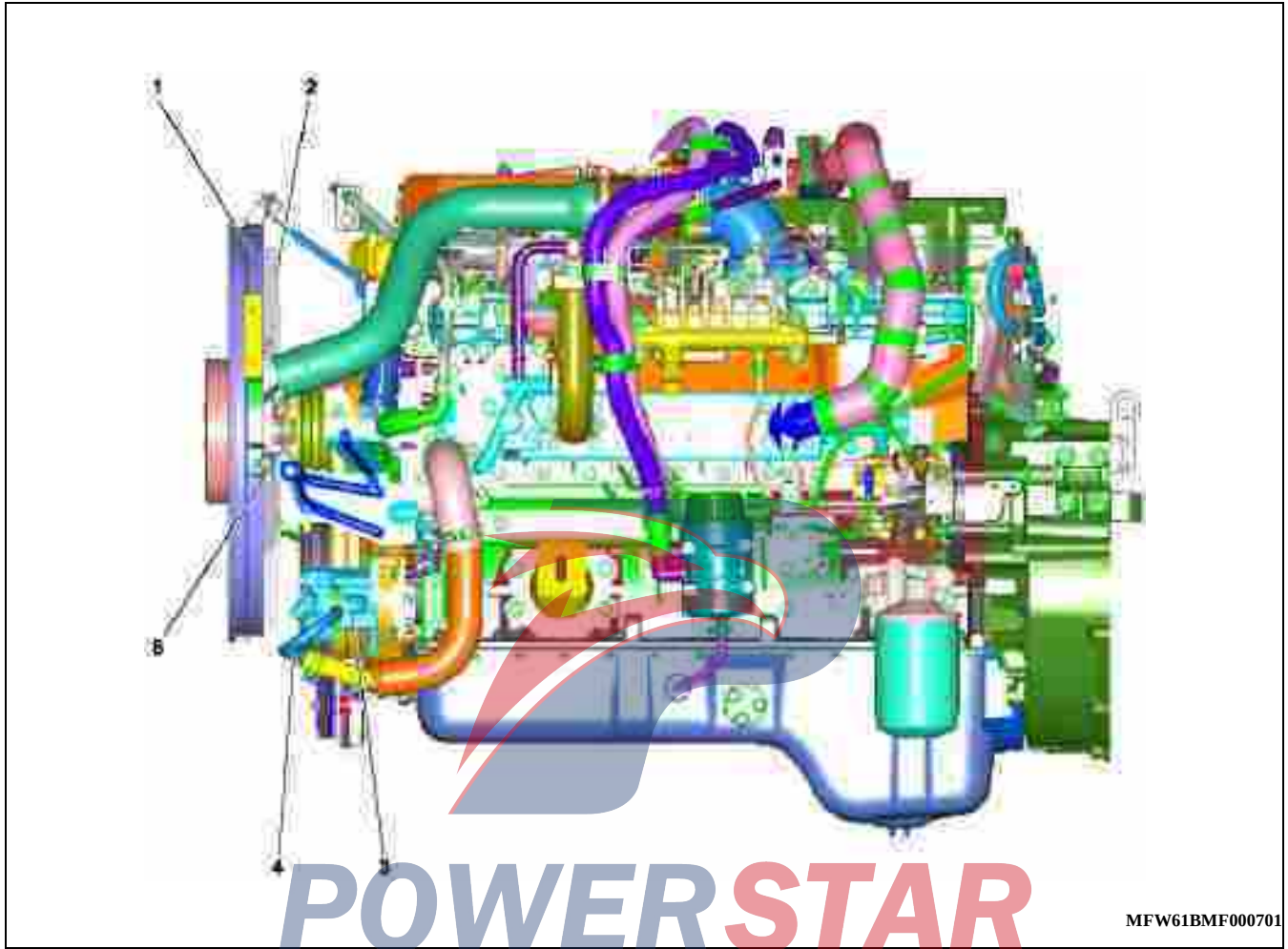
- Check engine oil quantity and cooling water quantity.
- The fuel system is completely degassed.
- Check the parking brake lever for engagement.
- Remove rags and other objects that may be caught by the engine and wipe off any fuel, oil, or other dirt on the engine.
- Make sure the DTC indicator is off.
- Make sure the gear lever is in neutral and tell other people to start the engine around you before starting the engine.
- Make sure that no part of the engine leaks or leaks.

- Make sure there are no fuel leaks, leaks or leaks at all pipe connections.
- Make sure that no abnormal sound is emitted from each part.
- It is believed that there is no exhaust gas leakage and the exhaust gas is colorless.
- After stopping the engine, check the oil level and cooling water to ensure that they are within the specified levels.
- Through test drive, there is no problem.



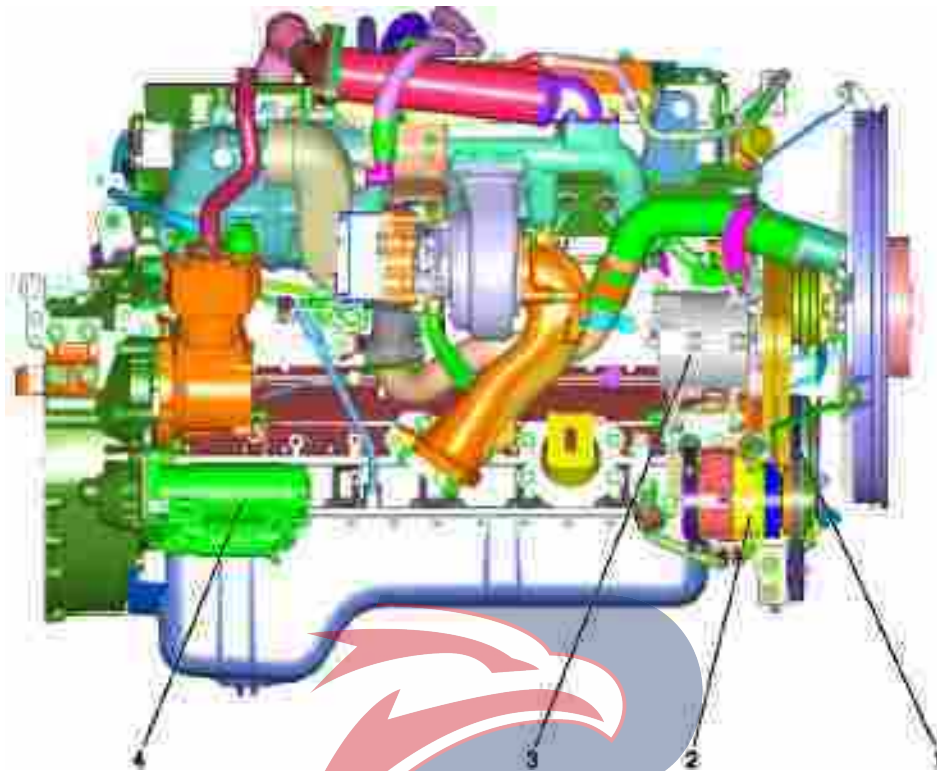
Engine accessories

Part



Legend

- | | |
|------------------------|------------------------|
| 1. Fan cover assembly | 4. Power steering tube |
| 2. Cooling fan | 5. Fan drive device |
| 3. Power steering pump | |
-



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Legend

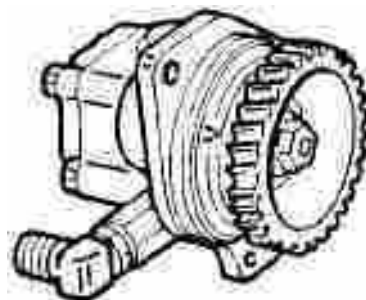
- | | |
|--------------------------------|------------------|
| 1. V-belt | 3. Generator |
| 2. Air conditioning compressor | 4. Starter motor |

Tear down

1. Remove the fan cover assembly.
2. Remove the cooling fan.
3. Remove the fan drive.
4. Remove the V-belt (fan, air conditioning compressor).
5. Remove the air conditioning compressor.
6. Disconnect the generator.
7. Remove the power steering hose.
8. Remove the power steering pump.

Note:

Choke the pump inside to prevent the chipping ingress.



MFW41BSH001501

9. Disconnect the starter.

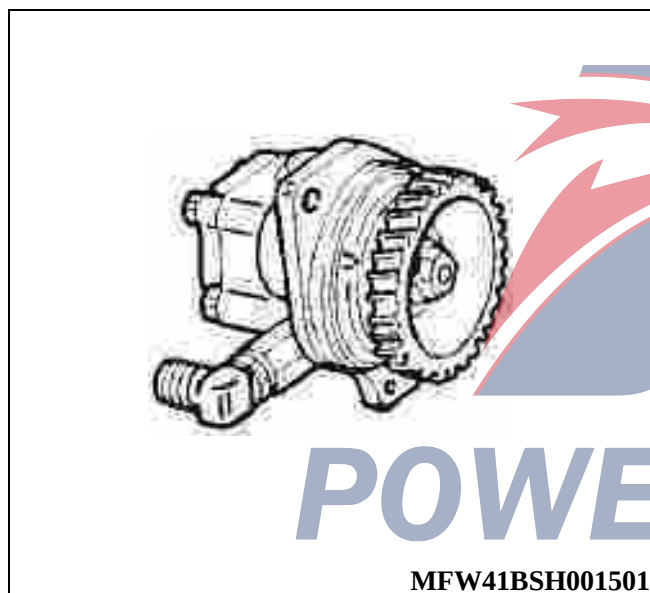
Reassemble

1. Install starter motor.

Tightening torque	N m (kgf m/lb ft)
82 (8.4 / 61)	

2. Install power steering pump.

Tightening torque	N m (kgf m/lb ft)
50 (5.1 / 37)	



3. Install power steering tubing.

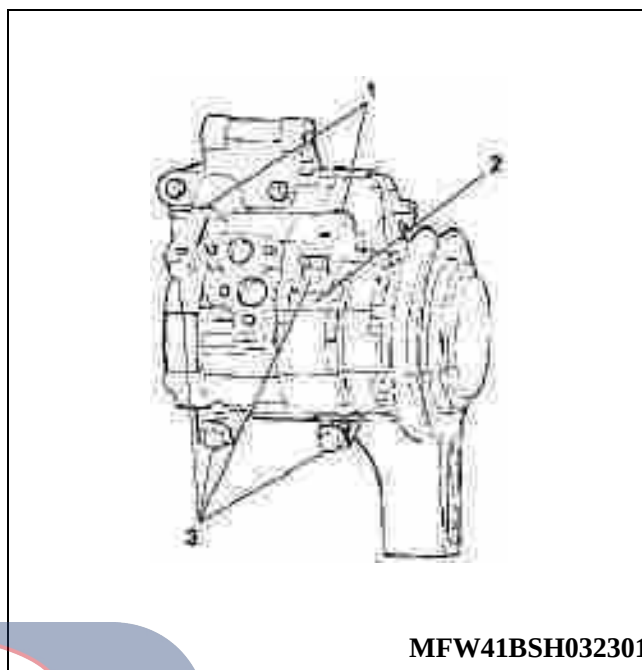
4. Install a generator.

5. Install the air conditioning compressor (2) and tighten the bolts (3).

Note:

Do not forget to insert the spacer (1) between the bracket and the air conditioner compressor.

Tightening torque	N m (kgf m/lb ft)
40 (4.1 / 30)	



6. Adjust the tension of the V-belt.

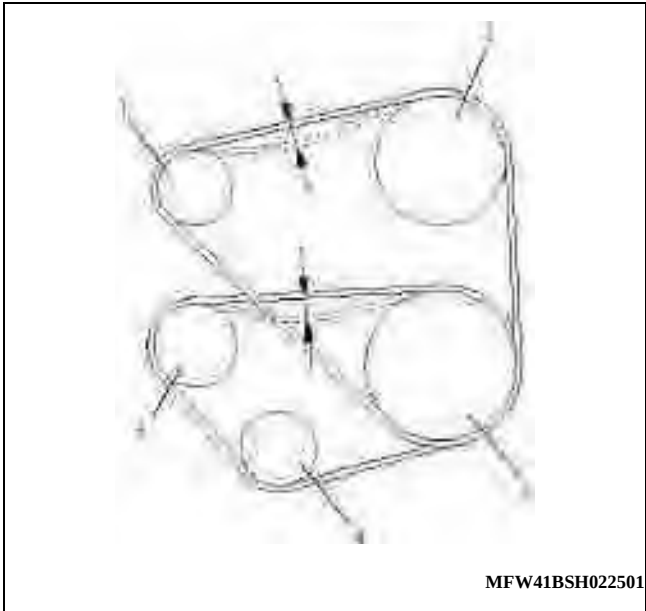
- The joint bolts at the top / bottom of the motor are loosened to adjust the adjusting bolts so that the deflection of the middle part of the belt is within the standard range with the force of about 98 cattle (10 kilogram force /72 pound force).
- If you need to adjust the air conditioning belt, loosen the idler nut and adjust the adjustment bolts.

Note:

After completing the adjustment, fully tighten the bolts.

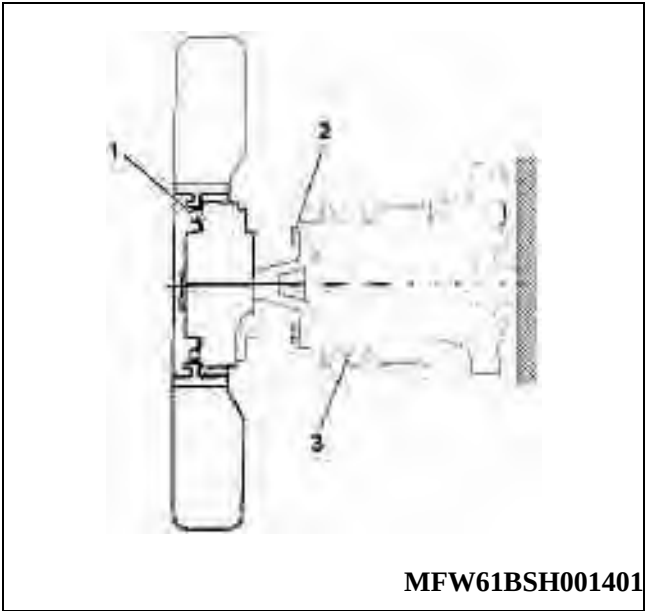
V- Belt bending amount: 10-15 mm

(0.3937—0.5906 inch)



Legend

- 1. Generator
- 2. Water pump
- 3. Crankshaft
- 4. Idler wheel
- 5. Air conditioning compressor



Legend

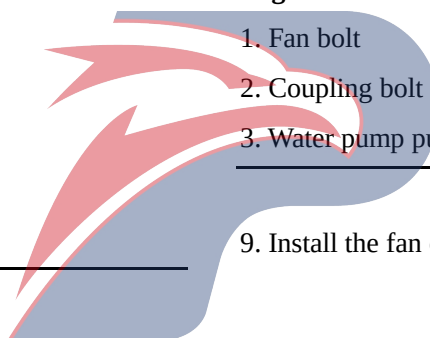
- 1. Fan bolt
- 2. Coupling bolt
- 3. Water pump pulley

7. Remove the harness connector.

Tightening torque	N m (kgf m/lb ft)
49 (5.0 / 36)	

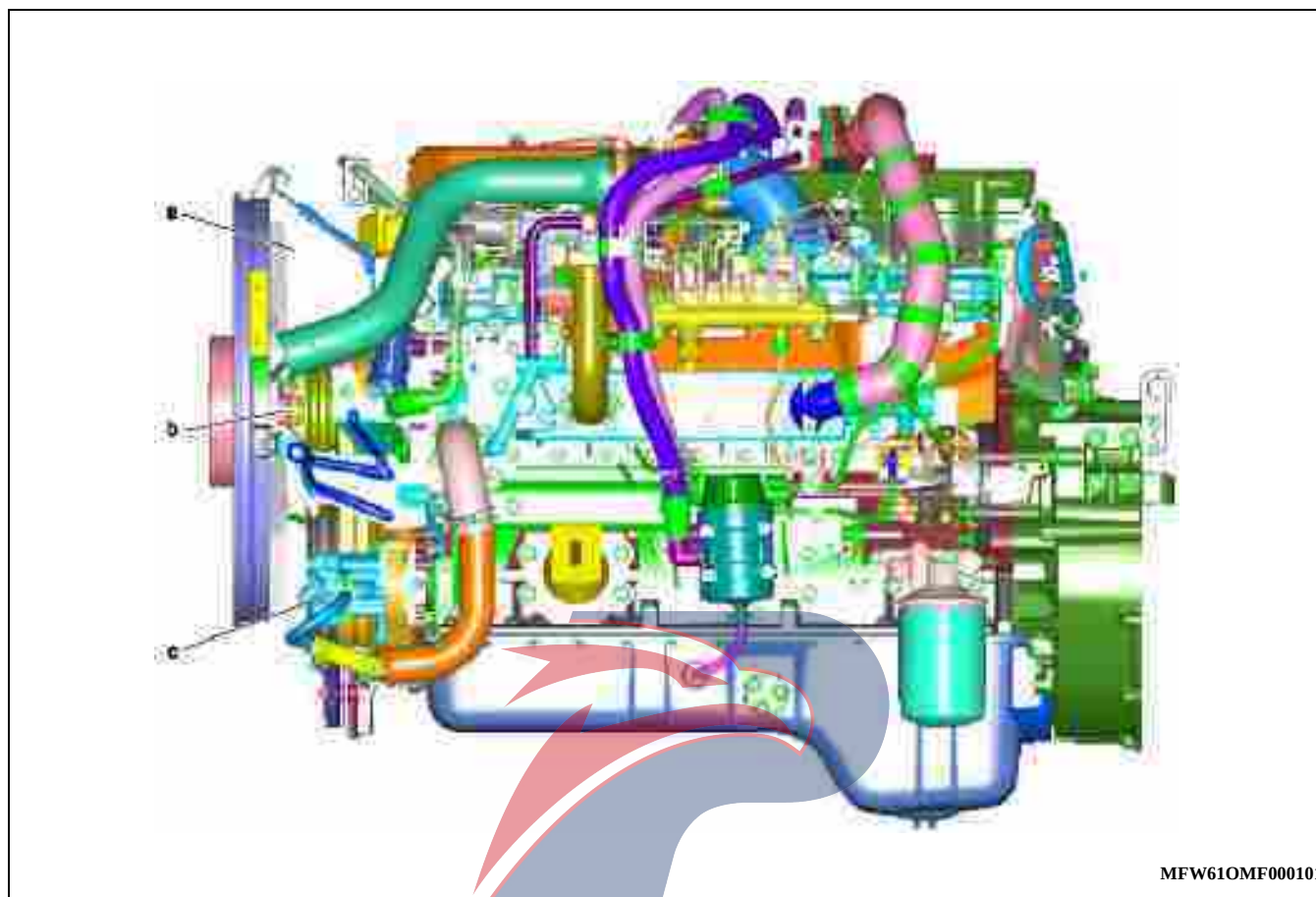
8. Put on a cooling fan.

Tightening torque	N m (kgf m/lb ft)
20 (2.0 / 14)	



POWERSTAR

Tightening torque table



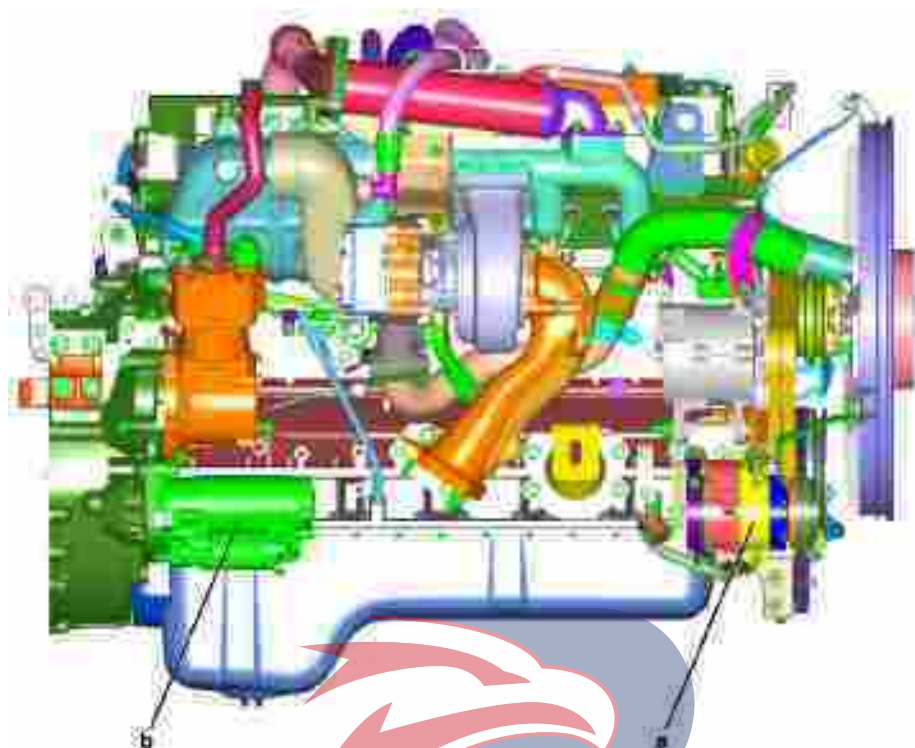
Legend

(a) 20 N m (2.0 kgf m/14 lb.ft)

(c) 50N m (5.1 kgf m/37 lb.ft)

(b) 49 N m (5.0 kgf m/36 lb.ft)

POWERSTAR



MFW61OMF000201

Legend

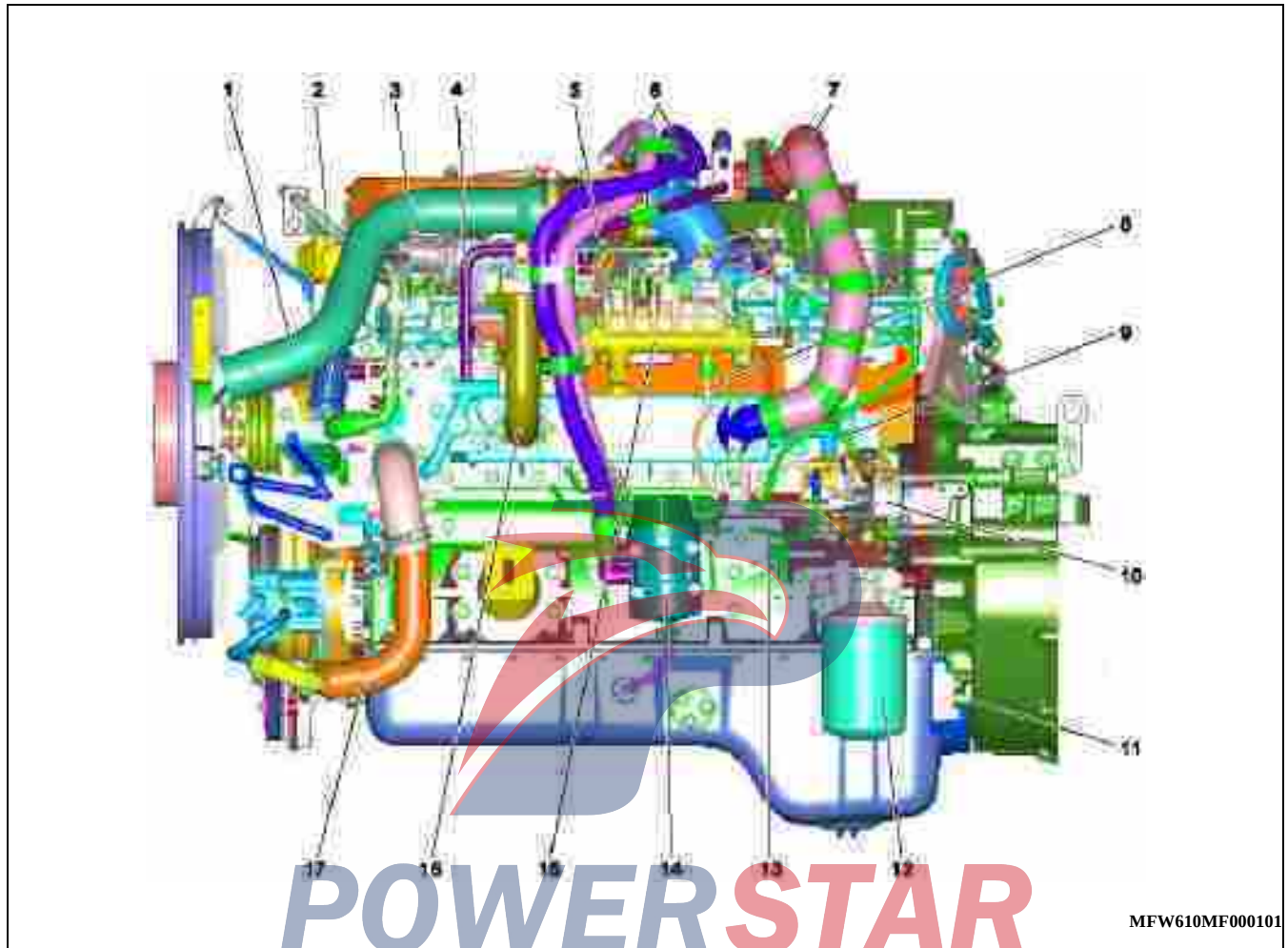
(a) 40 N m (4.1 kgf m / 30 lb. ft)

(b) 82 N m (8.4 kgf m/61 lb. ft)

POWERSTAR

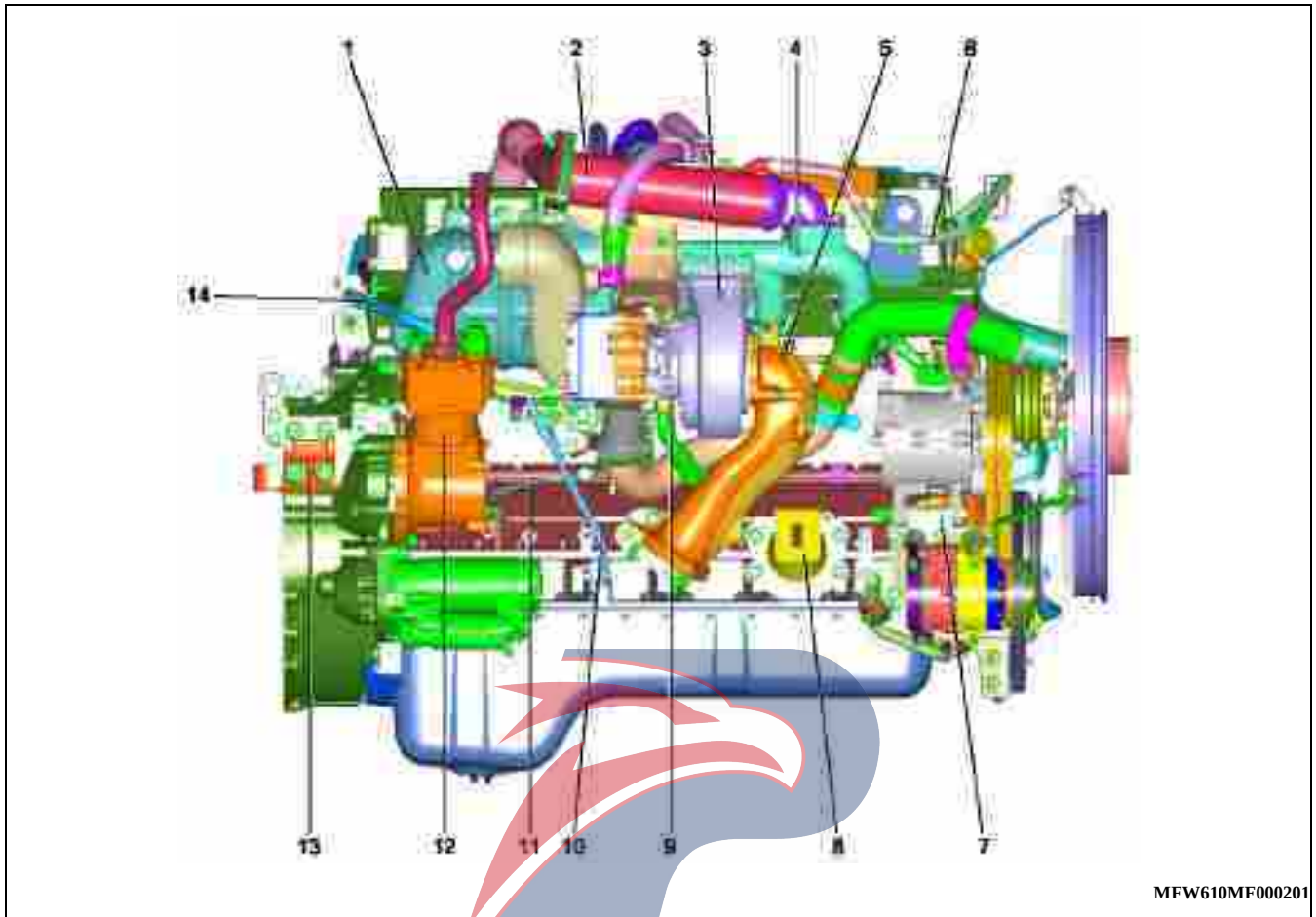
Engine exterior parts

Part



Legend

- | | |
|---------------------------|--------------------------------|
| 1. Intake pipe | 10. Feed pump |
| 2. EGR cooler feed pipe | 11. Crankshaft position sensor |
| 3. EGR valve | 12. Oil filter |
| 4. EGR cooler return pipe | 13. Fuel suction pipe |
| 5. Spray tube | 14. Oil separator |
| 6. Oil separator hose | 15. Common rail device |
| 7. EGR pipe | 16. EGR valve tube |
| 8. Pressure feed pipe | 17. Water pipe |
| 9. Drainage pipe | |
-

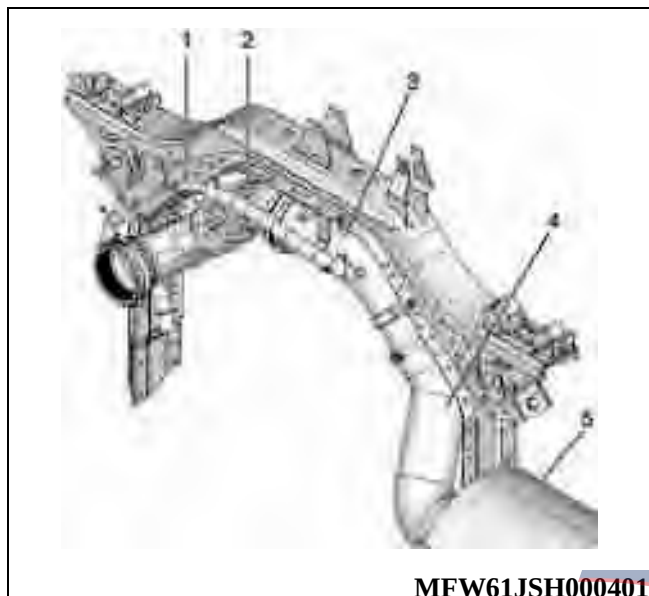


Legend

- | | |
|-------------------------|----------------------------------|
| 1. heat protector | 8. Ft engine supporting feet |
| 2. EGR cooler | 9. Return pipe |
| 3. Turbocharger | 10. Catheter |
| 4. Exhaust manifold | 11. A/C compressor oil feed pipe |
| 5. Turbine joint | 12. Air compressor |
| 6. EGR cooler feed pipe | 13. Rr engine feet |
| 7. Generator bracket | 14. Air duct |

Tear down

1. Remove the suction and intake pipes from the vehicle.



Legend

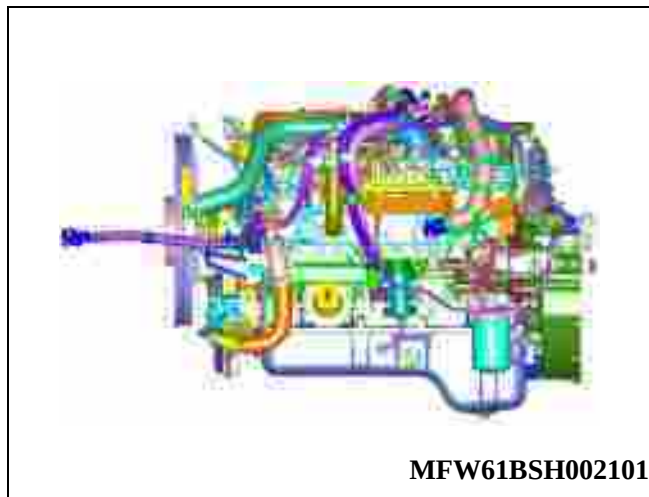
1. Straw
2. Intake pipe
3. Intake pipe
4. Intake pipe
5. Air filter

2. Remove the engine harness.

- Remove the harness for each sensor and electrical component.

Note:

Before removal, the position of the clamp should be clear (by painting, etc.).

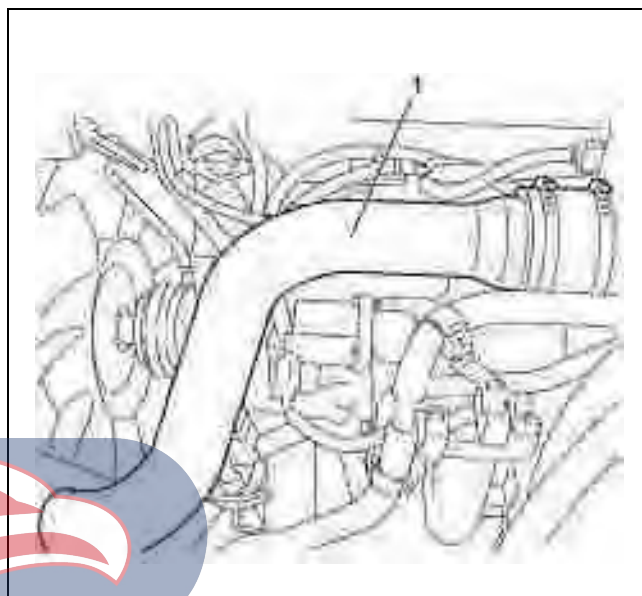


3. Remove the intake pipe.

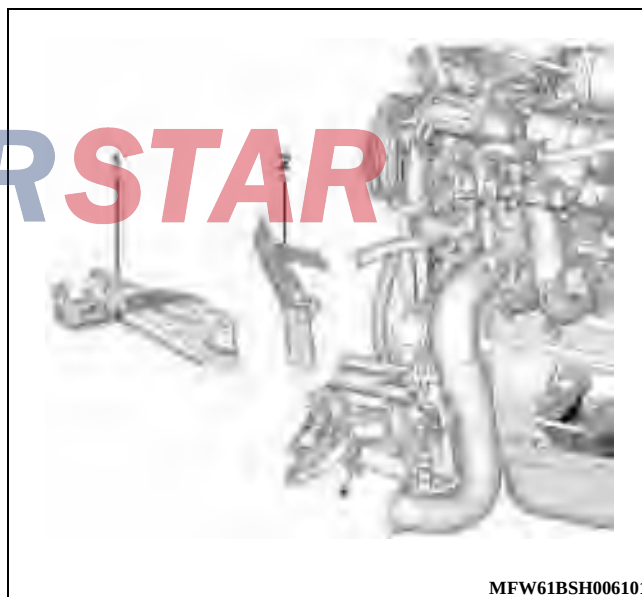
- Loosen the hose clamp, remove the nut from the lower part, and remove the intake pipe (1).

Note:

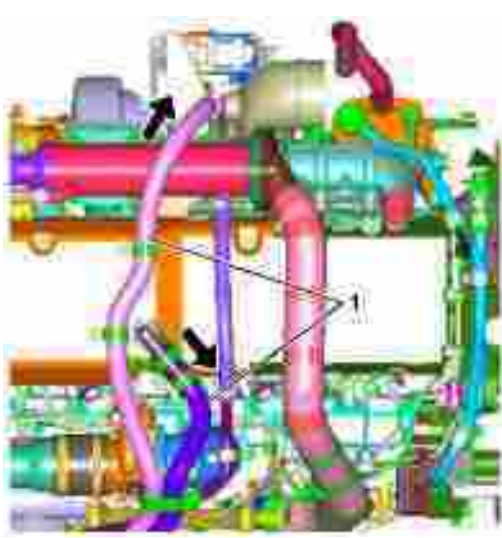
It should be sealed after removal to prevent debris from entering.



4. At the same time, remove the left fan guide (1) and engine control cable bracket (2).

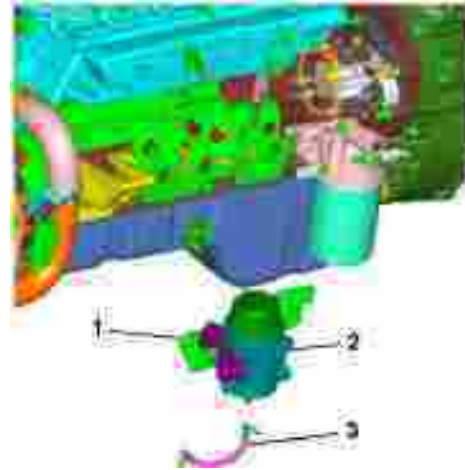


5. Remove the oil hose (1).



MFW61BSH005601

6. Keep the bracket connected while removing the oil separator.



MFW61BSH003601

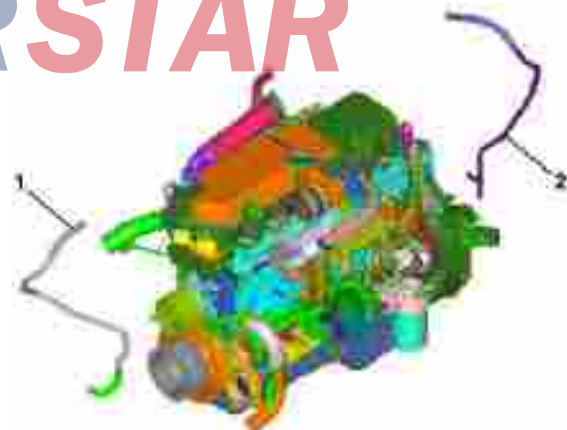
Legend

- 1. Bracket
- 2. Oil separator
- 3. Oil drain hose



MFW61BSH005701

7. Remove the water pipe (EGR cooler).

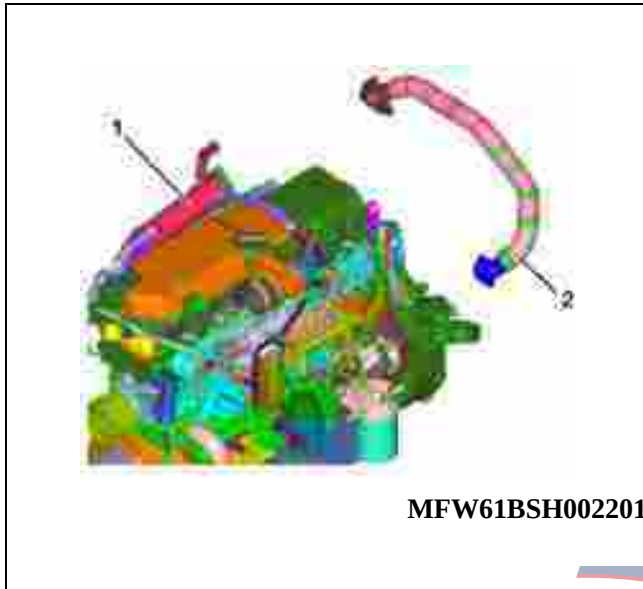


MFW61BSH002301

Legend

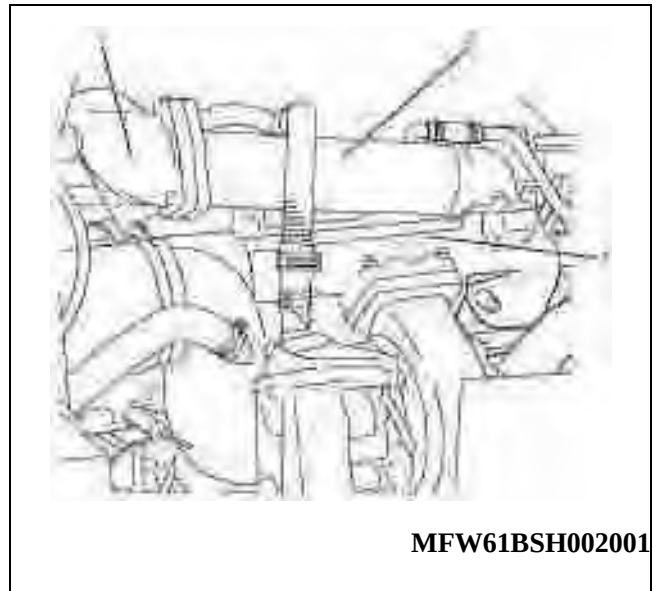
- 1. EGR cooler feed pipe
- 2. EGR cooler return pipe

8. Remove the EGR pipe.



Legend

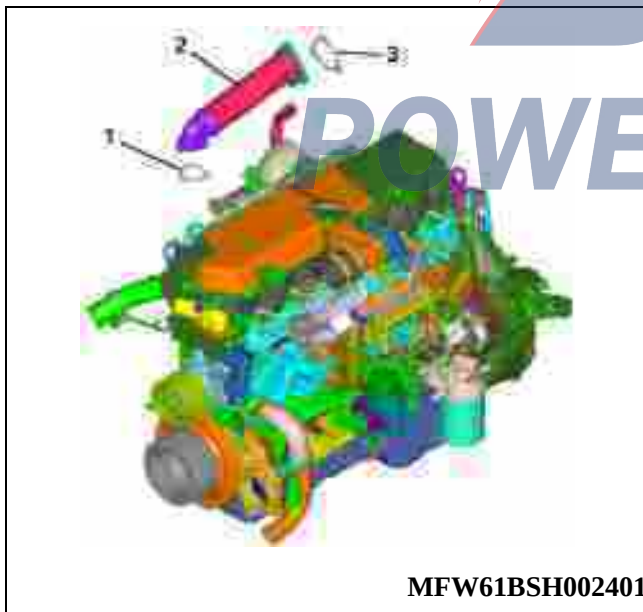
- 1. EGR cooler
- 2. EGR pipe



Legend

- 1. EGR pipe
- 2. EGR cooler
- 3. Exhaust manifold

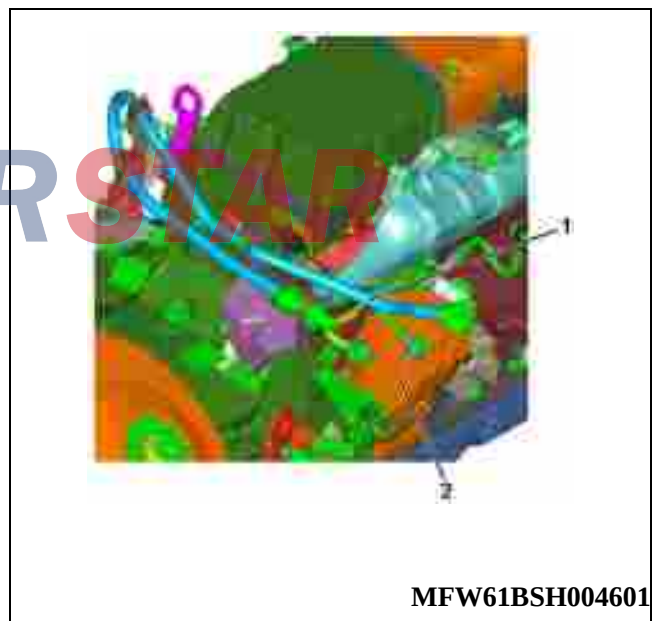
9. Remove the EGR cooler.



Legend

- 1. Gasket
- 2. EGR cooler
- 3. Gasket

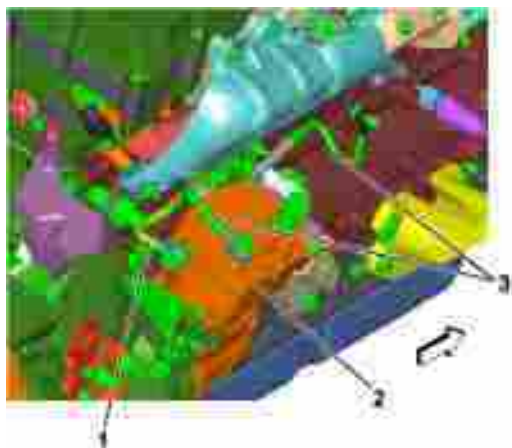
10. Remove the air duct (air compressor).



Legend

- 1. Feeding tube
- 2. Air compressor

11. Remove the air compressor hose (1).



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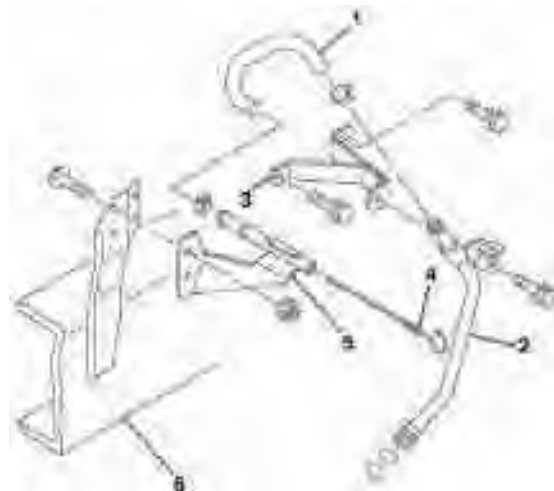
Legend

Water pipe (inlet water)

2. Compressor

3. Water pipe (return to turbine)

13. Remove the measuring tube.



MFW61BSH005301

Legend

1. Rubber tube

2. Catheter (engine side)

3. Bracket

4. Liquid level meter

5. Conduit (frame side)

6. Frame

12. Remove the air compressor.

- Please do not forget to take off the oil inlet pipe.

Note:

After removal, the interior of the air compressor should be sealed to prevent debris from entering.



MFW61BSH005501

14. Remove the turbine return line.

15. Remove the turbine intake pipe (cylinder block side).

16. Remove the turbine inlet (air compressor side).

17. Remove the turbine return pipe (cylinder block side).

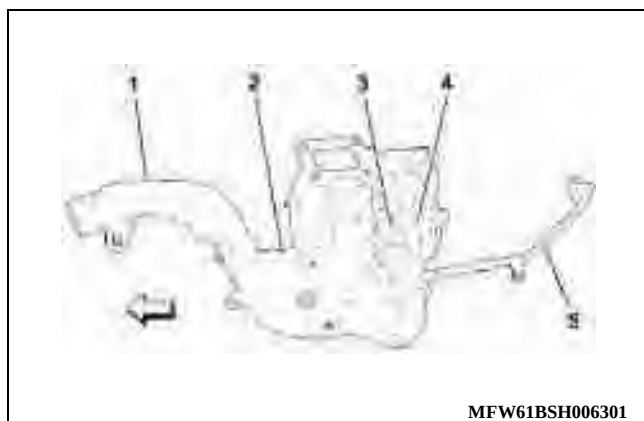
18. Remove the connector.

19. Remove the turbocharger.

- For the turbocharger, remove the intake pipe and the turbine return pipe and install the turbine intake pipe section.

Note:

Block intake and cooling water inlets and outlets to prevent debris from entering. At the same time, the compressor and turbine sides should be sealed in the same way.

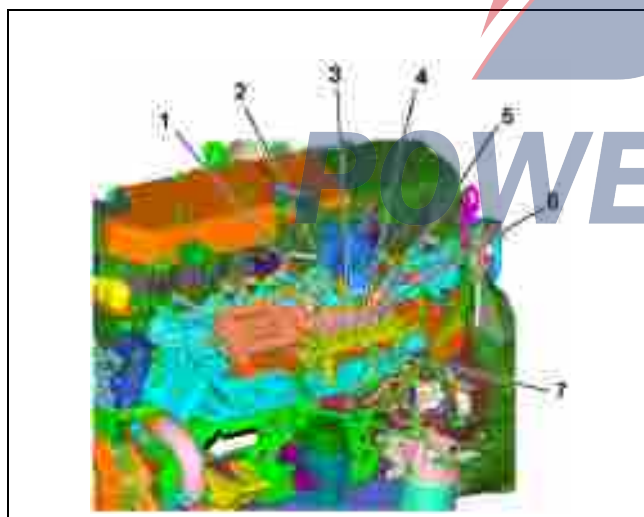


Legend

1. Inlet pipe
2. Turbine return pipe
3. Connector
4. Turbo inlet pipe
5. Turbine inlet pipe

20. Remove the thermal protector on the exhaust manifold.

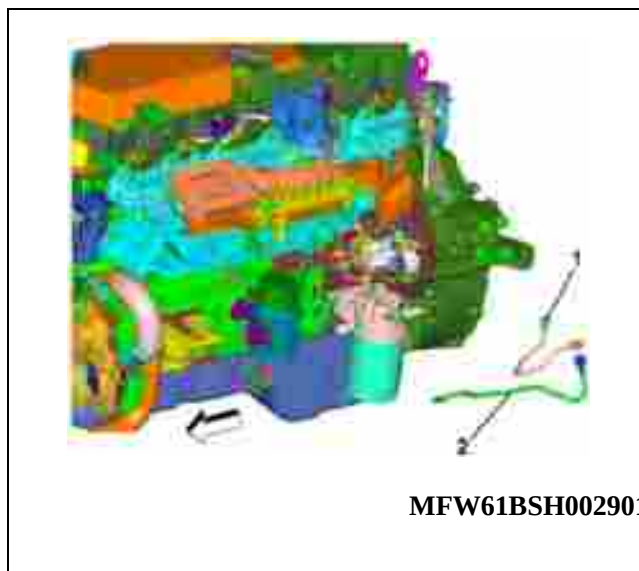
21. Remove the spray tube.



Legend

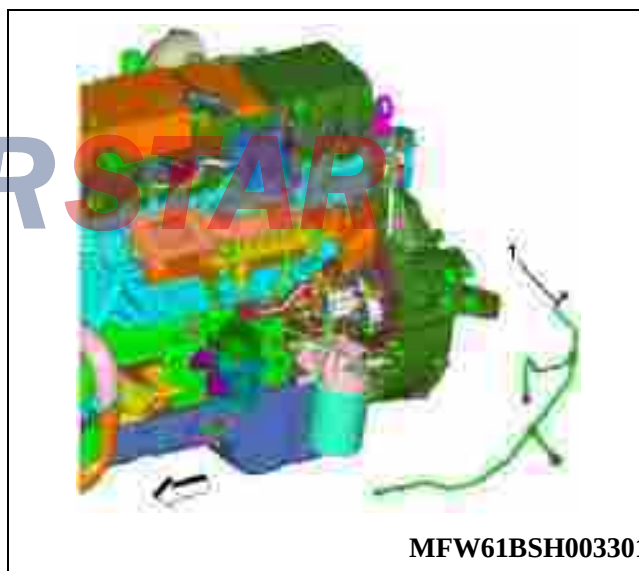
- 1 .No. 1 ejector tube
- 2 .No. 2 ejector tube
3. No. 3 ejector tube
4. No. 4 ejector tube
- 5 .No. 5 ejector tube
6. No. 6 ejector tube
7. Common rail equipment

22. Remove pressure feed pipe from common rail device and feed pump.



Legend

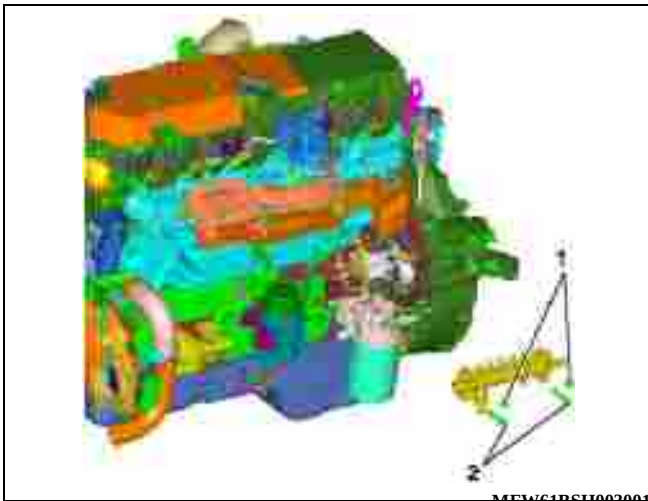
1. Pressure pipe
2. Fuel suction pipe
23. Remove the suction tube from the feed pump.
24. Remove the fuel drain tube (1).



25. Remove the common rail device.

Note:

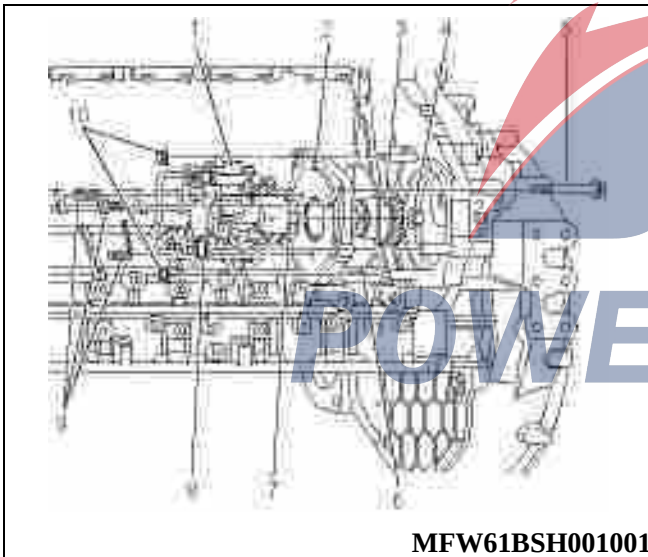
Move the sensor to prevent it interfering with other objects; choke it to prevent the chipping entering.



Legend

1. Nuts
2. Bolt

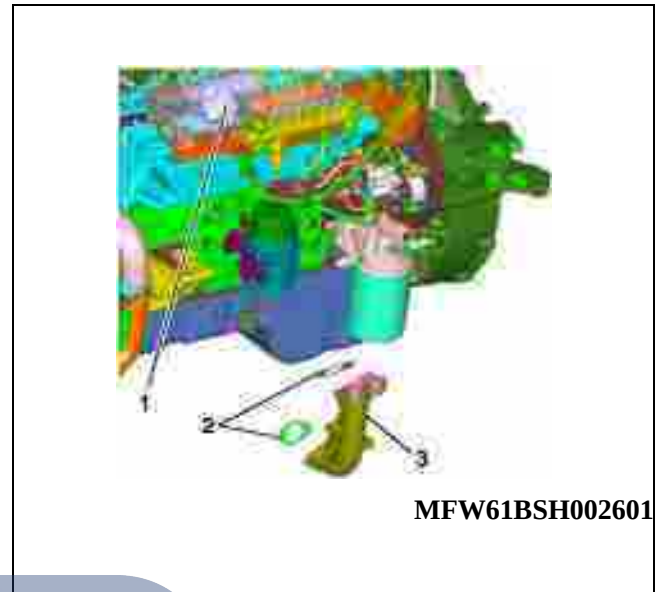
26. Remove the bracket and feed pump as a unit.



Legend

1. Charging pump
2. Bracket
3. Gear
4. Nuts
5. Bolt
6. O-ring
7. O-ring
8. Nuts
9. Bolt
10. Nuts

27. Remove the exhaust gas recirculation valve from the oil cooler.



Legend

1. EGR valve
2. Gasket
3. EGR valve

28. Remove the EGR valve from the pilot valve connector.



Legend

1. EGR valve
2. Gasket
3. EGR pilot valve connector

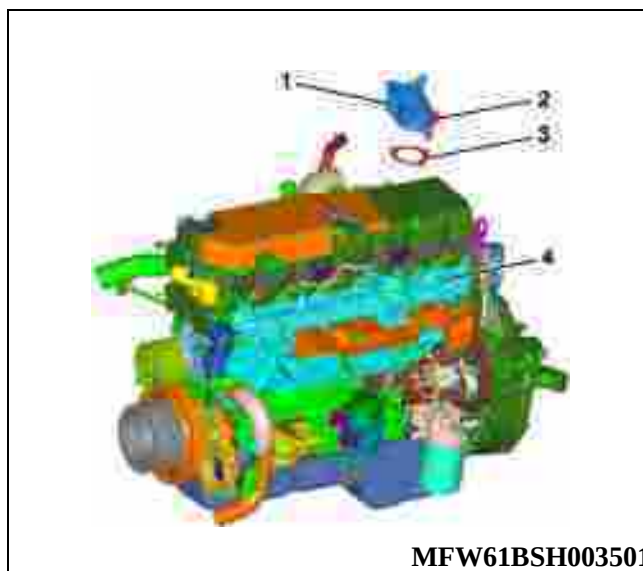
29. Remove the EGR guide valve connector from the inlet cover.



Legend

1. EGR pilot valve connector
2. Gasket

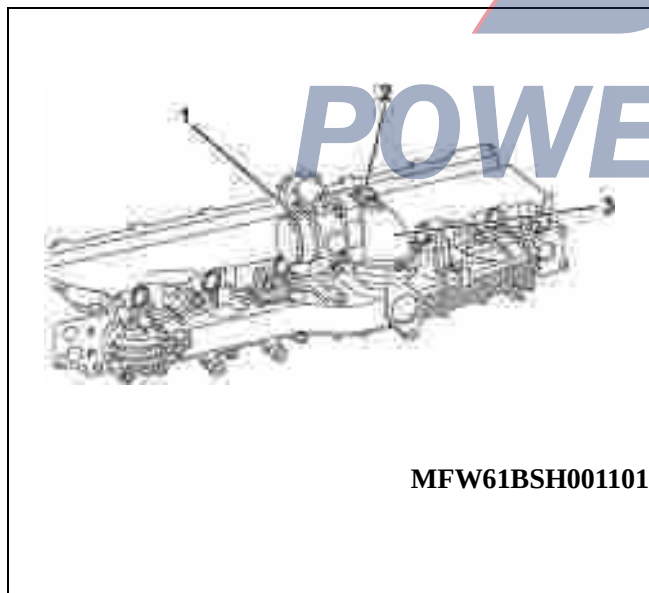
31. Remove the intake pipe from the inlet cover.



Legend

1. Inlet pipe
2. Bracket
3. Gasket
4. Inlet cover

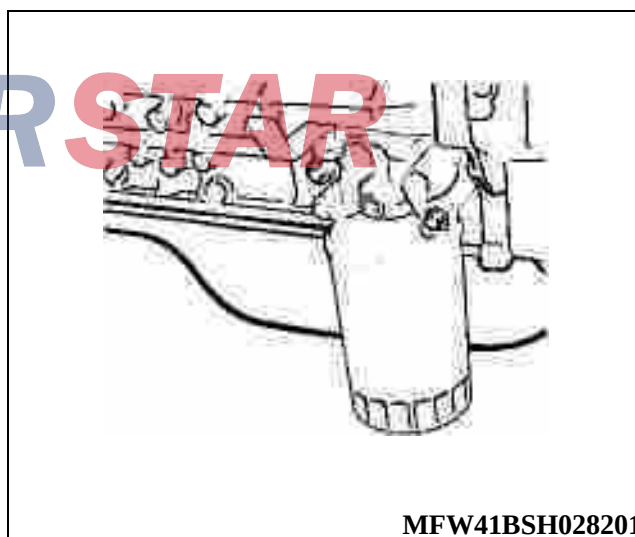
30. Remove the intake throttle.



Legend

1. Air intake throttle
2. Boost sensor
3. Inlet pipe

32. Remove the oil filter.



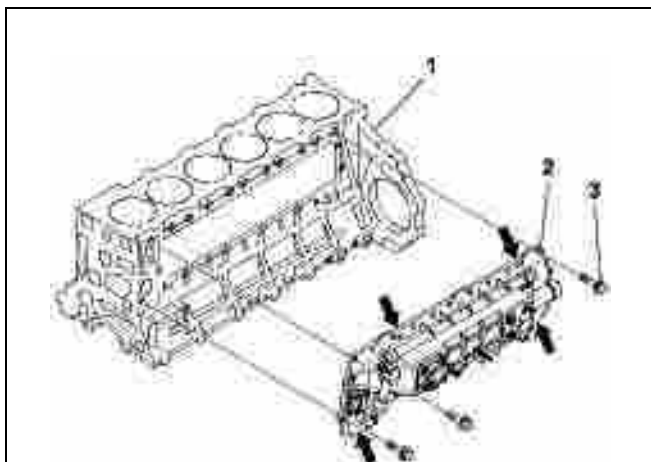
33. Remove the water pipe from the oil cooler.

34. Remove the crankshaft position sensor from the flywheel housing.

35. Remove the noise shield from the cylinder block.

36. Remove the oil cooler.

- Remove the oil cooler coupling bolt and insert the removed bolt into the replacement hole. Tighten it evenly and remove it.

**Legend**

- 1. Cylinder block
- 2. Oil cooler
- 3. Bolt

37. Remove the hydraulic gauge from the cylinder.

38. Remove the generator bracket from the cylinder block.

39. Remove the front engine feet from the cylinder block.

40. Remove the rear engine feet from the flywheel housing.

Reassemble

1. Install the rear engine feet and tighten them to the specified torque.

Tightening torque	N m (kgf m/lb ft)
	76 (7.7 / 56)

2. Install the front engine feet and tighten them to the specified torque.

Tightening torque	N m (kgf m/lb ft)
	44 (4.5 / 33)

3. Install the noise cap assembly.

4. Install oil pressure switch.

5. Install the generator bracket and tighten it to the specified torque.

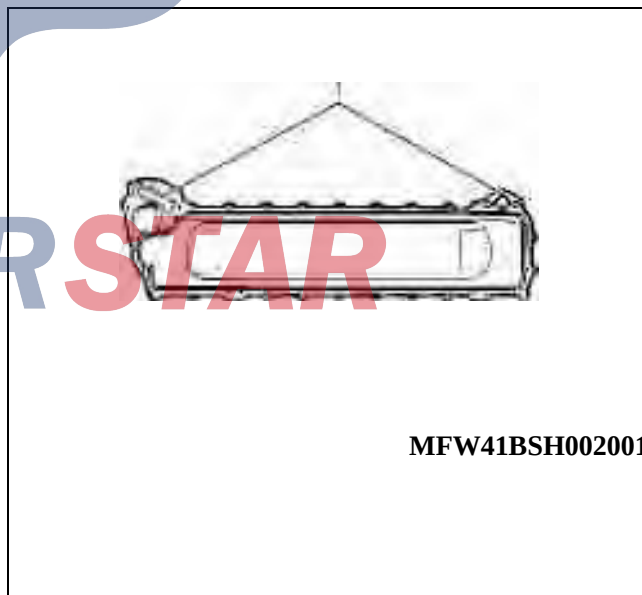
Tightening torque	N m (kgf m/lb ft)
	24 (2.4 / 17)

6. Install an oil cooler.

- Align the oil cooler reservoir with a drop of liquid filler (ThreeBond 1207C or 1216C adhesive) approximately 2-3 mm (0.0787-0.1181 inch) wide, then insert and install O-ring (1).

Note:

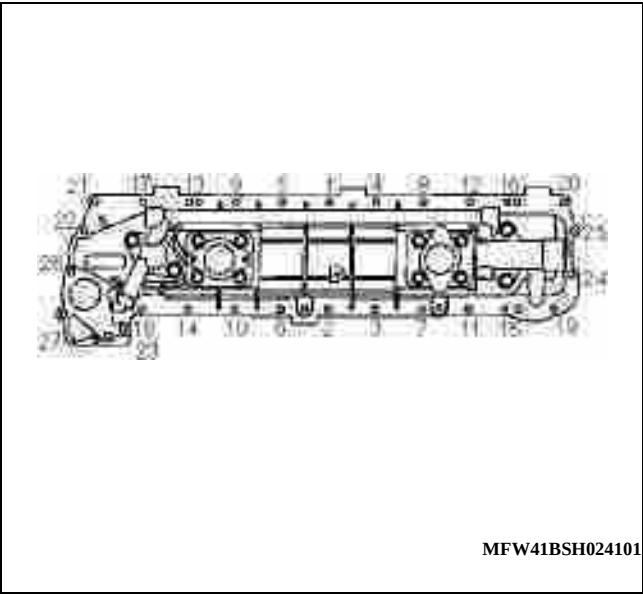
- If the liquid packing remains in the O-ring groove, it is difficult to insert the O-ring and therefore any remaining liquid packing on the O-ring should be removed.
- Load it within 7 minutes of using liquid filling.
- When installing the oil cooler on the oil cylinder, temporarily insert the M8 bolt into the holes of the two fronts and rear connecting bolts to install the oil cooler.



MFW41BSH002001

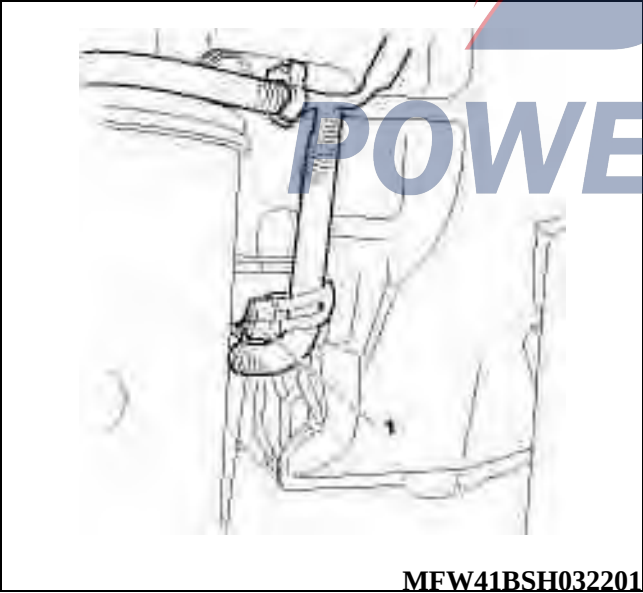
- Tighten the oil cooler in the order shown.

Tightening torque	N m (kgf m/lb ft)
	25 (2.5 / 18)



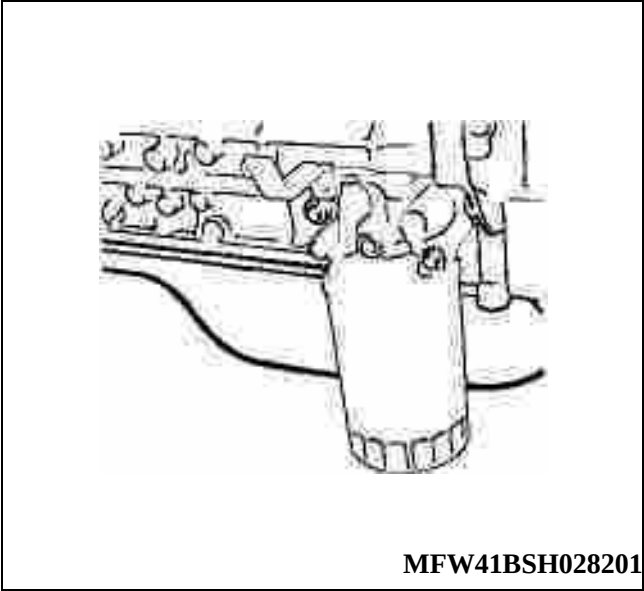
- 7. Install the tubing on the oil cooler.
- 8. Wear a noise protection cover.
- 9. Install the crankshaft position sensor (1).

Tightening torque	N m (kgf m/lb ft)
6 (0.6 / 52)	



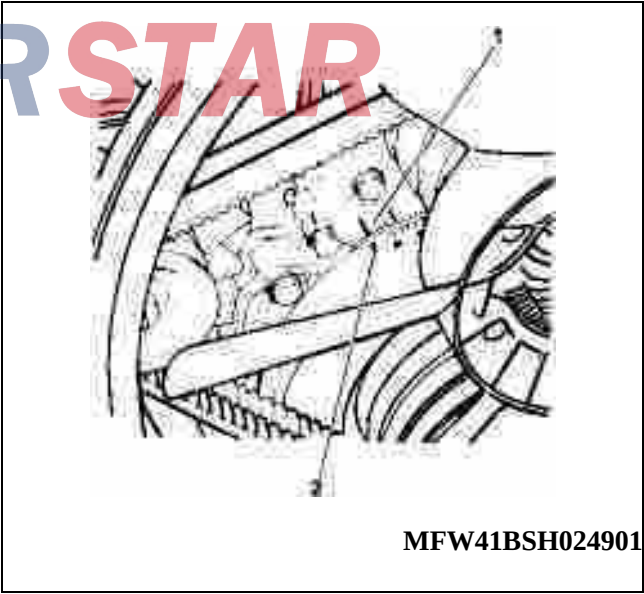
- 10. Install the oil filter
- Make sure the O-ring does not come out.

Tightening torque	N m (kgf m/lb ft)
38 (3.9 / 28)	

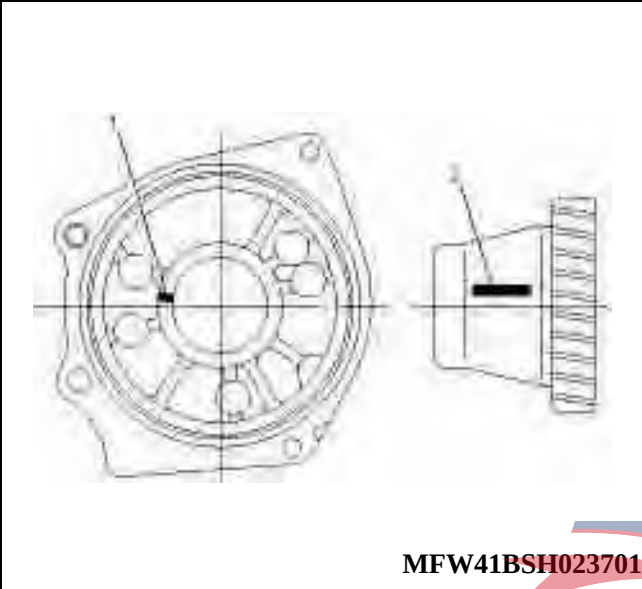


- 11. Install feed pump.
- Rotate the engine forward to align the pointer (1) with the vibration absorber top dead center TDC (0°) mark (2.)

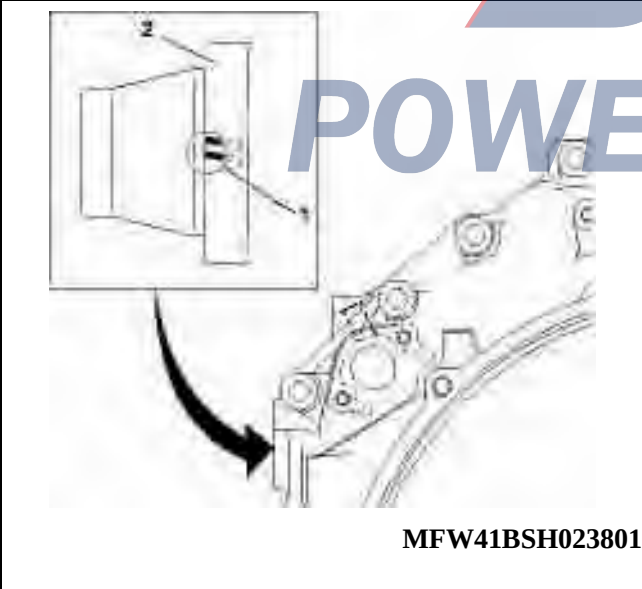
Note:
Make sure No. 1 cylinder is in the compressed dead point.



- Align the alignment mark (notch) on the gear (2) with the bracket (1) before assembling the feed pump.

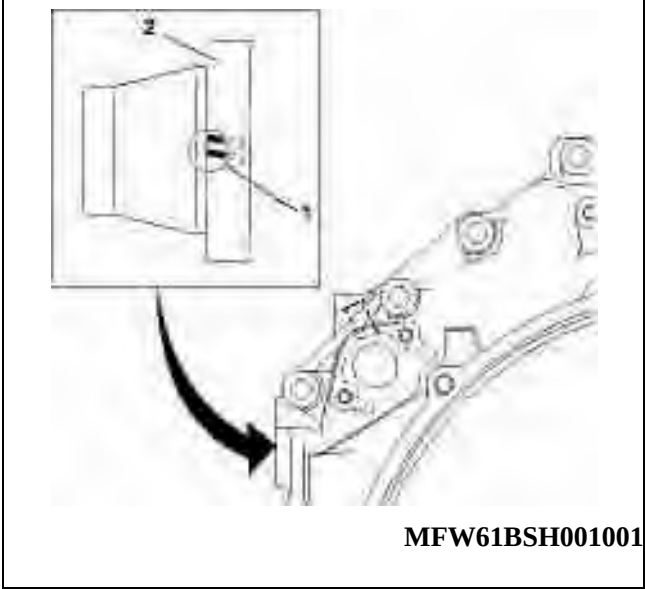


- Insert the feed pump horizontally into the cylinder block.
- When observe the gear wheel from the sensor holes on the side of the flywheel housing, you can see that the white paint is aligned with the mark (as shown below).



- Tighten the feed pump with a nut.

	Tightening torque	N m (kgf m/lb ft)
Nut	50 (5.1 / 37)	
Nut	77 (7.9 / 57)	
Bolt	77 (7.9 / 57)	

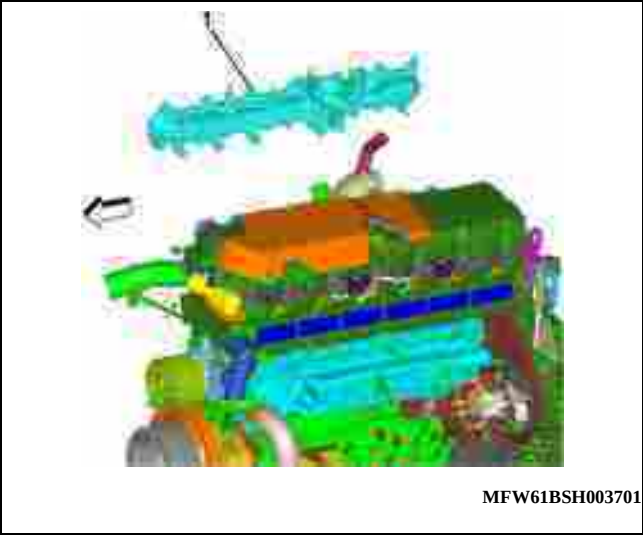


Legend

- Charging pump
- Bracket
- Gear
- Nuts
- Bolt
- O-ring
- O-ring
- Nuts
- Bolt
- Nuts

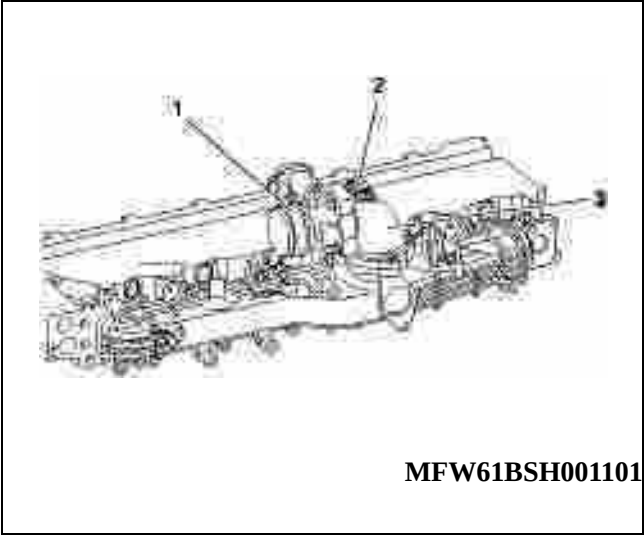
- Make sure again that the timing mark (crank damper) line alignment mark is aligned (0°) and that the feed pump gear alignment mark is also aligned.
 - If the mark is misaligned, reassemble the relevant part.
 - After checking the timing, the brake should be mounted on the flywheel housing.
12. Install the entrance cover (1).

Tightening torque	N m (kgf m/lb ft)
25 (2.5 / 18)	



13. Install intake and intake throttles.

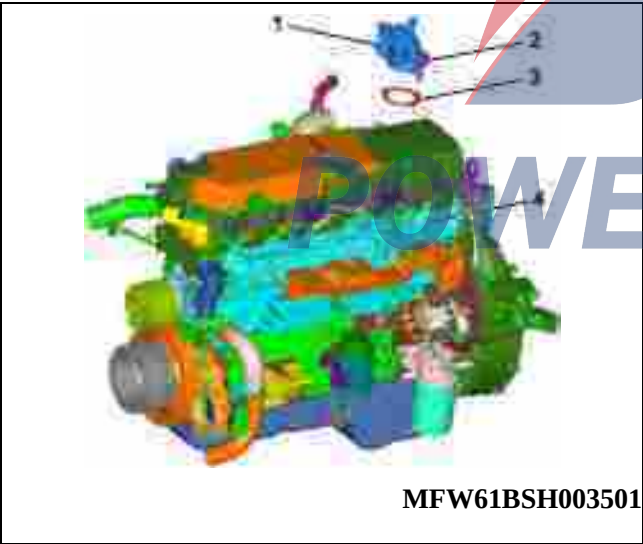
	Tightening torque	N m (kgf m/lb ft)
Intake pipe	25	(2.5 / 18)
Supercharging sensor	2	(0.2/17 lb. inch)
Air intake throttle	24	(2.4 / 17)



Legend

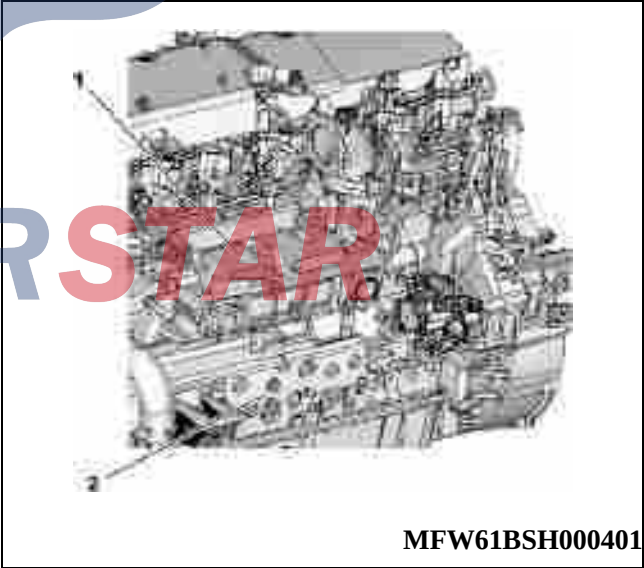
- 1. Air intake throttle
- 2. Boost sensor
- 3. Inlet pipe

14. Install the intake manifold on the oil cooler rubber septum.



Legend

- 1. Inlet pipe
- 2. Bracket
- 3. Gasket
- 4. Inlet cover



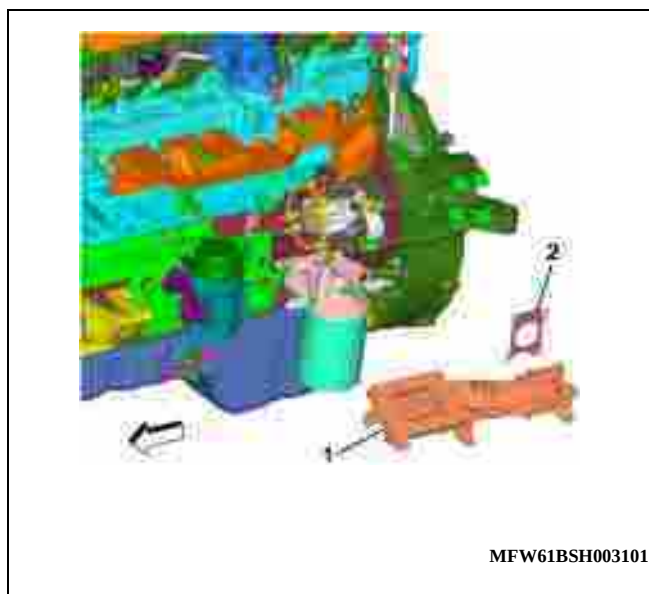
Legend

- 1. Rubber gasket
- 2. Noise protection cover

15. Temporarily connect the pilot valve fitting (1) to the inlet cap.

- Install the spacer (2) on the side of the inlet cover.

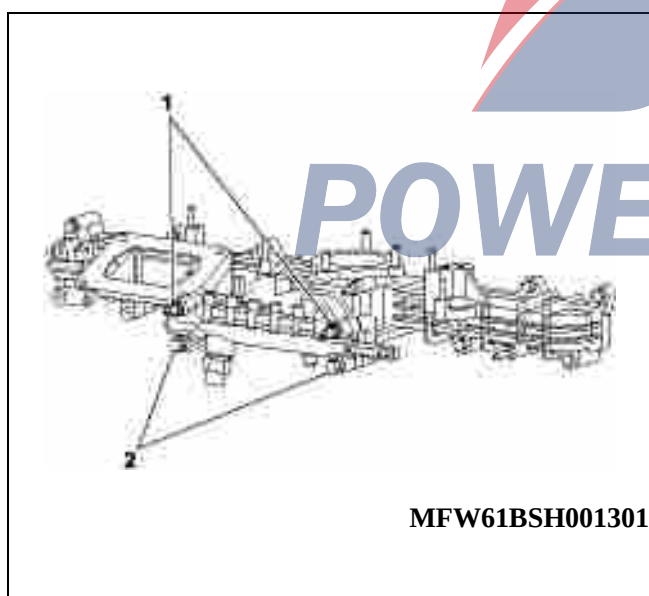
Tightening torque	N m (kgf m/lb ft)
25	(2.5 / 18)



16. Install common rail equipment.

- Temporarily tighten (1) nuts → (2) bolts
- Finally tighten (2) bolts → (1) nuts

Tightening torque	N m (kgf m/lb ft)
	25 (2.5 / 18)

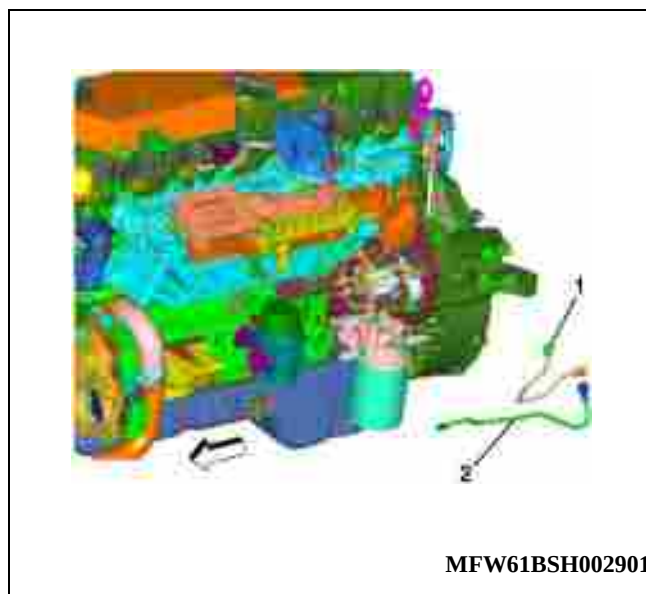


Legend

1. Nuts
2. Bolt

17. Install pressure delivery pipe.

Tightening torque	N m (kgf m/lb ft)
Feed pump side	44 (4.5 / 33)
Common rail device side	44 (4.5 / 33)

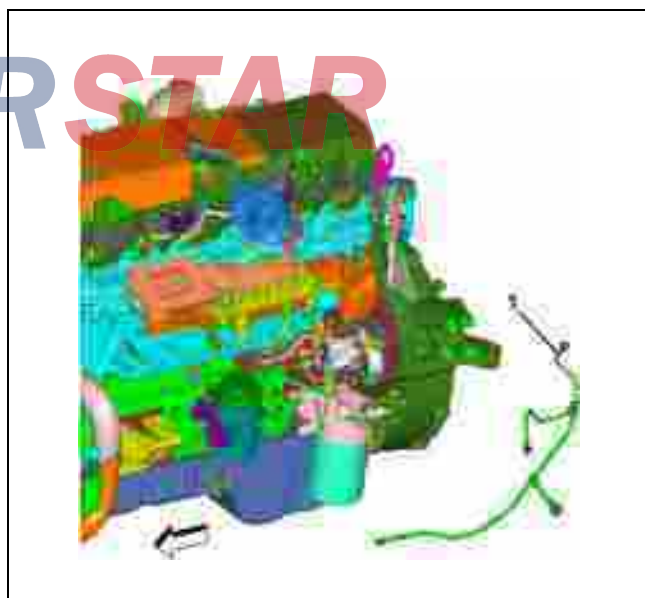


Legend

1. Pressure pipe
2. Fuel suction pipe

18. Install the fuel discharge pipe (1).

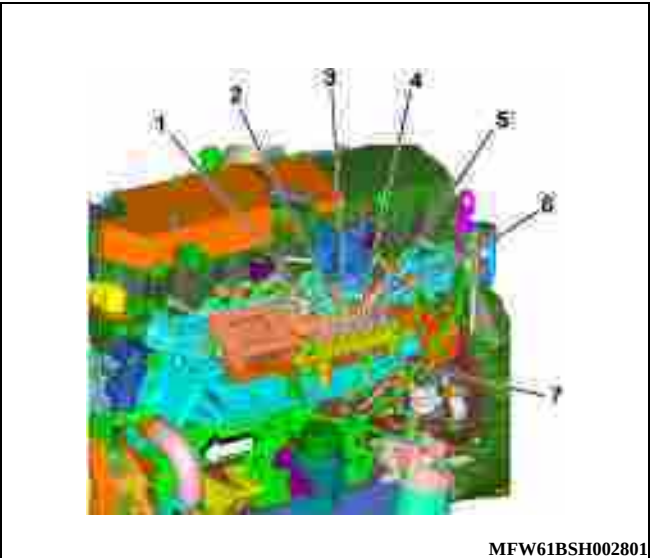
Tightening torque	N m (kgf m/lb ft)
	10 (1.0 / 87)



19. Place the suction tube on the feed pump and secure it in place with a pipe clamp.

20. Install the spray tube and tighten the sleeve nut.

Tightening torque	N m (kgf m/lb ft)
Injector side	30 (3.1 / 22)
Common rail device side	25 (2.6 / 19)

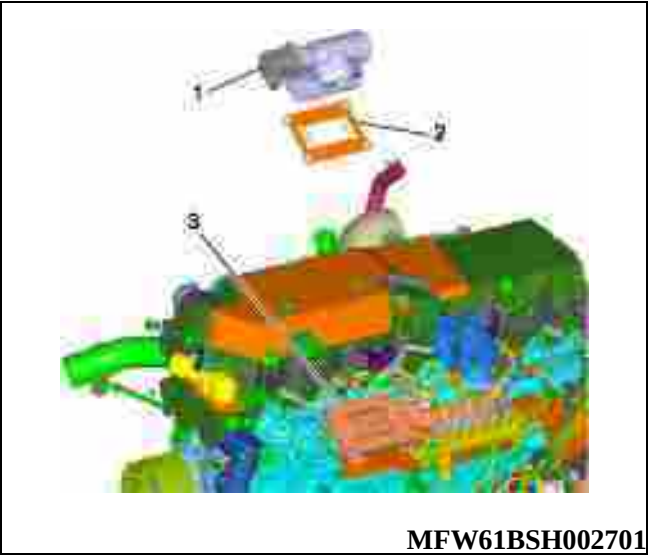


Legend

- 1. No. 1 ejector tube
- 2. No. 2 ejector tube
- 3. No. 3 ejector tube
- 4. No. 4 ejector tube
- 5. No. 5 ejector tube
- 6. No. 6 ejector tube
- 7. Common rail equipment

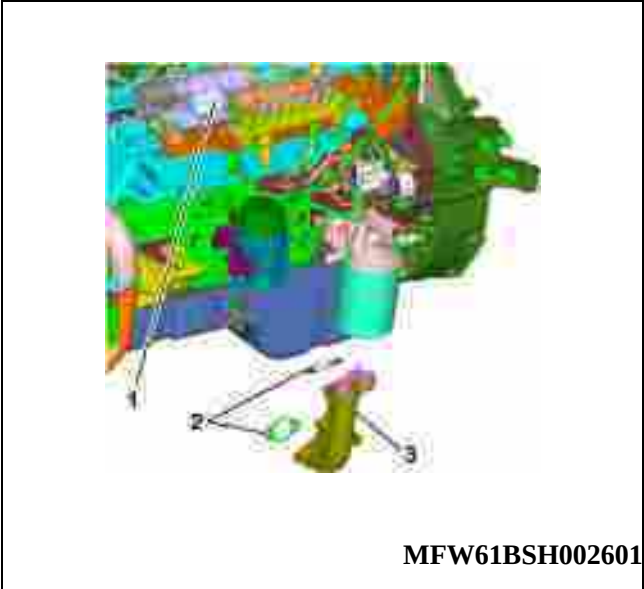
21. Install EGR valve (1) to the pilot valve joint temporarily.

- Install washer (2) on pilot valve connector side.



22. Install the EGR valve tube temporarily on the EGR valve and the oil cooler.

- Install the gasket to the EGR valve side and oil cooler side.



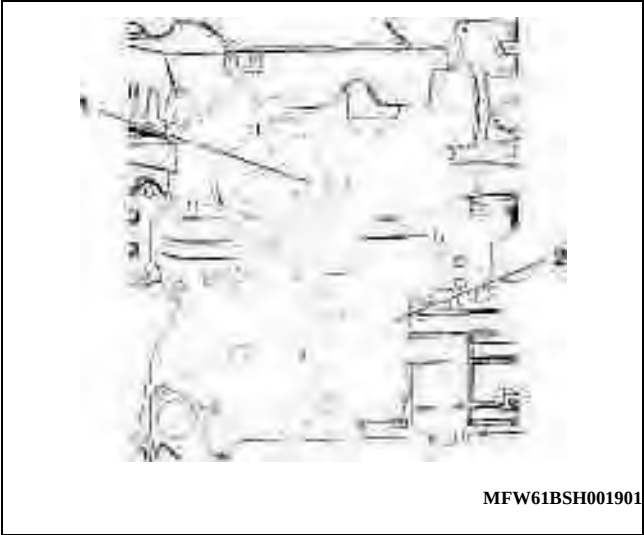
Legend

- 1. Pilot valve connector
- 2. Gasket
- 3. EGR valve

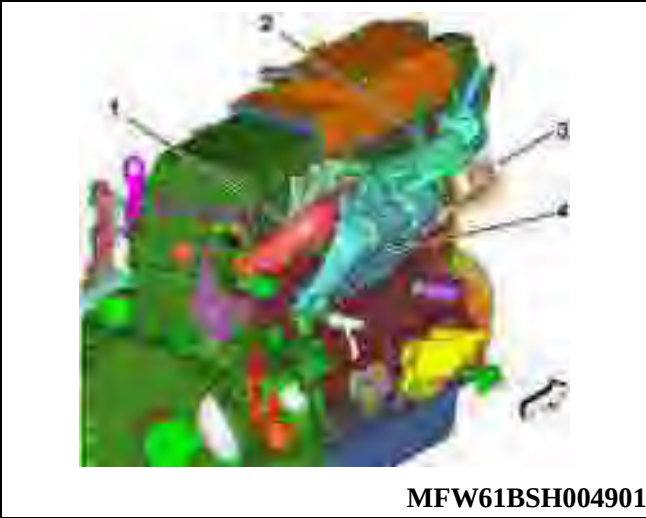
23. Connect the EGR valve.

- At this point, fully tighten the valve (2) and the EGR valve (1).

Tightening torque	N m (kgf m/lb ft)
EGR valve	46 (4.7 / 34)
EGR valve tube (EGR valve side)	46 (4.7 / 34)
EGR valve tube (Oil cooler side)	22 (2.2 / 16)



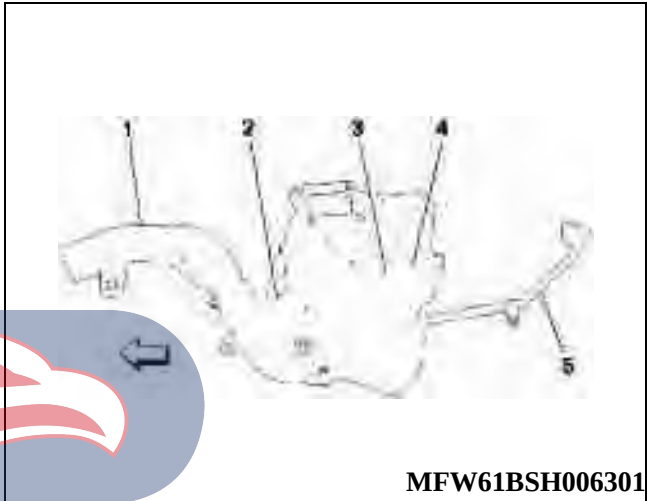
24. Install a thermal protector.



Legend

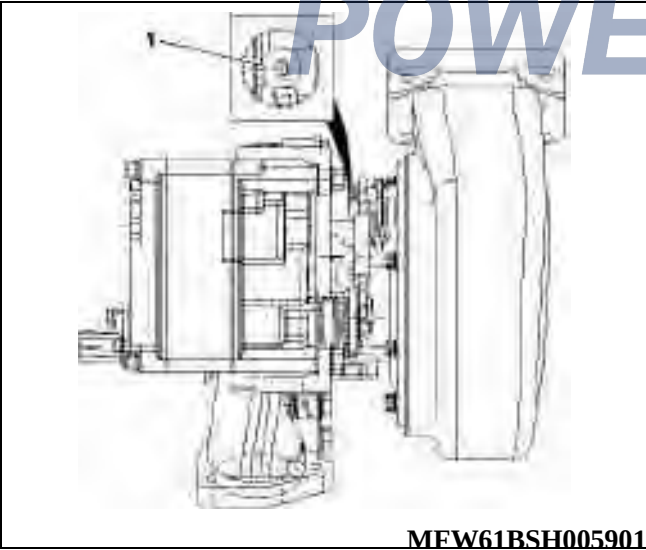
- 1. Exhaust manifold (back)
- 2. Exhaust manifold (front)
- 3. Thermal protector (front)
- 4. Thermal protector (back)

Tightening torque	N m (kgf m/lb ft)
Turbine return pipe (2)	41 (4.2 / 30)
Connector(3)	41 (4.2 / 30)
Turbine inlet pipe (4)	34 (3.5 / 25)
Turbine inlet pipe (5)	41 (4.2 / 30)



25. Install the turbocharger

- Pour the appropriate amount of engine oil into the turbocharger fill hole (1) and gently rotate the turbine shaft to lubricate the bearings.



- Before installing the turbocharger, install the intake pipe (1) and the turbine return (2) intake pipe (5) and install the turbine intake pipe (4).

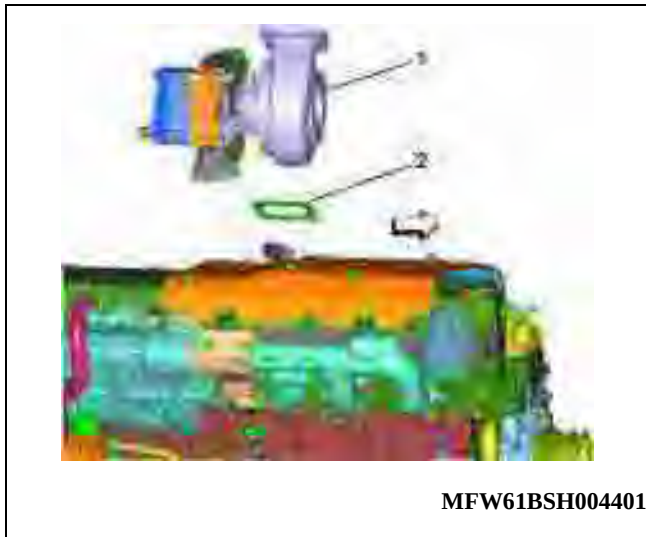
- The turbine water supply pipe (2) is vertically mounted in the cylinder block (1).



- Install the turbocharger (1) and insert the spacer (2) in between.

Tightening torque	N m (kgf m/lb ft)
	52 (5.3 / 38)

Tightening torque	N m (kgf m/lb ft)
Inlet pipe(1)	22 (2.2 / 16)



26. Install the inlet pipe.

- Install the pipe on the cylinder block.

Note:

Do not use spacers.

Tightening torque	N m (kgf m/lb ft)
41 (4.2 / 30)	

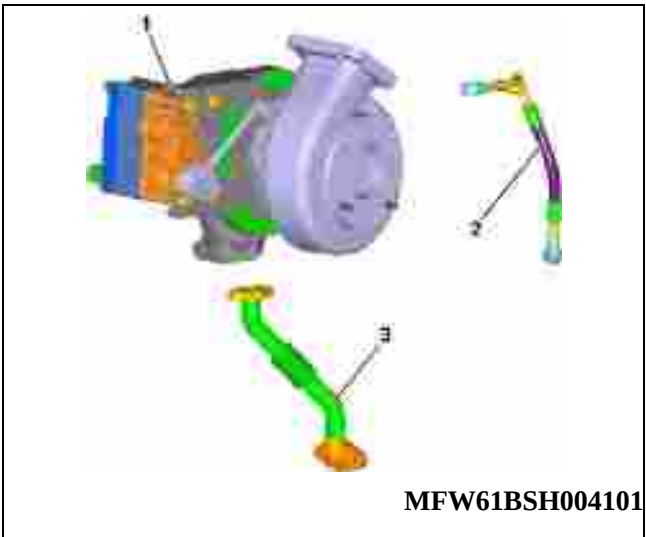
27. Put on the return pipe.

- The tubing is installed on the turbocharger and cylinder body.

Note:

Do not use spacers.

Tightening torque	N m (kgf m/lb ft)
19 (1.9 / 14)	

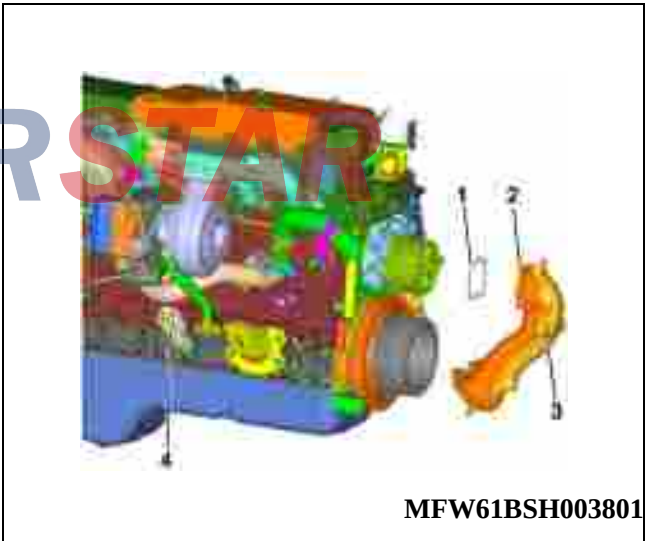


Legend

1. Turbocharger
2. Inlet pipe
3. Return pipe

28. Install exhaust fittings.

- Install the connector on the turbocharger and insert the spacer into the middle.



Legend

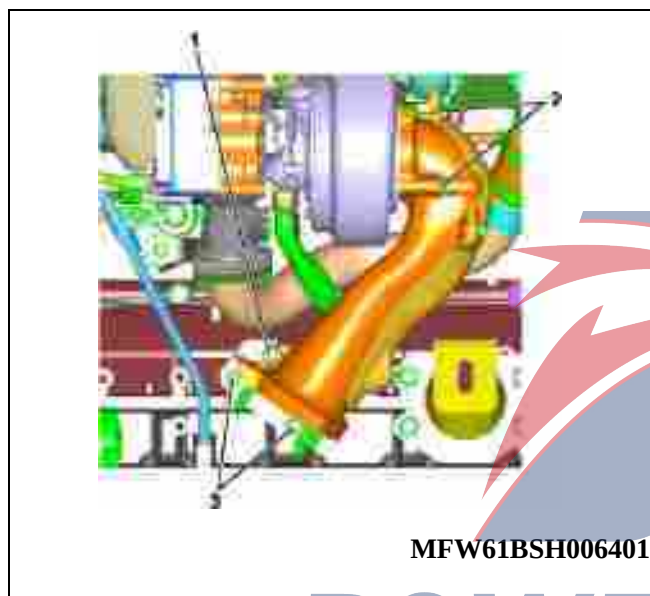
1. Gasket
2. Connector
3. Thermal protector
4. Bracket

a. Installation procedure:

- After temporarily grouping the exhaust unit and the bracket, tighten the installation procedure.

- b. Temporarily tighten the exhaust connector and bracket (1).
- c. Temporarily tighten the bracket and cylinder (3).
- d. Temporarily tightened in the exhaust pipe connector and turbocharger (2).
- e. Tighten it again.

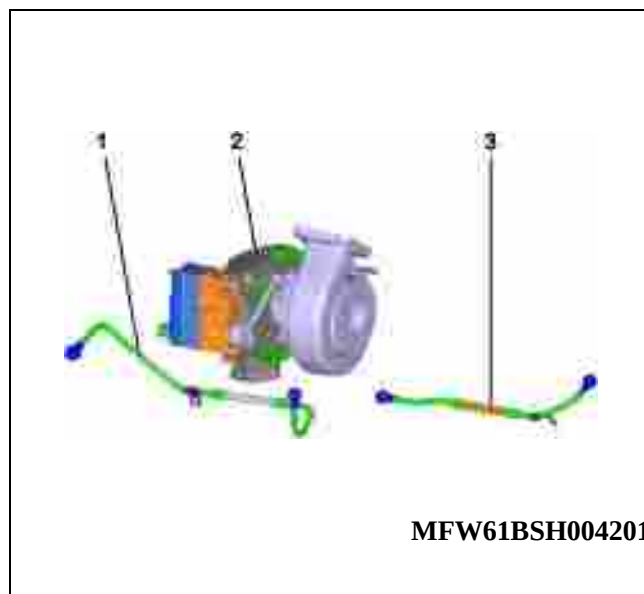
Tightening torque	N m (kgf m/lb ft)
	26 (2.7 / 20)



29. Install the return pipe.

Note:

Do not use spacers.



Legend

1. Water inlet pipe
2. Turbocharger
3. Water return pipe

30. Install the inlet pipe.

- Install the air tube on the air compressor.

Tightening torque	N m (kgf m/lb ft)	
	Cylinder head side	Air compressor side
Feed tube	—	50 (5.1 / 37)
Return pipe	41 (4.2 / 30)	—

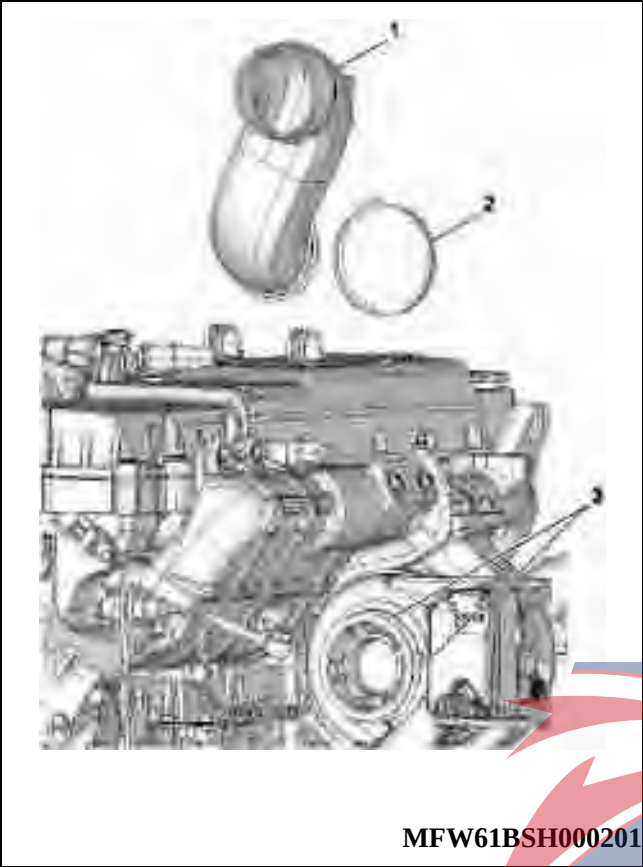
Note:

Do not use spacers.

31. Install the intake pipe.

- Position the protruding part (3) of the air intake pipe (1) with the turbocharger and then tighten it with a turbine V-belt (2).

Tightening torque	N m (kgf m/lb ft)
	16 (1.6 / 12)

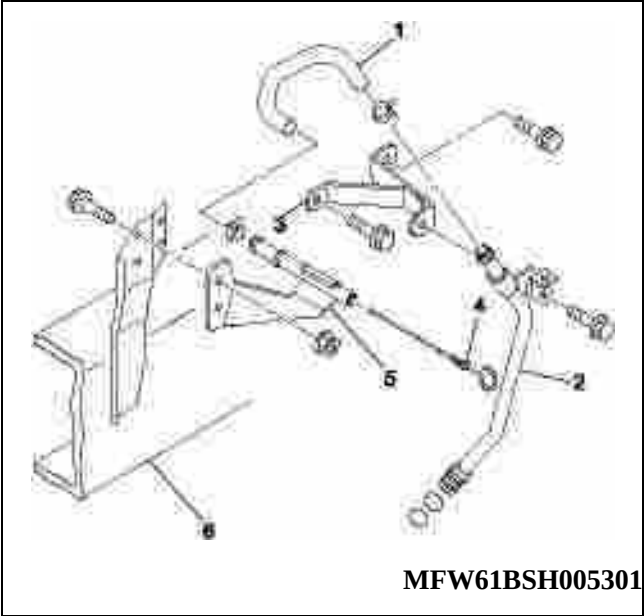


32. Install the catheter.

- Install the O-ring on the catheter and push it in until it contacts the cylinder block hole.

Note:

Be careful not to cut O-rings.



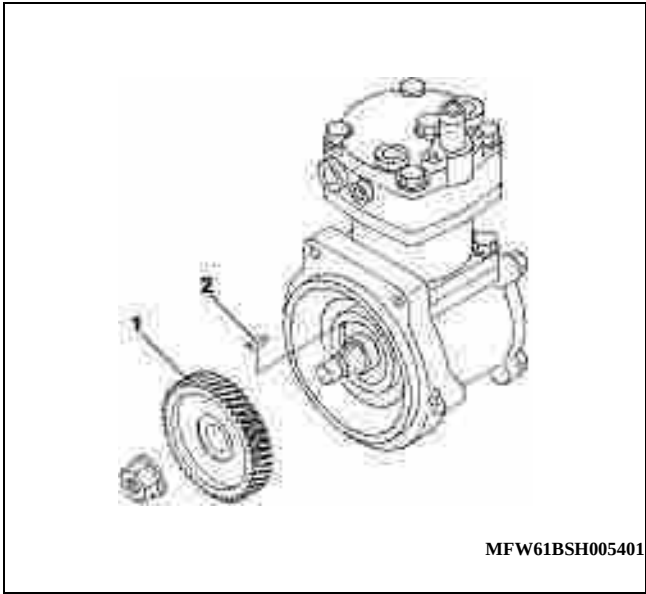
Legend

- 1. Hose
- 2. Catheter
- 3. Bracket
- 4. Liquid level meter
- 5. Bracket
- 6. Frame

33. Install the air compressor assembly on the flywheel housing.

- Screw the fixing pin into the compressor shaft and tighten the coupling nut of the drive gear to the specified torque.

Tightening torque	N m (kgf m/lb ft)
147 (15 / 108)	

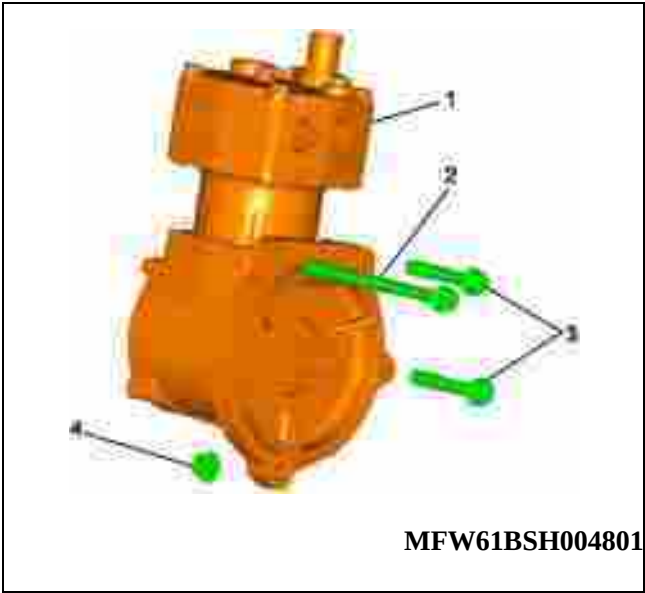


Legend

- 1. Gear
- 2. Fixed pin

- Apply an equal amount of liquid filler to the three points on the outer circumference of the O-ring and stick them in the slots of the air compressor.
- Three bond No. 1208 adhesive.

Tightening torque	N m (kgf m/lb ft)
73 (7.4 / 54)	

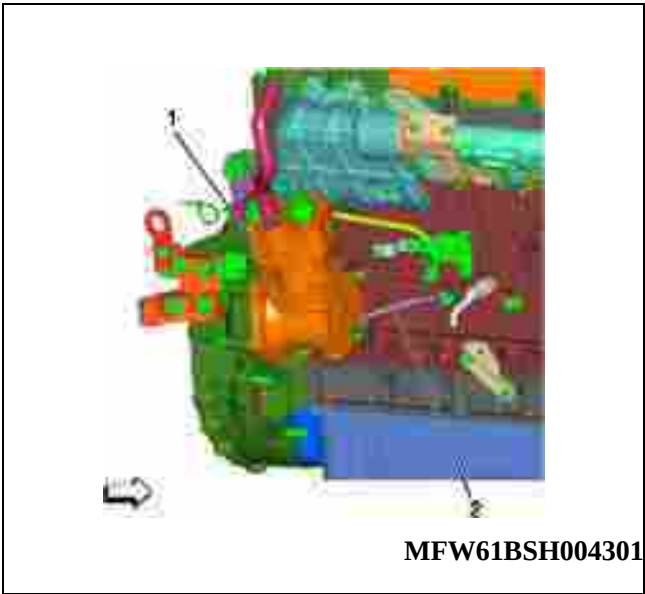
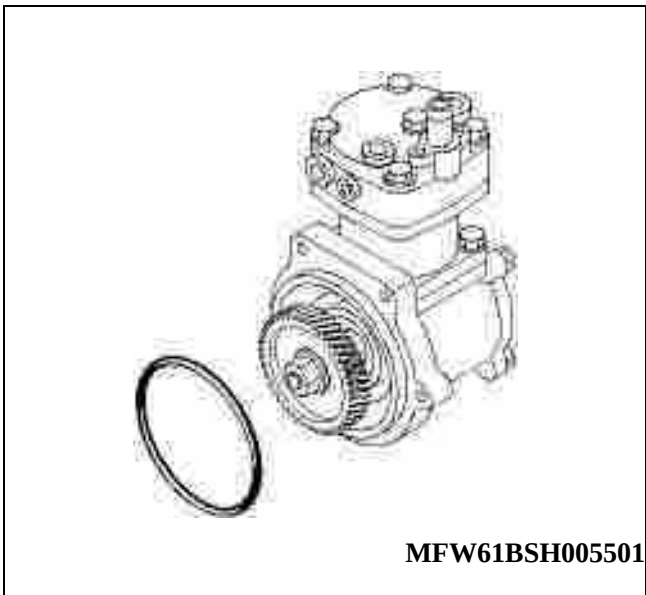


Legend

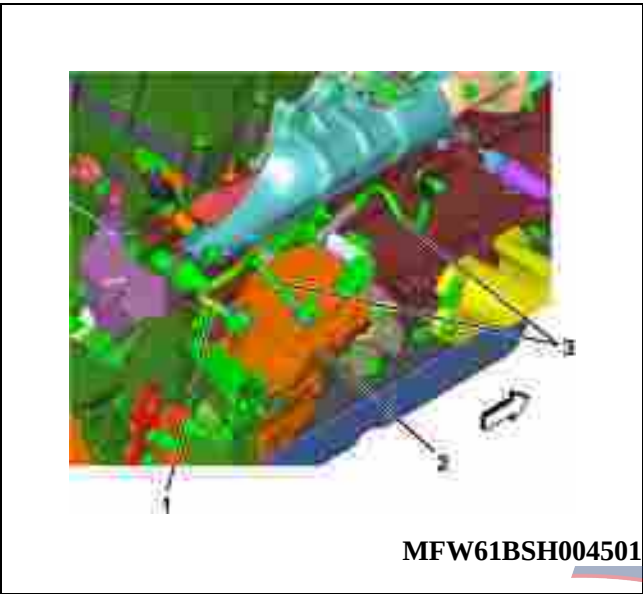
- 1. Compressor
- 2. Bolt (Long)
- 3. Bolt
- 4. Nuts

- Install the fuel supply pipe (2) on the compressor (1).

Tightening torque	N m (kgf m/lb ft)
Cylinder side	41 (4.2 / 30)
Air compressor side	26 (2.7 / 20)



34. Install the air compressor hose (1).



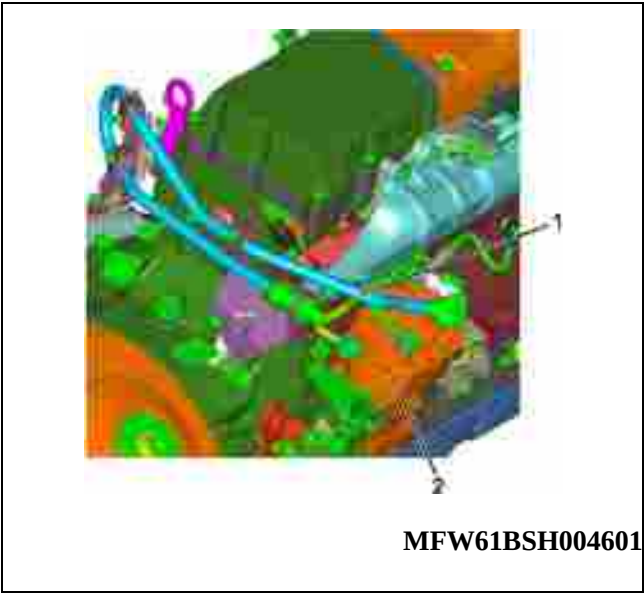
Legend

- 1. Water pipe (inlet water)
- 2. Compressor
- 3. Water pipe (return to turbine)

35. Install the air duct (air compressor).

- Temporarily install the clamp and install the feed tube and governor tube.

Tightening torque		N m (kgf m/lb ft)
Feed side	(Dry)	64 (6.5 / 47)



Legend

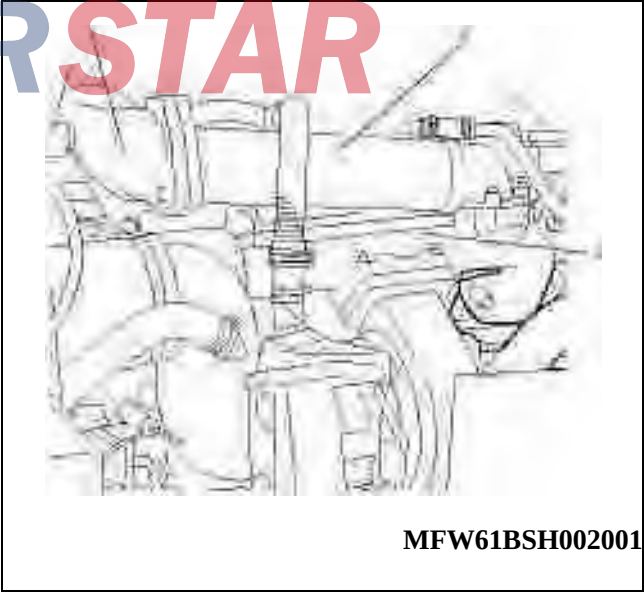
- 1. Feeding tube
- 2. Air compressor

36. Connect EGR cooler (external).

- Secure the gasket to the side of the exhaust manifold and install it.

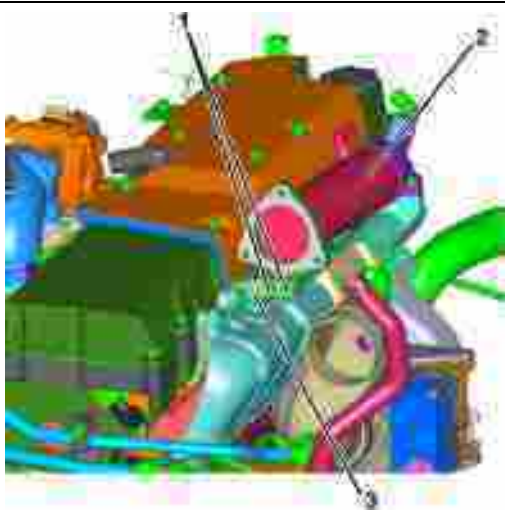
Note:

Do not use spacers.



Legend

- 1. EGR pipe
- 2. EGR cooler
- 3. Exhaust manifold

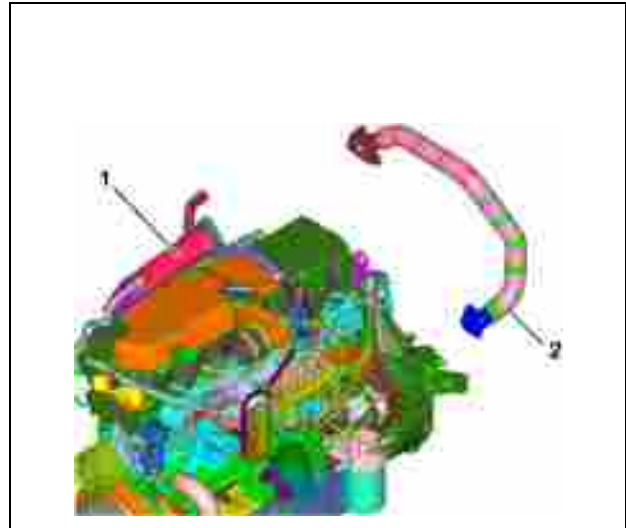


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Legend

1. Bolt
2. EGR cooler
3. Bracket

37. Connect EGR pipe.

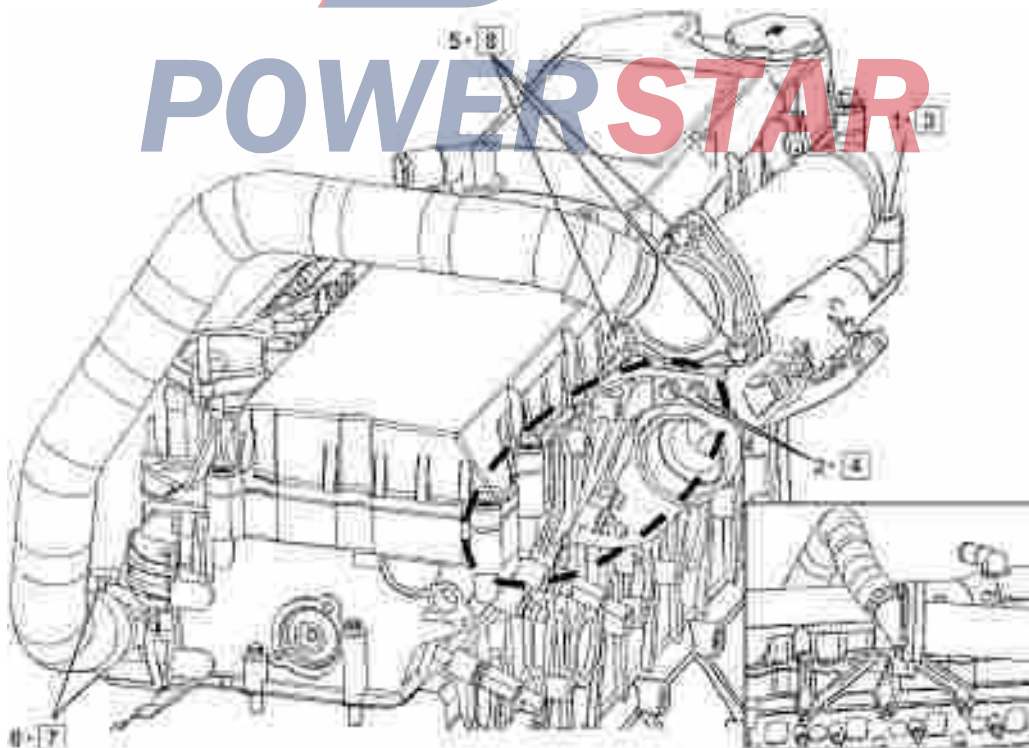


Legend

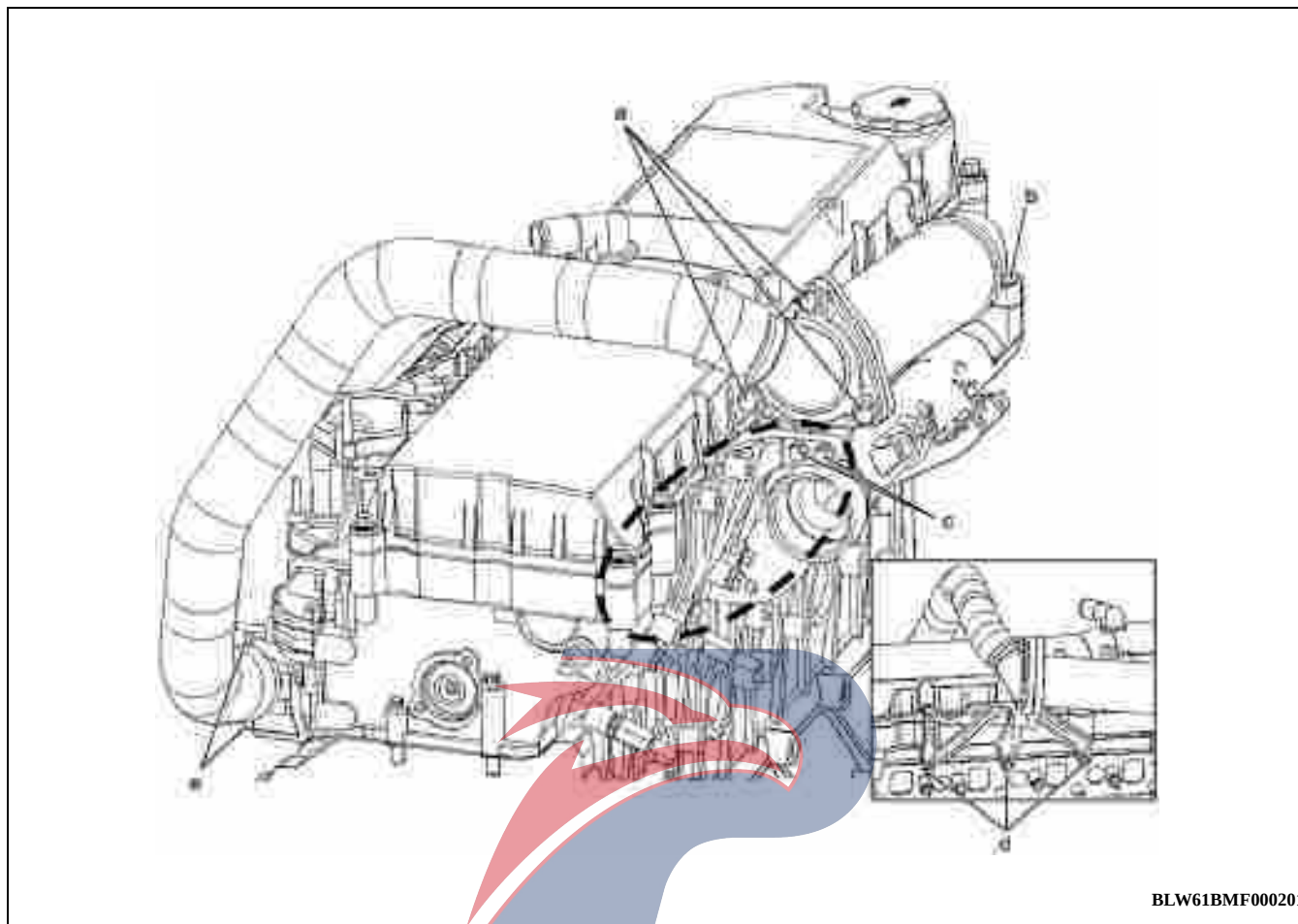
1. EGR cooler
2. EGR pipe

38. EGR cooler and EGR pipe tightening program.

- Numbers indicate temporary procedures.
- The number in the box indicates the tightening process.



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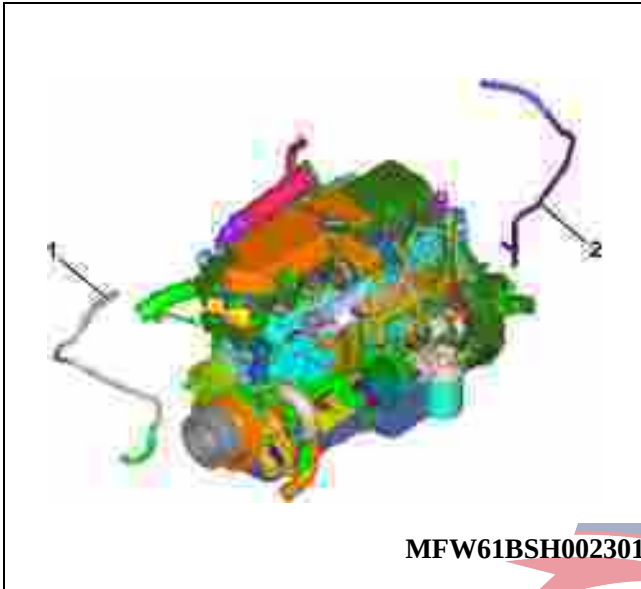
Legend

(a) 46 N m (4.7 kgf m/34 lb.ft)
(b) 28 N m (2.9 kgf m/21 lb ft)
(c) 25 N m (2.5 kgf m/18 lb.ft)

(d) 28N m (2.9 kgf m/21lb.ft)
(e) 25 N m (2.5 kgf m/18 lb ft)

-
- Bolt (b) cannot be recycled.

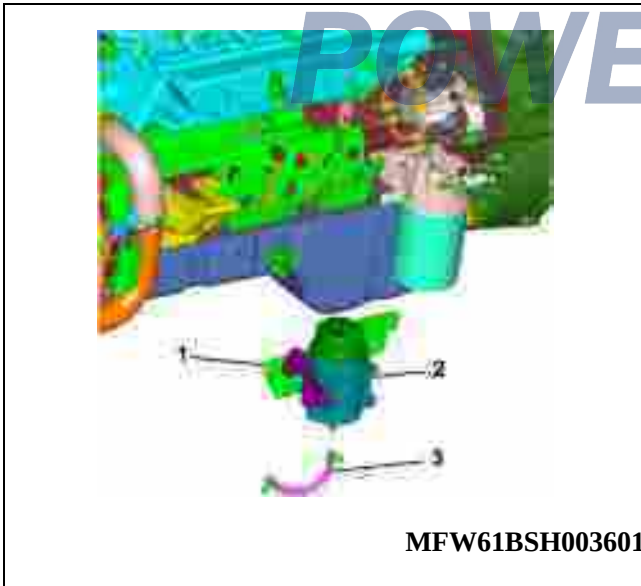
39. Install EGR cooler hoses and secure with pipe clamps.



Legend

- 1. Water pipe
- 2. Water pipe (out)

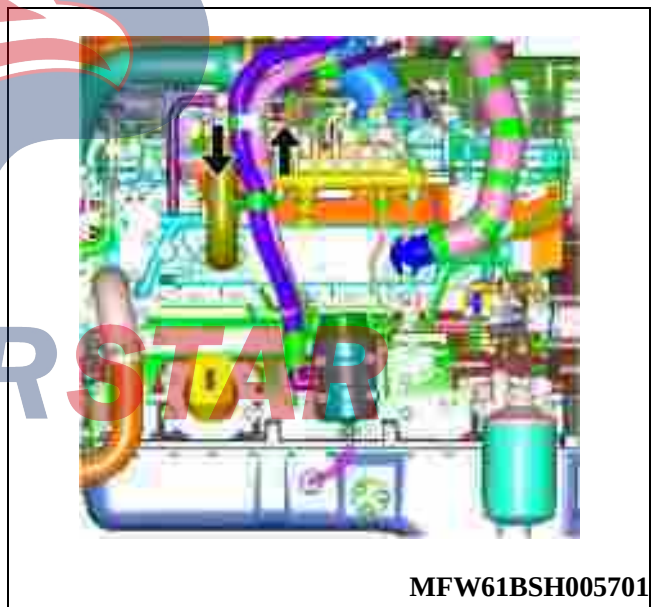
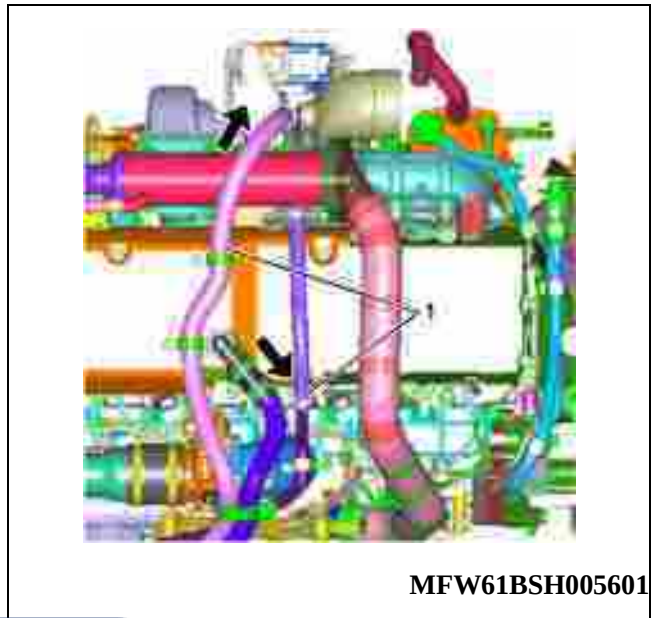
40. Install the oil separator so that the bracket remains attached to the cylinder block.



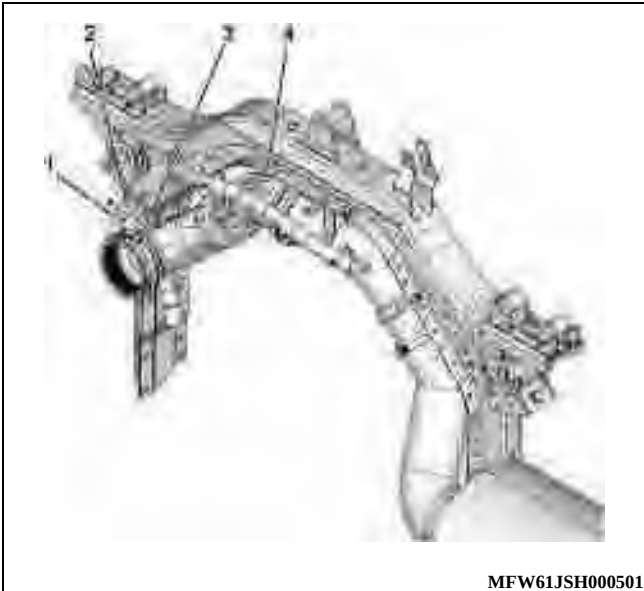
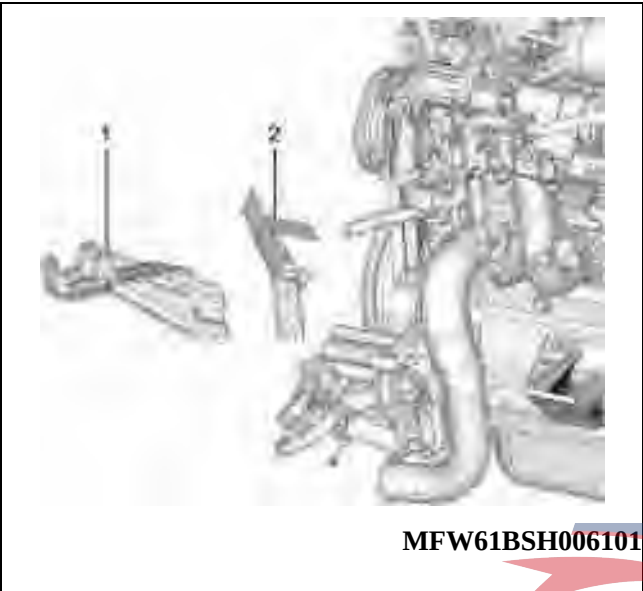
Legend

- 1. Bracket
- 2. Oil separator
- 3. Oil drain hose

41. Install the oil separator hose (1).

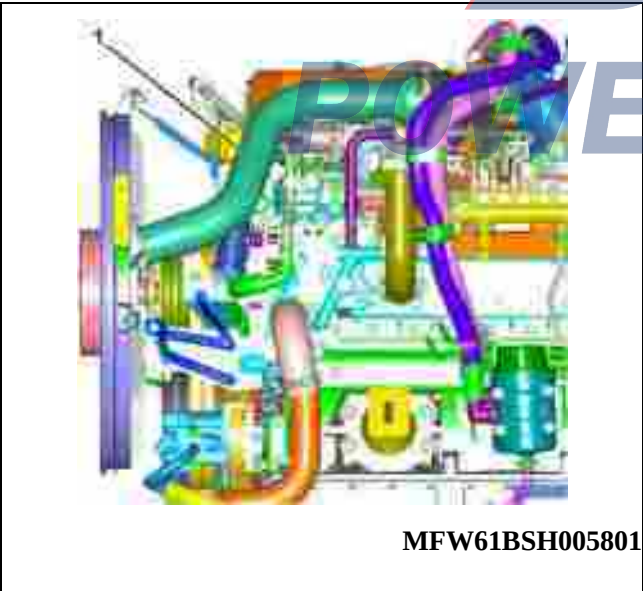


42. In addition, install the left fan guide rod stay (1) and engine control coil holder (2).



43. Install the intake pipe (1).

Tightening torque	N m (kgf m/lb ft)
18 (1.8 / 13)	

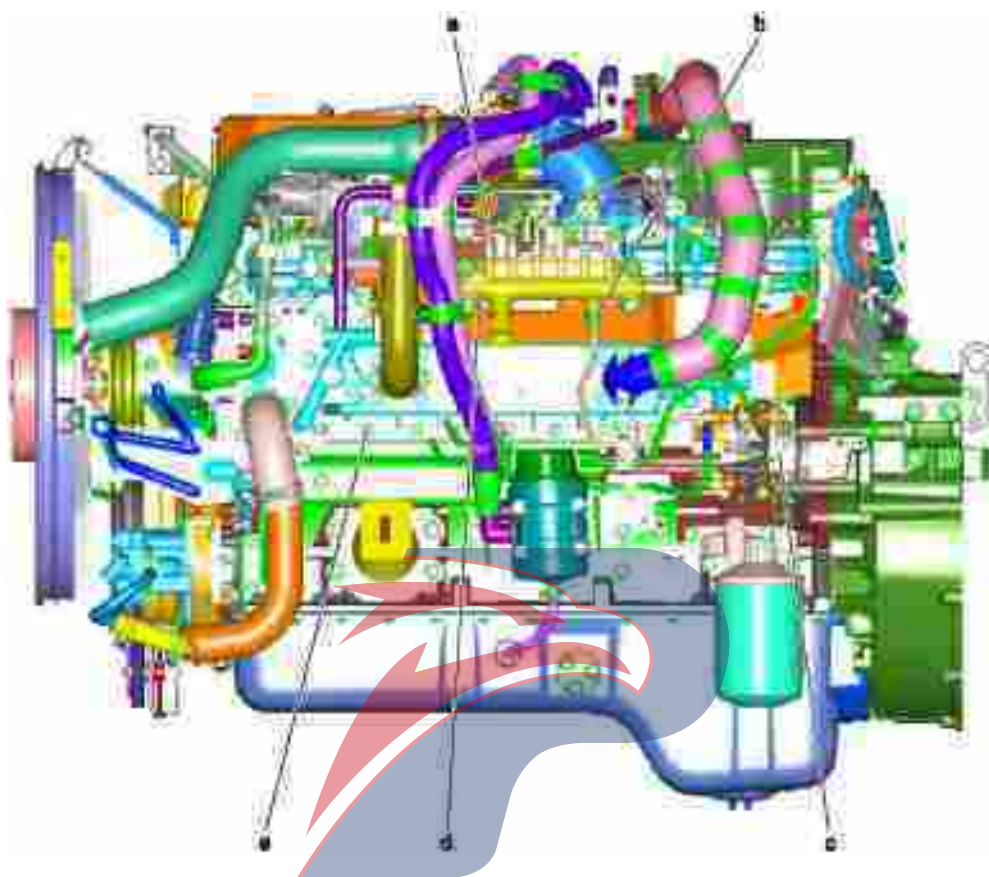


44. Connect the engine belt to each joint and install the clamps to the designated location.

45. Insert the air inlet pipe (4) and suction hose (1) into the engine and tighten with the clamp (2) (3).

Tightening torque	N m (kgf m/lb ft)
7 (0.7 / 61)	

Tightening torque table

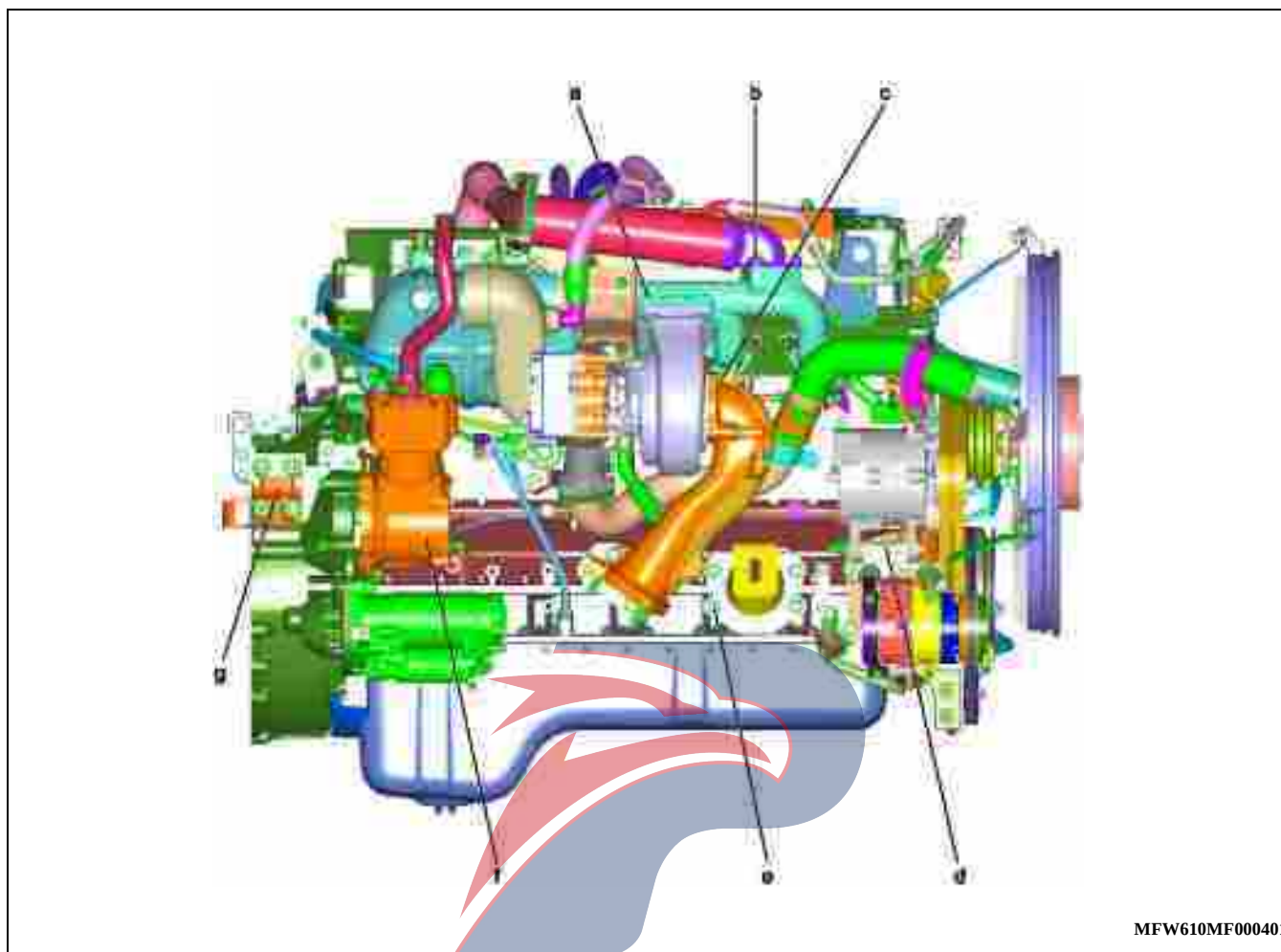


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POWERSTAR

Legend

- | | |
|---|---------------------------------|
| (a) 30 N m (3.1 kgf m/22lb ft) injection side | (d) 25N m (2.5 kgf m/18lb ft) |
| 25 N m (2.5 kgf m/18 lb.ft) C/R side | (e) 25 N m (2.5 kgf m/18 lb.ft) |
| (b) 44 N m (4.5 kgf m/33 lb.ft) | |
| (C) 50 N m (5.1 kgf-m / 37 lb ft) | |
| 77 N m (7.9 kgf m/57 lb- ft) | |



Legend

(a) 52 N m (5.3 kgf m/38 lb ft)

(b) 28 N m (2.9 kgf m/21 lb ft)

(c) 26 N m (2.7 kgf m/20 lb ft)

(d) 24 N m (2.4 kgf m/17 lb ft)

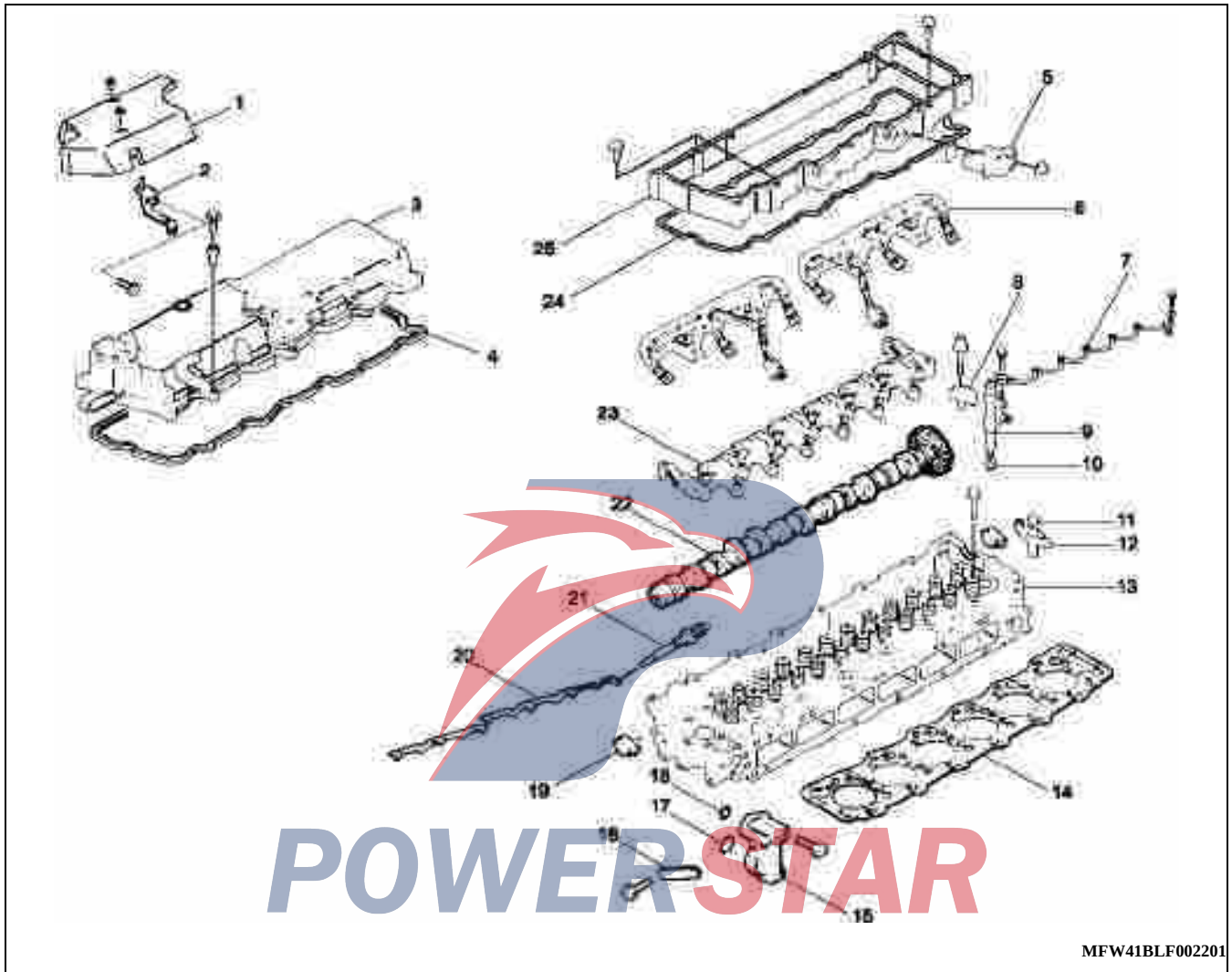
(e) 44 N m (4.5 kgf m/33 lb ft)

(f) 73 N m (7.4 kgf m/54 lb ft)

(g) 76 N m (7.7 kgf m/56 lb ft)

Cylinder head

Part



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Legend

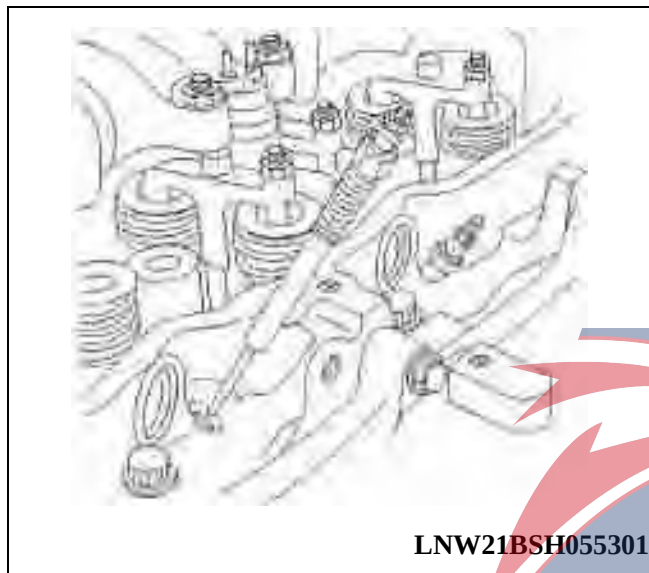
- | | |
|----------------------------|--------------------------|
| 1. Anti noise cover | 14. Cylinder head gasket |
| 2. Bracket | 15. Water pipe |
| 3. Top cover | 16. Plate |
| 4. Gasket | 17. O-ring(Water pipe) |
| 5. Connector | 18. O-ring |
| 6. Wire harness assembly | 19. Rubber stopper |
| 7. Ejector discharge tube | 20. Glow plug connector |
| 8. Ejector clamp | 21. Electric glow plug |
| 9. Ejector assembly | 22. Camshaft |
| 10. Gasket | 23. Rocker arm |
| 11. Bridge cover | 24. Gasket |
| 12. Bridge frame | 25. Under the box |
| 13. Cylinder head assembly | |

Tear down

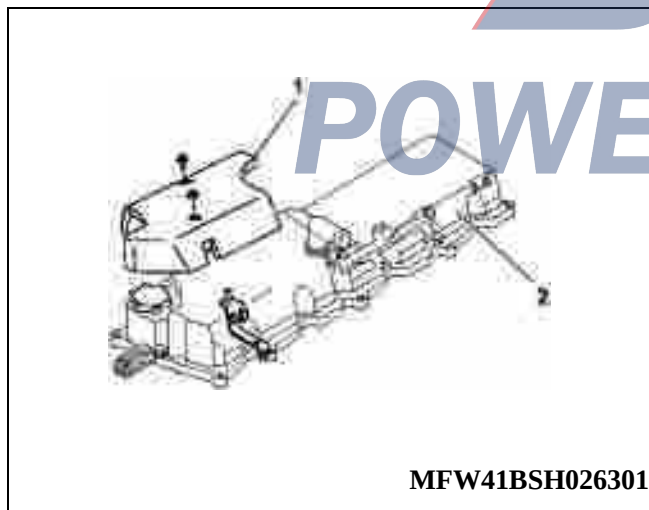
Note:

Seal inlet holes, feed pumps, and other locations to prevent debris from entering.

1. Remove the spray tube.
2. Remove the glow plug connector.
3. Remove the glow plug.



4. Remove the noise protection cover.



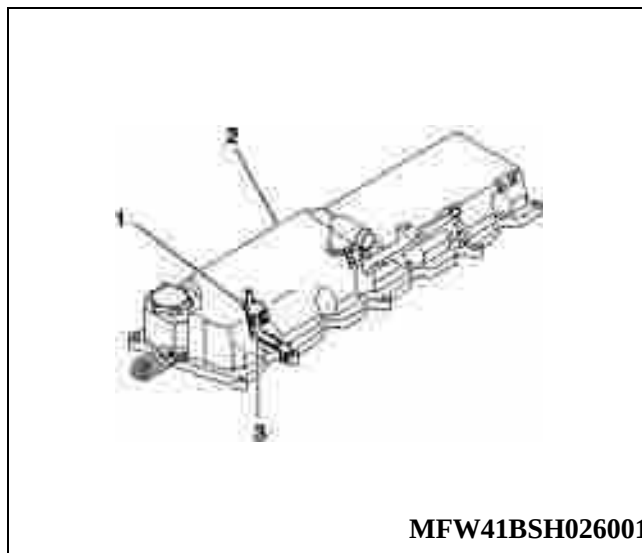
Legend

1. Anti noise cover
2. Top cover

5. Remove the noise cap bracket.

Note:

If the cradle is not damaged, do not remove the cradle.



Legend

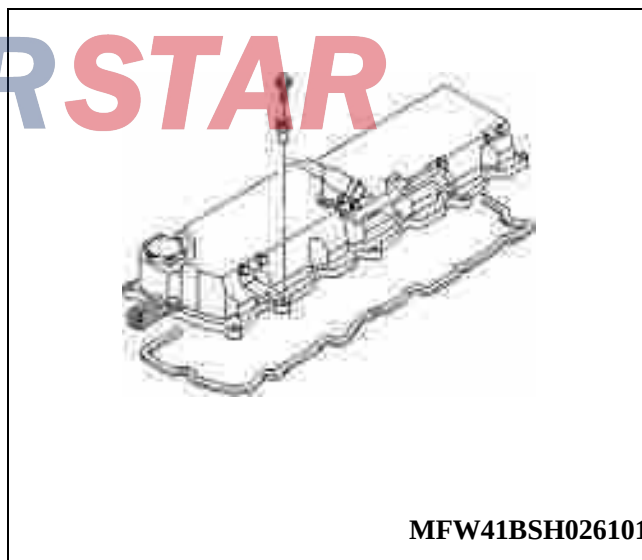
1. Bracket
2. Top cover
3. Bolt

6. Remove the top cover.

Note:

Make sure the collar inside the top cover is not lost.

7. Remove the gasket.

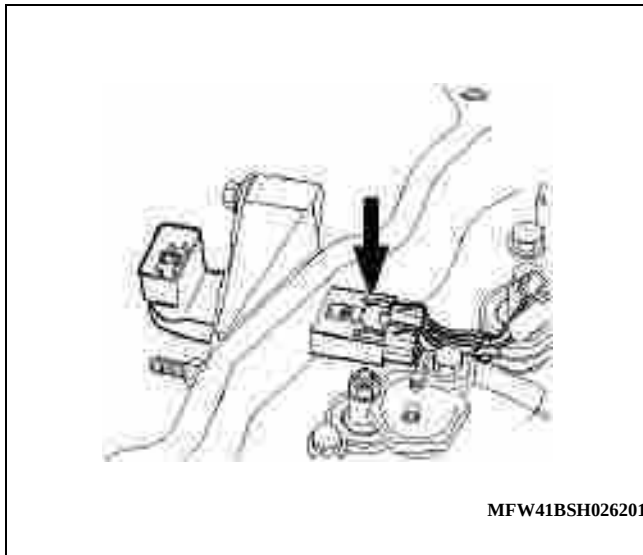


8. Remove the harness assembly.

- Use your fingers to push the arrow part of the connector to remove it.

Note:

Do not pull the wire or use a screwdriver to pry up the connector.

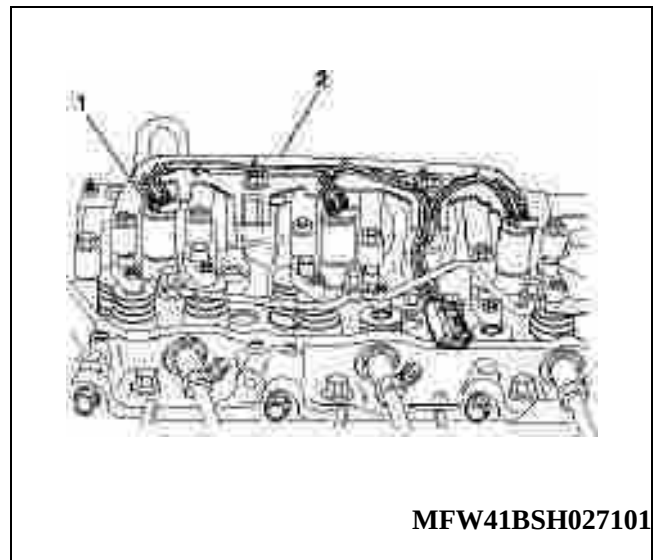


- Remove the injector harness terminal.

Note:

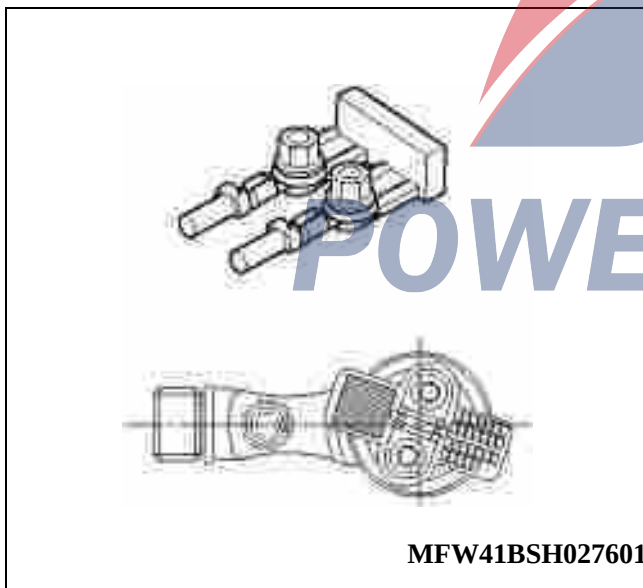
Since the terminal block is of a single type, the nut should be loosened in 2-3 steps.

- Remove the harness holder so that the harness remains attached to the harness holder.



Legend

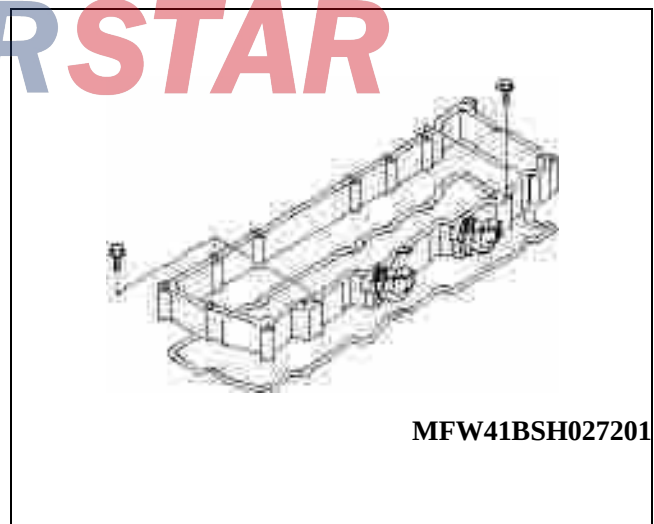
- 1. Harness terminal
- Harness bracket



- 9. If the connector does not malfunction, do not remove it.

- 10. Remove the lower cover.

- 11. Remove the gasket.

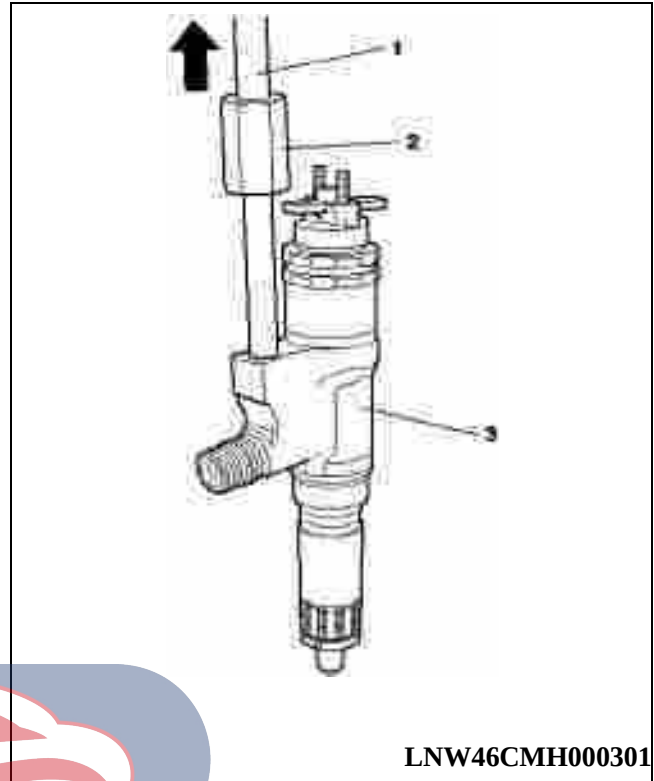
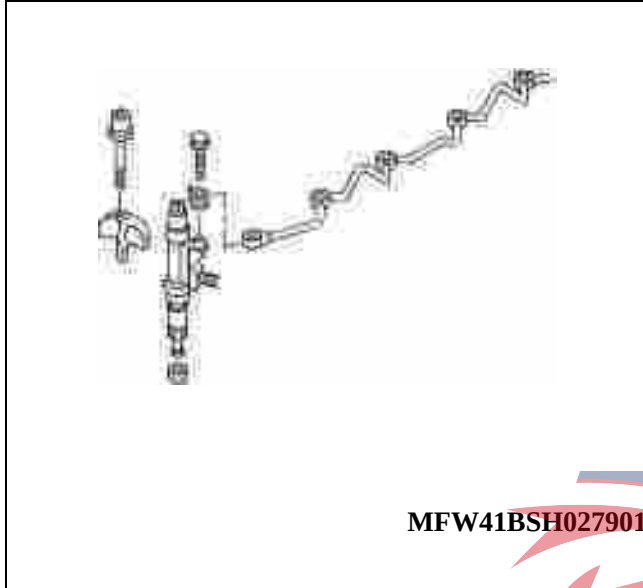


- 12. Remove the rubber stopper.

- 13. Remove the ejector exhaust pipe.

- 14. Remove the injector clip.

- Use a torque wrench (T-40) to loosen the injector clamping bolts.



15. Remove the injector assembly.

- Release the clamp and remove the sprayer with the replacement tool.
- If it is difficult to remove the injector, set the injector removal tool (5-8840-2826-0) on the injector, screw the bolt into the drain connector, and use the small slide hammer (5-8840-0019-0) Pull up the syringe.

Legend

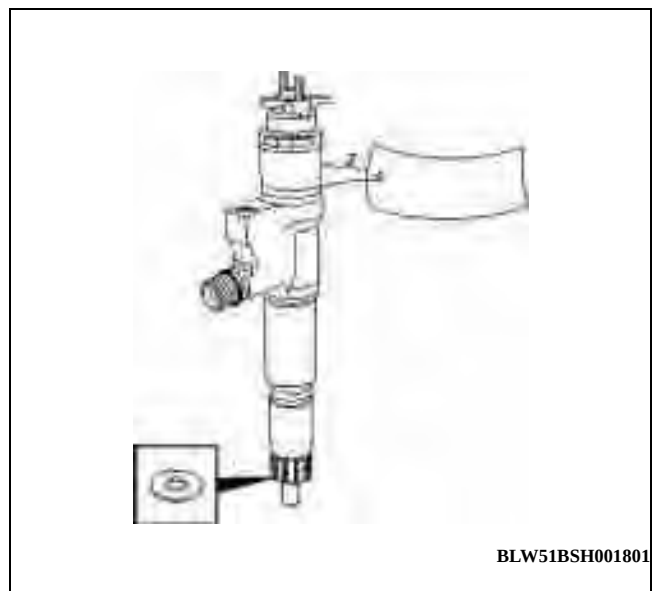
1. Small sliding hammer
2. Syringe removal tool
3. Ejector assembly

Note:

- The cylinder number should be indicated on the removed injector before depositing the injector.
- Be careful not to hit the nozzle against a hard surface.

Note:

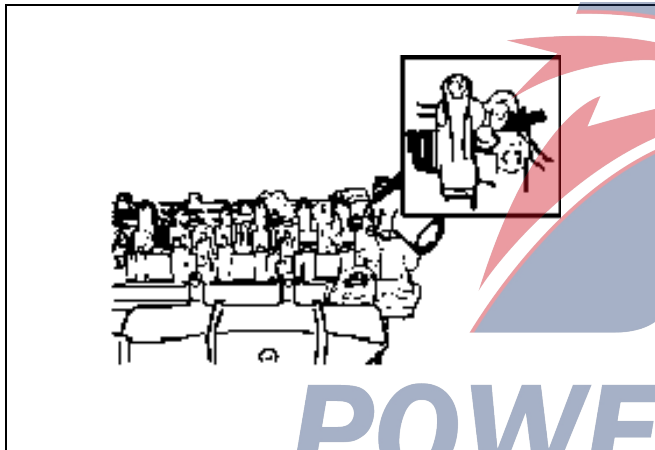
- When using this special tool to remove the injector, be sure not to remove the injector sleeve.
- Each ejector's identification code board is unique, so it should be affixed with a nameplate before storage to ensure that it will not be used for another ejector.
- Injectors and glow plugs should be extended first to prevent damage.



16. Remove the gasket.
17. Remove the plate.
18. Remove the water pipe.
19. Remove the O-rings.
20. Remove the O-ring (pump).
21. Remove rocker assembly.
- At the same time, loosen the cam rocker shaft brackets and rocker bracket bolts on both sides and remove them.

Note:

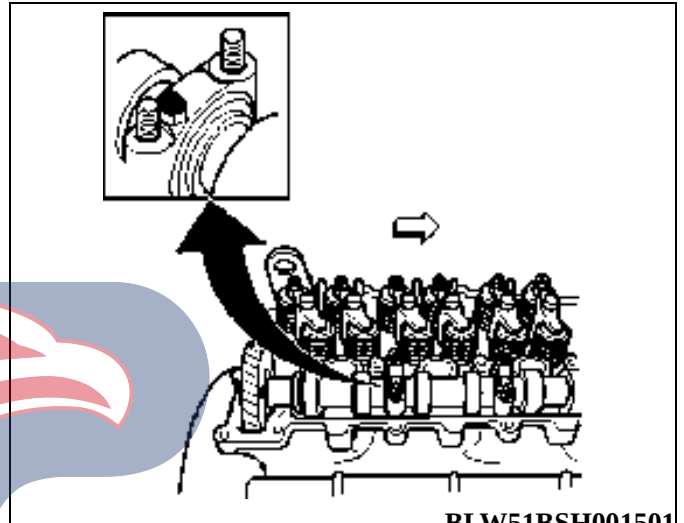
If the bolts at the ends of the camshaft rocker bracket (as indicated by the arrow) have been removed, the rocker arm and bracket may pop up. Therefore, they should not be dismantled



22. Remove the bridge and bridge cover.
- Please follow the installation sequence.
23. Remove the camshaft.

Note:

After checking the removed bearing, temporarily remove the removed bearing cap and bearing to avoid confusion with other parts.

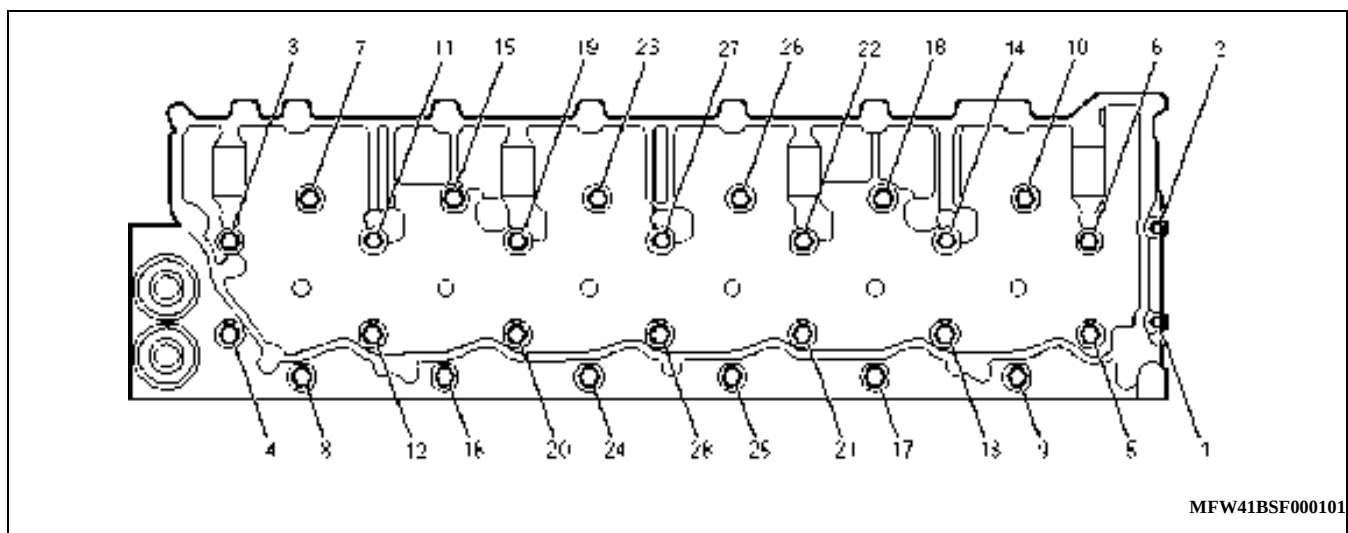


24. Remove the cylinder head assembly.

- Loosen and remove the connecting bolts from the outside according to the numerical sequence in the figure.

Note:

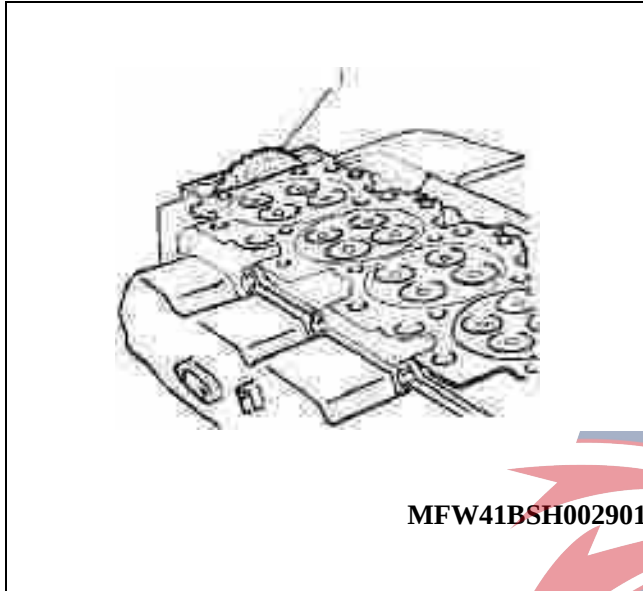
Before lifting the cylinder head, make sure all bolts have been removed.



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Note:

After removing the cylinder head, idler C (1) will protrude from the bottom of the cylinder head. Make sure the gear is not damaged.

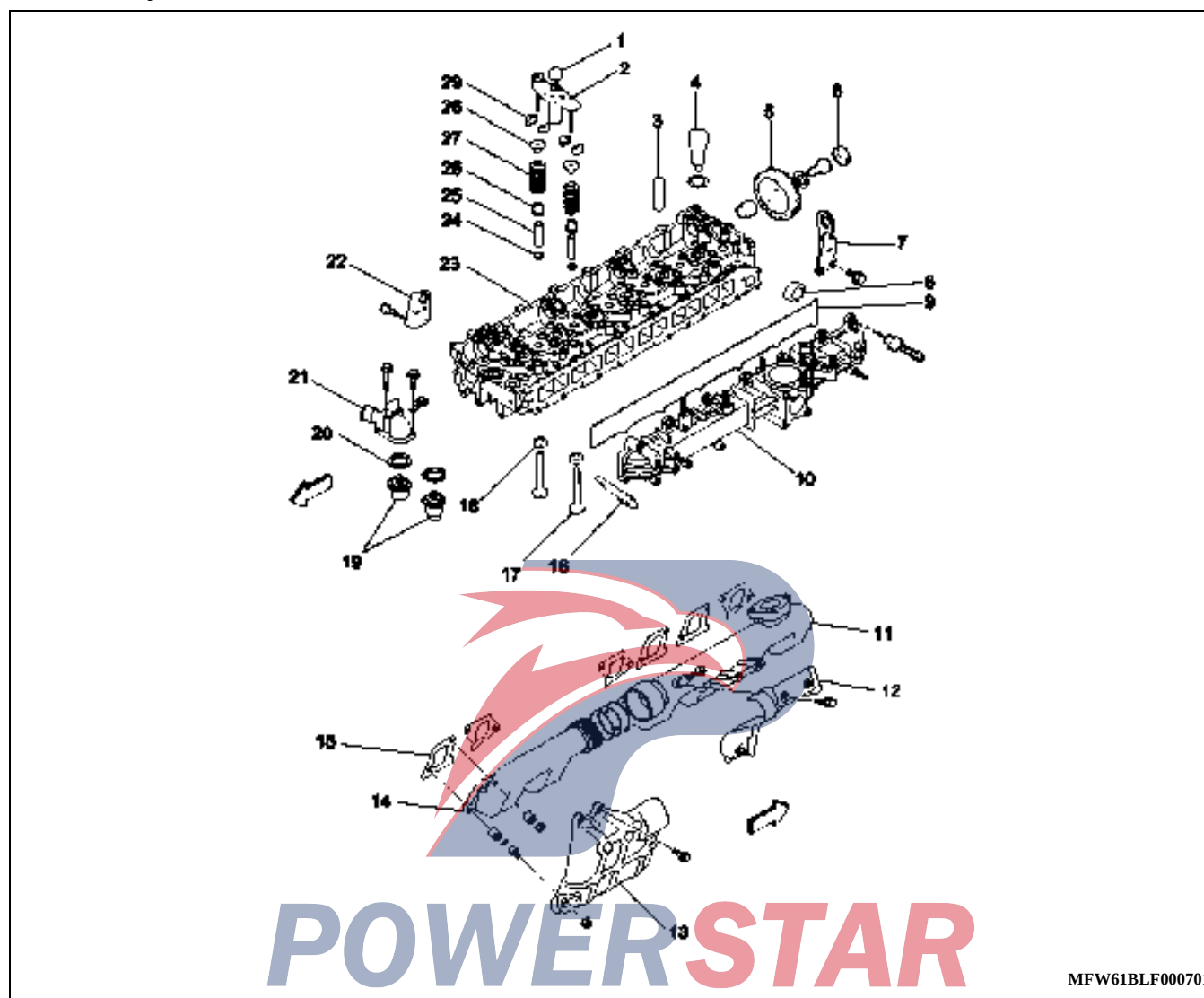


25. Remove the cylinder head gasket.

- Check for gas or water leaks.

POWERSTAR

Disassembly



MFW61BLF000701

Legend

- | | | | |
|-----|----------------------|-----|-----------------------|
| 1. | Bridge cover | 16. | Double thermal sensor |
| 2. | Bridge | 17. | Valve |
| 3. | Bridge Guide | 18. | Valve seat insert |
| 4. | Nozzle cover | 19. | Thermostat |
| 5. | Idler gear C | 20. | Gasket |
| 6. | Idler gear C cover | 21. | Outlet pipe |
| 7. | Engine hanger | 22. | Engine hanger |
| 8. | Oil seal | 23. | Cylinder head |
| 9. | Gasket | 24. | Spring seat |
| 10. | Entrance cover | 25. | Valve guide |
| 11. | Exhaust manifold(Ft) | 26. | Valve pad seal |
| 12. | Heat protector | 27. | Valve spring |
| 13. | Heat protector | 28. | Upper spring seat |
| 14. | Exhaust manifold(RR) | 29. | Shedding axle |
| 15. | Gasket | | |

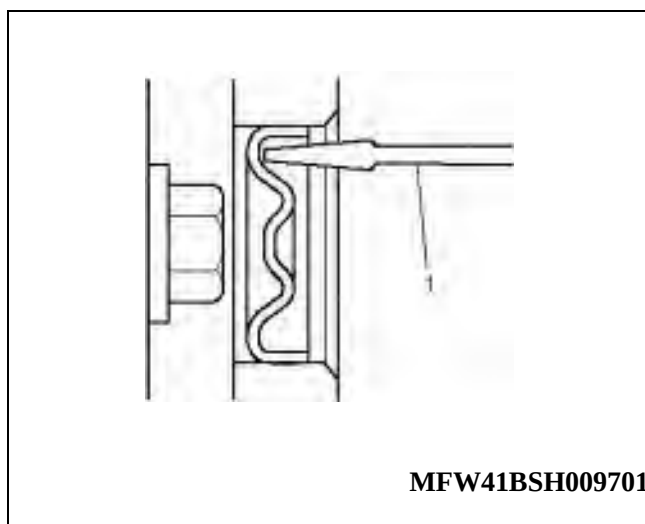
Note:

When the cylinder head assembly is placed on the ground, it is placed on a wooden strip to prevent the idler C protruding from the bottom from being damaged.

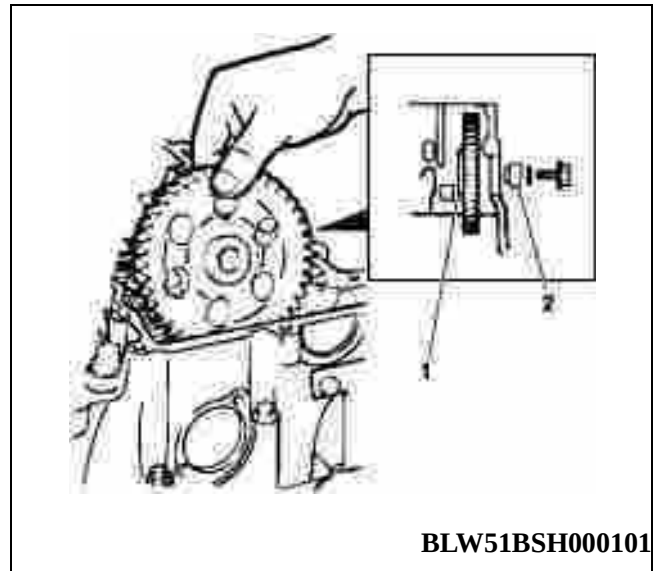
1. Remove the bridge cover.
2. Remove the bridge.
3. Remove the air intake cover.
4. Remove the gasket.
5. Remove the thermal protector.
6. Remove the exhaust manifold.
7. Remove the gasket.
8. Remove the bracket.
9. Remove the thermal protector.
10. Remove the engine hanger.
11. Remove the dual thermal sensor.
12. Remove the outlet tube.
13. Take out the incubator.
14. Remove the gasket.
15. Remove the idler C cover.
- Tap the outer circumference of the seal cup with the tip of a screwdriver (1), then flip the cup and pull it out with a tool such as pliers (as shown).

Note:

Do not damage the insert part of the lid.



16. Remove the idler C cover.
- Pull out the idler shaft and remove the gear.

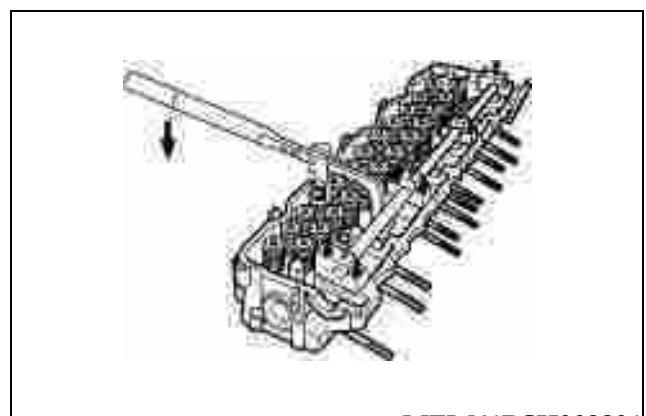


Legend

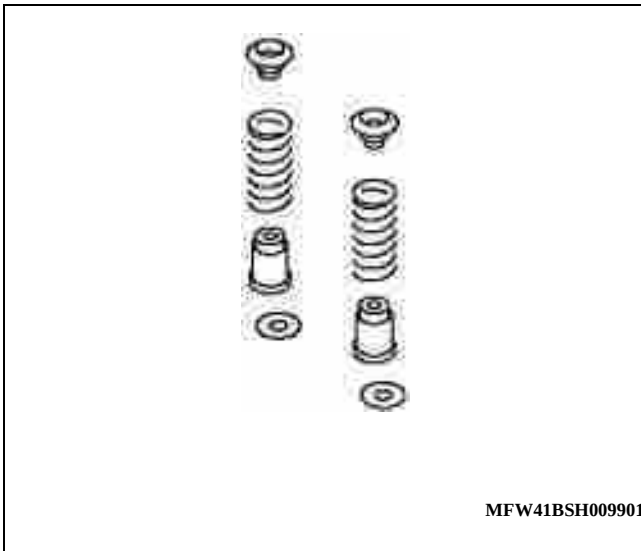
1. Idler gear C
2. Axis

17. Remove the opening collar.

- Place the cylinder head on a wooden board.
- Remove the split ring using the valve spring replacement tool (5-8840-2621-0) and the shaft (8-9439-6862-0) and remove the spring seat, spring, seal and valve. At this time, the spring seat on the lower side of the spring should also be removed.
- Attach the label numbered by cylinder to the removed valve.

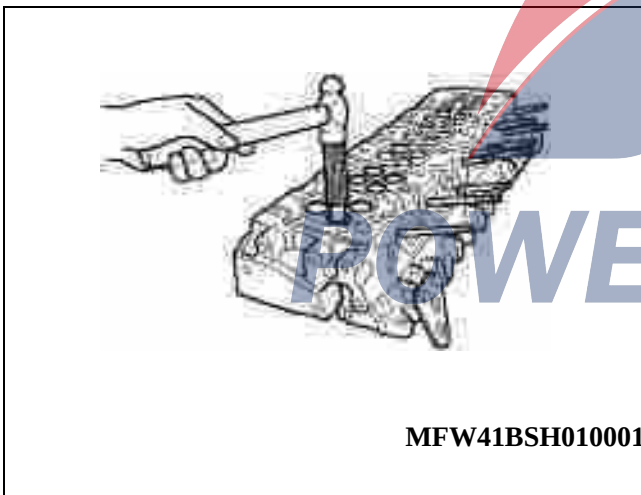


18. Remove the upper seat.
19. Remove the valve spring.
20. Remove the valve.
21. Remove the stem seal.
22. Remove the valve spring seat.



23. Remove the valve guide.

- Insert the valve guide changer (5-8840-2628-0) at the bottom of the cylinder head into the valve guide and pull it out with a hammer.

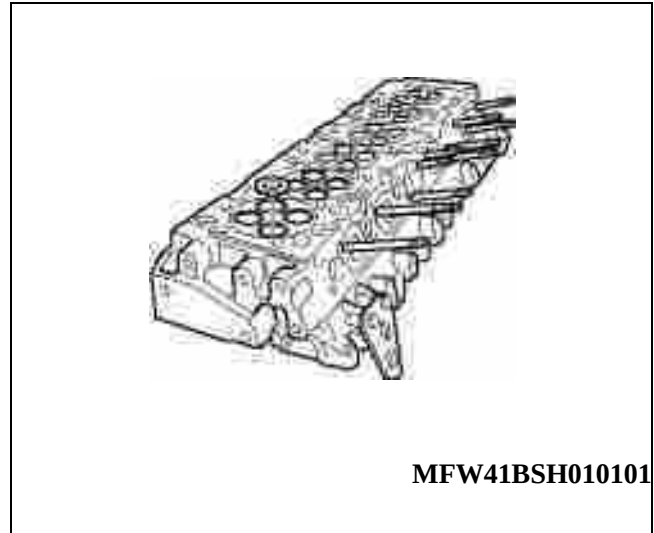


24. Remove the seat plug-in

- Use a welding torch to burn off the two parts of the insert's inner surface (700°C / 1292°F - 800°C / 1472°F) and let it cool for 3-5 minutes. Then use a screwdriver.

Caution:

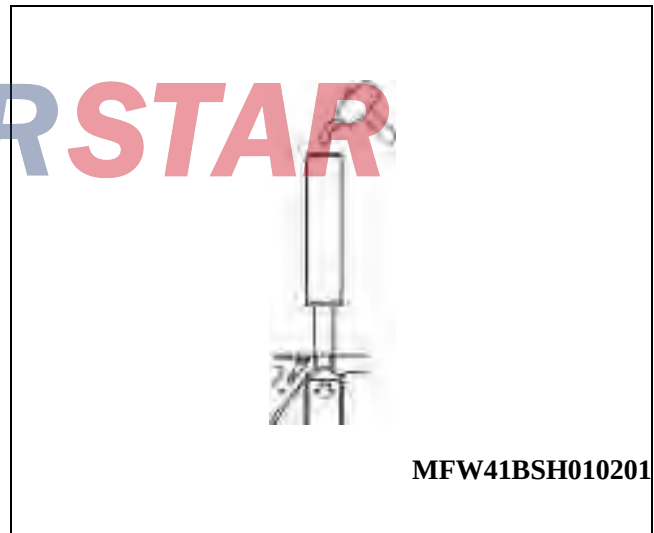
Do not accelerate cooling particularly. Do not damage the cylinder head when withdrawing.



25. Remove the nozzle sleeve. Use the nozzle sleeve removal tool (5-8840-2623-0) against the bottom of the cylinder head and pull out the sleeve by tapping.

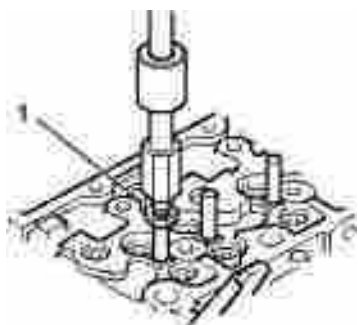
Note:

- Make sure that there is no damage to the bottom of the cylinder head.
- Do not use the extracted nozzle sleeve again.



26. Remove the bridge guide.

- Draw the electric welding bridge guide rod head nut (1) out with the sliding hammer.
- If it is not worn, please do not pull it out.



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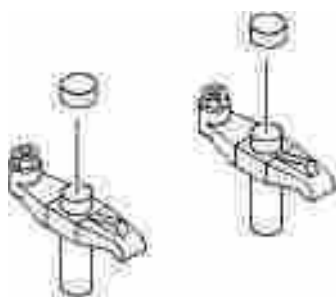
27. Remove the oil seal. Use a cart to hold the inside of the cylinder head and pull the seal out by tapping.

Inspection

1. Check the bridge cover.

- Check that the cover and rocker arm adjust the striking surface of the bolt to determine wear or damage.
- If wear is greater than 0.1 mm (0.004 inch) or linear wear occurs, replace it.

Bridge cover wear limit	mm(inch)
0.1(0.004)	



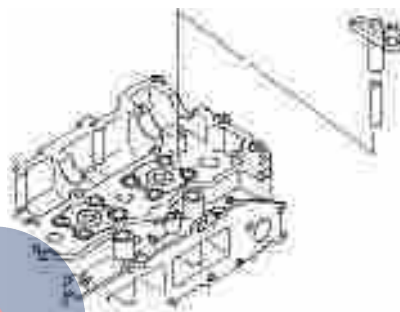
MFW41BSH010501

2. Check the bridge cover.

Make sure there is no blockage between the bridge and the bridge rails.

- Measure the gap between the bridge and the bridge rails.

Clearance		mm(inch)
Assembly standards	Use restrictions	
0.020(0.0008)–0.056(0.0022)	0.1(0.004)	

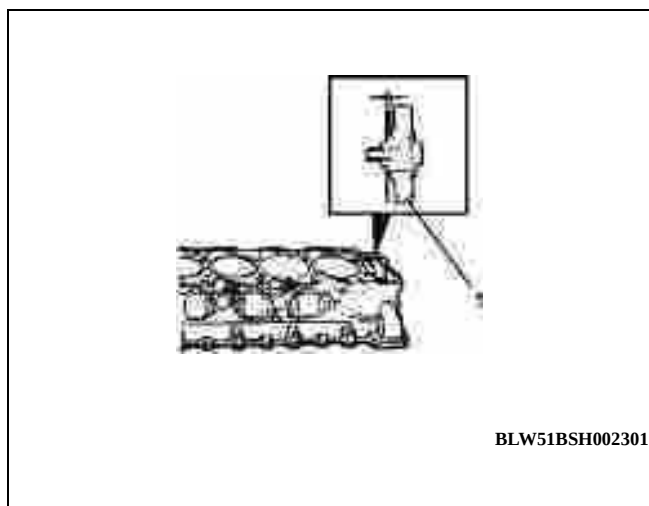


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3. Measure the axial clearance of the idler gear C(1).

- Measure the axial clearance and shaft diameter of idle gear C.

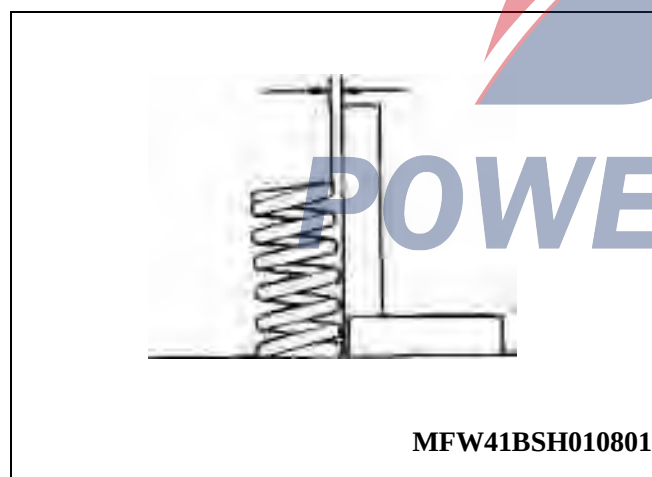
Axial clearance and shaft diameter		mm(inch)
Nominal size	Assembly standards	Use restrictions
—	0.09(0.0035)–0.144(0.0057)	0.2(0.0079)
30(1.1811)	—	29.85(1.172)



4. Measure the valve spring.

- Measure the verticality of the valve spring.

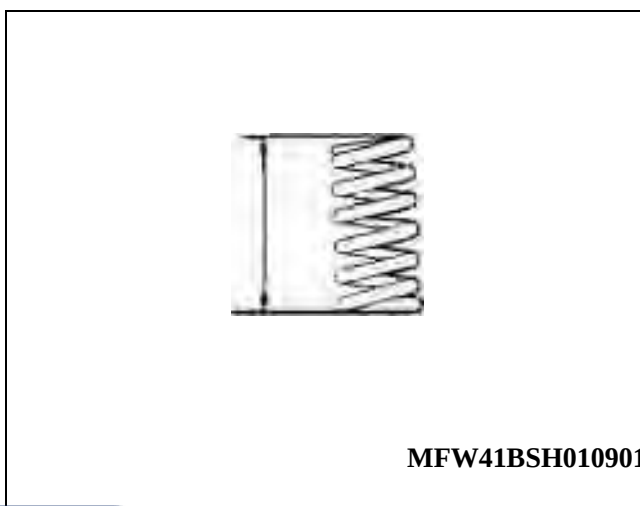
Verticality mm(inch)		
	Assembly standards	Use restrictions
Intake	—	2.6(0.1024)
Exhaust	—	3.0(0.1118)



- Measure the free length of the valve spring.

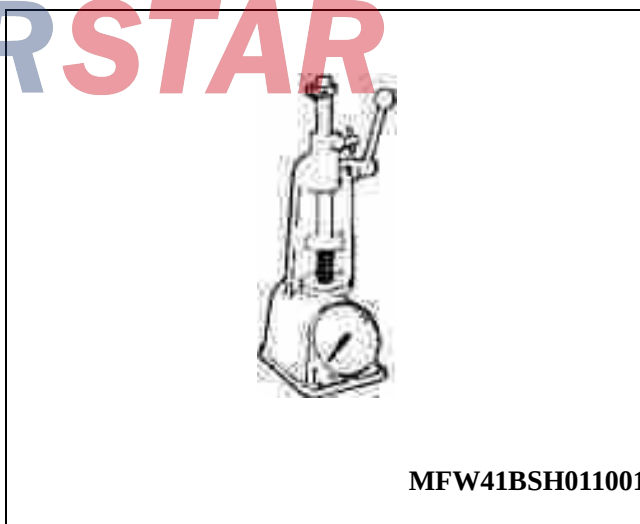
Free length mm(inch)		
	Nominal size	Use restrictions
Intake	59.9(2.3583)	58.7(2.311)

Free length mm(inch)		
Exhaust	68.1(2.681)	66.7(2.626)



- Measure the load on the valve spring.

Load / extra length (kgf / lb ft)		
	Assembly standards	Use restrictions
Intake	333(34.0/246)/46mm(1.811inch)	309(31.5/227.8)/46mm(1.811inch)
Exhaust	383(39.0/282.5)/46mm(1.811inch)	329(33.5/242.3)/46mm(1.811inch)



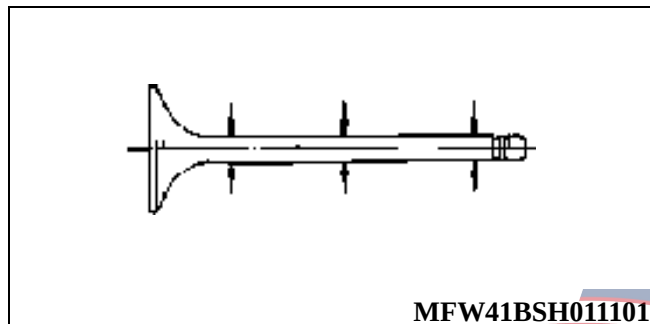
5. Check the valve.

- Measuring rod wear.

- Measured in three places shown in the figure.

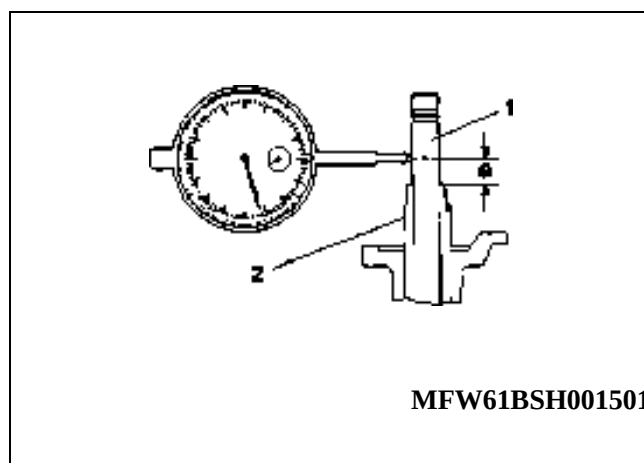
Stem outside diameter		mm(inch)
Nominal size	Use restrictions	
8(0.315)	7.88(0.310)	

- If the stem wear exceeds the limit, replace the valve and valve guide.



- Insert the valve into the valve guide and measure the looseness of the valve located at (a) 10 mm (0.3937 in) from the top of the guide.

Valve guide and stem clearance			mm(inch)
	Assembly standards	Use restrictions	
Intake valve	0.039(0.0015) — 0.072(0.0028)	0.2 (0.0079)	
Exhaust valve	0.064(0.0025) — 0.096(0.0038)	0.25 (0.0098)	

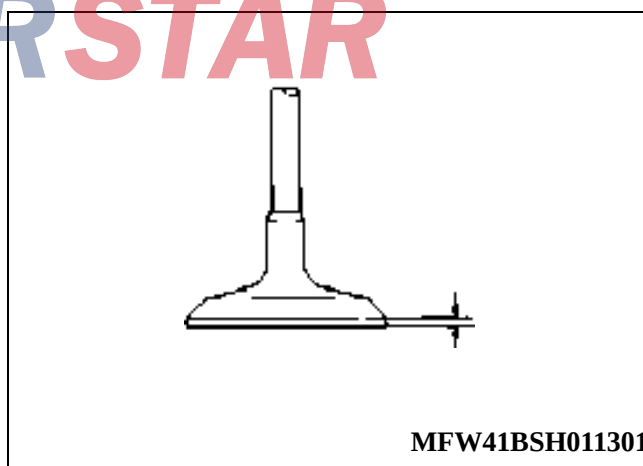


Legend

- Valve
- Valve guide rod

- Measure the thickness of the valve.

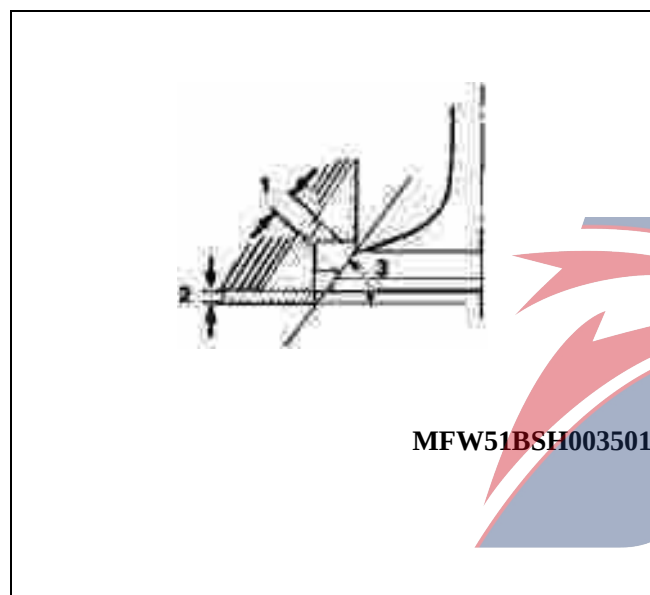
Valve thickness			mm(inch)
	Nominal size	Use restrictions	
Intake valve	1.71(0.0673)	1.3 (0.0511)	
Exhaust valve	1.75(0.0689)		



- Check and repair the seat impact surface.

- Apply the lead paint to the valve plug and check the width of the impact surface of the valve seat. Also check for impact and ensure that the impact location is at the center of the valve plate surface.

Valve impact surface valve impact width	mm(inch)
	Assembly standards
Intake valve	2.98(0.1173)
Exhaust valve	2.83(0.1114)



Legend

1. Shock width
2. Cave volume
3. Seat angle

- If there is any defect on the valve impact width or impact surface, the seat dressing tool matching with the stem diameter should be used to repair.

Seat angle	(°)
Intake valve	30°
Exhaust valve	45°

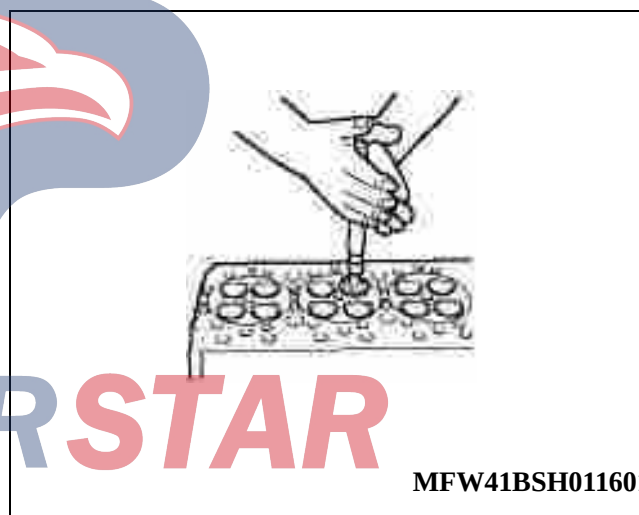
Valve stem diameter	mm(inch)
	8(0.315 inch)

- After the valve seat has been ground, measure the amount of valve sag to ensure it is within the usable range.

Valve cave volume	mm(inch)	
	Nominal size	Use restrictions
Intake valve	0.9(0.0394)	2.5(0.0984)
Exhaust valve	1.3(0.0512)	2.8(0.1102)

Valve sliding fit

- Apply the sealant on the impact surface of the valve and make sufficient sliding fit between the valve seat and the valve impact surface.

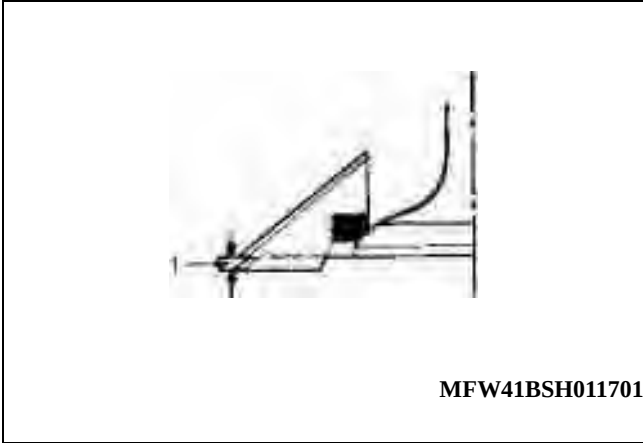


- After the valve slides, clean the cylinder head and valve.

6. Check valve seat insert.

- Measure the number of valve grooves (1).
- Insert the new valve into the valve guide and measure the dimensions from the bottom of the cylinder head to the valve surface.

Valve cave volume(1)	mm(inch)	
	Nominal size	Use restrictions
Intake valve	0.9(0.0354)	2.5(0.0984)
Exhaust valve	1.3(0.0512)	2.8(0.1102)



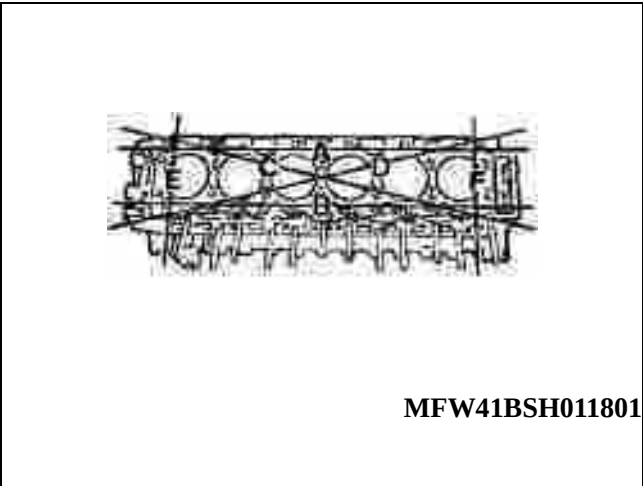
7. Check the cylinder head.

- Measure cylinder head deformation.
- Remove carbon deposits, do not damage cylinder head machining surfaces, valve seats, inserts, etc.
- Place a ruler on the mounting surface of the cylinder head and use a thickness gauge to measure the amount of deformation in each of the illustrations A to F.

Deformation quantity	mm(inch)
Use restrictions	0.2(0.0079)

Note:

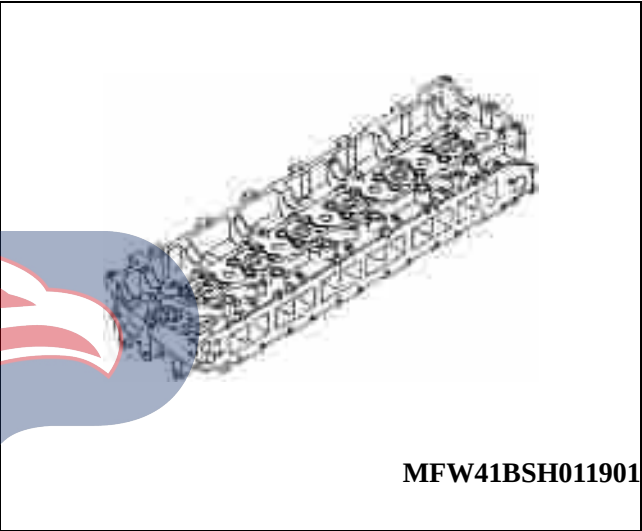
It cannot be repaired because the back seat of the timing gear will change.



- Check the cylinder head for cracks and damage.
- Use a magnetic flaw detector or dye penetration test (if necessary).

Note:

When magnetizing with a magnetic flaw detector, be sure to degauss after the test is completed. (Remanence less than 5G)



- In the hydrostatic test, apply 490 kPa (5 kg / cm / 71 psi) of water for 3 minutes and check for leaks in all parts of the cylinder head.

Note:

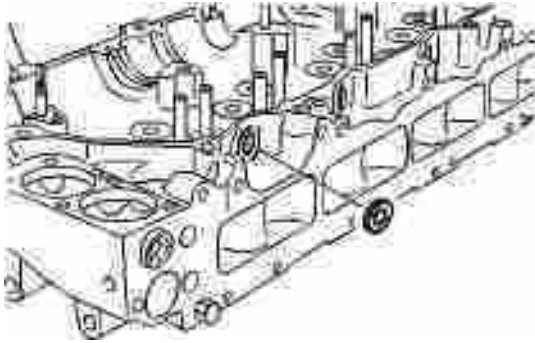
When performing a hydraulic test, the nozzle holder should remain connected.

Assembling

1. Install the oil seal on the cylinder head.
- The syringe insert can be installed using the oil seal installation tool (5-8840-2627-0).
 - Beat the seal unevenly to prevent it inclined.

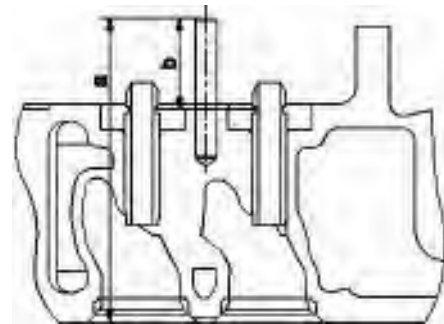
Note:

Be careful not to damage your lips.

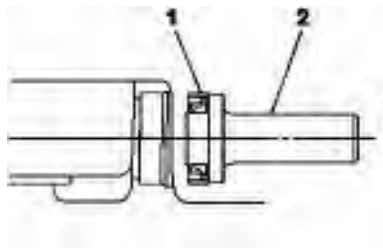


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- Apply oil to the outer periphery of the guide bar and use the bridge guide mounting tool (5-8840-2626-0) to tap into the depth of the cylinder head hole. (Reference: The height of the guide bridge above the cylinder head is approximately 37.5 mm (1.4763 in) (b)).



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MFW41BSH012101

Legend

- (a) 122.5 mm (4.8228 inch)
(b) 37.5 mm (1.4763 inch)

Legend

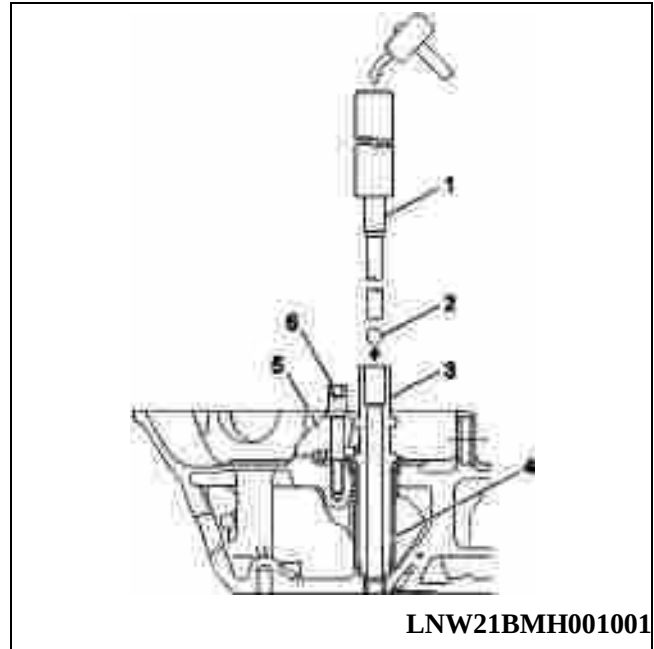
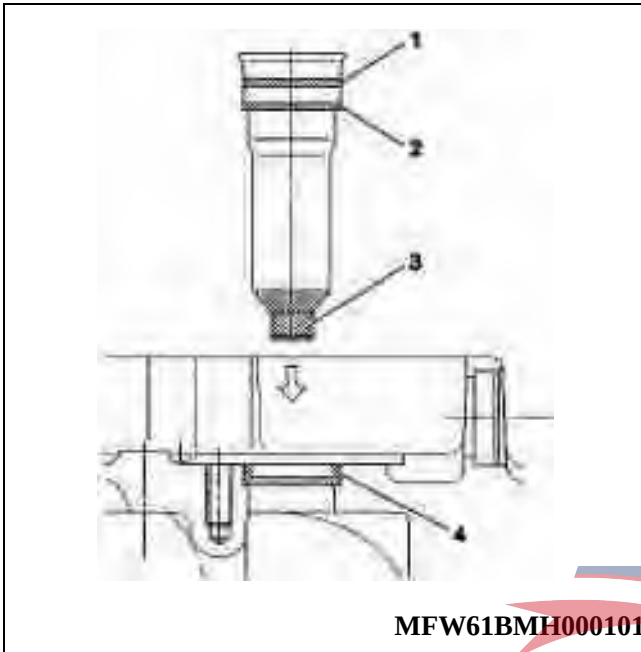
1. Oil seal
2. Oil seal installation tool

3. Install the nozzle sleeve.

- Install O-rings (1) and (2) on the nozzle housing and apply oil.
- Apply a sealant (Lactate brand LT620 adhesive) to the conical section (3) of the nozzle sleeve.

2. Install the bridge rails.

- Apply oil (4) to the nozzle sleeve mounting side.



Legend

1. Punching
2. Ball
3. Bushing
4. Nozzle sleeve
5. Clamping
6. Bolt

- Install the nozzle sleeve installation tool (5-8840-2624-0) on the cylinder head and install the nozzle sleeve.
- Push the nozzle sleeve with the guide sleeve so that it is firmly seated at the bottom of the cylinder head.
- After inserting the guide sleeve, fix it with clips and bolts.
- Insert the ball (bearing steel ball 9.525mm / 0.375 inch) into the guide sleeve and flush the ball with the punch and hammer.

4. Install seat inserts.

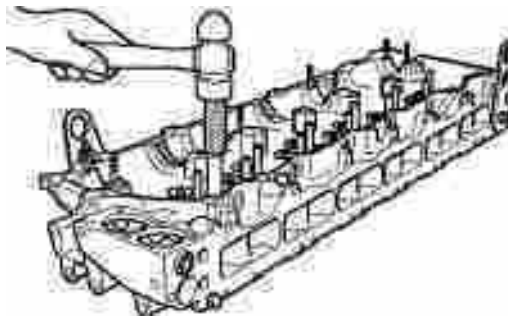
Note:

- Carefully remove the chips from the threaded part of the cylinder head and insert the sleeve.
- Lift the cylinder head so that the ball comes out from the bottom.

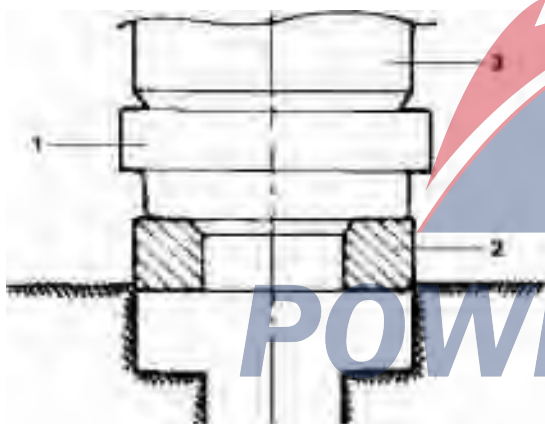
- Carefully remove carbon deposits, rust, etc. from the mounting surface of the plug and press the plug with the press.



MFW41BSH012401



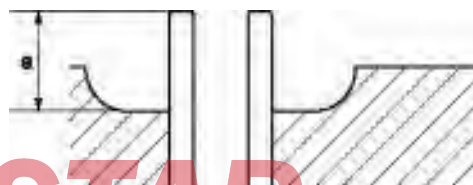
MFW41BSH0182501



LNW21BSH018601

Size of the valve guide surface (a) starting from the upper surface of the cylinder head.

19 mm (0.748 inch)



LNW71BSH001701

Legend

- Trolley
- Plug-in
- Press machine

5. Install the valve guide.

- Apply engine oil around the guide bar. Then, insert the guide ring into the valve guide replacement tool (5-8840-2628-0) and insert the guide rod, then strike the guide ring until it contacts the cylinder head surface.

Note:

Replace the valve guide with the valve as a whole.

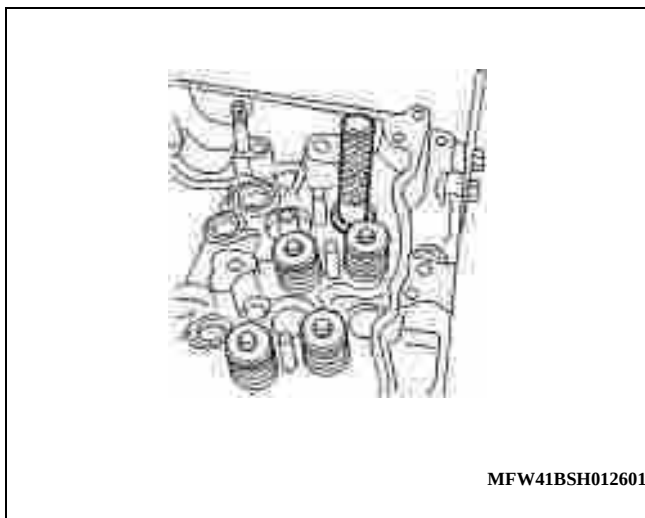
6. Install the spring seat.

7. Install the stem seal.

- Fill the oil seal and tap the valve guide using the mounting tool (5-8840-2625-0).

Note:

Check the valve seal for deformation or cracks and make sure the spring does not come off.



8. Install the valve.

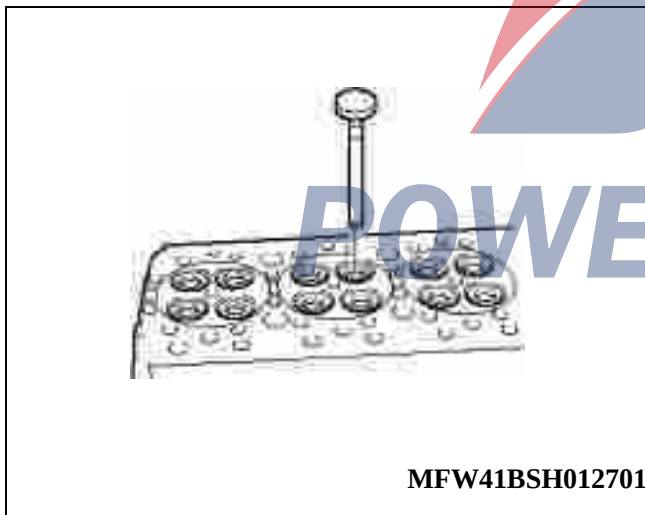
- Apply engine oil to the stem and insert the cylinder head.

Note:

Check valve components.

"E" mark: Exhaust

"I" mark: intake



9. Install the valve spring.

- Place the cylinder head on a wooden board.
- Install the valve spring so that one side of the marking material is facing the cylinder head surface.

	Identification paint
Inlet valve spring	Blue
Inlet valve spring	Red

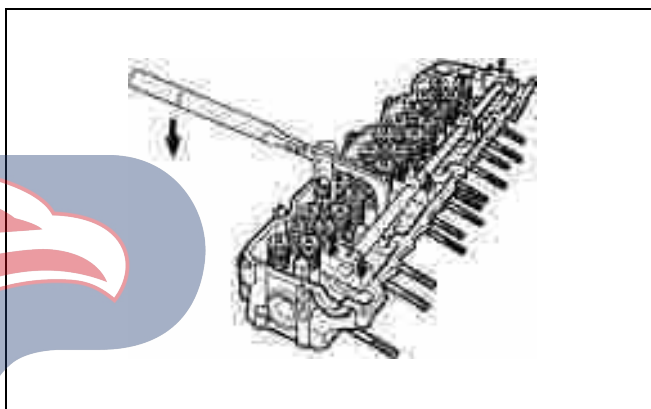
10. Put on the spring seat.

11. Install an open collar.

- Use the valve spring pressure tool (5-8840-2621-0) and shaft (8-9439-6862-0) to push in the valve spring and install the split ring.
- After installation, lightly tap the stem tip with a rubber hammer to secure the split ring in place.

Note:

Make sure that the split ring is firmly inserted into the groove.



12. Install the idle gear C cover.

- Set the engine to the top dead center of the No. 1 cylinder.
- In the inner periphery of the idler C, the idler's C-axis (outer circumference, in contact with the washer) and the washer are coated with engine oil. The shaft is mounted on the idler C and then the idler C is mounted on the upper part of the cylinder head. Do not drop the gear, otherwise it will damage the gear coating.

The convex surface of the shaft should face the cylinder side.

- Apply molybdenum disulfide lubricant to the threaded part of the C bolt and the surface of the base of the idler gear.

Temporarily tighten the idler C to the base (hand tightening).

- For reference: Check torque is 0.23 N m (0.0235 kgf m / 2.04 lb ft)
- After tightening the bolts, check that the idle gear C is turning correctly.

d. Fix the cam bearing to the bottom of the cylinder head. Apply oil to the inner circumference of the bearing.

e. Install the camshaft. Align the white paint gear teeth (the alignment marks with the top upper side. From the rear surface of the engine, alignment marks should be on the right side.

f. Camshaft bearings are mounted on the camshaft covers 1, 2, 3 and 4, and then the engine oil is coated in the bearing.

g. Install No. 2 and No. 3 shaft cover. The part marked with an arrow is fixed on the Ft side of the engine.

Temporarily install the rocker assembly (do not tighten the mounting bolts on the rocker bracket).

Install and tighten No. 2, No. 3 and No. 4 camshaft caps.

Do not fix No.1 camshaft cover (Ft).

Do not tighten the rocker arm fixing bolt of the No. 4 cover.

Tightening torque

Apply oil to threaded parts and base surface	
Cover bolt	27 N m (2.8 kgf-m / 20 lb ft)
Cover nut	27 N m (2.8 kgf-m / 20 lb ft)

- It is not allowed to tighten the M8 bolt (two fastening nuts) of the No. 4 cover.

h. Put the clamp on the camshaft gear and apply the torque on the camshaft by weight to fix the rotating gear C on the engine's rotating gear B and camshaft gear.

Apply torque with the clamp and fully tighten the idler C bolts.

Tightening torque

Apply molybdenum disulfide lubricant on threaded parts and base surface	
Idler gear C bolt	77 N m (7.2 kgf m/52 pounds ft)

- Do not change the C-axis position of the idle gear when tightening. (The oil and shaft washers allow the bolt seat surface to slide and the shaft does not move).

i. Delete the clip and make sure that the kickback is "0".

Manually fix idler C to suppress its movement. Rotate the camshaft gear with one finger, and use a dial gauge to confirm the camshaft backlash as "0".

Place the dial type indicator gage near the meshing pitch circle in the tangential direction. The reading of dial indicator should be 0.05 millimeters (0.0020 inches) or less.

- For reference: The camshaft gear torque is 1 N m or less at kickback.

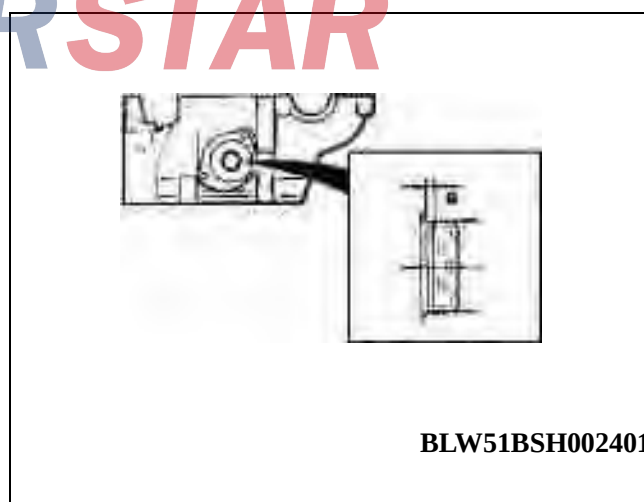
- If the clearance is not "0", the idle gear C is released and returns to step (h). Install the clamp on the camshaft gear and reposition the idler C.

Do not turn the crank in this step otherwise the coating of the idle gear C will fall off.

j. Unscrew the bolt and nut (Rr) of cam cap 4, then tighten the rocker assembly.

13. Install the idle gear C cover.

- Apply liquid filler (San key brand 1386 or Lactate brand adhesive 962T) over the entire perimeter of the insertion hole and tap the lid so that the edge of the cylinder bevel hypotenuse is 0.5 (0.0197) - 1.5 mm (0.0591 inch) (as the picture shows).



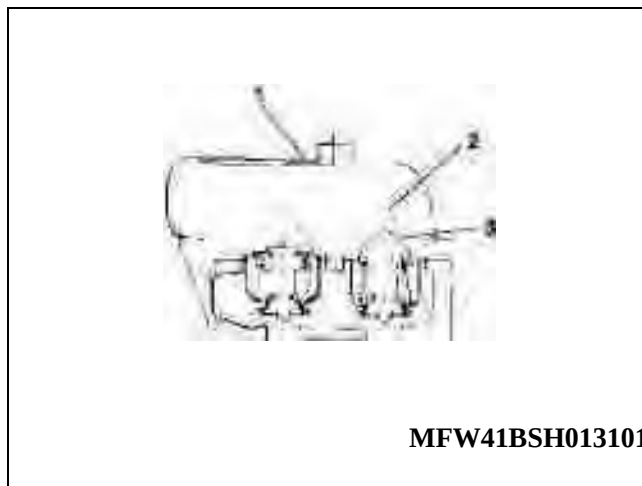
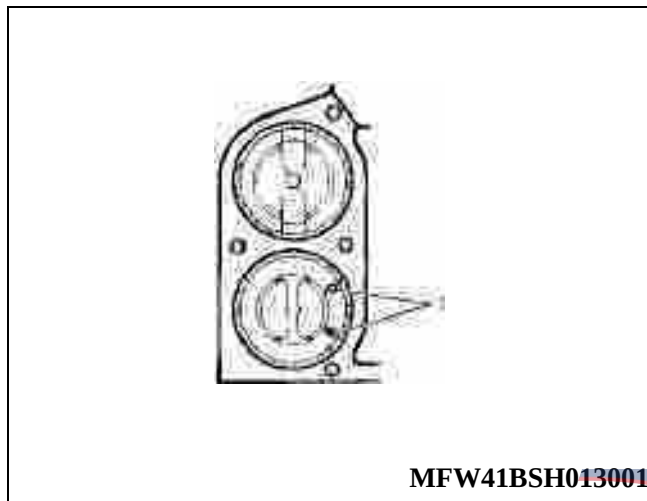
Legend

(a) 0.5(0.0197)—1.5mm(0.0591 inch)

14. Install the gasket.

15. Install the thermostat.

- Install a rubber gasket in the flange section.
- Align the position of the swing valve (1) as shown.



Legend

1. Outlet pipe
2. Anti-connection wrong plug
3. Incubator with rocking valve

16. Install the outlet pipe.

- Install the incubator and then install the outlet tube as shown.

Tightening torque	N m (kgf m/lb ft)
	24 (2.4 / 17)

17. Install dual thermal sensors.

- Secure the plug in place as shown (1).

Tightening torque	N m (kgf m/lb ft)
	25 (2.5 / 18)

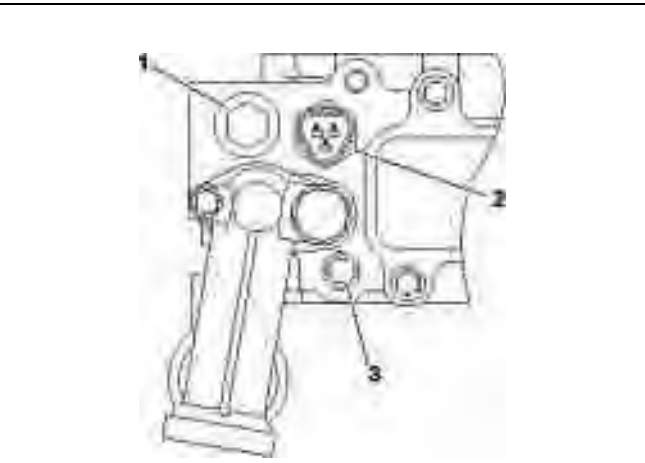
Note:

If the thermostat is installed in the wrong position, the outlet pipe cannot be installed.

- Secure the two-way thermal sensor in place (2) as shown.

Tightening torque	N m (kgf m/lb ft)
	20 (2.0 / 14)

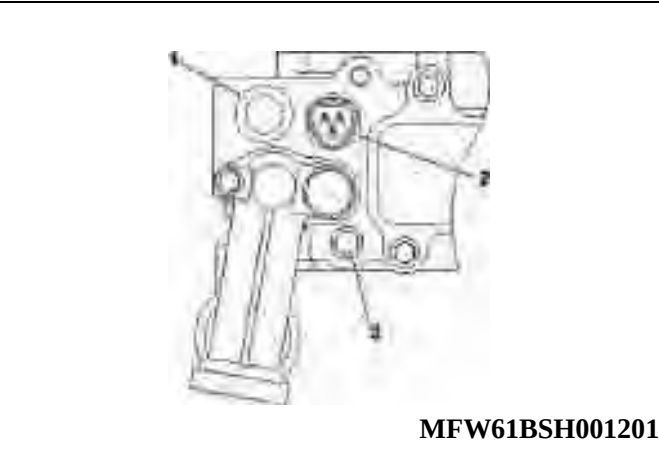
- Apply 1207C or Lactate 271 adhesive to threaded parts and bond.



Legend

- 1. Plug
- 2. Dual thermal sensor
- 3. Plug

- Apply 1207C or Lactate 271 adhesive to threaded parts and bond.
- Install the plug switch in the position shown in the illustration.
- The mesh processing method handles the new plug switch thread portion. When using the plug switch again, coat it with THREE BOND or LOCTITE 271 bonding agent and install it to the position shown in the figure.



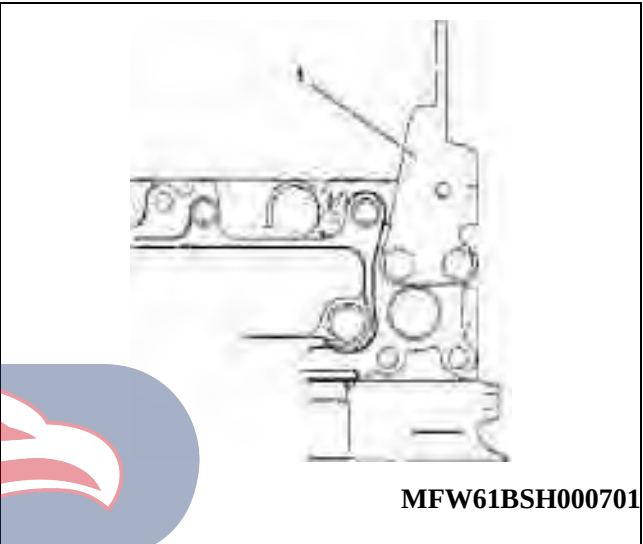
MFW61BSH001201

Legend

- 1. Plug
- 2. Dual thermal sensor
- 3. Plug

18. Install the rear engine pylon on the cylinder head.

Tightening torque (M12)	N m (kgf m/lb ft)
94 (9.6 / 69)	



MFW61BSH000701

Legend

- 1. Engine hanger (back)

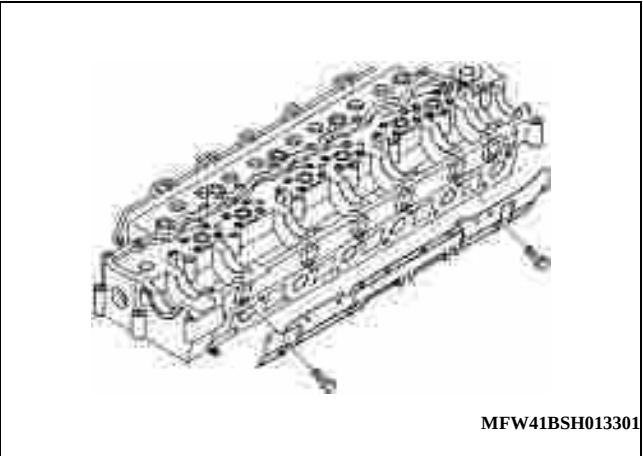
19. Install a thermal protector.

20. Mounting brackets.

Tightening torque	N m (kgf m/lb ft)
24 (2.4 / 17)	

Note:

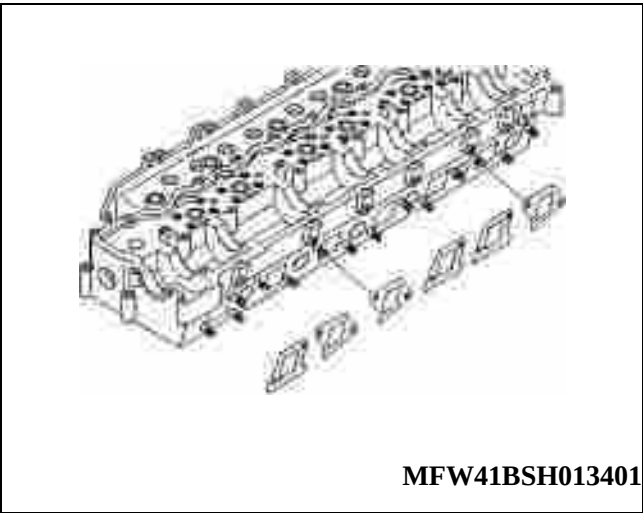
Install the exhaust manifold before installing it.



MFW41BSH013301

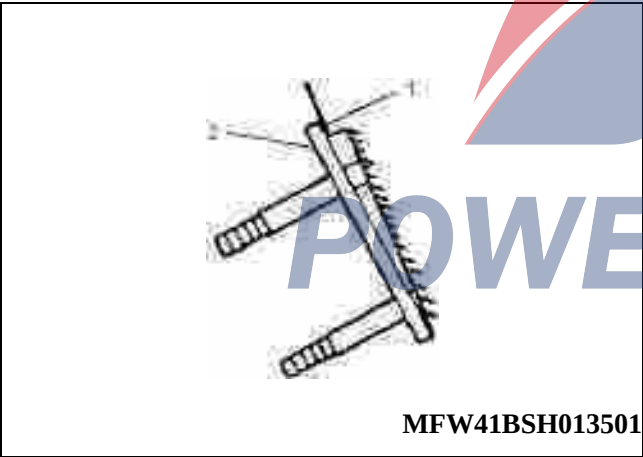
21. Install exhaust manifold gasket.

- The protruding part of the gasket should face down.



Note:

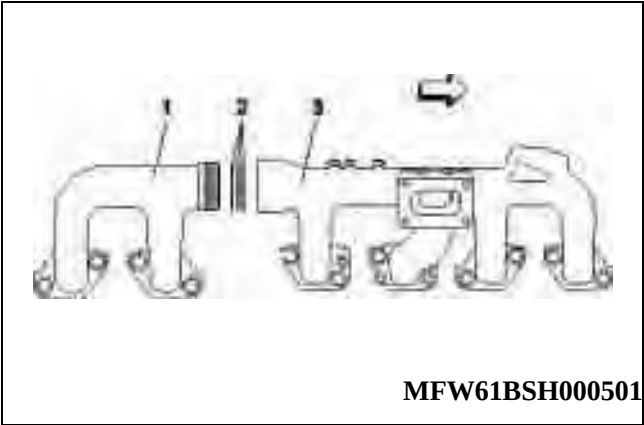
Make sure the gasket is firmly attached to the cylinder head. If it is not installed correctly, it will run to the side of the cylinder head casting surface (as shown).



Legend

1. Casting surface
2. Gasket

- Rotate the seal ring's positioning hole 120° to fit it on the rear exhaust manifold.



Legend

1. Exhaust manifold (rear)
2. Seal ring
3. Exhaust manifold (front side)

- Install the exhaust manifold.

Exhaust nuts (10 pieces)

Tightening torque	N m (kgf m/lb ft)
55 (5.6 / 41)	

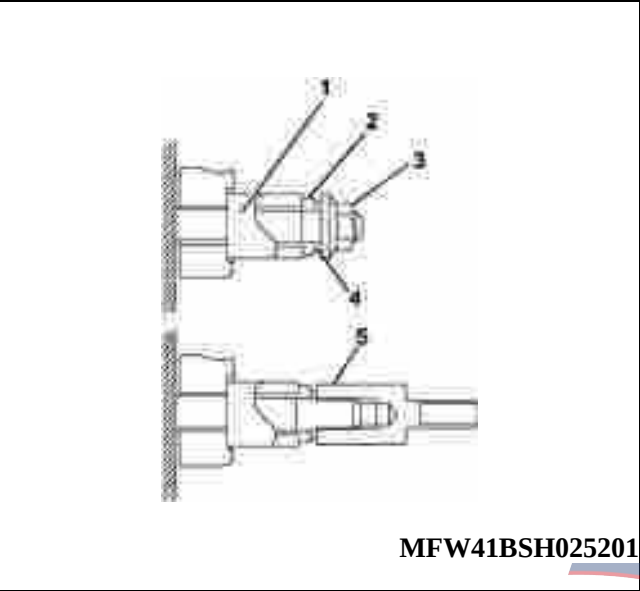
Exhaust isolation nut (2 pieces)

Tightening torque	N m (kgf m/lb ft)
34 (3.5 / 25)	

- Assemble the isolation tube, flat washer, and conical coil spring washer (as shown), then tighten the exhaust manifold.

22. Install the exhaust manifold.

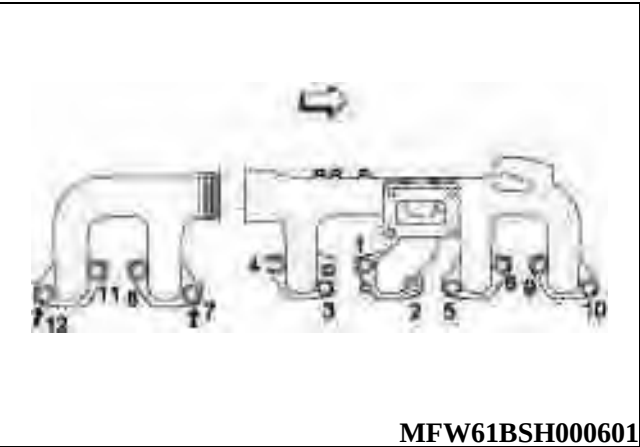
- Only the isolation nut can be used again.



Legend

1. Isolation tube
2. Flat washer
3. Nuts
4. Tapered spring washer
5. Isolation nut

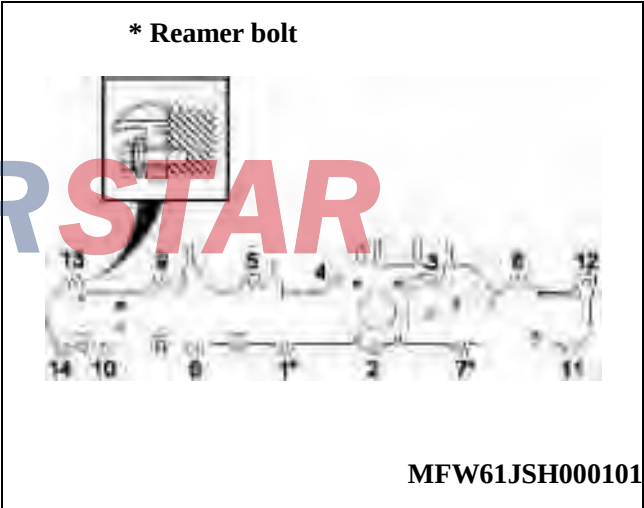
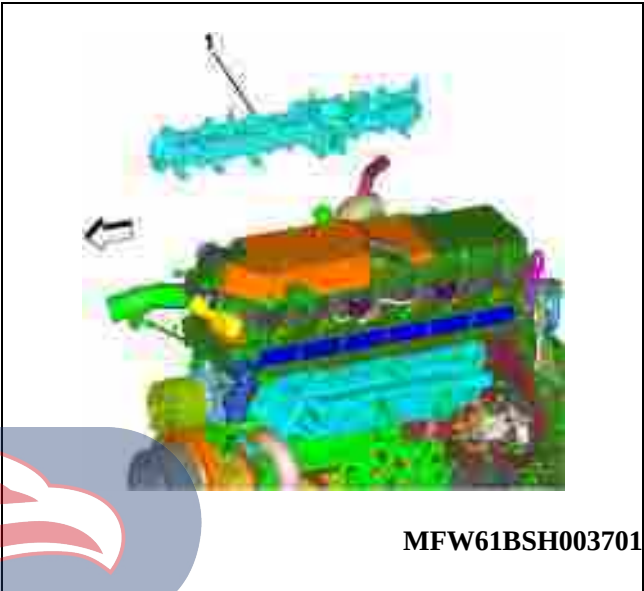
- Tighten the isolation nuts in position 7 and 12 shown in the figure (shown by the arrowhead).
- Tighten them in the order shown.



23. Install a thermal protector.
24. Install the entrance cover.

- Install gaskets on the inlet cover (1) and tighten them in the order shown in the diagram.

Tightening torque	N m (kgf m/lb ft)
25 (2.5 / 18)	

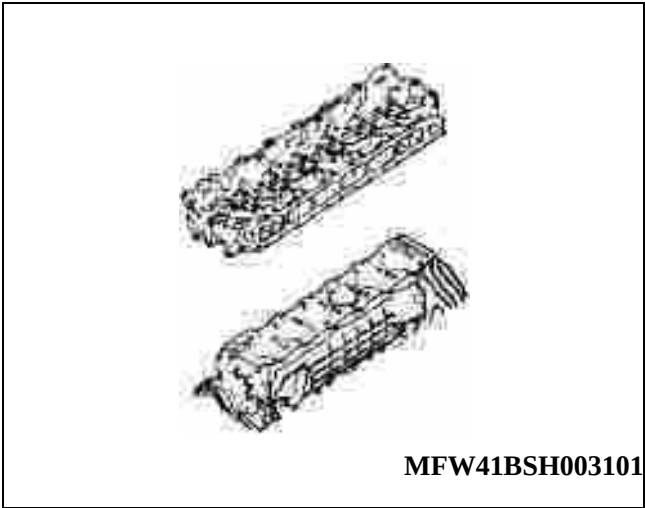
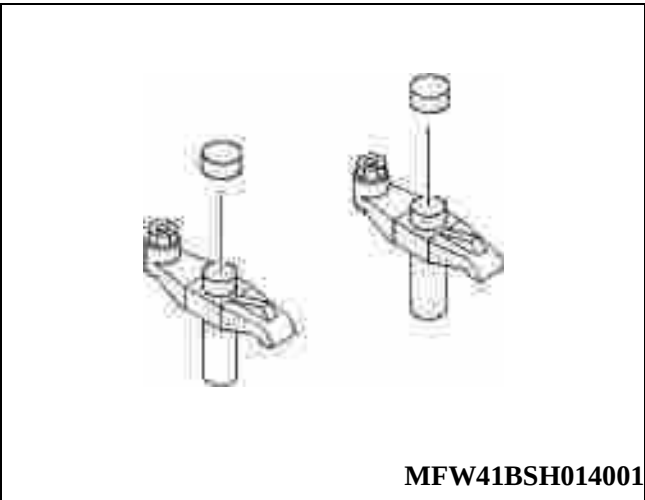


25. Apply engine oil to the bridge and install it.

Note:

Move up and down to ensure it moves smoothly.

26. Apply engine oil to bridge cover and install.



Reassemble

1. Install cylinder head gasket.

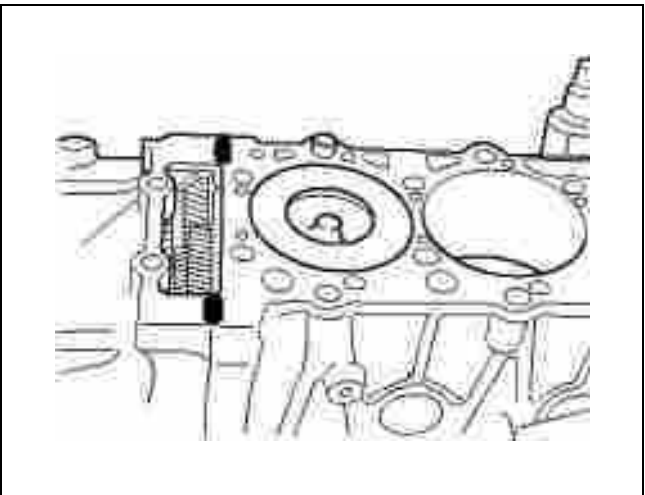
- Approximately 2 mm (0.0787 inch) thick, approximately 3 mm (0.1181 inch) wide liquid filler (Three Bond 1207C adhesive) is applied to the aligned side of the cylinder block and flywheel housing, and then the cylinder head gasket is installed. .

- The screw part and the supporting surface of the M14 bolt are coated with molybdenum disulfide lubricant, and the engine oil is coated on the thread part of the M10 bolt, and then the screws are tightened at a certain angle according to the specified torque and the order shown in the diagram.

Tightening torque	N m (kgf m/lb ft)
M14: 98(10.0/72)→147(15.0/108)→Angle method 30°—60°	
M10: 38(3.9/28)	

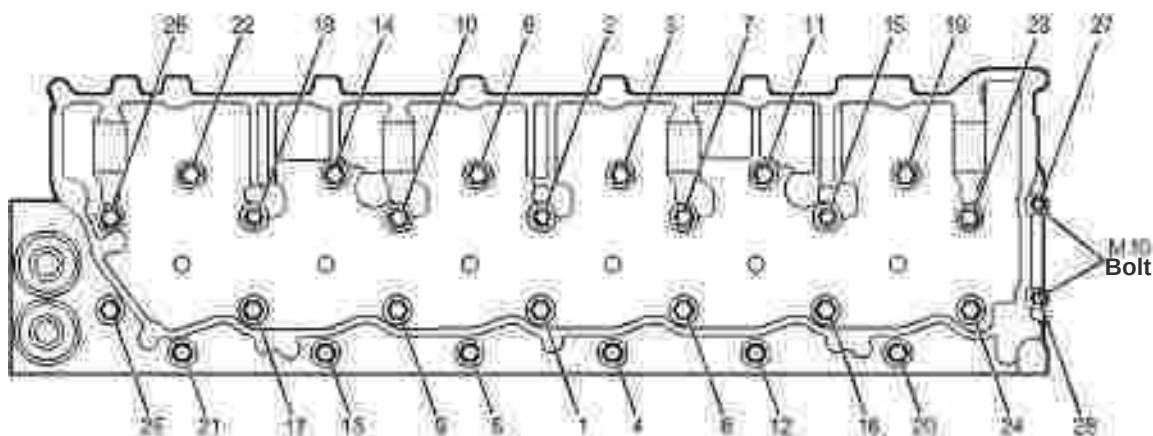
Note:

Remove deposits, carbon deposits and sealants on the upper and lower surfaces of the cylinder block.



2. Install the cylinder head assembly.

- Install a wooden pin on the cylinder block, align the bolt holes in the cylinder head, and gently open it.



MFW61BMF000701

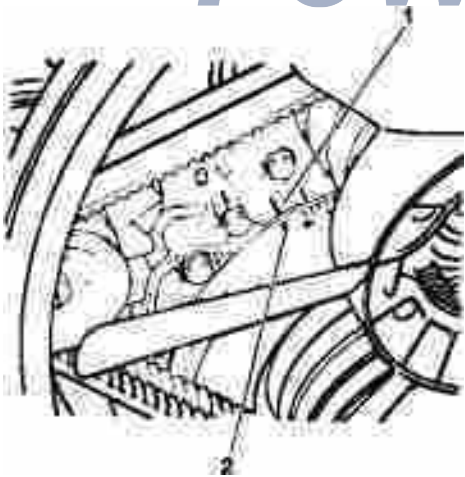
Note:

To check the thickness of M14 bolts, turn the tool in the tightening direction.

Tightening torque check	N m (kgf m/lb ft)
142 (14.5 / 105)	

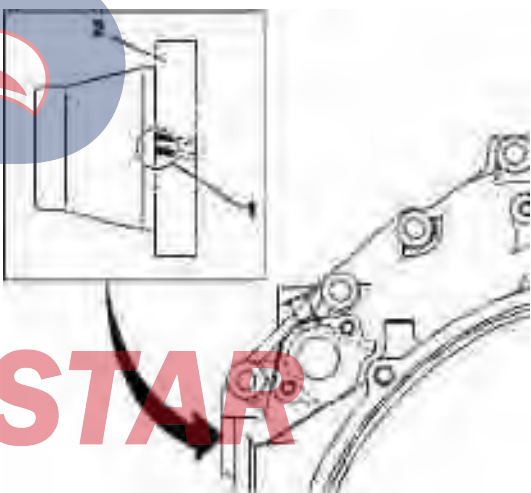
3. Install the camshaft.

- Rotate the crankshaft forward to align the dead center (2) on the crankshaft pulley with the needle (1).



MFW41BSH024901

- At this time, on the upper inclined surface of the feed pump gear, it is believed that the white paint alignment mark (as shown) can be seen from the center hole on the flywheel housing side (No. 1 cylinder compresses the dead point at the top).



MFW41BSH023801

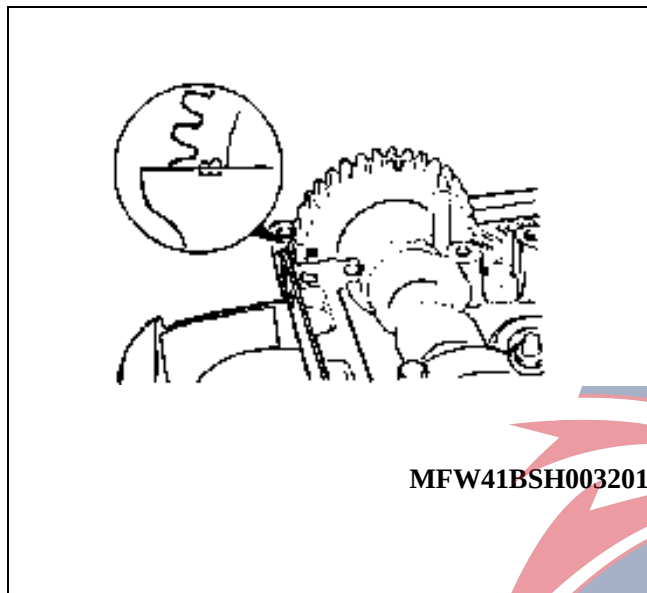
Legend

1. Center hole
2. Supply pump gear

- Clean the camshaft bearing and installation section, install the bearing and coat the engine oil to the bearing rotation plane.
- The idler gear C meshes with the camshaft gear so that the camshaft gear B is aligned with the top surface of the cylinder head (as shown), and then the camshaft is gently mounted.

Note:

- Make sure that the camshaft bearing is not damaged.
- Make sure the camshaft gear B mark is on the top of the cylinder head.
- If the camshaft gear B mark is not aligned with the top of the cylinder head, the B mark should be realigned.



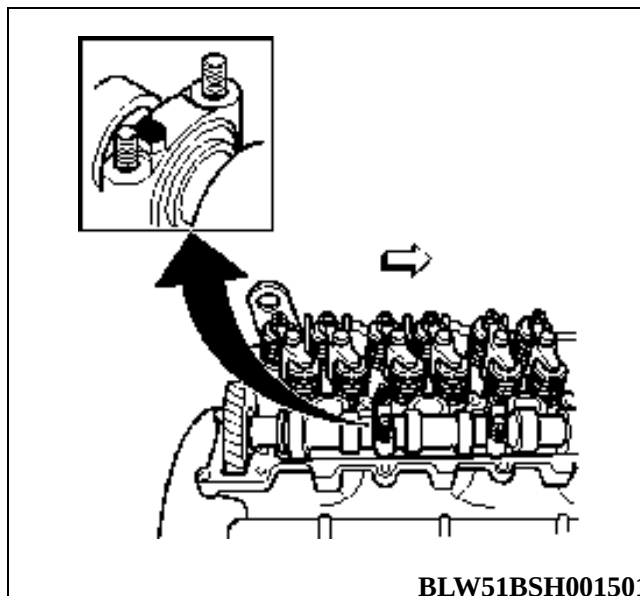
- Clean the camshaft bearing and shaft cover, assemble the bearing and the shaft cover, then apply oil on the rotating surface.
- Point the top of the camshaft bearing cap toward the arrow at the front of the engine and install them in numerical order. Tighten the sides of the shaft cover with the rocker arm.

Note:

Point the arrow at the top of the shaft cover to the front of the engine and make sure to install them in numerical order.

- Apply oil to threads and base sections and tighten evenly from the center.

Tightening torque	N m (kgf m/lb ft)
	28 (2.9 / 20)



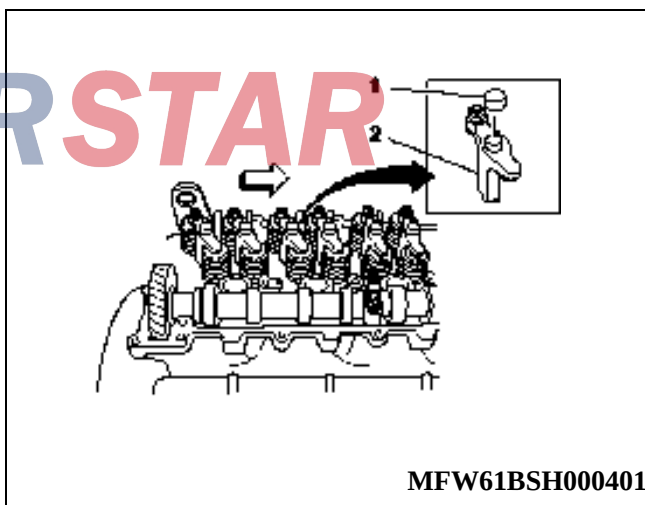
- Pull out the gear positioning pin. Be careful not to let it fall into the engine.

Note:

Do not forget to remove the gear positioning pin.

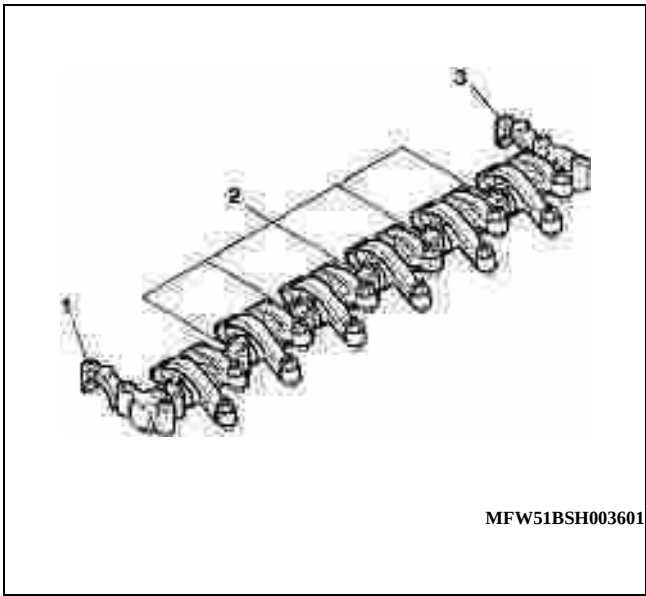
4. Install the bridge (2) and the bridge cover (1).

- Apply engine oil to the bridge and bridge cover and install.



5. Install rocker assembly.

- Clean the camshaft bearings and camshaft rocker shaft brackets, assemble the bearings and brackets, and then apply oil to the rotating surface.
- Tighten the rocker arm adjusting screw fully and coat the engine oil to the roller.
- Install the rocker assembly



Legend

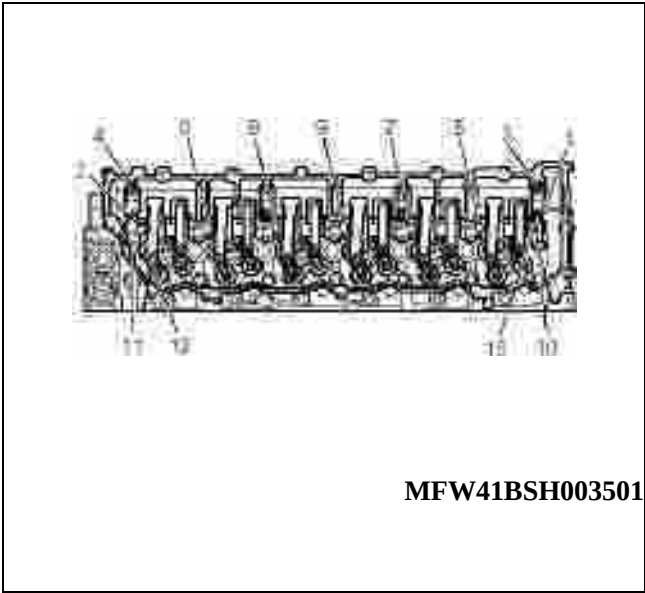
- 1. Front cam bracket
- 2. Rocker bracket
- 3. Rear cam bracket

- Tighten them in the numerical order shown in the figure.

Tightening torque	N m (kgf m/lb ft)
Cam rocker shaft bracket	27 (2.8 / 20)
Rocker bracket	56 (5.7 / 41)

Note:

- For camshaft bearing mounting parts, apply engine oil to stud parts, nut threads and bearing surfaces and tighten.
- For more information about valve clearance adjustments, see Valve Clearance Adjustment.

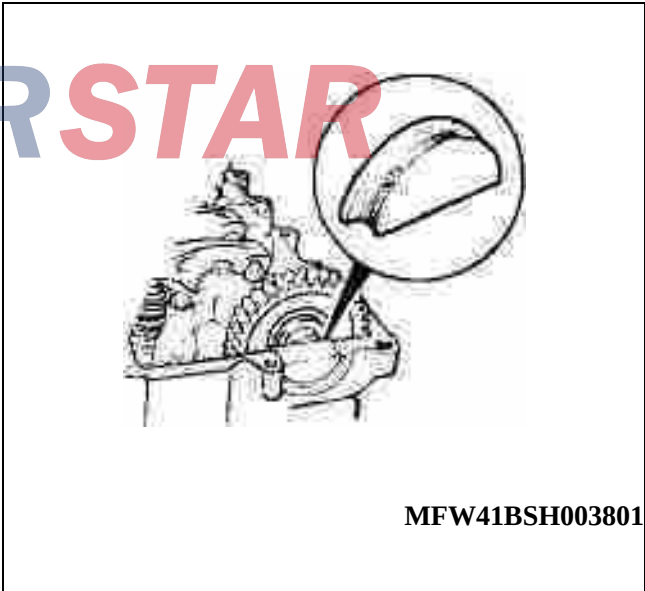


6. Install rubber stoppers.

- A liquid filler (Three Bond brand 1207B adhesive) was applied to the rubber stopper and mounted on the cylinder head.

Note:

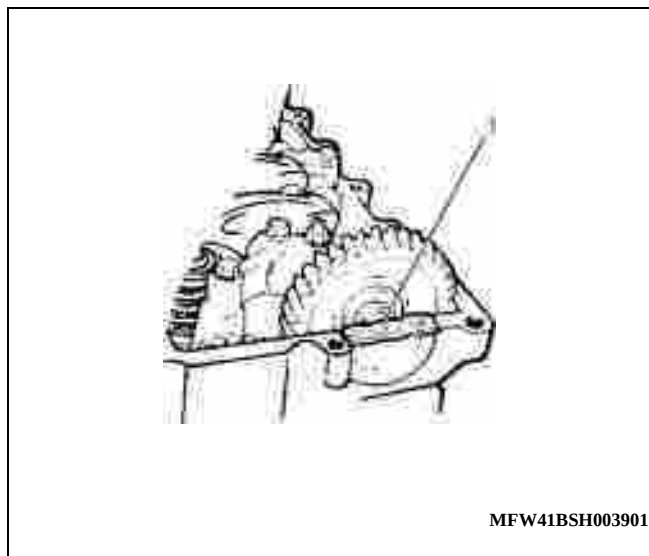
If the rubber plug has cracks, do not use it again.



- Apply liquid filler (1) (Sankey brand 1207B adhesive) on top of the rubber stopper.

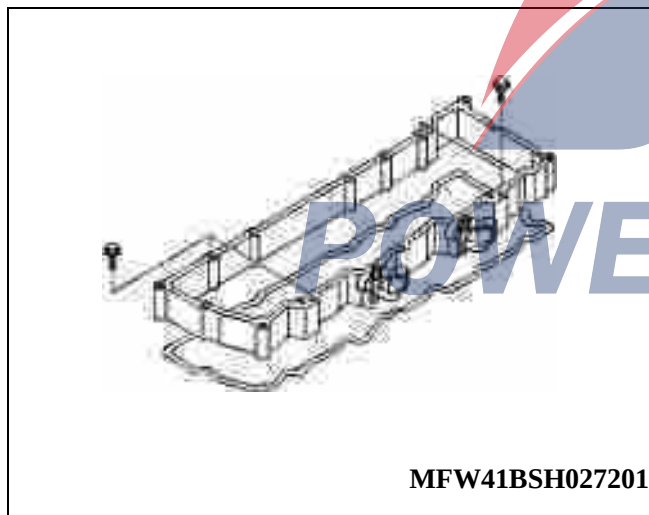
Note:

Cover the lid and the lower lid within 7 minutes of using liquid packing.



7. Insert the gasket into the lower cover groove and attach it to the cylinder head.

Tightening torque	N m (kgf m/lb ft)
13 (1.3 / 113)	

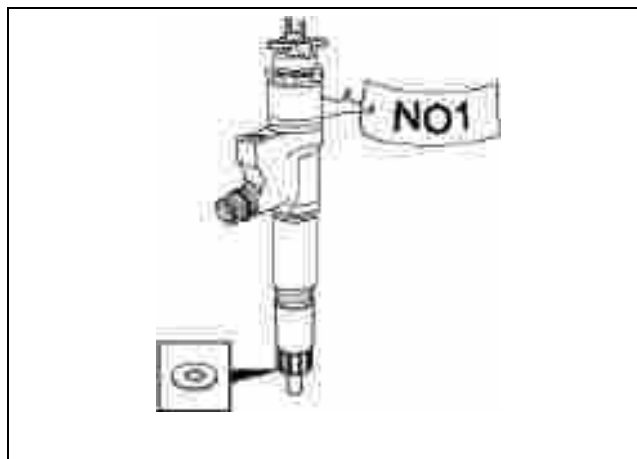


8. Install the injector.

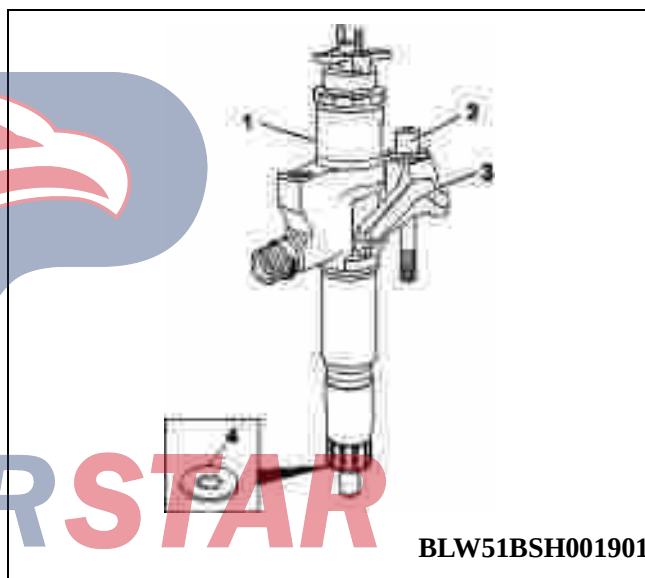
- Insert the gasket into the injector tip and insert it into the cylinder head.
- Insert the injector fixture at the same time.

Note:

- Each injector's injector identification plate is unique. Do not confuse it with other injector ID boards.
- Write the identification code in ECM with the scanning tool when replacing the injector assembly. For writing procedures, see "Engine Control Section."



9. Apply molybdenum disulfide lubricant to the thread and base part of the syringe clamp bolt and temporarily tighten it.



Legend

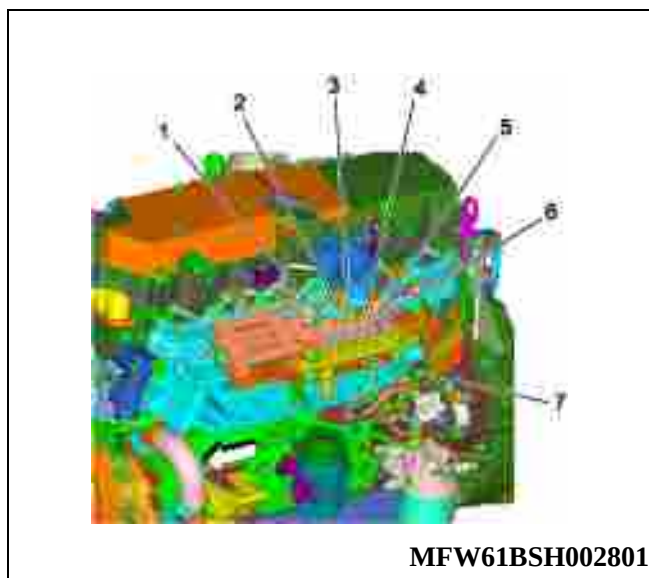
1. Ejector assembly
2. Clamping bolt
3. Clamping
4. Gasket

10. Install the injector pipe.

- Apply a light coat of oil to the sampler sleeve nut and install.
- Tighten the nut lightly with a wrench until it comes into close contact with the ejector and common rail side. Then tighten the nut with the specified torque.

Tightening torque	N m (kgf m/lb ft)
31 (3.2 / 23)	

- Install the fixture in the specified position.



Legend

1. No. 1 ejector tube
2. No. 2 ejector tube
3. No. 3 ejector tube
4. No. 4 ejector tube
5. No. 5 ejector tube
6. No. 6 ejector tube
7. Common rail equipment

11. Temporarily tighten the injector clamp with the torque wrench (T-40) with the specified torque.

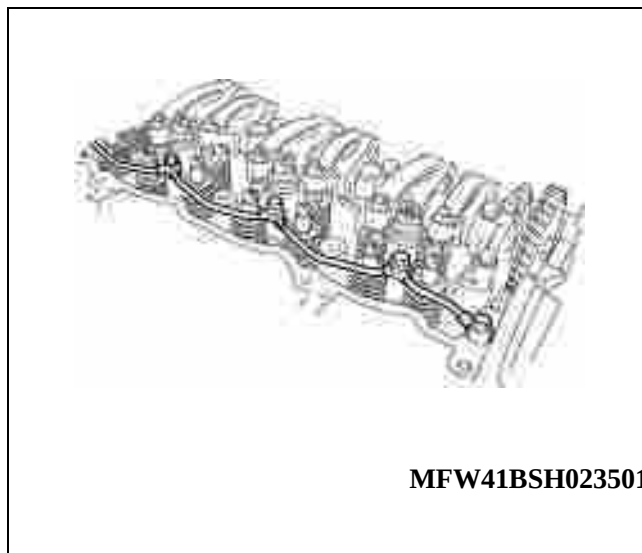
Tightening torque	N m (kgf m/lb ft)
30 (3.1 / 22)	

12. The leak-off tube is installed on the ejector.

Tightening torque	N m (kgf m/lb ft)
12 (1.2 / 104)	

Note:

- Be sure to use a new gasket.
- Check again to make sure you didn't forget to tighten any parts.



13. Install the syringe harness assembly.

- Install the harness holder so that the harness remains connected.

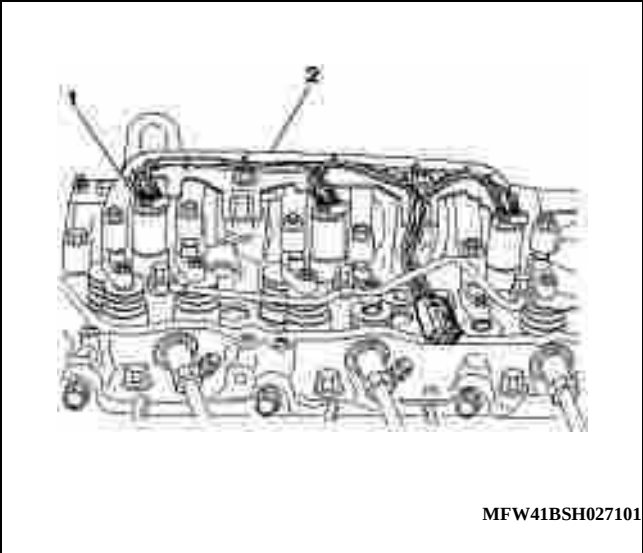
Tightening torque	N m (kgf m/lb ft)
22 (2.2 / 16)	

- Install the terminal nut on the injector and tighten it to the specified torque.

Tightening torque	N m (kgf m/lb ft)
2 (0.2 / 17)	

Note:

- Since the terminals are independent devices, each nut should be tightened 2-3 turns to fully tighten.
- Tightening the nuts may damage the terminals. Therefore, do not over-tighten them.

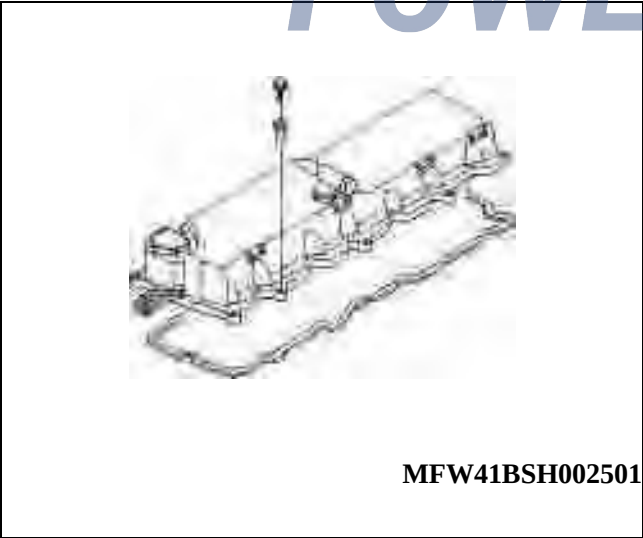


Legend

1. Harness terminal
Harness bracket

14. Install the gasket.
- Clean the top cover gasket connection groove and gasket, then attach the top cover to the gasket.

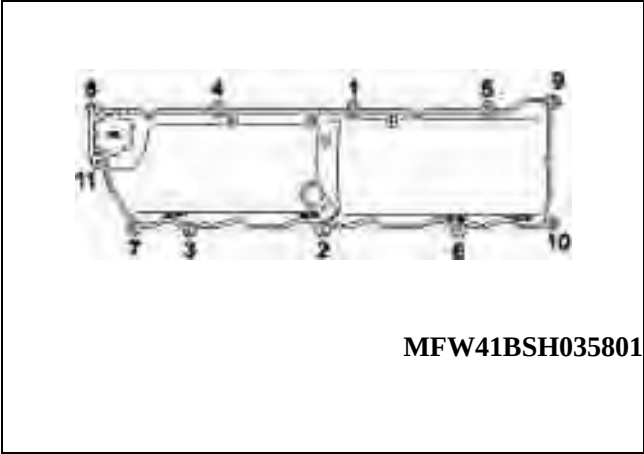
Note:
Make sure the collar does not come out.



15. Put on the top cover.
- After temporarily tightening all bolts, tighten them in the order shown.

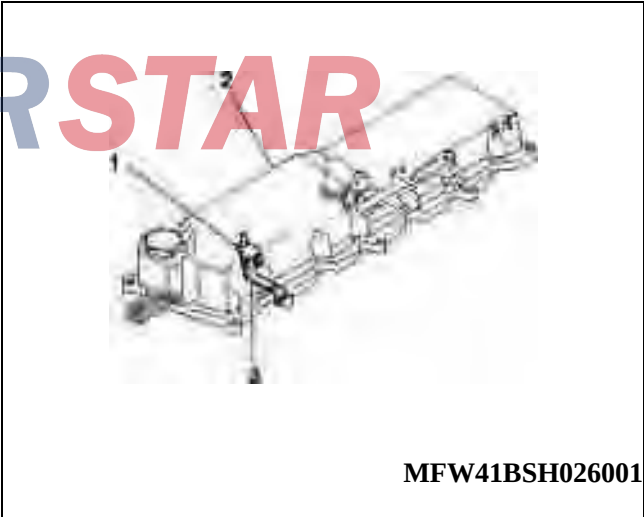
Tightening torque	N m (kgf m/lb ft)
13 (1.3 / 113)	

Note:
Make sure that the upper surface of the cylinder head is free of debris.



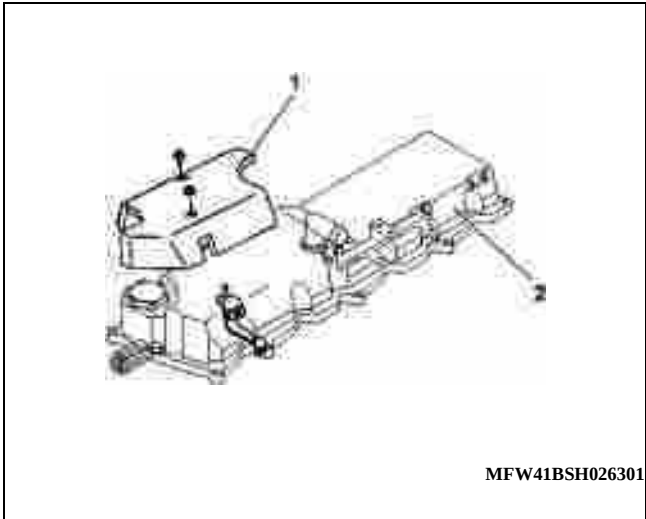
16. Mounting brackets.
- Install the noise cap bracket (1) on the cylinder head (2).

Tightening torque	N m (kgf m/lb ft)
20 (2.0 / 14)	



17. Wear a noise protection cover.
- Install a noise-proof cover (1) on the top cover (2) and tighten the nuts.

Tightening torque	N m (kgf m/lb ft)
20 (2.0 / 14)	



18. Install glow plugs.

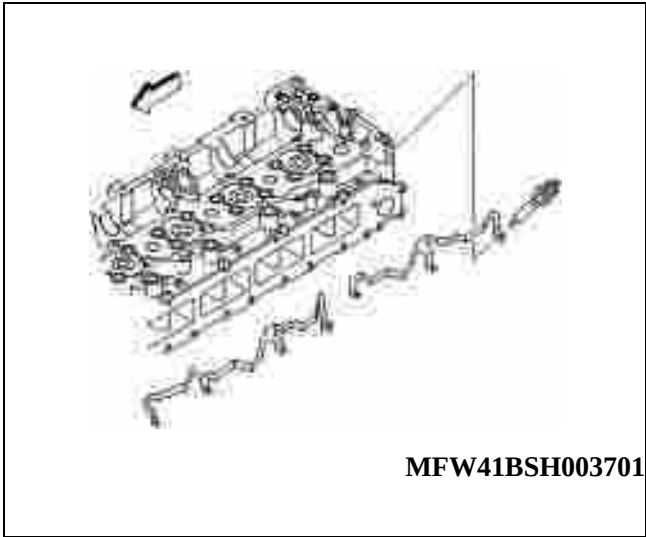
Tightening torque	N m (kgf m/lb ft)
20 (2.0 / 14)	

Note:

Be careful not to over-tighten the glow plugs as they may damage the glow plugs.

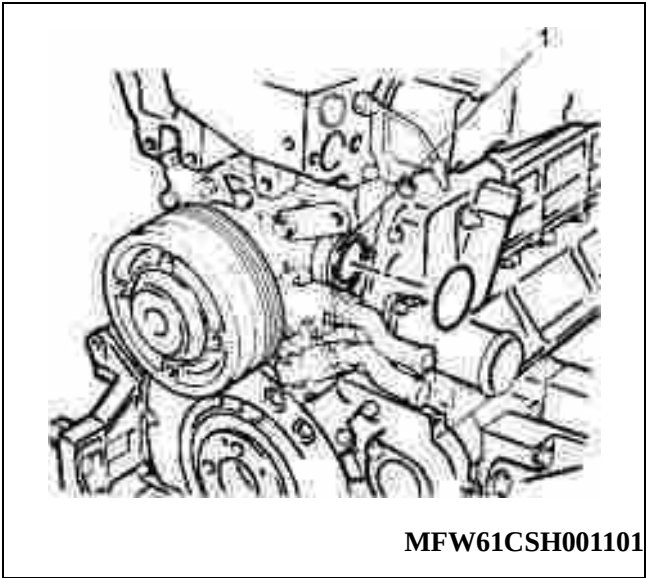
19. Install glow plug connectors.

Tightening torque	N m (kgf m/lb ft)
1 (0.1 / 9)	



20. Install O-rings.

- Install O ring on the pump.
- Apply soapy water around the O-ring (1).

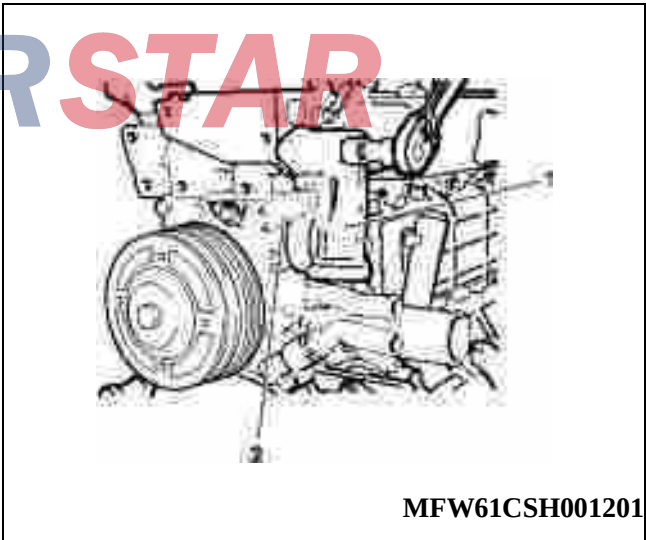


21. Install the water pipe

- Install the o-ring on the water pipe, but be careful not to damage the pump o-ring.

Tightening torque		N m (kgf m/lb ft)
Water pipe	No. 4 issue	ø81.851(19)1.9(14)
	M16:	103(10.5/76)

22. On the plate.



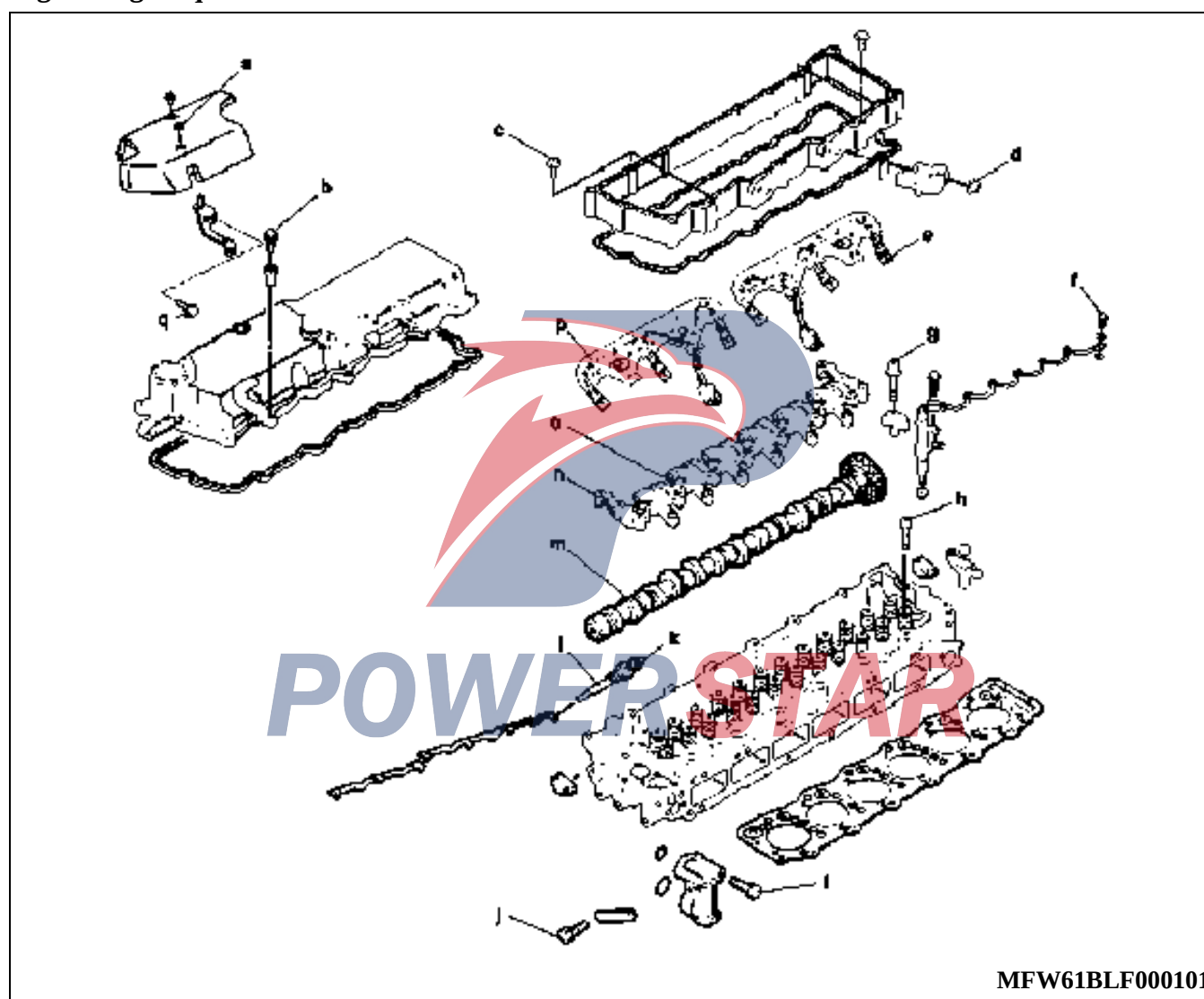
Legend

1. Water pipe
2. Plate

- Tighten the hose and install it.

Tightening torque		N m (kgf m/lb ft)
Board piece		M8: 19(1.9/14)
		M10: 38(3.9/28)

Tightening torque table



Legend

- | | |
|-------------------------------------|---|
| (a) 20 N m (2.0 kgf m / 14 lbs. ft) | (h) M14: 98 N m (10 kgf-m/72 lb ft)→147 N m |
| (b) 13 N m (1.3 kgf m/113 lb.ft) | (15 kg-m/108 lb·ft)→Angle 30°-60° |
| (c) 13 N m (1.3 kgf m/113 lb.ft) | M10: 38 N m (3.9 kgf-m / 28 lb ft) |
| (d) 20 N m (2.0 kgf m / 14 lbs. ft) | (i) M8: 19 N m (1.9 kgf-m / 14 lb ft) |
| (e) 2N m (0.2 kgf m/17lb.ft) | M16: 103 N m (10.5 kgf-m / 76 lb ft) |
| (f) 12N m (1.2 kgf m/104lb.ft) | (j) M8: 19 N m (1.9 kgf-m / 14 lb ft) |
| (g) 30 N m (3.1 kgf m/22 lb ft) | M10: 38 N m (3.9 kgf-m / 28 lb ft) |
| | (k) 1 N m (0.1 kgf m/9 lb.ft) |

(l) 20 N•m (2.0 kgf m / 14 lbs. ft)

(m) 27 N m (2.8 kgf-m / 20 lb ft)

(n) 27 N m (2.8 kgf-m / 20 lb ft)

(o) 56 N m (5.7 kgf m/41 lb ft)

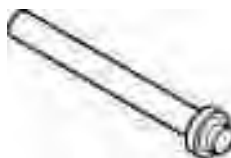
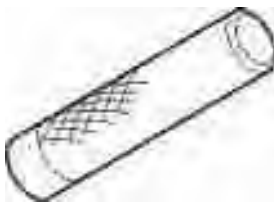
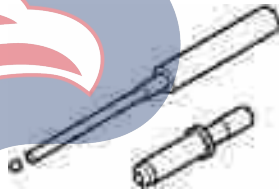
(p) 22 N m (2.2 kgf m/16 lb ft)

(q) 20 N m (2.0 kgf m / 14 lb ft)



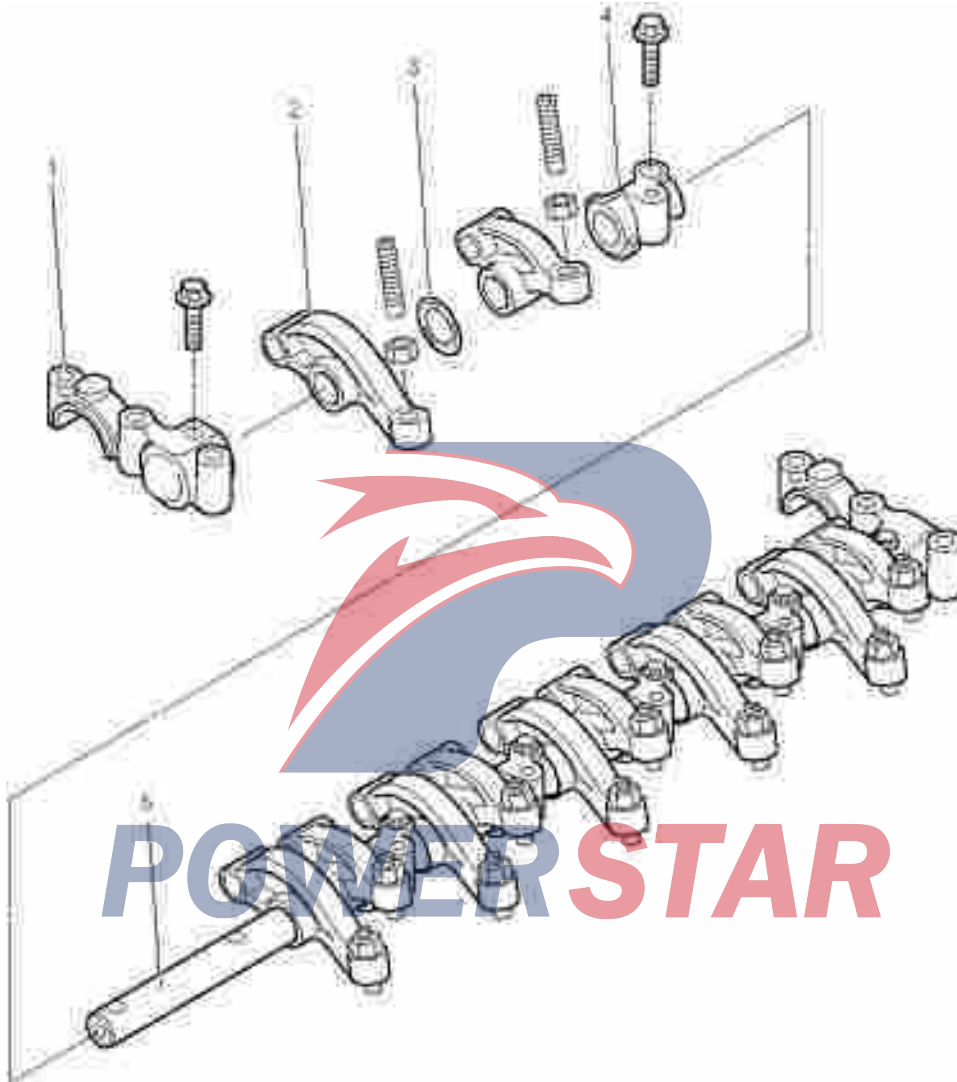
Special tools**Cylinder head**

Graphic representation	Tool number/description
 5884028260	5-8840-2826-0 Syringe Removal Tool
 5884000190	5-8840-0019-0 Small push hammer
 5884026210	5-8840-2621-0 Valve Spring Replacement Tool
 8943968620	8-9439-6862-0 axis
 5884026280	5-8840-2628-0 Valve Guide Dresser
 5884026230	5-8840-2623-0 Syringe Sleeve Removal Tool

Graphic representation	Tool number/description
 5884026270	5-8840-2627-0 Oil Seal Installation Tool
 5884026260	5-8840-2626-0 Bridge Orientation Adjustment Tool
 5884026240	5-8840-2624-0 Syringe Sleeve Setting Tool
 5884026250	5-8840-2625-0 Stem Seal Setting Tool

Rocker arm

Part



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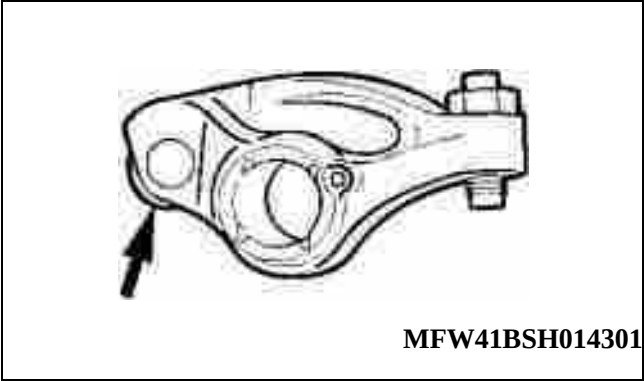
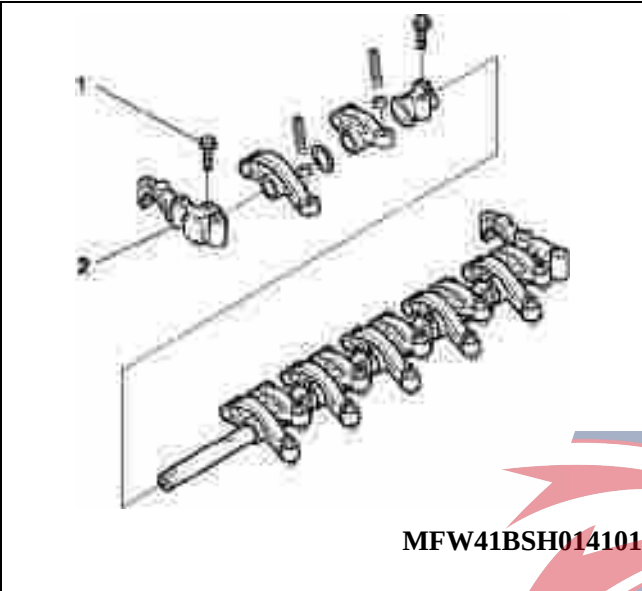
Legend

- | | |
|----------------------|--------------------|
| 1. Front cam bracket | 4. Bracket |
| 2. Rocker arm | 5. Rocker arm axis |
| 3. Washers | |

Disassembly

1. Remove the front and rear cam brackets.

When pressing the front cam bracket (2), remove the rocker arm bolts (1), and then remove the rocker arms, washers, and brackets one by one.

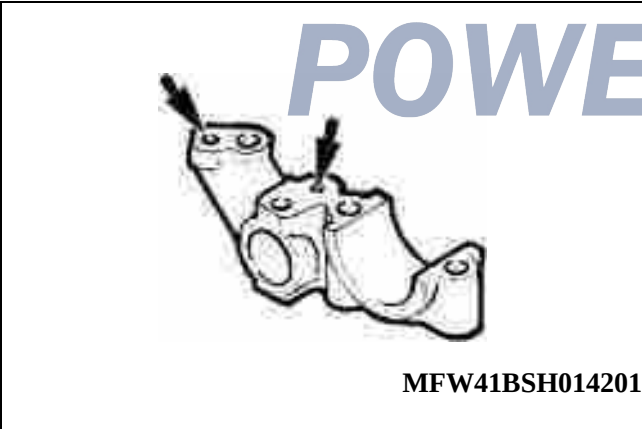


- When measuring the distance between the roller pin and the roller, use a rope and other objects to pass between the rocker arm and the roller (as shown in the figure), pull the rope in the direction of the arrow, and measure the rocker arm and roller when the roller is in the pressed state. The distance between (A size). Then mark the measuring point and pull out the rope, push the drum backwards and measure the distance between the rocker arm and the marking position (dimension B) on the drum.

Make sure that the measurement value of a dimension minus B is within the standard range.

Inspection

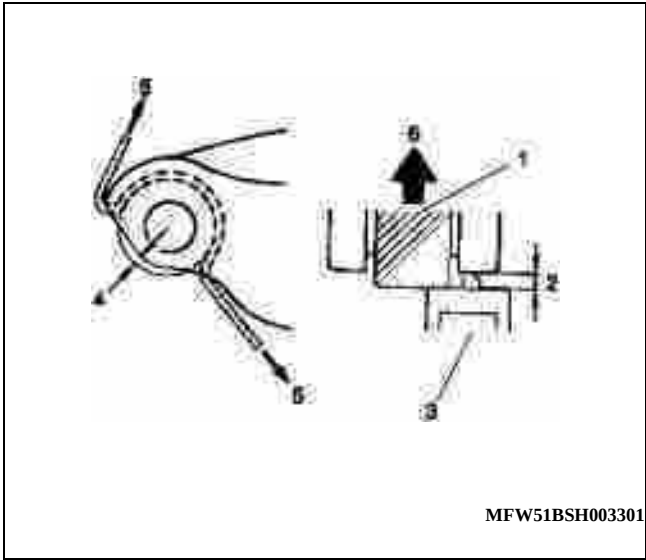
1. After the inspection, check the cam bracket oil hole for clogging.



Clearance between roller pin and roller		mm(inch)
Assembly standards	Use restrictions	
0.068(0.0027) — 0.097(0.0038)	0.5(0.0197)	

2. Check the rocker roller section for wear or damage.

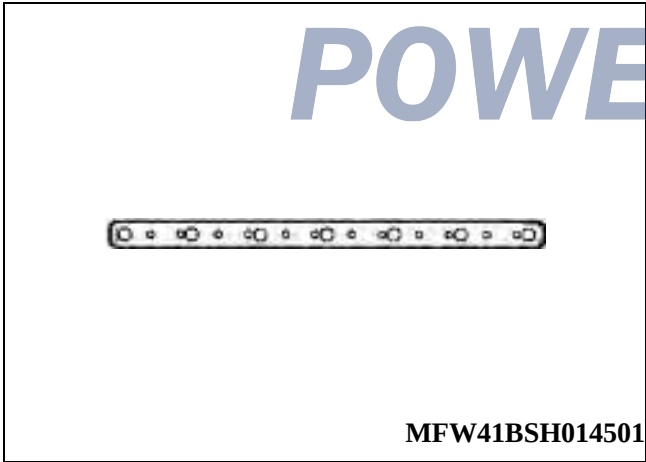
- Check the roller surface for damage.



Legend

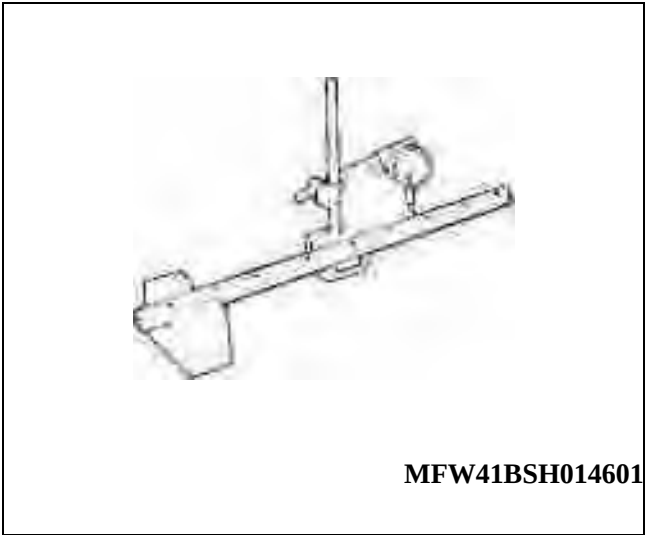
- 1. Roller
- 2. B size
- 3. Slide caliper
- 4. A size
- 5. Direction of pulling out the string
- 6. Push in the direction of the roller

3. Check that the rocker shaft oil hole is blocked.



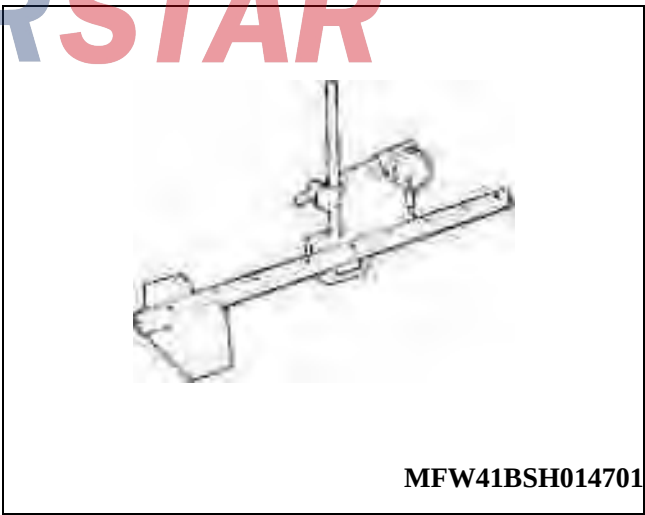
- Measure the swing of the rocker shaft.

Swing	mm(inch)
Use restrictions	0.3(0.0118)



- Measure the rocker shaft diameter and the clearance between the rocker shaft and the rocker arm.

mm(inch)			
	Nominal size	Assembly standards	Use restrictions
Rocker arm and rocker clearance	—	0.01(0.0004)—0.056(0.0022)	0.2(0.0079)
Rocker shaft diameter	22(0.8661)	—	21.85(0.8602)

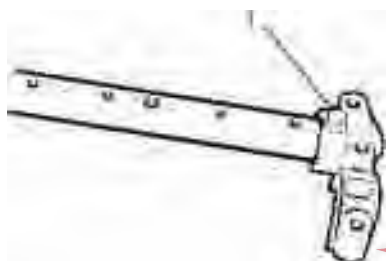


Assembling

1. Mount the rear cam bracket on the rocker shaft. Tighten the coupling bolt (1) temporarily to prevent it sliding out.

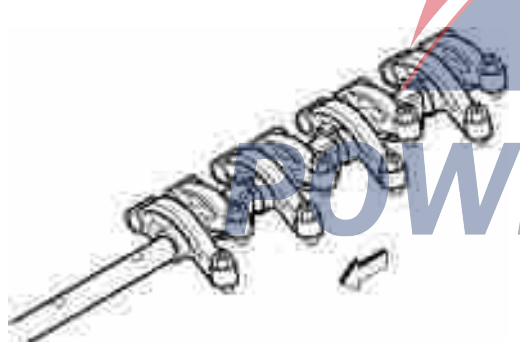
Note:

When installing the cylinder head, fully tighten the bolts.



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2. Install rocker arm, washer and bracket (as shown).



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- Apply engine oil to the rocker bushing (2) and the roller part (1) and install the rocker arm.



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3. Install the front cam bracket.

- Attach the rocker arm to the front cam carrier.
- Temporarily tighten the connecting bolts so that the front cam bracket does not slide out.

Note:

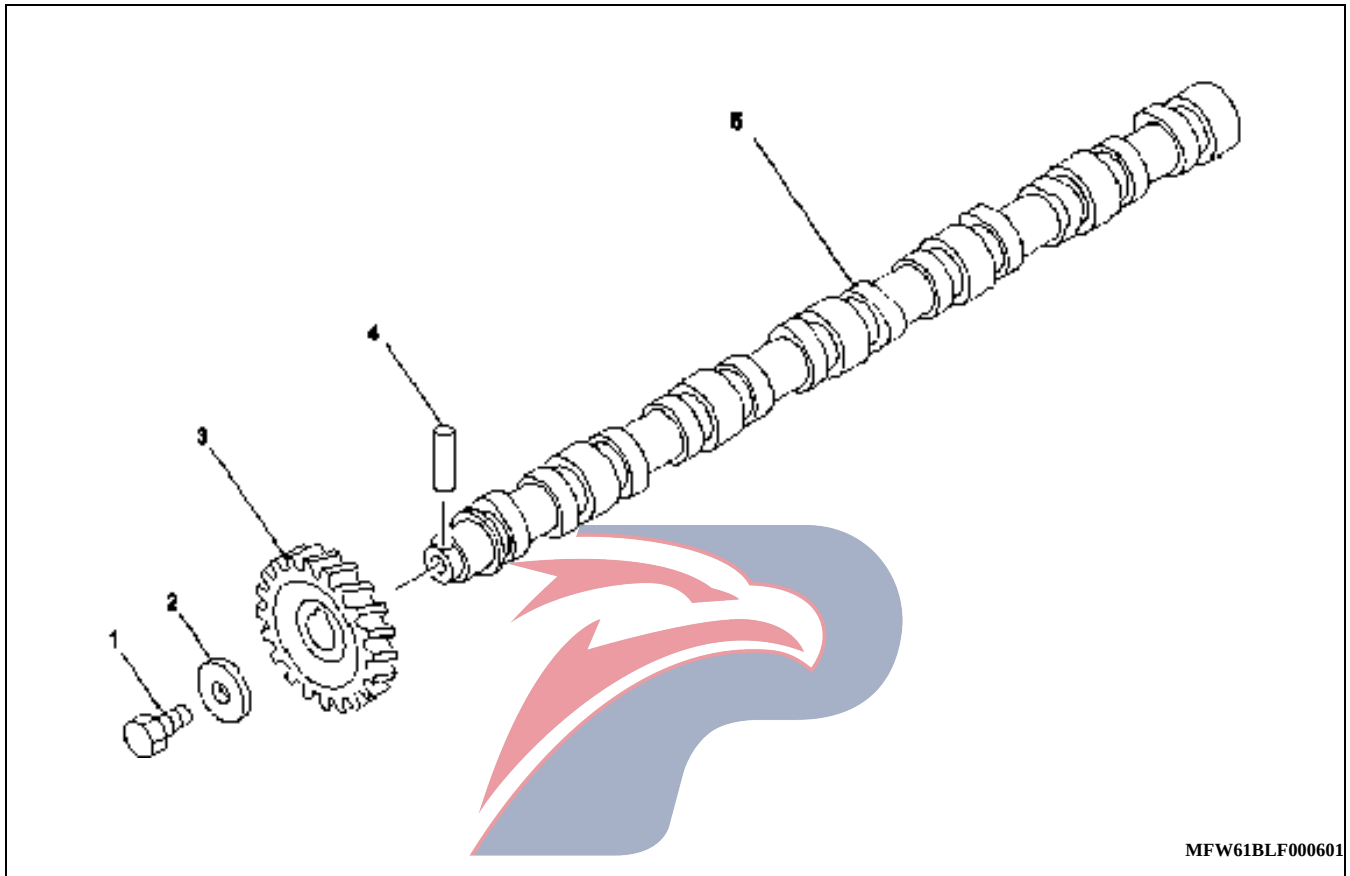
When installing the cylinder head, fully tighten the bolts.



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Camshaft

Part



Legend

- | | |
|-----------|--------------|
| 1. Bolt | 4. Fixed pin |
| 2. Gasket | 5. Camshaft |
| 3. Gear | |

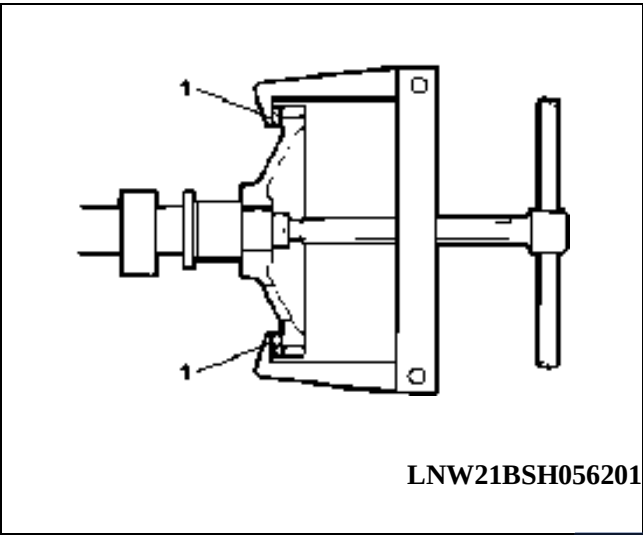
Disassembly

1. Remove the camshaft gear.
 - Use a vise to clamp the hexagonal part of the camshaft through aluminum joint and remove it.

Note:

Be careful not to damage the camshaft cam portion and the journal portion.

- Remove the camshaft gear coupling bolts, insert the sawdust into the extraction tool, and remove the camshaft gear.



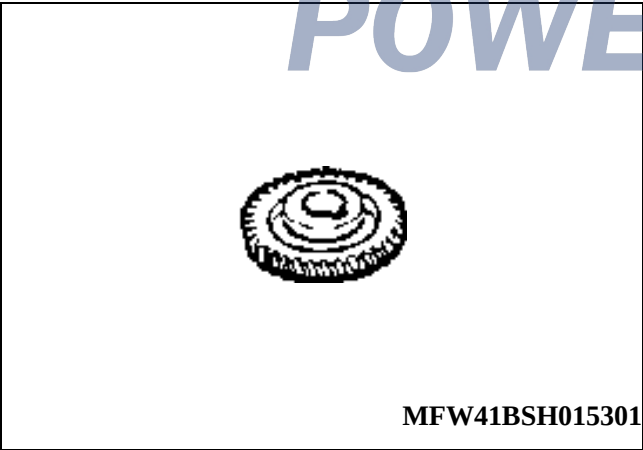
Legend

1. Sawdust

2. Remove the thimble.

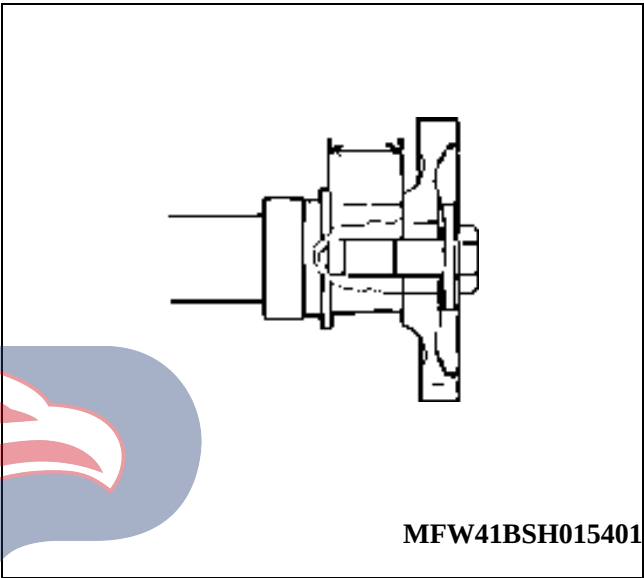
Inspection

1. Check the gear teeth, the phenomenon of detachment, the tilt of the tilt, etc.



2. Measure the distance from the camshaft thrust surface to the camshaft gear.

Distance between camshaft thrust surface and camshaft gear	mm(inch)
Assembly standards	28.015(1.1030)– 28.055(1.1045)



3. Visually check the camshaft and the degree of oscillation.

- Check the journal section and cam section for wear and replace the defective section.
- Fix the No. 1 and No. 7 journal portions with V-blocks. Place a dial indicator on No. 4 journal. Gently rotate the camshaft for one revolution and measure the swing.

Camshaft degree	mm(inch)
Use restrictions	0.05(0.0020)

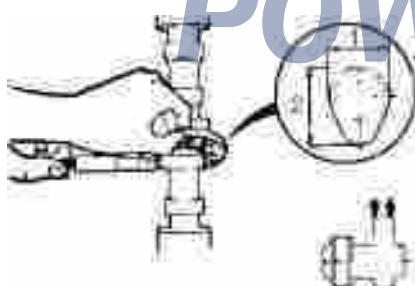


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4. Measure the cam lift.

- Check the cam surface for damage and then measure the cam head.
- Use 1-2 in the figure to make measurements. Use oil to repair minor damage or wear.

Cam lift (2-1)		mm(inch)
	Assembly standards	Use restrictions
Intake	7.851(0.3091)	7.351(0.2894)
Exhaust	9.491(0.3737)	9.040(0.3559)

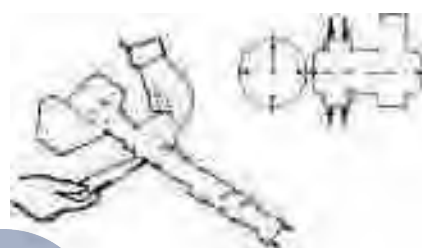


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5. Measure the cam log.

- Use a micrometer to measure the position of the display.

Diameter		mm(inch)
Nominal size		Use restrictions
40(1.5748)		39.85(1.5689)

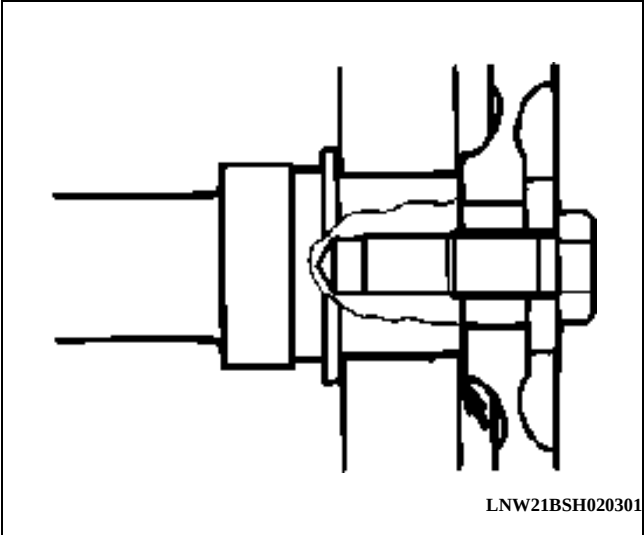
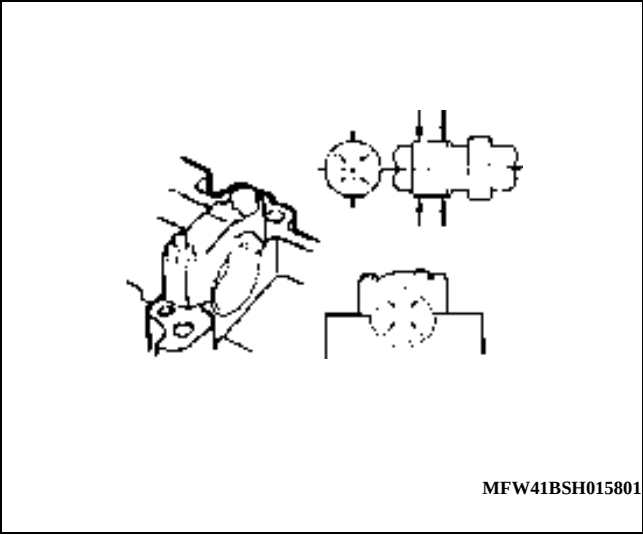


MFW41BSH015701

6. Measure journal and bearing clearances.

- Measure the inner diameter of the camshaft bearing with an aperture meter.
- Read the difference between bearing inner diameter and journal diameter. If the gap between oil films exceeds the limit, replace the bearing.

Journals and bearing gaps		mm(inch)
Assembly standards		Use restrictions
0.030(0.0012)–0.145(0.0057)		0.15(0.0059)



Assembling

7. Put on the thimble.

8. Install camshaft gears.

- The convex side of the central convex portion of the camshaft gear is located on the camshaft side, aligned with the top pin, and then the camshaft gear is assembled.

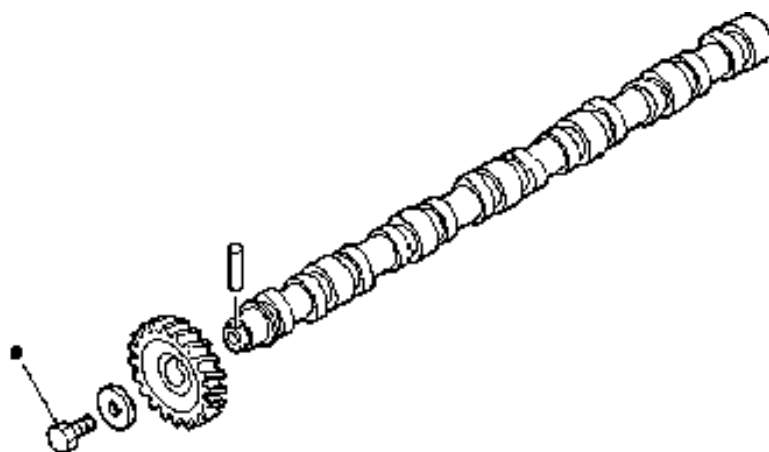
Tightening torque	N m (kgf·m/lb ft)
142 (14.5 / 105)	

Note:

- Carefully operate and tighten the gear so as not to damage the cam portion and the journal portion.
- The hexagonal portion of the camshaft is clamped with a vise via a snap ring (egg aluminum joint).



Tightening torque table



MFW61BMF000201

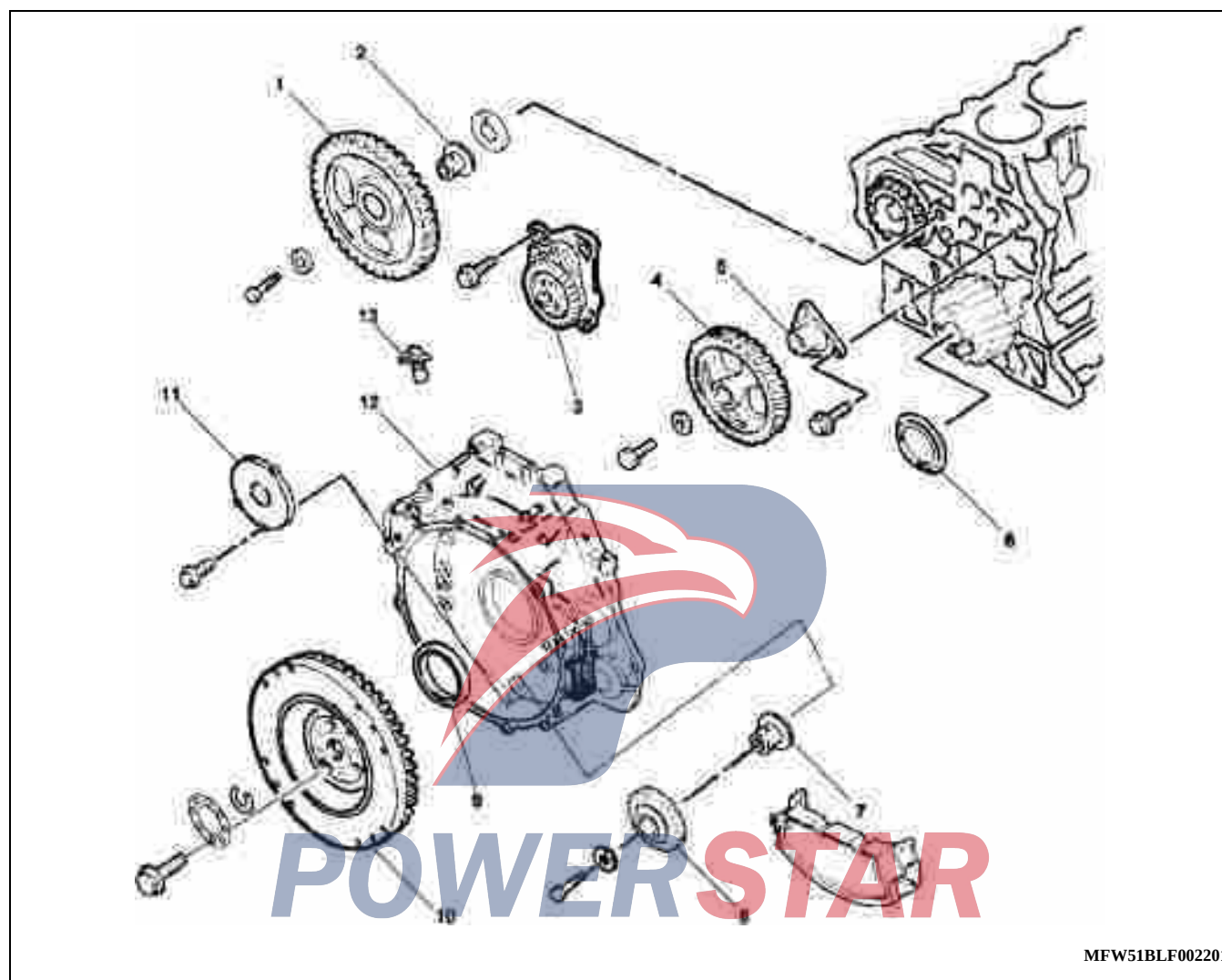
Legend

(a) 142 N m (14.5 kgf m / 105 lb. ft)

POWERSTAR

Timing gears, flywheels and flywheel housings

Part



Legend

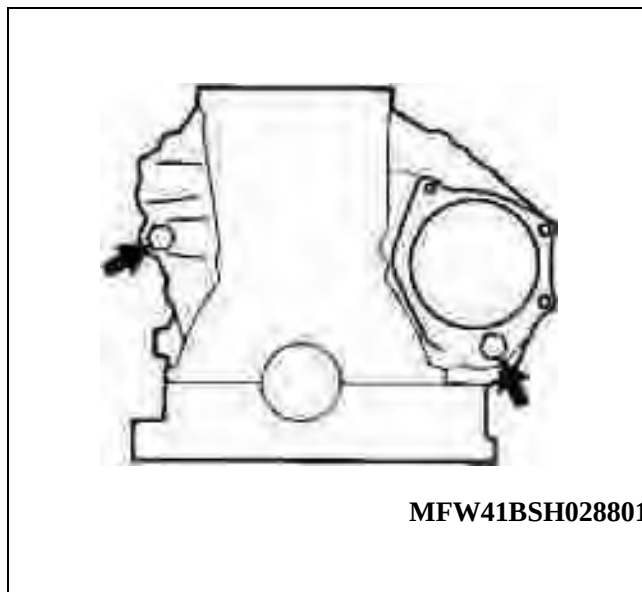
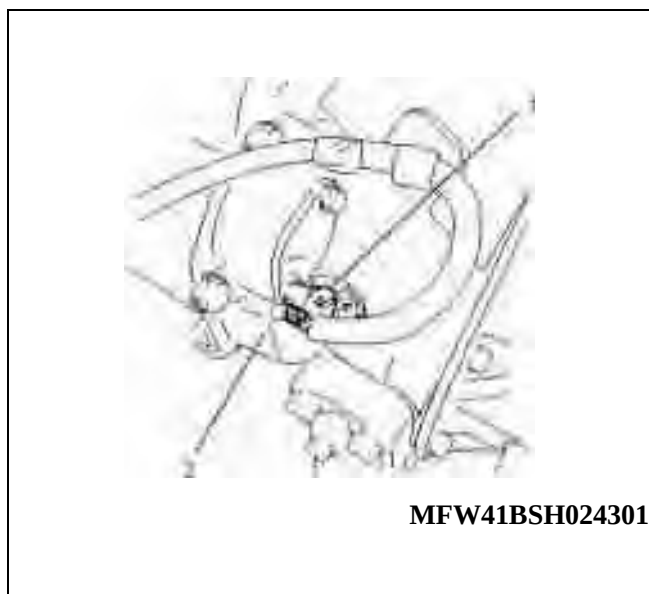
- | | | | |
|----|--------------------------------|-----|----------------------------|
| 1. | Idle speed A | 8. | Air compressor idler |
| 2. | Idle gear A axis | 9. | Oil seal |
| 3. | Oil pump | 10. | Flywheel |
| 4. | Idler gear B | 11. | Cover |
| 5. | Idler gear B axis | 12. | Flywheel housing |
| 6. | Oil retaining ring | 13. | Crankshaft position sensor |
| 7. | Air compressor idle gear shaft | | |

Tear down

1. Remove the engine components from the car.
 - Refer to "Engine Assembly".
2. Remove the cylinder head.

- Refer to "Cylinder Head".
3. Remove the oil pan.
 - Refer to "Oil pan" in the lubrication system.
 4. Remove the crankshaft position sensor (CKP).

- Remove the flywheel before removing the flywheel.



10. Remove the oil seal.

Legend

1. Crankshaft position sensor
2. Flywheel shell

5. Remove the flywheel.
6. Remove the cover.
7. Remove the air compressor idler.
8. Remove the air compressor idler shaft.
9. Remove the flywheel housing.

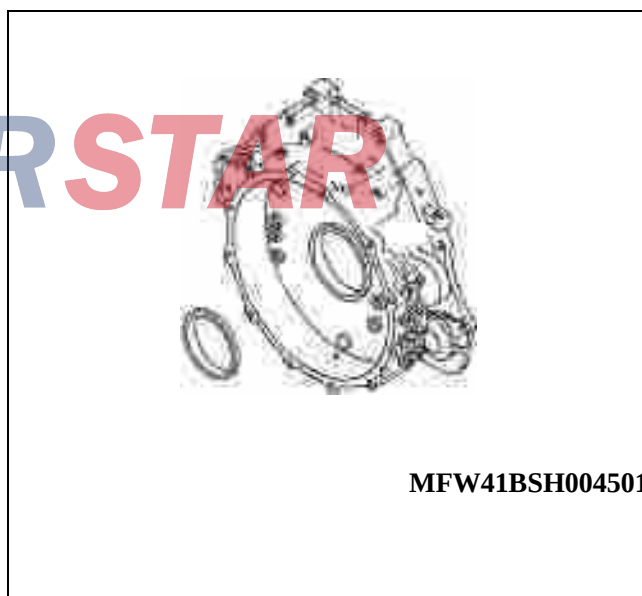
Note:

- As shown, the bolts fasten the flywheel housing from the cylinder side. Please don't forget to delete them.
- Attach the idler gear A-axis and flywheel housing with three M8 bolts. Please don't forget to delete them.
- After removing the air compressor idler, M14 bolts are still connected. Please don't forget to remove it.

- Remove the flywheel housing oil seal and handle it carefully without damage.

Note:

- Be sure to replace it when you delete it.
- Replace the oil retainer and oil seal as a whole.

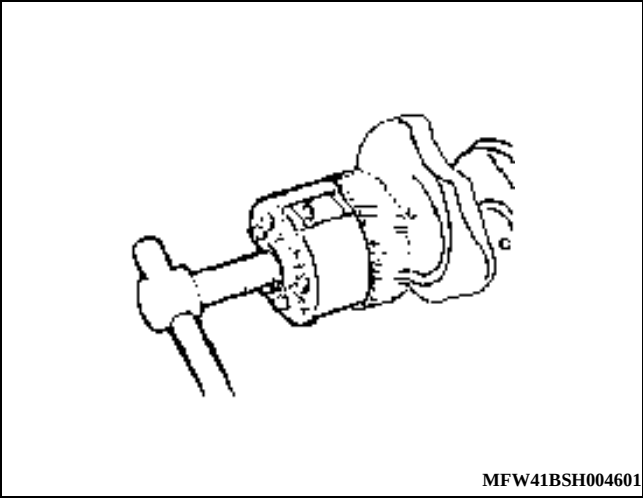


11. Remove the slinger and measure the clearance of each idler.

- Use a mudguard replacement tool (8-9439-6858-0) to remove the slinger.

Note:

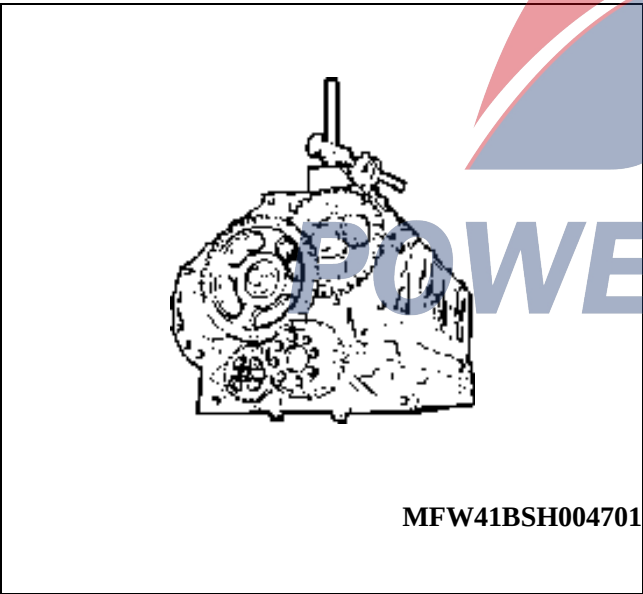
Replace the oil retainer and oil seal as a whole.



a. Measure back flushing.

- Measure the gap of each idler.

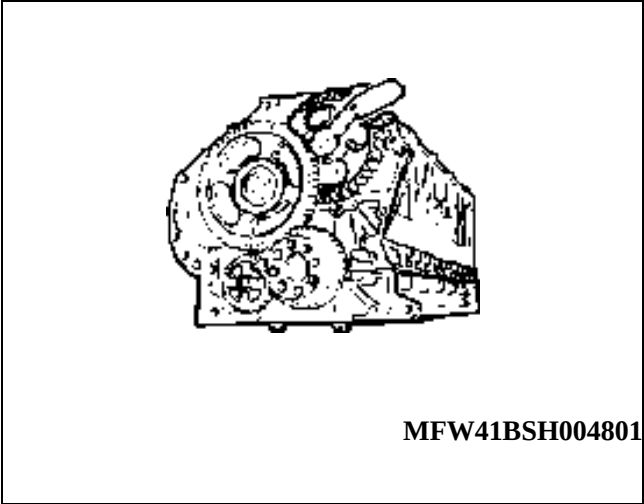
Fear	mm(inch)
Assembly standards	Use restrictions
0.1(0.0040)—0.17(0.0067)	0.3(0.0118)



b. Measured axial clearance.

- Measure the axial clearance of each idler.

Axle clearance	mm(inch)
Assembly standards	Use restrictions
0.08(0.0031)—0.14(0.0055)	0.2(0.0079)



12. Remove the idle gear A.

13. Remove the idler A shaft.

14. Remove the oil pump.

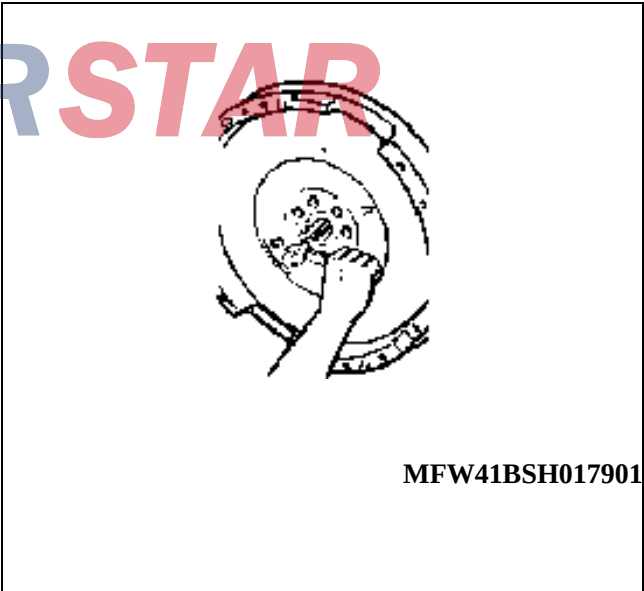
15. Remove the idler B cover.

16. Remove the idler B cover axis.

Disassembly

1. Remove the buckle from the flywheel.

2. Remove the guide bearing using the guide bearing replacement tool (9-8521-0157-0).

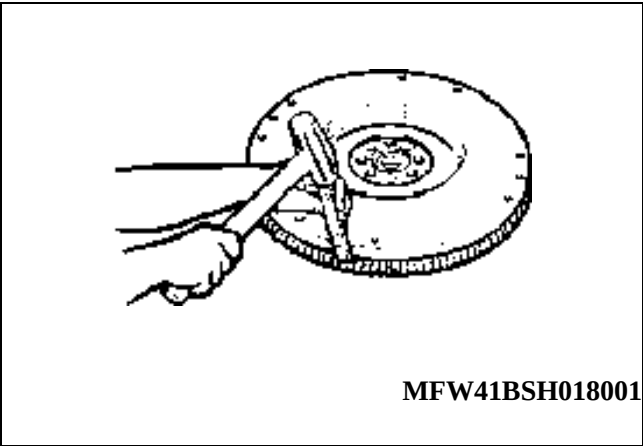


3. Remove the flywheel starter gear from the flywheel.

- Place some of the wood chips on the flywheel, place the mounting strip on the edge surface of the flywheel starter gear, and remove it by tapping it.

Note:

First check if the gear of the flywheel starter has been damaged; if there is no problem, do not remove it.



- Check the flywheel for cracks and shock marks and measure the degree of wear. If an abnormality occurs, it should be repaired or replaced.

mm(inch)			
		Nominal size	Use restrictions
Distance between clutch plate friction side and crankshaft mounting face	14 inch	42.45(1.6713)	41.45(1.6319)
	15 inch	42.46(1.6717)	41.46(1.6323)

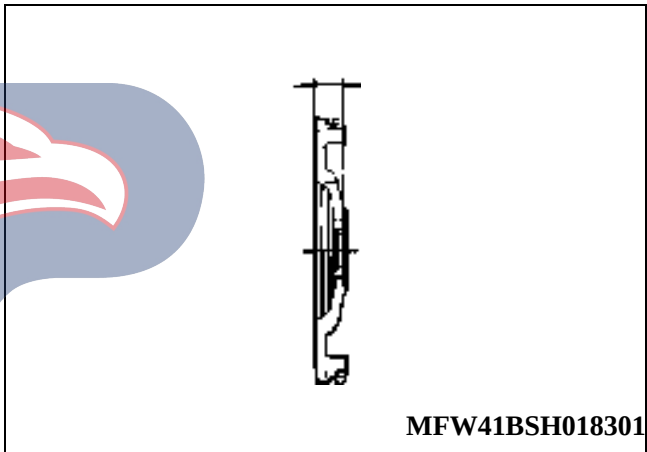
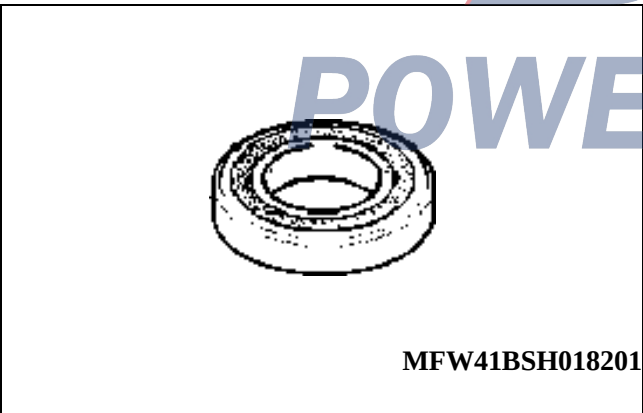
Inspection

1. Guide bearing.

- Check the guide bearing for wear and abnormal noise.

Note:

Do not clean it since it is grease sealed.

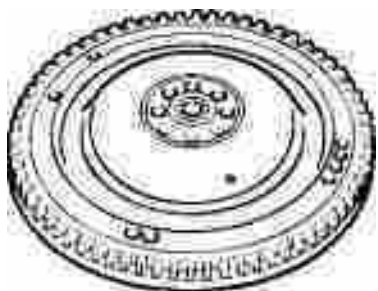


3. Check flywheel starter gear wear.

Note:

Check the tooth surface; if it is damaged or excessively worn, replace the flywheel starter gear.

2. Flywheel.



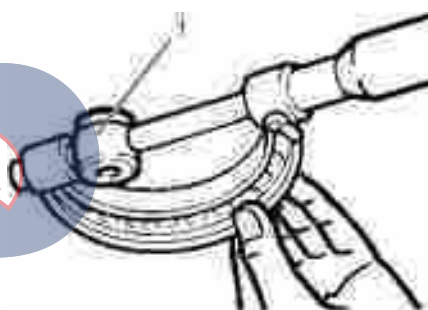
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mm(inch)		
	Nominal size	Use restrictions
Idler A, B axis and gear clearance	0.02(0.0008)– 0.061(0.0024)	0.2(0.0079)
Idler gear C-axis and gear clearance	0.020(0.0008)– 0.062(0.0024)	0.2(0.0079)
Air compressor idler shaft gear clearance	0.020(0.0008)– 0.062(0.0024)	0.2(0.0079)

4. Check teeth, disengage and tilt.



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MF41BSH019601

5. Measure the diameter of each idler shaft (1) and the gap between the gear shaft and idler.

mm(inch)		
	Nominal size	Use restrictions
Idler gear shafts A, B	29.959(1.1795)– 29.98(1.1803)	29.9 (1.1772)
Idle gear shaft C outer circumference	29.959(1.1795)– 29.980(1.1803)	29.9(1.1772)
Air compressor idler shaft periphery	29.959(1.1795)– 29.980(1.1803)	29.9 (1.1772)

Assembling

1. Install the flywheel starter gear on the flywheel.

- Place the flywheel starter gear on the metal grid and use the gas burner to evenly heat the outer surface of the gear so that the larger side of the bevel (1) faces the flywheel side and connects it.

Note:

- The shrink fit temperature of ring gear is below 200°C (392°F).
- After shrink-fitting, make sure that the ring gear is firmly fixed in the flywheel.



MFW41BSH019401

- Coat the inner and outer parts of the idling gear with oil, and then tighten the bolts through the thrust ring.

Tightening torque	N m (kgf m/lb ft)
	95 (9.7 / 70)

Note:

After installation, make sure that it rotates when you gently push by hand.

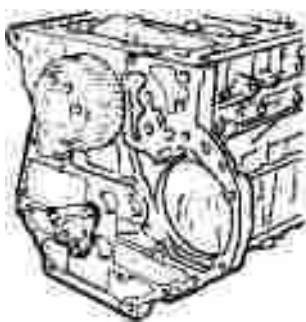
2. Mount the guide bearing on the flywheel.
3. Install the buckle on the flywheel to the flywheel.

Reassemble

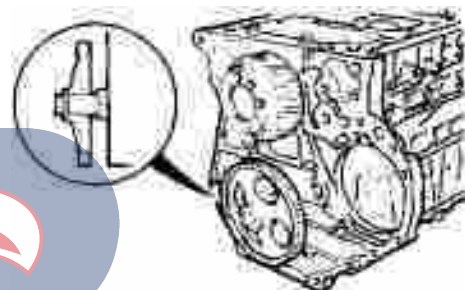
Remove the gasket completely from the flywheel housing and the cylinder block mounting surface.

1. Install the idle gear B shaft.
 - Install the oil hole on the shaft toward the side of the cylinder head.

Tightening torque	N m (kgf m/lb ft)
	31 (3.2 / 23)



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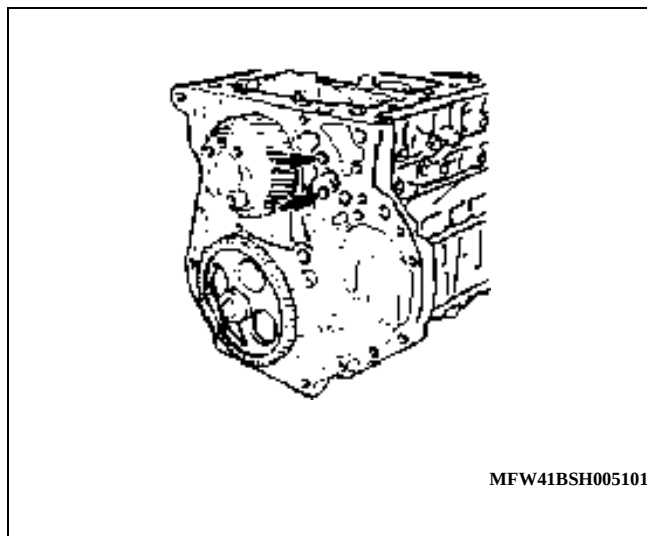
3. Put on the oil pump.

- Apply the engine oil to the gear insert part of the oil pump.
- Remove the liquid filler from the pump mounting surface and apply oil to the crankcase and cylinder liner.

Note:

Do not allow oil to enter the pump's mounting surface.

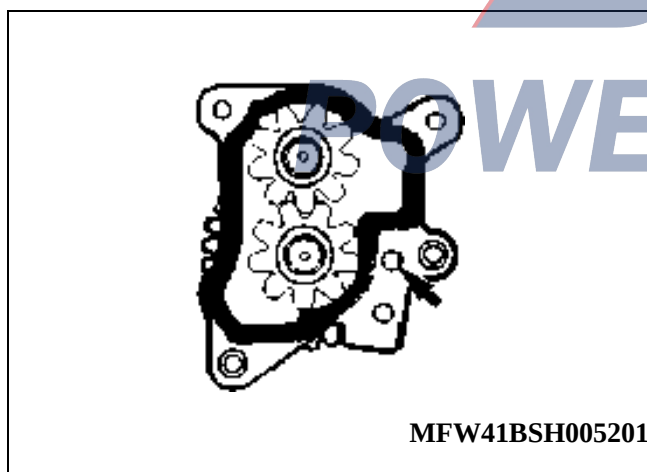
2. Install the idle gear B.



- Apply liquid filler (San key brand TB 1141E adhesive) to the oil pump mounting surface (as shown).

Note:

- Load it within 20 minutes of using liquid filling.
- Do not allow liquid to enter the hole in the gear or arrow section.
- Make sure that the driven gear does not come off.



- Align the wooden pins and install the oil pump.

Tightening torque	N m (kgf m/lb ft)
	24 (2.4 / 17)

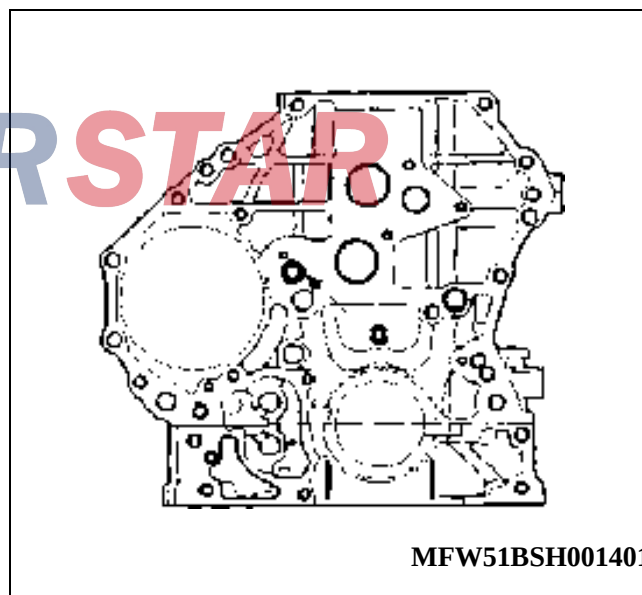
Note:

- There is no alignment mark on the gear.
- Install the oil pump before installing idler A.



4. Install the idler A-axis.

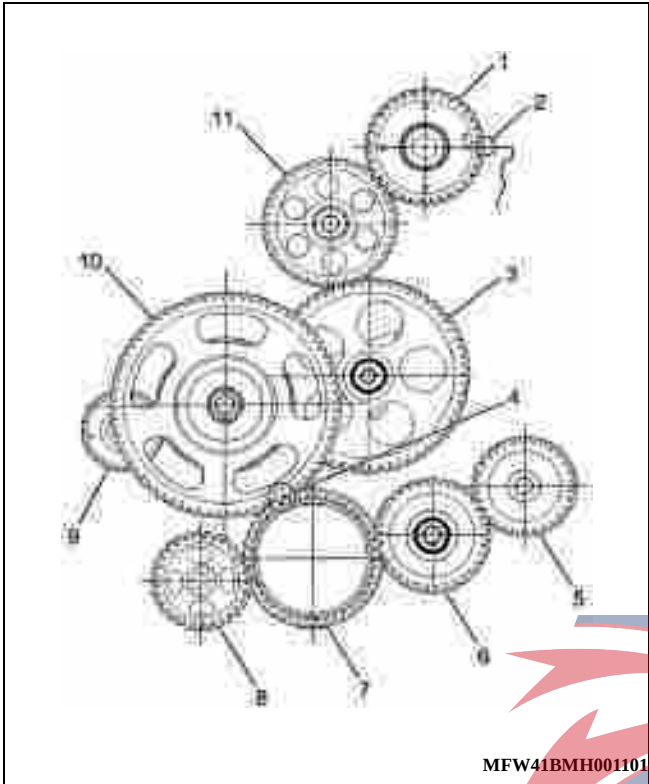
- Apply engine oil to the internal gear and external gear section.
- Insert the A-axis of the idler into the arrow section.



- Insert the gear into the shaft and insert it into the collar.

Note:

- Turn the crankshaft and place cylinder number 1 at top dead center.



Legend

- 1. Camshaft gear
- 2. Positioning
- 3. Idler gear B
- 4. Positioning
- 5. Air compressor gear
- 6. Air compressor idle gear
- 7. Crank gear
- 8. Oil pump gear
- 9. Supply pump gear
- 10. Idler gear A
- 11. Idler gear C

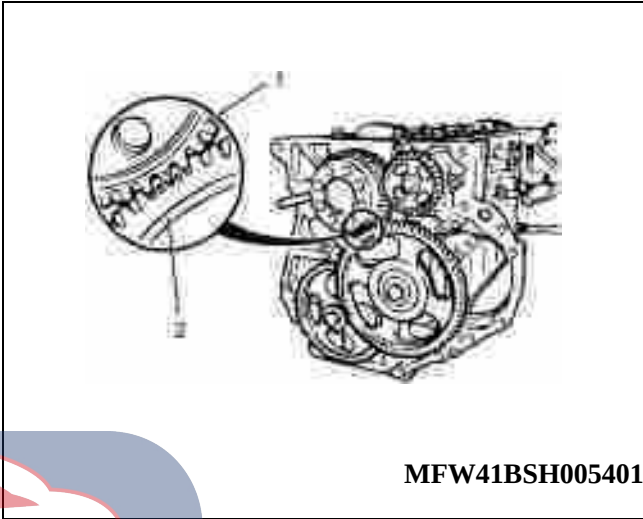
5. Install idler gear A.

- Align and install the shaft pin with the cylinder locating pin hole so that the L mark of the crankshaft gear (1) aligns with the O mark of the idler A (2).
- Apply molybdenum disulfide lubricant to the threads and base of bolts and washers, and tighten them.

Tightening torque	N m (kgf m/lb ft)
133 (13.6 / 98.4)	

Note:

- After installation, make sure that the marks are aligned with each other.
- After installing the flywheel housing, install the shaft with 3 bolts.

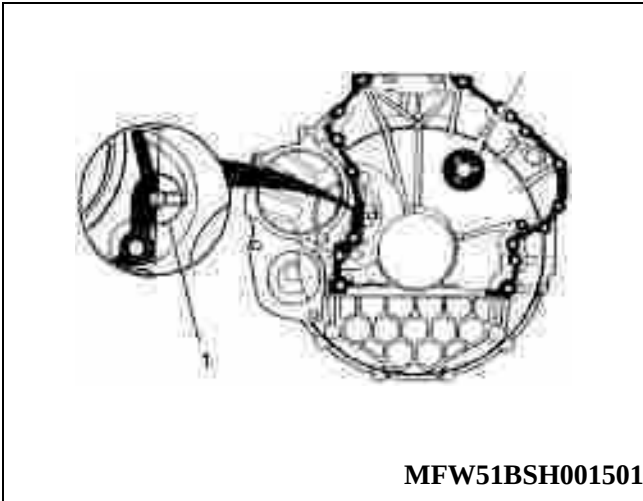


6. Install the flywheel housing.

- Apply a liquid filler (San key brand 1207C or 1201 glue) to the flywheel housing (as shown).

Note:

- The arrowhead section in the figure is coated with liquid filler.
- Apply the sealant within 7 minutes after applying the sealant.



Legend

- 1. Do not use packaging on this part.

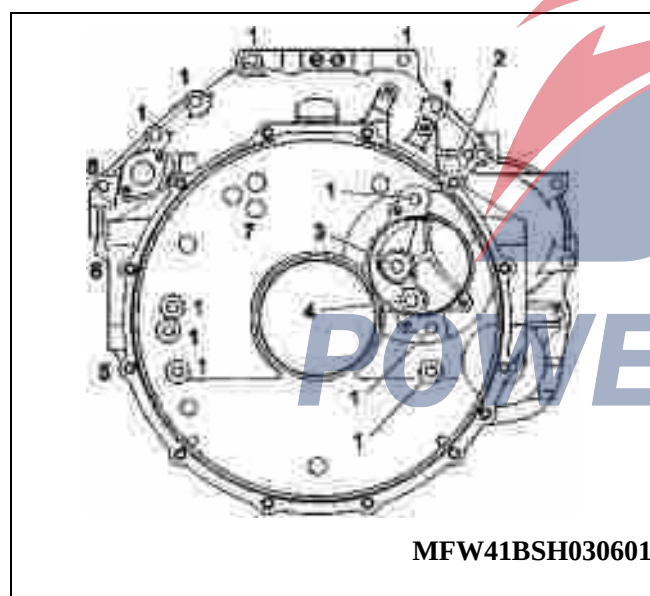
- Install the flywheel housing.
- First tighten the top pin position.

Tighten the rest in any order.

Tightening torque	N m (kgf m/lb ft)
73 (7.4 / 54)	
1: 96(9.8/71)	
8: 96(9.8/71)	
2: 86(8.8/64)	
4、3: 119(12.1/87.5)	
7: 26(2.7/20)	
5: 53(5.4/39)	

Note:

- Tighten the 5, 2 and 6 sections of the pattern from the side of the cylinder.

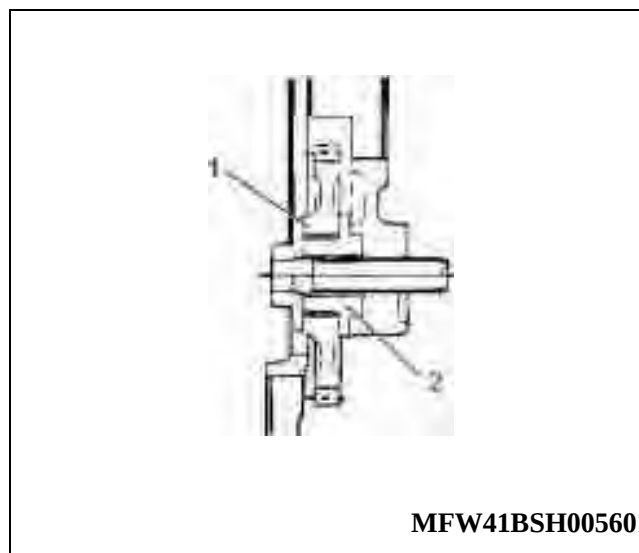


7. Install the air compressor idler shaft (2).

8. Install the air compressor idler (1).

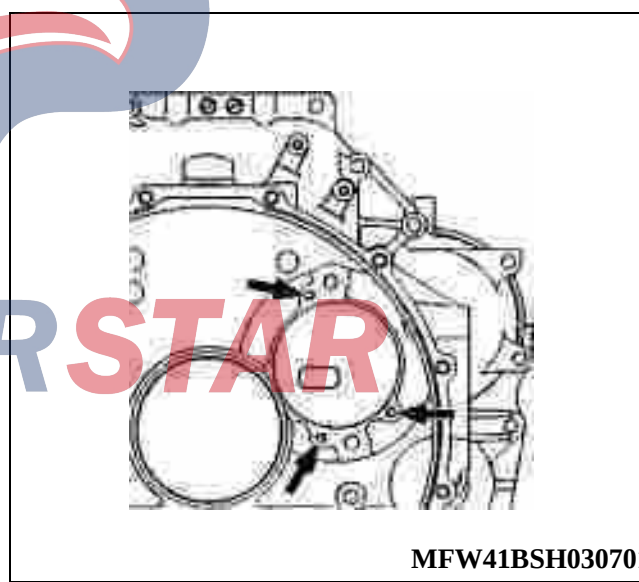
- Apply engine oil to the inside and outside of the idler and insert it into the shaft for installation.
- Tighten the bolts on the thrust ring (2).

Tightening torque	N m (kgf m/lb ft)
119 (12.1 / 87.5)	



9. Insert the O-ring and attach it to the lid.

Tightening torque	N m (kgf m/lb ft)
18 (1.8 / 13)	



10. Install the oil guide on the crankshaft.

- The oil retaining rings and oil seals should be installed separately with different sets of adjusting tools (8-9439-6856-0).
- Make sure the nozzle is installed in the correct direction.

Product name	Oil retaining ring		Oil seal	
	Front	Rear	Front	Rear
Front connector	○	—	○	—
Rear connector	—	○	—	○
Sleeve ring	○	○	—	—

Product name	Oil retaining ring		Oil seal	
	Front	Rear	Front	Rear
Sleeve seal	—	—	○	○
Bolted	○	○	○	○
Bolt center	○	○	○	○
Bolt PTO	—	○	—	○

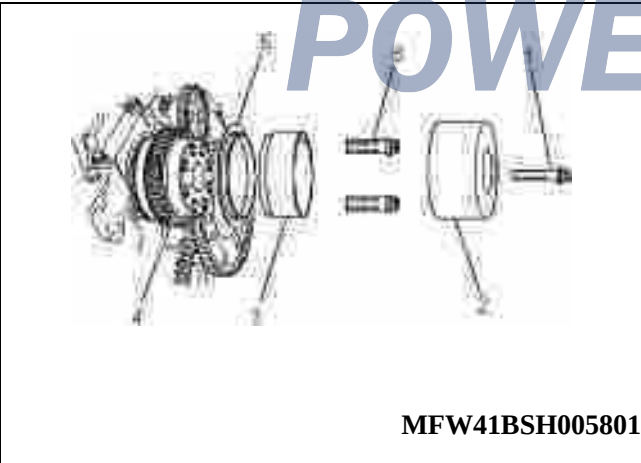
- Insert the slinger into the tip of the connector and secure the connector to the crankshaft.
- Cover the connector with the sleeve and tighten the center bolt until the sleeve contacts the connector.

Note:

- Since the treated part of 4 screws of the oil deflector ring can rotate the seal lip, operate carefully to avoid damaging it.
- The installation position of the slinger is set by four screw processing sections and is differentiated according to the color.

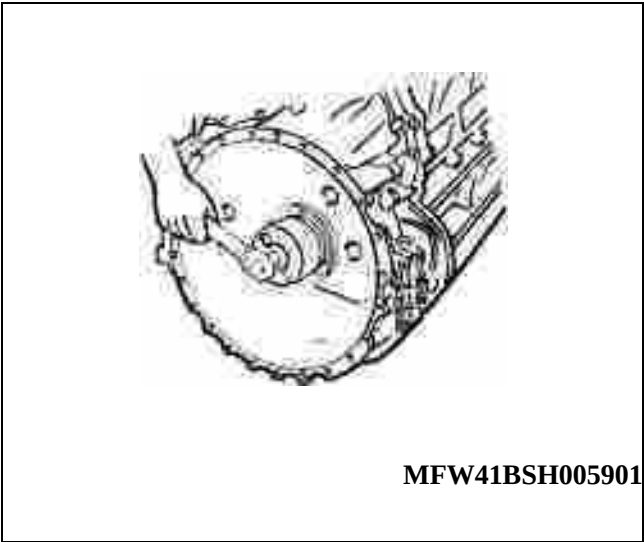
The front of the engine... Silver (the fourth groove from inside to outside along clockwise direction).

The rear side of the engine ... Graysgray (the fourth groove clockwise from the inside to the outside).



Legend

1. Center bolt
2. Sleeve
3. Connector (back)
4. Crankshaft
5. Rear retaining ring
6. Joint coupling bolt

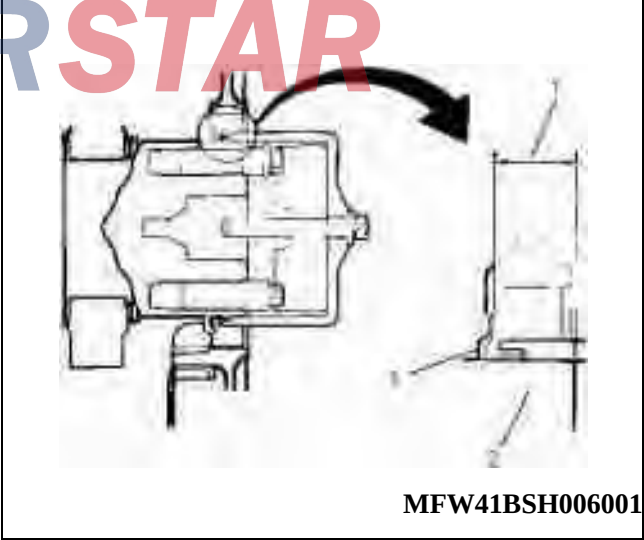


- Measure the outer eight equals and make sure that the slinger is embossed and meets the required dimensions.

Snap ring size (back)	mm(inch)
Assembly standards	16.7(0.6575)– 17.3(0.6811)

Note:

Measure the working surface from the edge surface of the crankshaft to the edge of the seal.



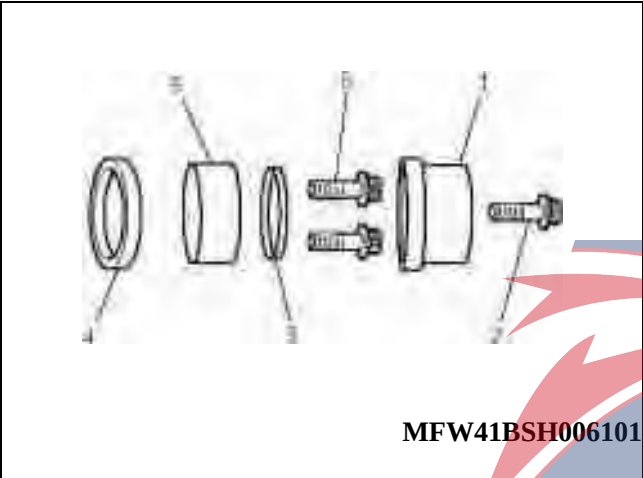
1. Ring size
2. Crankshaft
3. Oil retaining ring

11. Put on the oil seal.

- Be sure to apply engine oil to the lip portion before pressing the seal and install it with the setting kit (8-9439-6856-0).

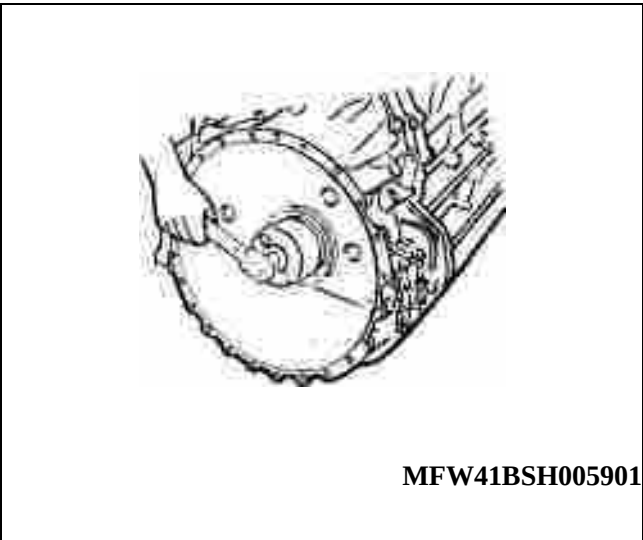
Note:

Do not allow the oil to come into contact with the outer circumference of the seals and felt parts.



Legend

1. Sleeve
2. Center bolt
3. Connector ring
4. Oil seal
5. Connector (back)
6. Connector bolt

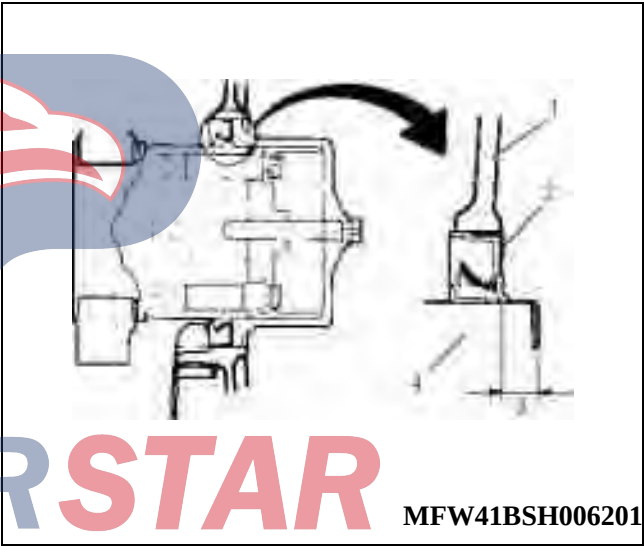


- Measure the outer eight equals and make sure that the slinger is embossed and meets the required dimensions.

Seal size (back)	mm(inch)
Assembly standards	7.35(0.2894)– 7.65(0.3012)

Note:

- If the damage to the rear axle of the slinger is removed, the inner surface of the slinger should be coated with liquid filler (San key brand 1207C glue) and installed.
- If there is oil leakage after installation, the oil should be wiped off.



Legend

1. Flywheel shell
2. Oil retaining ring
3. Oil seal size
4. Crankshaft

12. Put on the flywheel.

- Align and install the top pin with the flywheel.
- Apply molybdenum disulfide lubricant to the threads and base of the connecting bolts. Then assemble and tighten the washers in the order shown.
- Insert the guide bearing and place the retaining ring.

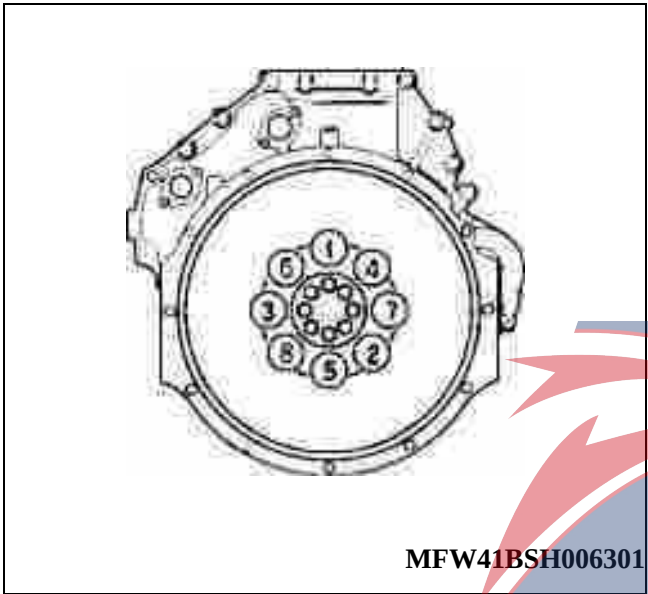
Tightening torque	N m (kgf m/lb ft)
First time:	78 (8.0/58)

Tightening torque	N m (kgf m/lb ft)
Second time: Angle method 60°→60°→90°	

Note:

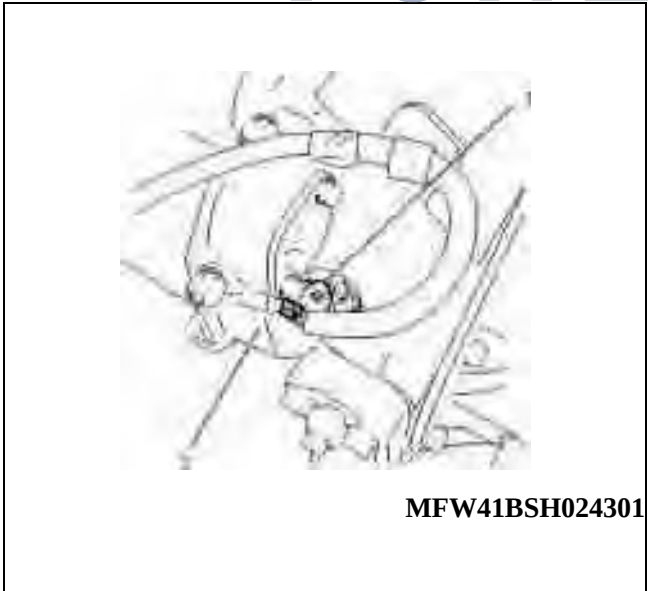
Ensure to check all the bolts and tighten every one.

The tightening sequence is the same.



13. Install the crankshaft position sensor (CKP).

Tightening torque	N m (kgf m/lb ft)
6 (0.6 / 52)	

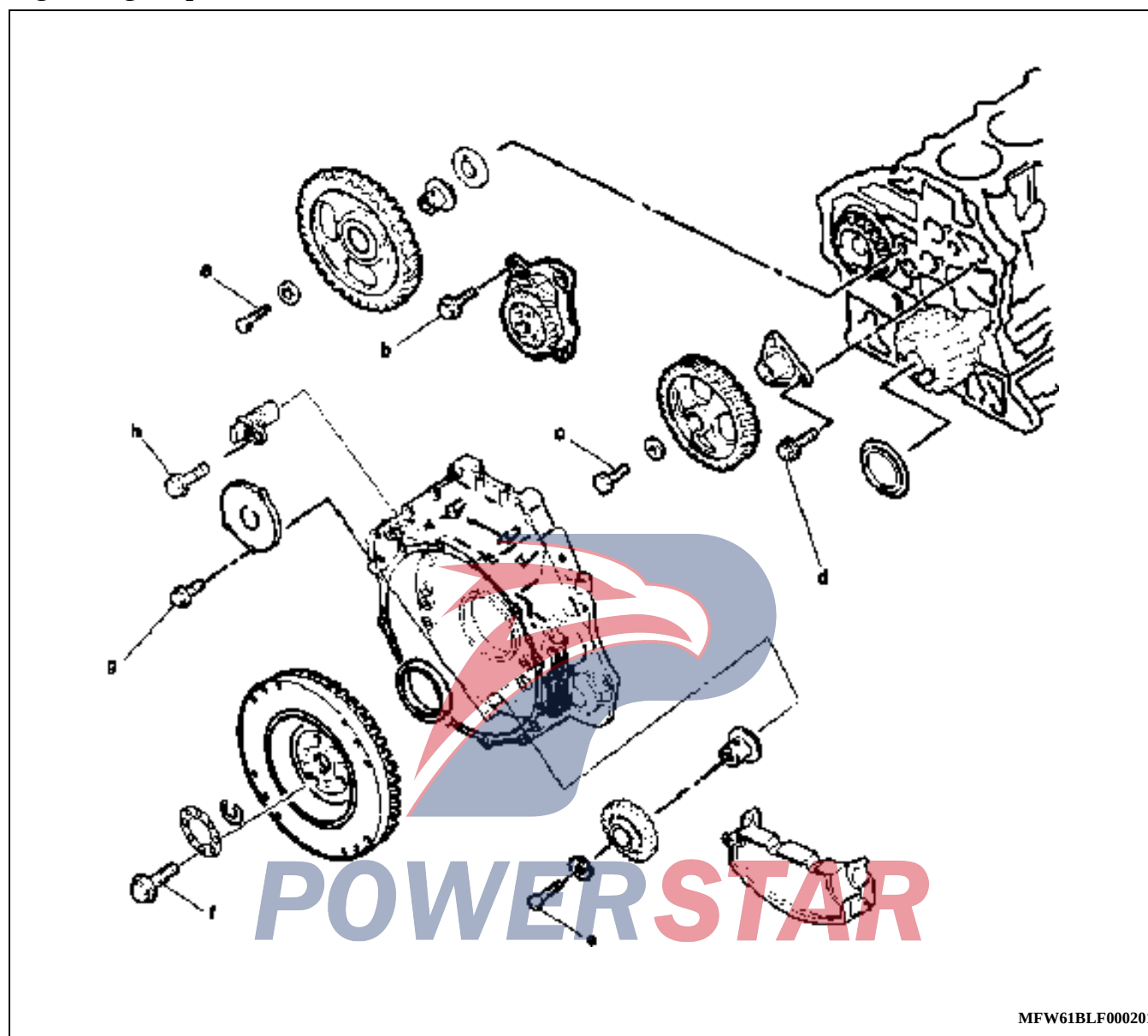


Legend

- 1. Crankshaft position sensor
- 2. Flywheel shell

- 14. Put on the oil pan.
 - Refer "Oil pan" in the lubrication system section.
- 15. Install the cylinder head.
 - Refer to "Cylinder Head".
- 16. Put the engine assembly into the car.
 - Refer to "Engine Assembly".

Tightening torque table



Legend

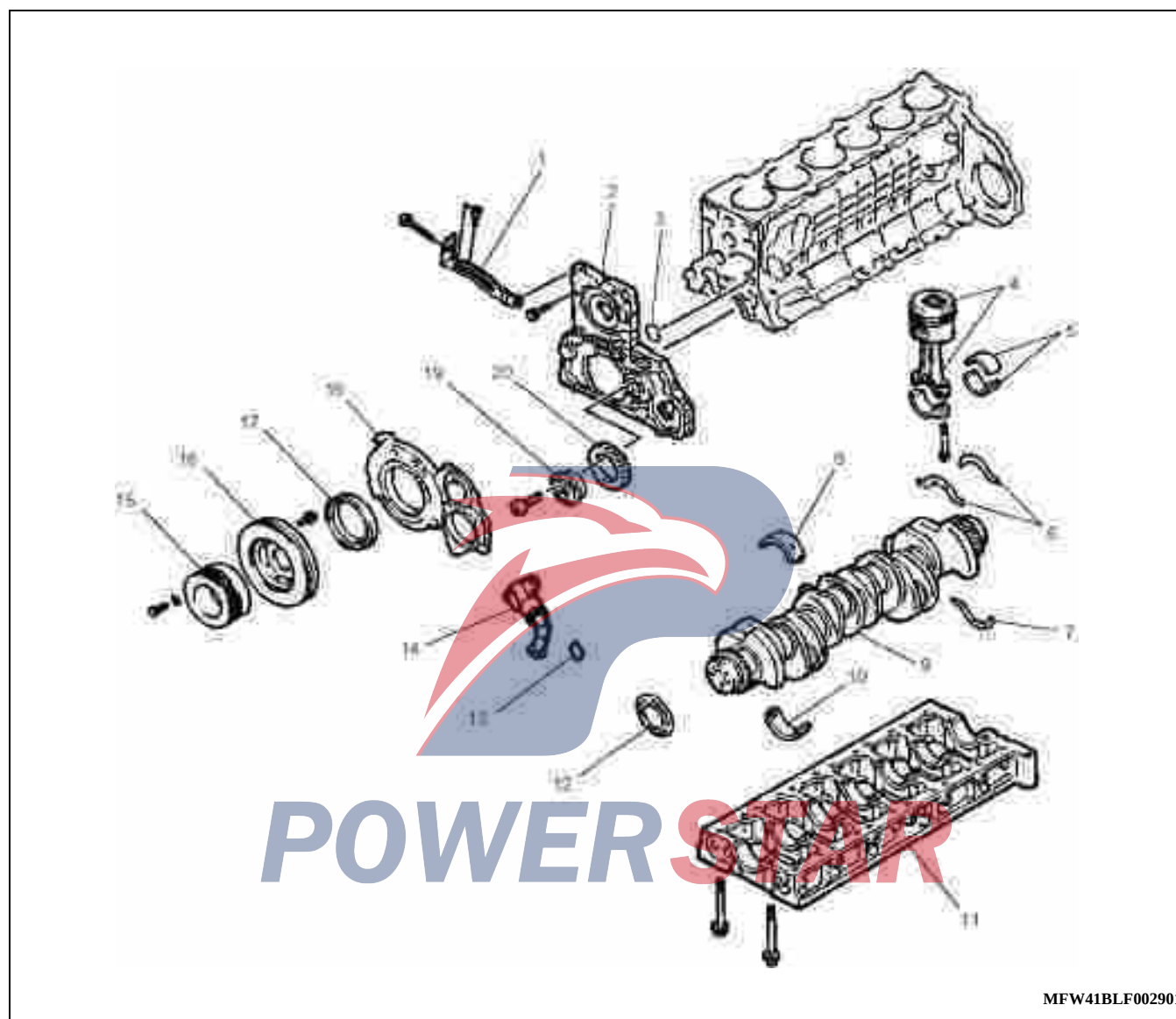
- | | |
|--|--|
| (a) 133 N m (13.6 kgf m / 98.4 lb. ft) | (f) 78 N m (8.0 kgf m/58 lb ft)→Angle60°→60°→90° |
| (b) 24 N m (2.4 kgf m/17 lb ft) | (g) 18 N m (1.8 kgf m/13 lb ft) |
| (c) 95 N m (9.7 kgf m/70 lb ft) | (h) 61 N m (0.6 kgf m/52 lb.ft) |
| (d) 31 N m (3.2 kgf m/23 lb ft) | |
| (e) 119 N m (12.1 kgf m/87.5 lb ft) | |

Special tools

Graphic representation	Tool number/description
 8943968580	8-9439-6858-0 slip ring dress up tool
 8943968560	8-9439-6856-0 Oil Seal Kit
 8943968180	8-9439-6818-0 crankshaft gear pull out tool
 8943968190	8-9439-6819-0 crankshaft gear setting tool

Crankshaft pulley, front cover and crankshaft

Part



MFW41BLF002901

Legend

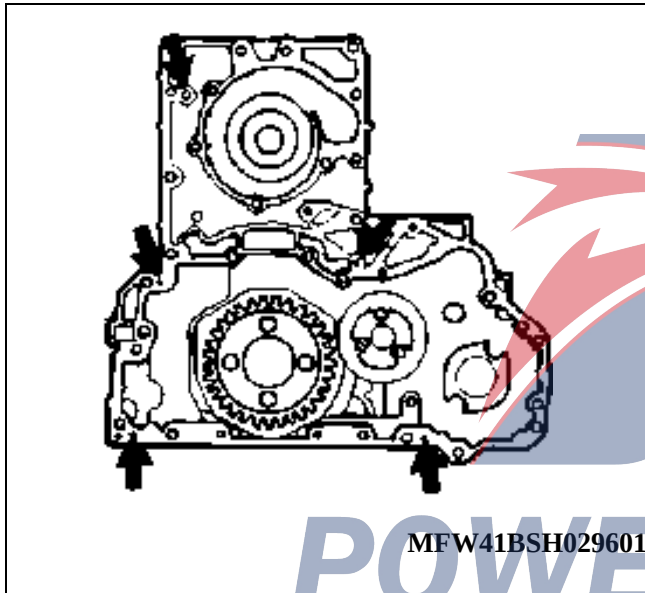
- | | |
|-----------------------------|----------------------------|
| 1. Adjusting plate | 11. Crankcase |
| 2. Gear box | 12. Oil retaining ring |
| 3. O-ring | 13. O-ring |
| 4. Pistons, connecting rods | 14. Water pipe |
| 5. Connecting rod bearing | 15. Crankshaft belt pulley |
| 6. Thrust bearing | 16. Shock absorbers. |
| 7. Thrust bearing | 17. Oil seal |
| 8. Crankshaft bearing | 18. Gear box cover |
| 9. Crankshaft | 19. Idler gear axis |
| 10. Crankshaft bearing | 20. Idler gear |

Tear down

1. Remove the engine components from the car.
Refer to "Engine Assembly".

2. Remove the cylinder head.
Refer to "Cylinder Head".

3. Remove the oil pan. Refer to "oil pan".
4. Remove the crankshaft pulley.
5. Remove the water pipe.
6. Remove the O-rings.
7. Remove the adjustment plate.
8. Remove the gearbox cover.
9. Remove idler and idler gear shaft.
10. Remove the transmission.
- Screw the bolts into the gearbox replacement holes (over 2 positions) and tighten and remove the gearbox evenly.



11. Remove the O-rings.
12. Remove the oil seal.
- When removing the gearbox cap oil seal, be careful not to damage the press fit surface.

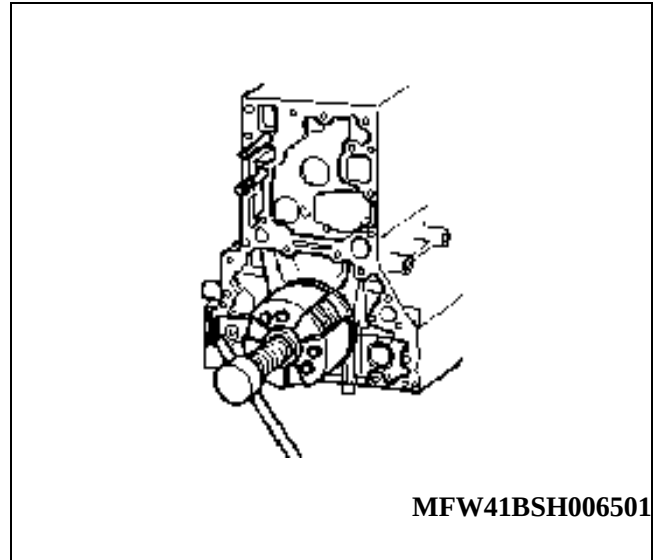
Note:

Replace the oil retainer and oil seal as a whole.

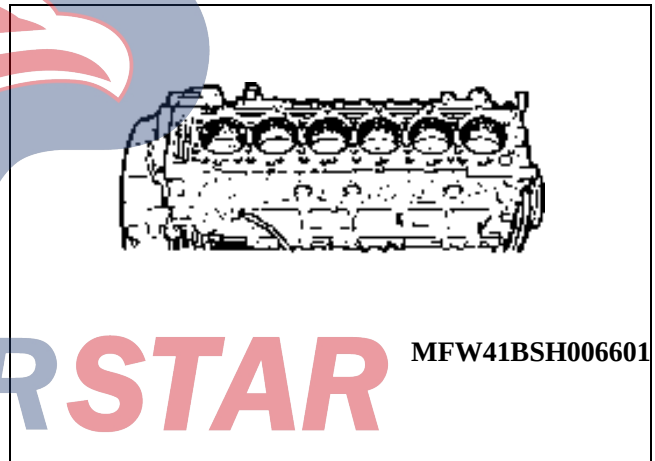
13. Remove the fender.
- Use a mudguard replacement tool (8-9439-6858-0) to remove the slinger.

Note:

Replace the oil retainer and oil seal as a whole.



14. Remove the piston rod.
- Remove the carbon deposit on the top surface of the bushing with a tool such as a scraper.



- Turn the crankshaft and move the cylinder piston to bottom dead center and remove the cylinder head.
- Turn the crankshaft and move the cylinder piston to the top dead center position. Use a handle such as a hammer to push the bearing out of the bottom of the connecting rod to avoid damaging the connecting rod.

Note:

- Be careful not to damage the injection hole and cylinder liner when ejecting the connecting rod.
- The cylinder cover, bearing and connecting rod are separately stored according to the original location of the cylinder.



MFW41BSH006701

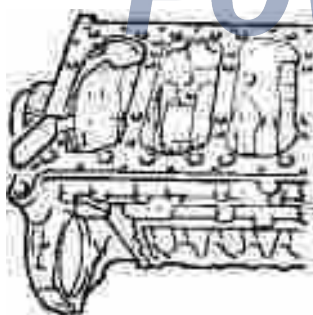
15. Remove the connecting rod bearing.

16. Measure the crankshaft axial clearance.

- The measurement shall be carried out with No.7 bearing part.

Horizontal clearance:		mm(inch)
Assembly standards	Use restrictions	
0.04(0.0016)–0.205(0.0081)	0.34 (0.0134)	

- If the measured value exceeds the available limit, replace the thrust bearing.



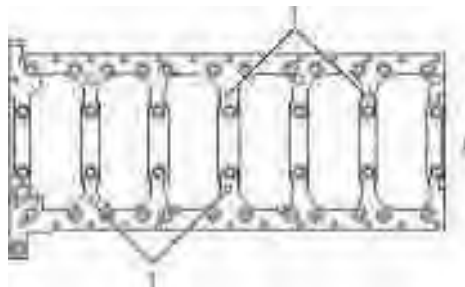
MFW41BSH006801

17. Remove the crankcase.

- Remove crankcase coupling bolts from outside to inside.
- Screw the bolt into the crankcase replacement hole (1) (4 places), tighten and remove the crankcase.

Note:

After removing the oil pump, remove the crankcase.



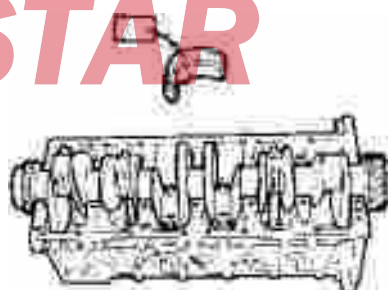
MFW41BSH006901

18. Remove the thrust bearing.

19. Remove the crankshaft bearing.

Note:

- When removing the crankcase, do not allow the bearings to come out.
- Be sure to identify the number of cylinders through the bearings.



MFW41BSH007001

20. Remove the Crankshaft.

- Lift the crankshaft horizontally to remove it.

Note:

If the end thrust bearing is lifted, it will be easier to lift.

21. Remove the thrust bearing.

22. Remove the crankshaft bearing.

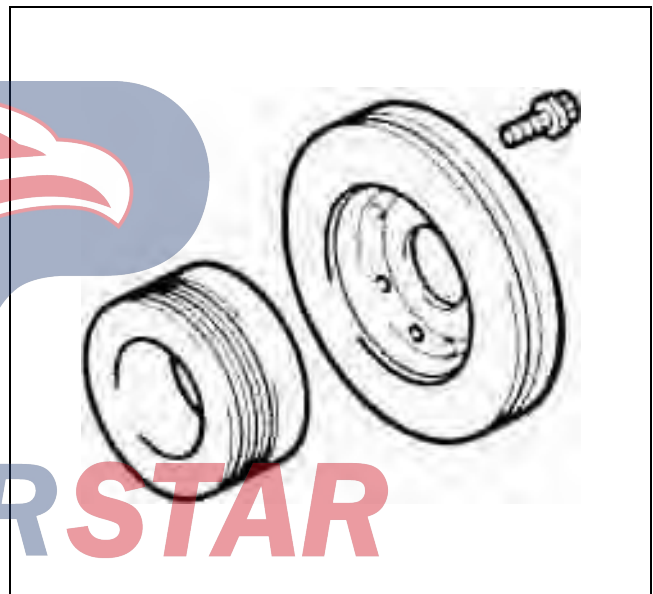
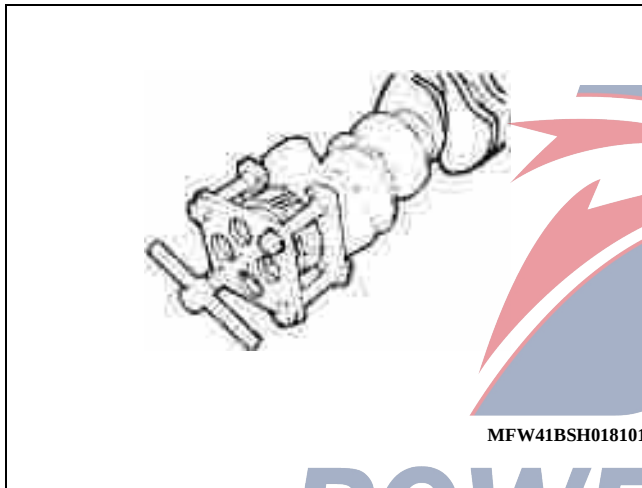
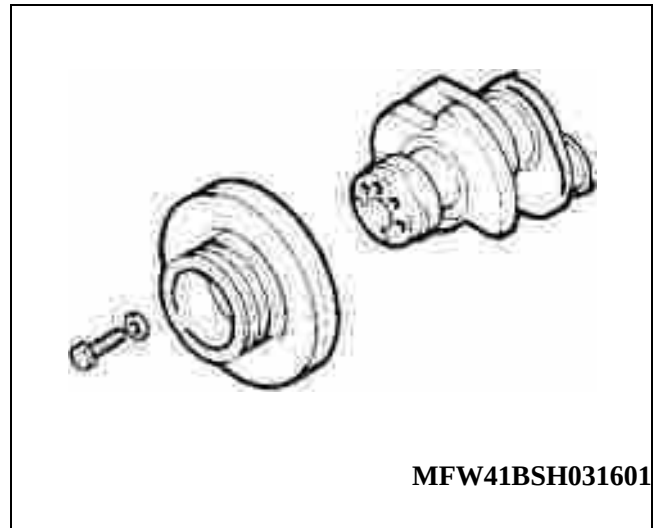
Note:

Be sure to identify the number of cylinders through the bearings.

Disassembly

Remove the gear.

Use a gear extraction tool (crankshaft gear extraction tool: 8-9439-6818-0) to remove it.



Inspection

1. Check the vibration absorber.

- Check if the damper cracks and falls off.

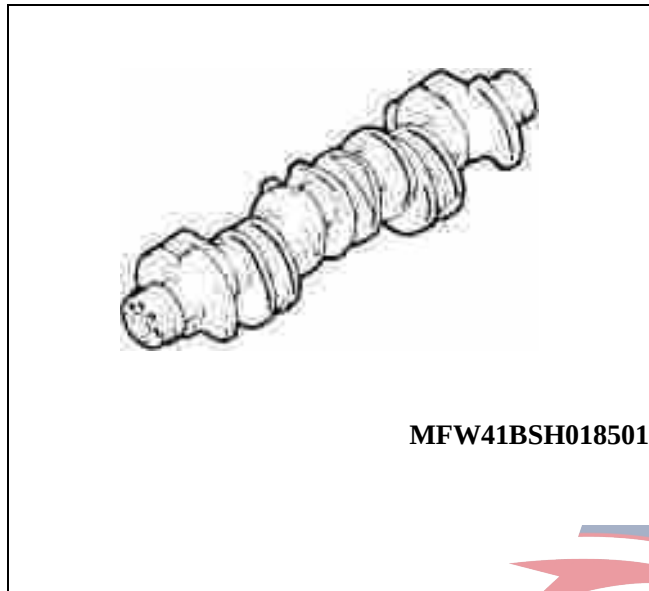
Note:

- If a fault is found, it should be replaced because a shock absorber may damage the crankshaft.
- If the shock absorber fails, check the crankshaft and flywheel bolts.

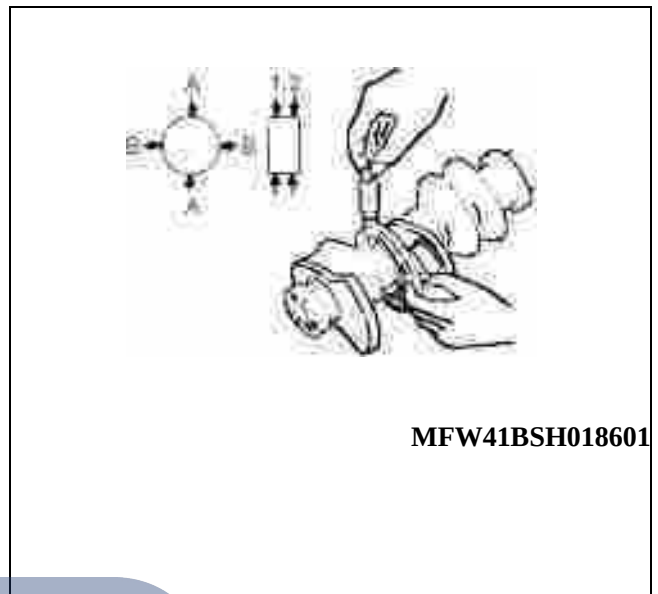
2. Check the crankshaft.

- Check if the crankshaft is cracked and the bearing impact surface is damaged.

- Use a magnetic flaw detector or dye penetration test (if necessary).



- For the log and fixed pin positions, four separate measurements should be made (as shown).

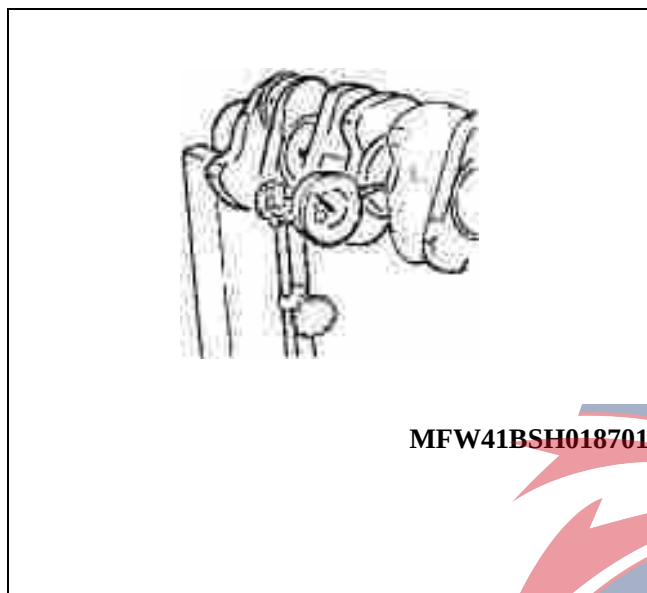


- b. Measure the degree of wear on the crankshaft journals and pins.

Shaft and fixing pin wear		mm(inch)	
	Nominal size	Assembly standards	Use restrictions
Outer diameter	82(3.2283)	No. 4 issue $\phi 81.851(3.2225)-81.871(3.2233)$	0.05(0.0020)
		Non-fourth bearing journal $\phi 81.881(3.2237)-81.901(3.2244)$	
Fixing pin outside diameter	73(2.8740)	$\phi 72.902(2.8702)-72.922(2.8709)$	0.05(0.0020)

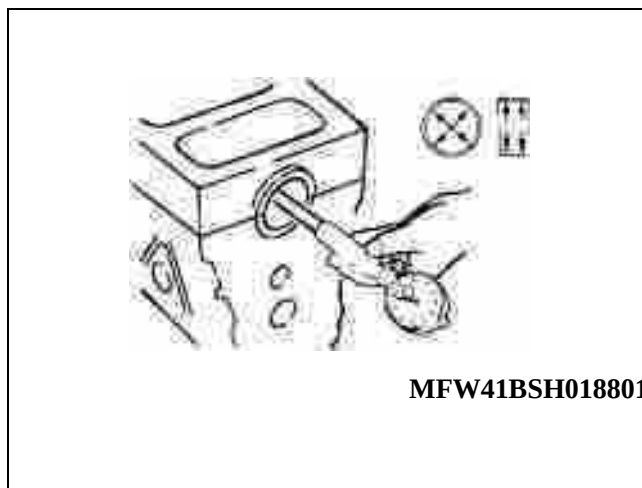
- c. Measure crankshaft swing.

- Secure the No. 1 and No. 7 journal sections with a crankshaft calibration tool or V-block. Set the dial indicator on the No. 4 journal of the crankshaft. Gently rotate the crankshaft position and measure the range of the dial indicator.

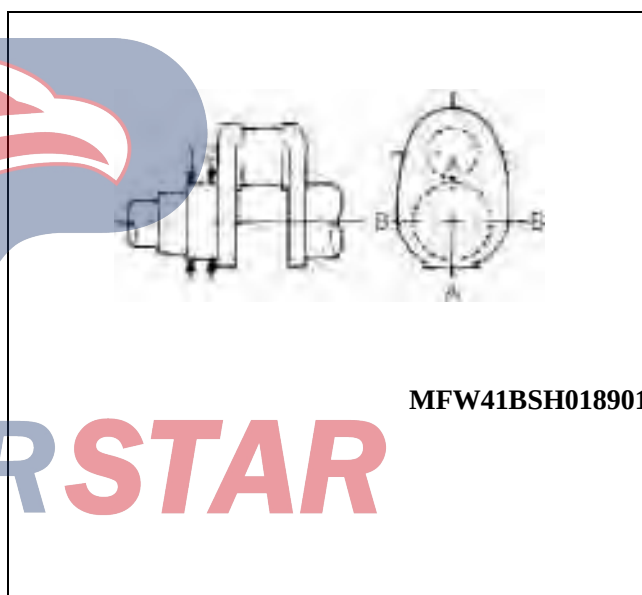


MF41BSH018701

- The outer diameter of the crankshaft is measured with a micrometer in the figure.



MF41BSH018801



MF41BSH018901

Crankshaft deflection		mm(inch)
Assembly standards	Use restrictions	
0.06(0.0024)	0.45(0.0177)	

- d. Measure the clearance between the bearing and the crankshaft journal.

Tightening torque	N m (kgf m/lb ft)
M14: 98(10.0/72)→132(13.5/98)→	
Angle Method 30° → -60°	

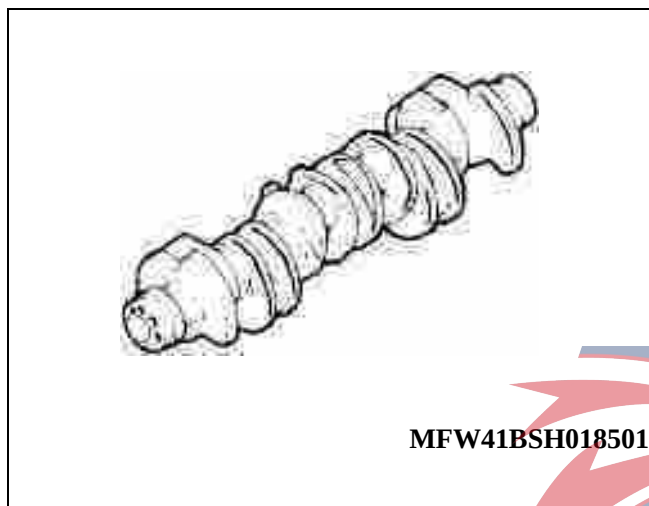
- Measure the bearing bore (when installed in crankcase) and crankshaft journal outside diameter and then work out the gap.
- Clean the bearing and bearing mounting surfaces, install the bearings, and tighten the crankcase according to the specified torque. Measure the inner diameter of the bearing in the four positions shown in the figure.

	mm(inch)	
	Assembly standards	Use restrictions
Clearance between crank bearing and crankshaft journal	0.093 (0.0037) - 0.124 (0.0049) for No. 4 journal 0.063 (0.0025) - 0.094 (0.0037) for non-No. 4 journal	0.14 (0.0055)

If you use a plastic gauge to make measurements, see the Overview section.

The presence of soft nitride layer

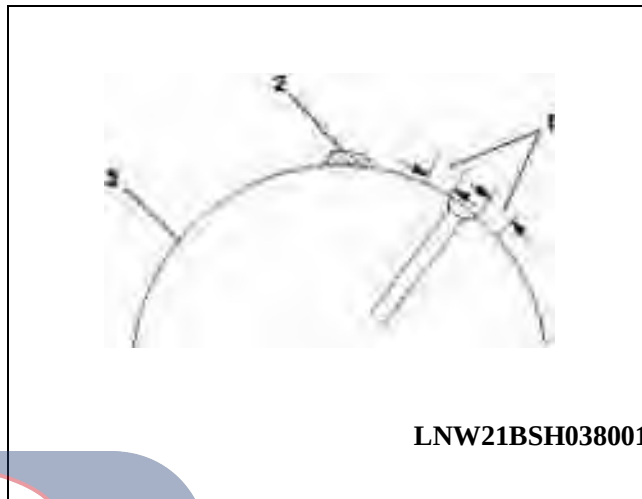
- Clean the crankshaft. Use an organic solvent, etc. to remove oil from the inspection area (except 10 mm (0.3937 in) from the oil hole).



Check whether there is a soft nitrating layer.

- Use a syringe or glass rod to inject the test solution into the test section and keep the test section flat to prevent the test solution from overflowing.

* Check solution: 5-10% cupric ammonium chloride dehydrate solution using distilled water as solvent.



Legend

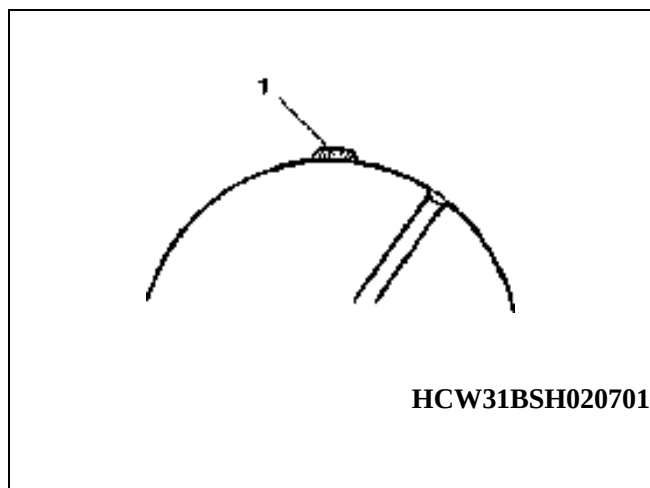
1. Do not check in a 10 mm (0.3937 inch) hole.
2. Check solution
3. Crankshaft

Decision

- If there is **no change** in the inspection part after 30-40 seconds, the crankshaft can be used again.

POWER STAR

- If the test solution (1) (light blue) becomes transparent after 30-40 seconds and then becomes copper, the crankshaft cannot be used again.



After treatment check

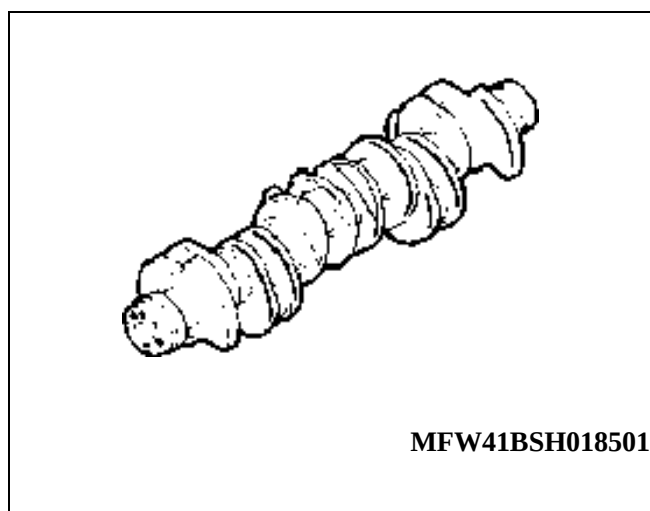
- Because the test solution is highly corrosive, wipe it off after inspection and clean the parts with water, steam, etc.

Note:

The crankshaft is nitro carburized and therefore cannot be repaired.

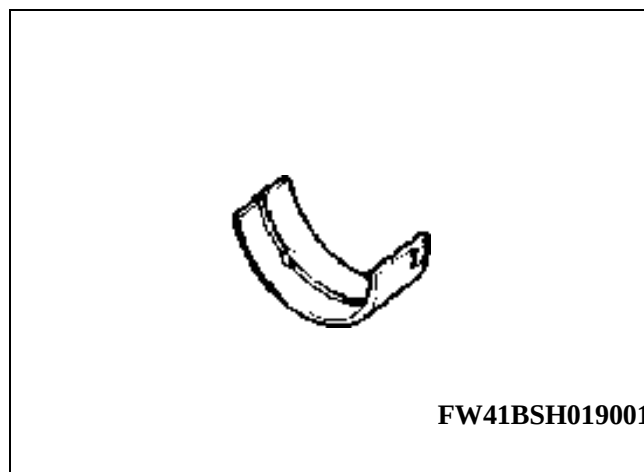
Caution:

Do not allow the test solution to come in contact with your eyes, hands, clothing, etc. If the liquid is accidentally splashed into someone's eyes, rinse the eyes with plenty of water immediately and receive immediate treatment.



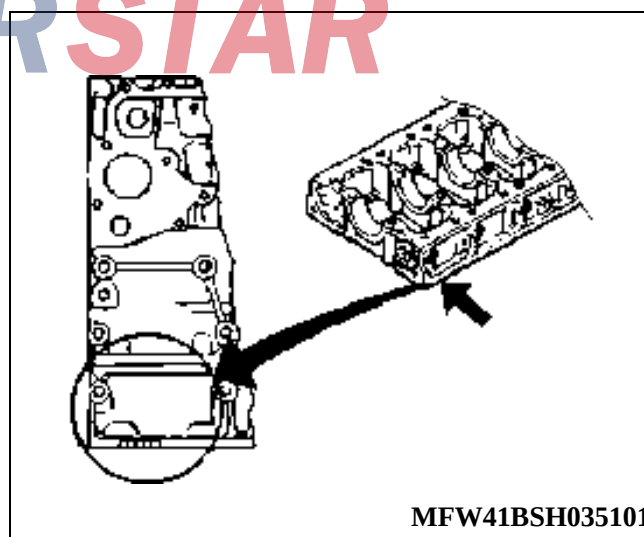
3. Check the bearings.

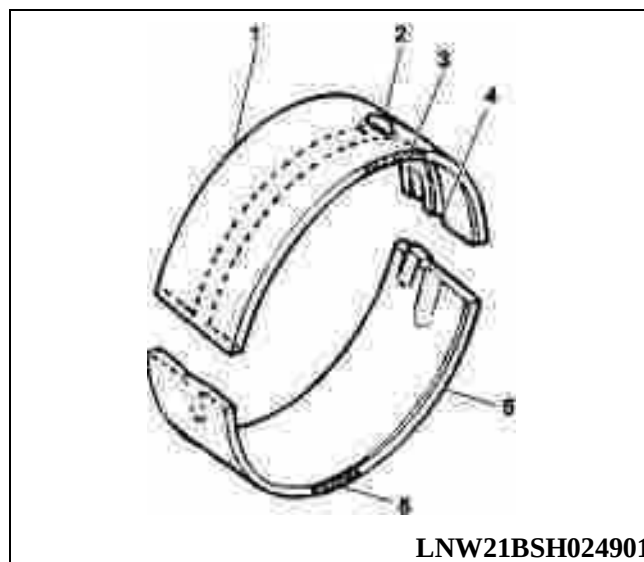
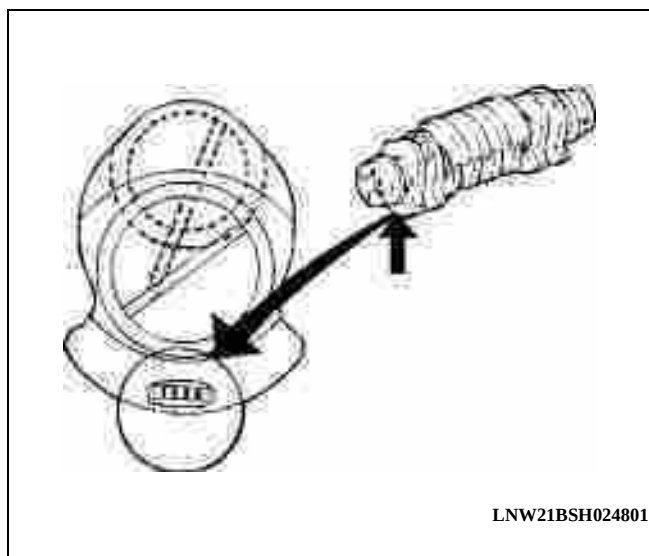
Inspect the bearing for wear, damage, shedding, and impact.



4. Crankshaft bearing selection program:

- When installing a new bearing or replacing an old bearing, select the proper bearing refer to the bearing selection table to.
- Check the bore diameter of the cylinder bore and the diameter of the crankshaft journal. Select a new bearing and install it.
- From the reading direction, magazine numbers are 1 to 7 from left to right.





Legend

1. Upper bearing
2. Oil holes
3. Level color
4. Oil grooves
5. Reduce the bearing
6. Level color

Note:

When installing, you should understand the difference in bearing configuration.

Bearing selection table

Combination of grades			Oil film clearance	mm(inch)
Cylinder level	Crankshaft neck level	Bearing identification color	No. 1,2,3,5,6,7	No.4
1	1	Black	0.065(0.0026) — 0.092(0.0036)	0.095(0.0037) — 0.122(0.0048)
1	2	Brown	0.063(0.0025) — 0.089(0.0035)	0.093(0.0037) — 0.119(0.0047)
2	1	Blue	0.067(0.0026) — 0.094(0.0037)	0.097(0.0038) — 0.124(0.0049)
2	2	Black	0.065(0.0026) — 0.091(0.0036)	0.095(0.0037) — 0.121(0.0048)

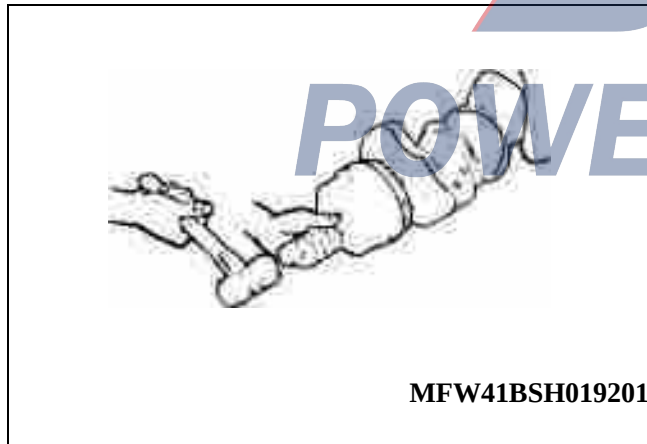
5. Check crankshaft gear teeth for looseness, wear, etc.



Assembling

Install crank gears.

- Heat the gear to 175°C (347°F).
- Position the timing gear toward the flywheel side and insert the preheated gear before cooling. After the gear cools down, align the crankshaft rear side fixing pin with the gear groove and insert the setting tool (No.: 8 - 9439 - 6819-0) until it contacts the shaft collar.



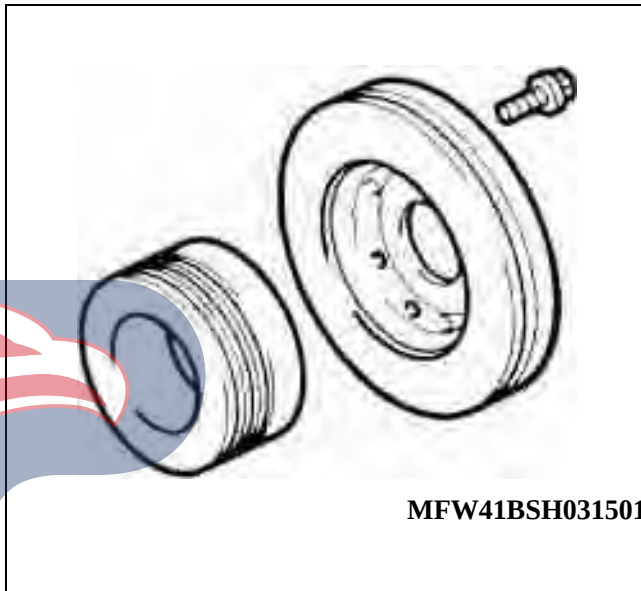
Install shock absorbers.

- Install a shock absorber on the crankshaft pulley.

Tightening torque	N m (kgf m/lb ft)
	48 (4.9 / 35)

Important note:

- Tighten the bolts diagonally.
- Do not allow lubricant or lubricant to come in contact with the shock absorber.



Reassemble

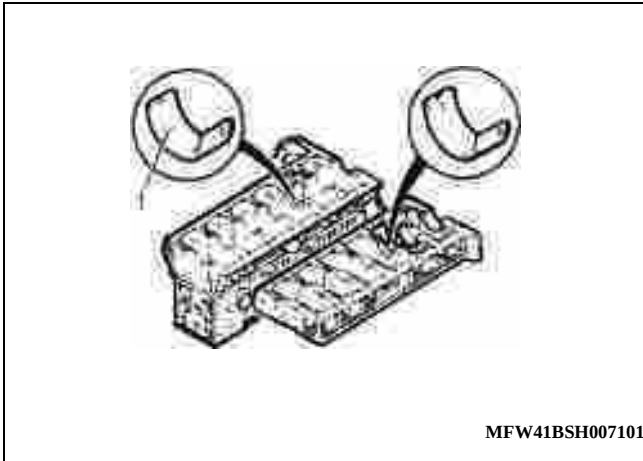
Clean all parts carefully before assembly.

1. Install the crankshaft bearing .

- If you need to install a new bearing, refer to the bearing selection table when selecting a new bearing (see Inspection section).
- Mount the bearing on the cylinder block and crankcase.
- Apply a lot of oil to the bearing.

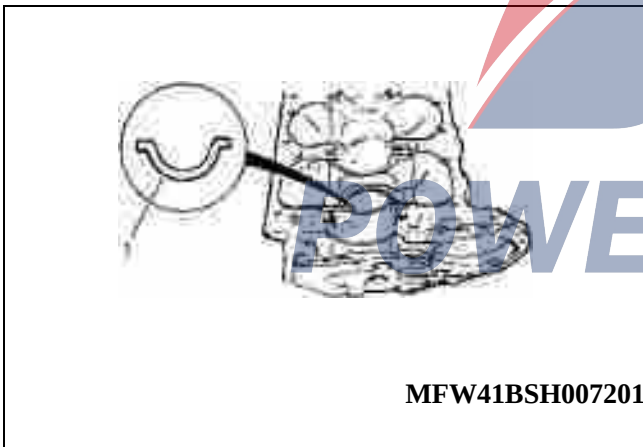
Note:

- The entire periphery of the cylinder block-side bearing has grooves and oil holes (1). The crankcase side bearing has a groove but no oil hole.
- Wear the bearing on the back under dry conditions.



2. Install thrust bearings.

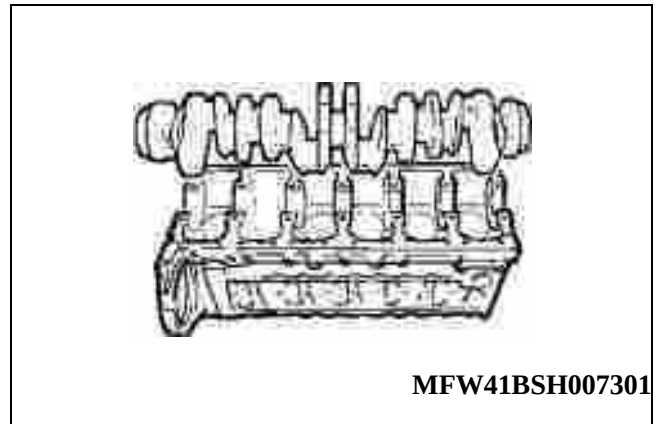
- Install the thrust bearing on the cylinder block.
- Apply oil to the bearing's rotating surface so that the oil groove (1) faces the crankshaft's rotating surface, aligns with the journal's No.7 cylinder block in the groove, and installs the bearing. At this time, applying a thin layer of grease on the surface of the roller bearing can make the operation smoother and eliminate the appearance of the phenomenon.



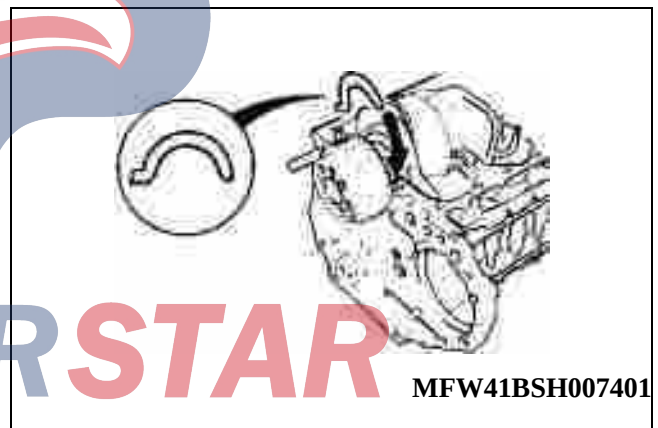
3. Install the crankshaft.

- Apply engine oil to the crankshaft journal, raise the crankshaft horizontally, and gently lower the crankshaft.

- Bring the crankshaft closer to the No.7 journal side so that it contacts the thrust bearing.



- Install the thrust bearing on the cylinder block.
- Apply oil to the rotating surface of the bearing so that the oil groove faces the crankshaft and aligns with the outer groove of the No.7 cylinders block and push it by hand.



4. Install the crankshaft bearing (lower part).

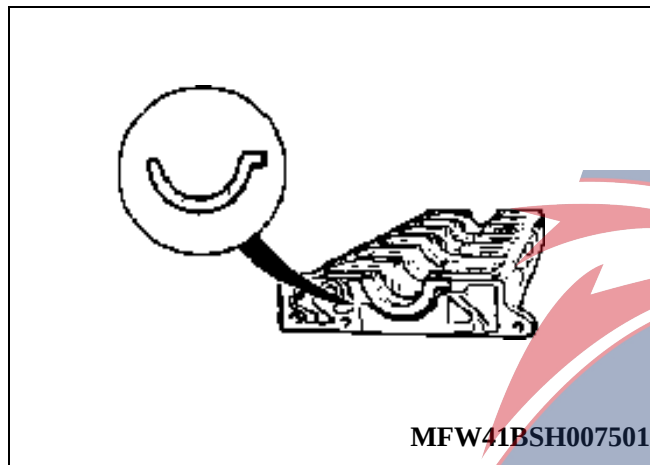
5. Install thrust bearings.

- Install the thrust bearing outside the crankcase.

- Apply oil to the rotating surface of the bearing so that the oil groove faces the rotating surface of the crankshaft and aligns with the outer groove of the No. 7 journal of the crankshaft, and then install the bearing. At this time, apply a thin layer of grease on the mounting surface of the crankshaft bearing, which can make the operation more smooth and eliminate the appearance of falling.

Note:

Always use a thrust bearing outside the crankcase.

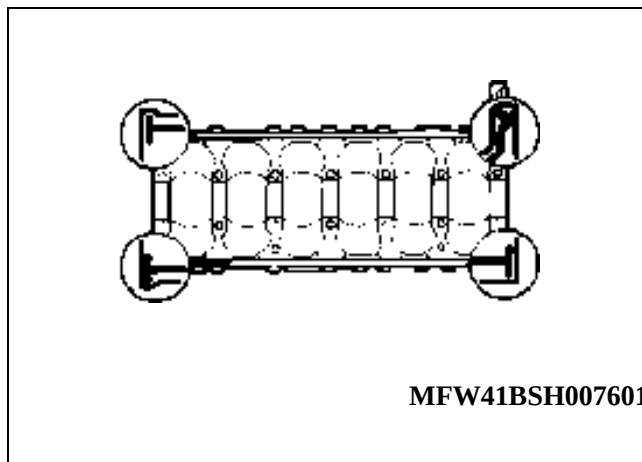


6. Install the crankcase.

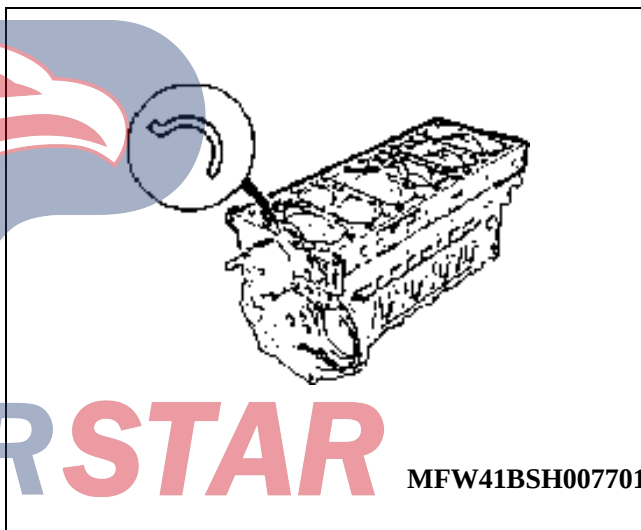
- Align the cylinder groove with a drop of liquid filler (No. 3 1207C or 1216C adhesive) approximately 3 mm (0.1181 inch) to 4 mm (0.1575 inch) wide.

Note:

- Use a filler to prevent the ball from cracking inside.
- Load it within 7 minutes of using liquid filling.



- Apply oil to the rotating surface of the bearing.
- Make the thrust bearing oil hole point to the cylinder body gear, install it gently and prevent the thrust bearing falling off.



- Tighten the crankcase.
- Apply molybdenum disulfide lubricant to the threads and base of the M14 bolts and install them in the position shown in the picture.
- Do not apply molybdenum disulfide lubricant or engine oil to M10 bolts.

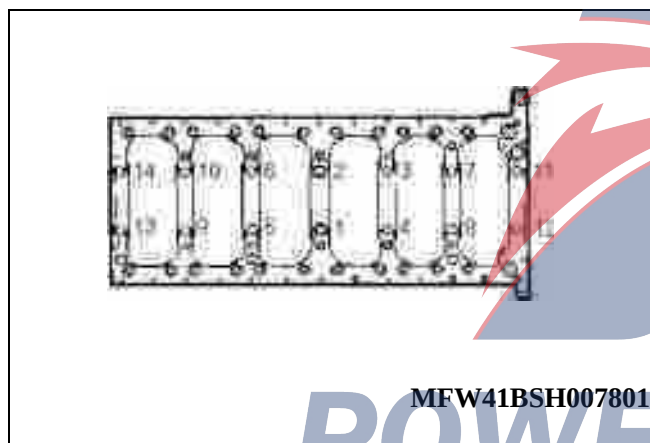
- Tighten the M14 bolt and install M10 bolt. (No fixed order)

Tightening torque	N m (kgf m/lb ft)
M14	98(10/72)→ 132(13.5/98)→Angle method 30°—60°
M10	37 (3.8 / 27)

- Clean spilled liquid packing.

Note:

Make sure the crankshaft can rotate smoothly.

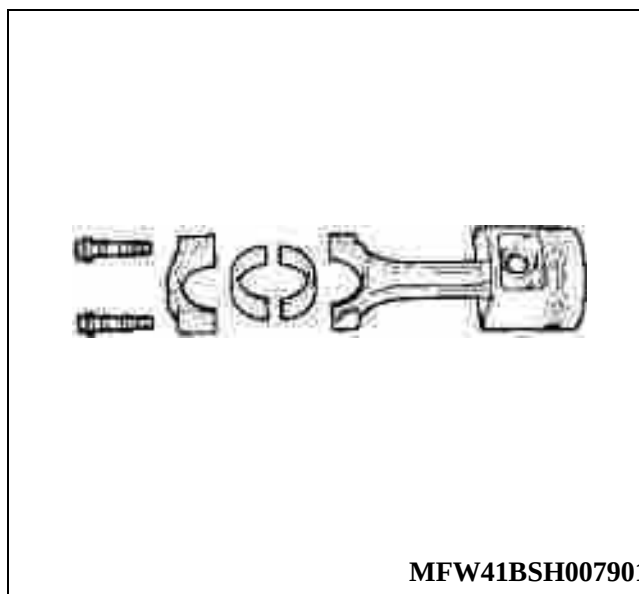


7. Install the connecting rod bearing.

- If you need to install a new bearing, refer to the bearing selection table when selecting a new bearing (see Inspection section).
- Clean the connecting rod and cover, align the groove, and install the bearing.
- Apply oil to the rotating surface of the bearing.

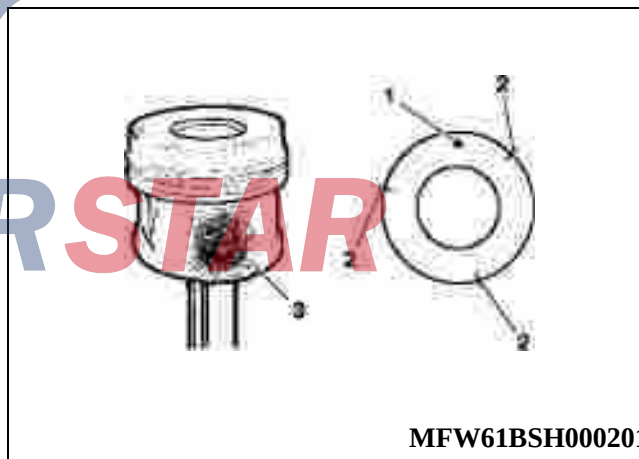
Note:

Wear the bearing on the back under dry conditions.



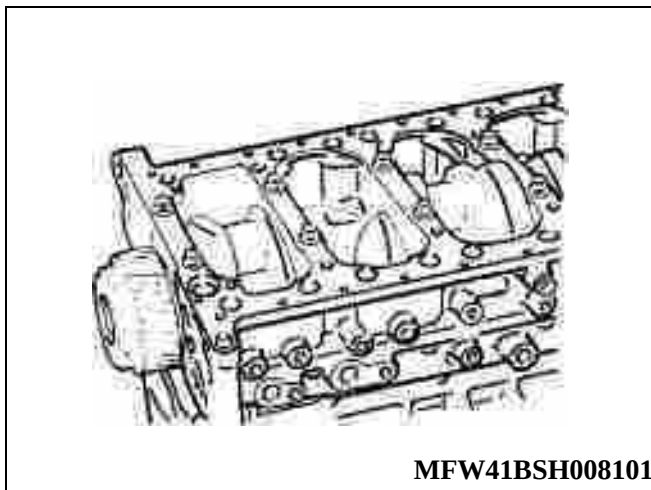
8. Install the piston rod.

- Apply oil over the entire circumference of the piston ring and arrange them separately (as shown) to avoid side pressure.
- As shown, apply molybdenum disulfide lubricant or engine oil to the piston.

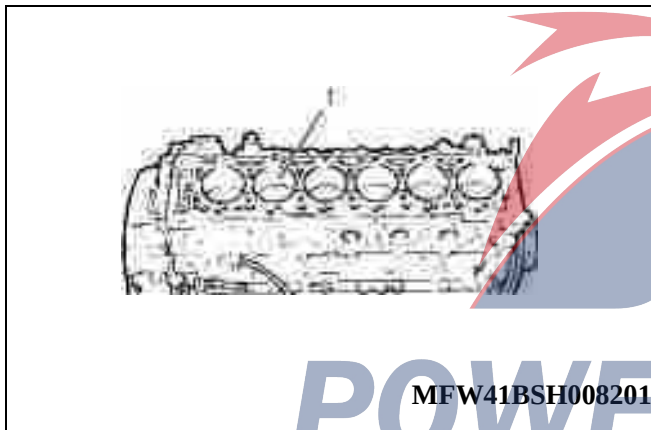
**Legend**

1. Front mark
2. Alignment section
3. Molybdenum disulfide lubricant

- Move the crank pin to the bottom dead center of the cylinder where the rod is installed.



- Apply engine oil to cylinder liner (1).



- Check the cylinder number of the connecting rod so that the front marked part of the piston head points to the front side of the cylinder and inserts it into the cylinder block.
- Use a piston to place the tool and use a handle such as a hammer to push the piston until it contacts the crank pin.

Note:

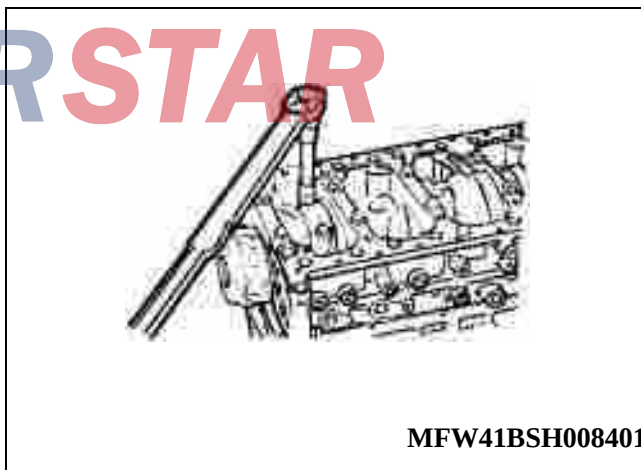
- When pushing in the piston, touch the mounted tool with the cylinder block. If it is not fully touched when pushed in, the piston ring will be damaged.
- Be careful not to damage the crank pin and spray hole.
- After inserting the piston rod, install a stopper to prevent the liner from falling off.



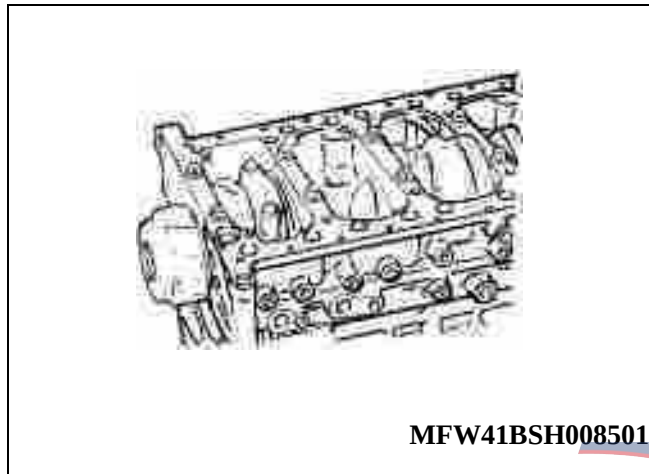
MFW41BSH008301

- Apply a lot of oil to the connecting rod bearings.
- Align the connection mark and cover mark, and then attach the cover.
- Molybdenum disulfide lubricant is applied to the thread and base of the bolt and then tightened alternately according to the specified torque.

Tightening torque	N m (kgf m/lb ft)
39 (4.0/29) → Angle Method 30° → 60°	



- Turn the crankshaft by hand to make sure it does not get stuck on any part and it can rotate smoothly. At the same time, make sure that the injection hole is not twisted and does not block other parts.

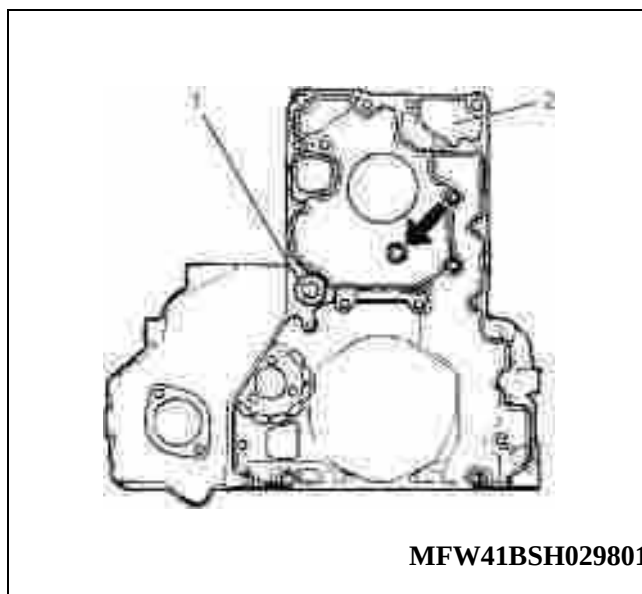


9. Install the gear box.

- Align the groove of the gearbox with a drop of liquid filler (San key brand 1207C or 1216C adhesive) about 2 mm (0.0787 inches) to 3 mm (0.1181 inches) wide.
- Assemble the O-ring (1).

Note:

- Do not forget to apply the liquid filler on the arrow in the figure.
- Load it within 7 minutes of using liquid filling.
- Alternatively, use THREE BOND 1208 or CH Dyne 290 bonding agent to prevent O-ring falling out.
- The engine is not equipped with an oil temperature valve. Because this valve may cause blockage and fire in the oil circuit

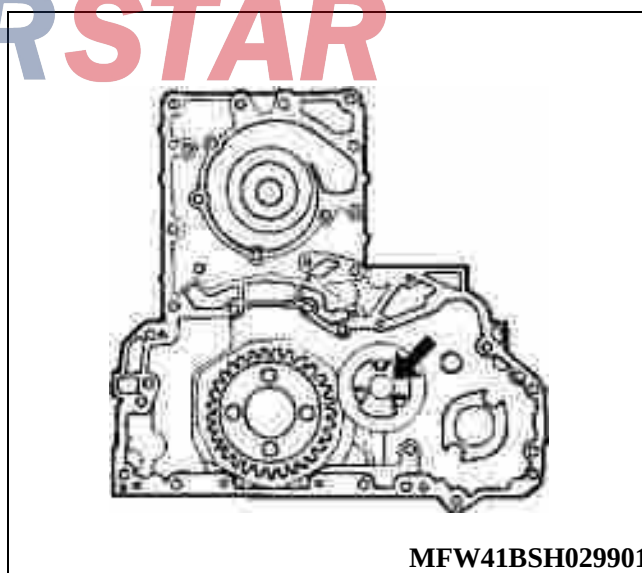


- Tighten the gearbox.

Note:

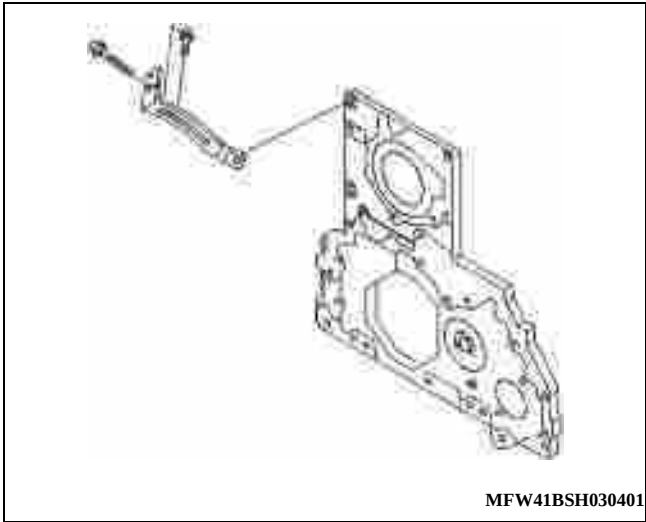
Remove liquid filler from the power steering idler shaft mounting holes.

Tightening torque	N m (kgf m/lb ft)
19 (1.9 / 14)	



- Tighten the adjustment plate evenly throughout the illustration.

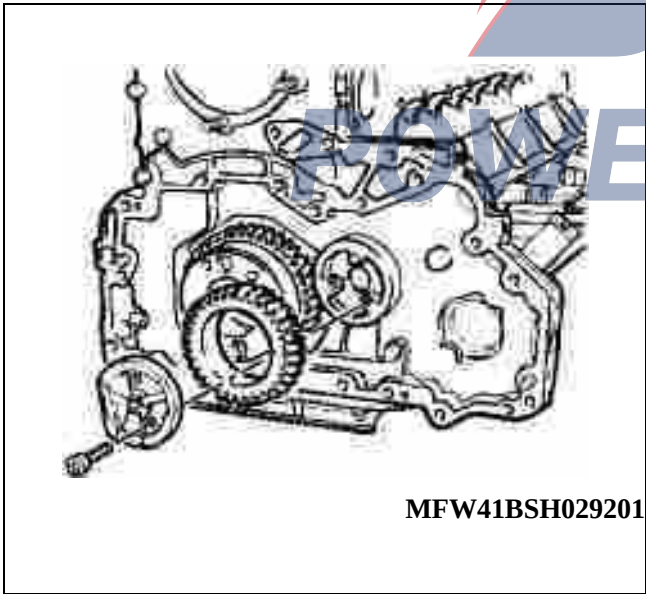
Tightening torque	N m (kgf m/lb ft)
19 (1.9 / 14)	



10. Install power steering pump idler.

Tightening torque	N m (kgf m/lb ft)
	19 (1.9 / 14)

- Rotate the rotating part of the idler gear and install it on the shaft.
- The notch portion of the shaft is positioned toward the crankshaft side and mounted on the gearbox.



Note:

Be careful not to damage the metal part of the idler.

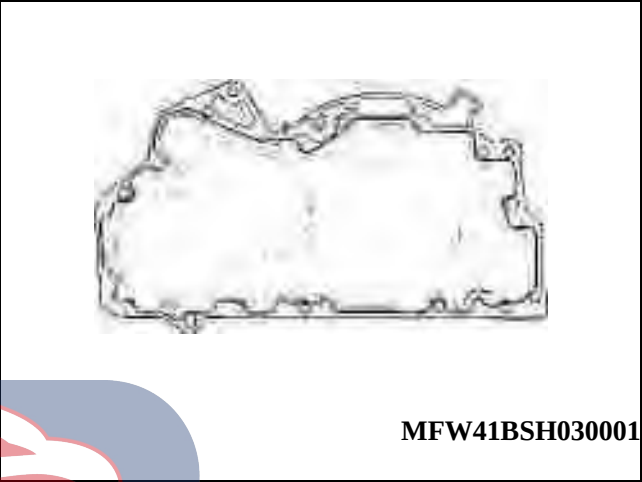
11. Install the idler shaft.

12. Install the transmission cover. Apply liquid filler on the gear box cover (Three Bond 1207C adhesive).

- Align the groove of the gearbox with a drop of liquid filler about 2 mm (0.0787 inches) to 3 mm (0.1181 inches) wide.

Note:

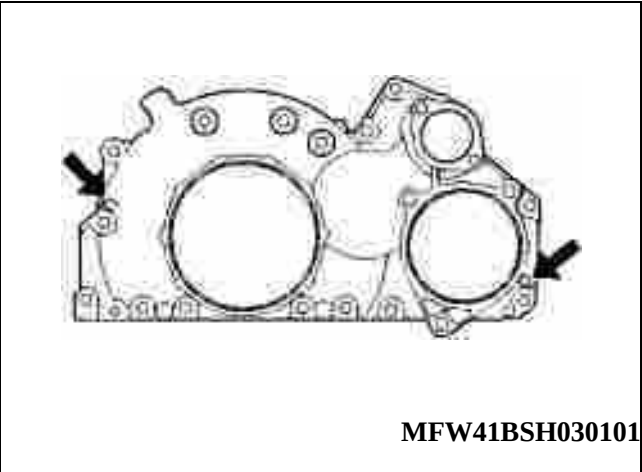
Load it within 7 minutes of using liquid filling.



- Tighten the transmission cover.

Tightening torque	N m (kgf m/lb ft)
	19 (1.9 / 14)

- First tighten the top pin and tighten it clockwise or counterclockwise.

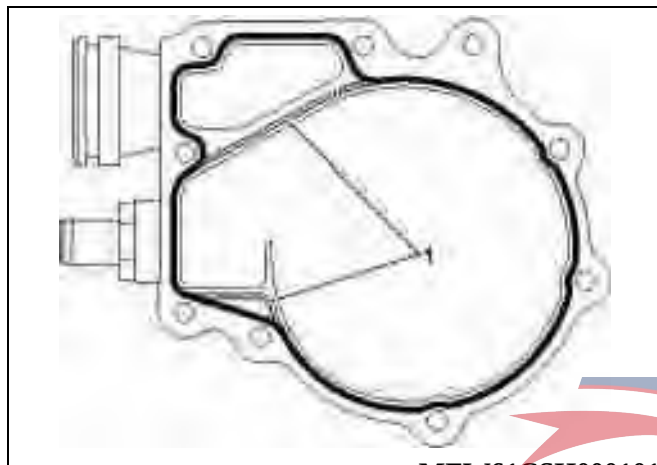


13. Install the water pump.

For more information, see cooling system.

Note:

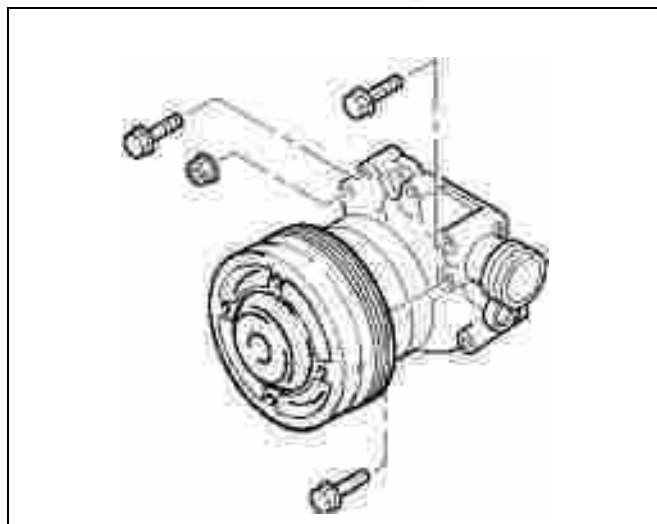
- Tighten the gearbox and install the pump (prevent liquid seals from drying out).
- Load it within 7 minutes of using liquid filling.

**Legend**

1. The filler cannot be used even if there is a groove

- Tighten the pump assembly.

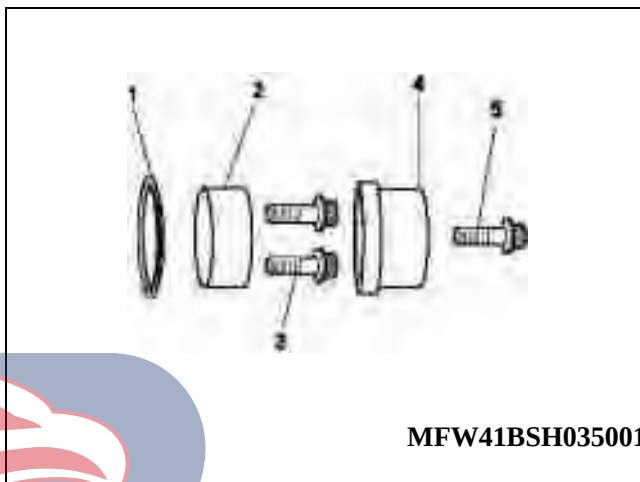
Tightening torque	N m (kgf m/lb ft)
19 (1.9 / 14)	

**Note:**

Remove liquid filler from the power steering idler shaft mounting holes.

14. Install the flinger on the front side of the crankshaft.

- Insert the slinger into the tip of the connector and secure the connector to the crankshaft. Use the set adjustment tool (8-9439-6856-0) to press the oil retaining ring according to the required size.
- Cover the connector with the sleeve and tighten the center bolt until the sleeve contacts the connector.



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Legend

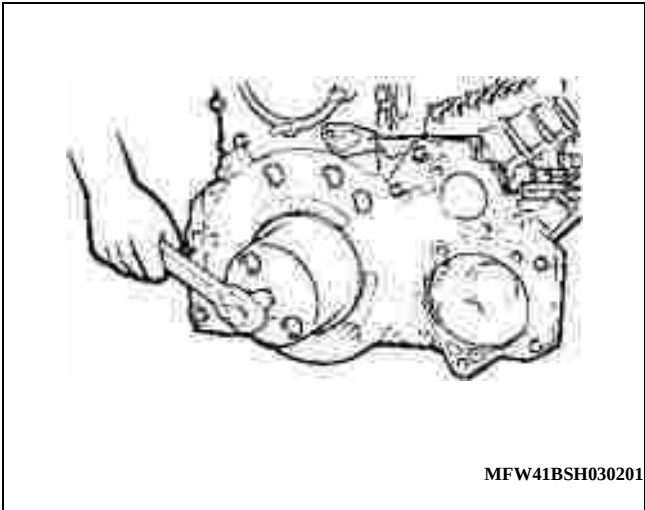
1. Oil retaining ring
2. Connector
3. Connector bolt
4. Sleeve
5. Center bolt

Note:

- Since the treated part of 4 screws of the oil deflector ring can rotate the seal lip, operate carefully to avoid damaging it.
- The installation position of the slinger is set by four screw processing sections and is differentiated according to the color.

The front of the engine... Silver (the fourth groove from inside to outside along clockwise direction).

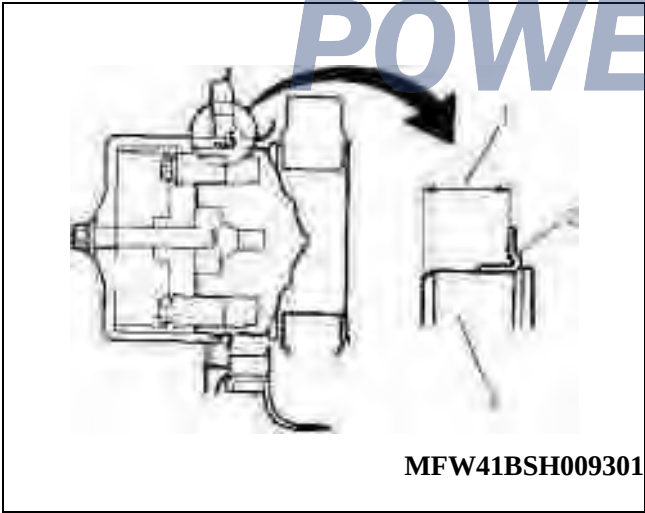
The rear side of the engine ... Grays (the fourth groove clockwise from the inside to the outside).



- Measure the distance between the crankshaft edge surface and the working surface of the oil seal lip.

Snap ring size (front side)	mm(inch)
Assembly standards	17.7(0.6968)–18.3(0.7205)

Note:
After pressing the oil retaining ring, make sure there is no deflection.



- Legend**
1. Ring size
 2. Oil retaining ring
 3. Crankshaft

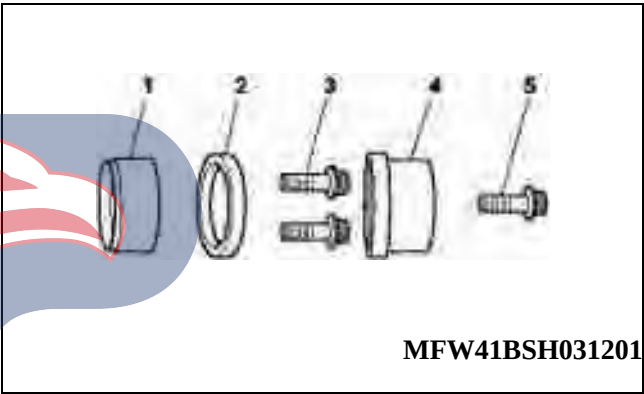
15. Put on the oil seal.

- Apply a thin layer of oil to the lip of the oil seal and install it on the crankshaft.
- Insert the adapter ring into the sleeve and press it together.

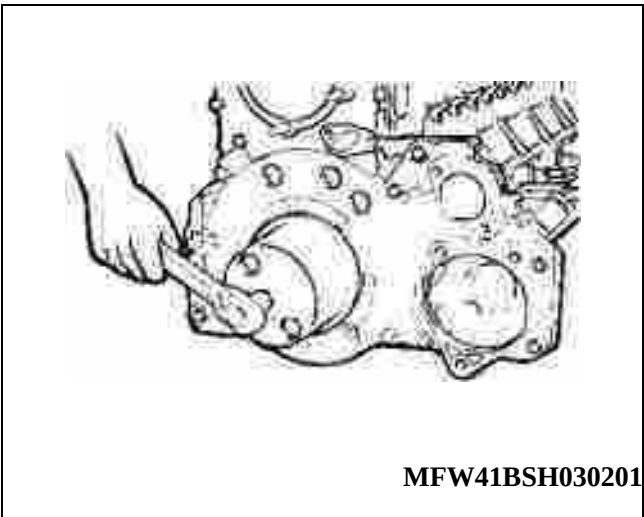
Note:
Be sure to apply oil to the lip of the oil seal.

Special tools

Complete set of tools: 8-9439-6856-0



- Legend**
1. Anterior joint
 2. Oil seal
 3. Connector bolt
 4. Sleeve
 5. Center bolt

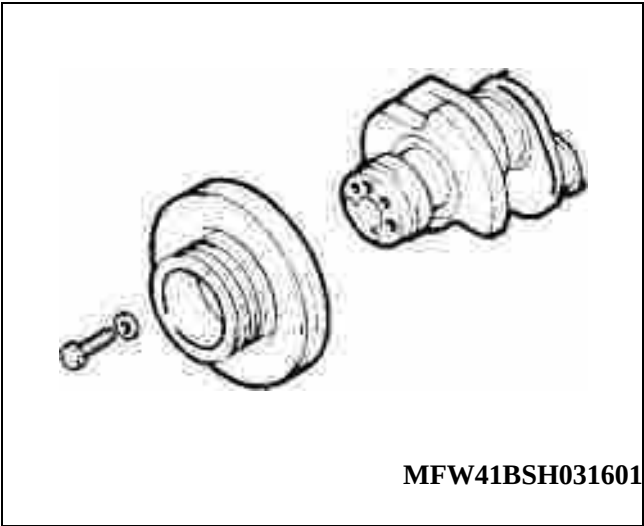
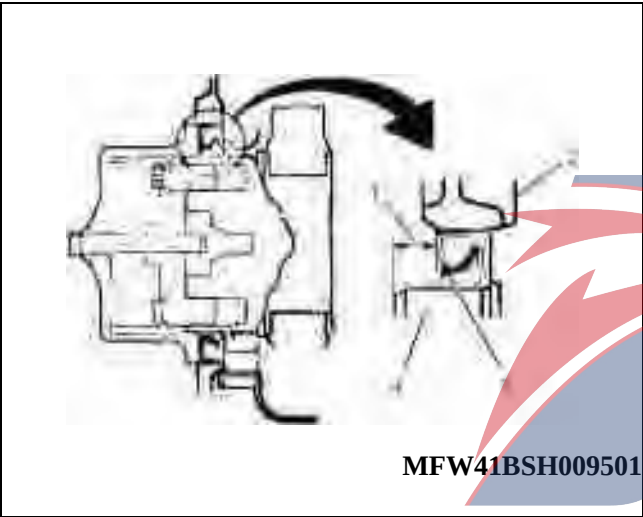


- Measure the distance between the crankshaft edge and the oil seal surface.

Oil seal size (front side)	mm(inch)
Assembly standards	8.35(0.3287)－8.65(0.3406)

Note:

Make sure the seal is compressed and not bent.



- 17. Put on the oil pan. Refer to "Oil pan".
- 18. Install the cylinder head. Refer to "Cylinder Head".
- 19. Put the engine assembly into the car. Refer to "Engine Assembly".

Legend

- 1. Oil seal size
- 2. Front cover
- 3. Oil seal
- 4. Crankshaft

POWERSTAR

16. Crankshaft belt pulley

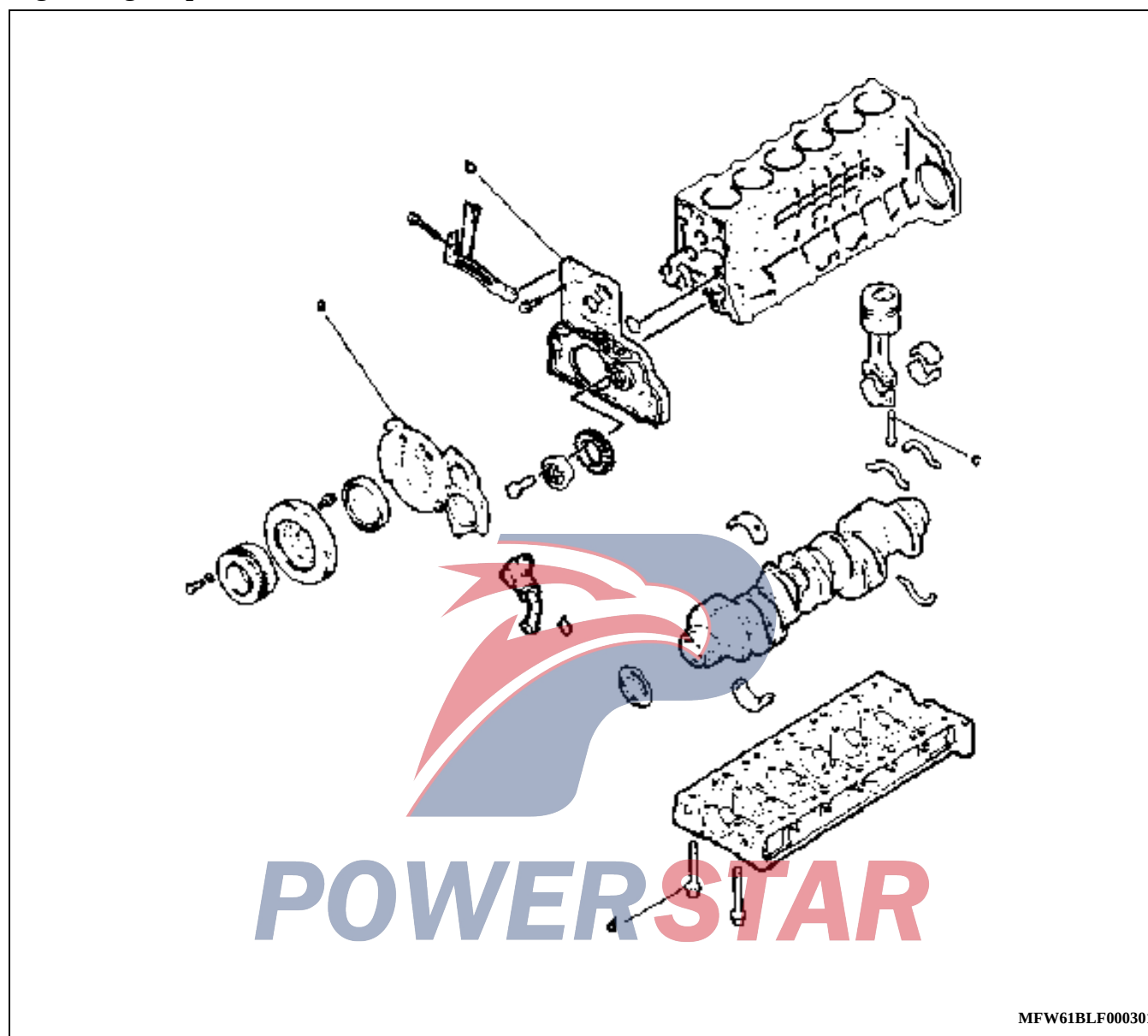
Align and install the top pin with the crankshaft pulley.

Tightening torque	N m (kgf m/lb ft)
200 (20.4 / 147.6)	

Note:

- Tighten the bolts diagonally.
- Do not allow oil or lubricant to come in contact with the crankshaft pulley damper.

Tightening torque table



(a) 19 N m (1.9 kgf m / 14 lbs. ft)

(b) 19 N m (1.9 kgf m/14 lb.ft)

(c) 39 N m (4.0 kgf m/29 lb ft)→Angle 60°→60°→-60°

(d) M14: 98 N m (10.0 kgf/m/72 lb ft)→132 N m (13.5 kgf/m/98 lb ft)→30°-60°

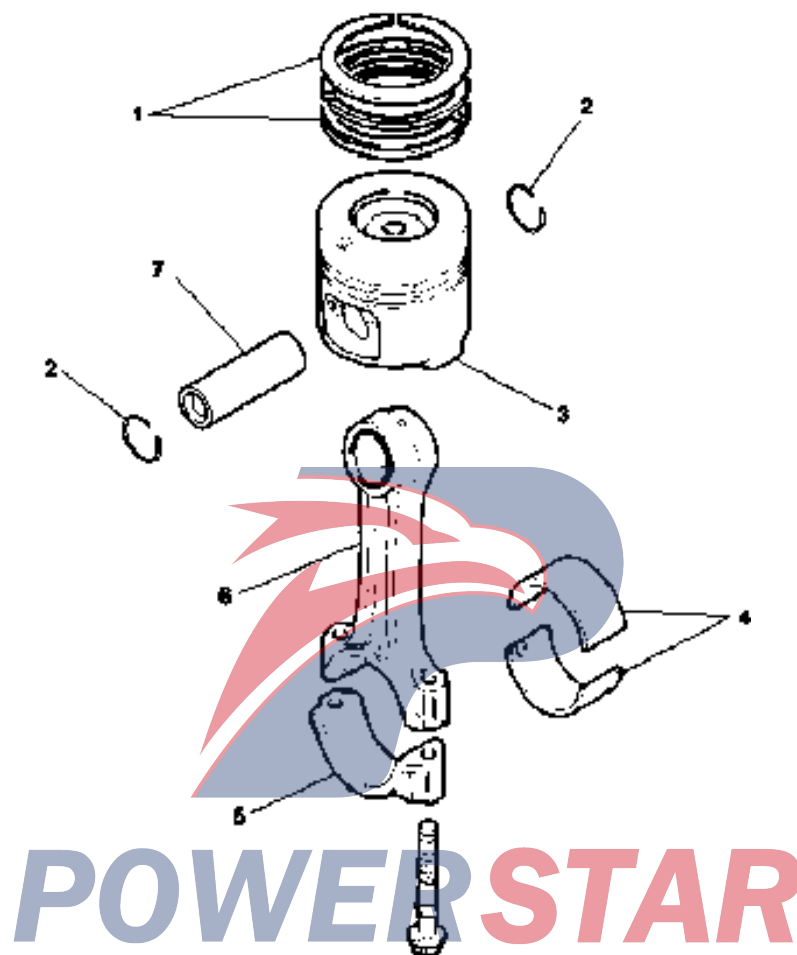
M10: 37 N m (3.8 kgf-m / 27 lb ft)

Special tools

Graphic representation	Tool number/description
 8943968580	8-9439-6858-0 slip ring dress up tool
 8943968560	8-9439-6856-0 Oil Seal Kit
 8943968180	8-9439-6818-0 crankshaft gear pull out tool
 8943968190	8-9439-6819-0 crankshaft gear setting tool

Piston and Connecting Rod

Part



BLW61BLF000101

Legend

- | | |
|----------------|-------------------------|
| 1. Piston ring | 5. Connecting rod cover |
| 2. Buckle | 6. Connecting rod |
| 3. Piston | 7. Piston pin |
| 4. Bearing | |

Tear down

1. Remove the oil pan.

Refer to "Crankshaft".

2. Remove the rocker shaft assembly.

Refer to the "rocker assembly".

3. Remove the camshaft assembly.

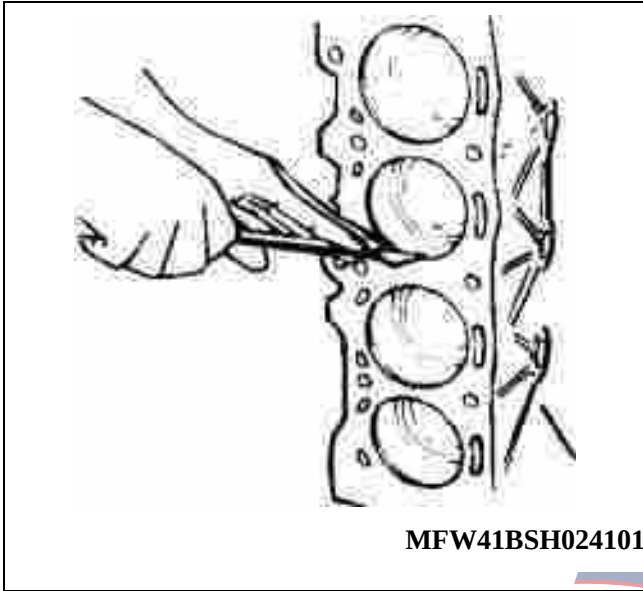
Refer to the 'camshaft assembly'.

4. Remove the cylinder head.

Refer to "Cylinder Head".

5. Remove the piston and link rod.

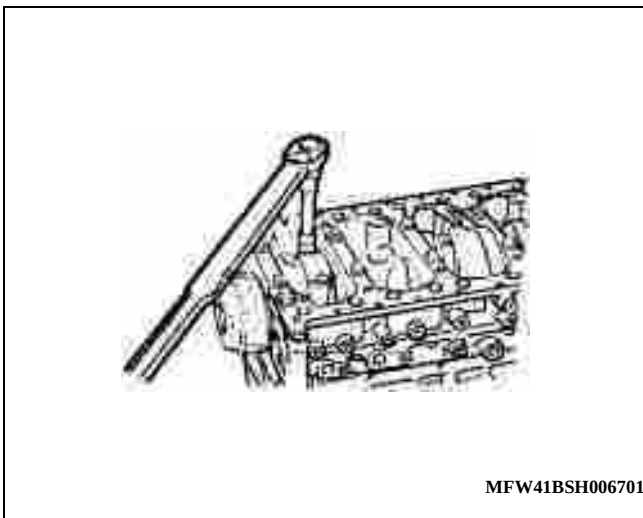
- Thoroughly remove the carbon deposit on the top surface of the bushing with a tool such as a scraper.



- Turn the crankshaft and move the cylinder piston to bottom dead center and remove the cylinder head.
- Turn the crankshaft and move the cylinder piston to the top dead center position. Use a handle such as a hammer to push the bearing out of the bottom of the connecting rod to avoid damaging the connecting rod.

Note:

- Be careful not to damage the injection hole and cylinder liner when ejecting the connecting rod.
- The cylinder cover, bearing and connecting rod are separately stored according to the original location of the cylinder.



Disassembly

1. Remove the bearing.

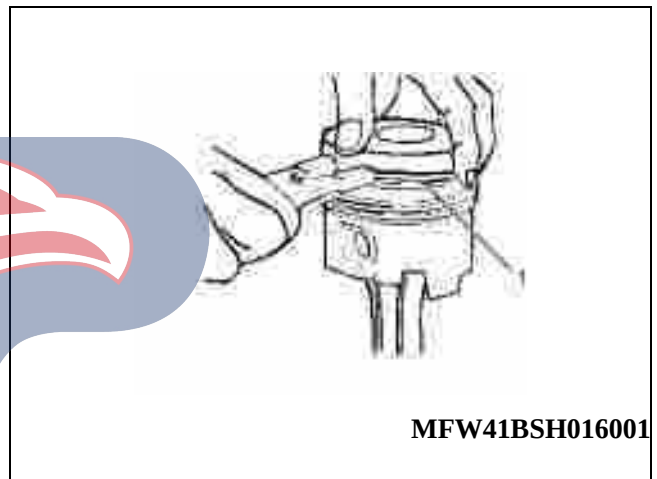
Note:

If you need to use the bearings again, they should be stored according to the cylinder number.

2. Remove piston ring (1) with replacement tool.

Note:

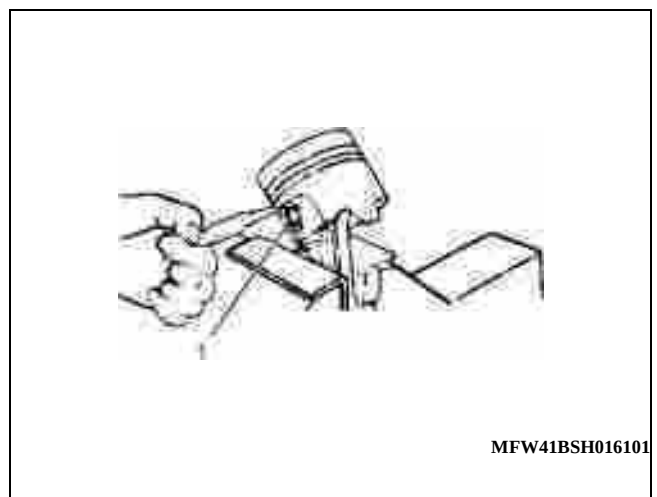
After disassembly save the bearing outer ring and other components according to the cylinder number.



3. Remove the buckle.
4. Remove the piston pin (1).

Note:

After disassembly save the bearing outer ring and other components according to the cylinder number.



Inspection

1. Measuring piston rings.

a. Check whether the piston ring is damaged and measure the clearance of the alignment hole.

- Insert the piston ring into the cylinder liner and press the piston head part together with the piston ring into the least worn part of the cylinder liner (above or below the cylinder liner). Then, insert the thickness gauge into the gap between the connecting rod alignment holes of the link and measure the gap.

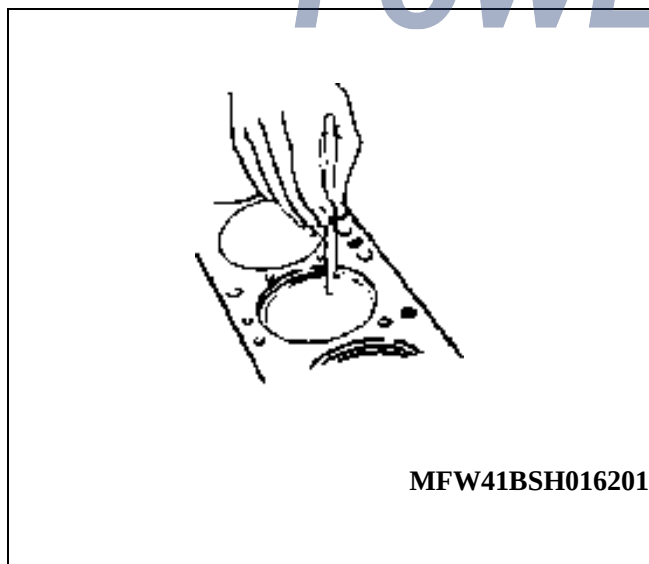
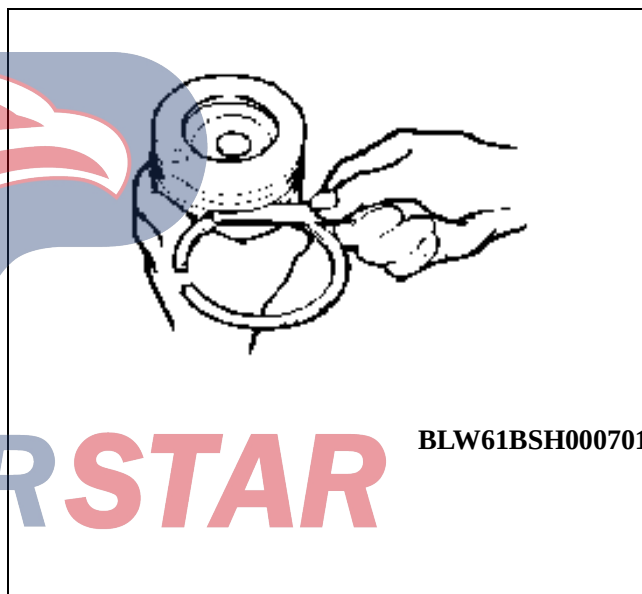
Note:

During the measurement, the coke on the alignment hole is completely removed so that the connecting rod is at right angles to the cylinder liner.

- Remove carbon deposits in the piston head and piston ring groove. Then insert the piston ring into the piston ring groove and measure the clearance with the thickness gauge.

Piston ring groove and piston ring clearance		mm(inch)
	Assembly standards	Use restrictions
Top piston ring	—	—
The second piston ring	0.030(0.0012) — 0.070(0.0028)	0.15(0.0059)
Oil ring	0.030(0.0012) — 0.070(0.0028)	0.15(0.0059)

Piston ring clearance		mm(inch)
	Assembly standards	Use restrictions
Top piston ring	0.27(0.0163) — 0.46(0.0181)	1.2(0.0472)
The second piston ring	0.37(0.0146) — 0.61(0.0240)	1.2(0.0472)
Oil ring	0.27(0.0168) — 0.56(0.0220)	1.2(0.0472)



b. Measures the gap between the piston ring groove and the piston ring.

c. The tensile strength of the piston ring is measured.

- Tensile strength was measured with a piston ring tensiometer.

Note:

- Use an installed ring to measure the oil ring.
- Measuring alignment hole clearance.

tensile strength		(kg f / lb ft)
	Assembly standards	Use restrictions
Top piston ring	26 — 34(2.65/19.17 — 3.47/25.1)	11.8 (1.2/8.68)

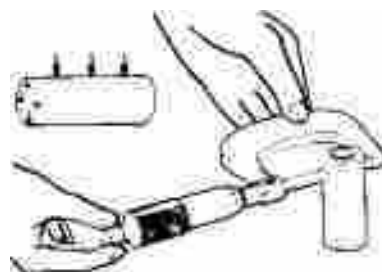
Tensile strength		(kg f / lb ft)
	Assembly standards	Use restrictions
The second piston ring	21.7—28.7(2.21/15.98 — 2.93/21.19)	5.9 (0.6 / 4.34)
Oil ring	30—40(3.06/22.13 — 4.08/29.51)	28 (2.9 / 21)

**MFW41BSH016401**

2. Measuring piston pin.

- Measure the outer diameter of the piston pin, the gap between the piston pin and the piston pin hole, and the gap between the piston pin and the connecting rod bush.
- Measure the six positions of the piston pin shown in the picture.

mm(inch)			
	Nominal size	Assembly standards	Use restrictions
Piston pin outside diameter	40(1.5748)	- 0.005(-0.0002) — 0.000(0. 000)	39.95(1.5728)
Clearance between piston pin and piston pin hole	—	0.009(0.0004)— 0.022(0.00087)	—
Clearance between piston pin and connecting rod bushing	—	0.012(0.00047) — 0.027(0.011)	0.05(0.002)

**MFW41BSH016501**

- At room temperature, insert the piston pin in the piston, ensure there is sufficient resistance and then gently push in the pin.
- If the piston pin is very loose at room temperature, replace the piston or piston pin.

**LNW21BSH009701**

2. Piston inspection and measurement.

a. Check the piston for vibration and cracks.

- Thoroughly remove carbon deposits or sludge from the piston before inspection and thoroughly clean it. Carefully inspect hard-to-see parts, such as piston rings, etc.
- Dye penetrates testing (if necessary).



BLW61BSH000601

- Clean bearing and bearing mounting surfaces and install bearings. Then apply molybdenum disulfide lubricant to the threaded part, tighten the connecting rod cover, and measure the connecting rod bearing with a bore gauge in four places as shown in the figure.

Bolt tightening torque	N m (kgf m/lb ft)
39 (4.0/29) → Angle Method 60° → -30 - 60°	

- The outer diameter of the crankshaft pin is measured with a micrometer in the figure.

b. Measure the outer diameter of the piston.

- Measure the piston pin at the point 85mm (3.3465 inch) from the piston crown and measure the piston outside diameter in right angle direction.

Piston outer diameter	mm(inch)
114.940(4.5252) — 114.989(4.5271)	

Internal clearance of piston and cylinder liner	mm(inch)
0.032(0.0013) — 0.11(0.0043)	

Note:

Replace piston or cylinder bushing if the clearance of piston liner exceeds the standard value.

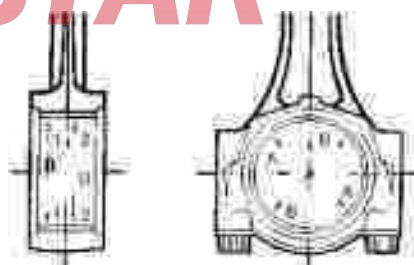
4. Measure connecting rod.

a. Measure the gap between the rod bearing and the crank pin.

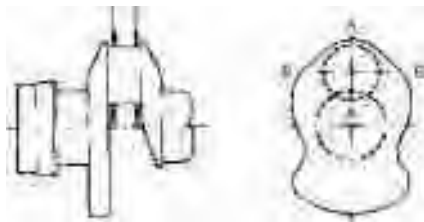
- Measure the inner diameter of the connecting rod bearing and the outer diameter of the crank pin.

Clearance between connecting rod bearing and crank pin		mm(inch)
Assembly standards		Use restrictions
0.037(0.0015) — 0.076(0.0030)		0.1(0.0039)

If you use a plastic gauge to make measurements, see the "How to use plastic gauges" section in the overview section.



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MFW41BSH016801

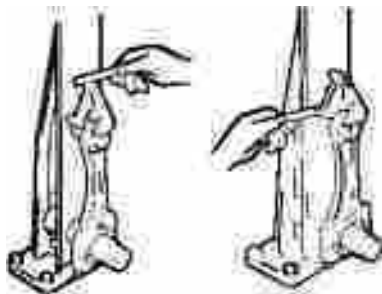
b. Cove measures the twist of the connecting rod.

- Use the rod calibration tool to measure the degree of distortion and parallelism of the smallest hole.

Note:

The measurement results are based on the bearing, so it should be assured that the bearing did not wear out.

Minimum side hole torsion degree and parallelism (100mm/3.937 inch)		mm(inch)
Assembly standards	Use restrictions	
0.05(0.002)	0.2(0.0079)	



MFW41BSH016901

c. Measure the axial clearance on the large side of the connecting rod.

- Connect the connecting rod to the crank pin and measure the axial clearance.

Axial gap between link rod big end and crankshaft		mm(inch)
Assembly standards	Use restrictions	
0.17(0.0067) — 0.3(0.0118)	0.35(0.0138)	



MFW41BSH017001

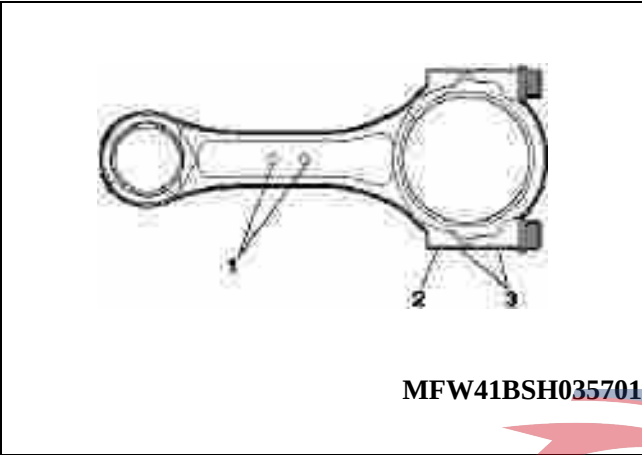
5. Check whether the connecting rod bearings are worn, damaged or detached.



MFW41BSH017101

6. Select the connecting rod bearing.

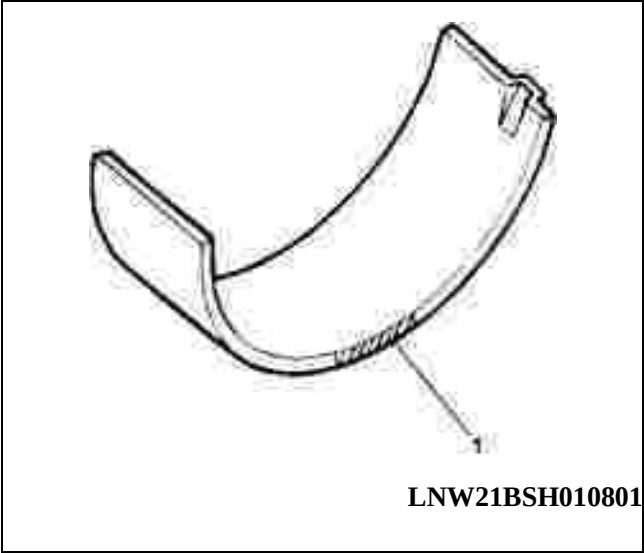
- When installing a new connecting rod or replacing an existing bearing follows the bearing selection table.
- Pay attention to choose the appropriate bearing according to the aperture grade of the larger side, and then install it.



Legend

1. Front/rear flag rose
2. Large edge aperture marking
3. The cylinder number

Combination of grades		mm(inch)
Linkage degree	Bearing Grade (Marker color)	Oil film clearance
A	Blue	0.037(0.0015)-0.076(0.003)
B	Brown	0.037(0.0015)-0.076(0.003)



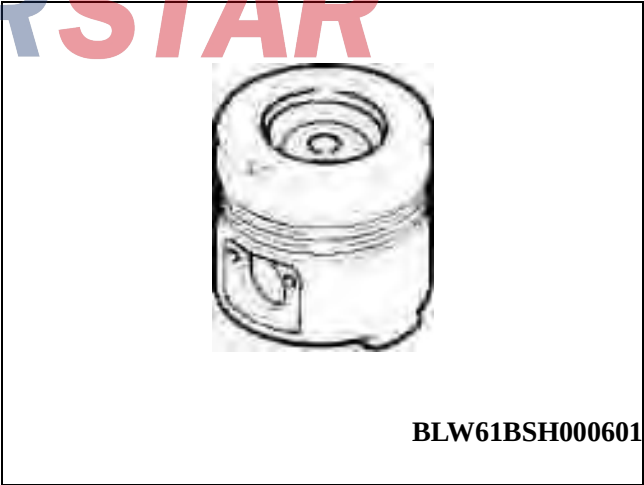
Legend

1. Bearing grade (identification color)

Assembling

The choice of piston

- When changing the piston, there is only one type of delivery, so there is no need to select the piston type.
- Pistons and bushings can provide packaging. In this case, it is not necessary to match the outer diameter of the bushing with the inner diameter level of the cylinder bore.



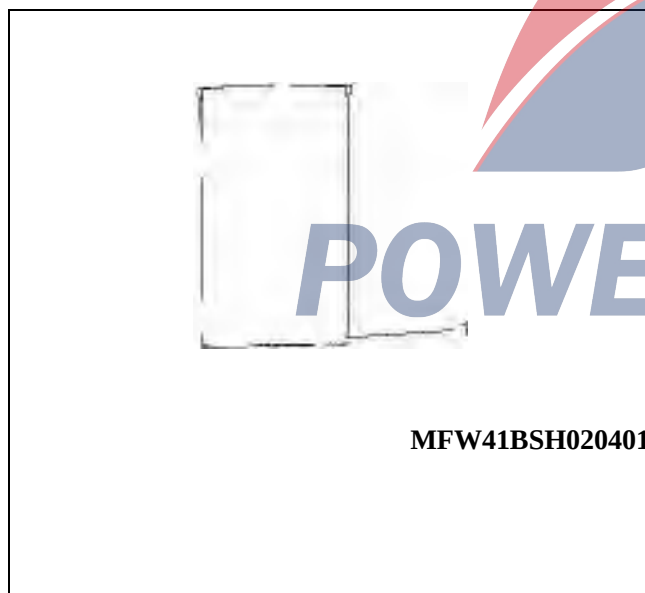
(For reference only) bushing horizontal selector

- The cylinder bore matches the outer diameter of the sleeve.

- Select bushings (1 and 2, 3, and 3) from the numbers printed on the top of the cylinder block and install them.

Combination of cylinder and sleeve outer diameter		mm(inch)
Cylinder body printing number	Bushings grade	
1、 2	1×(117.991/4.6449 — 118.000/4.6457)	
3	3×(118.001/4.6457 — 118.010/4.6461)	

Cylinder hole level		mm(inch)
	Hole diameter	
1、 2	118.001(4.6457)— 118.020(4.6464)	
3	118.021(4.6465)— 118.030(4.6468)	

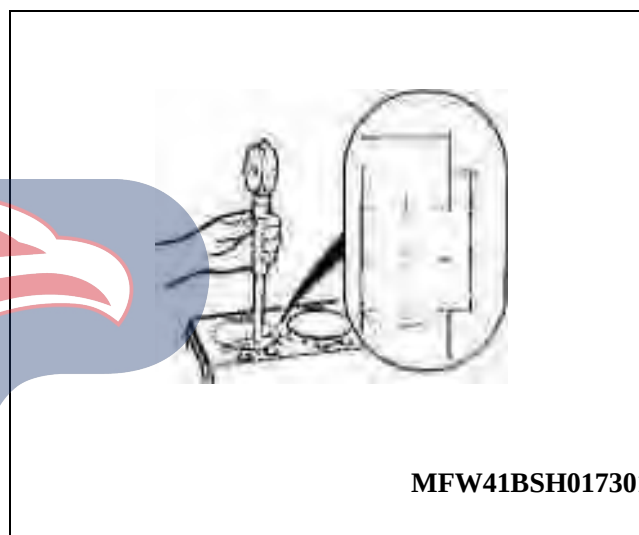


Legend

1. Level print position

- The measurement positions are three positions on the top surface of the cylinder block, and the four directions of each point are 45°, respectively, and a total of 12 measurement points are selected. The measurements are made at these locations, and then the average value is obtained.

Measuring position (starting from the top surface)			mm(inch)
A	B	C	
20(0.7874)	110(4.3307)	190(7.4803)	



Measurement procedure when using the piston again

- When using the piston again, measure the clearance between the piston and the bush.
- Measure the outer diameter of the piston at a position marked with grade. Remove any impurities such as carbon deposits on the bushing and measure it at the corresponding position on the inside diameter of the cylinder block. Check that the gap between the piston and the bushing is within the standard value range.

Grade selection measurement position (from the piston head long axis direction)	mm(inch)
85(3.3465)	

Piston and liner clearance (long axis direction)		mm(inch)
Assembly standards	0.032(0.0013)—0.11(0.0043)	



MFW41BSH017401



MFW61BSH001601

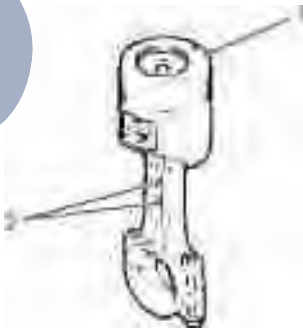
- (For reference only) Size of the supplied piston

	mm(inch)
Piston outer diameter (grade position)	114.944(4.5253)
	114.979(4.5267)

1. Install the piston on the connecting rod.
- Install the buckle to the side of the piston pin.
 - Apply engine oil to piston pin.
 - When assembled, the front mark (1) and the connecting rod mark (post mark) of the piston head should be reversed. Insert the piston pin completely and secure it with the snap.

Note:

Make sure that the buckle is securely installed in the groove. At this point, make sure that the opening of the buckle is vertical. Also make sure that the piston can move smoothly.



MFW41BSH017601

Legend

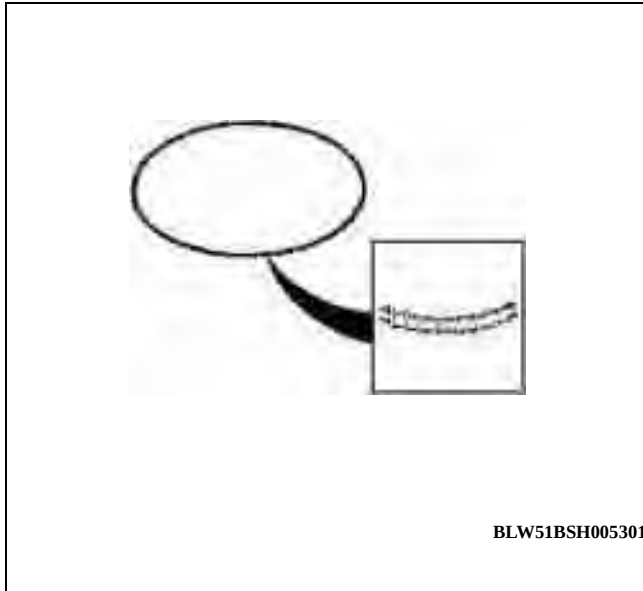
1. Front mark
2. Tooth-shaped mark

2. Put on the piston rings.

- Install the expansion ring into the piston so that they mesh with each other, thus eliminate the alignment gap, and then install the oil ring.

Note:

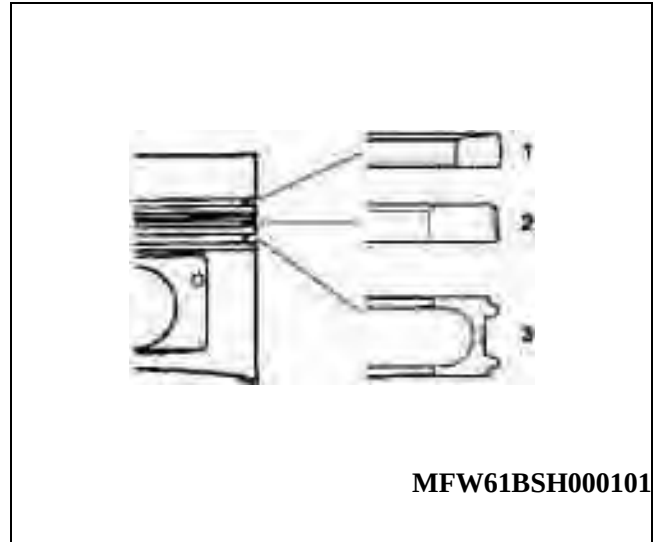
The alignment hole of expansion ring and that of buckles are in the reverse direction of 180°.



- When the 2T mark of the second piston ring faces upwards, use the piston ring mounting tool (1-8522-1029-0) for mounting.
- When the T1 mark on the top piston ring faces upwards, install using the piston ring mounting tool (1-8522-1029-0).
- Apply oil to the entire circumference of the piston ring and turn the piston ring to make sure that it can rotate smoothly without jamming.
- The first piston ring, the second piston ring and the oil ring connector should be able to rotate 120° directly.

Note:

- When using the piston ring again, make sure that it is not inserted backwards.
- There is no difference between the oil rings.



Legend

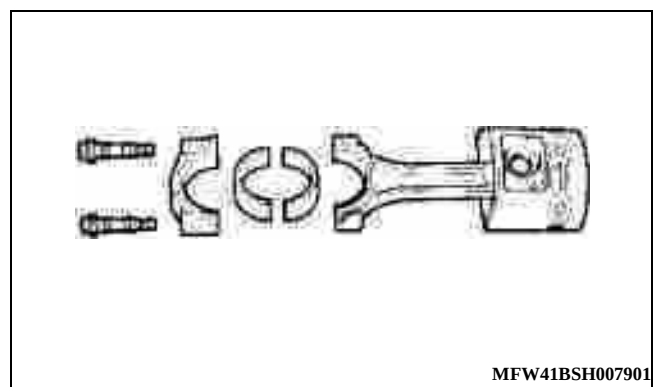
1. Top piston ring
2. Second piston ring
3. Oil ring

Reassemble

1. Install the connecting rod bearing.
 - When changing connecting rod or bearing, bearing should be replaced according to bearing replacement table (please refer to "Check").
 - Clean the connecting rod and cover, align the tip and install the bearing.
 - Apply oil to the rotating surface of the bearing.

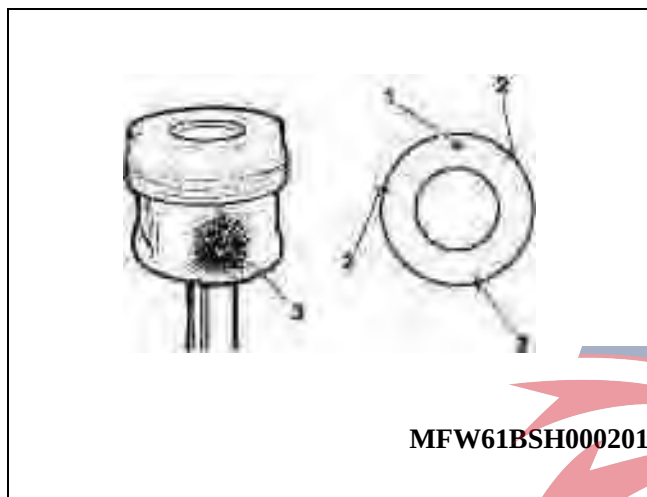
Note:

Wear the bearing on the back under dry conditions.



2. Install the piston and connecting rod assembly.

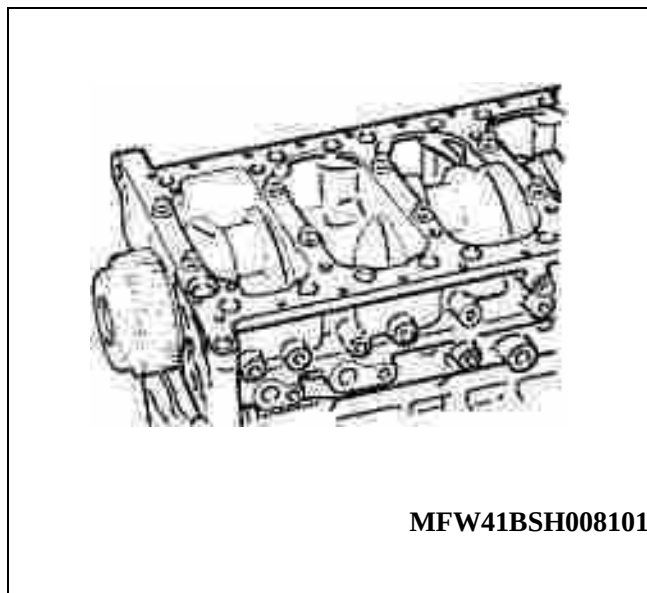
- Apply oil over the entire circumference of the piston ring to avoid lateral pressure on the aligned holes (as shown).
- Apply molybdenum disulfide lubricant or engine oil to the piston area shown in the figure.



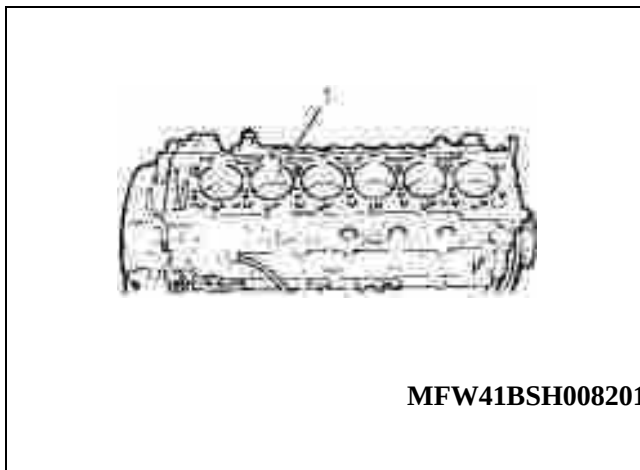
Legend

1. Front mark
2. Alignment hole
3. Coater molybdenum disulfide lubricant

- Move the crank pin to the bottom dead center of the cylinder and install the connecting rod.



- Apply oil to the cylinder liner.



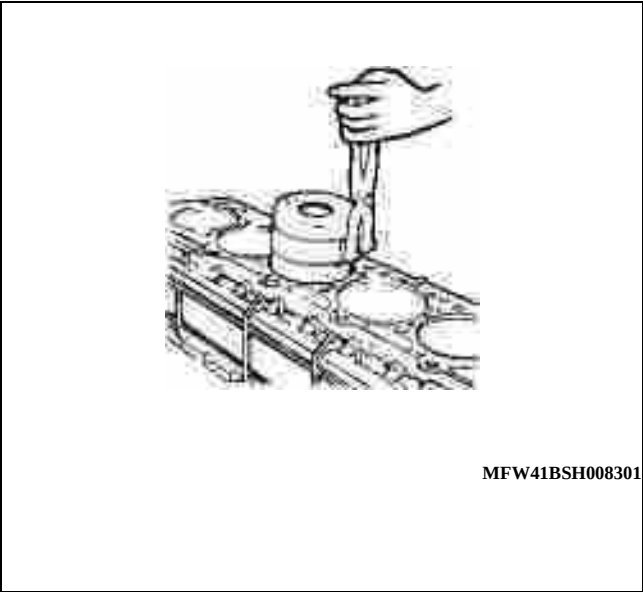
Legend

1. Oil coating

- Check the cylinder number of the connecting rod so that the front marked part of the piston head points to the front side of the cylinder and inserts it into the cylinder block.
- Use the shank of tools such as the piston setting tool and hammer to push the piston in until it contacts the crank pin.

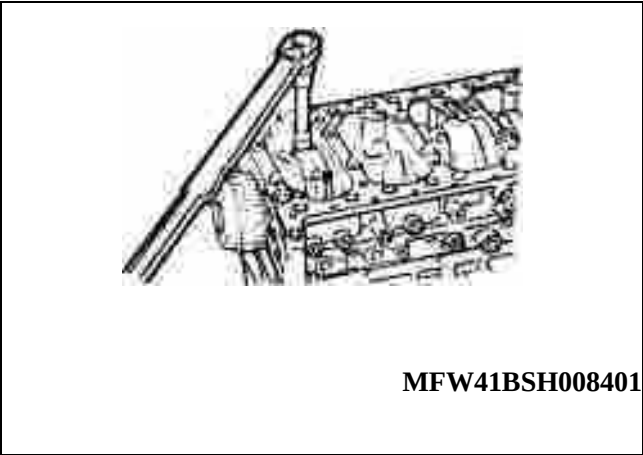
Note:

- When pushing in the piston, touch the mounted tool with the cylinder block. If it is not fully touched when pushed in, the piston ring will be damaged.
- Be careful not to damage the crank pin and spray hole.
- After inserting the piston rod, install a stopper to prevent the liner from falling off.

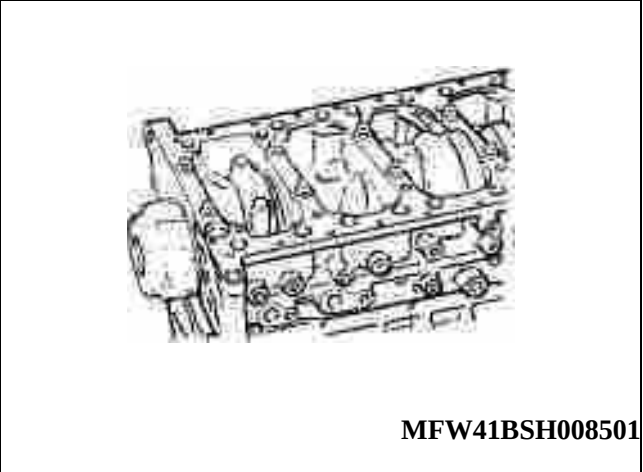


- Apply a large amount of oil to the bearing on the connecting rod cover.
- Align the link cover mark and connect it.
- Molybdenum disulfide lubricant is applied to the thread and base of the bolt and then tightened alternately according to the specified torque.

Tightening torque	N m (kgf m/lb ft)
39 (4.0/29) → Angle Method 60° → 30°	



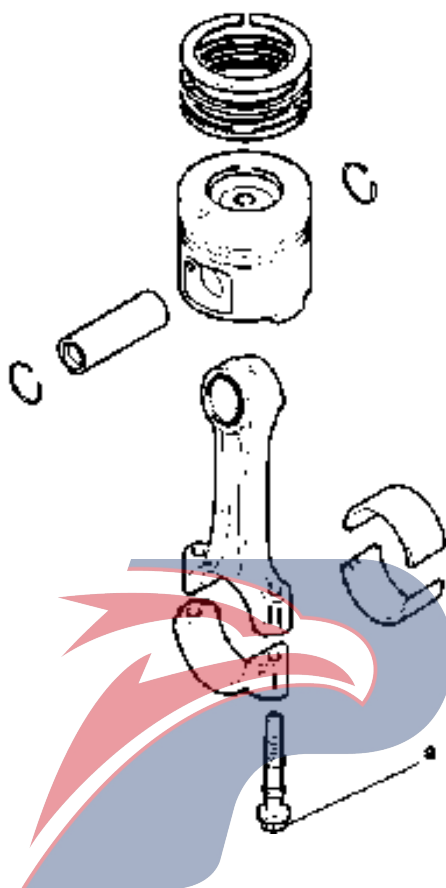
- Turn the crankshaft by hand to make sure it does not get stuck on any part and it can rotate smoothly. At the same time, make sure that the injection hole is not twisted and does not block other parts.



3. Install the cylinder head.
Refer to "Cylinder Head".
4. Install the camshaft.
Refer to the "Camshaft assembly".
5. Install rocker shaft.
Refer to the "Rocker shaft".
6. Put on the oil pan.
Refer to "Crankshaft".

POWERSTAR

Tightening torque table



MFW61BMF000101

POWERSTAR

Legend

(a) 39N m (4.0kgf m/29 lb ft) $\rightarrow$ Angle $60^\circ \rightarrow 30^\circ$

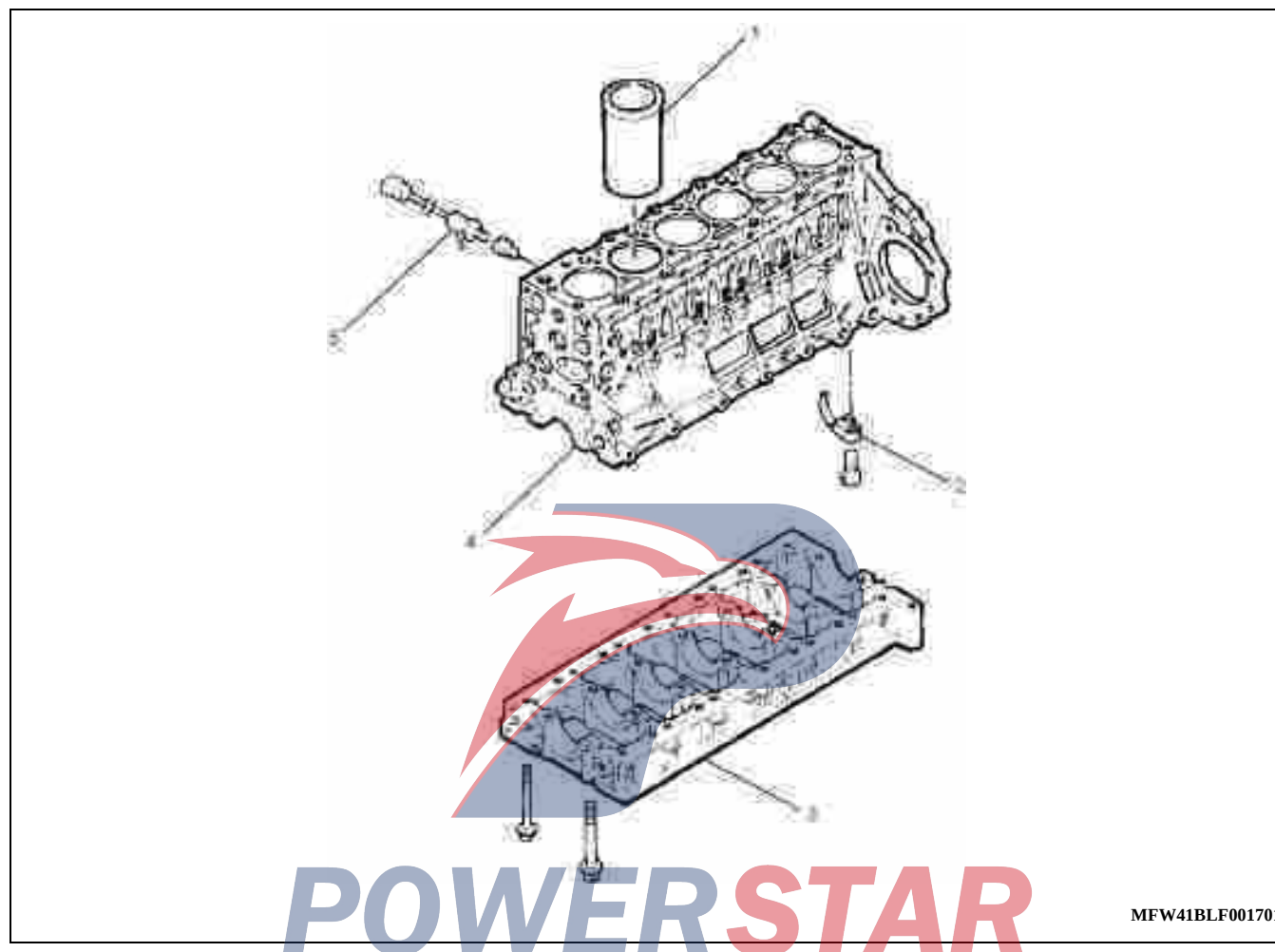
Special tools

Graphic representation	Tool number/description
 1852210290	1-8522-1029-0 Piston Ring Setting Tool



Cylinder block

Part



MFW41BLF001701

Legend

- | | |
|-------------------|------------------------|
| 1. Cylinder liner | 4. Cylinder block |
| 2. Syringe hole | 5. Oil pressure switch |
| 3. Crankcase | |

Disassembly

1. Remove the engine components from the car.
Refer to "Engine Assembly".
2. Remove the cylinder head.
Refer to "Cylinder Head".
3. Remove the feed pump, air compressor and oil cooler.
Please refer to "Engine Exterior Parts".

4. Remove the Crankshaft.

Refer to "Crankshaft".

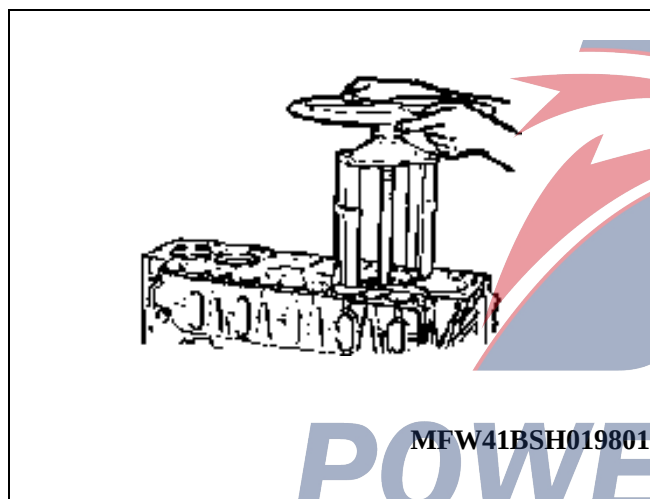
5. Remove the oil pressure switch.
6. Remove the injection hole.
7. Remove the cylinder liner.

Measure the inner circumference of the cylinder liner and determine if it is acceptable and remove it. Please refer to bushing wear section.

- If you cannot withdraw the cylinder liner by hand, perform the following procedure.
- Securely attach the retainer of the cylinder sleeve kit removal tool (5-8840-2397-0) to the lower edge of the sleeve and insert the turntable into the center bolt so that it does not engage with the collar portion of the upper sleeve. Tighten with bearing mounting nuts. Then fix a handle on the nut and turn the handle to pull out the sleeve.

Note:

When removing the bushing, be careful not to damage the surface of the cylinder bore.



MF41BSH019801

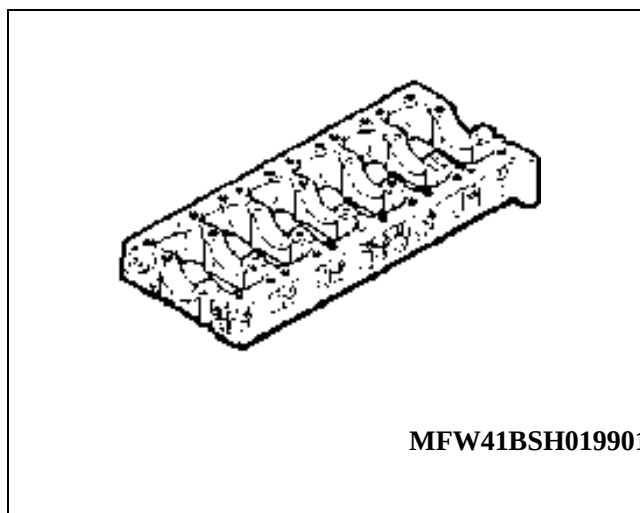
Inspection

1. Carefully remove deposits and other debris on the upper surface of the cylinder block.

Note:

Be careful not to damage the cylinder block.

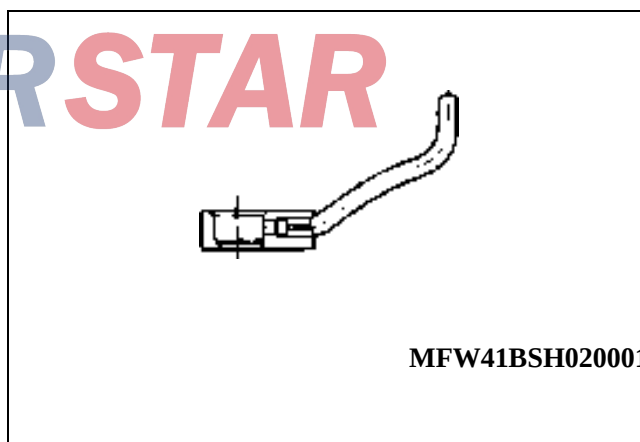
2. Carefully remove the liquid seal from the crankcase mounting surface.
3. Clean the cylinder block and crankcase.
4. Check the crankcase.
 - Check the crankcase for flaw and damage.
 - Use a magnetic flaw detector or dye penetration test (if necessary).



MF41BSH019901

5. Check the injection hole.
 - Check whether there is any oil hole blockage, pipe bending and so on, and check the operation of check valve.

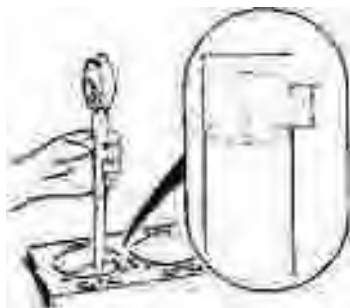
Check valve opening pressure	kPa (kg/cm ² /lb/in ²)/rpm
	150—200(1.5—2.0/21—28)



MF41BSH020001

6. Measure the degree of wear of the cylinder liner.
 - Measure the maximum wear degree within the 20 mm (0.7874 inch) distance of the upper surface of the bushing. Replace the bushing if the specified value is exceeded.

Maximum wear of cylinder liner	mm(inch)
Use restrictions	0.2(0.0079)



MFW41BSH019701

7. Check and measure the cylinder block.

- Measure the amount of deformation of the cylinder block.
- Place the scale diagonally on the surface of the cylinder block and measure at four locations using the thickness gauge (1).

Deformation of cylinder block		mm(inch)
Use restrictions		0.2(0.0079)

Note:

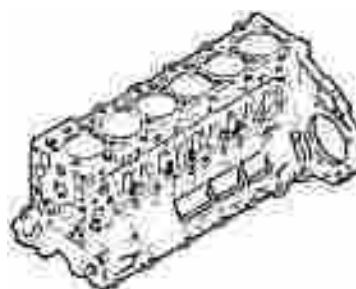
Because the gap of the timing gear changes, it cannot be repaired by polishing.



MFW41BSH020101

- Check the cooling pipes and lines between the cylinders for clogging or corrosion.

- Check the cylinder block for cracks and damage.
- Dye penetrates testing (if necessary).
- Check for damage or disconnection.
- In the hydrostatic test, apply 490 kpa (5 kg / cm / 71 psi) of water for 3 minutes and check for leaks in all parts of the cylinder head.



MFW41BSH020201

Assembling

1. Select the proper gasket grade and install it.

Caution:

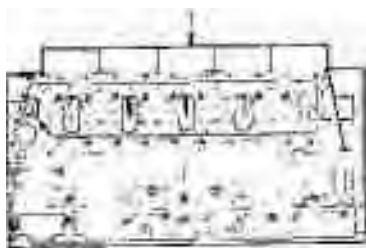
Do not use a thermal valve in the oil hole on the front side of the cylinder block. It may burn after installation.

A. Selection of cylinder liner grade.

- Match the cylinder bore diameter with the sleeve outer diameter class.
- Select bushings (1 and 2, 3, and 3) from the numbers printed on the top of the cylinder block and install them.

Caution:

- Due to the liner is thin; do not use the dropped liner.
- If the casing rating is not properly selected, insufficient cooling (gap size) may occur and combustion may occur.



MFW41BSH020301

Legend

1. Level print position

- The corresponding dimensions of the bushing grades are controlled and pressed on the lower surface of the bushing when the bushing is produced.

Cylinder body measurement position (from mm(inch) cylinder block upper surface)		
A	B	C
20(0.7874)	110(4.3307)	190(7.4803)

Cylinder body and sleeve outer diameter grade match			
Printed text	1	2	3
Bush outer diameter	1×		3×

Inner cylinder circumference, sleeve outer circumference grade			mm(inch)
Grade	(1).(1X)	(2).(1X)	(3).(3X)
Bush outer diameter	118.001(4.6457))— 118.010(4.6461))	118.011(4.6461))— 118.020(4.6464))	118.021(4.6465))— 118.030(4.6468))
Bush outer diameter	117.991(4.6453)—118.000(4.6457)		118.001(4.6457))— 118.010(4.6461))

Note:

- When the cylinder block height is determined, measurements are made at three locations on the upper surface of the cylinder block (a total of six locations) and the average value is determined.



MFW41BSH020501



MFW41BSH020401

Legend

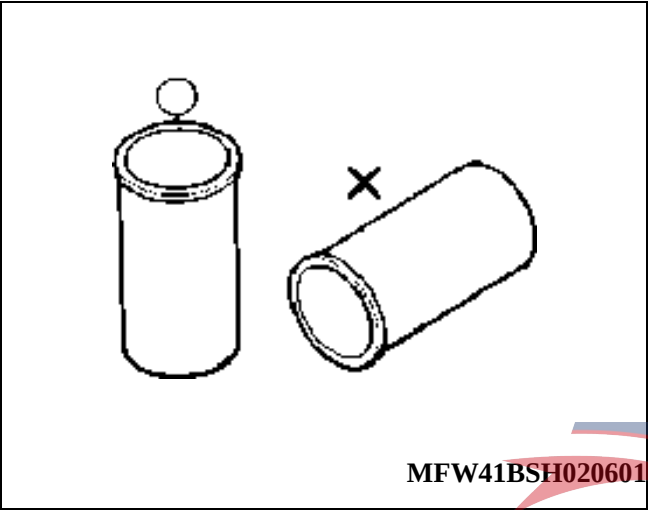
1. Grade Mark

2. Install the cylinder liner.

- Clean the cylinder block and bushings.
- Dry the cylinder and liner after drying.

Note:

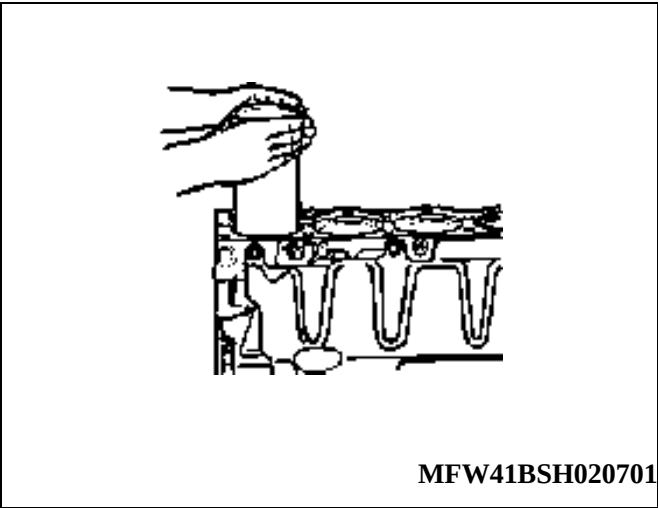
Be careful when handling liners because it is thin. Do not place the bushing traverse.



- A thin layer of engine oil is applied to the outer circumference of the cylinder liner and then mounted directly on the cylinder block.

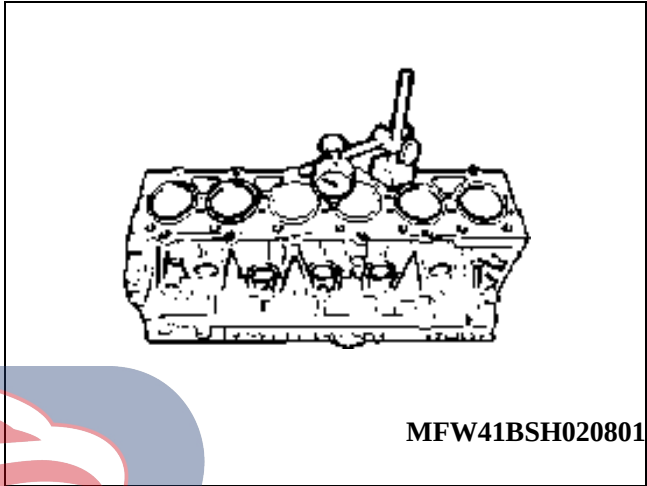
Note:

- Do not knock or apply excessive force when inserting the bushing.
- After inserting the bushing, prevent the bushing from coming out when the cylinder block reverses.



- Use a dial gauge to measure the amount of casing protrusion.

Assembly standards		mm(inch)
Casing quantity		0.06(0.0024) — 0.1(0.0039)
Displacement of bushing projections from adjacent bushings		0.03(0.0012)

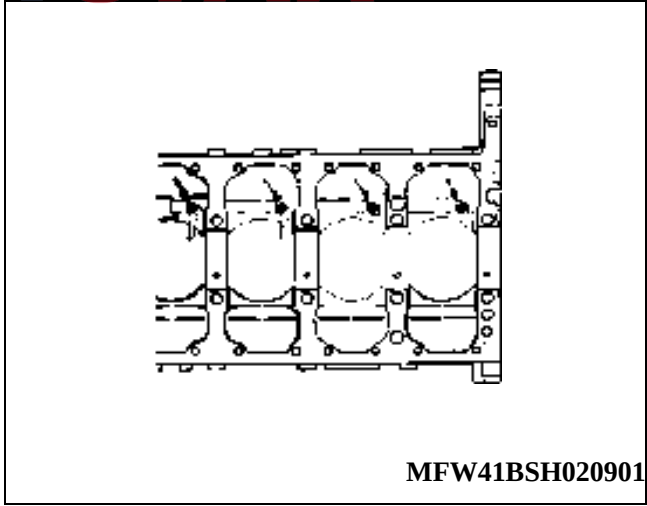


3. Install the injection hole.

Tightening torque	N m (kgf m/lb ft)
	21 (2.1 / 15)

Note:

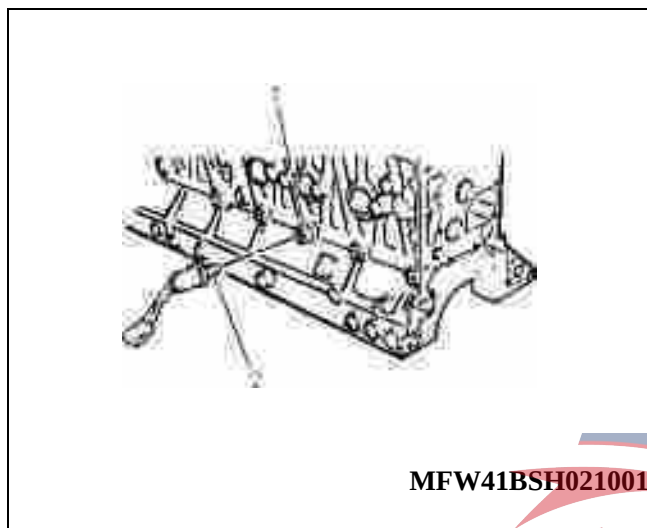
Do not bend the tip of the oil hole.



4. Install oil pressure switch.

- Install the connector to the cylinder block.
- At the end of the oil pressure switch, apply No. 1241 San key brand adhesive and attach it to the fitting.

Deformation of cylinder block		mm(inch)
Connector		118 (12.0 / 87)
Hydraulic switch		18 (1.8 / 13)

**Legend**

1. Connector
2. Hydraulic switch

5. Install the crankshaft.

Refer to "Crankshaft".

6. Install feed pumps, air compressors and oil coolers.

Please refer to "Engine Exterior Parts".

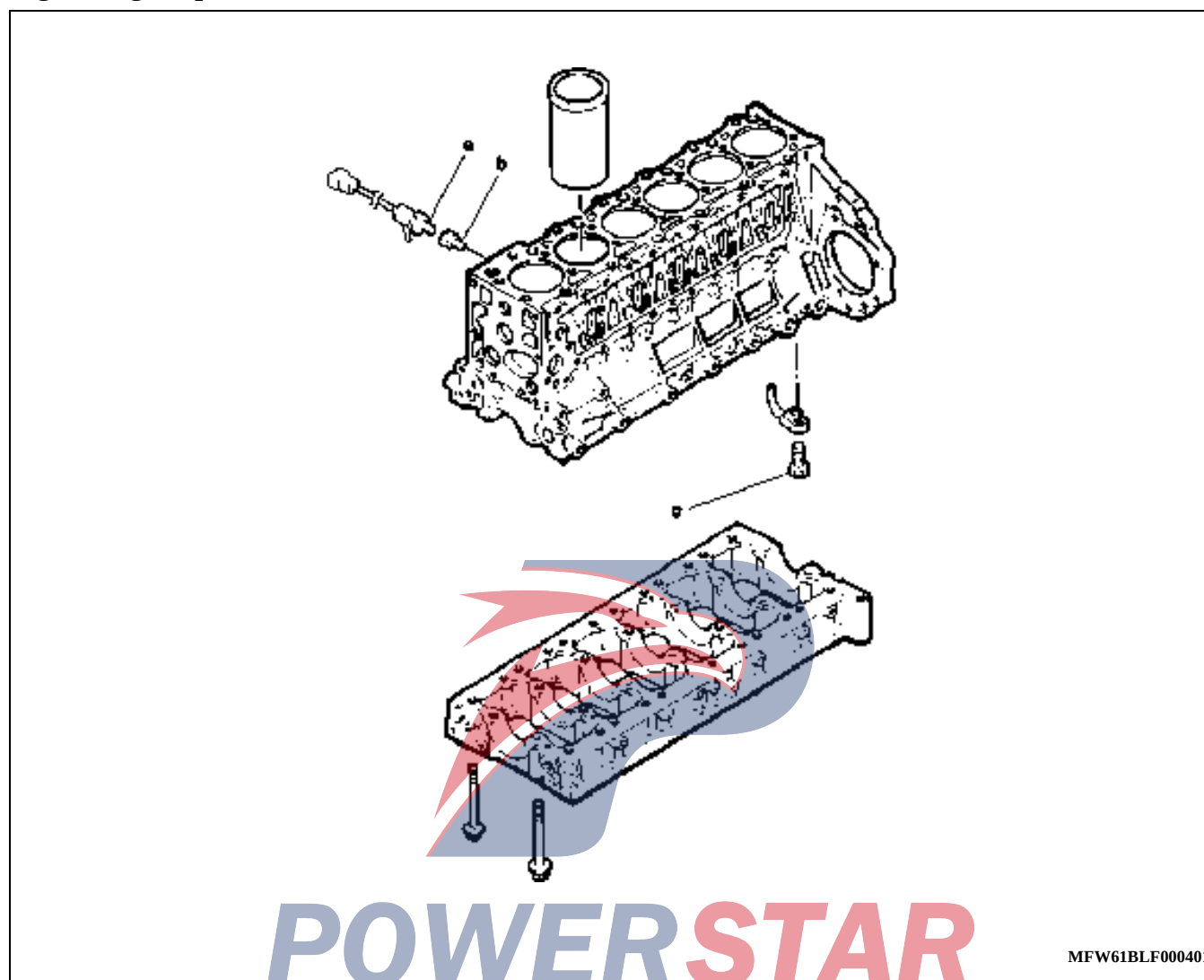
7. Put on the oil pan. Refer to "oil pan".

8. Install the cylinder head.

Refer to "Cylinder Head".

9. Put the engine assembly into the car.

Refer to "Engine Assembly".

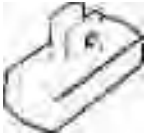
Tightening torque table**Legend**

(a) 18 N m (1.8 kgf m/13 lb ft)

(b) (c) 21 N m (2.1 kgf m/15 lb ft)

(c) 118 N m (12.0 kgf m/87 lb ft)

Special tools

Graphic representation	Tool number/description
 5884023970	5-8840-2397-0 Cylinder liner removal tool



Engine

Engine cooling

(6HK1)

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Disassembly	6B-20		
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Cooling system

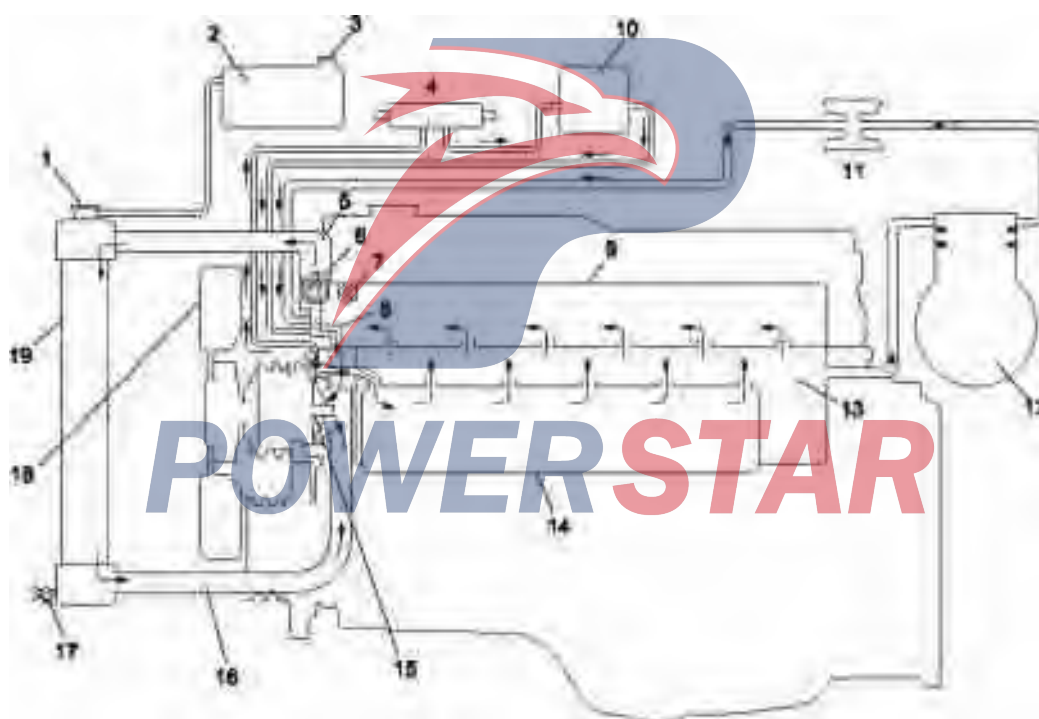
Precautions for maintenance

Warning:

Do not tighten or remove the radiator cap before the coolant cools down. Due to possible steam or hot water, severe burns may result. When opening the radiator cap, wait for the cooling water to cool, then cover the lid with a thick cloth and slowly rotate to release the pressure.

Description of function and operation

Cooling water circulation system diagram



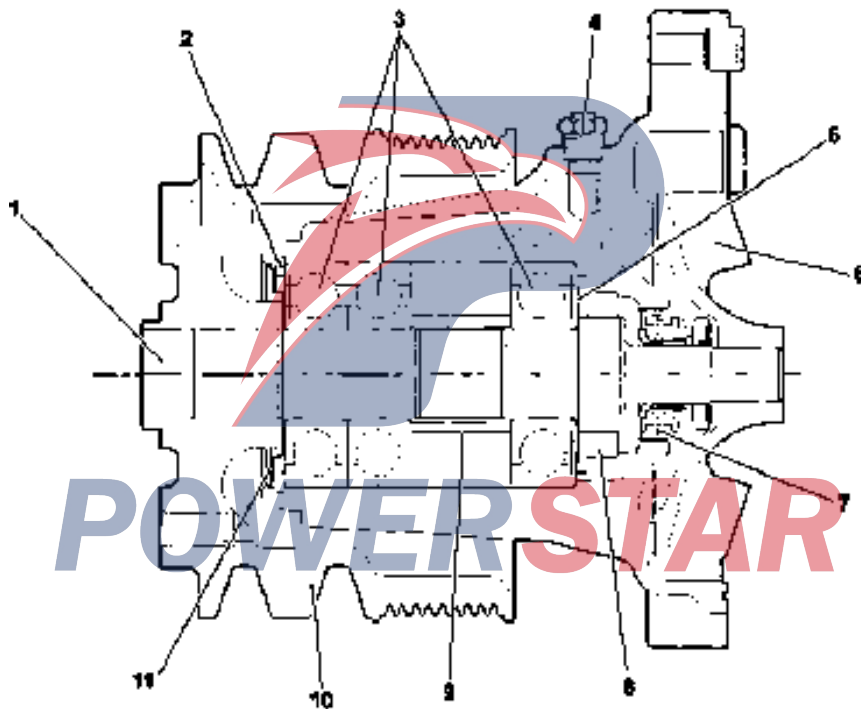
MFW41CLF000701

Legend

- | | |
|-----------------------------------|--------------------------|
| 1. Radiator cap | 5. Outlet pipe |
| 2. Water tank | 6. Incubator |
| 3. Reservoir water injection hole | 7. Double thermal sensor |
| 4. EGR cooler | 8. Bypass |

- | | |
|--------------------|-----------------|
| 9. Water jacket | 15. Water pump |
| 10. Heater core | 16. Below hose |
| 11. Turbocharger | 17. Drain plug |
| 12. Air compressor | 18. Cooling fan |
| 13. Water jacket | 19. Radiator |
| 14. Oil cooler | |

Pump structure diagram



MFW61CLF000101

Legend

- | | |
|------------------|--------------------|
| 1. Axis | 7. Sealing device. |
| 2. Buckle | 8. Oil seal |
| 3. Bearing | 9. Septum |
| 4. Grease nipple | 10. Pulley |
| 5. Washers | 11. Dust cover |
| 6. Impeller | |

Coolant inspection

1. Coolant concentration

- The freezing temperature varies with the concentration of long-lived coolant. The coolant concentration should be calculated at a temperature about 10°C (50°F) lower than the expected outside air temperature.
- Do not use coolant other than Isuzu's pure coolant.
- Coolant concentration calculation method: LLC (Isuzu specified product or equivalent)

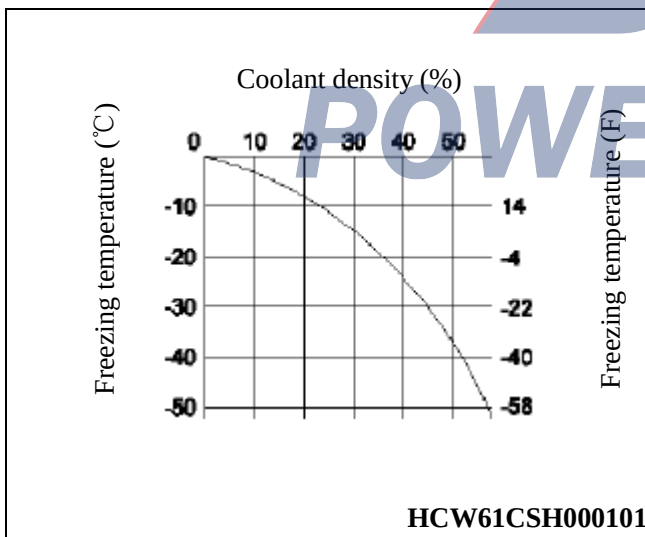
Coolant concentration (%) = 100 (coolant capacity) (l) / (coolant capacity (l) + water capacity (l))

Coolant capacity + Water capacity =

Coolant capacity = 29.0 liters (6.38 IMP)

- <Example> In areas where the minimum temperature reaches -10 °C (14 °F), the required coolant concentration is approximately 35%.
- $29.0(35/100) = 10.15(2.233)$

Dilute 10.15 liters (2.233 British gallons) of coolant with 18.85 liters (4.147 British gallons) of water to a total volume of 29.0 liters (6.38 British gallons), resulting in a coolant concentration of 35%.



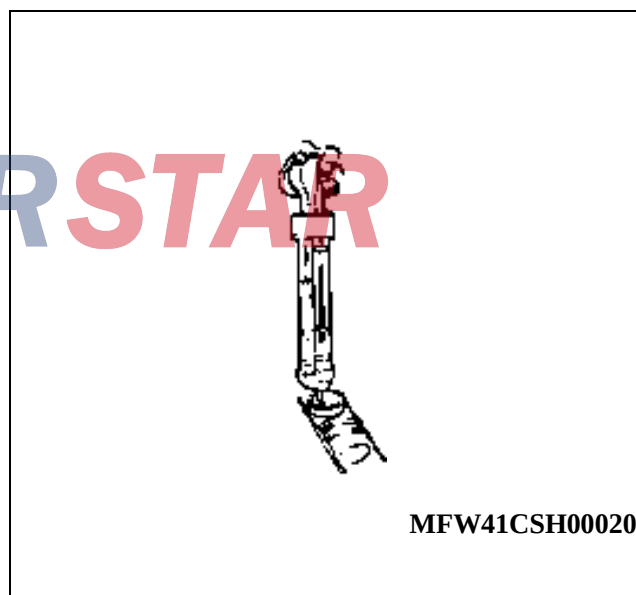
2. Coolant concentration measurement LLC (Isuzu specified product or equivalent)

- Measure the specific gravity and coolant temperature with an antifreeze hydrometer or thermometer.

If the temperature is high, special care should be taken because the cooling water may blow out after removing the cover.

- After measuring the temperature and specific gravity, the concentration should be calculated according to the following table.
- The depth of the container used should be greater than the length of the hydrometer.
- The temperature range of the cooling water to be tested is 0-50°C (32-122°F).

Freezing temperature roughly indicates °C (°F)	Mixing ratio (L/UK gallon)		Coolant concentration
	Coolant	Running water	
-18(-0.4)	10.2(2.244)	18.8(4.136)	35%
-25(-13)	12.2(2.684)	16.8(3.696)	42%
-35(-31)	14.5(3.19)	14.5(3.19)	50%
-40(-40)	15.4(3.388)	13.6(2.992)	53%



- The coolant range method

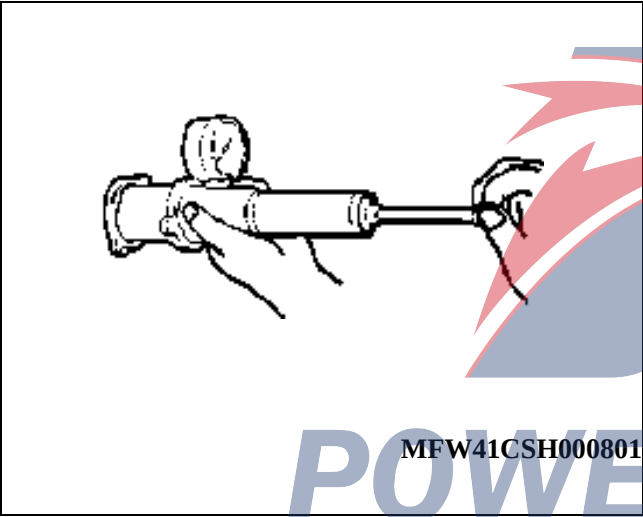
Use a coolant range to measure the concentration.

Radiator inspection

Radiator cover

- Check the spray valve opening pressure with the radiator cap tester. If the valve opening pressure during the inspection exceeds the standard value, the radiator cap should be replaced.

kPa (kg/cm2/lb/in2)/rpm	
Radiator cover, working pressure	Positive pressure=93—122 (0.9—1.2/13—17)
	Negative pressure=4.9 (0.05/0.7)



- Check the condition of the negative pressure valve at the center of the valve seat side. If the valve seat cannot run smoothly due to rust, dirt, etc., the radiator cap should be cleaned or replaced.

Radiator core

- Distortion of the heat sink fins can result in poor heat dissipation and overheating. Therefore, the fins must be repaired. Be careful not to damage the connector fitting during maintenance.
- Clear the dirt and chipping.

Radiator cleaning

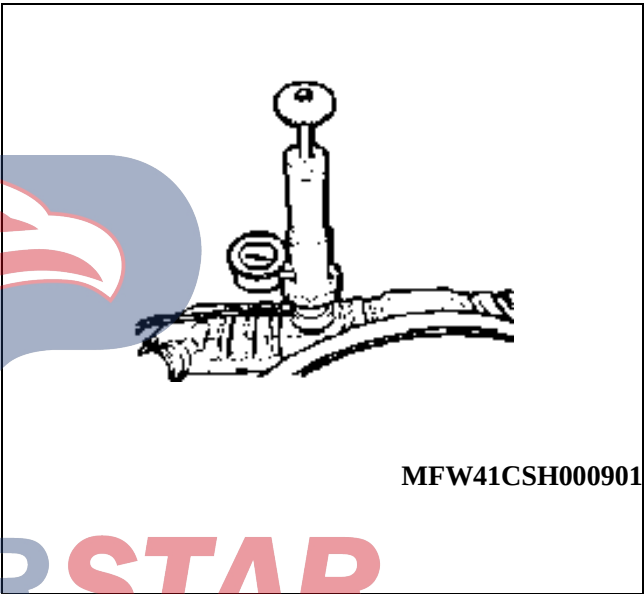
- Clean the inside of the radiator and water-cool the water channel. Remove all rust and dirt.

Cooling water leak check

- Apply 200 kpa (2.0 kg/ cm² / 28 psi) pressure to the radiator cap tester to inspect the following components for leaks. Also check radiator hoses and heater hoses for damage or deterioration, and check for loose hose clamps.

Check position

- Radiator assembly, pump assembly, radiator hose, heater hose



MFW41CSH000901

MFW41CSH000801

Engine Cooling Water Replacement Procedure

1. Make sure the engine is cool state.

Warning:

Do not tighten or remove the radiator cap before the coolant cools down. Due to possible steam or hot water, severe burns may result. When opening the radiator cap, wait for the cooling water to cool, then cover the lid with a thick cloth and slowly rotate to release the pressure.

2. Remove the radiator cap.

3. Open the radiator cooling water drain plug and drain the cooling water.

4. Close the drain plug.

5. Fill the radiator with water.

6. Check and clean radiator cap. If there are many abnormalities, replace the radiator cap.

7. Install the radiator cap firmly.

8. Fill tap water to the water tank until it reaches "MAX" mark.

9. Close the tank lid.

10. Start the engine, run it idling for 20min, then stop the engine and wait for engine cooling down to discharge the coolant.

11. Fill the radiator with coolant.

12. Install the radiator cap firmly.

13. Fill the tank with coolant to reach the "maximum" marking line.

14. Install the fuel tank cap.

15. Install the fuel tank cap and start the engine.

16. Wait for the engine to run for 2-3 minutes at normal idling speed and then stop the engine.

17. If the water level drops, once the coolant is cooled, remove the radiator cover and fill up the radiator.

18. Install the radiator cap firmly and then heat the engine at a speed of about 2000 r/minute.

19. If the needle of the coolant thermometer exceeds the middle point of the dial, let the engine run for about 5 minutes at normal idle speed and then stop the engine.

20. If necessary, once you have verified that the coolant has cooled, remove the radiator cap and fill the radiator. If coolant level is low, check if there is any leakage.

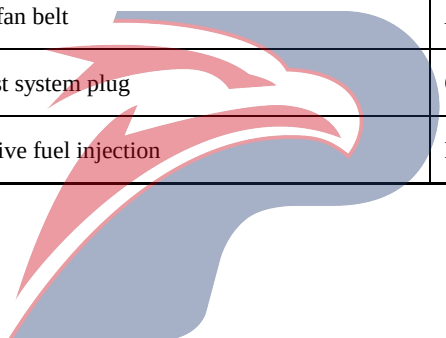
21. Install the radiator cap firmly.

22. Re-inject the coolant into the reservoir to reach the "maximum" marking line.



Phenomenon: The engine is too hot

Phenomenon	Probable causes	Correction method
Engine overheating	Insufficient cooling water	Add water.
	Incubator failure	Replace.
	Water pump failure	Replace.
	Blocking the heat sink	Clean or replace.
	Radiator cover failure	Replace.
	Insufficient engine oil or incorrect oil	Fill or replace engine oil.
	Cylinder head gasket failure	Replace.
	Loose fan belt	Adjust.
	Exhaust system plug	Clean or replace.
	Excessive fuel injection	Diagnosis of engine control systems.



POWERSTAR

Phenomenon: The engine is too cold

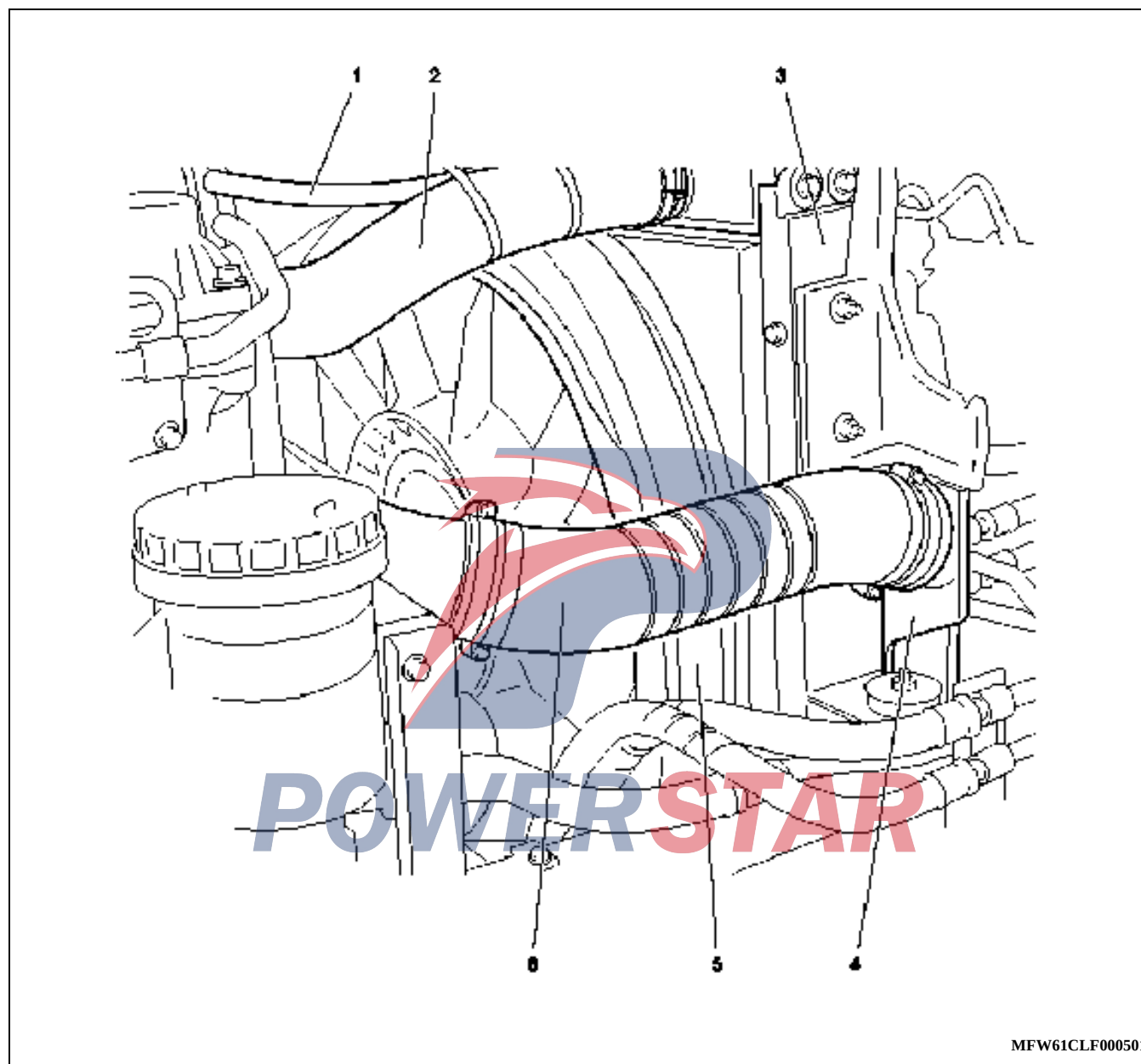
Phenomenon	Probable causes	Correction method
Engine is too cold	Incubator failure	Replace.

Specifications

Item		Specification
Cooling mode		Water-cooled preloaded forced circulation
Radiator		Colgate type
Water pump	Type	Multi V V-ribbed belt type
	Output (pump speed: 3300 rpm water: room temperature) (L / IMP / minute)	310.5(68.3)
Incubator opening temperature	With swing valve (°C/°F)	82/180, (fire trucks: 76.5 / 170)
	No shaking valve (°C/°F)	Level 1: 82/180, (Fire truck: 76.5/170), Level 2: 85/185 (Fire truck: 79.5/175.1)
Cooling water Liter (IMP)		TCN、TCC: 29/6.38 TCS: 30/6.6
Water tank capacity Liter (IMP)		4.0/0.88

Radiator

Part



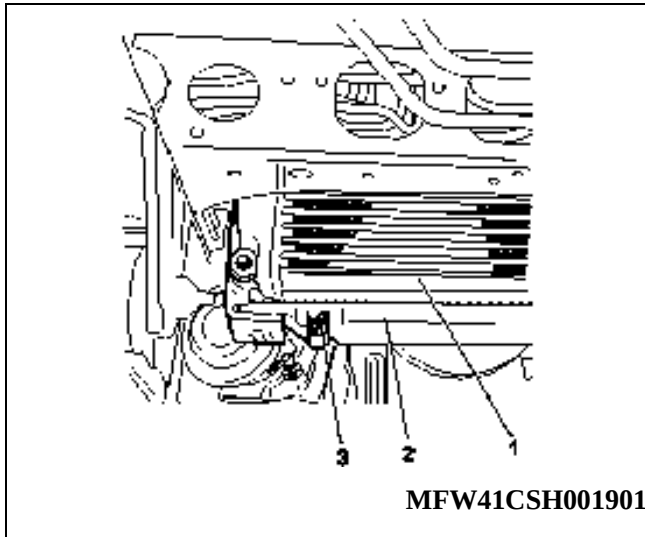
Legend

- | | |
|--------------------|------------------------------------|
| 1. Soft water pipe | 4. Intercooler |
| 2. Upper hose | 5. Fan guide rod |
| 3. Radiator | 6. Turbine output intercooler hose |

Tear down

1. Discharge cooling water from the radiator.

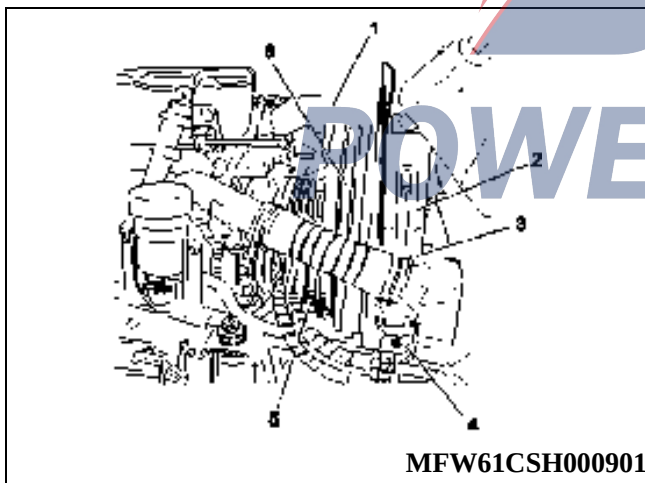
- Remove the drain plug and drain the cooling water from the radiator.



Legend

1. Intercooler
2. Radiator
3. Release plug

2. Remove the intercooler hose.

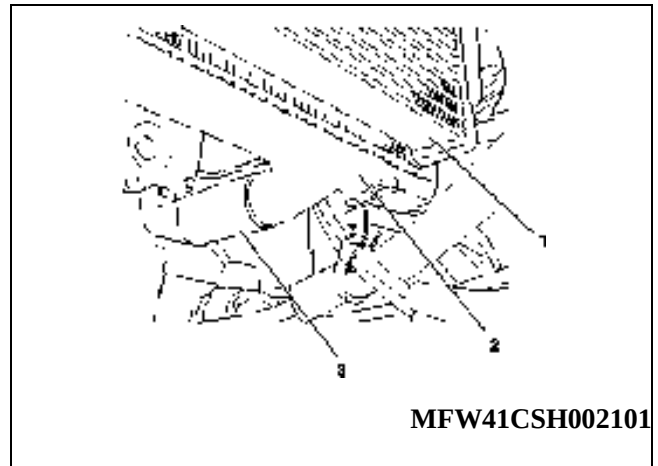


Legend

1. Upper hose
2. Radiator
3. Intercooler hose
4. Radiator mounting bracket
5. Fan guide bracket
6. Horn mouth

3. Remove the upper hose

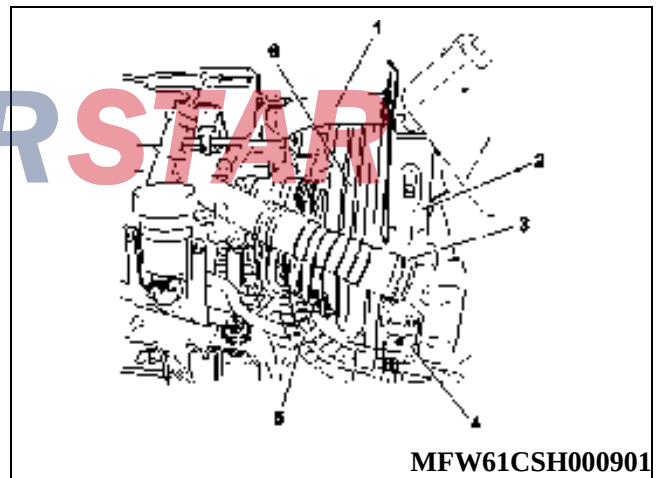
4. Remove the lower hose



Legend

1. Intercooler
2. Radiator
3. Lower hose
5. Soft water pipe.
6. Remove the fan.

- Remove the fan guide bracket from the next position and remove the fan guide so that the fan connector stays connected.

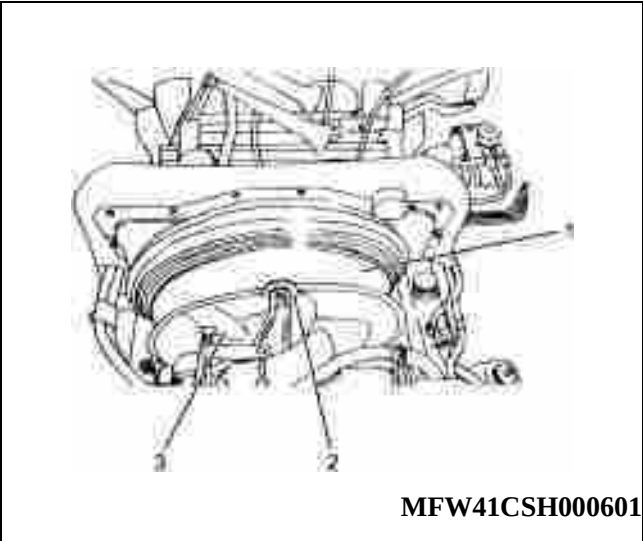


Legend

1. Upper hose
2. Radiator
3. Intercooler hose
4. Radiator mounting bracket
5. Fan guide bracket
6. Bell Guide

7. Remove the heat sink assembly.

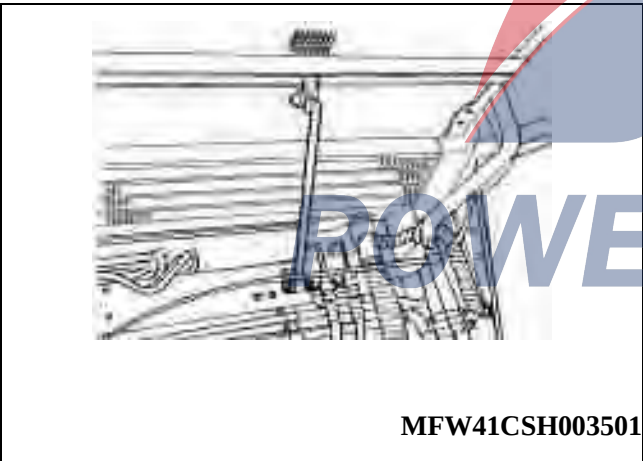
- Remove the rubber part of the radiator.



Legend

1. Fan guide
2. Fan guide bracket
3. Fan

- Remove radiator ribs.



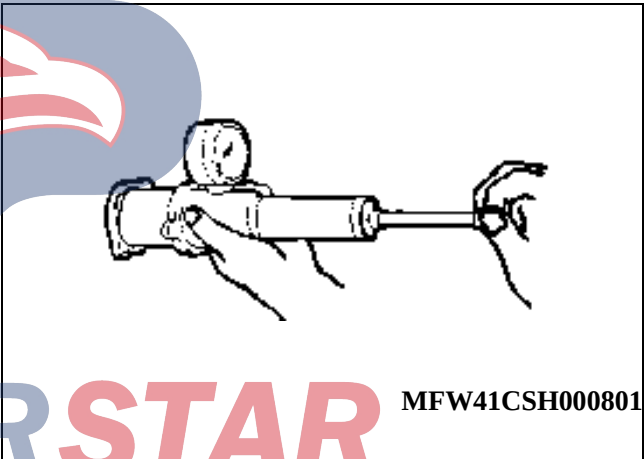
- Remove the heat sink assembly.
- Be careful not to damage the heat sink fins.

Inspection

Radiator cover

- Check the spray valve opening pressure with the radiator cap tester. If the valve opening pressure during the inspection exceeds the standard value, the radiator cap should be replaced.

kPa (kg/cm ² /lb/in ²)/rpm	
Radiator cover working pressure	Positive pressure = 93-122 (0.90—1.20/13—17)
	Negative pressure = 4.9 (0.05/0.7)



- Check the condition of the negative pressure valve at the center of the valve seat side. If the valve seat cannot run smoothly due to rust, dirt, etc., the radiator cap should be cleaned or replaced.

Radiator core

- Distortion of the heat sink fins can result in poor heat dissipation and overheating. Therefore, the fins must be repaired. Be careful not to damage the connector fitting during maintenance.
- Clear the dirt and chipping.

Radiator cleaning

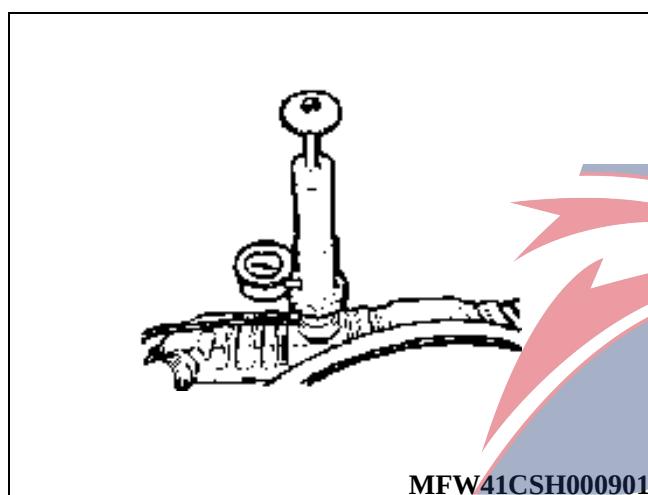
- Clean the inside of the radiator and water-cool the water channel. Remove all rust and dirt.

Cooling water leak check

- Apply 200 kpa (2.0 kg / cm² / 28 psi) pressure to the radiator cap tester to inspect the following components for leaks. Also check radiator hoses and heater hoses for damage or deterioration, and check for loose hose clamps.

Check position

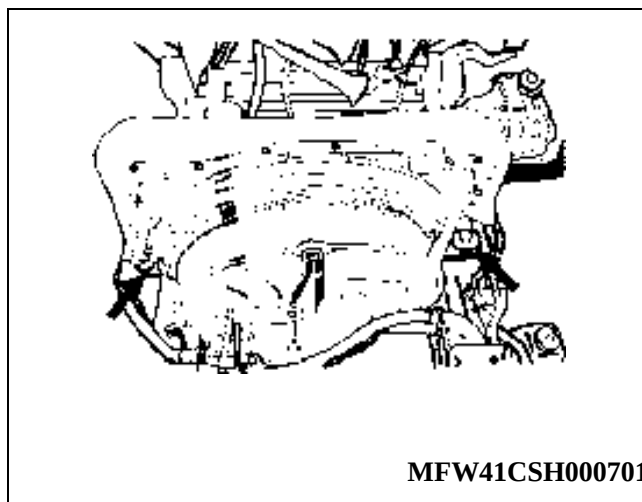
- Radiator assembly, pump assembly, radiator hose, heater hose

**Reassemble**

1. Install the heat sink assembly.

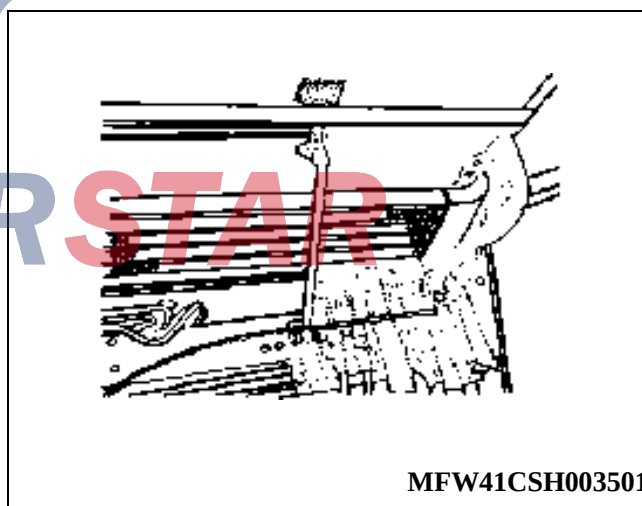
Tightening torque	N m (kgf m/lb ft)
	55 (5.6 / 41)

- When installing the heat sink, be careful not to damage the heat sink.



- Install the radiator bracket.

Tightening torque	N m (kgf m/lb ft)
	18 (1.8 / 13)

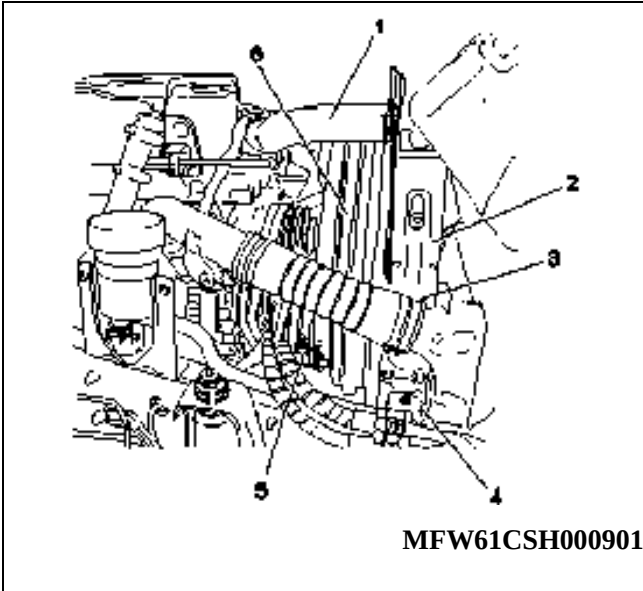
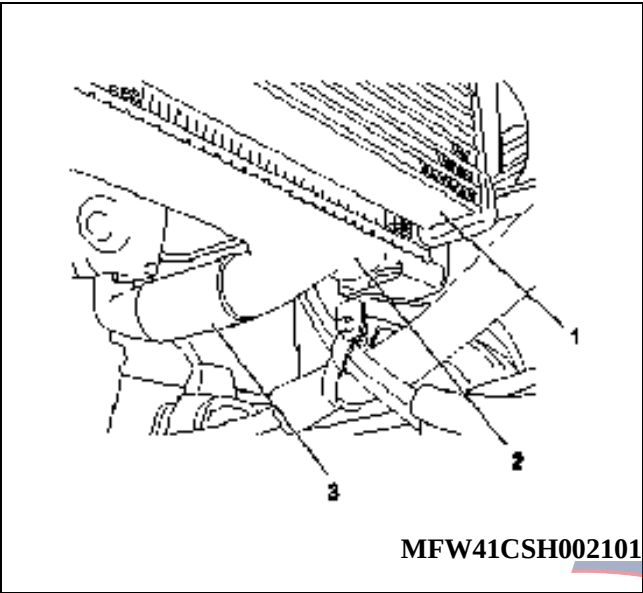


2. Install the fan.

Specification	Tightening torque	N m (kgf m /lb ft)
TCN	7 (0.7/61 lb. inch)	
TCC、TCS	20 (2.0 / 14)	

3. Install a soft water hose.

4. Install the upper and lower hoses.



Legend

Intercooler components

- 2. Radiator assembly
- 3. Radiator lower hose

Legend

- 1. Upper hose
- 2. Radiator
- 3. Intercooler hose
- 4. Radiator mounting bracket
- 5. Fan guide bracket
- 6. Bell Guide

5. Put on the hose.

Note:

Make sure that the clamp bolt head does not reach the top.

7. Close the drain plug and inject the cooling water.

6. Install intercooler hose.

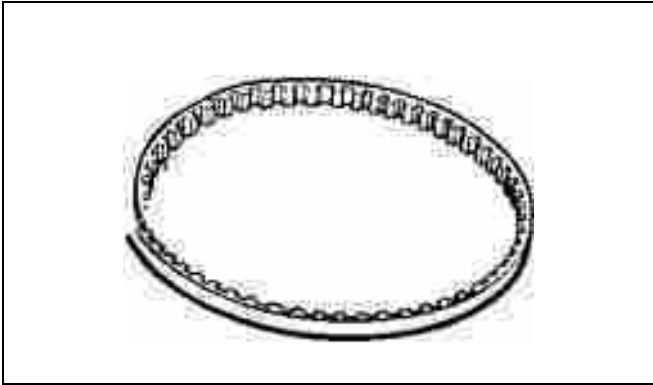
- Tighten the clamps.

Tightening torque	N m (kgf m/lb ft)
6 (0.6 / 52)	

Fan belt

Inspection

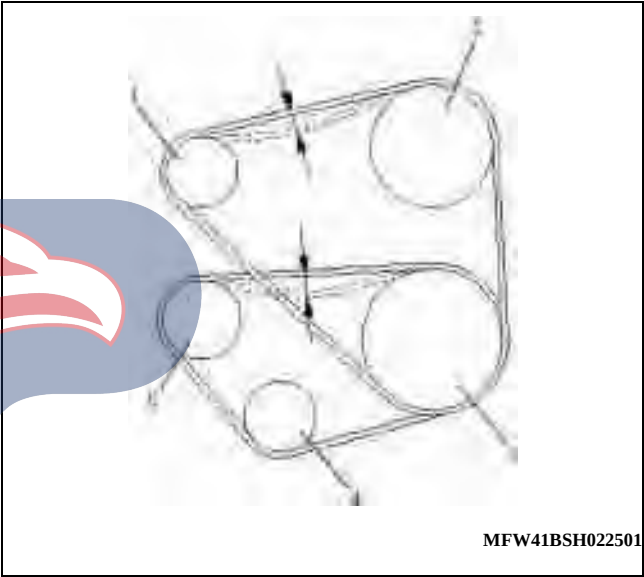
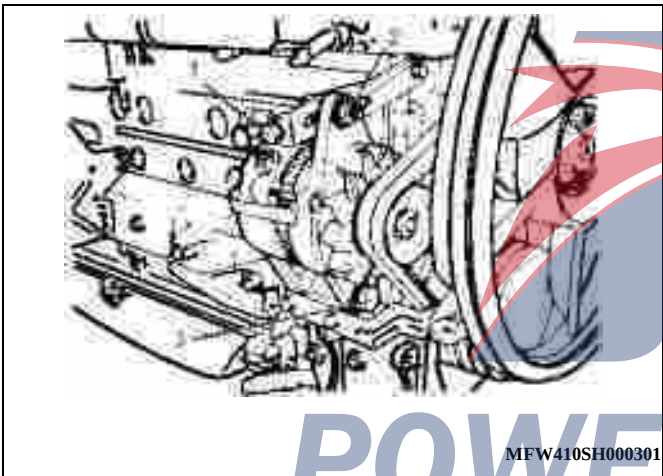
Check the V-belt for wear.



Caution:

After the adjustment is completed, the bolts should be fully tightened.

V- belt bending amount	mm(inch)
10—15/0.3937—0.5906	



Legend

1. Adjustment bolts
2. Coupling nut
3. Install the bolts.

Legend

1. Generator pulley
2. Water pump pulley
3. Crankshaft pulley
4. Idler wheel
5. Air-conditioning compressor pulley

Adjusting the tension of the generator drive belt

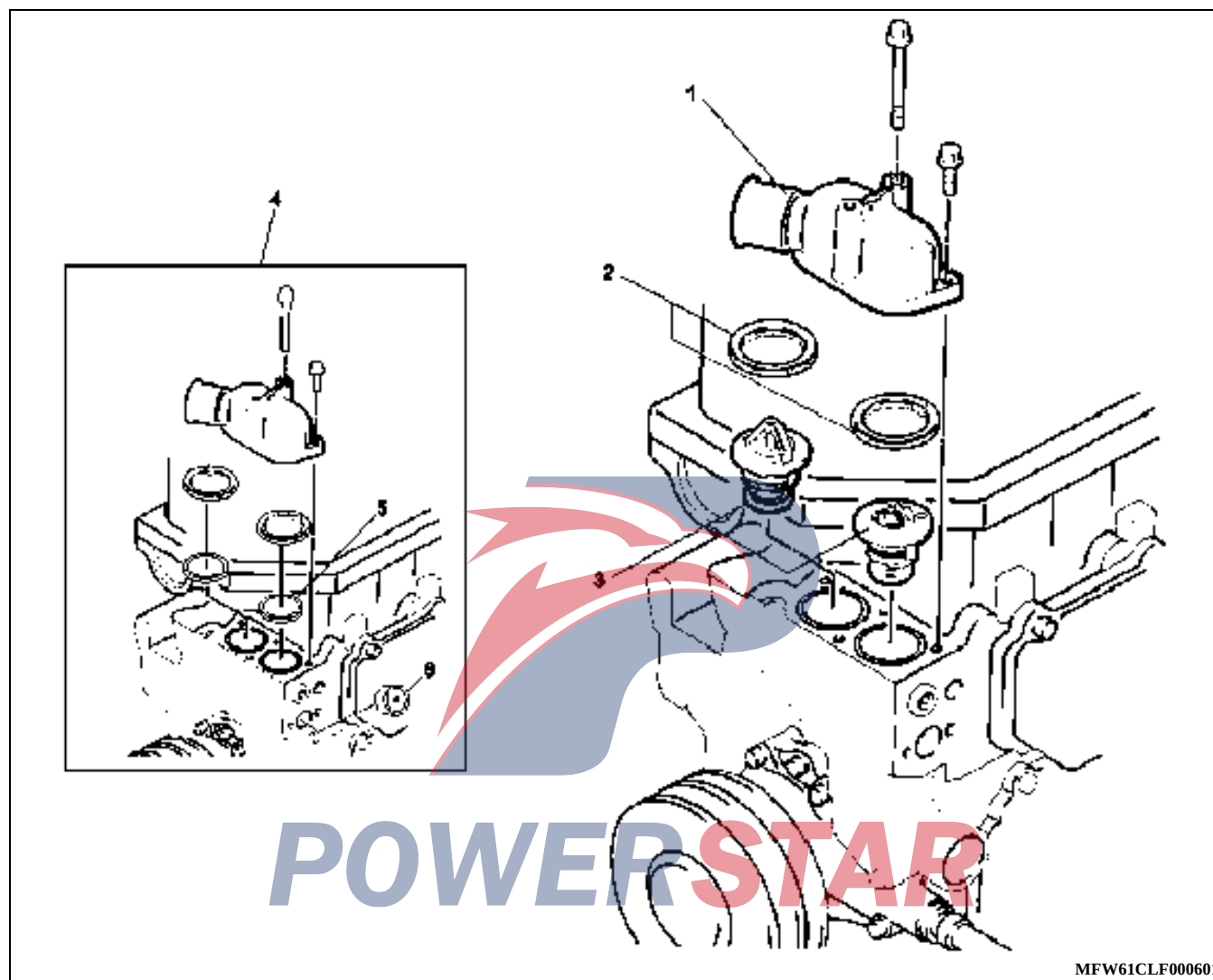
- The joint bolts at the top / bottom of the motor are loosened to adjust the adjusting bolts so that the deflection of the middle part of the belt is within the standard range with the force of about 98 cattle (10 kilogram force /72 pound force).

Note:

After the adjustment is completed, the bolts should be fully tightened.

Thermostat

Part



Legend

- | | |
|----------------|-----------------------|
| 1. Outlet line | 4. Without thermostat |
| 2. Gasket | 5. Spacer |
| 3. Incubator | 6. Sealing cup |

Tear down

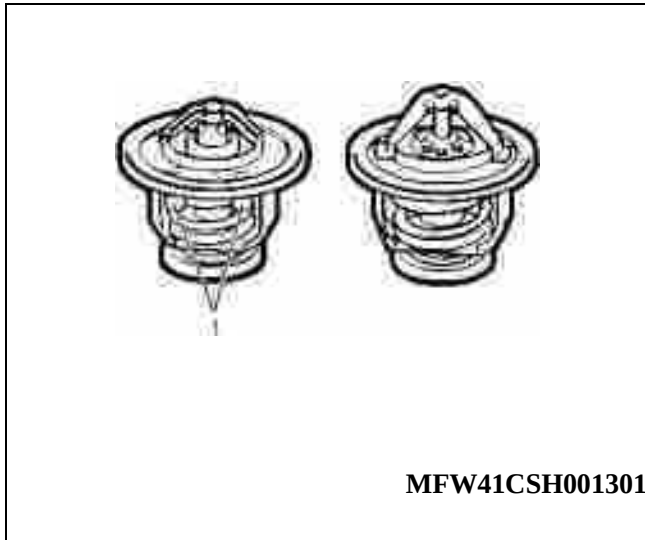
1. Discharge coolant. Prepare a container and control the discharge speed so that the cooling water does not spill onto the oven housing.
2. Remove the radiator suction hose from the radiator side.
3. Remove the outlet tube.

4. Take out the incubator. (When the septum is removed, the incubator has no specifications)

Inspection

1. Check the water outlet for corrosion.
2. Check the thermostat.

- Check the thermostat box assembly, spring, shaft, swing valve and plugging section for damage.



Legend

1. Shake the valve

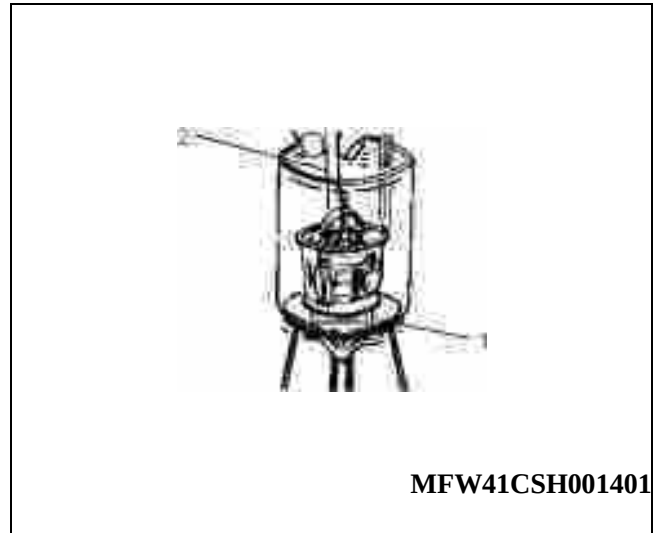
- Check the operation of the thermostat.
- The incubator was immersed in water and the water temperature was gradually raised to measure its initial temperature, full temperature and head.

	°C(°F)/mm(inch)
Start initial temperature	80-84 (176-183) {fire fighting trucks: 76.5 (169.7)}
Full open temperature	> 95(203){Fire Trucks: 90(194)}
The degree of openness	>8(0.3150)

- Due to the high heat capacity of the wax oven, there are some differences in the operation of the valve as the water temperature changes.

Note:

In order to avoid direct heating of the incubator, the bottom of the container should be covered with a flat plate or should be lifted. At this point, stirring should continue to maintain a constant water temperature.



Legend

1. Sawdust
2. Stir bar

3. Check the gasket for cracks and damage.

Reassemble

1. Install the gasket.

Note:

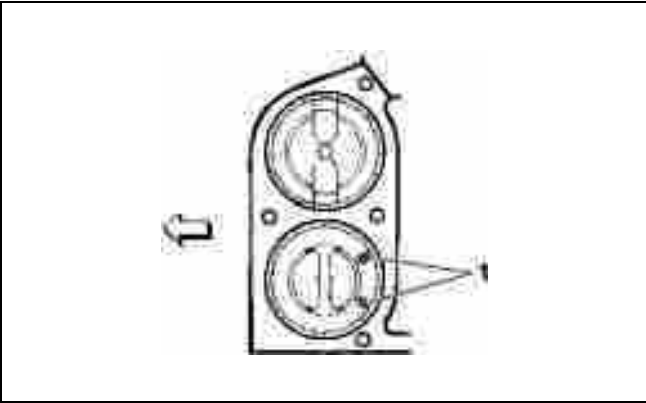
When the gasket leaks, it should be replaced.

2. Install the thermostat. (Install septum)

- Install the gasket on the thermostat and install it (as shown).

Note:

Do not install wrong or wrong directions.



Legend

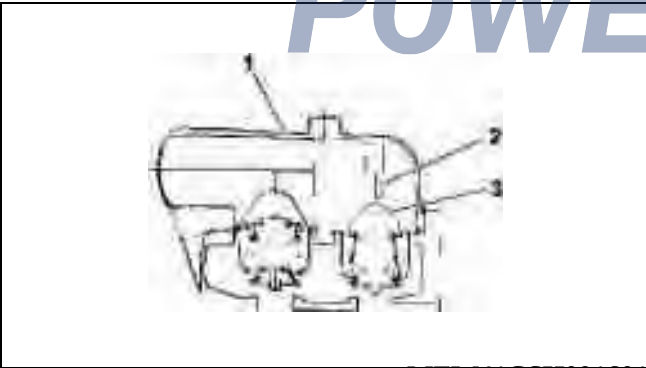
1. Shake the valve position

3. Install the outlet pipe.

Tightening torque	N m (·kgf m /lb ft)
24 (2.4 / 17)	

Note:

If the thermostat is installed in the wrong position, the outlet tube cannot be installed.



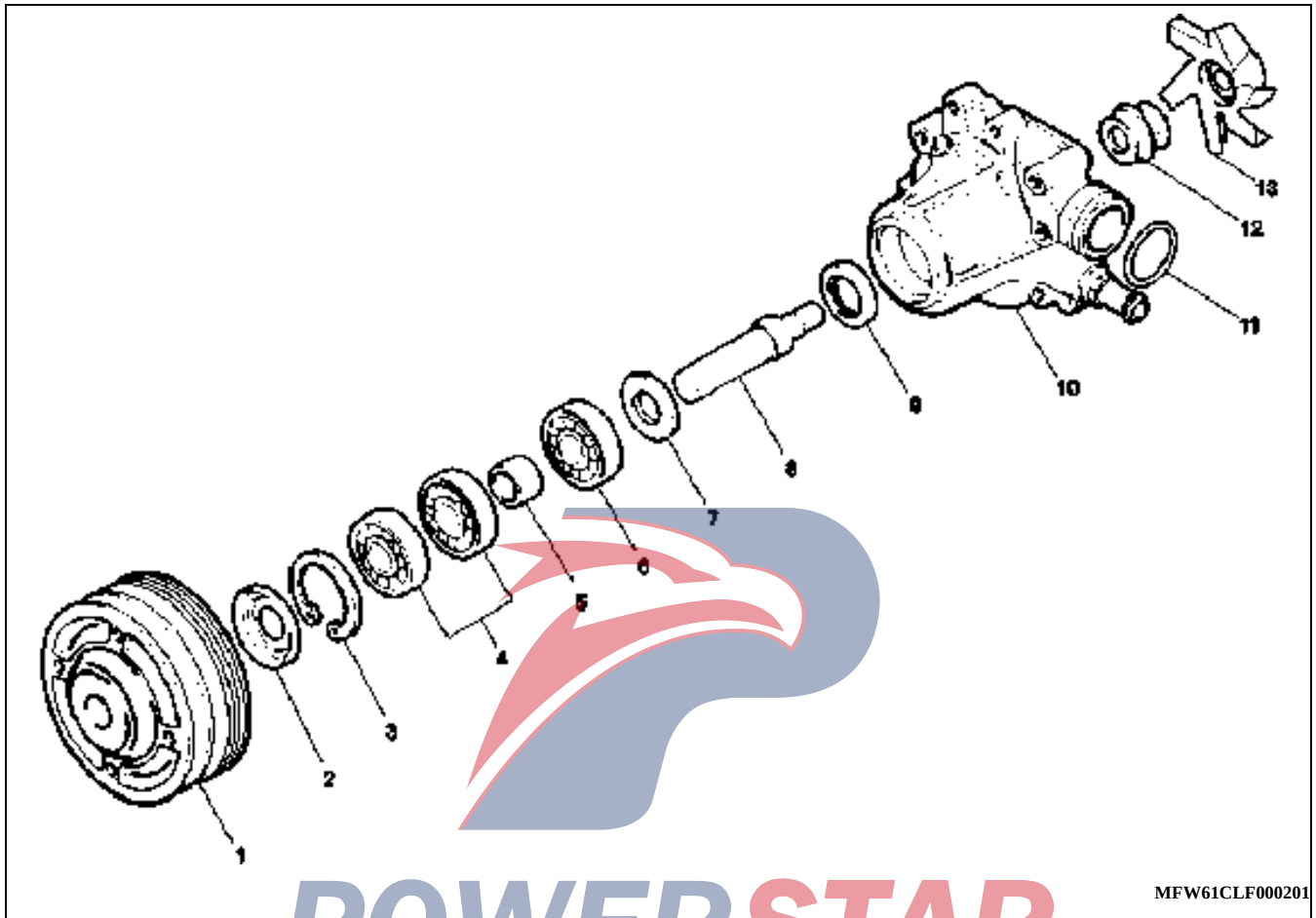
Legend

- 1. Outlet pipe
 - 2. Positioning plug
 - 3. Incubator
-

- 4. Install the radiator suction pipe.
- 5. Fill coolant.

Water pump

Part



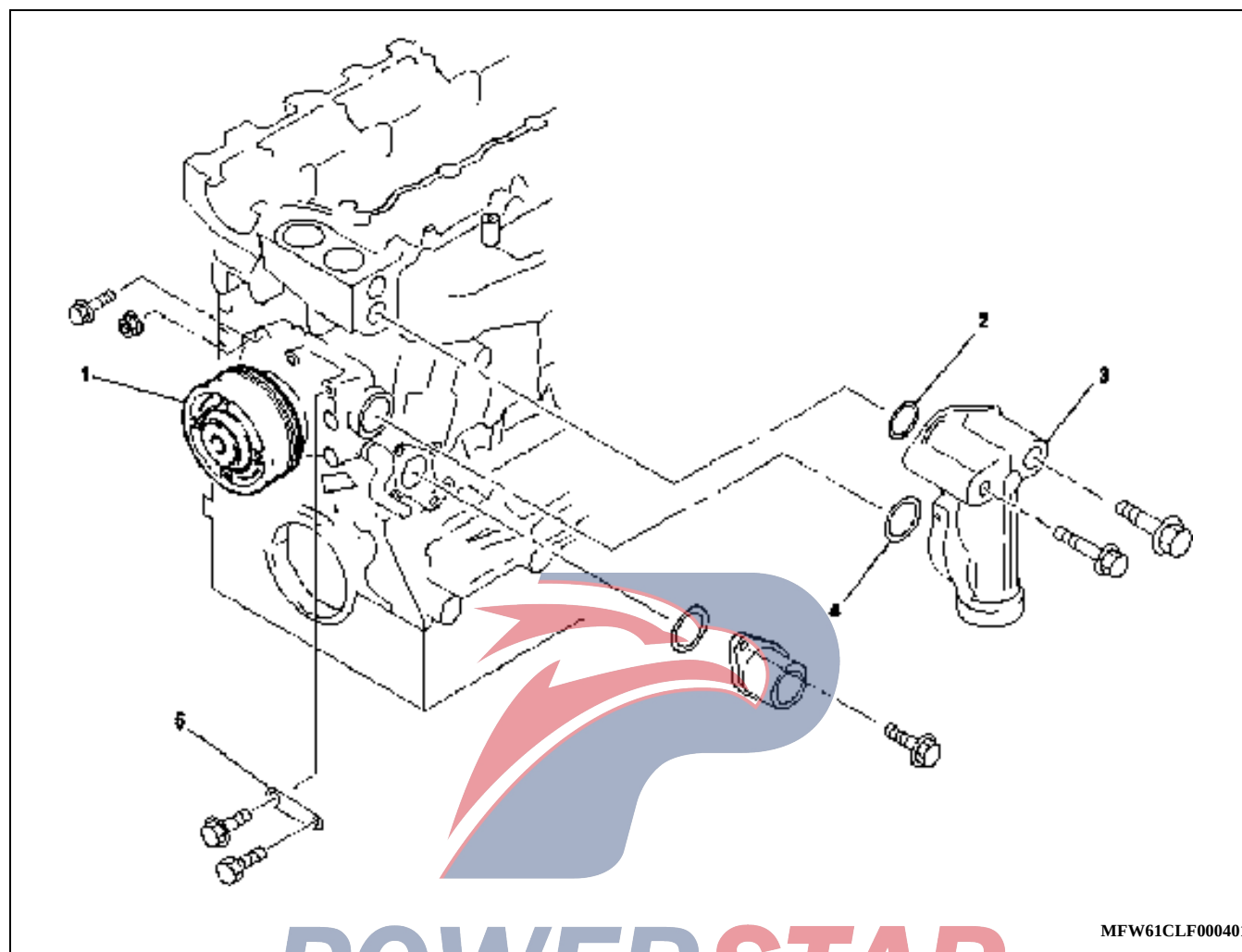
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Legend

- | | |
|---------------|---------------------|
| 1. Pulley | 8. Axis |
| 2. Dust cover | 9. Oil seal |
| 3. Buckle | 10. Water pump body |
| 4. Bearing | 11. O-ring |
| 5. Septum | 12. Sealing device. |
| 6. Bearing | 13. Impeller |
| 7. Washers | |

■ Tear down

- | | |
|---------------------------|------------------------|
| 1. Remove the plate. | 4. Remove the O-rings. |
| 2. Remove the water pipe. | 5. Remove the pump. |
| 3. Remove the O-rings. | |



Legend

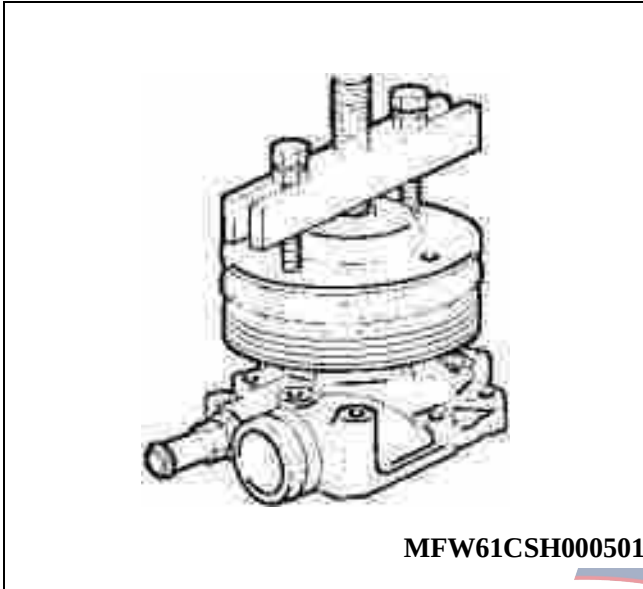
- | | |
|------------------------|------------------------|
| 1. Water pump assembly | 4. O-ring (water pump) |
| 2. O-ring (water pipe) | 5. Sheet bar |
| 3. Water pipe | |

Disassembly

After installing the pump set, if the leakage flow of the suction hole exceeds 10 ml/100 hours, it should be removed and adjusted.

1. Remove the impeller.

2. Use a pull-out tool to remove the pulley (unplug tool: 1-8521-0050-0).

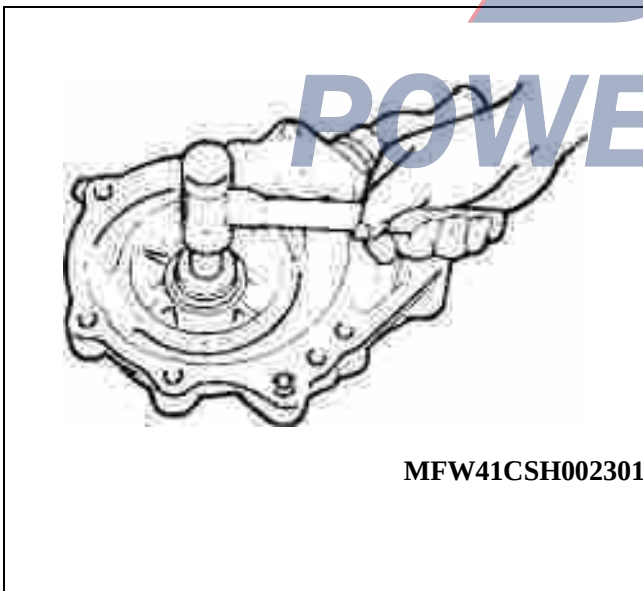


3. Remove the dust cover.

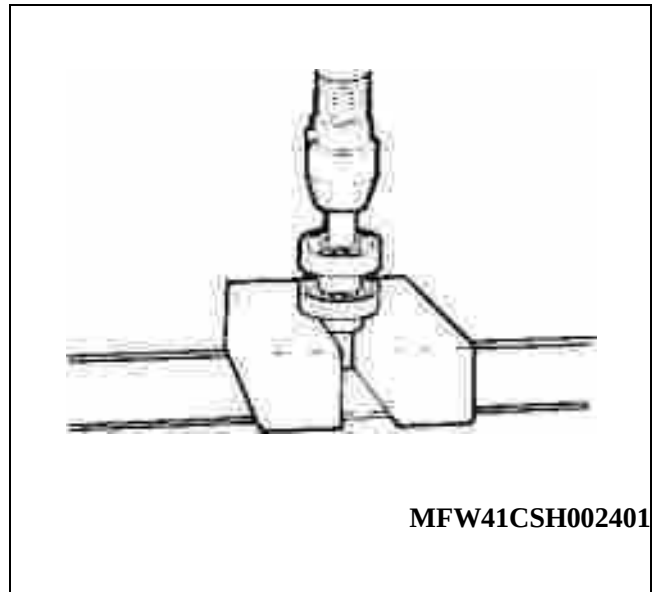
4. Remove the buckle.

5. Remove the shaft.

- Remove the retaining ring from the side of the pulley and gently tap the shaft with a hammer on the side of the impeller to remove it.



6. Use a press to remove the bearing.



7. Remove the septum.

8. Remove the washer.

9. Push the sealing device out of the pulley side and remove it.

10. Roll out the oil seal from the side of the impeller and remove it.

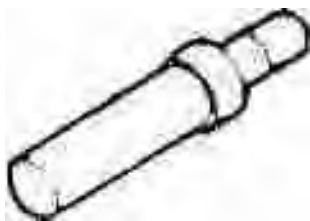
11. Remove the O-rings.

12. Only applies to pump body.

Inspection

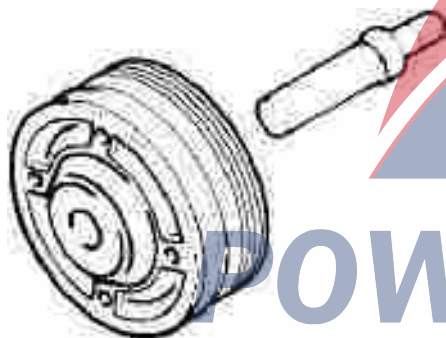
1. Check the impeller for wear and so on.
2. Check the pulley for wear.
3. Check the dust cover for deformation and damage.
4. Check the buckle for damage.
5. Check the axis.

- Check the shaft seal surface for wear or damage.



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- Measure the inner diameter of the insertion portion of the pulley shaft and the outer diameter of the axle. Check that whether the degree of interference is within the standard value.



MFW61CSH000401

- Check whether the ball and seat parts are worn for abnormal looseness.
- Rotate the bearing by hand to check if the movement is smooth and if there is abnormal noise.



MFW41CSH002701

7. Check the septum for damage.
8. Check the gasket for damage.
9. Change a new sealed device.
10. Put on a new oil seal.
11. Install the new O-ring.
12. Check whether the water pump body is rusty and damaged.

Assembling

1. Fix the water pump body.
2. Install O-rings.
3. Put on the oil seal.

- When the pump body is arranged, the part of the lip part is facing outwards, and then the oil seal is installed.
- Apply oil to the perimeter of the seal and insert its lips outward. Be careful not to deform it.
- Apply grease (BESCO L-3 or BESCO L-2 grease) to the inner surface of the lip.

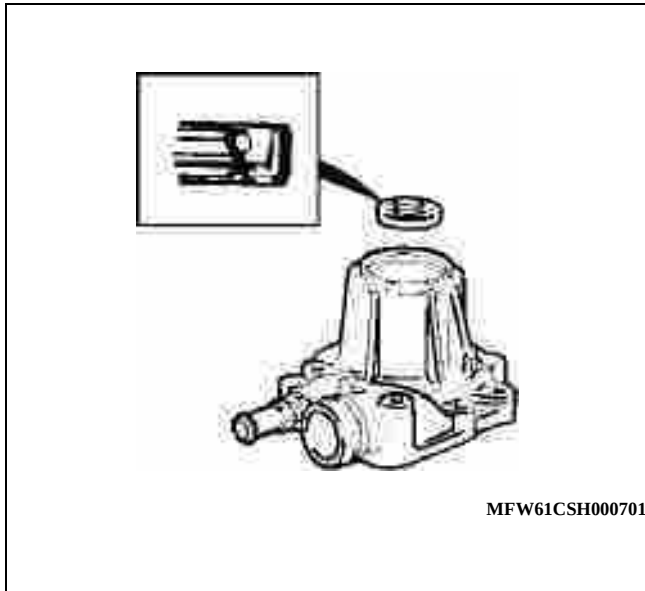
Note:

Oil seals must not be used again.

6. Check the bearings.

- Check the bearing for shedding, wear and abnormal noise.
- Check for scratches and separate from the spherical gap over the entire circumference of the race.

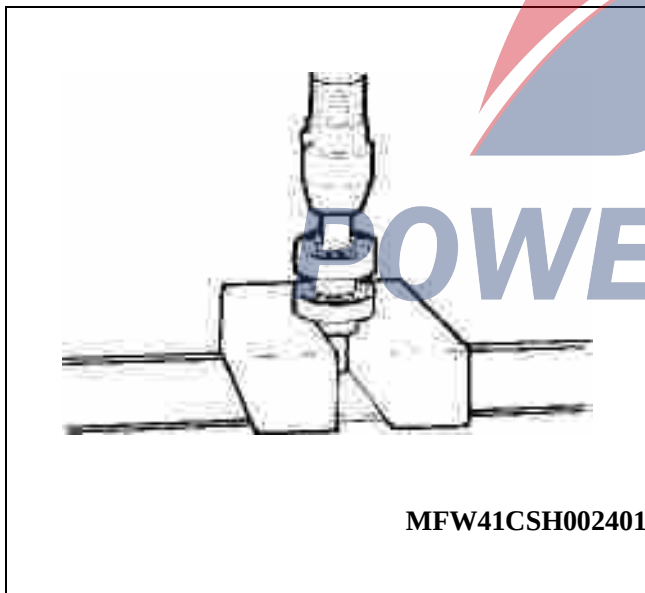
Interference between pulley and shaft		mm(inch)
Assembly standards		0.048(0.0019)–
		0.078(0.0031)



4. Install the washer.

5. Install the bearing.

- Apply engine oil to the shaft.
- Install the washers, bearings, washers and bearings on the shaft in turn and press them together with a press.



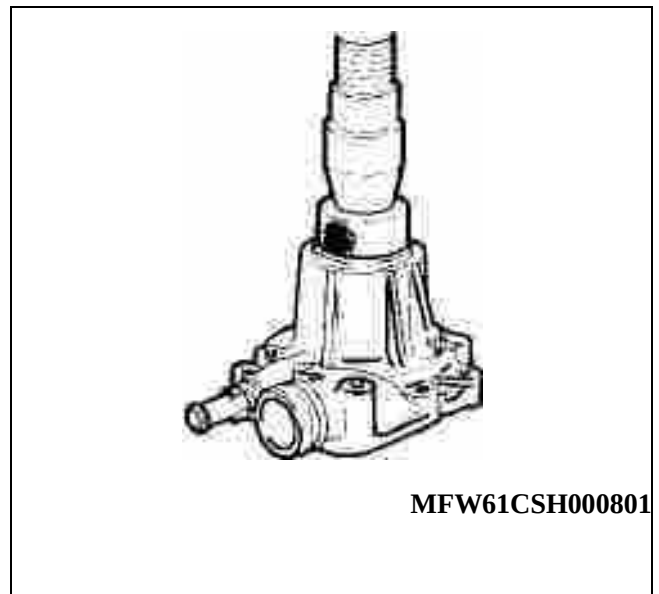
6. Put on the septum.

7. Install the shaft on the pump body.

- Inject grease into the bearing (BESCO L-3 or BESCO L-2 grease).
- Install the shaft on the pump body, place the tube on the bearing outer ring and press it with a press.

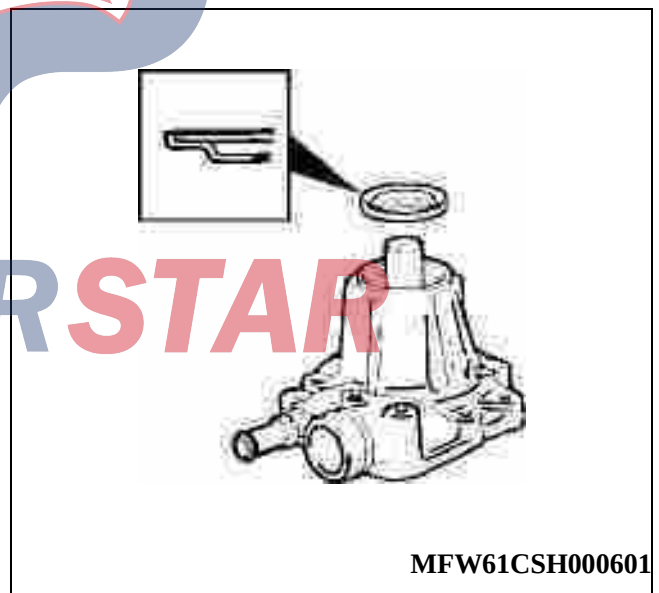
Note:

Do not break the oil seal.



8. Attach a buckle.

9. Install the dust cover on the pump body in the direction shown in the figure.

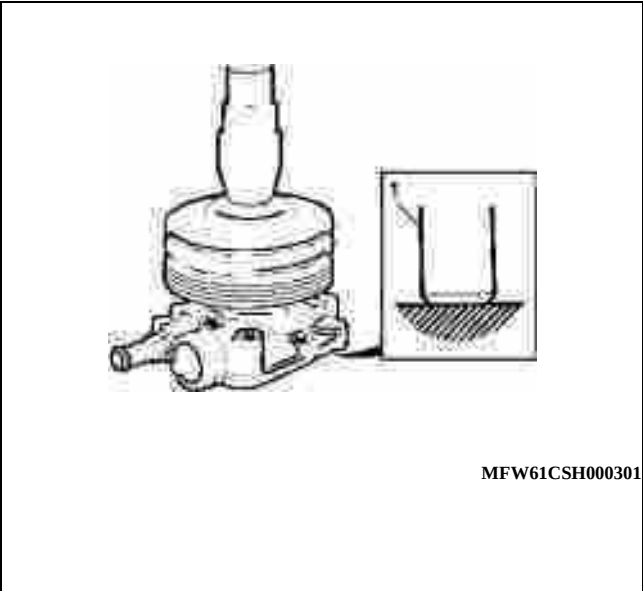


10. The belt pulley is installed on the shaft.

- Press the pulley with a press, taking care not to damage the dust cover.

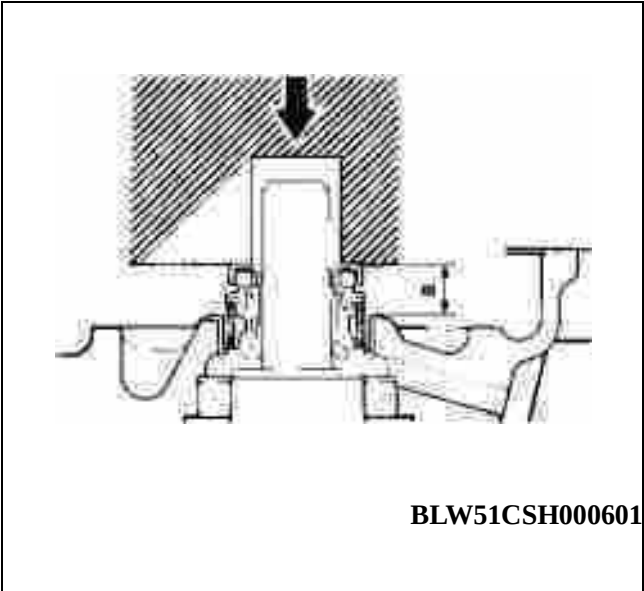
Note:

To prevent damage to the bearings, the tip of the impeller side shaft should be pressed tight.



Legend

1. Impeller insertion section



Legend

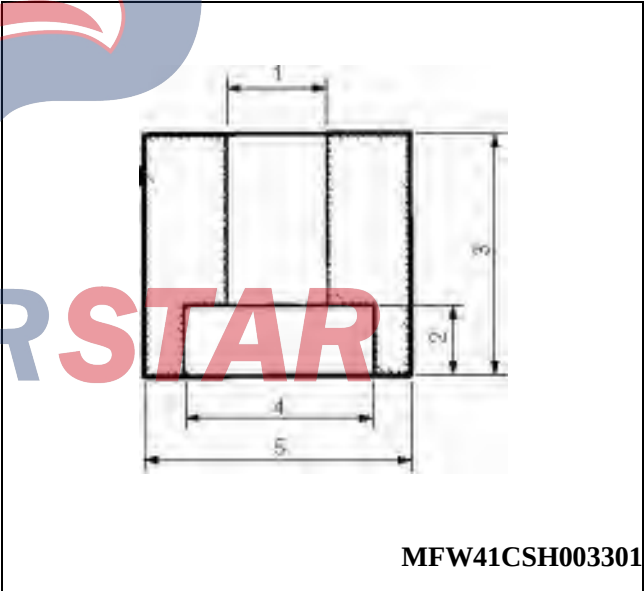
(a) 11.0—11.6 mm(0.4331—0.4567 inch)

11. Install the sealing device.

- Install the sealing device on the pump body.
- Press the seal with a press (seal setting tool: 1-8522-0047-0) until its flange portion contacts the pump body.

Sealing device size	mm(inch)
Assembly standards	11(0.4331)— 11.6(0.4567)

- The sealing device has a liquid filling that burns on the pump body and the impact surface of the shaft.
- If you need to make sealing device setting tool, please refer to the left figure.



Legend

1. $\Phi 17$ (0.6693 inch)
2. 11.3 ± 0.1 (0.4449 $\pm$ 0.0040 inch)
3. $\Phi 50$ (1.9685 inch)
4. $\Phi 33$ (1.2992 inch)
5. $\Phi 44$ (1.7323 inch)

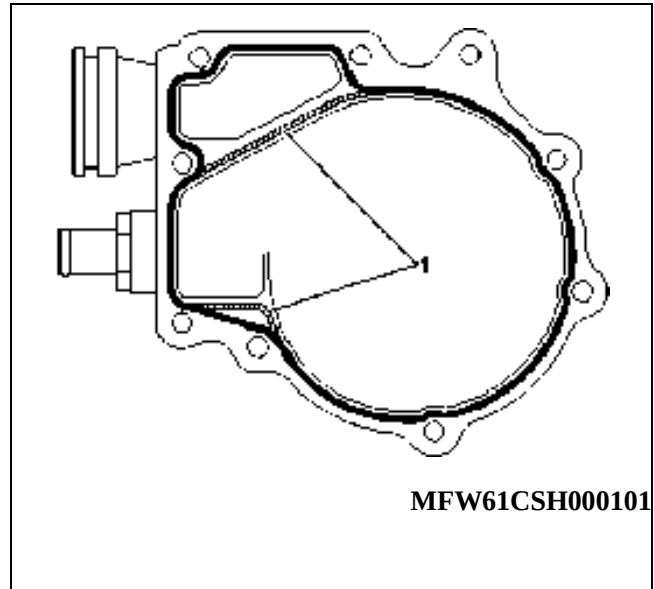
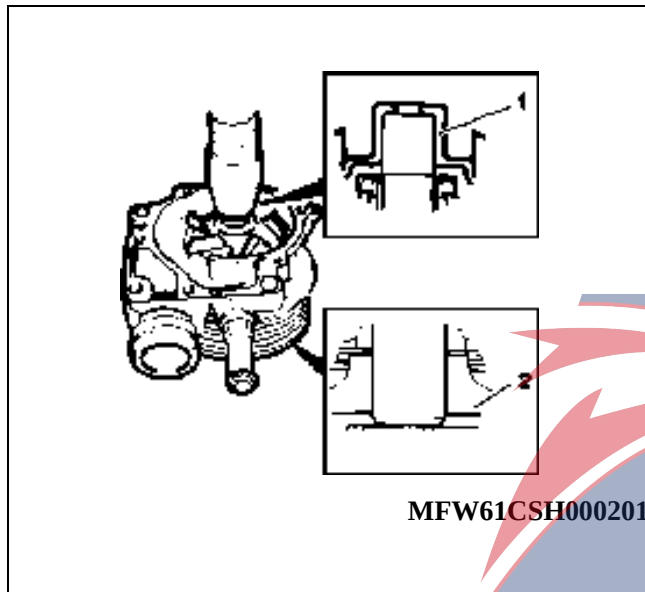
12. Install the impeller.

- Press with a press until the tip (1) is at the same height as the impeller edge surface (2).

Note:

In order to prevent damage to the bearing, it should be pressed with the tip of the shaft on the pulley side.

Body and impeller clearance	mm(inch)
0.4—1.2(0.0157—0.0472)	

**Legend**

1. The filler cannot be used even if there is a groove

- Install the pump assembly.

Tightening torque	N m (kgf m/lb ft)
19 (1.9 / 14)	

13. Check after reassembly.

- Turn the pump body and make sure it can rotate smoothly.
- Use a pressure equal to 0.1MPa (14 psi) for the pressure test and check for leaks in the seal.
- Temporarily install the bolts of the board connection section (1).

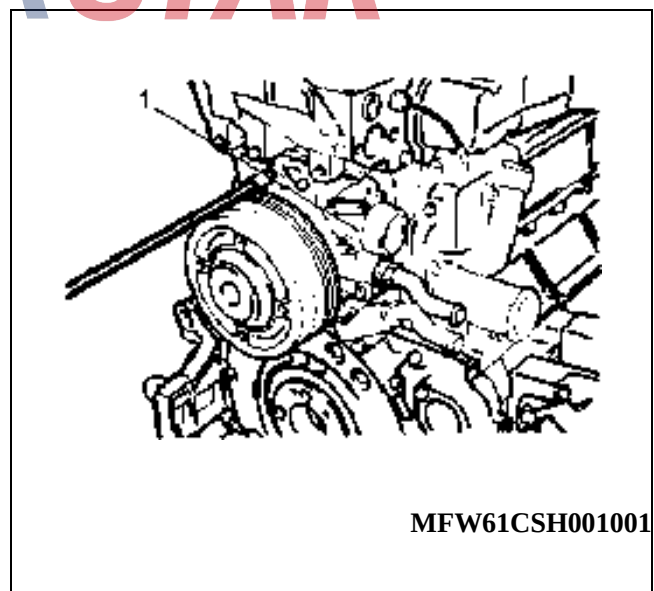
Reassemble

1. Install the pump assembly.

- Evenly spread a liquid filling with a width of about 2-3 millimeters (0.0787 to 0.1181 inches) (three bond 1207C adhesive) (as shown).

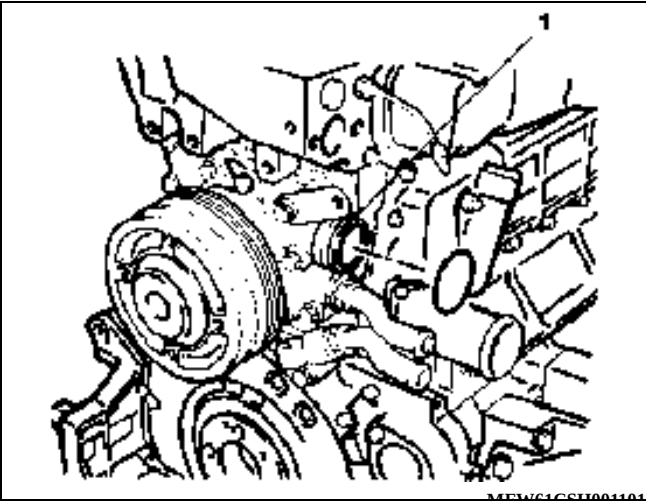
Note:

Load it within 7 minutes of using liquid filling.



2. Install O-rings (Water pump).

- After installing the O-ring, apply soapy water around the O-ring.



Legend

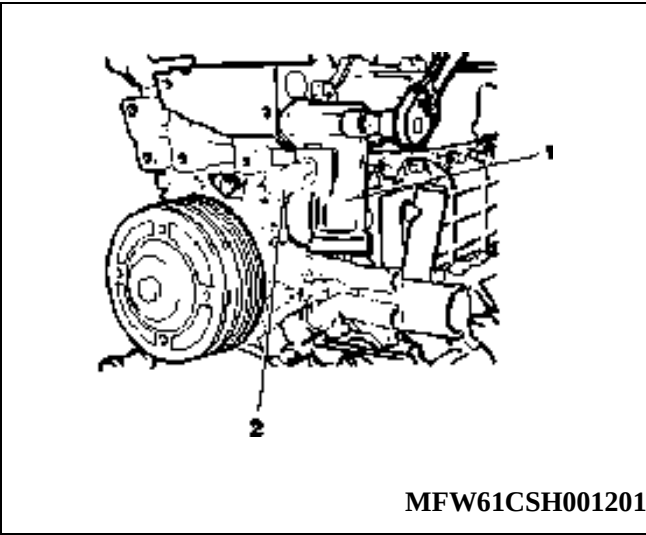
1. Soap water

3. Install O-rings (water pipes).

4. Install the water pipe

- Carefully press in the water pipe so as to prevent damaging the O-ring.

Tightening torque		N m (kgf m /lb ft)
Water pipe	M8:	19(1.9/14)
	M16:	103(10.5/76)



Legend

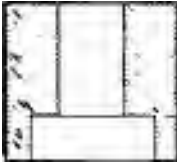
1. Water pipe

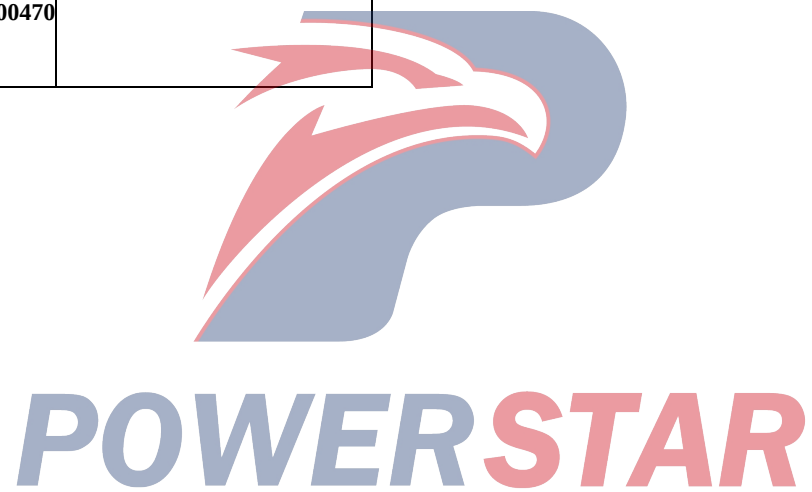
2. Plate

5. On the plate.

Tightening torque		N m (kgf m /lb ft)
Board piece	M10:	36(3.9/28)
	M8:	19(1.9/14)

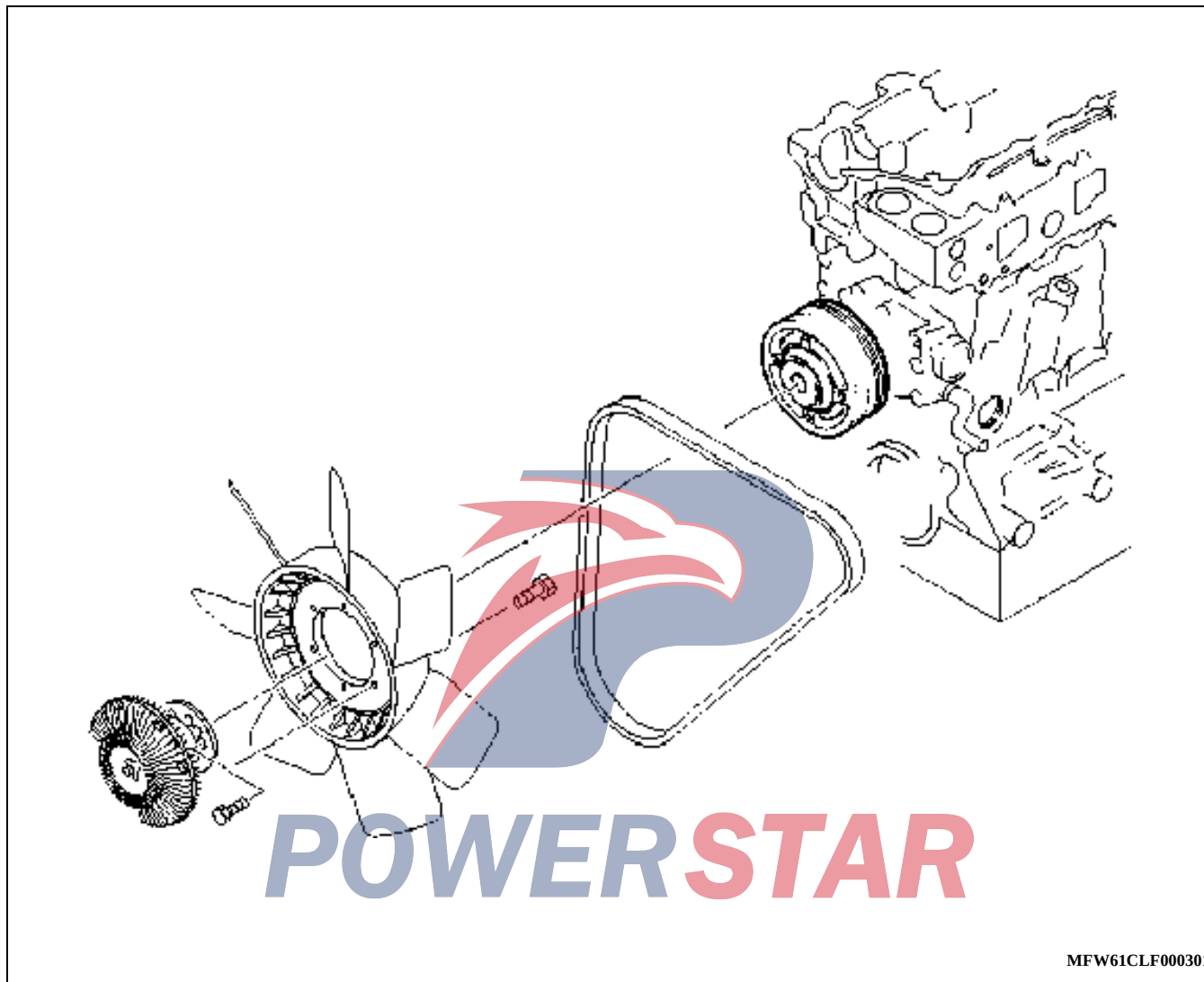
Special tools

Graphic representation	Tool number/description
 1852100500	1-8521-0050-0 Pulley puller
 1852200470	1-8522-0047-0 Seal Setting Tool



Fan drive device

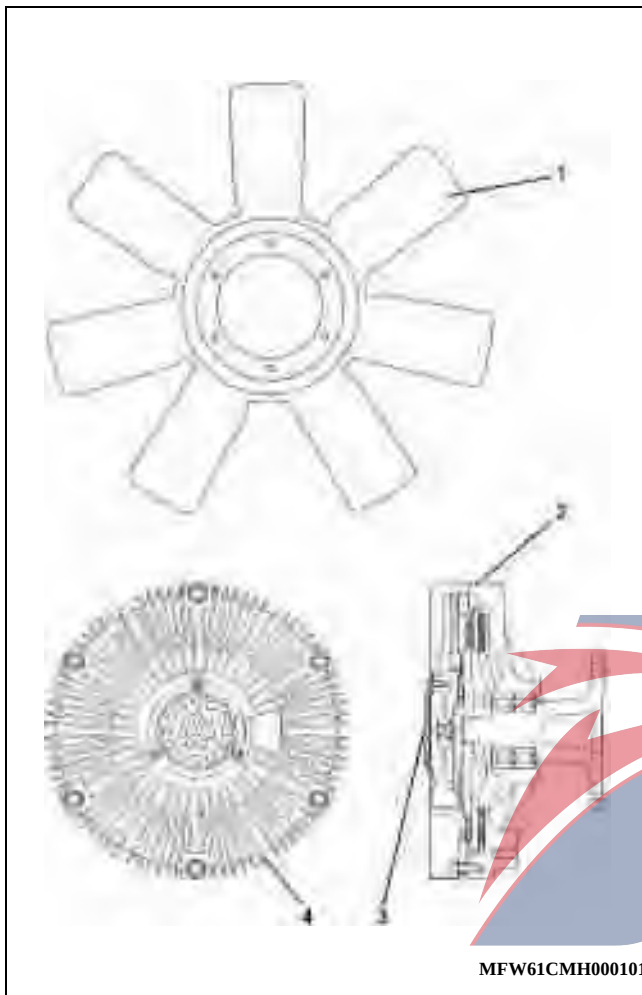
Inspection



MFW61CLF000301

1. Fan shaft coupling

- Check for abnormal noise and leaks.
- Turn by hand to check for any abnormal noise.



Legend

- 1. Cooling fan
 - 2. Fan drive
 - 3. Bimetallic strip
 - 4. Cooling fins
-

POWERSTAR

Engine

Engine fuel

(6HK1)

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Test	6C-20		
Empty fuel tank	6C-21		

Fuel System

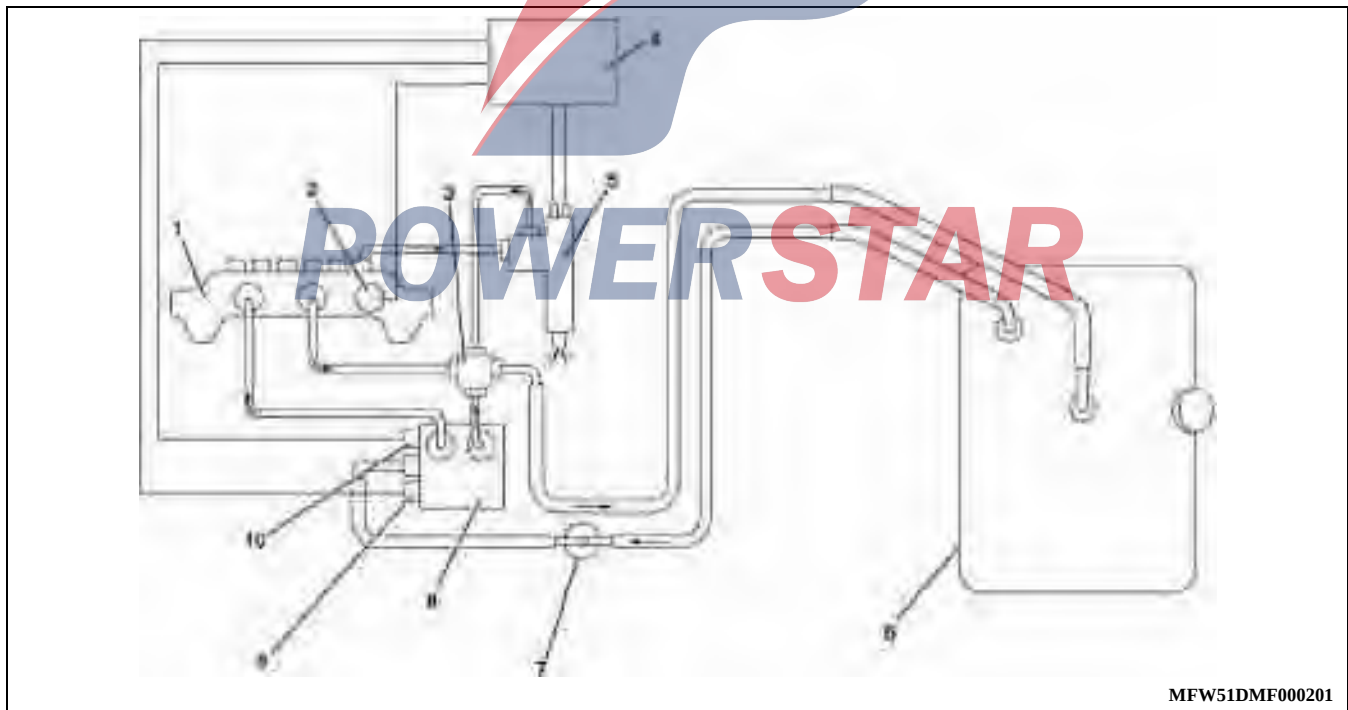
Precautions for maintenance

The pore space and gap of the fuel line inside the fuel system injector are strictly fabricated according to the highly precise specification. These parts are very sensitive to dust or debris, and dust or debris entering these parts can damage the fuel lines. Therefore, as much as possible, dust or debris should be prevented from entering these parts.

Each injector's injector identification plate is unique. Do not mix with other sprayer identification plates.

When replacing the injector assembly, use the scan tool to sculpt the ID on the ECM. For the engraving process, please refer to "Engine Control Section". Also, do not use water repellent in the fuel tank because the water repellent will cause diesel to absorb moisture and cause rust.

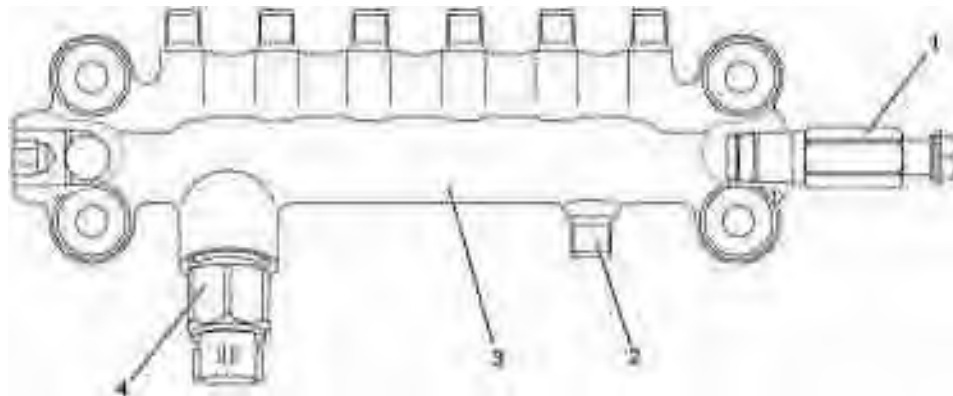
Fuel system diagram



Legend

- | | |
|--------------------------------|--|
| 1. Common Rail | 6. Fuel tank |
| 2. Common rail pressure sensor | 7. Fuel Filter Assembly (Precipitation Separator Assembly) |
| 3. Drainage pipe | 8. Supply pump |
| 4. ECM | 9. Air intake control valve |
| 5. Injector | 10. Fuel temperature sensor |

Common Rail



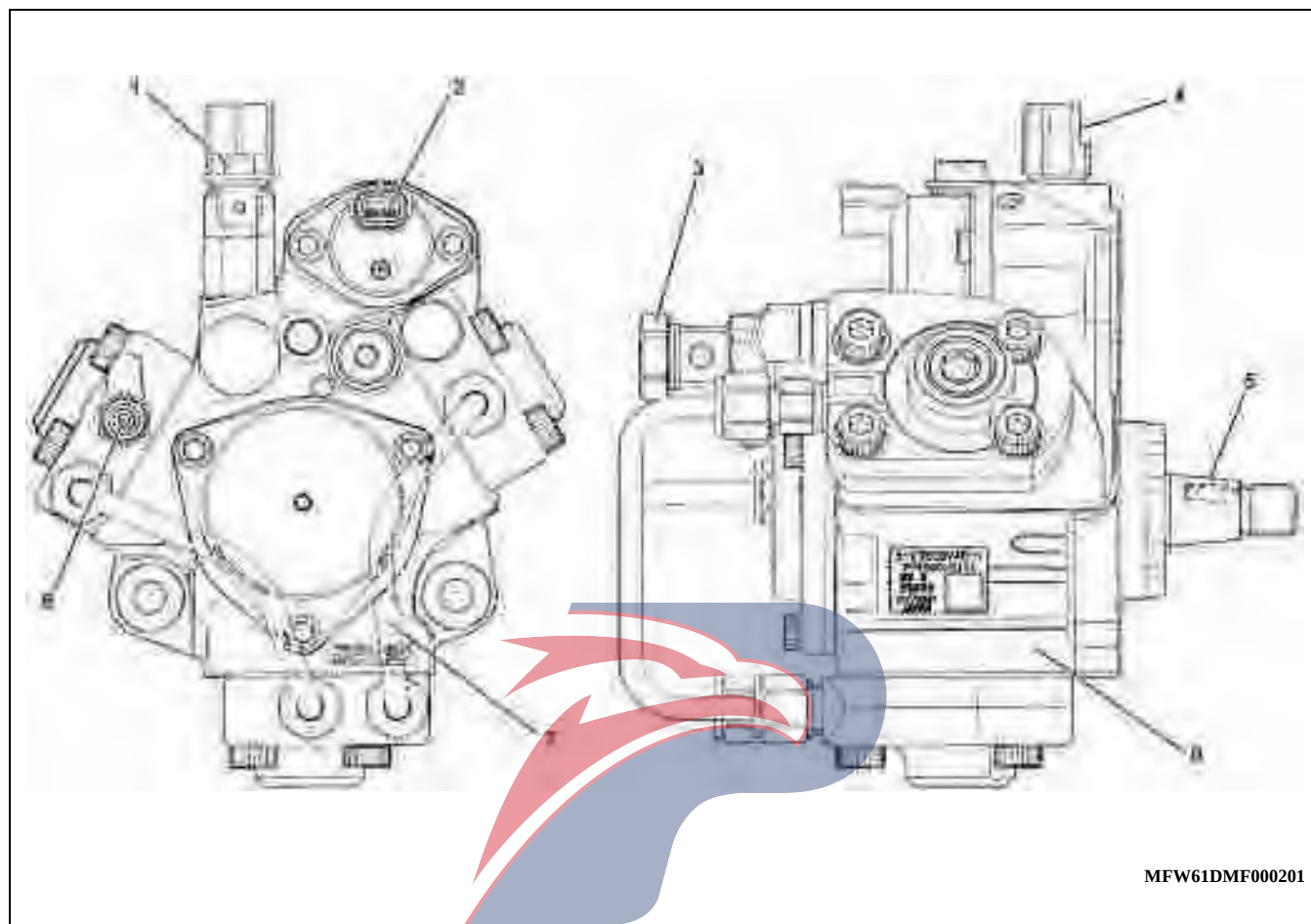
MFW61DMF000301

Legend

- | | |
|---------------------|-------------------------|
| 1. Pressure limiter | 3. Common rail assembly |
| 2. Fuel import | 4. Pressure Sensor |

POWERSTAR

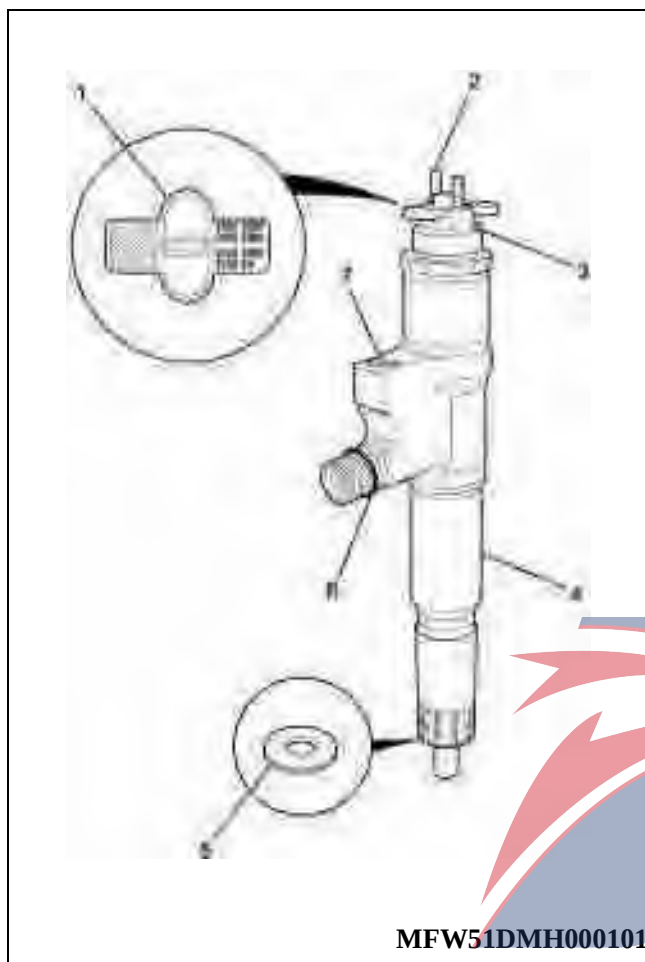
Oil feed pump



Legend

- | | |
|----------------------------|------------------------------|
| 1. Fuel outlet (backflow) | 5. Camshaft |
| 2. Suction control valve | 6. Supply pump assembly |
| 3. Fuel inlet | 7. High pressure tube |
| 4. Fuel temperature sensor | 8. High pressure fuel outlet |
-

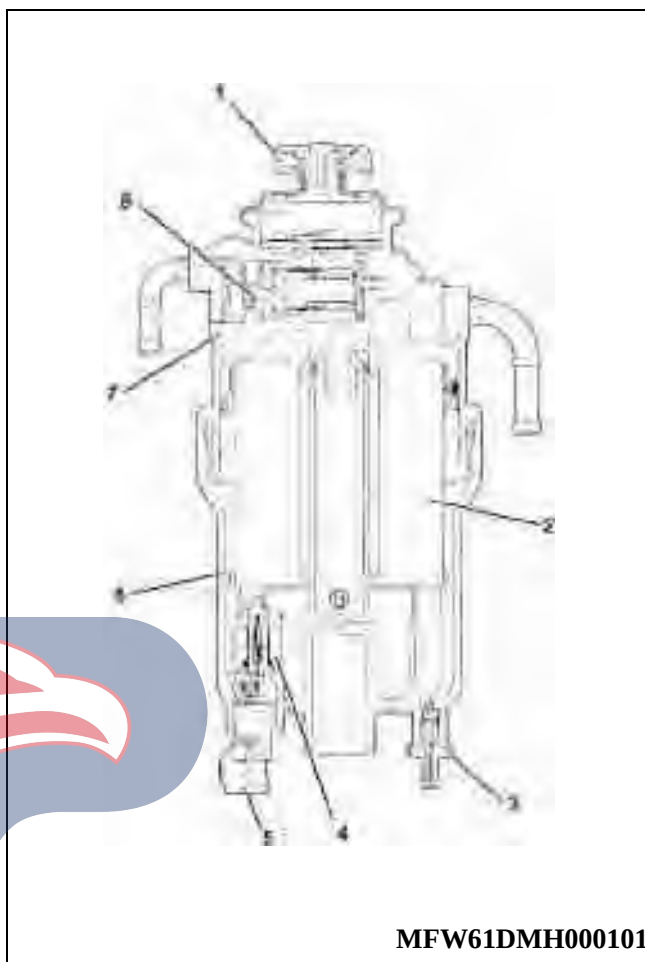
Injector



Legend

1. Identification label
2. Terminal position
3. Eject drive section
4. Ejector assembly
5. Gasket
6. O-ring
7. Drain pipe connection section

Fuel filter assembly



Legend

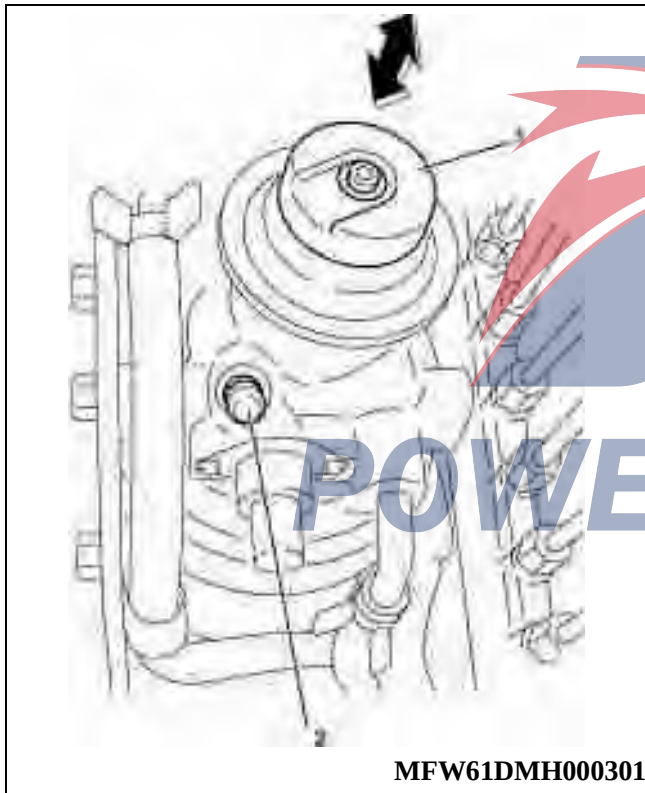
1. Starter pump
2. Fuel filter element
3. Release plug
4. Floater
5. Switch connector
6. Housing
7. Main body assembly
8. Control valve

Functional check:

Vent

1. Loosen the fuel filter air removal plug.
2. Move the priming pump and feed the fuel under pressure. Make sure that the fuel mixture with air bubbles flows out of the degassing plug.
3. After the air bubbles stop flowing out, tighten the vent plug.
4. After tightening the vent plug, move the priming pump until it weighs enough or moves about 150 times.
5. Clean spilled fuel.
6. Start the engine. Make sure there is no fuel leak. If the engine fails to start, repeat the procedure from step 1.

- A vinyl hose was inserted at the top of the drain plug to prepare a container for receiving water.
- Loosen the drain plug.
- In order to discharge water, start the pump several times.
- After draining, tighten the plug.
- Run the pump several times to ensure that there is no fuel leak.
- Make sure the warning light in the chart goes out.



Legend

1. Starter pump
2. Bleeding plug

Water removal

If the precipitation separator exceeds the prescribed amount of water, the warning light will light up. In this case, water is removed according to the following procedures.

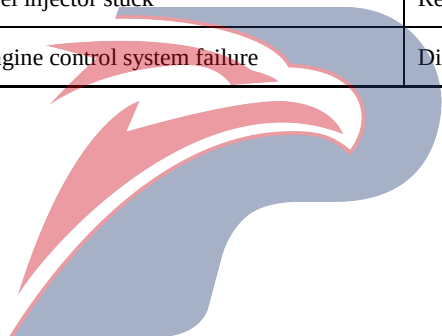
Failure symptoms list

- It is very difficult to start
- Idle oscillation
- Insufficient power output
- The maximum speed of the engine is too slow
- The engine can't stop
- The exhaust is blue or black



Phenomenon: Startup difficulties

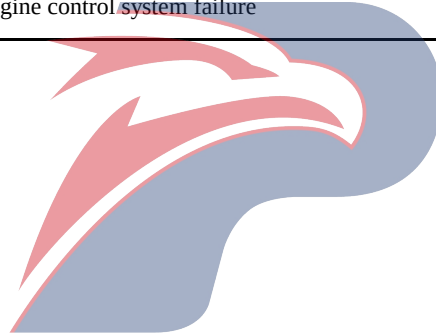
Phenomenon	Probable causes	Corrective measures
It is very difficult to start	Fuel tank is empty	Add fuel tank.
	Fuel tank exhaust plug	Clean tank/inflator.
	Whether there is air in the fuel system	Fresh air.
	The fuel line is blocked or damaged. Loose connection of fuel line	Repair or replace the fuel line. Re-adjust the connection.
	Fuel filter plugged	Replace the ink cartridges.
	Feed pump failed	Replace oil supply pump.
	Turn on the regulator	Replace oil supply pump.
	Fuel injector stuck	Replace the syringe.
	Engine control system failure	Diagnose the engine control system.



POWERSTAR

Phenomenon: Oscillation when idle

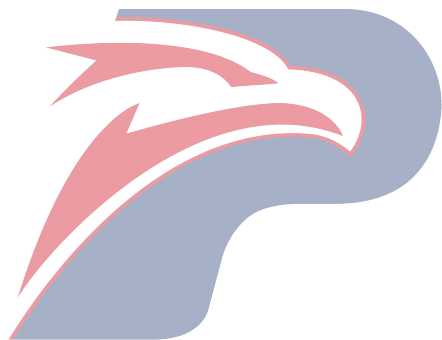
Phenomenon	Probable causes	Corrective measures
Idle oscillation	Whether there is air in the fuel system	Remove air from the fuel system.
	Blockage or leakage of the fuel system	Repair or replace the fuel system.
	Whether there is water in the fuel system	Replace fuel.
	Fuel filter plugged	Replace the ink cartridges.
	Fuel injector stuck	Replace the syringe.
	Syringe opening valve pressure is too low, poor injection conditions	Replace the syringe.
	The operation of the fuel pump is incomplete	Replace oil supply pump.
	Engine control system failure	Diagnose the engine control system.



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Phenomenon: Insufficient power output

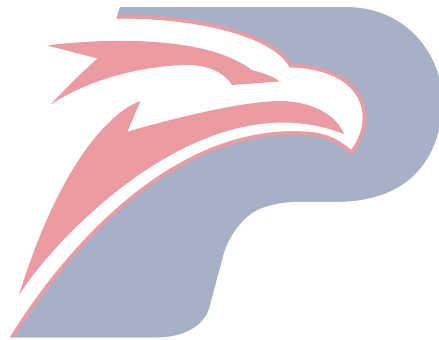
Phenomenon	Probable causes	Corrective measures
Insufficient power output	Water tank/ventilator plug	Clean tank/inflator.
	Whether there is air in the fuel system	Remove air from the fuel system.
	Blockage or leakage of the fuel system	Repair or replace the fuel system.
	Whether there is water in the fuel system	Replace fuel.
	Fuel filter plugged	Replace the ink cartridges.
	Fuel injector stuck	Replace the syringe.
	Engine control failed	Diagnose the engine control system.



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Phenomenon: The engine cannot stop

Phenomenon	Probable causes	Corrective measures
The engine can't stop	Engine control system failure	Diagnose the engine control system.



POWERSTAR

Phenomenon: The exhaust is blue or black

Phenomenon	Probable causes	Corrective measures
The exhaust is blue or black	Syringe opening valve pressure is too low, poor injection conditions	Replace the syringe.
	Engine control system failure	Diagnose the engine control system.

Specifications**Fuel pump specifications**

	Engine type	6HK1-TC
Item		
Fuel injection		EFI system (common rail type)
Oil feed pump		HP—4(Denso)
Pump's maximum emissions	m^3 / BP	490
Rotation direction		Left (from the drive side)
Diameter of diver	mm(inch)	8.5(0.335)×3
Opening pressure (regulating valve)	kPa (kg/cm ² /lb/in ²)/rpm	500 (5.1 / 73)
Feed pump	cm^3/rev	Secondary cycloid 4.11

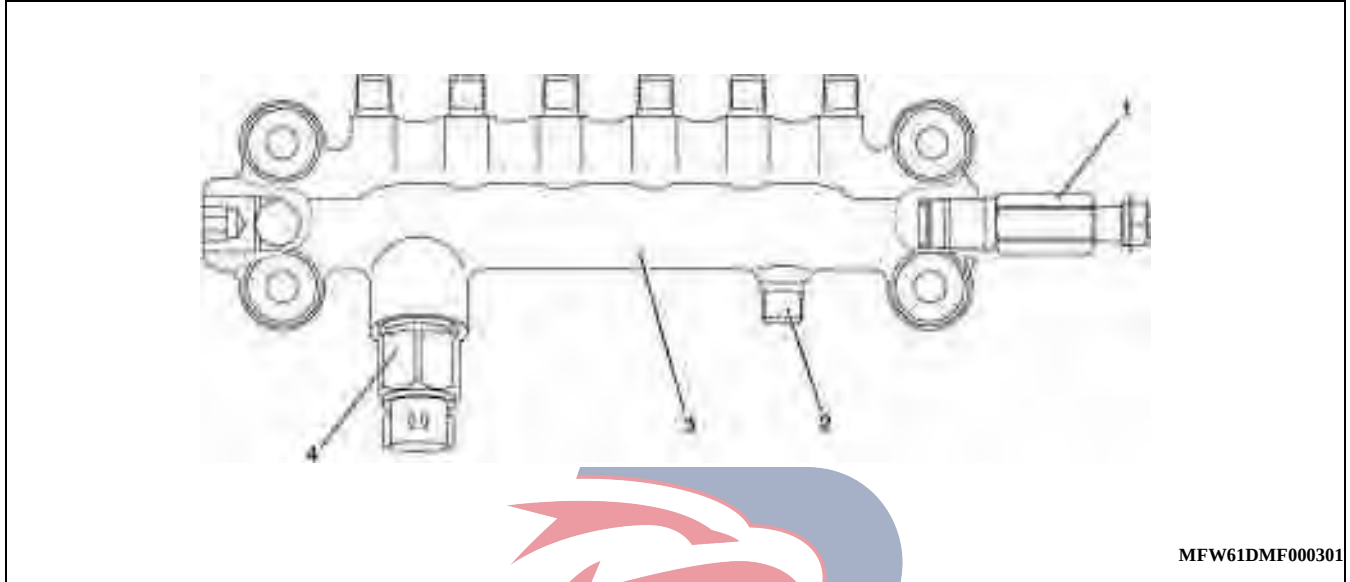
Injector specifications

Item	Specification
Injector	DLL—P
Nozzle size, mm (inches)	TCN,C 0.14(0.0055)—7/TCS 0.16(0.0063)—7

Fuel pressure sensor (common rail)

Part

Common Rail



Legend

- | | |
|---------------------|--------------------|
| 1. Pressure limiter | 3. Common Rail |
| 2. Fuel inlet | 4. Pressure Sensor |

Maintenance precautions

Maintenance precautions

- Pour cleaning fluid into the steam cleaner and carefully clean the sensor and the area around the common rail.
- Thoroughly remove moisture with compressed air.
- Before starting work, make sure that the impurities have been completely removed.
- After replacing any sensor, use the scan tool to clear the fault code and check that the sensor is operating properly.

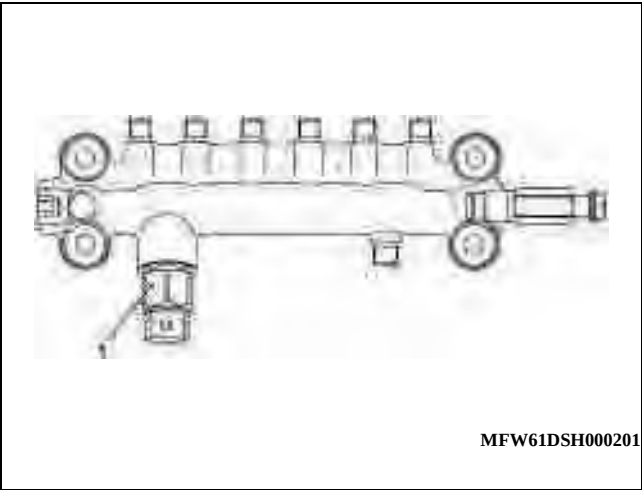
Note:

Do not use the removed sensor.

Tear down

Fuel pressure sensor

- Remove the fuel pressure sensor (1) with a socket wrench (27 mm / 1.063 inch).



Installation

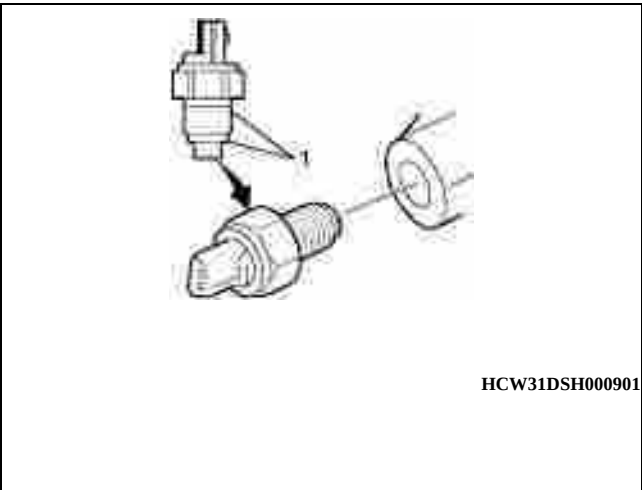
Fuel pressure sensor

- Grease the threads and base of sensor (1) and install the sensor.
- Install the connection terminals so that they do not interfere with each other.

Tightening torque	N m (kgf m/lb ft)
98 (10.0 / 72)	

Note:

Do not let impurities stick to the line. In addition, make sure there is no impurity in the installation area, then install the sensor.



Pressure restrictor (common rail)

Tear down

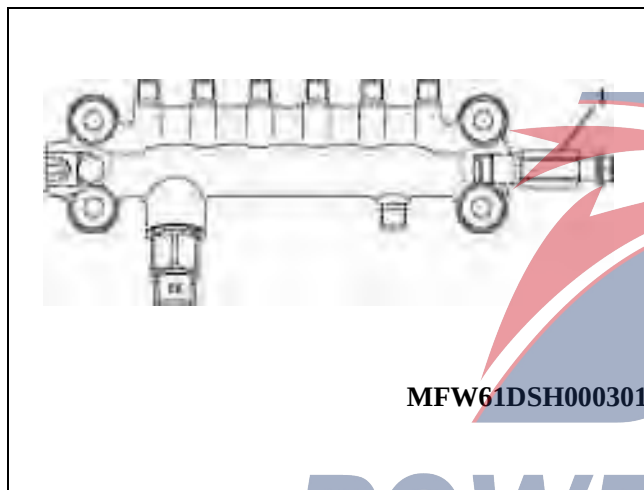
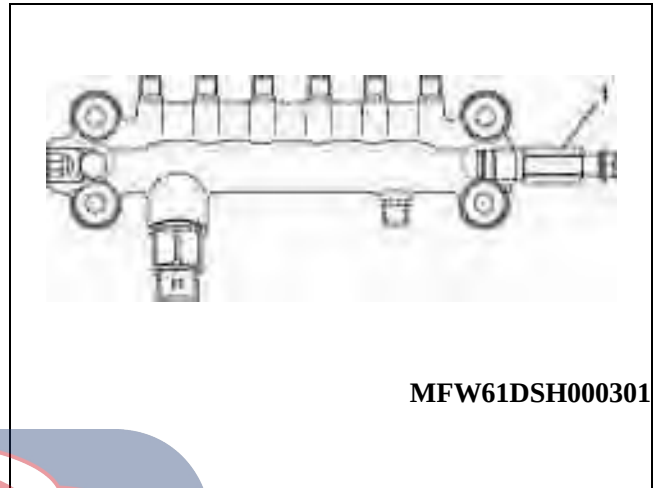
Pressure limiter

- Use a socket wrench (19mm/0.748 inch) to remove the lower limit.

Note:

- Do not use the removed sensor or gasket again.
- Do not damage the surface of the base when removing the gasket.

- Tighten the restrictor with a sleeve with a width of 19 mm (0.748 in.).



Legend

1. Voltage limiter (the width of opposite sides of sleeve 19 mm /0.748 inch).

Legend

1. Voltage limiter (the width of opposite sides of sleeve 19 mm /0.748 inch).

Installation

Pressure limiter

- Use a copper gasket to install the upper pressure limiter (1).

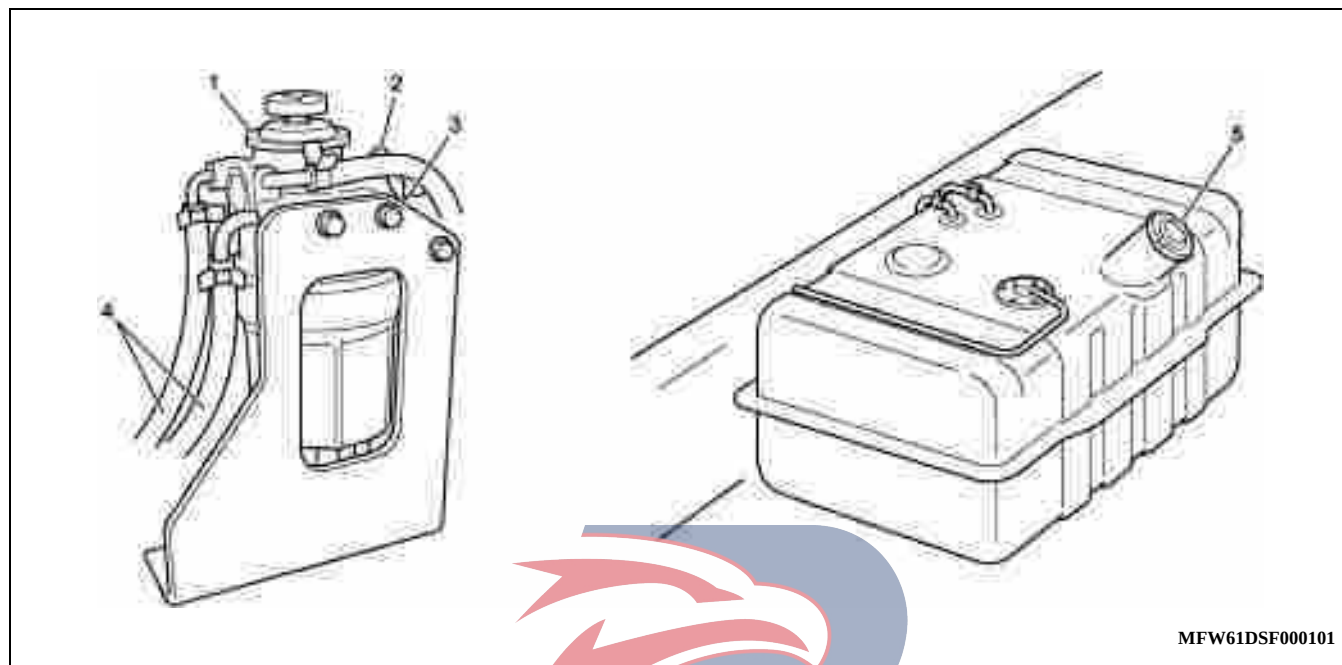
Tightening torque:	N m (kgf m/lb ft)
	172 (17.5 / 127)

Note:

- Make sure that the installation area is free from impurities.

Fuel filter assembly

Part



Legend

- | | |
|------------------------------|-------------------|
| 1. Fuel filter assembly | 4. Fuel hose |
| 2. Vents | 5. Oil filler cap |
| 3. Fuel filter coupling bolt | |

POWERSTAR

• Refer to the "Fuel System"

Tear down

1. Remove the filler cap.
2. Disconnect the sediment separator switch connector.
3. Pull out the tubing and return the hose from the fuel filter assembly and plug it with a cork to prevent fuel leakage.
4. Remove the fuel filter assembly.

Reassemble

1. Install the fuel filter assembly.
2. Install the oil hose and return hose.
3. Connect the settle separator switch connector lug again.
4. Install the filler cap.
5. Fresh air.

Fuel filter replacement procedure

Tear down

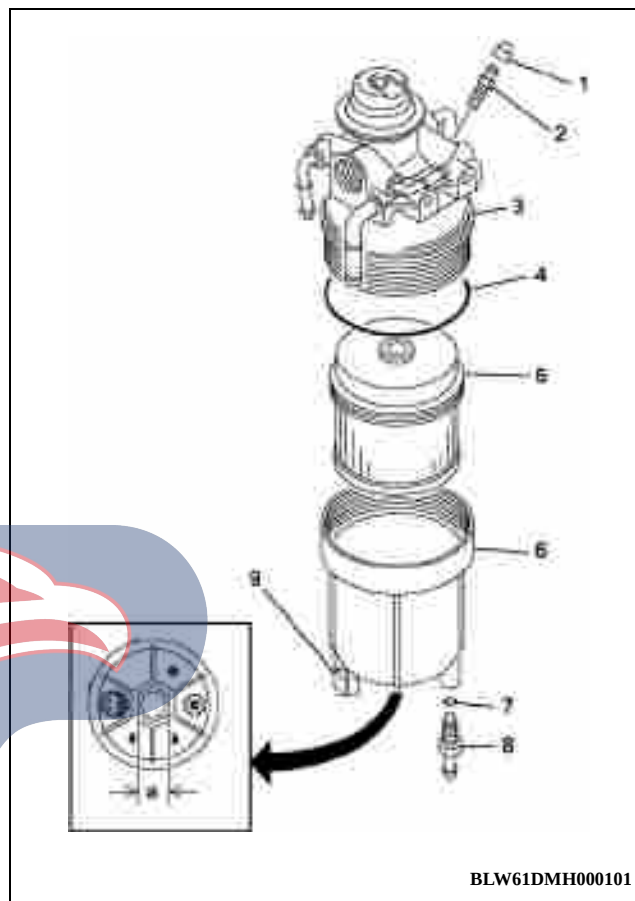
1. Remove the fuel from the filter housing.
 - a. Release the drain plug at the bottom of the filter cover.
 - b. When the ventilation device on the main body is released, the fuel leaks from the drain plug.
2. Remove the harness of the water level alarm switch.
3. Rotate the filter cover counter-clockwise with a sleeve (double sleeve width: 29 mm (a) filter wrench (5-8840-0253-0)) to remove it from the main body.

After removing the filter cap, make sure that the float on the inner bottom can move up and down smoothly.

If any impurities, sludge or other contaminants are found on the bottom of the filter element, the dirt should be completely removed.

4. Pull down on the filter element and remove the O-ring on the body.

If impurities adhere to the inner surface of the main body, clean it with a clean cloth.



POWERSTAR

Legend

1. Filter cover
2. Vents
3. Filter body
4. O-ring
5. Fuel filter element
6. Filter cover
7. O-ring
8. Release plug
9. Switch connector

Reassemble

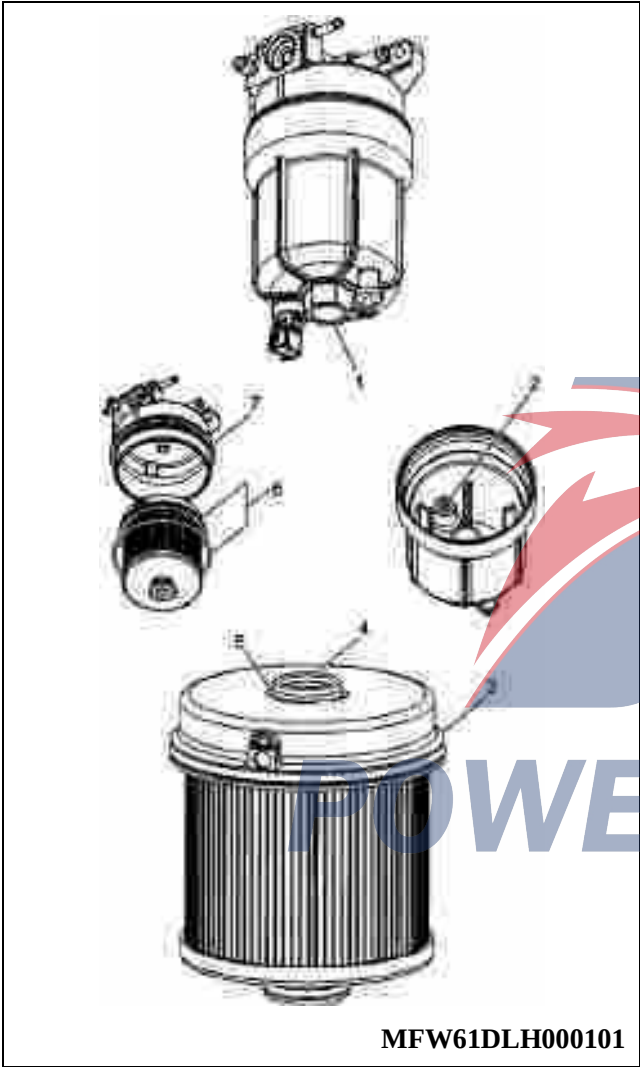
1. Put the O-ring combined with the filter element into the groove of the extracorporeal ring.

O-rings must be handled carefully to avoid damaging the threaded parts of the body.

2. Apply a thin layer of fresh diesel to the gasket inside and outside the filter element, and then insert the gasket until it contacts the gasket.

Note:

Be careful not to allow any foreign matter to enter the four holes (clean side) next to the inner gasket.



3. Apply a thin layer of fresh diesel to the inner ring of the filter cover (or O-ring part of the body side). Carefully rotate the filter element clockwise until it touches the human body. Jamming the O-ring must be avoided.

The filter element not contacting the filter element cover means the element is not completely inserted.

Rotate the filter element and push it in again.

4. Tighten the filter housing with a torque wrench.

Tightening torque	N m (kgf m /lb ft)
33 (3.4 / 25)	

5. Tighten the filter drain plug and reconnect the water level alarm switch connector.

Tightening torque	N m (kg/m/lb. inch)
0.4 (0.04 / 3.5)	

After starting the engine, check for loose parts.

Legend

- 1. Bush leaves
- 2. Floater
- 3. External washer
- 4. Inner gasket
- 5. Clean the side hole
- 6. Filter element
- 7. O-ring groove

Special tools

Graphic representation	Tool number/description
 5884002530	 5-8840-0253-0 Fuel Filter Wrench



Water sedimentation switch

Inspection

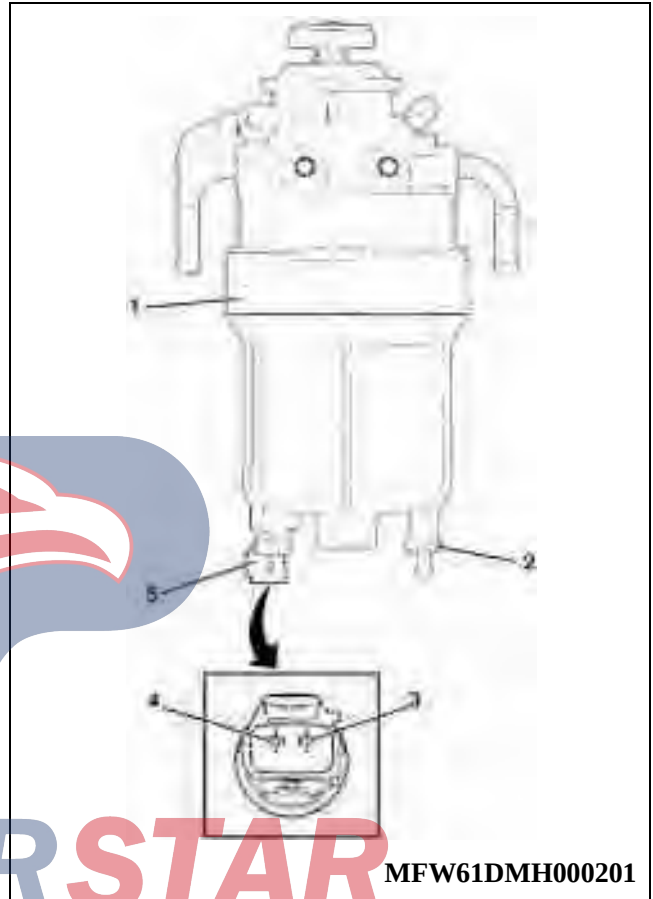
1. Check the water separator switch.

- Remove the fuel filter assembly from the bracket.
- Connect the separator switch terminal and open the key switch.
- Turn over the fuel filter assembly (move the float); if the warning light on the desktop lights up, it is in a normal state.

2. If the alarm light is not lit in step 1, please check the following items.

- Turn over the fuel filter assembly. If there is electrical conductivity between the terminals, it means the switch is in normal condition.
- Remove the sediment separation switch connector, connect the vehicle side connection terminal, and then open the start switch. At this point, if the precipitation separator warning light is on, the circuit is in a normal state.

If a fault is found after the inspection, replace the switch section and repair the incorrect connection and disconnection between circuits.

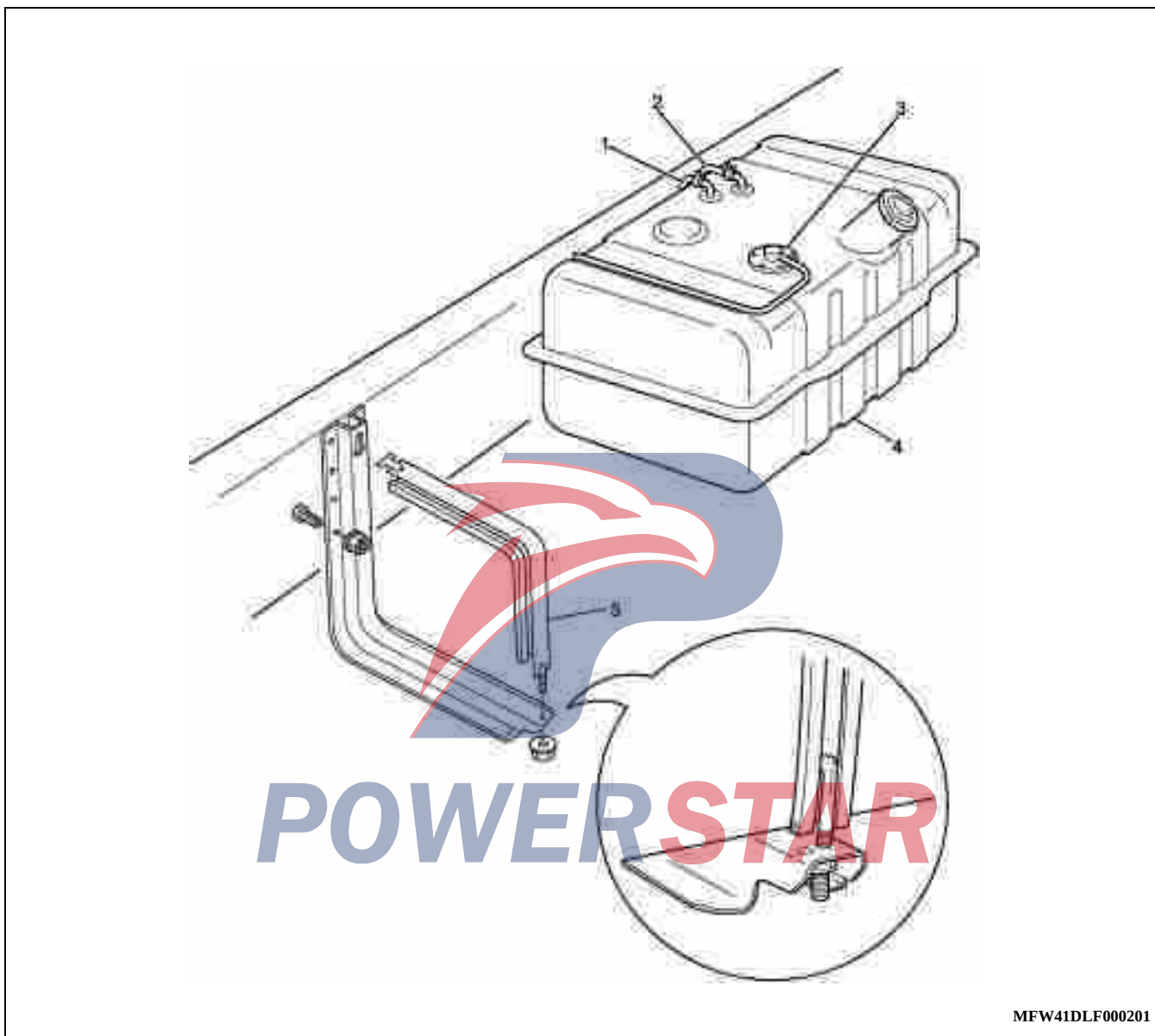


Legend

1. Fuel filter assembly
 2. Release plug
 3. Driving switch terminal
 4. Driving switch terminal
 5. Water separator switch assembly
-

Fuel tank

Part



MFW41DLF000201

Legend

- | | |
|-------------------------|-------------------------|
| 1. Fuel return hose | 4. Fuel tank |
| 2. Fuel supply hose | 5. The bundle of straps |
| 3. Fuel meter equipment | |

Tear down

1. Remove the battery ground cable.
2. Release the fuel cap.
3. Remove the fuel from the drain plug.
4. After removing the fuel, tighten the drain plug to the specified torque strength.

Tightening torque	N m (kgf m /lb ft)
20 (2.0 / 14)	

5. Remove the fuel gauge assembly connector.
6. Remove the fuel supply hose.

7. Remove the fuel return hose.

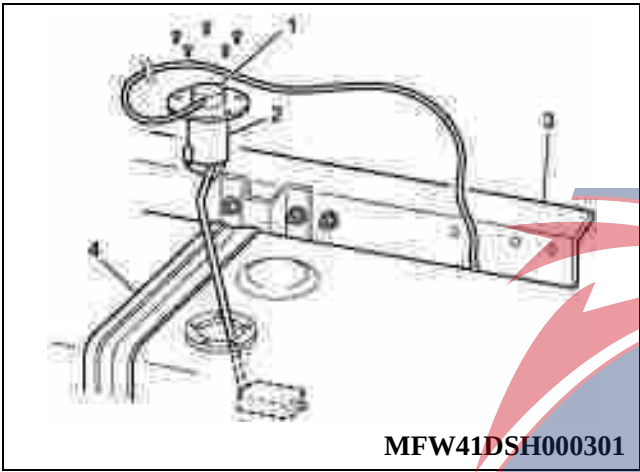
- Insert the hose so it does not leak. Fix it to the side of the rack and point it up.

8. Remove the tank strap and remove the rack side of the strap.

9. Pull out the tank.

Note:

If the fuel tank cannot be pulled out, remove the bracket and pull out the fuel tank from the bottom.



Legend

1. Fuel gauge device connector
2. Fuel gauge device
3. Side beam
4. The bundle of straps

Reassemble

1. Install the tank.

Note:

- When removing the tank, if the bracket has been removed, mount the bracket to the rack with the specified torque strength.
- When supporting the tank with the bracket, make sure that the rubber part fixed to the tank is firmly supported by the bracket.

Tightening torque	N m (kg/m/lb. inch)
-------------------	---------------------

55 (5.6 / 41)

2. Tie the strap to the tank and tighten the nut with the specified torque strength.

Tightening torque	N m (kg/m/lb. inch)
12 (1.2 / 104)	

3. Install fuel return hose.

4. Install fuel supply hose.

- Insert the hose into a 25 mm (0.9843 inch) or larger pipe.

5. Connect the fuel gauge connector.

6. Fill fuel to the fuel tank and tighten the filler cap.

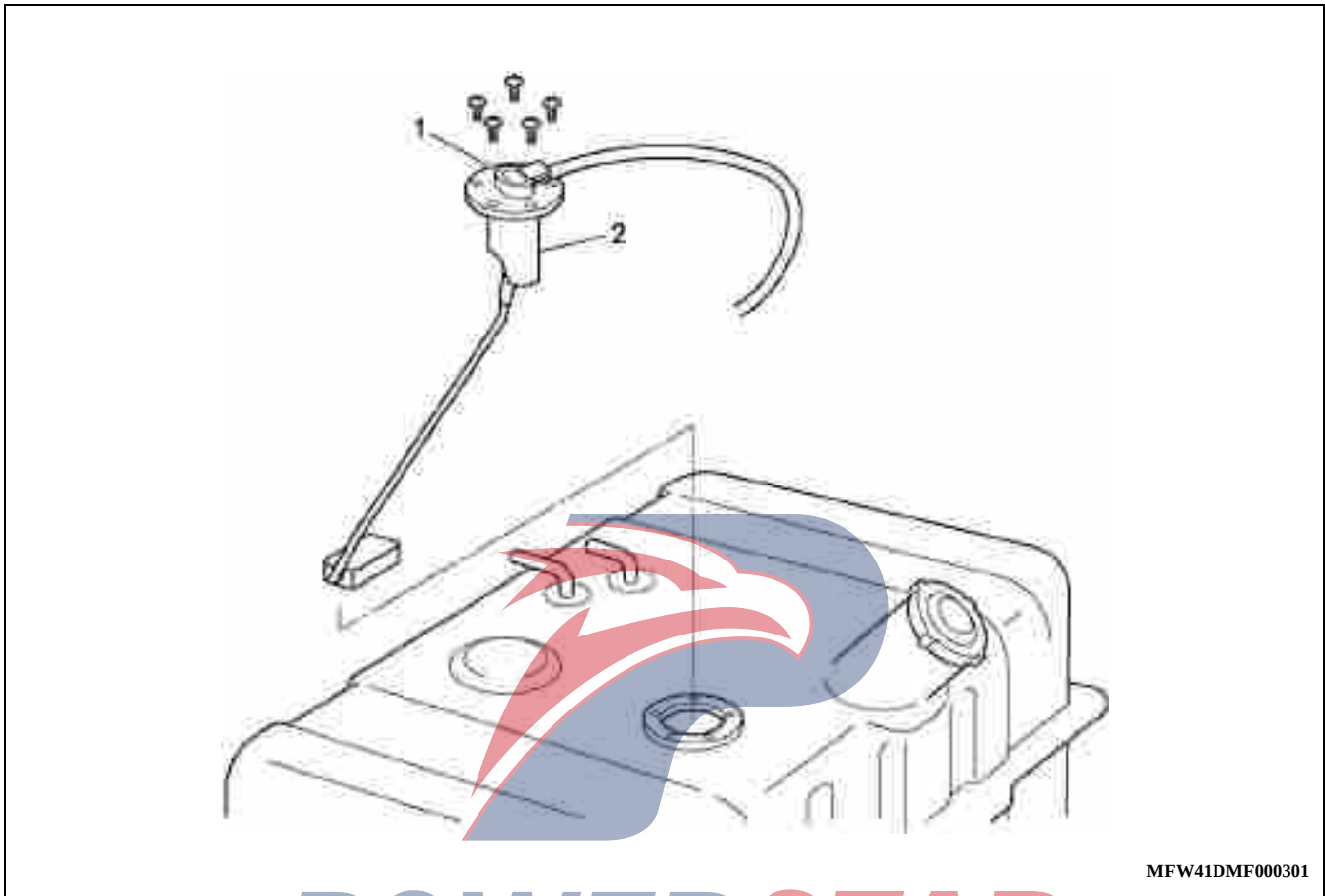
7. Connect the battery ground cable.

8. After starting the engine, check all hose connections to make sure there is no fuel leakage.

POWERSTAR

Fuel meter

Part



Legend

1. Fuel gauge device connector

2. Fuel meter equipment

Tear down

1. Remove the terminal from the fuel gauge device.
2. Remove the connecting screw and remove the power meter unit.

Note:

- When disassembling and replacing the fuel gauge unit, make sure that there is no interference from the surroundings and do not deform the arm.
- Remove the fuel gauge assembly and cover it with objects to prevent debris from entering the tank.

Inspection

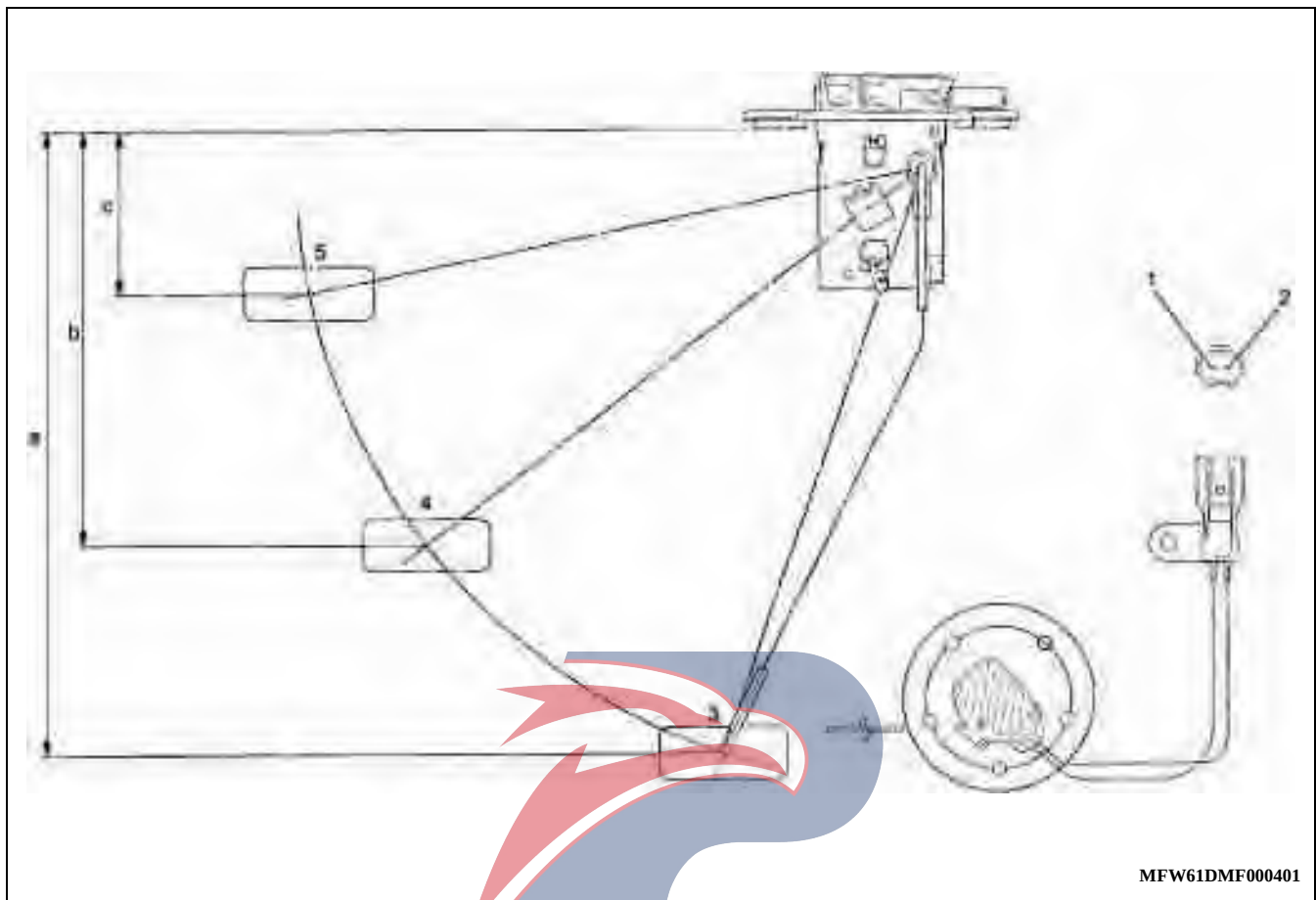
The internal resistance varies with the position of the float (fuel level) of the fuel tank unit, which in turn

moves the fuel gauge pointer.

1. When moving the float from "E" (3) to "F" (5), check the resistance between the connection terminals (1)–(2)
2. When the float is in the "E" (3) position, make sure that the residual alarm indicator lights up.

If an abnormality occurs, replace the defective part.

Float position	E(3)	1/2(4)	F(5)
Standard volume resistance (Ω)	150	50	0
Size from the top of the tank, mm (inches)	a 296(11.65)	B 167.8(6.61)	c 30.4(1.2)



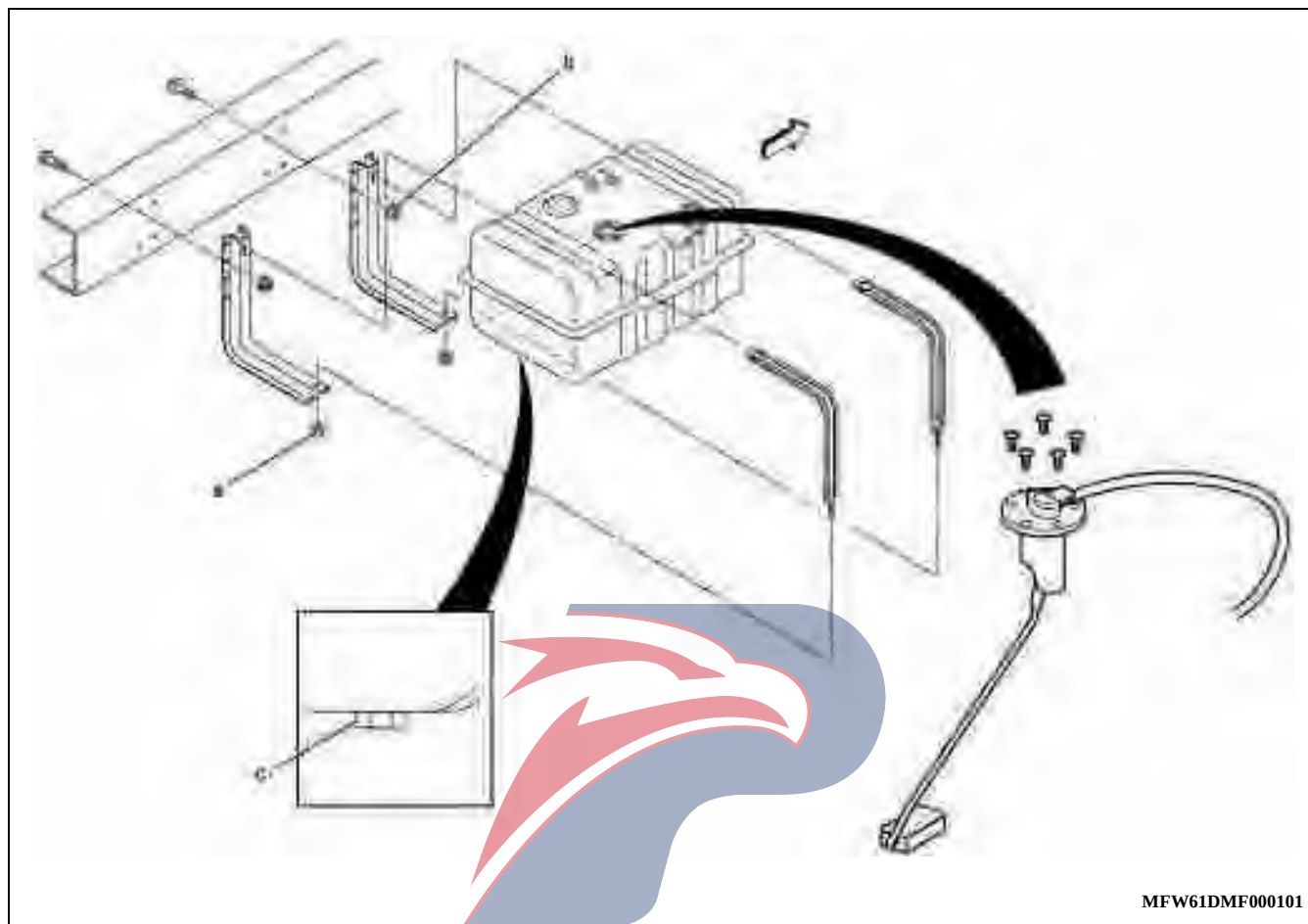
Reassemble

Reassemble in the reverse order of the removal procedure.

1. Install a fuel gauge device.
2. Connect the connector to the fuel gauge unit.

POWERSTAR

Tightening torque table



Legend

(a) 12N m (1.2 kgf m /104lb.ft)

(C) 20 N m (2.0 kgf m / 14 lb ft)

(b) 55N m (5.6 kgf m/14 lb.ft)

POWERSTAR

Engine

Engine electrical part

(6HK1)

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Phenomenon: Overcharging	6D-28		

Start system

Precautions for maintenance

When performing electrical system maintenance, make sure that the start switch, headlight switch, etc. are in the open position. Before starting maintenance work, be sure to disconnect the negative battery terminal.

Description of function and operation

Start circuit

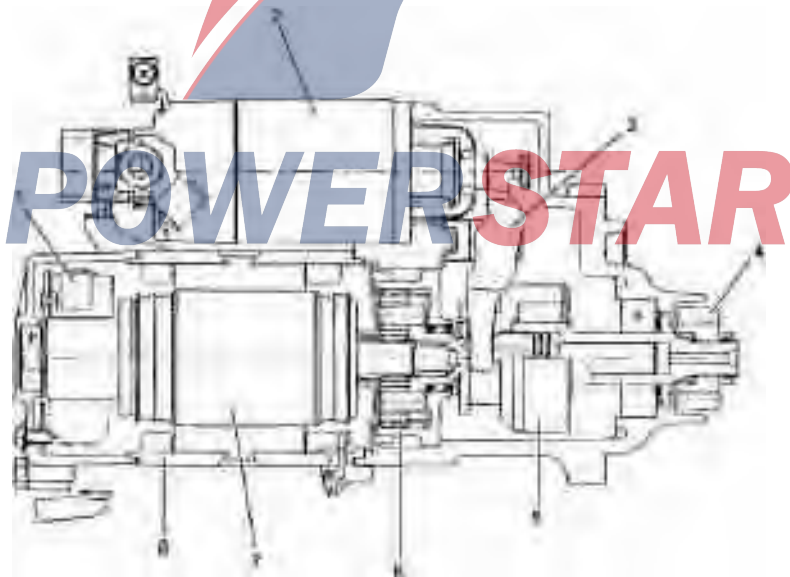
The start-up system consists of battery, starter, start switch, suppression switch (A / T), start-up relay, etc. The startup circuit diagram shows the wiring of these major components.

Initiator

The starter moves magnetically and has a boundary meshing deceleration starter. When the start switch is connected, the magnetic switch assembly contacts will close and the armature will start rotating. At the same time, the plunger is retracted by the rod assembly, the pinion gear is pushed out of the front and meshes with the ring gear, and the ring gear starts to rotate, thereby rotating the crankshaft and starting the engine. After the engine starts, if the start switch is turned on, the plunger will return, the pinion and ring gear will be disengaged, and the armature will stop rotating. If the engine speed is higher than the pinion, the pinion will rotate in reverse. However, because the pinion is idle under the action of the clutch, the armature is not driven.

24V 5.0KW

Start motor

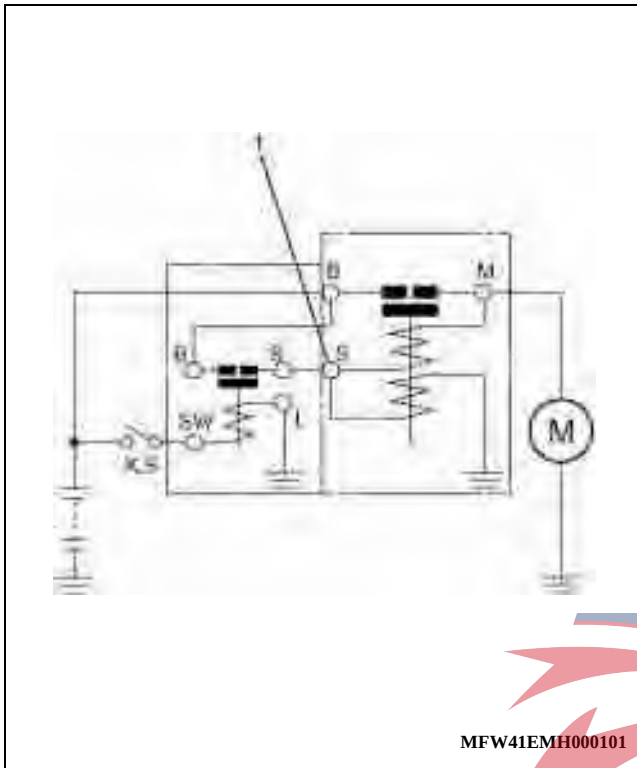


MFW51EMF000401

Legend

- | | |
|-----------------------------|---------------|
| 1. Electric brush holder | 5. Clutch |
| 2. Magnetic switch assembly | 6. Inner gear |
| 3. Rod assembly | 7. Armature |
| 4. Pinion | 8. Yoke |

Schematic diagram



Legend

1. Instantaneous current: 308.4A (24V 20°C / 68°F);
holding current: 15.7A (24V / 20°C / 68°F)

POWERSTAR

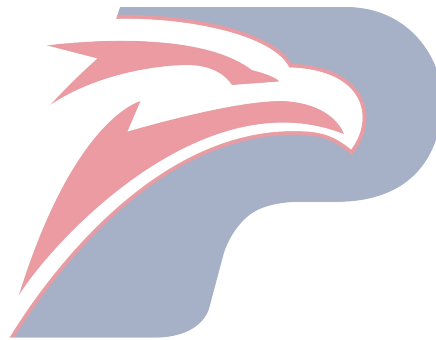
Failure symptoms list

- The starter does not run even if the start switch is turned on
- The pinion moves forward but cannot mesh with the ring gear
- The pinion rotates to engage the motor but does not rotate the engine
- Pinion can mesh but cannot rotate engine
- After the engine is started, the starter can't stop even if the starting switch is off



Phenomenon: The starter cannot operate even if the start switch is turned on

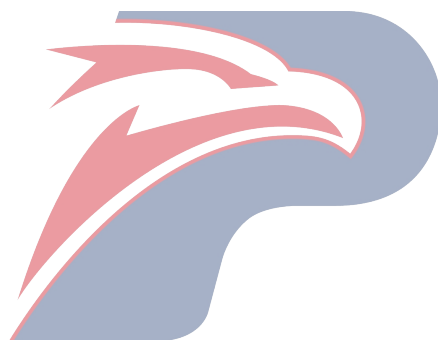
Phenomenon	Probable causes	Corrective measures
The starter does not run even if the start switch is turned on	The battery is fully discharged	Replacement battery.
	Line interruption or bad connection	Repair.
	Poor contact of start switch contact, or terminal not fully tightened	Repair or replace.
	Poor contact of start relay contact, or poor contact of inserted terminal	Repair or replace.
	Magnetic switch coil short-circuited or damaged	Replace.



POWERSTAR

Phenomenon: The pinion moves forward but cannot mesh with the ring gear

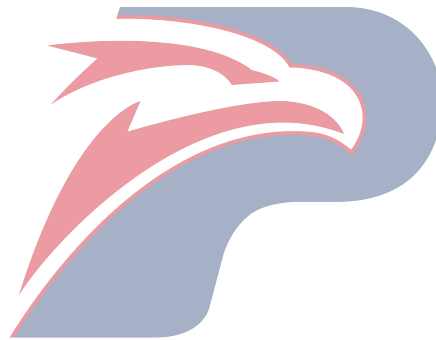
Phenomenon	Probable causes	Corrective measures
The pinion moves forward but cannot mesh with the ring gear	Low battery	Charge the battery.
	Pinion or ring gear tooth tip abrasion	Replace pinion or ring gear.
	The pinion shaft is dirty	Repair.



POWERSTAR

Phenomenon: The pinion rotates to mesh with the motor but does not rotate the engine

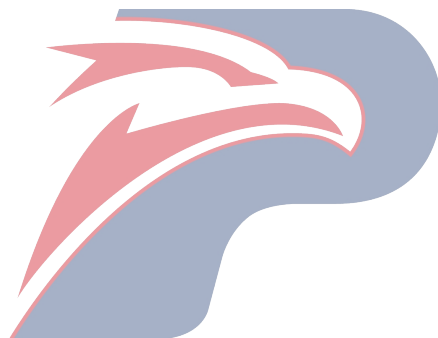
Phenomenon	Probable causes	Corrective measures
The pinion rotates to engage the motor but does not rotate the engine	Clutch slip	Replace the clutch or starter.
	Reduce gear damage	Replace the gear or starter.



POWERSTAR

Phenomenon: The pinion can mesh but cannot rotate the engine

Phenomenon	Probable causes	Corrective measures
Pinion can mesh but cannot rotate engine	Low battery	Charge the battery.
	Terminal connection in the circuit is not good	Repair.
	Armature or field coil short circuit	Replace the initiator.
	Dirt on the commutate	Repair or replace the starter.
	Brush spring weakened or brush worn	Replace brush springs or brushes.
	Bearing lock	Replace bearing or starter.
	Poor contact of magnetic switch contacts	Replace the magnetic switch.



POWERSTAR

Symptom: After starting the engine, the starter cannot stop even if the start switch is turned off.

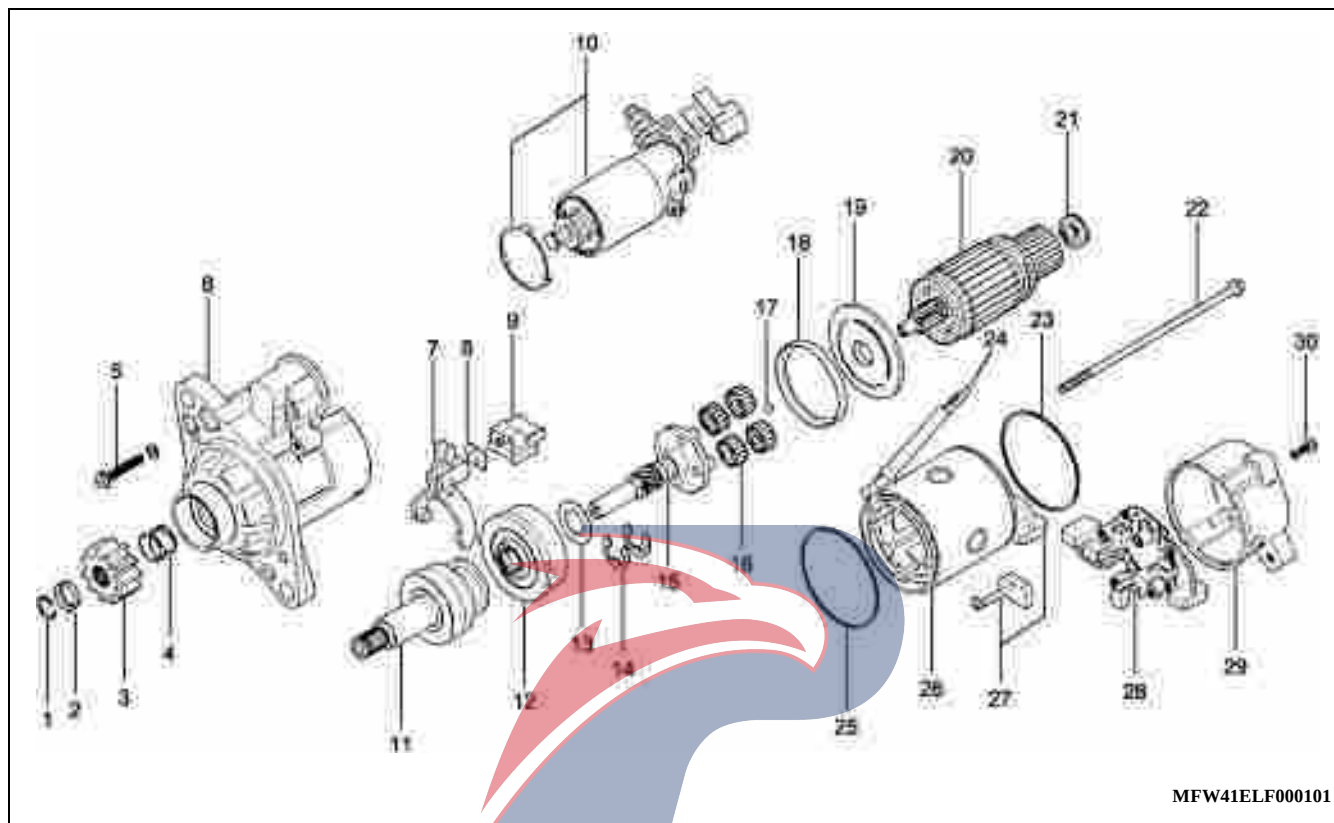
Phenomenon	Probable causes	Corrective measures
After the engine is started, the starter can't stop even if the starting switch is off.	Start switch contact return bad	Replace the start switch.
	Poor contact of starting relay	Replace start relay.
	Magnetic switch coil short circuit	Replace the magnetic switch.

Specifications

Model (manufacturer)	Mitsubishi
Rated value	
Voltage (V)	24
Output (kW)	5.0
Time (s)	30
Pinion tooth	11
Rotation direction (from the pinion side)	Clockwise
Weight (approx) (Kg /lb)	7.2(16.0)
No-load characteristic	
Voltage and current (V/A)	Maximum 23.5/85
Speed (rotate every minute)	Minimum 3300
Load characteristic	
Voltage and current (V/A)	18.5/400
Torque N m (kgf m/lb ft)	Minimum 26(2.6/19)
Speed (rotate every minute)	Minimum 1250
Locking characteristics	
Voltage and current (V/A)	Maximum 9/1400
Torque N m (kgf m/lb ft)	Minimum 88(9.0/65)

Start motor

Part



Legend

- | | |
|-----------------------------|--------------------------|
| 1. Buckle | 16. Planetary gears |
| 2. Stop ring | 17. Ball |
| 3. Pinion | 18. Gasket |
| 4. pinion spring | 19. Plate |
| 5. Screw | 20. Armature |
| 6. Front bracket | 21. Bearing |
| 7. Rod | 22. Through bolt |
| 8. Plate piece | 23. O-ring |
| 9. Gasket | 24. Wires with terminals |
| 10. Magnetic switch | 25. O-ring |
| 11. Over speed clutch | 26. Yoke assembly |
| 12. Inner gear | 27. Brush holder |
| 13. Gasket | 28. Brush holder |
| 14. E-shaped retaining ring | 29. Rear bracket |
| 15. Gear shaft | 30. Screw |
-

Disassembly

Note:

- Before dismantling the starter, the connecting part between the magnetic switch, the yoke, the front bracket and the rear bracket is identified so that they can be reassembled later.

- Remove the pinion first. If disassembly is performed without first disassembling the pinion, the pinion will not be removed, and subsequent disassembly operations cannot continue. However, the motor section and the magnetic switch can be removed even without removing the pinion.

1. Remove the buckle.
2. Remove the stop ring.
3. Remove the pinion.

- Normally, if the pinion or snap ring is mounted on the front bracket side, the front bracket will be damaged. Therefore, the stop ring should be tapped when the pinion leaves the starting position.

- Connect the starter to the battery as shown. With SW1 and SW2 closed, if the pinion moves forward to the starting position, the armature will rotate.

After that, switch SW2 is disconnected and the armature stops.

Note:

If you turn on the power, heat will be generated, so remove the pinion within 10 seconds.



HCW31ESH000201

Legend

1. Buckle
2. stop ring
3. Pinion

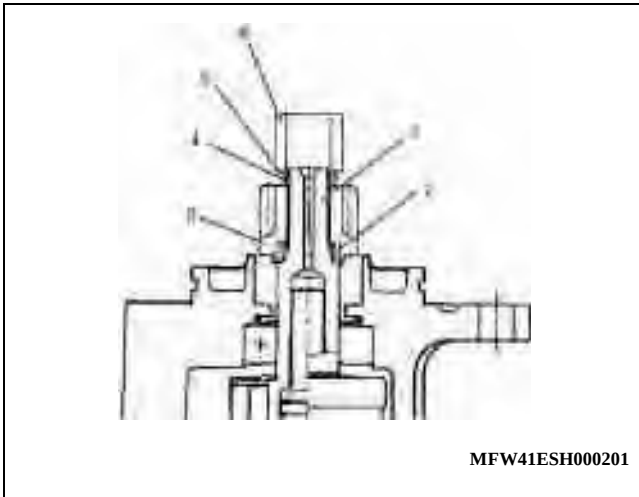


HCW51ESH001701

- Use a tube of exactly the same size as the stop ring against the stop ring, strike the tube with a hammer and move the stop ring to the pinion side. Then remove the retaining ring and remove the stop ring, pinion and pinion springs.

Note:

Do not use the buckle again.



Legend

1. Clutch shaft
2. pinion spring
3. Pinion
4. Buckle
5. Stop the ring
6. Pipeline

4. Remove the pinion spring.
5. Remove the conductor with the terminal.
6. Remove the screw.

7. Remove the magnetic switch.

- Remove the M-side nut of the magnetic switch and remove the wire with the terminal.

Then remove the magnetic switch fastening screw and remove the magnetic switch.

Note:

Tighten the screw tightly, so loose it with a vibrating wrench.

8. Remove the screw.

9. Remove the through bolts.

- Remove brush holder fixing screws and bolts.

Then remove the rear bracket. Remove the O-rings at the same time.

Note:

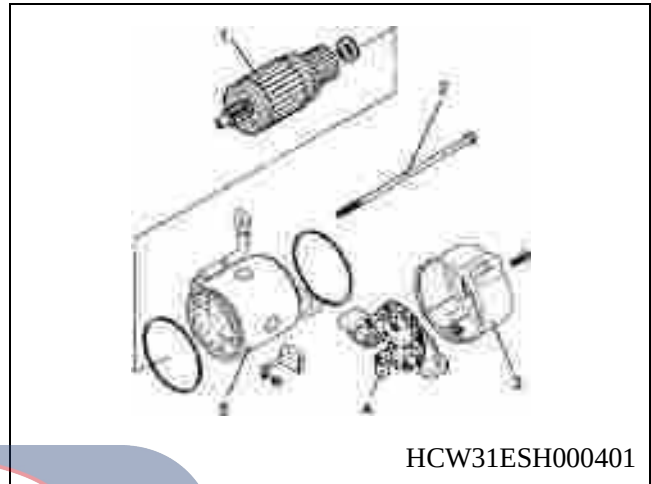
Do not use the O-ring again.

10. Remove the rear bracket.

11. Remove the yoke assembly.

12. Remove the O-rings .

- Use a sleeve (diameter 38 mm / 1.4961 inch) for the armature commutated. The brush is then slid toward the sleeve while the sleeve is used to support the brush and remove the yoke assembly. Remove the O-rings at the same time.



Legend

1. Armature
2. Through bolt
3. Rear bracket
4. Brush holder
5. Yoke assembly

13. Remove the brush holder.

- Pull up the brush spring and remove the brush from the brush holder. Then remove the brush holder.

14. Remove the armature.

15. Remove the ball.

- Remove the armature.

The balls should also come out at the same time, but sometimes they stick to grease and cannot come out. Pullers are usually used to remove the bearings.

16. Remove the plate.

17. Remove the gasket.

18. Remove the gasket.

19. Remove the plate.

- Remove the plate of the planetary gear section and remove the washer.

- Then remove the spacers and plates from the strut support section.

20. Remove the planet gears.

- Remove the four parts of the planetary gear.
- As a whole, the internal gear, gear shaft, overrunning clutch and lever are pulled out from the front bracket.

21. Remove the pole.

Note:

Please remember the direction of pole.

22. Remove the E-shaped retaining ring.

- Remove the two E-rings. Rotate the overdrive clutch spine gear by one pitch and then remove it from the gear shaft assembly.

Note:

Once the E-shaped baffle ring is removed, do not use it again.

23. Remove the internal gear.

24. Remove the washer.

- Remove the internal gear. At this time, remove the gasket that adjusts the clearance of the gear shaft along the axial direction.

25. Remove the gear shaft.

26. Remove the front bracket.

27. Remove the bearing.

28. Remove the brush.

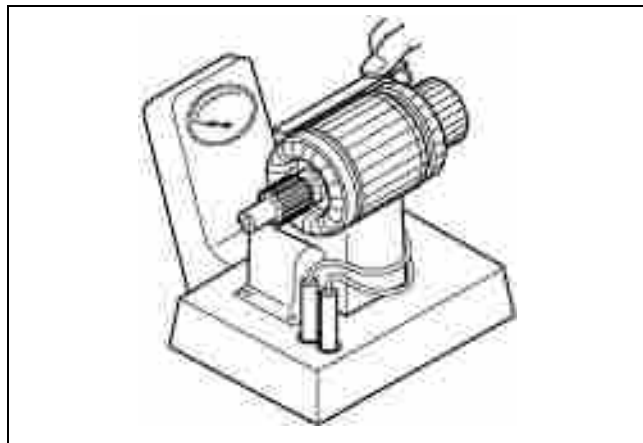
Inspection

1. Check the armature.

Short-circuit coil test

- Check the armature coil for inside short circuit through the short circuited coil test.

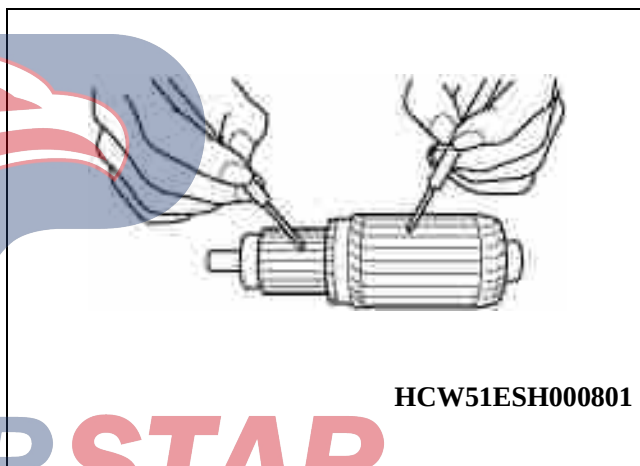
If an internal short circuit is found in the armature, it should be replaced.



HCW51ESH000701

Grounding examination

- Use a circuit tester to check the armature coil ground (insufficient insulation). If the armature is grounded, it should be replaced.

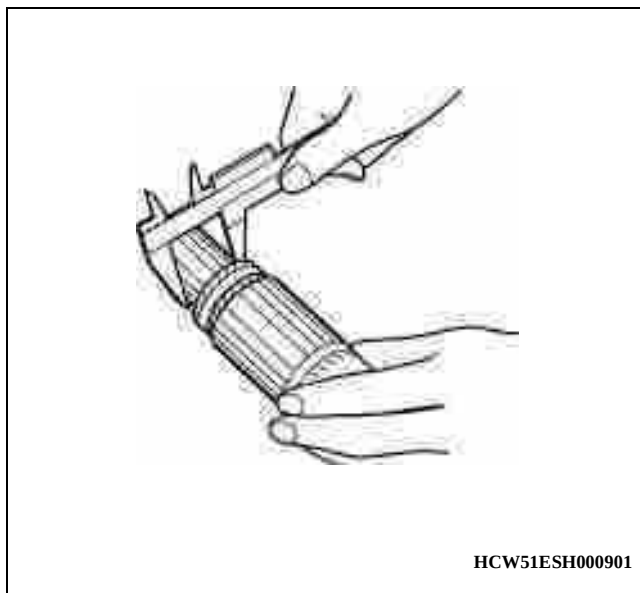


HCW51ESH000801

Commutator check

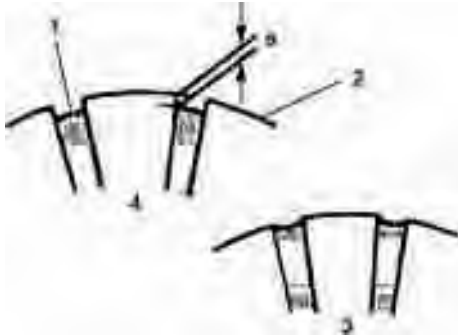
- Check the commutated outer ring for wear.

If the armature is damaged beyond its limits, it should be replaced.



HCW51ESH000901

- Measure the thickness of the commutated insulation die (called "undercut"). Remove brush debris and repair the malfunction.



HEW61ESH001201

Legend

1. Insulating mold
2. Commentator board
3. Poor
4. Fine

Gear check

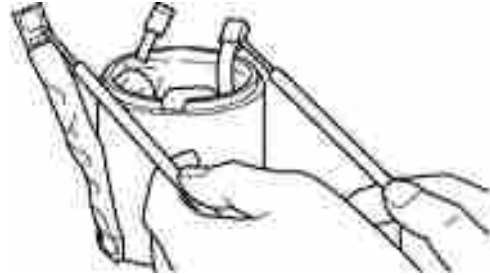
- Check whether the gear parts are damaged and worn. If the armature fails, it should be replaced.

Bearing inspection

- Check the bearing for unusual noise, looseness, and grease leakage. If the bearing fails, it should be replaced.
2. Check the excitation coil.

Conductivity check

- Use a circuit tester to check for broken coils. If disconnected, replace the yoke assembly.

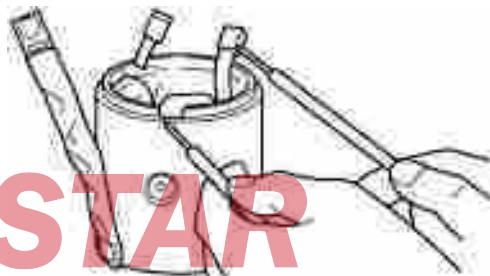


HCW51ESH001101

Grounding examination

- Check the coil ground (bad insulation) with a circuit tester.

If it is found to be grounded, repair or replace the yoke assembly.



HCW51ESH001201

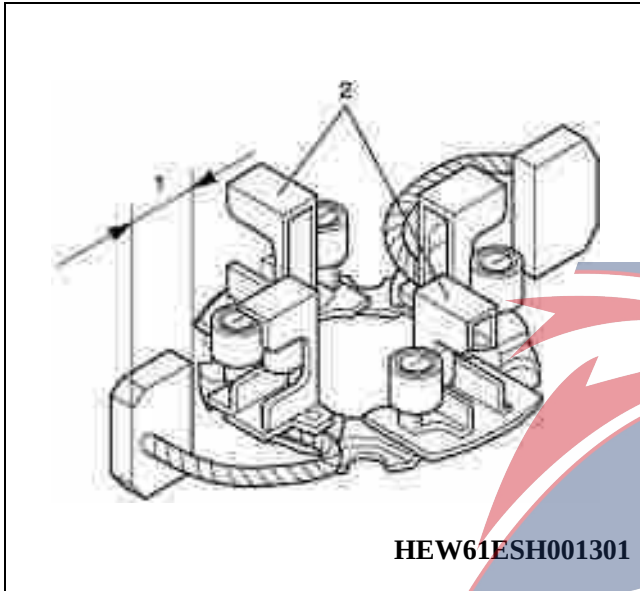
3. Check the brush.

- Check the wear length of the brush. If the wear of the brush is approaching its limit, replace the brushes on the brush holder side of the brush holder assembly. Just replace the brush on the side of the yoke assembly.
- Make sure the brush can move smoothly in the brush holder.

Also check the brush spring pressure.

- Use a circuit tester to check the grounding of the brush holder (insufficient insulation). However, the brush holder should be cleaned before checking.

Use one side of the circuit tester probe against the brush plate (metal part) and place the other probe on the insulating brush block (2). If there is no conductivity, it means poor insulation. The brush holder assembly should be replaced.



Legend

1. Length of the brush
2. Insulation brush holder

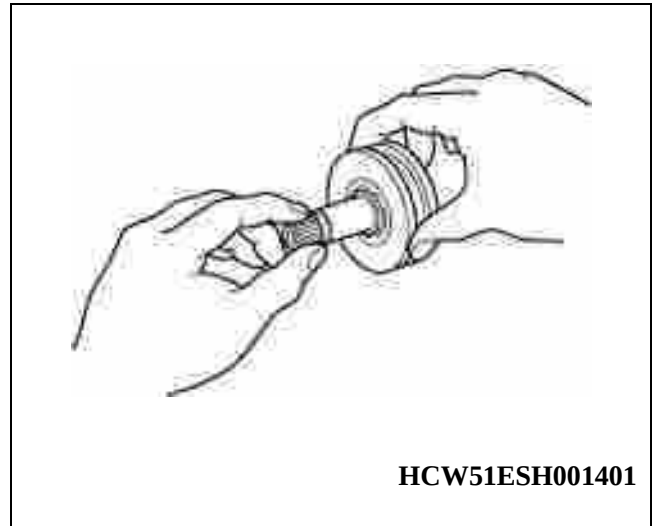
4. Over speed clutch and pinion. Check the speed clutch.

- Fix the cover of the overrunning clutch by hand; when turning the pinion by hand, it should be able to rotate slowly in one direction and have some resistance, but it cannot rotate in the opposite direction at all. If this is not the case, replace the overrunning clutch.

Check the pinion. If it is damaged or worn, it should be replaced.

Note:

The speed clutch is full of grease. Therefore, the grease may rush out if cleaned with a cleaning oil or a non-grease cleaner. Therefore, just wipe the dirt off the over speed clutch with a cloth.



5. Check the front bracket.

- Check the bearings and seals; if there is a problem, replace the front bracket assembly (including bearings and oil seals).

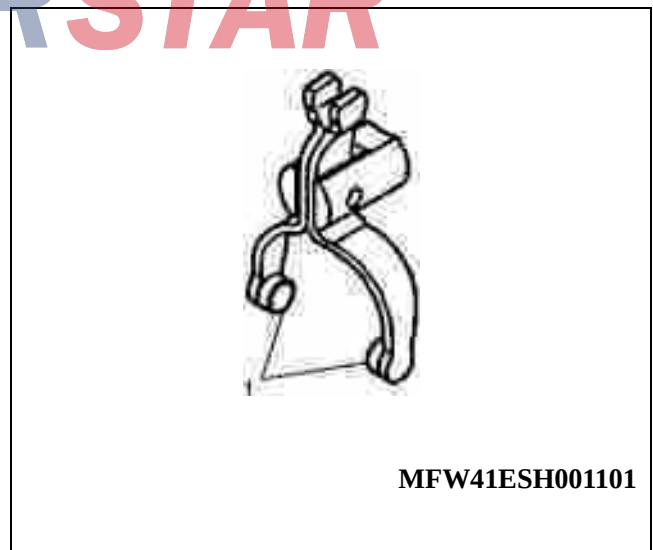
6. Check the internal gear and planetary gears.

- Check gear teeth; if abnormal wear or damage is found, replace internal gear or planet gear.

7. Check the rod.

- If frictional wear occurs with the overrunning clutch, the protruding position of the pinion is incorrect.

To adjust the protruding position of the pinion, see the assembly section.

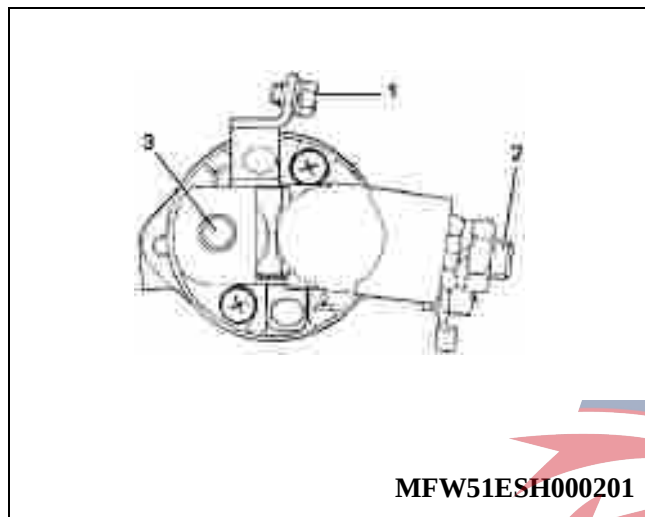


Legend

1. Wear part

8. Check the magnetic switch.

- Check the electrical conductivity between the M terminal and the main body. If there is no conductivity, replace the magnetic switch. Also make sure there is no conductivity between the B and M terminals. If there is electrical conductivity, replace the magnetic switch.



Legend

1. Terminal S
2. Terminal B
2. Terminal M

Assembling

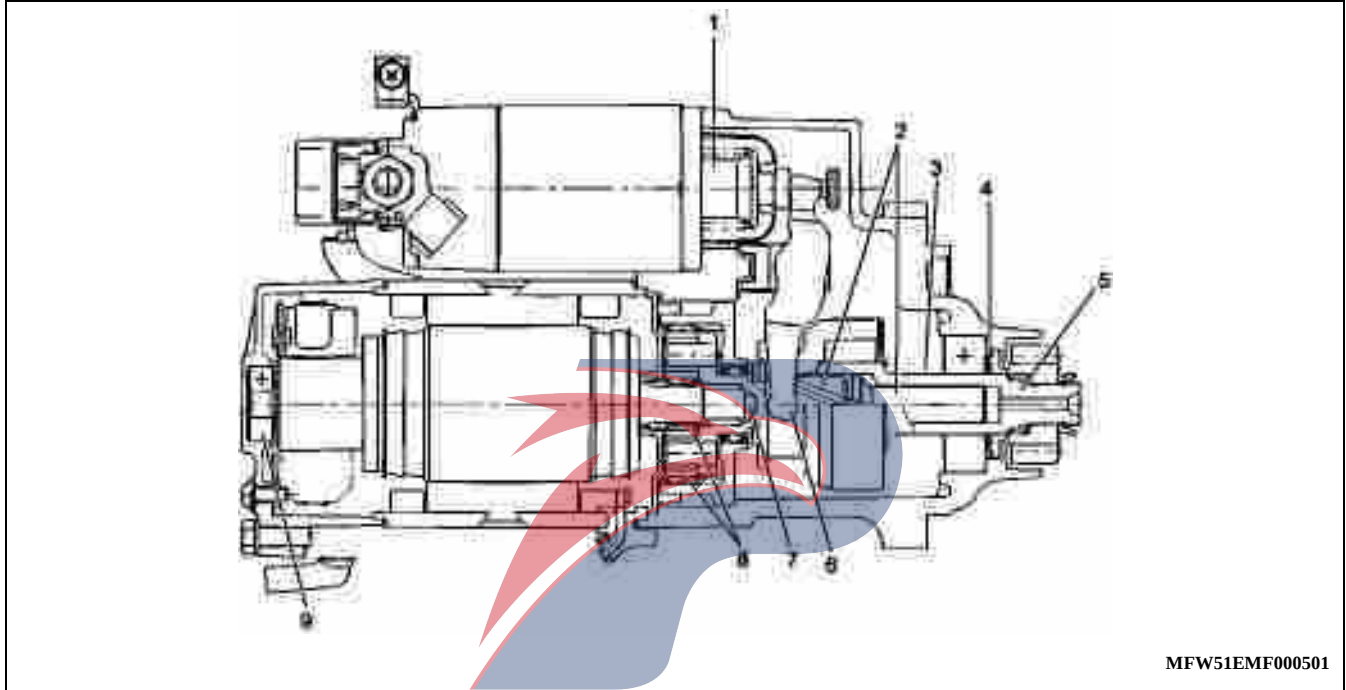
Assembly can be done in the reverse order according to the disassembly procedure. However, the following considerations must be observed.

Note:

Replace the following parts with new ones:

- Buckle (for securing snap rings)
- E-shaped retaining ring
- O-rings (gaskets on each side of the washer and each bolt connection)

1. The following parts require lubrication. We recommend using MULTEMP 0A-171 grease (produced by Nippon Synthetic Oils Co., Ltd.) in Parts 1, 2, 3, 4, 5 and 9 of the figure. Use grind able RAG650 grease in parts 6, 7 and 8 of the figure. (Made by Dow Corning Corporation)



MFW51EMF000501

Legend

- | | |
|---|---|
| 1. Plunger surface (small amount) | 6. Pole and rotating part of speeding clutch |
| 2. Gear shaft spines and rotating parts | 7. Sleeve bearing and ball |
| 3. Clutch shaft rotation section | 8. Armature shaft gears, planetary gears, internal gears |
| 4. Oil seal | 9. Armature bearing outer peripheral surface (a small amount) |
| 5. The bottom of the pinion | |

2. Tightening torque

Torque strength is based on the torque strengths listed in the attached table.

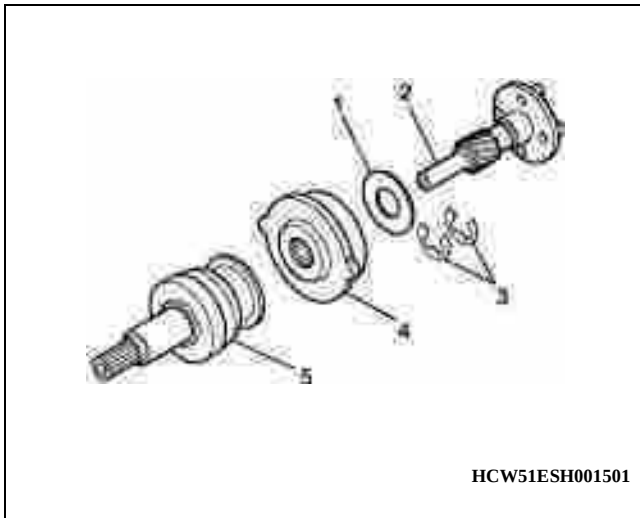
3. Axial clearance of armature axis

- There is no need to adjust. Insert the ball into the end of the shaft.

4. Axial clearance of gear shaft

- Insert a washer between the internal gear and the gear shaft to automatically maintain the specified clearance.

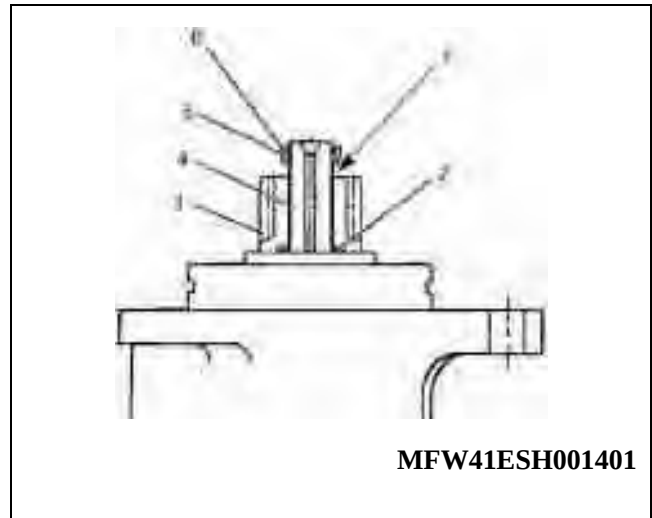
- Lead the gear shaft through the washer, inner gear and overdrive clutch. Then move the overrunning clutch by a splined pitch, make sure that the overrunning clutch does not slip, then install the E-ring.



Legend

1. Gasket
 2. Gear shaft
 3. E-shaped retaining ring
 4. Internal gear
 5. Over speed clutch
-
5. Pinion connection

- Before installing the magnetic switch on the front bracket, first install the pinion. Then, the pinion spring, the pinion gear, and the stop ring are mounted one after another on the clutch shaft. Finally, the buckle is securely fitted into the clutch shaft slot. Then use a gear puller to smoothly pull the stop ring to the side of the stop ring and fix both sides.



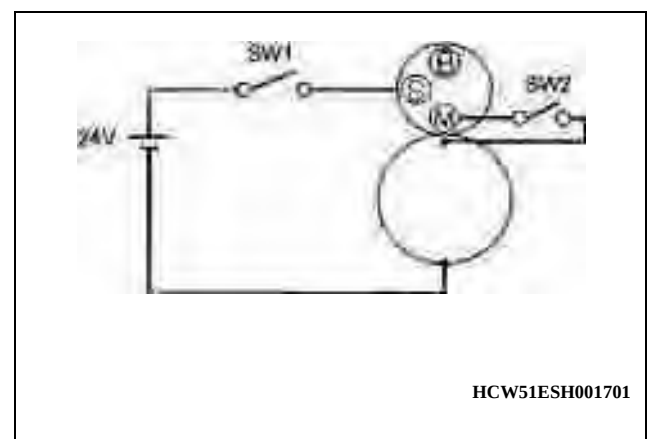
Legend

1. Push the pinion down and hook the tip of the puller
 2. pinion spring
 3. Pinion
 4. Clutch shaft
 5. Buckle
 6. stop ring
-

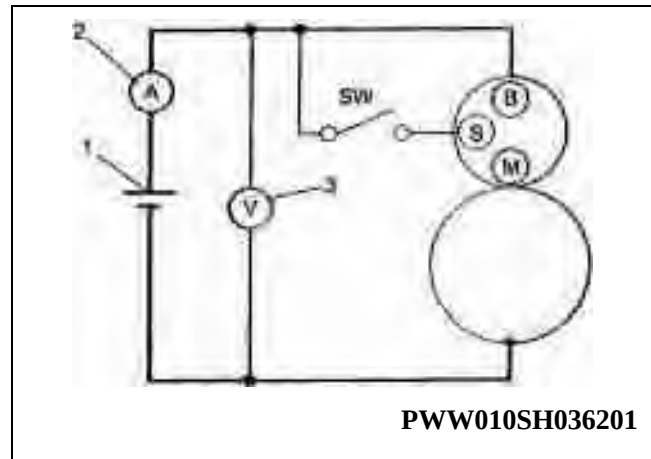
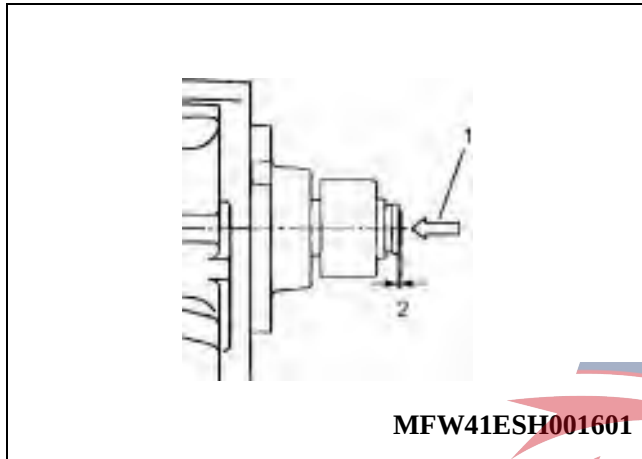
6. Check the pinion protruding position

After assembling the starter, check the pinion protruding position. The measurement and adjustment procedures are as follows.

- Connect the starter to the battery as shown. With SW1 and SW2 closed, if the pinion moves forward to the starting position, the armature will rotate. After that, switch SW2 is disconnected and the armature stops.



- Gently push the end of the crankshaft (pushback) and measure the crankshaft movement (return dimension). If the value is in the range of 0.5-2.0 mm (0.0197-0.0787 inches), it is normal. If the measured value is out of this range, replace the number of spacers between the magnetic switch and the front bracket (2 or less) (0.5 mm / 0.0197 in.), or replace the lever.



Legend

1. Battery
2. Ammeter
3. Voltmeter

Legend

1. Manually push the clutch shaft
 2. Axial movement
7. No-load test

- After adjusting the protruding position of the pinion, connect the starter as shown.

Note:

Use an appropriate thickness of wire and fully tighten the connection.

- Open read speed current and voltage compare with maintenance standards.

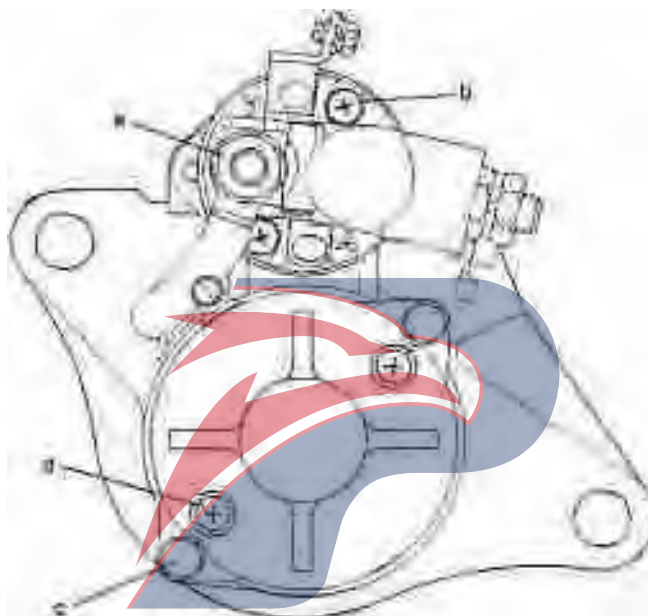
If these values are different from the maintenance standard values, please remove them and recheck them.

8. Maintenance standard

Item			Standard or extreme
No-load characteristic	Voltage	(V)	< 23.0
	Electric current	(Ampere)	85
	Speed	(r/min)	>3300
Commutator	Outside diameter	(mm/inch)	38.7 (1.5236)
	Size reduction limit	(mm/inch)	38.1 (1.5000)
Commutator	Undercut limit	(mm/inch)	0.8(0.0315)
		(mm/inch)	0.2(0.0079)
Electric brush	Length	(mm/inch)	23(0.9055)
	Size reduction limit	(mm/inch)	12(0.4724)
Brush spring	Pressure	(kg f / lb ft)	33-45(3.4-4.6/25-33)
	Limit	(kg f / lb ft)	20 (2.0/14)

Item	Standard or extreme
Pinion protruding position (mm/inch)	0.5—2.0 (0.0197—0.0787)

Tightening torque table



MFW61EMF000101

Legend

(a) 23 N m (2.3 kg f m / 17 lbs ft)

(b) 6 N m (0.6 kg f m/52 lb ft)

(c) 10 N m (1.0 kg f m/87 lb ft)

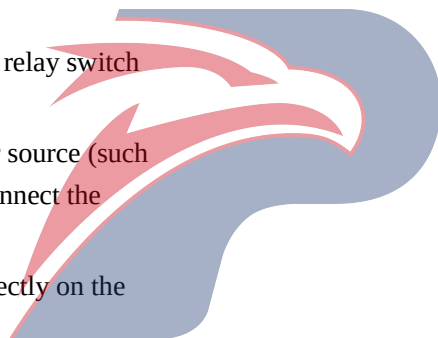
(d) 3 N m (0.3 kgf m/26 lb.ft)

Charging system

Precautions for maintenance

Please pay attention to the following when operating the generator.

- Connecting the positive and negative terminals of the battery can damage the generator diodes.
- Do not disconnect the battery terminals and charging circuit while the engine is running.
- Each terminal should be wired according to the terminal signal.
- Do not carry out Mega test over the semi-conductor elements.
- Always disconnect the battery terminals (-) when inspecting the unit.
- Do not disconnect /connect the battery relay switch while the engine is running.
- When charging with an external power source (such as during fast charging), be sure to disconnect the battery terminals first.
- Do not use steam cleaners or water directly on the equipment.
- After the wiring, tighten the terminal B and terminal E to the following torque.



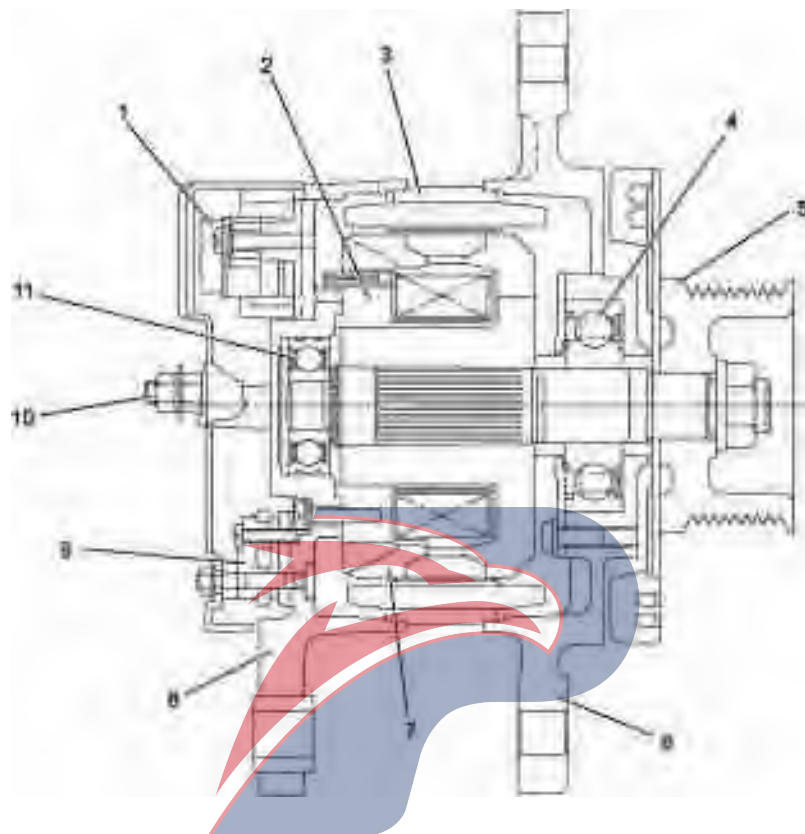
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50 ampere Specification	Terminal B	7 N·m (0.7 kgf m / 61 lbs.)
	E terminal	4 N·m (0.4 kgf m / 35 lbs.)
60 ampere Specification	Terminal B	7 N·m (0.7 kgf m / 61 lbs.)
	E terminal	4 N·m (0.4 kgf m / 35 lbs.)
90 ampere Specification	Terminal B	7 N·m (0.7 kgf m / 61 lbs.)
	E terminal	4 N·m (0.4 kgf m / 35 lbs.)

- For details on the adjustment of the V-belt tension, refer to the belt adjustment section.

Description of function and operation

Charging circuit

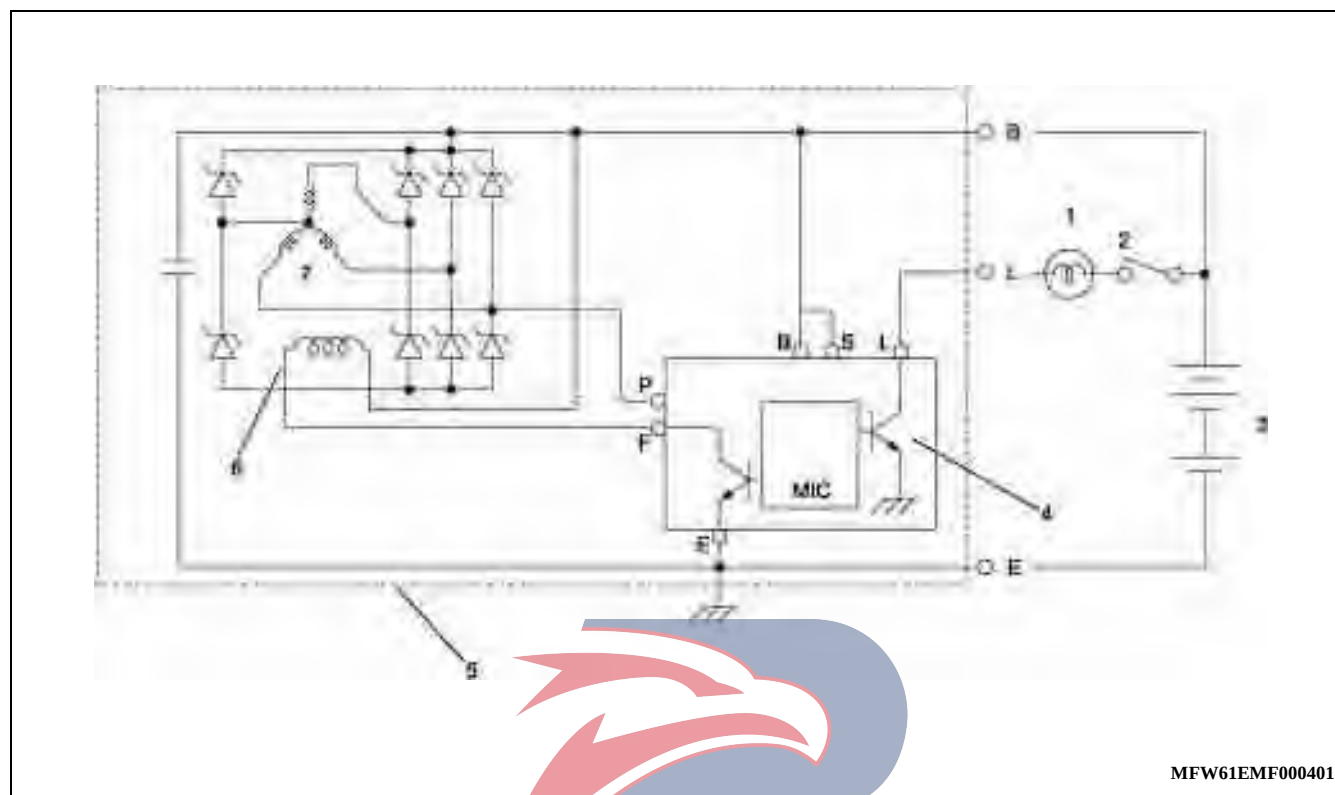


MFW61EMF000301

POWERSTAR

Legend

- | | |
|-----------------------|-------------------|
| 1. Pressure regulator | 7. Rotor |
| 2. Excitation coil | 8. Rear bracket |
| 3. Stator | 9. Rectifier |
| 4. Ball bearings | 10. Terminal B |
| 5. Pulley | 11. Ball bearings |
| 6. Front bracket | |

50A / 60A / 90A Specifications**Legend**

1. Charge indicator
2. Key switch
3. Battery
4. IC regulator

5. Generator

6. Excitation coil

7. Stator coil

POWER STAR

Specifications

Rated output	(V/A)	24—50	24—60	24—90
Specified rotary speed	(rotate every minute)	5000	5000	5000
Regulator type		IC type		
Adjust voltage	(V)	28.5 ± 0.5		
Quality	(kg f / lb ft)	9.9(21.83)	9.9(21.83)	18.3(40.35)

Functional check

General vehicle inspection

The charge warning light can indicate the operation of the charging system. When the start switch is on, the warning light is on; if the warning light goes off after the engine starts, the operation is normal. If the warning light indicates a fault, or if the battery is suspected to be undercharged or overcharged, troubleshoot according to the list.



Failure symptoms list

Can't charge at all

Insufficient charging

Arbitrary charges

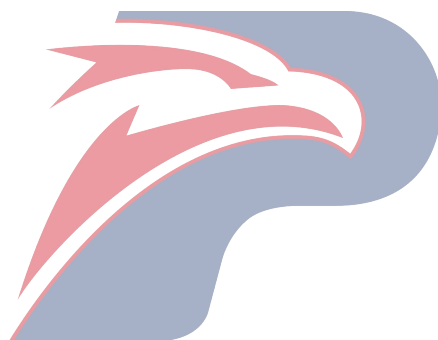
Instable charging current

Generator emits abnormal noise



Phenomenon: Can't charge at all

Phenomenon	Probable causes	Corrective measures
Can't charge at all	Battery failure	Replace
	Line/Ampere meter wire break, short circuit and joint part disconnection	Repair
	Generator coil damage, grounding, short circuit and diode damage	Replace
	Regulator failure	Replace
	Bad pressure regulator terminal connection	Repair

**POWERSTAR**

Phenomenon: undercharge

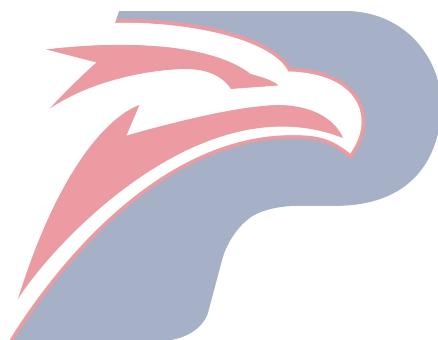
Phenomenon	Probable causes	Corrective measures
Insufficient charging	Battery failure	Replace
	Broken wire, short circuit, disconnected	Repair
	Generator drive belt loose	Adjust the belt tension
	Stator coil layer short circuit	Replace
	Diode failure	Replace
	A certain terminal connection failure	Repair
	Regulator failure	Replace
	Bad pressure regulator terminal connection	Repair



POWERSTAR

Phenomenon: Overcharging

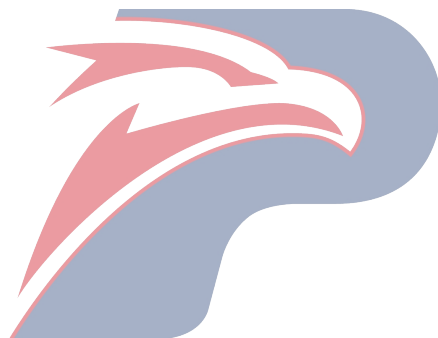
Phenomenon	Probable causes	Corrective measures
Arbitrary charges	Regulator failure	Replace



POWERSTAR

Symptom: Instable charging current

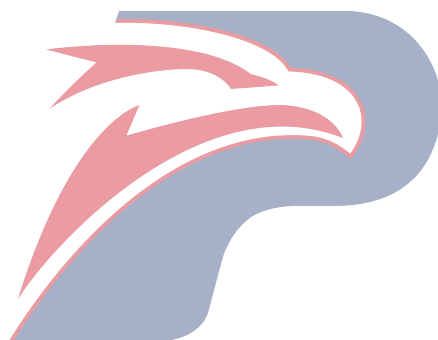
Phenomenon	Probable causes	Corrective measures
Instable charging current	Intermittent contact or partial disconnection of the wire	Repair
	Generator drive belt loose	Adjust the belt tension
	Stator coil short circuit, partially damaged	Repair
	Poor contact in each terminal	Repair
	Regulator failure	Replace
	Bad pressure regulator terminal connection	Repair



POWERSTAR

Phenomenon: Generator emits abnormal noise

Phenomenon	Probable causes	Corrective measures
Generator emits abnormal noise	Belt relaxation	Adjust the belt tension
	Bearing failure	Replace bearing
	Rotor core and stator core contact each other	Repair
	Diode failure	Replace
	Stator coil short circuit, partially damaged	Replace

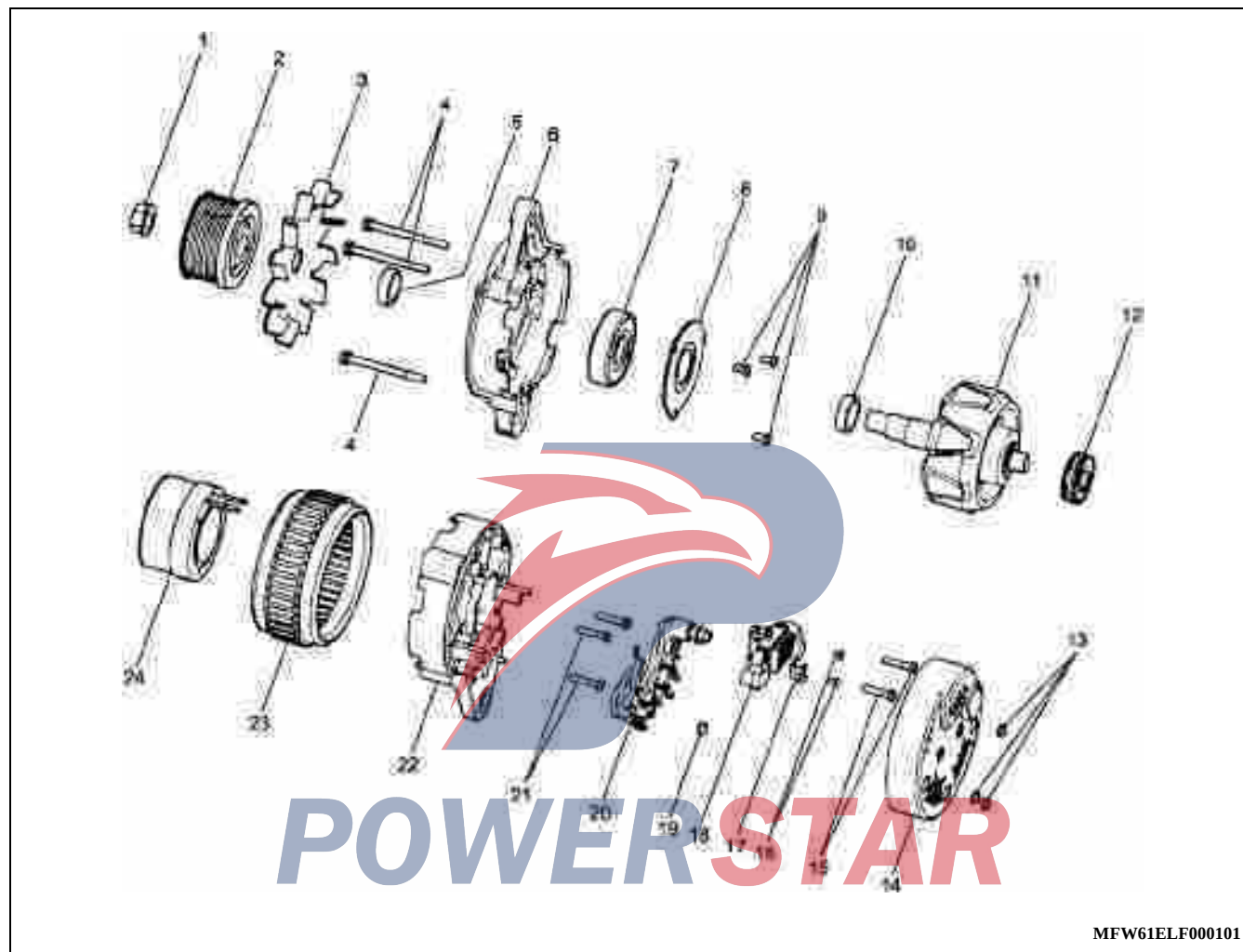


POWERSTAR

Generator

Part

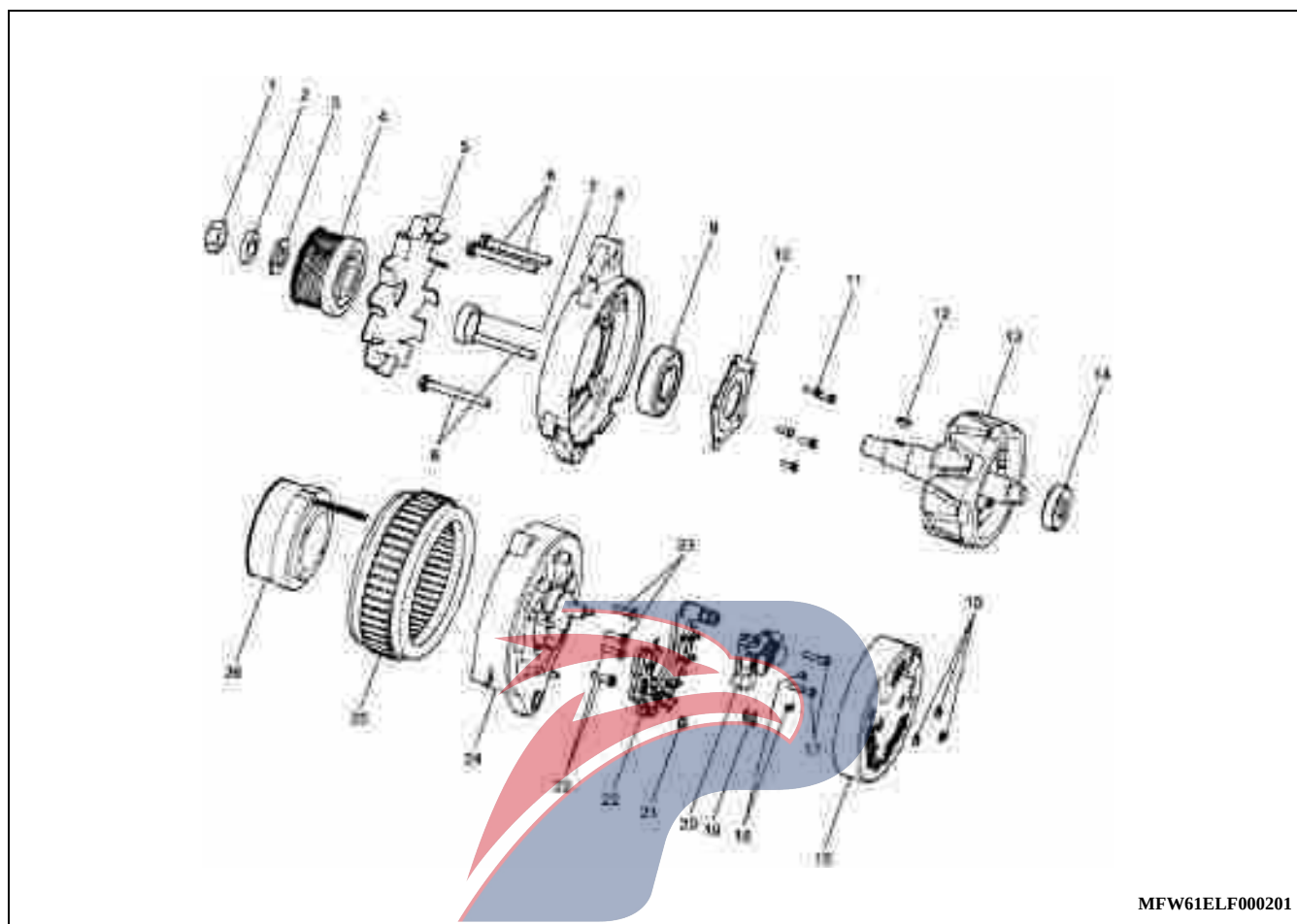
50A•60A Specifications



MFW61ELF000101

Legend

- | | |
|-------------------------------------|---|
| 1. Nut | 13. Nut (for tightening the lid) |
| 2. Pulley | 14. Cover |
| 3. Fan | 15. Screw (for installing pressure regulator) |
| 4. Through bolt | 16. Screws (for tightening the regulator) |
| 5. Axes ring | 17. Cramp |
| 6. Front bracket | 18. Pressure regulator |
| 7. Front bearing | 19. Nut (for fixing the rectifier) |
| 8. Guard ring | 20. Rectifier |
| 9. Screws (for tightening the cage) | 21. Screw (for fixed excitation coil) |
| 10. Axes ring | 22. Rear bracket |
| 11. Rotor | 23. Stator |
| 12. Rear bearing | 24. Excitation coil |

90A specifications**Legend**

- | | |
|--------------------------------------|---|
| 1. Nuts | 14. Rear bearing |
| 2. Spring washer | 15. Nut (for tightening the lid) |
| 3. Flat washer | 16. Cover |
| 4. Pulley | 17. Screw (for installing pressure regulator) |
| 5. Fan | 18. Screws (for tightening the regulator) |
| 6. Through bolt | 19. Clamping |
| 7. Axes ring | 20. Pressure regulator |
| 8. Front bracket | 21. Nut (for fixing the rectifier) |
| 9. Front bearing | 22. Rectifier |
| 10. Guard ring | 23. Screw (for fixed excitation coil) |
| 11. Screws (for tightening the cage) | 24. Rear bracket |
| 12. Semicircular drill bit | 25. Stator |
| 13. Rotor | 26. Excitation coil |

Disassembly**Note:**

Before and after removal, the front and rear brackets are marked in alignment for reassembly.

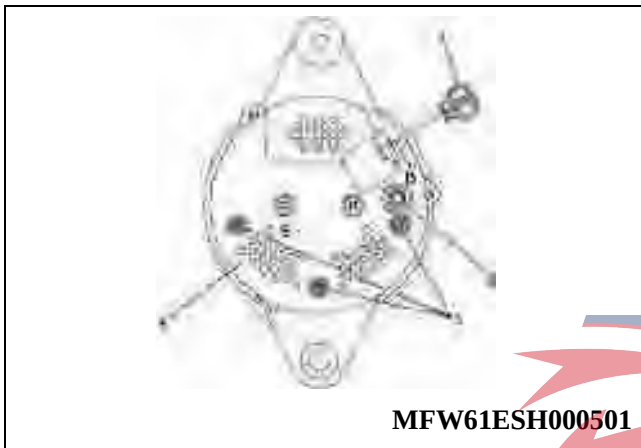
1. Remove the through bolts.

- Remove 3 bolts (50 A and 60 A).
- Remove 4 bolts (90 amps).

2. Remove the cover from the rear side.

- Remove the 3 nuts.
- Remove the 2 nuts (90A) on the B terminal.

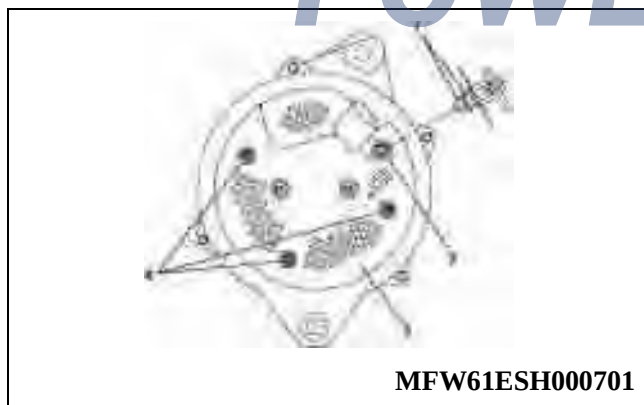
50A•60A Specifications



Legend

1. Connector
2. Terminal B
3. Nuts
4. Cover

90A specifications



Legend

1. Nut
2. Terminal B
3. Cover
4. Nuts

3. Remove the B terminal (90 amps).

4. Separate the front and back sides.

- Before separation, warm up the bearing housing with a tool such as a dryer and remove it.
- When separating, use a press to remove the hole left by the through hole bolt removal.

Note:

Be careful not to damage the stator coil.

5. Remove the pulley nut by knocking.

6. Remove spring washer and flat washer (90 amps).

7. Remove the pulley.

8. Remove the fan.

9. Remove the semi-circular drill bit (90 amps).

10. Remove the collar.

11. Remove the rotor from the front bracket assembly.

- When removing the rotor, use a tool such as a press.
- Do not place the rotor in a vise or deformation will occur.

12. Remove the collar (50 amps and 60 amps) from the rotor.

13. Remove the bearing fixing screws of the front bracket assembly.

14. Remove the front bearing.

- If you need to remove the front bearing, remove the retainer assembly screw and push the bearing out with a suitable sleeve.

Note:

Remove the bearing only when necessary. Once the bearing is removed, it must not be used again.

15. Remove the stator.

- Remove the stator wire and remove the stator from the rectifier.
- Remove the stator wires at seven locations and remove the stator (90 amps) from the rectifier.

Note:

When disassembling the stator wire, use the soldering iron (not more than 5 seconds) in a short time to prevent the rectifier from overheating and damaging the diode.

16. Remove the excitation coil terminal wire.

- Remove the converter cable and 2 screws (M4).

- Remove the magnetic flux at the connection point between the conductor and the rectifier.

Note:

When removing the excitation coil terminal wire, use the soldering iron for a short time (not more than 5 seconds) to avoid overheating the rectifier.

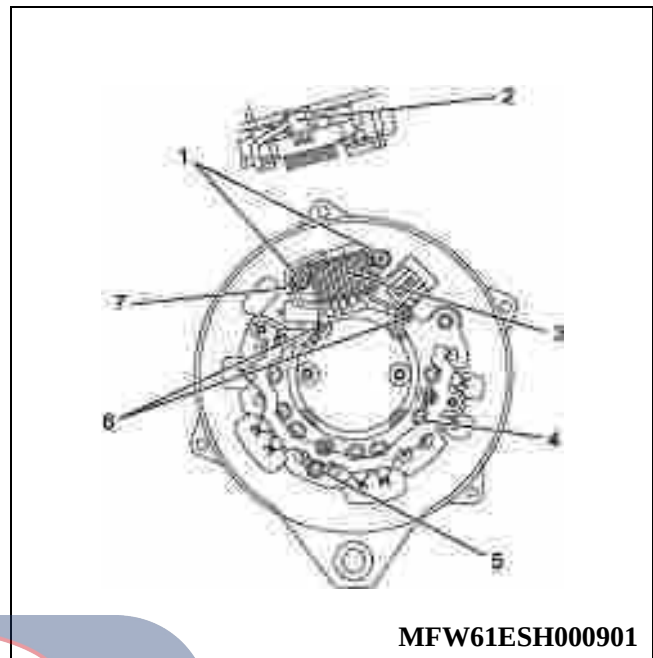
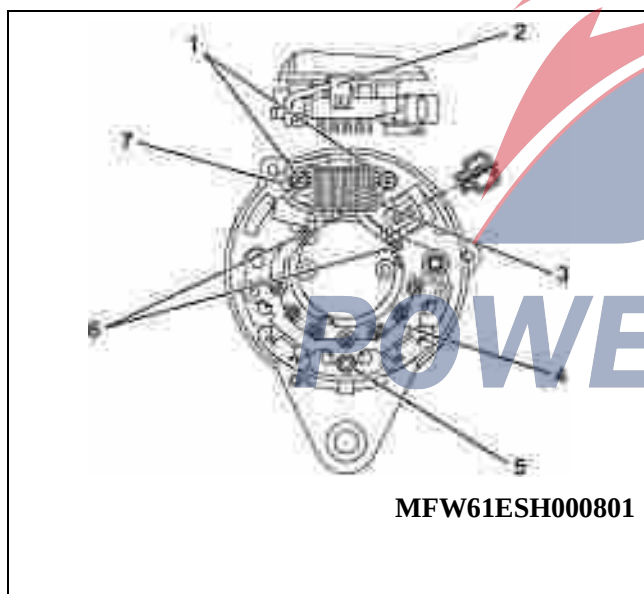
- Remove the wire from the clamp on the side of the pressure regulator.

17. Remove the adjuster mounting screws.

18. Remove the regulator.

19. Remove the pressure regulator retaining nut (in one place).

20. Remove the rectifier.

90A specifications**50A and 60A specifications****Legend**

1. Screw
2. Welding parts
3. Pressure regulator
4. Rectifier
5. Nut
6. Screw
7. Jaws

Legend

1. Screw
2. Welding parts
3. Pressure regulator
4. Rectifier
5. Nuts
6. Screw
7. Clamping

21. Remove the 3 mounting screws of the excitation coil.

22. Remove the 4 mounting screws (90A) of the excitation coil.

23. Remove the excitation coil.

24. Disconnect the rear bearing.

- If it is necessary to remove the rear bearing, use a press.

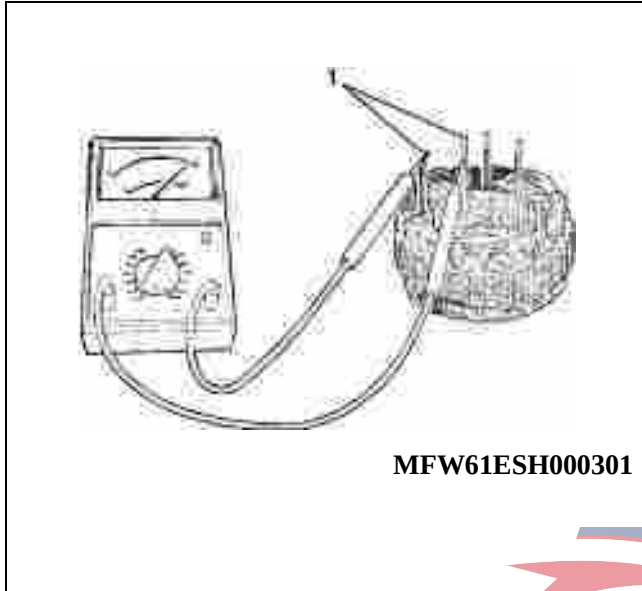
Note:

Remove the rear bearing only when necessary. Once the rear bearing is removed, it shall not be used again.

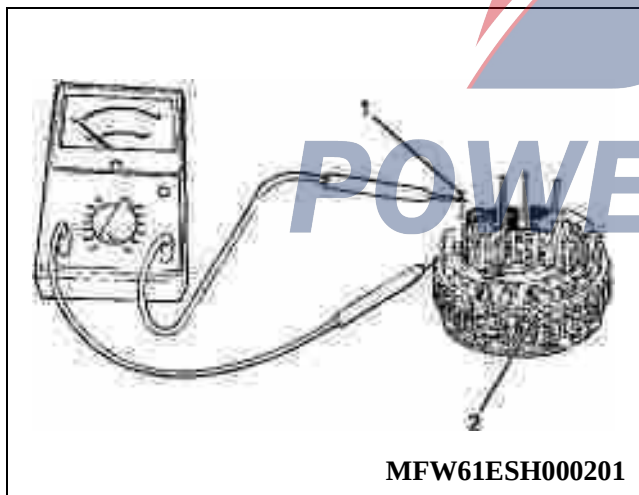
Inspection**Generator inspection and maintenance****Check the stator**

- Visually inspect for damage, loose connections and discoloration of the winding wires.

- Use a circuit tester to check the electrical conductivity between the two conductors of the stator coil (1). If there is no conductivity, replace the stator.



- Use a circuit tester to check the electrical conductivity between the stator coil wire (1) and the core wire (2). If there is electrical conductivity, the stator should be replaced.

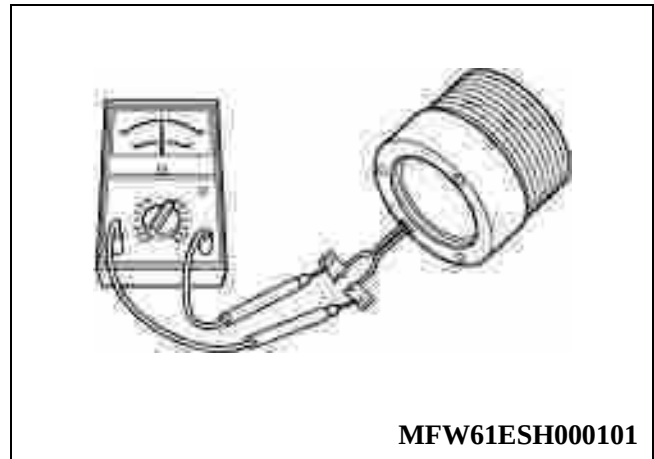


Check rotor

- Check the bearing for unusual noise, looseness, and grease leakage (if there is a small amount of grease, wipe it off). If you find a fault, replace the bearing. If the bearing is mounted on a press-in shaft, it is easier to check the bearing for unusual noise and looseness than to check the bearing.

Check the coil assembly

- Use a circuit tester to measure the resistance between the terminals of the terminals. If the resistance exceeds the standard value, replace the coil assembly.



Check the rectifier assembly

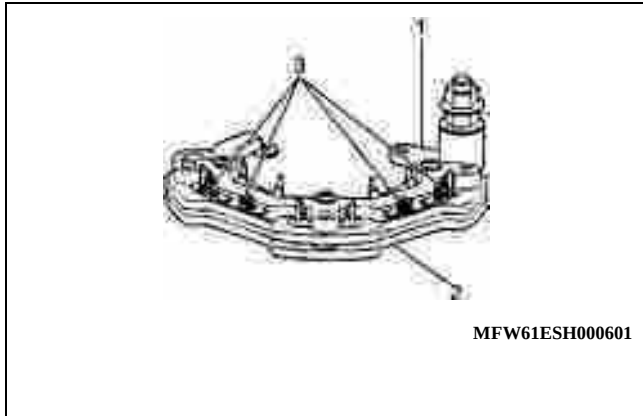
- Use a circuit tester to check the conductivity between the four terminals of the cathode diode.

If both directions are conductive, the positive diode will short circuit. In this case, replace the rectifier assembly.

If there is no conductivity in both directions, it means that the positive diode is disconnected. In this case, replace the rectifier assembly.

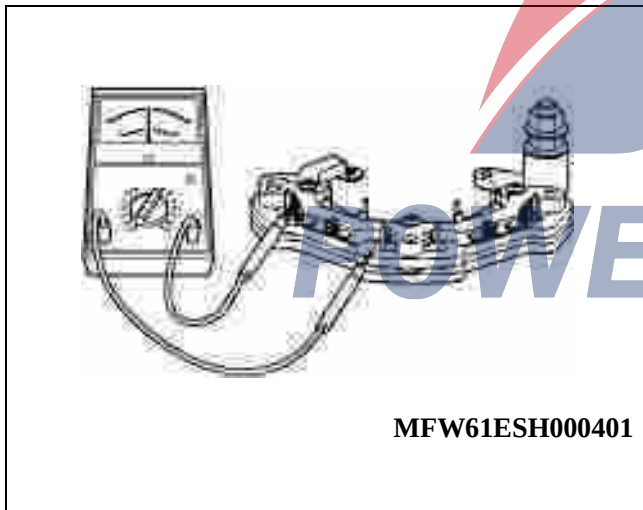
- Use a circuit tester to check the conductivity between the four terminals of the negative diode.

If there is conductivity in both directions, the negative diode will short circuit. In this case, replace the rectifier assembly. If both directions are not conducting, the negative diode is open-circuited. In this case, replace the rectifier assembly.



Legend

1. Positive thermal synchronization
2. A part of negative pole thermal synchronization
3. Diode termination



Check the front bearing

- Check the bearing for unusual noise, looseness, and grease leakage (if there is a small amount of grease, wipe it off). If you find a fault, replace the bearing. If the bearing is mounted on a press-fit bracket, it is easier to check the bearing for unusual noise and looseness than to check the bearing.

Check pressure regulator

- The circuit tester cannot be used to check the regulator. After reassembling the alternator, test the alternator to make sure that the voltage regulator can control the alternator voltage.

Assembling

Install it in the sequence reverse to the removal. Here is only the most important point. Please note the next point.

- The bearings are filled with grease and sealed so the bearings do not need to be lubricated.

Note:

Do not lubricate the resin bearings on the rotor shaft. If there is oil in the rear housing part, it may cause the rear bearing to rotate together, so the oil must be completely wiped clean.

- Use high temperature flux (230 °C / 446 °F melting point specification).

During the welding, quickly complete the operation (within 5 approximately) with the 180 - 270W electric iron to prevent the rectifier overheating.

- For the tightening torque of each part, refer to the tightening torque table.

Note:

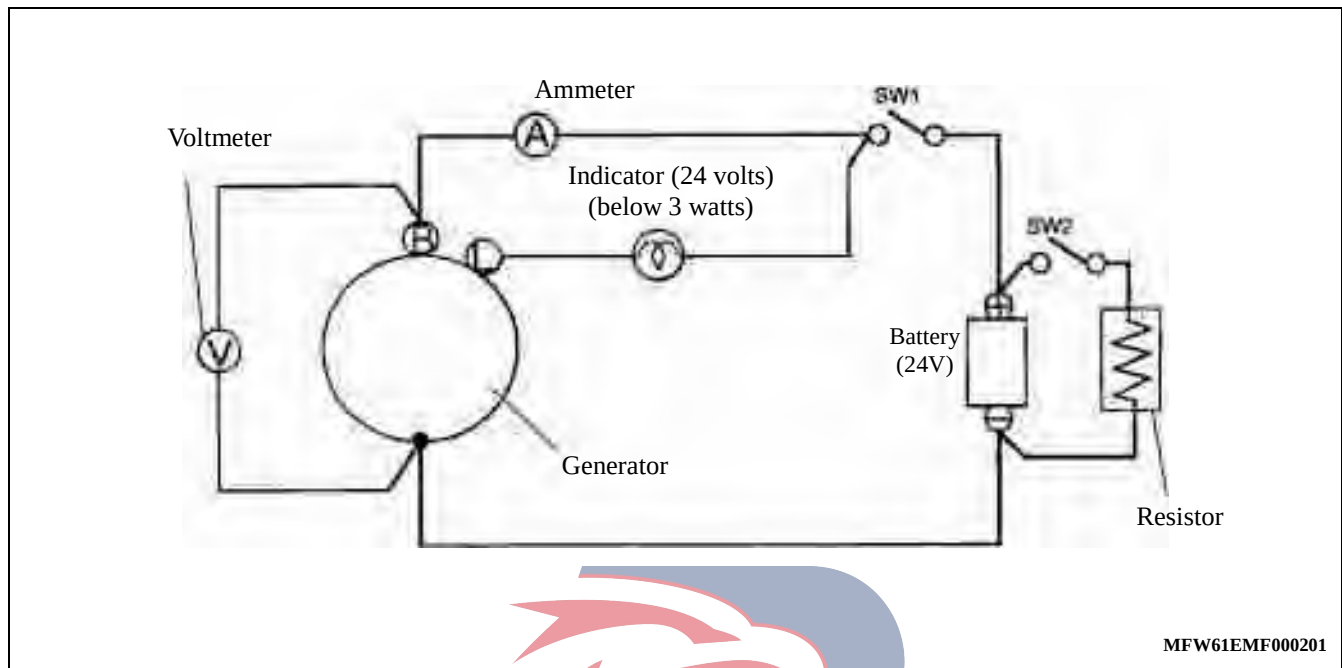
Be careful not to over-tighten.

Tightening torque		N m (kg/m/lb. inch)
Pulley nut	(50 A and 60 A specifications)	137 (14.0 / 101 lb ft)
	(90A specifications)	245 (25.0 / 181 lb ft)
Through the bolt	(50 A and 60 A specifications)	6 (0.6 / 52)
	(90A specifications)	14 (1.4 / 122)
Bearing cage		5 (0.5 / 43)
Excitation coil assembly	(50 A and 60 A specifications)	3 (0.3 / 26)
	(90A specifications)	5 (0.5 / 43)
Rectifier		3 (0.3 / 26)
Pressure regulator		3 (0.3 / 26)
Regulator and rectifier connection screws		2 (0.2 / 17)
Cover		3 (0.3 / 26)
Terminal B nut		7 (0.7 / 61)

- The rear bracket, stator assembly and front bracket rotor assembly are installed.

- Due to the tight fit between the rear brackets, the bearing housing portion of the rear bracket should be preheated to 50-60°C (122-140°F) and the bearings and brackets installed.
- After reinstalling the generator, turn the pulley by hand to ensure that the rotor can rotate smoothly.



Bench test:**Generator bench test****Voltage regulation**

- Connect the circuit as shown above.

Use ammeters rated at 100 amps and voltmeters rated at 30 volts.

Use a variable resistor with a rated current of 50 amps or more.

- Turn off switch 1. Cut off switch 2.

At this point, the indicator will light up.

If the indicator light fails to be turned on, check the voltage of terminal L.

If the voltage is close to the battery voltage, check the generator.

If the voltage is low, the indicator or indicator circuit turns on.

- Gradually increase the generator speed. Make sure that the indicator does not go off until it reaches approximately 1,300 rpm. Observe the voltage. If the voltage cannot be kept below 30 volts, the IC regulator may be damaged. If the voltage cannot exceed the battery voltage, the generator cannot generate electricity.

Check the generator again.

If the indicator does not light, check the generator

again.

- Let the generator run around 5,000 rpm and read the ammeter.

If the reading is 5 amps or less, read the voltmeter instructions. This is the adjusted voltage.

If the ammeter indicates 5 amps or more, continue charging until the current drops below 5 amps or replace the battery with a fully charged battery.

If the adjusted voltage exceeds the specified range, replace the regulator.

Output current

The output current test can determine whether the generator outputs the rated current.

- After checking the voltage adjustment, reduce the generator speed and stop the generator.
- Turn off switch 1. The indicator light should light up.
- Set the variable resistor to the highest value (minimum current) and turn switch 2 off.

- Increase the speed of the generator, keep the speed at 5000 rpm, and adjust the variable resistance to maximize the output current.

In this state, the generator is operated for 15 minutes to improve the generator temperature.

After 15 minutes, set the generator speed to match the specified speed (2,500 and 5,000 rpm). Then adjust the variable resistor in this state to maximize the output current.

If the indication of ampere meter is within the specified range at this time, the generator is operating normally.

If the instructions are out of specification, the generator should be re-inspected as described in the "Inspection" section.

Specification

50 amp specifications

	Speed	2500 rpm	5000 rpm
Output characteristics (at high temperatures)	Voltage	27 V	27 V
	Electric current	42 amps or more	50 amps or more
Excitation coil resistance (20°C/68°F)		8.3—9.1Ω	
Regulator adjustment voltage		28-29V	
Rotation direction (viewed from the side of the pulley)		Clockwise	

60A specifications

	Speed	2500 rpm	5000 rpm
Output characteristics (at high temperatures)	Voltage	27 V	27 V
	Electric current	45 amps or more	60 amps or more
Excitation coil resistance (20°C/68°F)		8.3—9.1Ω	
Regulator adjustment voltage		28-29V	
Rotation direction (viewed from the side of the pulley)		Clockwise	

90A specifications

	Speed	2500 rpm	5000 rpm
Output characteristics (at high temperatures)	Voltage	27 V	27 V
	Electric current	80 amps or more	90 amps or more

Excitation coil resistance (20°C/68°F)	6.3—7.5Ω
Regulator adjustment voltage	28-29V
Rotation direction (viewed from the side of the pulley)	Clockwise

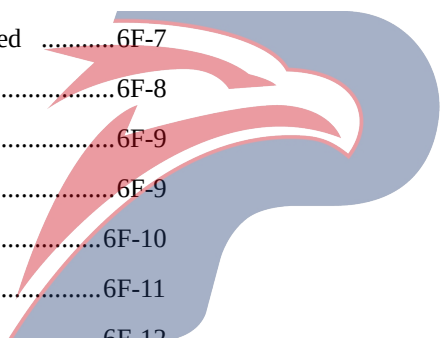
Engine

Engine Exhaust System

(6HK1)

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POWERSTAR

Exhaust system

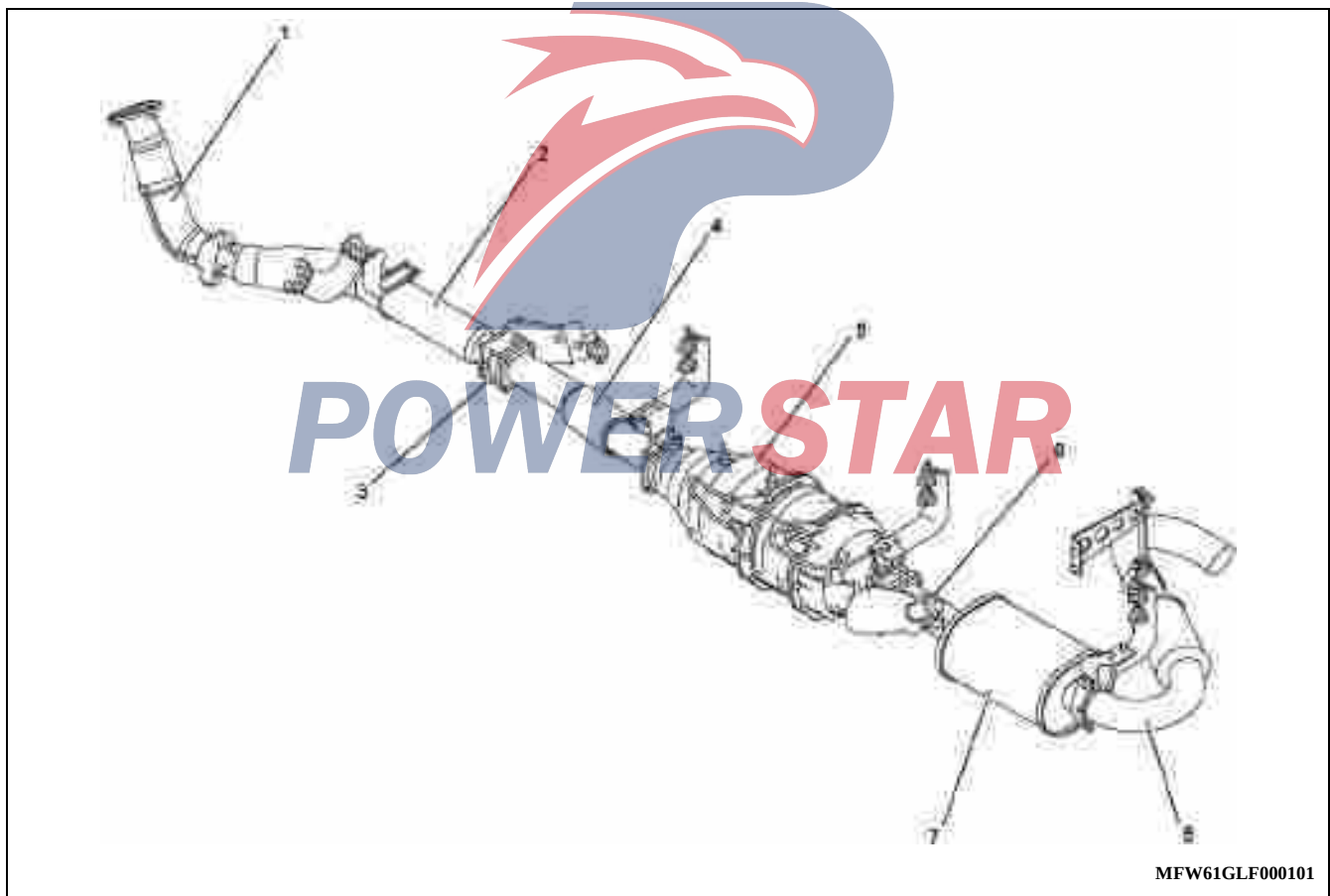
Precautions for maintenance

Since vibration and loosening in the exhaust system are usually caused by misalignment during system assembly, all bolts should be temporarily tightened until all parts are positioned. Finally, tighten all bolts one by one from the engine side to the rear tube side. .

Description of function and operation

The exhaust brake valve assembly regulates exhaust pressure to assist braking and perform functions such as the engine's rapid warm-up system (QWS). The Diesel Particulate Processor (DPD) is a unit consisting of a filter and an oxidation catalyst for cleaning the exhaust gas. The sound deafening is achieved by the rear muffler in the tail pipe.

The main components include exhaust brake valve assembly, front exhaust pipe A, B, C, DPD, exhaust throttle valve assembly, throttle muffler exhaust pipe, muffler and rear exhaust pipe (including rear muffler).



Legend

1.Front exhaust pipe A

2.Front exhaust pipe B

3. Exhaust brake valve assembly

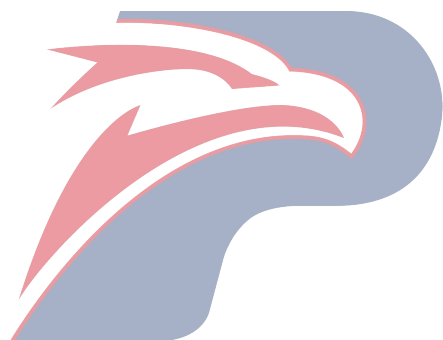
4 Front exhaust pipe C

5. Diesel Particle Processor (DPD)

7. Muffler

6. Exhaust throttle valve assembly

8. Rear exhaust pipe



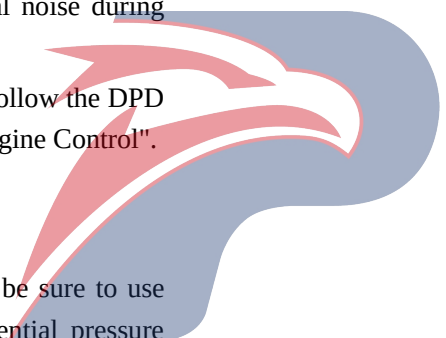
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Functional check:

- When checking or repairing the exhaust system, make sure there is sufficient clearance between the ground and the main body.
- Damage to the main panel caused by heat or vibration may cause the exhaust gas to leak into the room.
- Check the joints for damage or looseness and make sure that there is no leakage of exhaust gas.
- Check the clips and rubber for deterioration, wear and damage.
- If pipes, DPD or mufflers are damaged or have dents, they should be repaired or replaced.
- Check for signs of dents, damage, holes or cracks due to corrosion, and check for unusual noise during driving.
- If a DPD related to DPD is detected, follow the DPD check and repair in "Troubleshooting Engine Control".

Caution:

Before dismantling the DPD assembly, be sure to use the scan tool to check the DPD differential pressure status and detailed inspection/maintenance. For more information, please refer to "Engine Control Section."



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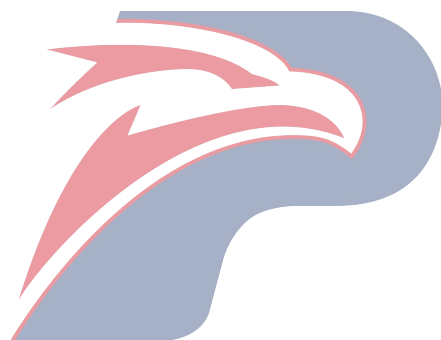
Failure symptoms list

- Exhaust system vibrates and rattles
- Exhaust system plug
- Exhaust gas leakage, noise



Phenomenon: Exhaust system vibration and rattle

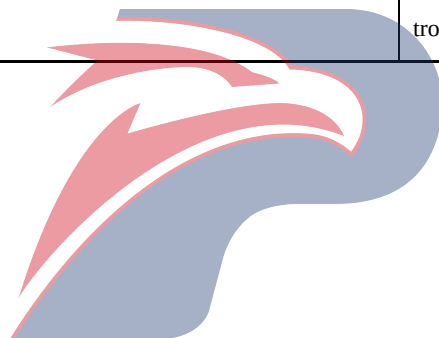
Phenomenon	Probable causes	Corrective measures
Exhaust system vibrates and rattles	Loose or inappropriate parts	Align and re-adjust the connection. Also check the hangers, connection brackets and fixtures for damage.



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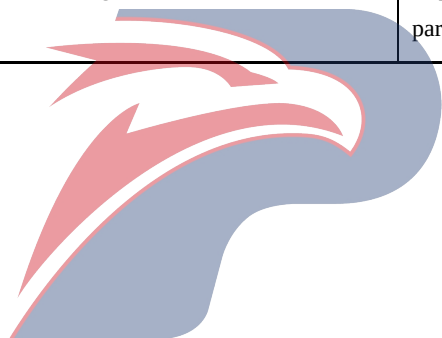
Phenomenon: The exhaust system is blocked

Phenomenon	Probable causes	Corrective measures
Exhaust system plug	Exhaust brake valve and exhaust throttle valve open	Refer to the Engine Control System section to check the electrical system and then fix the fault. Check whether the exhaust brake valve and exhaust throttle valve shaft are impeded by movement; if faults are found, they should be replaced.
	Exhaust pipe plugging	Repair or replace the exhaust pipe.
	DTCs related to DPD are detected	Check and repair according to the contents related to DPD in "engine control - troubleshooting".

**POWERSTAR**

Phenomenon: Exhaust leakage, noise

Phenomenon	Probable causes	Corrective measures
Exhaust gas leakage, noise	Bad connection or inappropriate cooperation	Align and re-adjust the connection.
	Exhaust pipe or muffler damaged or burned	Replace damaged parts.
	The exhaust system components (clamps, pipes and mufflers) have poor performance.	Replace the needless components.
	Internal damage of muffler	Replace the muffler.
	Internal damage of DPD	Replace the damaged part (oxidation catalyst part/filter).



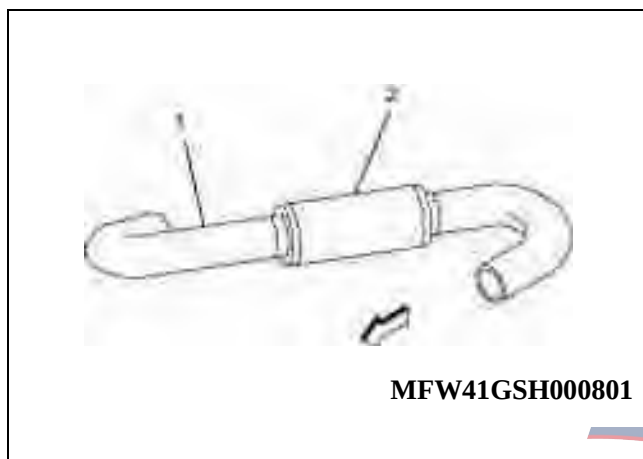
POWERSTAR

Exhaust pipe

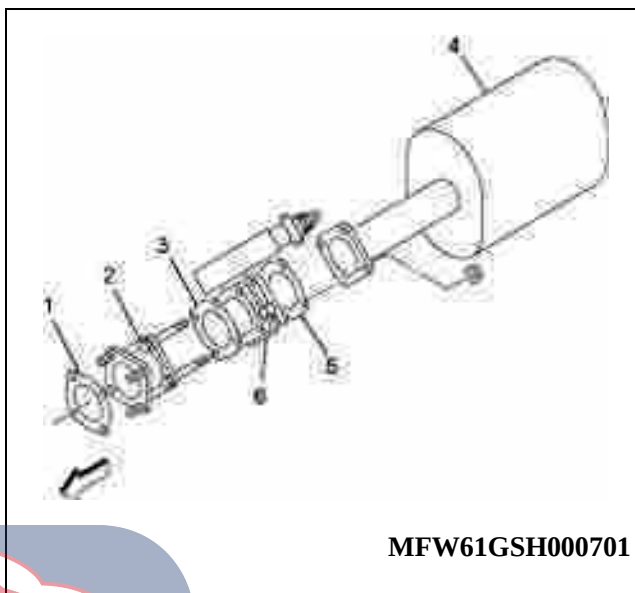
Tear down

1. Remove the rear exhaust pipe.

- Remove the clamp and remove the tail pipe.



6. Remove the exhaust connector.



Legend

1. Rear exhaust pipe
2. Rear muffler

2. Disconnect the exhaust throttle.

3. Remove the muffler bracket.

4. Remove the muffler .

5. Remove the exhaust throttle valve assembly.

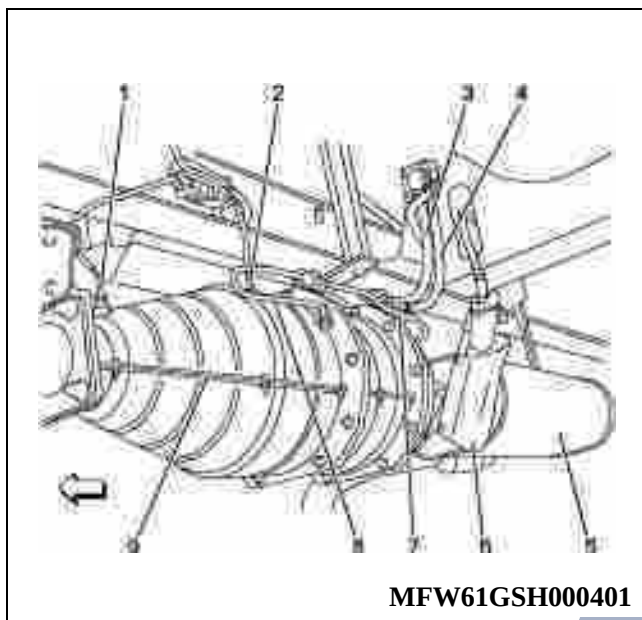
Legend

1. Gasket
2. Exhaust pipe joint
3. Gasket
4. Muffler
5. Gasket
6. Exhaust throttle valve assembly

7. Remove the 2 DPD exhaust temperature sensor connectors.

8. Remove 2 DPD pressure hoses.

9. Remove the DPD assembly from the front tube.



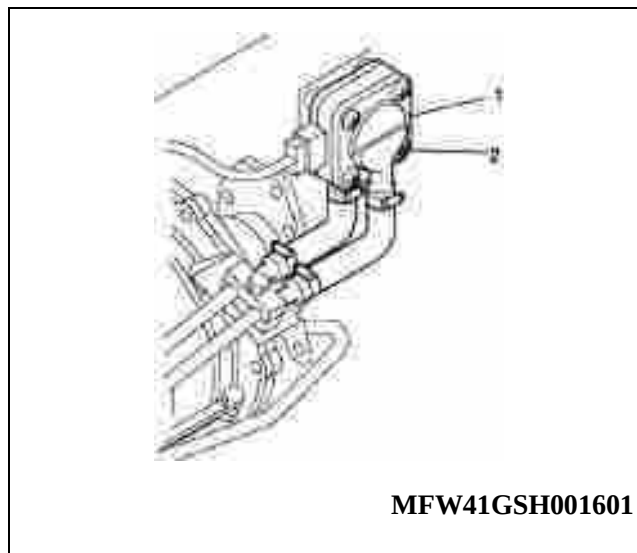
Legend

1. Exhaust temperature sensor 2 (catalyst inlet)
2. Exhaust temperature sensor 1 (filter inlet)
3. Pressure differential hose (front)
4. Pressure differential hose (rear)
5. Exhaust muffler
6. Exhaust throttle valve assembly
7. Differential pressure tube (rear)
8. Differential pressure tube (front)
9. DPD assembly

Note:

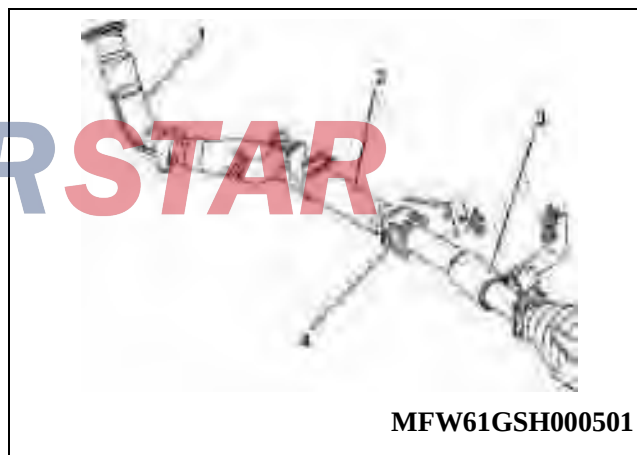
If you really need to remove the differential pressure sensor assembly, pay attention to the following points.

- Do not loosen the 4 sensor fixing screws (2). If they are loose, replace the sensor assembly.
- Do not use the sensor assembly again after it is detached.



Legend

1. Differential pressure sensor assembly
2. Fastening screws
10. Remove front exhaust pipes A, B, and C.
 - Remove the clamp and remove the front exhaust pipe.
 - Remove the exhaust brake valve assembly.



Legend

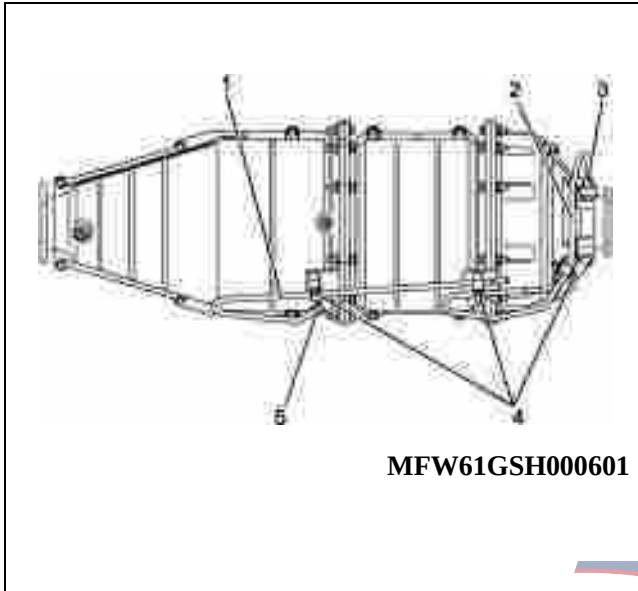
1. Front exhaust pipe A
2. Front exhaust pipe B
3. Front exhaust pipe C
4. Exhaust brake valve assembly

Disassembly

Remove the DPD assembly.

1. Loosen the sleeve nut fixing the differential pressure tube.

2. Remove the differential clamp (3 positions).
3. Remove the differential pressure tube.



Legend

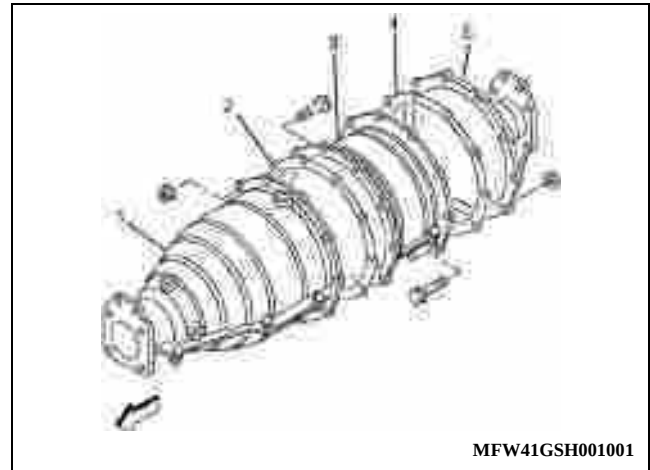
1. Differential pressure tube (front)
2. Differential pressure tube (rear)
3. Turnbuckle (rear)
4. Pipe clamp
5. Turnbuckle (front)

4. Remove the DPD assembly.

- Loosen the fastening nut of the oxidation catalyst section and remove the oxidation catalyst section.
- Loosen the fastening nut of the filter section and remove the filter section from the rear cover.

Note:

Do not open or detach the DPD assembly.



Legend

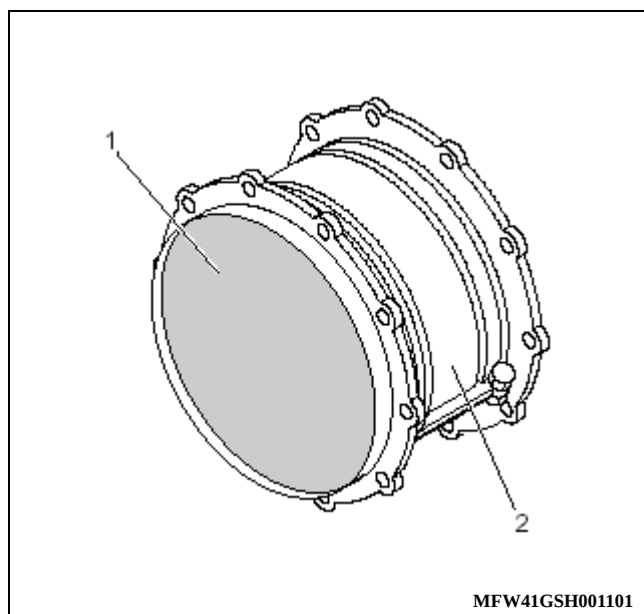
1. Oxidation catalyst section
2. Gasket
3. Filter part
4. Gasket
5. Back cover

Inspection

1. Check the relevant part of the exhaust pipe externally.
 - Damage to the main panel caused by heat or vibration may cause the exhaust gas to leak into the room.
 - Check the joints for damage or looseness and make sure that there is no leakage of exhaust gas.
 - Check the clips and rubber for deterioration, wear and damage.
 - Check the pipe line and muffler for dent and crack due to corrosion and repair or replace the damaged section.
2. Check the relevant parts of the DPD.
 - Visually inspect the filter for melt damage, other damage or clogging.
 - Illuminate from the other side of the filter. If the light can penetrate and is visible, the filter part should be replaced.

Note:

Before dismantling the DPD assembly, be sure to use the scan tool to check the DPD differential pressure status and detailed inspection/maintenance. For more information, please refer to "Engine Control Section."



Legend

1. Filter

2. Filter cover

3. If replacing the differential pressure sensor assembly, take the following measures (on the vehicle).

- Turn the key from OFF to ON (no need to start the engine).
- The DPD light is lit and goes out about 30 seconds later.
- After the DPD indicator lamp is off, disconnect the key switch more than 15s.

4. Check the oxidation catalyst.

- Visually inspect the catalyst section for melt damage, other damage or clogging.

Assembling

Assemble the DPD assembly.

1. Insert gaskets (2 pieces) between the filter and rear cover and tighten the bolt and nut temporarily.

Note:

- After the DPD is disassembled, do not use the old gaskets, bolts and nuts again (using gaskets and fixing kits).
2. Insert spacers (2 pieces) into the filter section and oxidation catalyst section and temporarily tighten the nuts and bolts.

Note:

- Check the connection of the filter section, back cover, filter section and oxidation catalyst section and tighten the two tube carriers.

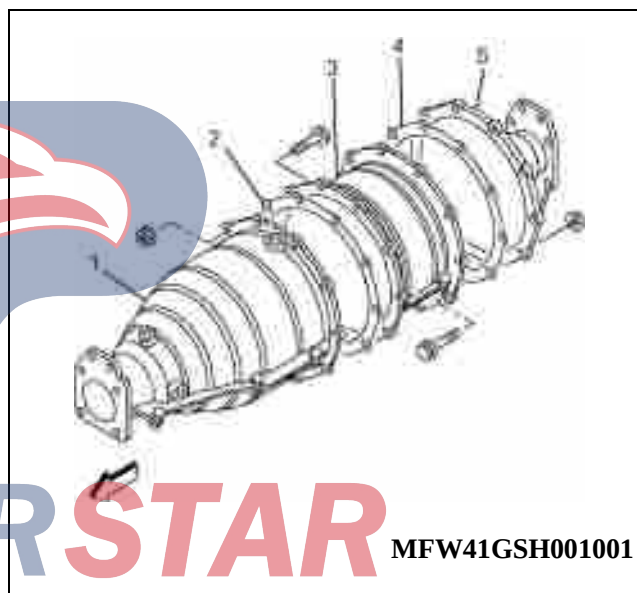
3. Tighten nuts and bolts to the specified torque.

- Tighten each bolt and nut twice.

Tightening torque	N m (kgf m/lb ft)
	25 (2.5 / 18)

Note:

After tightening bolts and nuts, make sure to tighten them again with the same torque (tightening twice).



Legend

1. Oxidation catalyst section

2. Gasket

3. Filter part

4. Gasket

5. Back cover

4. Insert the protruding part (front part) (rear part) of the pressure tube into the connecting part and tighten the sleeve nut by hand.

Note:

- Remove the old flame retardant on the sleeve nut and apply a new flame retardant (NEVER-SEEZ high temperature stainless steel).

5. Fix the differential pressure tube with a pipe clamp.

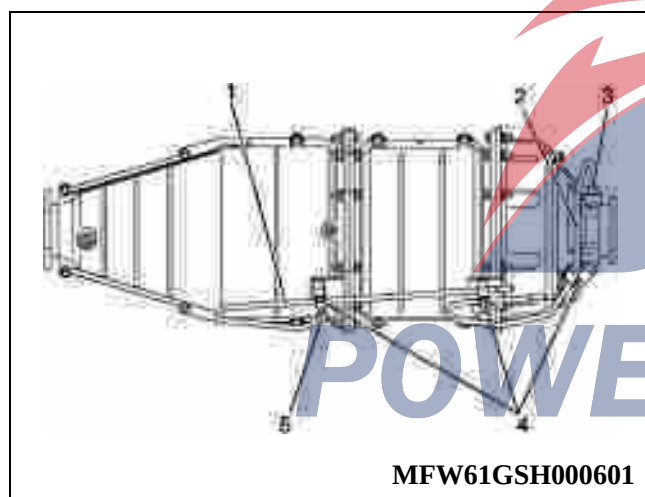
Tightening torque	N m (kg/m/lb. inch)
10 (1.0 / 87)	

6. Tighten the differential pressure tube sleeve nut according to the specified torque.

Tightening torque	N m (kgf m /lb ft)
Front sleeve nut	35 (3.6 / 26)
Rear sleeve nut	

Note:

- If you do not strictly adhere to the differential pressure connection procedure, the nut may loosen. Therefore, be sure to follow this procedure when connecting the pipes.



Legend

1. Differential pressure tube (front)
2. Differential pressure tube (rear)
3. Turnbuckle (rear)
4. Pipe clamp
5. Turnbuckle (front)

Reassemble

Note:

- Since vibration and loosening in the exhaust system are usually caused by misalignment during system assembly, all bolts should be temporarily tightened until all parts are positioned. Finally, tighten all bolts one by one from the engine side to the rear tube side. .

- Do not use the removed washer or nut again.

1. Before installing the exhaust pipe A.

- The front exhaust pipe A is mounted on the exhaust manifold and tightened with prescribed torque strength.

Tightening torque	N m (kgf m/lb ft)
67 (6.8 / 49)	

Note:

Tighten the front exhaust pipe A in the same way to ensure that there is no air leakage.

2. Install the front exhaust pipe B.

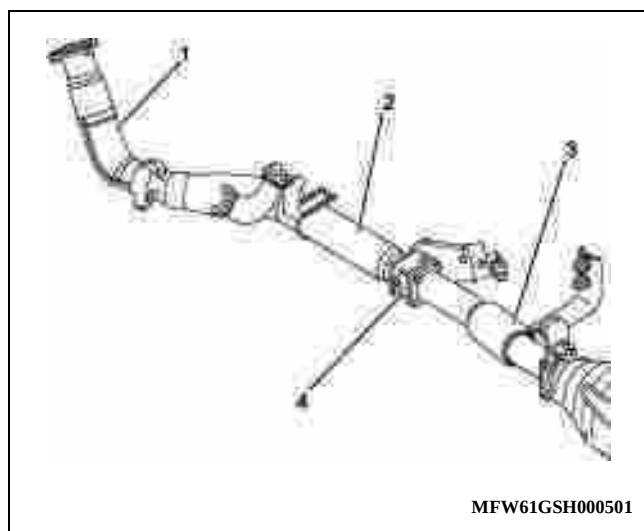
- The front exhaust pipe B is mounted on the front exhaust pipe A and tightened with a prescribed torque strength.

Tightening torque	N m (kgf m/lb ft)
41 (4.2 / 30)	

3. Install exhaust brake valve assembly.

- The exhaust brake valve assembly and the front exhaust pipe C are mounted on the front exhaust pipe B and fastened with predetermined torque strength.

Tightening torque	N m (kgf m/lb ft)
29 (3.0 / 22)	

**Legend**

1. Front exhaust pipe A
2. Front exhaust pipe B
3. Front exhaust pipe C
4. Exhaust brake valve assembly

4. Install the DPD assembly to the head tube.

Tightening torque	N m (kgf m/lb ft)
	83 (8.5 / 61)

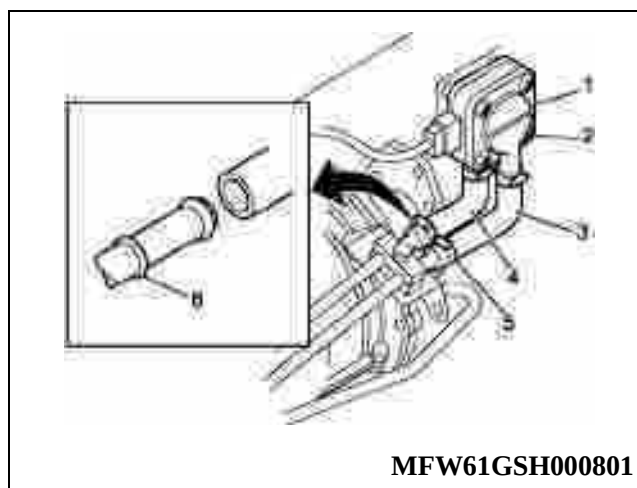
5. Install the differential pressure hose on the DPD differential pressure hose.

- Insert the differential pressure hose in the pipe crown firmly.
- Fix the hose clamp firmly.

Note:

- Do not confuse the front and back direction
- Do not use a broken hose.
- Do not tighten or loosen the fastening screw of the differential pressure sensor assembly.
- If a leakage occurs in the coupling parts, the sensor may malfunction.

After starting the engine, make sure there is no air leakage.

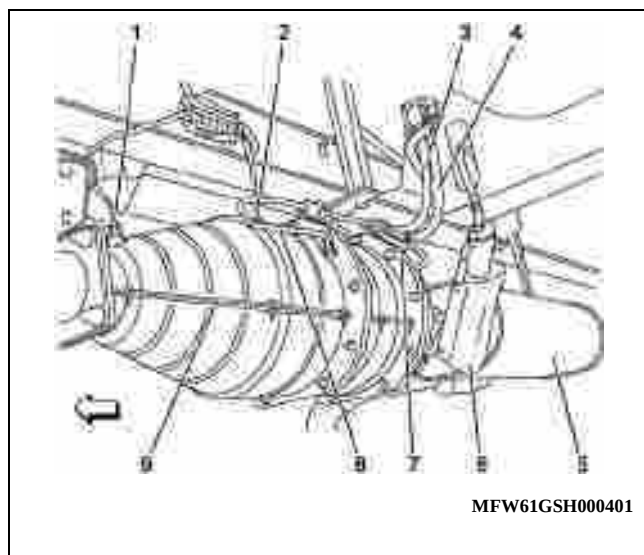
**Legend**

1. Differential pressure sensor assembly
2. Fastening screws
3. Pressure differential hose (rear)
4. Pressure differential hose (front)
5. Pressure differential hose clamp
6. Crown

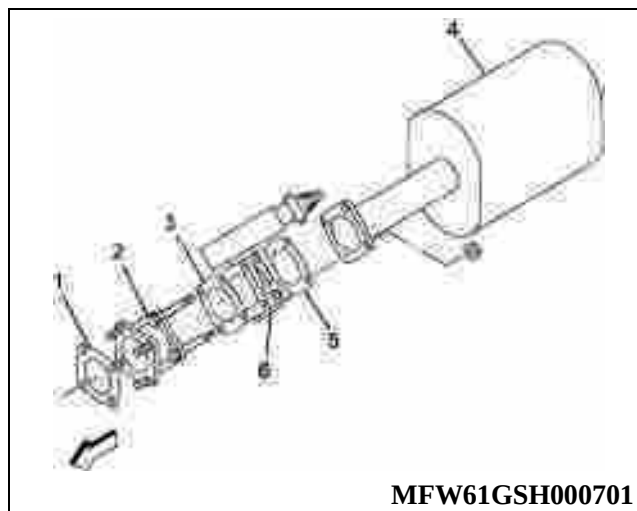
6. Install DPD exhaust temperature sensor connector.

Note:

Do not confuse the front and back direction



9. Install the air pipe to the exhaust throttle valve cylinder.



Legend

1. Exhaust temperature sensor 2 (catalyst inlet)
2. Exhaust temperature sensor 1 (filter inlet)
3. Pressure differential hose (front)
4. Pressure differential hose (rear)
5. Exhaust muffler
6. Exhaust throttle valve
7. Differential pressure tube (rear)
8. Differential pressure tube (front)
9. DPD assembly

Legend

1. Gasket
2. Exhaust pipe joint
3. Gasket
4. Muffler
5. Gasket
6. Exhaust throttle valve

10. Install the rear exhaust pipe.

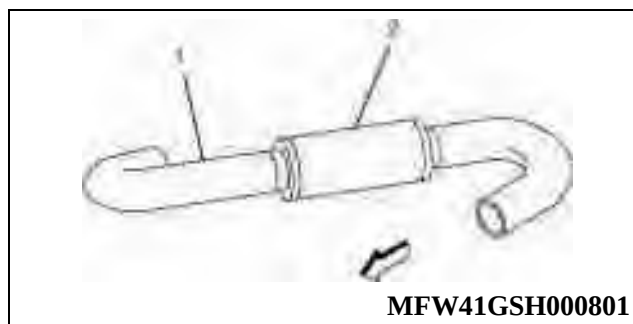
- Tighten the clamps.

7. Install the exhaust connector on the DPD.

8. Insert the exhaust throttle valve assembly into the exhaust port and install the muffler.

Tightening torque	N m (kgf m/lb ft)
41 (4.2 / 30)	

Tightening torque	N m (kgf m/lb ft)
18 (1.8 / 13)	



Legend

1. Rear exhaust pipe
2. Rear muffler

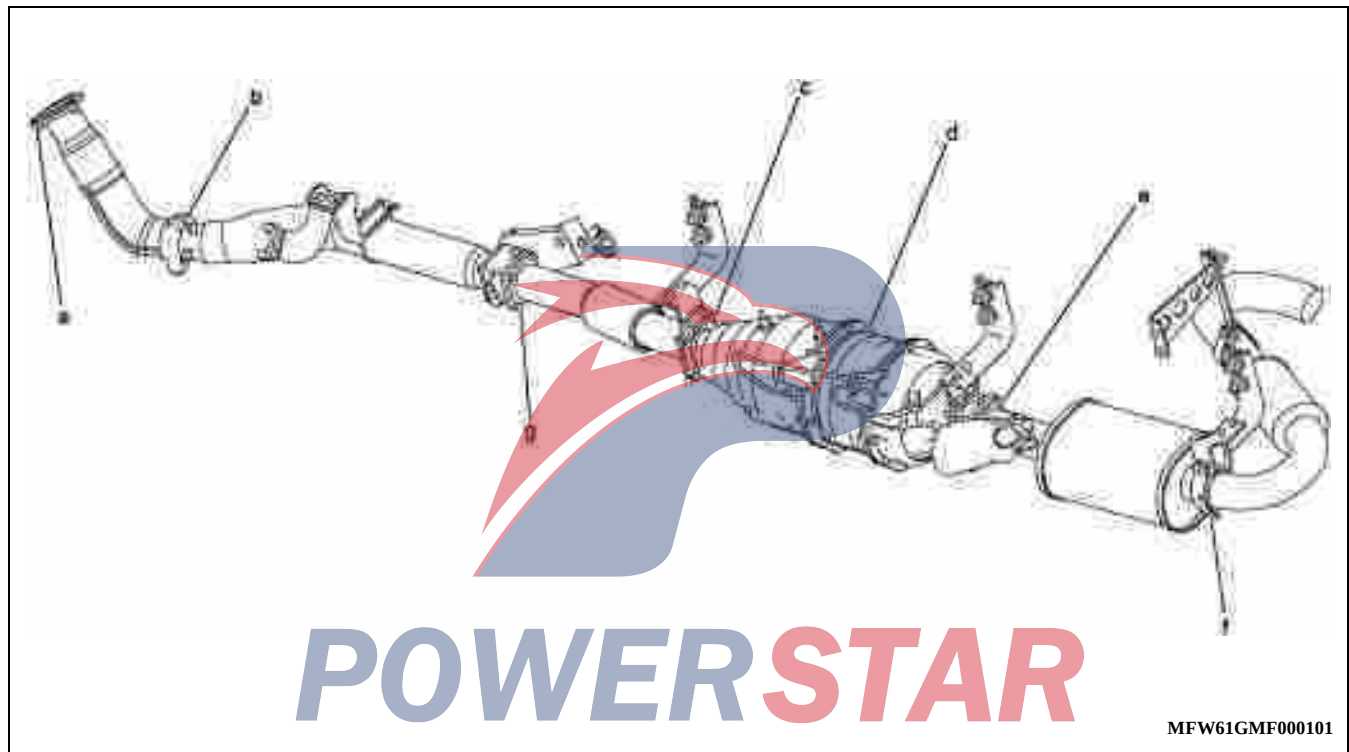
Note:

Start the engine after assembling the components. Make sure the coupling parts are free of air leaks or vibrations. If the DPD assembly has been disassembled, the following leak check should be performed carefully.

- Allow the engine to warm up sufficiently.

- Allow the engine to reach maximum speed. In this case, spray soapy water to the connection part of the differential pressure hose and sleeve nut, and make sure there is no air leakage.

Tightening torque table



Legend

(a) 67 N m (6.8 kgf m / 49 lbs. ft)

(b) 41N m (4.2 kgf m/30 lb.ft)

(C) 83 N m (8.5 kgf-m / 61 lb ft)

(d) 25 N m (2.5 kgf m / 18 lbs. ft)

(e) 41N m (4.2 kgf m/30 lb.ft)

(f) 18 N m (1.8 kgf m/13 lb ft)

(g) 29 N m (3.0 kgf m/22 lb ft)

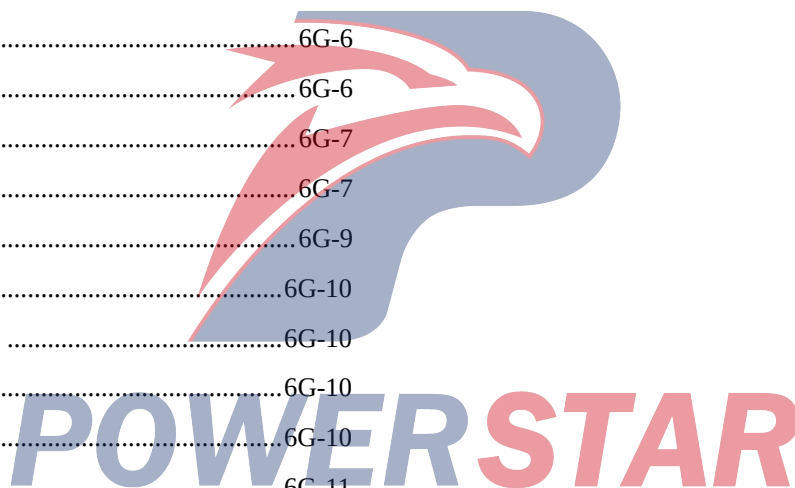
Engine

Engine lubrication

(6HK1)

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Lubrication system

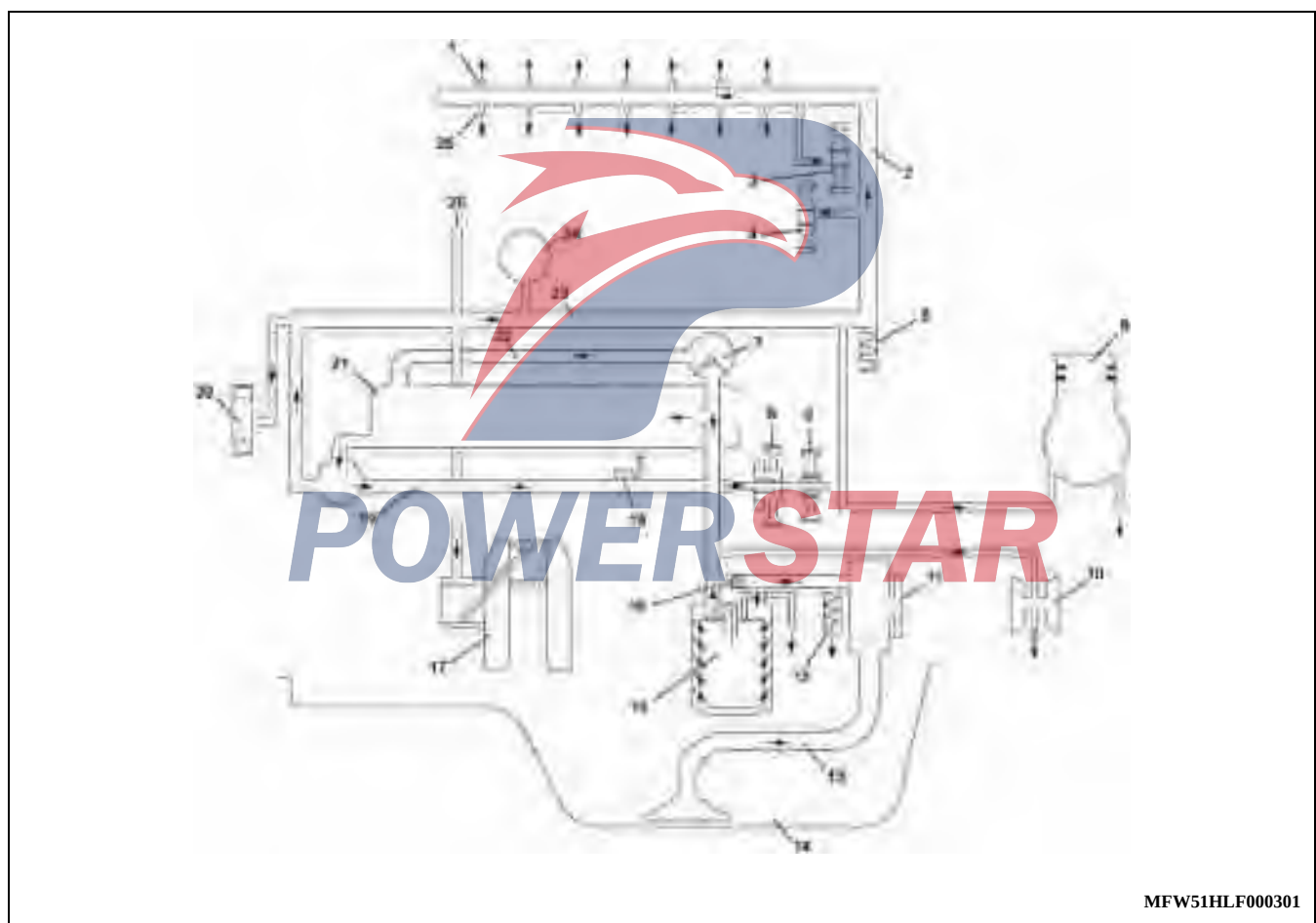
Precautions for maintenance

- When using a liquid packing unit for each disassembly, use a scraper to thoroughly remove all old gaskets from the parts and their mating parts. Use cleaning cloths and other tools to clean the parts to ensure that there are no oil, water, dirt or other adherents. When assembling, apply the specified type of liquid filler to each part.

- The coating quantity of liquid packing should not be too much or too little. Must be appropriate amount. Excessive liquid filling may cause blockage of passages or injection ports, thus causing combustion. The lack of liquid packing may cause oil leakage or water leakage.
- Start or finish the application so that the liquid fillers overlap each other.

Description of function and operation

Lubrication system diagram



Legend

- | | |
|--------------------------|----------------------|
| 1. Rocker arm | 5. Oil release valve |
| 2. Cylinder head channel | 6. Air compressor |
| 3. Idler gear C | 7. Oil thermostat |
| 4. Idler gear B | 8. Idler gear A |

9. PTO Idler (for PTO gear specifications only)

10. Turbocharger

11. Oil pump

12. Oil release valve

13. Oil filter network

14. Pan

15. Full flow oil filter

16. Bypass valve

17. Crankshaft

18. Injector

19. Special injection ports

20. Idler rotor (front)

21. Oil cooler

22. Bypass oil circuit (oil cooler)

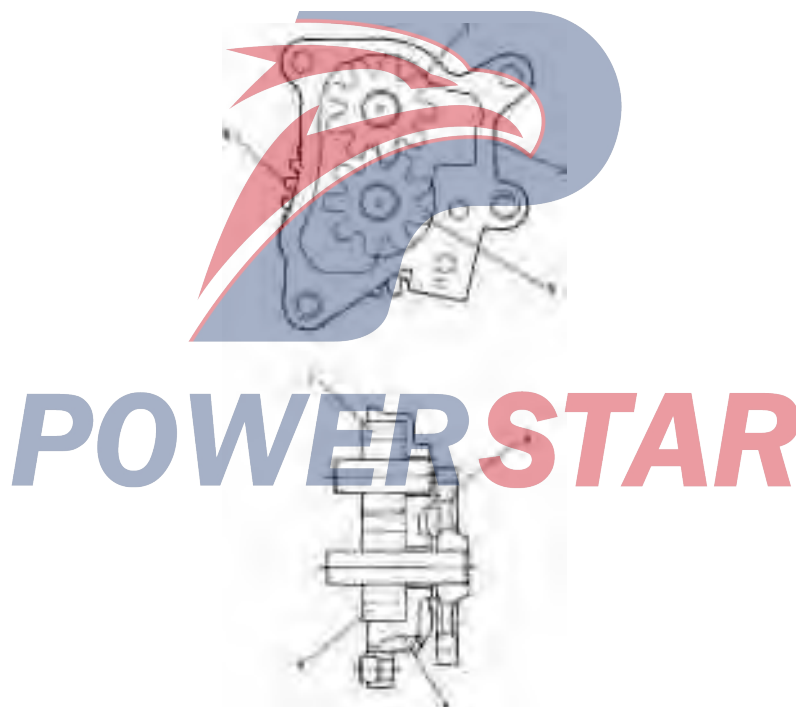
23. Major Oil Gallery

24. Hydraulic switch

25. PTO Gear Oil Supply (for PTO Gear Specifications Only)

26. Camshaft

Oil pump structure



MFW41HLF001001

Legend

1. Driven gear assembly

2. The main body of the oil pump

3. Driving gear assembly

4. Oil pump driving gear

5. The main body of the oil pump

6. Driving gear assembly

7. Driven gear assembly

8. Oil pump drive

Functional check:**Oil pressure inspection**

1. Check the engine oil level. If there is insufficient fuel, more motor oil should be added.
2. Remove the oil pressure switch.
3. Put on the hydraulic gauge.
4. Allow the engine to warm up sufficiently.
5. Measured at 1400 rpm, the oil pressure indicator exceeded 210kpa (2.1 kg/cm²/30 lb/in²).
6. Stop the engine.
7. Remove the hydraulic gauge.
8. Install oil pressure switch.
9. Start the engine and check for oil leaks.

Note:

Check the oil for abnormal dirt or fuel or water. If an abnormality is found, investigate the cause and take appropriate measures to correct and replace the oil.

Engine oil used

- Grade (API): CD Level

Viscosity (SAE) 10W-30 Biscoe clean super oil (DPD oil)

Note:

In this engine, be sure to use one of the engine oils listed above.

Engine oil quantity

- Park the vehicle on a level surface. After stopping the engine for at least 30 minutes, check the engine oil level with an oil level gauge. If the engine oil level is between the highest and lowest mark of the oil level gauge, the oil level is normal.

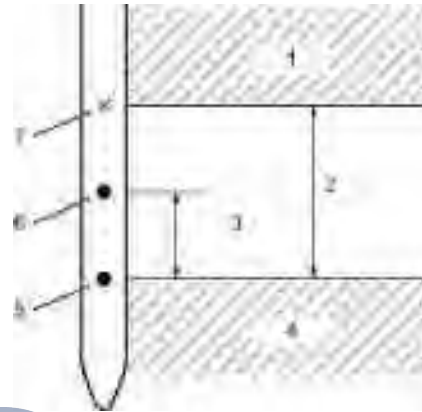
However, if the engine oil level is below the lowest mark, it should be filled. At the same time, the degree of contamination of the engine oil should be checked.

Specifications

Item	Specification
Lubrication system	Forced lubrication

Note:

If engine oil increases, it means that fuel or water may be mixed in. The reason must be investigated.



LNW41HSH000101

Legend

1. Need to change oil
2. Can be used
3. Supply oil to maximum oil level for refueling oil
4. Need to supply oil
5. Minimum oil level
6. Maximum fuel volume
7. The highest inspection level

Engine oil leak

- Check the oil seal, cylinder head casing and oil tray for oil leak.

Specifications

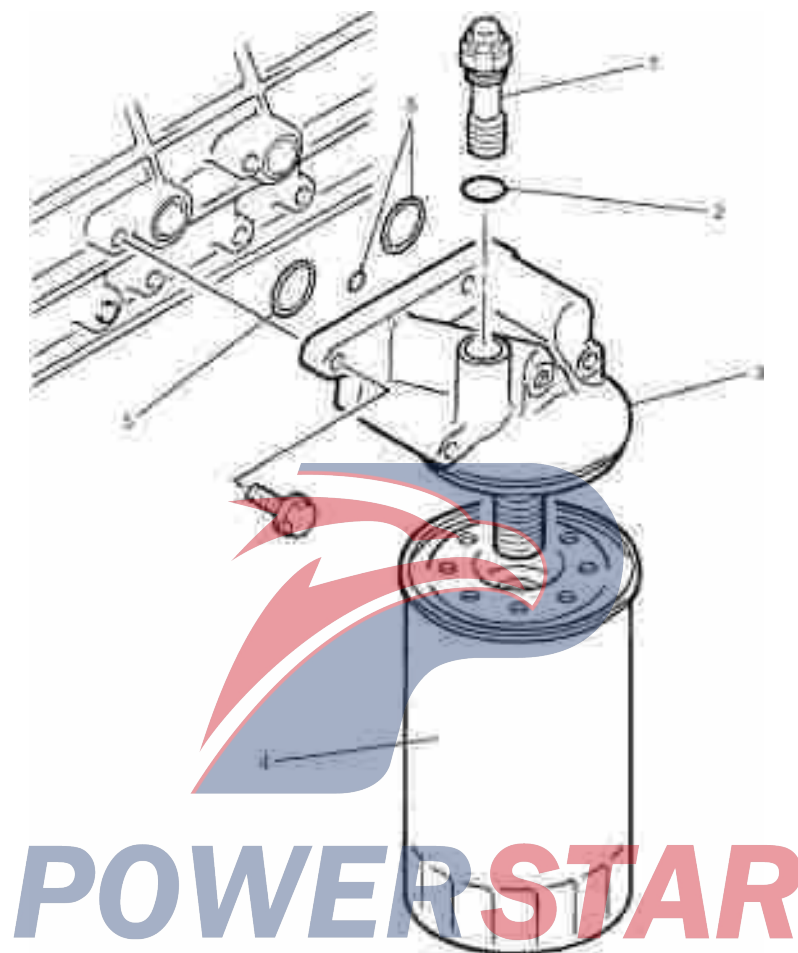
Item		Specification
	Type	Gear type
Oil pump	Output Pump speed: 4490 rpm Fuel : SAE #30 Oil temperature: 50°C / 122°F Output pressure: 390kpa (4 kg / cm ² / 57 lb / in ²) (L / IMP / minute)	115(25.3)
	Discharge valve opening pressure	kPa (kg/cm ² /lb/in ²)/rpm 780 (8 / 114)
Discharge valve opening pressure		kPa (kg/cm ² /lb/in ²)/rpm 490 (5.0 / 71)
Oil filter	Type	Full-flow paper, split paper, combined use
	Bypass valve opening pressure (inlet/outlet hydraulic pressure difference)	kPa (kg/cm ² /lb/in ²)/rpm 250 (2.5 / 36)
Oil cooler type		Water-cooled multi-board
Thermostatic valve (oil temperature thermostat)	Starting the temperature	(°C/°F) 90(194)
	Bypass valve opening temperature	(°C/°F) 95(203)
	By-pass valve lift	mm(inch) 5(0.1969)
Engine oil	Oil tray capacity	Liter (IMP) Maximum : 11.0 (2.42) and 16.0 (3.52) Minimum : 8 (1.76) and 13 (2.86)
	Total capacity	Liter (IMP) 13.5(2.97)/19.5(4.29)

Maintenance standard table

Item		Nominal size	Assembly standards	Use restrictions
Oil pump	Outer diameter of the driven shaft	(mm/inch) 16(0.63)	—	15.9(0.6259)
	Driven gear shaft, driven shaft and bushing clearance	(mm/inch) —	0.04—0.07(0.0016—0.0028)	—
	Body wall and gear tip clearance	(mm/inch) —	0.125—0.221(0.0049—0.0087)	—
	Body coupling surface and gear axial clearance	(mm/inch) —	0.064—0.109(0.0025—0.0043)	—
	Drive gear and shaft interfere with each other	(mm/inch) —	0.015—0.044(0.0006—0.0017)	—

Oil filter

Part



MF41HLF000501

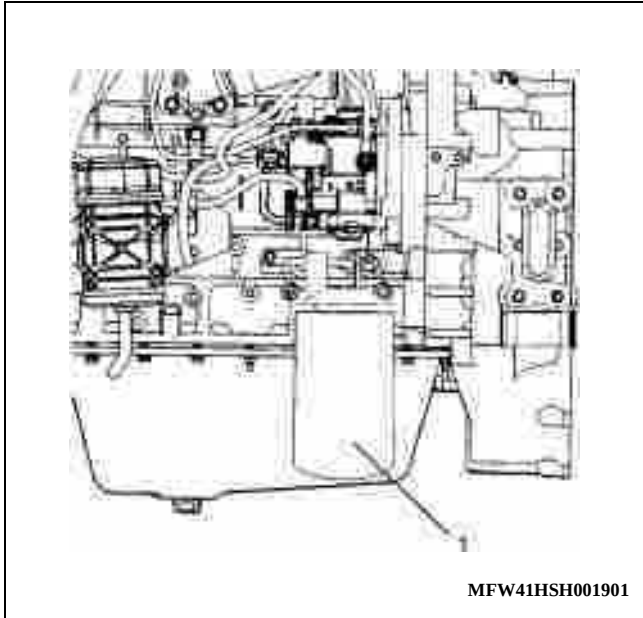
Legend

- | | |
|--------------------|---------------|
| 1. Bypass valve | 4. Oil filter |
| 2. O-ring | 5. O-ring |
| 3. Oil filter body | |

Disassembly

Disassembly

1. Remove the oil filter.
 - As the oil in the oil filter (1) accumulates to increase its weight, it should be supported with manual and oil filter wrenches (1-8522-1097-0) during disassembly.



2. Remove the bypass valve.
3. Remove the O-rings.
4. Remove the oil filter body.
5. Remove the O-rings.

Inspection

1. Check the oil filter.
 - If the oil filter is clogged, it should be replaced.
 - Check if the oil filter is deformed.
2. Check the bypass valve for clogging.



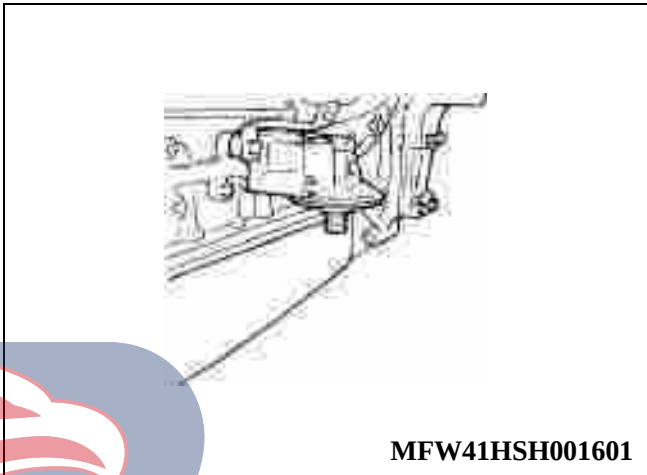
3. Replace the bypass O-ring.
4. Replace the oil filter body coupling O-ring.

Assembling

1. Install O-rings.

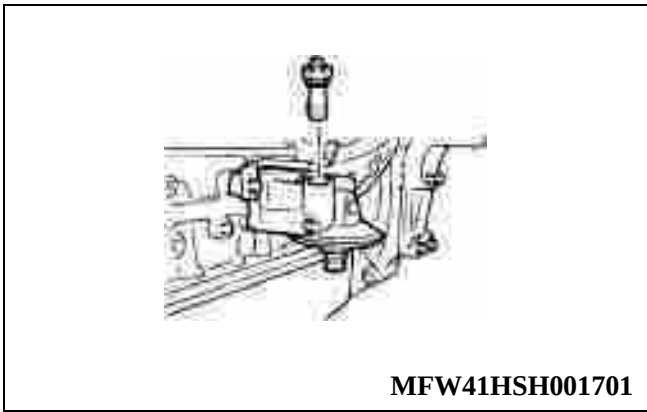
2. Install the O-ring on the oil filter and install it on the cylinder block.

Tightening torque	N m (kgf m /lb ft)
	38 (3.9 / 28)



3. Install O-rings.
4. Install the O-ring on the bypass valve and install it on the oil filter.

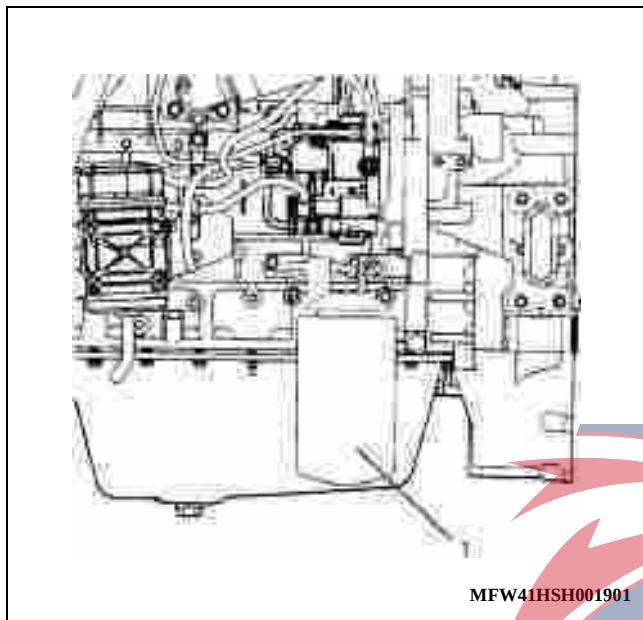
Tightening torque	N m (kgf m /lb ft)
	20 (2.0 / 14)



5. Install the oil filter (1).
 - Apply a thin layer of oil to the seal and secure it to the oil filter with screws. After the oil filter seal is in contact with the oil filter body, it should be tightened once. If the clip is short, tighten with an oil filter wrench (1-8522-1097-0).

Note:

- If the oil filter is tightened too much, the oil filter will distort and cause oil leakage.
- Thoroughly wipe off the oozing oil.



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Special tools

Graphic representation	Tool number/description
 1852210970	1-8522-1097-0 Oil Filter Wrench



Oil separator

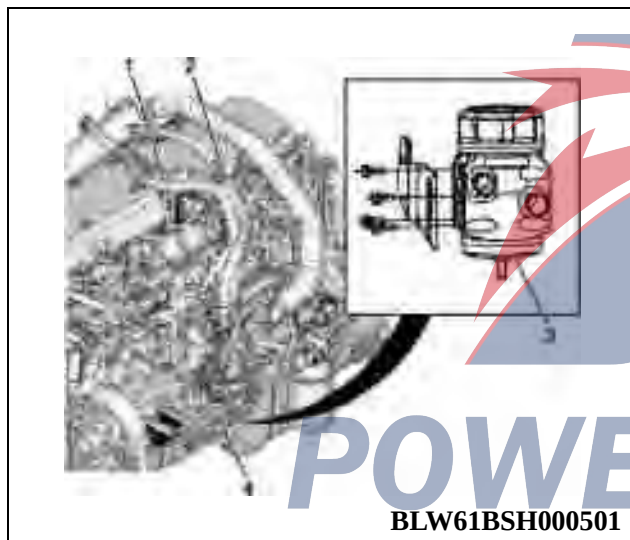
Maintenance precautions

Maintenance precautions

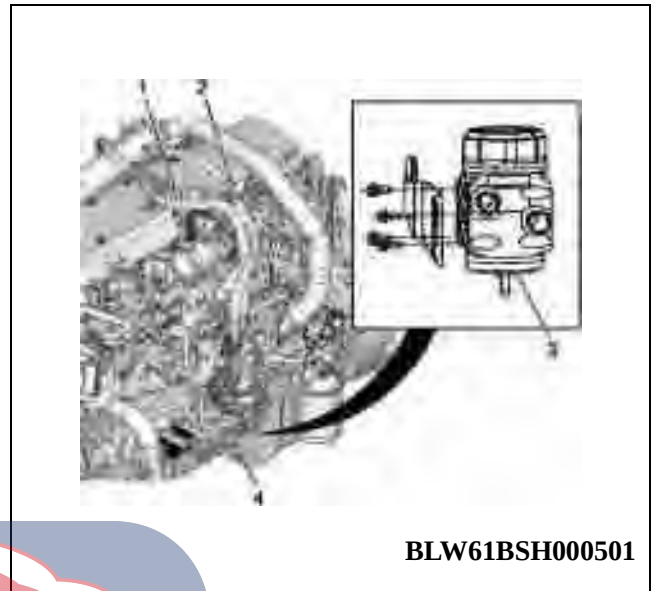
- The oil separator filter should be replaced once a year.

Disassembly

1. Air leak rubber hose
 - Disconnect the oil separator side rubber hose input and output.
2. Cover
3. Filter element
 - Remove the cap and remove the filter element.



3. Air leak rubber hose



Legend

1. Oil separator hose (out)
2. Oil separator hose (in)
3. Oil separator
4. Oil drain hose

Legend

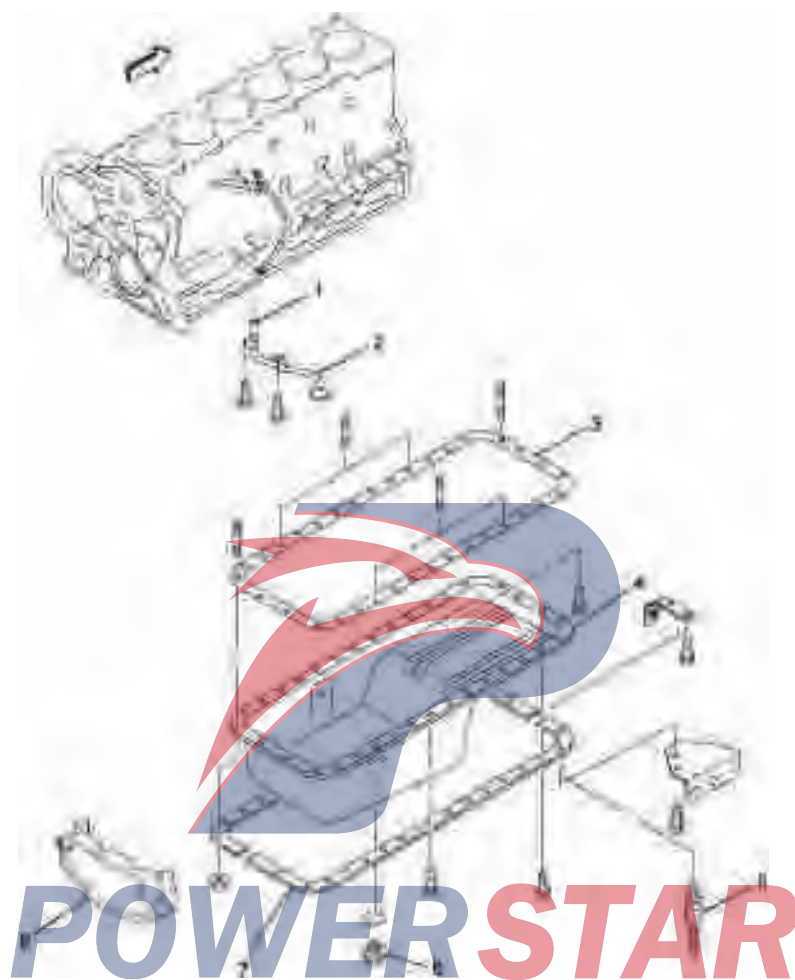
1. Oil separator hose (out)
 2. Oil separator hose (in)
 3. Oil separator
 4. Oil drain hose
-

Assembling

1. Filter element
 - Install a new filter element.
2. Cover
 - Install the O-ring on the lid and fully tighten the lid.

Oil pan

Part



BLW61HLF000101

Legend

- | | |
|-----------------------|---------------------|
| 1. O-ring | 5. Stud bolts |
| 2. Oil filter network | 6. Release plug |
| 3. Gasket | 7. Rubber |
| 4. Pan | 8. Rubber diaphragm |

Tear down

1. Remove the insulator.
2. Remove the connecting bolt and release the bracket.
3. Remove the connecting bolts and loosen the rubber.
4. Remove the oil pan.
5. Remove the oil pan gasket.

6. Remove the oil filter.

7. Remove the O-rings.

Reassemble

1. Install O-rings.

2. Install the O-ring on the oil screen and place it on the crankcase.

Tightening torque	N m (kgf m/lb ft)
M8: 24(2.4/17)	
M10: 48(4.9/35)	



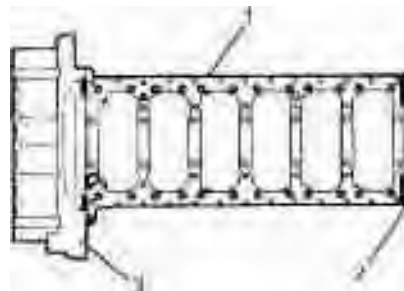
MFW41BSH004201

3. Put on the oil cover gasket.

- Apply a 2 mm (0.0787 in) thick, approximately 3 mm (0.1187 in) wide liquid filler (3 key 1207C) to the crankcase, flywheel housing, and transmission fitting to align the oil pan gasket with the studs. installation.

Note:

- Clean the mounting surface before applying the sealant.
- Apply the sealant within 7 minutes after applying the sealant.



MFW41BSH004301

Legend

1. Crankcase
2. Front gear box
3. Flywheel cover
4. Put on the oil pan.

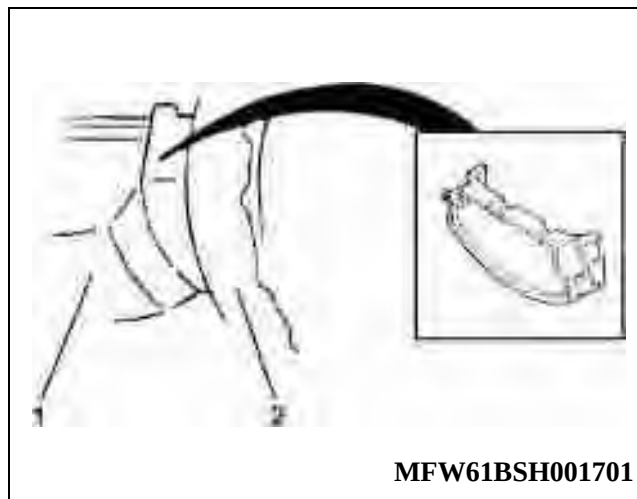
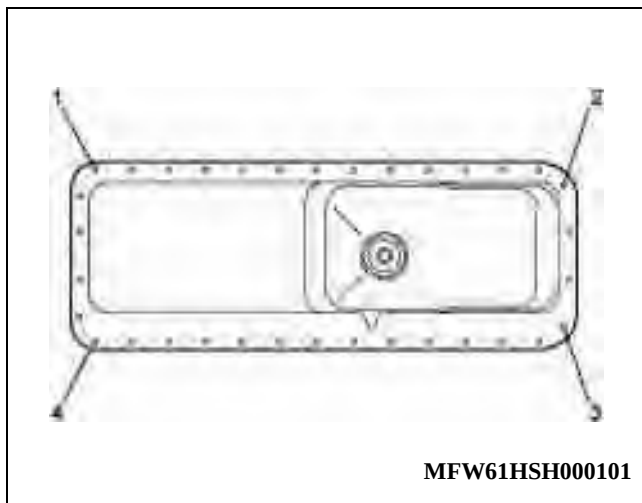
- When the oil pan is installed, the P/S pipe bracket (1) and the alternator bracket (2) will be tightened.



MFW61HSH000201

- In the tightening procedure, first tighten Point 1, 2, 3 and 4, then the other parts.

Oil pan tightening torque	N m (kgf m/lb ft)
	20 (2.0 / 14)

**Note:**

Check the oil drain plug thickness.

Drain plug tightening torque	N m (kgf m/lb ft)
	78 (8.0 / 58)

Legend

1. Pan
2. Flywheel cover

5. Mount the eraser.

- Install rubber on the board in advance.
- Also use an oil pan to tighten it.

6. Mounting brackets.

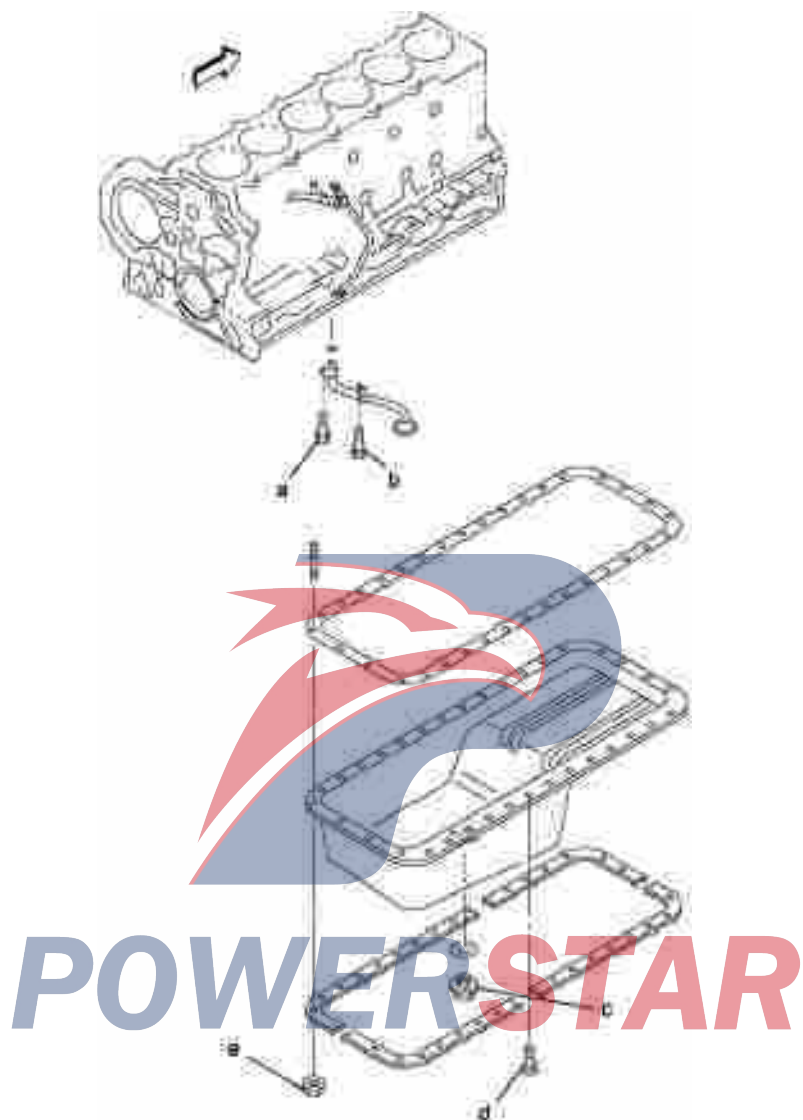
- Also use an oil pan to tighten it.

7. Install a rubber septum.

- Align the rubber septum radius with the oil pan cover, insert it, and secure the holes on both sides to the protruding part of the lid.

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Tightening torque table



MFW61BLF000501

Legend

(a) 24 N m (2.4 kgf m / 17 lbs. ft)

(b) 48N m (4.9 kgf m/35 lb.ft)

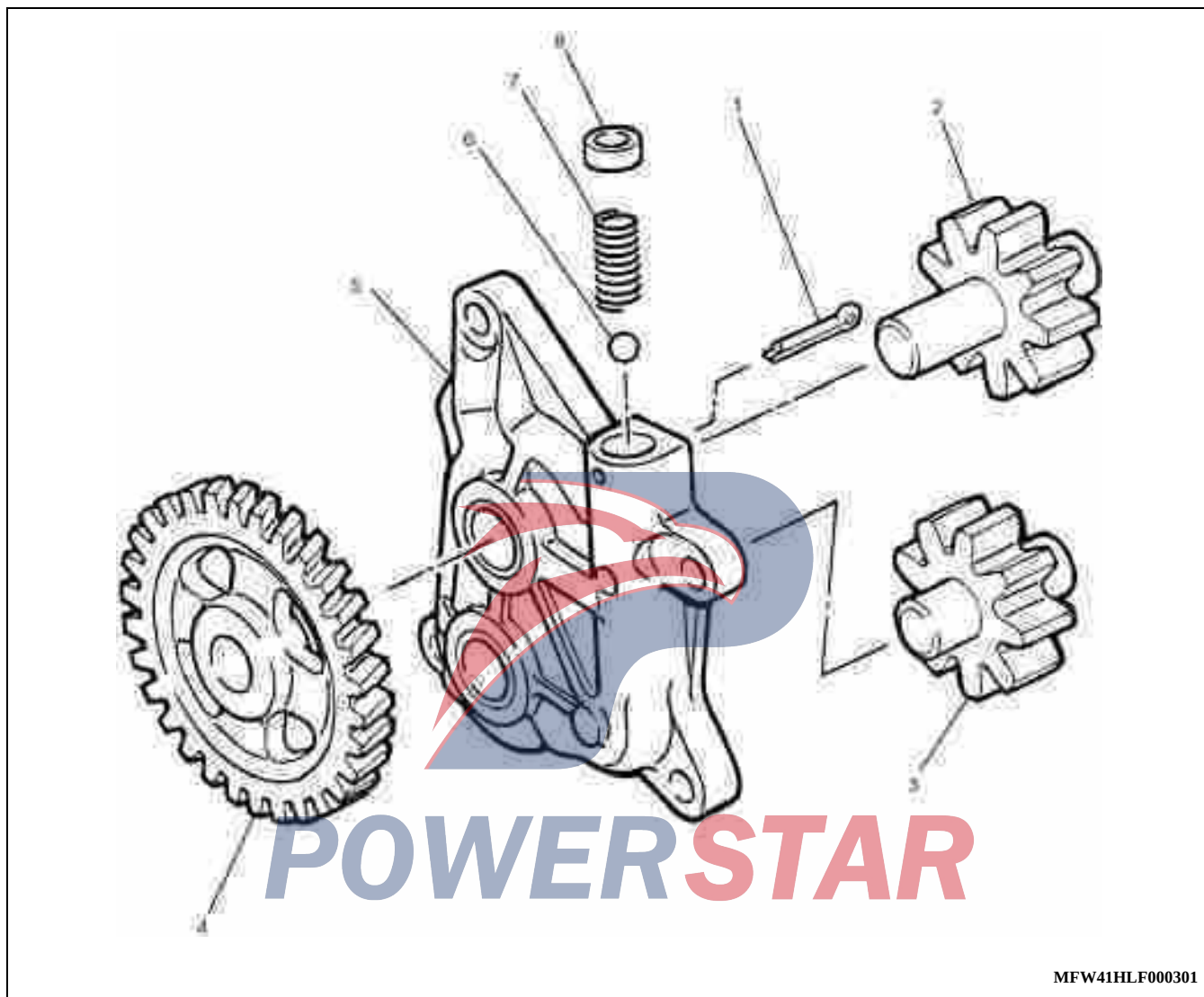
(C) 78N m (8.0 kgf-m / 58 lb ft)

(d) 20 N•m (2.0 kgf m / 14 lbs. ft)

(e) 20 N m (2.0 kgf m/314 lb.ft)

Oil pump

Part



MFW41HLF000301

Legend

- | | |
|---------------------|----------------------------------|
| 1. Split cotter pin | 5. The main body of the oil pump |
| 2. Driving gear | 6. Oil release valve |
| 3. Driven gear | 7. Spring |
| 4. Driving gear | 8. Spring seat |

Tear down

1. Remove the cylinder head assembly. Please refer to the cylinder head of the mechanical part of the engine.

2. Remove the air compressor.

Please refer to "Air Compressor" outside the engine.

3. Remove the fuel supply assembly.

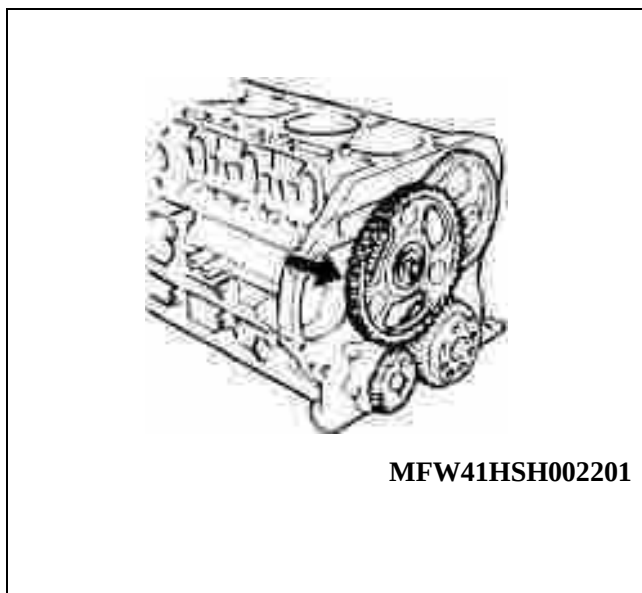
Please refer to the "supply pump" outside the engine.

4. Remove the oil pan.

5. Remove the flywheel cover.

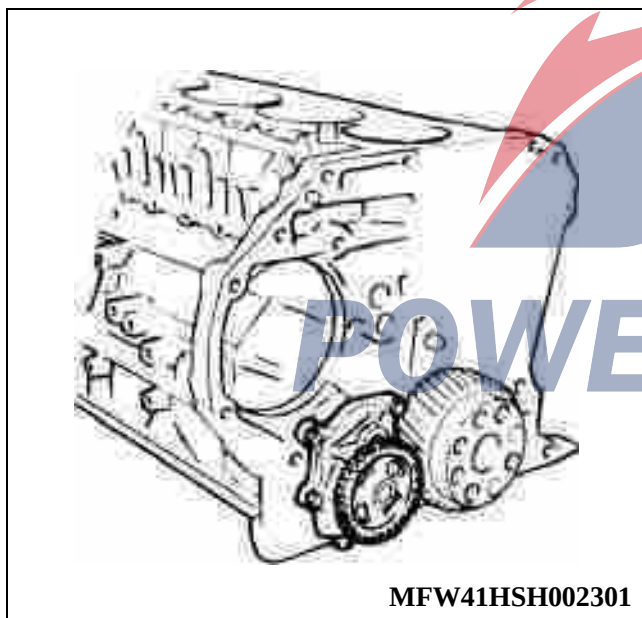
Please refer to the crankshaft of the mechanical part of the engine.

6. Remove the idle gear A.



7. Remove the idler A shaft.

8. Remove the oil pump assembly.



Disassembly

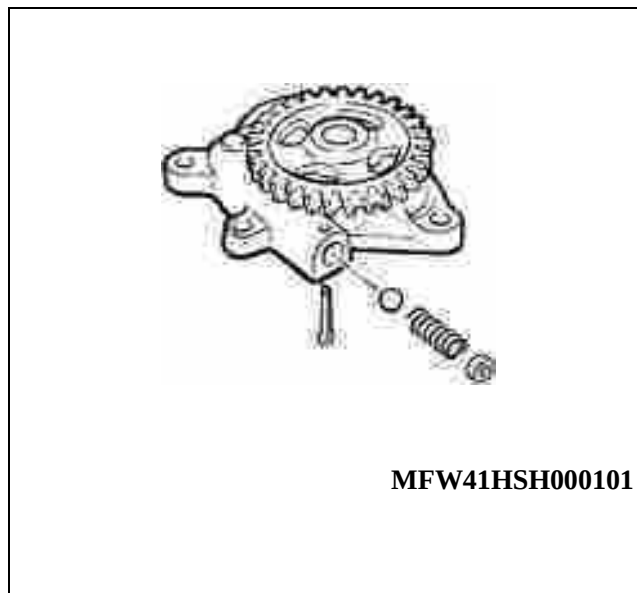
1. Remove the driven gear.

2. Remove the drain valve.

- Remove the split pin, and then withdraw the spring seat, spring and drain valve.

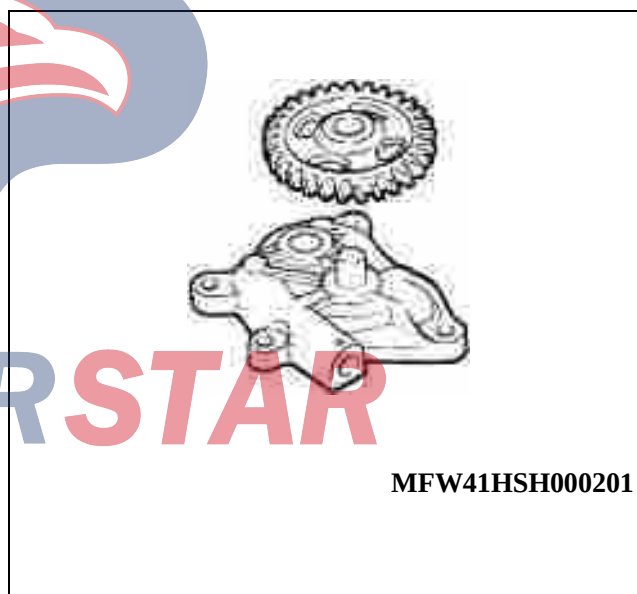
Note:

After removing the split pin, the spring seat and spring will spring out so do not let them fall off.



3. Remove the transmission gear.

- Use a gear puller to remove the drive gear.



4. The main body of the oil pump is an integral device.

Inspection

1. Measure the driven gear.

a. Measure the outer diameter of the driven gear shaft and the gap between the shaft and bushing.

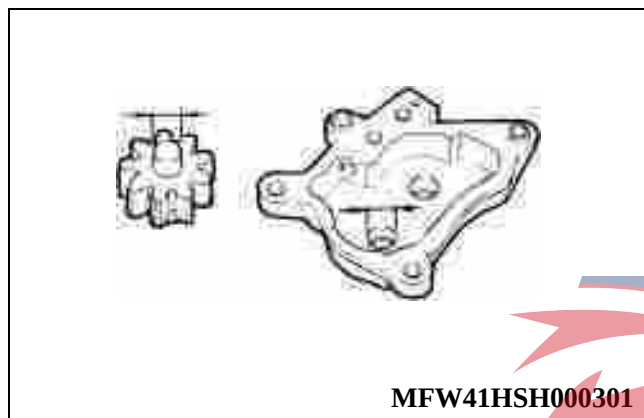
mm(inch)			
	Nominal size	Assembly standards	Use restrictions
Shaft outer diameter	16(0.6299)	—	15.9(0.626)
Assembly standards		mm(inch)	

Shaft and shaft clearance	0.04—0.07(0.0016—0.0028)
---------------------------	--------------------------

- The outside diameter and the gear side clearance should also be measured.

Note:

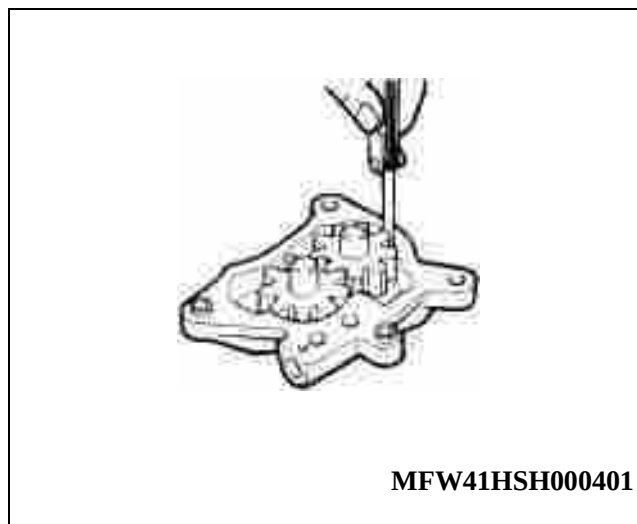
The inner diameter of the bushing is measured by the measuring method of the cylinder block side bushing.



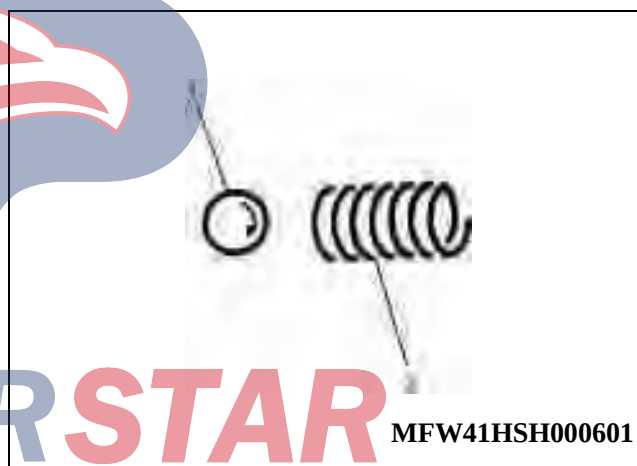
a. Measure the clearance between the pump body and the gear tips.

Assembly standards	mm(inch)
Clearance	0.125(0.049)—0.221(0.087)

- The same method is also used to measure the clearance on the drive gear side.



2. Check whether the spring seat is damaged.
3. Check the oil drain valve and spring for damage.

**Legend**

1. Oil release valve
2. Spring

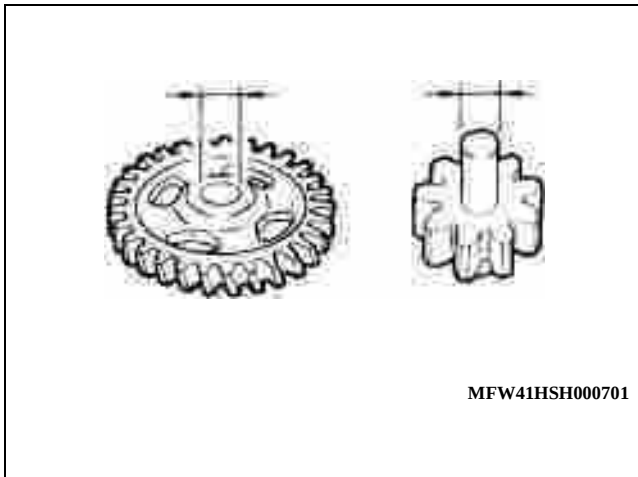
4. The measurement of the active gear is carried out.

- Measure the interference between the shaft and the drive gear.

Assembly standards	mm(inch)
interference	0.015(0.0006)—0.044(0.0017)

Note:

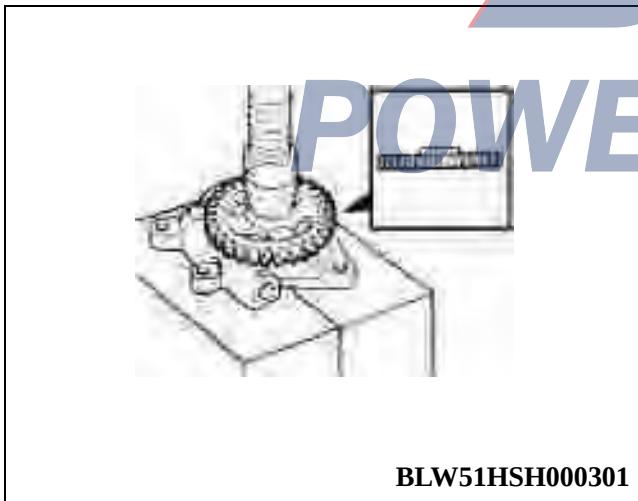
Check the gear teeth, the degree of separation, and the degree of tilt.



5. Check the pump body for damage.

Assembling

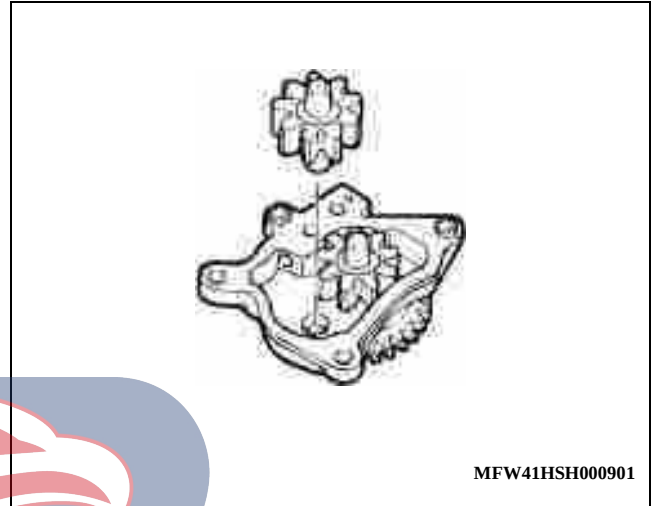
1. Install the pump body.
2. Put on driving gear.
 - Install the transmission gear on the pump body.
 - Lubricate the gear shaft, install it on the pump body, position the protruding part of the drive gear as shown, and press it together with the press.



3. Install the drain valve.
 - Insert the oil drain valve, spring and spring seat in turn and then choke them with the split pin.
4. Install driven gear.
 - Grease the driven gear and install it on the pump body.

Note:

- Make sure that the driven gear does not come off.
- Turn by hand to ensure that there is no abnormality.
- After installation, there is a certain degree of difference between the driven gear side and the cylinder block.



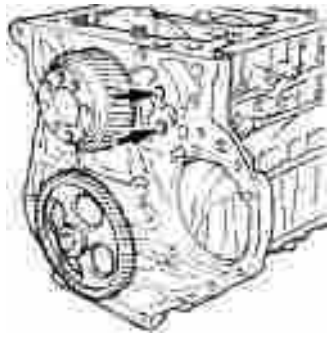
Reassemble

Remove the gasket completely from the flywheel housing and the cylinder block mounting surface.

1. Put on the oil pump.
 - The oil is applied in oil pump gear.
 - Remove the liquid filler from the pump mounting surface and apply oil to the crankcase and cylinder liner.

Note:

Do not allow oil to enter the pump's mounting surface.

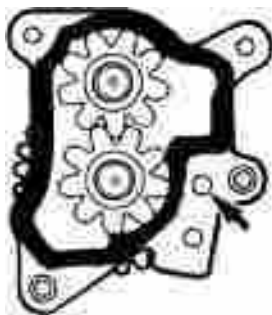


MFW41BSH005101

- As shown, apply a liquid filler (San key brand TB 1141E) to the oil pump mounting surface.

Note:

- Apply the sealant within 20 minutes after applying the sealant.
- Do not allow liquid packing to enter the hole in the gear or arrow section.
- Make sure that the driven gear does not come off.



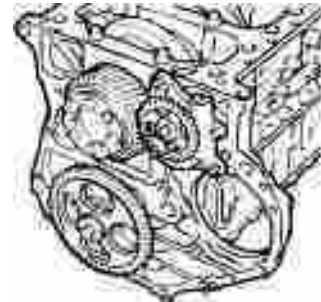
MFW41BSH005201

Align the nails and install the oil pump.

Tightening torque	N m (kgf m/lb ft)
24 (2.4 / 17)	

Note:

- There is no alignment mark on the gear.
- Install the oil pump before loading the idle gear A.



MFW41BSH005301

2. Install idler A-axis and idler A.

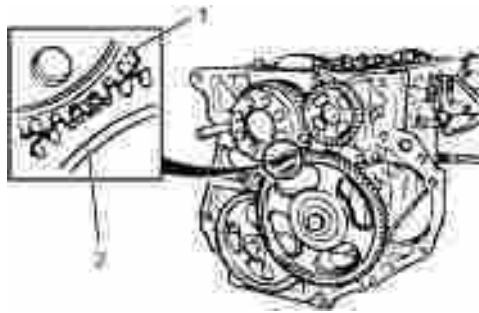
- Apply oil to the gear inside cycle and pushing section.
- Insert the gear into the shaft and fit into the collar.

Note:

Turn the crankshaft so that cylinder number 1 enters top dead center.

- Align the shaft positioning pin with the cylinder positioning pin hole, and align the "L" mark of the crank gear with the "O" mark of the idle gear A, and then install it.
- Apply molybdenum disulfide adhesive to the threads and base of the bolts and washers and tighten.

Tightening torque	N m (kg/m/ inch)
134 (13.6 / 98.4)	



MFW61BSH001801

Legend

1. Crankshaft gear
2. Idler gear A

Note:

- After installation, make sure that the marks are aligned with each other.
- After installing the flywheel cover, tighten the idler gear A shaft with 3 bolts.

3. Install the flywheel cover.

Please refer to the crankshaft of the mechanical part of the engine.

4. Put on the oil pan.

5. Install the oil supply pump assembly.

Please refer to the "supply pump" outside the engine.

6. Install an air compressor.

Please refer to "Air Compressor" outside the engine.

7. Install the cylinder head.

Please refer to the cylinder head of the mechanical part of the engine.

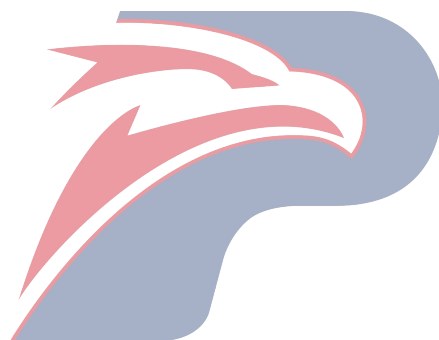
Engine

Engine Speed Control System

(6HK1)

Table of Contents

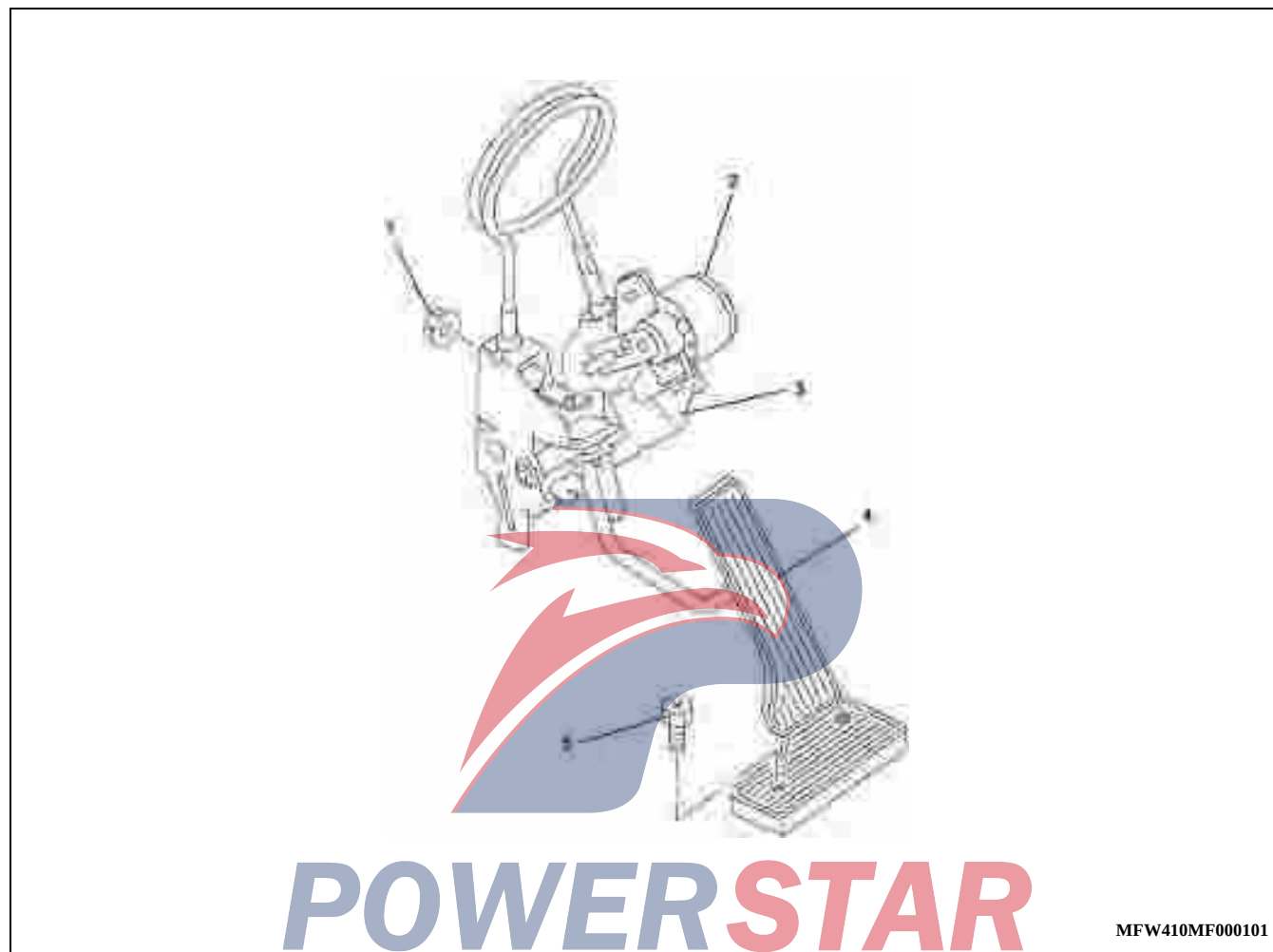
Throttle control components	6H-2
Part	6H-2
Disassembly	6H-2
Reassemble	6H-2



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Throttle control components

Part



Legend

- | | |
|--------------------------------------|----------|
| 1. Nuts | 4. Pedal |
| 2. Throttle sensor | 5. Bolt |
| 3. Control link and bracket assembly | |

Disassembly

1. Remove the retaining nut that controls the linkage and bracket assembly.
2. Remove the accelerator pedal fixing bolts.
3. Move the harness connector to the throttle sensor.

Note:

When removing the harness connector, remove the battery ground.

Reassemble

1. Install the harness connector.
2. Install accelerator pedal attachment bolts.
3. Install the control link and bracket assembly.

Note:

- After adjusting the accelerator pedal, always use the scan tool for the accelerator pedal test.
- For detailed information and adjustments for the throttle sensor, accelerator pedal, etc., refer to the Engine Control System section.

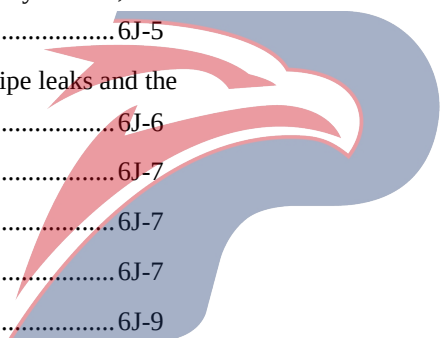
Engine

Suction

(6HK1)

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Part	6J-11
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Reassemble	6J-13



POWERSTAR

Turbo charging system

Precautions for maintenance

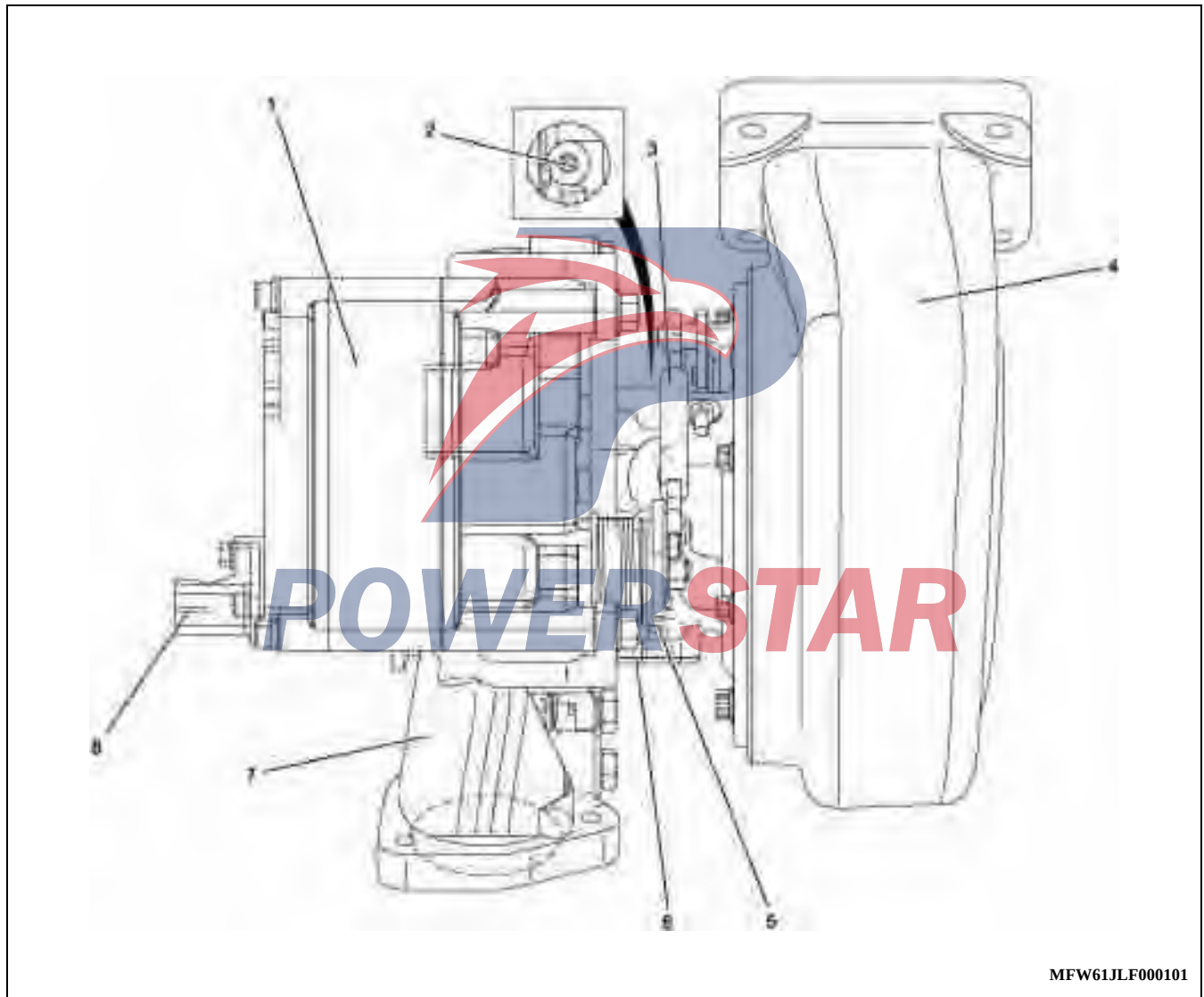
- The turbocharger is designed for high-speed (about 100,000 rpm) rotation. Therefore, as long as the wing, outer cover or other parts are damaged or bent slightly, or there are other faults, replace the relevant parts or turbine assembly.

- When doing maintenance, every effort should be made to prevent dust or debris from entering these parts.

Caution:

Loosen the bolts, nuts, and V-clamps of the turbocharger assembly. Do not remove any components.

Description of function and operation



Legend

- | | |
|--------------------------|------------------------------|
| 1. Actuator | 5. Oil outlet |
| 2. Oil inlet | 6. Central cover assembly |
| 3. Rod end assembly | 7. Compressor cover assembly |
| 4. Turbine hood assembly | 8. Connector |

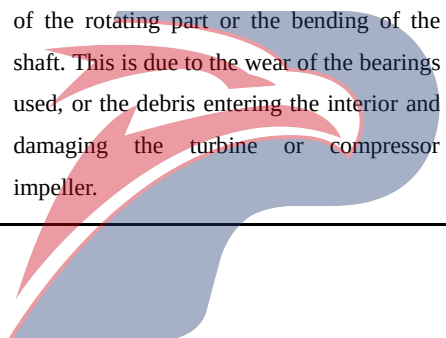
Failure symptoms list

- Failure or vibration
- Consider the turbocharger is normal as a whole while the output is low
- Exhaust or intake pipe leaks, exhaust is white



Phenomenon: Failure or vibration

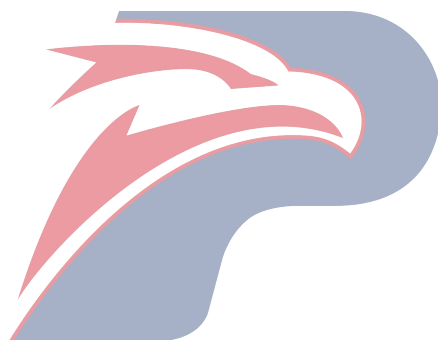
Phenomenon	Probable causes	Corrective measures
Failure or vibration	It is caused by the damage of the bearing or the rotating part and its peripheral contacts. Forgotten oil changes, debris entering the fuel supply line, plugging of the oil filter, or driving in the event of damage or repeated sudden starts and stops may result in bearing damage.	Check and repair
	If there is no abnormality in the lubrication system, but it is found that the rotating part is in contact with its periphery and causes damage, it may be due to the poor balance of the rotating part or the bending of the shaft. This is due to the wear of the bearings used, or the debris entering the interior and damaging the turbine or compressor impeller.	Check and repair



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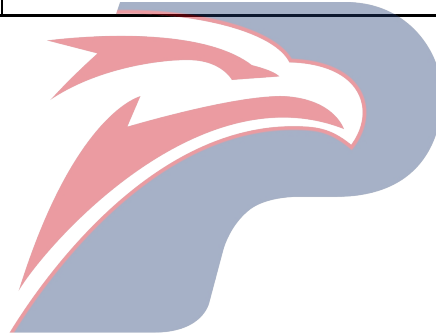
Phenomenon: Think turbochargers are usually normal, but production is low

Phenomenon	Probable causes	Corrective measures
Consider the turbocharger is normal as a whole while the output is low	Air leak of exhaust system	Repair or replace
	Exhaust resistance may be caused by carbon deposition or exhaust muffler deformation	Repair or replace
	Compressor output side air leak, air filter dirty	Clean or replace
	Compressor inside dirty	Cleaning
	The rotor shaft is dirty	Disassembly and cleaning

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Phenomenon: The exhaust pipe or intake pipe leaks and the exhaust gas is white

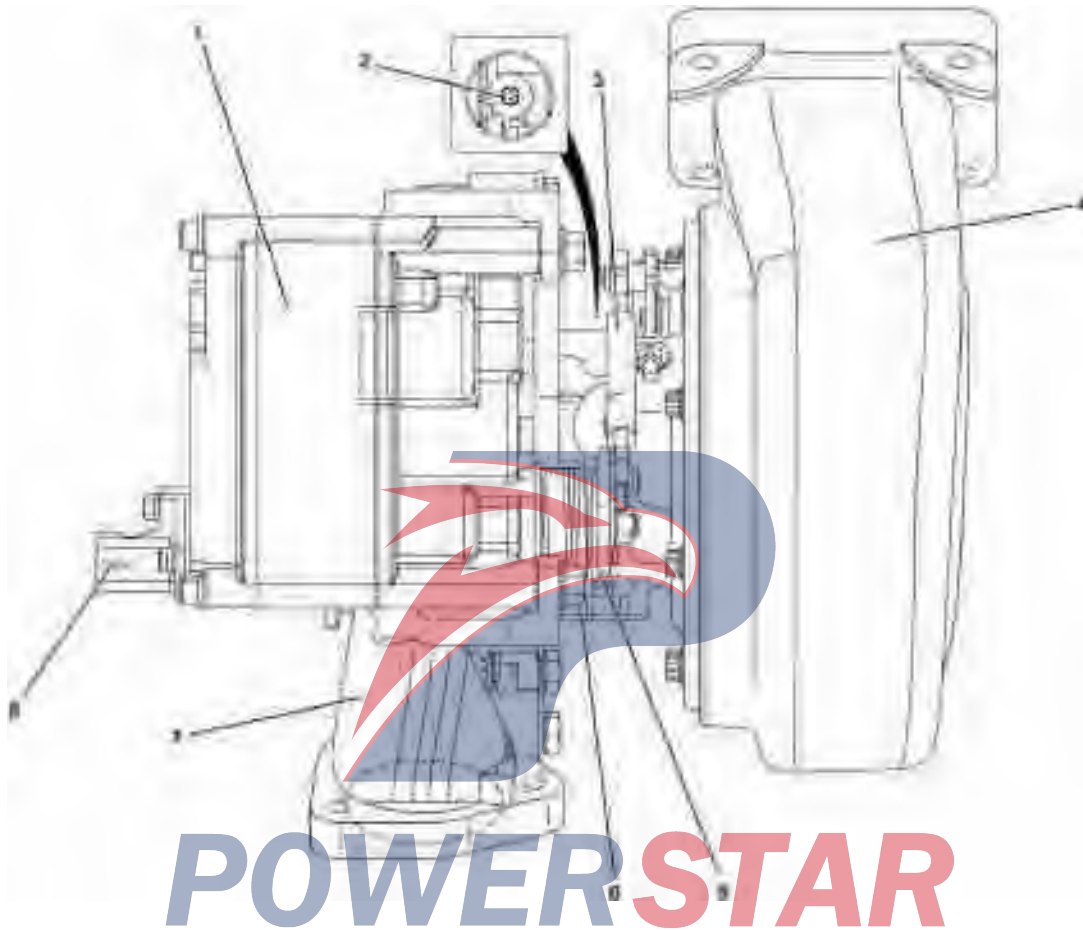
Phenomenon	Probable causes	Corrective measures
Exhaust pipe leaks or intake manifold exhaust is white	Due to the cause of the blockage, the air filter becomes dirty and leaks.	Clean or replace
	Clogging, extrusion, or deformation of tubing.	Repair or replace
	Turbine shaft partially worn or damaged	Repair or replace
	Closed PCV; aeration device plugging	Clean or replace
	Closed PCV; Inhalation on the turbo compressor side due to failure of the inflatable ring	Repair or replace



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Turbocharger

Part



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Legend

- | | |
|--------------------------|------------------------------|
| 1. Actuator | 5. Oil outlet |
| 2. Oil inlet | 6. Central cover assembly |
| 3. Rod end assembly | 7. Compressor cover assembly |
| 4. Turbine hood assembly | 8. Connector |

Inspection

Check and measure turbocharger parts and replace them only if an anomaly is found.

Note:

Oil sticking to the inner surface of the compressor port is not a fault. This is the lubricating oil of the turbine shaft, and the normal exhaust gas is not abnormal white

smoke.

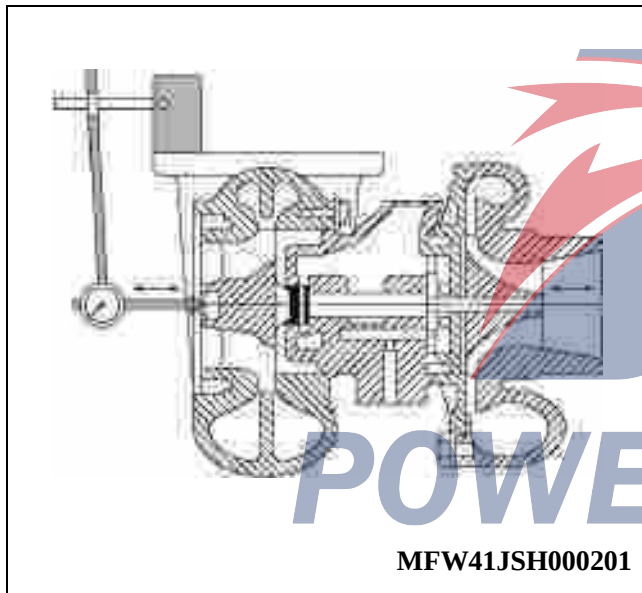
A. External inspection of components

1. Visually inspect each component for cracks, interference, damage, deformation, leaks, and leaks.
2. Rotate the turbine shaft by hand to see if it can rotate smoothly and check the axial and rotational clearances of the turbine shaft.

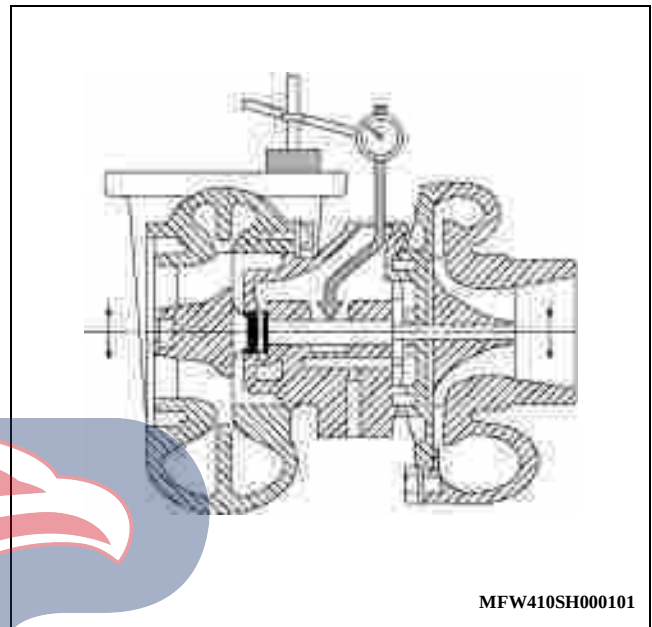
B. Measurement of the clearance of turbine shaft**1. Axial measurement of turbine shaft**

- A dial indicator is set on the shaft of the turbine shaft to zero the indicator.
- Rotate the turbine shaft back and forth to read the axial clearance. The gap should be within the following range.

Axle clearance	mm(inch)
6HK1-TCN	0.043(0.00169)–0.094(0.0037)
6HK1-TCC	
6HK1-TCS	

**Note:**

If the clearance of turbine shaft is not within the specified value range and a fault is found in the central cover assembly, the central cover assembly should be replaced.

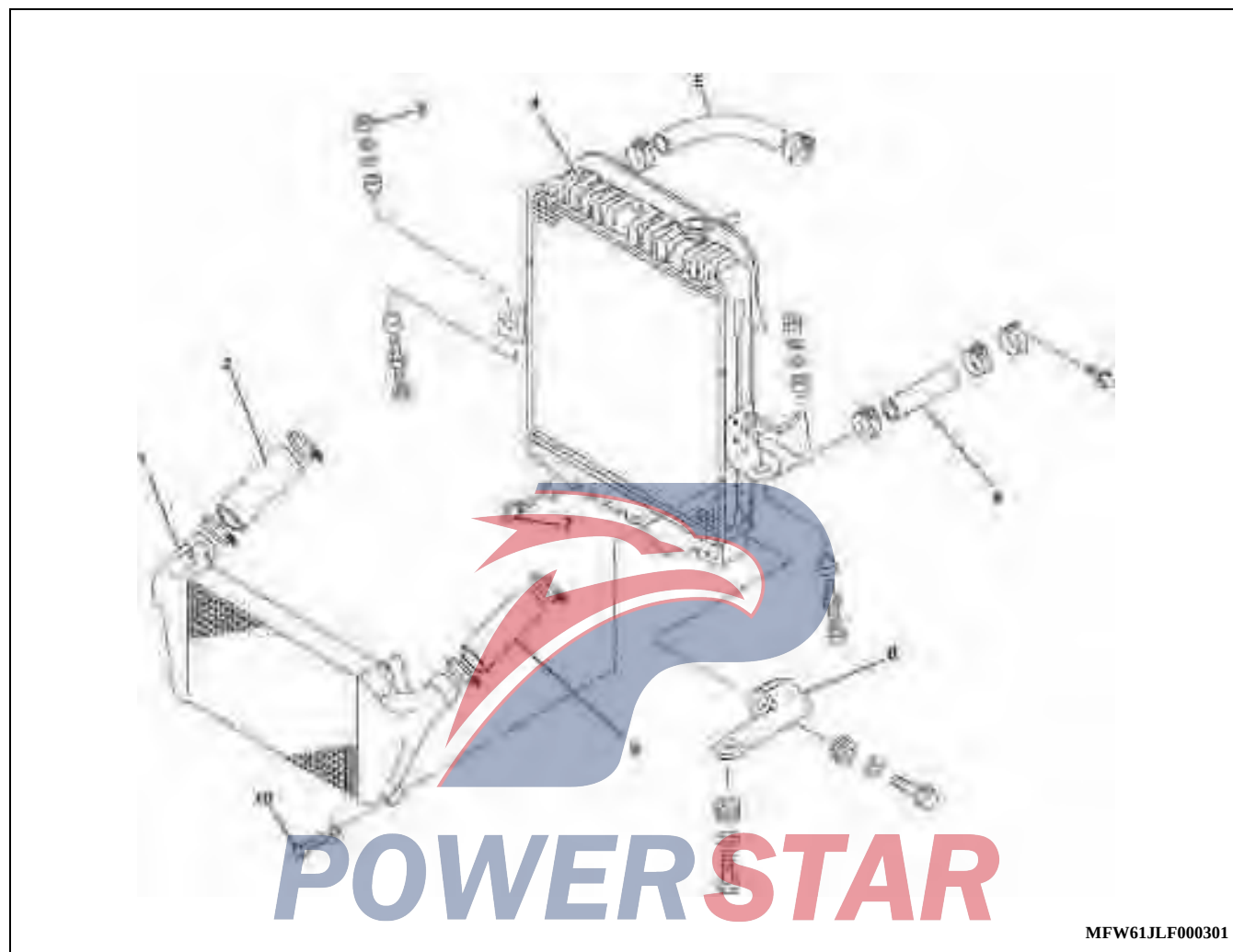
**2. Turbine shaft radial measurement**

- Set a dial indicator (including an offset pin) in the turbine shaft to zero the indicator.
- Rotate the turbine shaft up and down to read the radial clearance. The gap should be within the following range.

Radial clearance	mm(inch)
6HK1-TCN	0.086(0.00338)– 0.103(0.00406)
6HK1-TCC	
6HK1-TCS	

Intercooler

Part



Legend

- | | |
|--------------------------|--------------------------|
| 1. Intercooler | 6. Radiator hose (lower) |
| 2. Rubber hose (inside) | 7. Drainage cocks |
| 3. Nuts | 8. Radiator brace |
| 4. Radiator assembly | 9. Rubber hose (outside) |
| 5. Radiator hose (upper) | 10. Bolt |

Tear down

1. Remove the heat sink assembly.

Refer to the “Radiator”

2. Remove the four intercooler attachment bolts on the heat sink assembly and remove the intercooler.

Inspection

1. The distortion of intercooler fins may reduce the heat dissipation effect and consequently cause the engine performance to fall. Therefore, the fins must be repaired. Be careful not to damage the connector fittings during repairs.

2. Check the dirt in the intercooler. If it is particularly dirty, it should be clean.

Reassemble

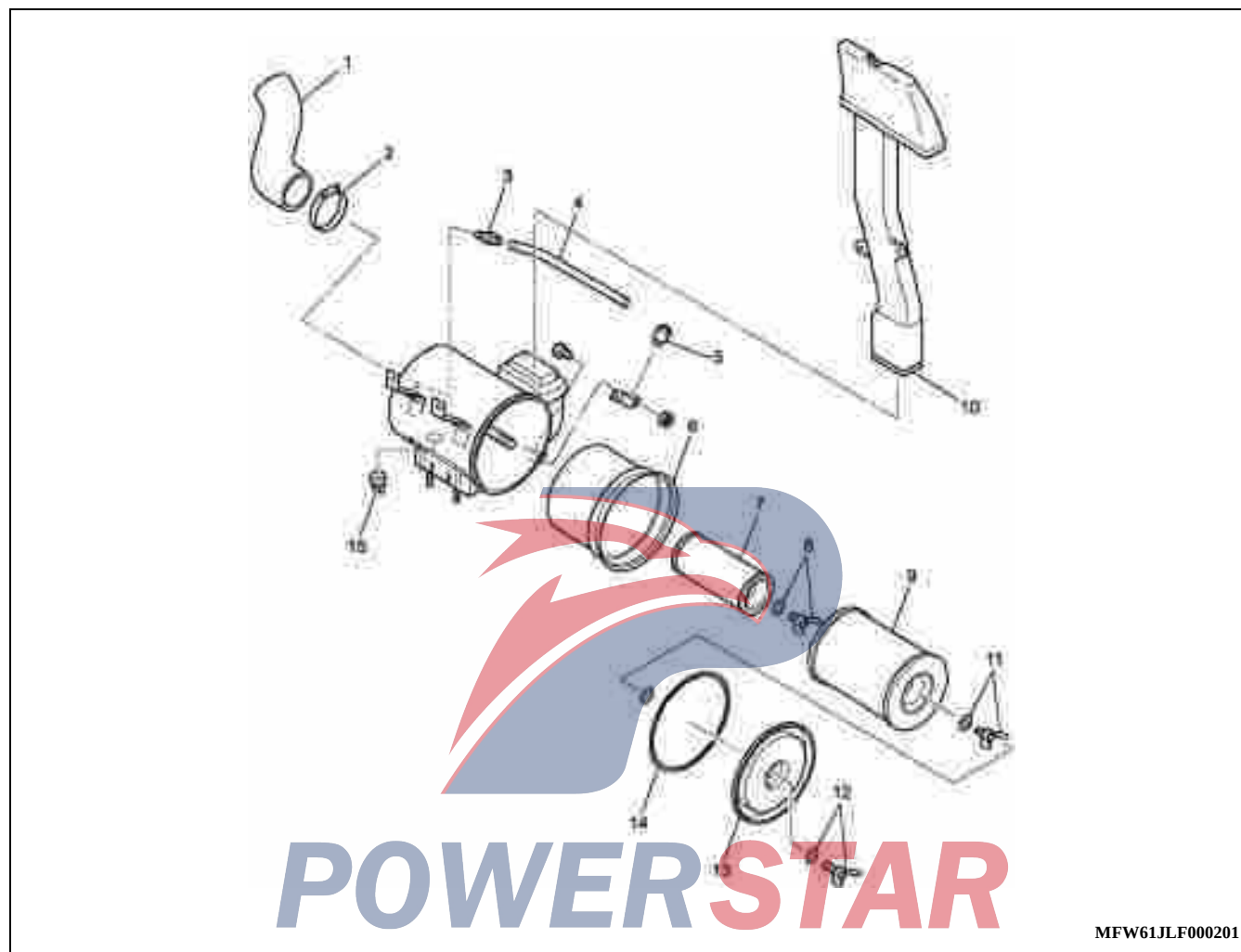
1. Install the intermediate cooler on the radiator.
2. Install the heat sink assembly.

Refer to the “Radiator”



Air cleaner element

Part



Legend

- | | |
|------------------------------|------------------------------|
| 1. Intake hose | 9. Filter (external) |
| 2. Clamping | 10. Vertical intake pipe |
| 3. Connector | 11. Wing nut, sealing washer |
| 4. Indicator hose | 12. Wing nut, sealing washer |
| 5. Indicator | 13. Air filter cover |
| 6. Adapter | 14. Gasket |
| 7. Filter element (internal) | 15. Drain valve |
| 8. Wing nut, sealing washer | |

Tear down

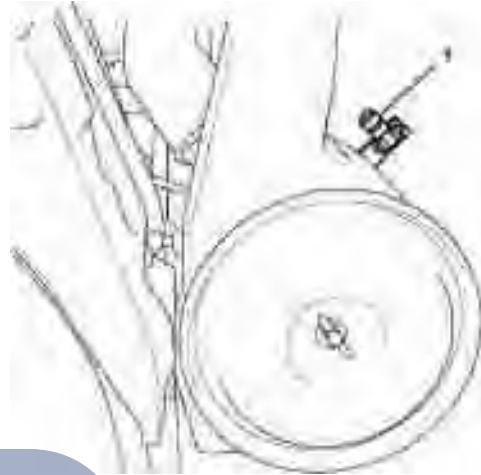
1. Tilt the cab.
2. Remove the air cleaner cover. The removal procedure varies according to the specifications.

- Use the thumbscrews to rotate the middle part of the screw to the left and remove it, and then remove the air cleaner cover.



MFW41JSH000801

- Visually inspect the dust indicator (1); if the back is red, press the button to return it to normal and check the filter element.



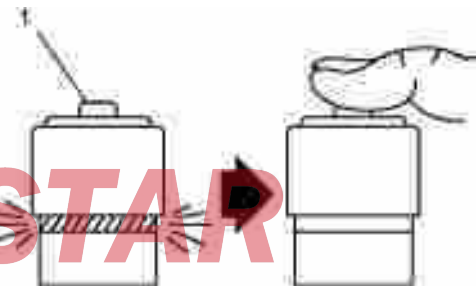
MFW61JSH000201

3. Remove the filter element.

- For thumb screw type, turn the middle part of the screw to the left and remove it, then remove the filter element. (Inner and External)



MFW41JSH001001



MFW41JSH002201

Inspection

1. Dust injector inspection

Legend

1. Button

2. Clean the filter element.

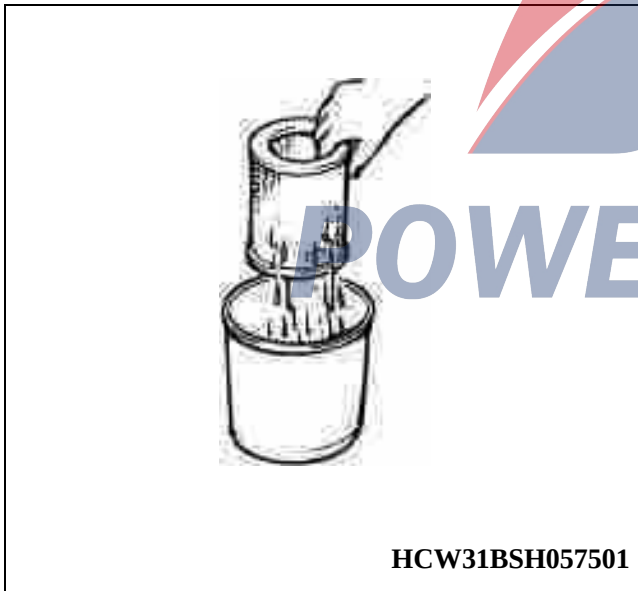
- If it is dirty and has dirt such as dry ash, blow the compressed air from the inside of the filter core when cleaning.

Note:

Do not clamp or tap the filter during cleaning. The pressure should not exceed 690kPa (7.0 kg f / cm² / 100 lb / in²).

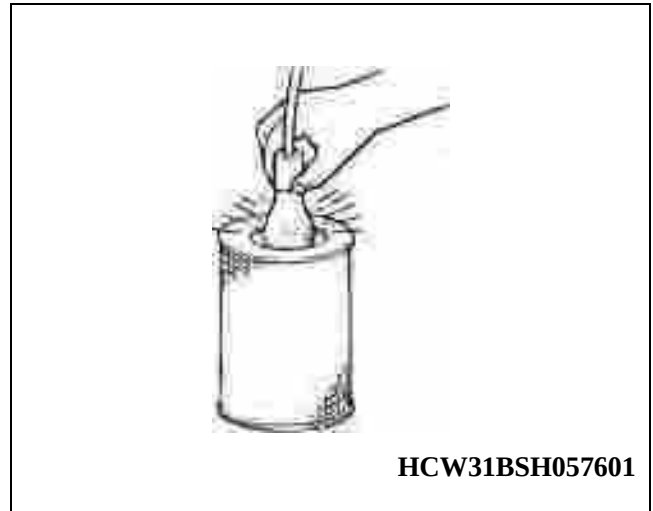


- If it is dirty and has dust (such as soot), put the filter in an aqueous solution of neutral detergent for about 20 minutes. Then rinse the filter element thoroughly with tap water. After washing, dry naturally without heating (2-3 days). Do not clamp or tap the filter during cleaning. Water pressure should not exceed 275kPa (2.8 kg/cm²/40 lb/in²).

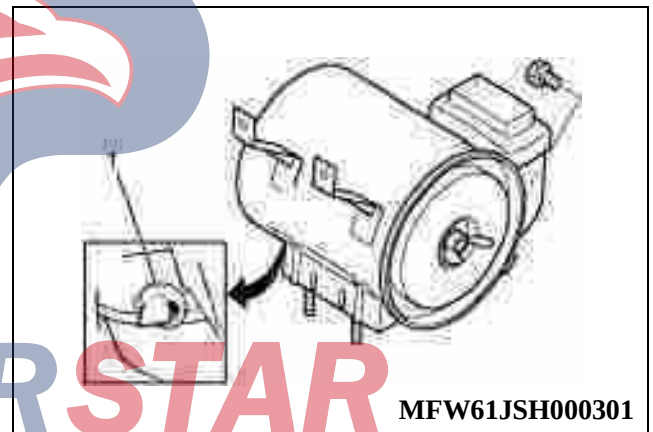


3. Check the filter for damage.

- Place a lamp in the filter and inspect the filter for damage or thinning.



4. Check whether the water removal valve (1) at the bottom of the air cleaner is clogged.



Reassemble

1. Install the filter element.

- For the wing screw type, align the center hole of the filter element with the attachment bolt in the box, insert the bolt, and then secure the bolt protruding from the center with the screw.

2. Install the air cleaner cover.

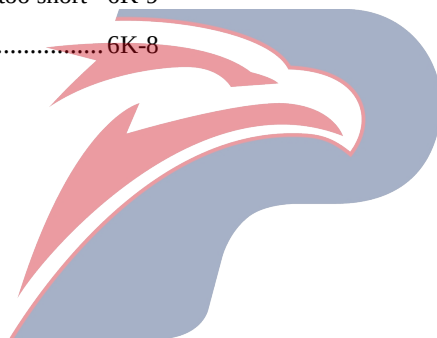
- For the thumbscrew type, align the center hole of the cover with the attachment bolt in the case, insert the bolt, and then turn the screw to the right to fix the bolt protruding from the center.
- For the clamp, push the three parts of the clamp and install the air cleaner cover.

Engine

Preheating system

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Phenomenon: Preheating system cannot operate	6K-9
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Specifications	6K-8



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Preheating system

Maintenance precautions

Overheating the glow plug can damage it.

Description of function and operation

The preheat system uses the QOS II system. At the same time, power supply control is performed by the ECM.



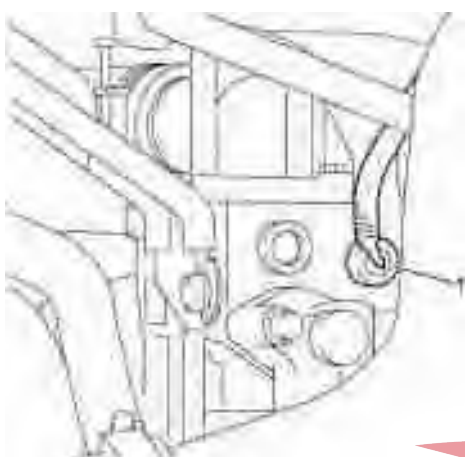
Functional check

QOS II check and correction

1. System check

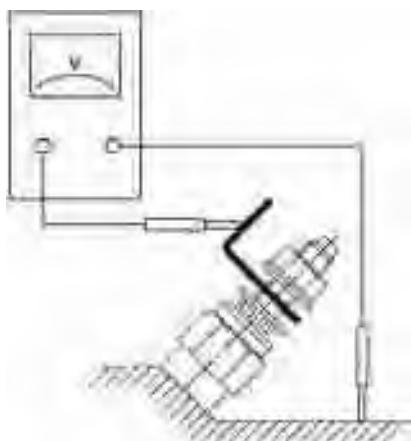
- Disconnect dual thermal sensor

(1) Connector



MFW61KSH000201

- A voltmeter is connected between the electric plug and the earthing of the engine.
- Turn on the ignition switch (do not start the engine) and check the following items.
 - The glow indicator lights up for 3.5 seconds.
 - The voltmeter indicates that the power supply voltage is approximately 18 seconds.
- If the check is abnormal, check the thermal relay, dual thermal sensor, and glow plug. If they are normal, check the engine control system.

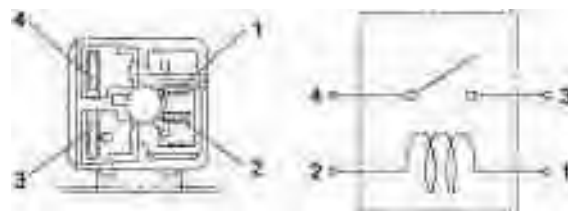


HCW31KSH000201

2. Inspection of the glow relay

- Use a circuit tester to measure the resistance of the glow relay (between 1 and 2).
- If the measurement shows that the resistance is out of specification, the glow relay should be replaced.

Resistance of glow relay = approx. 51.5 ohms



HCW31KSH000101

- Glow relay (1) connection position: the lower right side of the instrument panel.
- The graph shows the case of type RHD. The LHD relay group is in the opposite position, but the relay layout is the same.

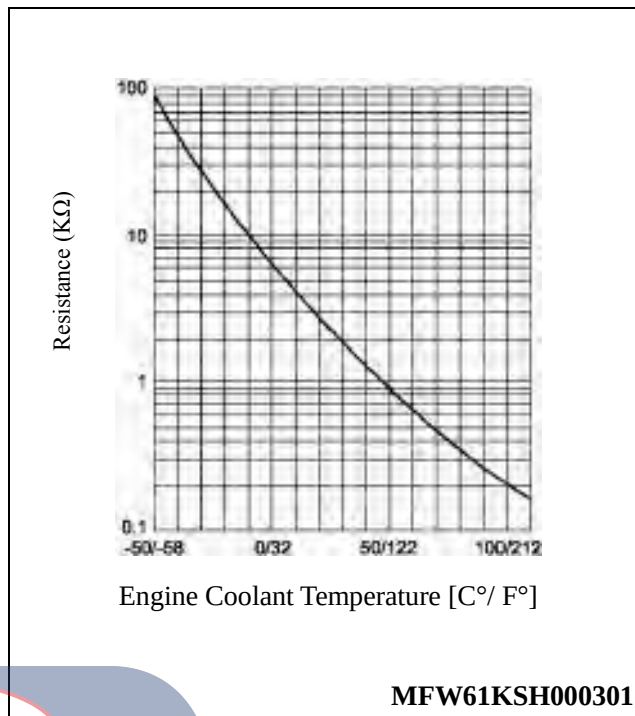
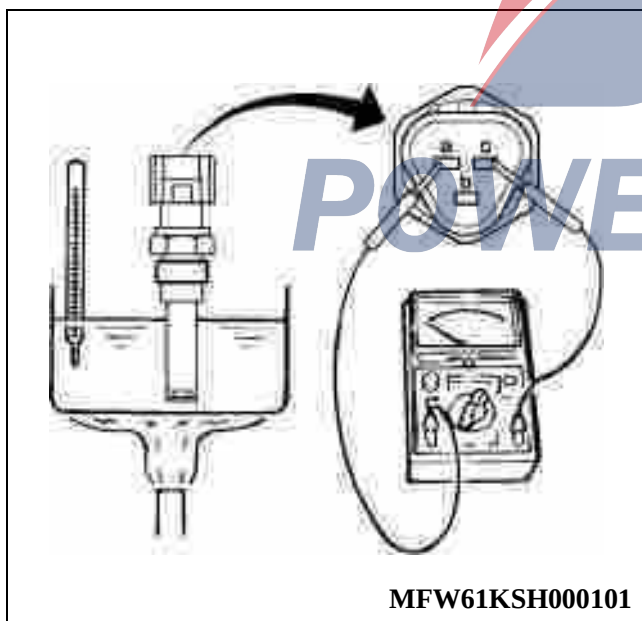


MFW41KSH000201



3. Check the dual heat sensor.

- The dual thermal sensor temperature sensor was immersed in water and changes in resistance and water temperature were considered to meet the conditions shown.



MFW61KSH000301

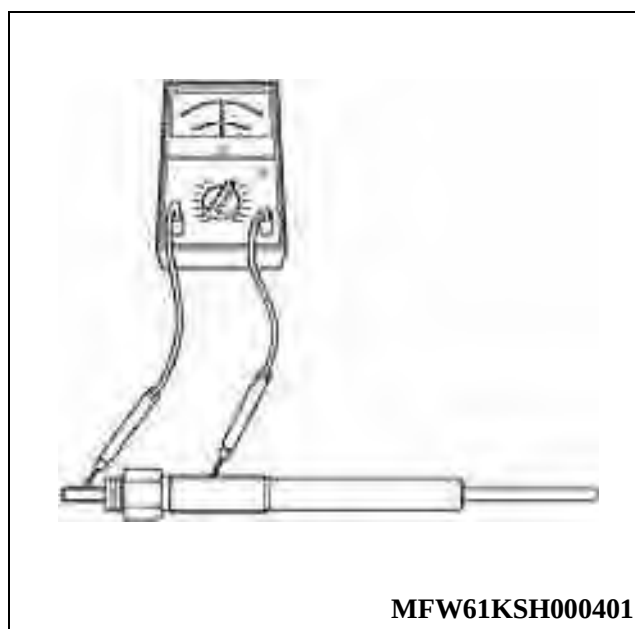
4. Electric glow plug inspection

- Use a circuit tester to measure the glow plug resistance.
- If the measurement shows that the resistance exceeds the specification, the glow plug should be replaced.

Glow plug resistance = approx. 6.0 Ω(normal temperature)

Note:

Use the same 6-piece glow plug made by the same manufacturer.



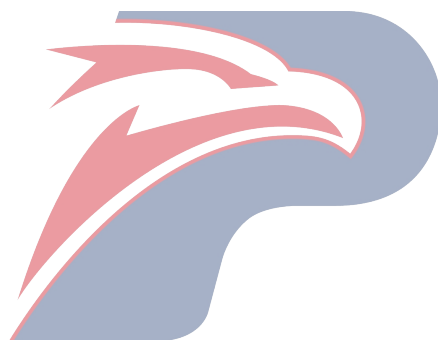
List of failure symptoms

- Glow indicator does not light up
- Preheat system does not work
- Preheating time is too long or too short



Symptom: The glow indicator does not light up

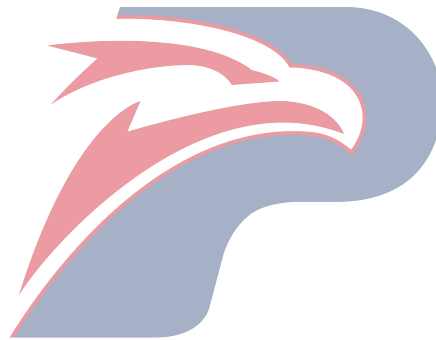
Phenomenon	Probable causes	Corrective measures
Glow indicator does not light up.	Slow fuse blown	Replace the slow break fuse.
	Blown fuse	Replace the fuse.
	Indication valve burnt	Replace the indicator valve.



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Phenomenon: Preheating system cannot operate

Phenomenon	Probable causes	Corrective measures
Preheat system does not work.	Slow fuse blown	Replace the slow break fuse.
	Light emitting relay connector failed to connect	Reinstall or replace the glow relay.

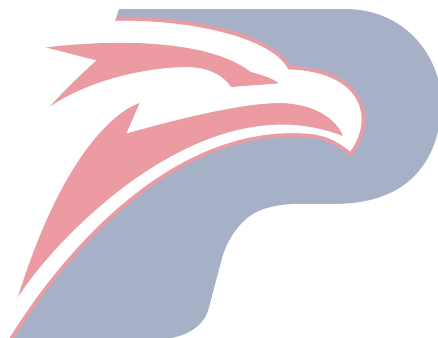
**POWERSTAR**

Phenomenon: Preheating time is too long or too short

Phenomenon	Probable causes	Corrective measures
Preheating time is too long or too short	Thermal switch failure, open or short circuit	Replace the thermal switch. Repair the circuit.
	Glow plug failure	Replace or tighten glow plugs.

Specifications

Item	Model
Preheating equipment model	Glow plug QOS timer
Electric glow plug rated voltage and current	20.5V—4.6A

**POWERSTAR**

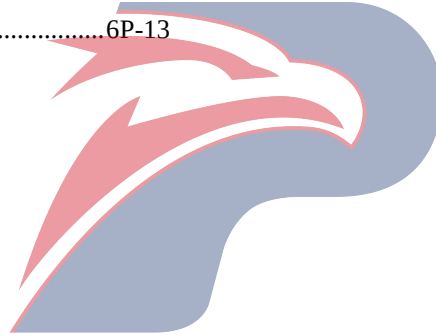
Engine

Power output mechanism (PTO)

(6HK1)

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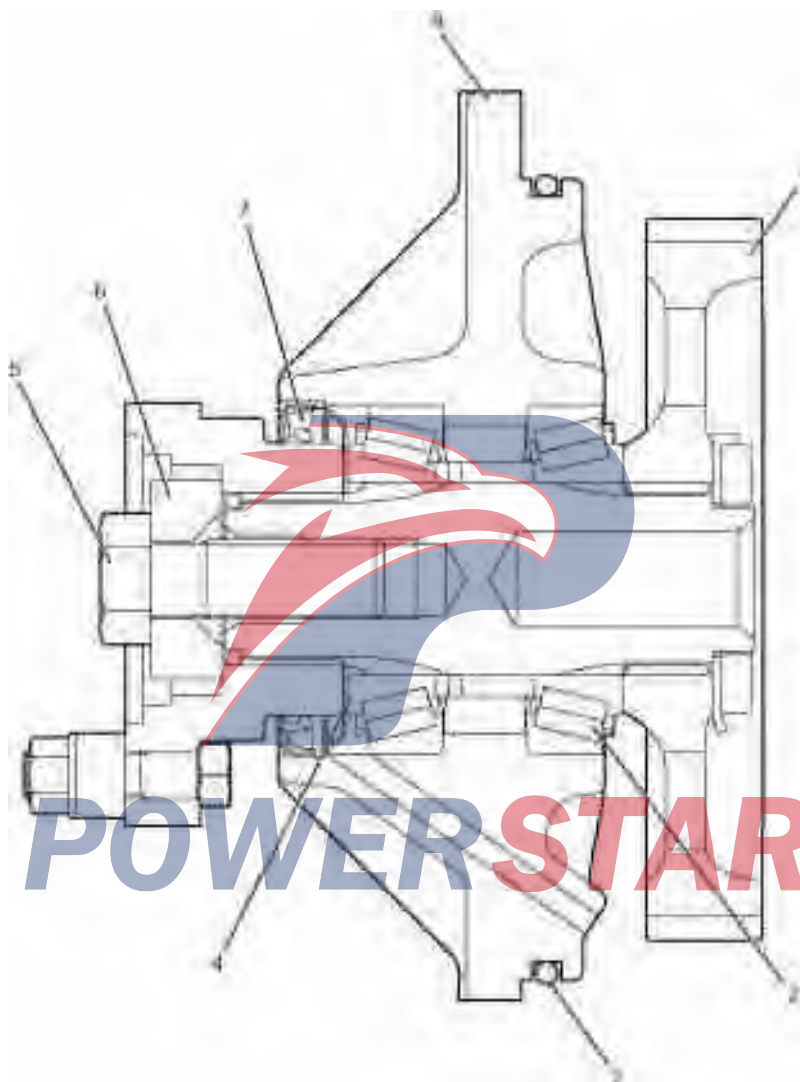


POWERSTAR

Flywheel PTO

Description of function and operation

Flywheel PTO structure

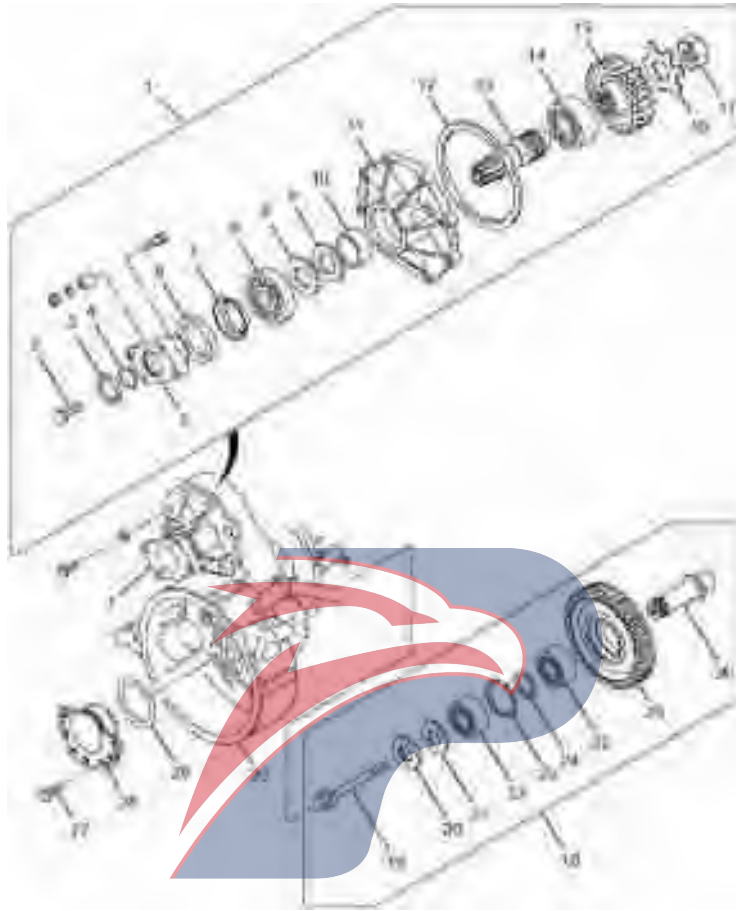


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Legend

- | | |
|------------|------------------|
| 1. Gear | 5. Shaft bolt |
| 2. Bearing | 6. Washers |
| 3. O-ring | 7. Oil seal |
| 4. Bearing | 8. Bearing cover |

Part



POWERSTAR

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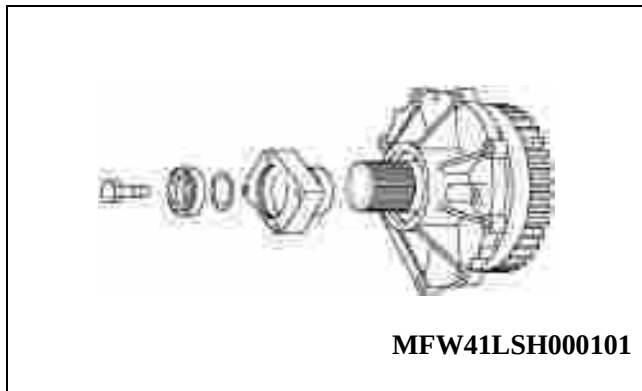
Legend

- | | |
|-------------------------------|-------------------------|
| 1. Power output gear assembly | 16. Washers |
| 2. Bolt | 17. Nuts |
| 3. Washers | 18. Idler gear assembly |
| 4. O-ring | 19. Bolt |
| 5. Flange | 20. Nuts |
| 6. Oil seal | 21. Lock washer |
| 7. Oil slinger | 22. Idler rotor bearing |
| 8. Bearing | 23. External gasket |
| 9. Gasket | 24. Inner gasket |
| 10. Septum | 25. Idler gear |
| 11. Bearing cover | 26. Idler gear spindle |
| 12. O-ring | 27. Bolt |
| 13. Axis | 28. Cover |
| 14. Bearing | 29. O-ring |
| 15. PTO gear | 30. Flywheel cover |

Disassembly

1. Remove the PTO gear assembly.

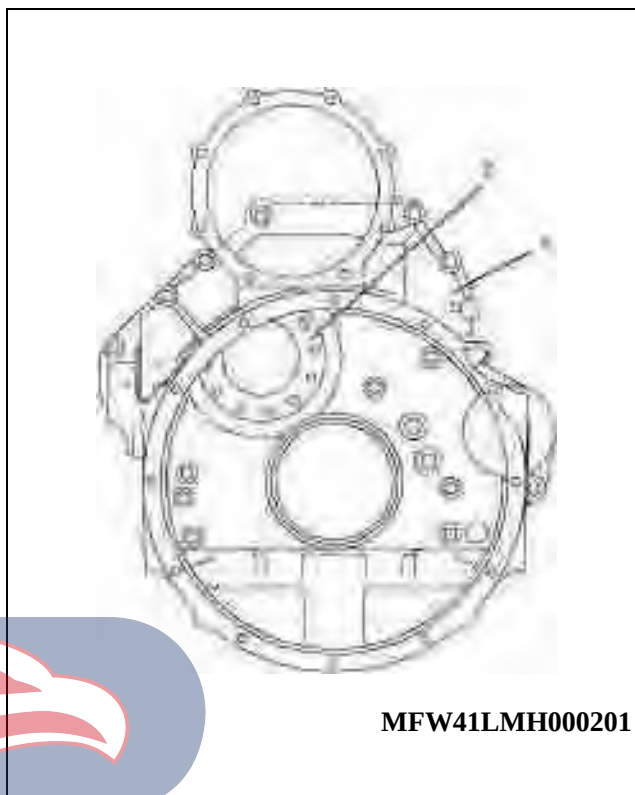
- Remove the oil pipe.
- Remove the PTO shaft bolt and remove the flange.
- Remove the bearing cover assembly.



2. Remove the PTO idler.

- Remove the PTO idler cover.

- Remove the flywheel cover.



Legend

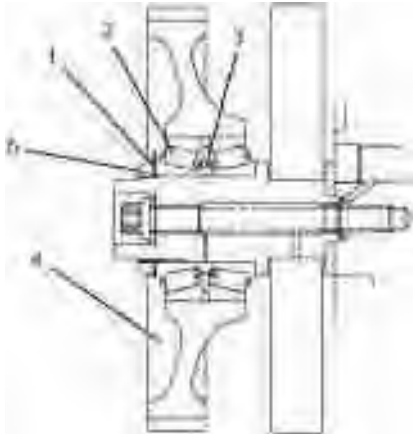
1. Flywheel cover

2. Cover

- Remove the PTO idler shaft nut.
- Pull out the inner edge of the tapered roller bearing and remove the PTO idler.

POWER STAR

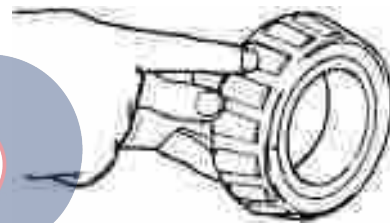
- Remove the inner and outer gaskets.



MFW41LSH002501



MFW41LSH000201



MFW41LSH000301

Legend

1. Washers
2. Bearing
3. Septum
4. PTO gear
5. Nuts

Inspection

1. Check PTO bearings.

- Check the bearing for shedding, wear and abnormal noise.
- Check for damage or detachment between the roller and the fixture and the entire inner edge.
- Check if the roller guard is loose due to wear.
- Rotate the bearing by hand to check if the movement is smooth and if there is abnormal noise.

2. Check the PTO gear.

- Check the tooth surface for damage and wear.
- Meanwhile check the spine for damage and abrasion.

POWERSTAR



MFW41LSH000401



MFW41LSH000501

3. Check the idler bearing.

- Check the bearing for shedding, wear and abnormal noise.
- Check for damage or detachment between the roller and the fixture and the entire inner edge.
- Check if the roller guard is loose due to wear.
- Rotate the bearing by hand to check if the movement is smooth and if there is abnormal noise.

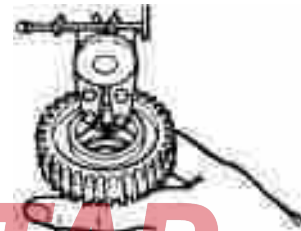


MFW41LSH000301

Assembling

1. Install idler bearings.

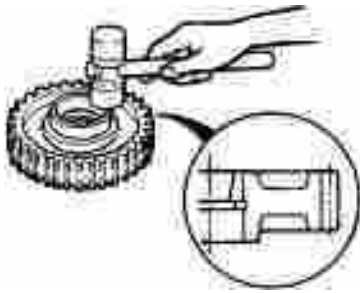
- Use pliers to insert the outer partition into the idler gear.
- Press in until the tapered roller bearing outer ring contacts the septum.



MFW41LSH000601

4. Check the gear at idling speed.

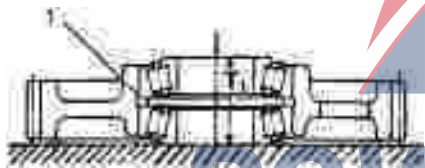
- Check the gear surface for damage and wear, and whether the inner diameter is damaged.



MFW41LSH000701

2. Select internal gasket.

- Measure the axial clearance of the idler bearing. Install the inner edge of the idler bearing without the inner spacer, and then measure the axial depth T1 of the bearing on the platform.



MFW41LSH000801

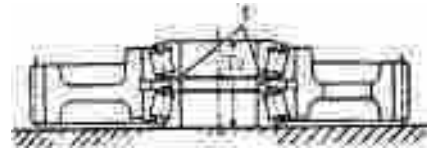
Legend

1. External gasket

Install the inner shim and measure the axial depth T2 of the bearing in the same way.

Select a spacer to measure T2 and T1.

Axle clearance	mm(inch)
$T2 - T1 = 0 - 0.05 (0 - 0.0020)$	



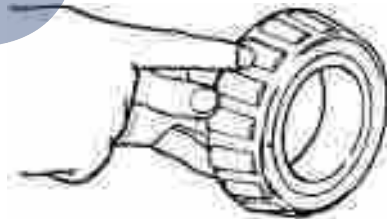
MFW41LSH000901

Legend

1. Inner gasket

3. Install the PTO idler.

- Press the inside of the front face into the spindle.



MFW41LSH000301

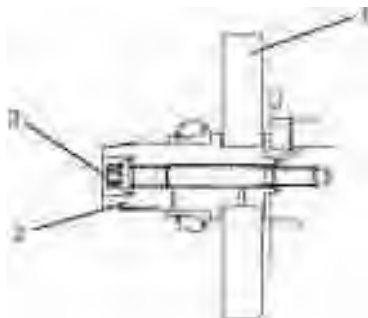
- Align the spindle (2) with the idler A (1) and install it. (Positioning with crankshaft gear)
- Apply molybdenum disulfide adhesive to the threaded portion and base portion of the bolt (3) and tighten.

Tightening torque

N m (kgf m/lb ft)

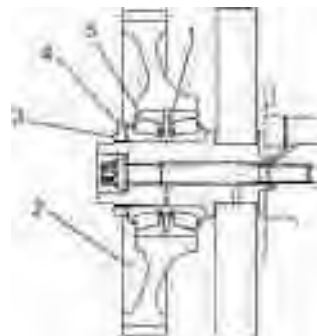
49 (5.0/36) → Angle Method 90° → -120°

- Apply oil to the bearing.



MFW41LSH001001

- The axial clearance value is very important. Always adjust to within the specified value.
- Bend the tip of the lock washer.



MFW41LSH001101

Legend

1. Idler gear A
2. Spindle
3. Bolt

- Install the PTO idler (2) on the spindle.
- Insert the inner space (1) and the cock into the rear inner side (5).
- Tighten the nut (3) with the lock washer (4).

Tightening torque	N m (kgf m/lb ft)
196 (20 / 145)	

- After tightening the nut, measure the axial clearance of the PTO idler.

If gaps exceed the specified value, the diaphragm should be replaced and readjusted.

Axle clearance	mm(inch)
0—0.05(0—0.0020)	
Start friction torque	N m (kg/m/lb. inch)
0.2 (0.02 / 1.4466)	

Note:

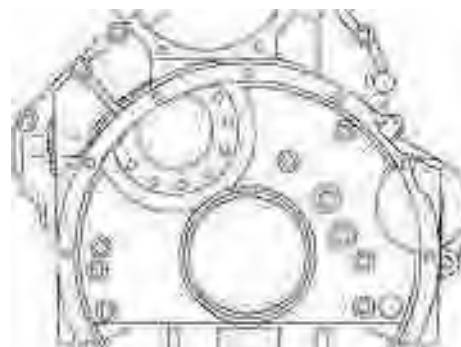
- Apply the engine oil to the bearing and install it.

Legend

1. Inner gasket
2. PTO idle gear
3. Nuts
4. Lock washer
5. Inside

4. Install the flywheel cover.

- For details on the flywheel housing connection procedure, see the flywheel housing connection section.



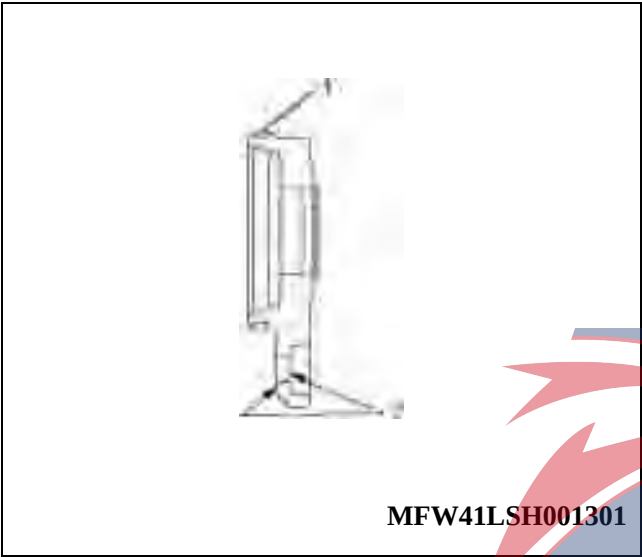
MFW41LSH001201

1. Install the idler shaft cover.
- Attach the O-ring to the lid.

- Apply liquid filler (San key brand 1207C or 1201) to the flange surface and the bolt base.
- Position the guide pins, etc. and presses them together.

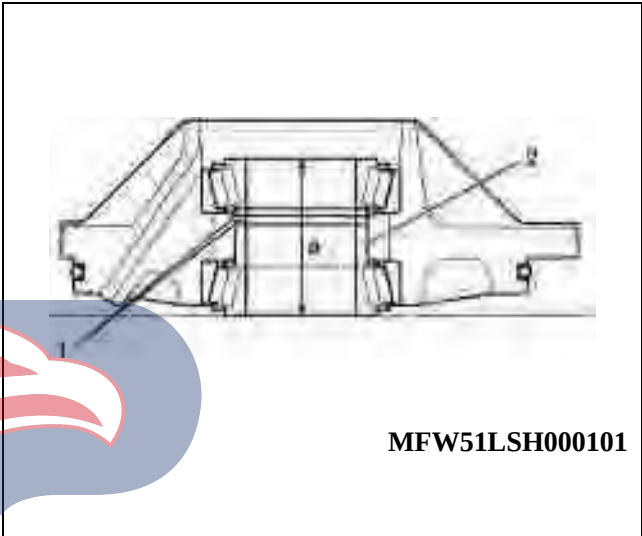
Note:

Do not beat the shaded portion of the cover when pressing it in.



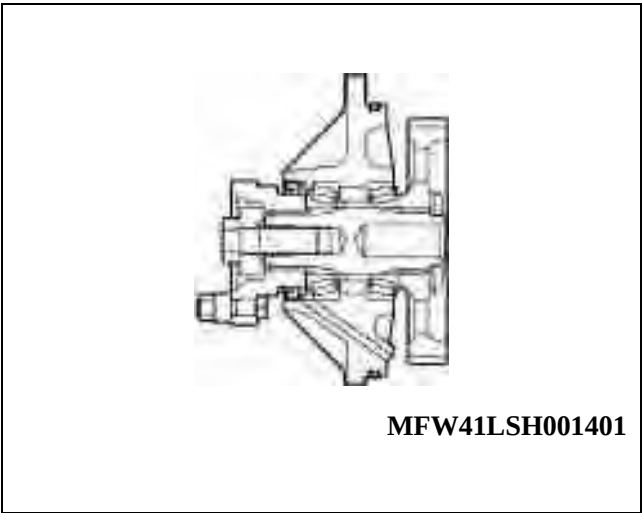
- Gaskets and washers are installed in bearing housings (a-b) in the same way as idler gears, and gaskets are selected.

mm(inch)
a—b=0.01—0.07(0.0004—0.0028)



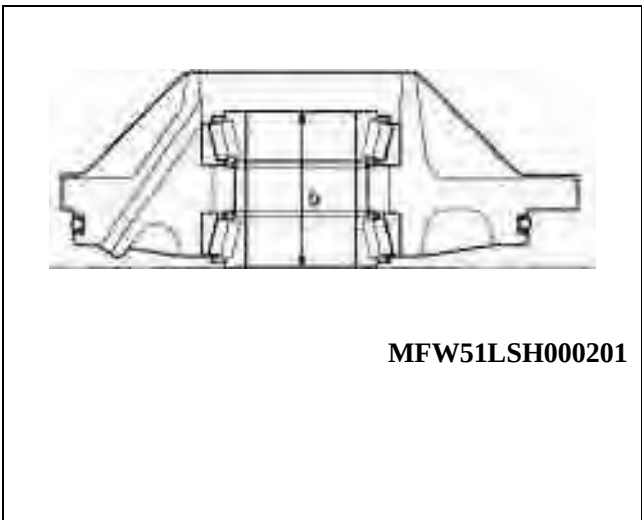
Legend

1. O-ring
2. coating surface for Three bond filler
6. Assemble the PTO gear assembly.



Legend

1. Gasket
2. Septum



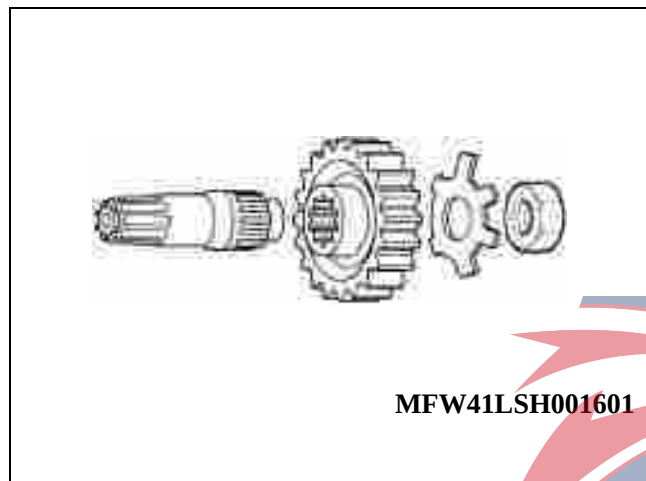
- a. Select a spacer.
- Screw the bearing outer race into the bearing seat.

- b. Install the shaft in the gear.

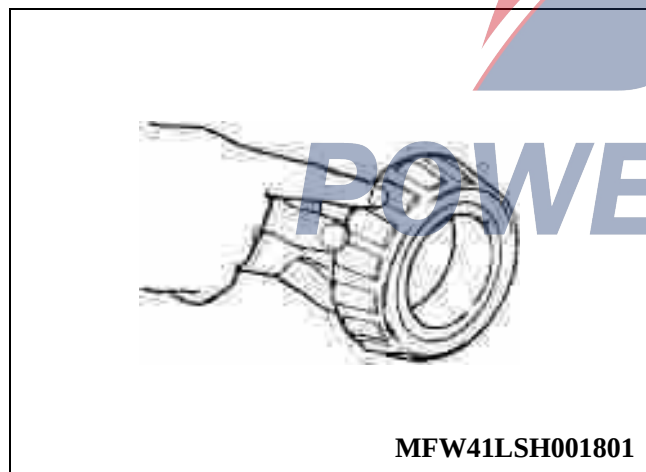
- Tighten the nut through the lock washer to the required torque.

Nut tightening torque	N m (kgf m /lb ft)
294 (30.0 / 217)	

- Bend lock washers after assembly.



- c. Install the front inner shaft into the shaft.

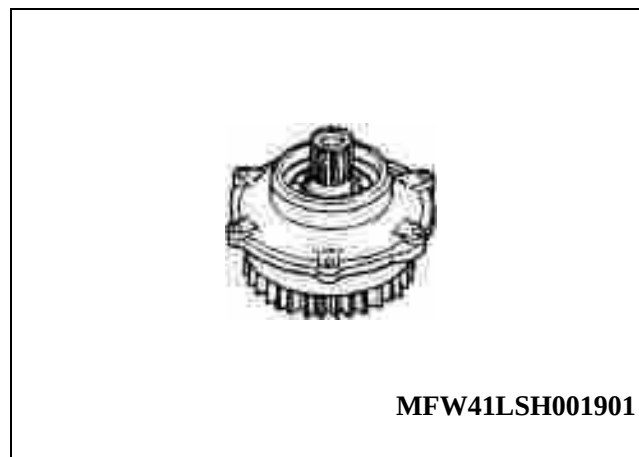


- d. Install the shaft assembly in the bearing housing.

- e. Insert the previously selected shim and shim.

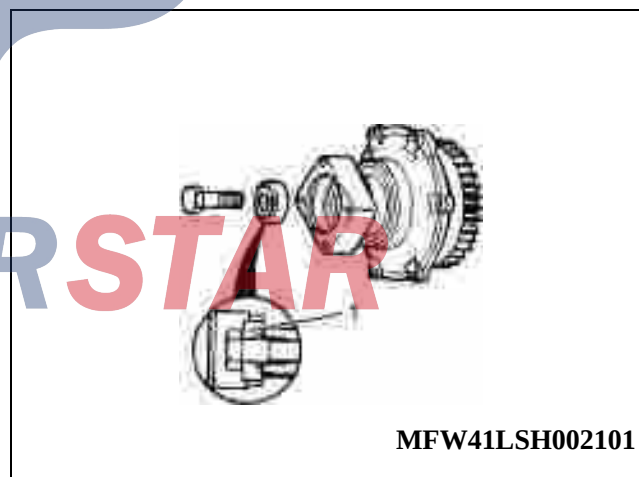
- f. Press the inner side of the rear side into the shaft.

- g. Insert the flinger.



- h. Install the flange and tighten the bolt to the specified torque strength.

Tightening torque	N m (kgf m/lb ft)
343 (35 / 253)	



Legend

1. Washers

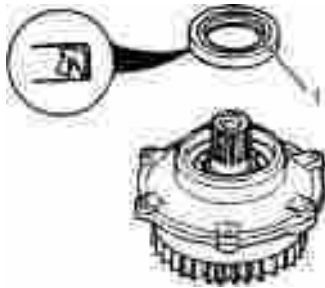
- i. Measure the axial clearance of the shaft.

- Install O-rings.

Axle clearance	mm(inch)
0.01—0.07(0.0004—0.0028)	

Note:

- Apply the engine oil to the bearing and install it.
 - The axial clearance value is very important. Always adjust to within the specified value.
- After reaching the specified gap, loosen the bolt and take out the flange.
 - Seal the oil into the bearing housing.
- The bearing cap is coated with KE42 (silicone rubber) manufactured by Shin-Etsu Chemical Co., Ltd.
 - Apply a thin layer of oil to the lip of the oil seal and press it together.

**MF41LSH002101****MF41LSH002001****Legend**

1. Washers

n. Bend the lock washer to fit the PTO gear tightening nut.

7. Install an O-ring on the bearing housing.

- Apply a thin layer of oil.

Legend

1. Oil seal

- Temporarily install 4 flange bolts and insert the shaft.
- Re-tighten the flange bolts to the specified torque strength.

Tightening torque	N m (kgf m /lb ft)
	343 (35 / 253)

**MF41LSH002201**

- Install the PTO gear assembly on the flywheel housing.

Bolt tightening torque	N m (kgf m /lb ft)
	42 (4.3 / 31)



MFW41LSH002301

9. Put on the tube.

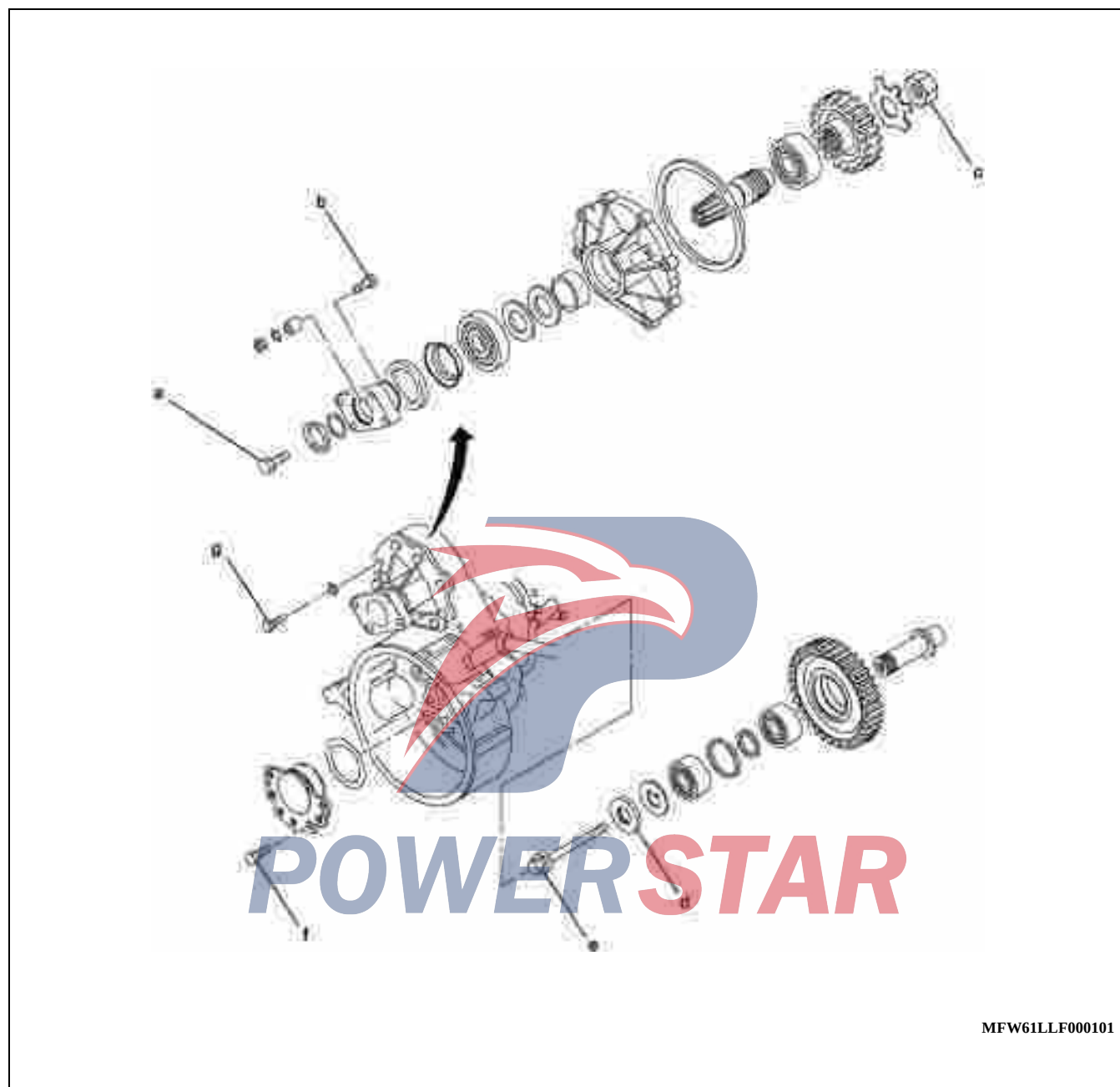
- Install the clips.



MFW41LSH002401

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Tightening torque table



Legend

(a) 343 N m (35 kgf m / 253 lbs. ft)

(b) 88N m (9.0 kgf m/65 lb.ft)

(C) 294 N m (30 kgf-m / 271 lb ft)

(d) 196 N•m (20 kgf m / 145 lbs. ft)

(e) 49N m (5.0 kgf m/36 lb ft)→Angle 90°→120°

(f) 28N m (2.9 kgf m/21 lb ft)

(g) 42 N m (4.3 kgf m/31 lb ft)

Engine
Engine Control System
(6HK1 [Euro 4 specification])
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Fuel injection system description	错误!未定义书签。
Exhaust gas recirculation (EGR) system description	错误!未定义书签。
Turbocharger description	错误!未定义书签。
Diesel particulate trap (DPD) system description	错误!未定义书签。
Special tools and equipment	错误!未定义书签。
Special tools and equipment	错误!未定义书签。

Specification

Temperature and resistance

Engine coolant temperature and resistance

°C	°F	Ω
Temperature and resistance (approximate)		
110	230	160
100	212	200
90	194	260
80	176	350
70	158	470
60	140	640
50	122	880
40	104	1250
30	86	1800
20	68	2650
10	50	4000
0	32	6180
-10	14	9810
-20	-4	16000
-30	-22	27000

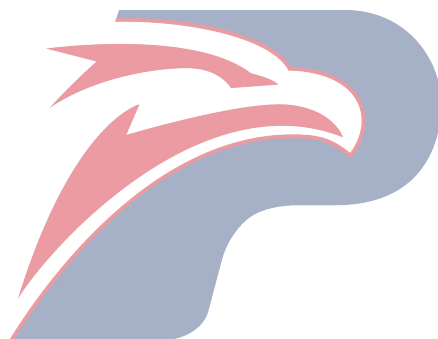
Intake air temperature and resistance

°C	°F	Ω
Temperature and resistance (approximate)		
90	194	240
80	176	320
70	158	430
60	140	590
50	122	810
40	104	1150
30	86	1650
20	68	2430
10	50	3660
0	32	5650
-10	14	8970
-20	-4	14700

Fuel temperature and resistance

°C	°F	Ω
Temperature and resistance (approximate)		
110	230	140
100	212	180
90	194	240
80	176	310
70	158	420
60	140	580

50	122	810
40	104	1150
30	86	1660
20	68	2450
10	50	3700
0	32	5740
-10	14	9160
-20	-4	15000
-30	-22	25400



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Exhaust temperature and resistance

°C	°F	Ω
Temperature and resistance (approximate)		
1000	1832	115 - 127
950	1742	130 - 143
900	1652	147 - 163
850	1562	167 - 187
800	1472	192 - 219
750	1382	226 - 258
700	1292	267 - 309
650	1202	333 - 360
600	1112	406 - 442
550	1022	504 - 552
500	932	640 - 709
450	842	835 - 937
400	752	1130 - 1280
350	662	1590 - 1850
300	572	2360 - 2830
200	392	6230 - 7830
150	302	11900 - 16300
100	212	27600 - 41200
50	122	82000 - 137000

Altitude and atmospheric pressure

Altitude (meters)	Elevation(inch)	Atmospheric pressure (kPa)
Contact your local weather station or refer to other information to determine the altitude in your area.		
4267	14000	56-64
3962	13000	58-66
3658	12000	61-69
3353	11000	64-72
3048	10000	66-74
2743	9000	69-77
2438	8000	71-79
2134	7000	74-82
1829	6000	77-85
1524	5000	80-88
1219	4000	83-91
914	3000	87-95
610	2000	90-98
305	1000	94-102
0	0 Sea level	96-104
-305	-1000	101-105

Diagnostic Fault Code (DTC) Type Definition

Emission-related diagnostic trouble codes

Operation when setting DTC - category A

- If the diagnosis is run but not passed, the engine controls module (ECM) lights up the fault indicator lamp (MIL).
- The engine control module records the operation when the diagnosis failed. The engine control module stores this information in the "Freeze Fault state/Fault Record ".

Operation when setting DTC - Class A

- During the second successive ignition cycles, the engine control module lights up the fault indicator when the diagnosis is not carried out and failed.
- The engine control module records the operation when the diagnosis failed. If the diagnostic fails for the first time, ECM will store this information in "fault record". If the diagnosis is not passed in the second consecutive ignition cycle, the engine control module will record the operation at the time of the fault and store this information in the "freeze fault state" and update the "fault record".

Fire Extinguishing Indicator / Condition for Clearing Diagnostic Trouble Codes - Class A or B

- After diagnostic running in 3 consecutive ignition cycles is performed and passed, the engine control module will turn off the fault indicator lamp.
- When the diagnosis is run and passed, the current diagnostic failure code will be cleared.
- After 40 consecutive warm-up cycles, if no failure condition is reported, the history diagnostic trouble code will be cleared.
- Use the fault diagnosis tester to turn off the fault light and clear the diagnostic trouble code.

Diagnostic trouble codes that have nothing to do with emission**Operation when setting DTC - Class C**

- When the diagnostic operation is not passed, the engine control module lights up " (SVS)" as soon as possible.
- The engine control module records the operation when the diagnosis failed. The engine control module stores this information in the "failure record".

Conditions to turn off "service vehicle soon" indicator lamp/clear DTC - category C

- The engine control module will extinguish the SVS indicator after diagnosing and running in 1 continuous ignition cycles.
- When the diagnosis is run and passed, the current diagnostic failure code will be cleared.
- After 40 consecutive warm-up cycles, if no failure condition is reported, the history diagnostic trouble code will be cleared.
- Use the fault diagnostic instrument to extinguish the "repair vehicle" indicator as soon as possible and remove the diagnostic fault code.

Operation when setting DTC - Class D

- The engine control module will not turn on the fault indicator or the "repair vehicle as soon as possible" indicator.
- The engine control module records the operation when the diagnosis failed. The engine control module stores this information in the "failure record".

Clear Diagnostic Trouble Code Condition - Level D

- When the diagnosis is run and passed, the current diagnostic failure code will be cleared.
- After 40 consecutive warm-up cycles, if no failure condition is reported, the history diagnostic trouble code will be cleared.
- Use fault diagnosis instrument to remove diagnostic fault code.

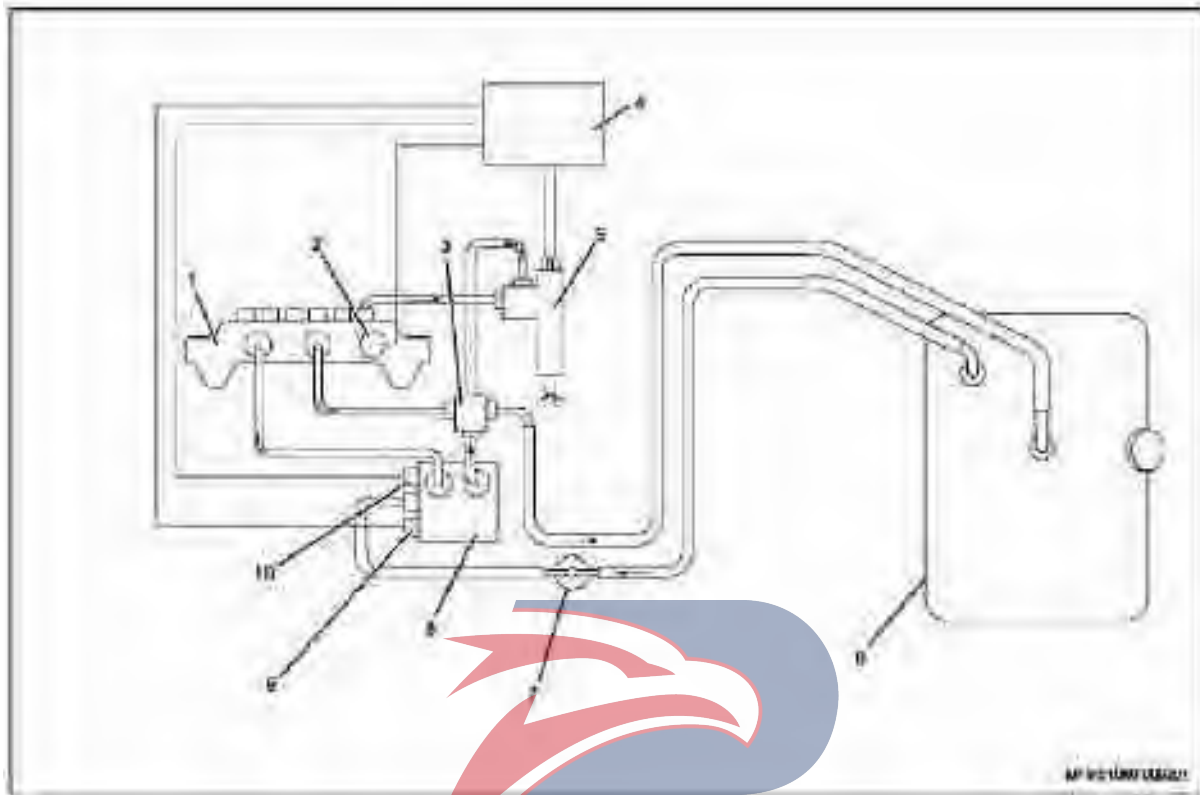
DTC list

Diagnostic trouble code	Flash code	Diagnostic trouble code type	Diagnosis fault code description information
P0016	16	A	The relationship between crankshaft position and camshaft position
P0045	33	A	Turbocharger Pressure Regulation Electromagnetic Circuit
P0079	38	A	Exhaust valve control solenoid circuit low
P0080	38	A	Exhaust valve control solenoid circuit high
P0087	225	A	Fuel distribution pipe/ System pressure is too low
P0088	118	D	Fuel distribution pipe/ System pressure is too high (first phase)
P0088	118	A	Fuel distribution pipe/system pressure too high (second stage)
P0089	151	A	Hydraulic regulator performance
P0091	247	A	Low oil pressure regulator control circuit
P0092	247	A	High oil pressure regulator control circuit
P0093	227	A	Fuel system leak detected
P0101	92	B	Measuring range / performance of mass air flow sensor circuit
P0102	91	A	Low input of mass air flow sensor circuit
P0103	91	A	Air mass flow sensor circuit high input
P0107	32	A	Intake Manifold Absolute Pressure Sensor Circuit Low Input
P0108	32	A	Intake Manifold Absolute Pressure Sensor Circuit High Input
P0112	22	A	Intake air temperature sensor circuit low
P0113	22	A	Intake air temperature sensor circuit high
P0116	23	B	Measuring range / performance of engine coolant temperature sensor circuit
P0117	23	A	Engine Coolant Temperature Sensor circuit low
P0118	23	A	Engine Coolant Temperature Sensor circuit high
P0122	43	A	Throttle position sensor circuit voltage low
P0123	43	A	Throttle position sensor circuit voltage high
P0182	211	A	Fuel temperature sensor circuit low
P0183	211	A	Fuel temperature sensor circuit high
P0192	245	A	The circuit of FRP sensor is low.
P0193	245	A	The circuit of FRP sensor is high.
P0201	271	A	Fuel injector circuit disconnect - cylinder 1
P0202	272	A	Fuel injector circuit disconnect - cylinder 2
P0203	273	A	Fuel injector circuit disconnect - cylinder 3
P0204	274	A	Fuel injector circuit disconnect - cylinder 4
P0205	275	A	Fuel injector circuit disconnect - cylinder 5
P0206	276	A	Fuel injector circuit disconnect - cylinder 6
P0217	542	D	Engine coolant over temperature status
P0219	543	D	Engine speeding
P0234	42	A	Turbocharger super boost state
P0299	65	B	Turbocharger supercharge low
P0335	15	A	Crankshaft position sensor circuit
P0336	15	A	Measuring range/performance of crankshaft position sensor circuit
P0340	14	A	Camshaft position sensor circuit
P0341	14	A	Camshaft position sensor circuit range/performance
P0380	66	C	Glow plug circuit
P0381	67	C	Glow plug indicator lamp circuit
P0401	93	A	Insufficient EGR flow detected
P0404	45	A	EGR control circuit range / performance
P0409	44	A	EGR sensor circuit
P0426	143	C	Catalyst temperature sensor circuit measuring range /performance - sensor 1
P0427	48	A	Catalyst Temperature Sensor Circuit Low - Sensor 1
P0428	48	A	Catalyst Temperature Sensor Circuit High - Sensor 1
P042B	145	C	Catalyst temperature sensor circuit measuring range /performance - sensor 2
P042C	49	C	Catalyst Temperature Sensor Circuit Low - Sensor 2
P042D	49	C	Catalyst Temperature Sensor Circuit High - Sensor 2
P0500	25	B	Speed sensor
P0502	25	A	Low input of the vehicle speed sensor circuit
P0503	25	A	High input of the vehicle speed sensor circuit
P0560	155	A	System voltage
P0563	35	C	System voltage high
P0601	53	A	Internal control module memory check error

P0602	154	C	Control module programming error
P0604	153	A	Internal control module random access memory (RAM) error
P0606	51	A	Engine control module processor
P060B	36	A	A/D processing performance of internal control module
P0638	61	B	Control range / performance of throttle actuator
P0641	55	A	Sensor reference voltage 1 circuit
P0650	77	A	Fault Indicator (MIL) Control Circuit
P0651	56	A	Sensor reference voltage 2 circuit
P0651	56	A	Sensor reference voltage 5 circuit
P0685	416	C	Engine control module power relay control circuit disconnect
P0687	416	C	Engine control module power relay control circuit is high
P0697	57	A	Sensor reference voltage 3 circuit
P0697	57	A	Sensor reference voltage 4 circuit
P1093	227	A	Fuel distribution pipe pressure is too low
P1261	34	A	Injection nozzle positive pressure control circuit group 1
P1262	34	A	Injection nozzle positive pressure control circuit group 2
P1404	45	A	EGR position failure (closed position error)
P1404	45	A	EGR position failure (known position error)
P1455	132	A	Over accumulation of PM
P1471	149	A	DPD reactivation insufficient
P1621	54	A	Long-term memory performance of control module
P1664	76	C	Service Vehicle Soon indicator lamp control circuit
P1669	75	C	DPD indicator lamp control circuit
P2122	121	B	Pedal position sensor 1 circuit low input
P2123	121	B	Pedal position sensor 1 circuit high input
P2127	122	B	Pedal position sensor 2 circuit low input
P2128	122	B	Pedal position sensor 2 circuit high input
P2138	124	B	Correlation between pedal position sensor 1 and sensor 2 voltage
P2146	158	A	Power supply voltage circuit of injection nozzle group 1
P2149	159	A	Power supply voltage circuit of injection nozzle group 2
P2227	71	B	Atmospheric pressure sensor circuit measuring range/performance
P2228	71	A	Atmospheric pressure sensor circuit is low
P2229	71	A	Atmospheric pressure sensor circuit is high
P242F	131	A	DPD restriction
P2452	142	A	DPD differential pressure sensor circuit
P2453	141	A	DPD differential pressure sensor circuit measuring range/scope
P2454	47	A	DPD differential pressure sensor circuit is low
P2455	47	A	DPD differential pressure sensor circuit high
P2456	47	A	Known position of DPD differential pressure sensor
P2458	139	A	DPD reactivation persisting
P253A	28	C	PTO sensor circuit
P256A	31	C	Engine idle selector sensor
U0073	84	B	The control module communication bus is disconnected.
U0110	87	A	Lose communication with VNT system

Schematic and wiring diagrams

Fuel system wiring diagram

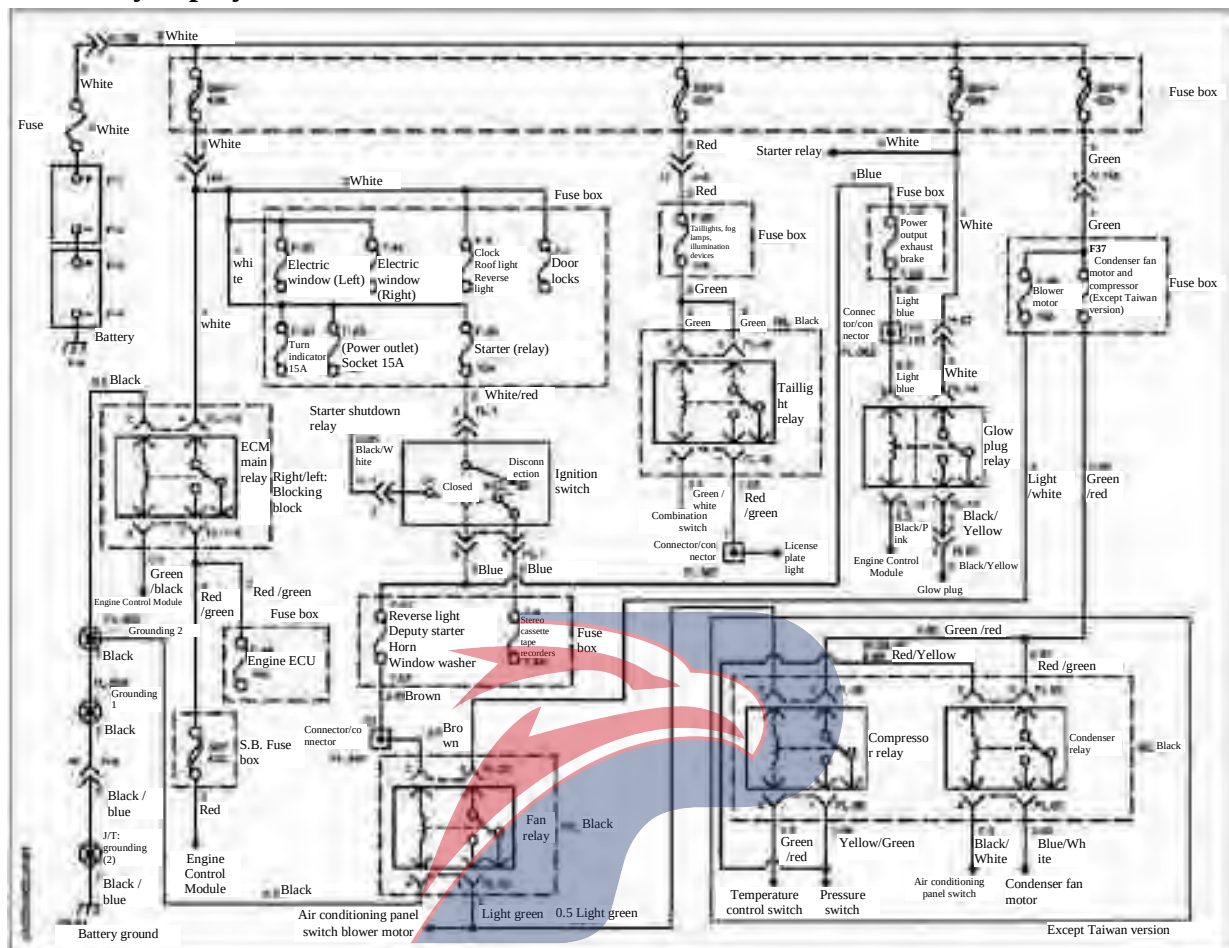


Key

- 
1. Fuel distribution pipeline
 2. Fuel rail pressure (FRP) sensor
 3. Drainage pipe
 4. Engine Control Module (ECM)
 5. Nozzle
 6. Fuel tank
 7. Fuel filter (with water separator)
 8. Supply pump
 9. Fuel temperature (FT) sensor
 10. Fuel Distribution Pipe Pressure (FRP) Regulator

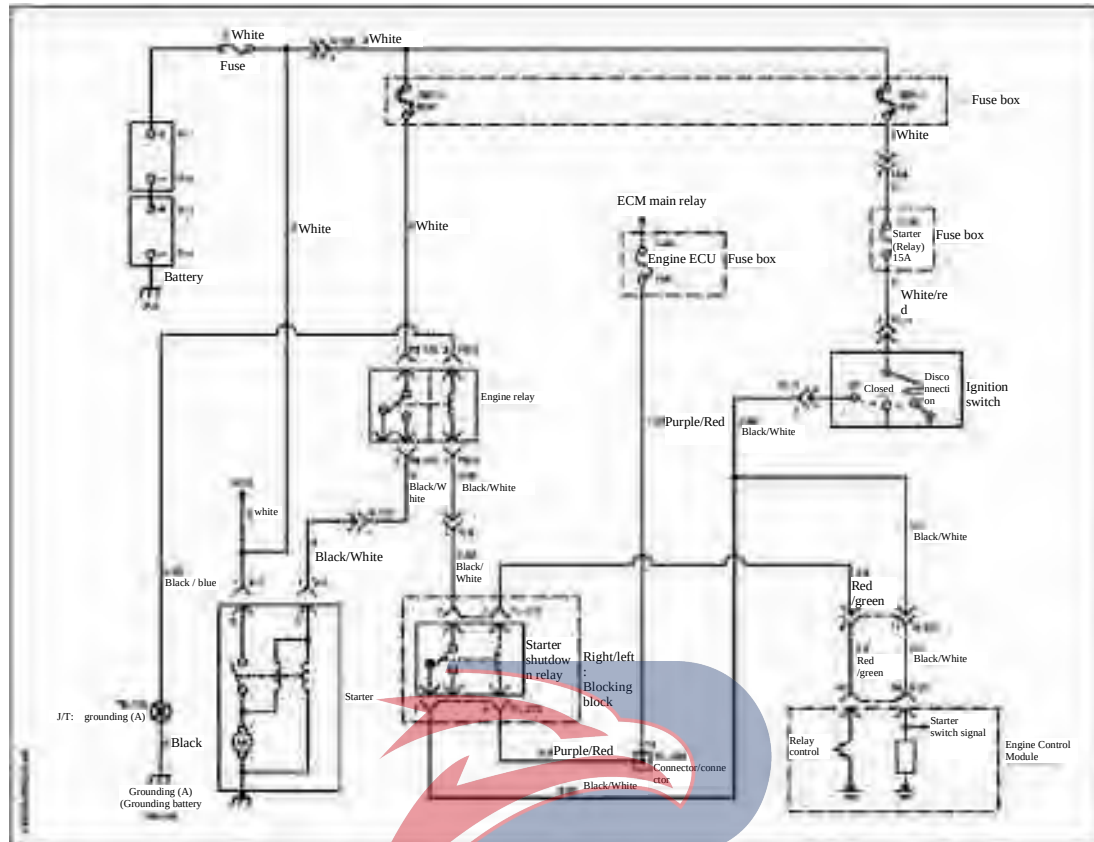
Schematic diagram of engine control system

Electricity deployment



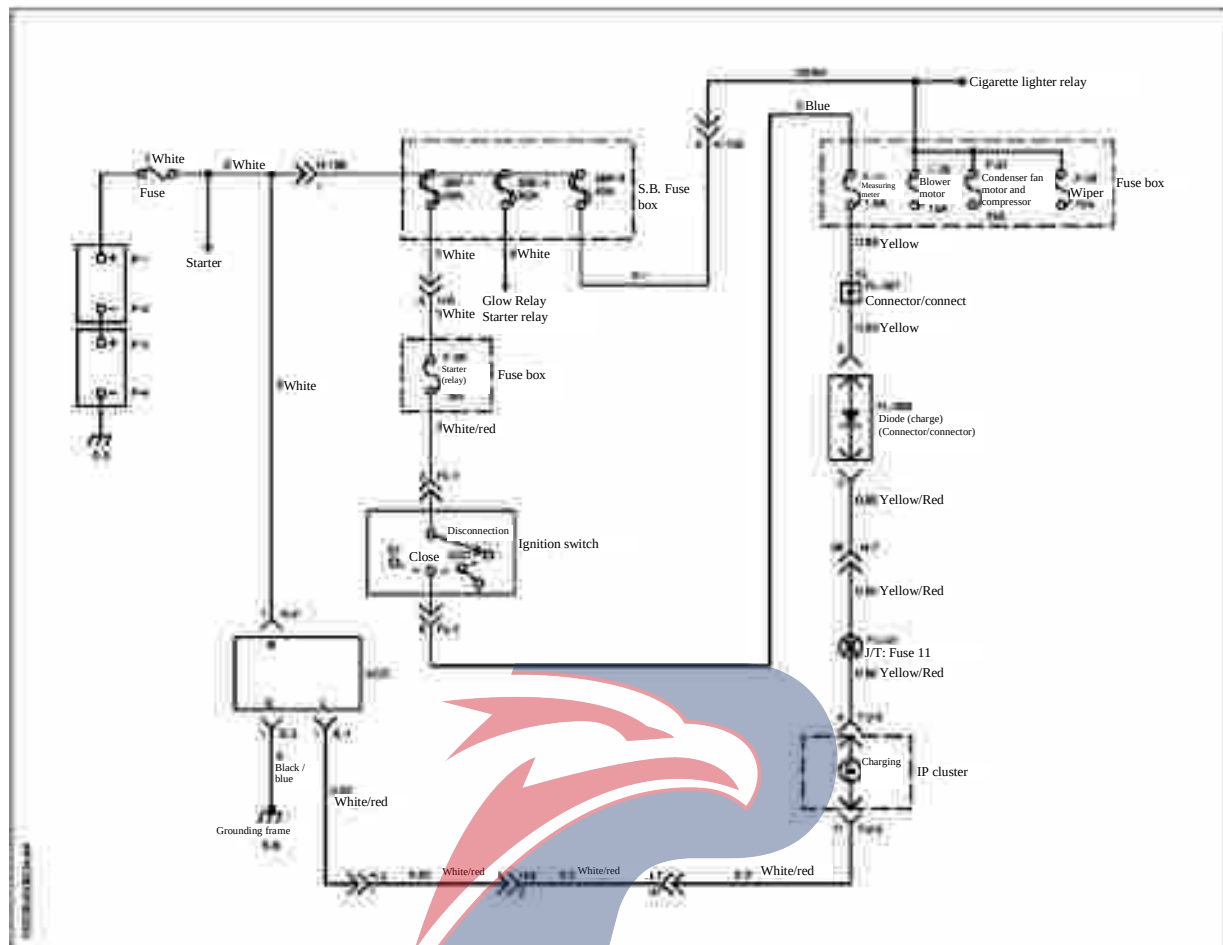
POWERSTAR

Starting system



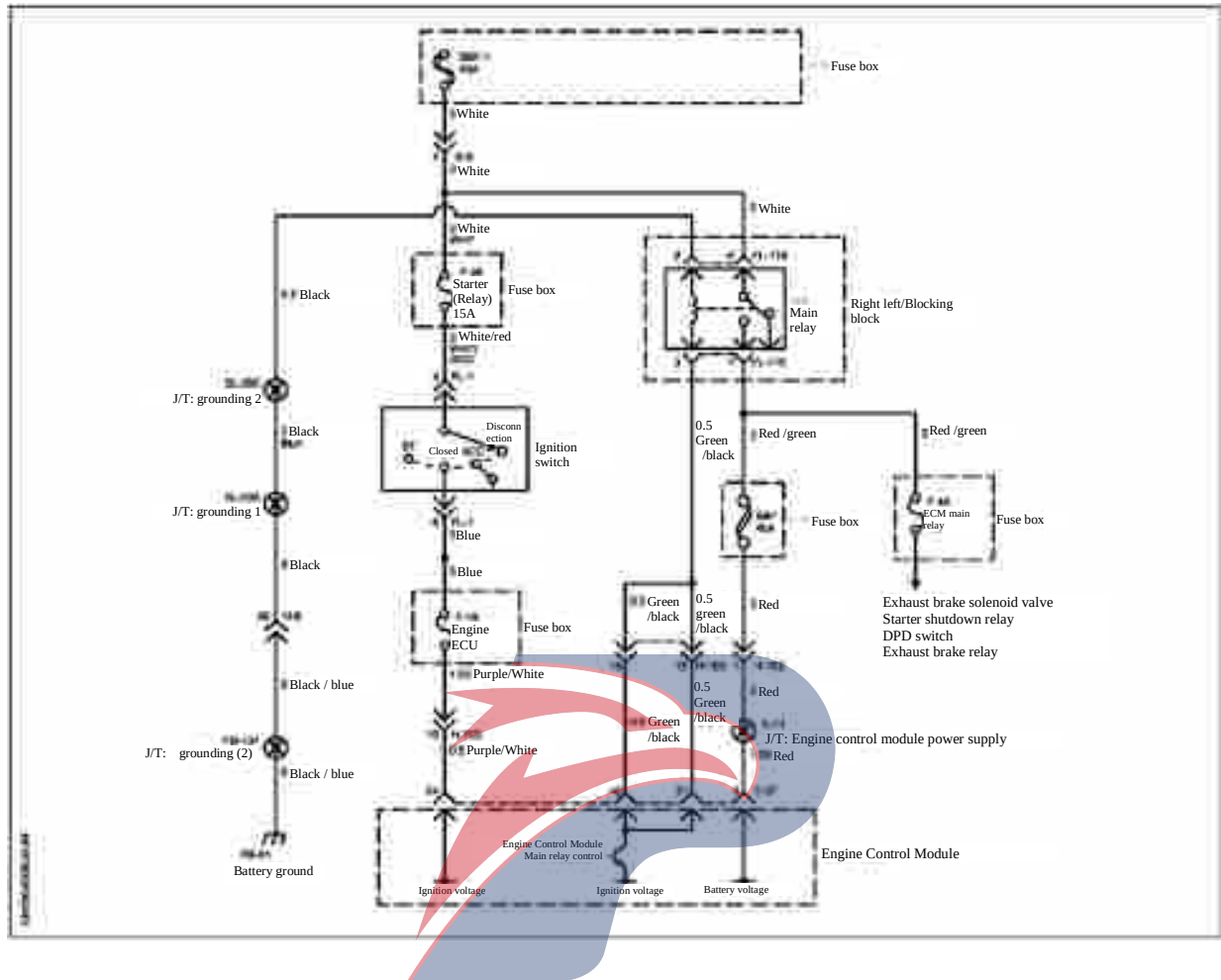
POWERSTAR

Charging system



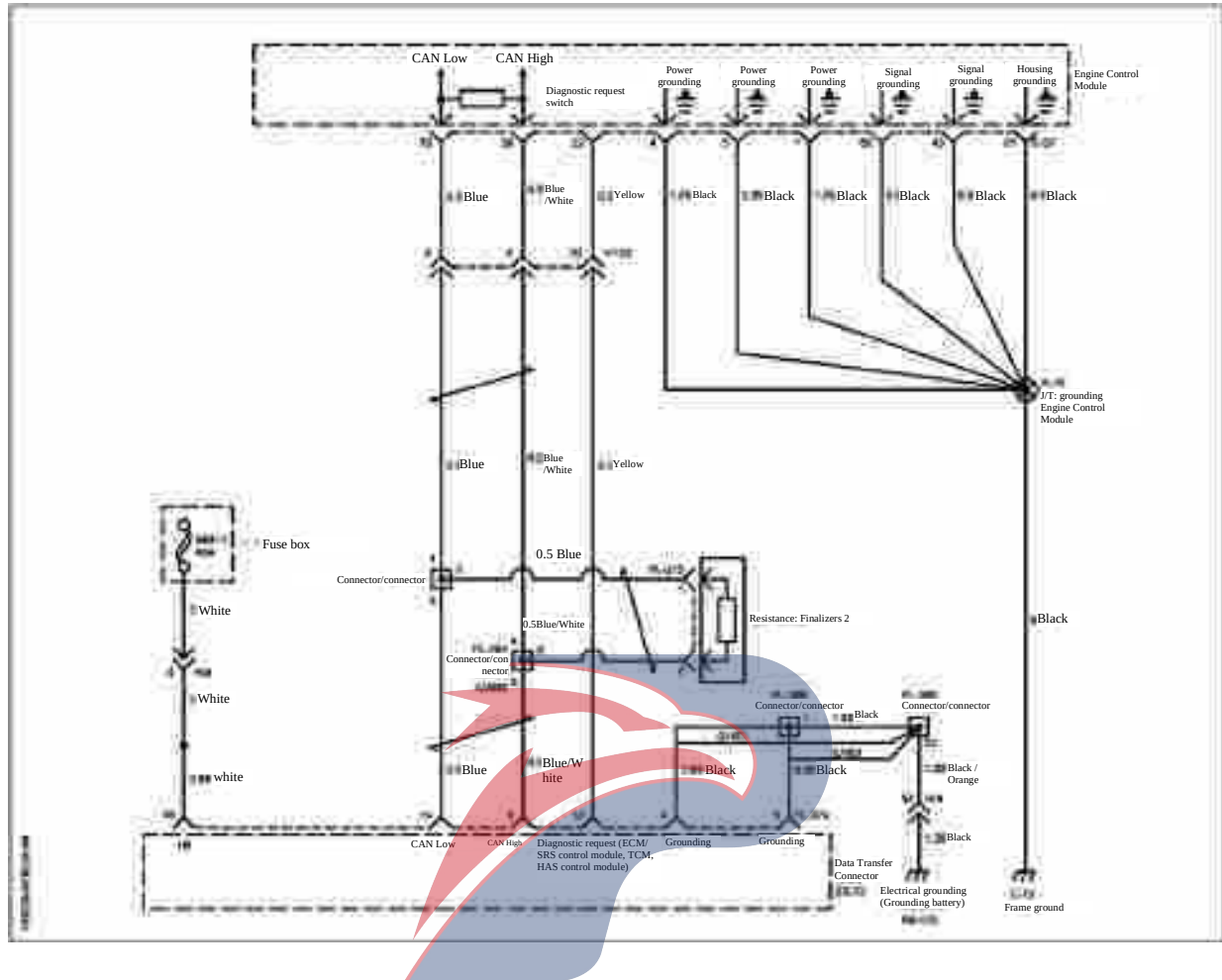
POWERSTAR

Engine control module power supply



POWERSTAR

Engine control module grounding and data transfer connector (DLC)



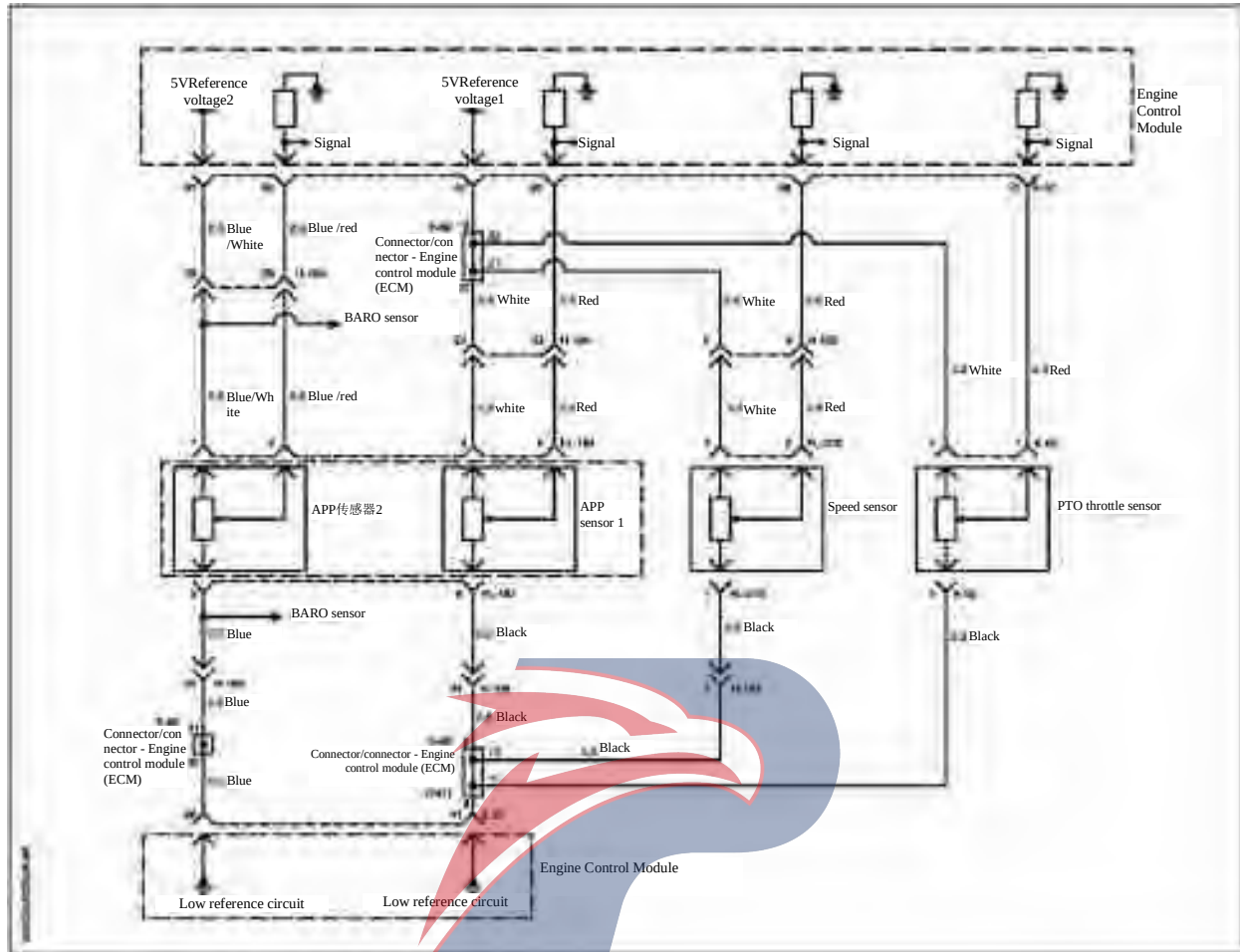
POWERSTAR

The diagram illustrates the electrical wiring for a vehicle, showing the connection of various components to a central power source (battery) and ground. Key components and their connections include:

- Fuse Box:** Contains a measuring meter and a tachometer. The tachometer is connected to the engine's tachometer output (0.5 Pink).
- Engine Control Module (ECM):** Receives inputs from the tachometer (0.5 Pink), speed sensor (0.5 Yellow), and engine coolant temperature (ECT) sensor (0.5 Red / Blue). It controls the engine's operation.
- Sensors:**
 - Thermometer:** Connected to the engine coolant temperature (ECT) sensor (0.5 Red / Blue).
 - Tachometer:** Connected to the engine's tachometer output (0.5 Pink).
 - Speed sensor:** Connected to the ECM (0.5 Yellow).
 - Oil pressure switch:** Connected to the ECM (0.5 Red / Blue).
 - Check engine:** Connected to the ECM (0.5 Red / Blue).
 - Thermoelectric wire:** Connected to the ECM (0.5 Red / Blue).
 - Exhaust brake:** Connected to the ECM (0.5 Red / Blue).
 - Parking brake:** Connected to the ECM (0.5 Red / Blue).
- Wiring Colors:** The diagram uses various wire colors to identify different circuits, including Yellow, Red, Pink, Black, Green, Brown, Blue, and Purple. A legend at the bottom right lists the wire colors and their corresponding functions.
- Grounding:** The system is grounded to the battery (Electrical grounding (Grounding battery)).

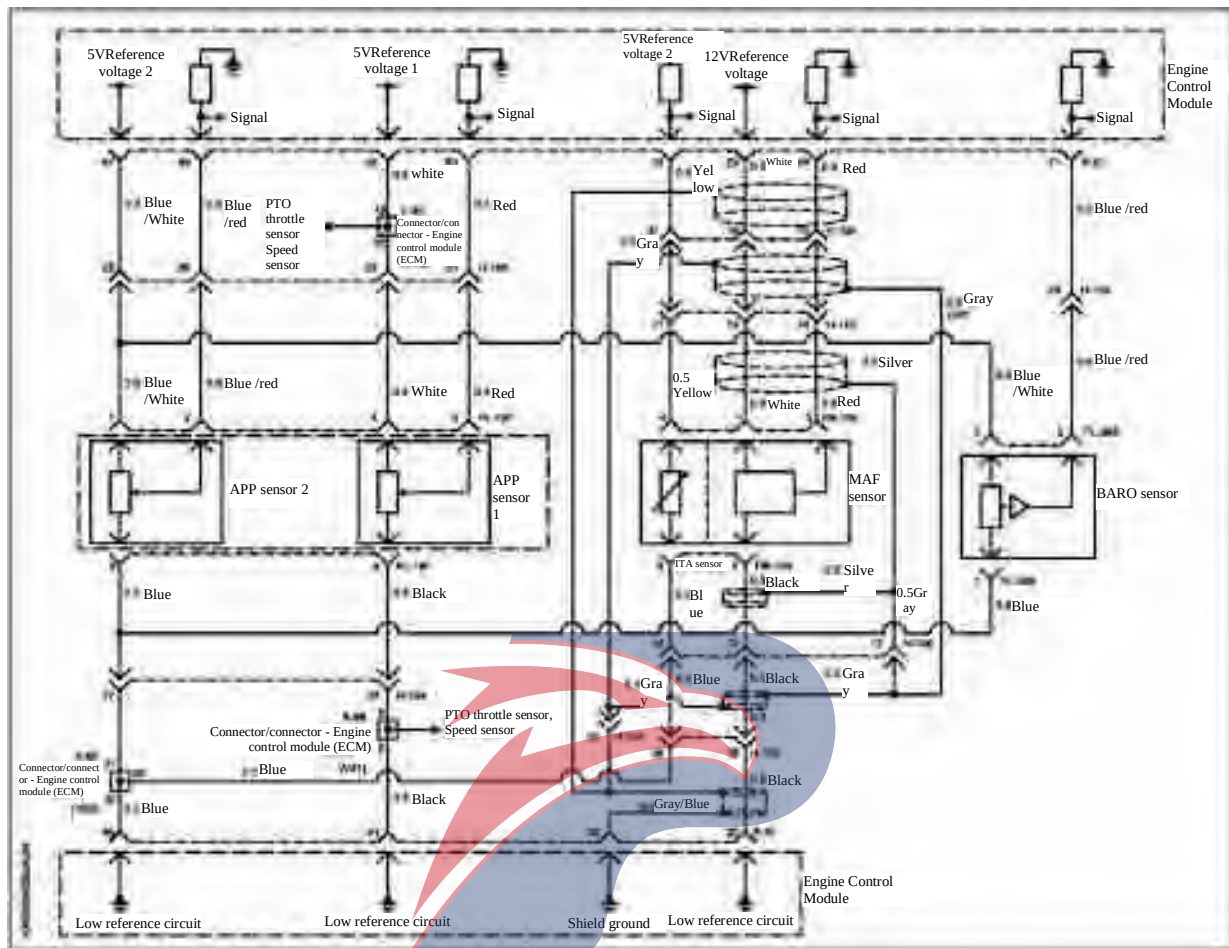
POWERSTAR

Throttle speed sensor and PTO throttle sensor



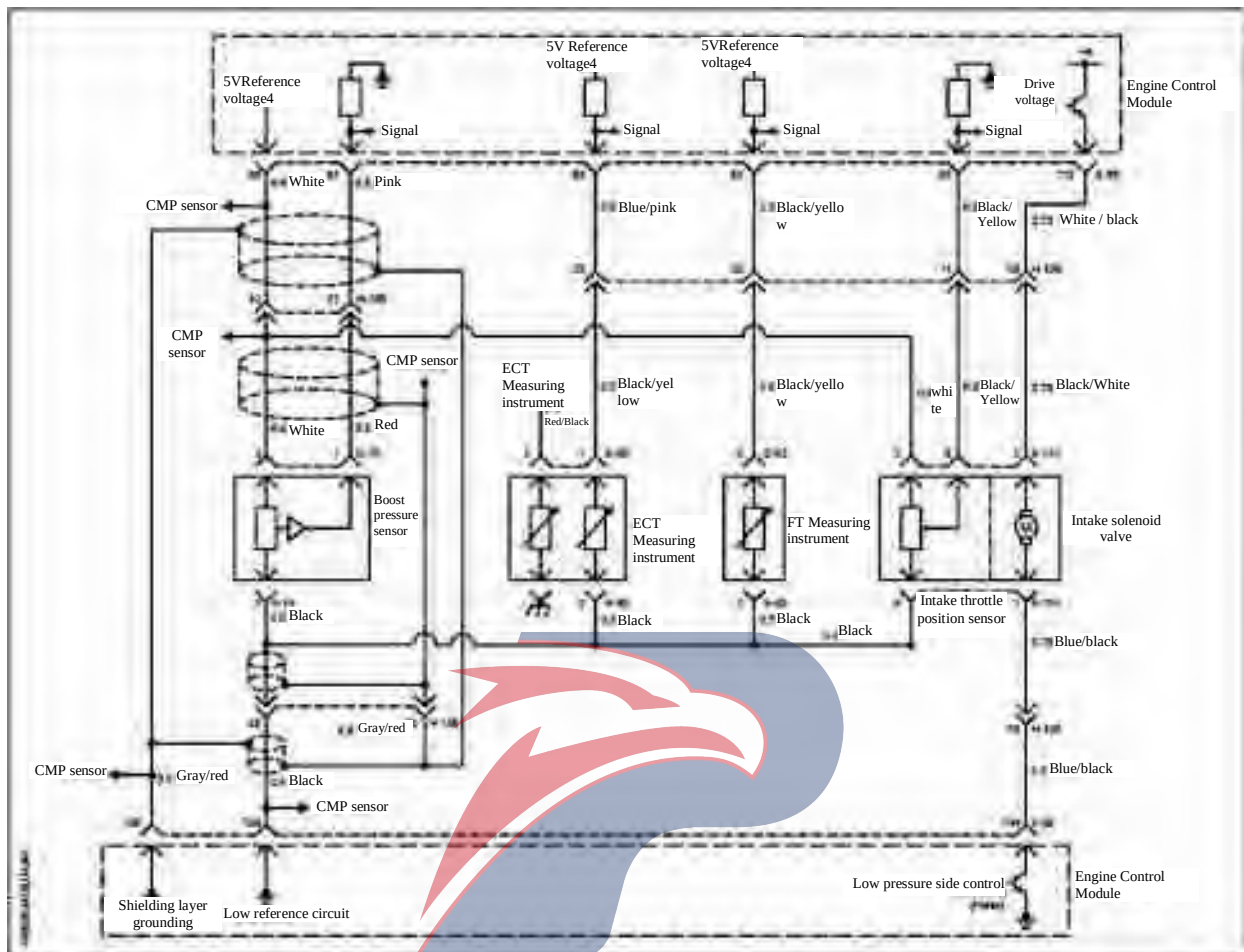
POWERSTAR

Accelerator pedal position (APP) sensor, Inlet air temperature (IAT) sensor, and atmospheric pressure (BARO) sensor



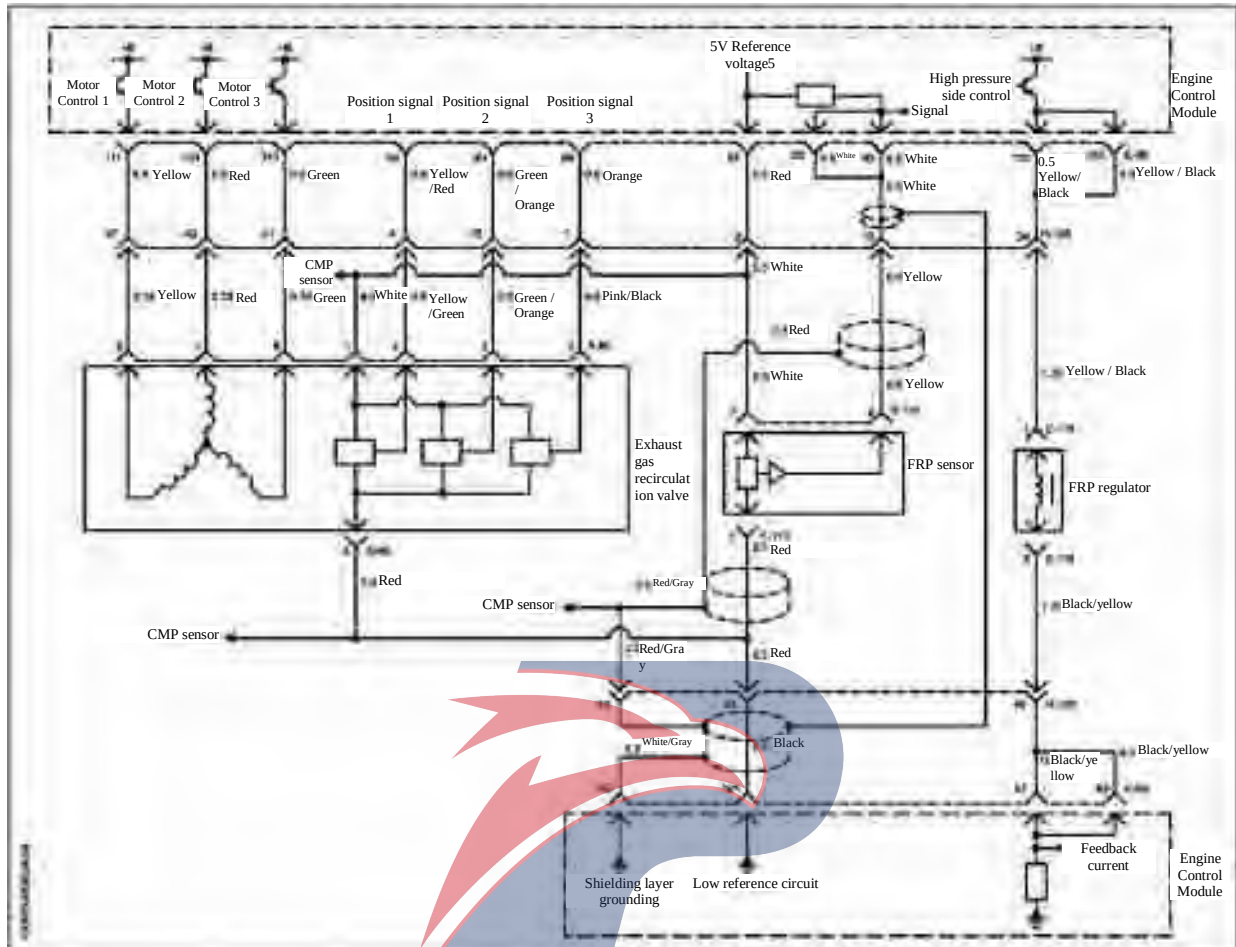
POWERSTAR

Intake throttle sensor, boost pressure sensor, engine coolant temperature (ECT) sensor and fuel temperature (FT) sensor



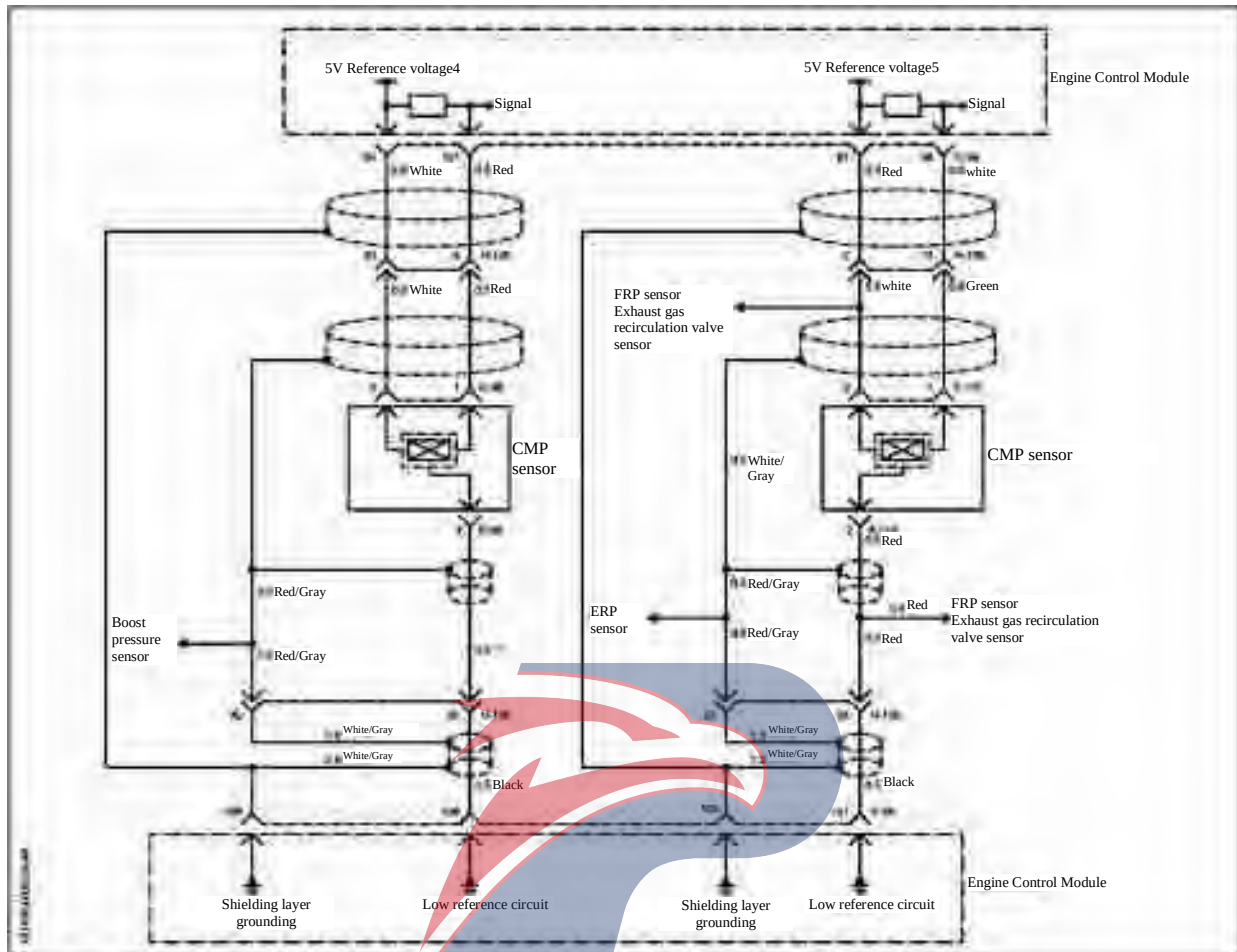
POWERSTAR

Exhaust gas recirculation (EGR) valve, FRP sensor and FRP regulator



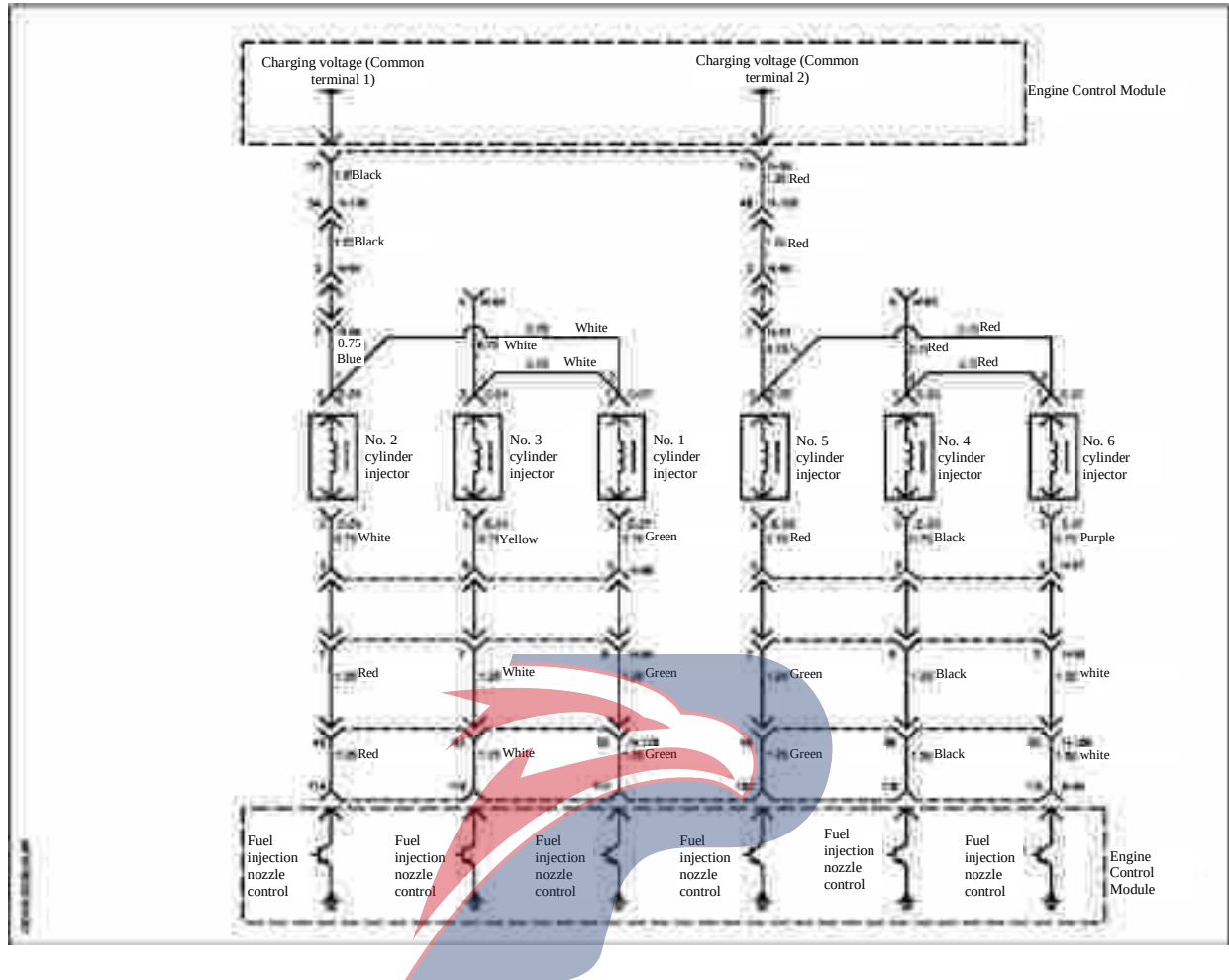
POWERSTAR

Crankshaft (CKP) and Camshaft (CMP) Sensors



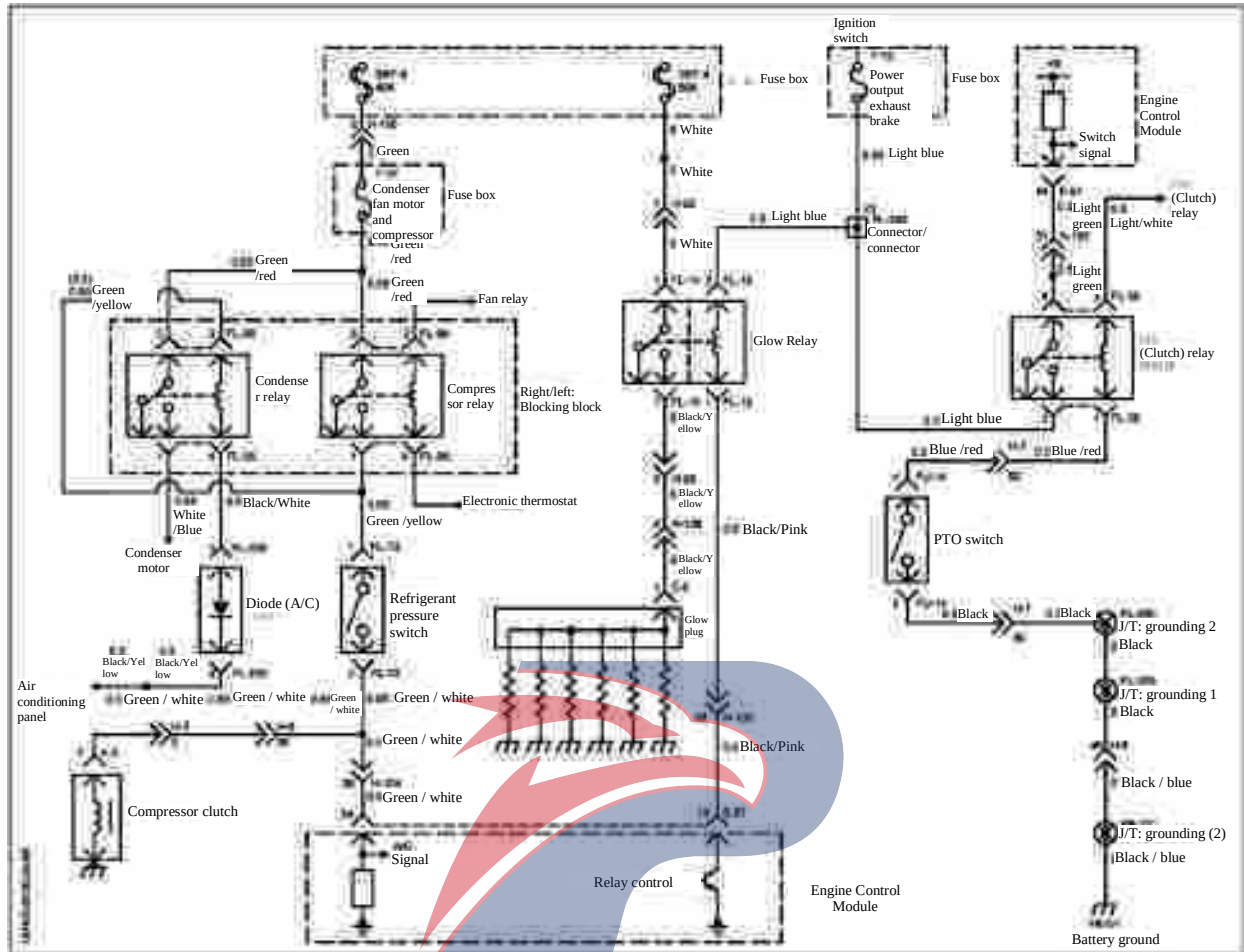
POWERSTAR

Oil spray nozzle.



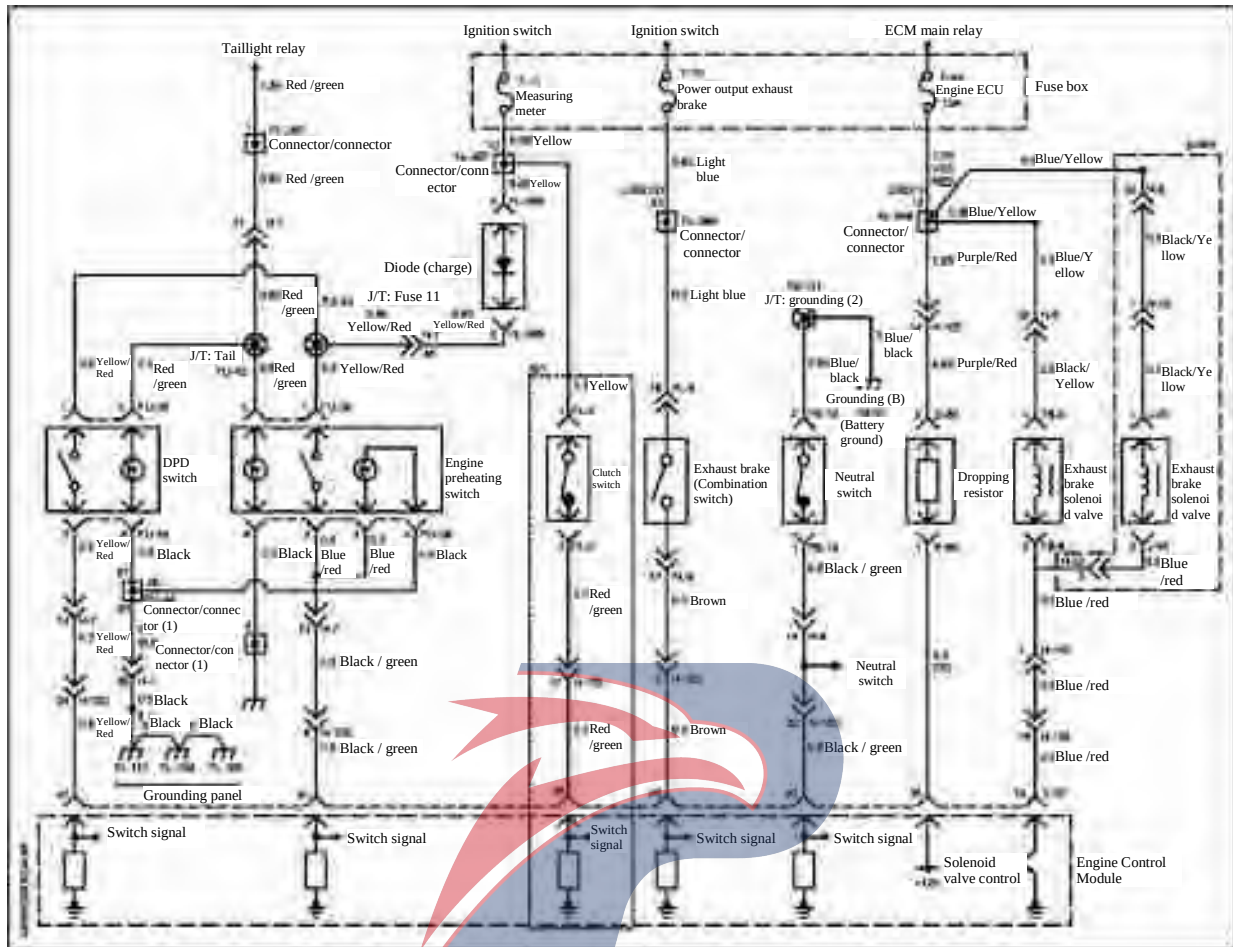
POWERSTAR

The glow control device, the A/C control device and the PTO switch



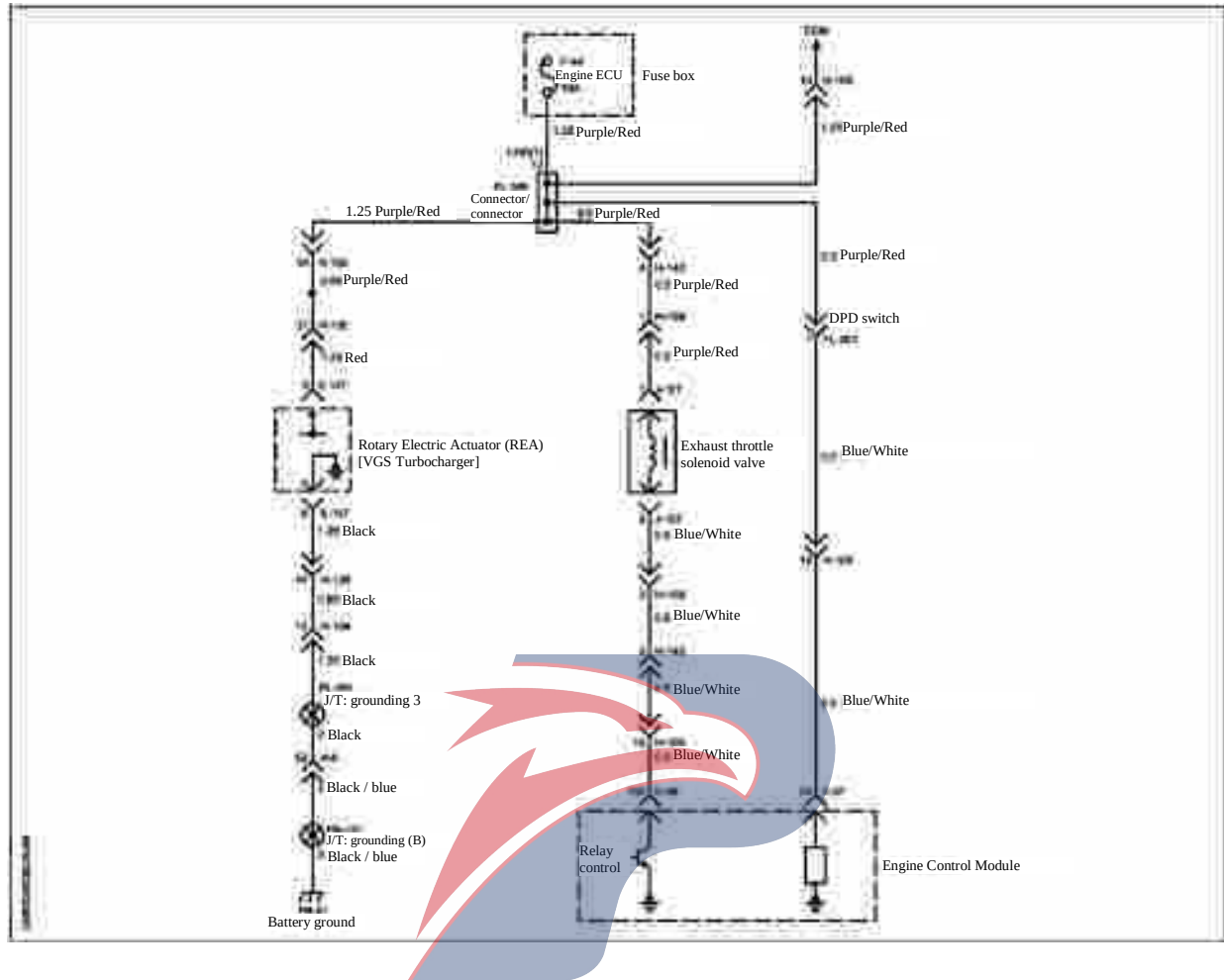
POWERSTAR

Switch, exhaust brake magnet valve and voltage dropping resistor



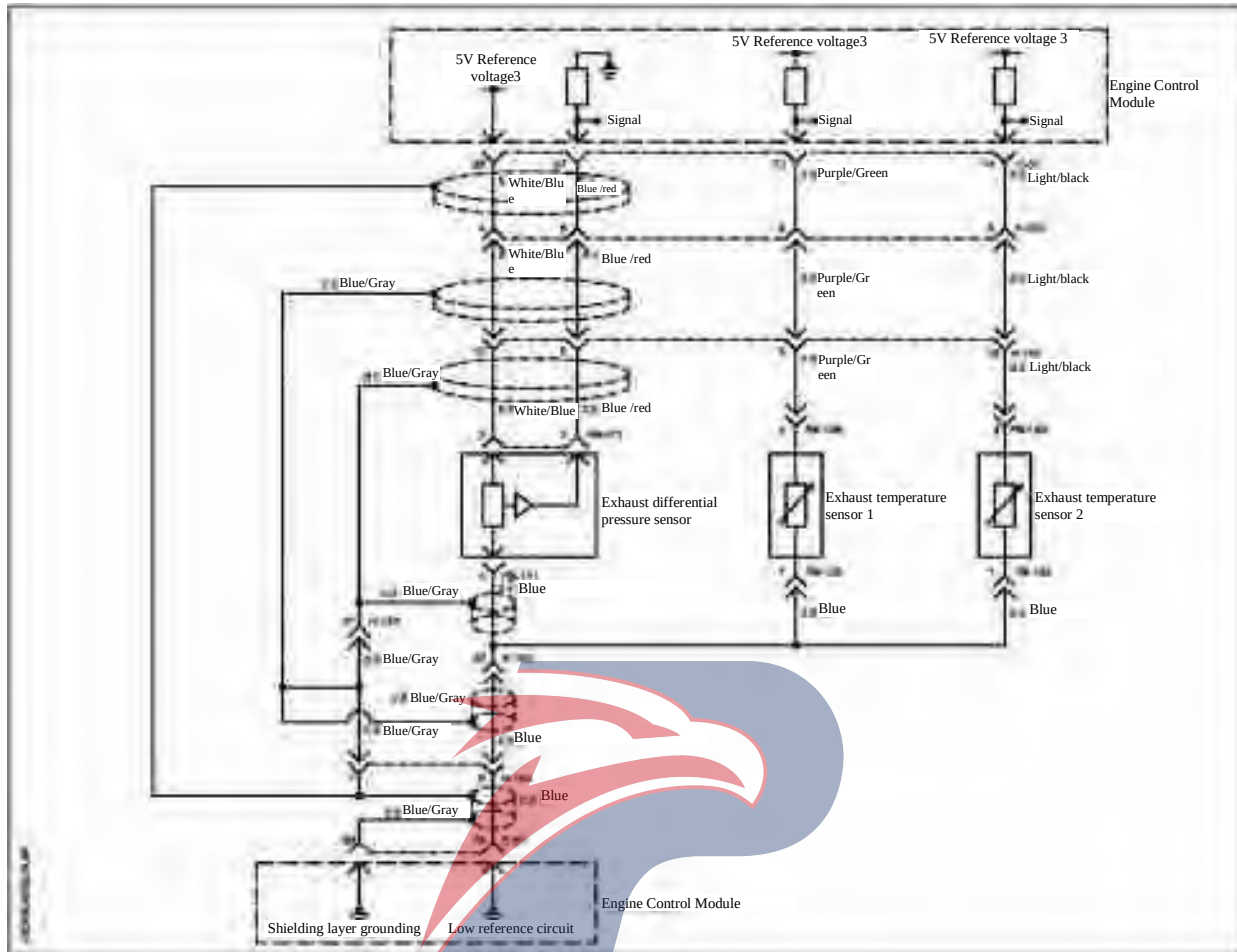
POWERSTAR

Exhaust throttle and VGS turbocharger



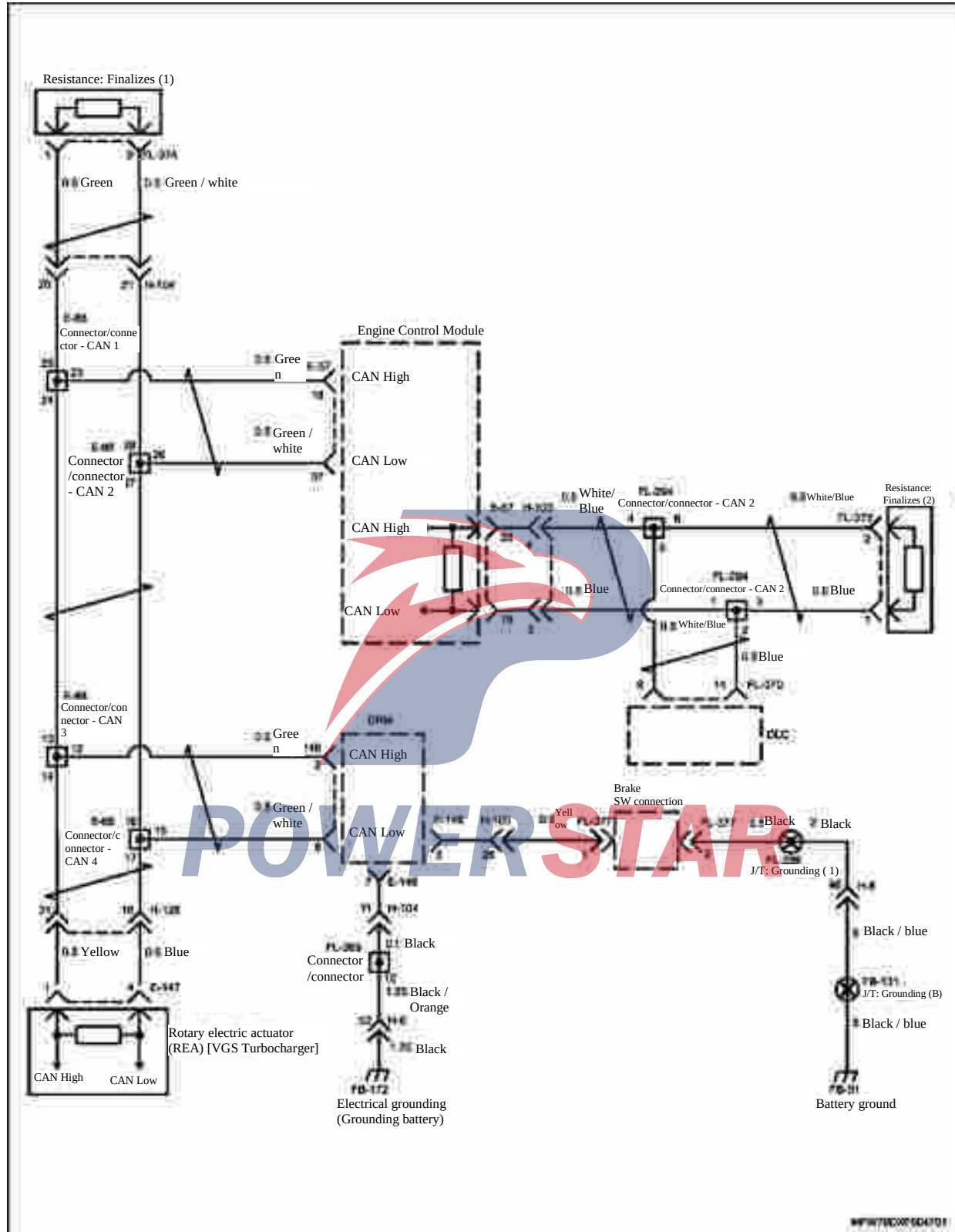
POWERSTAR

Exhaust differential pressure and exhaust temperature sensors

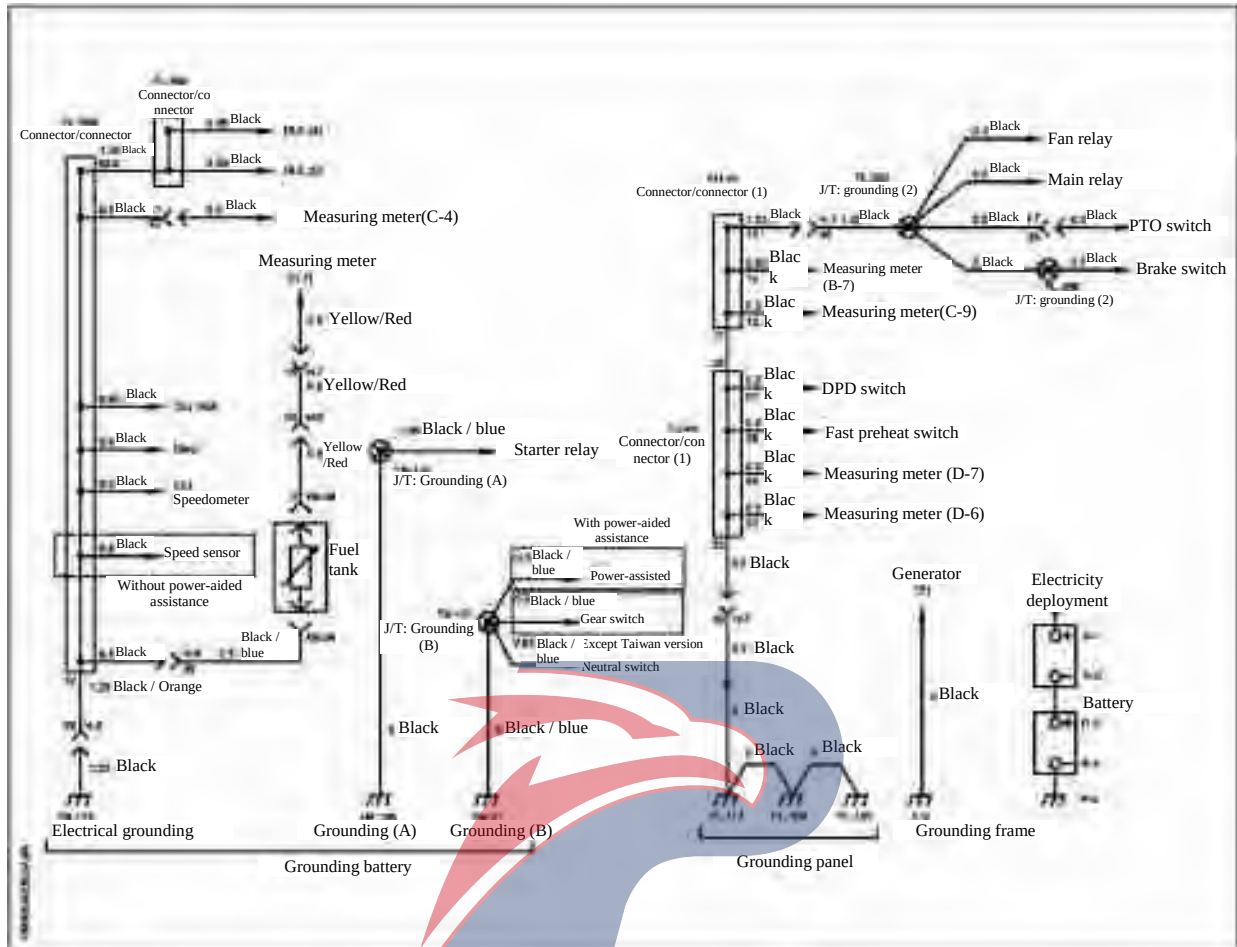


POWERSTAR

Controller Area Network (CAN)



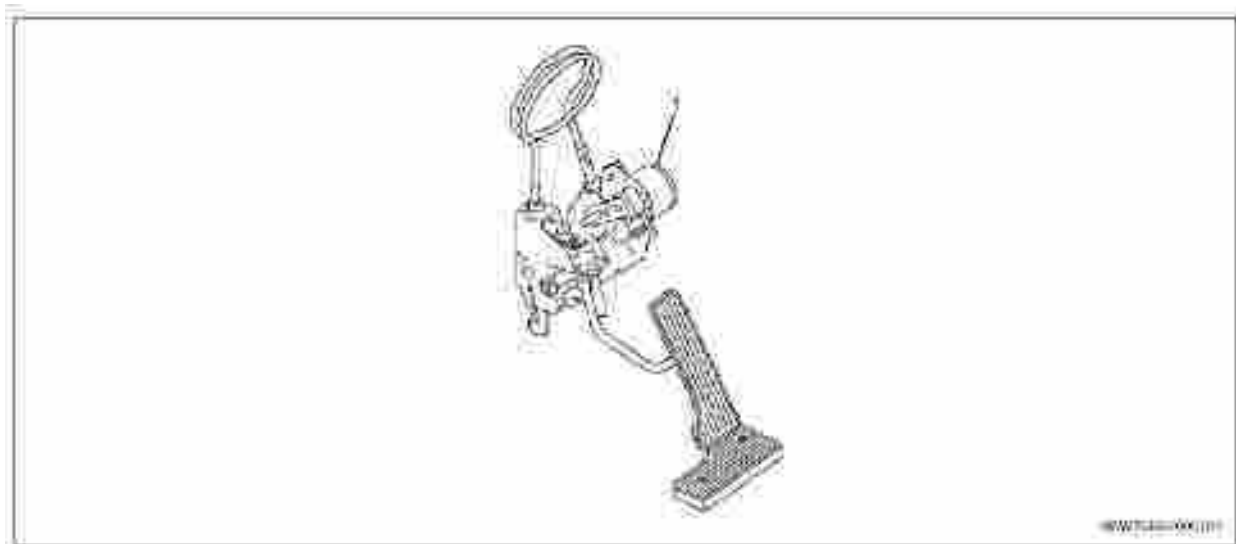
Grounding arrangements



POWERSTAR

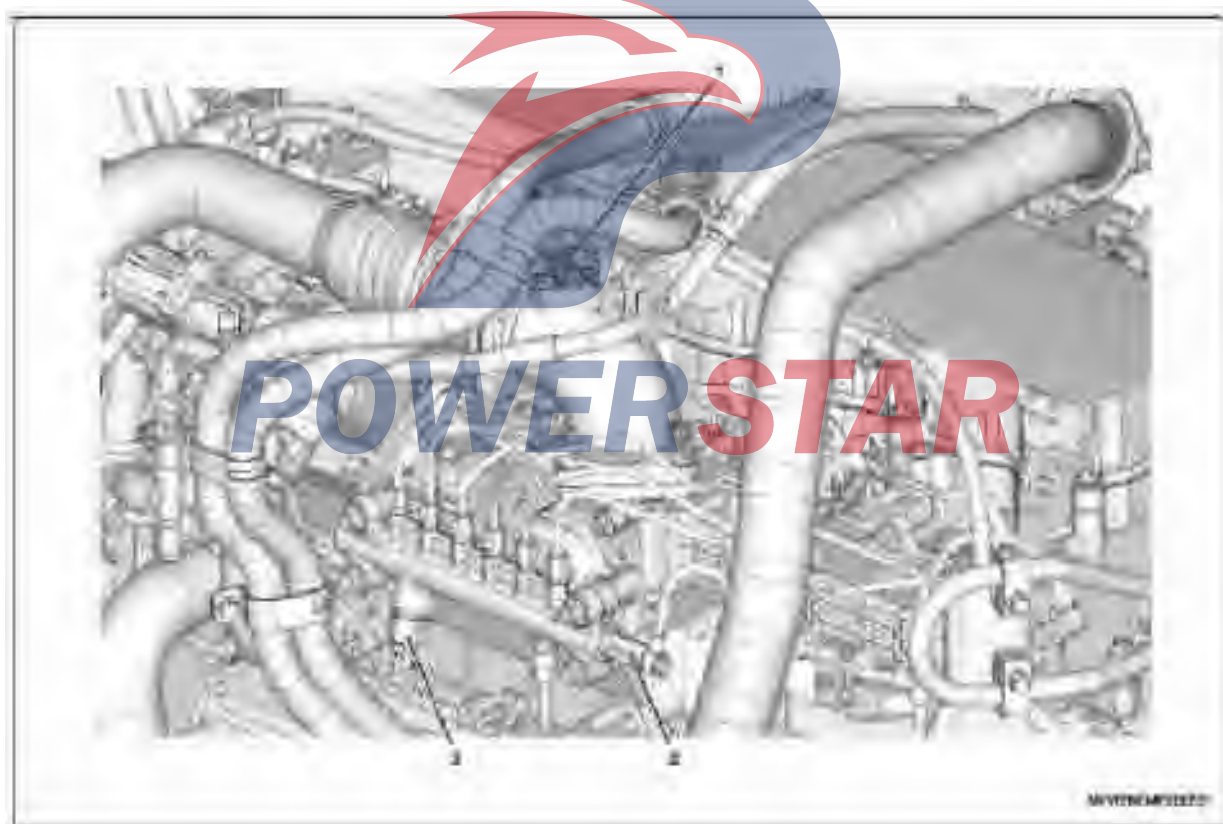
Part positioning map

Components View of Engine Control System



Key

1. Accelerator Pedal Position (APP) sensor

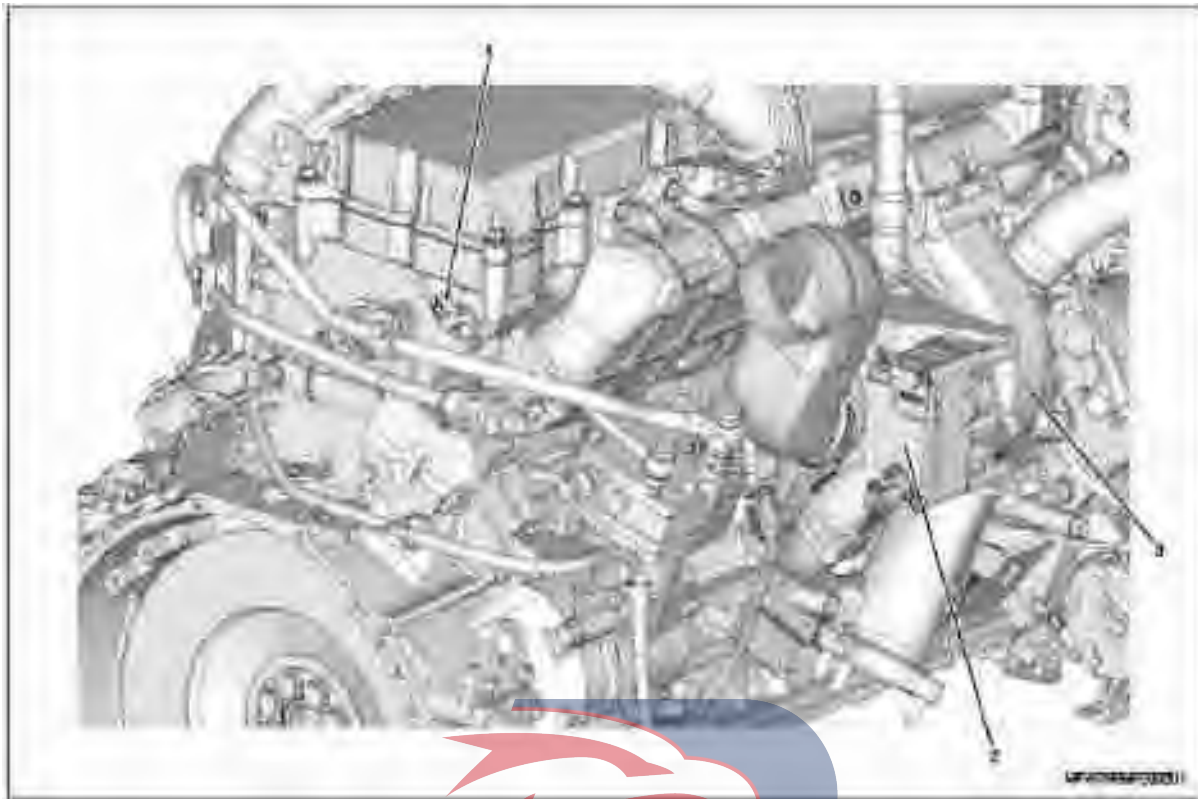


Key

1. Boost pressure sensor

2. Pressure limiting valve

3. Fuel rail pressure (FRP) sensor

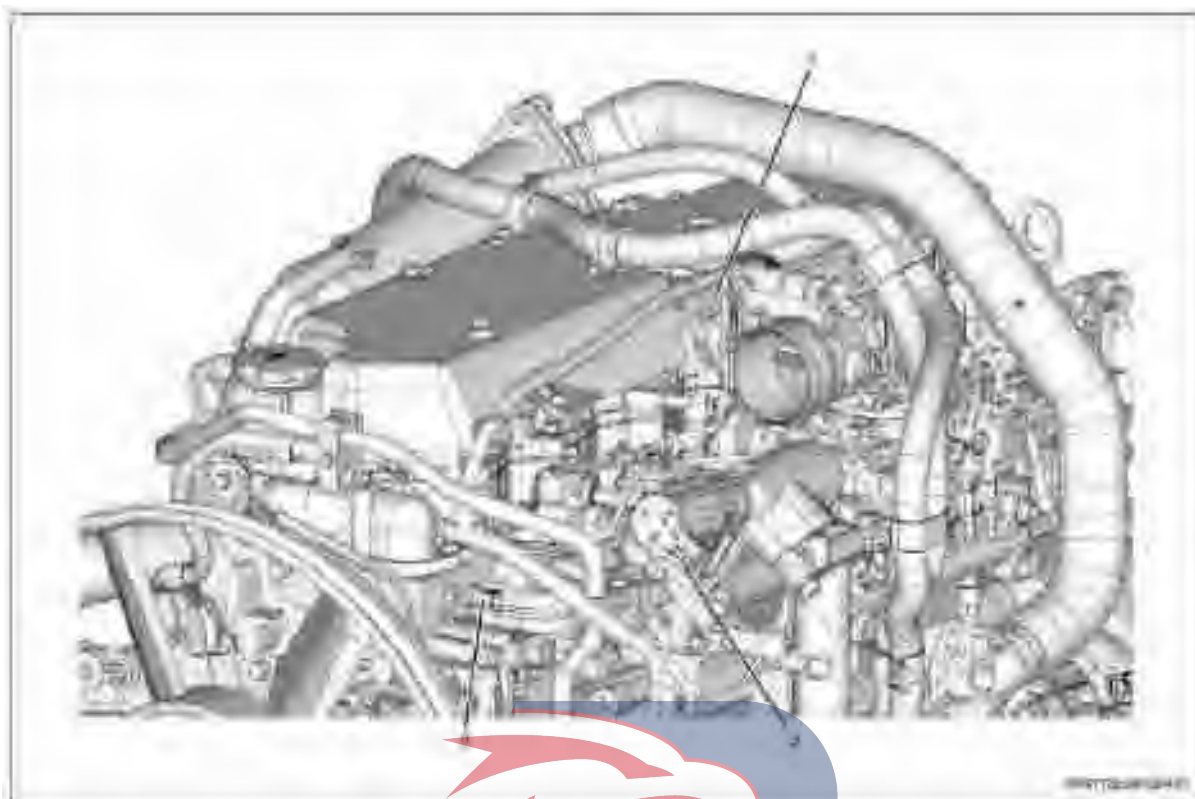


Key

- 1. Camshaft position (CMP) sensor
- 2. Rotary Electric Actuator (REA)

3. Turbocharger

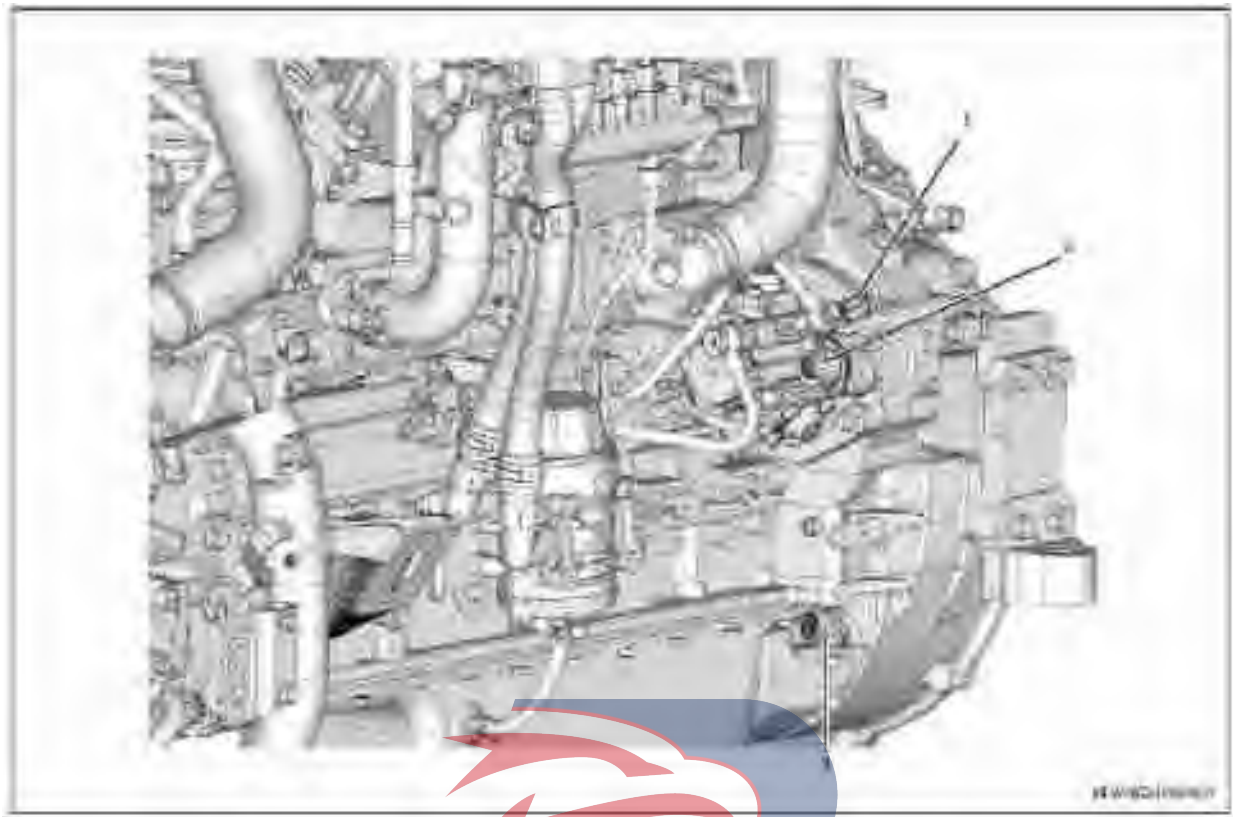
POWERSTAR



Key

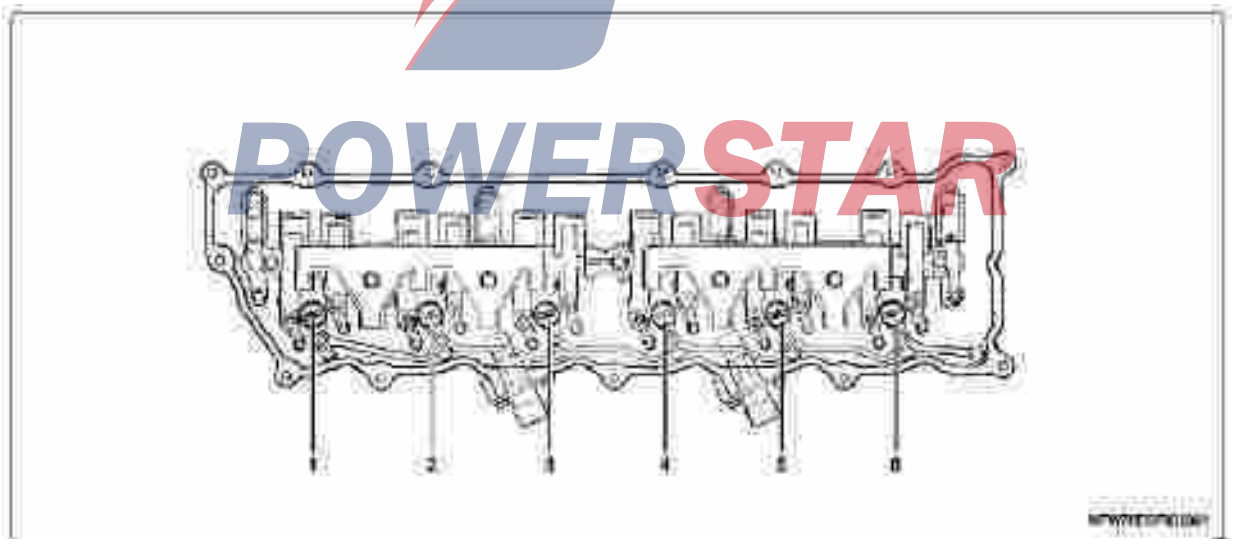
- 1. Air intake throttle
- 2. Exhaust gas recirculation (EGR) valve
- 3. Engine coolant temperature (ECT) sensor

POWERSTAR



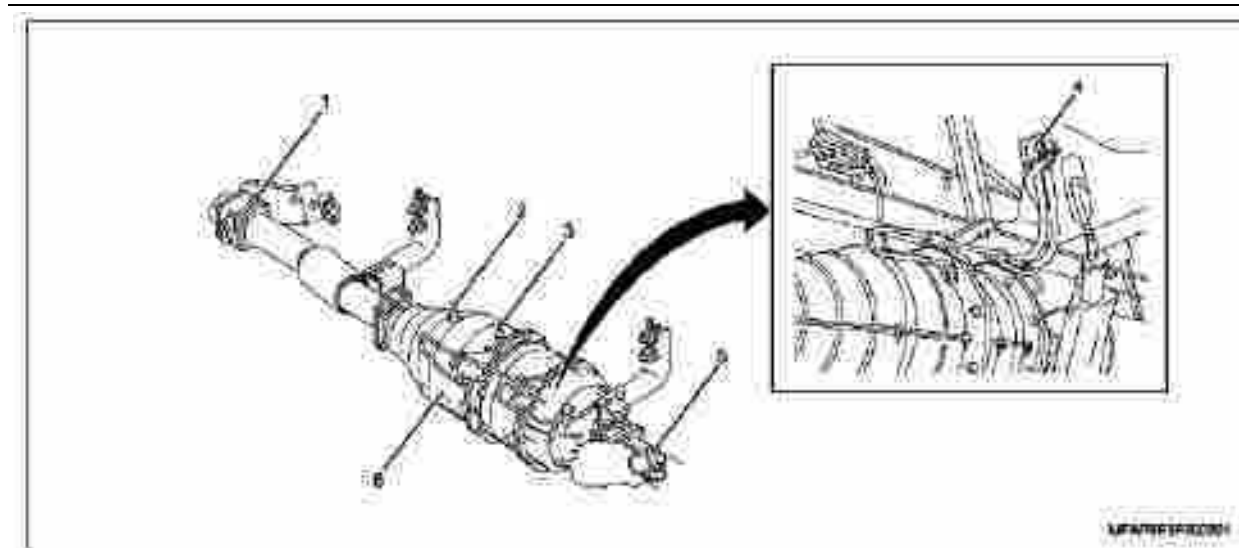
Key

- | | |
|--|-------------------------------------|
| 1. Fuel temperature (FT) sensor | 3. Crankshaft position (CKP) sensor |
| 2. Fuel Distribution Pipe Pressure (FRP) Regulator | |



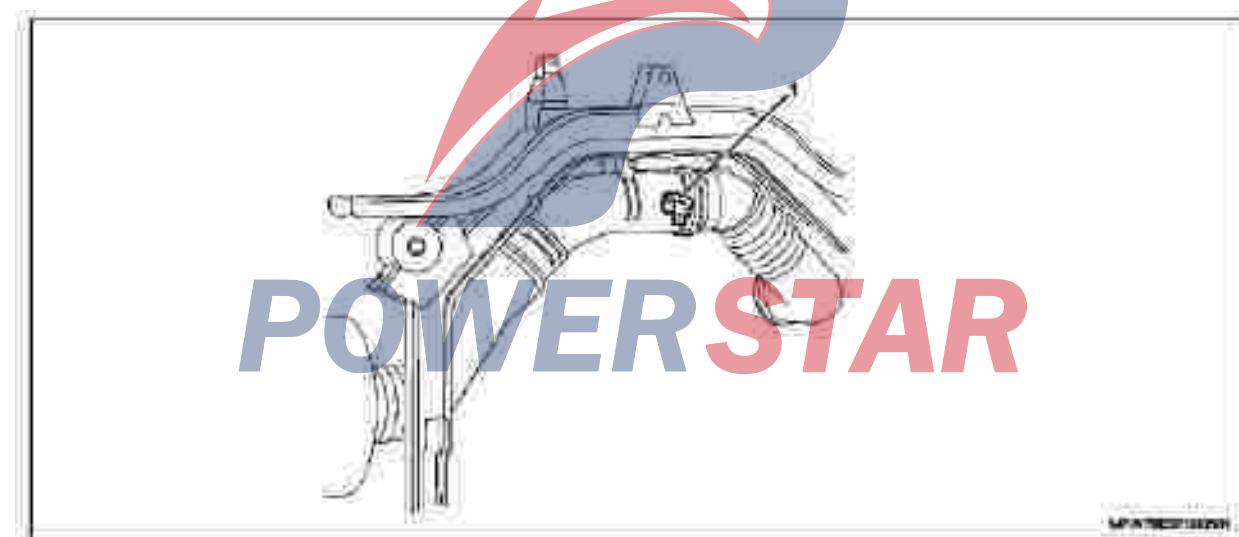
Key

- | | |
|----------------------------|----------------------------|
| 1. No. 1 cylinder injector | 4. No. 4 cylinder injector |
| 2. No. 2 cylinder injector | 5. No. 5 cylinder injector |
| 3. No. 3 cylinder injector | 6. No. 6 cylinder injector |



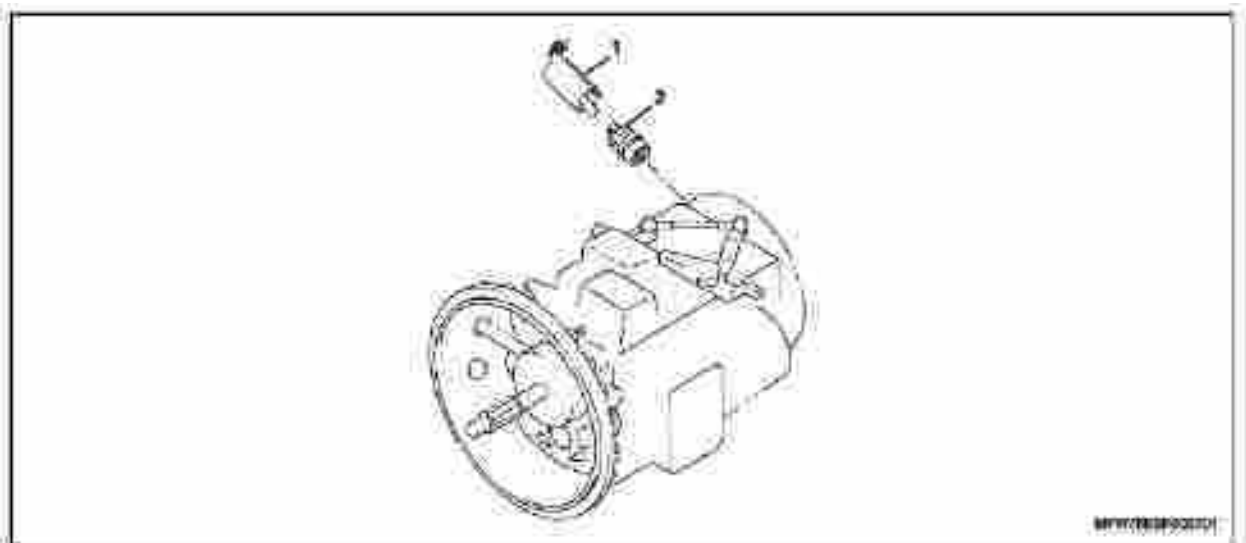
Key

- | | |
|--|---|
| 1. Exhaust brake valve
(in front of oxygen catalyst) | 4. Exhaust differential pressure sensor |
| 3. Exhaust temperature sensor 1 (in front of the filter) | 5. Exhaust throttle |
| 6. DPD assembly | |



Key

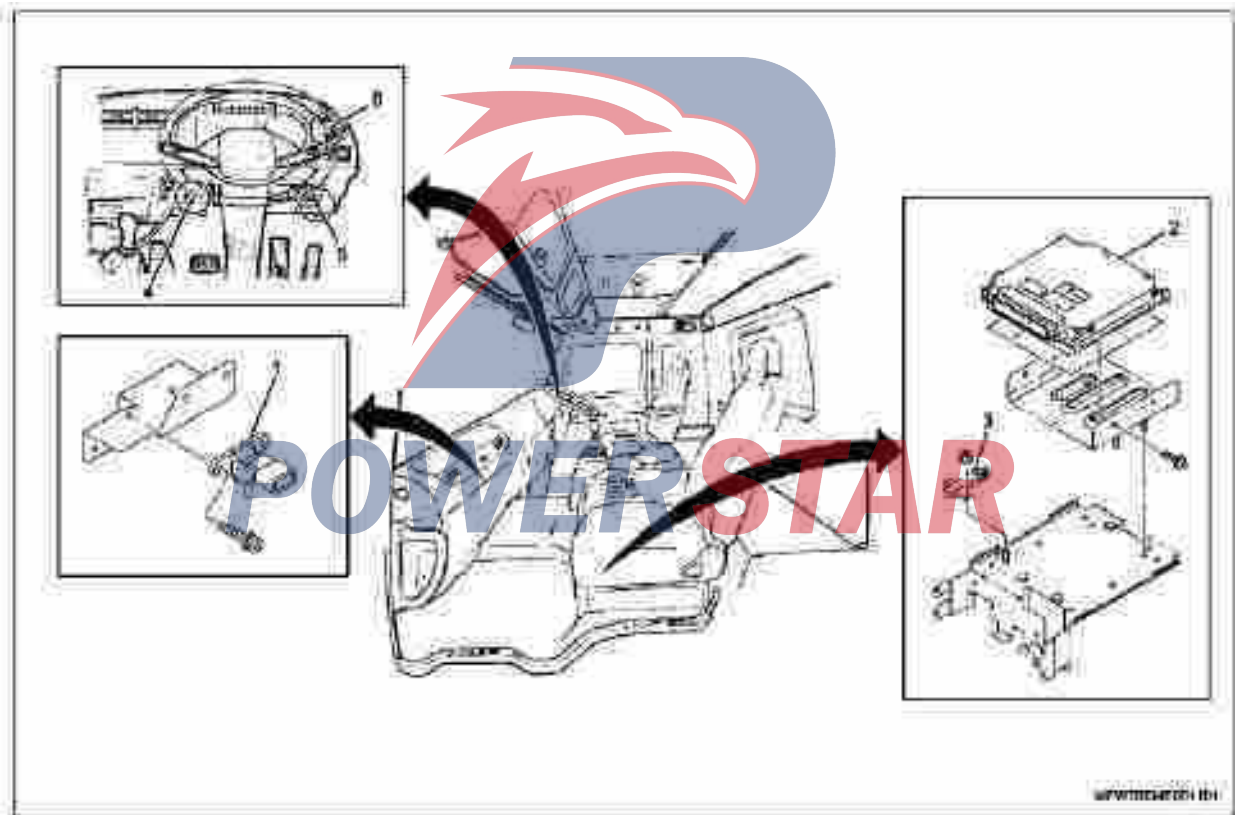
1. MAF sensor / IAT sensor



Key

1. Sensor cover

2. Speed sensor shaft(VSS)



Key

1. Atmospheric pressure (BARO) sensor

2. Engine Control Module (ECM)

3. Step-down resistor

4. Controller LAN resistance 2

5. Speed sensor control device

6. Controller Area Network Resistance 1 (IP Cluster)

ECM connector end view

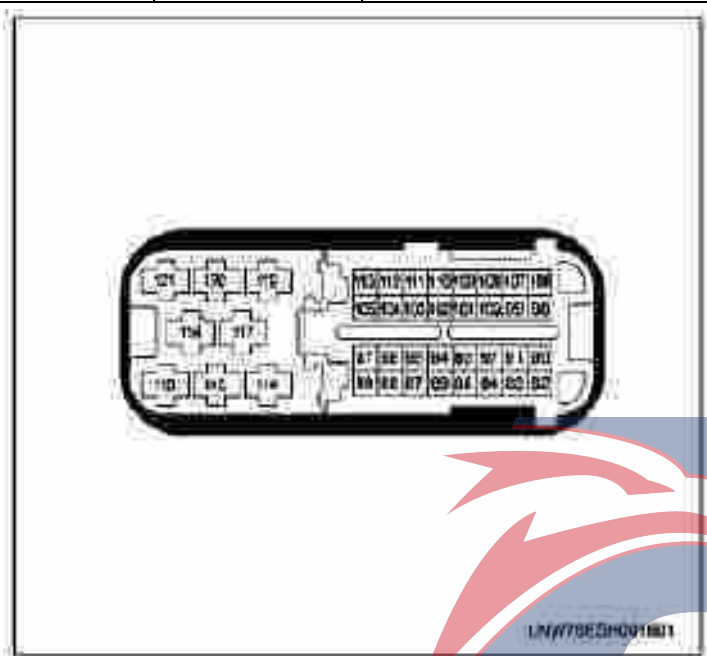
Engine Control Module (ECM)



Connector number		E-57
Connector color		Black
Test adapter number		J-35616-64A
Pin No.	Wire color	Pin function
1	Black	ECM signal ground
2	Red	Battery voltage
3	Black	ECM signal ground
4	Black	ECM signal ground
5	Red	Battery voltage
6	Brown/white	Malfunction Indicator Lamp (MIL) Control
7	Yellow/black	Exhaust brake light control
8	Red/black	Engine speed signal output to tachometer
9	Purple /red	DPD indicator control
10	Black/Pink	Glow plug relay control
11	Black/Green	Luminous control
12	-	Not used
13	-	Not used
14	Red/Green	Starter shutdown relay
15	Blue/Red	Exhaust brake solenoid valve control
16	-	Not used
17	Green	Service Vehicle Soon (SVS) indicator lamp control
18	Green	High signal of controller LAN
19	Pink/Blue	Speed sensor signal
20	Black	APP sensor 1 shielding layer is grounded
21	Green/black	Engine Control Module Main Relay
22	Black	Low reference voltage of MAF sensor
23	White	MAF sensor 12 volt reference voltage
24	Purple /white	Ignition voltage
25	-	Not used
26	Red/Green	Clutch pedal switch signal
27	-	Not used
28	Blue /white	DPD regenerative switch signal

Connector number		E-57
Connector color		Black
Test adapter number		J-35616-64A
29	-	Not used
30	-	Not used
31	-	Not used
32	-	Not used
33	-	Not used
34	Green /white	A/C signal
35	Purple	Dropping resistor
36	-	Not used
37	Green /white	Low signal of controller LAN
38	-	Not used
39	Gray/Blue	APP sensor 2 and MAF sensor shielding layer grounding
40	Green/black	ECM main relay control
41	Black	low reference voltage of the APP sensor, throttle acceleration sensor and PTO throttle sensor
42	White	APP sensor, throttle increase sensor and PTO throttle sensor 5V reference voltage
43	Black	Engine control module signal ground
44	Blue/Orange	PTO mode switch signal
45	Brown	Exhaust brake switch signal
46	Black /white	Starter switch signal
47	Yellow/Red	DPD switch signal
48	Blue/black	Parking brake switch signal
49	-	Not used
50	Black/Green	Neutral switch signal
51	Blue/Green	Engine Preheat Switch signal
52	Yellow	Diagnostic request switch
53	-	Not used
54	-	Not used
55	-	Not used
56	-	Not used
57	-	Not used
58	Blue /white	High signal of controller LAN
59	Gray/Blue	Shielding layer grounding of exhaust differential pressure sensor
60	Blue	Low reference voltage of APP sensor 2, BARO sensor and IAT sensor
61	Blue /white	APP sensor and BARO sensor 5V reference voltage
62	Black	Engine control module signal ground
63	Red	APP sensor 1 signal
64	Blue/Red	APP sensor 2 signal
65	-	Not used
66	Red	Throttle speed sensor signal
67	Blue/Red	Exhaust differential pressure sensor signal
68	-	Not used
69	Red	MAF sensor signal
70	Red	PTO Throttle Sensor Signal
71	Blue/Red	BARO sensor signal
72	Yellow	IAT sensor signal
73	Blue/Green	Exhaust temperature sensor 1 signal
74	Light Green/black	Exhaust temperature sensor 2 signal
75	-	Not used
76	-	Not used
77	-	Not used

Connector number		E-57
Connector color		Black
Test adapter number		J-35616-64A
78	Blue	Low signal of controller LAN
79	Blue	Low reference voltage of exhaust differential pressure sensor, exhaust temperature sensor 1 and exhaust temperature sensor 2
80	Blue /white	5 volt reference voltage of the Exhaust differential pressure sensor
81	Black	Grounding of engine control module housing

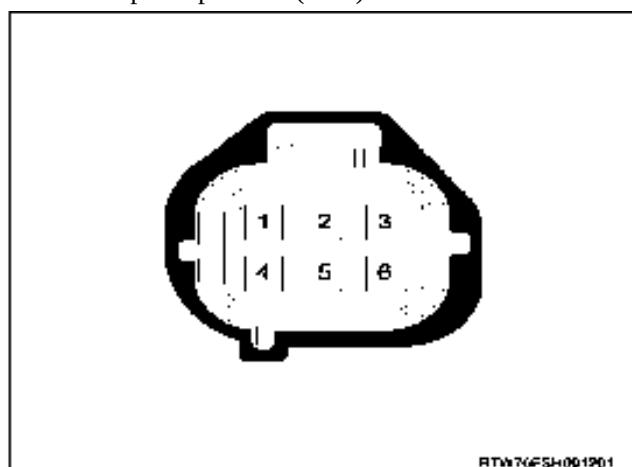


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Connector number		E-56
Connector color		Black
Test adapter number		J-35616-64A
Pin No.	Wire color	Pin function
82	White	FRP sensor signal
83	Red/Blue	FT sensor signal
84	Blue/Pink	ECT sensor signal
85	Black / Yellow	Intake throttle position sensor signal
86	-	Not used
87	Red	CMP sensor, FRP sensor and EGR position sensor 5 volt reference voltage
88	-	Not used
89	Blue/Red	low pressure side of the FRP regulator
90	White	FRP sensor signal
91	Pink	Signal circuit of booster pressure sensor signal
92	-	Not used
93	Green/Orange	EGR Position sensor signal 2
94	Yellow/Green	EGR position sensor signal 1
95	White	CKP sensor, intake throttle sensor station sensor and supercharge pressure sensor 5V reference voltage
96	-	Not used
97	Blue/Red	FRP regulator low control
98	White	CMP sensor signal
99	Orange	EGR Position sensor signal 3
100	Gray /white	CAM sensor and FRP sensor shielding layer GND
101	Black	Low reference voltage of the CMP sensor, FRP sensor and EGR position sensor
102	Blue /white	Exhaust throttle solenoid valve
103	Red	EGR motor control 2
104	Blue/black	Intake throttle low pressure side control
105	Yellow/black	FRP regulator high pressure side control
106	-	Not used
107	Red	CKP sensor signal
108	Gray/Red	CKP sensor and boost pressure sensor shielding layer grounding
109	Black	Low reference voltage of CKP sensor, intake throttle position sensor, FT sensor, ECT sensor and boost pressure sensor low reference voltage
110	Green	EGR motor control 3
111	Yellow	EGR motor control 1
112	White /black	Intake throttle drive voltage
113	Yellow/black	FRP regulator high pressure side control
114	Red	No. 2 cylinder injection nozzle control
115	White	No. 4 cylinder injection nozzle control
116	Red	Common terminal 2 (No.4, No.5 and No. 6 cylinders) nozzle charging voltage
117	Green	No. 3 cylinder injection nozzle control
118	Black	No. 6 cylinder injection nozzle control
119	White	No. 1 cylinder injection nozzle control
120	Green	No. 5 cylinder injection nozzle control
121	Black	Common terminal 1 (No.1, No.2 and No.3 cylinder) injection nozzle charging voltage

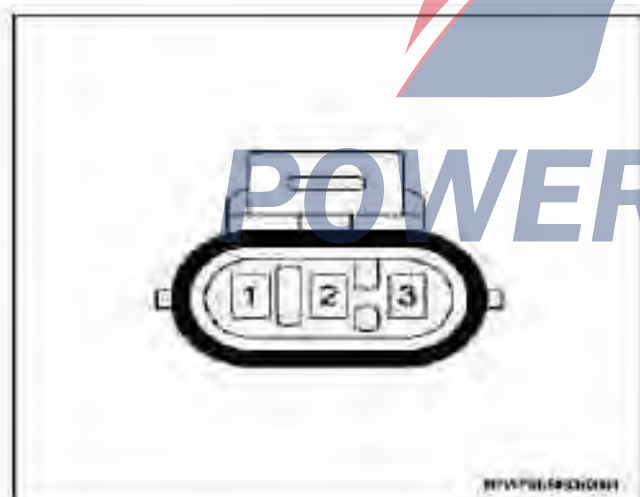
Engine control system connector end view

Accelerator pedal position (APP) sensor



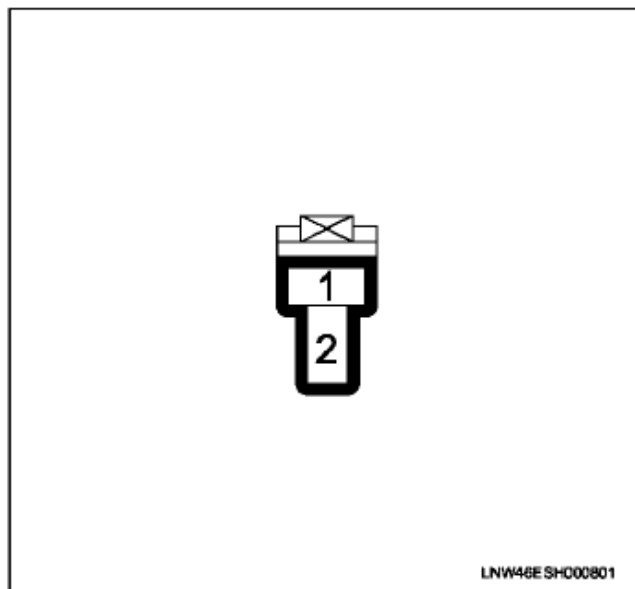
Connector number		FL-167
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue /white	25 volt reference voltage of APP sensor
2	Blue/Red	APP sensor 2 signal
3	Blue	APP sensor 2 low reference voltage
4	White	15 volt reference voltage of APP sensor
5	Red	APP sensor 1 signal
6	Black	APP sensor 1 low reference voltage

BARO sensors

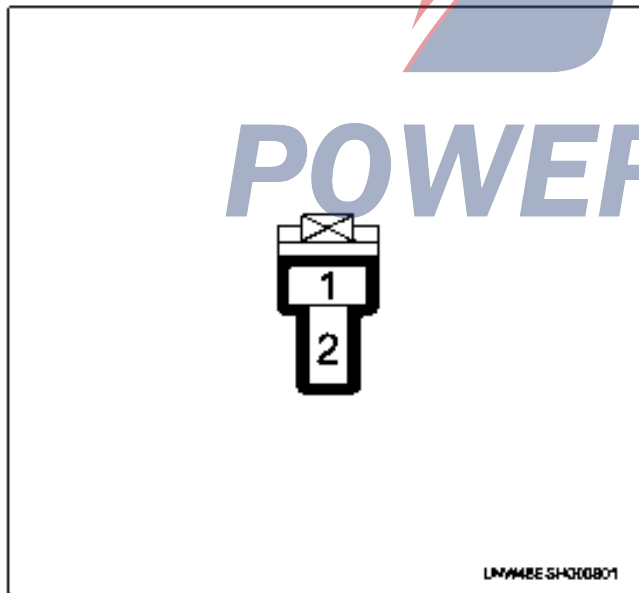


Connector number		FL-269
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue	Low reference voltage of sensor
2	Blue/Red	Sensor signal
3	Blue /white	Sensor 5V reference voltage

Booster pressure sensor



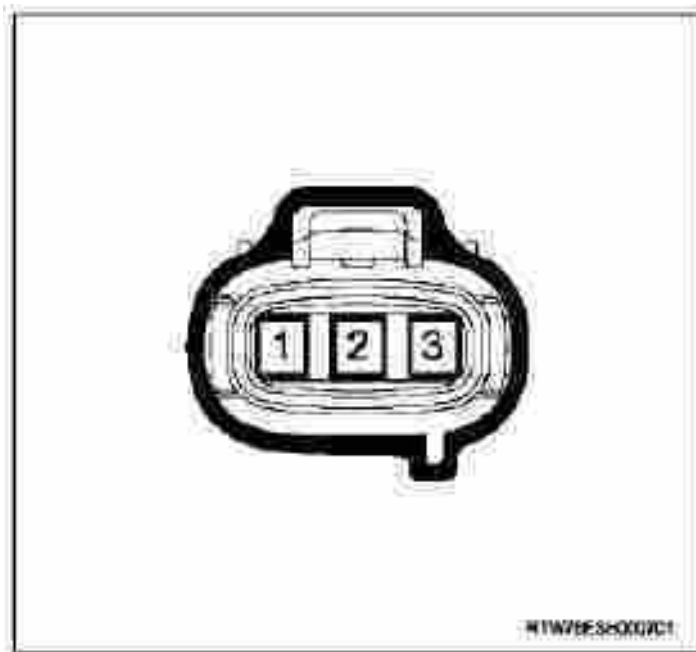
Connector number		E-75
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Red	Sensor signal
2	Black	Low reference voltage of sensor
3	White	Sensor 5V reference voltage



Clutch switch

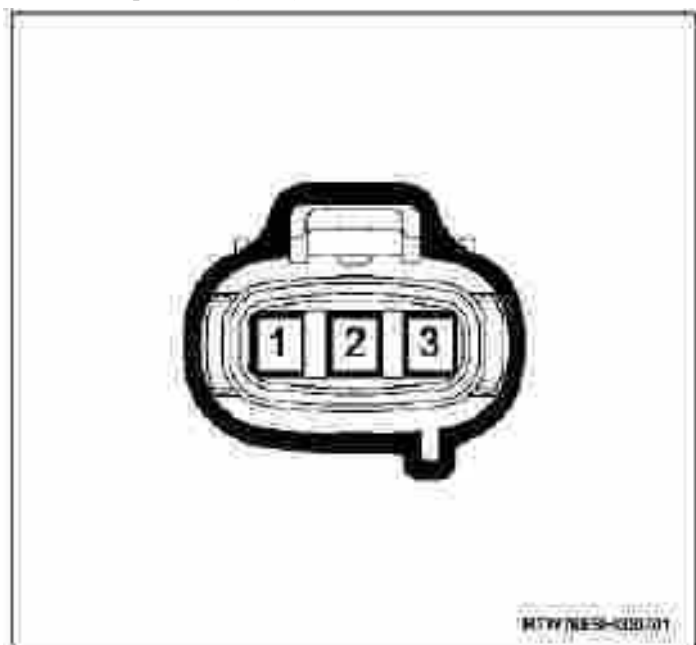
Connector number		FL-2
Connector color		White
Test adapter number		J-35616-42
Pin No.	Wire color	Pin function
1	White /Green	Ignition Voltage feeder
2	Yellow	Switch signal

Camshaft position (CMP) sensor



Connector number		E-112
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Green	Sensor signal
Connector number		E-112
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
2	Red	Low reference voltage of sensor
3	White	Sensor 5V reference voltage

Crankshaft position (CKP) sensor



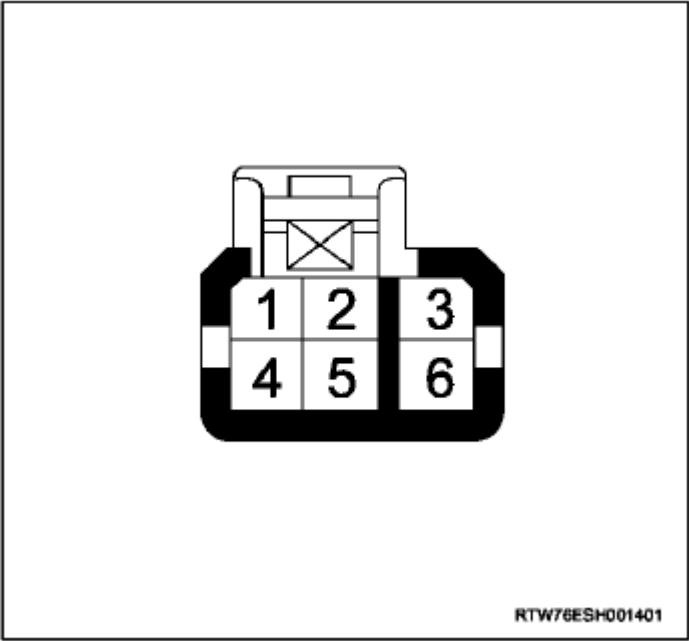
Connector number		E-98
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Red	Sensor signal
2	Black	Low reference voltage of sensor
3	White	Sensor 5V reference voltage

Data Transfer Connector (DLC)



Connector number		FL-370
Connector color		Light blue
Test adapter number		J-35616-2A
Pin No.	Wire color	Pin function
1	-	Not used
2	-	Not used
3	-	Not used
4	Black	Grounding
5	Black	Grounding
6	Blue /white	High signal of controller
7	Light blue	Keyword Serial Data (HSA Control Module)
8	-	Not used
9	-	Not used
10	-	Not used
11	-	Not used
12	Yellow	Diagnostic request switch (ECM HAS control module)
13	-	Not used
14	Blue	Controller LAN is low
15	-	Not used
16	Red / Yellow	Battery voltage

DPD switch



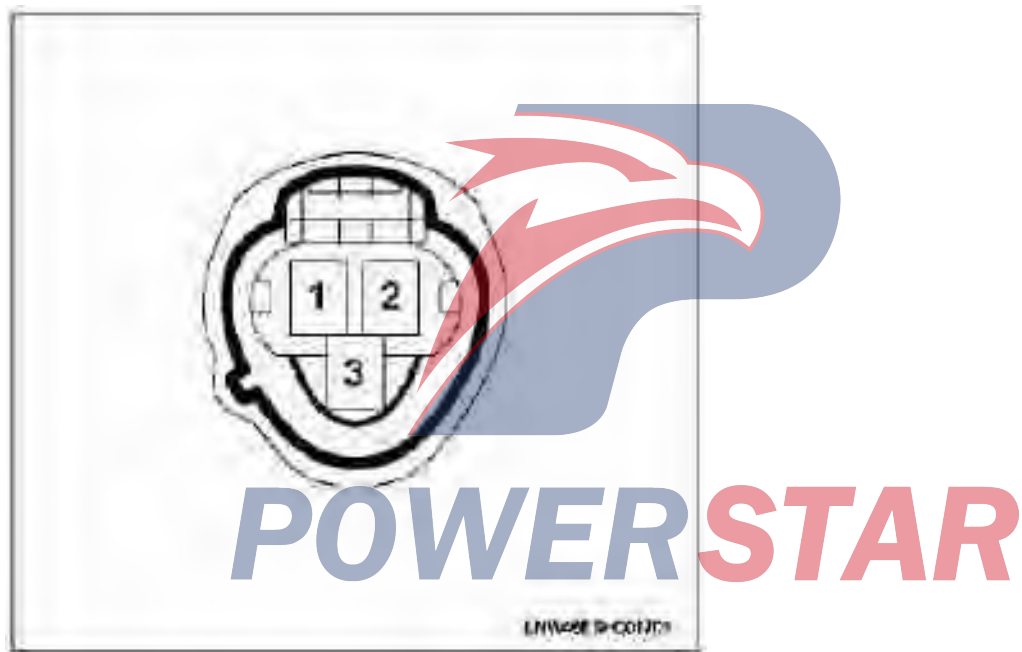
Connector number		FU-36
Connector color		Blue
Test adapter number		J-35616-4A
Pin No.	Wire color	Pin function
1	Yellow/Red	DPD switch signal
2	Black	Lighting lamp ground
3	Yellow/Red	DPD Switch Voltage Feeder
4	-	Not used
5	-	Not used
6	Yellow/Green	Lamp voltage feed

Exhaust gas recirculation valve



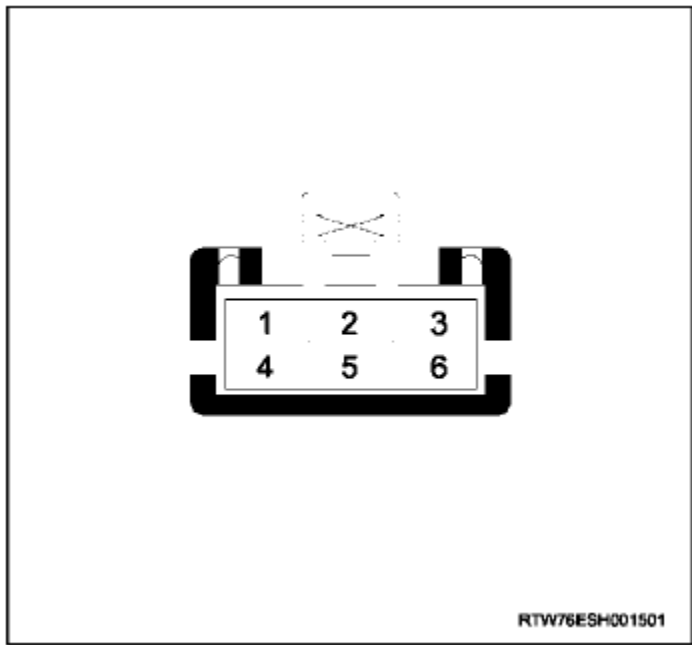
Connector number		FU-80
Connector color		Black
Test adapter number		J-35616-64A
Pin No.	Wire color	Pin function
1	White	Position sensor 5V reference voltage
2	Pink /black	Position sensor signal 3
3	Green/Orange	Position sensor signal 2
4	Yellow/Green	Position sensor signal 1
5	Red	Position sensor low reference voltage
6	Green	Motor Control 3
7	Red	Motor Control 2
8	Yellow	Motor Control 1

Engine coolant temperature (ECT) sensor



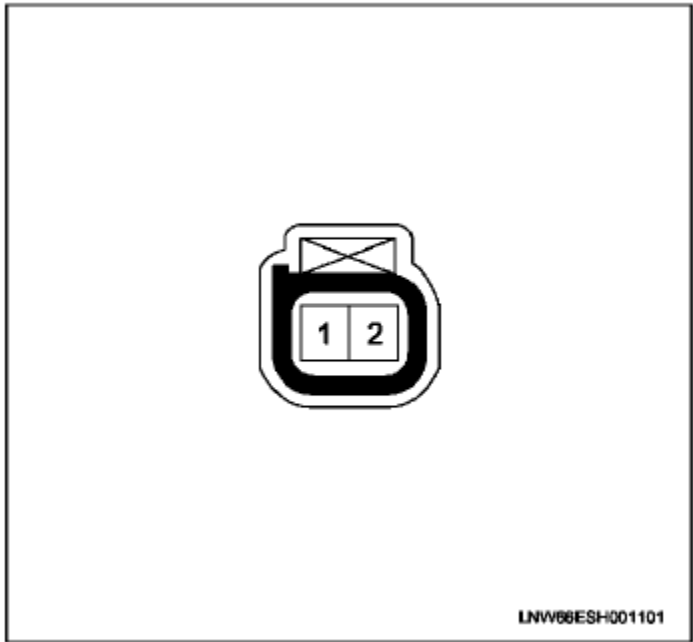
Connector number		E-90
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue/Red	Sensor signal
2	Black	Low reference voltage of sensor
3	Red/Blue	Measuring instrument signal

Engine preheating switch



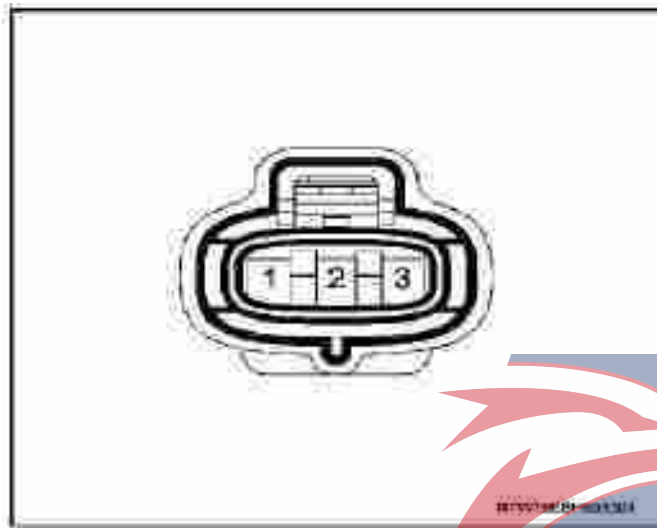
Connector number		FU-39
Connector color		Gray
Test adapter number		J-35616-42
Pin No.	Wire color	Pin function
1	Blue/Green	Lighting lamp ground
2	Black	Engine Preheat Switch signal
3	Yellow/Red	Engine preheating switch voltage feed
4	Black	LED ground
5	Blue/Green	The indicator lamp voltage feeds through the engine preheating switch signal
6	Red/Green	Lamp voltage feed

Exhaust brake solenoid valve



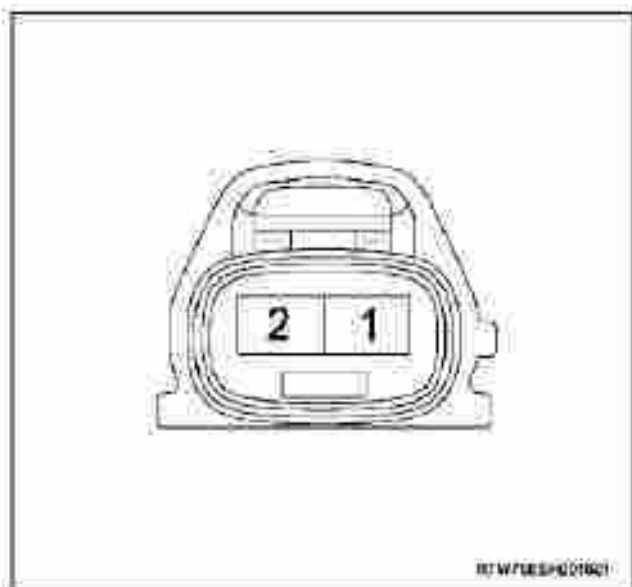
Connector number		FB-6, J-45
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue / Yellow	Battery voltage feed
2	Blue/Red	Solenoid valve control

Exhaust differential pressure sensor



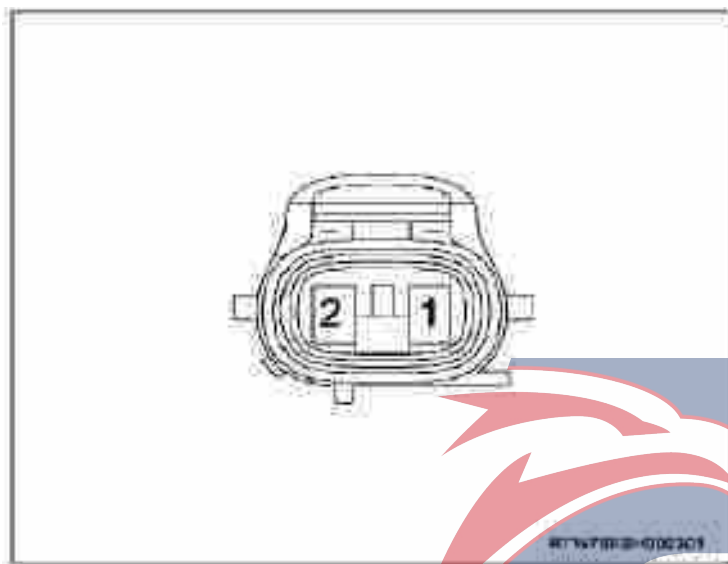
Connector number		FB-171
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue	Low reference voltage of sensor
2	Blue/Red	Sensor signal
3	Blue /white	Sensor 5V reference voltage

Exhaust temperature sensor 1



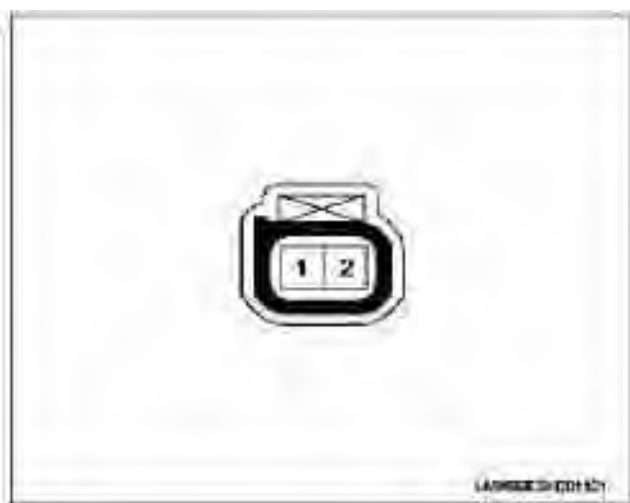
Connector number		FB-128
Connector color		Light grey
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue	Low reference voltage of sensor
2	Blue/Green	Sensor signal

Exhaust temperature sensor 2



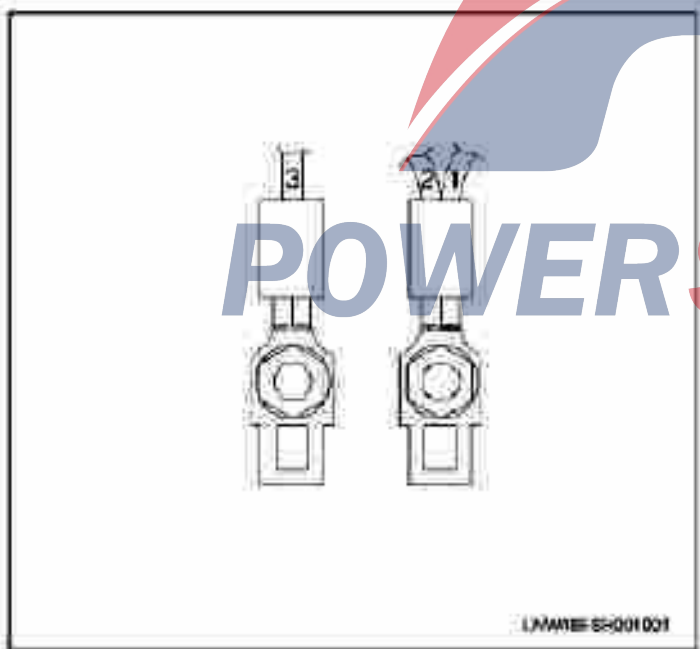
Connector number		FB-129
Connector color		Dark grey
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Blue	Low reference voltage of sensor
Connector number		FB-129
Connector color		Dark gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
2	Light Green/black	Sensor signal

Exhaust throttle solenoid valve



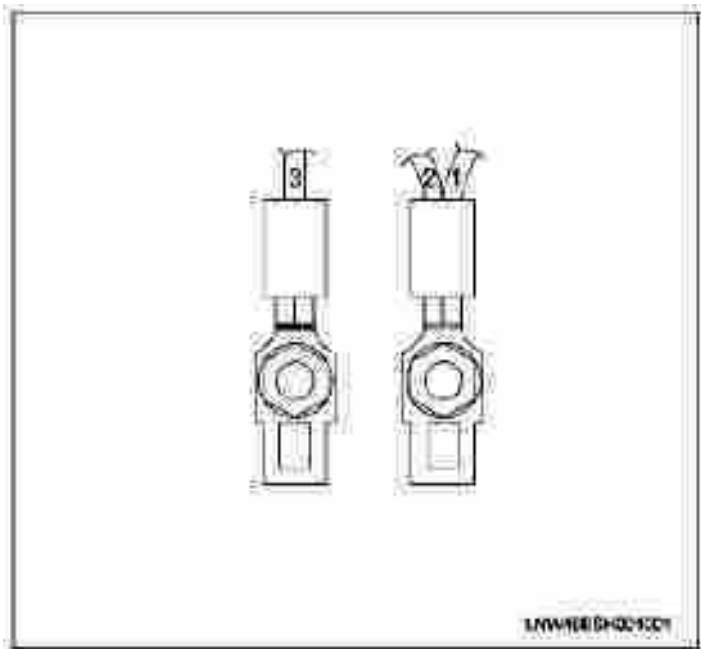
Connector number		J-127
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Purple /red	Battery voltage feed
2	Blue /white	Solenoid valve grounding

No. 1 cylinder injector



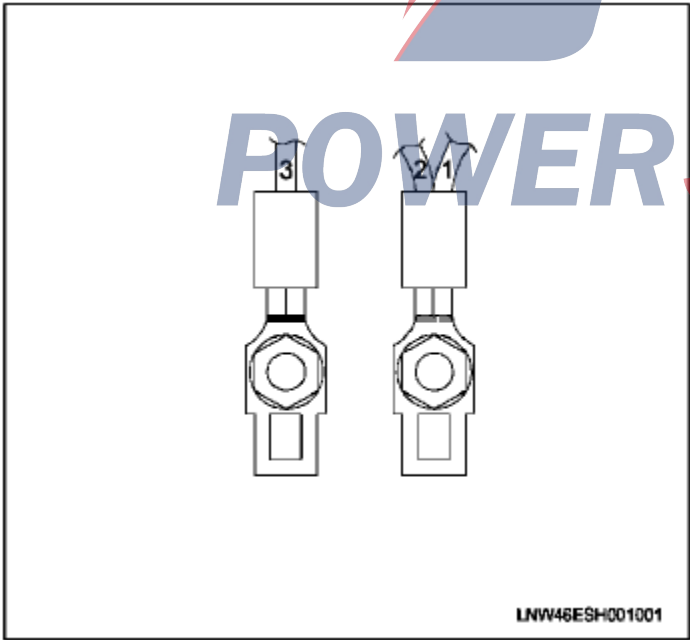
Connector number		E-27
Pin No.	Wire color	Pin function
1	White	Charging voltage(Common terminal 1)
2	White	Charging voltage(Common terminal 1)
3	Green	Solenoid valve control

No. 2 cylinder injector



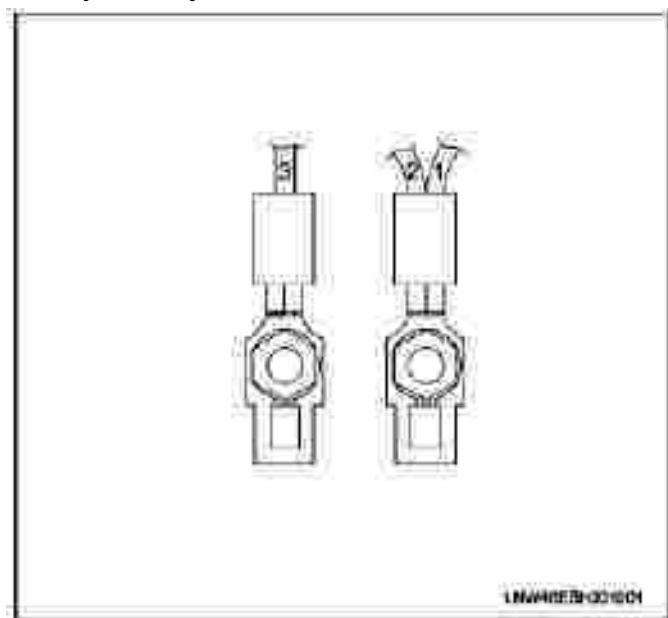
Connector number		E-29
Pin No.	Wire color	Pin function
1	White	Charging voltage(Common terminal 1)
2	White	Charging voltage(Common terminal 1)
3	Blue	Solenoid valve control

No. 3 cylinder injector



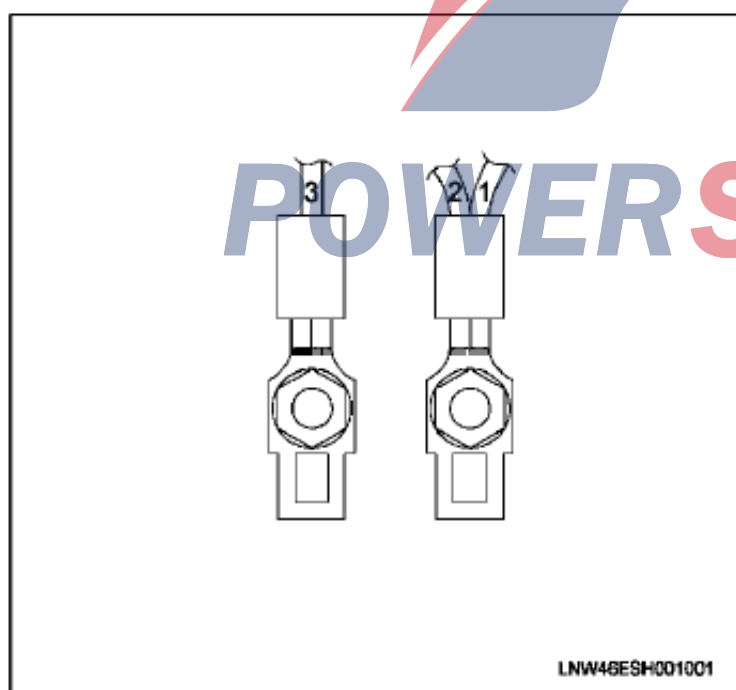
Connector number		E-31
Pin No.	Wire color	Pin function
1	White	Charging voltage(Common terminal 1)
2	White	Charging voltage(Common terminal 1)
3	Yellow	Solenoid valve control

No. 4 cylinder injector



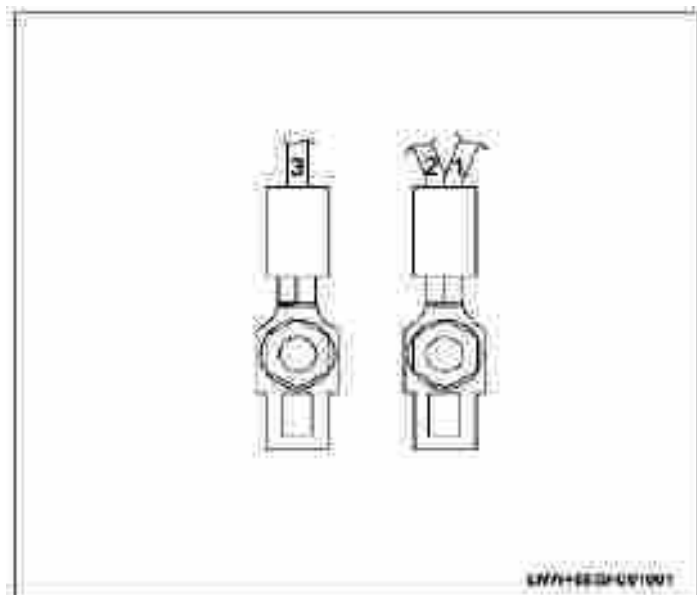
Connector number		E-33
Pin No.	Wire color	Pin function
1	Red	Charging voltage(Common terminal 2)
2	Red	Charging voltage(Common terminal 2)
3	Black	Solenoid valve control

No. 5 cylinder injector



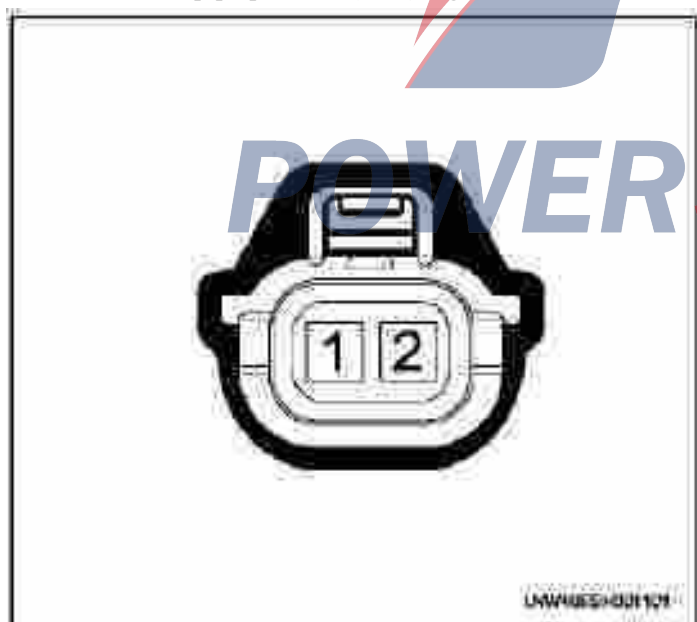
Connector number		E-35
Pin No.	Wire color	Pin function
1	Red	Charging voltage(Common terminal 2)
2	Red	Charging voltage(Common terminal 2)
3	Black	Solenoid valve control

No. 6 cylinder injector



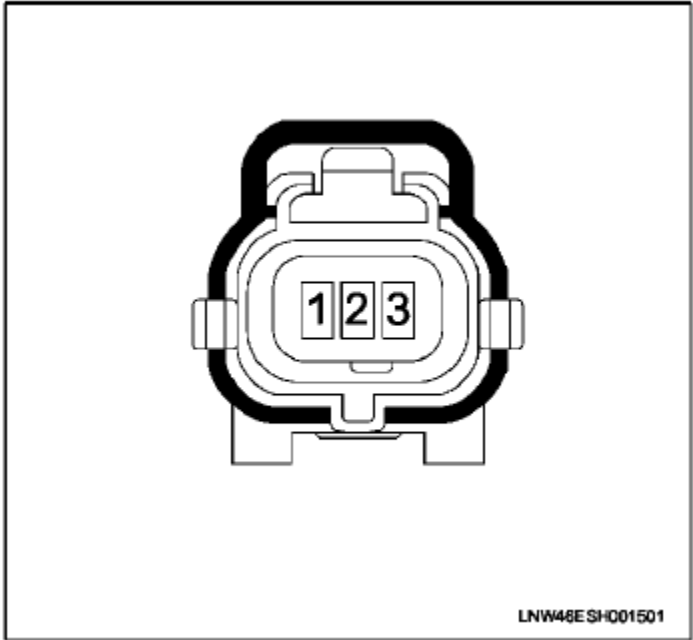
Connector number		E-37
Pin No.	Wire color	Pin function
1	Red	Charging voltage(Common terminal 2)
2	Red	Charging voltage(Common terminal 2)
3	Black	Solenoid valve control

Fuel distribution pipe pressure (FRP) regulator



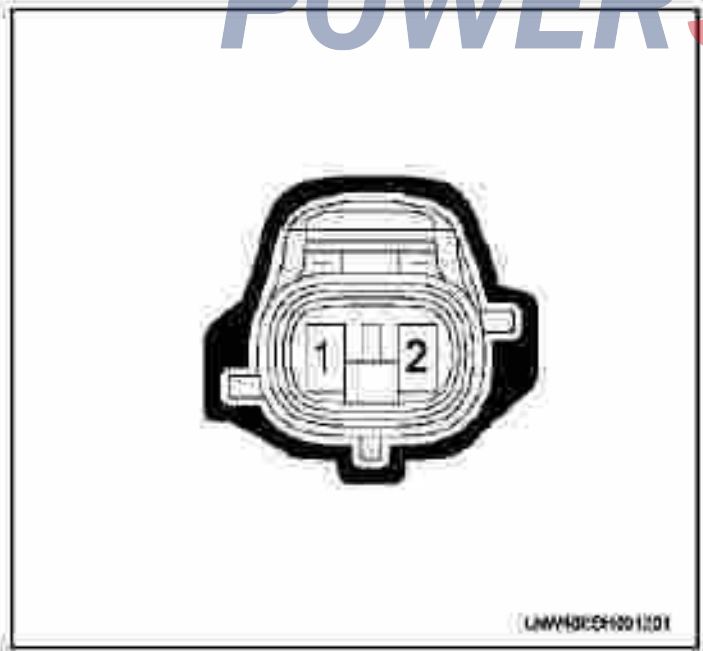
Connector number		E-116
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Yellow/black	High pressure side control (PWM)
2	Blue/Red	Low pressure side

Fuel distribution pipe pressure (FRP) regulator



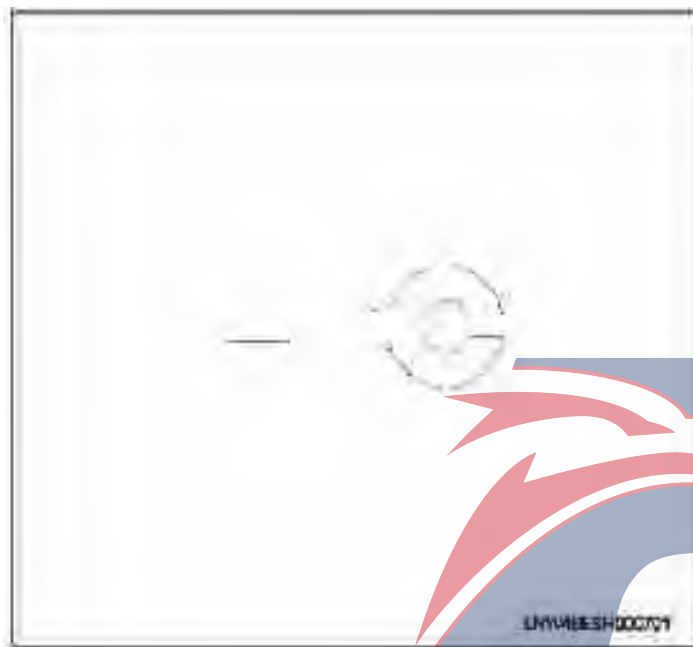
Connector number		E-113
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Red	Low reference voltage of sensor
2	Yellow	Sensor signal
3	White	Sensor 5V reference voltage

Fuel Temperature (FT) Sensor



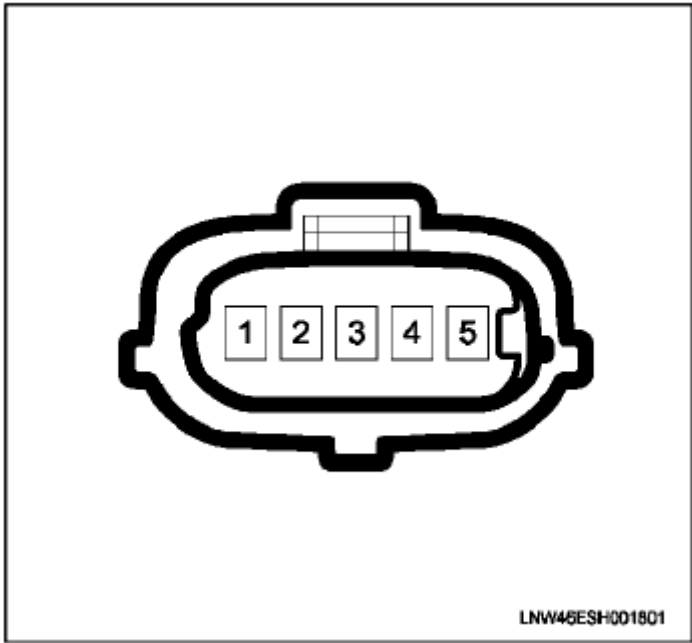
Connector number		E-93
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Black	Low reference voltage of sensor
2	Red/Blue	Sensor signal

Glow plug



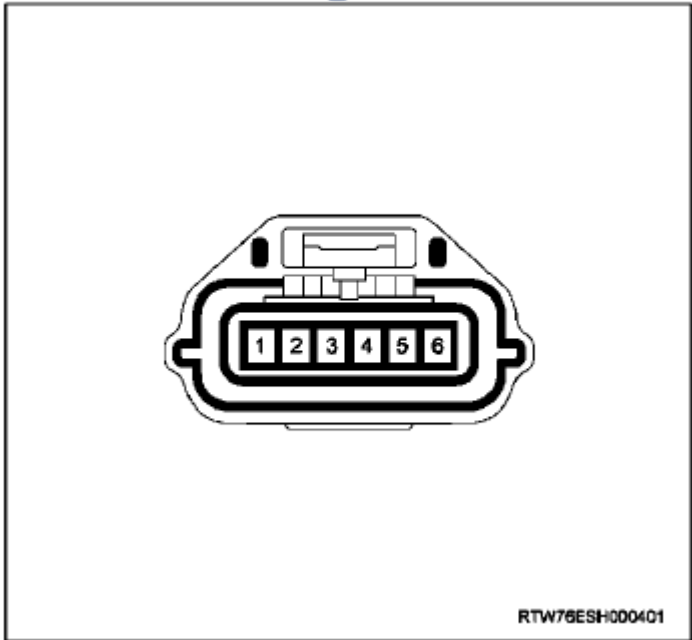
Connector number		E-6
Connector color		Silver color
Pin No.	Wire color	Pin function
1	Black / Yellow	power supply

Air Mass Flow (MAF) Sensor/Inlet Temperature (IAT) Sensor



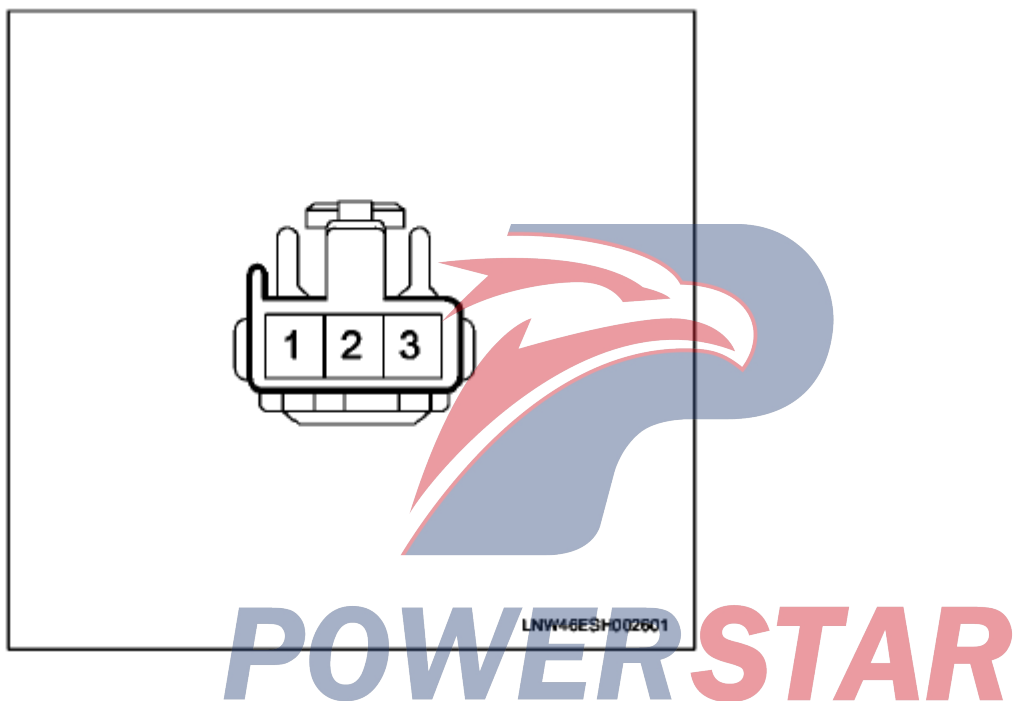
Connector number		FB-124
Connector color		Black
Test adapter number		J-35616-64A
Pin No.	Wire color	Pin function
1	White	MAF sensor 12 volt reference voltage
2	Black	Low reference voltage of MAF sensor
3	Red	MAF sensor signal
4	Yellow	IAT sensor signal
5	Blue	Low reference voltage of IAT sensor

Air intake throttle



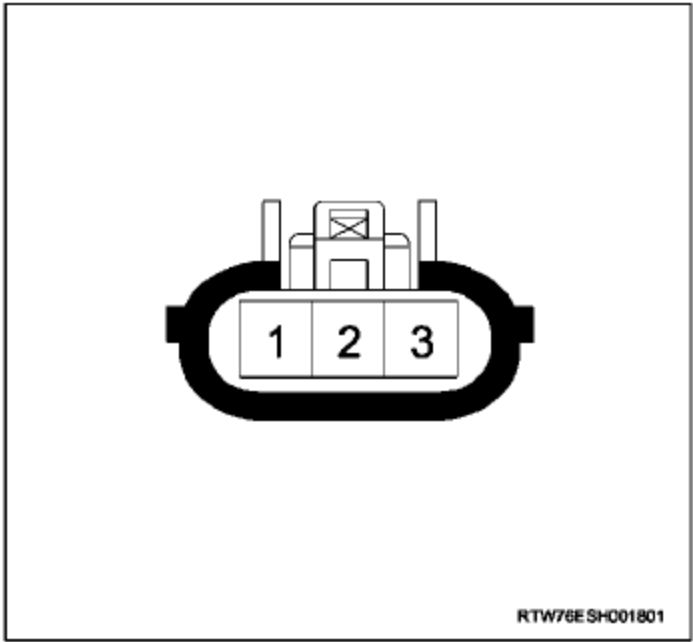
Connector number		E-114
Connector color		Black
Test adapter number		J-35616-64A
Pin No.	Wire color	Pin function
1	Blue/black	Solenoid valve low-voltage side control (PWM)
2	White /black	Solenoid valve driving voltage
3	-	Not used
4	Black	Position sensor low reference voltage
5	White	Position sensor signal
6	Black / Yellow	Position sensor 5V reference voltage

Throttle speed sensor



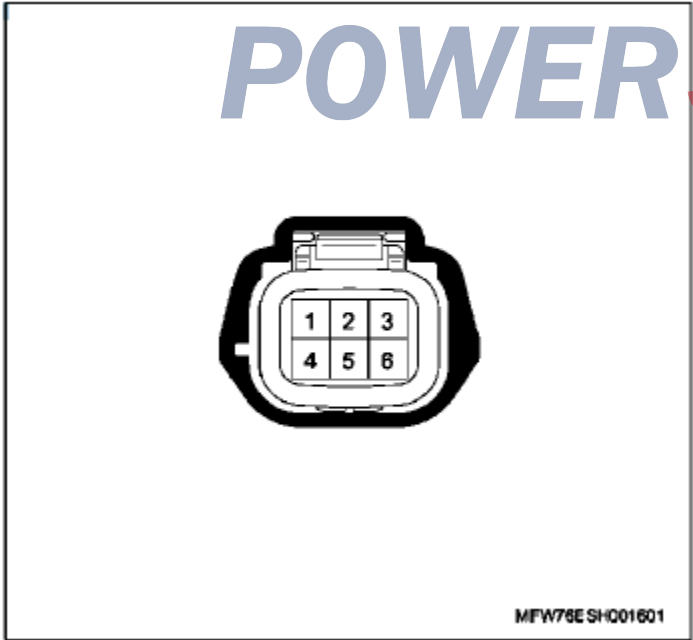
Connector number		FL-136
Connector color		White
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Black	Low reference voltage of sensor
2	Red	Sensor signal
3	White	Sensor 5V reference voltage

PTO throttle sensor



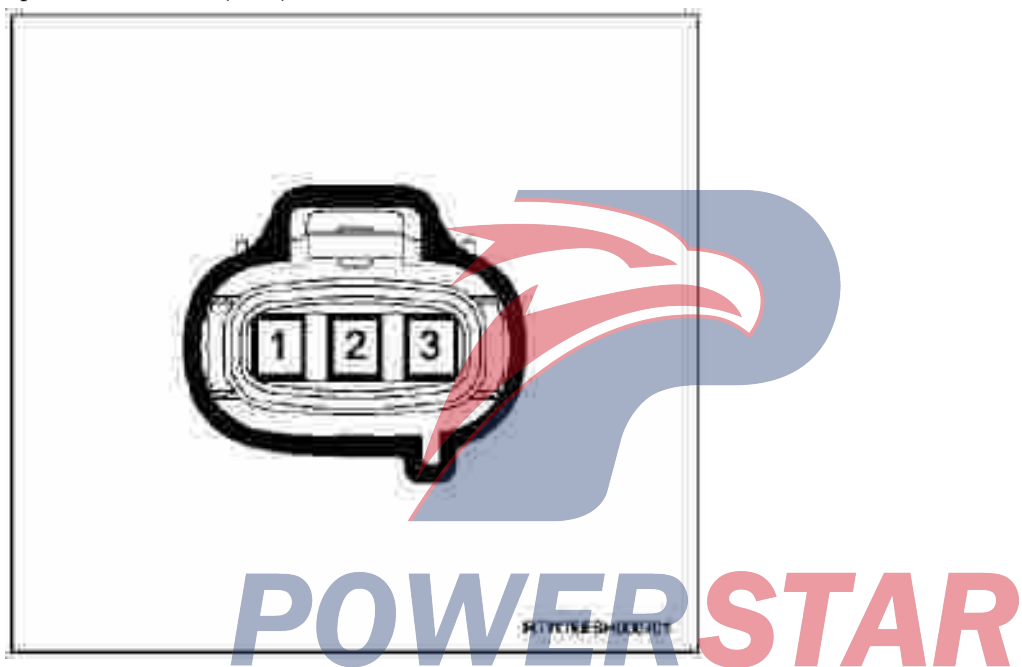
Connector number		J-55
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Red	Sensor 5V reference voltage
2	Brown	Sensor signal
3	Pink /black	Low reference voltage of sensor

Rotary Electric Actuator (REA)



Connector number		E-147
Connector color		Black
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Yellow	High signal of controller
2	-	Not used
3	Red	Voltage feeder
4	Blue	Controller LAN is low
5	-	Not used
6	Black	Grounding

Speed sensor shaft (VSS)



Connector number		FB-23
Connector color		Gray
Test adapter number		J-35616-33
Pin No.	Wire color	Pin function
1	Yellow	Ignition Voltage feeder
2	Black	Low reference voltage of sensor
3	White	Sensor signal

Diagnostic information and program

Engine control system check

Engine control system check		Inspector's name		
Customer name		Model and model year		
Driver's name		Chassis number		
Vehicle introduction date		Engine number		
License number		Odometer reading		Km/mile

Problem symptoms	☐ The engine does not run	☐ Engine crankshaft cannot rotate	☐ No initial combustion	☐ Incomplete combustion
	☐ Hard start	☐ Slow engine crankshaft rotation	☐ Others(	
	☐ Idle speed is wrong	☐ Idle speed abnormality ☐ Bad idling ☐ Others(	☐ High idle speed (RPM)	☐ Low idle speed (RPM)
	☐ Driving performance poor	☐ Pause, rotary speed drop, unstable rotary speed ☐ Insufficient power, slow response or atony ☐ Others(	☐ Surge	☐ Sudden extinguishing
	☐ Engine stall	☐ Shortly after start ☐ During A/C operation ☐ Others(	☐ After depressing the accelerator pedal ☐ After releasing the accelerator pedal	☐ From N to D gear
	☐ Others(	☐ Black smoke ☐ Abnormal combustion noise ☐ Long regeneration time of DPD ☐ Others(	☐ White smoke ☐ Frequent manual regeneration of DPD	☐ Poor fuel economy

Date of problem occurrence			
Problem occurrence frequency		☐ Often ☐ Others(☐ intermittent (... times/day/month) ☐ Only once	
Situations when the problem appears	Weather	☐ Sunny ☐ Various/Other (☐ Cloudy ☐ Rain ☐ Snow	
	Outdoor temperature	☐ Heat (approximate value) ☐ Any temperature ☐ Warm ☐ Cool ☐ Cold (approximate value)	
	Position	☐ Highway ☐ Downhill ☐ Suburbs ☐ Urban area ☐ Undulating roads ☐ Others(☐ Uphill)	
	Load conditions	☐ Overload (about... tons) ☐ Others (about ... tons) ☐ No load	
	Engine temperature	☐ Cold ☐ Others(☐ Preheating ☐ After preheating ☐ Any temperature)	
	Engine operation	☐ Start ☐ Shortly after start (minutes) ☐ Idling ☐ Drive ☐ Constant speed ☐ Idling ☐ Deceleration ☐ A/C switch closed/open ☐ Acceleration ☐ Others(	
	Fuel flow	☐ Full ☐ Above 1/2 ☐ Below 1/2 ☐ Almost empty	
	Fuel concentration		
Situation of fault indicator lamps, " Service Vehicle Soon (SVS)" indicator lamp or DPD indicator lamp		☐ Keep open ☐ open intermittently ☐ Not open ☐ Blinking	
DTC or flash code	Current diagnostic code or flash code	☐ No ☐ Code P or code U Number()	
	Historical diagnostic trouble code or flash code	☐ No ☐ Code P or code U Number()	
Other special cases			
LNW78EXF008201			

POWERSTAR

Diagnostic starting point - Engine control system

Start the system diagnostics from "diagnostic system inspection - engine control system". The "Diagnostic System Check - Engine Control System" will provide the following information:

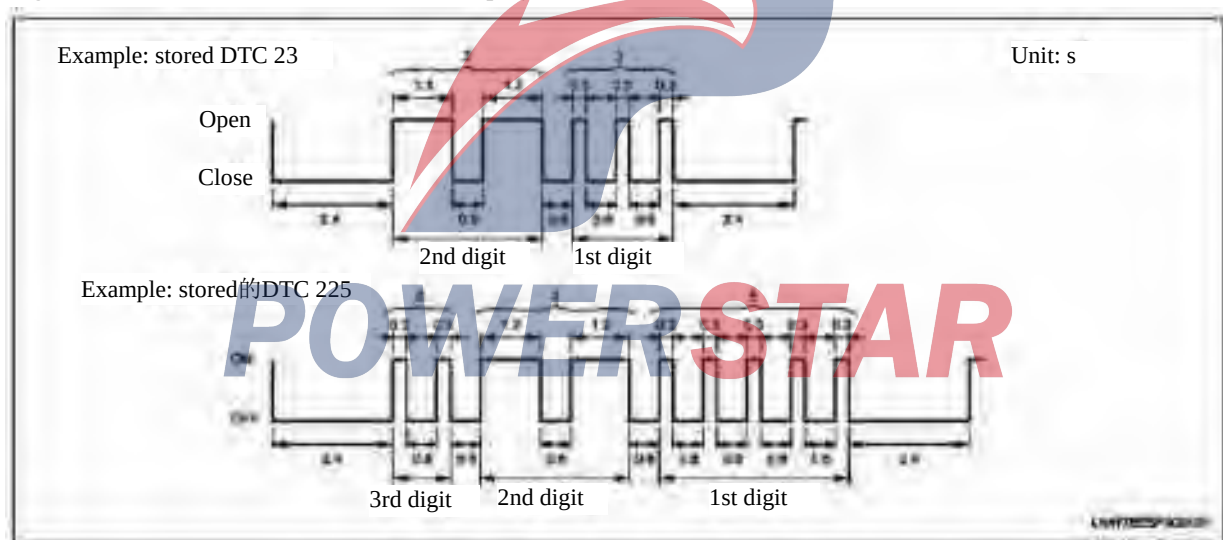
- Identification of control modules that issue commands to the system;
- Control module's ability to communicate through serial data circuits.
- Identification of the stored DTC and its status.

Use the diagnostic system check - engine control system to determine the correct system diagnostics and program location.

Precautions: The list of engine control systems must be used to confirm customer complaints. You need to understand the correct (normal) operating status of the system and confirm that the customer's complaint is a system failure.

Reading flash diagnostic trouble code (DTC)

The connector communicating with ECM is the data link connector (DLC). When the data transfer connector diagnostic test terminal is grounded, diagnostic trouble code stored in the engine control module memory can be read with a hand-held fault diagnosis tester (such as the Tech 2 plugged into the data transfer connector) or by counting the number of "SVS" indicator lamps flashing. The data transmission connector terminal "12" (through the diagnosis request) is pulled "low" (grounded) by connecting to the data transmission connector terminal "4" as the ground line. Once the terminal "12" is connected to the terminal "4", the ignition switch is turned on and the engine is turned off. The "Service Vehicle Soon" indicator lamp will show three current diagnostic trouble codes and history diagnostic trouble codes. If multiple diagnostic trouble codes are stored in the engine control module memory, the diagnostic trouble codes will be output in numerical order and each trouble diagnosis code will be displayed three times. As long as the diagnostic code is short enough, the flash diagnostic fault code will continue to be displayed.



Diagnostic System Check - Description of Engine Control System

The "Diagnostic System Check-Engine Control System" systematically describes the method of identifying the situations caused by the failure of the electronic engine control system. The diagnosis of driving performance should start with "diagnostic system check". The "Diagnostics System Check" will instruct the maintenance personnel to take the next logical step for fault diagnosis. Understanding and using diagnostic tables correctly can reduce diagnostic time and avoid errors when completing components.

Test Description

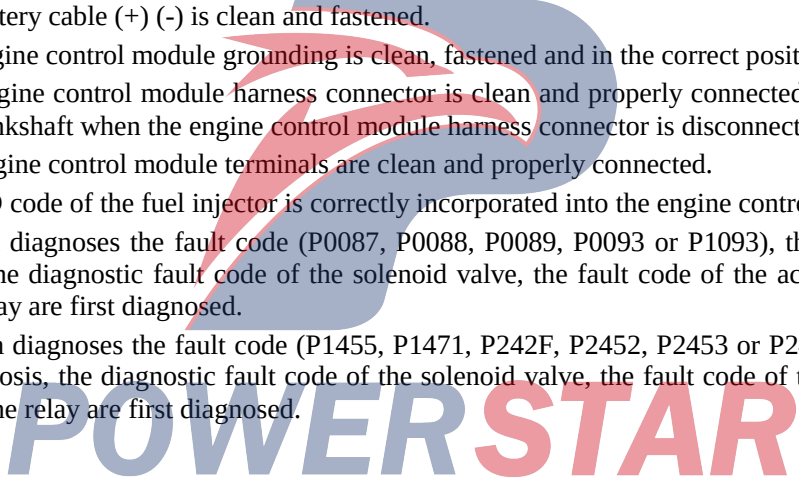
The following number indicates the number of the steps in the diagnosis table.

2. Insufficient communication may be due to a partial or entire failure of serial data circuit. 5. The diagnostic trouble code with a beginning of U indicates that some other modules are not communicating. 8. If there is another module that sets the diagnostic trouble code, see Diagnostic Trouble Code List. The list of diagnostic trouble codes guides you in the proper diagnostic procedure. If the control module stores multiple diagnostic trouble codes, the diagnostic trouble codes are diagnosed in the following order.

- Diagnosing a plurality of diagnostic trouble codes belonging to the category of diagnostic trouble codes at the component level ((e.g. sensor diagnostic trouble code, solenoid diagnostic trouble code, actuator diagnostic trouble code and relay diagnostic trouble code) according to the numbering sequence. Unless otherwise stated in the diagnostics table, the diagnosis begins with the lowest numbered diagnostic trouble code.

Important notes for Diagnostic System Check - Engine Control System:

Diagnostic System Check - Engine Control System

- If no driving performance problem, do not perform the diagnosis unless other procedures guide you to do so.
 - Before you diagnose, you should search for applicable service bulletins.
 - Do not clear the diagnostic trouble code unless instructed by the diagnostic procedure.
 - If there is a problem in the starting system, please refer to the Starting System section of the engine parts.
 - Make sure that the battery is full of electricity.
 - Make sure the battery cable (+) (-) is clean and fastened.
 - Make sure the engine control module grounding is clean, fastened and in the correct position.
 - Make sure the engine control module harness connector is clean and properly connected. Do not attempt to turn the engine crankshaft when the engine control module harness connector is disconnected.
 - Ensure that the engine control module terminals are clean and properly connected.
 - Ensure that the ID code of the fuel injector is correctly incorporated into the engine control module.
 - If the fuel System diagnoses the fault code (P0087, P0088, P0089, P0093 or P1093), the fault code of the sensor diagnosis, the diagnostic fault code of the solenoid valve, the fault code of the actuator and the fault diagnosis of the relay are first diagnosed.
 - If the DPD system diagnoses the fault code (P1455, P1471, P242F, P2452, P2453 or P2458), the fault code of the sensor diagnosis, the diagnostic fault code of the solenoid valve, the fault code of the actuator and the fault diagnosis of the relay are first diagnosed.
- 

Step	Operation	Value	Yes	No
1	Install fault diagnosis instrument. Is the diagnostic scan tool turned on?	—	Go to Step 2	To "Fault diagnosis tester is not energized"
2	1. Turn on the ignition and turn off the engine. 2. Attempt to establish communication with engine control module Can the fault diagnosis instrument communicate with the engine control module?	—	Go to Step 3	To "The troubleshooter cannot communicate with the controller LAN device"
3	Try to start the engine. Does the engine start and idle?	—	Go to Step 4	To "engine start but not run"
4	Select the diagnostic fault code display function for the following control modules: • Engine control module Does the fault diagnosis tester display any diagnostic trouble codes?	—	Go to Step 5	Go to Step 9
5	Does the scan tool display a diagnosis trouble code with a beginning of U or a diagnosis trouble code for other module communication trouble?	—	To "Applicable Diagnostic Trouble Codes"	Go to Step 6
6	Does the fault diagnosis tester display the engine control module diagnostic trouble code P1621?	—	To "Applicable Diagnostic Trouble Codes"	Go to Step 7
7	Does the fault diagnosis tester display the engine control module diagnostic trouble code P0563?	—	To "Applicable Diagnostic Trouble Codes"	Go to Step 8
8	Is there any other fault code that has not been diagnosed by the controller?	—	To "Applicable Diagnostic Trouble Codes"	Go to Step 9
9	1. Check for the following situations. 2. Refer to the applicable symptom diagnosis list: • Hard start • The idling speed is poor, unstable or inaccurate and stall • High idle speed • Sudden flameout • Surge • Insufficient motivation, slow response or weakness • Pause, speed decline, rotational speed instability • abnormal combustion noise • Fuel economical efficiency low Excessive black smoke • Excessive white smoke • Frequent manual regeneration of the DPD system • Long regeneration time of DPD system Did you identify the problem and correct it?		Normal system	For "discontinuous state"

Data list of fault diagnosis tester

The engine diagnostics instrument data list contains all engine-related parameters on the diagnostic tool. A certain parameter may appear in any data list. In certain cases, it can appear for many times or in more than one data list so as to combine the relevant parameters. The engine diagnosis tool data list can only be used after determining the following conditions:

The "engine control system -diagnostic system check" has been completed.

•The vehicle diagnosis system runs normally.

•The diagnostic value of a normally operating engine can be used to compare with the engine you are diagnosing. The values in the Engine Diagnostic Instrument Data List represent the data values of the engine during normal operation. This service manual only describes the following parameters used in diagnosis.

Diagnostic instrument parameters	Displayed unit	Typical data values when the engine is idling	Typical data values at 1400 rpm
Operating conditions: engine idle or 1400 rpm / engine coolant temperature range of 75 degrees Celsius to 85 degrees Celsius (i.e., 167 degrees Fahrenheit to 185 degrees Fahrenheit)/accelerator pedal constants/neutral or sea level parking/vehicle/DPD will not be regenerated			
Engine speed	r/min	About 525 rpm	About 1400 rpm
Expected idle speed	r/min	525 rpm	525 rpm
Calculated engine load	%	Below 25%	Below 20%
Coolant temperature	°C/°F	75°C~85°C/167°F~185°F	75°C~85°C/167°F~185°F
Engine Coolant Temperature Sensor	V	0.4-0.6V	0.4-0.6V
Intake air temperature	°C/°F	20°C~40°C/68°F~104°F	20°C~40°C/68°F~104°F
intake temperature sensor	V	1.4-2.3V	1.4-2.3V
Fuel temperature	°C/°F	20°C~60°C/50°F~140°F	20°C~60°C/50°F~140°F
Fuel temperature sensor	V	0.8-2.3V	0.8-2.3V
MAF sensor (mass air flow)	g /Cylinder	Higher than 0.8g / cylinder at sea level	Higher than 0.7 g / cylinder at sea level
MAF sensor (mass air flow)	V	Higher than 1.5V at sea level	Higher than 2.3V at sea level
Atmospheric pressure	kPa/lb/inch2	About 100kPa/14.5 psi at sea level	About 100kPa/14.5 psi at sea level
Atmospheric pressure sensor	V	About 2.3V at sea level	About 2.3V at sea level
Expected turbocharger position	%		
Desired boost pressure	kPa/lb/inch2		
Boost pressure	kPa/lb/inch2	About 100kPa/14.5 psi at sea level	Below 115kPa/16.7 psi at sea level
Booster pressure sensor	V	About 1.0V at sea level	Less than 1.3V at sea level
Expected fuel distribution pipe pressure	Megapascal/lb/square inch		
Fuel distribution pipe pressure	Megapascal/lb/square inch		
Fuel distribution pipe pressure sensor	V	1.4-1.6V	Higher than 2.4 volts
FRP regulator command (fuel distribution pipe pressure)	%	30~45%	25~35%
Fuel flow specified by the FRP regulator	mm ³ /s		
Feedback current of the fuel distribution pipe pressure regulator (fuel distribution pipe pressure)	Ma	900-1100 Ma	800-1100 Ma
Accelerator pedal position	%	0%	23~30%
APP sensor 1 (accelerator pedal position)	V	0.3-0.7V	1.2-1.6V
APP sensor 2 (accelerator pedal position)	V	4.1-4.5V	3.4-1.6V
Expected EGR location	%	0%	Below 35%
EGR position 1:	%	0%	Below 35%
EGR position 1:	Low/High	Low or High	Low or High
EGR position 2:	Low/High	Low or High	Low or High
EGR position 3:	Low/High	Low or High	Low or High
Air intake throttle solenoid valve command	%		
Desired intake throttle position	%		
Intake throttle position	%		
Intake throttle position sensor	V	Higher than 3.9 volts	Higher than 3.9 volts

Throttle speed sensor	V	0.3-0.6V	0.3-0.6V
Fuel compensation cylinder 1	mm ³	-5.0 to 5.0 cubic millimeters (change)	0.0 mm ³
Fuel compensation cylinder 2	mm ³	-5.0 to 5.0 mm ³ (change)	0.0 mm ³
Fuel compensation cylinder 3	mm ³	-5.0 to 5.0 mm ³ (change)	0.0 mm ³
Fuel compensation cylinder 4	mm ³	-5.0 to 5.0 mm ³ (change)	0.0 mm ³
Fuel compensation cylinder 5	mm ³	-5.0 to 5.0 mm ³ (change)	0.0 mm ³
Fuel compensation cylinder 6	mm ³	-5.0 to 5.0 mm ³ (change)	0.0 mm ³
Fuel pump state	Unknown/known	have already known	have already known
Pressure feedback model of fuel distribution pipe	Wait mode/feedback mode/shutdown mode/waiting for restart	Feedback mode	Feedback mode
Engine running status	Shutdown/ignition on/crankshaft rotation/operation	Running	Running
Engine running time	Time (Hours:Minutes:Seconds)	Variation	Variation
Speed	Km/h; mph	0 km/h; 0 mph	0 km/h; 0 mph
Starter switch	Close/disconnect	Disconnection	Disconnection
Ignition Switch	Close/disconnect	Closed	Closed
Ignition voltage	V	22.0-30.0V	22.0-30.0V
Battery voltage	V	22.0-30.0V	22.0-30.0V
Clutch pedal switch	Use/release	Loosen	Loosen
Neutral switch	Neutral/in normal state	Neutral gear	Neutral gear
Parking brake switch	Use/release	Loosen	Loosen
Exhaust brake switch	Close/disconnect	Disconnection	Disconnection
Exhaust brake valve command	Close/disconnect	Disconnection	Disconnection
Exhaust brake cut request from ABS	Inactivated/activated	Not activated	Not activated
Exhaust brake cut request from TCM	Inactivated/activated	Not activated	Not activated
ASR designated APP (accelerator pedal position)	%	100%	100%
PTO switch	Close/disconnect	Disconnection	Disconnection
PTO remote throttle	%	0%	0%
PTO remote throttle sensor	V	Less than 0.4 volts	Less than 0.4 volts
Engine preheating switch	Close/disconnect	Disconnection	Disconnection
A/C signal	switch	OFF	OFF
Refrigerating machine switch	Close/disconnect	Disconnection	Disconnection
Glow Relay Command	switch	OFF	OFF
Malfunction indicator lamp (MIL)	switch	OFF	OFF
The distance when the fault indicator lights up	Km/mile	0 Km/0 mile	0 Km/0 mile
Fault indicator is the engine running time when the light is on	Minute	0	0
Total Engine Over speeding Events	count	Variation	Variation
Total number of engine over temperature events	count	Variation	Variation
Total number of fuel over-temperature events	count	Variation	Variation
Total number of intake air over-temperature events	count	Variation	Variation
DPD mode	Variation	30	30
DPD accumulation state	0/ 1/ 2/ 3/ 4/ 5	0,1 ,2,3 or 4	0,1 ,2, 3 or 4

DPD distance state	0/ 1/ 2/ 3/ 4	0,1 ,2, or 3	0,1 ,2, or 3
Incomplete regeneration state of DPD	0/ 1/ 2	0 or 1	0 or 1
DPD is not completely regenerated	0/ 1/ 2	0 or 1	0 or 1
DPD reactivation switch	Close/disconnect	Disconnection	Disconnection
Differential pressure	kPa/lb/inch ²	Variation	Variation
Exhaust differential pressure sensor	V	Variation	Variation
Exhaust temperature 1	°C/°F	Variation	Variation
Exhaust temperature sensor 1	V	Variation	Variation
Exhaust temperature 2	°C/°F	Variation	Variation
Exhaust temperature sensor 2	V	Variation	Variation
Exhaust throttle state	switch	OFF	OFF

Data definition of fault diagnosis instrument

This information will help with emissions or driving performance issues. This displayed information can be seen when the vehicle is running. The "diagnostic system check engine control system" is usually executed first. The Diagnostic System Check will confirm whether the system is running normally.

Engine speed

This parameter shows the crankshaft speed calculated by the engine control module based on the input signal sent by the crankshaft position (CKP) sensor or camshaft position (CMP) sensor.

Expected idle speed

This parameter shows the idle speed required by the engine control module. The engine control module will change the expected idle speed according to the engine coolant temperature and other input signals.

Calculated engine load

This parameter shows the engine load in percentage calculated according to the input signal sent by each engine sensor to the engine control module. When the engine is with small load or no load idling, the fault diagnosis tester will show a lower percentage. When the engine is running at high speed and heavy load, the diagnostic tool will show a high percentage.

Coolant temperature

This parameter shows the engine coolant temperature calculated by the engine control module by using the engine coolant temperature (ECT) sensor signal. When the engine coolant temperature sensor signal voltage is high, the fault diagnostic tester will display a low temperature; when the engine coolant temperature sensor signal voltage is low, the fault diagnostics tester will display a high temperature.

Engine Coolant Temperature Sensor

This parameter shows the voltage signal sent from the engine coolant temperature (ECT) sensor to the engine control module. The engine coolant temperature sensor has a range of value: when the temperature is high, the display voltage is low; when the temperature is low, the display voltage is high.

Intake air temperature

This parameter shows the intake air temperature calculated by the engine control module using the intake air temperature (IAT) sensor signal. When the intake air temperature sensor signal voltage is high, the fault diagnosis tester will display a low temperature; when the intake air temperature sensor signal voltage is low, the fault diagnostic display will show a high temperature.

Intake temperature sensor

This parameter shows the voltage signal sent from the intake air temperature (IAT) sensor to the engine control module. The intake air temperature sensor has a range of value: when the temperature is high, the display voltage is low; when the temperature is low, the display voltage is high.

Fuel temperature

This parameter shows the fuel temperature calculated by the engine control module using the fuel temperature (FT) sensor signal. When the fuel temperature sensor signal voltage is high, the fault diagnosis tester will display a low temperature; when the fuel temperature sensor signal voltage is low, the fault fault diagnosis tester will display a high temperature.

Fuel temperature sensor

This parameter shows the voltage signal sent from the fuel temperature (FT) sensor to the engine control

module. The fuel temperature sensor has a numerical range: when the temperature is high, the display voltage is low; when the temperature is low, the display voltage is high.

MAF (mass air flow)

This parameter shows the air flow into the engine calculated by the engine control module based on the mass air flow (MAF) sensor input signal. When the engine speed is high, the diagnostic tool will show a high value; when the engine speed is low, the diagnostic tool will show a low value.

MAF sensor (mass air flow)

This parameter shows the voltage signal sent from the mass air flow (MAF) sensor to the engine control module. The mass air flow sensor has a range of value: when the engine speed is low, the display voltage is low; when the engine speed is high, the display voltage is high.

Atmospheric pressure

This parameter shows the barometric pressure (BARO) calculated by the engine control module using the barometric pressure sensor signal. At high altitudes, troubleshooting will show a low pressure.

Atmospheric pressure sensor

This parameter shows the voltage signal sent from the BARO sensor to the engine control module. The atmospheric pressure sensor has a range of value: the display voltage is low at high altitudes, and medium pressure at sea level.

Desired turbocharger position

This parameter shows the turbocharger nozzle position desired by the engine control module based on the current boost pressure.

Expected boost pressure

This parameter shows the boost pressure desired by the engine control module based on the current driving conditions. The required boost pressure can be compared to the actual boost pressure to determine sensor accuracy or turbocharger control problems.

Boost pressure

This parameter shows the boost pressure in the intake manifold calculated by the engine control module using the boost pressure sensor signal. When the engine load is low, the fault diagnosis tester will show that the booster pressure is low. When the engine load is high, the fault diagnosis tester will show high boost pressure. Note: The actual boost pressure is determined by subtracting the atmospheric pressure from the actual reading.

Booster pressure sensor

This parameter shows the voltage signal sent from the boost pressure sensor to the engine control module. The boost pressure sensor has a range of values: when the boost pressure is low (at idle or low engine load), the displayed voltage is low; when the boost pressure is high (high engine load), the displayed voltage is high.

Fuel distribution pipe pressure

This parameter shows the fuel distribution pipe pressure calculated by the engine control module using the fuel distribution pipe pressure (FRP) sensor signal. When the engine load is small, the diagnostic instrument will show low pressure; when there is a lot of engine load, the diagnostic instrument will show high pressure.

Fuel distribution pipe pressure sensor

This parameter shows the voltage signal sent from the Fuel Distribution Pipe Pressure (FRP) sensor to the engine control module. The fuel distribution tube sensor has a range of values: when the fuel distribution pipe is low, the display voltage is low; when the fuel distribution pipe is high, the display voltage is high.

Fuel flow assigned by fuel distribution pipe (fuel distribution pipe pressure)

This parameter shows the fuel flow assigned to the fuel distribution pipe by the fuel distribution pipe pressure (FRP) regulator.

Fuel distribution pipe pressure regulator command (fuel distribution pipe pressure)

This parameter shows the fuel distribution pipe pressure (FRP) regulator control duty ratio calculated based on the input signal sent by each engine sensor to the engine control module. When the control state of the fuel distribution pipe pressure regulator is opened (fuel supply in the fuel distribution pipe is increases), the fault diagnostic meter will displays the low percentage. When the control state of the fuel distributor pressure regulator is off (the fuel supply of the fuel distribution pipe decreases), the fault diagnosis tester will display a higher percentage.

Fuel distribution pipe pressure regulator feedback (fuel distribution pipe pressure)

This parameter shows the fuel distribution pipe pressure (FRP) regulator control feedback current measured by the engine control module. When the control state of the fuel distribution pipe pressure regulator is opened

(fuel supply in the fuel distribution pipe is increases), the fault diagnosis tester will display a higher current. When the control state of the fuel distribution pipe pressure regulator is closed (fuel supply in the fuel distribution pipe is reduced), the fault diagnosis tester will display a higher current.

Accelerator pedal position

This parameter shows the accelerator pedal angle calculated by the engine control module using the accelerator pedal position (APP) sensor signal. The fault diagnosis instrument will display 0%~100% linearly according to pedal operation.

APP sensor 1 (accelerator pedal position)

This parameter shows the voltage signal sent from the accelerator pedal position (APP) sensor 1 of the APP sensor assembly to the engine control module. The APP sensor 1 has a range of values: the display voltage is low when the accelerator pedal is not depressed, and the display voltage is high when the accelerator pedal is fully depressed.

APP sensor 2 (accelerator pedal position)

This parameter shows the voltage signal sent from the accelerator pedal position (APP) sensor 2 of the APP sensor assembly to the engine control module. The APP sensor 2 has a range of values: the display voltage is high when the accelerator pedal is not depressed, and the display voltage is low when the accelerator pedal is fully depressed.

Expected EGR location

This parameter shows the EGR position desired by the engine control module based on the current driving conditions. This required EGR position can be compared to the actual EGR position to determine the accuracy of the sensor or the solenoid valve control problem.

EGR position

This parameter shows the EGR valve position calculated by the engine control module using the EGR position sensor signal. When the EGR valve is closed, the fault diagnosis tester will show a low percentage; when the EGR valve is open, the fault diagnosis tester will show a high percentage.

EGR position 1:

This parameter shows the EGR valve position calculated by the engine control module using separate three EGR position sensor signals. When the EGR valve is closed, the fault diagnosis tester will show a low percentage; when the EGR valve is open, the fault diagnosis tester will show a high percentage.

EGR position 1:

This parameter shows the state of the EGR position sensor 1 input to the engine control module. When the position sensor is turned on, the fault diagnosis tester will display "Low." The fault diagnostic meter will display HIGH when the position sensor is off.

EGR position 2:

This parameter shows the state of the EGR position sensor 2 input to the engine control module. When the position sensor is turned on, the fault diagnosis tester will display "Low." The fault diagnostic meter will display HIGH when the position sensor is off.

EGR position 3:

This parameter shows the state of the EGR position sensor 3 input to the engine control module. When the position sensor is turned on, the fault diagnosis tester will display "Low." The fault diagnostic meter will display HIGH when the position sensor is off.

Air intake throttle solenoid valve command

This parameter shows the load ratio of the intake throttle solenoid valve control calculated according to the input signals from each engine sensor to the engine control module. When the control state of the intake throttle solenoid valve is on, the fault diagnosis tester will display a lower percentage. When the control state of the intake throttle solenoid valve is off, the diagnostic scan tool will display a high percentage.

Desired intake throttle position

This parameter shows the intake throttle position desired by the engine control module based on the current driving conditions. The desired intake throttle position can be compared to the actual intake throttle position to determine sensor accuracy or solenoid valve control problems.

Intake throttle position

This parameter shows the intake throttle position calculated by the engine control module using the intake throttle position sensor signal. Fault diagnostics will show a low percentage when the intake throttle is closed; the diagnostic scanner will show a higher percentage when the intake throttle is open. Note: If the solenoid

valve is closed by command, the intake throttle position will be displayed to be above 100%.

Intake throttle position sensor

This parameter shows the voltage signal sent from the intake throttle position sensor to the engine control module. The intake throttle position sensor has a range of values: when the intake throttle is closed, the display voltage is low; when the intake throttle is open, the display voltage is high.

Throttle speed sensor

This parameter shows the voltage signal sent from the throttle acceleration sensor to the engine control module. The accelerator speed sensor has a numerical range: when the accelerator speed sensor rotates counterclockwise, the displayed voltage is low; when the accelerator speed sensor rotates clockwise, the displayed voltage is high.

Fuel compensation cylinder 1 to fuel compensation cylinder 6

This parameter shows the adjustment of the fuel quantity of each cylinder calculated by the engine control module when the engine speed is low. If the amount of fuel decreases, the fault diagnosis tester will show a negative value. If the amount of fuel increases, the fault diagnosis tester will show a positive value. If the cylinder shows a value that is too high or too low, it means that the injector may be malfunctioning, the cylinder performance is weak or slightly stuck, the cylinder size is wrong, or the injector ID code is wrong.

Fuel pump state

This parameter shows that the state of the fuel supply pump is being understood. Unknown means the initialization state of replacing with the new engine control module or resetting of the adjusted value. After warming up the engine, run it idling to observe. Known indicates that the understanding process has been completed.

Pressure feedback model of fuel distribution pipe

This parameter shows the state of the fuel distribution pipe pressure feedback to the engine control module. The wait mode indicates that the ignition switch is in the closed position. The feedback mode indicates that the engine crankshaft is rotating or the engine is running. The closing mode indicates that the ignition switch is at the disconnect position.

Engine mode

This parameter shows the engine state. Ignition On indicates that the ignition switch is in the closed position. Cranking indicates that the engine crankshaft is rotating. Running indicates that the engine is running. Off indicates that the ignition switch is in the off position.

Engine running time

This parameter shows the elapsed time since the engine was started. The fault diagnosis instrument will display the time in hour, minute and second.

Speed

This parameter shows the vehicle speed calculated by the engine control module using the vehicle speed sensor (VSS) signal. When the vehicle speed is low, the fault diagnosis instrument will display low value; when the vehicle speed is higher, the fault diagnosis instrument will display high value.

Starter switch

This parameter shows the state of the starter switch input to the engine control module. When the ignition switch is in the closed position, the fault diagnosis tester will display ON.

Ignition Switch

This parameter shows the state of the starter switch input to the engine control module. When the ignition switch is in the closed position, the fault diagnosis tester will display ON.

Ignition voltage

This parameter shows the ignition voltage measured in the ignition supply circuit of the engine control module. This voltage is applied to the engine control module when the ignition switch is in the closed position.

Battery voltage

This parameter shows the battery voltage measured in the main relay switching voltage supply circuit of the engine control module. This voltage is applied to the engine control module when the engine control module main relay is energized.

Clutch pedal switch

This parameter shows the state of the clutch pedal switch input to the engine control module. When the clutch pedal is pressed, the fault diagnosis device will display "Applied". If semi-automatic transmission is installed, the input status of the shift clutch from TCM will be displayed.

Neutral switch

This parameter shows the state of the neutral switch input to the engine control module. The fault diagnostic meter will display neutral if the transmission is in parking gear or neutral gear.

Parking brake switch

This parameter shows the state of the parking brake switch input to the engine control module. When the parking brake lever is pulled, the scanning tool shows "Applied".

Exhaust brake switch

This parameter shows the state of the exhaust brake switch input to the engine control module. On indicates that the exhaust brake switch is in a closed state, and the exhaust brake solenoid valve and the intake throttle valve can be energized according to the driving conditions. Off indicates that the exhaust brake switch is open and the exhaust brake will not be closed.

Exhaust brake valve command

This parameter shows the command state of the exhaust brake solenoid control circuit. On indicates that the engine control module is grounding the exhaust brake solenoid valve so that the vacuum pressure reaches the exhaust brake valve.

Exhaust brake cut request from ABS

This parameter shows the state of the exhaust braking cut request sent by the electric hydraulic control module (EHCU) (ABS module) to the engine control module. Active indicates that the EHCU has released the exhaust brake according to the command.

Exhaust brake cut request from TCM

This parameter shows the state of the exhaust brake cut-off request sent by the transmission control module (TCM) to the engine control module. Active indicates that the EHCU has released the exhaust brake according to the command.

ASR designated APP (accelerator pedal position)

This parameter shows the control angle of the accelerator pedal calculated by the electro-hydraulic control module (EHCU) (ABS module) when the anti-skid regulator (ASR) system is in operation. When the engine control module is not controlled to reduce fuel injection, the diagnostic scanner will display 100%. If the controls the ECM to reduce the injection volume and consequently reduces the driving wheel revolution, the fault diagnostic meter will displays the low percentage.

PTO switch

This parameter shows the state of the PTO switch input to the engine control module. When the PTO switch is operated by a lever or a button, the fault diagnosis device will be "On".

PTO remote throttle

This parameter shows the angle of the PTO remote throttle sensor calculated by the engine control module using the PTO remote throttle sensor signal. The PTO remote throttle angle has a range of values:when the throttle sensor is not operating, a low percentage is displayed; when the throttle sensor is operating, a high percentage is displayed.

PTO remote throttle sensor

This parameter shows the voltage signal sent from the PTO remote throttle sensor to the engine control module. The PTO remote throttle sensor has a numerical range:when the throttle sensor is not working, the display voltage is low; when the throttle sensor operates, the display voltage is high.

Engine preheating switch

This parameter shows the state of the engine preheating switch input to the engine control module. On indicates that the engine preheating switch is in the closed state, and the exhaust brake solenoid valve and the intake throttle valve can be energized according to the engine condition.

A/C signal

This parameter shows the state of the A/C compressor engagement. On indicates that the engine control module receives the starting signal of the A/C compressor. Off indicates that the engine control module does not receive the starting signal of the A/C compressor.

Refrigerating machine switch

This parameter shows the state of the refrigeration compressor engagement. On indicates that the engine control module receives the refrigeration compressor starting signal. Off indicates that the engine control module does not receive the refrigeration compressor starting signal.

Glow Relay Command

This parameter shows the command state of the glow relay control circuit. On indicates that the engine control module is grounding the glow relay control circuit so that the voltage reaches the glow plug.

Malfunction indicator lamp (MIL)

This parameter shows the command state of the fault indicator lamp (MIL) control circuit. When the fault detection instrument is on, the fault indicator should be in the "On" state. When the fault detection instrument is off, the fault indicator should be in the "OFF" state.

The distance when the fault indicator lights up

This parameter shows the mileage since the fault indicator lamp (MIL) lights up.

Fault indicator is the engine running time when the light is on

This parameter shows the cumulative time of the closing of the ignition switch since the fault indicator lamp (MIL) was turned on. The fault diagnosis tester will display the time in minutes.

Total Engine Over speeding Events

This parameter shows the count of engine over speed events. If any diagnostic trouble code is cleared, the count will show zero.

Total number of engine over temperature events

This parameter shows the count of engine over-temperature events. If engine coolant temperature exceeds 110°C (230°F), the counting will be activated. If any diagnostic trouble code is cleared, the count will show zero.

Total number of fuel over-temperature events

This parameter shows the count of fuel over-temperature conditions. If fuel temperature exceeds 110°C (230°F), the counting will be activated. If any diagnostic trouble code is cleared, the count will show zero.

Total number of intake air over-temperature events

This parameter shows the count of the intake air over-temperature condition. If intake air temperature exceeds 600°C (140°F), the counting will be activated. If any diagnostic trouble code is cleared, the count will show zero.

DPD mode

This parameter shows the DPD state. This parameter mainly shows the operation request or state sent to the user during the regeneration event. 1 or 50 indicates that a diagnostic trouble code has been set and DPD regeneration is not allowed. 30 indicate that it is in the normal operation of the engine.

DPD accumulation state

This parameter shows the state of PM accumulation. It shows the cumulative increase in filter PM as the parameter increases from 0 to 5. Determine the reactivation type manually with this parameter and DPD distance state parameter.

DPD trip status

This parameter shows the distance state since the last regeneration. It shows that when the parameter is increased from 0 to 4, the cumulative distance increases without a regeneration event. Use this parameter and the DPD accumulation status parameter to manually determine the regeneration type.

Incomplete regeneration state of DPD

This parameter shows the DPD regeneration error status. 1 or 2 indicates the frequency of uncompleted regeneration events in a predetermined event.

DPD is not completely regenerated

This parameter shows the DPD regeneration error status. 1 or 2 indicates how often the filter was not cleaned after the regeneration event.

Distance since last DPD reactivation

This parameter shows the mileage since the last regeneration was completed.

DPD reactivation switch

This parameter shows the state of the DPD regeneration switch input to the engine control module. On indicates that the DPD regeneration switch is in the closed state, and it is allowed to enter the regeneration process according to the accumulation of PM.

Differential pressure

This parameter shows the exhaust differential pressure calculated by the engine control module using the exhaust differential pressure sensor signal. When the PM accumulation is small, the fault diagnosis tester will show that the exhaust differential pressure is low; when the PM is accumulation is large, the fault diagnosis

tester will show that the exhaust differential pressure is high.

Exhaust differential pressure sensor

This parameter shows the voltage signal sent from the exhaust differential pressure sensor to the engine control module. The exhaust differential pressure sensor has a numerical range: the displayed voltage is low when the differential pressure is low (for example, when PM accumulation is low), and the displayed voltage is high when the differential pressure is high (for example, when PM accumulation is high).

Exhaust temperature 1

This parameter shows the exhaust temperature before the filter calculated by the engine control module by using the exhaust temperature sensor 1 signal. When the signal voltage of exhaust temperature sensor 1 is high, the fault diagnostic tool will display a low temperature; when the signal voltage of exhaust temperature sensor 1 is low, the fault diagnostic display will show a high temperature.

Exhaust temperature sensor 1

This parameter shows the voltage signal sent from the exhaust temperature sensor 1 to the engine control module. The exhaust temperature sensor 1 has a range of values: when the temperature is high, the display voltage is low; when the temperature is low, the display voltage is high.

Exhaust temperature 2

This parameter shows the exhaust temperature before the oxygen catalyst calculated by the engine control module using the exhaust temperature sensor 2 signal. When the signal voltage of exhaust temperature sensor 2 is high, the fault diagnostic tool will display a low temperature; when the signal voltage of exhaust temperature sensor 2 is low, the fault diagnostic display will show a high temperature.

Exhaust temperature sensor 2

This parameter shows the voltage signal sent from the exhaust temperature sensor 2 to the engine control module. The exhaust temperature sensor 2 has a range of values: when the temperature is high, the display voltage is low; when the temperature is low, the display voltage is high.

Exhaust throttle state

This parameter shows the command state of the exhaust throttle solenoid control circuit. On indicates that the engine control module is grounding the exhaust throttle solenoid valve so that the vacuum pressure reaches the exhaust throttle.

Output control of diagnostic scan tool

Output control of diagnostic scan tool	Explanation:
Fuel pressure control	The purpose of this test is to check if the fuel distribution pipe pressure changes from 30 to 80 megapascals (4,350 to 11,600 psi) when the command is issued. If the differential pressure of the fuel distribution pipe is large, you can consider whether the fuel supply pump, fuel distribution pipe pressure (FRP) regulator, pressure limiting valve or other fuel line has failed.
Ignition injection control	The purpose of the test is to check whether the igniting fuel injection operation was carried out when the on / off command was issued. If engine noise does not change when the command is turned off, consider whether the injector is malfunctioning.
Injection Time Control	The purpose of the test is to check whether the main injection time changes within the range of -5 to 5°C A when the delay / advance command is sent.
Positive drive of fuel injection nozzle	The purpose of this test is to check if the injection nozzle is operating normally when commanded to turn on. If there is no ticking sound (electromagnetic valve operation noise), interruption noise or abnormal noise when commanded to turn on, consider whether the injection nozzle has a fault.
Gas cylinder balance test	The purpose of this test is to check if the injection nozzles operate when commanded to turn on/off. If engine speed does not change when the command is turned off, consider whether the injector is malfunctioning.
Air intake throttle solenoid valve control	The purpose of this test is to check if the intake throttle valve is operating according to commands. If the position is very different, you can consider whether the foreign material restricts the operation of the valve, whether the deposit is excessive or whether the valve is faulty.
EGR magnet valve control	The purpose of the test is to check whether the EGR valve operates normally according to instructions. If the position is very different, you can consider whether the foreign material restricts the operation of the valve, whether the deposit is excessive or whether the valve is faulty.
Turbocharger solenoid valve control	The purpose of this test is whether the turbocharger nozzle control solenoid valve operates as according to the commands. If the solenoid valve operates abnormally, consider whether the foreign material restricts the operation of the actuator, whether there is excess sediment, whether the connection is damaged, whether the solenoid valve is faulty, or whether the position sensor is malfunctioning.
Light emitting relay control	The purpose of the test is to check whether the glow relay operates when commanded to turn on. If no power is applied when commanded to turn on, it may be considered whether the circuit or glow relay fails.

Exhaust brake control	The purpose of this test is to check if the exhaust brake control solenoid valve is operating when commanded to open. If no power is applied when commanded to turn on, it may be considered whether the circuit or the solenoid valve fails.
Exhaust throttle control	The purpose of this test is to check if the exhaust throttle relay operates when commanded to turn on. If no power is applied when commanded to turn on, it may be considered whether the circuit or the exhaust throttle relay fails.
Help Oil supply pump understand program reset	The purpose of this function is to reset the adjustment value of the fuel supply pump. Precautions: When replacing the fuel pump or engine, or installing an engine control module removed from another vehicle, the fuel supply pump must be re-understood.
DPD status reset	The purpose of this function is to reset the DPD state parameters. Precautions: The DPD state must be reset before using the fault diagnosis device to regenerate.
DPD regeneration control	The purpose of this test is for forced regeneration of the DPD filter. Precautions: Do not perform this test if the differential pressure exceeds the threshold.
DPD slow regeneration control	The purpose of this test is for forced regeneration of DPD filters in the event of over accumulation of PM. Compare this very slow regeneration with another one. Precautions: Do not perform this test if the differential pressure exceeds the threshold. The engine oil must be replaced after slow regeneration of DPD has been completed.
Output control of diagnostic scan tool	Explanation:
Fuel pressure control	The purpose of this test is to check if the fuel distribution pipe pressure changes from 30 to 80 megapascals (4,350 to 11,600 psi) when the command is issued. If the differential pressure of the fuel distribution pipe is large, you can consider whether the fuel supply pump, fuel distribution pipe pressure (FRP) regulator, pressure limiting valve or other fuel line has failed.
Ignition injection control	The purpose of the test is to check whether the igniting fuel injection operation was carried out when the on / off command was issued. If the engine noise is not changed when the closing order is issued, the fuel injector can be considered if there is any fault.
Injection Time Control	The purpose of the test is to check whether the main injection time changes within the range of -5 to 5°C A when the delay / advance command is sent.
Positive drive of fuel injection nozzle	The purpose of this test is to check if the injection nozzle is operating normally when the turn-on command is issued. If the tick sounds, the interruption noise or the abnormal noise is not being produced when an opening command is issued (the noise of the solenoid valve operation), the failure of the nozzle can be considered.
Gas cylinder balance test	The purpose of this test is to check if the injection nozzle operates when an on/off command is sent. If the engine speed is not changed when the closing command is issued, it can be considered whether the injection nozzle has malfunctioned.
Air intake throttle solenoid valve control	The purpose of this test is to check if the intake throttle valve is operating according to commands. If the position is very different, you can consider whether the foreign material restricts the operation of the valve, whether the deposit is excessive or whether the valve is faulty.
EGR magnet valve control	The purpose of the test is to check whether the EGR valve operates normally according to instructions. If the position is very different, you can consider whether the foreign material restricts the operation of the valve, whether the deposit is excessive or whether the valve is faulty.
Turbocharger solenoid valve control	The purpose of this test is whether the turbocharger nozzle control solenoid valve operates as according to the commands. If the solenoid valve operates abnormally, consider whether the foreign material restricts the operation of the actuator, whether there is excess sediment, whether the connection is damaged, whether the solenoid valve is faulty, or whether the position sensor is malfunctioning.
Light emitting relay control	The purpose of this test is to check if the glow relay is operating when the turn-on command is issued. If there is no power when the turn-on command is issued, it may be considered whether the circuit or the glow relay fails.
Exhaust brake control	The purpose of this test is to check if the exhaust brake control solenoid valve is operating when the turn-on command is issued. If no power is applied when the turn-on command is issued, it may be considered whether the circuit or the solenoid valve fails.
Exhaust throttle control	The purpose of this test is to check if the exhaust throttle relay operates when the turn-on command is issued. If no power is applied when the turn-on command is issued, it may be considered whether the circuit or the exhaust throttle relay fails.
Help Oil supply pump understand program reset	The purpose of this function is to reset the adjustment value of the fuel supply pump. Precautions: When replacing the fuel pump or engine, or installing an engine control module removed from another vehicle, the fuel supply pump must be re-understood.
DPD status reset	The purpose of this function is to reset the DPD state parameters. Precautions: The DPD state must be reset before using the fault diagnosis device to regenerate.
DPD regeneration control	The purpose of this test is for forced regeneration of the DPD filter. Precautions: Do not perform this test if the differential pressure exceeds the threshold.
DPD slow regeneration control	The purpose of this test is for forced regeneration of DPD filters in the event of over accumulation of PM. Compare this very slow regeneration with another one. Precautions: Do not perform this test if the differential pressure exceeds the threshold. The engine oil must be replaced after slow regeneration of DPD has been completed.

The fault diagnostic meter is not powered

Circuit description

The DLC is a standard 16-cavity connector. The connector design and location are regulated by industry standards and require the following:

- Positive pressure of fault diagnostic power supply battery at terminal 16
- Grounding the fault diagnosis instrument at terminal 4

Circuit/system test failure detector is not energized

- Perform signal ground of the common port at terminal 5. The fault detector is energized and the ignition device is kept turned off. However, some modules will not be able to communicate unless the ignition is on.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module connector end view.

Step	Operation	Value	Yes	No
1	Precautions: Before using this table, ensure the fault diagnostic meter and 24V adapter can work normally in another vehicle. 1. Turn off the ignition. 2. Check the SBF1 (40A) slow break fusible link in the glove box. Is the SBF4 (40A) slow-motion fuse disconnected?	—	Go to Step 2	Go to Step 3
2	Replace the SBF1 (40A) slow blow fuse. If the fuse continues to open, repair the ground short circuit on one of the circuits fed through the SBF1 (40A) slow blow fuse, or replace the shorted accessory. Is maintenance completed?	—	Go to Step 7	—
3	1. Check each circuit at the data transfer connector for terminal unscrewing, extension or missing. 2. Repair the terminals if necessary. Did you find the problem and complete the repair?	—	Go to Step 7	Go to Step 4
4	Connect a test lamp between the +B circuit of the data transfer connector (FL-370 pin 16) and the known good ground. Is the test light bright?	—	Go to Step 6	Go to Step 5
5	Service the open circuit of the battery power circuit connected to the data transfer connector. Is maintenance completed?	—	Go to Step 7	—
6	1. Test each ground pin circuit of the data transfer connector (FL-370 pin 4 and pin 5) for open circuit or high resistance condition. 2. Repair the circuit and clean or tighten the floor as required. Did you identify the problem and correct it?	—	Go to Step 7	For "discontinuous state"
7	1. Connect the fault diagnostic device to the DLC. 2. Try to enable the fault diagnostic meter. Is the diagnostic scan tool turned on?	—	Normal system	Go to Step 1

Scanner cannot communicate with controller LAN devices

Circuit description

The engine control module communicates with the diagnostic scan tool via a connection to controller area network (CAN). The engine control module and data logging module (DRM) communicate with each other through separate controller LAN connections that do not affect the communication of the diagnostic scan tool.

Diagnostic help

The following conditions can result in loss of controller-LAN serial data communication between the scan tool and the engine control module.

- The controller LAN serial data circuit is cut off.
- Controller LAN serial data circuit grounded short circuit
- Controller LAN serial data circuit voltage short circuit
- Internal conditions in the connector or module on the controller LAN serial data circuit that causes the controller LAN serial data circuit voltage to be short-circuited or grounded shorted

Schematic reference: Schematic diagram of engine control system

Connector end view reference: Engine control system connector end view or engine control module connector end view circuit/system test fault diagnosis tester can not communicate with the controller LAN device

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	Precautions: Make sure that the CANdi module does not fail. The CANdi module LED will flash during normal operation. If there is a problem, the LED will continue to be on or off. 1. Install the fault diagnostic meter 2. Turn on the ignition and turn off the engine. 3. Attempt to establish communication with engine control module Can the fault diagnosis instrument communicate with the engine control module?	—	For "discontinuous state"	Go to Step 3
3	1. Check for continuity of data transfer connector (DLC) (FL-370 pins 6 and 14), poor connections, and corrosion. 2. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 18	Go to Step 4
4	1. Check whether there are discontinuity, bad connection and corrosion in the connector of the engine control module. 2. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 18	Go to Step 5
5	1. Check the SBF1 (40A) slow blow fuse, SBF (40A) slow blow fuse. If the fuse is disconnected, replace it and retest. If the fuse continues to open, repair the short circuit to ground on each circuit fed by the fuse. 2. Turn off the ignition. 3. Disconnect the engine control module harness connector. 4. Turn on the ignition and turn off the engine. 5. Connect a test lamp to the ground and check the voltage at the ignition voltage and power supply circuit of the engine control module (E-57 pin 24). Is the test lights bright?	—	Go to Step 7	Go to Step 6
6	Repair the open circuit of the ignition voltage circuit connected to the engine control module. Is maintenance completed?	—	Go to Step 18	—
7	1. Turn off the ignition. 2. If connected, disconnect the fault diagnosis tester from the data transfer connector. 3. Measure the low and high resistance of the controller's local area network by probing the data transfer connector (FL-370 pin 6 and pin 14). Is the resistance within the specified value (the parallel resistance of the 120 Ω resistor and the 120 Ω controller LAN resistor 2 in the engine control module should be 60 Ω)?	50~70 Q	Go to Step 9	Go to Step 8
8	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Inspect the harness connector of engine control module (E-57 pin 58 and	—	Go to Step 18	Go to Step 17

	pin 78) for intermittent connections, poor connections, and corrosion. 4. Test controller LAN low serial data circuit and high serial data circuit between engine control module (E-57 pin 58 and pin 78) and data transfer connector (FL-370 pin 6 and pin 14) for the following conditions: • Open circuit • Short circuit to each other • High resistance 5. Repair the connection or circuit if necessary. Did you identify the problem and correct it?			
9	1. Test the controller LAN low serial data circuit and the high serial data circuit between data circuit data transfer connector (FL-370 pin 6 and pin 14) and engine control module (E-57 pin 58 and pin 78) or resistor 2 (B-375 pin 1 and pin 2)) for the following conditions: • Short to ground • Voltage short circuit 2 Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 18	Go to Step 10
10	Caution: An open circuit in the ground circuit of the engine control module will not cause loss of communication. 1. Check the grounding of the engine control module for corrosion and tightening conditions. 2. If necessary, clean or fasten the ground. Did you identify the problem and correct it?	—	Go to Step 18	Go to Step 11
11	1. Turn off the ignition. 2. If disconnected, reconnect the harness connector of engine control module. 3. Replace engine control module main relay with engine relay or known good relay. 4. Turn on the ignition and turn off the engine. 5. Attempt to establish communication with engine control module Can the fault diagnosis instrument communicate with the engine control module?	—	Go to Step 16	Go to Step 12
12	1. Turn off the ignition. 2. Remove the main relay of engine control module. 3. Turn on the ignition and turn off the engine. 4. Check the voltage circuit connected to the ECM main relay (FL-116 pin 3 and pin 4) with the test lamp. 5. Turn on the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 18	Go to Step 13
13	1. Reinstall the engine control module main relay. 2. When the main control relay of the engine control module hears a click, turn the ignition switch on and off. Wait 7 seconds during the conversion. Does the ECM main relay tick when the ignition switch is on or off?	—	Go to Step 15	Go to Step 14
14	Repair the main relay ground circuit of the engine control module between the engine control module main relay (FL-116 pin 2) and the cabinet ground terminal (FB-51) in the following cases: • Open circuit • Bad connection or high resistance at main control relay or ground terminal of engine control module Is maintenance completed?	—	Go to Step 18	—
15	1. Test whether there are the following situations in the battery voltage circuit between the engine control module (E-57 pin 2 and pin 5) and the main relay of the engine control module. • Open circuit • The engine control module or the main relay of the engine control module has bad connection or high resistance. 2 Repair the circuit if necessary.	—	Go to Step 18	Go to Step 17

	Did you identify the problem and correct it?			
16	Replace the engine control module main relay Whether to complete the replacement?	—	Go to Step 18	—
17	Important note: The engine control module change must be programmed and understood. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 18	—
18	1. Turn off the ignition. 2. Reconnect all previously disconnected fuses, relays or wiring harness connectors. 3. Turn on the ignition and turn off the engine. 4. Attempt to establish communication with engine control module Can the fault diagnosis instrument communicate with the engine control module?	—	Normal system	Go to Step 3

Engine starts but does not run

Explanation:

The "Engine Starts but Fails to Run" diagnostic table systematically describes the identification method for cases that cause the engine to fail to start. The diagnostic sheet will guide maintenance personnel to perform proper system diagnostics. The diagnostics table assumes that the following conditions are met:

- The battery is full of electricity and the terminals are clean and fastened.
- The crankshaft rotation speed of the engine is normal.
- The fuel tank is rich in fuel.
- No fuel leakage occurred in the fuel line.
- There is no air in the fuel line.
- Cleaner (air and fuel) is clean.
- Fuse and slow blow fuse are normal.

Diagnostic help

- If you suspect there is an intermittent situation, see "Intermittent Situations" in this section.

Test Description

The numbers below indicate the step number of the circuit / system table.

5. If the fuel distribution circuit between the engine control module and the fuel distribution pipe pressure (FRP) regulator is short-circuited to the low-voltage side circuit, the fuel line pressure regulator feedback current is approximately 300mA or less (and normal Current compared to).

Schematic reference: Schematic diagram of engine control system

Connector end view reference: Engine control system connector end view or engine control module connector end view

Circuit / system test engine starts but cannot run

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To System Diagnostic Check - Engine Control System
2	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3. Rotate the engine crankshaft in the specified time. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Does the fault diagnosis tester display any diagnostic trouble codes that caused the ignition to fail?	15 s	To "Applicable Diagnostic Trouble Codes"	Go to Step 3

Step	Operation	Value	Yes	No
3	1. Turn off the ignition. 2. Wait 2 minutes for the fuel pressure to drain from the fuel distribution tube. 3. Turn on the ignition and turn off the engine. Do not start the engine. 4. Observe the fuel distribution pipe pressure parameters with the fault diagnosis tester. Does the fault diagnosis tester display the specified value?	0MPa (0 psi)	Go to Step 4	Go to Step 6
4	Caution: If the vehicle runs out of fuel, the air may enter the fuel system. 1. Make sure the fuel in the tank is adequate and the fuel is of good quality (take a sample). 2. Observe the pressure of the fuel distribution pipe on the fault diagnosis tester when the engine crankshaft turns 5 seconds. During the rotation of the crankshaft, is the fault diagnosis instrument displaying a value higher than the specified value?	25MPa (3,600 psi)	Go to Step 9	Go to Step 5
5	Observe the fuel gauge pressure regulator feedback parameters on the diagnostic tool when the engine crankshaft turns for 5 seconds. During the rotation of the crankshaft, is the fault diagnosis instrument displaying a value higher than the specified value?	800mA	To "Fuel System Check" in the "Fuel System" section	Go to Step 8
6	1. Turn off the ignition. 2. Disconnect the harness connector of pressure sensor for fuel distribution tube. 3. Inspect the fuel distribution pipe pressure sensor harness connector (E-113 pin 1, pins 2 and 3) for intermittences, bad connections and corrosion. 4. Disconnect the engine control module harness connector. 5. Inspect the harness connector of engine control module harness connector (E-56 pin 82, pin 87, pin 90, and pin 101) for intermittent connections, poor connections, and corrosion. 6. Test for high resistance in each circuit. 7. Repair the connection or circuit if necessary. Did you identify the problem and correct it?		Go to Step 11	Go to Step 7
7	Replace the fuel distribution pipe pressure sensor. Whether to complete the replacement?	—	Go to Step 11	—
8	Repair the short to ground between the engine control module (E-56 pin 89 and pin 97) and the pressure regulator of fuel distribution tube (E-116 pin 2). Is maintenance completed?	—	Go to Step 11	—

Step	Operation	Value	Yes	No
9	1. Check whether the readings are normal when the adjusted higher according to the input signals of the following sensors: Use the data list of diagnostic scan tool or a known good vehicle to determine the nominal value. • Engine coolant temperature sensor • BARO sensors • Booster pressure sensor • Intake throttle position sensor 2. Repair the circuit or replace the sensor if necessary. Did you identify the problem and correct it?	—	Go to Step 11	Go to Step 10
10	1. Other possible causes of unstated conditions: • Engine mechanical timing • The incorrect installation of the flywheel causes the crankshaft position (CKP) sensor to be incorrectly synchronized with the engine. Disconnect the sensor and try to start the engine for confirmation. • Blockage solids that seriously inhibit the intake and exhaust • Poor performance of cylinder compression. • There is water pollution or petrol pollution in the fuel. 2. Repair if necessary. Did you identify the problem and correct it?	—	Go to Step 11	—
11	1. Reconnect all previously disconnected harness connectors. 2. Turn off the ignition for 30 seconds. 3. Try to start the engine. Does the engine start and continue running?	—	Go to Step 12	Go to Step 2
12	Use fault diagnostic instrument to observe the diagnose fault code information. Is there any undiagnosed diagnostic fault code?	—	To "Diagnostic Trouble Code List"	Normal system

Diagnostic Trouble Code P0016 (flash code 16)

Circuit description

The crankshaft position (CKP) sensor is located on the flywheel cover. The sensor rotor is mounted on the flywheel. There are 56 notches spaced 6 apart and a 30° bracket less span. The no-support-crossing parts can be detected by TDC. The camshaft position (CMP) sensor is installed to the cylinder head behind the camshaft gear to detect the total 5 holes' number per engine cycle (the four holes are separated by 90° evenly and the reference hole is on the camshaft gear surface). By detecting the span portion of the bracket from the crankshaft position sensor and detecting the reference hole from the camshaft position sensor, the engine control module determines the No. 1 cylinder compression diagnostic trouble code to ensure that they are correlated. If the engine control module detects that the two signals are not synchronized, this diagnostic trouble code will be set.

The operation of this diagnostic trouble code

- Diagnostic trouble codes P0335, P0366, P0340 and P0341 are not set.
- The ignition switch is closed.
- Detection of the signal pulse of the crankshaft position sensor.

Circuit / system test diagnostic fault code P0016

- Camshaft sensor signal pulse are detected.

Setting situation of this diagnostic trouble code

- During engine works, the engine control module detects that there is no synchronization between the crankshaft position sensor signal and the camshaft position sensor signal.

What to do when setting this diagnostic trouble code

- If the diagnostic runs but fails, the engine control module illuminates the fault indicator. See Diagnostic Trouble Code Type Definitions to implement the action taken when setting up diagnostic trouble codes - Class A.

Conditions for clearing this DTC

- See definitions of diagnostic trouble code type to understand the conditions for turning off fault indicators/clearing diagnostic trouble codes - class A.

Diagnostic help

- The fault code is caused by incorrect mechanical timing, and the incorrect mechanical timing may be caused by incorrect installation of timing gear, flywheel or camshaft.

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3. Start the engine. If the engine is not started, turn the engine crankshaft for 10 seconds. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Is P0335, P0036, p0340 or P0341 also set?	—	To "Applicable Diagnostic Trouble Codes"	Go to Step 3
3	1. Check the crankshaft position (CKP) sensor and camshaft position (CMP) sensor for the following conditions: • Physical damage to the sensor • The sensor is installed insecurely or inappropriate. • Air gap is too large • The foreign matter enters the sensor and sensor rotor or camshaft gear. • Physical damage to the sensor rotor or camshaft gear • Sensor rotor or camshaft gears are not installed properly or are not suitable 2. Check the engine timing of the incorrectly installed timing gear or camshaft. Caution: The diagnostic trouble code may be set if the flywheel locating pin is missing or pushed in or the flywheel is incorrectly installed. 3. Repair or replace as necessary. Is maintenance completed?	—	Go to Step 4	—
4	1. If disconnected, reconnect all previously disconnected harness connectors. 2 Use the scan tool to clear the diagnostic trouble code. 3. Turn off the ignition for 30 seconds. 4. Start the engine. If the engine is not started, turn the engine crankshaft for 10 seconds. 5. Operate the vehicle under the conditions required to run this diagnostic trouble code. It is also possible to run the vehicle according to the condition observed from the frozen fault state / fault record. Is diagnostic trouble code causing ignition failure?	—	Go to Step 2	Go to Step 5
5	Use fault diagnostic instrument to observe the diagnose fault code information. Is there any undiagnosed diagnostic fault code?	—	To "Diagnostic Trouble Code List"	Normal system

To diagnostic trouble code U0110 (flash code 87)**Circuit description**

The engine control module and the rotary electric actuator (REA) convey the control and diagnosis information through the CAN local communication bus. REA is inside the turbocharger. The engine control module monitors the CAN operation status by obtaining a constant message flow from each module. If the engine control module receives the expected message from REA, this diagnostic fault code will be set.

The operation of this diagnostic trouble code

- Diagnostic trouble codes U0073 are not set.
- The battery voltage is higher than 20 volts.
- The ignition switch is on.

Setting situation of this diagnostic trouble code

- The engine control module detects that no CAN bus message from REA has been received.

What to do when setting this diagnostic trouble code

- If the diagnostic runs but fails, the engine control module illuminates the fault indicator. See definition of DTC type for action to take when setting diagnostic trouble codes - class A.

Conditions for clearing this DTC

- See the DTC type definition for the conditions for turning off the fault indicator lamp/clearing the diagnostic trouble code - Class A.

Diagnostic help

- If you suspect there is an intermittent situation, see "Intermittent Situations" in this section.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Circuit / system test diagnostic fault code U0110

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3 Start the engine. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Is the diagnostic trouble code U0073or P0045 also set?	—	Suitable for diagnosing fault codes	Go to Step 3
3	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Measure the resistance in series with the CAN low and high circuits (E-169 pin 18 and 37). Is the resistance within the specified range (the parallel resistance of 120 ohm resistor and 120 ohm can resistor 1 in REA shall be 60 ohms.)?	50-70 ohm	Go to Step 8	Go to Step 4

4	1. Maintain the connection between the DMM and the engine control module. 2. Disconnect resistor 1 from the harness connector (resistor 1 is located behind the instrument). Is the resistance within the specified range?	110-130 ohm	Go to Step 5	Go to Step 6
5	Measure the resistance of resistor 1. Is the resistance within the specified range?	110-130 ohm	Go to Step 7	Go to Step 14
6	1. Turn off the ignition. 2. Disconnect the data recording module (DRM) harness connector (the DRM is on the right side of the passenger seat bracket). 3. Turn on the ignition and turn off the engine. 4. Connect a DMM between the CAN low level circuit (E-147 pin 37) and a known good ground. 5. Connect a DMM between the CAN high circuit (E-147 pin 18) and a known good ground. 6. Test CAN low level circuit and high level circuit (wherever the voltage reading is not within the specified value range) between the engine control module (E-37 pin 18 and pin 37) and REA (E-147 pin 15 and pin 16) to check for open circuit or high resistance. 7. Repair the circuit or connection if necessary. Did you identify the problem and correct it?	1.5-3.5V	Go to Step 17	Go to Step 13
7	1. Test whether there is an open circuit or high resistance in the CAN low level circuit and the high level circuit between resistor 1 (FL-374 pin 1 and pin 2) and REA. 2. Check the resistance 1 harness connector (FL-374 pin 1 and pin 2) for intermittent and poor connections and corrosion. 3. Repair the circuit or connection if necessary. Did you identify the problem and correct it?	—	Go to Step 17	To "Diagnostic Help"
8	1. Connect a DMM between a known good ground and the ignition voltage feed circuit (E-147 pin 3) connected to REA. 2. Turn on the ignition and turn off the engine. Is the DMM voltage higher than the specified value?	20.0V	Go to Step 9	Go to Step 10
9	Connect the circuit for DMM ignition voltage feeding and the ground circuit (E-147 pin 3 and pin 6) in series. Is the DMM voltage higher than the specified value?	20.0V	Go to Step 12	Go to Step 11
10	Repair open circuit or high resistance between F-44 (15A) fuse and REA. Check F-44 (15A) fuse first. Is maintenance completed?	—	Go to Step 17	
11	Repair open circuit or high resistance between REA and ground terminal (FB-51). Clean or compress the floor if necessary. Is maintenance completed?	—	Go to Step 17	
12	1. Turn off the ignition. 2. Disconnect the REA harness connector. 3. Inspect the REA harness connector (E-147 pin 1, pin 3, pin 4 and pin 6) for breaks, poor connections, and corrosion. 4. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 17	Go to Step 16

Step	Operation	Value	Yes	No
13	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Inspect the harness connector of engine control module (E-57 pin18 and pin 37) for intermittent connections, poor connections, and corrosion. 4. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 17	Go to Step 15
14	Replace resistor 1. Whether to complete the replacement?	—	Go to Step 17	—
15	Important note: The changes to the engine control module must be programmed and learned. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 17	—
16	Replace the turbocharger. (REA inside the turbocharger) Whether to complete the replacement?	—	Go to Step 17	—
17	1. Reconnect all previously disconnected harness connectors. 2 Use the scan tool to clear the diagnostic trouble code. 3. Turn off the ignition for 30 seconds. 4 Start the engine. 5. Operate the vehicle under the conditions required to run this diagnostic trouble code. It is also possible to operate the vehicle under conditions observed from a frozen fault condition/fault log. Is diagnostic trouble code causing ignition failure?	—	Go to Step 3	Go to Step 18
18	Use fault diagnostic instrument to observe the diagnose fault code information. Is there any undiagnosed diagnostic fault code?	—	To "List of Diagnostic Trouble Codes "	Normal system

POWERSTAR

EGR control system inspection

Explanation:

EGR system recycles a part of exhaust gas into the intake manifold and consequently reduces the emission of NOx. The EGR control system uses electronic control system to ensure both driving performance and low emission. The control current from the engine control module will run the solenoid coil to control the lift of the EGR valve. Moreover, a EGR position sensor is also provided at the rear of the spiral ring, so that the actual valve lift amount is fed back to the engine control module for more precise control. EGR control will start working when the engine speed, engine coolant temperature, intake air temperature and barometric pressure conditions are met. Then, the valve opening degree is calculated based on the engine speed and the target fuel injection amount. According to the opening condition of the valve, determine the driving task of the solenoid valve and start the valve accordingly. It is equipped with intake throttle to acquire sufficient intake manifold differential gas pressure and consequently ensure the EGR air flow.

EGR control operation

- The engine coolant temperature (ECT) is between 70 °C(158 °F) and 100 °C (212 °F).
- The inlet air temperature (IAT) is higher than 0°C(32°F).
- Atmospheric pressure (BARO) exceeds 90 thousand PA (13psi).

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Test of circuit / system test EGR control system

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3. Turn on the ignition and turn off the engine. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Do you want to set the diagnostic trouble code to "engine control module suppresses EGR control "when "setting the action taken when the DTC sets" in the special code state?	—	See Applicable Diagnostic Trouble Codes	Go to Step 3
3	1. Check for the following conditions: • Loss or damage of EGR valve gaskets • EGR valve bonding • Between any exhaust manifold and intake manifold, EGR gas leakage occurs in any EGR channel. • There is a blocked or broken EGR channel between the exhaust manifold and the EGR valve • There are any types of restrictions on the emission system • Air purifier elements are obstructed and the air duct between air purifier and intake manifold is blocked or broken • Any air intake leaks • Is there any moisture enters into the induction system • Any dirt or objects blocking inlet of the MAF sensor • The MAF sensor is skewed or slow • The engine coolant temperature (ECT) sensor is skewed. Please refer to the temperature and resistance meter to test the ECT sensor at various temperature levels in order to assess the possibility of sensor deflection. • BARO sensor skew. The external atmospheric pressure is determined based on the height and position specified in the barometer. Please refer to "Altitude and Atmospheric Pressure" • Intake throttle bonding 2. Repair if necessary.	—	Go to Step 13	Go to Step 4

	Did you identify the problem and correct it?			
4	1. Place the gear in the neutral position and set the parking brake. 2. Start engine and warm up (allow engine coolant temperature to reach at least 75 degrees Celsius [167 degrees Fahrenheit]). 3. Accelerate the engine several times (between idling and W.O.T.; accelerator pedal full stroke) while observing the required EGR position and parameter of EGR position by using a diagnostic scan tool. Is the EGR position parameter within the specified range?	$\pm 5\%$	Go to Step 5	Go to Step 6
5	1. Use the diagnostic scan tool for EGR solenoid control. 2. While observing the EGR position, command the desired EGR position to increase or decrease. Is the EGR position parameter quickly within the specified range?	$\pm 5\%$	Normal system	Go to Step 6
6	1. Remove the EGR valve assembly from the engine. 2. Check the EGR valve for the following conditions: • Obstruction of EGR valves due to foreign bodies • Excess sediment at the valve • Valve shaft bending 3. Repair or replace as necessary. Did you identify the problem and correct it?	—	Go to Step 9	Go to Step 7
7	1. Turn off the ignition. 2. Disconnect the EGR valve harness connector. 3. Check the EGR valve harness connector (E-80 pins 1, 2, 3, 4, 5, 6, 7 and 8) for intermittence, poor connections, and corrosion. 4. Disconnect the engine control module harness connector. 5. Inspect the harness connector of engine control module (E-56 pin 87, 93, 94, 99, 101, 103, 110 and 111) for intermittent connections, poor connections, and corrosion. 6. Test for high resistance in each circuit. 7. Repair the connection or circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 9	Go to Step 8
8	Replace the EGR valve. Whether to complete the replacement?	—	Go to Step 9	—
9	Reconnect all previously disconnected components or harness connectors. Complete the operation or not?	—	Go to Step 4	—

Glow control system inspection

Explanation:

The glow control system includes an engine control module, a glow relay, a glow indicator, and a glow plug. When the coolant temperature of the engine is low, the glow control system starts to run, so as to make the engine start easier. The engine control module commands the glow relay to turn on for a certain amount of time with the ignition on and the engine off. After warm-up, the glow plug is energized for a period of time while the engine is running.

Glow control operation

- According to the engine coolant temperature, the glow indicator lights up for 1.5 seconds to about 10 seconds. Light up for 1.5 seconds at -5 degrees Celsius (23 degrees Fahrenheit) or higher.
- During the preheating stage, the glow relay energized 3 seconds to about 15 seconds according to the engine coolant temperature.
- After preheating, depending on the engine coolant temperature, the glow relay is energized for 3 seconds to approximately 30 seconds.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Circuit / system test glow control system check

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine	—	Go to Step 2	To Diagnostic

	control system"?			System Check - Engine Control System
2	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3. Turn on the ignition and turn off the engine. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Is the diagnostic trouble code P0117, P0118, P0380 or P0381 also set?	—	See Applicable Diagnostic Trouble Codes	Go to Step 3
3	1. Turn off the ignition. Make sure that the metal bus connecting the switch battery voltage supply terminal (E-6) and all glow plug is firm. 3. Turn on the ignition and turn off the engine. 4. Connect a test lamp between the metal bus (glow plug power supply E-6 end) and the known good ground. 5. Use the diagnostic scan tool for glow relay control. 6. Command the relay to open and observe the test lamp Is the test light only turned on when the command is turned on?	—	Go to Step 4	Go to Step 5
4	1. Turn off the ignition. 2. Remove the metal bus from the glow plug. 3. Test the resistance of each glow plug between the glow plug connector and the known good ground. Ensure that all measurements are recorded, and such measurements should be carried out promptly so that the engine temperature does not change between measurements. Is the resistance within the mutual specified range?	1 ohm	Normal system	Go to Step 15
5	1. Turn off the ignition. 2. Replace the glow relay with a starter relay or a known good relay. 3. Turn on the ignition and turn off the engine. 4. Connect a test lamp between the metal bus (electric plug power E-6 connector) and the known good ground. 5. Use the diagnostic scan tool for glow relay control. 6. Command the relay to open and observe the test lamp Is the test light only turned on when the command is turned on?	—	Go to Step 13	Go to Step 6

Step	Operation	Value	Yes	No
6	Check the SBF4 (60A) slow-motion fuse in the chassis side fuse and relay box. Is the SBF4 (60A) slow-motion fuse disconnected?	—	Go to Step 7	Go to Step 8
7	Replace the SBF4 (60A) slow blow fuse. If the slow blow fuse continues to open, repair the short to ground on one of the circuits fed through the slow blow fuse, or replace the short-circuited accessory. Is maintenance completed?	—	Go to Step 16	—
8	1. Turn off the ignition. 2. Remove the glow relay. 3. Use a test lamp (connected to a known good ground) to probe the relay battery voltage feed circuit (FL-14 pin 1). Is the test light bright?	—	Go to Step 9	Go to Step 10
9	1. Use a test light (connected to a known good ground) to detect the glow plug voltage supply circuit (FL-15 pin 1). 2. Turn on the ignition and turn off the engine. Is the test light bright?	—	Go to Step 12	Go to Step 11
10	Repair the open circuit or high resistance between SBF4 (60A) slow blow fuse and glow relay (FL-14 pin 1). Is maintenance completed?	—	Go to Step 16	—
11	Repair the open circuit or high resistance between the glow relay (FL-15 pin 1) and the glow plug (E-6 terminal). Is maintenance completed?	—	Go to Step 16	—
12	Important note: If there is a short circuit between the battery voltage supply circuit and the voltage source, the glow plug may be burned. Repair the short circuit of battery or ignition voltage between the glow relay (FL-15 pin 1) and the glow plug (E-6 terminal). Is maintenance completed?	—	Go to Step 16	—
13	1. Remove the glow relay. 2. Check whether there is interruption and bad connection on each glow relay terminal. 3. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 16	Go to Step 14
14	Replace the glow relay. Whether to complete the replacement?	—	Go to Step 16	—
15	Replace the proper glow plug. Whether to complete the replacement?	—	Go to Step 16	—
16	1. Reconnect all previously disconnected components, relays, fuse or wiring harness connectors. 2. Turn off the ignition for 30 seconds. 3. Turn on the ignition and turn off the engine. 4. Connect a test lamp between the metal bus (electric plug power	—	Go to Step 4	Go to Step 2

	E-6 connector) and the known good ground. 5. Use the diagnostic scan tool for glow relay control. 6. Command the relay to open and observe the test lamp Is the test light only turned on when the command is turned on?			
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Engine preheating control system inspection

Explanation:

The engine preheating system consists of an engine control module, an intake throttle, an exhaust brake valve, an exhaust brake solenoid valve and an engine preheating switch. When the coolant temperature of the engine is low, the engine preheating system will start to operate, and the engine will be preheated. The engine control module commands the intake throttle solenoid valve and the exhaust brake solenoid valve to close the valves according to the input signal of the engine coolant temperature and preheating requirements.

Engine preheating control operation

- Turn on the engine preheating switch.
- Engine operation.
- Do not depress the accelerator pedal.
- The engine coolant temperature is below 65 degrees to 70 degrees (149 °F to 158 °F).
- Vehicles are not running.

Test Description

The numbers below indicate the step number of the circuit / system table.

6. Observe the position of accelerator pedal with the fault diagnostic meter. The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation.

7. Use a fault diagnosis instrument to observe the speed of the vehicle. The speed parameter should indicate the speed correctly.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Check of warm-up control system for circuit/system test engine

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	Is the exhaust brake valve operating correctly?	—	Go to Step 3	To "Exhaust Brake Control System Inspection"
3	1. Install the fault diagnostic meter 2. Turn off the ignition for 30 seconds. 3. Turn on the ignition and turn off the engine. 4 Use the fault diagnosis tester to monitor the diagnostic trouble code information. Is the diagnostic trouble code P0117, P0118, P0122, P0123, P0638, P0641, P0651 or P0697 also set?	—	See Applicable Diagnostic Trouble Codes	Go to Step 4

4	1. Parking. 2.Ensure that the throttle speed control knob is at the lowest position. (completely counter clockwise) 3. Cool the engine if necessary (cool the engine coolant to at least 65°C [149°F]). 4 Start the engine. 5. Turn on the engine preheat switch. Is the air intake throttle valve and exhaust brake valve closed when opened?	—	Go to Step 5	Go to Step 8
5	Turn off the engine preheating switch. Whether or not to delete the engine preheating control?	—	Go to Step 6	Go to Step 16
6	1. Turn on the engine preheat switch. 2. Step down the accelerator pedal. Whether or not to delete the engine preheating control?	—	Go to Step 7	Observe accelerator pedal position parameters to diagnose accelerator pedal position sensor.
7	Driving vehicle. (Raise driving wheel when necessary) Whether or not to delete the engine preheating control?	—	Normal system	Observe the speed parameter. For the diagnosis of the speed sensor, see P0500.
8	Use diagnostic tools to observe the engine preheat switch parameters. Is the fault diagnosis tester indicated as "ON"?	—	Go to Step 9	Go to Step 10
9	Use the diagnostic scan tool to observe the command parameter for exhaust brake valve. Is the fault diagnosis tester indicated as "ON"?	—	The problem must be in the solenoid valve, the exhaust brake valve or the vacuum pressure tube. To "Exhaust Brake Control System Inspection"	Engine control module does not allow engine preheating control. Refer to "operation of engine warm-up control"
10	1. Turn off the ignition. 2. Remove the glass cover of the cluster around the instrument panel (IP) completely and disconnect the engine preheating switch. 3. Disconnect the harness connector of engine preheat switch. 4. Connect a test lamp between the voltages feed circuit of the switch wire harness (FU-39 pin 4) and the known good ground. 5 Start the engine. Is the test lamp on?	—	Go to Step 11	Go to Step 12
11	When crossing the switch wire connector between the pin 1 and the pin 2 of the FU-39 to have jumper connection with the jumper of 3-A fuse, the fault diagnosis instrument is used to observe the parameters of engine preheating switch. Does the scan tool indicate "on" when the circuit is bridged and "off" when the circuit is not bridged?	—	Go to Step 14	Go to Step 13

12	Repair open circuit or high resistance between F-11 (7.5A) fuse and switch (FU-39 pin 1). Is maintenance completed?	—	Go to Step 20	—
13	1. Test the signal circuit between the engine control module (pin 51 of E-57) and the switch (pin 2 of FU-39) to check for open or high resistance. 2 Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 15
14	1. Inspect the harness connector of switch (FU-5 pin A5) for intermittent connection and poor connection. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 18
15	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Inspect the harness connector of engine Control Module (E-57 pin 51) for intermittent and poor connections. 4. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 19
16	1. Turn off the ignition. 2. Completely remove the glass cover surrounding the IP cluster and disconnect the engine preheating switch. 3. Disconnect the harness connector of engine preheat switch. 4 Start the engine. Whether or not to delete the engine preheating control?	—	Go to Step 18	Go to Step 17
17	1. Test the signal circuit between the engine control module (E-57's pin 51) and the switch (FU-39 pin 2) to check whether there is a short circuit in a battery or the ignition voltage. 2 Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 19
18	Repair or replace the engine preheating switch. Is maintenance or replacement completed?	—	Go to Step 20	—
19	Important note: The changes to the engine control module must be programmed and learned. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 20	—
20	Reconnect all previously disconnected components, fuses or harness connectors. Complete the operation or not?	—	Go to Step 4	—

Exhaust brake control system inspection

Explanation:

The exhaust brake control system includes an engine control module, an intake throttle valve, an exhaust brake valve, an exhaust brake solenoid valve and an exhaust brake switch. The engine control module commands the intake throttle solenoid valve and the exhaust brake solenoid valve to close the valves according to the operation conditions of the vehicle and the input signal of the exhaust brake switch.

Exhaust brake control operation

- The exhaust brake switch is turned on.
- Engine operation.
- Do not depress the accelerator pedal.
- Don't step on the clutch pedal.
- The speed of the vehicle is greater than the pre set range.

Test Description

The numbers below indicate the step number of the circuit / system table.

4. Observe the position of accelerator pedal with the fault diagnostic meter. The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation.

8. If the exhaust brake solenoid command parameter indicates "off," use the scan tool to observe the vehicle speed, throttle pedal position, and clutch switch parameters to allow exhaust brake control. For diagnostic information, please refer to the engine control schematic diagram and the data list of the fault diagnosis instrument.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Circuit / system test exhaust brake control system Checklist (Table 1 of 2 tables)

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	1. Check that the dashboard (IP) cluster is available. 2 Start the engine. 3. Turn the exhaust brake switch on and off. Is the exhaust brake indicator on and off at each switching transition?	—	Go to Step 3	To Step 2 of Table 1
3	1. Drive the vehicle so that the exhaust brake "opens" long enough. 2. Open the exhaust brake switch. 3. Release the accelerator pedal. Is the exhaust brake fully used (if the vehicle speed is reduced by exhaust brake, but engine valve noise occurs, the intake throttle may not be fully closed)?	—	Go to Step 4	Go to Step 6
4	1. Rein put the exhaust brake control. 2. Step up the accelerator pedal while performing the exhaust brake control. Is the exhaust brake control removed?	—	Go to Step 5	Observe accelerator pedal position parameters to diagnose accelerator pedal position sensor.
5	1. Rein put the exhaust brake control. 2. Step down the clutch pedal while performing the exhaust brake control. Is the exhaust brake control removed?	—	Normal system	Go to Step 8

Step	Operation	Value	Yes	No
6	1. Parking. 2. Install the fault diagnostic meter 3. Turn off the ignition for 30 seconds. 4. Turn on the ignition and turn off the engine. 5 Use the fault diagnosis tester to monitor the diagnostic trouble code information. When the "action taken when DTC is set" is set in the special code state, set the diagnostic trouble code to "engine control module suppression exhaust brake control"?	—	See Applicable Diagnostic Trouble Codes	Go to Step 7
7	1. Rein put the exhaust brake control. 2. Trigger the snapshot function to acquire and store engine parameters during exhaust brake control. 3. Review the exhaust brake valve command parameters. When the exhaust brake is fully operational, does the exhaust brake valve command parameter show "ON"?	—	Go to Step 17	The engine control module does not allow exhaust brake control. Please see Exhaust Brake Control Operation.
8	1. Install the fault diagnostic meter 2. Turn on the ignition and turn off the engine. 3. When the clutch pedal is fully depressed and released, use the scan tool to observe the clutch pedal switch parameters. When stepping on the clutch pedal, does the malfunction indicator indicate "Applied"? Does the scan tool show "released" when the clutch pedal is released?	—	Go to Step 17	Go to Step 9
9	1. Check to make sure the clutch pedal is adjusted correctly. When the pedal is released, the plunger should always be inserted, and the clutch pedal should not be prevented from fully ascending. 2. Adjust the clutch switch if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 10
10	1. Turn off the ignition. 2. Disconnect the clutch pedal switch harness connector. 3. Turn on the ignition and turn off the engine. 4. Connect a test lamp between the ignition voltage feed circuit of the online beam (FL-2 pin 1) and the known good ground. Is the test lamp on?	—	Go to Step 11	Go to Step 12
11	When crossing the switch wire connector between the pin 1 and the pin 2 of the FL-2 to have jumper connection with the jumper of 3-A fuse, the fault diagnosis instrument is used to observe the parameters of engine preheating switch. Does the scan tool indicate "Released" when the circuit is bridged and "Applied" when the circuit is not bridged?	—	Go to Step 14	Go to Step 13
12	Repair open circuit or high resistance between F-11 (7.5A) fuse and switch (FL-2 pin 1). Is maintenance completed?	—	Go to Step 28	—

Step	Operation	Value	Yes	No
13	1. Test the signal circuit between the engine control module (E-57 pin 26) and the switch (FU-39 pin 2) for the following conditions: • Open circuit • Short circuit of battery or ignition voltage • High resistance 2. Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 15
14	1. Inspect the harness connector of switch (FU-5 pin A5) for intermittent connection and poor connection. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 16
15	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Inspect the harness connector of engine Control Module (E-57 pin 26) for intermittent and poor connections. 4. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 27
16	Replace the clutch switch. Whether to complete the replacement?	—	Go to Step 28	—
17	1. Use the diagnostic scan tool for exhaust brake control. 2. Open and close the solenoid valve command. Does the solenoid valve act according to the command?	—	Problems arise in solenoid valves, exhaust brake valves or vacuum pressure pipes. Please refer to the diagnostic chart for the exhaust system chapter.	Go to Step 18
18	1. Turn off the ignition. 2. Disconnect the exhaust brake solenoid harness connector. 3. Turn on the ignition and turn off the engine. 4. Connect the test lamp between the ignition voltages feed circuit (FB-6, J-45 pin 1) and a known good ground. Is the test lamp on?	—	Go to Step 19	Go to Step 21
19	1. Pass the test light through the exhaust brake solenoid harness connector (FB-6, J-45 pins 1 and 2). 2. Use the diagnostic scan tool for exhaust brake control. 3. Give open and close commands to solenoid valve. Under each command, does the test light turn on and off?	—	Go to Step 24	Go to Step 20
20	Is the test lamp constantly on under each command?	—	Go to Step 23	Go to Step 22

Step	Operation	Value	Yes	No
21	Repair the open circuit or high resistance between the F-44 (15A) fuse and the solenoid valve (FB-6, J-45 pin 1). Is maintenance completed?	—	Go to Step 28	—
22	1. Test whether the control circuit between the engine control module (E-57 pin 15) and the solenoid valve (FB-6, J-45 pin 2) has the following conditions: • Open circuit • Short circuit of battery or ignition voltage • High resistance 2. Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 25
23	1. Test the control circuit between the engine control module (E-57 pin 15) and the solenoid valve (FB-6, J-45 pin 2) for a short to ground. 2. Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 28	—
24	1. Check the solenoid harness connector (pins 1 and 2 of the FB-6 and J-45) for intermittent connections and poor connections. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 26
25	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Inspect the harness connector of engine Control Module (E-57 pin 15) for intermittent and poor connections. 4. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 28	Go to Step 27
26	Replace the exhaust brake solenoid valve. Whether to complete the replacement?	—	Go to Step 28	—
27	Important note: The changes to the engine control module must be programmed and learned. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 28	—
28	Reconnect all previously disconnected components, fuses or harness connectors. Complete the operation or not?	—	Go to Step 3	—

Circuit / system test exhaust brake control system Checklist (Table 2 of 2 tables)

Step	Operation	Value	Yes	No
1	1. Install the fault diagnostic meter 2. Start the engine. 3. Turn on and off the exhaust brake switch when observing the exhaust brake switch parameters using the diagnostic tool. Is the diagnostic device turned on and off during each switch transition?	—	Go to Step 2	Go to Step 3

Step	Operation	Value	Yes	No
2	Is the exhaust brake light always on at each switch?	—	Go to Step 10	Go to Step 11
3	1. Turn off the ignition. 2. Disconnect the combination switch harness connector. 3. Connect a test light between the voltage feeding circuit of the combination switch harness (FL-9 pin 18) and the known good ground. 4 Start the engine. Is the test lamp on?	—	Go to Step 4	Go to Step 5
4	When crossing the switch wire connector between the pin 17 and the pin 18 of the FU-39 to have jumper connection with the jumper of 3-A fuse, the fault diagnosis instrument is used to observe the parameters of exhaust brake switch. Does the scan tool indicate "on" when the circuit is bridged and "off" when the circuit is not bridged?	—	Go to Step 7	Go to Step 6
5	Repair open circuit or high resistance between F-13 (7.5A) fuse and switch (FL-9 pin 18). Check F-13 (7.5A) fuse first. Is maintenance completed?	—	Go to Step 20	—
6	1. Test the signal circuit between the engine control module (E-57 pin 45) and the switch (FL-9 pin 17) for the following conditions: •Open circuit • Short circuit of battery or ignition voltage • High resistance 2 Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 8
7	1. Check for intermittence and bad connection at the switch harness connector (pins 17 and 18 of FL-9). 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 9
8	1. Inspect the harness connector of engine Control Module (E-57 pin 45) for intermittent and poor connections. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 19
9	Repair or replace the combination switch. Is maintenance or replacement completed?	—	Go to Step 20	—
10	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Turn on the ignition and turn off the engine. Is the exhaust brake indicator lamp off?	—	Go to Step 19	Go to Step 16

Step	Operation	Value	Yes	No
11	1. Turn off the ignition. 2. Disconnect the engine control module harness connector. 3. Dismantle the F-44 (15A) fuse supplying power to the exhaust brake indicator. 4. Turn on the ignition and turn off the engine. 5. Measure the voltage generated by the exhaust brake indicator lamp control circuit of the engine control module harness connector (E-57 pin 7) connected to the known good ground. Is the voltage below the specified value?	1V	Go to Step 12	Go to Step 17
12	1. Turn off the ignition. 2. Reinstall the F-44 (15A) fuse again. 3. Turn on the ignition and turn off the engine. 4. Connect a jumper with a 3-amp fuse between the harness connector of engine control module (E-57 pin 7) and a known good ground. Is the exhaust brake indicator lamp on?	—	Go to Step 15	Go to Step 13
13	1. Test the control circuit between the engine control module (E-57 pin 7) and the IP cluster (FU-5 pin A5) to check whether there is an open circuit or a high resistance. 2. Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 14
14	1. Inspect the harness connector of cluster cable (FU-5 pin A5) for intermittent connection and poor connection. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 18
15	1. Inspect the harness connector of engine Control Module (E-57 pin 7) for intermittent and poor connections. 2. Repair the connector if necessary. Did you identify the problem and correct it?	—	Go to Step 20	Go to Step 19
16	Repair the short to ground between the engine control module (E-57 pin 7) and the cluster of instrument panel (FU-5 pin A5). Is maintenance completed?	—	Go to Step 20	—
17	Repair the battery or ignition voltage short circuit between the engine control module (E-57 pin 7) and the instrument panel cluster (FU-5 pin A5). Is maintenance completed?	—	Go to Step 20	—
18	Repair or replace the dashboard cluster. Is maintenance or replacement completed?	—	Go to Step 20	—
19	Important note: The changes to the engine control module must be programmed and learned. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 20	—
20	Reconnect all previously disconnected components, fuses or harness connectors. Complete the operation or not?	—	To Step 1 of Table 1	—

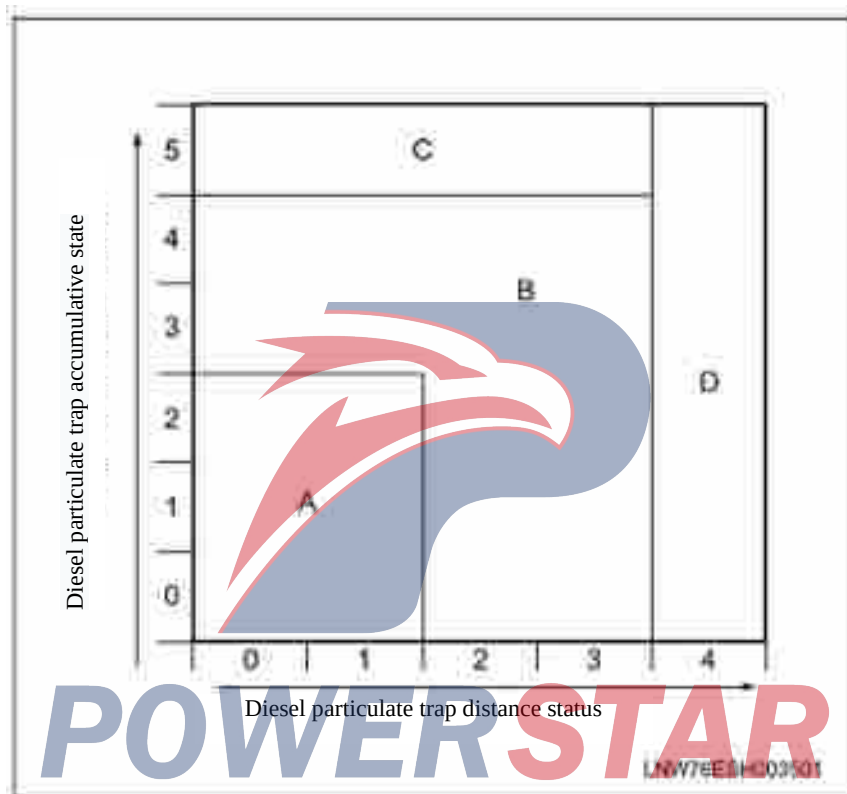
Diesel particulate trap (DPD) control system inspection

Explanation:

The engine control module detects the accumulation of pm from the exhaust differential pressure sensor connected to the chassis near the diesel particulate trap (DPD) assembly or from mileage. The exhaust differential pressure sensor is a sensor that changes the voltage according to the differential pressure of the exhaust gas before and after the filter. When specific PM cumulates or specific mileage lengths are reached,

automatic regeneration will be initiated. If for some reason automatic regeneration cannot be completed, the driver is required to manually regenerate by flashing the DPD lamp on the dashboard cluster. Once regeneration begins, it must be completed within a period of time. After the regeneration is completed, the purge degree of the filter is judged by monitoring the differential pressure of the exhaust. If excessive PM accumulation is detected, regeneration cannot be regenerated because uncontrolled combustion may occur during regeneration.

By comparing the diesel particulate trap cumulative state parameter with the diesel particulate trap distance state parameter; the diesel particulate trap state table is used to determine the type of diesel particulate trap filter regeneration process. To use this table, use the troubleshooter to observe both parameters and draw their readings on the diesel particle trap status chart. The crossover range is expressed as A, B, C or D (for example, the diesel particulate trap cumulative state is 3 and the diesel particulate trap distance state is 1. The range is B.)



Range A: general range

Scope B: The DPD lights flicker, but manual regeneration can be done.

Scope C: The exhaust differential pressure range is too high, and manual regeneration is prohibited.

Range D: The distance state is abnormal, and manual regeneration is prohibited.

Test Description

The numbers below indicate the step number of the circuit / system table.

6. 8. 9. If no regeneration is initiated, any diagnostic failure code that suppresses regeneration control can be set. If no diagnostic trouble code is set, use the scan tool to observe the vehicle speed, accelerator pedal position, neutral switch, speed sensor and PTO switching parameters for DPD regeneration control. For diagnostic information, please refer to the engine control schematic diagram and the data list of the fault diagnosis instrument.

Schematic reference: engine control system schematic

Connector end view reference: engine control system connector end view or engine control module (ECM) connector end view.

Circuit/System Test, DPD Control System Check

Step	Operation	Value	Yes	No
1	Have you conducted the "diagnostic system check - engine control system"?	—	Go to Step 2	To Diagnostic System Check - Engine Control System
2	1. Check that the dashboard (IP) cluster is available. 2. Turn ignition on for 30 seconds and keep the engine off. Is the DPD indicator turned on and then turned off?	—	Go to Step 3	For the DPD indicator lamp circuit diagnosis, refer to P1669
3	1. Check the engine oil level. Make sure that the amount of fuel is within the specified range. 2. Install the fault diagnostic meter 3 Start the engine. 4. Use the fault diagnosis tester to observe the DPD switch parameters. Is the diagnostic device turned on and off during each switch transition?	—	Go to Step 4	Go to Step 15
4	1. Check for the following conditions: • Whether the exhaust differential pressure hose or pipe is incorrectly routed or disconnected. Important note: The exhaust differential pressure hose has a direction for installation. For proper installation, see section Exhaust system. • Is the exhaust differential pressure hose or pipe blocked, broken or twisted? • Whether the exhaust differential pressure sensor is misplaced or tilted. For proper installation, see section Exhaust system. • If the exhaust differential pressure sensor is physically damaged. • Whether there is any contaminant or object blocking the inlet of the differential pressure sensor? • Exhaust differential pressure sensor is tilted or slow • Whether there is any loss or damage of exhaust system gaskets? • Exhaust leaks from exhaust ping, gaskets, DPD components, exhaust temperature sensors or exhaust differential pressure hoses or pipes • Has the exhaust system been modified? • If the exhaust throttle or exhaust brake valve is jammed, please refer to the exhaust system section for diagnostic information. • Is the exhaust temperature sensor tilted or slow • Is the air purifier element obstructed and the air duct between the air purifier and the intake manifold is blocked or broken? • Whether the intake system has been modified • Are there any air intake leaks • Whether there is any dirt or object blocking (MAF) sensor inlet? • Whether or not a MAF sensor is skewed or slow 2. Repair or replace as necessary. Did you identify the problem and correct it?	—	Go to Step 14	Go to Step 5
5	Start the engine and let it idle for 30 seconds. Is the DPD indicator flashing?	—	Go to Step 6	Go to Step 7

Step	Operation	Value	Yes	No
6	Press DPD switch to enable the manual DPD reactivation. Does the reactivation start?	—	Go to Step 10	Engine control module does not allow DPD regeneration control. See Test Description and Diagnostic Trouble Code List
7	Observe the DPD accumulating status and DPD distance status with the fault diagnostic meter and plot the readings on the DPD table. Is the cross range in A or B?	—	Go to Step 8	Go to Step 9
8	Important note: Reset the DPD state before using the fault diagnosis device to regenerate. Use the diagnostic scan tool for DPD regeneration. Does the reactivation start?	—	Go to Step 10	Engine control module does not allow DPD regeneration control. See Test Description and Diagnostic Trouble Code List
9	Important note: Reset the DPD state before using the fault diagnosis device to regenerate. Important note: engine oil must be replaced after slow regeneration of DPD. Use the fault diagnosis tester for slow DPD regeneration. Does the reactivation start?	—	Go to Step 10	Engine control module does not allow DPD regeneration control. See Test Description and Diagnostic Trouble Code List
10	Is the regeneration completed?	—	Go to Step 11	Go to Step 13
11	1. Observe the parameter of exhaust temperature 1 by using the diagnostic scan tool. Make sure the temperature is less than 100 °C (212°F). If the temperature exceeds 100 degrees Celsius (212 degrees Fahrenheit), increase the speed by 1000 rpm (no load). (Turn the throttle speed control knob clockwise) 2. When the parameter of exhaust temperature 1 reaches 150 °C (302 °F), accelerate the engine to full throttle (full stroke of accelerator pedal) and read the pressure parameter of exhaust differential. Is the exhaust differential pressure parameter below the specified value?	3.3kPa	Go to Step 12	The zoom should finish. Please see the appropriate procedures in the exhaust system section.
12	Does the exhaust differential pressure exceed the specified value?	1.0kPa	Normal system	For the diagnosis of differential pressure that is too low, see Diagnostic Trouble Code P2452

Step	Operation	Value	Yes	No
13	Note: if the diagnostic scan tool displays the following message "A failure that cannot be subjected to mandatory regeneration has been detected", refer to the applicable diagnostic trouble code. 1. Remove the DPD filter from the DPD component. For information on dismantling, please see Section of Exhaust System. Check for the following conditions: •Whether there is any damage, rupture or melting of (DPD) filter? •Whether oxygen catalysts are damaged, cracked, melted or over deposited. • Oxygen catalyst metamorphism 2. Test engine cooling system for overheating. For the test information, refer to engine cooling system diagnostics. 3. Check for fuel system problems. See the fuel system checklist and the appropriate procedure in the Fuel System section. 4. Repair or replace as necessary. Complete the operation or not?	—	Go to Step 14	—
14	Reconnect all previously disconnected components or harness connectors. Complete the operation or not?	—	Go to Step 5	—
15	1. Turn off the ignition. 2. Remove the cover glass completely surrounding the IP cluster and disconnect the DPD switch. 3. Disconnect the DPD switch harness connector. 4. Connect a test lamp between the voltages feed circuit of the DPD switch wiring harness (FU-36 pin 2) and the known good ground. 5 Start the engine. Is the test light bright?	—	Go to Step 16	Go to Step 17
16	When crossing the switch wire connector between the pin 1 and the pin 2 of the FU-36 to have jumper connection with the jumper of 3-A fuse, the fault diagnosis instrument is used to observe the parameters of DPD switch. Does the scan tool indicate "on" when the circuit is bridged and "off" when the circuit is not bridged?	—	Go to Step 19	Go to Step 18
17	Repair open circuit or high resistance between F-11 (7.5A) fuse and switch (FL-36 pin 6). Check F-11 (7.5A) fuse first. Is maintenance completed?	—	Go to Step 23	—
18	1. Test the signal circuit between the engine control module (E-57 pin 47) and the switch (FU-36 pin 2) for the following conditions: •Open circuit • Short circuit of battery or ignition voltage • High resistance 2 Repair the circuit if necessary. Did you identify the problem and correct it?	—	Go to Step 23	Go to Step 20

Step	Operation	Value	Yes	No
19	1. Inspect the harness connector of switch (FU-36 pin 1 and pin 2) for intermittent connection and poor connection. 2. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 23	Go to Step 21
20	1. Check the engine control module (E-57 pin 47) to see whether there is intermittent, bad connection and corrosion in the harness connector. 2. Repair the connection when necessary. Did you identify the problem and correct it?	—	Go to Step 23	Go to Step 22
21	Repair or replace the DPD switch. Is maintenance or replacement completed?	—	Go to Step 23	—
22	Important note: The changes to the engine control module must be programmed and learned. Replace the engine control module. Please refer to engine control module replacement. Whether to complete the replacement?	—	Go to Step 23	—
23	1. Reconnect all previously disconnected components, fuses, or harness connectors. 2 Start the engine. 3. Use the fault diagnosis tester to observe the DPD switch parameters. Is the diagnostic device turned on and off during each switch transition?	—	Go to Step 4	Go to Step 15

Symptoms - Engine Control System

Symptoms - Engine Control System

Important preliminary inspection before starting

Before using these symptom tables, perform a diagnostic system check - engine control system and confirm that the following conditions are true:

- Engine control module and malfunction indicator lamp (MIL) / SVS indicator light is working normally.
- Fault diagnosis instrument data in normal working range. See the data list of diagnostic scan tool in this section.
- Verify the problems reporting for repair by customers and find corresponding symptoms in the following table. Check the item under this symptom.

Visual check and physical examination

Many symptomatic procedures require careful visual and physical examinations, and this step is very important. Save valuable time because visual and physical inspections can correct problems without further inspection Guarantee:

- Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct.
- The vacuum hose is not cracked or kinked and is properly connected. Thoroughly check for any type of leak or restriction.
- The intake pipe is not broken or damaged.
- The exhaust pipe is not broken or damaged.
- The engine wiring harness and terminals are connected correctly and are not pressurized or cut.

Intermittent failure

Important note: If there is an intermittent condition, check that if the electrical components are not installed properly. After sale, check if there are additional electrical equipment, lights and car phone. Ensure there is no

additional after-service equipment connected to CAN or other serial data circuit.

Important note: This failure may or may not turn on the MIL/SVS light or store a diagnostic trouble code. The wrong electrical connection or wiring is the most possible causes for intermittence problem. Visually and physically visually inspect the suspicious connector to see if the following conditions exist:

- The two halves of the connector do not match with each other
- Unfixed terminal
- Damage or misalignment of terminals

Correct or replace the connector terminals in the problem circuit to ensure correct contact voltage. Remove the terminal from the connector body and check the terminal cable connector for damage. Connect the DMM to the doubtful circuit and drive the vehicle on the road. Abnormal reading in case of failure well proves that there is a fault in the circuit being monitored. To help detect intermittent malfunctions, use the diagnostic scan tool. The Tech 2 troubleshooter contains the following features:

- Triggers a snapshot function so that engine parameters can be obtained and stored in the event of a fault. Review the stored information to see the specific operating conditions that caused the failure.
- Freeze Frame/Failure Record can also be used to help determine discontinuity. Audit and obtain information related to the intermittent diagnostic trouble code being diagnosed in the frozen fault state/fault record. Drive the vehicle under the conditions where the diagnostic trouble code was initially set.
- Use the plotting function of the fault diagnosis tester to plot the required data parameters. This stored information is checked to help determine intermittent problems. For more information, see the user guide for fault diagnostics.

The data recording module (DRM) is used to help detect intermittent situations. In the event of a diagnostic trouble code, DRM can store the engine log data. DRM storage can store three log data. If more than three storage areas are set, the oldest log data will be overwritten. However, if the same diagnostic fault code is set up within eight hours, the diagnostic failure code will not be stored in the DRM storage.

When a faulty vehicle performance event occurs (rather than a diagnostic trouble code event), the manual triggering function saves the log data through any operation of the driver. If the driver presses and releases the manual trigger switch once, this is the trigger operation and the log data before and after the trigger will be stored in the DRM memory. If there is space in the DRM store, log data will be stored in this space. However, if more than 3 are stored, the eldest log data will be covered. For more information, see the DRM user guide.

Important note: If there is an intermittent problems during startup, stop and test for diagnostic trouble codes related to the anti-theft system of vehicle. Test whether there is improper installation of electrical components (such as lights, car phones, etc.). Any of the following conditions can cause intermittent MIL / SVS indicators (diagnostic trouble codes are not stored):

- The engine control module is loose or dirty. Please see the schematic diagram of the engine control system.
- There is an intermittent short to ground in the MIL/SVS lamp circuit.
- The electrical system caused by the failure of the relay or the engine control module to drive the solenoid valve or switch can interfere with the electrical components and cause severe surges. Typically, this problem occurs when the faulty component runs
- There is any diode open.

Important note: The following symptom tables contain a series of possible causes of the symptoms. The order of these programs is not important. If the diagnostic program readings do not show any problems, check the simplest logic sequence or understand the most likely first cause. To determine if a particular vehicle uses a particular system or component, see the engine control system schematic for application information. Use the following table to diagnose the following symptoms reported by the customer:

- Discontinuity
- Hard start
- The idling speed is poor, unstable or inaccurate and stall
- High idle speed
- Sudden flameout
- Surge
- Insufficient motivation, slow response or weakness
- Pause, speed decline, rotational speed instability
- Abnormal combustion noise
- Fuel economical efficiency low

Excessive black smoke

- Excessive white smoke
- Frequent manual regeneration of the DPD system
- Long regeneration time of DPD system



Intermittent situation

Inspection	Operation
Definition: This failure has not yet occurred, but the historical diagnostic code indicates that the failure has occurred. or The customer reported the fault for repair, but if the fault is not related to the diagnostic trouble code, the fault symptom cannot be reproduced at present.	
Preliminary inspection	Before you start, please see Symptom - Engine Control System.
Harness connection	When the harness/connector moves, many intermittent open or short circuits are caused by vibration, engine torque and collision/road inequality. Perform appropriate procedures to test for such failures from the following items - Move related connectors and wiring when monitoring appropriate fault diagnostic instrument data. - When parts are turned on or off according to the order, use the fault diagnosis instrument to detect whether the connectors and wiring are moving. Observe the operation of the component. - Let the engine run and monitor the operation of the engine when moving the relevant connector and wiring. If the harness or connector movement affects the displayed data, component/system operation, or engine operation, check and repair the harness/connector as necessary.
Electrical connection or wiring	Poor electrical connection, terminal voltage or wiring problems can cause the most serious intermittent problems. Check whether there are the following cases: - Whether the two parts of the connector are malformed or improper in the installation position of the terminal in the connector body. - Unfit Terminal shape or terminal damage. Check for a poor terminal voltage. - The terminal cable is connected poorly, including curly terminal and terminal outside of insulating layer. This requires removing the terminal from the connector body. - Corrosion / water intrusion. The insulation layer perforation or damage will allow the moisture to enter the wiring and result in the conductor corrosion without any visible evidence. check the swelling and stiffness parts of wires in suspicious circuits - The wires in the insulating layer were broken. - The harness is extruded, cut or rubbed during wiring. - Make sure the wiring is not in contact with the hot exhaust component.
Control module power supply, grounding element power supply and grounding.	Poor power supply or ground connection will cause extensive symptoms. - Test all power circuits of control module. Many vehicles have multiple circuits to power the control module. It may also be necessary to individually test other components in the systems that have separate power circuits. Check the module/component connector, the connector on the fuse, and any intermediate connectors between the power supply and the module/component. The test lamp or DMM can indicate there is voltage or not, but cannot test whether the circuit can bear the sufficient current. Ensure that the circuit can carry the current required for the operation of the components. - Test all control module grounding circuits and system grounding circuits. The control module may have multiple ground circuits. Other components in the system may also have separate ground points that need to be tested. Check that the ground point connector is clean and tight. When appropriate, check the fittings on the assembly and check the fittings in the assembly where appropriate. Ensure that the circuit can carry the current required for the operation of the components.
Inspection	Operation
Temperature sensitivity	- Discontinuity may occur when the temperature of components / joints reach normal operating temperature. This can happen only when the component/connector is cold or only when the component/connector is at a high temperature. - Freezing fault status, fault records or snapshot data can help detect this intermittent condition when appropriate. - If the intermittent condition is related to heat, the relationship between the inspection data and the followings: -High ambient temperature - Heat generated by the hood / engine - Heat generated by the circuit due to poor connection or high electrical load. - Conditions higher than general load conditions, traction, etc. - If intermittent conditions are related to low temperatures, check the following data: -Low ambient temperature - at very low temperature, ice may be formed in joints or components. Check whether there is water vapor entry. - This situation occurs only at cold start. -The situation disappeared after the heating of the vehicle. - Information from customers can help determine whether a failure in a certain mode is related to temperature.

Electromagnetic interference (EMI) and electronic noise	Some electrical components/circuits are very sensitive to electromagnetic interference or other types of electrical noise. Check for the following conditions: • Some incorrect harnesses may be too close to high voltage/high current devices (such as injection elements, motor, generator, etc.). These component circuits may cause electrical noise and interfere with general circuit operation. • The electrical system interference caused by the failure relay or the engine control module to drive the solenoid valve or switch may cause intense surges. In general, this problem occurs when the faulty component is running • Improper installation of accessories not assembled in the factory or additional accessories (such as lights, two-sound-channel radio, loudspeaker, motor, remote starter, alarm system, car phone, etc.). These accessories may generate discharge-related faults when used, but will not fail when not in use. • Test whether the circuit of any diode is open. Some relays may contain clamping diodes. • Test whether there is a bad rectifier bridge which can cause AC noise into the electrical system.
Improper programming of engine control module	• Inadequate programming of engine control modules is rare: - The engine control module from another vehicle is installed. - A revised software / calibration document is issued for the vehicle. Important note: do not use the same software / calibration files already existing in the engine control module to reprogram the engine control module. This is not an effective maintenance for any type of driving performance problem. • Check and verify that the correct software / calibration procedure is contained in the engine control module. If you find improper programming, use the latest software/calibration program to reprogram the engine control module.
Reproduce fault status	• If the previous test is unsuccessful, then try to copy and / or acquire the fault state. • At that time, freezing fault status / fault record data contains conditions when the diagnostic fault code is set. - Audit and record freeze fault status / fault record data. - Drive vehicles in the same condition as close to the frozen failure status / fault record data as possible, and drive the vehicle under the condition of running the fault code. Please refer to Diagnostic Trouble Code Operating Conditions in the supporting text of diagnostic trouble code under diagnosis. • Another way is to connect DMM to suspicious circuits and drive vehicles. Anomalous readings on the DMM when a problem occurs can help identify the problem.
Inspection	Operation
Troubleshooting Snapshot	A fault diagnostic device can be set up to snapshot the available parameters through serial data. The snapshot function will record real-time data for a long time. Recorded data can be replayed and analyzed. The fault diagnosis tester can draw a single data chart, and can also draw combined data charts for comparison. The snapshot function can be triggered manually when symptoms are detected, or set up in advance to trigger this function when the diagnostic fault code is set. The abnormal values obtained from the recorded data can indicate the system or component that needs further investigation. For more information, please refer to the user guide for fault diagnostics.
DRM storage	Use data stored in DRM memory or use manual triggering. In the event of a diagnostic trouble code, DRM can store the engine log data. When a faulty vehicle performance event occurs (rather than a diagnostic trouble code event), the manual triggering function saves the log data through any operation of the driver. If the driver presses and releases the manual trigger switch once, this is the trigger operation and the log data before and after the trigger will be stored in the DRM memory. For more information, see the DRM user guide.

Hard start

Inspection	Operation
Definition: The engine crankshaft rotates normally, but it does not start for a long time. The vehicle can finally run or immediately stalls after starting.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Ensure that the driver's starting program is correct. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check whether the wiring harness connector is correct. • Check fuel type and quality. • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for

	high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged. • Whether CMP sensor is tightened and whether the camshaft gear is not damaged.
Fuel system inspection	2. Check whether the followings appear on the fuel System: See the fuel system checklist and the appropriate procedure in the Fuel System section. • Whether there is air in the fuel system; • Whether there is water pollution in fuel; • Whether there is paraffin or ice in the fuel. • Whether the fuel filter is blocked or not; • Is there any external fuel leakage or high engine oil level? • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; • Whether the fuel line between the fuel tank and the fuel pump is broken, twisted, tight, cracked and blocked? • Whether the fuel tank outlet hose is blocked; • The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. • Fuel supply pumps operation. Note: The fuel supply pump must be synchronized with the engine and the adjusted value must be sent to the engine control module. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the feedback current of the FRP regulator on the fault diagnostic instrument.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or there are holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking.
Inspection	Operation
exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Whether there is a bad cylinder compression; • Whether there is improper mechanical timing (timing gear); • Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn.
Electrical system inspection	Check the engine electrical system for the following conditions. Please see the chapter of the electrical system of the engine. • The operation of the glow plug control system; see the "preheating control system check" in this section. • Whether the starting speed is low; • Whether the battery is weakened.

Poor, unstable, or inaccurate idle and stall

Inspection	Operation
Definition: Rough idling of engine. If it is very serious, the engine or the vehicle will tremble. The speed of the engine may change at idle speed. Both of these faults can be severe enough to stall the engine.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Remove air purifier, check whether it is dirty or check whether air duct is blocked or restricted. Replace if necessary. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check whether the wiring harness connector is correct. • Check fuel type and quality. • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section.

	•Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. •Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe FRP parameter at idling speed in neutral position. After preheating, the fuel distribution pipe pressure should always be higher than 25MPa (3600 psi). • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged. •Whether CMP sensor is tightened and whether the camshaft gear is not damaged.
Fuel system inspection	Check whether the fuel system has the following conditions: see the fuel system checklist and appropriate procedures in the fuel system section. •Whether there is air in the fuel system; • Whether there is water pollution in fuel; •Whether there is paraffin or ice in the fuel. •Whether the fuel filter is blocked or not; • Is there any external fuel leakage or high engine oil level? • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; •Whether the fuel line between the fuel tank and the fuel pump is broken, twisted, tight, cracked and blocked? • Whether the fuel tank outlet hose is blocked; •The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. • Fuel supply pumps operation. Note: The fuel supply pump must be synchronized with the engine and the adjusted value must be sent to the engine control module. •Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed. •Observe the feedback current of the FRP regulator on the fault diagnostic instrument.
Inspection	Operation
Intake system check	Check the air intake system for the following conditions: •Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. •Whether there is a bad cylinder compression; • Whether there is improper mechanical timing (timing gear); •Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn. • Basic engine parts (such as camshaft, cylinder head, piston, etc.) are not used properly.
Other inspection	• The electromagnetic interference (EMI) on the reference circuit may cause engine to stall. The fault diagnosis tester can usually detect electromagnetic interference by monitoring the engine speed. Suddenly increasing the speed, but actually changing the speed of the engine is so small that there is electromagnetic interference. If there is a problem, check the route arrangement of the high voltage element (for example, the fuel injection solenoid valve wiring near the sensor circuit). • Whether the engine is installed wrong; •Whether or not the crankshaft pulley is inappropriate; • Whether the engine and air conditioning compressor are improper; • Output voltage of generator; • Whether the EGR system is operating correctly. See the Check of EGR Control System in this section. •Air conditioner operation.

High idle speed

Inspection	Operation
Definition: The idle speed of engine is greater than normal speed (regardless of engine coolant temperature or DPD system regeneration).	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Check whether the wiring harness connector is correct. • Use the fault diagnosis tester to compare engine speed and tachometer speed on the instrument panel (IP) cluster. • Check the condition of the air conditioning. • Check fuel type and quality. • Check the engine oil level. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe FRP parameter at idling speed in neutral position. After preheating, the fuel distribution pipe pressure should always be higher than 25MPa (3600 psi). • Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation. • Observe speed sensor. In the fully counterclockwise direction, the speed sensor parameter should be less than 0.6 volts. If not, check the high resistance or sensor skews in the low reference circuit. • Observe the PTO remote throttle sensor. The reading of PTO remote throttle sensor should be less than 0.4 volts at the lowest position. If not, check the high resistance or sensor skews in the low reference circuit.
Fuel system inspection	Check whether the fuel system has the following conditions: See the Fuel System section. • Oil spray nozzle. Disassemble the injector and perform a visual inspection. (the mouth of the nozzle may be damaged)

Sudden flameout

Inspection	Operation
Definition: After the engine starts to crank, the engine usually becomes more pronounced as the engine load increases. When idling, low speed or full acceleration occurs due to insufficient fuel, the exhaust gas will continuously crackle, causing the engine to suddenly shut down.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Check whether the wiring harness connector is correct. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Observe whether MAF parameters are skew or MAF sensor is slow. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation. • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged.
Fuel system inspection	2. Check whether the followings appear on the fuel System: See the fuel system checklist and the appropriate procedure in the Fuel System section. • Whether there is air in the fuel system; • Whether there is water pollution in fuel;

	•Whether there is paraffin or ice in the fuel. •Whether the fuel filter is blocked or not; • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; •Whether the fuel line between the fuel tank and the fuel pump is broken, twisted, tight, cracked and blocked? •The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. •Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Intake system check	Check the air intake system for the following conditions: •Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Other inspection	• The electromagnetic interference (EMI) on the reference circuit may cause engine to stall. The fault diagnosis tester can usually detect electromagnetic interference by monitoring the engine speed. Suddenly increasing the speed, but actually changing the speed of the engine is so small that there is electromagnetic interference. If there is a problem, check the route arrangement of the high voltage element (for example, the fuel injection solenoid valve wiring near the sensor circuit).

Surge

Inspection	Operation
Definition: The engine power will change when the throttle is stable or when cruising. When the accelerator pedal position does not change, it feels that the speed will rise or fall.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Make sure the driver understands the operation of the air conditioning compressor. • Use the fault diagnosis instrument to ensure that the vehicle speed parameter reading matches the vehicle speedometer readings. •Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. •Check whether the wiring harness connector is correct. •Check fuel type and quality. •Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. •Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Observe whether MAF parameters are skew or MAF sensor is slow. •Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). •Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation.
Fuel system inspection	2. Check whether the followings appear on the fuel System: See the fuel system checklist and the appropriate procedure in the Fuel System section. •Whether there is air in the fuel system; • Whether there is water pollution in fuel; •Whether there is paraffin or ice in the fuel. •Whether the fuel filter is blocked or not; • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; •Whether the fuel line between the fuel tank and the fuel pump is broken, twisted, tight, cracked and blocked? • Whether the fuel tank outlet hose is blocked; •The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. • Fuel supply pumps operation. Note: The fuel supply pump must be synchronized with the engine and the adjusted value must be sent to

	the engine control module. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. • Whether the control solenoid valve of turbocharger nozzle is jammed; see turbocharger in the engine machinery chapter.
Inspection	Operation
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Other inspection	• Check whether the EGR system is running correctly. See the Check of EGR Control System in this section. • Check the condition of the air conditioning. • Check whether there is any deformation tire that may cause surge in the fixed vehicle speed range.

Underpowered, response lag or powerless response

Inspection	Operation
Definition: Engine power below expected value. When you do not fully depress the accelerator pedal, there is little or no acceleration.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Compare vehicles with similar devices. Ensure that there are practical problems with the vehicle. • Remove air purifier, check whether it is dirty or check whether air duct is blocked or restricted. Replace if necessary. • Whether to change the tire size? • Is the vehicle overloaded? • Check whether the clutch is slipping. • Check the brake resistance. • Check whether gearshift shift mode and speed reduction operation are appropriate. • Check fuel quality (sixteen alkanet index) • Check the oil level and quality of engine oil. • Use the fault diagnosis instrument to ensure that the vehicle speed parameter reading matches the vehicle speedometer readings. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation. • Observe the boost pressure and atmospheric pressure (BARO) when the ignition device is on and the engine is shut down. These two parameters should be 7kPa (1 psi).
Inspection	Operation
Fuel system inspection	Check whether the fuel system has the following conditions: see the fuel system checklist and appropriate procedures in the fuel system section. • Whether there is air in the fuel system; • Whether there is water pollution in fuel; • Whether there is paraffin or ice in the fuel. • Whether the fuel filter is blocked or not;

	• Is there any external fuel leakage or high engine oil level? • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; • Whether the fuel line between the fuel tank and the fuel supply pump is broken, twisted, tight, cracked and blocked? • Whether the fuel tank outlet hose is blocked; • The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. • Fuel supply pump operation. Note: the fuel supply pump must be synchronized with the engine, and the adjustment value must be sent to the engine control module. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter. • Whether the control solenoid valve of turbocharger nozzle is jammed; see turbocharger in the engine machinery chapter.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Whether there is a bad cylinder compression; • Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn.
Other inspection	• Check whether the EGR system is running correctly. See the Check of EGR Control System in this section. • Check whether the engine is overheated. See Engine Cooling section. • Check the condition of the air conditioning.

Suspension, speed drop, speed instability

Inspection	Operation
Definition: Nothing happens when the accelerator pedal is depressed. This fault may occur at any vehicle speed. This failure is usually more pronounced when it is started for the first time after parking. If this fault is severe enough, it will cause the engine to stall.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Compare vehicles with similar devices. Ensure that there are practical problems with the vehicle. • Remove air purifier, check whether it is dirty or check whether air duct is blocked or restricted. Replace if necessary. • Check whether gearshift shift mode and speed reduction operation are appropriate. • Check fuel quality (sixteen alkanet index) • Check the oil level and quality of engine oil. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Observe whether MAF parameters are skew or MAF sensor is slow. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full

	travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation. • Observe the boost pressure and atmospheric pressure (BARO) when the ignition device is on and the engine is shut down. These two parameters should be 7kPa (1 psi). • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged.
Inspection	Operation
Fuel system inspection	2. Check whether the followings appear on the fuel System: See the fuel system checklist and the appropriate procedure in the Fuel System section. • Whether there is air in the fuel system; • Whether there is water pollution in fuel; • Whether there is paraffin or ice in the fuel. • Whether the fuel filter is blocked or not; • Is there any external fuel leakage or high engine oil level? • If the fuel leaks from the fuel pressure limiter valve and the fuel injectors; • Whether the fuel line between the fuel tank and the fuel pump is broken, twisted, tight, cracked and blocked? • Whether the fuel tank outlet hose is blocked; • The presence of any foreign matter in the fuel tank may cause a blockage in the sensor of the fuel delivery line. • Fuel supply pumps operation. Note: The fuel supply pump must be synchronized with the engine and the adjusted value must be sent to the engine control module. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter. • Whether the control solenoid valve of turbocharger nozzle is jammed; see turbocharger in the engine machinery chapter.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Is there a cylinder compression failure? • Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn.
Other inspection	• Check whether the EGR system is running correctly. See the Check of EGR Control System in this section. • Check whether the engine is overheated. See Engine Cooling section. • Check the condition of the air conditioning.

Abnormal combustion noise

Inspection	Operation
Definition: Mild to severe snoring, usually louder when accelerating. The engine emits a sharp metal knock that changes as the throttle opens.	
Preliminary inspection	• Diagnostic System Check - Engine Control System • Ensure that there are practical problems with the vehicle. • Check whether there is smoke related to combustion noise. • Check fuel quality (sixteen alkanet index) • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high

	resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged.
Fuel system inspection	• If there is too much smoke, check whether the open nozzle is stuck. Check if fuel leaks into the combustion chamber. • Check the fuel injector, remove the nozzle and conduct visual inspection. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis device to drive the injector nozzle. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Whether there is a bad cylinder compression; • Whether basic engine parts (such as camshaft, cylinder head, piston, etc.) are used properly. • Check whether there is too much fuel entering the combustion chamber.
Other inspection	• Check other possible causes such as unsightly install parts and brackets and unstable clutch buffer springs.

Fuel economical efficiency low

Inspection	Operation
Definition: The measured fuel economy during the actual road test was significantly lower than expected. In addition, fuel economy is also significantly lower than the value shown in the actual road test of the vehicle.	
Preliminary inspection	• Remove air purifier, check whether it is dirty or check whether air duct is blocked or restricted. Replace if necessary. • Check the driver's driving habits. • Has the air conditioning or defroster mode been always "on"? • Is the tire pressure correct? • Whether to change the tire size? • Is the vehicle overloaded? • Whether the acceleration is too fast and too frequent? • Check whether the clutch is slipping. • Check the brake resistance. • Check the tension of the transmission belt. • Check whether gearshift shift mode and speed reduction operation are more appropriate. • Check fuel quality (sixteen alkanet index) • Check the oil level and quality of engine oil. • It is recommended that the owner fill the fuel pipe and re check the fuel economy. • Check whether the odometer is running correctly. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on.
Fuel system inspection	Check whether the fuel system has the following conditions: See the Fuel System section. • Fuel type and quality. • Check if the fuel is leaking.
Check the cooling system	Check that the cooling system meets the following conditions, see chapter Cooling system. • Engine coolant grade. • Whether the engine thermostat is always turned on or in the wrong heat range? • Whether the engine cooling fan is always on.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited;

	• Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter. • Whether the control solenoid valve of turbocharger nozzle is jammed; see turbocharger in the engine machinery chapter.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.
Inspection	Operation
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Whether there is a bad cylinder compression; • Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn.

Excessive black smoke

Inspection	Operation
Definition: White smoke on load, idle or hot or cold start.	
Preliminary inspection	• Ensure that there are practical problems with the vehicle. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Remove air purifier, check whether it is dirty or check whether air duct is blocked or restricted. Replace if necessary. • Check fuel quality (sixteen alkanet index) • Check the oil level and quality of engine oil. • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (9°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Observe whether the MAF parameters are inclined to the sensor or slow down the MAF sensor. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Observe the boost pressure and atmospheric pressure (BARO) when the ignition device is on and the engine is shut down. These two parameters should be 7kPa (1 psi).
Fuel system inspection	Check the fuel system for the following conditions. Please refer to section Fuel system. • Oil spray nozzle. Disassemble the injector and perform a visual inspection. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis instrument for pre injection control. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. • Check whether the air mass flow sensor is restricted or damaged. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter.
Exhaust system check	Check the exhaust system for restriction. See the section of the exhaust system, please.

Inspection	Operation
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Check whether there is poor performance on cylinder compression; • Whether there is improper mechanical timing (timing gear); • Whether the valve clearance is not appropriate; • The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn. • Whether there is excess fuel into the combustor.
Other inspection	• Whether the EGR system is operating correctly. See the Check of EGR Control System in this section. • Whether there is too much air leakage.

Excessive white smoke

Inspection	Operation
Definition: White smoke on load, idle or hot or cold start.	
Preliminary inspection	• Ensure that there are practical problems with the vehicle. • Check whether the grounding of the engine control module is clean, fastened, and whether the location is correct. • Check fuel quality (sixteen alkanet index) • Check the programming nozzle ID code for each cylinder. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices".
Sensor check	Check whether the engine control sensor has the following conditions. See the data list of diagnostic scan tool in this section. • Compare engine coolant temperature and intake temperature (IAT) and fuel temperature (FT) on a cooler. If the difference between the refrigerator temperature readings exceeds 5°C (-12.78°F), check each circuit for high resistance or sensor skew. Note: The air mass flow (MAF) sensor is heated, causing the IAT to be higher than normal intake air temperature if the ignition switch is turned on. • Let the engine shut down and observe the parameters of FRP sensor. After the engine has been stopped for at least 1 minute, the FRP sensor reading should be between 0.9 and 1.0 volts after the key is in the open position and the engine is off. If not, check each circuit for high resistance or sensor skew. • Observe the FRP and the required FRP parameters between the idle and throttle opening (W.O.T.) (full travel of the throttle pedal). The parameter of the fuel rail should be within the range of $\pm 5\text{MPa}$ ($\pm 725\text{ psi}$). • Observe the accelerator pedal position (APP). The accelerator pedal location parameter should be modified from 0% to 100% linearly according to the accelerator pedal operation. • Observe the boost pressure and atmospheric pressure (BARO) when the ignition device is on and the engine is shut down. These two parameters should be 7kPa (1 psi). • Whether CKP sensor is tight and whether the rotor of the sensor is not damaged.
Fuel system inspection	• If there is too much smoke, check whether the open nozzle is stuck. Check if fuel leaks into the combustion chamber. • Oil spray nozzle. Disassemble the injector and perform a visual inspection. • Use the fault diagnostic apparatus to perform cylinder balancing test. • Use the fault diagnosis instrument for pre injection control. • Observe the fuel compensation of each cylinder on the fault diagnosis tester at idle speed.
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. • Check whether the air mass flow sensor is restricted or damaged. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Whether the turbocharger is leaking. Please refer to "turbocharger inspection" in engine machinery chapter.
Exhaust system check	If there is excessive white smoke during regeneration of DPD filter, whether oxygen catalyst is likely to deteriorate or damage.

Inspection	Operation
Check the Engine Mechanical	Check the engine mechanical system for the following conditions. Please see the section of the engine. • Whether there is a bad cylinder compression; • Whether there is improper mechanical timing (timing gear); • Whether the valve clearance is not appropriate;

	• The valve spring is broken or lack of spring force; • Whether the convex corners of camshaft are worn. • Whether thermostat works (adhesive, cannot shut down) • Whether there is excess fuel into the combustor.
Electrical system inspection	• Operation of glow plug control (warming up) system; refer to “check of preheating control system” in this section.

Frequent manual reactivation of DPD system

Inspection	Operation
Definition: Compared to before, manual regeneration is frequent carried out. The driver can be regenerated manually by flashing the DPD indicator frequently or frequently (automatic regeneration cannot be done often).	
Preliminary inspection	• Check the quality of fuel injection. • Check the oil level and quality of engine oil. • Check whether the odometer is running correctly. • If excessive black smoke appears, please refer to the “excessive black smoke” table in the symptom diagnosis table. • Whether excess ashes are accumulated in the DPD filter. • Check the data list of the fault diagnosis instrument in this section. • Inspection of “service notices”. Check that the owner’s driving habits/conditions meet the following conditions: It is recommended that owners use temporary regeneration when necessary. Under normal operation: • Is idle time too long? • Whether to drive cars at low speed? • Whether to drive on a long downhill road? • Whether it is a small cargo delivery car or a garbage truck? • Whether the running time of PTO is too long? Under the regeneration conditions: • Is idle time too long? • Whether to drive cars at low speed? • Is PTO running too often/time is too long? • Is the engine running time very short? • Is the vehicle running time very short? • Whether to step down the accelerator pedal frequently?
Sensor check	Check that the engine control sensor associated with the DPD system control meets the following conditions. See the data list of diagnostic scan tool in this section. • Whether the exhaust differential pressure hose or pipe is incorrectly routed or disconnected. Note: the exhaust differential pressure hose has the directivity of installation. For proper installation, see section Exhaust system. • Is the exhaust differential pressure hose or pipe blocked, broken or twisted? • Whether the exhaust differential pressure sensor is misplaced or tilted. For proper installation, see section Exhaust system. • If the exhaust differential pressure sensor is physically damaged. • Whether there is any contaminant or object blocking the inlet of the differential pressure sensor? • Observe whether the exhaust differential pressure parameters have sensor bias or whether the exhaust differential pressure sensor is slow. • Is the exhaust temperature sensor skewed or slow Use the temperature and resistance meter to test the exhaust temperature sensor at all temperature levels in order to assess the possibility of sensor deflection. • Observe whether the MAF parameters are inclined to the sensor or slow down the MAF sensor. • Observe the boost pressure and atmospheric pressure (BARO) when the ignition device is on and the engine is shut down. These two parameters should be 7kPa (1 psi).
Fuel system inspection	Detect the problem of fuel system. See the fuel system checklist and the appropriate procedure in the Fuel System section.
Inspection	Operation
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited; • Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. • Whether the intake system has been modified. • Whether MAF sensors are limited or damaged, or whether there are any contaminants or objects clogging.

	Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter.
Exhaust system check	Check the exhaust system for the following conditions: refer to the chapter of exhaust system. • Whether there is any missing or damaged gasket for exhaust system? • Exhaust leaks from exhaust piping, gaskets, DPD components, exhaust temperature sensors or exhaust differential pressure hoses or pipes • Has the exhaust system been modified? • Check whether the exhaust throttle valve or exhaust brake valve is stuck. See the Exhaust System chapter for diagnostic information. • Whether excess ashes are accumulated in the DPD filter. • Whether the DPD filter is damaged, cracked, melted or excessively deposited. • Whether oxygen catalysts are damaged, cracked, melted or over deposited. • If there is excessive white smoke during regeneration, the oxygen catalyst may deteriorate or be damaged.

DPD system has long regenerative time

Inspection	Operation
Definition: The reactivation time is longer than the expected time or than the previous.	
Preliminary inspection	• Check the quality of fuel injection. • Check the data list of the fault diagnosis instrument in this section. • Inspection of "service notices". • Whether to misunderstand engine preheating control as DPD regeneration? Check that the owner's driving habits/conditions meet the following conditions: Under the regeneration conditions: • Whether manual regeneration starts from cold engines? • Is idle time too long? • Whether to drive cars at low speed? • Is PTO running too often/time is too long? • Is the engine running time very short? • Is the vehicle running time very short? • Whether to step down the accelerator pedal frequently? • Whether the ambient temperature is too high?
Sensor check	Check that the engine control sensor associated with the DPD system control meets the following conditions. See the data list of diagnostic scan tool in this section. • Whether the exhaust differential pressure hose or pipe is incorrectly routed or disconnected. Note: the exhaust differential pressure hose has the directivity of installation. For proper installation, see section Exhaust system. • Is the exhaust differential pressure hose or pipe blocked, broken or twisted? • Whether the exhaust differential pressure sensor is misplaced or tilted. For proper installation, see section Exhaust system. • If the exhaust differential pressure sensor is physically damaged. • Whether there is any contaminant or object blocking the inlet of the differential pressure sensor? • Observe whether the exhaust differential pressure parameters have sensor bias or whether the exhaust differential pressure sensor is slow. • Is the exhaust temperature sensor skewed or slow Use the temperature and resistance meter to test the exhaust temperature sensor at all temperature levels in order to assess the possibility of sensor deflection.
Fuel system inspection	Detect the problem of fuel system. See the fuel system checklist and the appropriate procedure in the Fuel System section.
Check the cooling system	Check if the cooling system is overheated. For test information, please refer to chapter Cooling system. • Engine coolant grade. • Whether or not the engine coolant is leaking. • Whether the cooling fan belt is slid. • Whether the cooling fan clutch works. • Whether the thermostat works. • Whether the pump works. • Whether radiator is clogged or not. • Has the cooling system with less cooling capacity changed?
Inspection	Operation
Intake system check	Check the air intake system for the following conditions: • Check whether air purifiers, intake pipes and pressurized air coolers are restricted, or have holes or leaks. • Whether the intake pipe of the turbocharger is limited;

	• Whether the intake valve is stuck; • Whether the intake manifold is restricted or leaking. • Whether the intake system has been modified. • Whether MAF sensors are limited or damaged, or whether there are any contaminants or objects clogging. Whether there is any wear or damage to turbocharger turbine wheel, shaft or compressor runner. Please refer to "turbocharger inspection" in engine machinery chapter.
Exhaust system check	Check the exhaust system for the following conditions: refer to the chapter of exhaust system. • Whether there is any missing or damaged gasket for exhaust system? • Exhaust leaks from exhaust pipe, gaskets, DPD components, exhaust temperature sensors or exhaust differential pressure hoses or pipes • Has the exhaust system been modified? • Check whether the exhaust throttle valve or exhaust brake valve is stuck. See the Exhaust System chapter for diagnostic information. • Whether the DPD filter is damaged, cracked, melted or excessively deposited. • Whether oxygen catalysts are damaged, cracked, melted or over deposited. • If there is excessive white smoke during regeneration, the oxygen catalyst may deteriorate or be damaged.
Other inspection	• Check whether the engine is overheated. See Engine Cooling section. • Check the condition of the air conditioning.

Maintenance guide

Replacement of ECM

Explanation:

The following steps A to E describe the general steps of engine control module replacement and reprogramming. This section further introduces Step A to Step E.

A. Manually record the injector identifier of the old engine control module.

B. Use the new engine control module to replace the old engine control module.

C. Use SPS to program the latest software and calibration procedures into the new engine control module.

D. Use the programming function of the fault diagnosis instrument to program the recorded fuel injector ID code and the vehicle identification code (VIN) into the engine control module.

E. Keep the engine run at idle speed under the condition of parking or neutral state, so that the fuel pump can re-understand the procedure again until the normal working temperature is reached.

A. Record the syringe ID code

The injection nozzles are expressed by hexadecimal characters (0-9 or A-F), which must be programmed to ECM so as to enable the designated cylinder to feed fuel to the engine. These characters are available at any of the following locations:

Get injector ID code data from engine control module

The ID code data of the fuel injector can be determined by using the fault diagnosis instrument. If there is no communication between the engine control module and the fault diagnosis tester, go to the next step.

1. Install the fault diagnostic meter
2. Turn on the ignition and turn off the engine.
3. Select diagnosis > appropriate vehicle identification number > 6HK1 > additional function > display nozzle code.
4. Record the 24-bite ID code of each injection nozzle.
5. After the recording is completed, close the diagnostic tool.
6. Turn off the ignition.

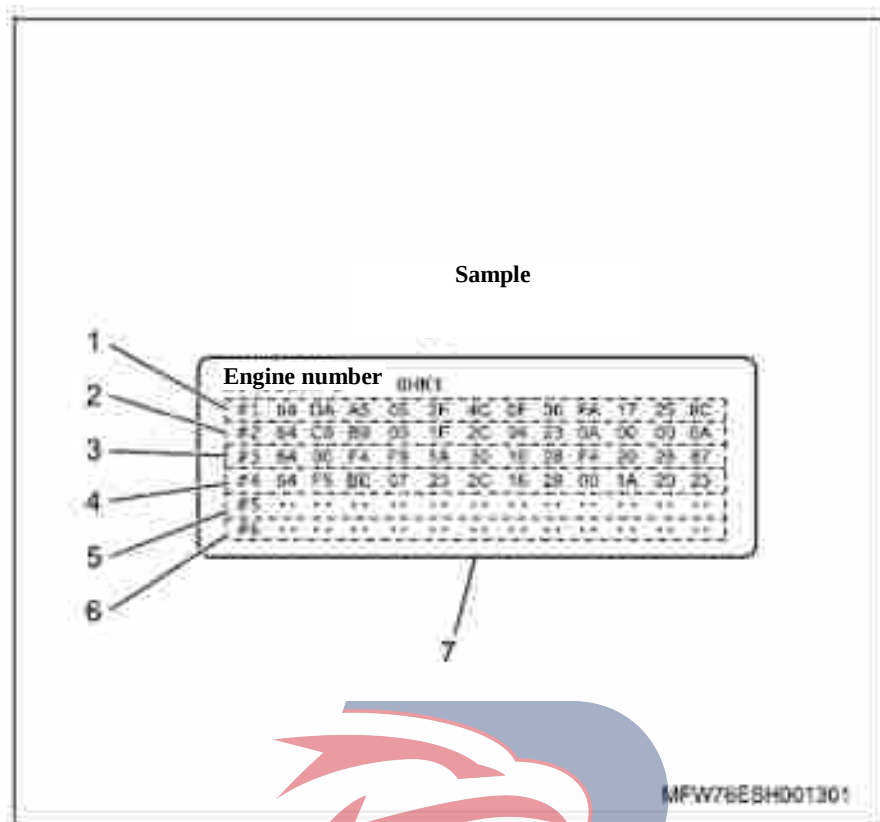
Use non-communication engine control module to obtain injector ID code data

If the fault diagnosis tester does not communicate, you must record the ID number of the injection nozzle from the factory-applied label on the cylinder head or from each injection nozzle's signboard.

Recorded on the cylinder head on the label.

Note: only when the nozzle has not been replaced before, do this step.

1. Record all the numbers for each cylinder on the label.

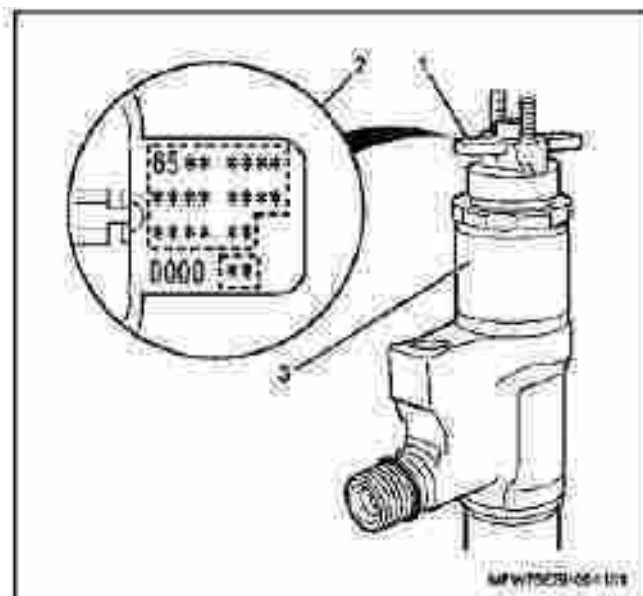


Key

1. No. 1 cylinder injection nozzle ID code
2. No. 2 cylinder injection nozzle ID code
3. No. 3 cylinder injection nozzle ID code
4. No. 4 cylinder injection nozzle ID code
5. No. 5 cylinder injection nozzle ID code
6. No. 6 cylinder injection nozzle ID code
7. Nozzle ID code label

Record from each injector

1. Remove the cylinder head.
2. Record 24 numbers on each syringe mark. The correct injector ID code sequence in the figure below is:
65 ** ** ** **

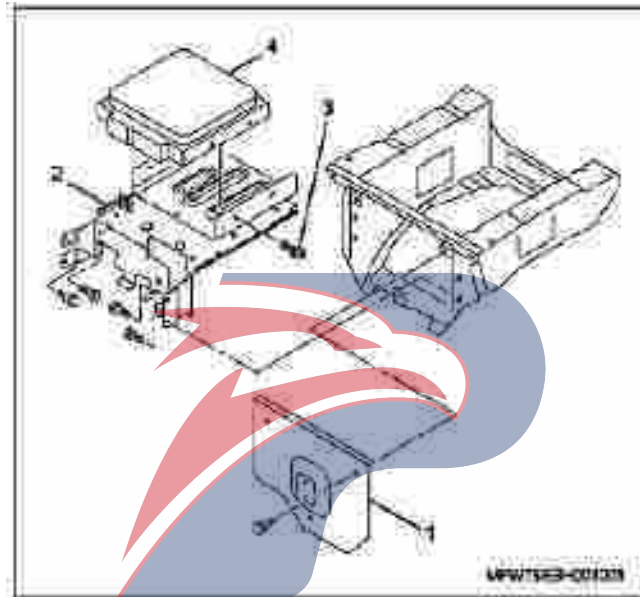


Key

1. Syringe nozzle sign
2. Nozzle ID code
3. Nozzle

B. Disassembly and installation**Disassembly procedure**

1. Disconnect the negative battery cable;
2. Remove the cover (1);
3. Disconnect the engine control module harness connector.
4. Remove the engine control module and bracket (2);
5. Loosen the screw (3) and remove the engine control module (4).

**Installation procedure:**

1. Install the engine control module (4) to the bracket (2) and tighten the fastening screw of engine control module (3);
2. Install the ECM and bracket (2);
3. Connect the engine control module harness connector;
4. Install the cover (1);
5. Connect the negative battery cable.

C. Programming software and proofreading program

Programmer latest software / proof reading program (if issued). See the SPS description and SPS (Remote Program) or SPS (Straight through Procedure) in this section.

D. Injector ID and VIN Codes

1. Install the fault diagnostic meter
2. Turn on the ignition and turn off the engine.
3. Select Diagnostics> Appropriate Vehicle ID> 6HK1> Programming> Replace Fuel Injector.
4. In order to obtain programming permission, the user will display information on the screen. Use the following procedure to obtain program approval from TIS 2000.
 - a. Connect the diagnostic tool to the terminal with TIS 2000 installed, and insert the latest Software and hardware key into the port;
 - b. Operate the fault diagnostic instrument and keep it in the title screen.
 - c. Open the TIS application;
 - d. Select "Security Access" on the main screen.
 - e. In the "diagnostic tool selection" screen, highlight "Tech 2" and click "Next";
 - f. Click "Close" on the "Security Access Enabled" screen.

- G. Close the diagnostic tool;
- h. Disconnect the scan tool from the terminal.
5. Install the fault diagnostic meter to the vehicle.
6. Turn on the ignition and turn off the engine.
7. Select Diagnostics> Appropriate Vehicle ID> 6HK1> Programming> Replace Fuel Injector.
8. Enter the 24-digit ID code for each injector.
9. Select VIN programming. If programming was performed in the SPS introduced earlier, check and verify the VIN on the screen. If it is not programmed or VIN is incorrect, enter the correct VIN.
10. After completing the programming, turn off the ignition switch for 30 seconds.
11. Start the engine and make it run at idle speed.
12. Check that the engine is operating correctly and there is no diagnostic trouble code. Please refer to diagnostic system check - engine control system if necessary.

E. Re-understand the oil supply pump

1. Install the fault diagnostic meter
2. Start the engine and make it run at idle speed until the engine coolant temperature is 65 °C (149°F) or higher, and the state parameters of the supply pump are observed by the fault diagnosis instrument. The fault diagnosis tester parameters change the state from "Not Learned" to "Learned."

SPS Description

The SPS allows technicians to program the control module through DLC. The information transmission circuit and the fault diagnosis instrument used by the DLC are the same as the serial data circuit used to obtain the diagnostic trouble code, display data, and clear fault code. The program can install software / proofreading programs that match a specific vehicle. Most control modules have two types of memories. Software/proofreading programs are stored in flash memory. The following are two types of storage:

- Electrically Erasable Programmable Read-Only Memory (EEPROM)

This type of memory allows programming of selected portions of the memory while the other portions remain unchanged.

The specific known data is stored in the EEPROM, for example:

- Vehicle identification code (VIN)
- Software / proofreading recognition code
- Control module security information
- Flash only memory - flash memory

Flash has more memory than before. During programming, all the information in this type of memory will be deleted and completely replaced by new information.

Maintenance programming method

The two methods of programming the engine control module are as follows:

- Remote programming
- Direct-connection program

For information about using any of these methods to program the engine control module, see SPS (remote program) or SPS (direct connection program).

• Before programming the control module

Important Note: Do not use a similar software/proof package to program the existing engine control module. This step is not a shortcut to correct driving performance. This is an invalid maintenance method. ECM programming is allowed only when the following conditions occur:

- When the maintenance procedure points out that the engine control module needs to be replaced.
- When the updated software / proofreading program is issued.

Before programming the engine control module, make sure that the following conditions are met:

- The fault diagnosis tester PCMCIA card is programmed with the latest software.
- The latest software is installed for TIS 2000.
- The hardware key is plugged into the computer port.
- Vehicle system voltage:

-There is no worrying problem in the charging system. All faulty charging system items must be repaired before programming the engine control module.

-The battery voltage is greater than 24 volts less than 30 volts. The battery must be fully charged before programming the engine control module.

The battery charger is not connected to the vehicle battery. Incorrect system voltage or battery charger voltage fluctuations may cause programming failure or damage to the engine control module.

- Turn off or disable any system that may impose load on vehicle batteries. Turn off or disable systems such as the following:

◇ Heating, ventilation, and air conditioning (HVAC) system

◇ Headlights

◇ Interior lamp

◇ Auxiliary equipment

• The ignition switch is in the proper position. The fault diagnosis tester prompts you to turn on the ignition device and keep the engine off. Do not modify the ignition switch position while programming unless otherwise directed.

• All tool joints are fastened:

- RS-232 cable

- DLC joint

Power supply voltage circuit

• Do not interfere with the tool harness during the programming process. If disturbed, programming may fail or the engine control module may be damaged.

• If you use a laptop without a power cord to perform a through programming procedure, make sure that the internal battery is fully charged.

SPS (remote program)

Note: Some modules will not accept SPS remote programs using 10MB PCMCIA card. If so, use 32MB PCMCIA card or SPS direct connection program. The remote SPS method includes 3 steps as follows:

1. Connect the fault diagnosis device to the vehicle and get information from the engine control module.
2. Connect the fault diagnostic instrument to the terminal and download the new confirmed file from the terminal to the fault diagnosis instrument storage.
3. Reconnect the scan tool to the vehicle and upload the new proofreader file to the engine control module.

Execute the remote program

1. Connect the fault diagnosis device to the vehicle and use the following procedure to get the engine control module information:

Note: Before programming, ensure that the engine control module is installed in the vehicle and the battery is fully charged.

a. Install the scan tool.

b. Turn on the ignition device and keep the engine off.

c. Select SPS > request information.

d. If the data stored in the fault diagnostic instrument is available, the existing data will be displayed on the screen. The scan tool will ask the user whether to "keep data" or "continue" or not in order to request new vehicle information from the engine control module. If there is no data in the scan tool, vehicle identification will begin immediately.

e. Select the vehicle instructions according to the VIN on the stamping or the VIN card attached to the vehicle and the instructions on the screen.

f. While getting information, the fault diagnosis instrument is receiving information from all modules. However, only the engine control module information is displayed on the screen.

g. Close all the attachments and press the "OK" key.

h. Check whether the correct VIN is displayed on the fault diagnostic device. If VIN is incorrect or there is no VIN, record the correct VIN.

2. Turn off the ignition.

3. Turn off the fault diagnosis tester and disconnect it from the vehicle.

4. Use the following procedure to send data from the terminal to the fault diagnosis tester.

Note: TIS only supports service programming using Tech 2 diagnostic scan tool.

- a. Connect the fault diagnosis instrument to the terminal.
- b. Opens the TIS application;
- c. On the home screen, select Service Programming System.
- d. Highlight the following information on the "Select Diagnostic Tool" and the Programming Process screen, and then clicks on "Next".

- Selection of diagnostic tools - Tech 2 (Select Diagnostic Tool- Tech 2)
- Select Programming Process) - determine whether or not the engine control module is being reprogrammed or whether the module is replaced with a new engine control module
- Select ECU location - vehicle (Select ECU Location - Vehicle)

E. Verify "Preparing for Communication" screen, then click " (Next)".

f. Verify the vehicle identification number (VIN) on the Verify Vehicle Identification Number screen and click next.

Note: if the engine control module is replaced by the new module, the vehicle identification number will not be displayed. Enter the correct VIN reading based on the VIN or attached VIN label on the vehicle. If you have an engine control module for another vehicle, enter the correct VIN in the same way.

g. If the instructions are displayed on the screen, highlight the " (Engine)" on the screen of the "Select System Type".

h. Complete the following information according to the maintenance sign on the "Validate Vehicle Data" screen until the "(Next)" is highlighted, and then "Next" is clicked.

- Model
- Model year
- Engine type
- Model mark
- Destination code
- Type of supply

I. Verify your choice on the "Summary" screen.

Note: if notices and proofreading documents are listed, please refer to the service notice and instructions before making maintenance programming.

Note: If the received message indicates that the selected proofreading program is exactly the current proofreading program in the engine control module, the "Cancel" is selected and the same download cannot be used for reprogramming.

j. Click " Report".

k. The transfer data screen does not appear until the progress bar reaches 100%.

5. Close the application and return to the TIS Application Selection screen after downloading.

6. Shut down the fault diagnosis instrument and disconnect from the terminal.

7. Transmit the data from the fault diagnostic instrument to the engine control module by using the following procedures.

a. Install the scan tool.

B. Turn on the ignition device and keep the engine off.

c. Select Service Programming System (SPS) > Programming ECU.

d. Close all attachments and press the "Continue" key.

e. "Programming" will appear before the progress bar reaches 100%.

Note: Some warning indicators may illuminate when programming the engine control module because the communication between the engine control module and other modules is interrupted. Clear the diagnostic trouble code in any module after programming.

f. After the fault diagnostic instrument displays "Programming Was Successful", press the "Continue" and exit the program.

8. Turn off the ignition.

9. Turn off the fault diagnosis tester and disconnect it from the vehicle.

SPS (Direct connection procedure)

Through-programming allows the fault diagnosis tester to remain connected to the terminals and to the vehicle throughout the programming process. When using delivery procedures, the vehicle must be close to the terminal.

Note: TIS only supports service programming using Tech 2 diagnostic scan tool.

1. Open the TIS application;
2. Select Service Programming System on the main screen.
3. Highlight the following information on the "Select Diagnostic Tool" and the Programming Process screen, and then clicks on "Next".

- Select "Fault diagnosis tester-Select Pass - (Diagnostic Tool-Select Pass - Thru)"
- Select the programming process - determine whether or not the engine control module is being reprogrammed or whether the module is replaced with a new engine control module
- Select ECU location - vehicle (Select ECU Location - Vehicle)

4. Complete all vehicle data on the "Prepare Communication/Determined Vehicle" screen until "Next" is highlighted, then click "Next".

5. Follow the instructions on the "Preparing Communications" screen and click Next.

Note: To reduce potential signal loss, the RS-232 cable must not exceed 25 feet in length.

6. Verify the vehicle identification number (VIN) on the Verify Vehicle Identification Number screen and click next.

Note: If the engine control module is replaced by the new module, the vehicle identification number will not be displayed. Enter the correct VIN reading based on the VIN or attached VIN label on the vehicle. If you have an engine control module for another vehicle, enter the correct VIN in the same way.

7. If the indication is displayed on the screen, highlight "Engine" on the "Select System Type" screen.

8. Fill in the following message according to the maintenance flag on the "Verify Vehicle Data" screen until "Next" is highlighted, then click "Next".

- Model
- Model year
- Engine type
- Model mark
- Destination code
- Type of supply

9. Verify your choice on the "Summary" screen.

Note: if notices and proofreading documents are listed, please refer to the service notice and instructions before making maintenance programming.

Note: If the received message indicates that the selected proofreading program is exactly the current proofreading program in the engine control module, the "Cancel" is selected and the same download cannot be used for reprogramming.

10. Click on "Report".

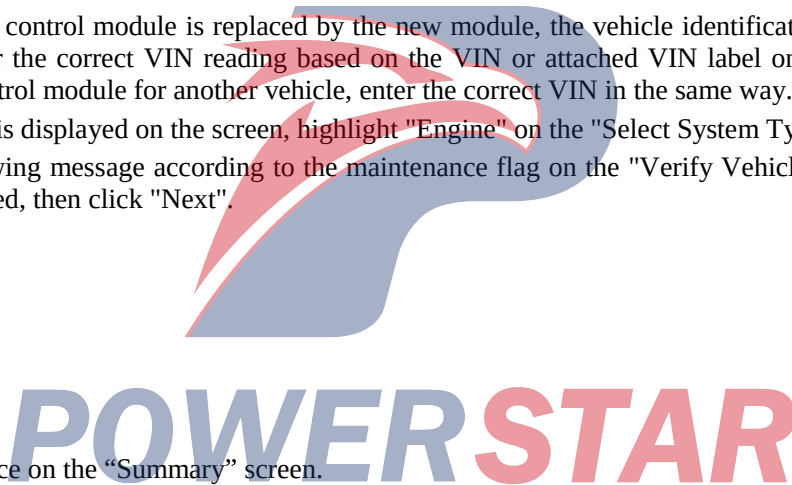
11. The transfer data screen will not appear until the progress bar reaches 100%.

Note: Some warning indicators may illuminate when programming the engine control module because the communication between the engine control module and other modules is interrupted. Clear the diagnostic trouble code in any module after programming.

12. Close the application and return to the TIS Application Selection screen after downloading.

13. Turn off the ignition.

14. Turn off the fault diagnosis tester and disconnect it from the vehicle.



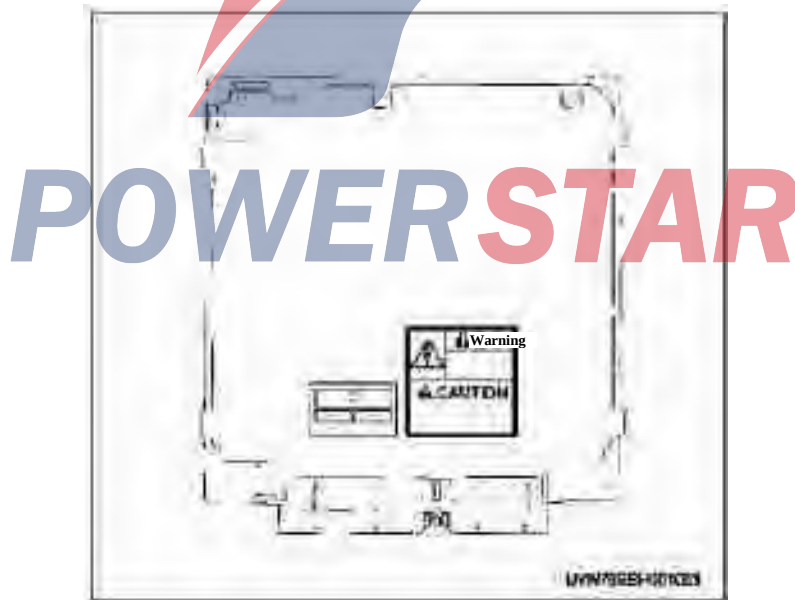
Description and use

Engine Control Module (ECM) Description

Maintenance and prevention of engine control module (ECM)



Important: The "!" symbol warns you of the danger of electric shock. To avoid electric shock and possible serious injury, do not touch the terminals. When disconnecting the harness connector, always turn off the ignition switch or disconnect the battery cable.



ECM is designed to withstand the normal current consumption relevant to the vehicle driving. Please avoid overloading any circuit. When there is an open circuit or a short circuit in the test, do not connect to the ground or apply any voltage to any engine control module circuit unless expressly stated that it is necessary to do so. In some cases, these circuits can only be tested with DMM. The engine control module should remain connected to the engine control harness.

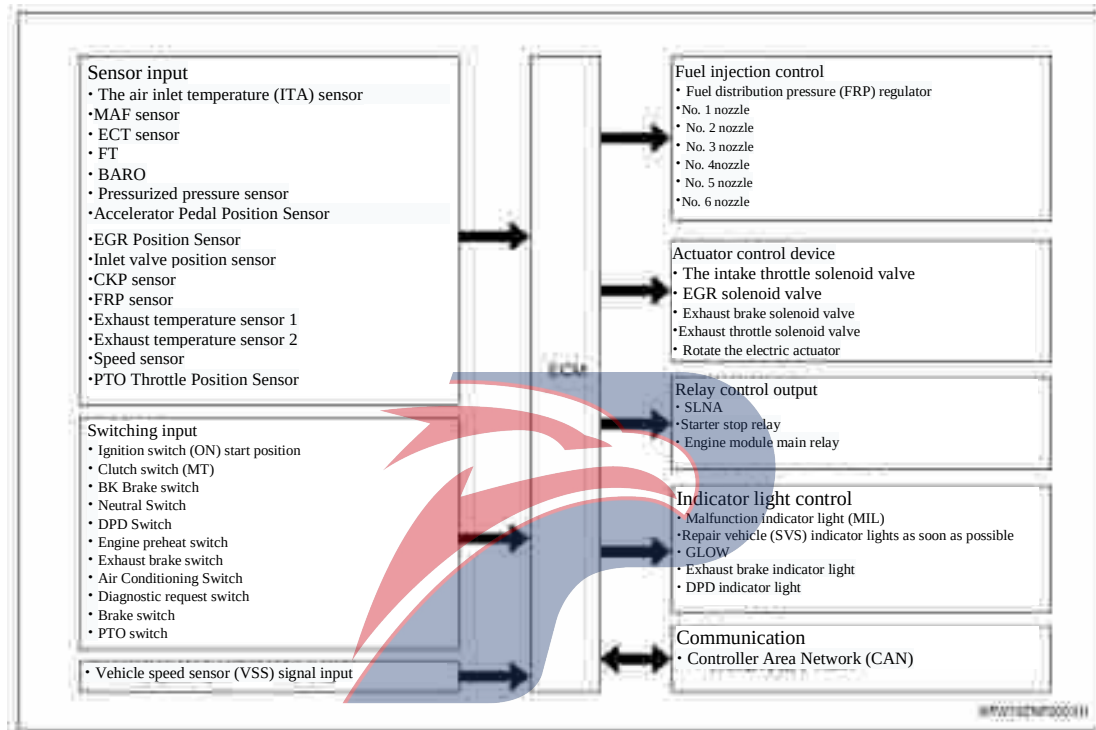
The engine control module is mounted inside the left engine side cover through a mounting bracket. The engine control module mainly controls:

- Fuel system control
- Exhaust gas recirculation (EGR) system control
- Control of turbocharger system
- DPD system control

- Preheating system control
- Exhaust brake system control
- Power take-off (PTO) system control
- Vehicle diagnosis for engine control

The engine control module continuously observes information from each sensor, controls the system that affects the vehicle performance, and executes the diagnostic function of the system. The engine control module can identify operational problems, alert the driver via a fault indicator (MIL) and store a diagnostic trouble code (DTC). Diagnostic trouble codes can identify system failures and help technicians repair.

Inputs and Outputs of Engine Control Module



Description of Engine Control Module Voltage

The engine control module provides buffer voltages to the switches and sensors. The engine control module can operate because the resistance value of the engine control module is very high, and the light may not be illuminated when connected to the circuit. The common purpose voltmeter cannot yield the precise reading since its input impedance is very low. Use a 10 mega ohm input impedance DMM to ensure accurate voltage readings. The input and / or output devices in the engine control module include analog to digital converters, signal buffers, counters and dedicated drivers. The engine control module can control most components with electronic switches that, when opened, can form a grounding circuit.

After-sales additional electrical and vacuum equipment

Additional electrical equipment and vacuum equipment installed after sale refers to any equipment connected to the vehicle electrical system or vacuum system installed on the vehicle after the vehicle leaves the factory. It is not allowed to use such equipment in vehicle design. Adding any aftermarket vacuum equipment to the vehicle is not allowed. Only at the battery and earthing parts, the additional electrical equipment after sale can be connected to the electrical system of the vehicle. Although these additional electrical equipments are installed in accordance with these guidelines, they may also cause power system failure. It may also include devices that are not connected to the electrical system of the vehicle, such as portable phones and audio. Therefore, the first step in diagnosing any power system failure is to remove all aftermarket additional electrical equipment from the vehicle. If the fault persists after removing these devices, you can diagnose the fault in the usual way.

Electrostatic discharge damage

Electronic components in the engine control module are usually designed with only a very low voltage. Electronic components are easily damaged by electrostatic discharge. In contrast, a person's electrostatic discharge even dies up to 4000 volts. There are several ways to generate static electricity. The most common

methods are friction and induction.

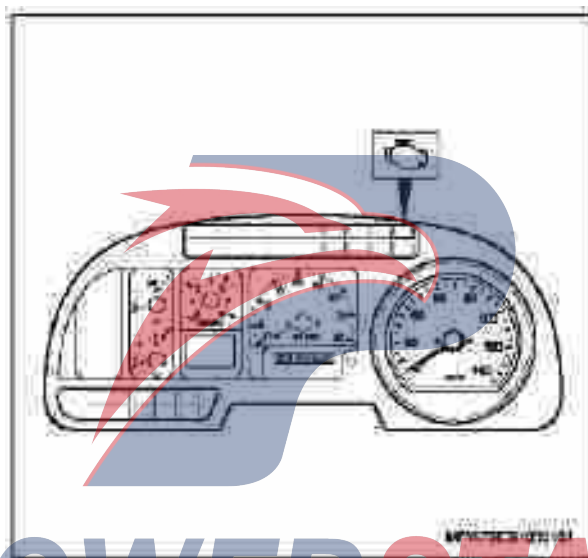
- An example of static electricity generated by friction is moving on the seat of a car.

Important note: in order to prevent potential ESD damage, please follow the following guidelines.

- Do not touch the engine control module connector pins or welding elements on the engine control module circuit board.
- Please do not open spare parts package unless you are ready to install parts.
- Before the parts are removed from the bag, the part bag is ground to the known good ground on the vehicle.
- If the part is handled when you slip through the seat, sit at the standing position or walk for a period of distance, touch the known good ground before installing the part.
- When a person wears a pair of very insulated shoes standing next to a highly charged object and then touches the ground immediately, it induces electrostatic induction. Charges with the same polarity will slowly disappear from charged persons with opposite polarity.

Malfunction indicator lamp (MIL) operation

Fault indicator is located in the instrument panel cluster. When the fault indicator is turned on by a command, the following symbol is displayed:



The fault indicator lamp indicates that an emissions-related fault (Class A or B) has occurred and the vehicle needs need to be maintained. The following is a list of operating modes for the fault indicator:

- When the engine is turned off and the ignition switch is turned on, the fault indicator lights up. This is a light bulb test to ensure that the fault light is on.

If there is no diagnostic fault, the fault indicator will be turned off after the engine starts.

- If the engine control module detects failure, the fault indicator will still be on after the engine starts. Whenever the ECM illuminating the vehicle MIL related to emission, DTC will be stored.

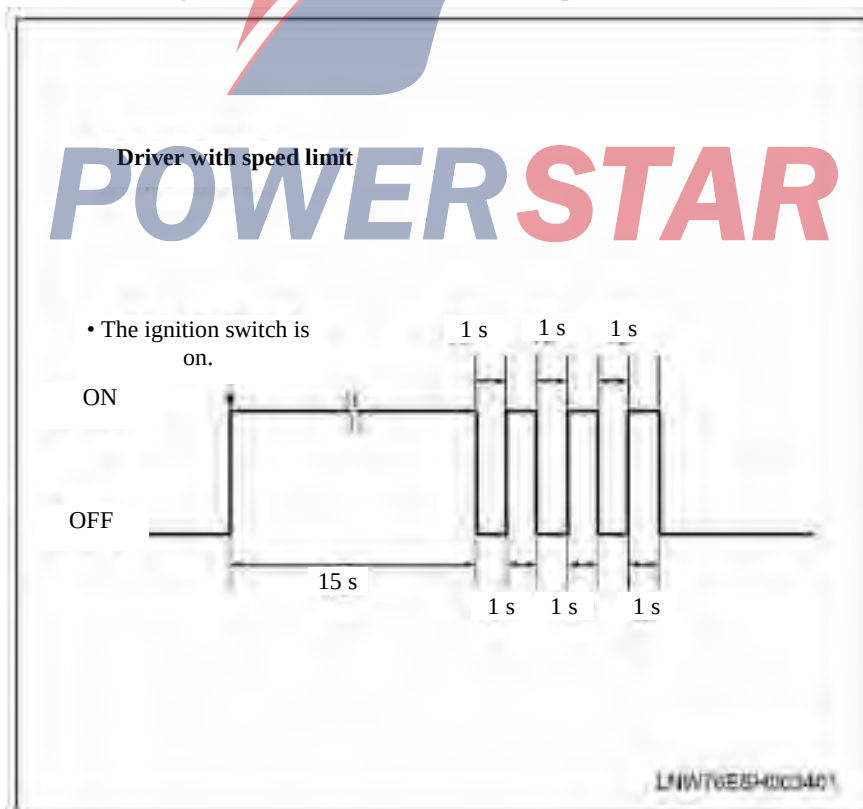
Service vehicle soon (SVS) indicator lamp working

SVS indicator is in the dashboard cluster. When the service vehicle soon lamp is on upon the command, the following symbols will occur:



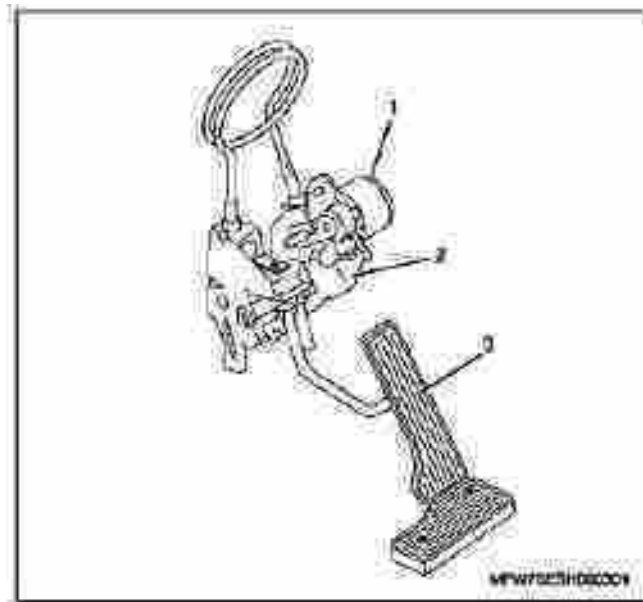
As soon as the vehicle lights are repaired, it indicates that a non-emission-related fault (Class C diagnostic fault code) has occurred and the vehicle needs to be repaired. The following is a list of operating modes of SVS indicator lamp:

- The SVS indicator is lit when the engine stalls and the ignition switch is on. This is a light bulb test to ensure that the lights on the vehicle light up as quickly as possible.
- When the engine is off and the ignition is on, the fault indicator lamp lights for about 15 seconds and flashes 3 times. This is a light bulb test (to ensure that the fault light can emit light) and a governor device installation test. (with speed limiter)
- If there is no diagnostic fault, the SVS indicator will be off after the engine is started.
- If the engine control module detects fault, the vehicle SVS indicator will still be on after the engine starts. Whenever the ECM illuminating the vehicle service indicator lamp unrelated to emission, DTC will be stored.



Description on engine control components

Accelerator pedal position (APP) sensor

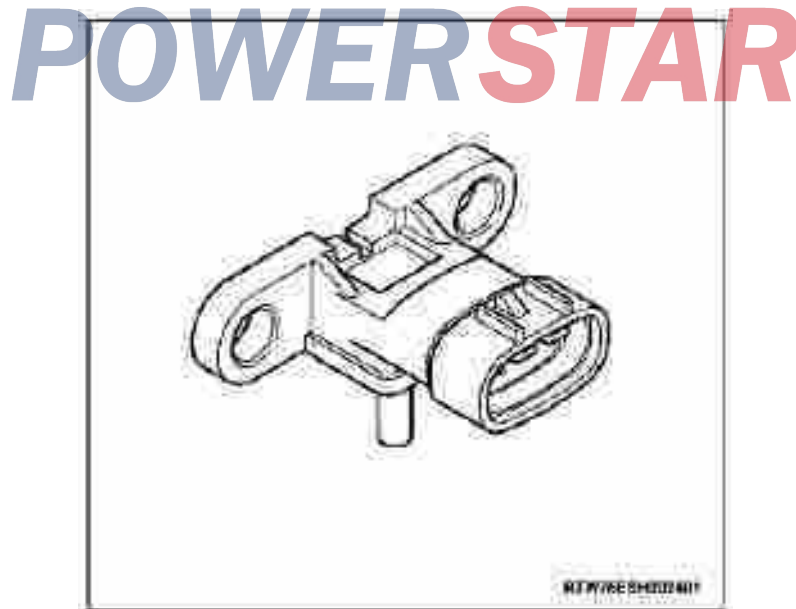


Key

1. Accelerator Pedal Position (APP) sensor
2. Accelerator pedal support
3. Accelerator pedal

The APP sensor is mounted on the accelerator pedal control assembly. The sensor consists of two separate sensors in the same housing. The engine control module uses the APP sensor to determine the required acceleration or deceleration. APP sensor is HALL element type. Each APP sensor sends a different signal related to the angular position change of the accelerator pedal to the engine control module on each signal circuit. When the APP sensor 1 is not in operation, the signal voltage is low and gradually increases when stepping down the pedal. The signal voltage is high when the APP sensor 2 is not in operation, and gradually decreases when the pedal is depressed.

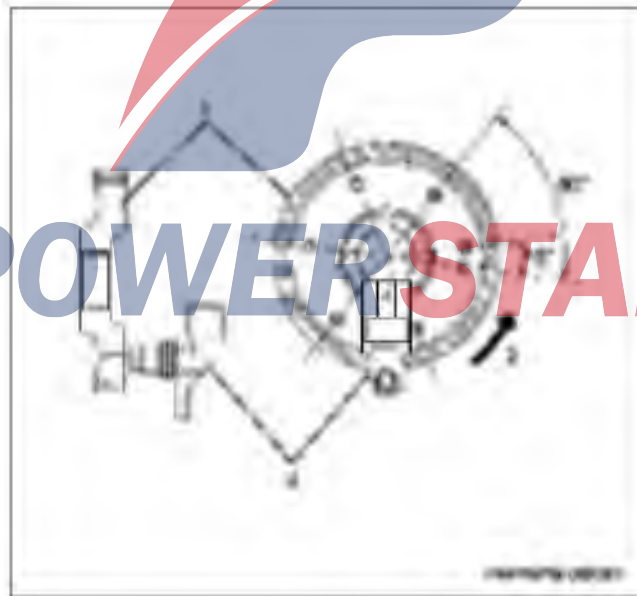
BARO sensors



The BARO sensor is located under the instrument panel (IP) cluster near the pedal support and is a sensor that changes the voltage as pressure changes. The BARO sensor sends signals to the engine control module related to atmospheric pressure changes in the signal circuit. The engine control module should detect a low signal voltage at low atmospheric pressure, such as high altitude position, and should detect a high signal voltage at high atmospheric pressure. The engine control module uses this voltage signal to calibrate the fuel injection volume and injection timing required for height compensation.

Booster pressure sensor

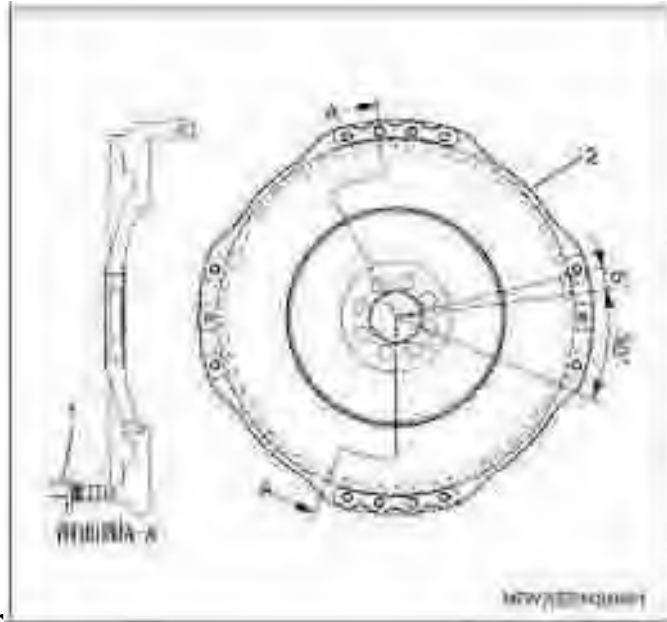
The boost pressure sensor is located in the intake duct and is a sensor that changes the voltage with the change of the air pressure inside the air duct. The boost pressure sensor sends signals to the engine control module on the signal circuit that is related to the change in air line pressure. The engine control module should detect low signal voltage at low pressure, such as low engine load, and high signal voltage should be detected at high pressure, such as high engine load.

Camshaft position (CMP) sensor**Key**

1. Timing sprocket teeth
2. Rotation direction
3. Camshaft position (CMP) sensor

The CMP sensor is mounted on the cylinder head at the rear of the camshaft gear and is used to detect the number of five total holes in each cycle of the engine (four holes are evenly spaced at 90 degrees, and the other reference hole is located on the camshaft gear surface). The CMP sensor is a magneto resistive element (MRE) type sensor that generates square wave signal pulses.

Section A-A



Key

Crankshaft position (CKP) sensor

1. Crankshaft position (CKP) sensor
2. Flywheel

The CKP sensor is located on the flywheel housing and the rotor of the sensor is fixed on the flywheel. There are 56 holes spaced 6 apart and a 30 unsupported span. The TDC detection can be performed with this bracket less part. The CKP sensor is a magneto resistive element (MRE) type sensor that generates a square wave signal pulse. The engine control module uses a CKP sensor to detect the no-bracket span and uses a camshaft position (CMP) sensor to detect the reference hole to determine cylinder No. 1 compression top dead center (TDC) to ensure interconnection.

Engine Coolant Temperature (ECT) Sensor

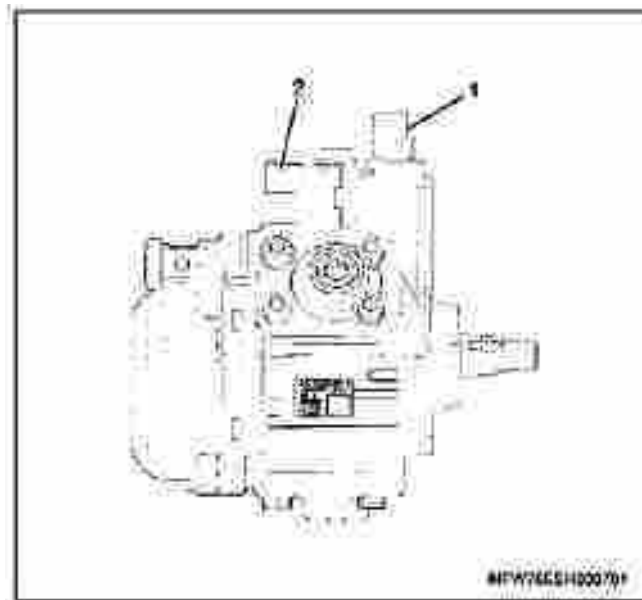


Key

1. Engine Coolant Temperature (ECT) Sensor

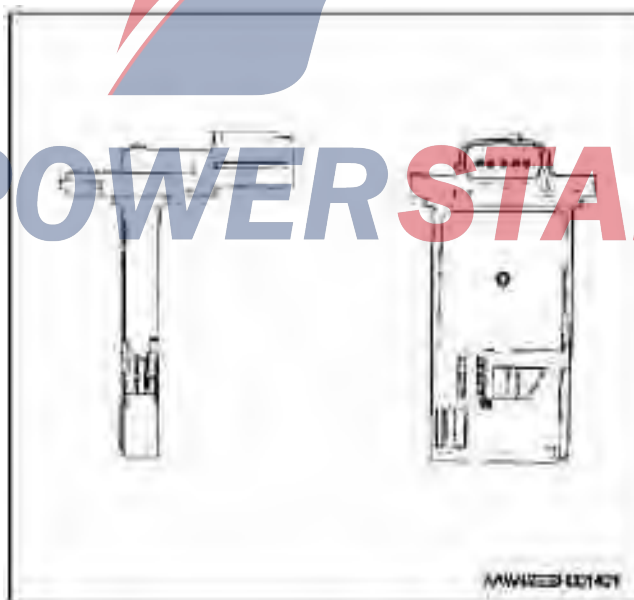
The ECT sensor is mounted inside the thermostat housing and is a variable resistor that measures the temperature of the engine coolant. When the engine coolant temperature sensor temperature is cold, the sensor resistance is high; when the engine coolant temperature increases, the sensor resistance decreases. In the case of a high sensor resistance, the engine control module detects that the signal circuit voltage is high; in the case of a low sensor resistance, the engine control module detects that the signal circuit voltage is low.

Fuel Temperature (FT) Sensor

**Key**

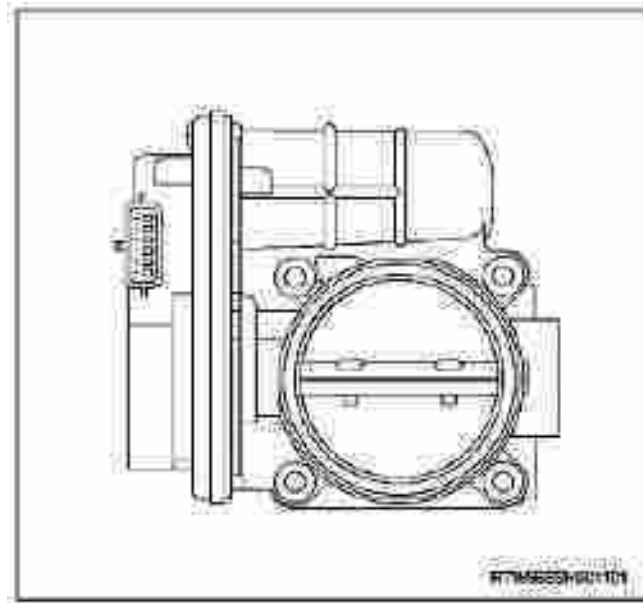
1. Fuel temperature (FT) sensor
2. Fuel Distribution Pipe Pressure (FRP) Regulator

The FT sensor is installed in the fuel supply pump and is a variable resistor that measures the temperature of the fuel entering the fuel supply pump. When the fuel temperature sensor is cold, the sensor resistance is high; when the fuel temperature increases, the sensor resistance decreases. In the case of a high sensor resistance, the engine control module detects that the signal circuit voltage is high; in the case of a low sensor resistance, the engine control module detects that the signal circuit voltage is low.

Intake air temperature (IAT) sensor

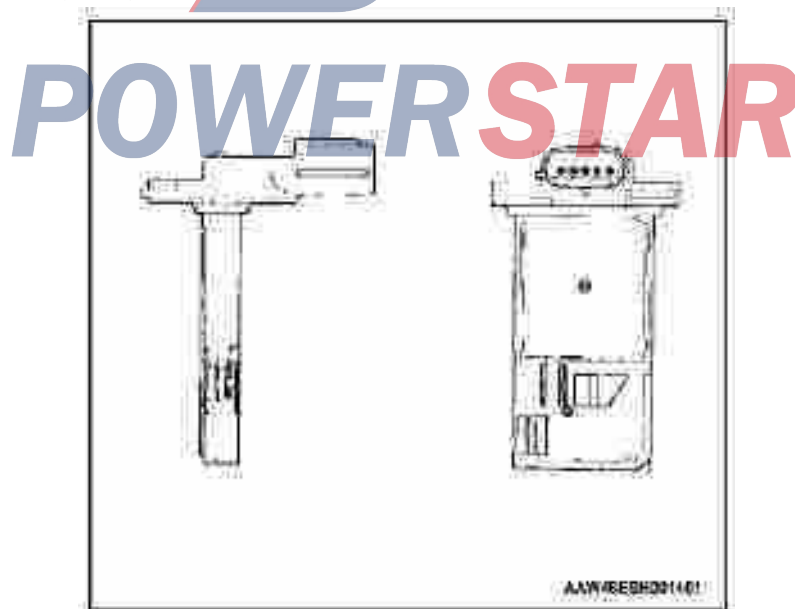
The IAT sensor is installed between the air purifier and the turbocharger; inside the mass air flow (MAF) sensor. The intake air temperature (IAT) sensor is a variable resistor that measures the temperature of air entering the engine. When the IAT sensor is cold, the sensor resistance is high. As air temperature increases, the sensor resistance decreases. In the sensor resistance is high, the engine control module detects a higher voltage signal circuit; in sensor resistance is lower, the engine control module detects a lower voltage signal circuit.

Air intake throttle



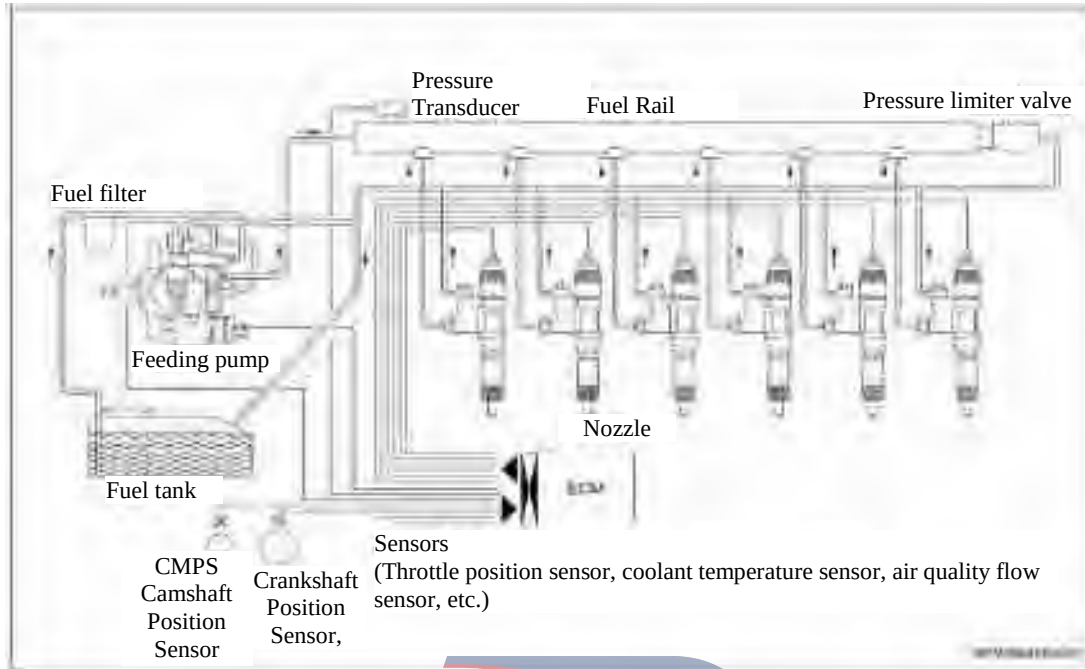
The intake throttle is located at the intake manifold inlet. The engine control module controls the opening of the intake throttle according to the engine operating conditions and controls the intake throttle by controlling the solenoid valve. The solenoid valve is controlled based on a pulse width modulation (PWM) signal from the engine control module. Changing the duty cycle from 0% to the appropriate percentage is the intake throttle opening angle control. To open the valve, increase the duty cycle. To close the valve, decrease the duty cycle. The position of the intake throttle is detected by the position sensor and transferred to the engine control module. The position sensor will send a signal to the engine control module on the signal circuit that is related to the change in the intake throttle position. In low opening or closed positions, the engine control module should detect low signal voltages. The engine control module should detect a high signal voltage when the opening is large.

Air Mass Flow Sensor (MAF)



The MAF sensor is an air flow meter for measuring the amount of air entering the engine and is installed between the air purifier and the turbocharger. A small amount of air entering the engine shows a deceleration or idling, and a large amount of air entering the engine shows acceleration or a high load condition. The MAF sensor assembly includes a MAF sensor element and an intake air temperature (IAT) sensor, both of which are exposed in the air flow to be measured. The MAF sensor element measures the partial air quality through the measuring tube on the sensor housing.

Fuel system description



The common rail system uses a cylinder housing called a fuel distribution tube to store pressurized fuel, and uses a fuel injector containing an electronically controlled solenoid valve to spray pressurized fuel in the combustion chamber. The fuel injection system (injection pressure, injection rate, and injection timing) is controlled by the engine control module so that the common rail system can be controlled independently of engine speed and load. This ensures that stable injection pressures are maintained at all times, especially when the engine speed range is low, so that the black smoke generated by a particular diesel engine during engine startup or acceleration will be significantly reduced. As a result, the discharged exhaust gas will become cleaner and greatly reduced, and achieve higher output.

1. High pressure control

- Start high pressure injection from a low engine speed range.
- Optimize control to minimize particulate matter and NOx emissions.

2. Fuel injection timing control

- Start the fine-tuned optimization control according to the operating conditions.

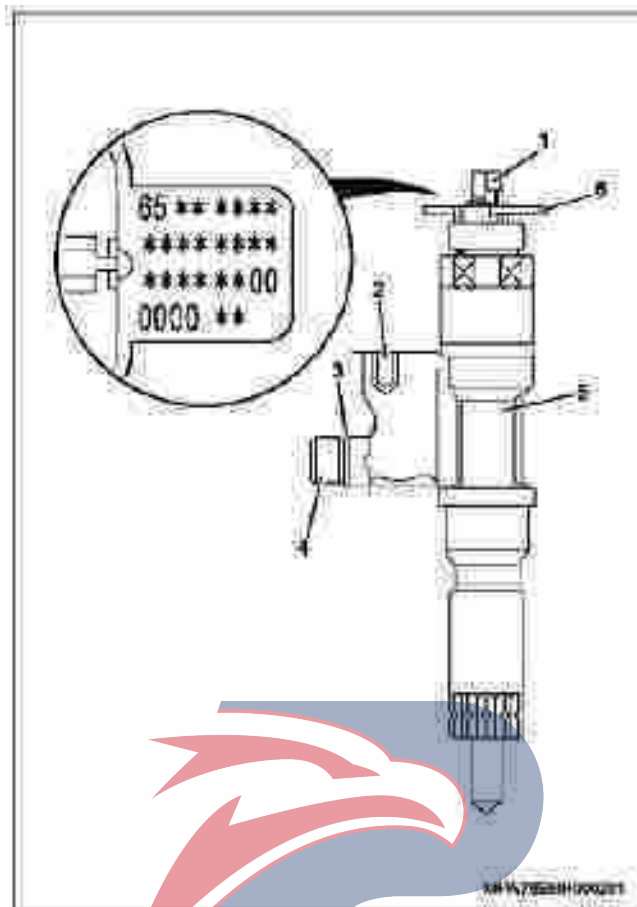
3. Injection rate control

- Pre-injection control with a small amount of injection before the main injection.

The fuel distribution pipe system mainly includes an oil supply pump, a fuel distribution pipe, an injector, and an engine control module.

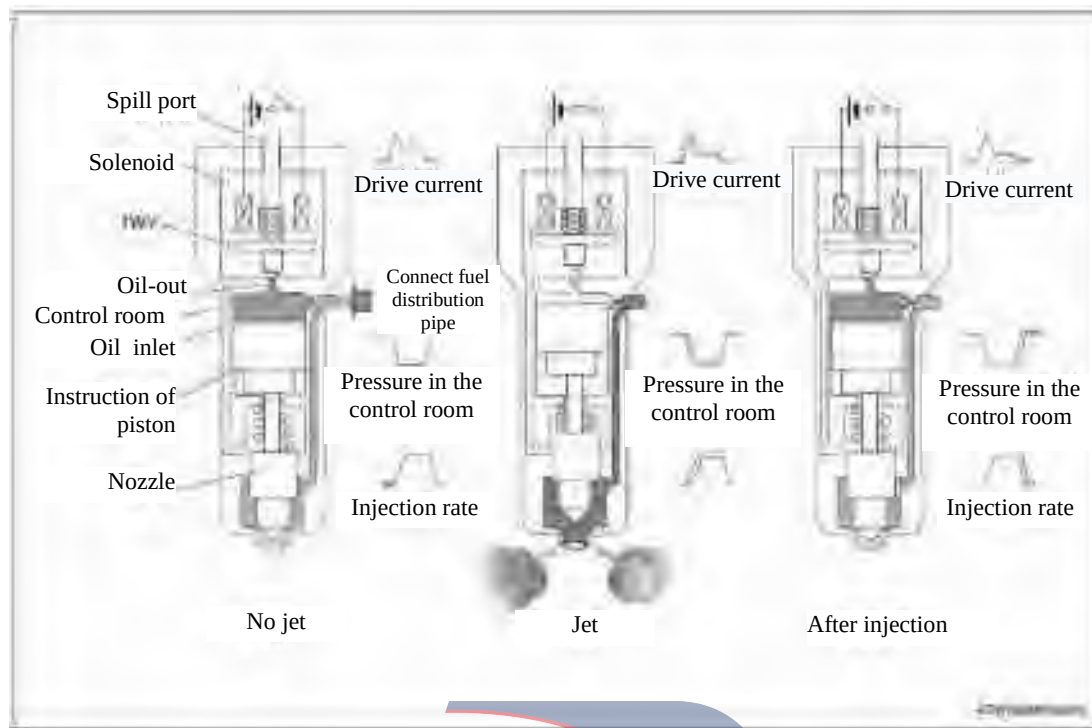
■ Fuel System Parts Description

Fuel injector

**Key****Wiring screw pile****2. Fuel leak hole****3. O-ring****4. Fuel inlet****5. Nozzle part number marking****6. Signage**

An electronically controlled injector controlled by an engine control module is used. Compared to conventional injectors, command pistons, solenoid valves, etc. have been added.

An ID code showing various nozzle characteristics is laser-marked on the placard, and these ID codes displayed in digital form (30 alphanumeric characters, but only 24 are displayed) are also laser-marked on the placard. The system uses injector flow rate information (ID code) to optimize the injection quantity control. When a vehicle is newly installed with a fuel injector, it is necessary to enter its ID code into the engine control module. Use the QR (Quick Response) code or injector flow rate (ID code) to increase the injector fuel injection accuracy. The use of these codes allows diffusion control of fuel injection over the entire pressure range, which helps to improve combustion efficiency and reduce exhaust emissions.



1) Non-injection

When the engine control module does not supply current to the solenoid, the two-way valve (TWV) will close the oil outlet through spring force. At this time, the fuel pressure applied to the front end of the nozzle is equal to the fuel pressure applied to the control room through the oil inlet. In terms of force competition in this state, the pressure on the upper surface of the command piston + the spring force of the nozzle spring is greater than the pressure at the front end of the nozzle, and therefore, the nozzle is pressed down and the injection hole is closed.

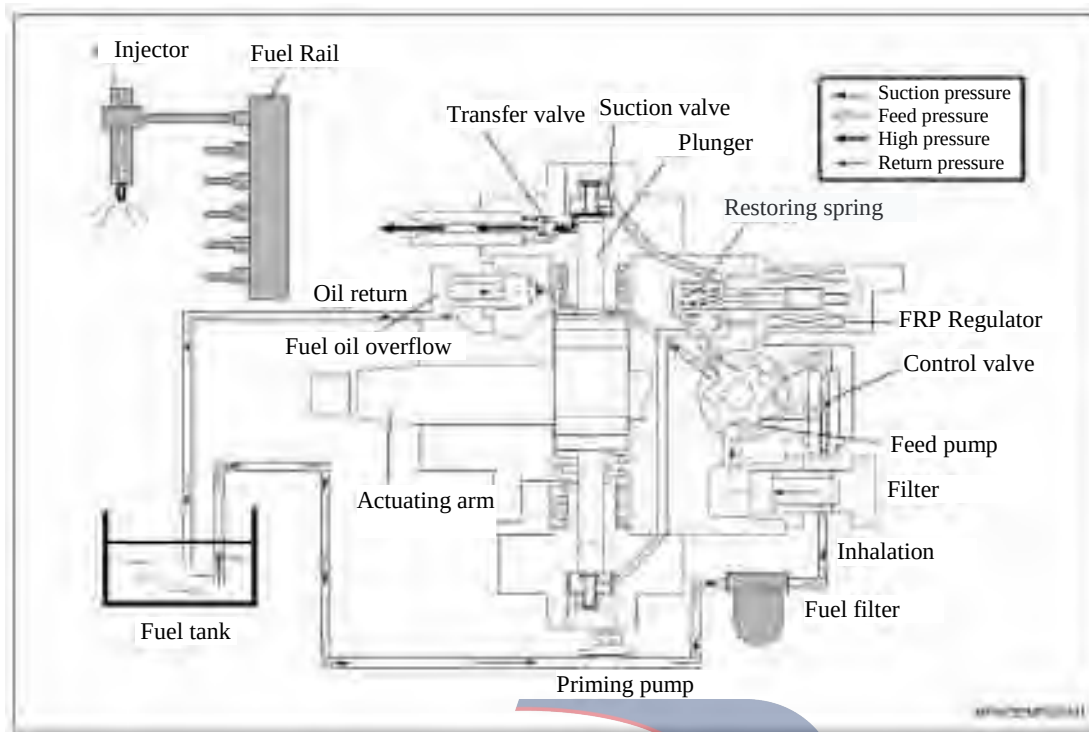
2) Start of injection

Pull up the TWV and open the oil outlet so that the fuel leaks into the oil return hole when the engine control module supplies power to the solenoid. Therefore, the nozzle and the command piston are pushed upward by the fuel pressure applied to the front end of the nozzle, and then the nozzle injection hole is opened, and the fuel is injected.

3) End of injection

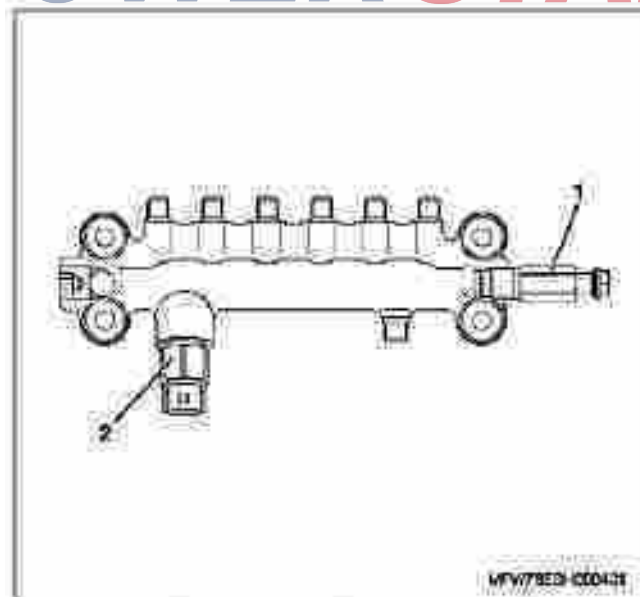
When the engine control module closes the current supplied to the solenoid, the TWV decreases to close the oil outlet. As a result, fuel cannot leak out of the control chamber, so that the fuel pressure in the control chamber suddenly rises, and the command piston presses the nozzle down to close the nozzle injection hole, resulting in the end of injection.

Supply pump



The fuel supply pump is the core of the common rail electronic fuel injection system. The installation position of the fuel supply pump is the same as that of the conventional injection type pump. The rotation speed ratio between the fuel supply pump and the crankshaft is 1:1. The fuel distribution pipe pressure (FRP) regulator and the fuel temperature sensor are part of the supply pump assembly. The fuel pump uses an internal supply pump (trochoidal) to withdraw fuel from the fuel tank. The supply pump pumps the fuel into the three piston chambers that are also located inside the supply pump. Fuel pumped into the chamber is regulated by an FRP regulator that is individually controlled by the current supplied by the engine control module. The maximum fuel flow will be generated when the solenoid is not supplied with current, and no fuel flow will be generated when the full current is supplied. As the engine rotates, these three plungers will generate high pressure in the fuel distribution tube. Since the engine control module controls the flow of fuel to the three piston chambers, it will control the amount and pressure of fuel supplied to the fuel distribution tubes. This will optimize performance, increase economics, and reduce NOx emissions.

Fuel distribution pipe (general distribution pipe)



Key

Pressure limiting valve

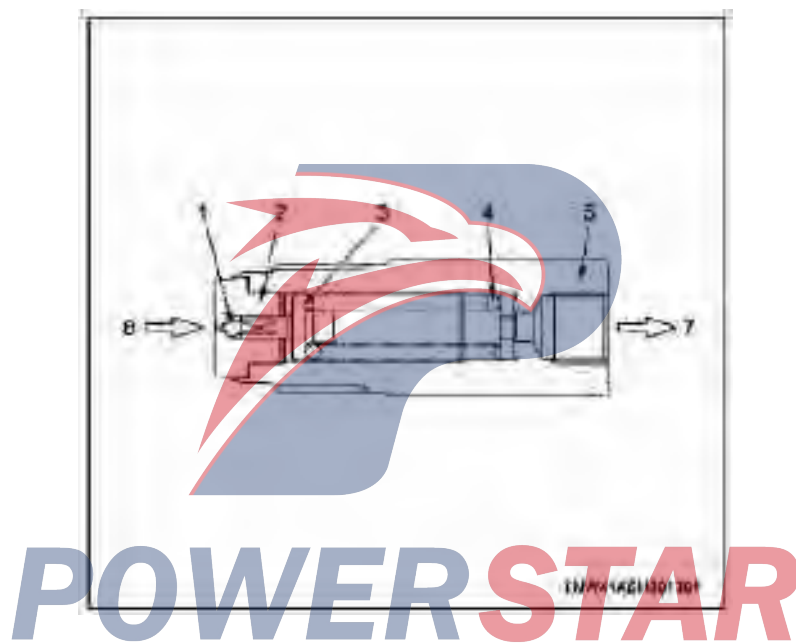
2. Fuel Distribution Pipe Pressure (FRP) Sensor

With the use of a common rail electronically controlled fuel injection system, a fuel distribution pipe is provided to store high pressure fuel between the supply pump and the injection nozzle. The pressure sensor and pressure limiter will be installed on the fuel distribution pipe. The pressure sensor detects the fuel pressure inside the fuel distribution pipe and sends the signal to the engine control module. Based on this signal, the engine control module will control the fuel pressure inside the fuel distribution pipe through the fuel distribution pipe pressure (FRP) regulator of the supply pump. When the fuel pressure inside the fuel distribution pipe is too high, the pressure limiter will mechanically open the valve to release the pressure.

Fuel distribution pipe pressure sensor

The fuel distribution pipe pressure (FRP) sensor is installed on the fuel distribution pipe to detect the oil pressure in the fuel distribution pipe, convert the oil pressure into a voltage signal, and send the signal to the engine control module. The engine control module will monitor the FRP sensor signal voltage. Higher pressure on the fuel distribution pipe will provide higher signal voltage; lower pressure on the fuel distribution pipe will provide lower signal voltage. The engine control module will calculate the actual fuel distribution pipe pressure (fuel pressure) from the voltage signal and apply the result to injection control and other control tasks.

Pressure limiting valve

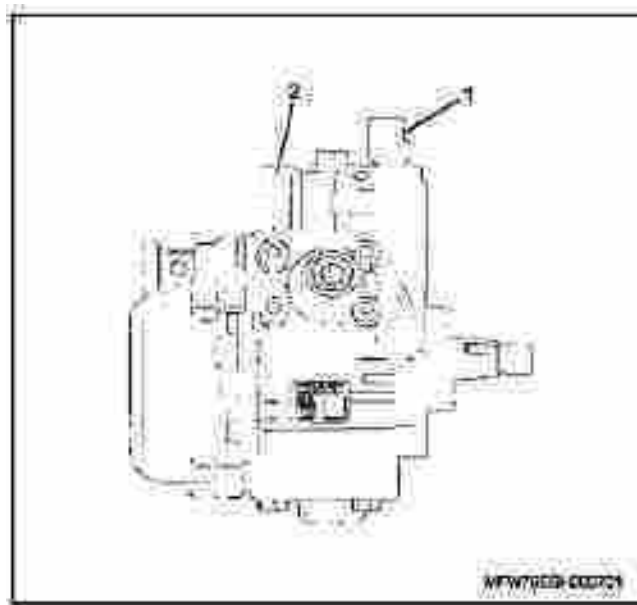


Key

1. Valve
2. Body
3. Valve guide
4. Spring
5. Shell
6. Fuel distribution pipe
7. Return pipe

If there is an abnormally high pressure, the pressure limiter will relieve the pressure by opening the valve. The valve will open when the pressure in the distribution tube reaches approximately 220MPa (32000 psi), and the valve will close when the pressure drops to approximately 50MPa (7250 psi). Fuel leaked through the pressure limiter will return to the fuel tank.

Fuel Distribution Pipe Pressure (FRP) Regulator



The engine control module will control the duty cycle of the linear fuel-distribution tube pressure (FRP) regulator (the length of time the current is applied to the FRP regulator) in order to control the amount of fuel supplied to the high-pressure plunger. Since only the amount of fuel required to reach the target distribution pipe pressure is extracted, the drive load of the supply pump drops. When current is supplied to the FRP regulator, a variable electromotive force will be generated according to the duty cycle, and the armature will be moved to the left. The armature moves the cylinder to the left and changes the opening of the fuel line to regulate the amount of fuel. As the FRP regulator closes, the return spring contracts, completely opening the fuel line and supplying fuel to the ram (full intake and full discharge). When the FRP regulator is opened, the spring force of the return spring moves the cylinder to the right and closes the fuel line (normally open). With the FRP regulator on/off, the fuel is supplied in an amount that complies with the actuation duty cycle, and the fuel is discharged through the plunger.

Fuel injection system description

Fuel injection control

The control will determine the fuel injection quantity by increasing the coolant temperature, fuel temperature, intake air temperature, atmospheric pressure, air mass flow rate, and some switching input information correction of the basic injection quantity will be calculated by the engine control module according to the engine operating condition (engine speed, accelerator pedal depression, and boost pressure sensor). More fuel ratios will indicate whether the engine load increases with the depression of the accelerator pedal at a constant engine speed.

In combination with atomized fuel high pressure injection, this control improves the exhaust gas and ensures proper fuel consumption. Compared to the traditional mechanical speed control, the electronic control system provides a higher degree of freedom of fuel injection control, resulting in a very high accelerator response (acceleration and pressure).

Fuel injection control begins

When the engine is started (the engine is started by turning the key switch to the "START" position until the push switch returns to the "ON" position), the optimum spray will be controlled based on the engine speed and coolant temperature information. The amount of oil. At low temperatures, the amount of fuel injected will increase. When the engine is fully started, the boost mode at start-up will be cancelled and the normal operating mode will be restored.

Idle speed control

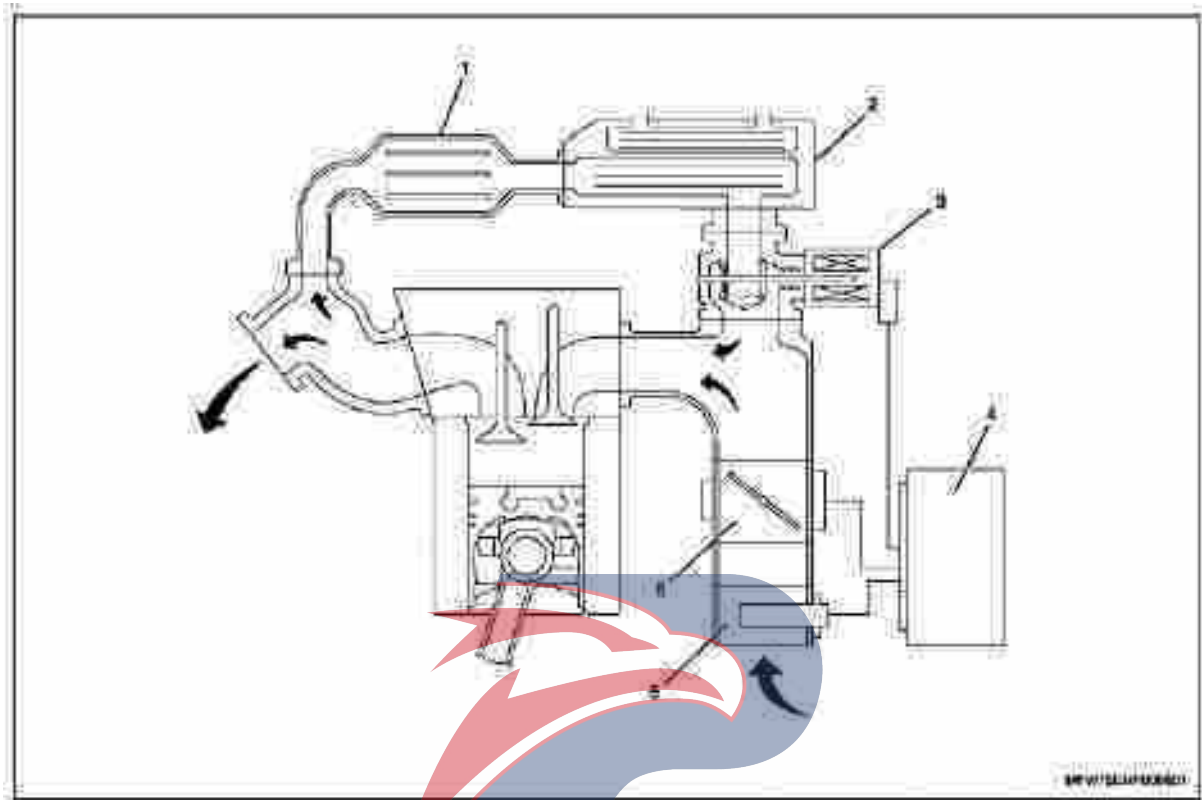
This control is performed so that stable idling can be achieved at any time (irrespective of long-term changes in the engine or changes in engine conditions). The engine control module will set the target idle speed and control the injection amount according to engine conditions (actual engine speed, coolant temperature, and engine load) to regulate the actual engine speed at the target idle speed, thereby ensuring stable idle speed.

Idling vibration control

This control is performed so as to reduce engine vibration caused by a change in torque between the cylinders due to a change in the injection amount of each cylinder or a change in the performance of the injector. The engine control module will correct the amount of fuel injected between the cylinders based on the rotation

signal from the crankshaft position (CKP) sensor. Typical corrections between cylinders range from ± 5 mm³.

Exhaust gas recirculation (EGR) system description



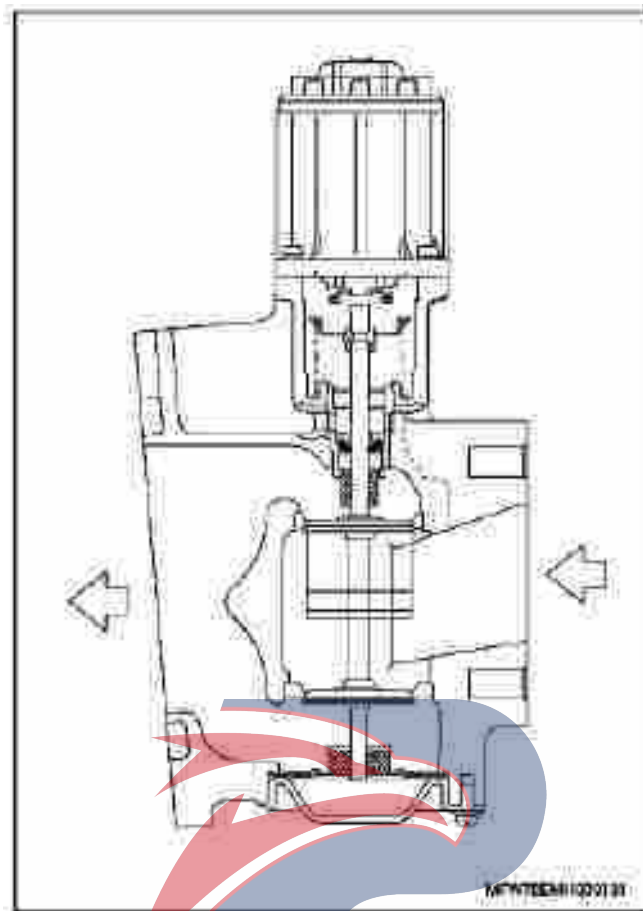
Key

1. Engine control module
2. Exhaust gas recirculation valve
3. Engine coolant outlet
4. Engine coolant inlet
5. Exhaust gas recirculation cooler
6. Intake throttle valve
7. MAF sensor

The EGR system recalculates a portion of the exhaust gas to the intake manifold, reducing nitrogen oxide (NO_x) emissions. The EGR control system uses an electronic control system to simultaneously ensure driving performance and low emissions. The control current from the engine control module runs the solenoid to control the lift of the EGR valve. The EGR position sensor is also provided at the rear of the motor, so that the actual valve lift amount is fed back to the engine control module for more accurate control of the EGR amount.

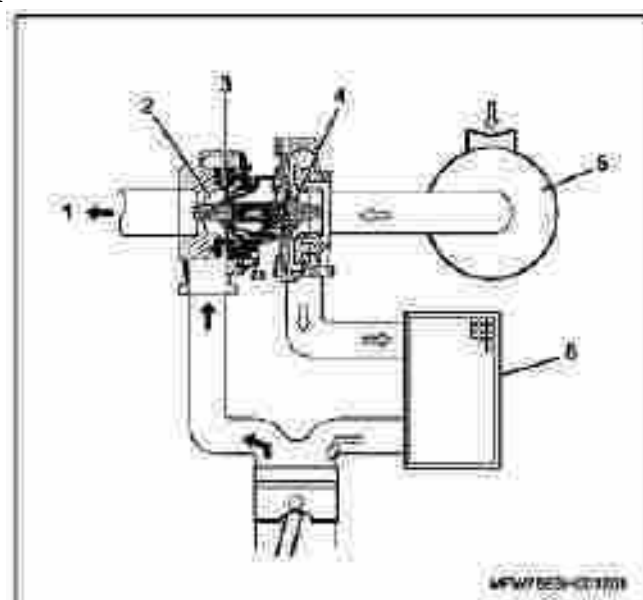
When the conditions of engine speed, engine coolant temperature, intake air temperature, and atmospheric pressure are satisfied, EGR control begins. Then, the valve opening degree is calculated based on the engine speed and the target fuel injection amount. Based on the opening degree of the valve, a solenoid coil driving task is provided, and the valve is driven accordingly.

Exhaust gas recirculation (EGR) valve



The EGR valve is mounted on the intake manifold. The engine control module controls the opening degree of the EGR valve according to the engine operating conditions, and controls the EGR valve by controlling the solenoid valve. The solenoid valve is controlled based on a pulse width modulation (PWM) signal from an engine control module. The position of the EGR valve is detected by the position sensor and transferred to the engine control module. The position sensor consists of three separate sensors in the same housing. The EGR position sensors 1, 2 and 3 are Hall element type sensors. The position sensor will send a high/low signal state to the engine control module on the signal circuit that is related to the change in EGR valve position.

Turbocharger description



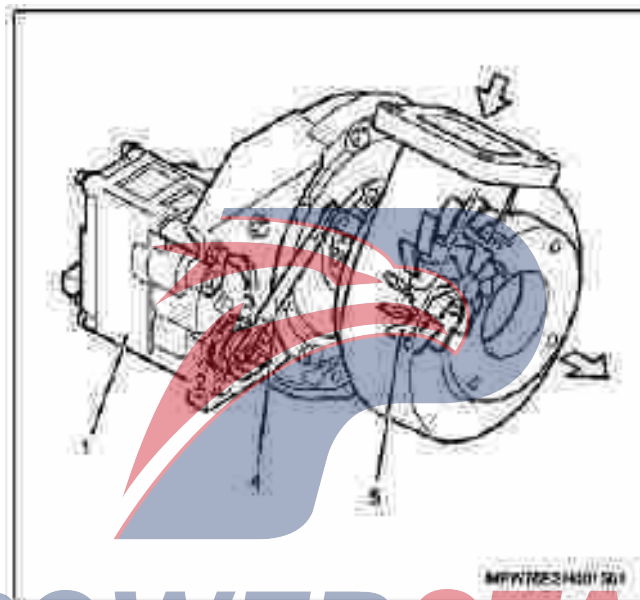
Key

Exhaust gas

2. Turbine wheel

3. Nozzle
4. The compressor impeller
5. Air purifier
6. Charge air cooler

Turbochargers are used to increase the amount of air entering the engine cylinders. This will cause the fuel injected to the cylinder to increase proportionately, resulting in increased power output, more complete combustion of the fuel, and more cooling of the cylinder head, piston, valves, and exhaust gas. This cooling effect will help to prolong engine life. The heat and pressure of the engine exhaust gas will be used to drive the turbine and the exhaust gas will flow directly to the turbine casing. The turbine housing serves as a nozzle to guide the axle wheel assembly. Since the compressor impeller is directly connected to the shaft, the compressor impeller rotates at the same speed as the turbine impeller. Clean air from the air purifier will be introduced into the compressor housing and the impeller. This air is compressed air, which is fed across the pipe to the engine intake manifold and then delivered to the cylinder.



Key

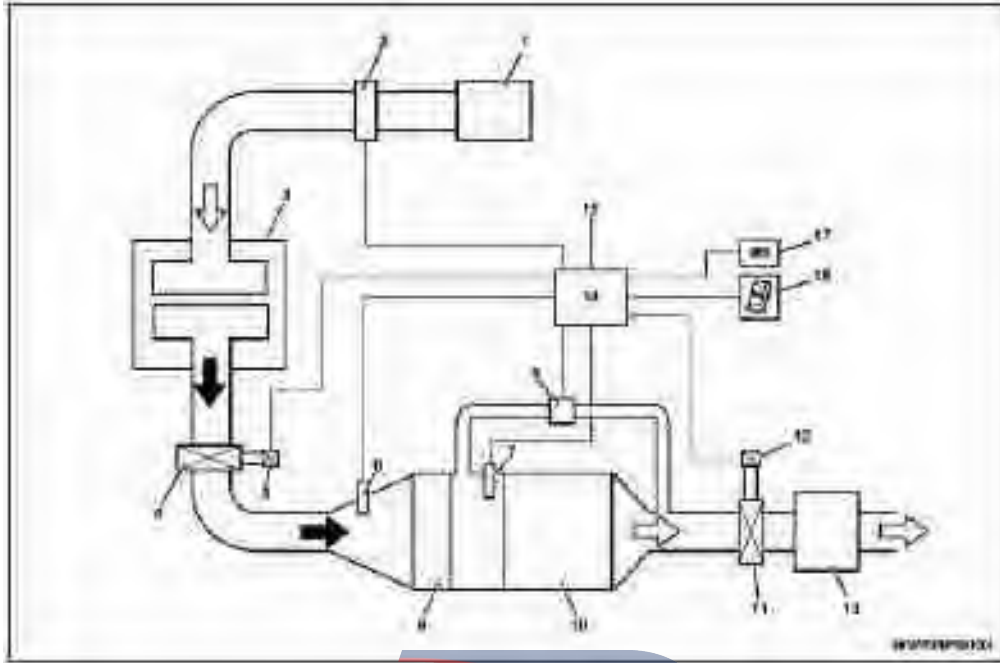
1. Rotary Electric Actuator (REA)
2. The nozzle's closing direction
3. The opening direction of the nozzle
4. Control lever
5. Nozzle

The position of the turbocharger nozzle is controlled by the rotary electric actuator (REA) according to instructions from the engine control module. The REA is located inside the turbocharger. When the engine is unloaded, the turbocharger nozzle is in the open position or no boost pressure conditions. When the engine is under load, the engine control module commands REA to close the turbocharger nozzle to increase boost pressure. The engine control module will change the boost pressure according to the load requirements of the engine.

Charge air coolers also contribute to the diesel engine's performance. Inlet air is input to the turbocharger compressor housing through the air purifier. The compressed air from the turbocharger then flows to a charge air cooler located in front of the radiator. The compressed air flows from the charge air cooler back to the intake manifold.

The charge air cooler is a heat exchanger that uses air flow to dissipate the air. The air temperature increases as the turbocharger air pressure increases. By pushing more air particles into the same space, lowering the intake air temperature will increase engine efficiency.

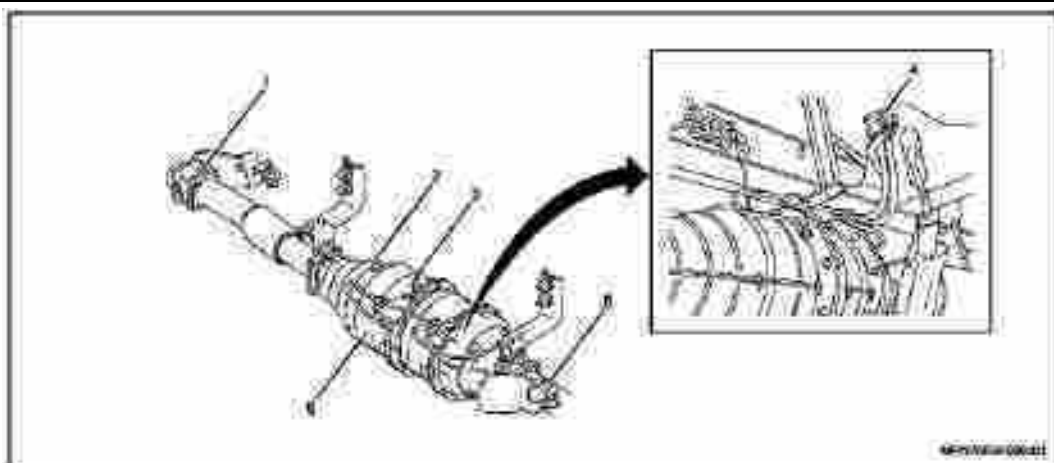
Diesel particulate trap (DPD) system description



Key

1. Air purifier
2. Air Mass Flow (MAF) Sensor
3. Engine assembly
4. Exhaust brake valve
5. Exhaust brake solenoid valve
6. Exhaust temperature sensor 2
7. Exhaust temperature sensor 1
8. Differential pressure sensor
9. Oxygen catalyst
10. DPD filter
11. Exhaust throttle
12. Exhaust throttle solenoid valve
13. Exhaust muffler
14. Engine Control Module
15. Other inputs
16. DPD switch
17. DPD indicator

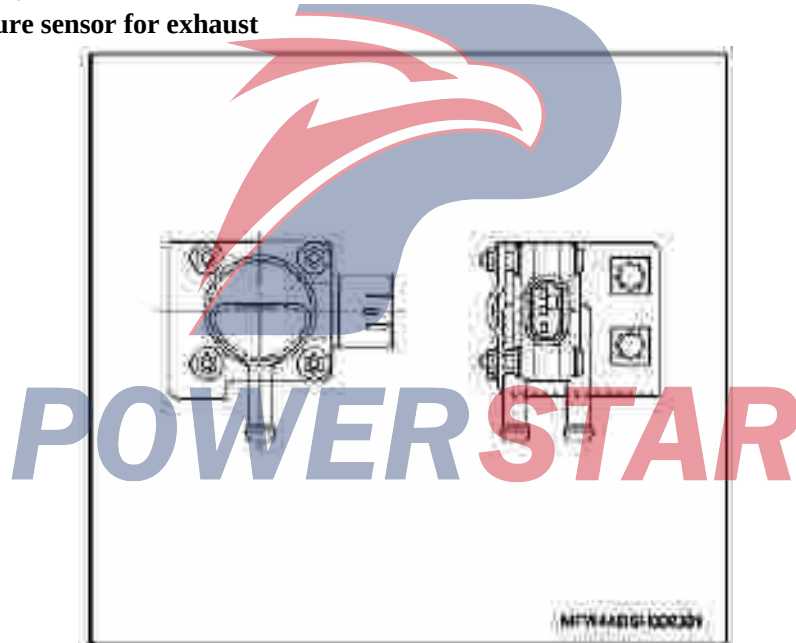
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Key

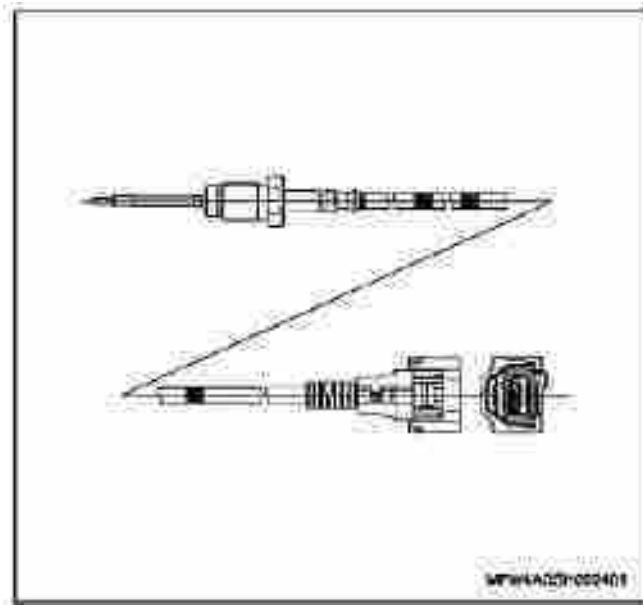
1. Exhaust brake valve
 2. Exhaust temperature sensor 2 (in front of the oxygen catalyst)
 3. Exhaust temperature sensor 1 (in front of the filter)
 4. Differential pressure sensor
 5. Exhaust throttle
 6. DPD assembly
-

DPD is a system that uses an oxidation catalyst to purify NO_x and HC and a filter that collects PM emissions from the engine. When PM accumulates in the filter, it will be regenerated. The PM detects the accumulation of PM from the exhaust differential pressure sensor or from mileage. When a specific PM accumulation amount or a specific mileage length is reached, automatic regeneration will start. If for some reason the automatic regeneration cannot be completed, the driver is required to manually regenerate through the DPD indicator on the cluster of flashing dashboards. During the regeneration process, the temperature inside the filter rises, and the accumulated PM will be burned off. In order to adjust to the optimum temperature, the engine control module will use the exhaust temperature sensor to detect the exhaust temperature and then control the injector and exhaust throttle to raise the temperature to burn the accumulated PM. Once regeneration begins, it must be completed within a certain period of time. After the regeneration is completed, the filter decontamination level is determined by monitoring the differential pressure of the exhaust gas. If excessive PM accumulation is detected, regeneration must not be started because uncontrolled combustion may occur during regeneration.

Differential pressure sensor for exhaust

The exhaust differential pressure sensor is connected to the chassis near the diesel particulate trap (DPD) assembly and is a sensor that changes the voltage according to the differential pressure of the exhaust gas before and after the DPD filter. The exhaust differential pressure sensor sends a signal to the engine control module on the signal circuit related to the differential pressure of the exhaust before and after the DPD filter. The engine control module should detect low signal voltages at low differential pressures (such as a small amount of PM build-up); high signal voltages should be detected at high differential pressures (such as large PM buildups).

Exhaust temperature sensor



Exhaust temperature sensors 1 and 2 are installed in the DPD housing and both sensors are variable resistors. Sensor 1 is used to measure the exhaust gas temperature before the DPD filter, and Sensor 2 is used to measure the exhaust gas temperature before the oxygen catalyst. When the exhaust temperature sensor is cold, the sensor resistance is high. As the exhaust temperature increases, the sensor resistance decreases. With higher sensor resistance, the engine control module detects higher signal circuit voltages. With low sensor resistance, the engine control module detects signal circuit voltages that are lower.

Exhaust throttle and exhaust brake valve

The exhaust throttle valve is part of the exhaust system after it is installed in the DPD housing. The engine control module controls the exhaust throttle relay and supplies power to the exhaust throttle solenoid valve according to the regeneration state of the DPD system. The exhaust throttle solenoid is energized to apply vacuum pressure to the diaphragm chamber to operate the exhaust throttle.

The exhaust brake valve is installed in front of the DPD housing and forms part of the exhaust system. The engine control module controls the exhaust brake solenoid valve based on the DPD regeneration state or the exhaust brake command signal. The exhaust brake solenoid is energized to apply vacuum pressure to the diaphragm chamber to operate the exhaust brake valve.

Special tools and equipment

Special tools and equipment

Schematic

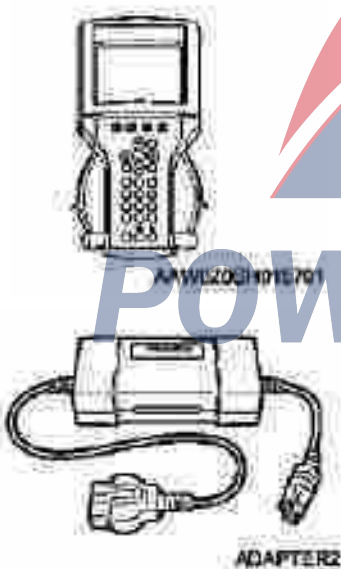
Tools number/Notes



5-8840-2835-0(J-35616-C)
Connector Test Adapter Kit
(with test light)



5-8840-0285-0(J-39200)
Digital Multimeter



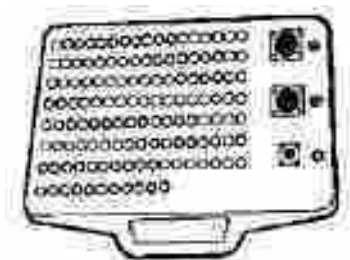
Tech2 Suite



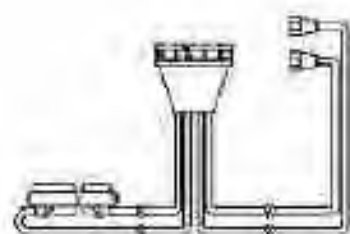
Tech 2 24V DC Adapter II



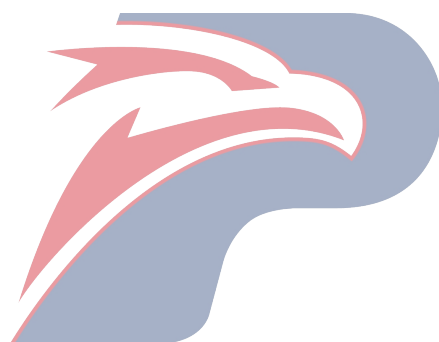
CAN-di module



Switch boxes



Adapter harness



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Engine control system

Precautions for maintenance

Use the loop test tool

If not specified in the diagnostic procedure, do not use the test lamp when performing driveline diagnostics on the powertrain. When using the probe as per the diagnostic procedure, use a connector to test the accessory 5-8840-0385-0.

General electric appliances sold in the market

The so-called general electric appliances sold on the market refer to electric appliances that are sold on the market and can be installed on vehicles after leaving the factory. There is no special consideration for the installation of the product during the design phase of the vehicle, so be careful.

Caution:

General-purpose electrical products sold in the market should be connected to circuits and power grounding wires that are not related to the electronic control system circuits.

Generally speaking, the general electrical parts sold on the market will sometimes cause functional failure of the electronic control system even if it can be installed. These include devices that are not installed on the electrical installation system of a vehicle, such as mobile phones and portable radios. Therefore, when diagnosing problems with the powertrain system, it is first necessary to confirm whether these general-purpose electrical equipment sold in these markets are installed, and if they are equipped with general-purpose electrical equipment sold in these markets, they should be taken from the vehicle. Remove it. If the problem is not resolved after removal, follow the general procedure for diagnosis.

Damage caused by electrostatic discharge

The electronic composition parts in the electronic control system are designed to work under very low voltage. Therefore, it is highly easy to be damaged due to ESD. Even the static electricity below 100V that cannot be sensed by the human body will damage some electronic composition parts. At the same time, in order to allow the body to sense the current of the electrostatic discharge, a voltage in excess of 4000V is required.

There are several ways to charge the body with static electricity. The most live test methods include friction method and induction method.

- Example of friction-charging: when the person moves on the seat transversely.

- Charge created by induction, for example, the person wearing the insulated shoes stands near the high electrified object, the instantaneous ground will occur sometimes. At this point, the charges of same polarity will flow out. At the higher opposite polarity, that person will be electrified. Static electricity can cause damage, so be careful when handling and using electronic components.

Caution:

To prevent damage from electrostatic discharge, follow the rules below.

- Do not touch ECM pins or the electronic parts soldered to the ECM circuit board.
- Do not unpack the parts unless the preparation of part installation is finished.
- Connect the packing and vehicle normal ground wire before taking the parts out of the packing.
- Ensure the parts to be installed is contacting the normal ground wire when moving the seat traverse, sitting from standing position or after walking for a certain distance and operating the parts.

Abbreviate table

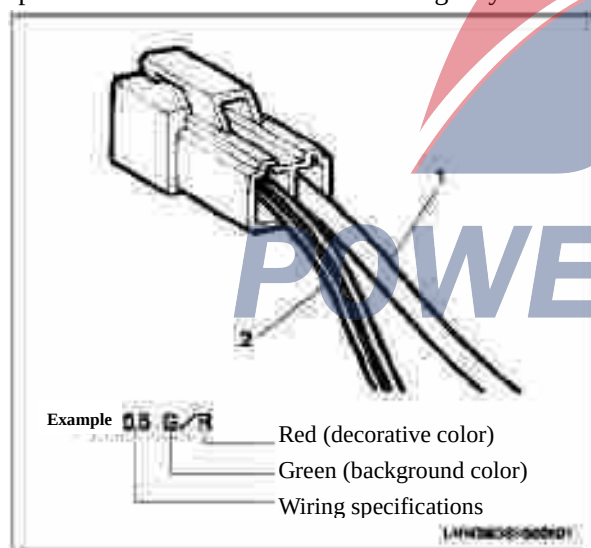
Abbreviation	Name	Meaning, remarks
A/D	Analog/Digital	Analog / Digital
APP	Accelerator Pedal Position	Accelerator pedal depression
ASR	Anti Slip Regulator	Detect wheel skidding and control engine output
CAN	Controller Area Network	A control unit communication
CKP	Crankshaft Position	Crankshaft (sensor mounted on flywheel housing) position
C/U	Control/Unit	Abbreviation of control unit
CMP	Camshaft Position	Cam (sensor mounted on the rear of the cylinder head) Position

Abbreviation	Name	Meaning, remarks
DLC	Data Link Connector	Scanning Tool Connection Connector (Other Names: Gage Connector)
DPD	Diesel Particulate Defuse	This refers to a continuous regenerative type PM post-treatment device that is composed of an oxidation catalytic converter and a filter and that can remove PM and Nox in the exhaust gas.
DTC	Diagnostic Trouble Code	It is the self-diagnosis code indicating the codes under various malfunction status
ECT	Engine Coolant Temperature	Engine coolant temperature
ECM	Engine Control Module	It is the engine control computer, which is the core of the engine control
ECU	Electronic Control Unit	Various control computers
E/E	Electrical/Electronic	Abbreviation for Electrical/Electronic Systems
EGR	Exhaust Gas Recirculation	The meaning of the exhaust gas in the cycle means that the gas discharged from the engine is again mixed with the inhaled gas to lower the combustion temperature, thereby reducing the Nox's system.
EHC	Electronic Hydraulic Control Unit	It means the actuator with built-in control unit and consisting of control unit and actuator to control ABS
EMI	Electro Magnetic Interference	Radio interference
Exh	Exhaust	Exhaust
F/B	Feed/Back	Abbreviation for feedback
FT	Fuel Temperature	It means the fuel temperature sensor is installed to the fuel feed pump
GND	Ground	Ground/Earth
GVW	Gross Vehicle Weight	The total weight of the vehicle
IAT	Intake Air Temperature	Suction temperature (outdoor air temperature)
ITP	Intake Throttle Position	Intake throttle switching value
J/C	Joint/Connection	Connection of each wiring
MAF	Mass Air Flow	Air intake
MIL	Malfunction Indicator Lamp	Warning lamp MIL (engine MIL)
PC	Pressure Control	Pressure Control / Common Rail Pressure

Abbreviation	Name	Meaning, remarks
PTO	Power Take Off	In addition to vehicles, power take-offs mounted on gearboxes and gearboxes to use other than the engine
PWM	Pulse Width Modulation	Pulse width modulated wave
QOS	Quick On Start	Warming up device
RP	Rail Pressure	Common rail inside pressure
SBF	Slow Blow Fuse	Delay type fuse for protecting circuits such as batteries and motors
SCV	Suction Control Valve	The valve that controls the fuel flow of the common rail is installed on the fuel pump.
SIG	Signal	Signal
VSS	Vehicle Speed Sensor	Sensors for detecting vehicle speed and controlling instruments and engines
W/S	Weld/Splice	No connector is used for each connection

About wiring color

All the wirings can be identified through the color of the cladding. In a certain electrical system, the wiring of the main circuit can be identified by monochromatic color. In addition, the wiring of the auxiliary circuit can be identified by the color of the decorative strip. In the wiring diagram, express the specification and color in the following way.



2. The color of the decorative strip

Name

1. Monochrome

Mark	Color difference	Mark	Color difference
BLK	Black	BRN	Brown
WHT	White	LT GRN	Light green

Mark	Color difference	Mark	Color difference
RED	Red	GRY	Grey
GRN	Green	PNK	Pink color
YEL	Yellow	LT BLU	Light blue
BLU	Blue	VIO	Purple
ORN	Orange		

Function and action description

- Self-diagnosis function.

On the engine control (common rail) system

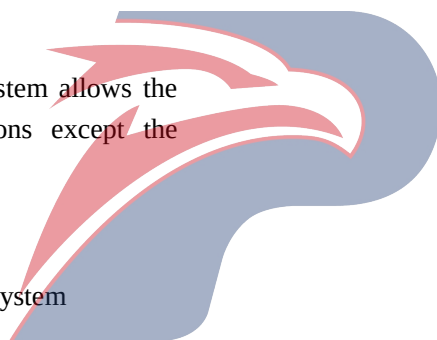
The summary and expansion of the engine control system

The engine control system refers to an electronic control system that always controls the engine to the best combustion state according to the driving conditions.

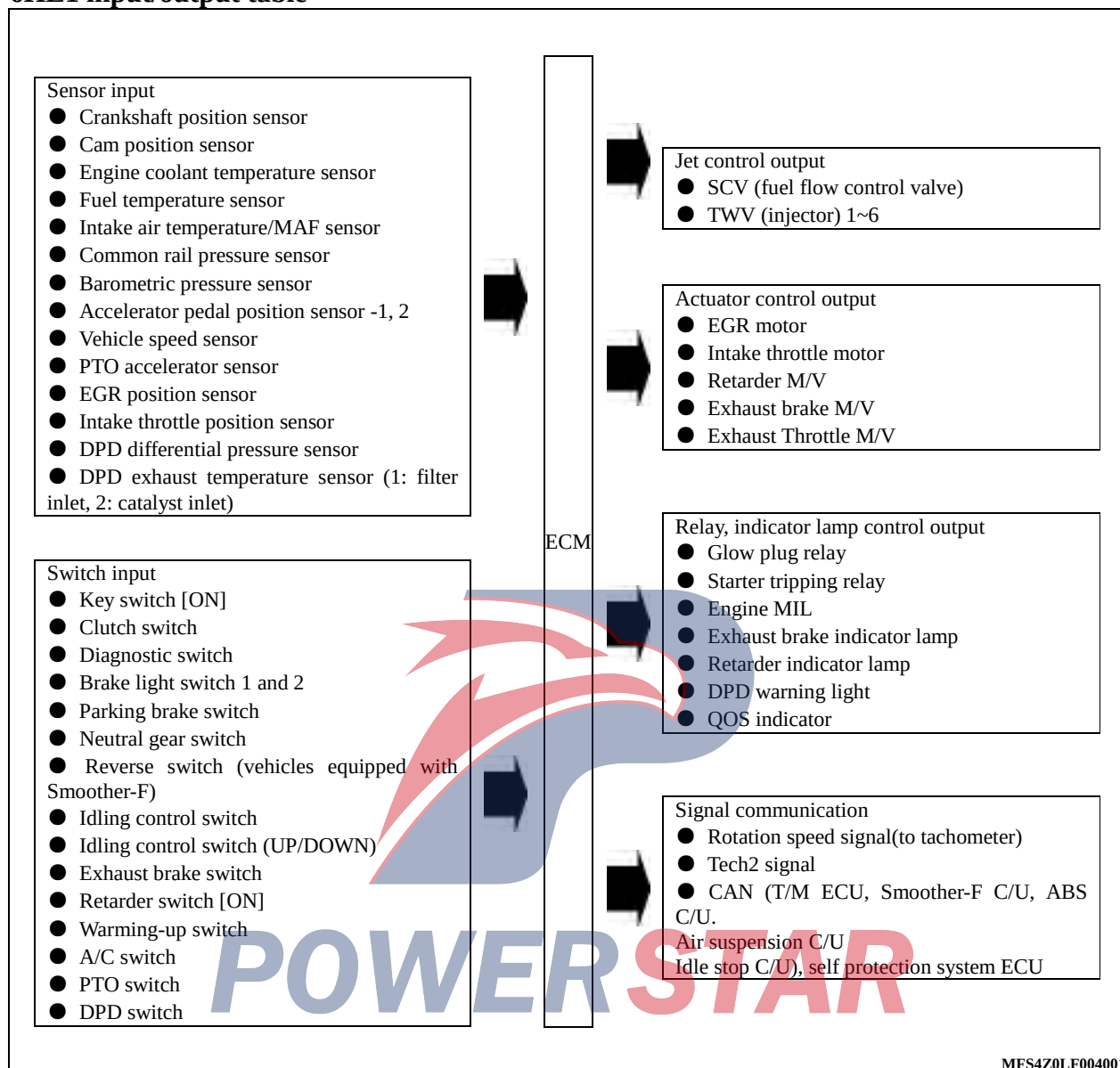
- EFI system (common rail type)
- EGR (exhaust gas recirculation)
- Idle speed control
- PTO control composition.

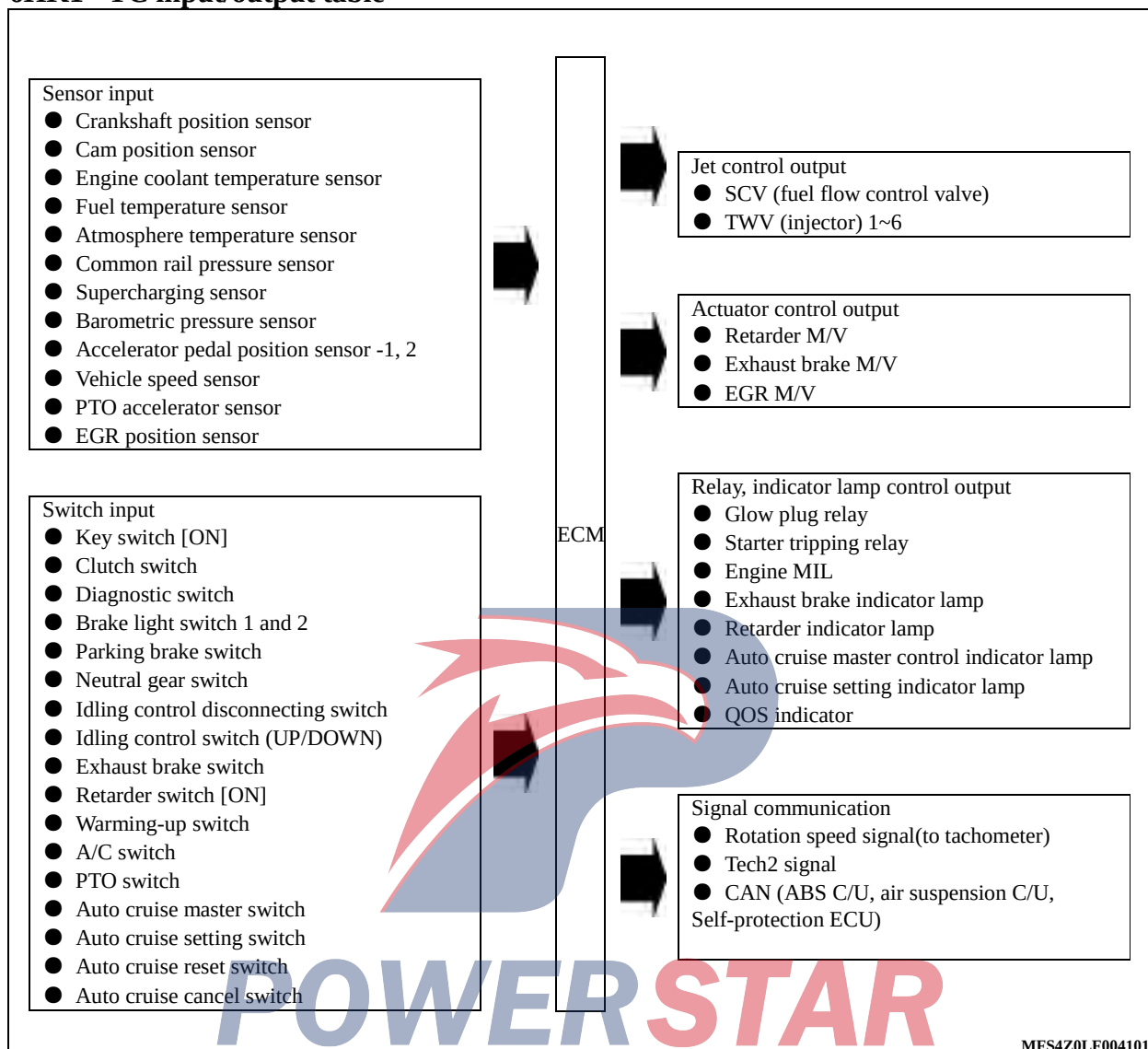
In addition, the engine control system allows the following system control functions except the engine control:

- QOS (heating device) system
- QWS.
- DPD (Diesel Particulate Filter) System
- Automatic cruise control.
- Auxiliary brake inter locking auto cruise control.
- Control of speed control device (function).
- Communication function and injection quantity control with Smoother-F control unit, ISS (idle stop & start) system, self-protection system, etc.
- Engine speed output fed to tachometer.



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6HL1 input/output table

6HK1 - TC input/output table**EFI system (common rail type)**

Engine Control Module (ECM:ECM acquires engine speed and engine load (from the output signals of various sensors), transfers the electronic signal to the fuel feed pump and injector according to this information and controls the fuel injection volume and injection time of each cylinder under the reasonable state.

Injection volume control

In order to optimize the injection quantity, the injector coil is mainly controlled based on signals of the number of engine revolutions and the accelerator opening so as to control the fuel injection amount.

Control of injection pressure

The injection pressure is controlled by controlling the fuel pressure in the common rail. The reasonable pressure inside the common rail can be worked out according to the engine speed and fuel injection volume. This pressure is controlled by controlling the fuel feed pump to inject the appropriate fuel volume and transport to the common rail.

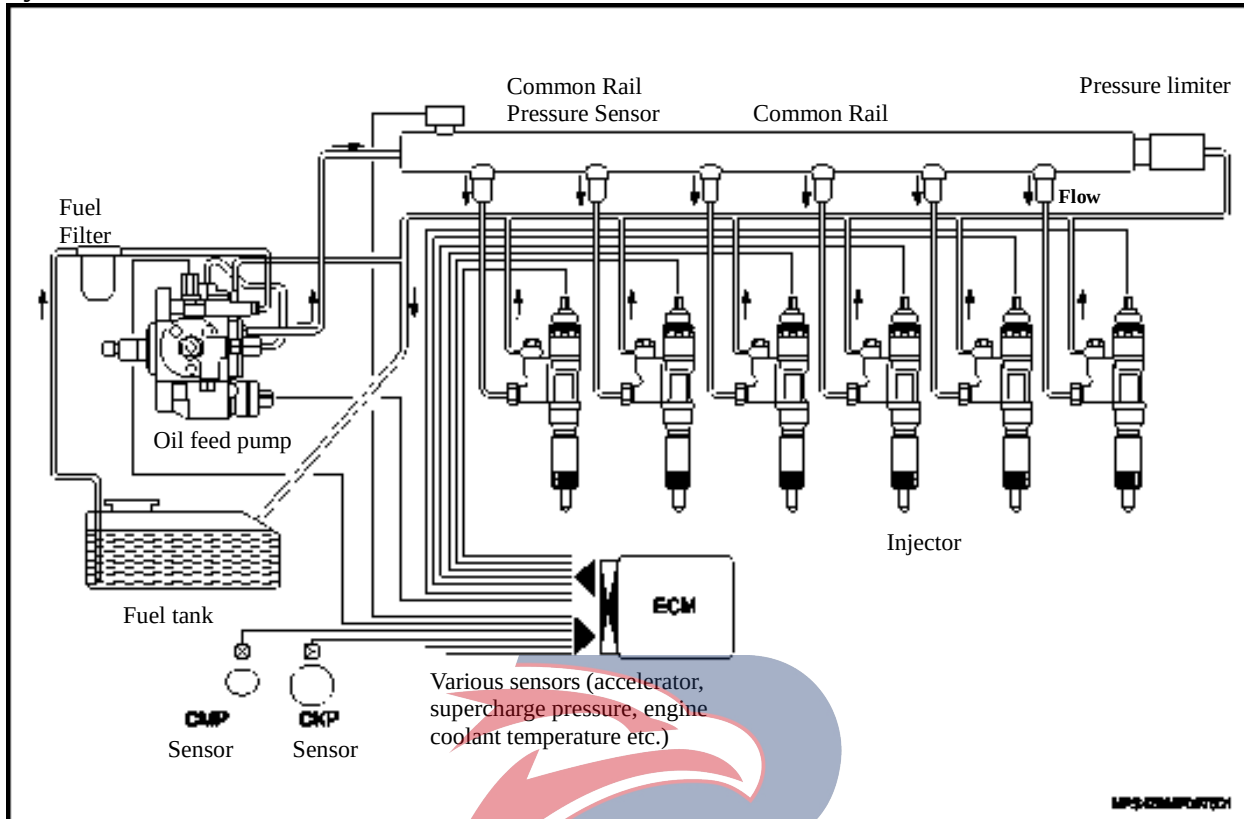
Injection time control

It has a function that can replace the timing injection, and appropriate fuel injection timing is mainly calculated based on the number of revolutions of the engine, the injection amount, and the like, and the timing injection is performed by controlling the injector.

Injection rate control

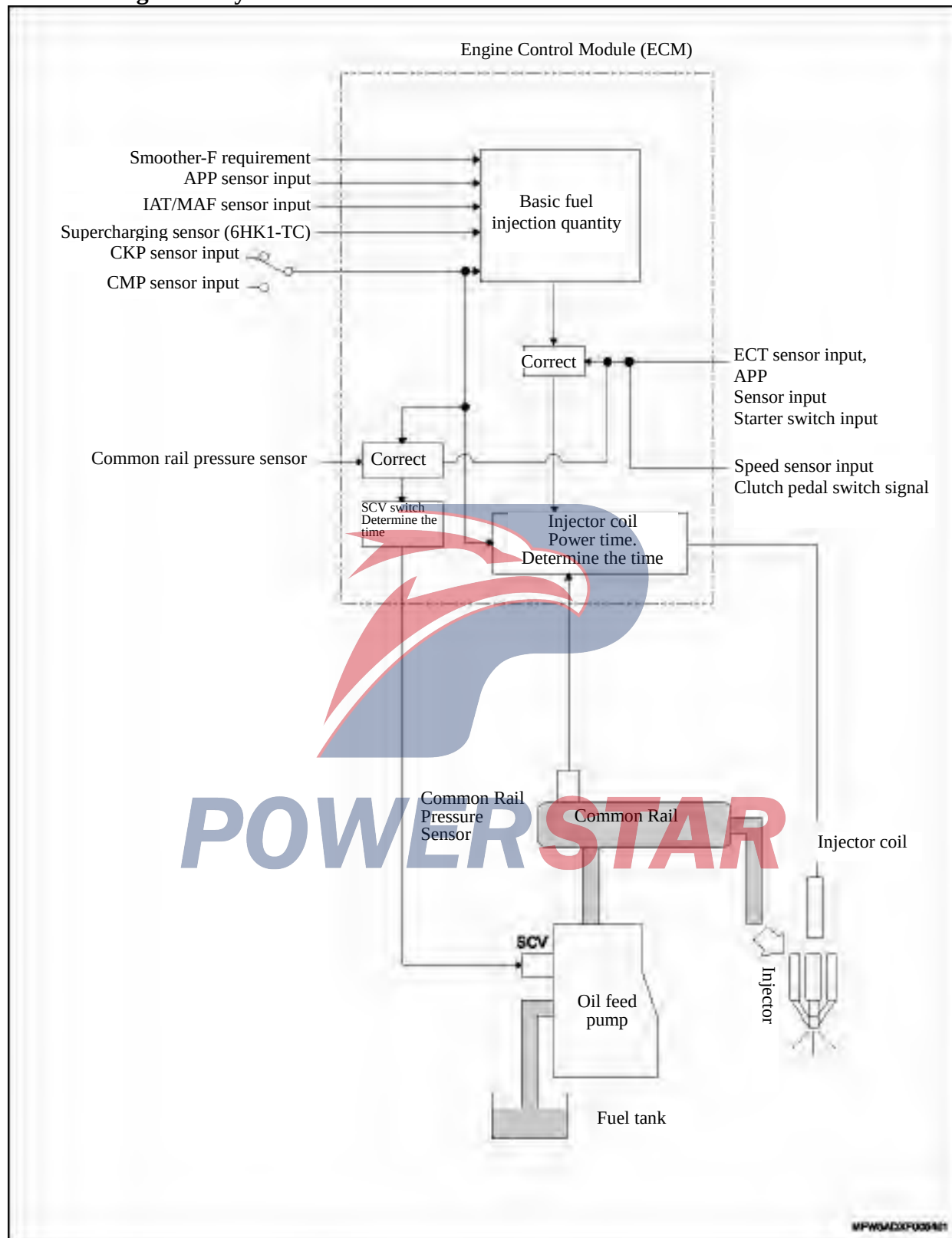
In order to increase the combustion efficiency in the cylinder, the fuel injection (direct injection) is first ignited, and the second injection (main injection) is performed at the time of ignition. Injection time and injection volume are controlled by controlling the syringe (syringe coil).

System Overview



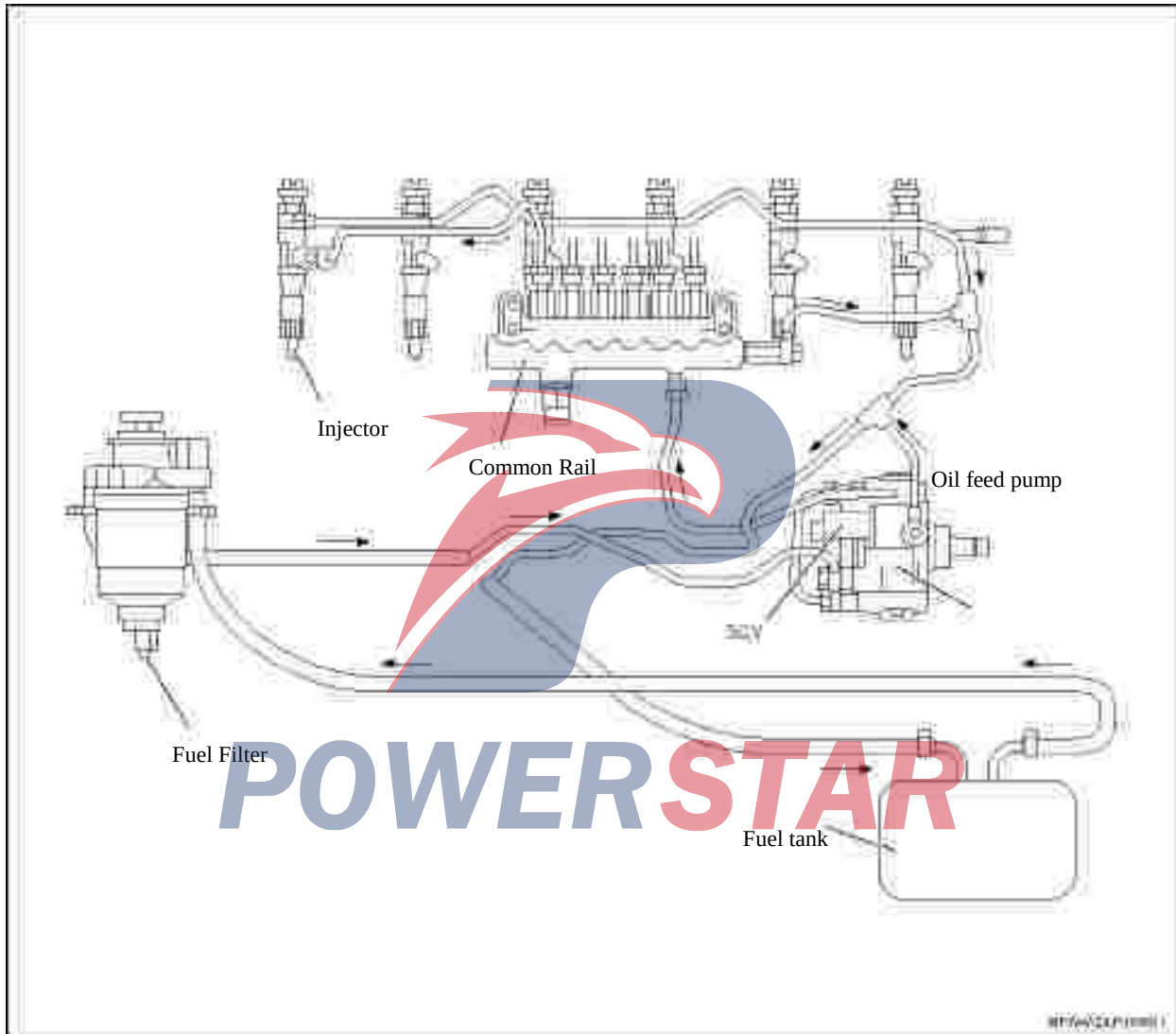
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Skeleton diagram of system control



Fuel System

The fuel is supplied to the fuel pump from the fuel tank, and then supplied to the common rail after being boosted by the oil pump. At this point, the signal from ECM will control the volume of fuel fed to the common rail through controlling the fuel flow control valve (SCV).



EGR (exhaust gas recirculation)

The EGR system is the abbreviation of the [Exhaust Gas Recirculation] system, which is to recirculate part of the gas in the intake manifold, and to reduce the combustion temperature by mixing the inhaled air into the inert gas, thus inhibiting the production of the nitrogen and oxygen compounds (NOx).

The amount of exhaust gas recirculation is controlled by the operation (switch) of the EGR valve between the exhaust manifold and the intake manifold. It determines the EGR volume according to the engine speed and engine load ratio (the fuel injection volume) to make the EGR valve work and consequently control the EGR volume.

The EGR gas channel is equipped with a cooling

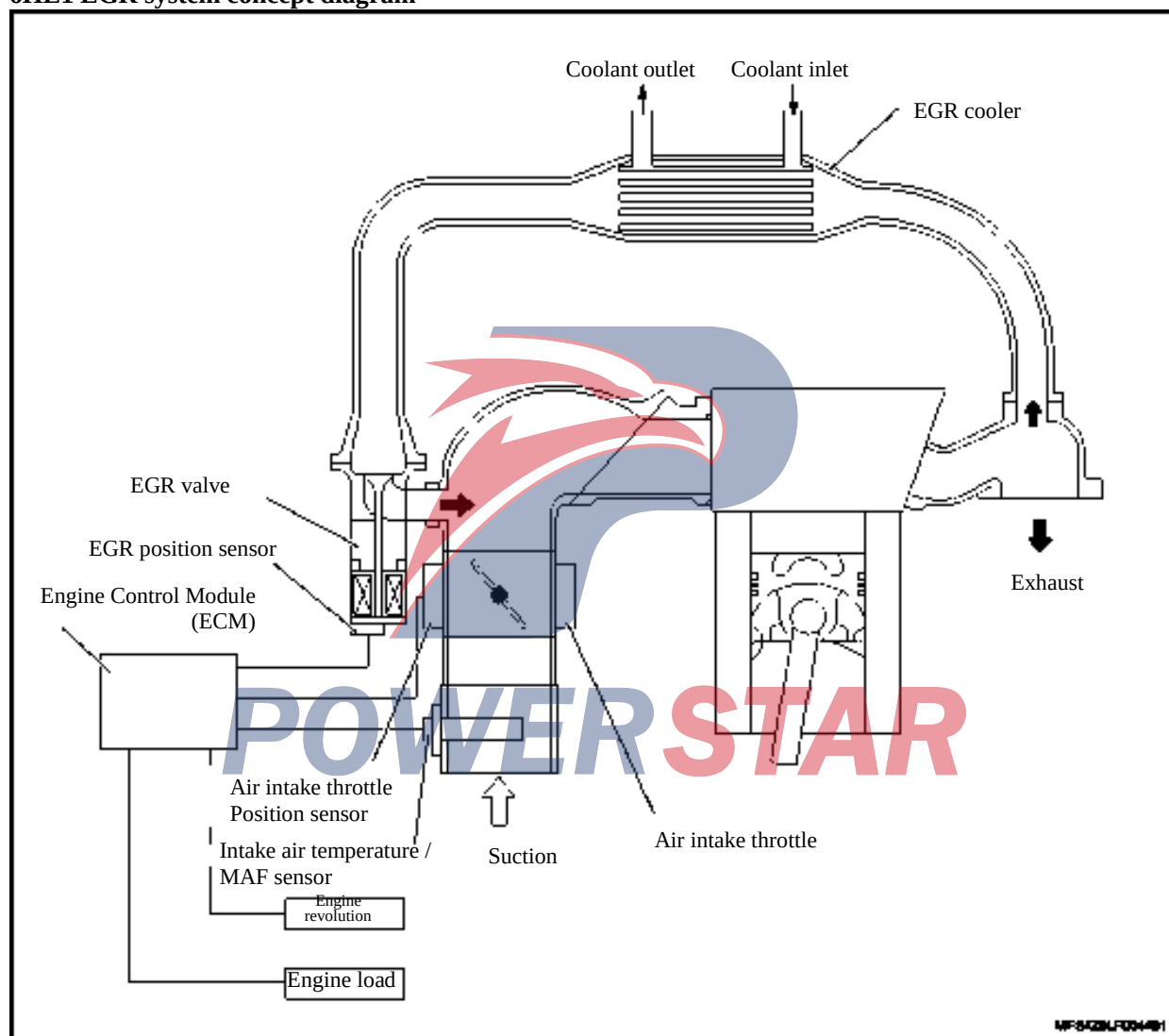
device (cooler), which can cool the high temperature EGR gas, and can reduce the combustion temperature to lower than the usual EGR temperature by mixing with the newly inhaled air, and will contribute to the reduction of the nitrogen oxides (exhaust gas recirculation).

The 6HK1-TC vehicle also uses a check valve on the EGR system to control the return flow of EGR gas and restricts EGR gas flow in only one direction, thereby increasing the amount of exhaust gas recirculation.

6HL1 EGR control

ECM will drive the motor to run according to the engine revolution and engine load and consequently control the EGR valve lift. Valve lift can be detected using the EGR position sensor.

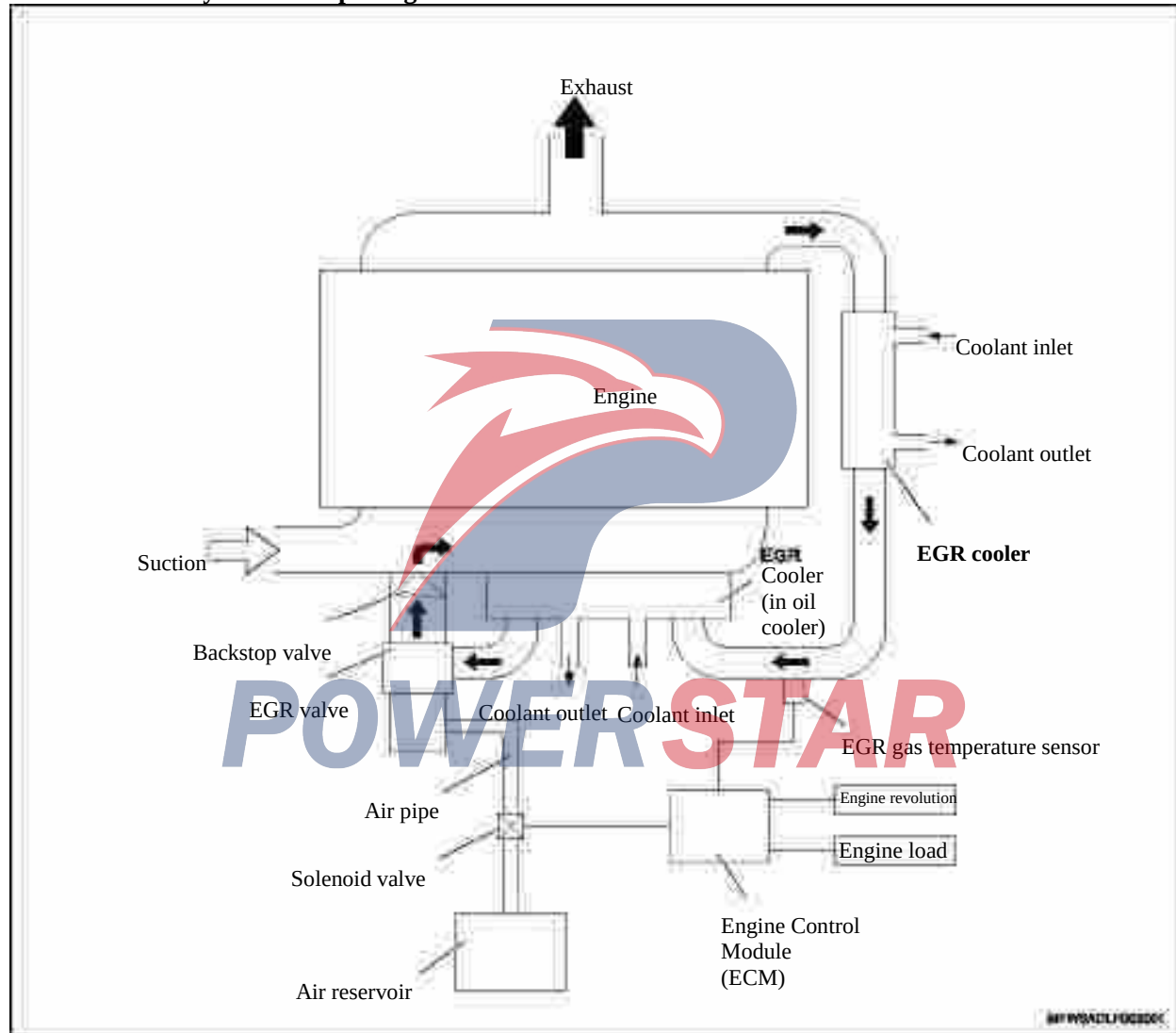
In addition, intake throttle control performed by the intake throttle valve is also performed at the same time, and the intake throttle valve is closed to perform control during EGR operation.

6HL1 EGR system concept diagram

6HK1-TC EGR control

The EGR valve is controlled by a solenoid valve (M/V) [ON] [OFF] to control the gas supply. ECM will bring the magnet valve to [ON] if the engine coolant temperature and other vehicle conditions are established and then EGR valve will open. In addition, an EGR gas temperature sensor is installed on the upstream side of the EGR valve, and the operating state of the EGR valve may be detected based on the temperature of the EGR gas.

6HK1-TC EGR system concept diagram



Idle speed control

During idle running, the minimum idle speed can be adjusted by operating [idle control switch]. In addition, by operating the idle speed control switch, the minimum idle speed can also be automatically adjusted to a reasonable value according to the engine cooling temperature.

In addition, this function is inactive in [PTO control] mode

Idling control switch

The idle speed control function can be switched by operating [idle control switch].

AUTO·Auto mode

Adjust the minimum idle speed to a reasonable value based on the engine coolant temperature.

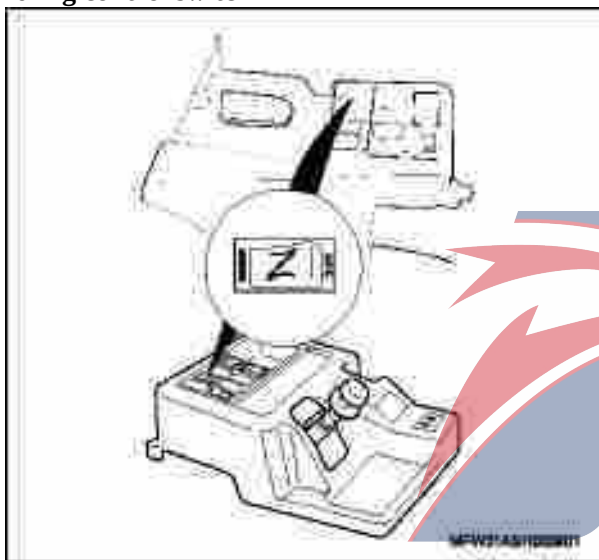
When the coolant temperature is low, the idle speed will increase. When the coolant temperature rises, the idle speed will decrease.

When using the idle speed control switch to adjust idle speed, you need to switch to manual mode.

MANU·Manual mode

The idling speed can be adjusted through the idling control switch.

As the lowest idle speed when switching to automatic mode, the number of rotations set in the manual mode is also valid.

Idling control switch**Idle speed control switch**

The idle speed can be adjusted by operating the [Idle Control Switch].

Mark:

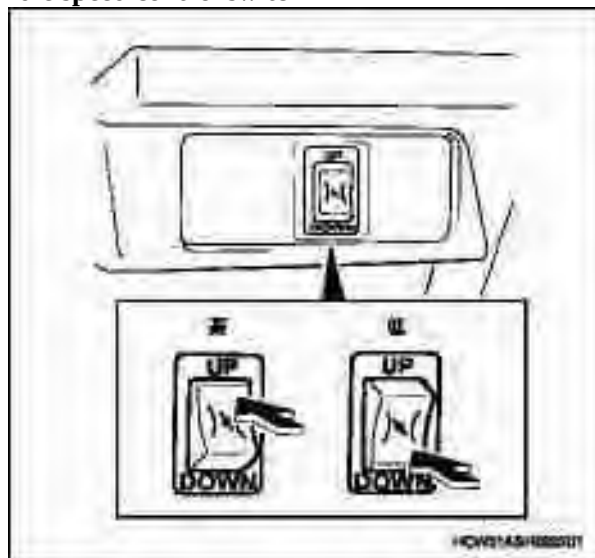
The maximum number of rotations varies depending on the vehicle specifications and the warm-up state of the engine.

UP

When the [UP] side of the switch is pressed, the number of engine revolutions increases, and the number of idling revolutions can increase.

DOWN

When the [DOWN] side of the switch is pressed, the number of revolutions of the engine is reduced, and the number of idling revolutions can be reduced. Nonetheless, it cannot reach the state in which the speed is lower than the minimum idling speed.

Idle speed control switch**PTO control**

The so-called PTO control refers to [governor characteristics switching] and [accelerator switching] depending on the use of rack installation. [Drive Accelerator] and [External Accelerator] can be used separately when an external accelerator is connected.

Switch of accelerator

It means the switching of driving accelerator (APP sensor input) and external accelerator (PTO accelerator sensor input).

After connecting an external accelerator, both the driving accelerator and the external accelerator can be used.

The conditions for using an external accelerator (external accelerator and drive accelerator are valid at the same time) require that all of the following conditions (AND conditions) be satisfied:

- Vehicle parked (the vehicle speed below 20km/h).
- The gear position is in the neutral position
- The accelerator pedal of the accelerator is at the idle position.
- No DTC related PTO.
- The number of engine revolutions is above 500 rpm..

Only when the above conditions (vehicle speed, gear, accelerator pedal, DTC, engine revolutions) are all established and PTO switch [ON] or external accelerator [ON], external accelerator opening will be activated.

The condition that the external accelerator becomes invalid and one of the following conditions holds (OR condition):

- DTC related PTO is detected.
- The sensor is abnormal.
- The gear is in normal state.
- Speed 25km / h or more
- PTO switch in [OFF] state and external accelerator is [shut down].

Governor characteristics switching (switching operation characteristics and operating characteristics)

The governor characteristics can be switched during input of the PTO switch to the engine control module (ECM) only when an external accelerator is connected. However, the [Accelerator Function] switch can only be used when the vehicle is parked.

The conditions for switching from the operating characteristics to the operating characteristics (conditions in which the operating characteristics can

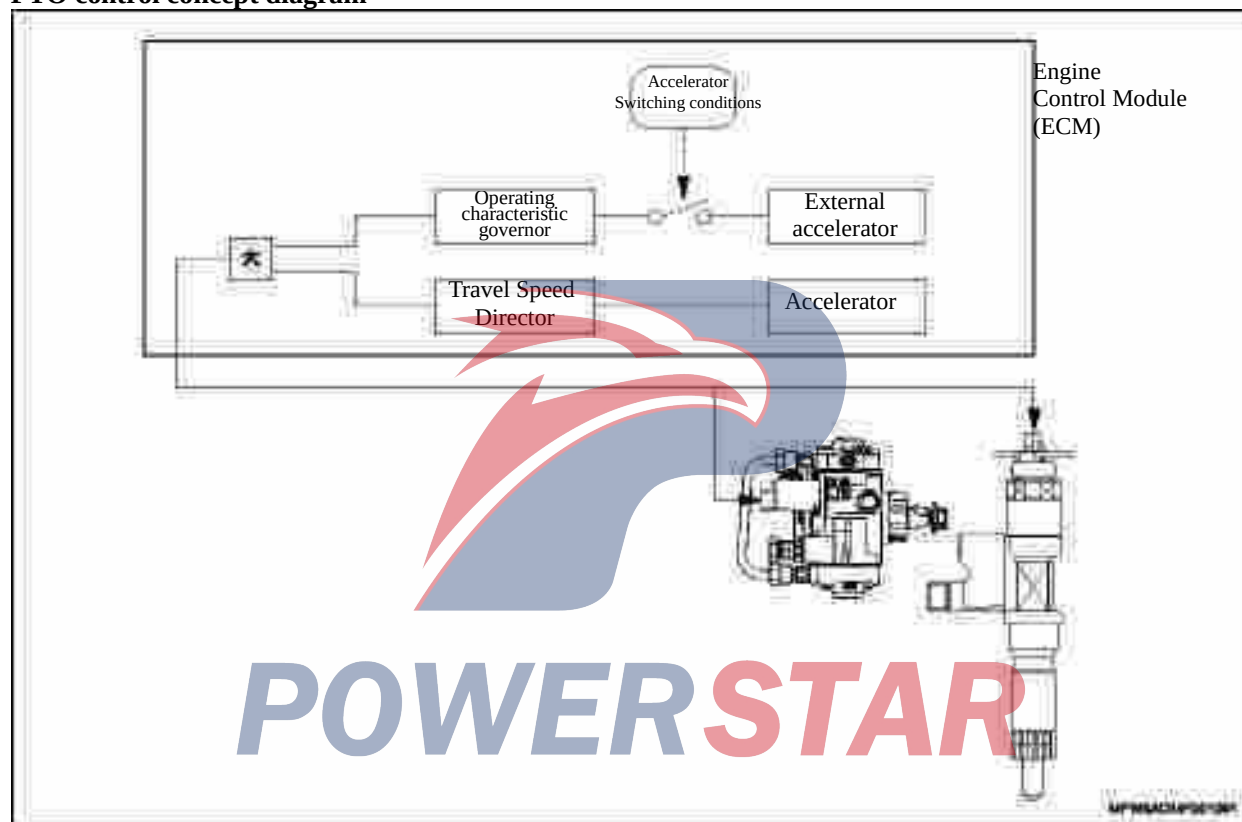
PTO control concept diagram

be used) require that all of the following conditions (AND conditions) be satisfied.

The governor's characteristics can be switched only when the accelerator conditions are met:

- It cannot become the operation characteristic unless the external accelerator is [ON].
- It cannot become the driving characteristic unless the driving accelerator is [ON].

In addition, ECM will judge the higher output side (the higher fuel injection volume side) and reflect to the engine.

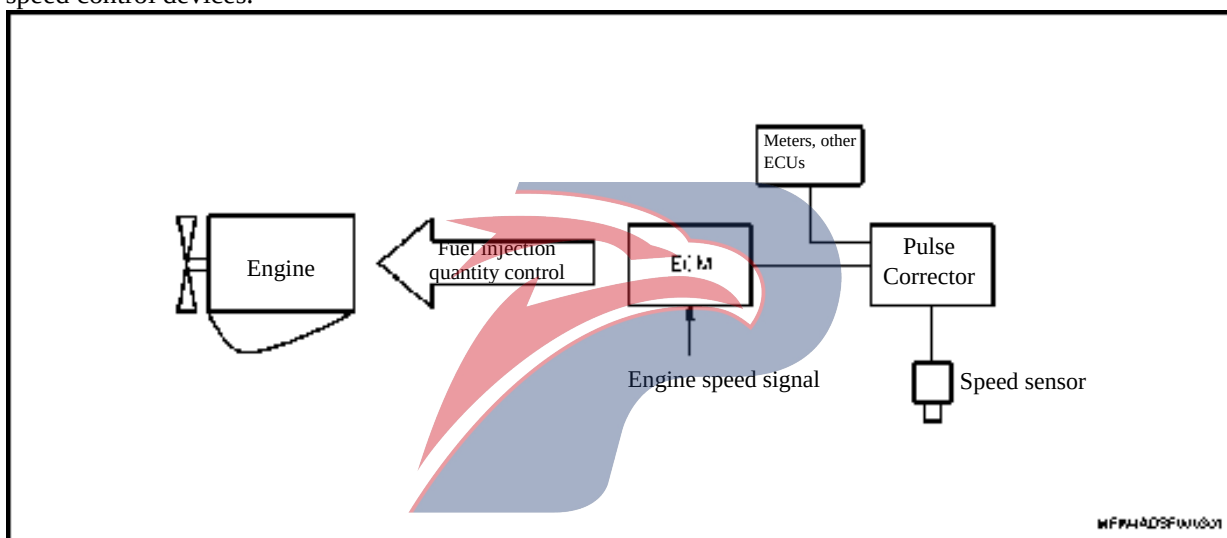


Speed control

The speed control means ECM controls the vehicle speed within the setting value (90km/h) according to the vehicle speed signal and engine speed and consequently controls the fuel injection volume.

The vehicle speed signal is transmitted to the pulse corrector by the vehicle speed sensor and the pulse corrector will correct this signal according to the vehicle conditions and transmit it to ECM and instruments. The ECM recognizes signals from the crankshaft position (CKP) sensor regarding the number of engine revolutions.

According to the "Road Transport Vehicle Law," vehicles that are fully loaded with vehicles of 8 tons or more or 5 tons or more must be equipped with speed control devices.



Communication function with ABS / ASR control unit and fuel reduction control

Communication with ABS / ASR control unit

ECM can carry out CAN communication with ABS/ASR control unit.

- ECM → ABS/ASR control unit (normally in transfer status)

ECM will take the current accelerator pedal depression volume as the ASR accelerator opening signal and transmit it to ABS/ASR control unit.

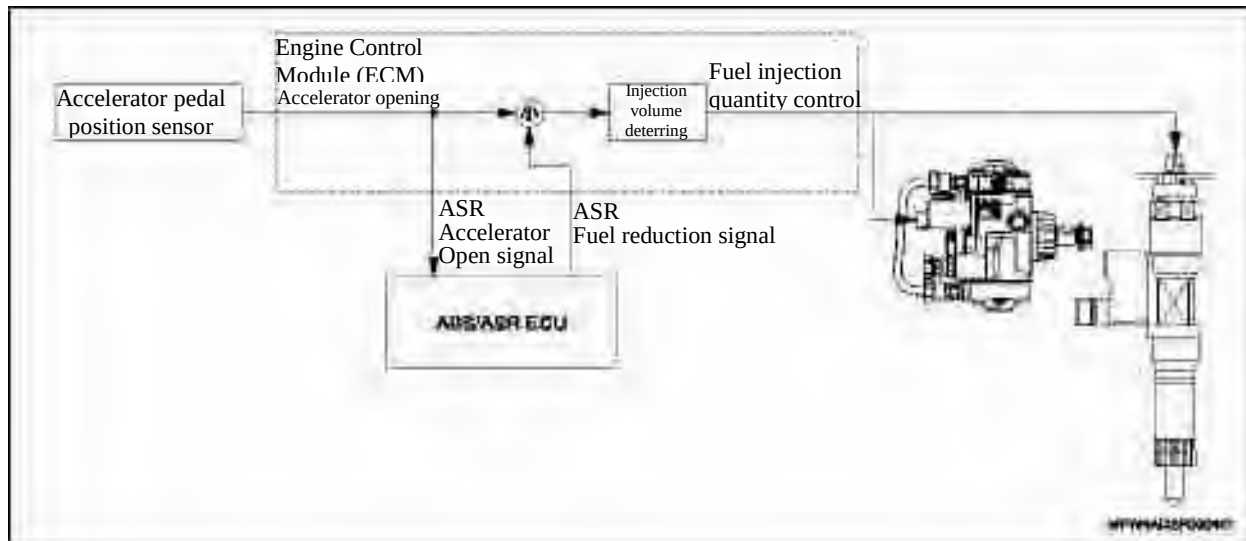
- ABS/ASR control unit → ECM (transfer allowed only in ASR control)

The ABS/ASR control unit transmits the accelerator down pressure controlled as the ASR fuel reduction signal to the ECM.

Fuel reduction control

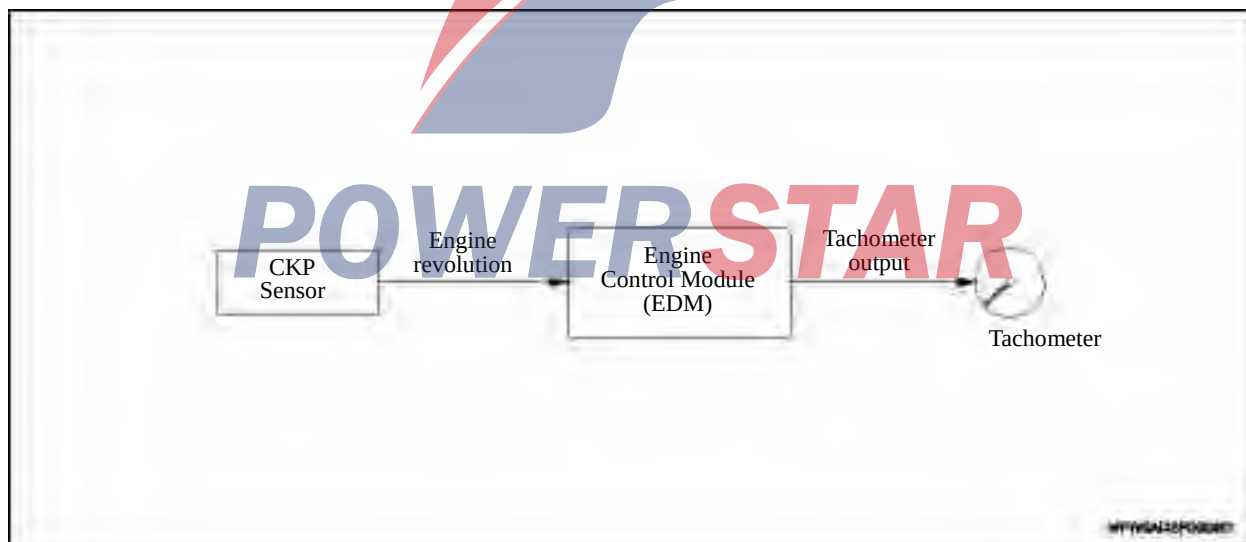
When detecting driving wheel slipping, ABS/ASR control unit will transfer the ASR fuel reducing signal from the ASR accelerator opening signal to ECM. According to the ASR fuel reduction signal,

the ECM performs throttling control of the fuel injection quantity and controls the output of the engine, thereby suppressing the slipping phenomenon of the driving wheel.



Engine speed output to tachometer

The engine control module (ECM) will generate the engine revolution pulse (the engine generates 4 pulses per revolution), which is used as the output to the tachometer. The tachometer can display the engine speed according to the engine speed pulse from ECM.



About control of exhaust brake

The exhaust brake is provided with valve inside the exhaust pipe. Closing the valve can increase the exhaust stroke resistance and enhance the engine brake effect. The action of the exhaust brake valve is controlled by air pressure. As a solenoid valve is provided to control the supply of the air pressure, the exhaust brake is controlled by the electronic control of the switch of the solenoid valve. Solenoid valve

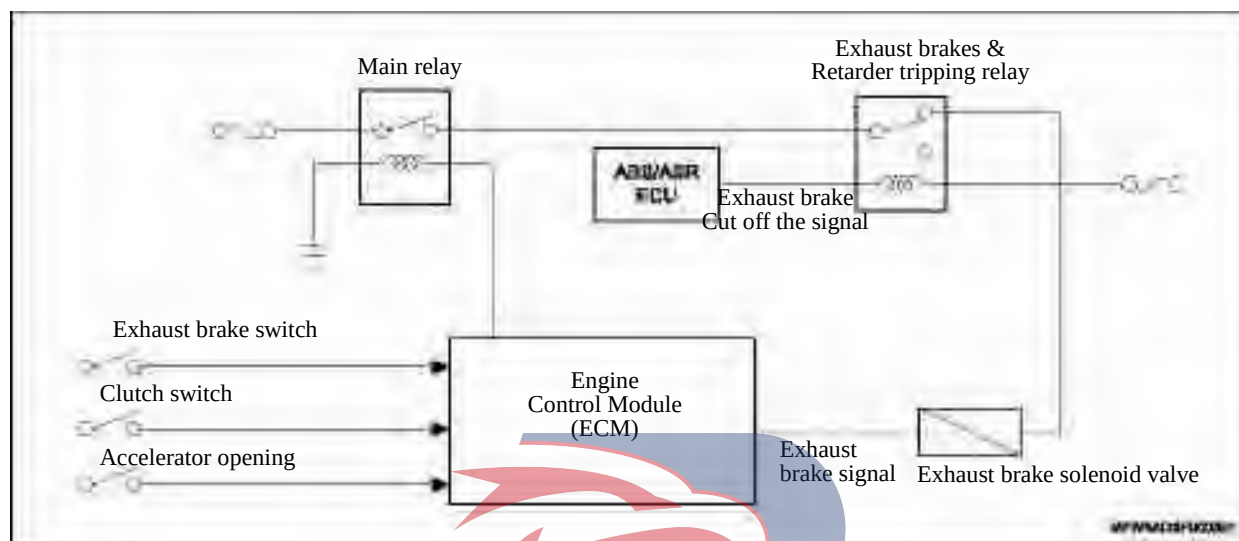
The engine control module (ECM) will be in the [ON]

state when the engine revolutions exceed 525 rpm and the exhaust brake operating conditions are all met.

Exhaust brake operating conditions

- Exhaust brake switches [ON].
- Accelerator pedal in idling position.

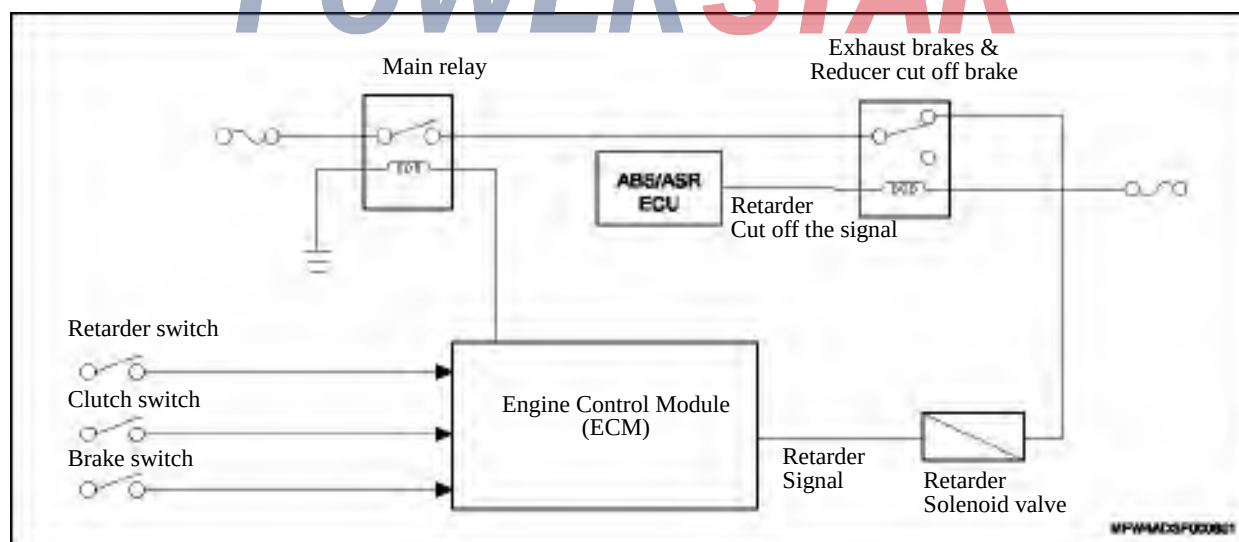
- Abnormal accelerator pedal position (APP) sensor, abnormal exhaust brake circuit, abnormal clutch switch, abnormal A/D conversion.
- Clutch pedal not depressed
- System voltage above 24V

**About retarder control**

The retarder uses the permanent magnet and control the propeller shaft rotation through the high magnetic force. When the working conditions of the retarder are all established, (ECM) will make the solenoid valve (ON) and retarder work.

Retarder operating conditions

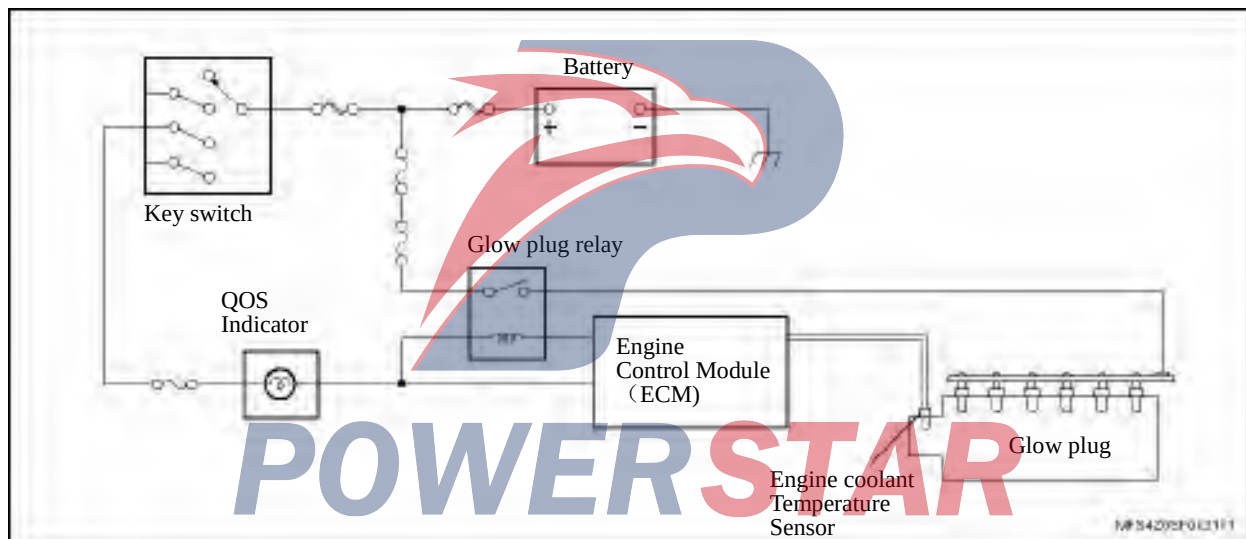
- Retarder switch [ON]
- Clutch pedal not depressed
- Accelerator pedal in idling position
- Retarder not switched off



Preheat control

QOS (Quick Start) System

QOS system can start with ease even if the temperature is low. In addition, it can reduce the white smoke and noise generated after the start. When the key switch is turned on, the ECM will detect the engine coolant temperature based on the signal from the engine coolant temperature (ECT) sensor and will change the warm-up time so that reasonable starting conditions are always obtained. . In addition, since it has a glow plug to continue the heating function, the idle speed after starting is also very stable. ECM decides the warming-up time according to the engine coolant temperature to drive the glow plug relay and QOS indicator lamp to work.



Warming up control

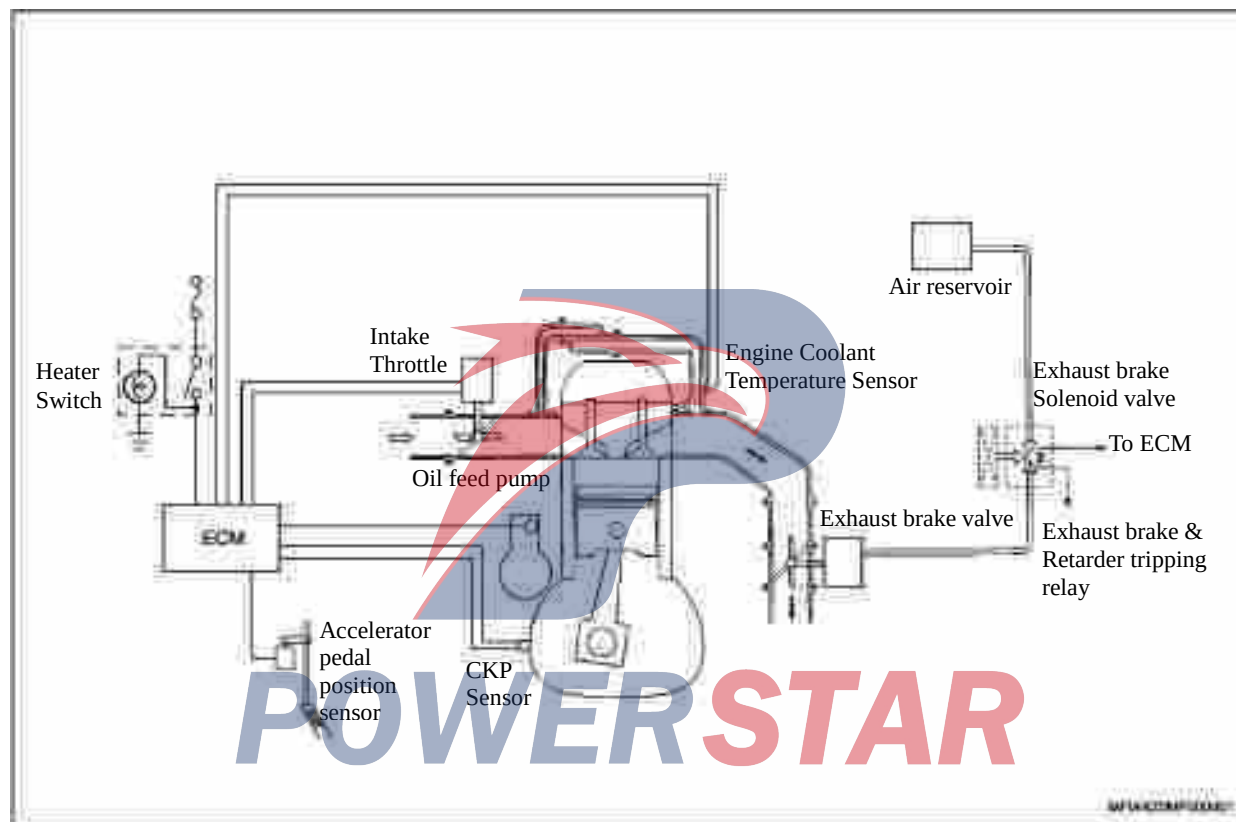
QWS (Quick Warming-up System)

While improving warm-up performance, QWS can also reduce the amount of white smoke produced at low temperatures. By operating each throttle, the engine load due to the exhaust pressure can be increased, and the cylinder can be used to transfer the energy generated by the engine to the engine coolant, thereby improving the warm-up performance. In addition, by increasing the fuel injection amount at the same time, it is also possible to increase the temperature of the exhaust gas and promote warm-up.

Operation of 6HL1 QWS

When the engine coolant temperature, engine speed, vehicle speed and accelerator pedal opening all reach the specified conditions, turn on the warming up switch and the ECM will switch on the exhaust brake magnet valve while controlling the intake throttle valve. In addition, controlling the SCV (supply pump) can also increase the amount of fuel.

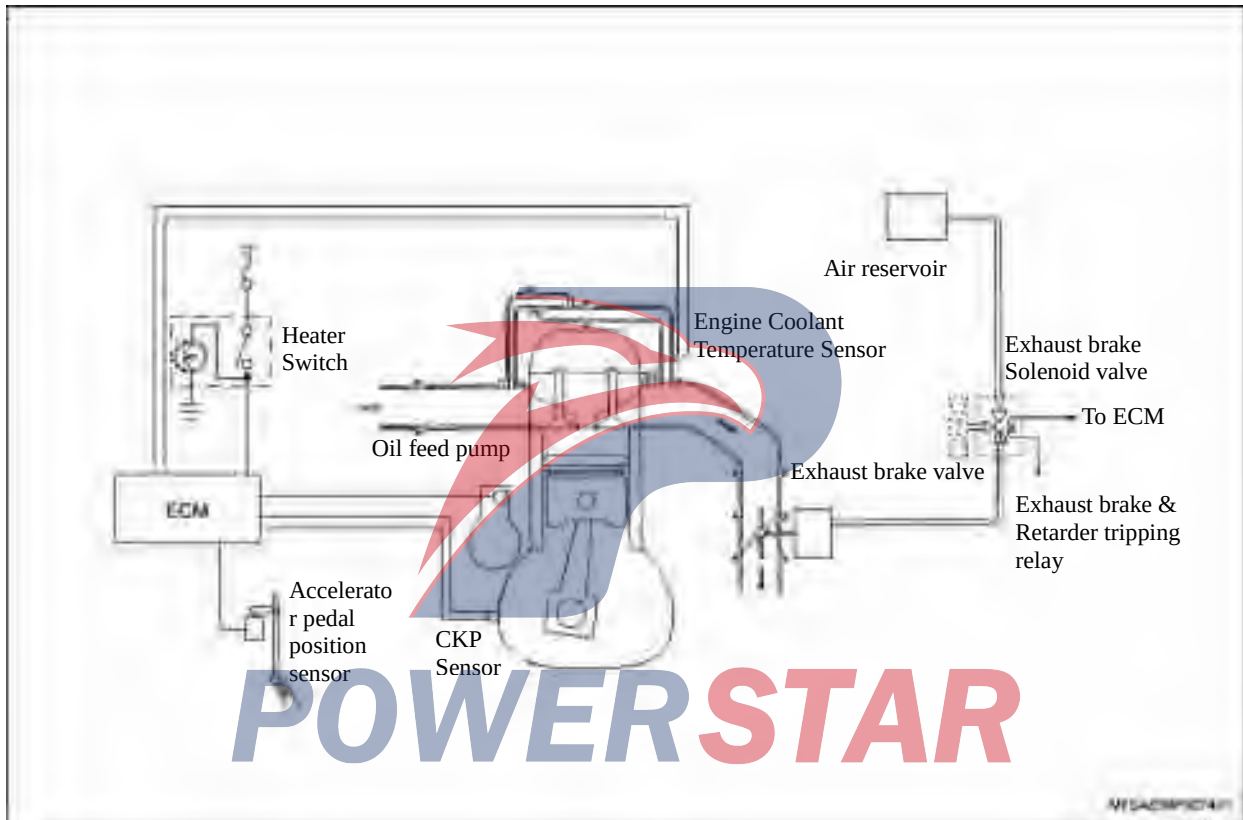
6HL1 QWS concept diagram



Operation of 6HK1-TC QWS

When the conditions such as engine coolant temperature, engine rpm, vehicle speed and accelerator pedal opening all reach the specified state, after the warm-up switch is turned on, the ECM will open the exhaust brake solenoid valve, and at the same time control the SCV (fuel supply pump) to increase the fuel quantity.

6HK1-TC QWS concept diagram



DPD (Diesel Particulate Defuse)

Control (6HL1 only)

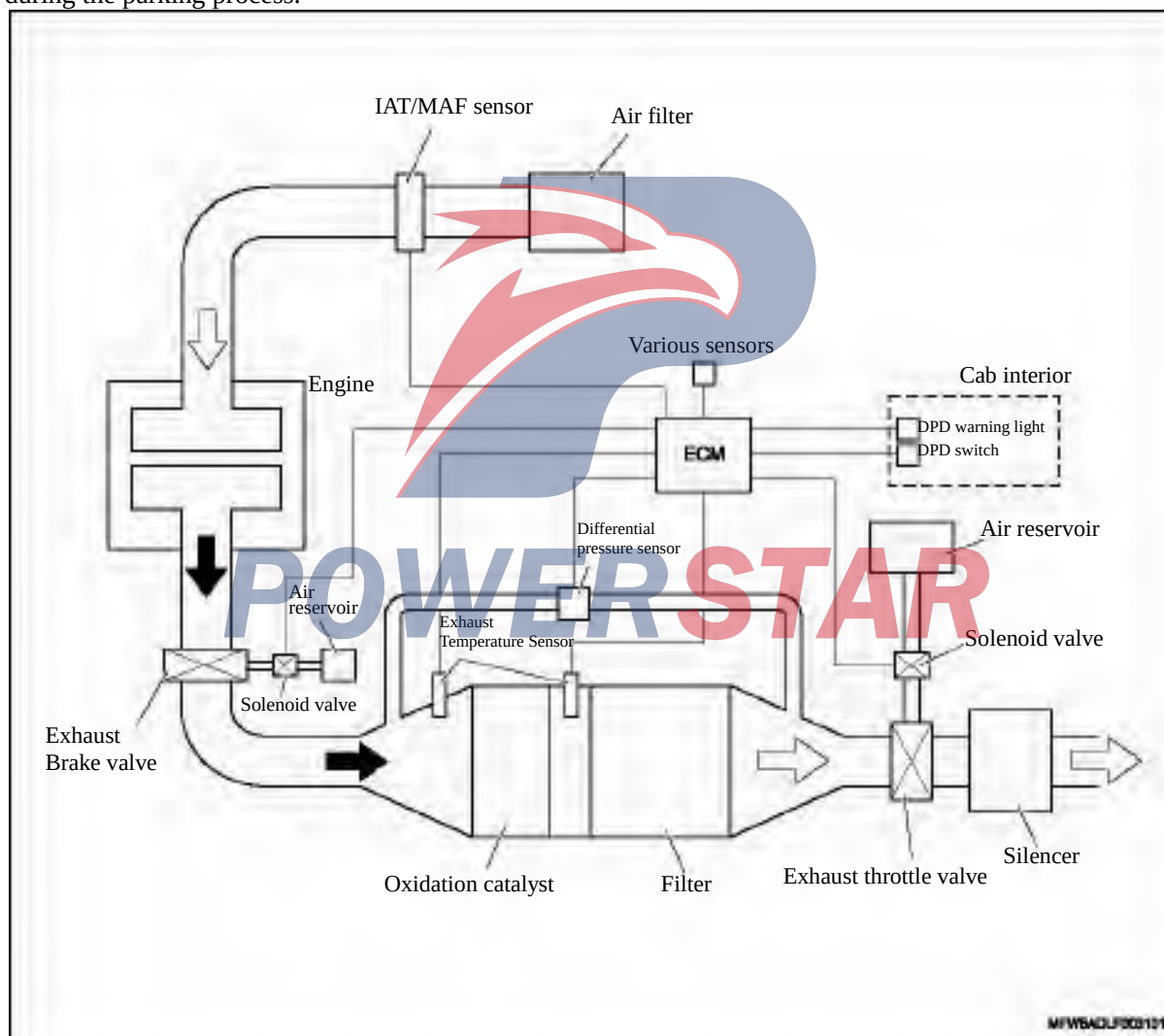
DPD system

DPD system catches PM (particle matters) to the filter to enable the auto reactivation (combustion) and purge the emission gas.

Jobs

DPD system can judge the PM deposit status in DPD according to the driving distance and DPD differential pressure before and after to carry out DPD reactivation (P, combustion).

When DPD is regenerated, ECM will adjust the temperature in the filter to the appropriate temperature. When the exhaust gas temperature sensor detects the temperature of the exhaust gas, it controls the fuel injection and the exhaust throttle to raise the temperature in the filter and perform PM combustion (automatic regeneration). DPD warning lamp will blink if the auto reactivation is not completed under the driving conditions. At this point, the DPD switch can be used for manual regeneration during the parking process.



Auto cruise control (for 6HK1-TC only)

This is a function that allows the vehicle to run at a set speed (in the range of 40 to 90 km/h without a speed limiter set to 100 km/h) without stepping on the accelerator pedal. When the cruise master control switch reaches the [ON] position (cruise master control indicator lights up) and the desired speed is reached, the operation [press → release] cruise

setting switch will become the auto cruise (cruise setting) indicator lights up. At this time, even if the accelerator pedal is released, it is possible to continue running at the desired speed.

Temporary acceleration function

During the auto cruise driving, when the accelerator pedal is depressed, the speed will increase temporarily. When the accelerator pedal is released, it can return to its original auto-cruise state even if it

Set deceleration function (reduce the setting speed of auto cruising)

During the auto cruise driving, pressing the cruise setting switch can reduce the speed. When decelerating to the required speed, after releasing the cruise setting switch, the set speed is updated to the vehicle speed.

In addition, short-term operation can also reduce the set speed (each operation speed can be reduced to 1km / h).

Set acceleration function (increases the setting speed of auto-cruise)

During the auto cruise driving, if the cruise reset switch is operated, the speed will increase. When the speed is accelerated to the required speed, after releasing the cruise reset switch, the set speed is updated to the current speed.

In addition, short-term operation can also increase the set speed (each operation speed can be reduced to 1km / h).

Automatic cruise discharge

The auto-cruise feature can be released under one of the following conditions.

- When the cruise master switch is set to the [OFF] position (Cruise master switch: OFF).
- When the vehicle speed is below 30km/h
- When the vehicle speed is lower than the set speed in the cruise mode by 40km/h or more..
- The cruise setting switch and cruise reset switch are operated at the same time (both cruise setting switch and reset switch are in ON position).
- Engine control system abnormal (refer to DTC).
- Engine stop running.
- Clutch pedal depressed (clutch switch: OFF).
- Foot brake depressed (brake switch -1: ON, or brake switch-2: ON).
- Exhaust brake lever operated (exhaust brake switch: ON).
- Retarder tie rod operated (retarder switch: ON).

Auxiliary brake linkage automatic cruise control

During the auto cruise driving, enable the auto interlocking of auxiliary brake (the exhaust brake and retarder) to restrain the speed increase function on the gradual downhill road. Its purpose is to reduce the driver's driving burden (foot brakes need to be depressed due to an increase in speed, etc.) and improve safety. .

In the case of turning to downhill under the auto cruise state, if the vehicle speed exceeds the cruise setting speed, the exhaust brake will work to restrain the speed rise. When the vehicle speed increases during the operation of the exhaust brake, the retarder operates on the basis of the exhaust brake to suppress the increase in speed. (The stop lamp will be on when the auxiliary brake works).

When the vehicle speed decelerates to the cruising set speed, the auxiliary brake can be released (separated in the order of 1 speed reducer and 2 exhaust brakes).

- Cruise cancel switch operated (cruise cancel switch:

OFF).

- Gear lever in neutral position (neutral gear switch: ON)

Reset function (restore to auto cruise state)

When the auto-cruise function is disabled, the auto-cruise state before release (the CRUISE SET indicator lights up) can be restored after the cruise reset switch [pressed → release].

However, if the auto cruise state is released according to the following conditions, it cannot restore the auto cruise state before the release.

The automatic cruising state needs to be set again.

- Cruise master switch in [OFF] position (cruise master switch: OFF).

- When the vehicle speed is below 40km/h.

When the vehicle speed is lower than the set speed in the cruise mode by 40km/h or more.

- The cruise setting switch and cruise reset switch are operated at the same time (both cruise setting switch and reset switch are in ON position).

- Engine control system abnormal (refer to DTC).

- Engine stop running. In the above description,

- Set acceleration function (increasing auto cruise setting speed)

During the auto cruise driving, if the cruise reset switch is operated, the speed will increase. When the speed is accelerated to the required speed, after releasing the cruise reset switch, the set speed is updated to the current speed.

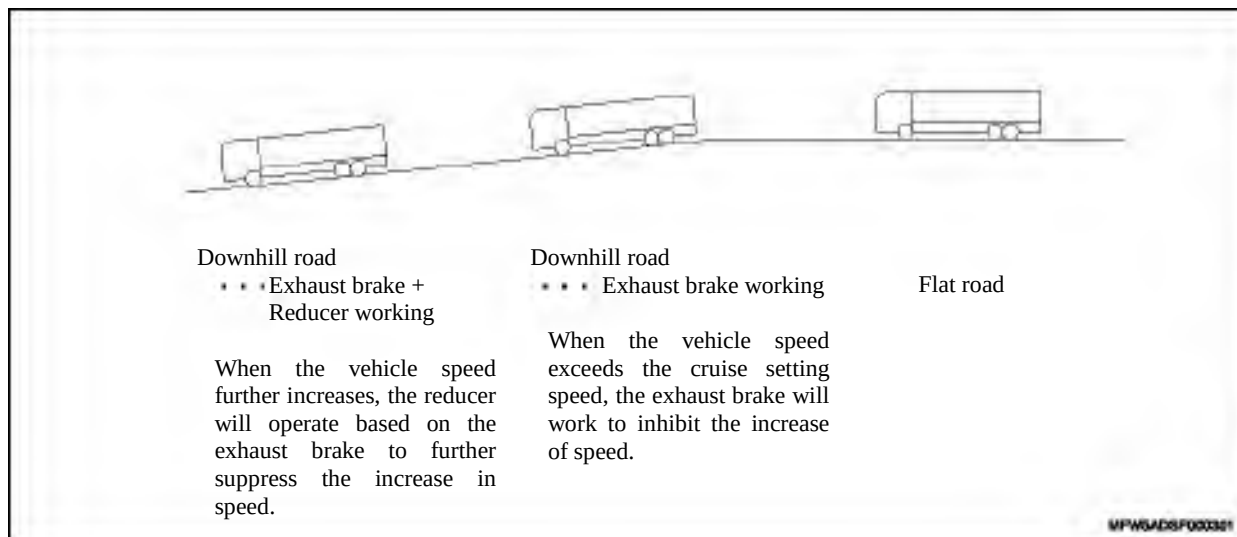
- Release of auto cruise

When operating cruise cancel (cruise cancel switch: OFF).

- Release of auto cruise

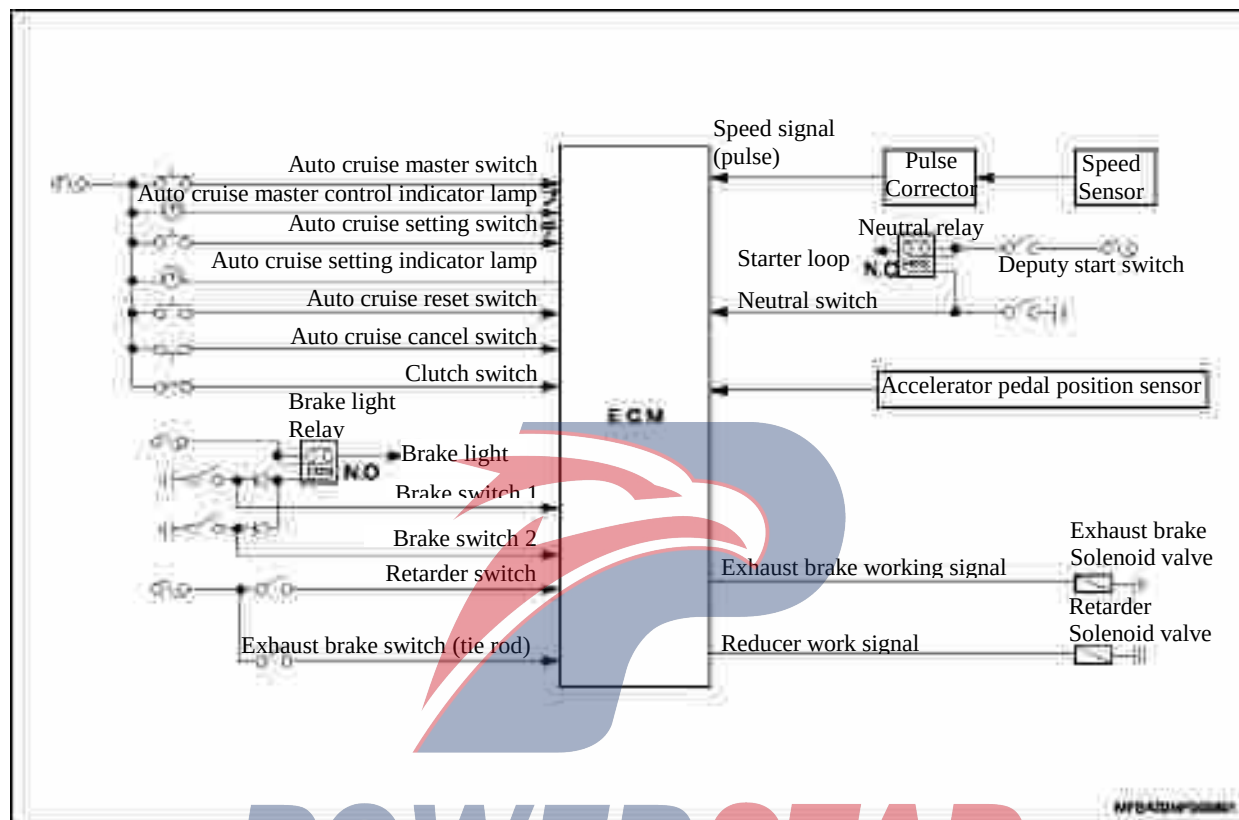
When the selector lever is in the neutral position (neutral switch: ON).

- Reset function (restore to auto cruise state)

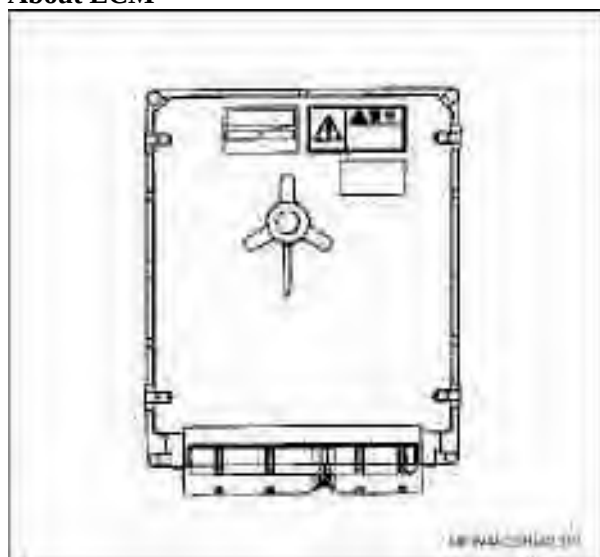
Auto cruise control sketch

During the auto cruise driving, if the auxiliary brake (the exhaust brake and retarder) works, it will transfer the operating signal to ECM to enable/disable the auxiliary brake (the exhaust brake and retarder).

Auxiliary brake interlocking auto cruise control sketch



Engine control unit (ECM) About ECM



The ECM is mounted on the seat base of the assistant seat in the cab. The ECM constantly monitors the various information transmitted from various sensors and controls the power train

systems. ECM can implement the system diagnostic function, test the system running problem, remind the driver by the engine MIL and store DTC. The DTC can identify the area where the problem has occurred and the after-sales personnel can perform maintenance operations.

ECM function

ECM can yield 5V voltage and supply power to the sensors and switches. Since the power is supplied through ECM resistance, the resistance is very high and the test lamp will not be illuminated even if connected in the circuit. In a special case, even if a normal voltmeter is used, because the resistance value is too low, a digital millimeter with minimum input impedance of 10 MΩ is required to be able to display the correct reading value. The ECM controls the output circuit by using a transformer and one of the devices to control the ground wire or power supply circuit. Special tool 5-8840-2691-0 is suitable. ECM can control the ground or power circuit through the transistor and one device and consequently control the output circuit.

ECM and components

ECM is designed to maintain the emission gas to the limit level and acquire the excellent control feel and fuel efficiency. ECM monitors the engine and vehicle performance through the crankshaft position (CKP) sensor and vehicle speed sensor (VS) etc.

ECM voltage description

The ECM will apply the reference voltage to various switches and sensors. The reason why ECM can enforce voltage like this is due to the very high resistance value of ECM, and in fact the enforced voltage of the circuit is very low, even if the test lamp is connected to the circuit, it will not light the lamp at times. Because the input impedance of voltmeter commonly used in maintenance factories is too low, sometimes the correct reading value cannot be displayed. If you use a 10M Ω digital millimeter (for example:5-8840-2691-0), you can get normal voltage readings.

The input/output unit of ECM consists of analog - digital converter, signal damper, counter and special drivers. ECM uses electronic switches to control almost all components.

Electronically Erasable Programmable Read-Only Memory Card (EEPROM)

EEPROM is the nonvolatile storage chip and welded to the ECM base plate. The EEPROM stores the program and verification information required by the ECM to control the operation of the powertrain.

Unlike conventional ROM, EEPROM cannot be replaced. When it is determined that a fault condition occurs in the EEPROM, the ECM should be replaced.

Precautions of ECM maintenance

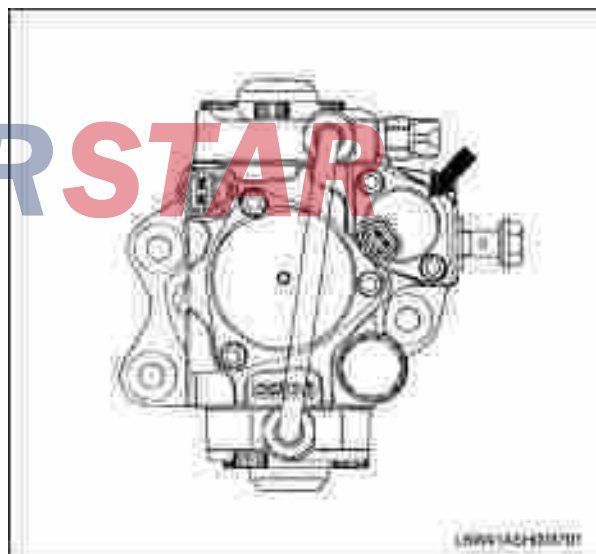
ECM is designed to withstand the general current consumption related to the vehicle running. The load of the circuit is not too large. When testing open or short circuit conditions, the ECM circuit should be grounded or not applied unless specifically required. In testing these circuits, digital millimeter 5-8840-2691-0 must be used.

Oil feed pump

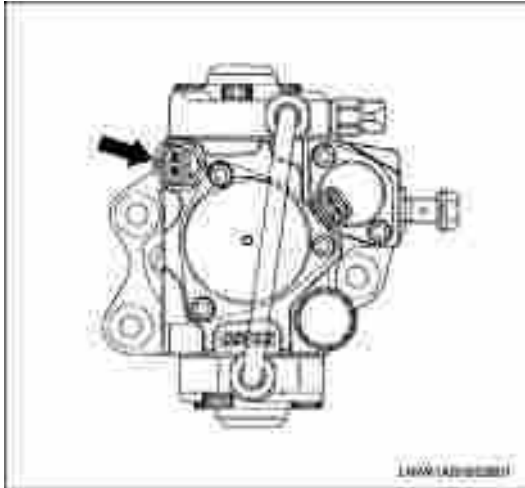


The oil supply pump uses the torque of the engine to increase the pressure of the fuel and feed it to the common rail. In addition, the SCV (fuel flow control valve), fuel temperature (FT) sensor, and fuel pump are installed on the fuel supply pump.

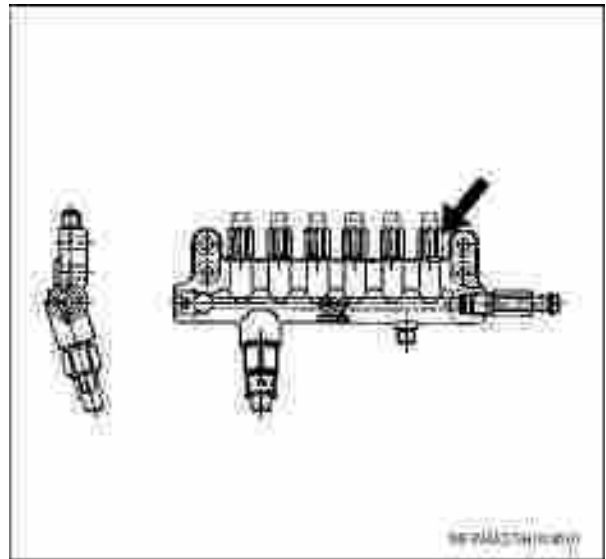
SCV (fuel flow control valve)



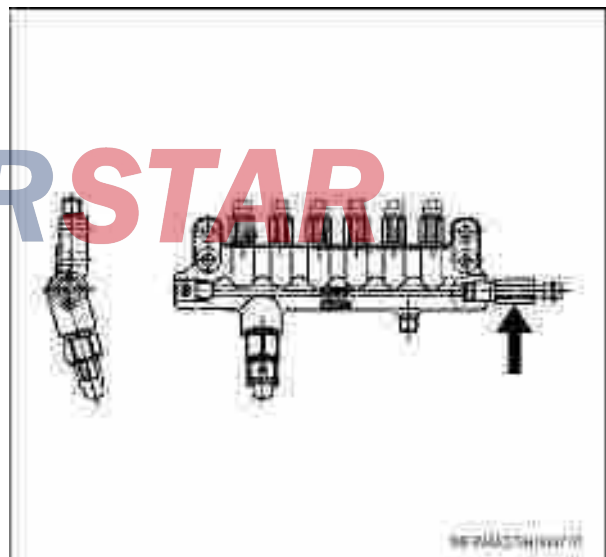
SCV (fuel flow control valve) is installed in the fuel feed pump to control the volume fed to the common rail (the injection volume). The engine control module (ECM) controls the time of energizing the SCV and also controls the fuel injection quantity.

Fuel temperature (FT)

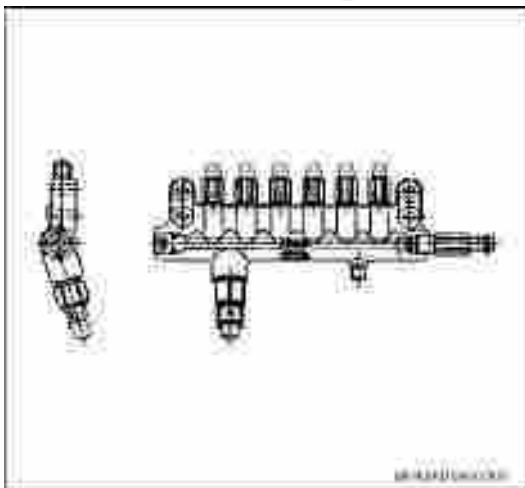
The fuel temperature (FT) sensor is mounted on the fuel pump, and the thermistor will adjust the resistance value according to the change of temperature. When the fuel temperature is high, the resistance value becomes smaller and vice versa. ECM can apply 5V voltage to FT sensor through the load resistance and work out the fuel temperature according to the voltage variation to control the fuel feed pump. When the resistance is small (the temperature is high), the voltage will decrease and the voltage will rise when the resistance is large (the temperature is low).

Flow regulator

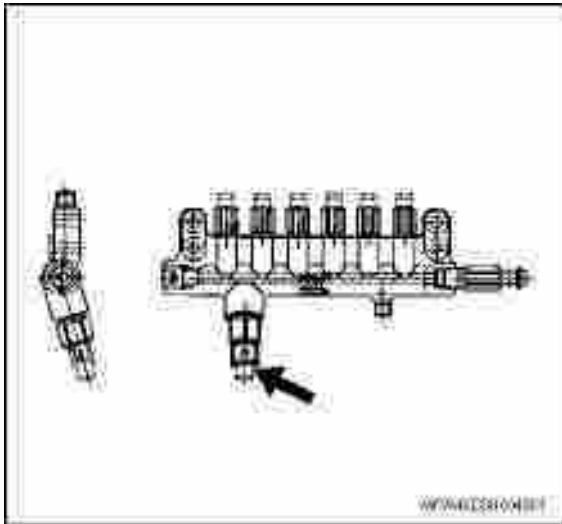
The flow regulator can control the common rail inside pressure pulse and prevent the injector injecting excessive fuel. When the flow regulator is operating, fuel will stop supplying to the injector.

Pressure limiter

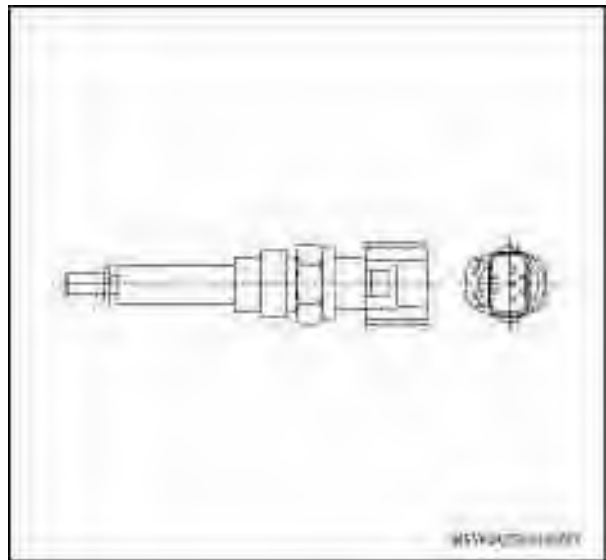
The pressure restrictor will work to release the common rail inside pressure when it becomes abnormally high

Common Rail

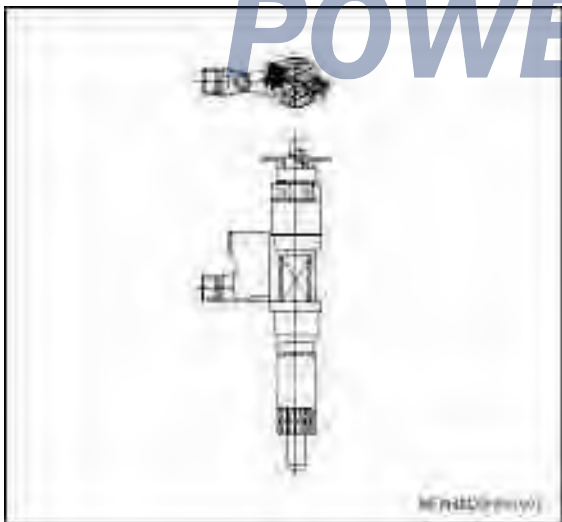
The common rail will receive fuel supplied from the fuel pump to maintain common rail fuel pressure and distribute fuel to each cylinder. The common rail is equipped with a common rail pressure sensor, a flow regulator, and a pressure limiter.

Common rail pressure sensor

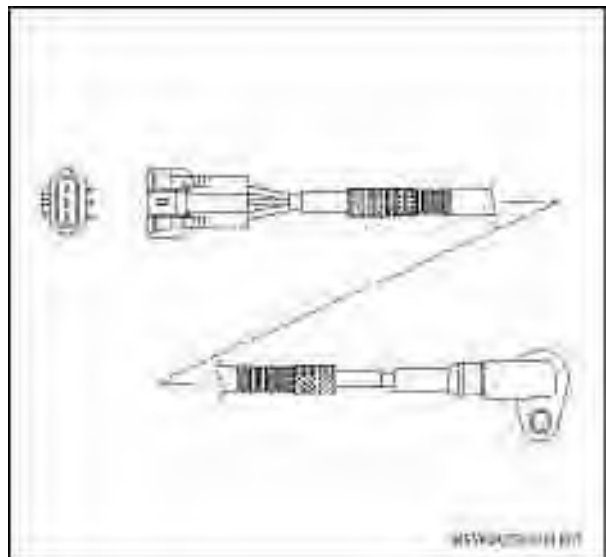
The common rail pressure sensor is installed to the common rail and can detect the fuel pressure inside, convert the pressure into voltage signal and transmit it to ECM. The higher the pressure, the higher the voltage value, and vice versa, the lower the pressure, the lower the voltage. The ECM can calculate the actual common rail pressure (fuel pressure) based on the voltage signal transmitted from the sensor for fuel injection quantity control.

Engine coolant temperature (ECT) sensor

The engine coolant temperature (ECT) sensor is mounted on the engine block and the thermostat body. The thermistor adjusts the resistance value according to the change of temperature. The resistance will be lower if the engine coolant temperature is high and high if the coolant temperature is low. The engine control module (ECM) applies a 5V voltage to the ECT sensor through a load resistor, and calculates the temperature of the engine coolant based on the change in voltage for various controls such as fuel injection quantity control. When the resistance is small (the temperature is high), the voltage will decrease and the voltage will rise when the resistance is large (the temperature is low).

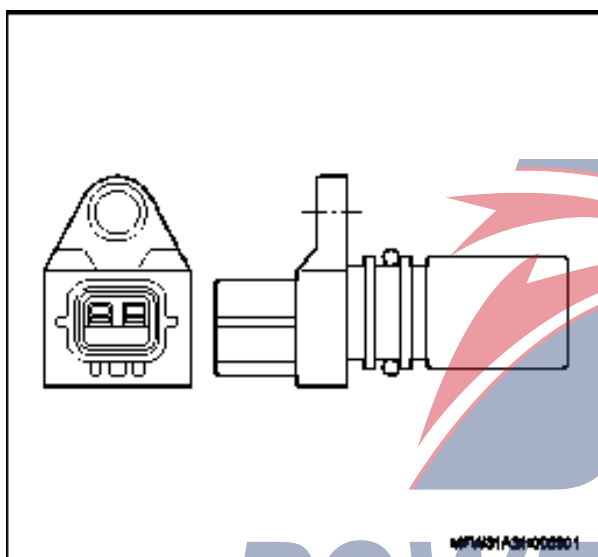
Injector

The injector is installed to the cylinder head and controlled by ECM to inject fuel. ECM inside can increase the injector drive voltage and feed to the injector. Through controlling the injector powering time, ECM can control the fuel injection volume and injection time.

Camshaft position (CMP) sensor

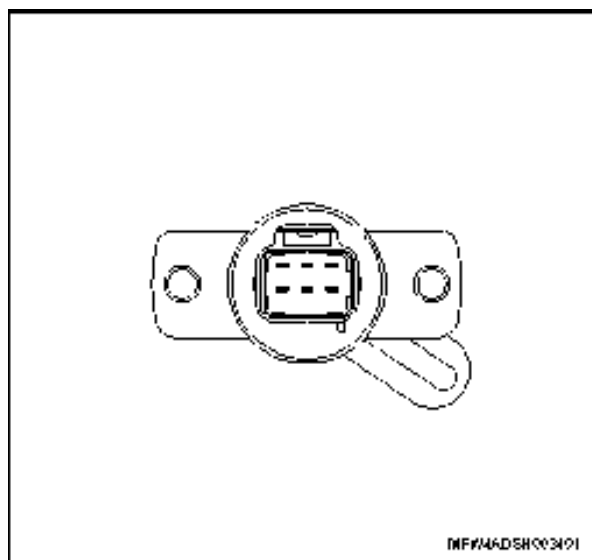
The cam position (CMP) sensor is mounted on the rear of the cylinder head and the cam portion of the camshaft generates a CMP signal through the sensor. The engine control module (ECM) will identify the cylinder and determine the crank angle based on the CKP signal and the CMP signal transmitted from the CMP sensor to calculate the fuel injection quantity control and engine revolving speed. These controls are usually based on CKP signals. When the CKP sensor is abnormal, it is based on the CMP signal.

Crankshaft position (CKP) sensor



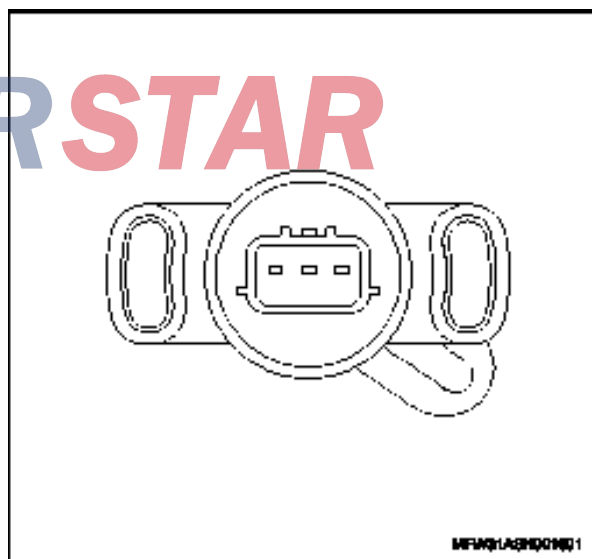
The crankshaft position (CKP) sensor is mounted on the flywheel housing, and the flywheel hole will generate the CKP signal through the sensor. The engine control module (ECM) will identify the cylinder and determine the crank angle based on the CKP signal and the CMP signal transmitted from the CMP sensor to calculate the fuel injection quantity control and engine revolving speed. These controls are usually based on CKP signals. When the CKP sensor is abnormal, it is based on the CMP signal.

Accelerator pedal position (APP) sensor

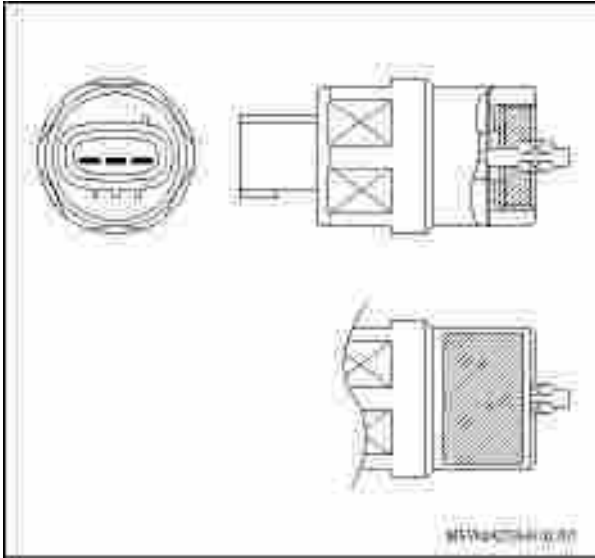
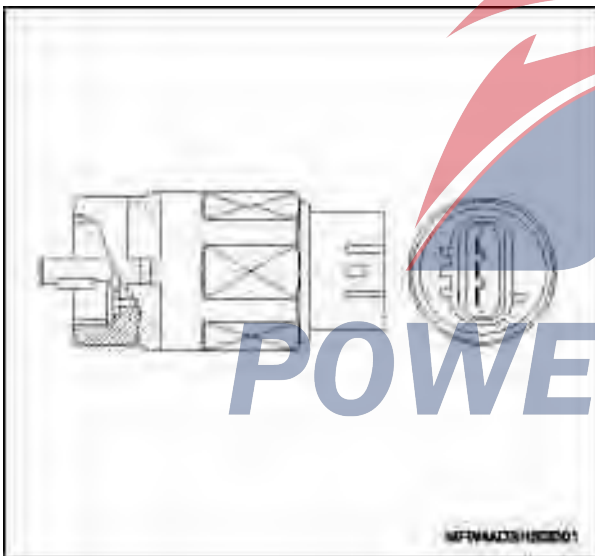


An accelerator pedal position (APP) sensor is mounted on the accelerator pedal and supplies a voltage signal corresponding to the angle of the accelerator pedal to the engine control module (ECM). ECM works out the accelerator position according to the voltage signal to control the fuel injection volume and other items.

PTO acceleration sensor



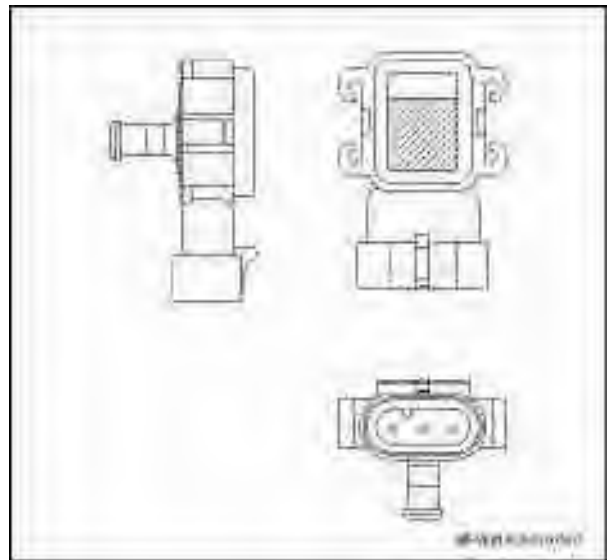
The PTO acceleration sensor is installed outside the cab as an external accelerator, and the voltage signal corresponding to the change of the PTO accelerator angle is supplied to the engine control module (ECM). ECM calculates the PTO accelerator's opening according to the voltage signal for fuel injection quantity control and other more control.

Vehicle speed (VS) sensor**Made by Yazaki****CALSONICKANSEI system**

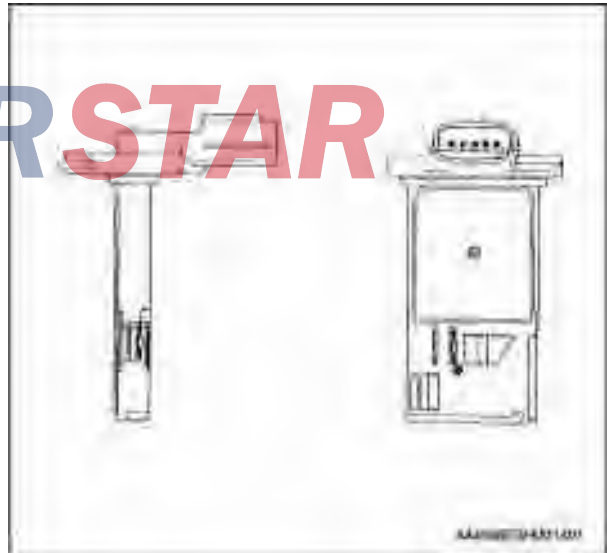
The vehicle speed (VS) sensor is installed to the transmission and equipped with HALL effect circuit. Through the interaction of magnetic field and the simultaneous motion of transmission output shaft magnet valve and shaft, this sensor can transmit the signal to the pulse corrector. ECM can receive the vehicle speed signal from the pulse corrector.

Caution:

When changing the vehicle speed sensor, note that vehicles with FS*, FT* and FV* must be equipped with Yazaki's vehicle speed sensor. Other models can be equipped with one of Yazaki and CALSONICKANSEI.

Barometric pressure sensor

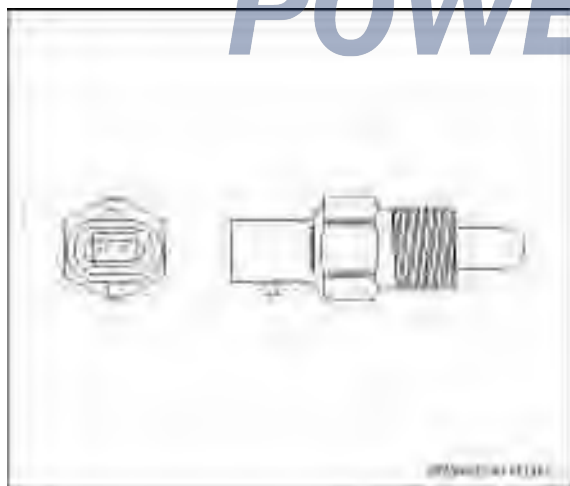
The atmospheric pressure sensor is installed in the instrument panel and it converts the atmospheric pressure into a voltage signal. ECM can work out the barometric pressure according to the voltage signal and correct the fuel injection volume (highland correction) according to the barometric pressure.

Inlet Air Temperature (IAT)/MAF (Mass Air Flow) Sensor (6HL1 only)

The Inlet Air Temperature (IAT)/MAF (Mass Air Flow) sensor is installed in the intake manifold. The IAF sensor detects the temperature of the inhaled air and the MAF sensor detects the amount of inhaled air. The IAT sensor is a thermistor (variable resistance) and the resistance value changes with temperature. The resistance is low when the intake air temperature (the ambient temperature) is high; on the other hand, when the intake air temperature is low, the resistance will be high. ECM can apply 5V voltage to IAT sensor through the load resistance and work out the intake air temperature according to the voltage variation to control the fuel injection etc. When the resistance is small (the temperature is high), the voltage will decrease and the voltage will rise when the resistance is large (the temperature is low).

The MAF sensor changes the output voltage based on the amount of air flowing through the sensor. The ECM will confirm the operating state of the engine based on the signal transmitted by the MAF sensor to calculate the fuel supply. When the air volume is large, it will show acceleration or high load state. When the air volume is small, it will show deceleration or idle state.

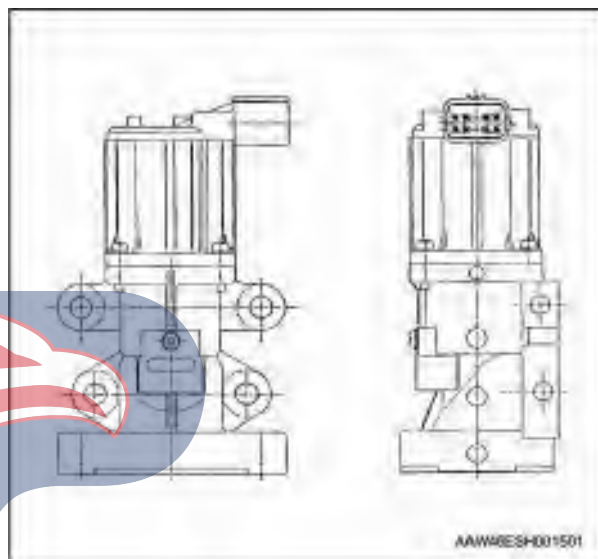
Intake air temperature (IAT) sensor (for 6HK1-TC only)



IAT sensor is installed inside the front grid and detects temperature of intake air to optimize the fuel injection volume control. The IAT sensor is a thermistor (variable resistance) and the resistance value changes with temperature. The resistance is low when the intake air temperature (the ambient temperature) is high; on the other hand, when the

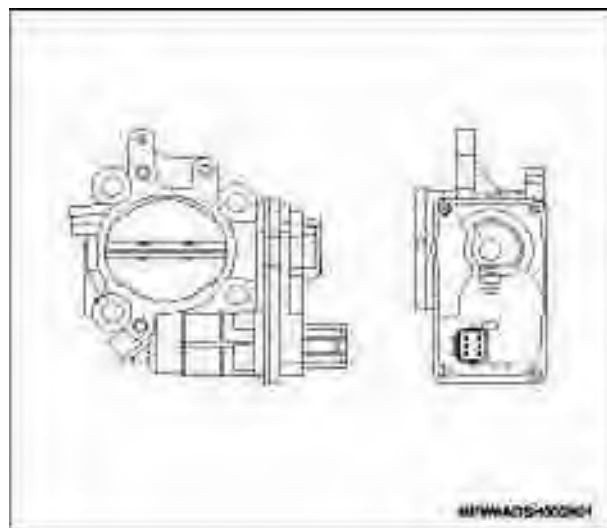
intake air temperature is low, the resistance will be high. ECM can apply 5V voltage to IAT sensor through the load resistance and work out the intake air temperature according to the voltage variation to control the fuel injection etc. When the resistance is small (the temperature is high), the voltage will decrease and the voltage will rise when the resistance is large (the temperature is low).

EGR position sensor (6HL1 only)

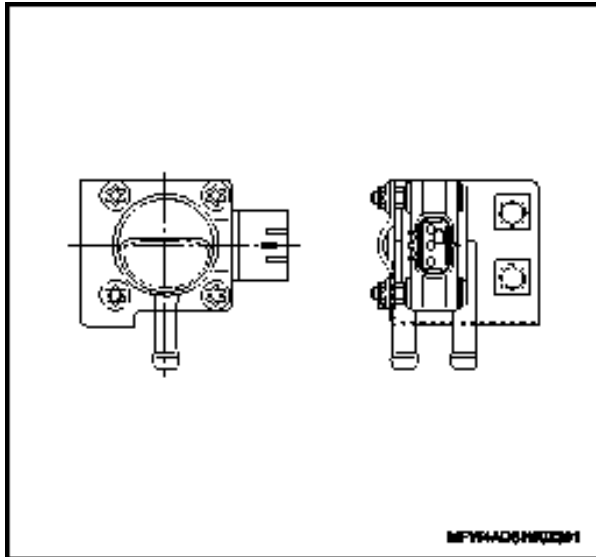


The EGR position sensor is installed in the EGR valve and can detect the valve lift amount of EGR.

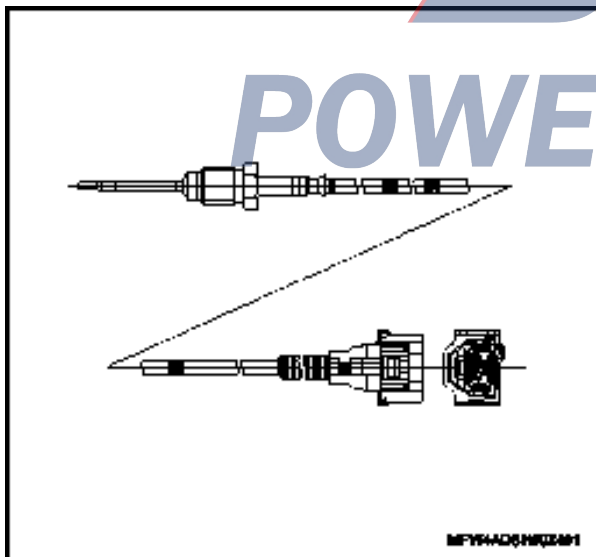
Intake Throttle Position (ITP) Sensor(Only 6HL1)



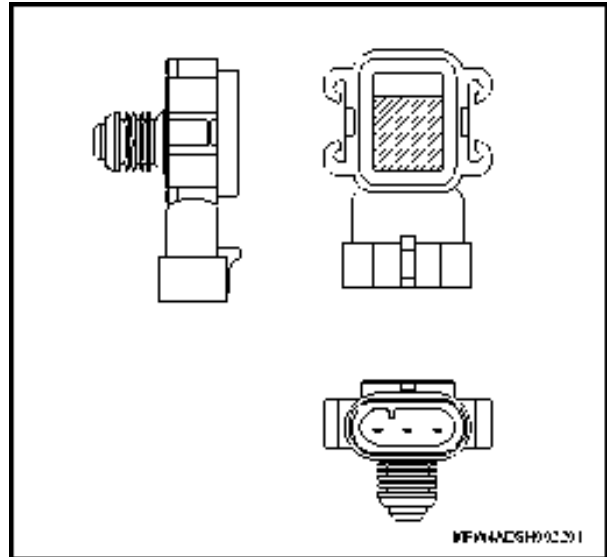
ITP sensor is installed to the intake throttle valve. ECM detects the valve position according to the ITP sensor signal to control the engine air intake.

Differential pressure sensor (6HL1 only)

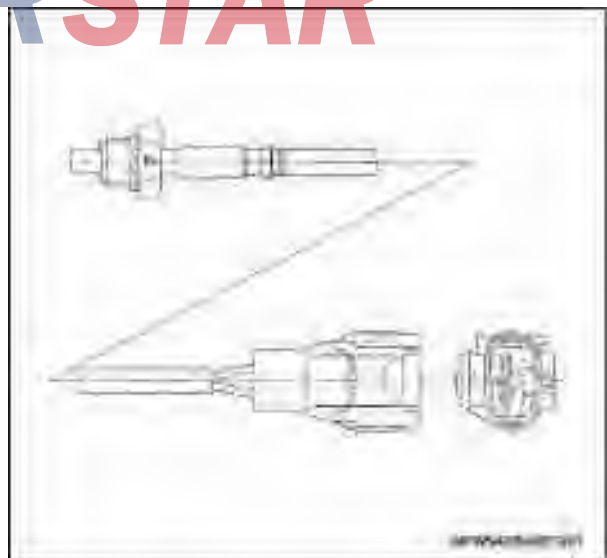
The DPD differential pressure sensor detects the pressure difference before and after the DPD through differential pressure tubes installed before and after the DPD. ECM can judge the reactivation time of DPD according to the signal from the DPD differential pressure sensor and determine the purification after the DPD reactivation.

DPD exhaust temperature sensor (for 6HL1 only)

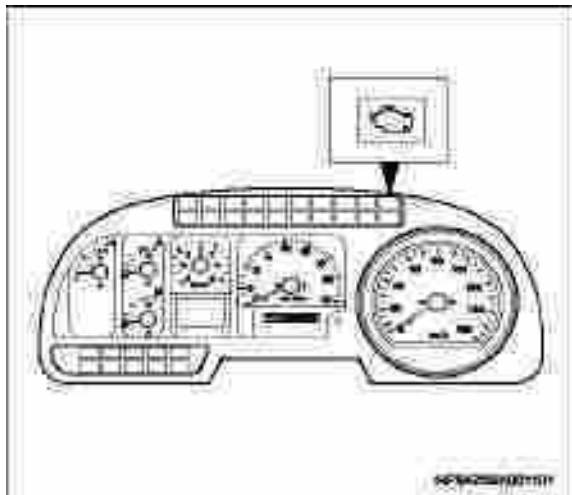
DPD exhaust temperature sensor is installed to the DPD oxidation type catalyst front and filter front. ECM can detect the DPD inside exhaust temperature according to the signal from the DPD exhaust temperature sensor and control the temperature of DPD reactivation.

Booster sensor (6HK1-TC only)

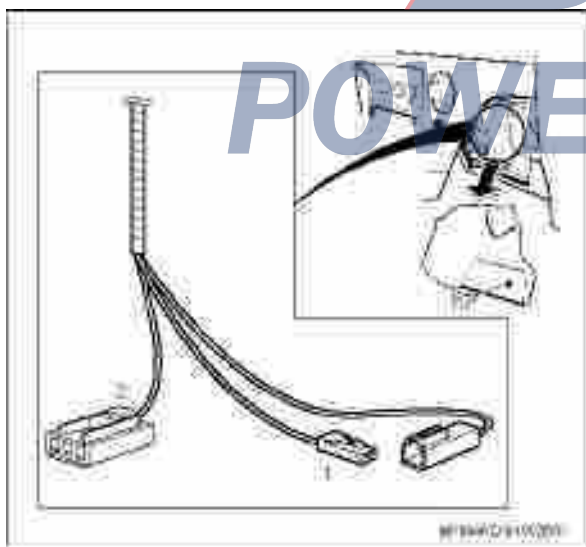
The supercharging sensor is installed to the back of cab. Between the boost sensor and the intake pipe, a pressure hose is used to detect boost (intake pressure), and the pressure is converted into a voltage signal for transmission to the engine control module (ECM). The higher the pressure, the higher the voltage, and the lower the voltage. ECM can work out the supercharging (the intake air pressure) according to the voltage signal from the sensor and control the fuel injection volume.

EGR gas temperature sensor (6HK1-TC only)

EGR gas temperature sensor is installed to the EGR valve upstream side. The sensor is thermistor type and the inside resistance varies along with the temperature.

Engine fault indicator lamp

The engine fault indicator is located in the dashboard to inform the driver of an abnormal condition in the engine or related system. The engine control module (ECM) detects an abnormal fault according to its self diagnosis function and then lights up the engine fault indicator. In addition, after the diagnostic joint is connected, the engine MIL will blink and the DTC detection status can be confirmed.

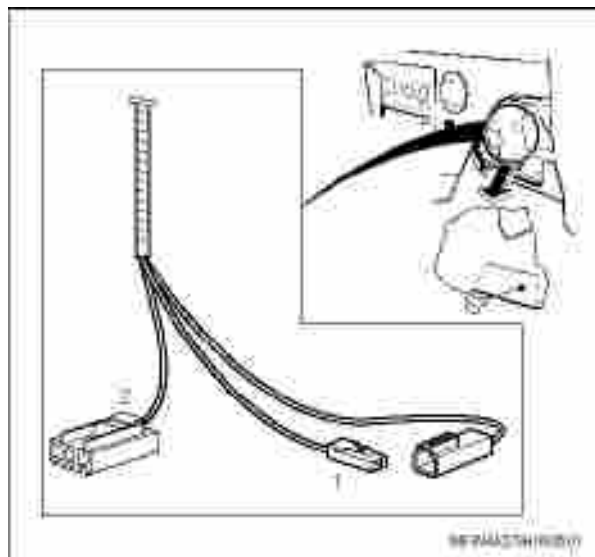
Data Link Connector (DLC)**Name**

Diagnostic connector (red)

2. Data Link Connector (DLC)

The data link connector (DLC) is installed inside the inspection cover and used for the communication and connection if external diagnostic device (the

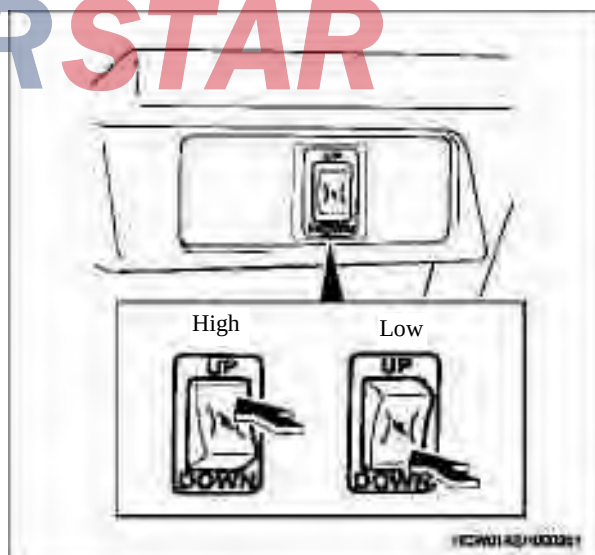
scanning tool) and various controllers.

Diagnostic connector**Name**

Diagnostic connector (red)

2. Data Link Connector (DLC)

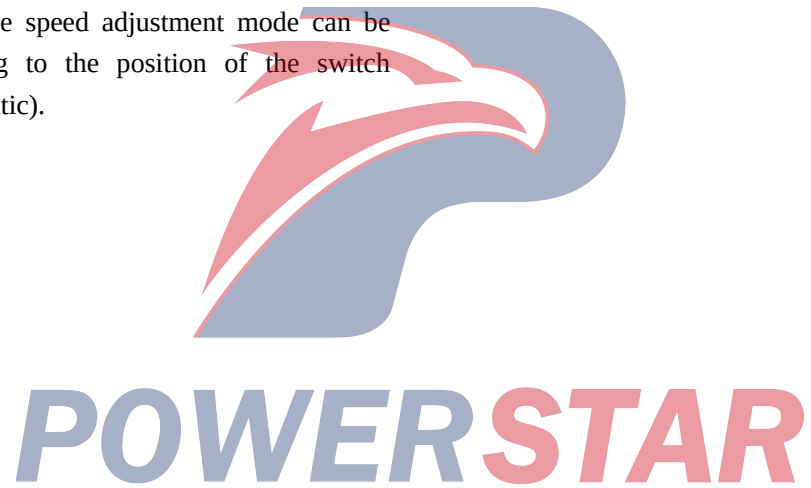
The diagnostic connector is installed in the inspection cover, and the engine fault indicator lamp lights will flicker DTC after the connection is finished.

Idle speed control switch

The idle control switch is installed next to the panel, and when the idle control switch is in the manual position, the number of idle speeds can be adjusted by the operation switch.

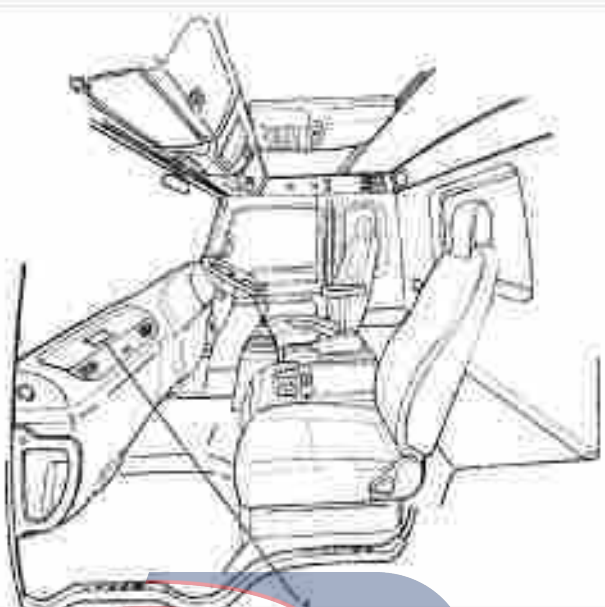
Idling control switch

The idle speed control switch is located next to the panel, and the idle speed adjustment mode can be selected according to the position of the switch (manual or automatic).



Parts configuration diagram

Fuse layout



**Caution**

When the HSA fuse is pulled out, the HSA will be lifted. When replacing the fuse, be sure to pull the parking brake lever and operate it.
Using an unspecified fuse can cause a fire and stop the device. Do not use it. (The maximum current of applied load of the standby power supply is 70% of rated current).

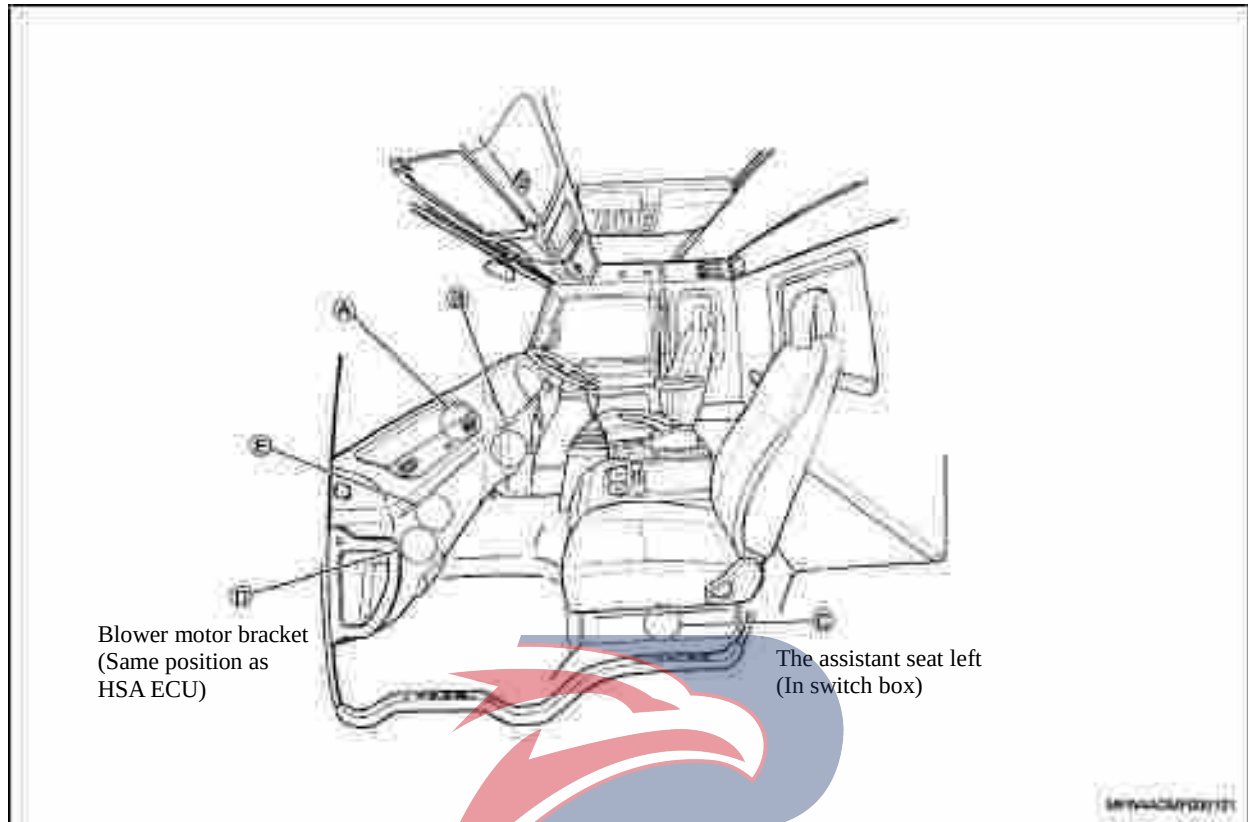
Fuse box

Headlight (up, left)	Micrograph, self-protection device, CAN converter indicator, instrument	Alternating wiring	Turn lights, tail lights, fog lights	Blower motor (air conditioning)	ABS control unit
Headlights (up, right)	Air Dryer, Reversing Light Horn, A/C(R), please refer to Out Group 1	Alternating wiring	Brake light	Condenser fan motor (A/C) Compressor(A/C)	ABS control unit
Head lamp (downward, left)	PTO exhaust brake Reducer head lamp corrector	Alternating wiring	Please refer to start the circuit Supporting-outside 2	Wiper	Stable period F
Head lamp (downward, right)	Caution Single lock Stabilizer F		Socket	Concealed type rearview mirror rearview mirror wiper	ABS (trailer)
Clock (rotation record sheet), indoor light, reversing light, power supply tilt	Low gear shift, body height adjustment	Supporting-outside 1	Strobe light	Cigarette lighter	ABS control unit
Door lock Audible alarm (customer)	Front drive Idle stop HI gear shift	Washing machine, automatic lighting control unit, Blower motor relay	Engine control	Rearview mirror heater, defroster	Outside group 2 Fuse
Light	Engine control	Collectable rearview mirror relay Ignition relay	Power window(left)	Standby power supply (key switch)	Main (only idling vehicles)
Body height adjustment	Trailer	Rotational light relay Power window (key ON)	Power window (RH)	Anti-folding device for braking of rotational light, tractor and single trailer	
Preloaded seat belts SHS Airbag	Radio, recorder, Air purifier, body high speed adjustment device	Audible alarm (customer button is on)	Round curtain (LH)	Standby power supply (Taillight)	
			Round curtain (RH)	Standby power supply (battery)	

Name

1. Fuse nameplate's stickup position (inside)

Relay configuration diagram



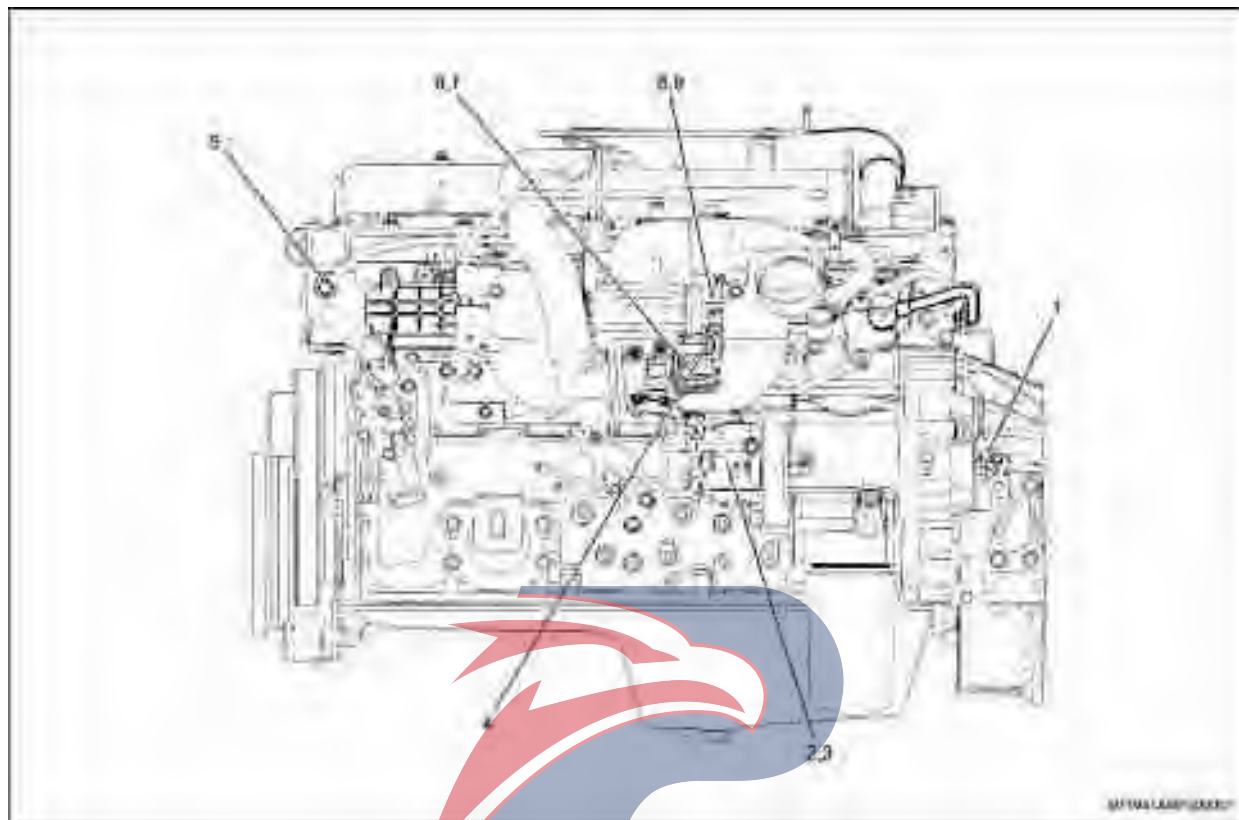
POWERSTAR

		Name	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	Remarks
A	1	PL-37 Fan relay	○	○	○	○	○	○	○	○	○	○	○	○	○	Fan motor
	2	PL-50 Fog lamp relay	○	○	○	○	○	○	○	○	○	○	○	○	○	Fog lamp
	3	PL-43 Folding rearview mirror relay	○	○	○	○	○	○	○	○	○	○	○	○	○	Vehicles equipped with electric folding rearview mirror
	4	PL-47 Headlamp relay (high)	○	○	○	○	○	○	○	○	○	○	○	○	○	Headlights (up)
	5	PL-51 Headlamp relay (low)	○	○	○	○	○	○	○	○	○	○	○	○	○	Headlight (down)
	6	PL-273 Starter tripping relay	○	○	○	○	○	○	○	○	○	○	○	○	○	
	7	PL-40 Taillight relay	○	○	○	○	○	○	○	○	○	○	○	○	○	
	8	PL-52 Brake light relay	○	○	○	○	○	○	○	○	○	○	○	○	○	
	9	PL-233 VGS relay	○	○	○	○	○	○	○	○	○	○	○	○	○	
	10	PL-20 Key on relay	○	○	○	○	○	○	○	○	○	○	○	○	○	Anti-aircraft lights
B	1	PL-27 Charging relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Condenser (air conditioning)
	2	PL-48 Condenser relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Charging relay (auto A/C)
	3	PL-25 A/C OFF relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Compressor (A/C)
	4	PL-96 Compressor relay	△	△	△	△	△	△	△	△	△	△	△	△	△	A/C off (auto A/C)
	5	PL-46 Neutral gear relay	△	△	△	△	△	△	△	△	△	△	△	△	△	A/T all vehicles
	6	PL-299 Smoother-F relay (start control)	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicle equipped with Smoother-F
	7	PL-304 Warming up indicator lamp tripping rel.	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
	8	PL-334 ABS relay (main)	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicle equipped with Smoother-F
	9	PL-306 Strobe light and dangerous breaker relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
	10	PL-213 ABS relay (main)	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicles with ABS / ASR or ABS
C	1	PL-132 Exhaust brake & reducer tripping relay	○	○	○	○	○	○	○	○	○	○	○	○	○	
	2	PL-209 Brake lock mechanism relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Brake locking mechanism
	3	PL-159 Parking relay	△	△	△	△	△	△	△	△	△	△	△	△	△	
	4	PL-81 Rack mounting taillight relay (spare)	△	△	△	△	△	△	△	△	△	△	△	△	△	Rack tail light circuit (optional item)
	5	PL-34 Working lamp & water pump power rel.	△	△	△	△	△	△	△	△	△	△	△	△	△	Working lamp & water pump power relay circuit (truck mixer)
	6	PL-112 Rearview mirror wiper relay (LH)	△	△	△	△	△	△	△	△	△	△	△	△	△	Rearview wiper (right)
	7	PL-360 Rearview mirror wiper relay (RH)	△	△	△	△	△	△	△	△	△	△	△	△	△	Rearview wiper (right)
	8	PL-307 Side PTU indicator lamp tripping relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
	9	PL-97 Rotational light relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Rotational light
	10	PL-188 Air suspension relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Height-adjustable air suspension vehicle
D	1	PL-300 Tachometer relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Rotation speed recorder equipped with duplicated record
	2	PL-124 PTO relay (front)	△	△	△	△	△	△	△	△	△	△	△	△	△	Front (mid cut) PTO
	3	PL-277 Reverse signal relay	△	△	△	△	△	△	△	△	△	△	△	△	△	DPD, with Smoother - F
	4	PL-299 Differential lock indicator circuit relay	△	△	△	△	△	△	△	△	△	△	△	△	△	
	5	PL-218 PTO relay (clutch)	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
	6	PL-298 PTO relay (neutral relay)	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicles with PTO
	7	PL-305 BO2 relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
	8	PL-196 Wheel parking relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicles equipped with electromagnetic clutch PTO
	9	PL-304 B.O. relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Rear wheel parking
	10	PL-30 Condenser relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Anti-aircraft lights
E	1	PL-95 Compressor relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Condenser (auto A/C)
	2	PL-45 Trailer information relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Compressor (auto A/C)
	3	PL-66 A/C tripping relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Manual shutdown A/C of vehicles equipped with idling stop function
	4	PL-38 Maximum temperature relay	△	△	△	△	△	△	△	△	△	△	△	△	△	All vehicles equipped with semi-trailer
	5	PL-64 Brake light relay (trailer information)	△	△	△	△	△	△	△	△	△	△	△	△	△	Maximum temperature (automatic air conditioning)
	6	PL-39 Low-light circuit relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicles equipped with idling stop function
	7	PL-53 Curtain open relay (left)	△	△	△	△	△	△	△	△	△	△	△	△	△	All vehicles equipped with semi-trailer
	8	PL-55 Idle stop & Starting system relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Electric round window open (left)
	9	PL-43 Trailer flash relay	△	△	△	△	△	△	△	△	△	△	△	△	△	Vehicles equipped with idling stop function
	10	PL-67 A/C tripping relay-2	△	△	△	△	△	△	△	△	△	△	△	△	△	All vehicles equipped with semi-trailer
F	1	PL-56 Curtains close relay (left)	△	△	△	△	△	△	△	△	△	△	△	△	△	Electric round window close (left)
	2	PL-45 Trailer Information Relay - 2	△	△	△	△	△	△	△	△	△	△	△	△	△	Manual shutdown A/C of vehicles equipped with idling stop function
	3	PL-58 Curtain open relay (parking)	△	△	△	△	△	△	△	△	△	△	△	△	△	All vehicles equipped with semi-trailer
	4	PL-59 Curtain open relay (right)	△	△	△	△	△	△	△	△	△	△	△	△	△	Electric round window open (right)
	5	PL-57 Curtains close relay (right)	△	△	△	△	△	△	△	△	△	△	△	△	△	Electric round window closed (right)
	6	Blank	△	△	△	△	△	△	△	△	△	△	△	△	△	
	7		△	△	△	△	△	△	△	△	△	△	△	△	△	
	8		△	△	△	△	△	△	△	△	△	△	△	△	△	
	9		△	△	△	△	△	△	△	△	△	△	△	△	△	
	10		△	△	△	△	△	△	△	△	△	△	△	△	△	

		Name	11A1	11A2	11A3	11A4	11A5	11A6	11A7	11A8	11A9	11A10	11A11	11A12	11A13	11A14	11A15	11A16	11A17	11A18	11A19	11A20	11A21	11A22	11A23	11A24	11A25	11A26	11A27	11A28	11A29	11A30	11A31	11A32	11A33	11A34	11A35	11A36	11A37	11A38	11A39	11A40	11A41	11A42	11A43	11A44	11A45	11A46	11A47	11A48	11A49	11A50	11A51	11A52	11A53	11A54	11A55	11A56	11A57	11A58	11A59	11A60	11A61	11A62	11A63	11A64	11A65	11A66	11A67	11A68	11A69	11A70	11A71	11A72	11A73	11A74	11A75	11A76	11A77	11A78	11A79	11A80	11A81	11A82	11A83	11A84	11A85	11A86	11A87	11A88	11A89	11A90	11A91	11A92	11A93	11A94	11A95	11A96	11A97	11A98	11A99	11A100	11A101	11A102	11A103	11A104	11A105	11A106	11A107	11A108	11A109	11A110	11A111	11A112	11A113	11A114	11A115	11A116	11A117	11A118	11A119	11A120	11A121	11A122	11A123	11A124	11A125	11A126	11A127	11A128	11A129	11A130	11A131	11A132	11A133	11A134	11A135	11A136	11A137	11A138	11A139	11A140	11A141	11A142	11A143	11A144	11A145	11A146	11A147	11A148	11A149	11A150	11A151	11A152	11A153	11A154	11A155	11A156	11A157	11A158	11A159	11A160	11A161	11A162	11A163	11A164	11A165	11A166	11A167	11A168	11A169	11A170	11A171	11A172	11A173	11A174	11A175	11A176	11A177	11A178	11A179	11A180	11A181	11A182	11A183	11A184	11A185	11A186	11A187	11A188	11A189	11A190	11A191	11A192	11A193	11A194	11A195	11A196	11A197	11A198	11A199	11A200	11A201	11A202	11A203	11A204	11A205	11A206	11A207	11A208	11A209	11A210	11A211	11A212	11A213	11A214	11A215	11A216	11A217	11A218	11A219	11A220	11A221	11A222	11A223	11A224	11A225	11A226	11A227	11A228	11A229	11A230	11A231	11A232	11A233	11A234	11A235	11A236	11A237	11A238	11A239	11A240	11A241	11A242	11A243	11A244	11A245	11A246	11A247	11A248	11A249	11A250	11A251	11A252	11A253	11A254	11A255	11A256	11A257	11A258	11A259	11A260	11A261	11A262	11A263	11A264	11A265	11A266	11A267	11A268	11A269	11A270	11A271	11A272	11A273	11A274	11A275	11A276	11A277	11A278	11A279	11A280	11A281	11A282	11A283	11A284	11A285	11A286	11A287	11A288	11A289	11A290	11A291	11A292	11A293	11A294	11A295	11A296	11A297	11A298	11A299	11A300	11A301	11A302	11A303	11A304	11A305	11A306	11A307	11A308	11A309	11A310	11A311	11A312	11A313	11A314	11A315	11A316	11A317	11A318	11A319	11A320	11A321	11A322	11A323	11A324	11A325	11A326	11A327	11A328	11A329	11A330	11A331	11A332	11A333	11A334	11A335	11A336	11A337	11A338	11A339	11A340	11A341	11A342	11A343	11A344	11A345	11A346	11A347	11A348	11A349	11A350	11A351	11A352	11A353	11A354	11A355	11A356	11A357	11A358	11A359	11A360	11A361	11A362	11A363	11A364	11A365	11A366	11A367	11A368	11A369	11A370	11A371	11A372	11A373	11A374	11A375	11A376	11A377	11A378	11A379	11A380	11A381	11A382	11A383	11A384	11A385	11A386	11A387	11A388	11A389	11A390	11A391	11A392	11A393	11A394	11A395	11A396	11A397	11A398	11A399	11A400	11A401	11A402	11A403	11A404	11A405	11A406	11A407	11A408	11A409	11A410	11A411	11A412	11A413	11A414	11A415	11A416	11A417	11A418	11A419	11A420	11A421	11A422	11A423	11A424	11A425	11A426	11A427	11A428	11A429	11A430	11A431	11A432	11A433	11A434	11A435	11A436	11A437	11A438	11A439	11A440	11A441	11A442	11A443	11A444	11A445	11A446	11A447	11A448	11A449	11A450	11A451	11A452	11A453	11A454	11A455	11A456	11A457	11A458	11A459	11A460	11A461	11A462	11A463	11A464	11A465	11A466	11A467	11A468	11A469	11A470	11A471	11A472	11A473	11A474	11A475	11A476	11A477	11A478	11A479	11A480	11A481	11A482	11A483	11A484	11A485	11A486	11A487	11A488	11A489	11A490	11A491	11A492	11A493	11A494	11A495	11A496	11A497	11A498	11A499	11A500	11A501	11A502	11A503	11A504	11A505	11A506	11A507	11A508	11A509	11A510	11A511	11A512	11A513	11A514	11A515	11A516	11A517	11A518	11A519	11A520	11A521	11A522	11A523	11A524	11A525	11A526	11A527	11A528	11A529	11A530	11A531	11A532	11A533	11A534	11A535	11A536	11A537	11A538	11A539	11A540	11A541	11A542	11A543	11A544	11A545	11A546	11A547	11A548	11A549	11A550	11A551	11A552	11A553	11A554	11A555	11A556	11A557	11A558	11A559	11A560	11A561	11A562	11A563	11A564	11A565	11A566	11A567	11A568	11A569	11A570	11A571	11A572	11A573	11A574	11A575	11A576	11A577	11A578	11A579	11A580	11A581	11A582	11A583	11A584	11A585	11A586	11A587	11A588	11A589	11A590	11A591	11A592	11A593	11A594	11A595	11A596	11A597	11A598	11A599	11A600	11A601	11A602	11A603	11A604	11A605	11A606	11A607	11A608	11A609	11A610	11A611	11A612	11A613	11A614	11A615	11A616	11A617	11A618	11A619	11A620	11A621	11A622	11A623	11A624	11A625	11A626	11A627	11A628	11A629	11A630	11A631	11A632	11A633	11A634	11A635	11A636	11A637	11A638	11A639	11A640	11A641	11A642	11A643	11A644	11A645	11A646	11A647	11A648	11A649	11A650	11A651	11A652	11A653	11A654	11A655	11A656	11A657	11A658	11A659	11A660	11A661	11A662	11A663	11A664	11A665	11A666	11A667	11A668	11A669	11A670	11A671	11A672	11A673	11A674	11A675	11A676	11A677	11A678	11A679	11A680	11A681	11A682	11A683	11A684	11A685	11A686	11A687	11A688	11A689	11A690	11A691	11A692	11A693	11A694	11A695	11A696	11A697	11A698	11A699	11A700	11A701	11A702	11A703	11A704	11A705	11A706	11A707	11A708	11A709	11A710	11A711	11A712	11A713	11A714	11A715	11A716	11A717	11A718	11A719	11A720	11A721	11A722	11A723	11A724	11A725	11A726	11A727	11A728	11A729	11A730	11A731	11A732	11A733	11A734	11A735	11A736	11A737	11A738	11A739	11A740	11A741	11A742	11A743	11A744	11A745	11A746	11A747	11A748	11A749	11A750	11A751	11A752	11A753	11A754	11A755	11A756	11A757	11A758	11A759	11A760	11A761	11A762	11A763	11A764	11A765	11A766	11A767	11A768	11A769	11A770	11A771	11A772	11A773	11A774	11A775	11A776	11A777	11A778	11A779	11A780	11A781	11A782	11A783	11A784	11A785	11A786	11A787	11A788	11A789	11A790	11A791	11A792	11A793	11A794	11A795	11A796	11A797	11A798	11A799	11A800	11A801	11A802	11A803	11A804	11A805	11A806	11A807	11A808	11A809	11A810	11A811	11A812	11A813	11A814	11A815	11A816	11A817	11A818	11A819	11A820	11A821	11A822	11A823	11A824	11A825	11A826	11A827	11A828	11A829	11A830	11A831	11A832	11A833	11A834	11A835	11A836	11A837	11A838	11A839	11A840	11A841	11A842	11A843	11A844	11A845	11A846	11A847	11A848	11A849	11A850	11A851	11A852	11A853	11A854	11A855	11A856	11A857	11A858	11A859	11A860	11A861	11A862	11A863	11A864	11A865	11A866	11A867	11A868	11A869	11A870	11A871	11A872	11A873	11A874	11A875	11A876	11A877	11A878	11A879	11A880	11A881	11A882	11A883	11A884	11A885	11A886	11A887	11A888	11A889	11A890	11A891	11A892	11A893	11A894	11A895	11A896	11A897	11A898	11A899	11A900	11A901	11A902	11A903	11A904	11A905	11A906	11A907	11A908	11A909	11A910	11A911	11A912	11A913	11A914	11A915	11A916	11A917	11A918	11A919	11A920	11A921	11A922	11A923	11A924	11A925	11A926	11A927	11A928	11A929	11A930	11A931	11A932	11A933	11A934	11A935	11A936	11A937	11A938	11A939	11A940	11A941	11A942	11A943	11A944	11A945	11A946	11A947	11A948	11A949	11A950	11A951	11A952	11A953	11A954	11A955	11A956	11A957	11A958	11A959	11A960	11A961	11A962	11A963	11A964	11A965	11A966	11A967	11A968	11A969	11A970	11A971	11A972	11A973	11A974	11A975	11A976	11A977	11A978	11A979	11A980	11A981	11A982	11A983	11A984	11A985	11A986	11A987	11A988	11A989	11A990	11A991	11A992	11A993	11A994	11A995	11A996	11A997	11A998	11A999	11A1000	11A1001	11A1002	11A1003	11A1004	11A1005	11A1006	11A1007	11A1008	11A1009	11A1010	11A1011	11A1012	11A1013	11A1014	11A1015	11A1016	11A1017	11A1018	11A1019	11A1020	11A1021	11A1022	11A1023	11A1024	11A1025	11A1026	11A1027	11A1028	11A1029	11A1030	11A1031	11A1032	11A1033	11A1034	11A1035	11A1036	11A1037	11A1038	11A1039	11A1040	11A1041	11A1042	11A1043	11A1044	11A1045	11A1046	11A1047	11A1048	11A1049	11A1050	11A1051	11A1052	11A1053	11A1054	11A1055	11A1056	11A1057	11A1058	11A1059	11A1060	11A1061	11A1062	11A1063	11A1064	11A1065	11A1066	11A1067	11A1068	11A1069	11A1070	11A1071	11A1072	11A1073	11A1074	11A1075	11A1076	11A1077	11A1078	11A1079	11A1080	11A1081	11A1082	11A1083	11A1084	11A1085	11A1086	11A1087	11A1088	11A1089	11A1090	11A1091	11A1092	11A1093	11A1094	11A1095	11A1096	11A1097	11A1098	11A1099	11A1100	11A1101	11A1102	11A1103	11A1104	11A1105	11A1106	11A1107	11A1108	11A1109	11A1110	11A1111	11A1112	11A1113	11A1114	11A111
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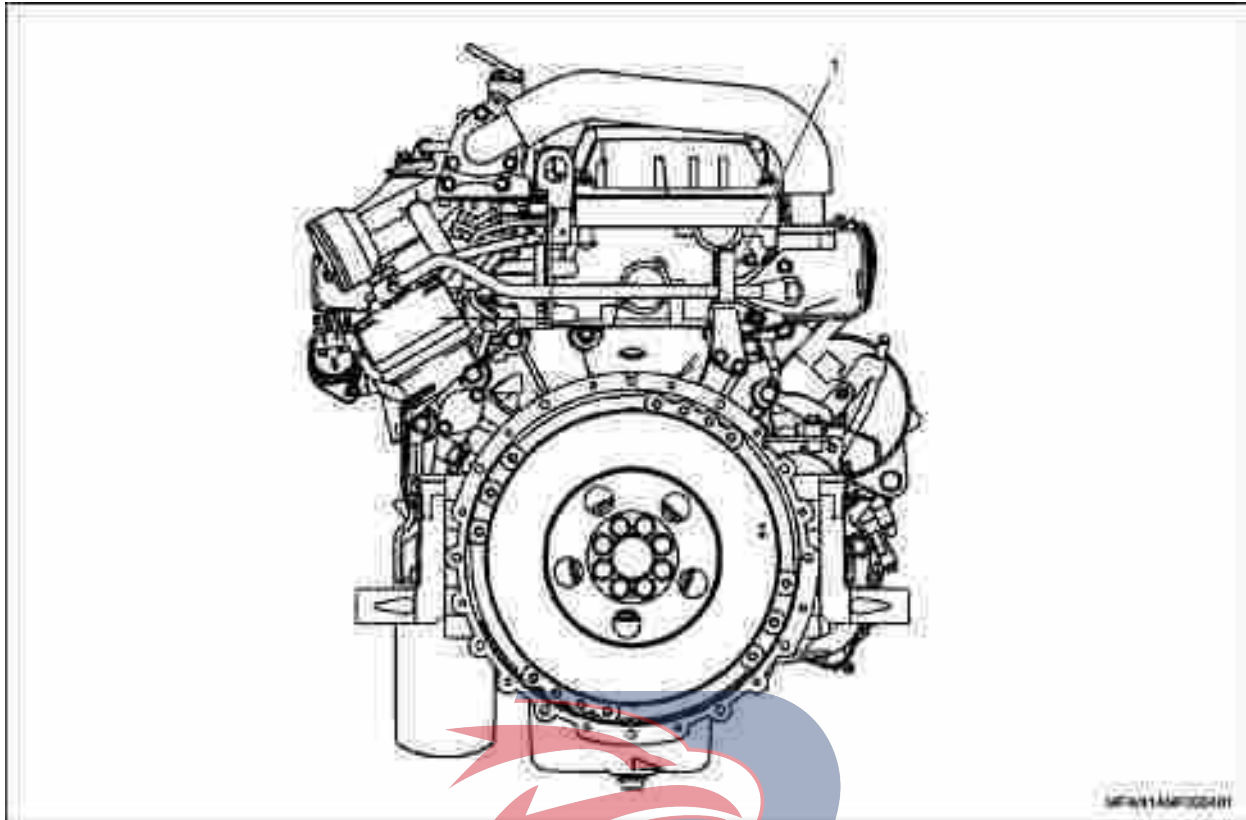
Engine components configuration diagram

6HL1



Name

- | | |
|--|--|
| 1. Crankshaft position (CKP) sensor | 6. EGR motor |
| 2. Fuel flow control valve (SCV) | 7. EGR position sensor |
| 3. Fuel temperature (FT) sensor | 8. Inlet throttle motor |
| Common rail pressure sensor | 9. Intake Throttle Position (ITP) Sensor |
| 5. Engine Coolant Temperature (ECT) Sensor | |
-

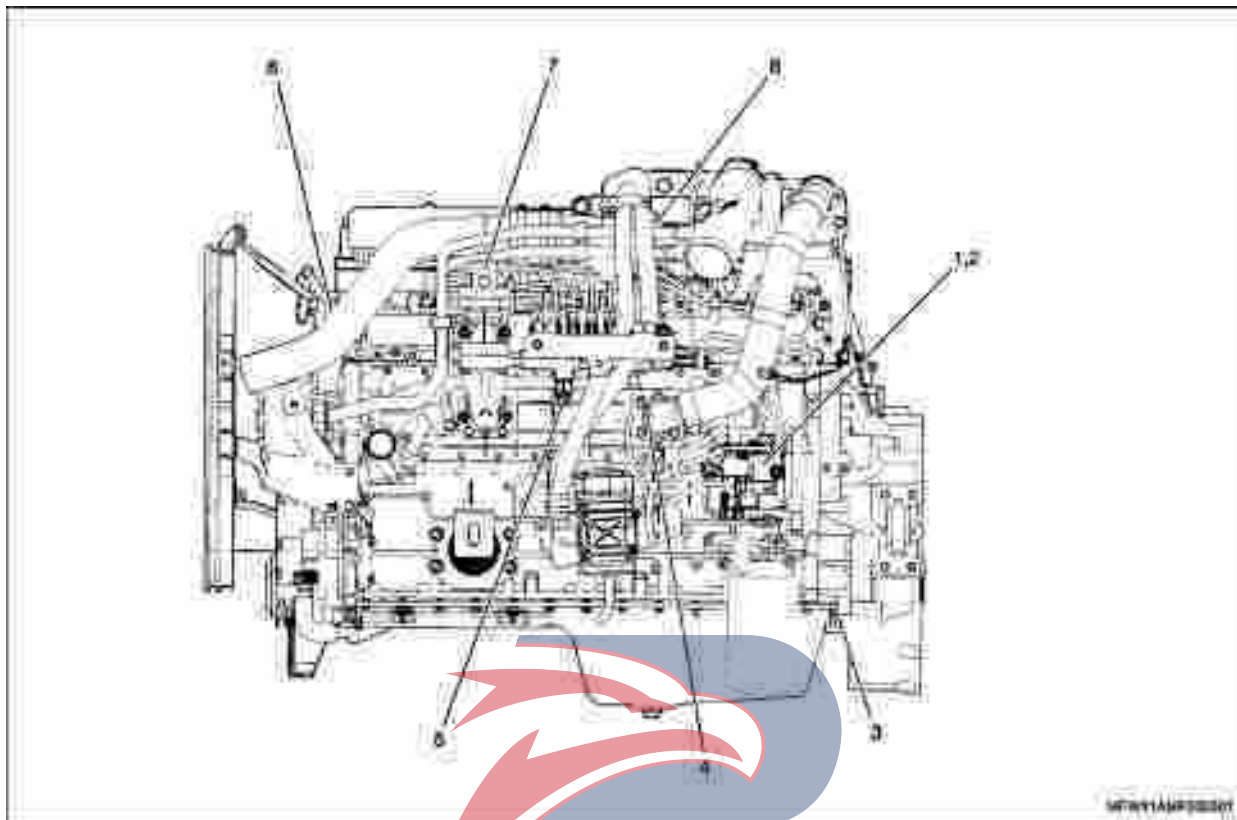


Name

1. Camshaft position (CMP) sensor

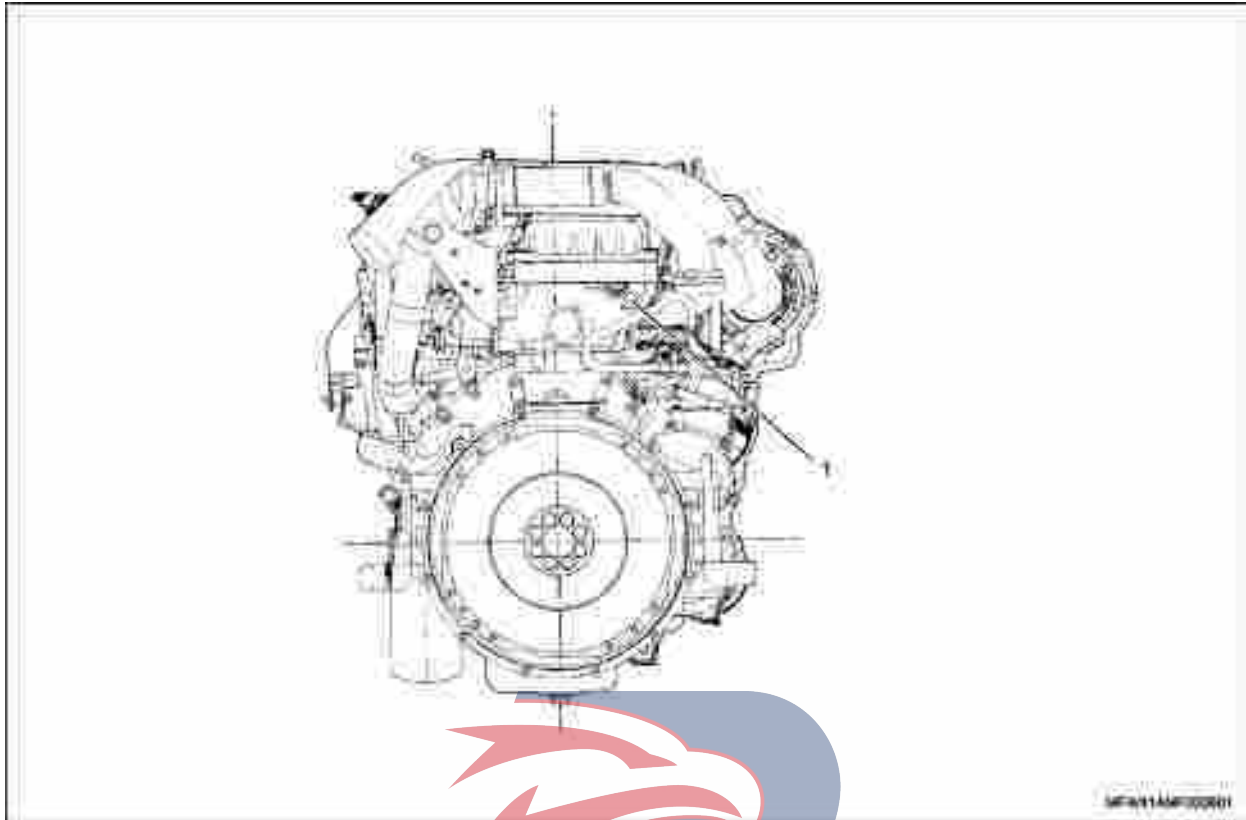
POWERSTAR

6HK1-TCP



Name

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|-------------------------------------|--|
| 1. Fuel flow control valve (SCV) | Common rail pressure sensor |
| 2. Fuel temperature (FT) sensor | 6. Engine Coolant Temperature (ECT) Sensor |
| 3. Crankshaft position (CKP) sensor | 7. GER valve |
| 4. EGR gas temperature sensor | 8. Supercharging sensor |



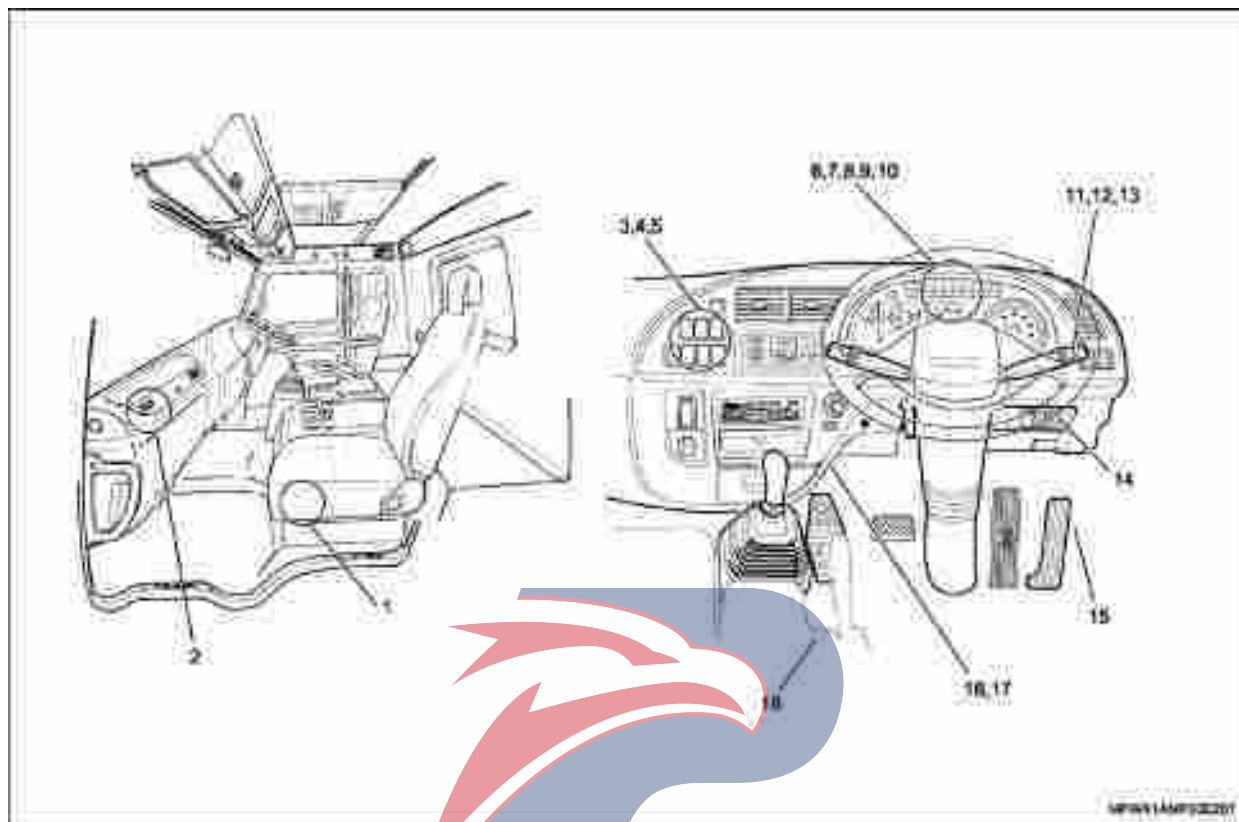
Name

1. Cam position (CMP) sensor

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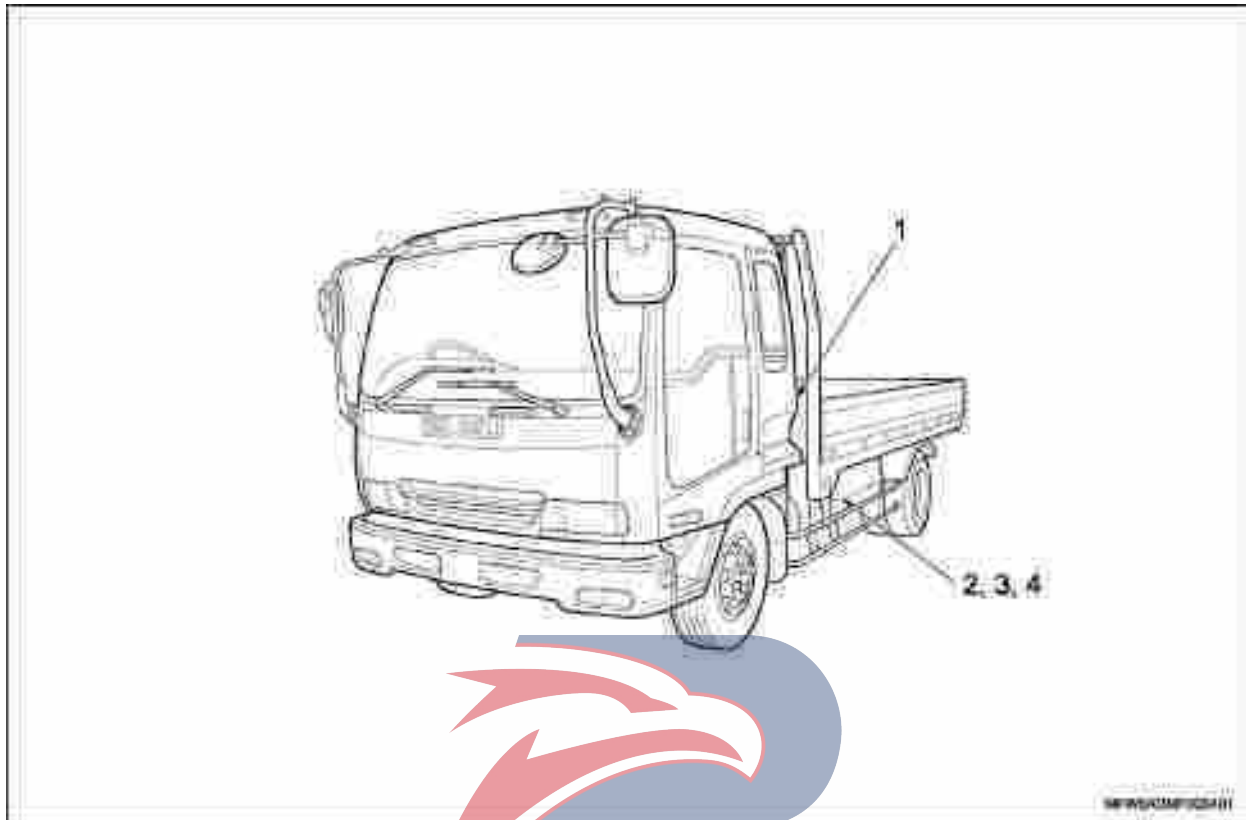
Configuration diagram of composition parts

Configuration diagram of composition parts 1



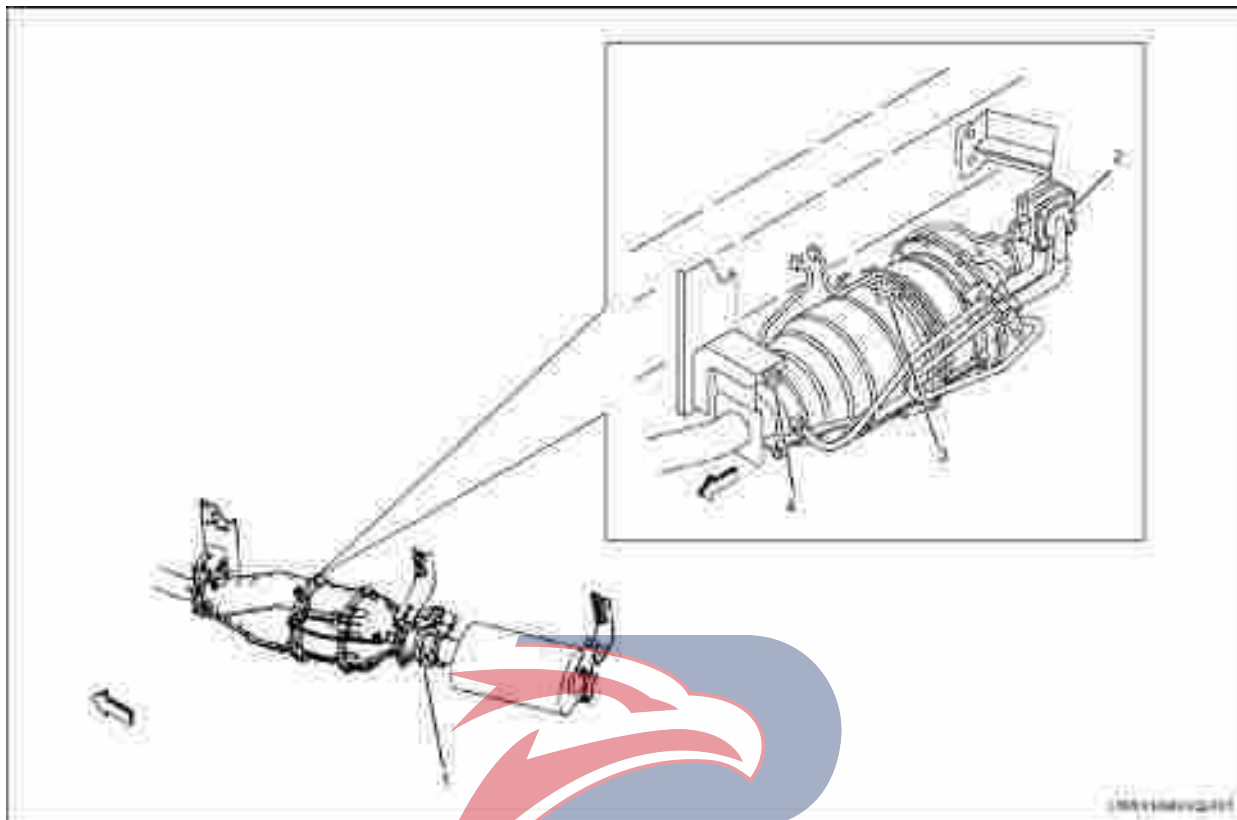
Name

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|---|---|
| 1. ECM (under the assistant) | 10. Automatic cruise setting indicator |
| 2. Atmospheric pressure sensor (in dashboard) | 11. Automatic cruise setting switch |
| 3. DPD switch | 12. Automatic cruise reset switch |
| 4. Thermal switch | 13. Auto cruise cancel switch |
| 5. Automatic cruise main switch | 14. Idle speed control switch |
| 6. Engine fault indicator | 15. Accelerator Pedal Position (APP) sensor |
| 7. DPD indicator | 16. Diagnostic connector |
| 8. QOS indicator | 17. Data link connector |
| 9. Automatic cruising main control indicator lamp | 18. Idle speed control switch |

Configuration diagram of composition parts 2**Name**

1. Intake air temperature (IAF) / MAF sensor (air purifier section)
2. Speed sensor (at the transmission)
3. Neutral switch (gearbox)
4. Reversing lamp switch (vehicle with Smoother-F) (transmission part)

Configuration diagram of composition parts 3



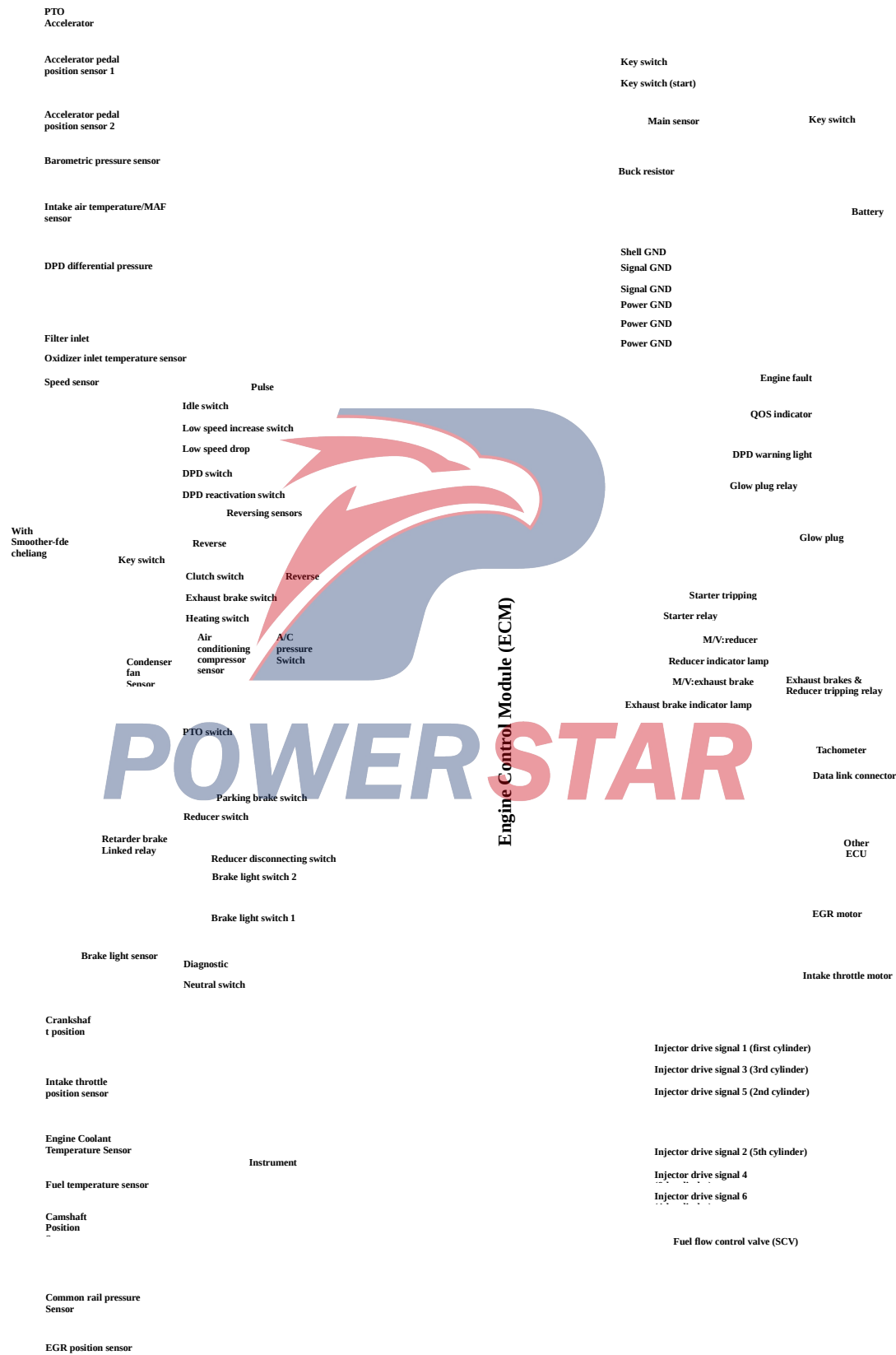
Name

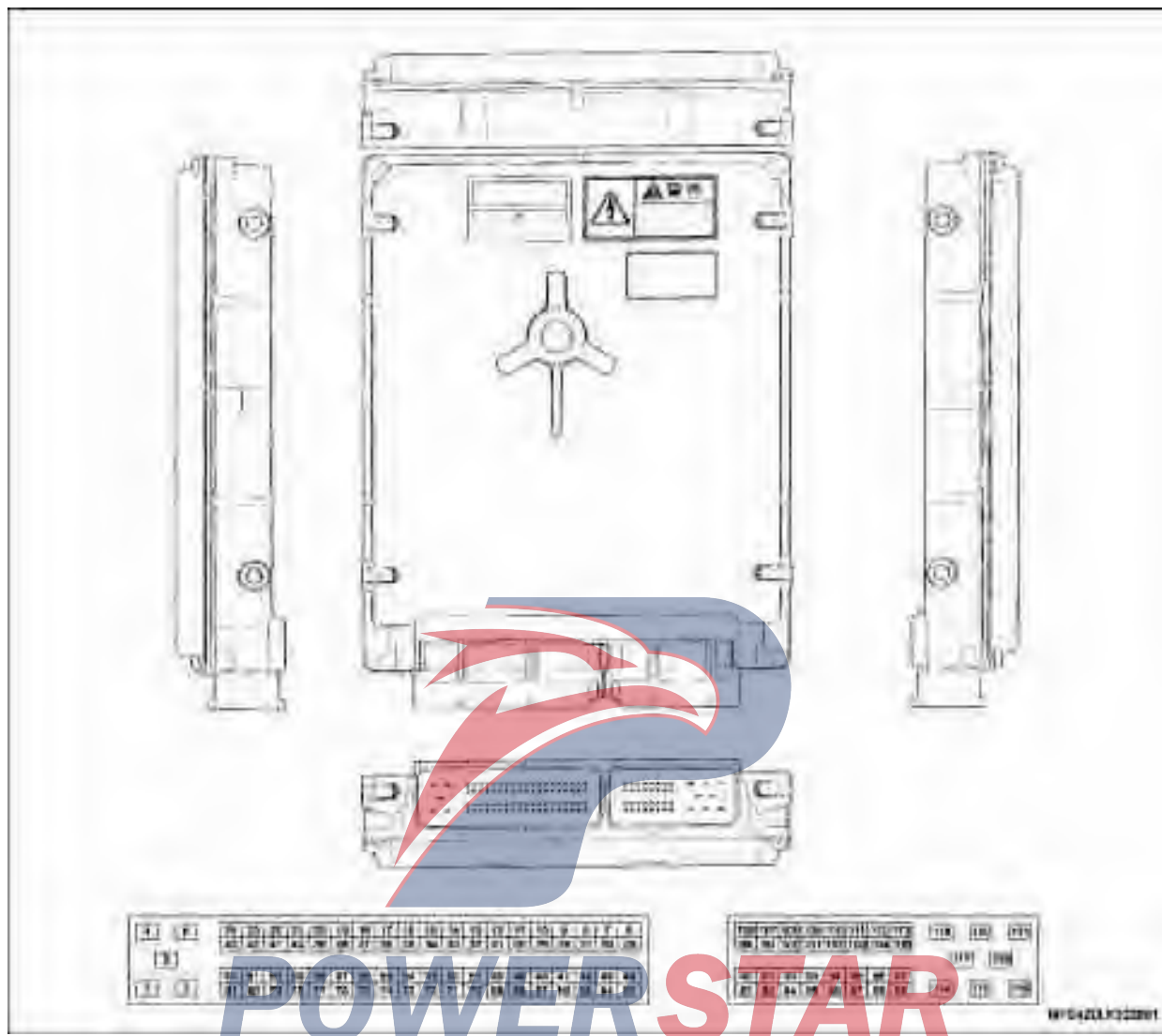
1. Exhaust throttle (forced temperature rise auxiliary device)
2. DPD differential pressure sensor
3. Before filter of exhaust temperature sensor 1
4. Exhaust temperature sensor 2 (before oxidation catalyst)

Circuit diagram

6HL1

Wiring diagram of ECM



The configuration of the engine control module (ECM) pin**81 Pin joint**

Pin No.	Terminal name	linkage
1	PG-POWER	GND
2	PS-+B	power supply
3	PG-POWER	GND
4	PG-POWER	GND
5	PS-+B	power supply
6	OS-DIAGL	Engine fault indicator lamp
7	OS-EXBL	Exhaust brake indicator lamp
8	OF-TACHO	Tachometer
9	OS-CRML	DPD light
10	OS-GLOWR	Glow plug relay

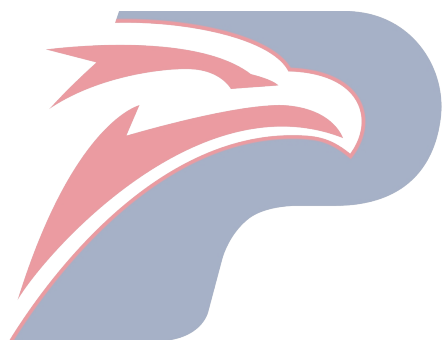
Pin No.	Terminal name	linkage
11	OS-GLOWL	QOS indicator
12	OS-RET	Retarder indicator lamp, retarder magnet valve
13	-	-
14	OS-STARTR	Retarder tripping relay
15	OS-EXB	Exhaust brake solenoid valve
16	-	-
17	-	-
18	CC-CAN-H	CAN-HIGH
19	IF-SPD	Speed sensor signal
20	-	-
21	OS-MAINR	ECM main relay
22	SG-12VRT	MAF sensor GND
23	SP-12V	MAF sensor power supply
24	IS-IGKEY	Key switch ON signal
25	-	-
26	IS-CRSET	DPD switch
27	IS-CRRES	Standby light switch (only for vehicles with smoother-F)
28	IS-CRREL	DPD reactivation switch
29	IS-IDLMNL	Idling control switch
30	IS-IDLUP	Increasing signal of Idle control switch
31	IS-IDLDWN	Decreasing signal of Idle control switch
32	-	-
33	-	-
34	IS-A/C	A/C switch
35	PS-MAFBAT	Varistor buck
36	-	-
37	CC-CAN-L	CAN-LOW
38	CC-KW2000	Data link connector
39	SG-SLD2	MAF sensor shielding
40	OS-MAINR	ECM main relay
41	SG-5VRT1	PTO Accelerator Sensor, Accelerator Pedal Position (APP) Sensor 1 GND
42	SP-5V1	PTO Accelerator Sensor, Accelerator Pedal Position (APP) Sensor 1 power supply
43	PG-SIGN	GND
44	IS-PTO	PTO switch
45	IS-EXB	Exhaust brake switch
46	IS-START	Key switch start signal
47	IS-CL	Clutch switch

Pin No.	Terminal name	linkage
48	IS-PBRK	Parking brake switch
49	IS-RET	Retarder switch
50	-	-
51	IS-QWS	Heating switch
52	IS-DIAG	Diagnostic connector
53	IS-LOWOIL	Neutral switch
54	IS-BRAKE1	Brake light switch 1
55	IS-BRAKE2	Brake light switch 2
56	-	-
57	-	-
58	-	-
59	SG-SLD	DPD differential pressure sensor shielding
60	SG-5VRT2	Accelerator pedal position (APP) sensor 2, barometric pressure sensor, Intake temperature (IAT) sensor GND
61	SP-5V2	Accelerator pedal position (APP) sensor 2, barometric pressure sensor power
62	PG-SIGN	GND
63	IA-ACCEL1	Accelerator pedal position (APP) sensor 1 signal
64	IA-ACCEL2	Accelerator pedal position (APP) sensor 2 signal
65	-	-
66	-	-
67	IA-EXHP1	DPD differential pressure sensor signal
68	-	-
69	IA-MAF	MAF sensor signal
70	IA-PTOPOS	PTO acceleration sensor signal
71	IA-BARO	Atmospheric pressure sensor signal
72	IA-IAT	Intake temperature (IAT) sensor signal
73	IA-THE1	DPD exhaust temperature sensor 1 (filter front) signal
74	IA-THE2	DPD exhaust temperature sensor 2 (before oxidation catalyst) signal
75	-	-
76	-	-
77	-	-
78	-	-
79	SG-5VRT3	DPD differential pressure sensor, DPD exhaust temperature sensor 1 (filter front), DPD exhaust temperature sensor 2 (before oxidation catalyst) GND
80	SP-5V3	Power supply of DPD differential pressure sensor
81	PG-CASE	GND

40 Pin joint

Pin No.	Terminal name	linkage
82	IA-PFUEL	Common rail pressure sensor signal
83	IA-THL	Fuel temperature (FT) sensor signal
84	IA-THW	Engine coolant temperature (ECT) sensor signal
85	IA-IDMPOS	ITP sensor signal
86	-	-
87	SP-5V5	Common rail pressure sensor, EGR position sensor power
88	-	-
89	IA-SCVLO	SCV-LOW
90	IA-PFUEL	Common rail pressure sensor signal
91	-	-
92	IS-EBMPOS3	EGR position sensor-3
93	IS-EBMPOS2	EGR position sensor-2
94	IS-EBMPOS1	EGR position sensor 1
95	SP-5V4	ITP sensor power
96	-	-
97	IA-SCVLO	SCV-LOW
98	IF-CAMHAL	Camshaft position (CMP) sensor power
99	AP-CANHAL	Camshaft position (CMP) sensor signal
100	SG-SLD5	Camshaft Position (CMP) Sensor, Common Rail Pressure Sensor Shielded
101	SG-5VRT5	Common rail pressure sensor, EGR position sensor GND
102	OS-CCS	DPD M/V
103	OM-EBM2	EGR motor drive 2
104	OM-IDM2	Intake throttle motor(-)
105	OP-SCVHI	SCV-HIGH
106	IF-CRANK-	Crankshaft position (CKP) sensor (-)
107	IF-CRANK +	Crankshaft position (CKP) sensor (+)
108	SG-SLD4	Crankshaft position (CKP) sensor shielding
109	SG-5VRT4	Intake throttle position (ITP) sensor, engine coolant temperature (ECT) sensor, Fuel Temperature (FT) Sensor GND
110	OM-EBM3	EGR motor drive 3
111	OM-EBM1	EGR motor drive 1
112	OM-IDM1	Intake throttle motor(+)
113	OP-SCVHI	SCV-HIGH
114	OS-INJ5	Injector 5 (2nd cylinder)
115	OS-INJ6	Injector 6 (46th cylinder)
116	OP-COM2	Ejector power supply 2 (4th cylinder, 5th cylinder, 6th cylinder)
117	OS-INJ3	Injector 3 (3rd cylinder)

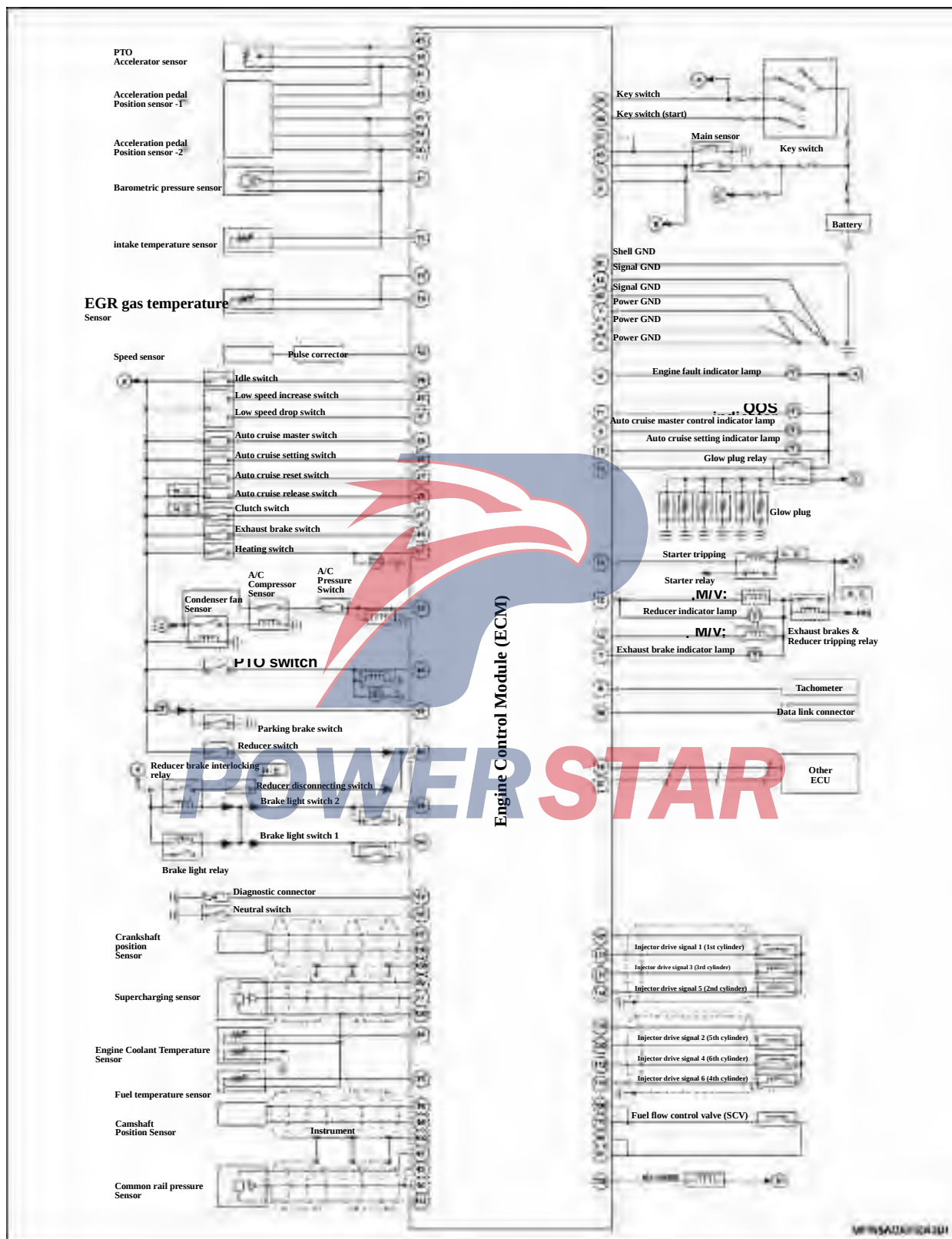
Pin No.	Terminal name	linkage
118	OS-INJ4	Injector 4 (6th cylinder)
119	OS-INJ1	Injector 1 (No.1 cylinder)
120	OS-INJ2	Injector 2 (5th cylinder)
121	OP-COM1	Ejector power supply 1 (first cylinder, second cylinder, third cylinder)

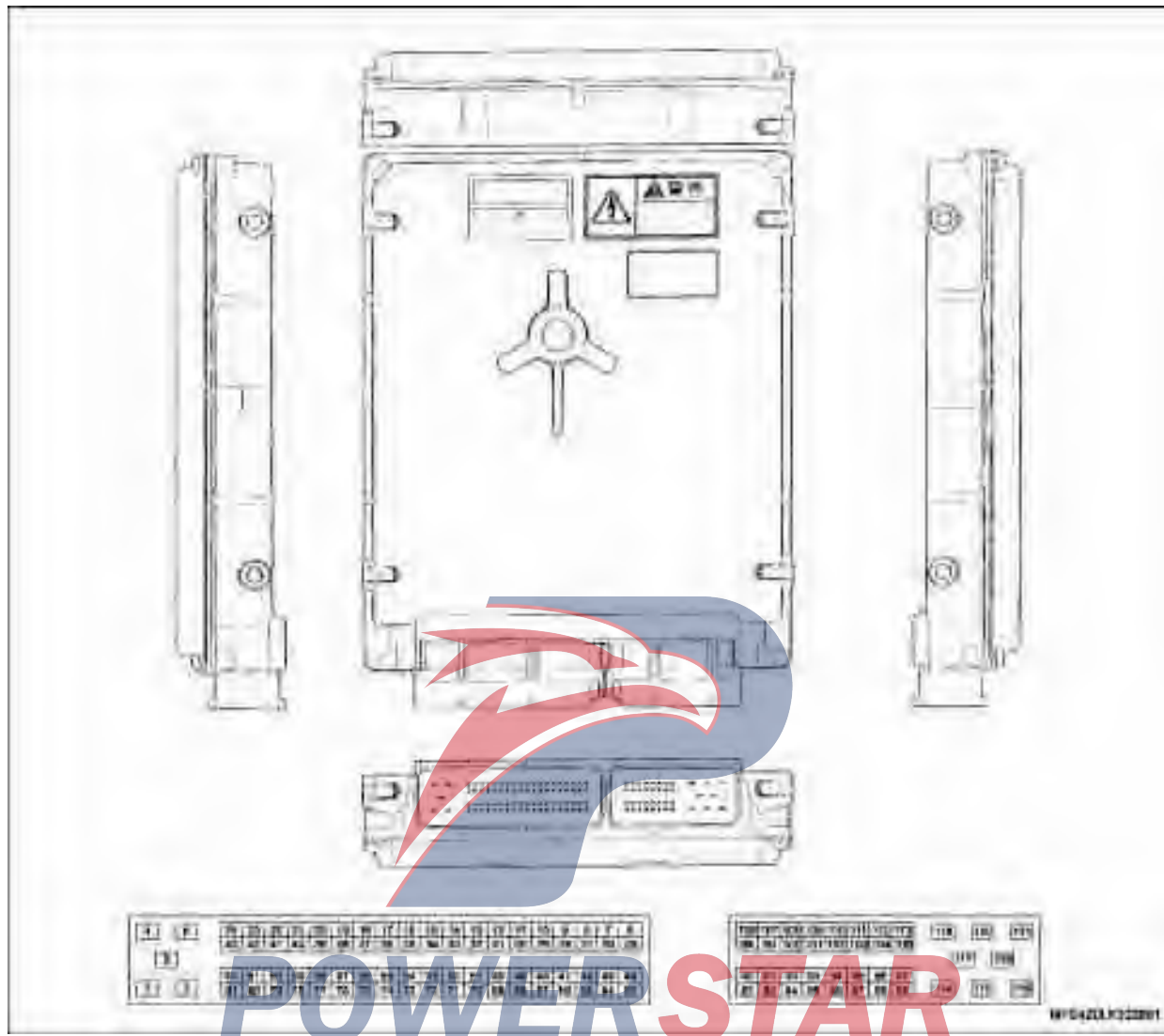


POWERSTAR

6HK1-TC

Wiring diagram of ECM



The configuration of the engine control module (ECM) pin**81 Pin joint**

Pin No.	Terminal name	linkage
1	PG-POWER	GND
2	PS-+B	power supply
3	PG-POWER	GND
4	PG-POWER	GND
5	PS-+B	power supply
6	OS-DIAGL	Engine fault indicator lamp
7	OS-EXBL	Exhaust brake indicator lamp
8	OF-TACHO	Tachometer
9	OS-CRML	Auto cruise master control indicator lamp
10	OS-GLOWR	Glow plug relay

Pin No.	Terminal name	Connection
11	OS-GLOWL	QOS indicator
12	OS-RET	Retarder indicator lamp, retarder magnet valve
13	OS-ECOR	Auto cruise setting indicator lamp
14	OS-STARTR	Starter disconnect relay
15	OS-EXB	Exhaust brake solenoid valve
16	-	-
17	-	-
18	CC-CAN-H	CAN-HIGH
19	IF-SPD	Speed sensor signal
20	-	-
21	OS-MAINR	ECM main relay
22	-	-
23	-	-
24	IS-IGKEY	Key switch ON signal
25	IS-CRM	Auto cruise master switch
26	IS-CRSET	Auto cruise setting switch
27	IS-CRRES	Auto cruise reset switch
28	IS-CRREL	Auto cruise cancel switch
29	IS-IDLMNL	Idling control switch
30	IS-IDLUP	Increasing signal of Idle control switch
31	IS-IDLDWN	Decreasing signal of Idle control switch
32	-	-
33	-	-
34	IS-A/C	A/C switch
35	-	-
36	-	-
37	CC-CAN-L	CAN-LOW
38	CC-KW2000	Data link connector
39	-	-
40	OS-MAINR	ECM main relay
41	SG-5VRT1	PTO accelerator sensor, accelerator pedal position (APP) sensor 1 GND
42	SP-5V1	PTO accelerator sensor, accelerator pedal position (APP) sensor 1 power supply
43	PG-SIGN	GND
44	IS-PTO	PTO switch
45	IS-EXB	Exhaust brake switch
46	IS-START	Key switch start signal
47	IS-CL	Clutch switch

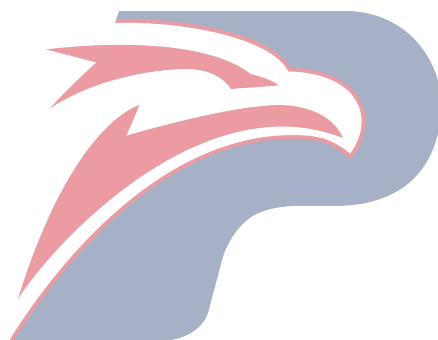
Pin No.	Terminal name	Connection
48	IS-PBRK	Parking brake switch
49	IS-RET	Retarder switch
50	-	-
51	IS-QWS	Heating switch
52	IS-DIAG	Diagnostic connector
53	IS-LOWOIL	Neutral switch
54	IS-BRAKE1	Brake light switch 1
55	IS-BRAKE2	Brake light switch 2
56	-	-
57	-	-
58	-	-
59	-	-
60	SG-5VRT2	Accelerator pedal position (APP) sensor 2, barometric pressure sensor, Intake temperature (IAT) sensor GND
61	SP-5V2	Accelerator pedal position (APP) sensor 2, barometric pressure sensor power
62	PG-SIGN	GND
63	IA-ACCEL1	Accelerator pedal position (APP) sensor 1 signal
64	IA-ACCEL2	Accelerator pedal position (APP) sensor 2 signal
65	-	-
66	-	-
67	-	-
68	-	-
69	-	-
70	IA-PTOPOS	PTO accelerator sensor signal
71	IA-BARO	Atmospheric pressure sensor signal
72	IA-IAT	Intake temperature (IAT) sensor signal
73	IA-THE1	EGR gas temperature sensor signal
74	-	-
75	-	-
76	-	-
77	-	-
78	-	-
79	SG-5VRT3	EGR gas temperature sensor GED
80	-	-
81	PG-CASE	GND

POWERSTAR

40 Pin joint

Pin No.	Terminal name	linkage
82	IA-PFUEL	Common rail pressure sensor signal
83	IA-THL	Fuel temperature (FT) sensor signal
84	IA-THW	Engine coolant temperature (ECT) sensor signal
85	-	-
86	-	-
87	SP-5V5	Common rail pressure sensor power supply
88	-	-
89	IA-SCVLO	SCV-LOW
90	IA-PFUEL	Common rail pressure sensor signal
91	IA-MAP	Boost sensor signal
92	-	-
93	-	-
94	-	-
95	SP-5V4	Power supply of booster sensor
96	-	-
97	IA-SCVLO	SCV-LOW
98	IF-CAMHAL	Camshaft position (CMP) sensor power
99	AP-CANHAL	Camshaft position (CMP) sensor signal
100	SG-SLD5	Camshaft Position (CMP) Sensor, Common Rail Pressure Sensor Shielded
101	SG-5VRT5	Common rail pressure sensor GND
102		
103		
104		
105	OP-SCVHI	SCV-HIGH
106	IF-CRANK-	Crankshaft position (CKP) sensor (-)
107	IF-CRANK +	Crankshaft position (CKP) sensor (+)
108	SG-SLD4	Crankshaft position (CKP) sensor shielding
109	SG-5VRT4	Booster sensor, engine coolant temperature (ECT) sensor, Fuel Temperature (FT) Sensor GND
110	-	-
111	-	-
112	-	-
113	OP-SCVHI	SCV-HIGH
114	OS-INJ5	Injector 5 (2nd cylinder)
115	OS-INJ6	Injector 6 (46th cylinder)
116	OP-COM2	Ejector power supply 2 (4th cylinder, 5th cylinder, 6th cylinder)
117	OS-INJ3	Injector 3 (3rd cylinder)

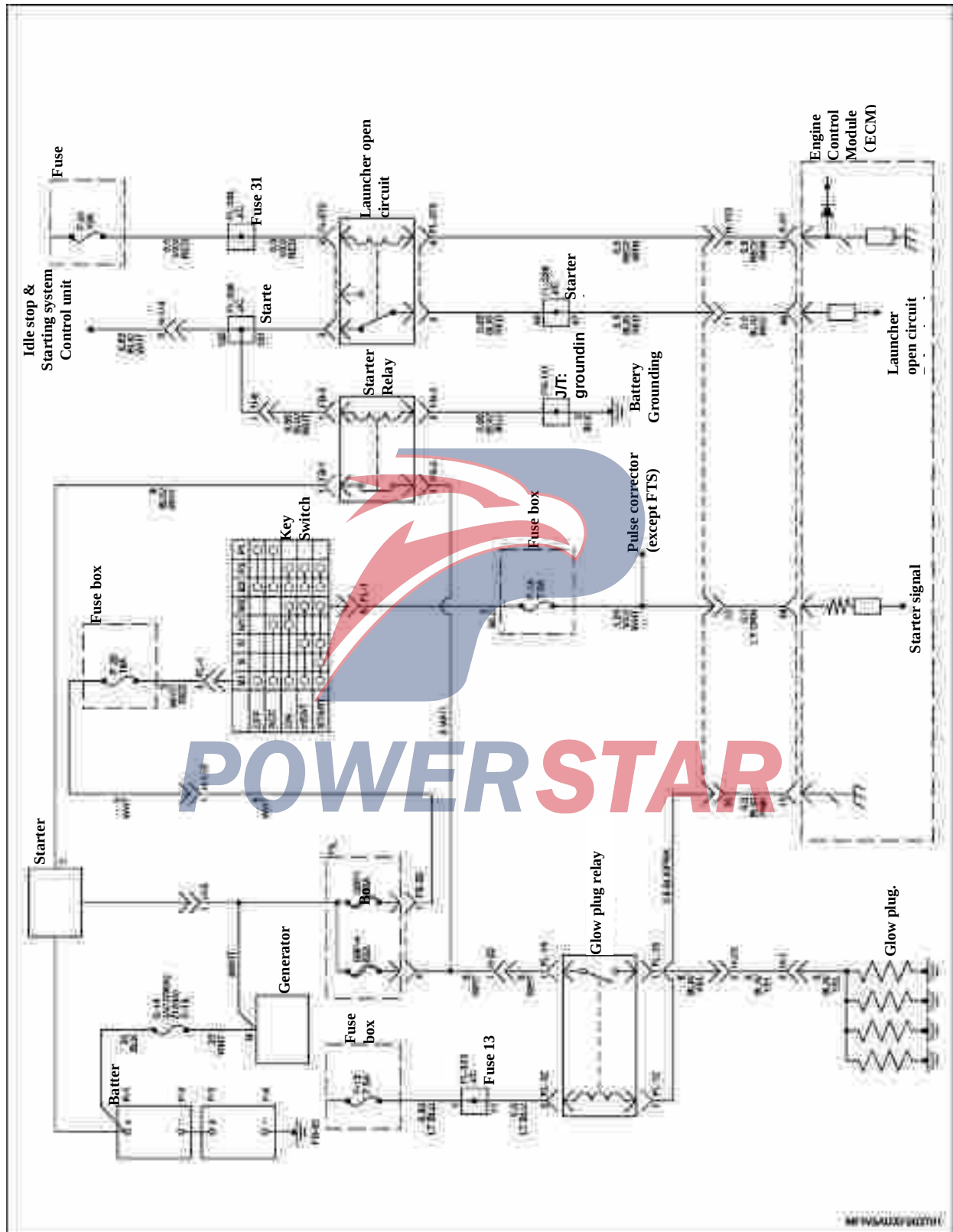
Pin No.	Terminal name	linkage
118	OS-INJ4	Injector 4 (6th cylinder)
119	OS-INJ1	Injector 1 (No.1 cylinder)
120	OS-INJ2	Injector 2 (5th cylinder)
121	OP-COM1	Ejector power supply 1 (first cylinder, second cylinder, third cylinder)



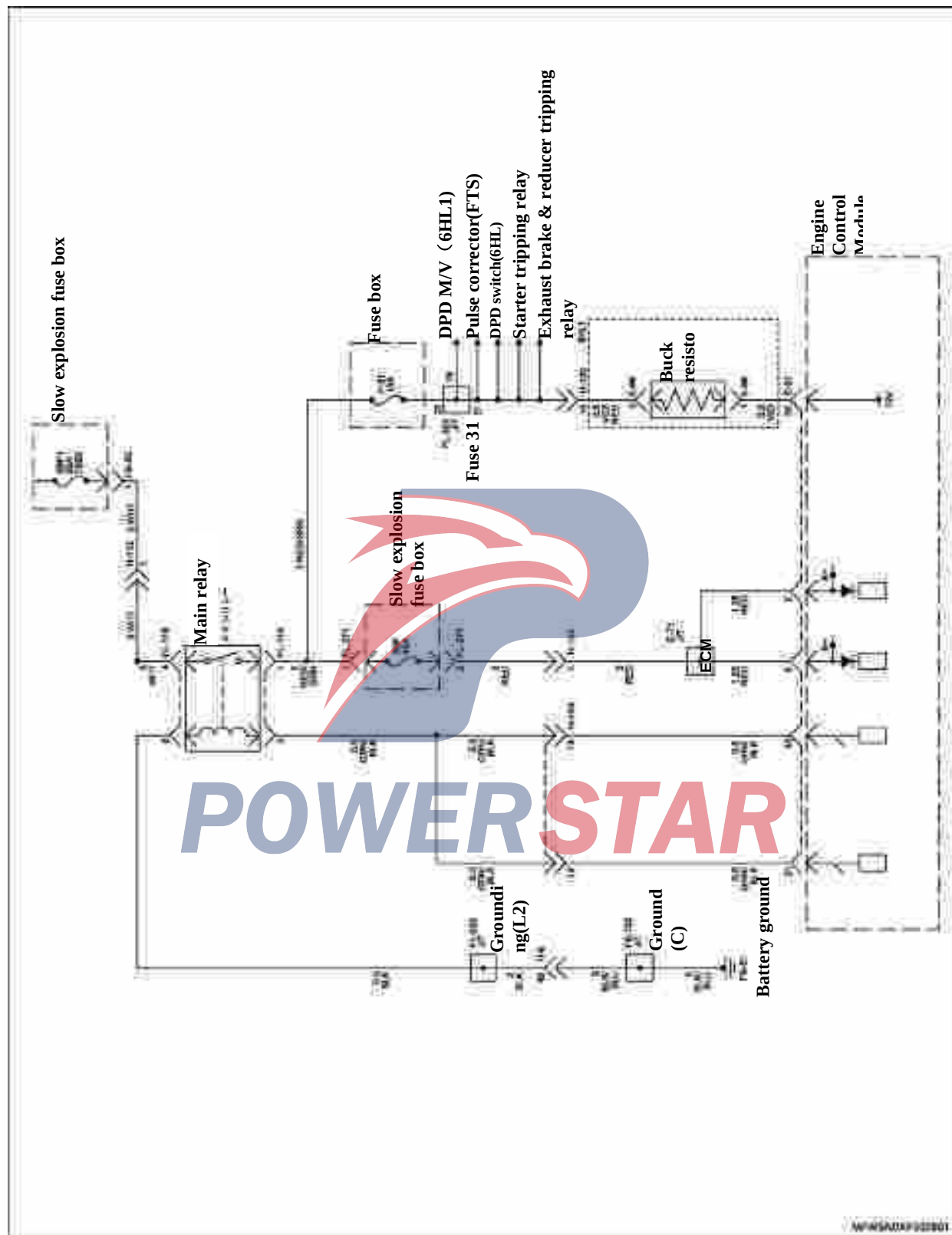
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6HL1 / 6HK1-TC loop diagram

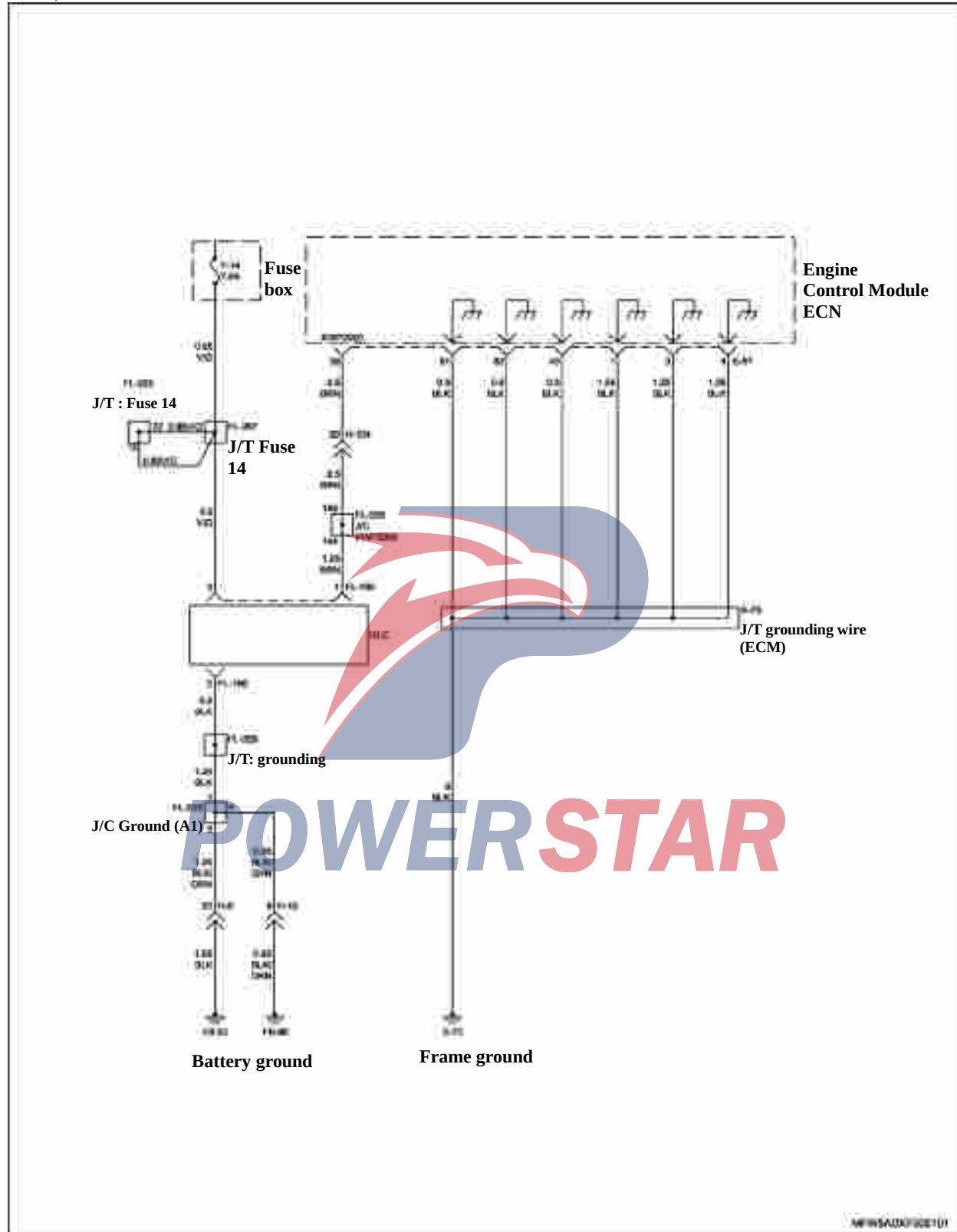
Starter, glow plug circuit



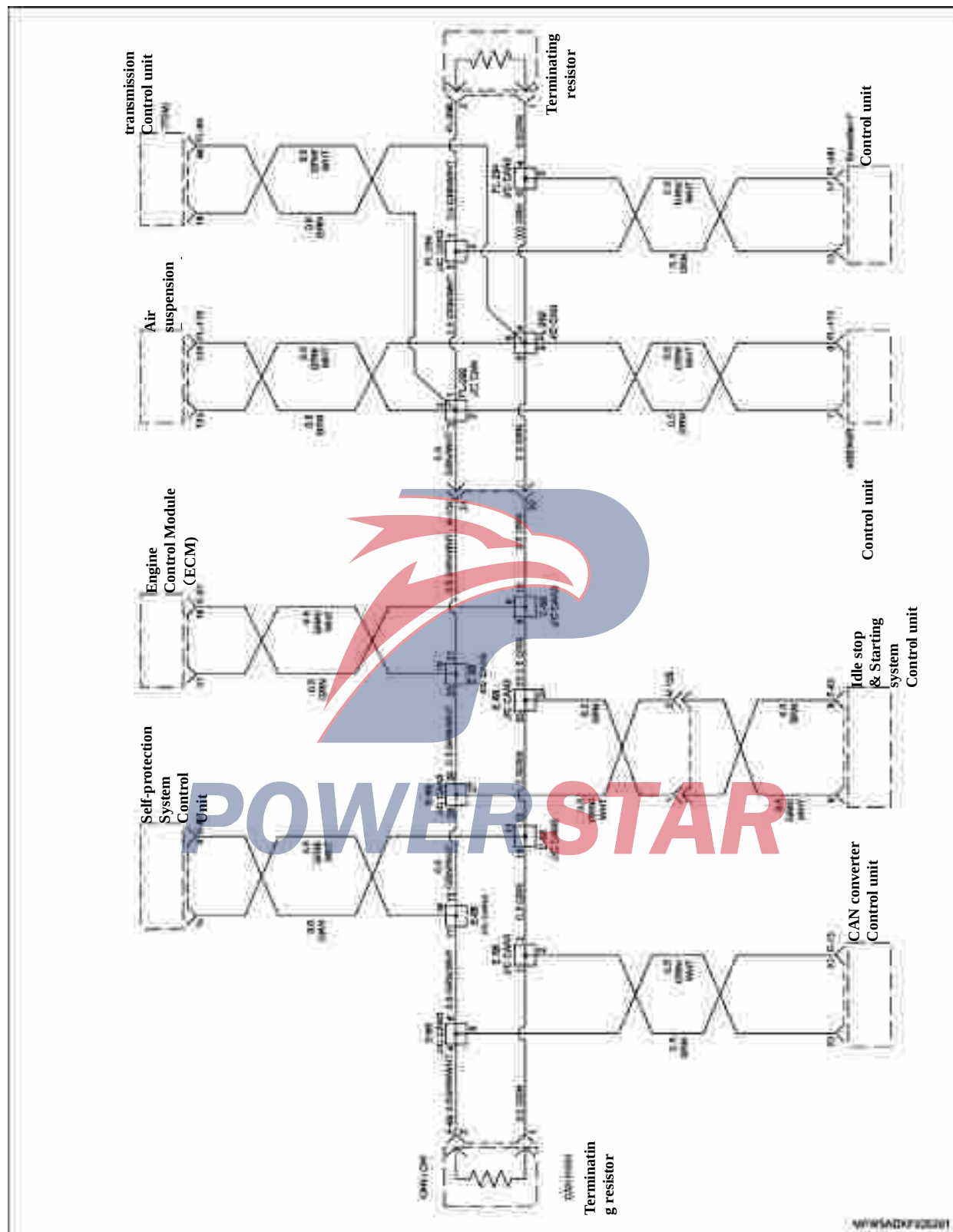
Main relay, dropping resistor (6HL1) circuit



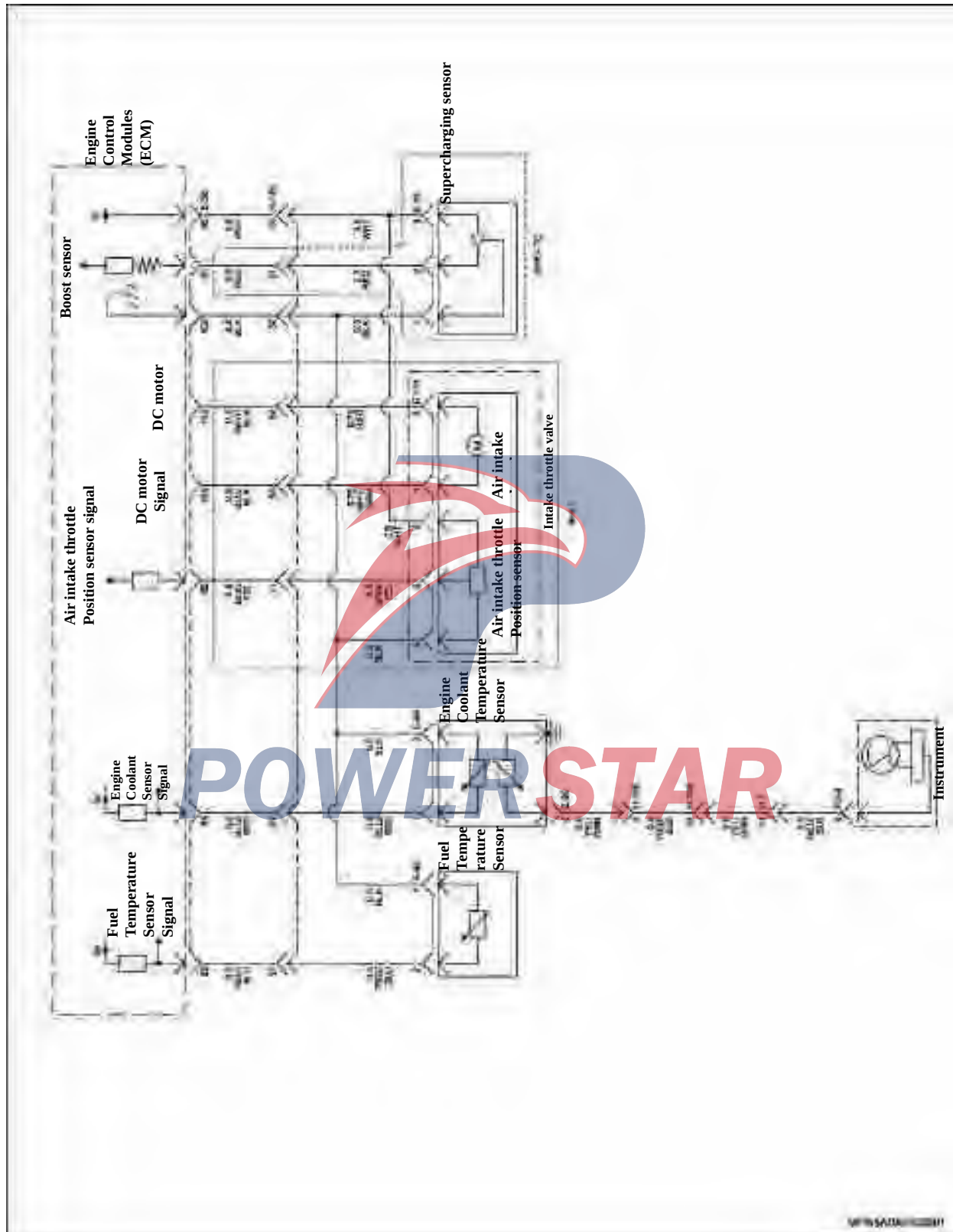
GND, DLC Circuit



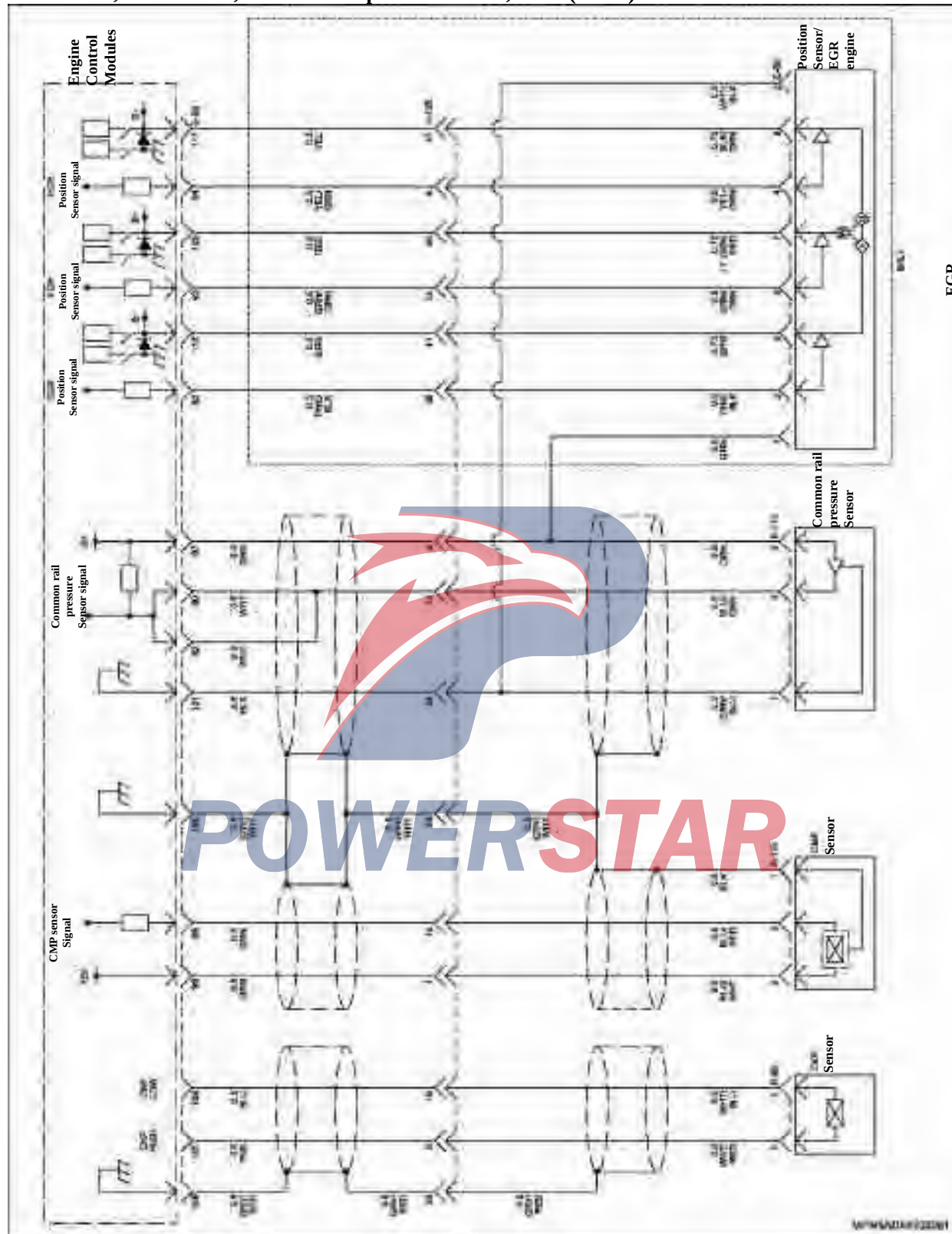
CAN circuit



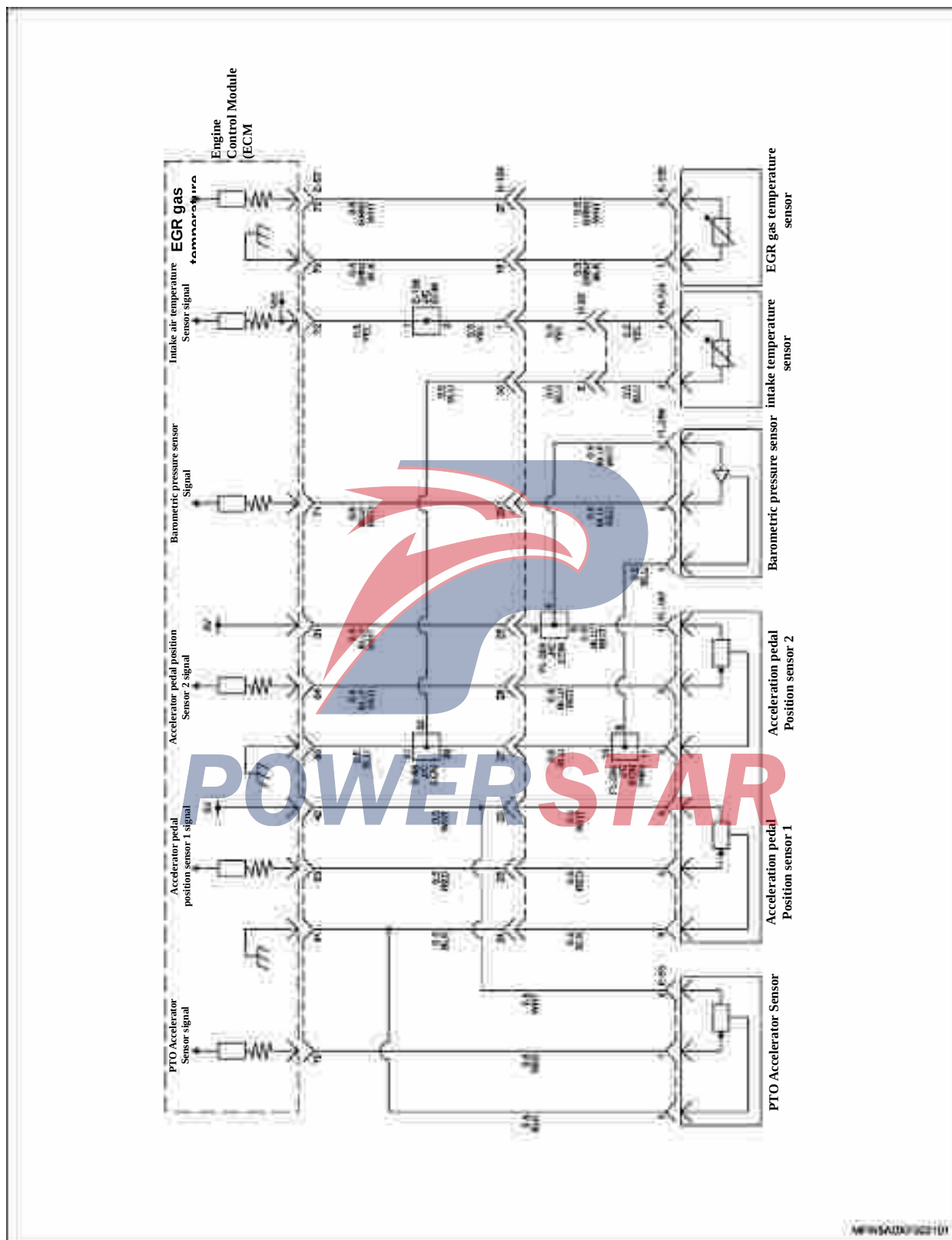
Fuel temperature sensor, engine coolant temperature sensor, intake throttle (6HL1), supercharging sensor (6HK1-TC) circuit



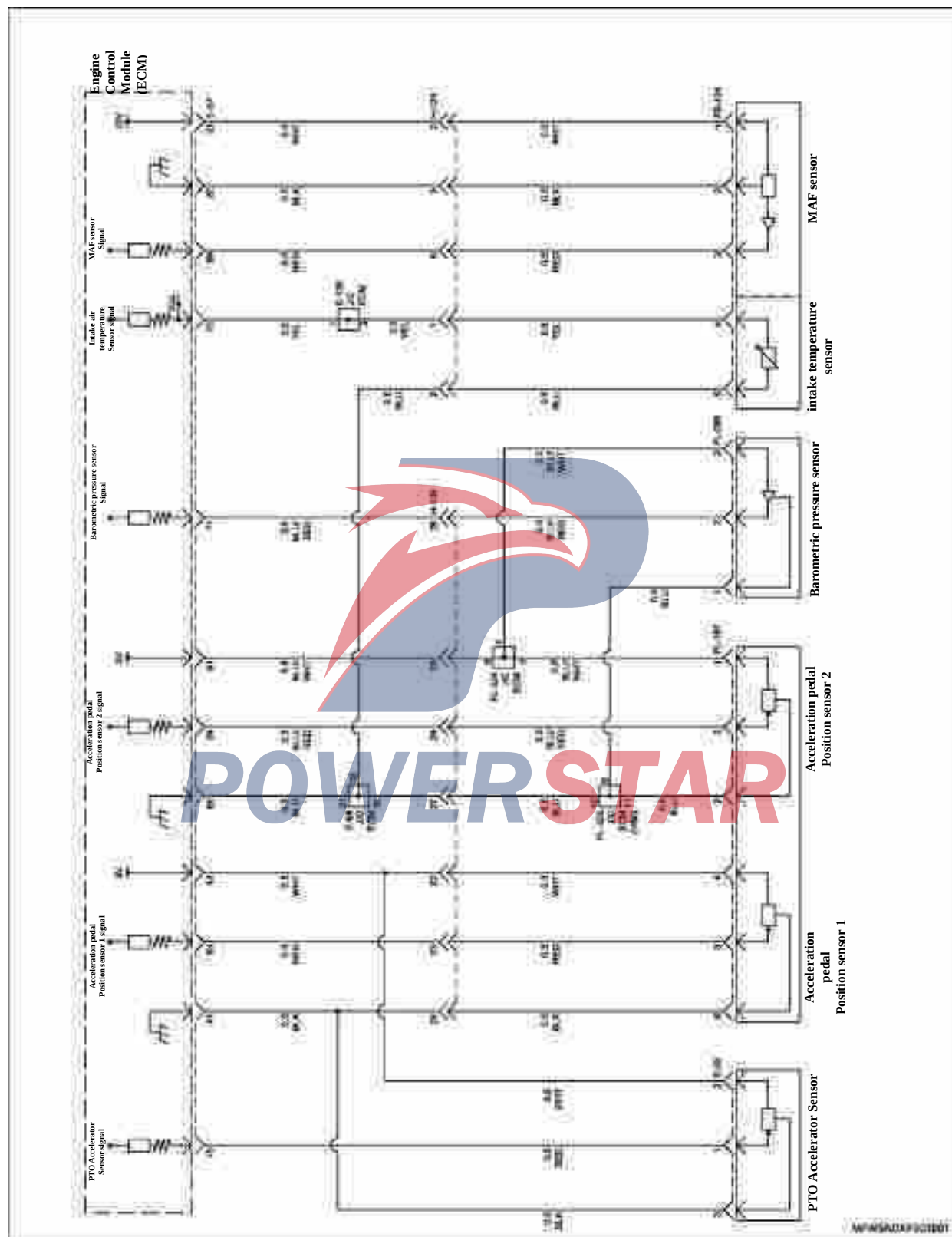
CKP sensor, CMP sensor, common rail pressure sensor, EGR (6HL1) circuit



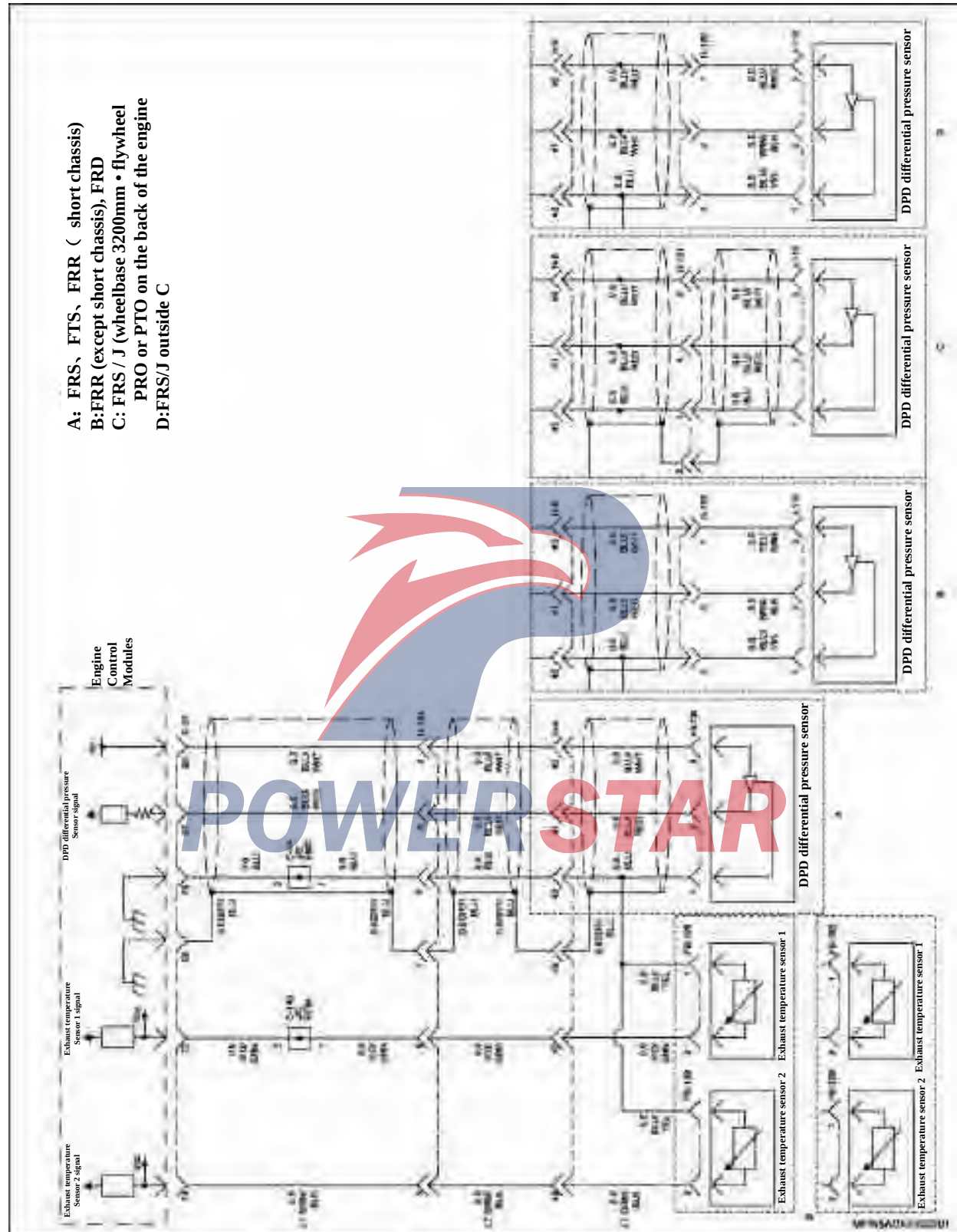
PTO accelerator sensor, accelerator pedal position sensor (1, 2), barometric pressure sensor, intake air temperature sensor, EGR gas temperature sensor circuit (for 6HK1-TC only)



PTO accelerator sensor, accelerator pedal position sensor (1, 2), barometric pressure sensor, intake air temperature sensor/MAF sensor circuit (for 6HL1 only)

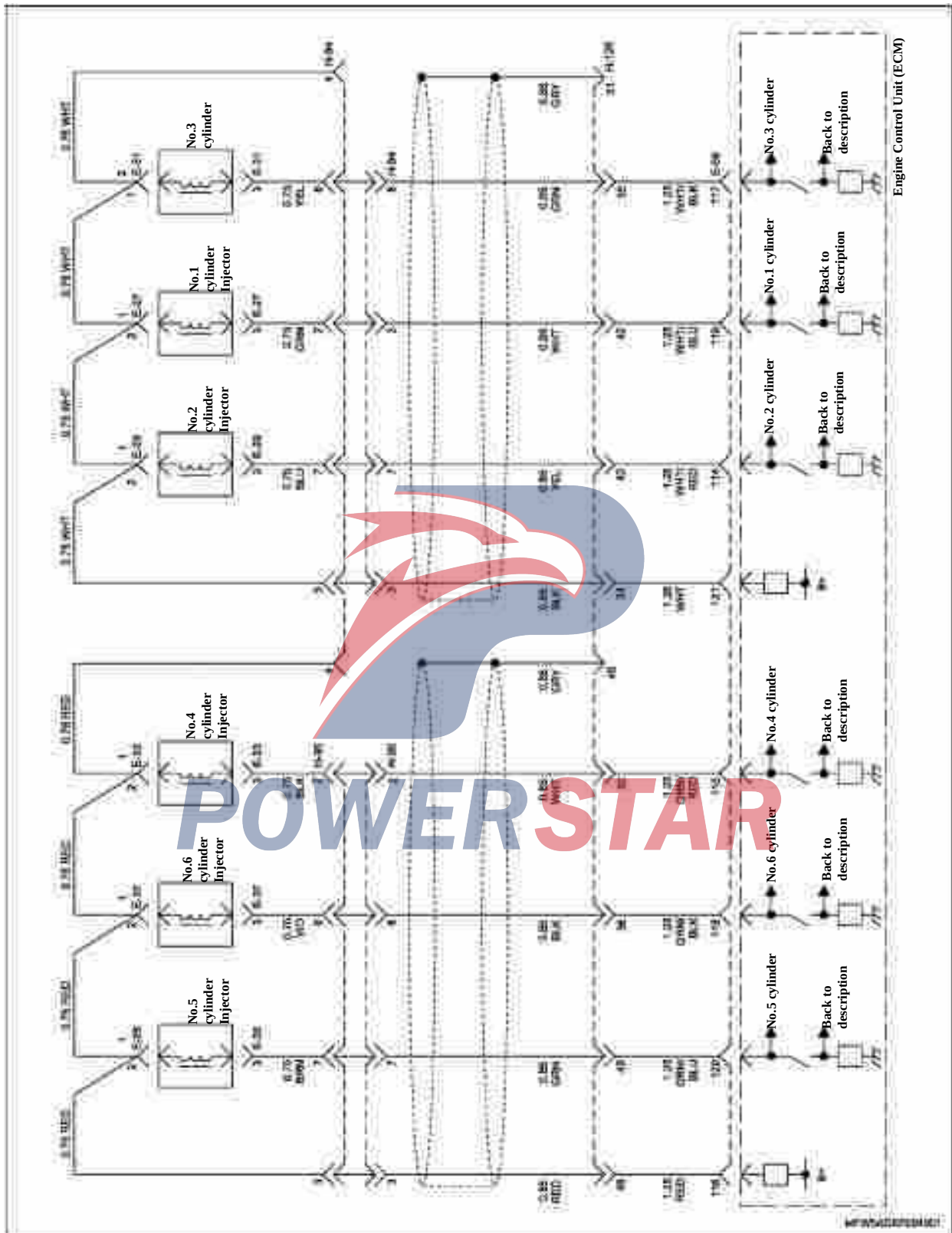


Exhaust temperature sensor (1, 2), DPD differential pressure sensor circuit (6HL1 only)

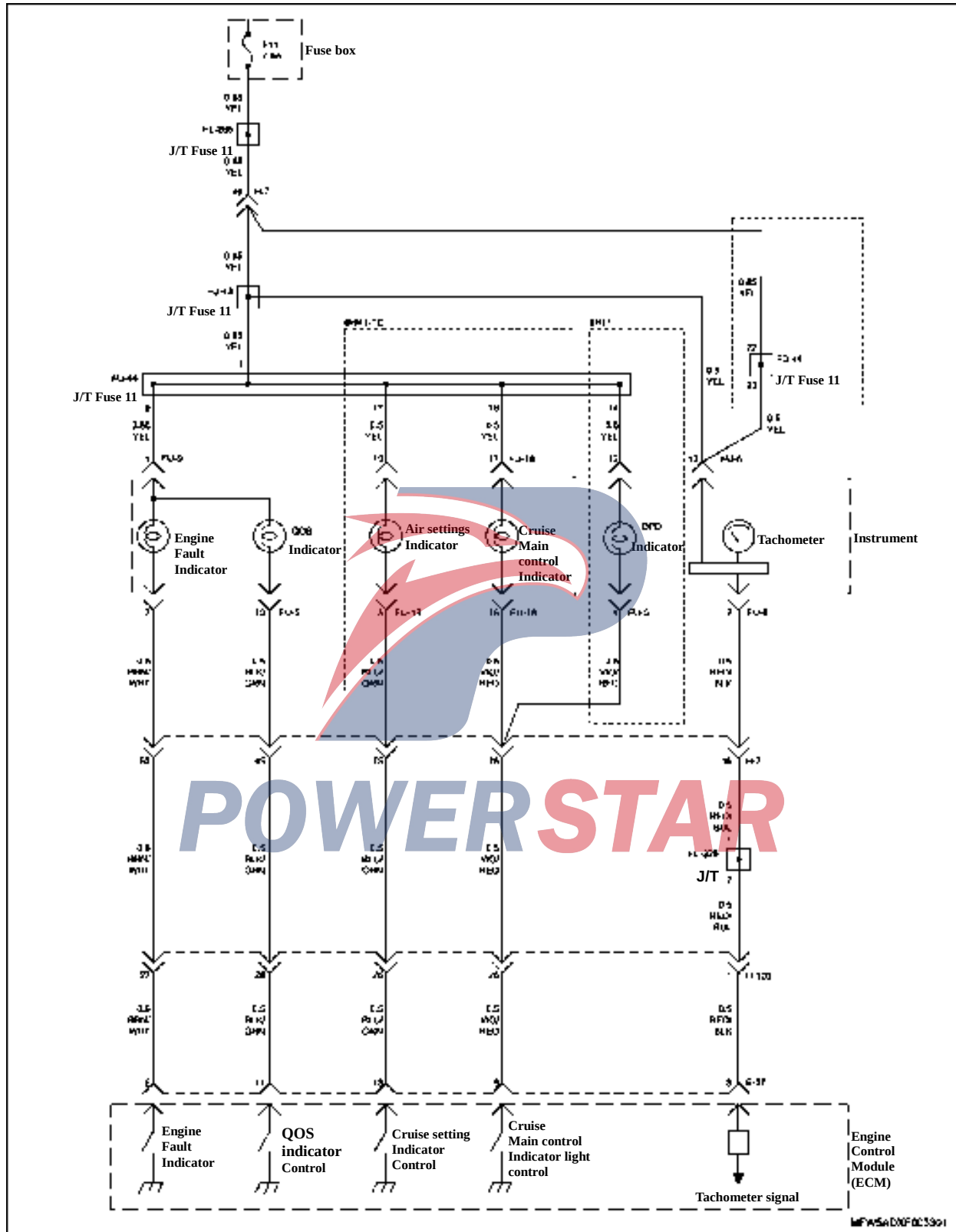


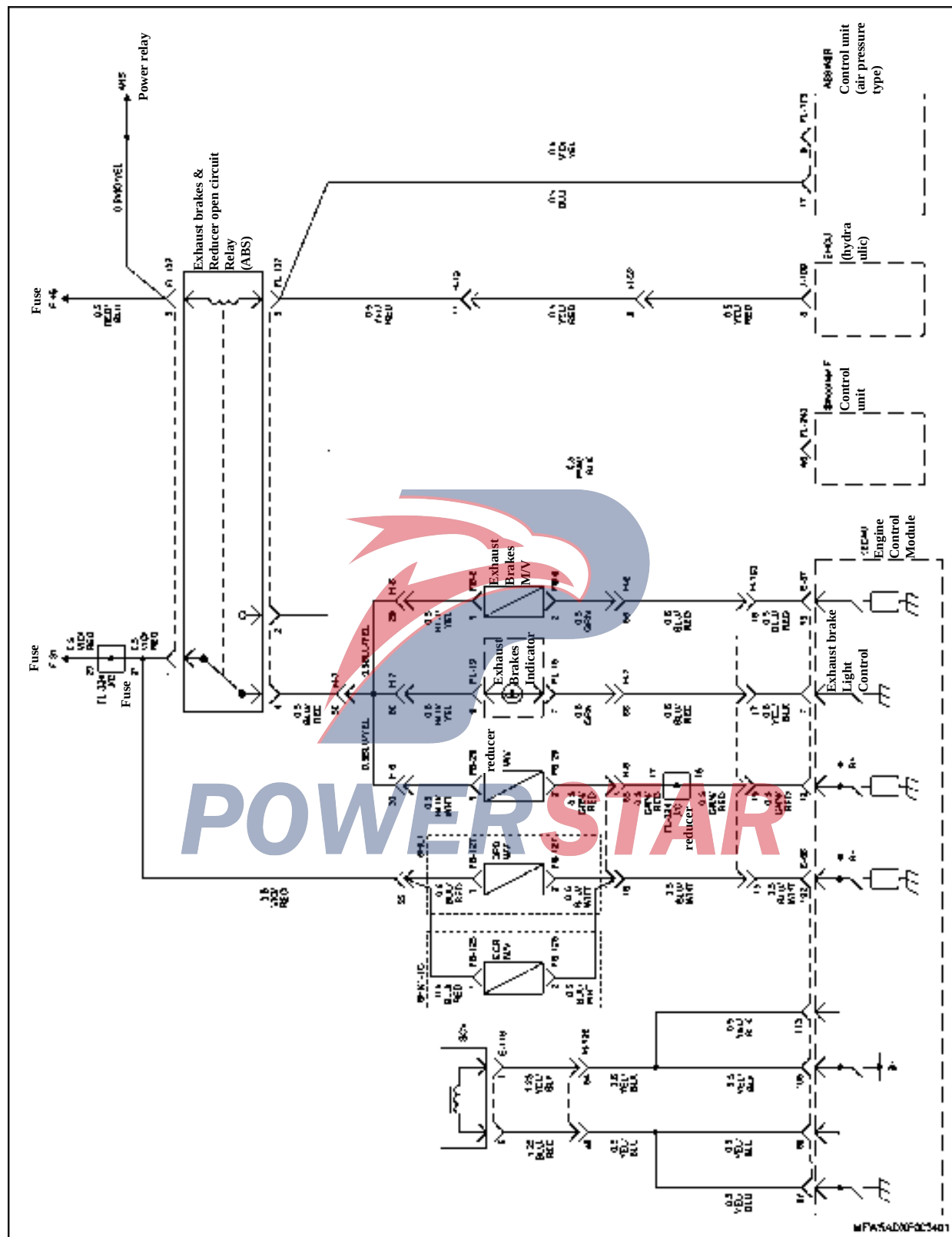
Outsi

Injector circuit

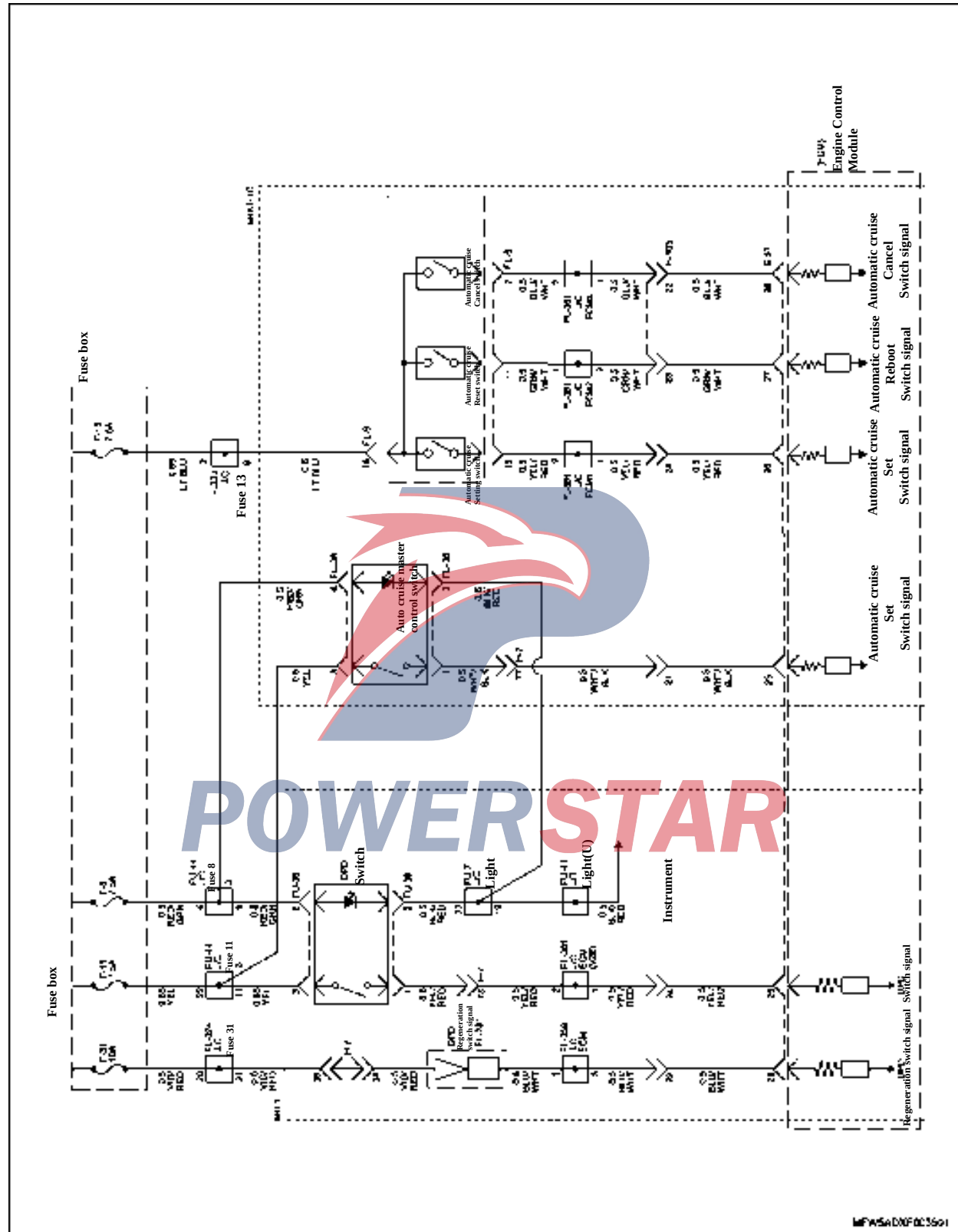


Instrument cycle

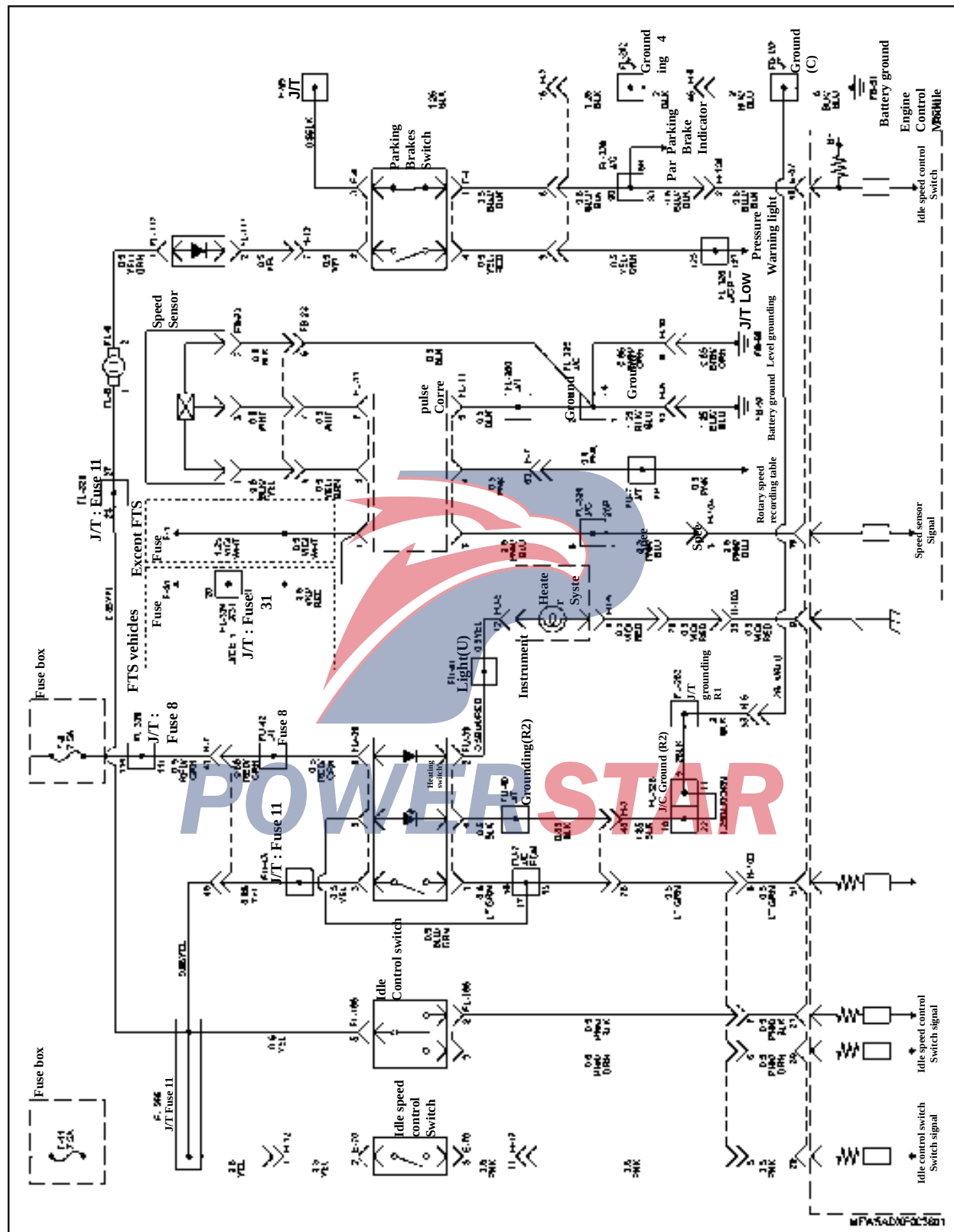




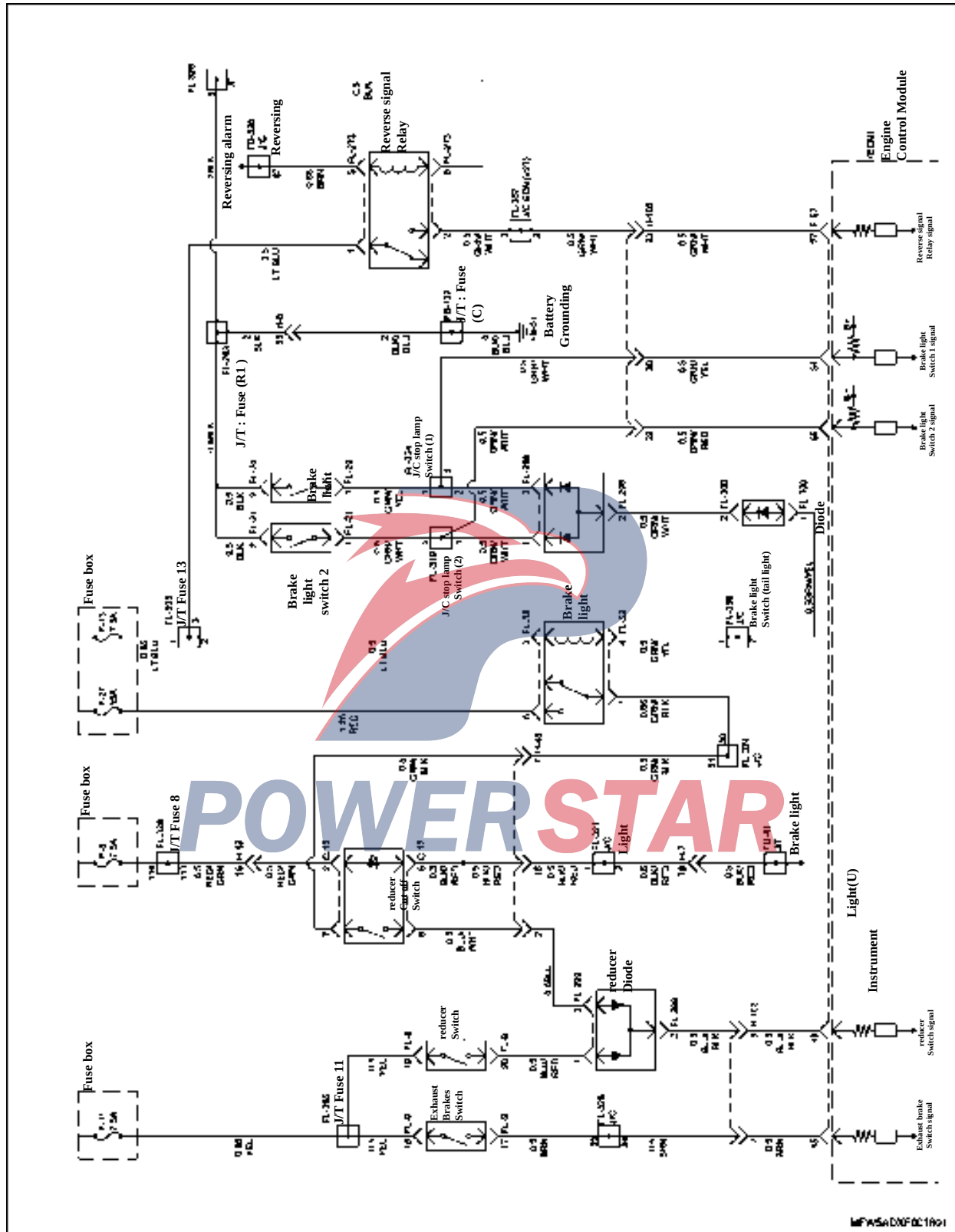
DPD switch (6HL1), auto cruise (6HK1-TC) circuit



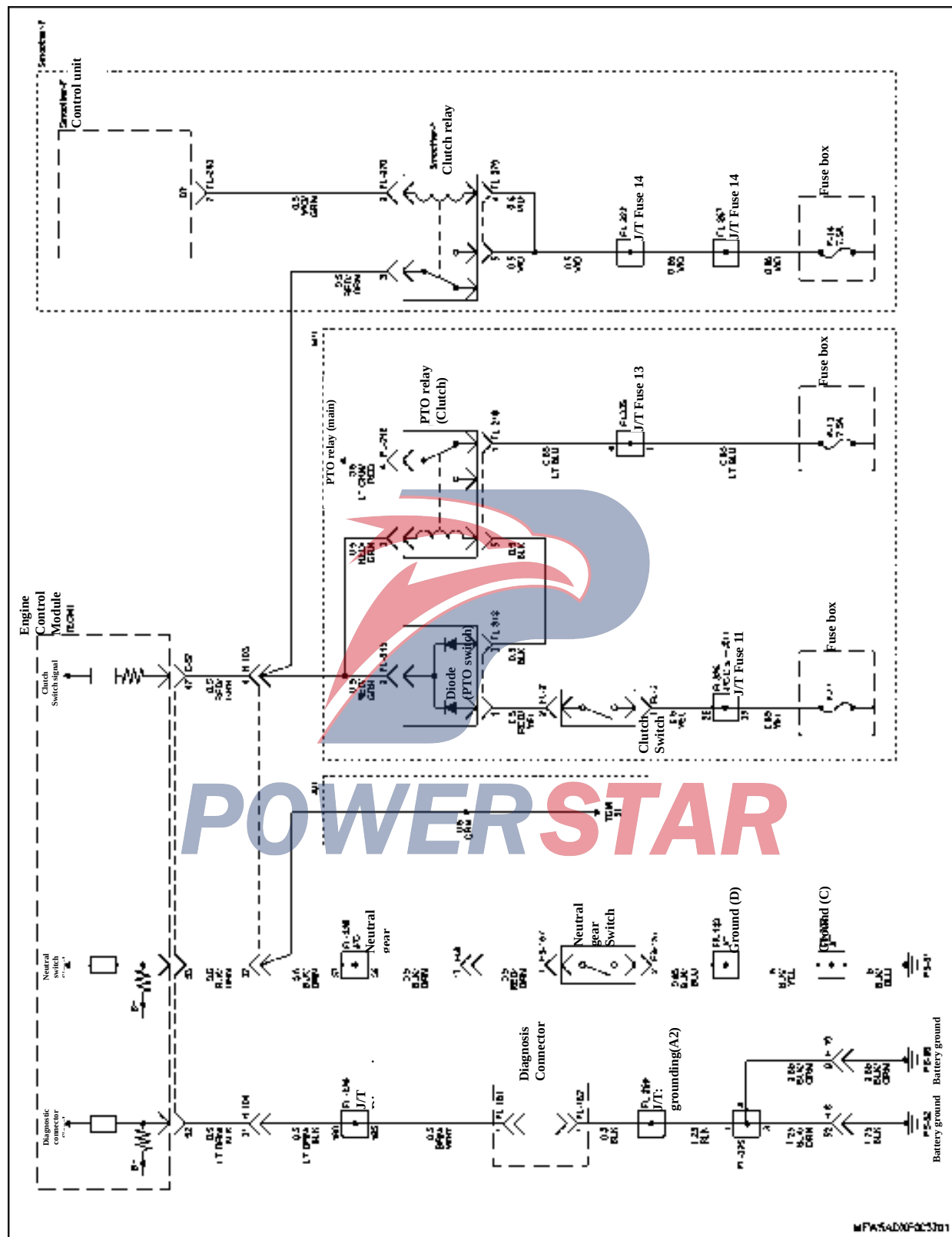
Parking brake, idle switch, idle speed control switch and idle speed control switch.



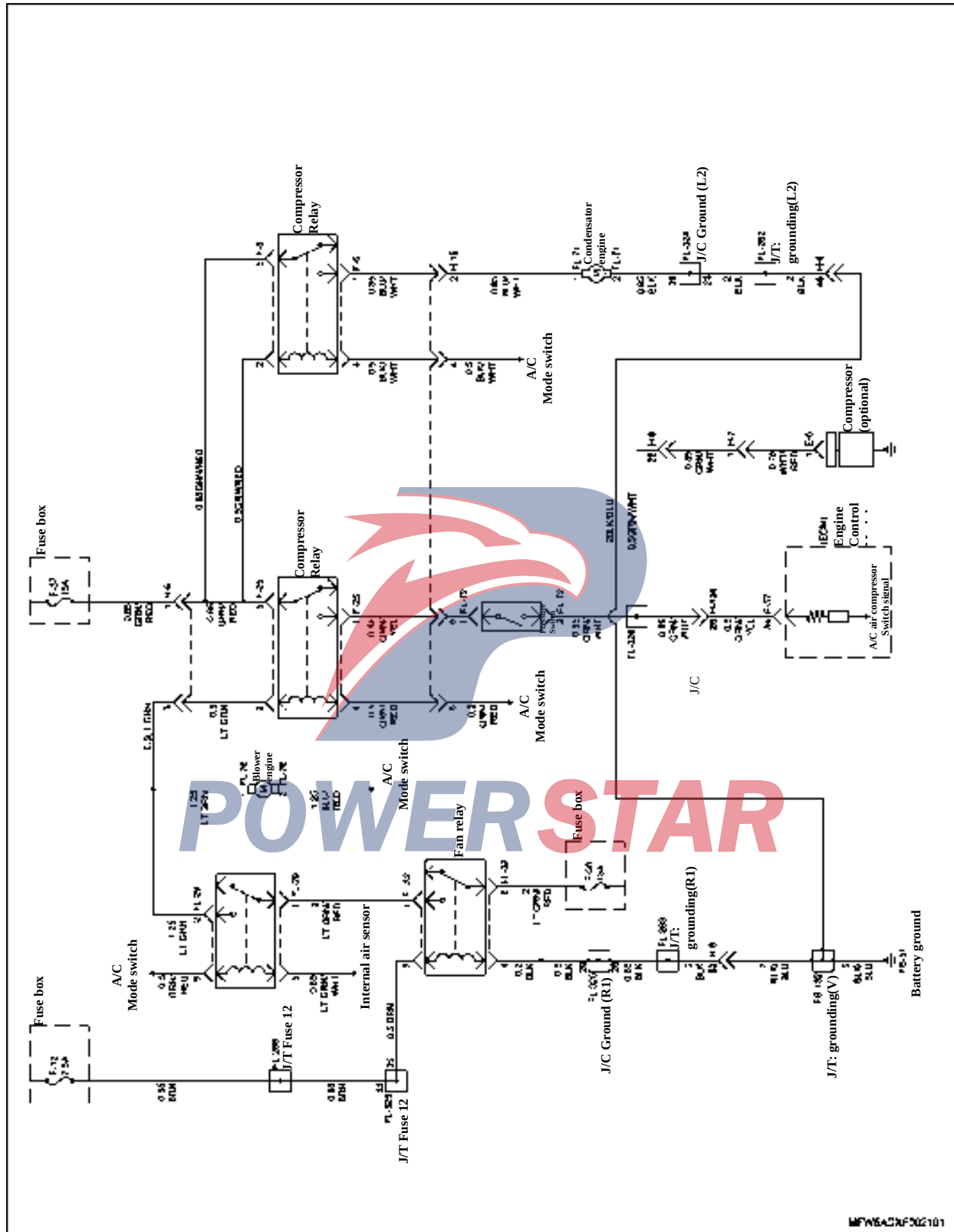
Brake light switch, retarder switch, exhaust brake switch circuit

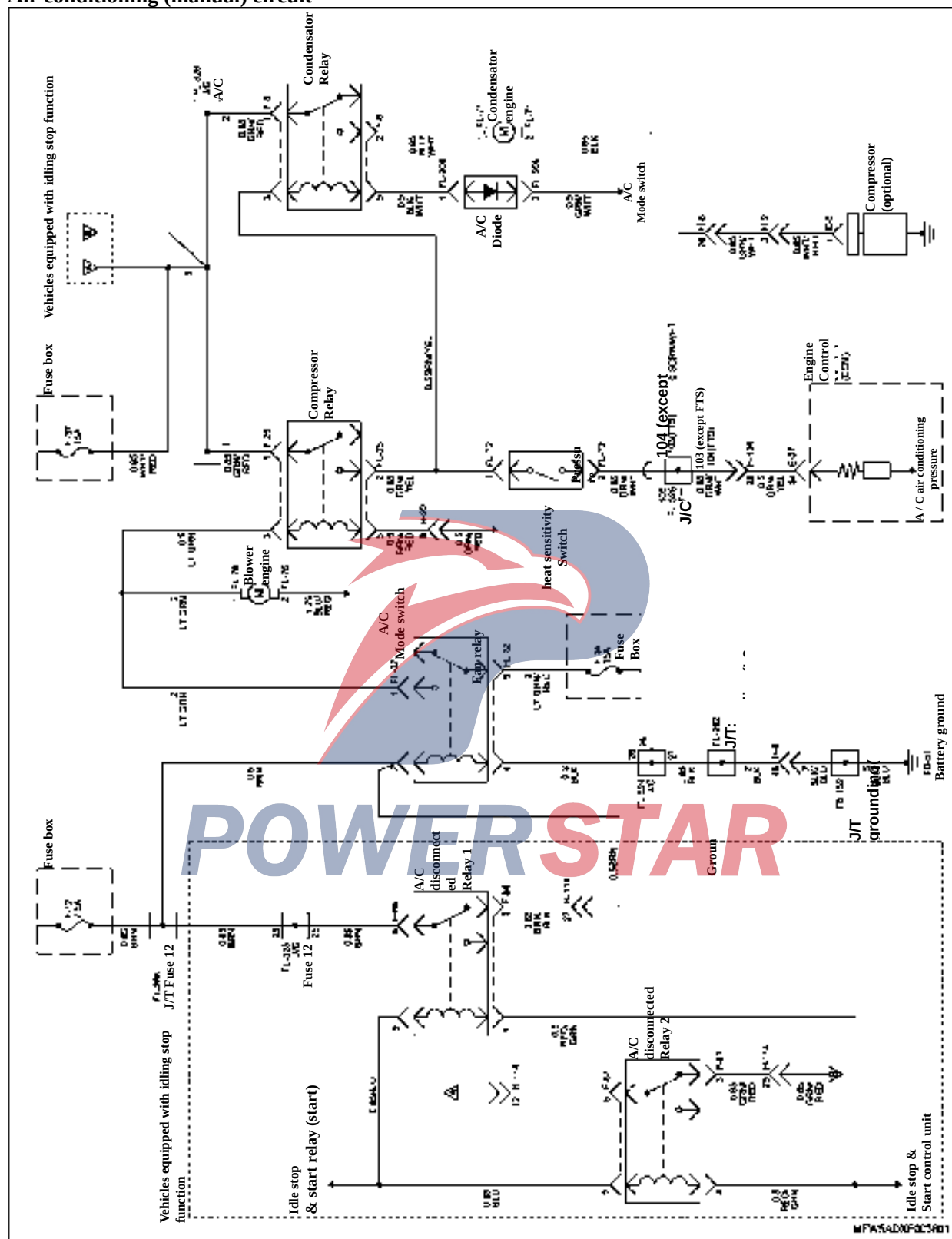


Clutch switch, neutral gear switch, diagnostic joint circuit

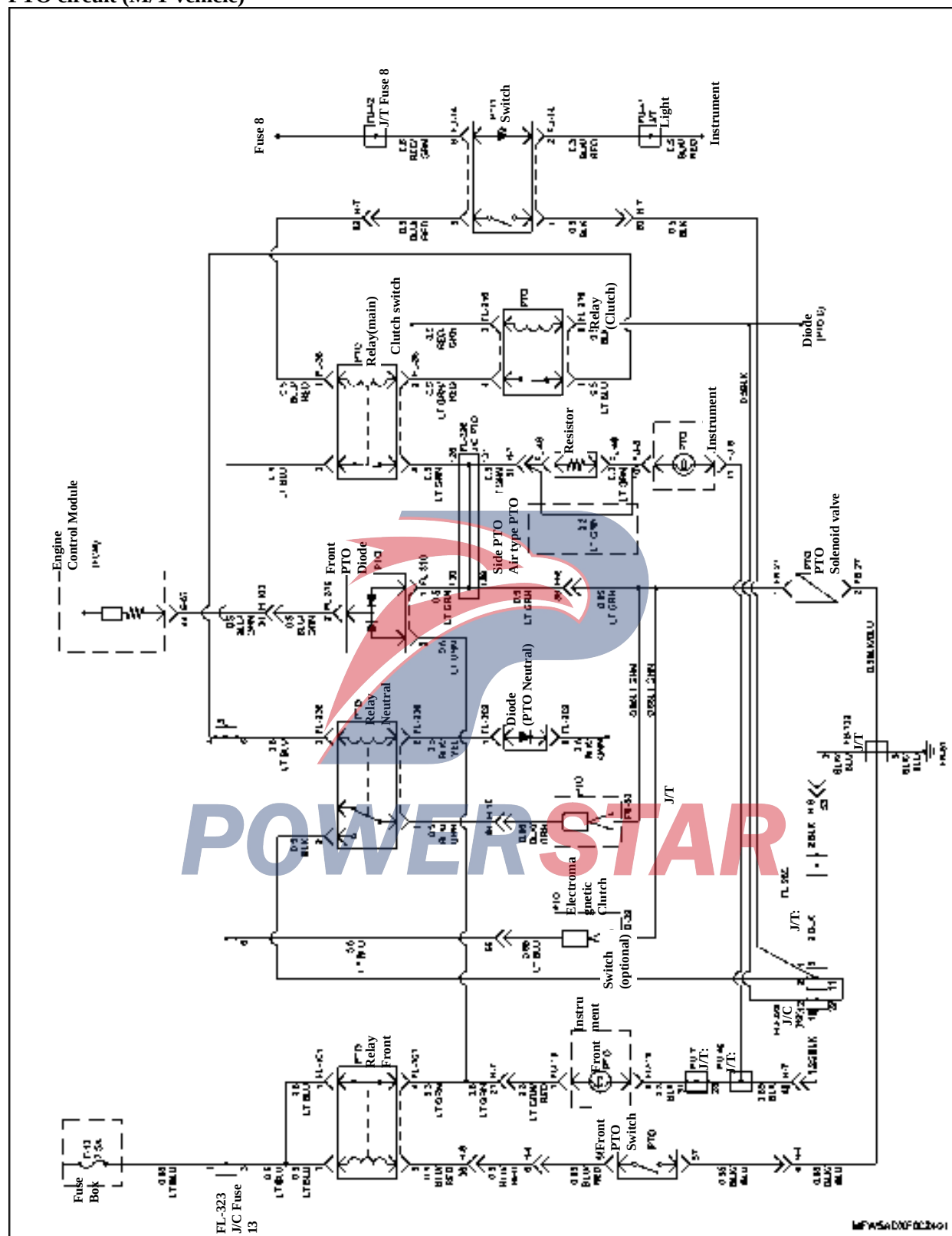


A/C (full auto) circuit

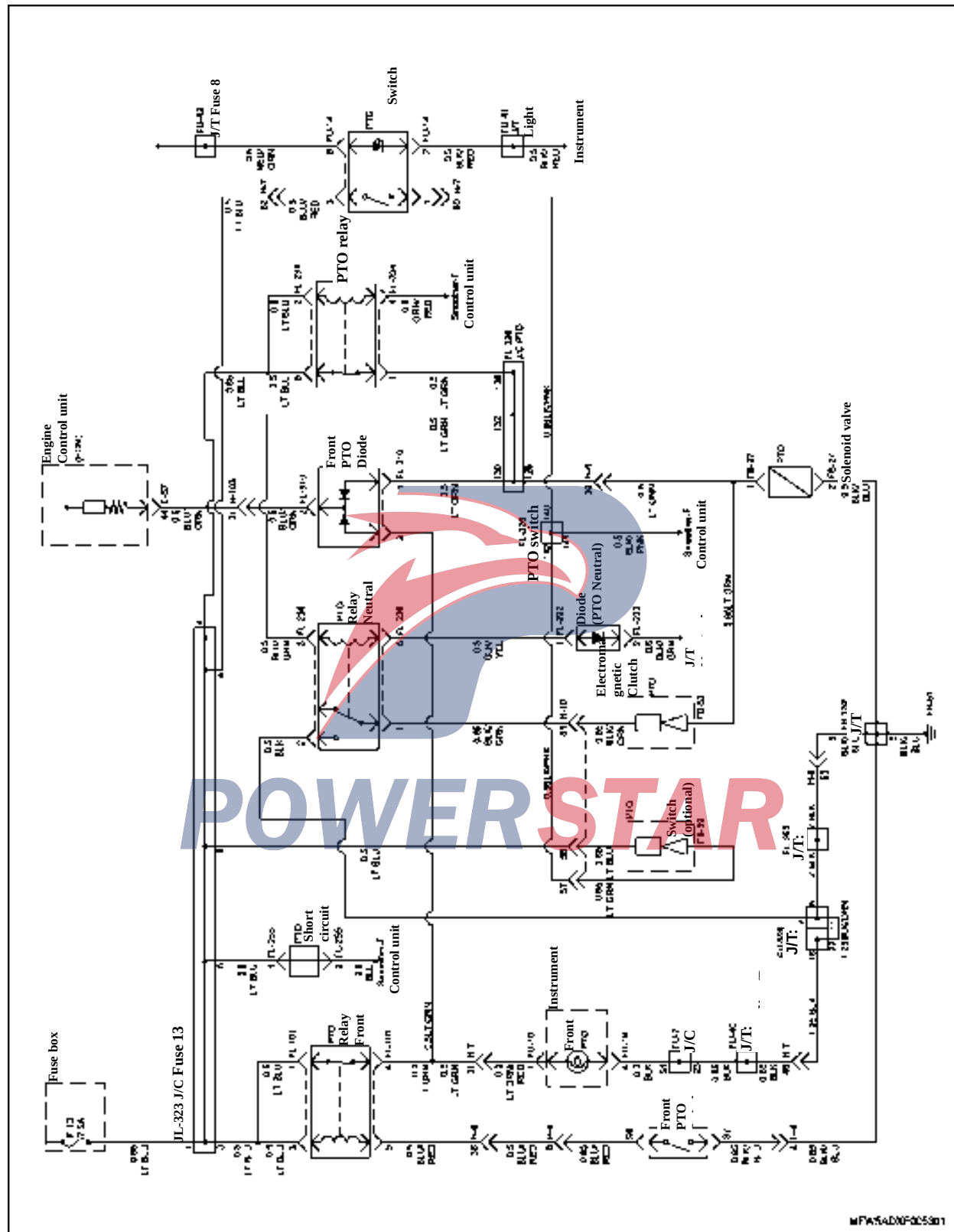




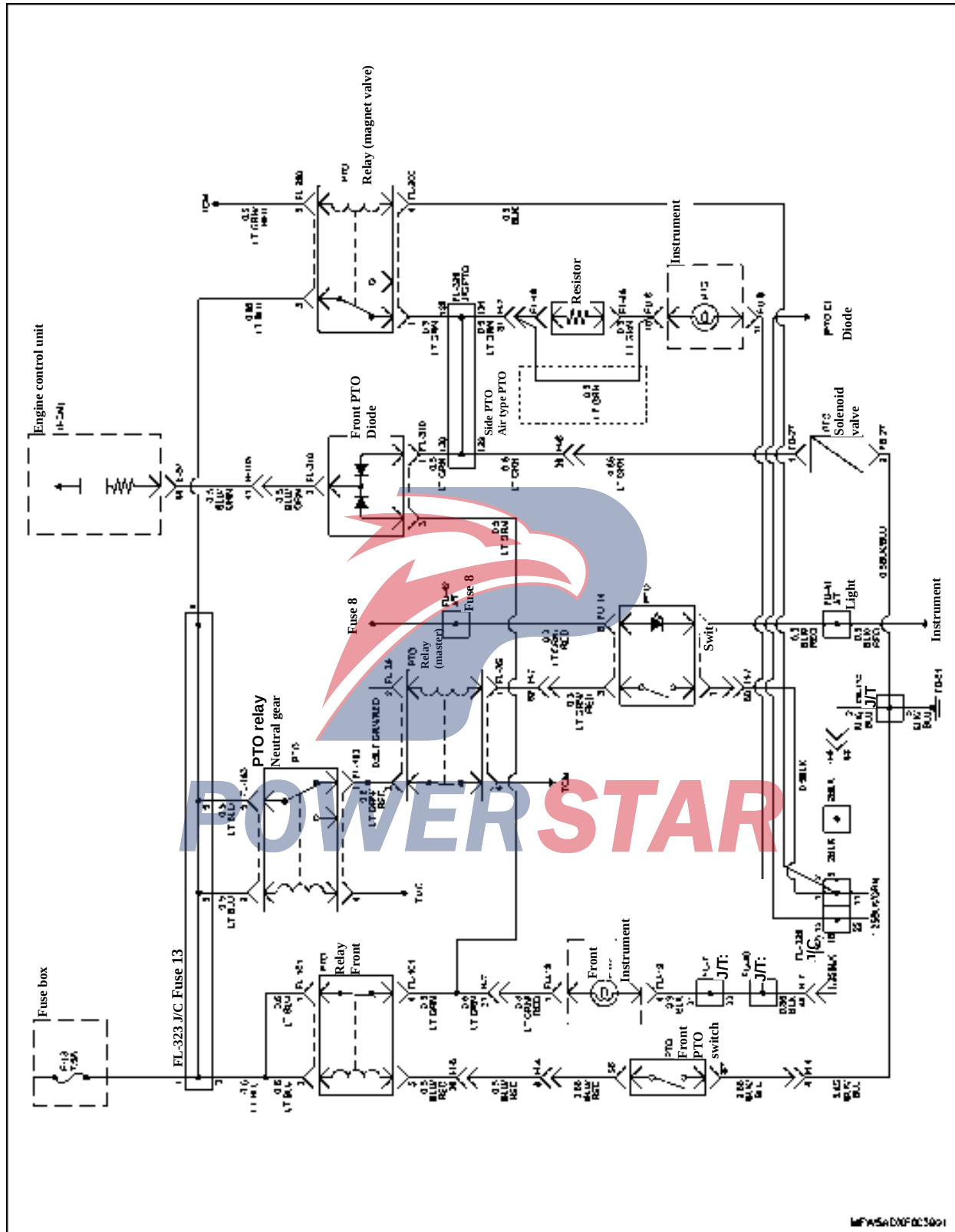
PTO circuit (M/T vehicle)



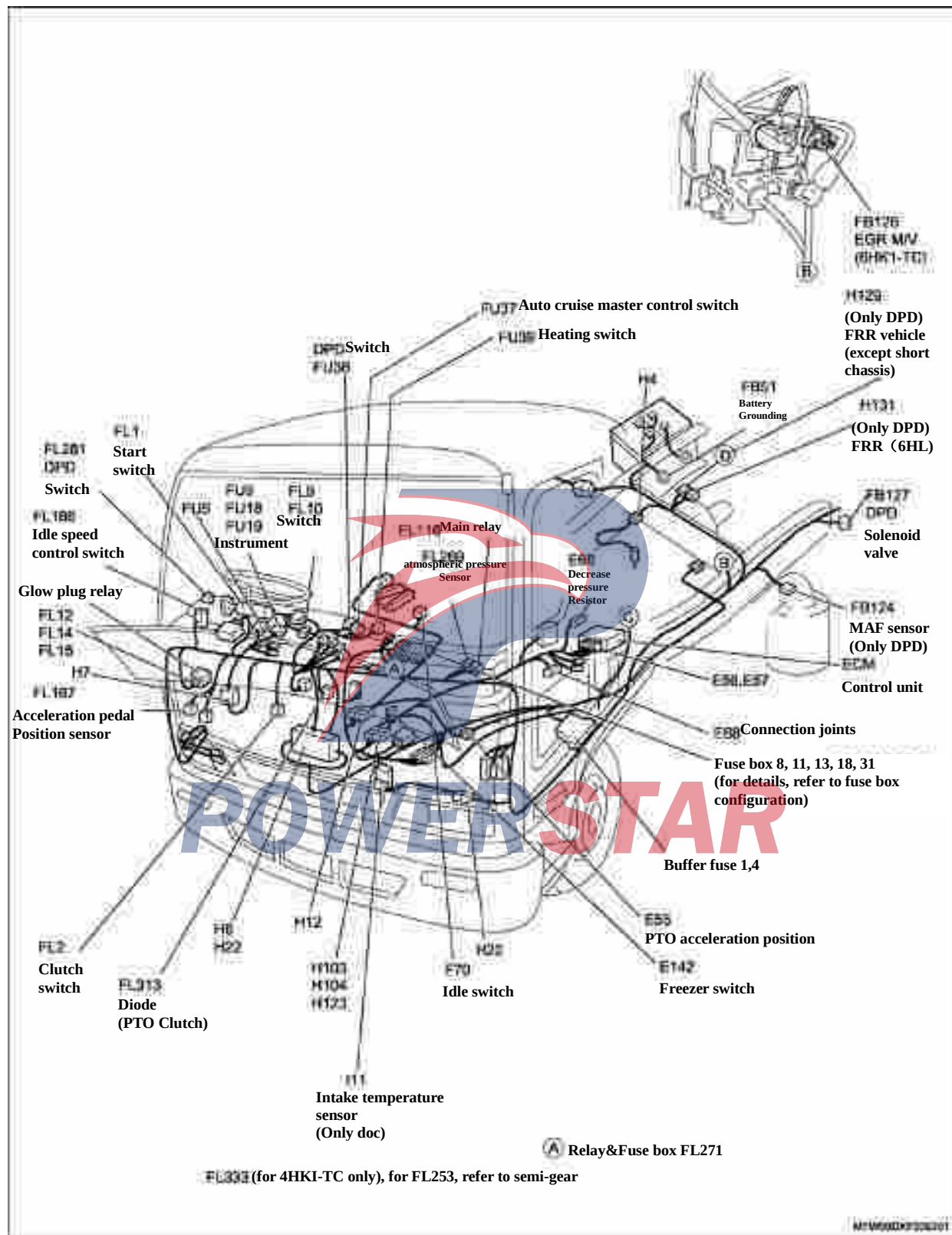
PTO circuit (vehicles equipped with Smoother-F)

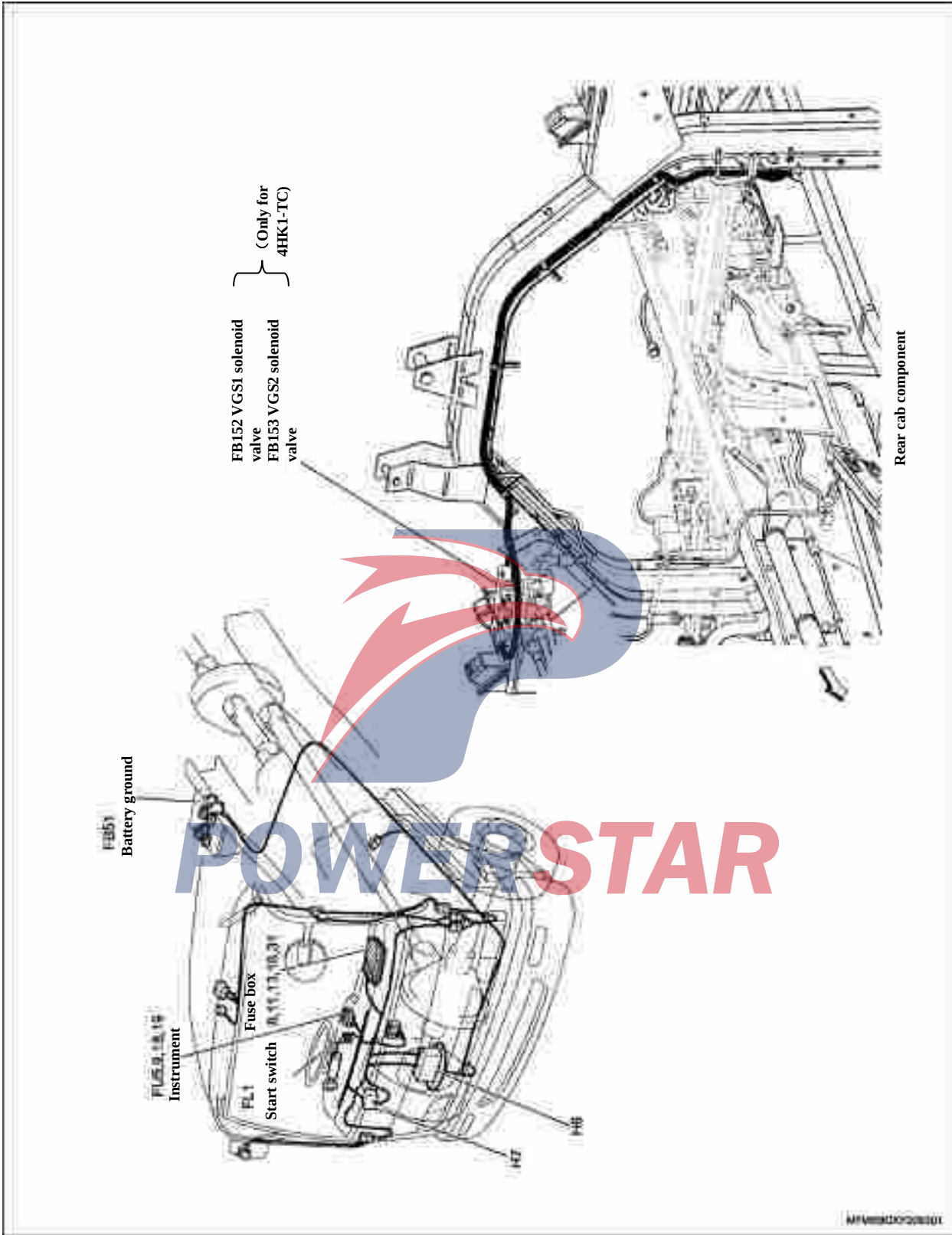


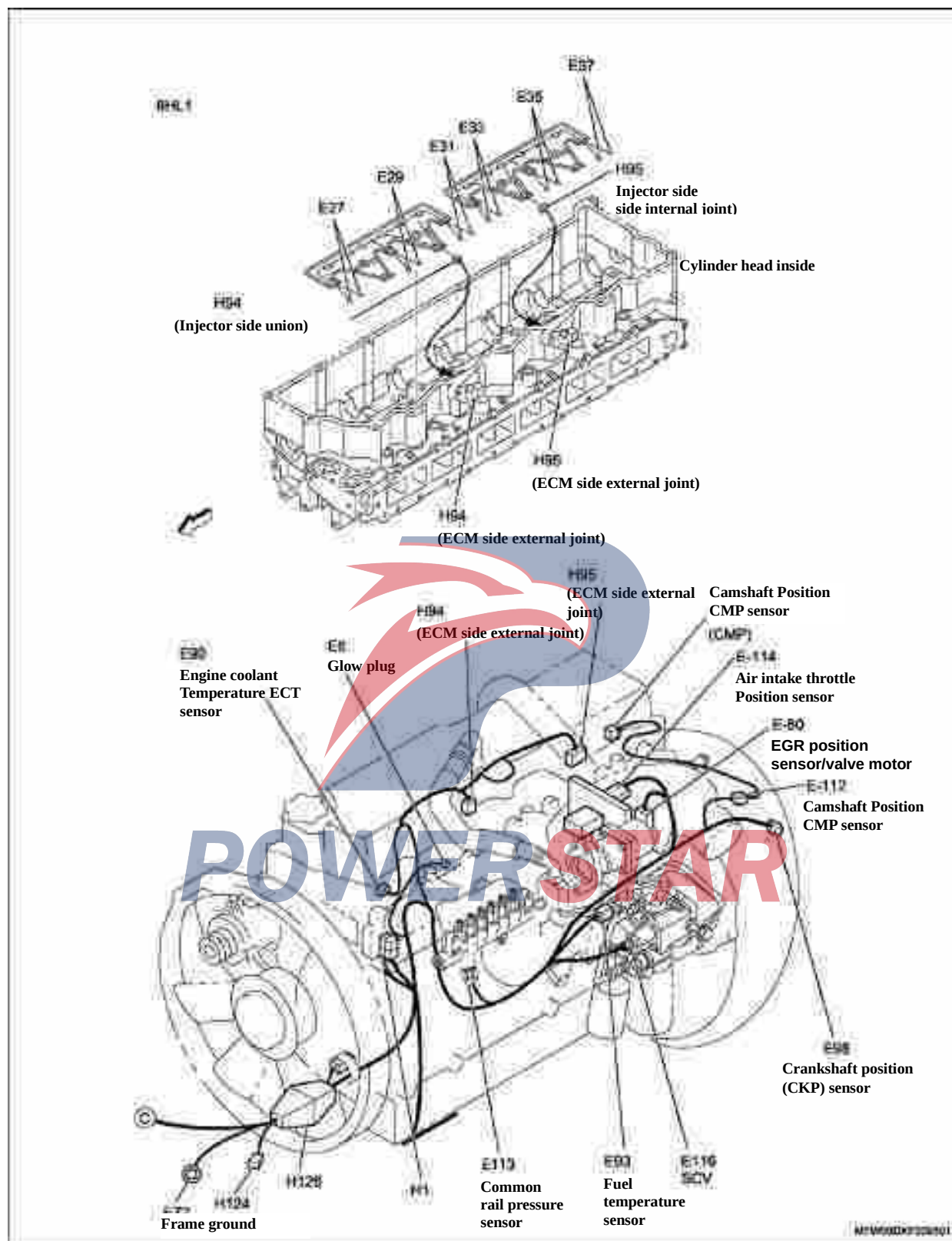
PTO circuit (A/T car)

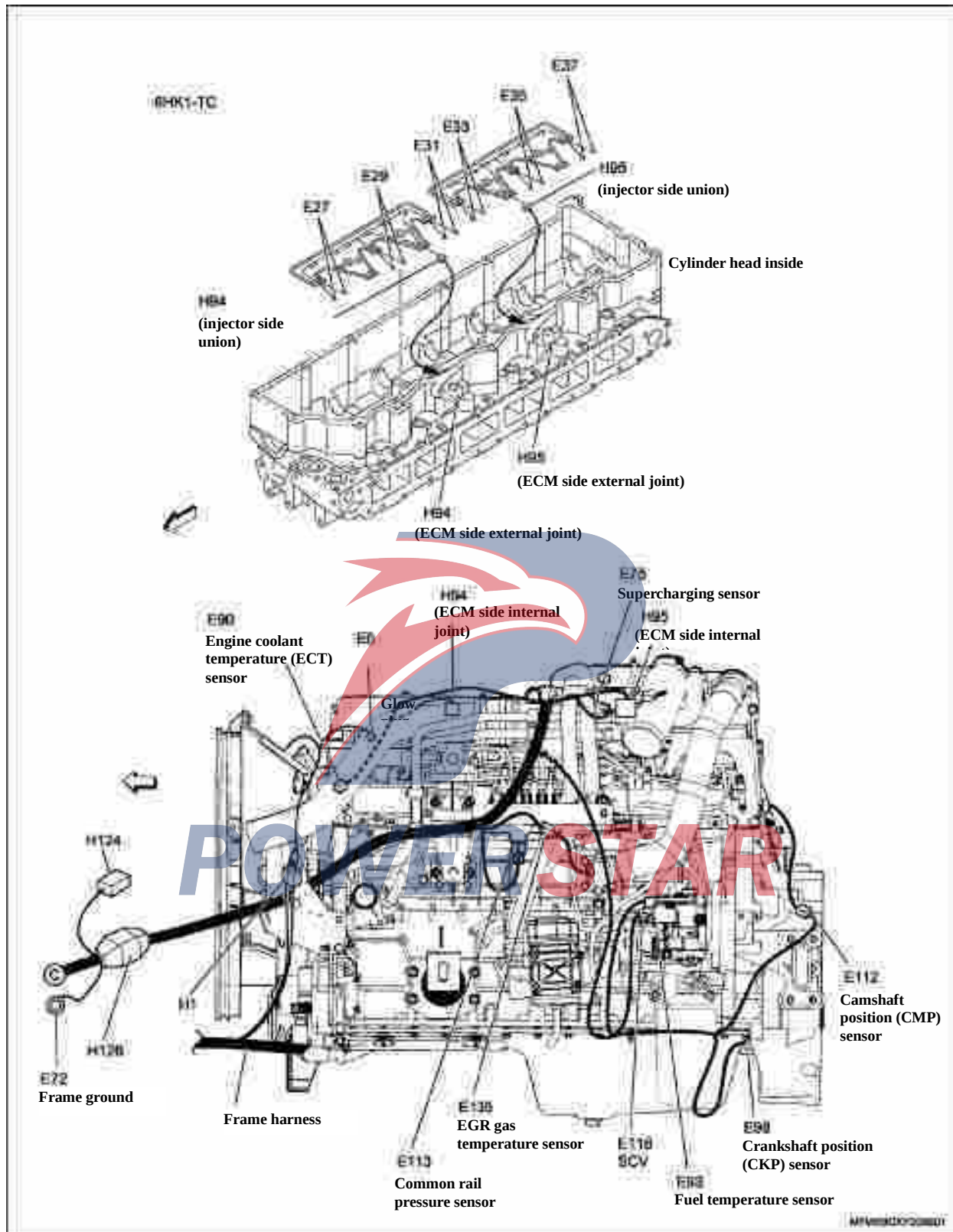


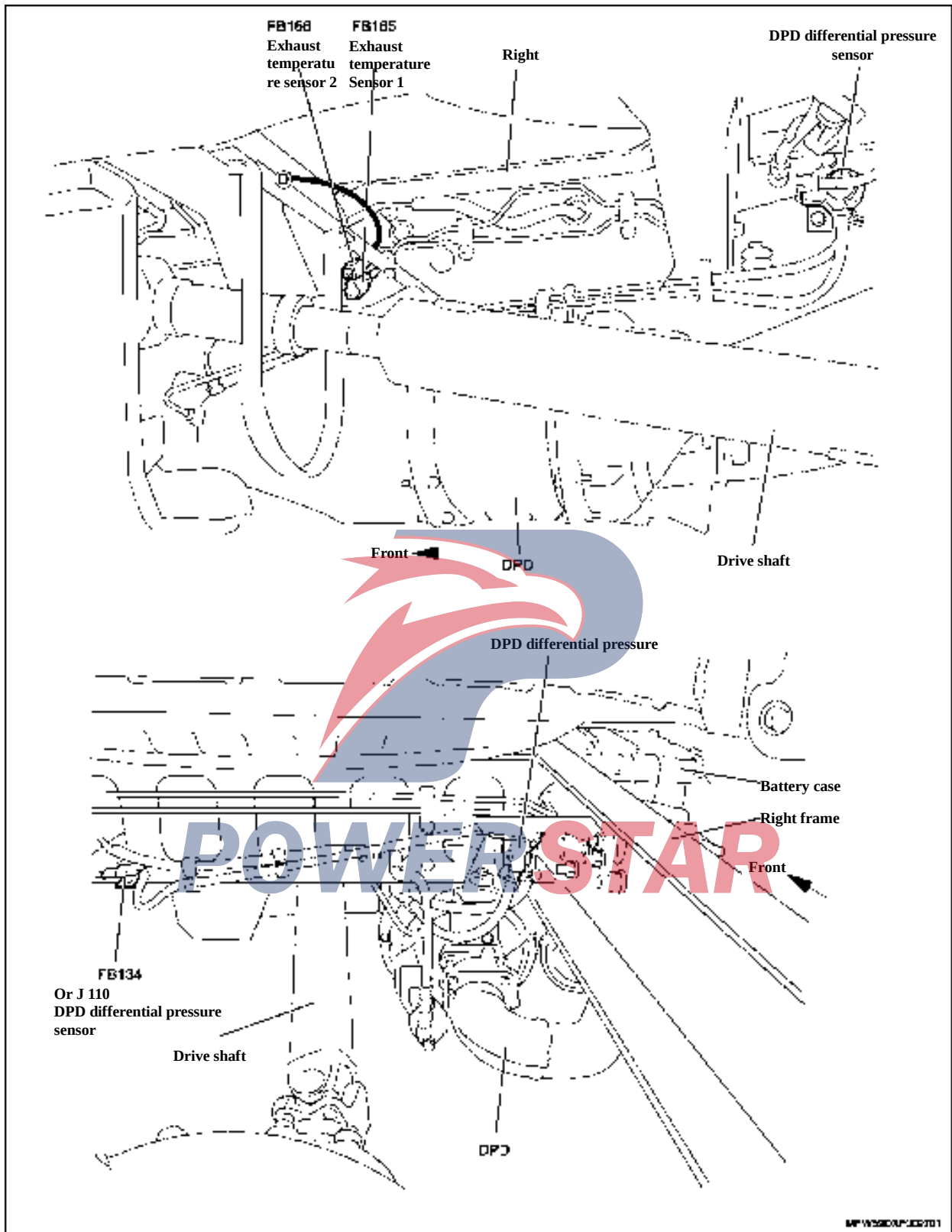
Part layout



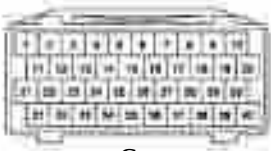
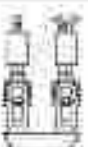
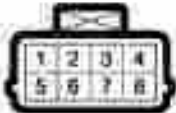
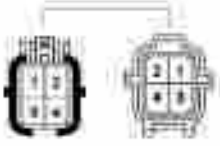




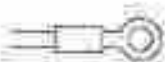


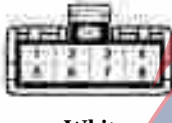
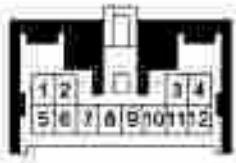


Connector list

No.	Joint interface	No.	Joint interface
E-55	 Grey 001-010	E-56	 Grey 040-001
E-27	 Silver color 1013-120	E-57	 Grey 300-000
E-20	 Silver color 1013-120	E-60	 Grey 002-000
E-31	 Silver color 1013-120	E-70	 Black 300-000
E-30	 Silver color 1013-120	E-72	 Black 100-012
E-35	 Silver color 1013-120	E-75	 Grey 1013-000
E-37	 Silver color 1013-120	E-80	 Brown 1013-070
E-55	 Grey 004-010	E-90	 Blue 003-100

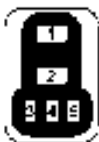
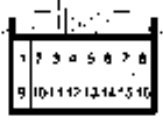
No.	Joint interface	No.	Joint interface
E-03	 Grey	E-06	 Black
E-08	 Brown	E-25	 Black
E-112	 Black	E-110	 Black
E-113	 Black	E-111	 Black
E-114	 Green	E-115	 White
E-116	 Brown	E-117	 White
E-135	 Grey	E-133	 Grey
E-1	 White	E-136	 Grey

No	Joint interface	No	Joint interface
FB-23	 Green	FB-53	 Transparent green
FB-27	 Grey	FB-58	 Transparent green
FB-29	 Grey	FB-90	 Black
FB-39	 Transparent green	FB-124	 Black
FB-39	 Transparent green	FB-125	 Grey
FB-50	 Black	FB-127	 Grey
FB-51	 Black	FB-128	 Grey
FB-52	 Black	FB-129	 Dark gray

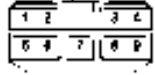
No.	Joint interface
FB-104	 Black
FB-107	 Black
FB-160	 Grey
FB-109	 Grey
FL-1	 White
FL-2	 Green
FL-9	 White
FL-11	 White

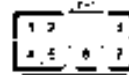
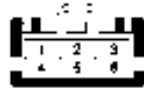
No.	Joint interface
FL-12	 White
FL-14	 Transparent green
FL-15	 Transparent green
FL-11	 Black
FL-20	 Grey
FL-1	 Grey
FL-29	 Black
FL-12	 Black

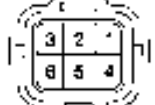
No.	Joint interface	No.	Joint interface
FL-35	 White	FL-117	 Black
FL-46	 Black	FL-187	 Black
FL-52	 Black	FL-169	 Black
FL-71	 Black	FL-161	 Red
FL-72	 White	FL-162	 Red
FL-73	 White	FL-163	 Black
FL-101	 Black	FL-167	 Grey
FL-116	 Black	FL-170	 Black

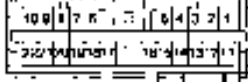
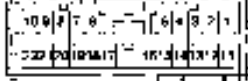
No.	Joint interface
FL-186	 White 006-004
FL-202	 Black 002-043
FL-209	 Black 004-021
FL-216	 Black 006-023
FL-220	 Black 005-012
FL-231	 Black 005-012
FL-236	 Black 005-012
FL-240	 Black 016-000

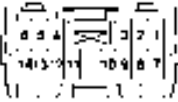
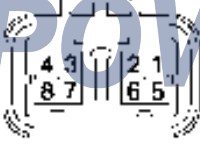
No.	Joint interface
FL-241	 Black 020-000
FL-243	 Black 016-016
FL-255	 White 002-022
FL-260	 Grey 003-000
FL-270	 Black 005-012
FL-272	 Black 005-012
FL-273	 Black 006-023
FL-281	 White 001-000

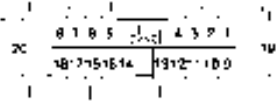
No.	Joint interface
FL-298	 Black 003-023
FL-299	 Black 003-024
FL-300	 Black 002-042
FL-310	 Black 003-024
FL-313	 Red 003-024
FL-3	 White 015-001
FL-8	 White 013-001
FL-9	 White 009-006

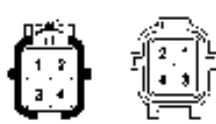
No.	Joint interface
FL-14	 White 006-024
FL-18	 White 017-001
FL-19	 White 007-002
FL-26	 White 006-024
FL-37	 White 006-024
FL-38	 White 006-024
G-12	 White 008-016
I-1	 Grey 002-101

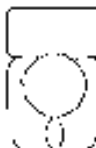
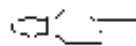
No.	Joint interface
H-1	 Grey 002-087
II-2 (9HL1)	 Grey 004-015
II-2	 Grey 008-009
H-2	 Grey 008-010
H-3	 Grey 002-001
H-3	 Grey 002-087
II-4	 Grey 006-001
H-4	 Grey 006-011

No.	Joint interface
F-12	 Black 022-001
II-12	 Black 102-000
II-15	 Black 022-001
F-15	 Black 022-002
I-20	 Black 008-009
II-20	 Black 008-010
II-22	 Blue 002-008
I-22	 Blue 002-077

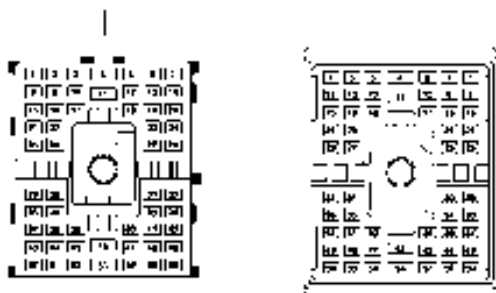
No.	Joint interface
H-32	 Black 008-009
H-32	 Black 008-009
H-48	 Black 014-003
H-48	 Black 014-004
H-91	 Grey 003-046
H-91	 Grey 008-047
H-15	 Grey 008-016
H-15	 Grey 008-047

No.	Joint interface
H-103	 Black 032-014
H-103	 Black 032-015
H-104	 Black 020-009
H-104	 Black 020-021
H-112	 Black 002-048
H-112	 Black 002-037
H-114	 Black 032-015
H-114	 Black 032-014

No.	Joint interface
7-123	 White
7-123	 White
7-124	 Black
4-124	 Black
4-124	 Grey
3-129	 Grey
7-131	 Black
4-139	 Black

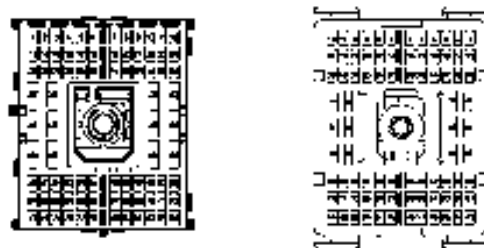
No.	Joint interface
J-110	 Black
T-1	 Silver
P-2	 Silver color
P-3	 Silver color
P-4	 Silver color
4-6	 Transparent green
4-7	 Transparent green

F-3 white



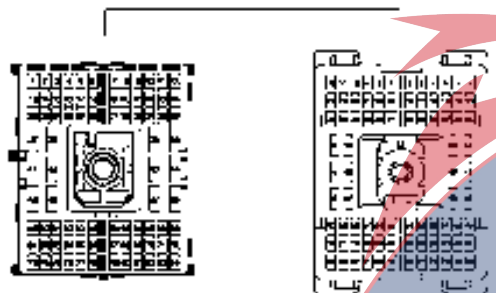
#C/N39CS401401

F-10 Grey



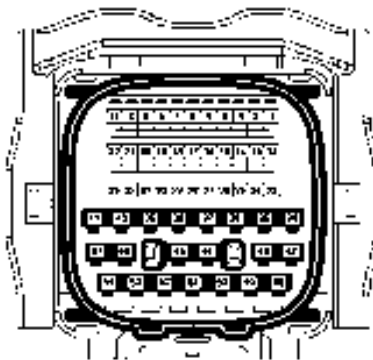
#C/N38CS401401

F-7 white



#C/N39CS401401

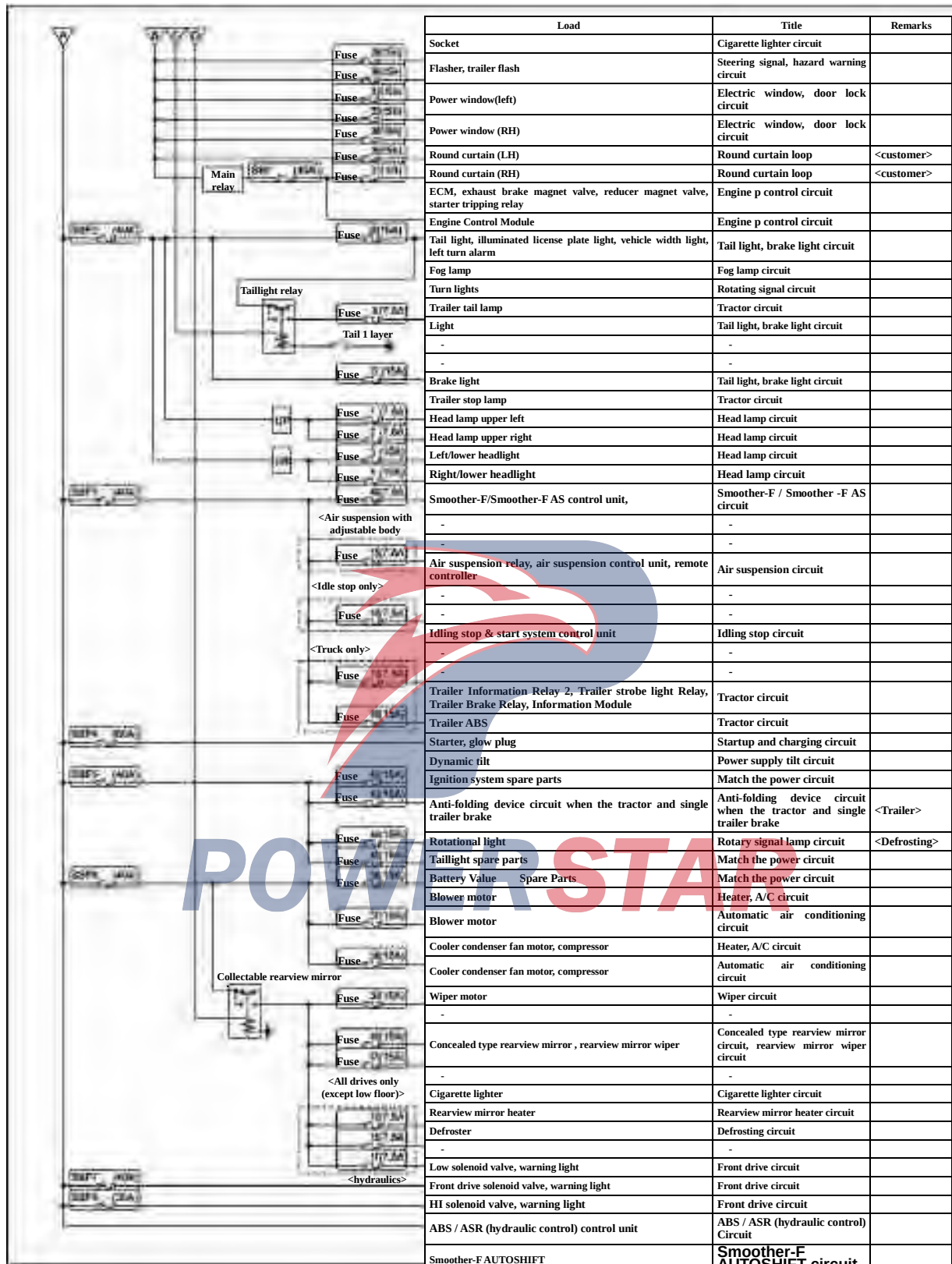
F-126 Black



UNW45A5F-100001

Power system diagram

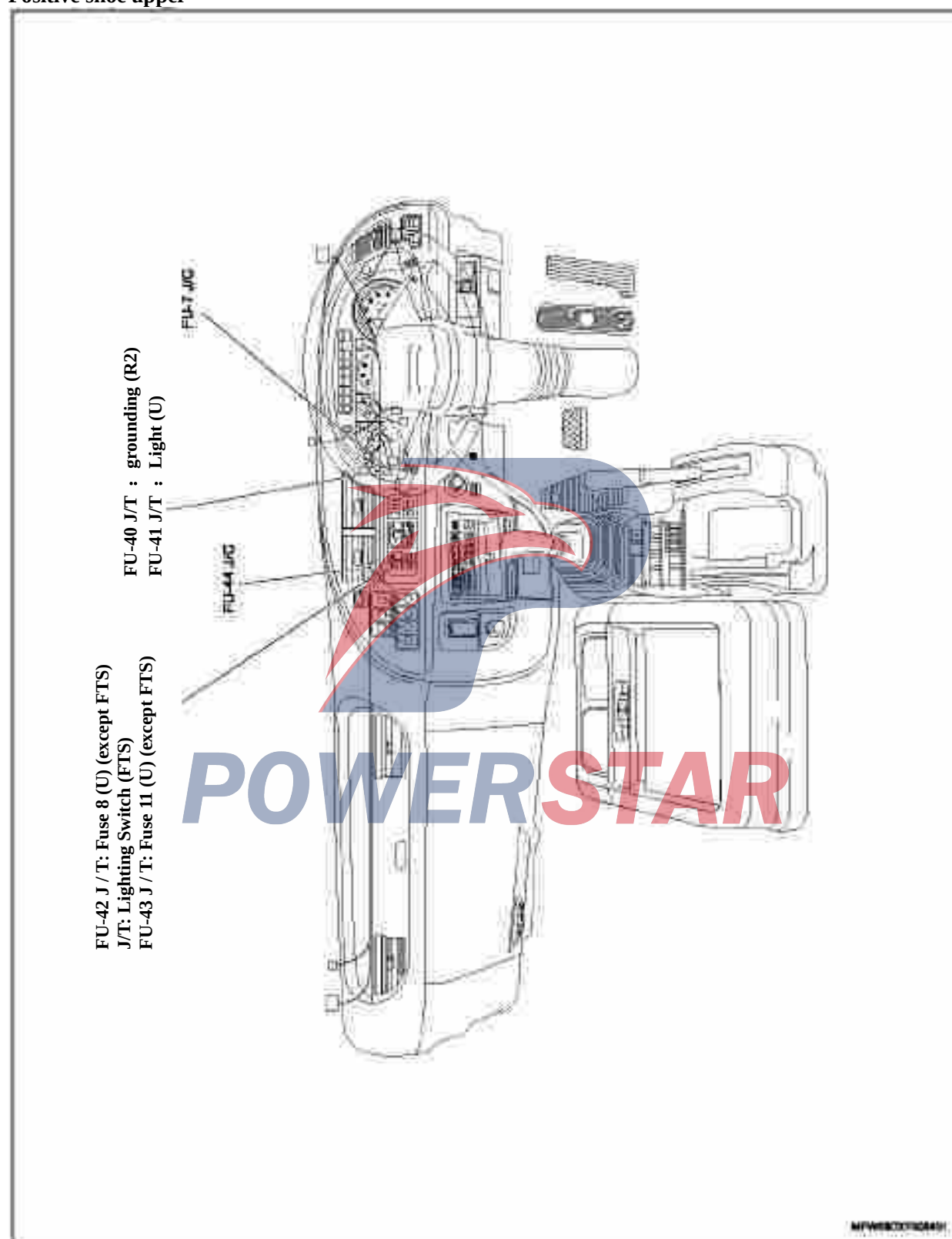




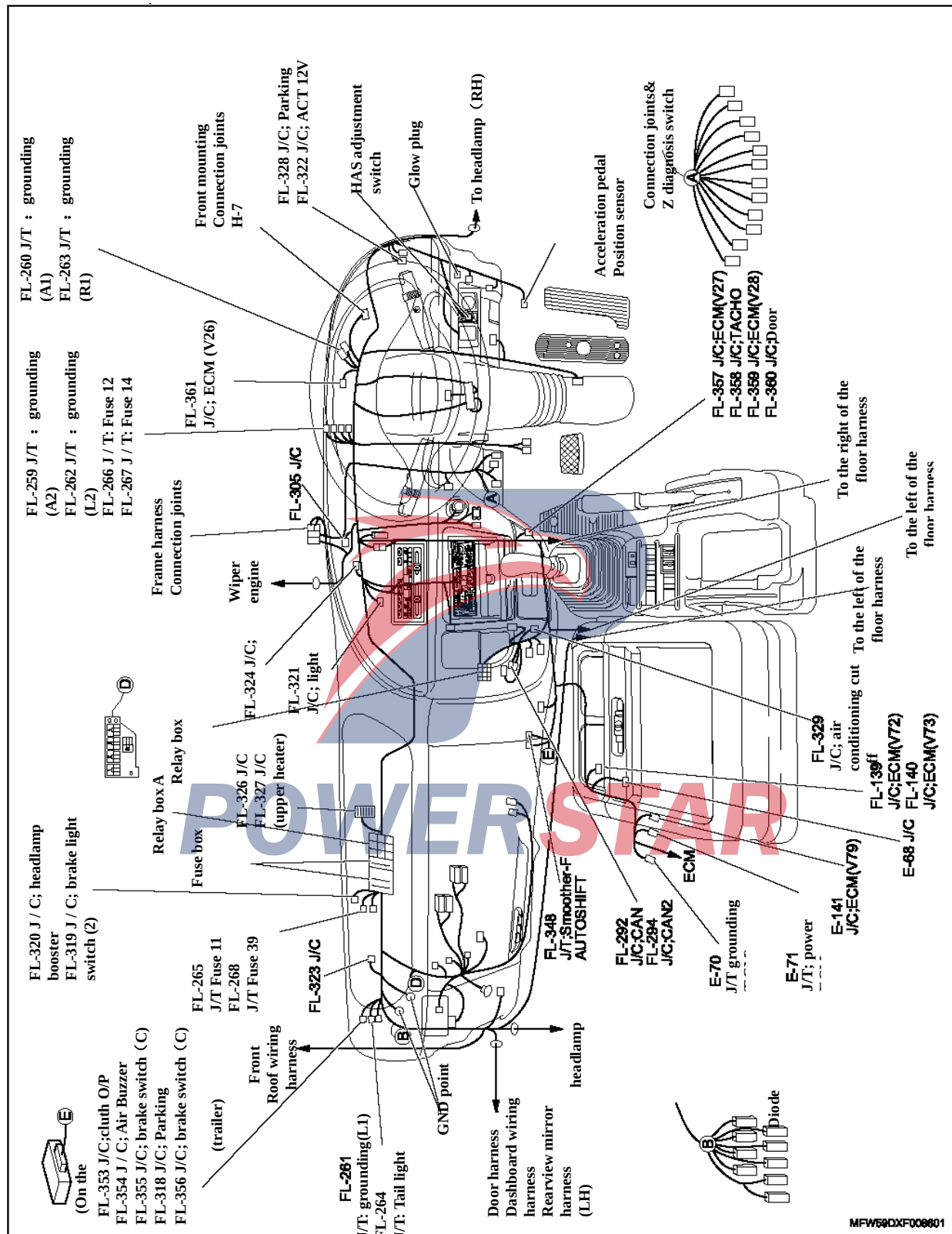
Connector/connector

Parts configuration diagram

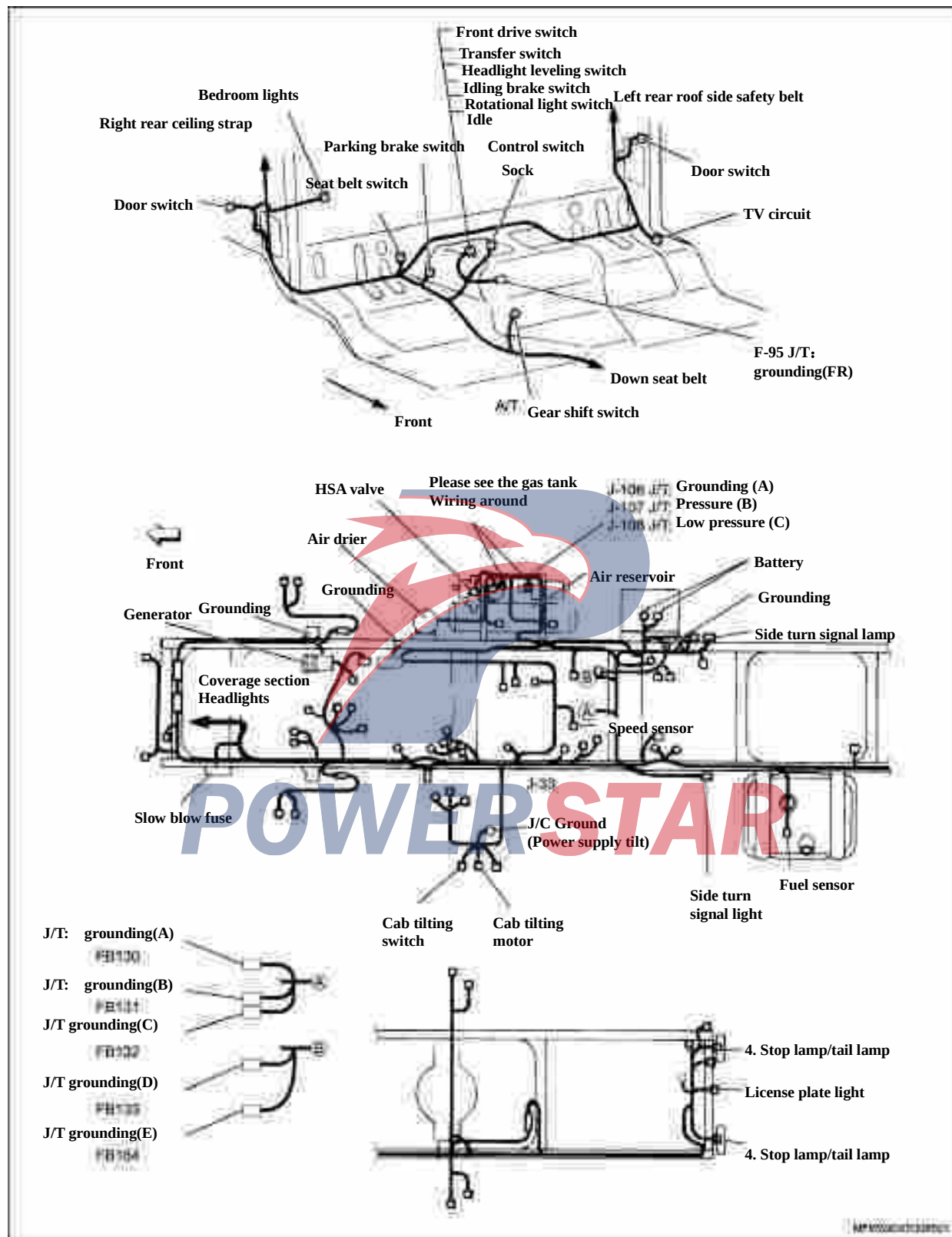
Positive shoe upper



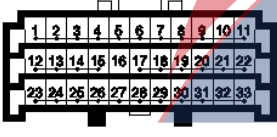
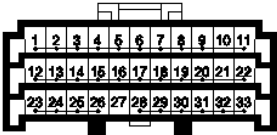
Front lower part



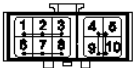
Frame section



Terminal configuration

J/C No.	J/C	Connector	Connector name
E-68		1-3, 4-6, 12-14, 15-17, 23-25, 26-28	J/C; CAN3
		8-11, 19-22	J/C; CAN(ECM)
		30-33	J/C; ECM(V60)2
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
FU-7		1-4	J/C; Speed 8P
		6-8	J/C; Fuse 5 (U)
		9-11	J/C; Separation device
		12-15	J/C; TACHO
		17-19	J/C; ECM(V51)
		20-22	J/C; Light(U)
		23-33	J/C; Grounding(R3)
		-	-
		-	-
		-	-
FL-44		1-4	J/C; Rearview mirror
		6-11	J/C; heater
		12-33	J/C; Fuse 8 (U)
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
FL-323		1-11	J/C; Fuse 13
		12-15	J/C; Fuse 14 (vehicle with standard cab)
		12-15	J/C; Head lamp (H-L) (vehicles & FTS with widened cab)
		17-22	J/C; Grounding (L1)
		23-33	J/C; Fuse 14
		-	-
		-	-
		-	-
		-	-
		-	-
FL-324		1-3	J/C; Automatic switch (1)
		4-6	J/C; ECM(V61)
		8-11	J/C; ECM(V60)-1
		12-14	J/C; Actuator signal
		15-17	J/C; reducer
		19-22	J/C; Fuse 31
		23-28	J/C; Grounding(L2)
		29-33	J/C; Brake light
		-	-
		-	-
FL-325		1-11, 12-22	J/C; Grounding(A1)
		23-26	J/C; Fuse 12
		28-33	J/C; Grounding(R1)
		-	-
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J/C No.	J/C	Connector terminal	Connector name
FL-326 Except FTS		1-4	J/C: Fuse 20
		6-11	J/C: Speed 25p
		12-22	J/C: Fuse 5
		23-28	J/C: Fuse 11
		29-33	J/C: Fuse 39
		34-37	J/C: Exhaust switch
		39-44	J/C: Strobe light (left)
		45-48	J/C: Starter 1
		50-55	J/C: Flashlight (right)
		56-66	J/C: Neutral gear
		67-69	J/C: Flash lamp switch (LH)
		70-72	J/C: Blinking lamp switch (RH)
		74-77	J/C: Idle brake master switch
		78-80	J/C: Door lock (locked) (vehicle with standard cab)
			J/C: Differential locking device (large vehicle)
		81-83	J/C: Door lock (unlocked) (vehicle with standard cab)
			J/C: Head lamp (H-R) (vehicles & FTS with widened cab)
		85-88	J/C: Reversing
		89-99	J/C: Parking
		100-102	J/C: Starter -2
		103-105	J/C: Compressor
		107-110	J/C: AGG (L)
		111-121	J/C: Fuse 8
		122-127	J/C: Low pressure
		128-132	J/C: PTO
		133-135	J/C: Tail lamp switch
		136-138	J/C: Headlight switch
		140-143	J/C: PTO switch (vehicle with standard cab)
			J/C: Headlight (L-R) (vehicle with cab)
		144-154	J/C: KWP2000
		155-158	J/C: Memory clear
		160-165	J/C: Diagnosis
166-168	Diode (Brake Locking Mechanism 2)		
169-171	Diode (Thermistor)		
172-175	Diode (interior light)		
177-179	Diode (charge)		
180-182	Diode (precursor)		
183-186	Diode (wheel parking)		
188-191	J/C: Brake lock master switch (vehicle with standard cab)		
193-196	J/C: Headlight (L-L) (vehicle with cab)		
FL-326 (FTS)		1-11	J/C: B.O.
		12-22	J/C: Fuse 5
		23-33	J/C: B.O.
		34-37	J/C: Fuse 11
		39-44	J/C: Flash lamp switch (LH)
		45-48	J/C: Starter 1
		50-55	J/C: Strobe light switch(right)
		78-80	J/C: Differential locking device
		81-83	J/C: Headlights (H-R)
		85-88	J/C: Speed 25P
		89-99	J/C: Parking
		100-102	Not used
		103-105	J/C: Compressor
		107-110	J/C: ACG (L)
		122-127	J/C: Low pressure
		128-132	J/C: PTO
		133-135	Not used
		136-138	Not used
		140-143	J/C: Headlights (L-R)
		144-154	J/C: KWP2000
155-158	Not used		
160-165	J/C: Diagnosis		
166-168	Diode (ABS)		
169-171	Diode (Thermistor)		
172-175	Diode (interior light)		
177-179	Diode (charge)		
180-182	Not used		
183-186	Diode (wheel parking)		
188-191	J/C: Headlights (L-L)		
193-196	Not used		
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J/C No.	J/C	Connector terminal	Connector name
FL-327		1-3	J/C; Key ON 1
		6-8	J/C; Key ON2
		4-5,9-10	J/C; SBE3
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
FL-328		1-11,12-22	J/C;Grounding(R2)
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
FL-329		1-4	J/C; Air conditioning cut off
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
		-	-
J-33		1-2,5-8	J/C; Ground (dynamic tilt)
		3-4,7-8	J/C; Ground (dynamic tilt)
		-	-
		-	-
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POWERSTAR

Troubleshooting method

System diagnostics

System diagnostics is a uniform method used when repairing all electrical/electronic (E/E) systems. E/E system malfunction is different from the common vehicle malfunction. It mostly occurs in the following steps.

1. The early stages of failure

- Failure is short-term and one-time failure. In many cases, customers do not know about failure. Customers at this stage will not be aware of their complaints and will not be able to reproduce the symptoms. Nonetheless, ECM sometimes will store the fault contents.

= Failure of the past.

2. Mid-term failure

- Although the failure is a short-term one-time failure, as long as the failure occurs under certain conditions, the failure occurs intermittently and repeatedly. The complaint contents (the malfunction description) in this stage are very definite while the fault conditions are not.

For this reason, the vehicle's diagnostic personnel can reproduce the fault by grasping the conditions.
= Intermittent faults (intermittent)

3. Stage of fault reality

- Faults often occur in a fixed manner. The content of customer complaints is very practical and very clear.

For this reason, the diagnostician of the vehicle can reproduce the fault. But sometimes there are more than 2 causes of failure.

= The current failure.

The diagnostic procedure is often used to solve problems in the E/E system and is the starting point for maintenance. In the following steps, the diagnosis will be described.

1. Confirm customer complaint contents

- In order to confirm the customer's complaint, the person responsible for vehicle diagnosis must understand the normal operation of the system.

2. Conduct check priority

- Visual inspection of the whole
- Reconfirm the repair record
- Detect abnormal conditions such as abnormal noise and odors.
- Use scan tools to collect information such as DTCs and data.

3. Check after-sales service information.

4. Refer to the manual "don't diagnose symptoms".

- "Symptom diagnosis" contains information about the system that is not displayed by the DTC (diagnostic trouble code). Therefore, according to "Symptom Diagnosis", you can confirm the normal operation of the system so that you have a reasonable diagnosis.

5. Please refer to the relevant records of engine parts.

When storing DTC (diagnostic trouble code)

Follow the prescribed DTC chart to perform correct maintenance.

No DTC situation

Select symptoms from symptom diagnosis. Follow the diagnostic steps to complete the repair. It can also be checked with reference to functional diagnosis.

No corresponding symptoms

1. Detailed investigation of complaints
2. Develop a diagnostic plan.
3. Use wiring diagram and working principle.

Technical assistance can be requested in a similar

situation where a maintenance resume can be obtained. Combine technical knowledge and available after-sales service information.

Intermittent faults (intermittent faults)

The usual contingent state is called intermittent failure. To solve these intermittent faults, the following steps should be taken.

1. Confirm DTC information and data display.
2. Assess the symptoms and status of customer complaints.
3. Efficient use of checklists or other methods to confirm the components of a circuit or electrical system.

When no fault is detected

The state indicates that it can be judged as the normal operation of the vehicle. The state stated by customers is also normal. The user's instructions can be compared and confirmed with other vehicles in normal operation. However, sometimes there are intermittent failures due to different states. Before returning the vehicle to the customer, please confirm the content of the complaint according to the status of the customer's declaration.

1. Reconfirm complaint contents

If it is not possible to fully detect or determine the situation of the complaint, it is necessary to re-diagnose and reconfirm the situation of the complaint. The content of customer complaints is sometimes as described in the case of "intermittent failures" mentioned above, and may also be intermittent or normal.

2. Repair and confirm

After the cause is determined, it should be repaired. And confirm the normal operation of the vehicle and whether it has repaired the symptoms. The confirmed contents include the road test confirmation and other confirmation methods for confirming whether the symptoms complained by the customer have been resolved and other methods of confirmation under the following states.

- Confirmation of the test in the state stated by the customer

- After diagnosing the diagnostic trouble code, confirm the diagnostic instrument data and reproduce the phenomenon set in the diagnostic trouble code to confirm that the diagnostic trouble code has been repaired.

Confirmation of vehicle maintenance

When carrying out the electronic control system maintenance, it is necessary to confirm the maintenance effect after vehicle maintenance. This process is very important; if ignored, the engine MIL may illuminate and the vehicle control may be abnormal after the vehicle is returned to the customer. In particular the intermittent fault should be reproduced and confirmed under the customer's complaint status.

Number	Item	Purpose	Method
1	DTC check	Confirm DTC display after maintenance	Clear previous DTC information. After idling, fully warm up and increase the engine speed to more than 200 rpm, then perform idle to ensure test conditions.
2	Idling after engine warm-up	To check whether the idling control is normal	Check the engine's idle speed after warming up and keep it near 530 to 550 rpm while the air conditioner is off. If detecting any fault, refer to the unstable idling description in the symptom diagnostics column.
3	Scan Tool Data - List Confirmation	Confirm based on engine control and communication state.	Scanning Tool Data - The watch list is monitored and confirmed using a representative data table. Confirm the representative value of scanning tool data list.
4	Restarting confirmation	Confirm whether the starting control is normal.	After confirming that the engine is warmed up and then started, if the crankshaft rotation time is less than 5 seconds, the engine operation after starting is stable.
5	Confirmation of the strong radio wave transmitter	When installing a radio wave transmitting unit such as a radio transceiver, it is necessary to confirm whether or not the interference radio wave is eventually transmitted.	Check whether the engine speed in idling state will change due to the ON and OFF of electric wave sending components such as transceiver. When it is found that there is a fault condition, it is necessary to contact the customer and the installation position of the radio wave transmission section and need to change the output.

Auxiliary confirmation of strong radio wave transmission equipment; when a fault phenomenon is found in the project, the users can be contacted for the following confirmation items as required.

- The antenna should be as far away as possible from the vehicle electronic system such as the control unit and sensor.
- The antenna should be installed more than 20cm away from the vehicle's electronic system, such as control modules and sensors.
- Do not twist the antenna with other harnesses. In addition, the antenna should be separated from other harness as long as possible.
- For the components to be installed later, the correct operation must be performed in strict accordance with the installation instructions.
- Cannot install high-power mobile communication devices.

Caution:

When using the on-board diagnostic (OBD) system to confirm maintenance, follow the steps below. Otherwise it may lead to unnecessary maintenance.

- Readjust the DTC-related scan tool data for diagnosis.
- Clear DTC content
- Run the vehicle while confirming the relevant scan tool data.

Non-OEM parts

All on-board diagnostics (OBD) system adjustments are made so that genuine parts can operate normally. Therefore, when installing general types of sensors, switches, etc., which are commercially available, it is possible to start an error diagnosis and an engine failure indicator.

When electronic devices sold on the market, such as mobile phones, audio and anti-theft devices, are improperly installed, the control system sometimes sends EMI (interference waves). As a result, erroneous sensor information may be generated and the engine failure indicator lamp will light up. When troubleshooting, make sure that all non-genuine parts installed on the vehicle are turned off and removed from the vehicle before performing diagnostics.

Vehicle maintenance and service failure

The sensitivity of the OBD system is very sensitive. When the vehicle maintenance is normal, the engine failure indicator lights up. In case of the oil change frequency insufficient, oil viscosity improper, oil filter and fuel filter clogging, crankcase settling adhesion, the vehicle fault may not be confirmed before the OBD system inspection.

Poor vehicle maintenance is not a "fault outside the vehicle," but the sensitivity of the onboard diagnostic (OBD) system inspection is very sensitive. Therefore, the vehicle maintenance plan must be further strictly observed.

Related system failure

Many on-board diagnostic (OBD) systems do not operate properly when the engine control module (ECM) detects a failure of the associated system or component parts.

Visual inspection of the engine compartment

During the diagnosis, the engine compartment should be carefully checked. Normal operation in accordance with the prescribed procedures often can solve the problem. When performing a visual inspection, follow the instructions below.

- Check all air tubes, etc., check if there is punching, cutting, disconnection, and whether the pipeline is installed correctly.
- Check hoses that is difficult to see hidden in other component parts
- Check that all wiring harnesses in the engine compartment are connected correctly, there are burned or worn parts, there is airtightness in the wiring harness, there is contact with the tip, there is exhaust manifold or high temperature wire harness and so on contact

The basic knowledge of the necessary tools**Important:**

If you do not have basic knowledge of the powertrain system during the diagnosis process, this could result in misdiagnosis or damage of the powertrain components. Therefore, if you do not have the basics, do not diagnose problems related to the powertrain. In order to make effective use of after-sales service manuals, it is necessary to understand basic tools such as scanning tools.

About on-board diagnostic tests

The bad diagnosis results should be confirmed as follows.

- The diagnostic test in the last ignition stroke was completed
- Qualified diagnostic tests during the current ignition stroke

During the diagnostic test, the identified symptoms do not exist.

When the diagnostic test fails, confirm and record the data according to the following.

- The diagnostic test in the last ignition stroke was completed
- The fault identified in the diagnostic test does not exist.
- There is a fault condition during the ignition stroke
- Operating conditions in the event of a failure

Ignition stroke

Since the ignition stroke is performed by the method in which the vehicle satisfies the prescribed diagnostic criteria, the ignition stroke is defined as the key ON, the operation, and the key OFF stroke.

About EMC OFF

About 15 seconds after the key switch is turned off, the ECM power will be completely turned off to a ECM OFF state.

Full diagnostics operation of the component monitor

In order for the engine to work properly, it is necessary to perform a full diagnosis of its components.

Input component

In order to confirm whether the circuit is damaged or not, its value is out of specification, please confirm the input device. It also includes checking and confirming its reasonableness. The so-called plausibility check confirmation means that the signal sent by the sensor is considered unreliable and indicates that a malfunction has occurred.

The input component is not a component defined below and contains the following sensors.

- Crankshaft position (CKP) sensor
- Cam position (CMP) sensor
- Engine coolant temperature (ECT) sensor;

Output component

The output component is used to diagnose whether the response of the control unit is reasonable. With the function monitor unavailable, if the composition parts can be approved, check the circuit for disconnection and numerical value exceeding the limit. The confirmed output component is not the one defined below and contains the following circuits.

- Fuel Flow Control Valve (SCV)
- Lights, relay control
- Solenoid valve

The general name of self-diagnosis

When diagnosis is used as a noun, the term "diagnosis" refers to any vehicle test carried out according to the vehicle diagnostic method. Diagnostic refers to tests performed on a system or component to determine whether a system or component operates according to its specifications. Includes the various diagnostic methods listed in the following list.

- Engine speed
- Engine coolant temperature
- Idle speed switch

Diagnostic execution program

The so-called diagnostic execution program means that not only the adjustment of the diagnostic program is specified but also its priority is specified, and the result will be recorded and displayed. The main items of the diagnostic execution program include the following.

- ON/OFF command of engine fault indicator
- DTC records and clears
- Record scan tool data
- Current status information related to each diagnosis

Diagnostic information

According to the diagnostic procedure, the fault condition of the circuit or component parts is detected. The chart is drawn on the condition that the vehicle can function normally when it is assembled, and it does not happen at the same time. Specific control features include continuous self-diagnosis. This diagnostic function supplements the diagnostic procedures described in this manual. The language of transmitting the reason for the poor function is the DTC system. DTC will be set and engine MIL will be on if the malfunction control unit is detected.

Malfunction indicator lamp (MIL)

A fault indicator (MIL) indicates a warning light such as an engine fault indicator. Basically, when the MIL lamp is lit, it means that an electronic control system such as an engine control module (ECM) has failed and a DTC (diagnostic trouble code) is detected.

2. Data Link Connector (DLC)

The communication device with the control unit is the data link connector (DLC). In addition, DLC is provided to connect the scanning tool. The general method of use of the scan tool is as follows.

- The ID of the stored DTC
- Read sequence data

About vehicle maintenance confirmation

Using an on-board diagnostic (OBD) system to confirm vehicle maintenance is very effective. After the repair, the diagnostic person can operate as follows.

1. Reconfirm and record the diagnosed DTC (diagnostic trouble code) and data of diagnostic scan tool or both.
2. Delete the DTC.
3. Make the vehicle running according to scan tool data.

4. The diagnostic DTC condition information, it will be repeatedly confirmed before the control unit completes the diagnostic test related to this DTC.

When using the on-board diagnostics (OBD) system to check the maintenance status, follow the above procedure. Failing to observe the requirements may result in the unnecessary repair in some cases.

About DTC

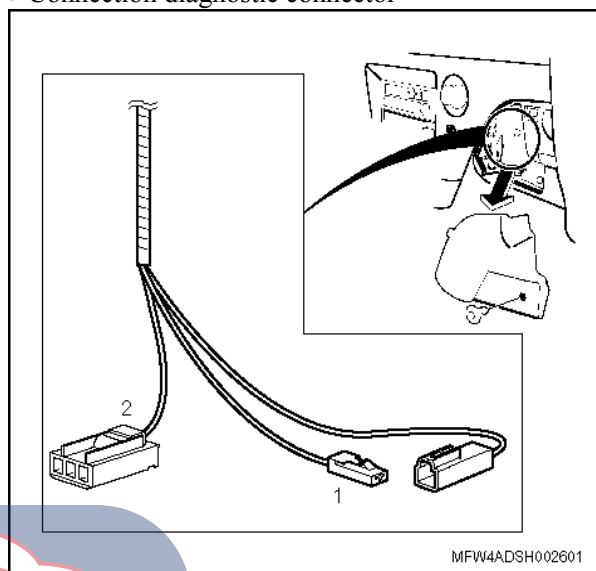
Every time the key switch is turned on, ECM makes most of the harness and components perform self-checking. When the system detects a fault, it will be stored in the memory of the ECM, and the feedback control will be carried out by the DTC. In addition, when the system generates an abnormal condition that affects the driving condition, the engine fault indicator on the instrument panel will light up to inform the driver.

Flashing diagnostic error code reading

The current and past fault codes (DTCs) stored in the ECM, when the diagnostic connector is connected, will cause the engine fault indicator to

light up and display the corresponding content.

- Open the key switch and check that the engine fault light is on. (Confirm light bulb)
- The key switch and engine should be turned on and off, respectively.
- Connection diagnostic connector



Name

Diagnostic connector (red)

2. Data Link Connector (DLC)

- Read engine failure indicator blinks
- The DTC will judge the content based on the DTC table.

When you install the scan tool, you can read the values by operating the scan tool.

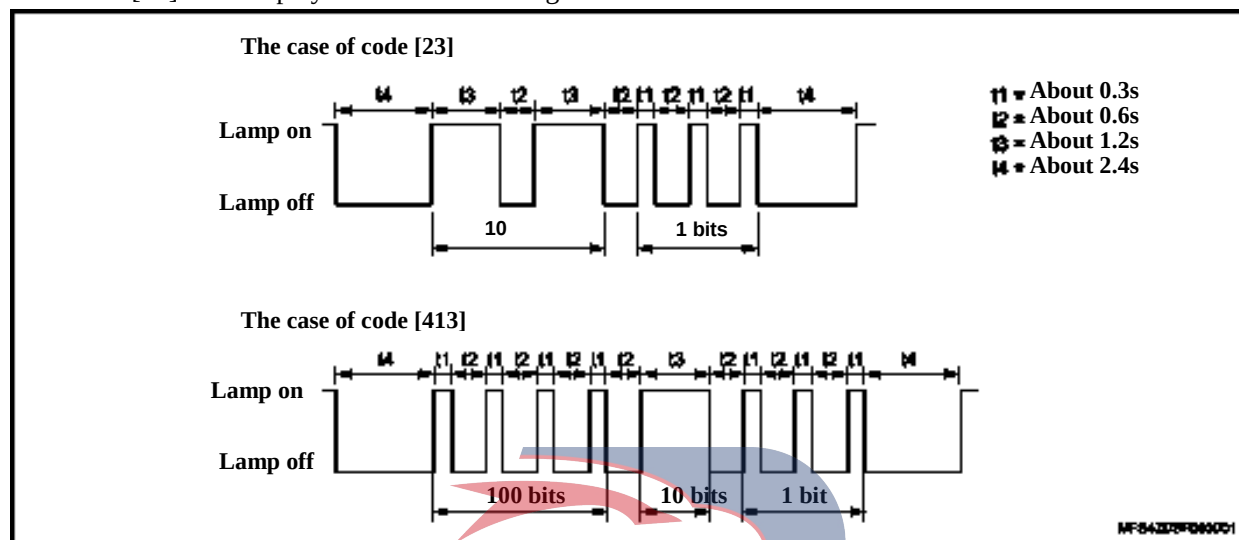
DTC not stored

The code [1] indicating the code starting display will be repeatedly displayed.

Stored the DTCs conditions

diagnosis connection.

The stored DTC will be displayed 3 times. If more than 2 DTCs are stored, each DTC will be displayed 3 times in the sequence of number from low to high. After the code shows a loop, the code will repeat from code [23]. The display will continue during the

**About DTC displayed by engine fault indicator light**

During the engine stop, the currently generated DTC and DTC are simultaneously displayed.

The engine will only show the DTC being generated now during start-up.

DTC removal

When an abnormality occurs in the system and the DTC is stored in the engine control module (ECM), the DTC memory does not disappear even if the faulty part is repaired. Therefore, perform forced liquidation as follows.

Speed pedal clearance by operation**Caution:**

※ Except for a part of the electrical system (ABS, Smoother-F, SRS, etc.), the DTC of the electrical system will be cleared once, so be careful when using it.

- The key switch should be in the ON state and the engine should be stopped

Diagnostic connector should be short-circuited

When the following operations are performed under this condition, the fault code will be cleared.

1. Depress the accelerator pedal within 1 to 3 seconds

2. Release the accelerator pedal within 1 to 3 seconds
3. Depress the accelerator pedal within 1 to 3 seconds
4. Release the accelerator pedal within 1 to 3 seconds
5. Depress the accelerator pedal within 1 to 3 seconds

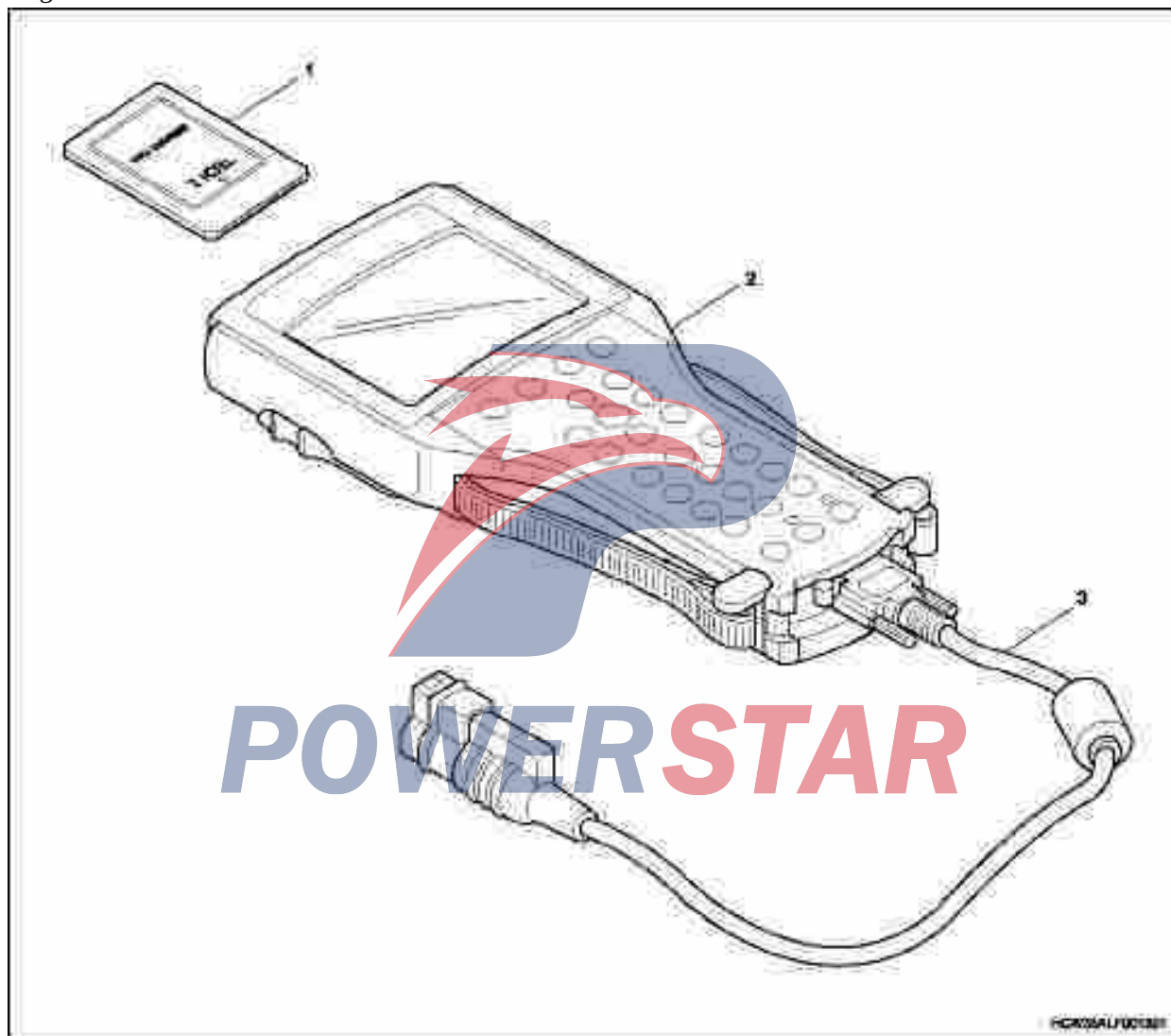
Clear with the scanning tool

When installing the scan tool, the DTC can be cleared by the scan tool's scan operation.

Use the scan tool for troubleshooting

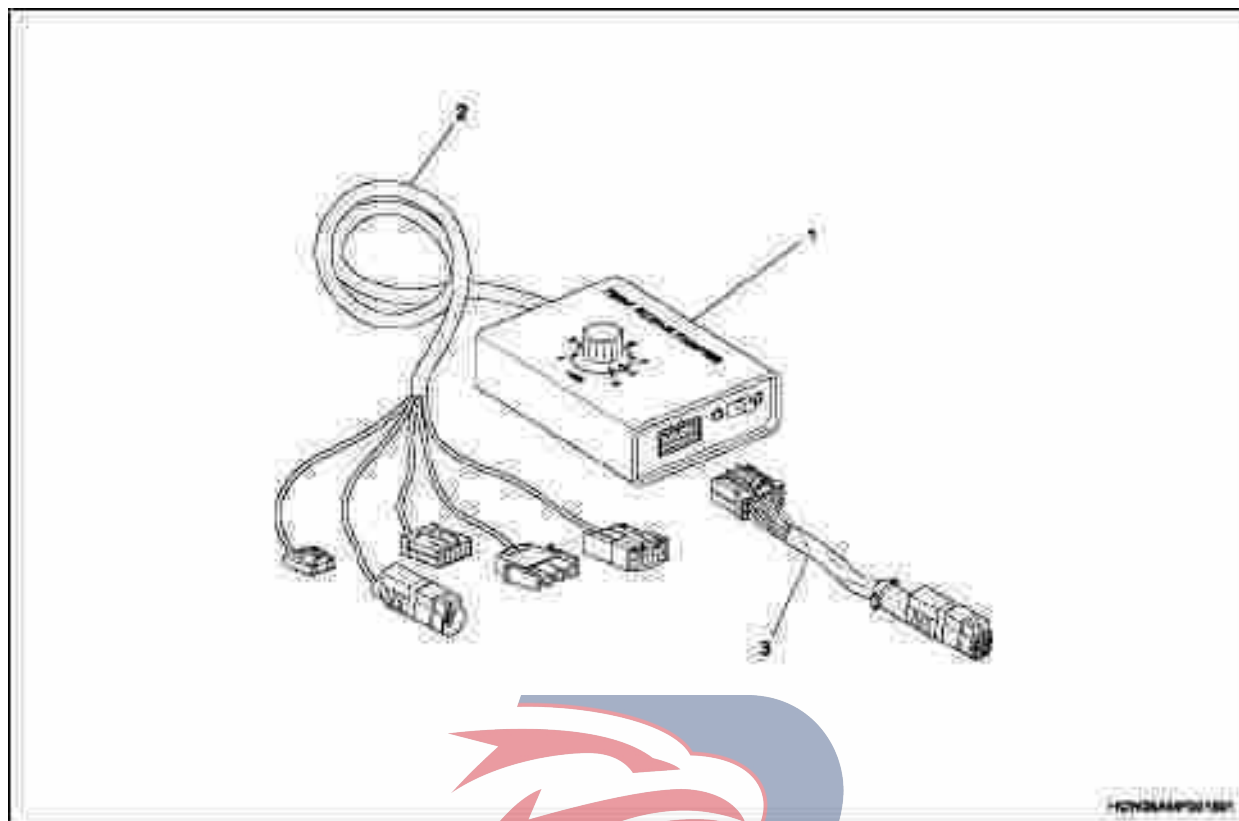
About the Scan Tool (Tech2)/Tech2 Adapter

This is a very effective tool for electrical fault diagnosis and system inspection of electronic control systems. Using the Tech2 adapter to connect it to the DLC on the vehicle, it can communicate with each control unit of the vehicle to perform various diagnostics and tests.



Name

- 1. PCMCIA card (dedicated) 3. DLC cable
- 2. Tech2 (host)

**Name**

1. Tech2 adapter

2. Adapter cable (for connection with vehicle DLC)

3. Adapter cable (for connecting Tech2 DLC cables)

Tech2 / Tech2 Adapter Features, Precautions

- Tech2 is a 12V power system. The 24V power supply cannot be used directly. The power supply for Tech2 is supplied through the Tech2 adapter. (The Tech2 adapter is equipped with the function to convert 24V power to 12V power.) Therefore, the Tech2 power supply cannot use the power supplied directly from the battery and cigarette lighter.
- Remove the PCMCIA card after the power is turned off.
- Tech2 has 2 print capacity snapshots.
- PCMCIA cards are very sensitive to magnetic and static electricity, so be careful when handling them.
- Tech2 can draw a snapshot of the printed chart.
- When selecting a menu, use the selection or function keys to select a menu, and then press the enter key.
- Press the EXIT button at any time to return to the home screen.
- To clear the diagnostic trouble code (DTC), open the application menu and select "Clear DTC info".

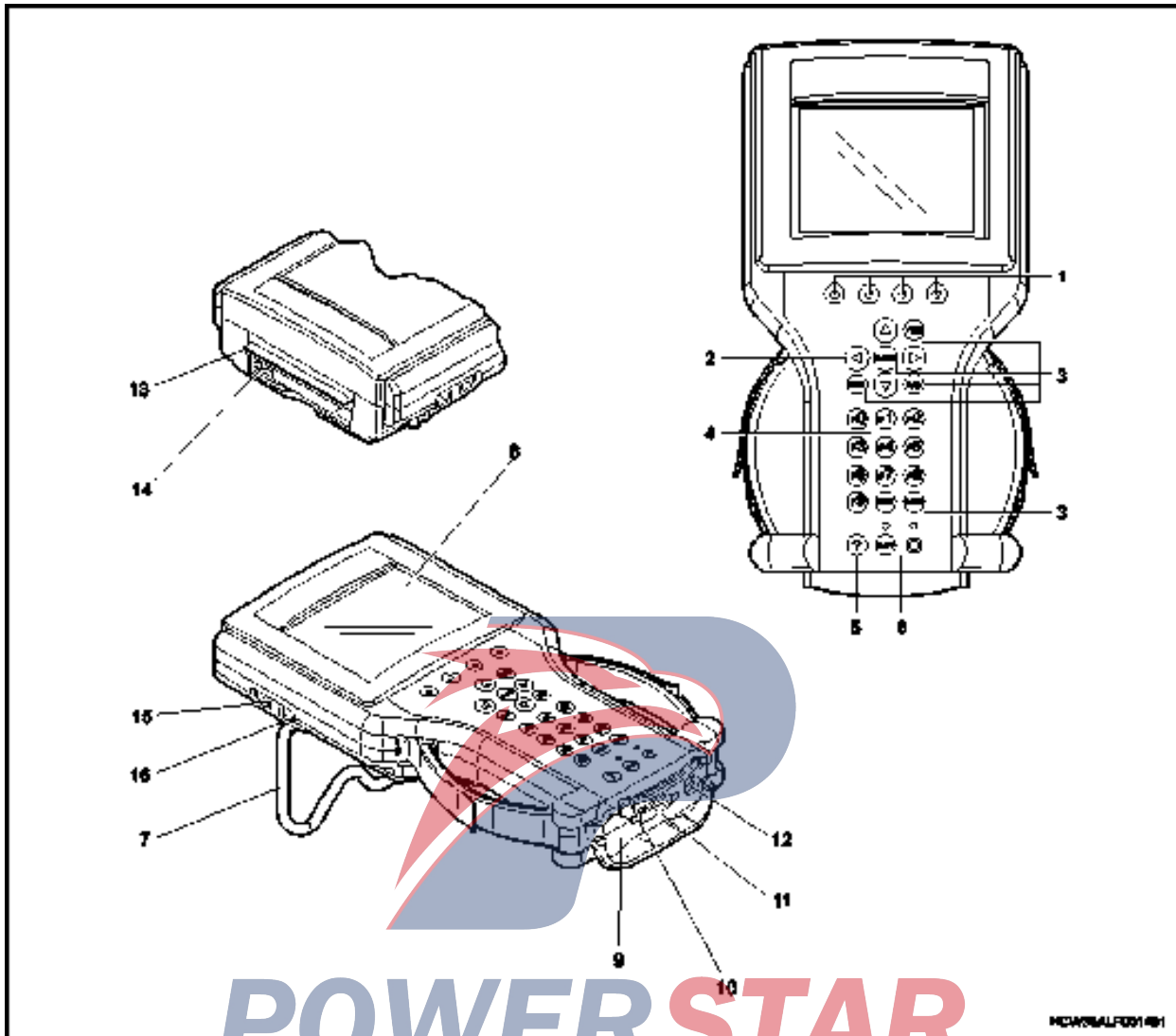
The Tech2 adapter is equipped with a DC-DC transformer (converts 24V to 12V). For details, refer to the operating instructions.

- The Tech2 adapter is equipped with selectors (each ECM signal) for switching signals.

Engine control option 3

- The Tech2 card is installed in the Tech2, and the DLC cable is installed in the Tech2 host and VCI (Vehicle Communication Interface). Second, the adapter cable should be installed on the Tech2 adapter. Connect the Tech2 and Tech2 adapters and connect the adapter cable to the DLC of the vehicle.

About the parts of Tech2



Name

- | | |
|--|---|
| 1. Software | 9. Vehicle communication interface (VCI) module |
| 2. Selection key (mobile key) | 10. VCI module holding rod |
| 3. Function key (YES, NO, ENTER, EXIT) | 11. DLC cable connector |
| 4. Function keys (F0 to F9) | 12. AC adapter connector |
| 5. Help key (?) | 13. PCMCIA card jack (with outside cap) |
| 6. Control keys (PWR / SHIFT) | 14. PCMCIA card unlock (delete) button |
| 7. Widening bracket | 15. For external communication (RS485 port) |
| 8. Monitor (LCD Screen) | 16. For external communication connector (RS232 port) |

1. Software

When Tech2 is operated, the screen upper will display the selection indication box. The software corresponds to its selection box. When the selection box on the screen is not displayed, it cannot be used.

2. Selection key (mobile key)

It is used to carry out switch of menu options and displays on the Tech2 screen. This is where the brightest place in the image displayed in the selection screen is selected.

3. Used for operation keys (YES, NO, ENTER, EXIT)

Tech2 operation confirmation, picture request / question answer, menu switch / move.

4. Function keys (F0 to F9)

Determine the menu function within the screen display. In addition, when the menu records "F0" or other, the displayed contents correspond to the keys.

5. Help key (?)

The screen for using the Tech2 function that is used when the button is displayed will be displayed.

6. Control keys (PWR / SHIFT)

PWR key is used to power ON/OFF Tech2. The SHIFT key is used to adjust the contrast of the picture. When the SHIFT button is activated, the LED (orange) indicator lights up. In addition, buttons and functions other than the selection keys will be locked at this time. The LED (wire) lamp will be on when the power is turned on.

7. Widening bracket

Used as parentheses. The operating angle of the bracket is 0 to 180°.

8. Monitor (LCD Screen)

It is a display screen with the function of contrast adjustment. It can display ECM information and various indication screens.

9. Vehicle Communication Connector (VCI Module)

Module that enables communication between the vehicle and the Tech2.

10. VCI module fixing bracket

The bracket can be used to mount/fix or remove the VCI.

11. DLC cable connector

When using the Tech2 in the car (when connected). Used to connect the DLC cable.

12. AC adapter connector

It is the joint to connect the attached AC adapter. In addition to vehicles, Tech2 can also be used.

13. PCMCIA card slot (with cover)

When inserting the PCMCIA card into the Tech2, open the lid. Insert the PCMCIA card with the cover open. When the Tech2 is turned on, it is forbidden to insert/extract a PCMCIA card.

14. PCMCIA card unlock (delete) button

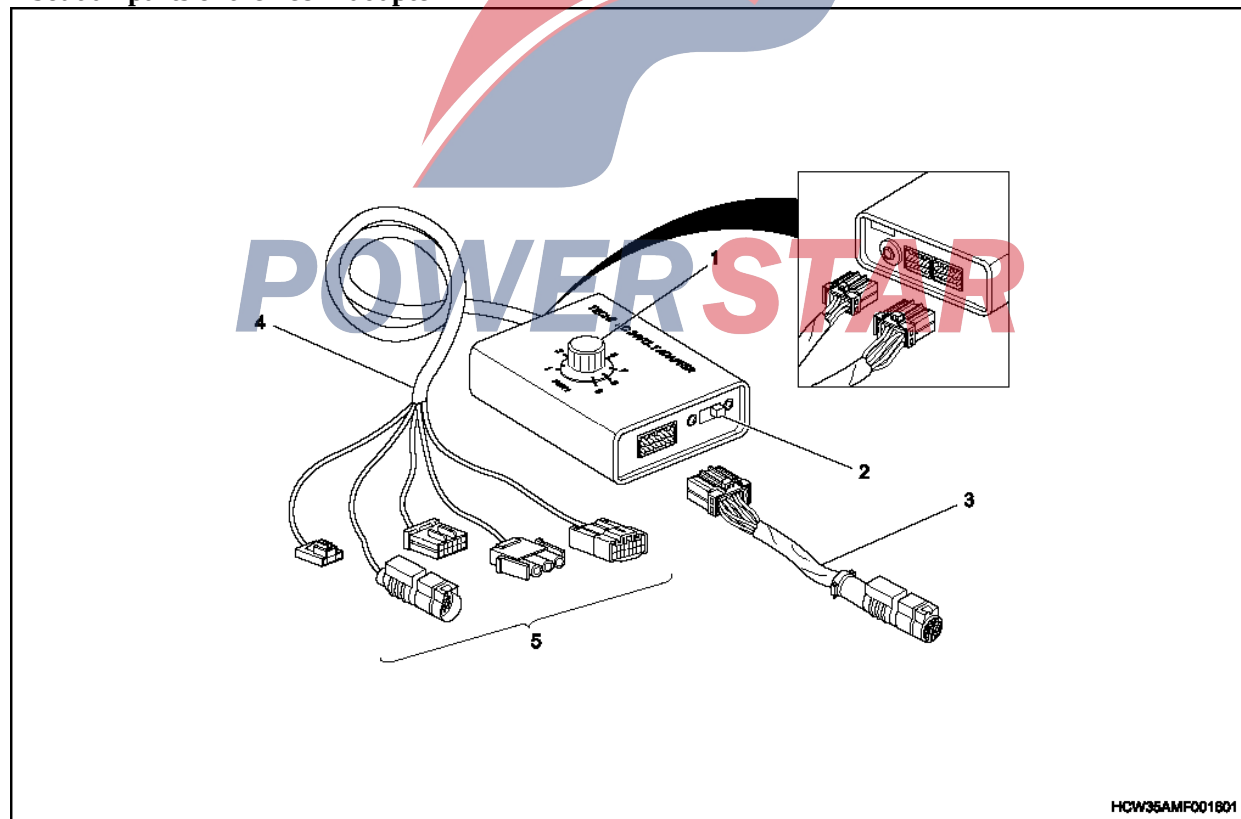
Used to remove the PCMCIA card from the Tech2. After pressing the button, the PCMCIA card will be removed. Disable the PCMCIA card removal operation when Tech2 is on.

15. External Communication Connector (RS485 Interface)

This is a connector for telephone line connection. It is not yet available on Tech2. This connector must not be connected to the telephone line.

16. External Communication Connector (RS232 Interface)

It is the connector for connecting Tech2 and other computers.

About all parts of the Tech2 adapter

HCW35AMF001801

Name

- | | |
|--|---|
| 1. Selector | 4. Connector used for Adapter cable (used for vehicle DLC cable connection) |
| 2. Select switch (signal switch) | 5. Vehicle DLC connector |
| 3. Adapter cable (for connecting Tech2 DLC cables) connector | |

1. Selector

In order to enable Tech2 to recognize signals of various control units and to transmit signals from Tech2 to control units, selectors are installed.

The installation position of the selector varies depending on the target system.

1: HSA (except vehicles with smoother-F)

3: Engine

Other: at discretion

2. Select switch (signal switch)

It is a switch used by Tech2 to read signals from ECM and transmit signals to ECM. (The position of the engine control switch is arbitrary)

Read Signal from ECM: Position of Switch 1

(Read DTC, data confirmation, etc.)

Signal from Tech2 to ECM: Switch 2 Location

(DTC clear command, various test instructions, programming, etc.)

3. Adapter Cable (Tech2 DLC Cable Connection) Connector

Install the cable that connects the Tech2 DLC cable and the Tech2 adapter.

4. Adapter cable (for vehicle DLC connection) connector

Install the cable connecting the Tech2 connector and the vehicle DLC.

5. Vehicle DLC Connector

It is the joint to connect the vehicle DLC.

DLC varies from the devices installed in the vehicle, therefore there are many types.

20 pin: IE (only for 6HH1, idle stop & start system, ABS/ASR) hydraulic control type

10 pins: HSA (except vehicles with smoother-F)

Engine (6HL1 / 6HK1-TC: Common Rail), SRS, ABS (Air Pressure Control), Smoother-F, Air Suspension, HSA (Vehicle with Smoother-F) A511 Transmission (AT), Idle Stop and Startup System, Self-protection System, DPD System (6HL1 Only)

3-pin (AMP system: mail): ZF gearbox (AT) (not for F **)

19. Pin: Allison AT (not for F **)

Connection method

1. Ensure the key switch is in [OFF] state.

2. Insert the Isuzu System PCMCIA card into the Tech2 host.

3. Install DLC cable to Tech2 host.

4. Install the adapter cable to Tech2 adapter.

5. Connect the DLC cable and adapter cable

6. Connect the adapter cable to the DLC.

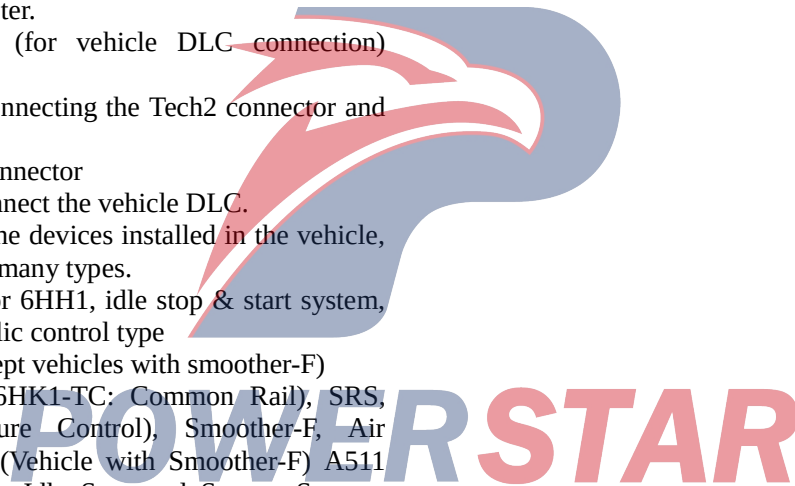
7. Open the key switch and press the Tech2 "PWR" key.

8. Confirm Tech2 display.



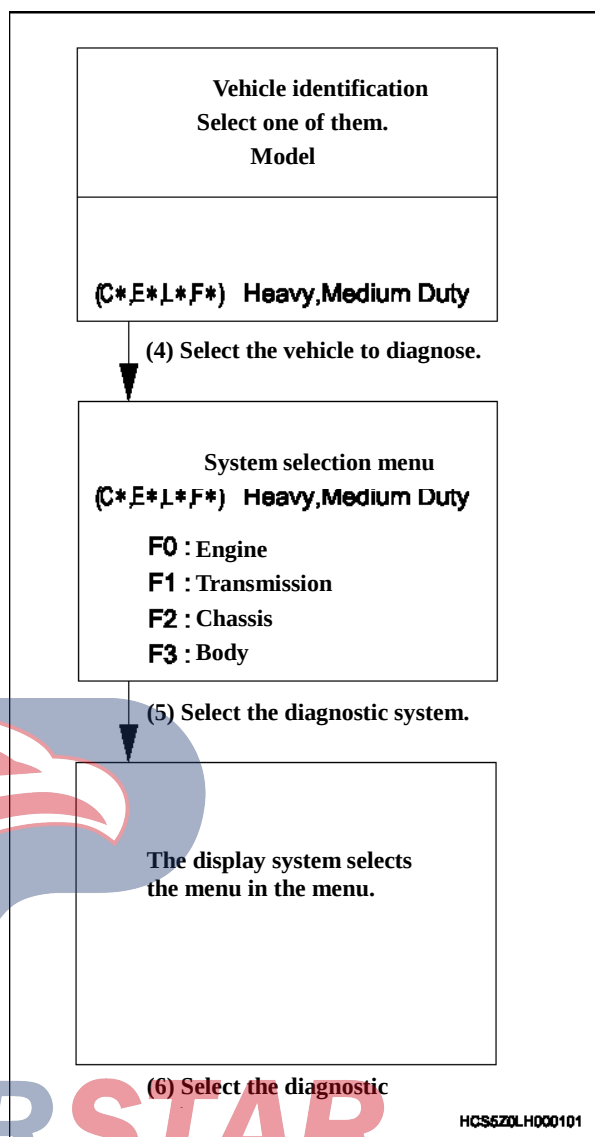
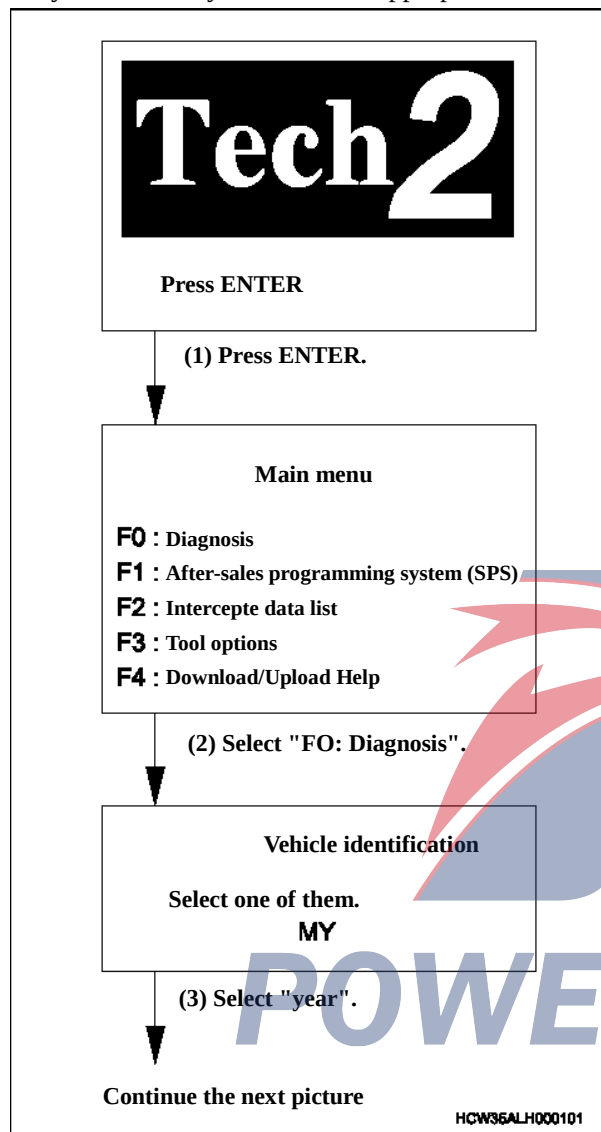
Caution:

PCMCIA card must be installed when Tech2 is not powered.



Steps

Select Diagnostics from the main menu and select the year, model, system, and the appropriate device.



Menu list

The following table shows the current Tech2 software display.

F0:Diagnostic Code (DTC)
F0:DTC display
F1:DTC removal
F1:Data display
F2:Snapshot Printing
F3:actuator test
F0:common rail system
F0:Common Rail Pressure Control
F1:Stop spraying for each injector
F2:Basic Injection Time Control
F3:pre-injection stopping

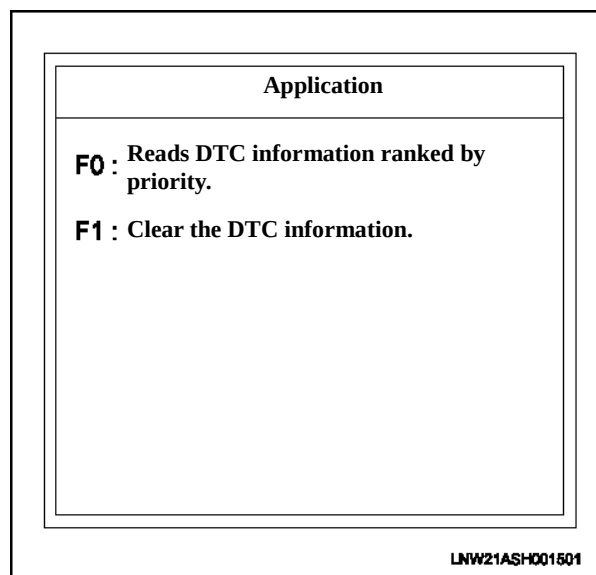
F4: Forced Injector Drive
F1: Electronic Device Control
F0: glow plug relay
F1: Exhaust brake control
F2: Retarder Control
F3: intake throttle control (6HL1)
F3: EGR control (6HK1-TC)
F4: exhaust and intake valve control (for 6HL1 only)
F5: EGR control (6HL1 only)
F6: Forced regeneration with DPD (Only 6HL1)
F4: Programming
F0: Q adjustment value correction
F0: Q adjustment value correction data uploading (Tech2)
F1: Q adjustment value correction data downloading (ECM)
F2: After-sales service model
F1: Syringe ID code
F0: Syringe ID code
F1: ID code registration
F2: ID code uploading (Tech2)
F3: ID code downloading (ECM)
F2: MAF correction (6HL1)
F0: MAF correction data upload (Tech2)
F1: MAF correction data downloading (ECM)
F2: After-sales service model
F2: Engine starting number (6HK1-TC)
F3: Engine starting number (6HL1)
F3: Number of starts after the starter replacement (6HK1-TC)
F4: number of starts after the starter replacement (6HL1)
F5: DPD reactivation data clearing (for 6HL1 only)

About each menu item

Confirm the DTC and data, tests and memory when selecting the menu.

DTC

Read, confirm and clear DTC.



The DTC is the communication language between the scanner and the vehicle ECM. If the DTC is roughly classified, the past DTC and the current DTC are included.

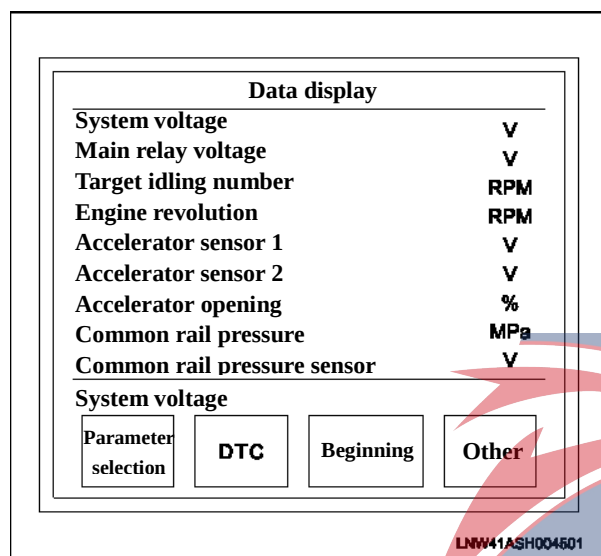
- DTC now: Indicates the fault that occurred during this ignition.
- Fault diagnosis code in the past: It indicates the fault phenomenon that occurred during the last or previous ignition.
- Multiple DTCs: Multiple DTCs are sometimes displayed at the same time. This is called multiple DTCs. This phenomenon occurs when faults overlap. In addition, when multiple sensors and switches share a single power supply or GND, the shared power supply and GND are disconnected. • In the event of a short circuit, the DTCs of the associated sensors and switches are multiplexed. If the DTC has multiple displays, it is necessary to check whether the shared power supply or the shared GND is disconnected or short-circuited. Please see Information of Clearing Vehicle's DTC on DTC Diagram.

Data display

ECM can display the identified and controlled status for every item. The Tech2 adapter selector switch should be at position 1 when reading data.

The value of the displayed item is the representative value specified in the Tech2 data list.

The content of the displayed item is described in data format.



Use this menu to display current data.

The display is synchronized with the status of the vehicle and the device.

- When the menu for displaying data cannot be selected:

Harness malfunction between the ECM and DLC.

Tech2 adapter select switch position incorrect.

- When the abnormally fixed is displayed:

[Example] When the number of engine revolutions and the data display value of each sensor are fixed and the actual status does not match. If when the

connector or harness is moved by hand, the display is released from the fixed state, this is because the terminal is incompletely connected, the pins are not completed, and the harness in the hand-moved part is line-broken or short-circuited. Therefore, maintenance is required.

- When the value is displayed as *:

There is a problem with Tech2 software

In data display, it is necessary to upgrade the PCMCIA card even if there are 1 *markings in it.

Snapshot printing

In order to find out the cause of the malfunction, it is necessary to perform snapshot printing and readjust the data before use.

- Snapshots Print a menu of recorded data lists and draws graphics.
- Use this mode to make the customer's complaint partially reappear. After the record, abnormal data can be found.
- The method of determining the trigger type can also record the instantaneous data displayed by the DTC.
- Recorded data can also be regenerated using home power.

Programming

Before replacing ECM, confirm the record information (data confirmation). Furthermore, it is necessary to upload the data to the Tech2. After replacing ECM, download to new ECM from Tech2.

Actuator test

Carry out various tests to check the movement and function of the parts of the body, so as to confirm whether they are normal.

When the instruction (command) is input from the test screen to ECM, the selection switch of the Tech2 adapter should be in the 2 position.

Caution:

When the conditions of the running state of the vehicle and the engine, the water temperature, the speed, the switch, the position of the gear and so on are not satisfied, and the DTC is detected, sometimes the tests cannot be carried out.

Item		Notes	How to use
F0:common rail system	F0:Common Rail Pressure Control	When performing UP / DOWN command, the value of [common rail pressure] should be in the range of -5Mpa to 5Mpa	When the value of [Common Rail Pressure] is not in the range of -5Mpa to 5Mpa, there is an abnormality in the fuel system, circuit, ECM, etc.
	F1:Stop spraying for each injector	When carrying out the ON/OFF command, the engine's sound should change, and the idling operation is unstable (idling disorder)	Even if the OFF command is given but the engine speed and vibration are not changed, it can be judged that the injector of the cylinder that has executed the stop command has stopped before the command is issued. There is abnormality on ejector, wire harness and ECM.
	F2:Basic Injection Time Control	When performing UP / DOWN command, confirm the change of main injection time	When the UP / DOWN command is issued, the injection time control is still not running. This is due to the abnormality of the CKP sensor harness and ECM.
	F3:pre-injection stopping	When releasing the pre - injection command, [pre-injection volume] should be around 0mm ³ /st	Although the pre-injection stop command was given, the value of [pre-injection amount] does not reach 0 mm ³ /st, and an abnormality occurs in the ECM or the like due to the short-circuit of the injector harness.
	F4:Forced Injector Drive	When the ON / OFF command is issued, check the operating sound of the injector	With the ON command issued, if the injector action sound is not heard, it means the corresponding injector, harness and ECM are abnormal.

POWERSTAR

Item		Notes	How to use
F1:device control	F0:glow plug relay	Confirm the operation of the glow plug relay when confirming the ON command	When the on command is issued, the glow plug relay, glow plug relay harness, and ECM are abnormal when the glow plug relay is not operating.
	F1:Exhaust brake control	When the ON command is issued, check the control status of the exhaust brake relay	When the ON command is issued, if the exhaust relay control status does not change, the exhaust brake solenoid valve, exhaust brake solenoid valve harness, ECM, etc. are abnormal.
	F2:Retarder Control	When the ON command is issued, check the operation status of the deceleration solenoid valve	When an ON command is issued and the speed reducer solenoid valve does not operate, the speed reducer solenoid valve, the reducer solenoid valve harness, ECM, etc.
	F3:Air intake throttle control (6HL1)	When executing the UP / DOWN command, confirm that the value of [inlet throttle opening] should be in the range of -3% to 3%.	When the value of [inlet throttle opening] is not in the range of -3% to 3%, this is caused by an abnormality such as intake throttle, circuit, ECM, etc.
	F3 EGR control 6HK1-TC	Check the temperature of EGR gas when confirming the ON / OFF command	During ON/OFF command, if EGR gas temperature does not change, it means the EGR magnet valve, EGR magnet valve harness, EGR valve and EGR are abnormal.
	F4:intake throttle control (6HL1)	When the ON / OFF command is issued, check the exhaust throttle control status.	When the ON / OFF command is issued, the control state of the exhaust throttle valve does not change due to abnormality of the DPD solenoid valve, DPD solenoid valve harness, ECM, etc.
	F5 EGR control (6HL1)	When executing the UP / DOWN command, confirm that the value of [EGR Position] is in the range of -3% to 3%.	If the value of [EGR position] is not in the range of -3% to 3%, abnormality occurs in GER valve, circuit, ECM, etc.
	F6:Forced regeneration with DPD (6HL1)	When the DPD cannot be used to regenerate the DPD, the Tech2 command can be used to forcibly regenerate the DPD	When DTC is set and DPD manual regeneration cannot be performed even if the DPD switch is pressed, the Tech2 command can be used to forcibly regenerate DPD.

Method for determining trigger type to draw a chart

About the making of chart

The snapshot dump data can be plotted into table.

Through the adjustment and confirmation of the form, the cause of the failure can be ascertained.

A method of drawing charts after displaying data

1. After the vehicle data is displayed in [Data Display] in the application menu, you can select [Point Search] under the screen.
2. After a period of time, press the [EXIT] button.
3. After the screen changes and [Continue] appears at the bottom of the screen, select [Continue].
4. When [Draw] is displayed under the screen, select [Draw].
5. After turning to the item selection screen, press the Enter key in combination with the item (you can select three items), and then click [Agree] under the screen.
6. A graphic display will appear on the screen.

1. Select [Data Display] in the application menu.
2. Please make an appropriate choice after displaying the item on the screen
3. After the snapshot print option screen is displayed, confirm the trigger method, and then select [Snapshot Print] under the screen.
4. When you are ready to flash in the lower right corner of the screen, select [Trigger] under the screen.
5. Confirm the trigger type.
6. After the screen changes and [Continue] appears at the bottom of the screen, select [Continue].
7. When [Draw] is displayed under the screen, select [Draw].
8. After the item select screen appears, select up to 3 items by pressing ENTER and then press [OK] on the screen.
9. The corresponding chart will be displayed on the screen.

Method for reproducing a chart of recorded data

1. In the picture after selecting to turn on Tech2 power [print data by sampling].
2. The title of the snapshot data is displayed on the screen.
3. Select the title of the snapshot data.
4. There will be data display on the screen. When observing the chart, please select the [Draw] under the picture.
5. After selecting the selection screen for this item, press the Enter key in combination with this item to select up to three items, and then press [Approve] at

6HL1

Data display item	Unit	Idle reference value (changes due to vehicle condition)
System voltage	V	22 ~ 30
Main relay voltage	V	22 ~ 30
Target Engine Revolution	RPM	525
Engine revolution	RPM	520 ~ 530
Accelerator sensor 1	V	0.3 ~ 0.7
Accelerator sensor 2	V	4.2 ~ 4.8
Accelerator opening	%	0
Common rail pressure	MPa	± 5
Common rail pressure sensor	V	1.46 ~ 4.59
Common rail pressure feedback mode	-	5
SCV load rate	%	About 30 ~ 50
SCV F/B	mA	About 900 ~ 1200
Engine water temperature sensor	V	0.4 ~ 1.4
Engine water temperature	°C	50 ~ 90
Intake air temperature sensor	V	1.3 ~ 4.1
Intake air temperature	°C	-10 ~ 40
Fuel temperature sensor	V	0.5 ~ 2.1
Fuel temperature	°C	30 ~ 70
Barometric pressure sensor	V	2
atmospheric pressure	kPa	100
Exhaust temperature sensor 1(filter front)	V	Change according to usage
Exhaust temperature sensor 2 (before oxidation catalyst)	V	Change according to usage
Exhaust temperature 1 (filter front)	°C	60 ~ 200
Exhaust temperature 2 (before oxidation catalyst)	°C	60 ~ 200
EGR position (BLDC motor)	%	± 3
EGR BLDC motor duty ratio	%	9 ~ 30
EGR BLDC position 1	ON/OFF	One of the polarity of 1,2,3 will be reversed (For example: 1=ON、2=ON、3=OFF etc.)

the bottom of the screen.

6. The chart will be displayed on the screen

Tech2 data reference

The function of data list is to confirm the vehicle and device status.

Compare each vehicle's data with basic values and use it for troubleshooting when the value of the occurrence is temporarily deviating from the reference or is always in a deviating state. (Tech2 display menus are sometimes subject to change without notice)

Data display item	Unit	Idle reference value (changes due to vehicle condition)
EGR BLDC position 2	ON/OFF	One of the polarity of 1,2,3 will be reversed (For example: 1=ON、 2=ON、 3=OFF etc.)
EGR BLDC position 3	ON/OFF	One of the polarity of 1,2,3 will be reversed (For example: 1=ON、 2=ON、 3=OFF etc.)
MAF sensor (air flow)	V	Changes due to the model
MAF(airflow)	g/cyl	0 ~ 0.2
MAF correction value		It varies with the models
ITP (intake throttle position) sensor	V	0.1 ~ 4.9
Intake throttle open	%	± 3
Intake throttle duty ratio	%	± 2 ~ ± 10
ASR control accelerator	%	-
Speed	km/h	0
First cylinder fuel injection correction	Mm ³ /st	± Below 5
Second cylinder fuel injection correction	Mm ³ /st	± Below 5
Third cylinder fuel injection correction	Mm ³ /st	± Below 5
Fourth cylinder fuel injection correction	Mm ³ /st	± Below 5
Fifth cylinder fuel injection correction	Mm ³ /st	± Below 5
Sixth cylinder fuel injection correction	mm ³ /st	± Below 5
Engine mode	-	3
QWS switch	ON/OFF	OFF
Start switch (On)	ON/OFF	OFF
Start switch (ST)	ON/OFF	OFF
Brake switch 1	ON/OFF	OFF
Brake switch 2	ON/OFF	OFF
Clutch switch	ON/OFF	OFF
Parking /neutral gear switch	ON/OFF	ON
Parking switch	ON/OFF	ON
Exhaust brake switch	ON/OFF	OFF
Exhaust brake relay control	ON/OFF	OFF
Retarder switch	ON/OFF	OFF
Retarder relay control	ON/OFF	OFF
PTO switch	ON/OFF	OFF
PTO accelerator opening	%	-13 ~ 0
PTO Accelerator Sensor	V	0 ~ 0.44
Idling manual switch	ON/OFF	OFF
Low speed increase switch	ON/OFF	OFF
Low speed drop switch	ON/OFF	OFF
A/C clutch switch	ON/OFF	-

Data display item	Unit	Idle reference value (changes due to vehicle condition)
ABS exhaust brake cut off request	YES/NO	NO
TCM exhaust brake cut off requirement	YES/NO	NO
Refrigerator switch	ON/OFF	-
Glow plug relay	ON/OFF	OFF
Glow plug indicator lamp	ON/OFF	OFF
Diagnostic switch	ON/OFF	-
DPD differential pressure (high resolution performance)	KPa	0 ~ 0.5
DPD switch	ON/OFF	OFF
DPD mode	-	20 or 30
DPD TRIP status	-	Changed due to driving conditions
DPD error status	-	Changed due to driving conditions
DPD alarm state	-	Changed due to driving conditions
DPD PM status	-	Changed due to driving conditions
Control state of exhaust throttle	ON/OFF	OFF
Q adjustment value resistance data 1	-	Different from the engine
Q adjustment value resistance data 2	-	Different from the engine
Q adjustment value resistance data 3	-	Different from the engine
Engine starting times	Count	Changed due to driving conditions
Engine starting number (after starter replacement)	Count	Changed due to driving conditions

6HK1-TC

Data display item	Unit	Idle reference value (changes due to vehicle condition)
System voltage	V	22 ~ 30
Main relay voltage	V	22 ~ 30
Target idling number	RPM	525
Engine revolution	RPM	520 ~ 530
Accelerator sensor 1	V	0.3 ~ 0.7
Accelerator sensor 2	V	4.2 ~ 4.8
Accelerator opening	%	0
Common rail pressure	MPa	± 5
Common rail pressure sensor	V	1.46 ~ 1.59
Common rail pressure feedback mode	-	5
SCV load rate	%	About 30 ~ 50
SCV feedback	mA	About 900 ~ 1200
Engine water temperature sensor	V	0.4 ~ 1.4
Engine water temperature	°C	50 ~ 90

Data display item	Unit	Idle reference value (changes due to vehicle condition)
Intake air temperature sensor	V	0.9 ~ 4.1
Intake air temperature	°C	-10 ~ 40
Fuel temperature sensor	V	V 0.5 ~ 2.1
Fuel temperature	°C	30 ~ 70
Barometric pressure sensor	V	About 2
atmospheric pressure	kPa	100
EGR gas temperature sensor	V	30 ~ 90
EGR valve control	ON/OFF	ON (after warm-up ※)
MAP (manifold absolute pressure) sensor	kPa	About 1
Manifold Absolute Pressure	kPa	100
ASR control accelerator	%	-
Speed	km/h	0
First cylinder fuel injection correction	Mm ³ /st	± Below 1/3
Second cylinder fuel injection correction	Mm ³ /st	± Below 3
Third cylinder fuel injection correction	Mm ³ /st	± Below 1/3
Fourth cylinder fuel injection correction	Mm ³ /st	± Below 3
Fifth cylinder fuel injection correction	Mm ³ /st	± Below 3
Sixth cylinder fuel injection correction	Mm ³ /st	± Below 1/3
Engine mode	-	3
QWS switch	ON/OFF	OFF
Start switch (On)	ON/OFF	OFF
Start switch (ST)	ON/OFF	OFF
Brake switch 1	ON/OFF	OFF
Brake switch 2	ON/OFF	OFF
Clutch switch	ON/OFF	OFF
Parking /neutral gear switch	ON/OFF	ON
Parking switch	ON/OFF	ON
Exhaust brake switch	ON/OFF	OFF
Exhaust brake relay control	ON/OFF	OFF
Retarder switch	ON/OFF	OFF
Retarder relay control	ON/OFF	OFF
PTO switch	ON/OFF	OFF
PTO accelerator opening	%	-13 ~ 0
PTO Accelerator Sensor	V	0 ~ 0.44
Idling manual switch	ON/OFF	OFF
Low speed increase switch	ON/OFF	OFF
Low speed drop switch	ON/OFF	OFF
Cruise master switch	ON/OFF	OFF

Data display item	Unit	Idle reference value (changes due to vehicle condition)
Cruise setting switch	ON/OFF	OFF
Cruise reset switch	ON/OFF	OFF
Cruise cancel switch	ON/OFF	OFF
A/C clutch switch	ON/OFF	-
ABS exhaust brake cut off request	YES/NO	NO
TCM exhaust brake cut off requirement	YES/NO	NO
Refrigerator switch	ON/OFF	-
Glow plug relay	ON/OFF	OFF
Glow plug indicator lamp	ON/OFF	OFF
Diagnostic switch	ON/OFF	-
Q adjustment value resistance data 1	-	Different from the engine
Q adjustment value resistance data 2	-	Different from the engine
Q adjustment value resistance data 3	-	Different from the engine
Engine starting times	Count	Changed due to driving conditions
Engine starting number (after starter replacement)	Count	Changed due to driving conditions

About distribution cabinet

It is a very effective tool for harness inspection. Rather than using its direct measurement of the vehicle's ECM pin, it is possible to use a power distribution cabinet that can use the digital millimeter 5-8840-2691-0 to measure voltage, conduction, and other conditions on the distribution cabinet.

Important

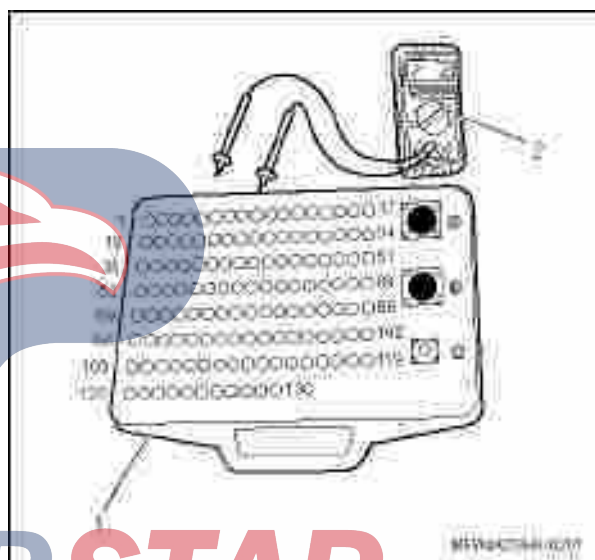
When checking or measuring the voltage, conduction, etc. of the ECM, use the power distribution cabinet and the adapter harness for the model.

Important:

Since the ECM may be damaged in some cases, be sure to check the pin position before measuring the ECM pin.

Mark:

The ECM pin number in the diagnostic procedure should match the pin number of the power distribution cabinet.



Name

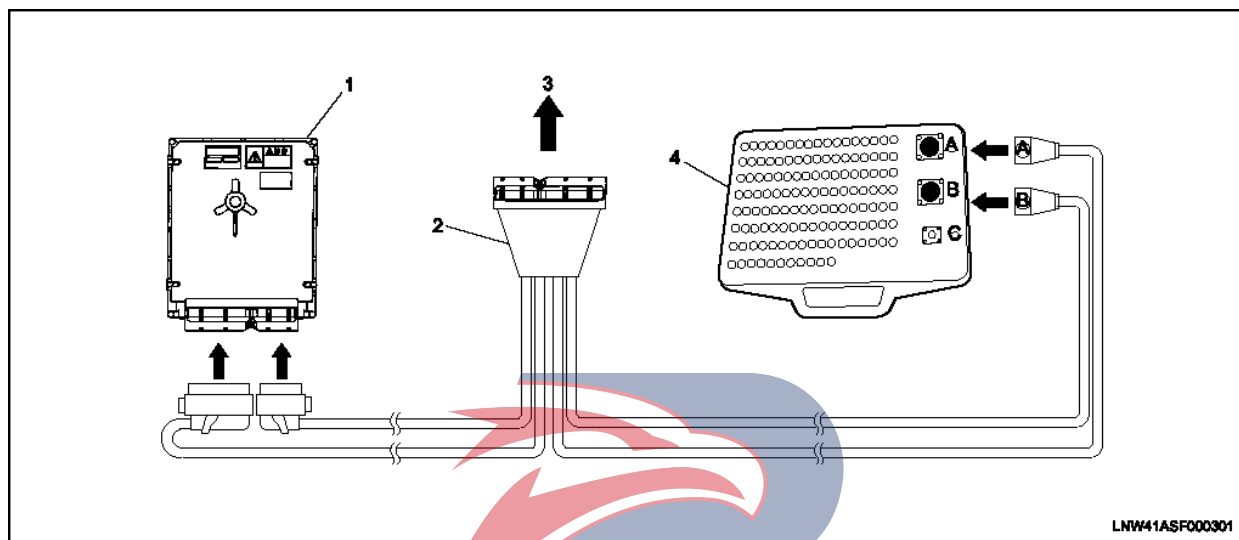
1. Distribution Cabinet
2. Digital multimeter

Distribution cabinet connection method

1. Remove the ECM harness connector and remove the ECM.

2. Connect the adapter harness of the distribution cabinet to the ECM wiring harness on the vehicle side.

3. Connect the distributing box adapter harness joint to the removed ECM and distributing box host.



Name

1. ECM

2. Distribution cabinet adapter harness

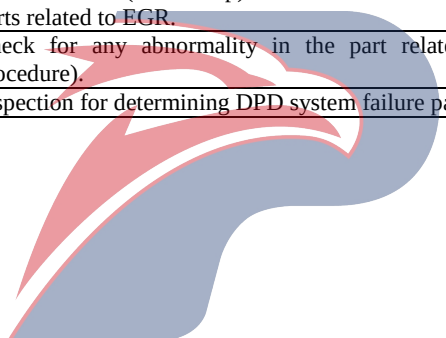
3. Connect to vehicle side ECM harness

4. Distribution cabinet

POWERSTAR

Functional Checklist

Inquiry	According to the customer's statement content, we should fully grasp the symptoms of failure and implement the correct diagnosis steps.
On-board diagnostic (OBD) system check	Check to determine an engine control system component failure (inspection step).
Engine fault indicator circuit system confirmation	When the key switch is set at the [ON] position, the engine fault indicator should be checked when it is not lighted.
Confirmation of engine MIL blinking circuit system	The inspection which should be made if engine MIL does not blink (always on) when the diagnostic joint is connected.
Confirmation of scanning tool power circuit system	The check that should be performed when the scan tool is not running.
Confirmation of scanning tool communication circuit system	If the scan tool and ECM cannot communicate, a check should be performed.
Start circulatory system confirmation	Inspection should be carried out when the starter does not run.
Confirm startup system	Check that the starter is running but the engine is not starting.
Fuel system confirmation	Check the components related to the fuel System for any abnormalities (check procedures).
Confirmation of intake system	Check the components related to the air intake system for any abnormalities (check procedures).
Exhaust system confirmation	Check the components related to the intake system for any abnormalities (check procedures).
Confirmation of EGR control system	It it the check (check step)in order to confirm the possibility of abnormal detection of parts related to EGR.
Confirmation of QOS control system	Check for any abnormality in the part related to the QOS (heating plug) (check procedure).
DPD functional check	Inspection for determining DPD system failure part (inspection steps).



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Inquiry

Inquiry

1. Use the engine control system inquiry form to fully grasp the contents stated by the customer.

Reference is not hard interrogation, but centralized inquiry for the fault phenomenon (fact) on the speculative fault system.

2. Correct judgment of fault system

Grasp the specific fault information according to the principle of 5W1H

For example: low temperature, start, normal, close to the engine, metal tone, etc.

Key points of inquiry

What	Failure phenomenon?
When	Date, time, frequency of occurrence
Where	Road conditions
In what state	Driving status, operating status, climate conditions
How (how to)	Feeling of the phenomenon?

Inquiry form of engine control system

When the maintenance shop picks up the car from the customer, it is necessary to confirm the "fault symptom" and "fault occurrence data" with the customer according to the engine control system survey questionnaire.

HCW01ASH011801

Name

1. Failure symptoms

2. Failure frequency Fault condition

Causes:

1. Symptoms of failure sometimes it is not possible to replicate at after-sales facilities.

2. The customer complains what may not be the real fault.

3. Failing to properly report the fault to the maintenance personnel will cause the maintenance personnel to waste.

• The questionnaires can play an auxiliary role in the vehicle diagnosis, maintenance and maintenance confirmation of after-sales service factories.

Inquiry form of engine control system

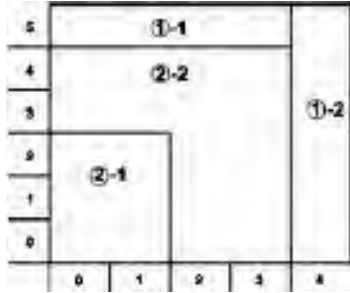
Inspector's name _____

Customer name		Model and year	
Driver's name		License number	
Vehicle incoming date		Engine type	
Registration number		Engine number	
		Engine form	
		Odometer display value	

Failure symptoms	Engine Don't start	☐ Engine crankshaft not operating ☐ No initial explosion ☐ First explosion but insufficient
	☐ difficult to start	☐ Engine crankshaft running slowly (more than 6 seconds) ☐ Others _____
	☐ idle instability	☐ Idle rotation abnormality (When fully warmed up, operate as required) ☐ Idle speed operation abnormality ☐ High speed (rpm) ☐ Low speed (rpm) ☐ Idle speed misalignment (more than the specified speed after full warm-up) ☐ Others _____
	☐ Poor handling	☐ Surge ☐ Explosion ☐ More smoke, white smoke and more engine vibration ☐ abnormal sound ☐ other _____
	☐ Engine flameout	☐ Immediately after start-up ☐ Release pedal ☐ After air conditioning is turned on ☐ After shifting ☐ Others _____
	☐ idling vibration	☐ Lateral vibration of the engine and longitudinal vibration of the body side (The longitudinal vibration of the engine is in principle very weak on the body)

Malfunction data		
The frequency of failure		☐ often ☐ occasionally (several times; day/month) ☐ only once ☐ Others _____
Fault status	Weather	☐ Sunny ☐ cloudy ☐ rainy ☐ snowy ☐ other _____
	temperature	☐ More than 30°C ☐ Before and after 20-30°C ☐ Before and after 10-15°C ☐ Less than 0°C (°C)
	Site	☐ Expressway ☐ Suburb ☐ Urban ☐ Uphill ☐ Downhill ☐ uneven road surface ☐ other _____
	Engine temperature	☐ Cooler ☐ Insulator ☐ Insulator ☐ Water temperature °C ☐ Oil temperature °C
	Operating conditions	☐ At startup ☐ After startup (minutes) ☐ Idle speed ☐ High speed idling ☐ During operation ☐ Constant speed ☐ At acceleration ☐ At deceleration ☐ A/C SW ON/OFF ☐ others _____

Engine fault indicator lamp	☐ Lighting ☐ Occasional lighting ☐ Off		
DTC	Past	☐ ☐	Now ☐ ☐
Fault history	☐ YY MM DD _____		
	☐ YY MM DD _____		

DPD	DPD warning light	☐ Blinking ☐ No flicker	DPD status	DPD PM Status 
	DPD switch	☐ Press ☐ Not pressed		
	DPD status	☐ ①—1 ☐ ①—2 ☐ ②—1 ☐ ②—2		
	DPD PM status	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5		
	DPD TRIP status	☐ 1 ☐ 2 ☐ 3 ☐ 4		
	DPD differential pressure	_____ kPa		

DPD TRIP Status

MFW41DXF002301

On-board diagnostic (OBD) system check

Inspection of on-board diagnostics (OBD) system

The inspection of the OBD system is a system for confirming a problem caused by an incomplete function of the engine control system, and is also a starting point for diagnosing all complaints about controllability. Proper use of this diagnostic procedure can shorten the time of diagnosis and prevent replacement of previously fault-free parts.

Test instruction

The numbers shown below indicate the step numbers in the table

4. Communication with the scan tool fails due to an abnormal communication circuit between the ECU and the scan tool.

7. When a DTC is detected, refer to the DTC list to enter the appropriate DTC diagnosis. When multiple DTCs are detected, they are diagnosed in descending order of DTC counts if there are no special requirements in the diagnostic program.

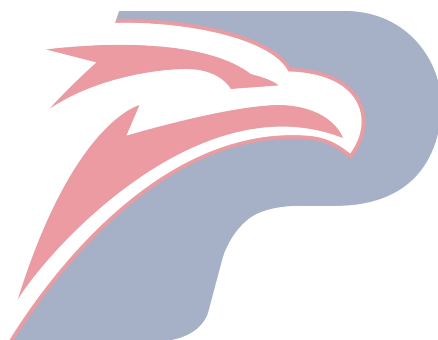
Considerations for inspection of OBD system

Important:

- If you do not handle the complaint, do not perform this diagnostic procedure as long as there are no special requirements for other items.
- Confirm the after-sales service information before starting diagnosis.
- Do not delete the DTC when there are no special requirements in the diagnostic program.
- If an abnormality is found in the engine starting system, refer to "Electrical System" in the Maintenance Manual "Engine".
- The battery is fully charged.
- Make sure the battery cable is normal and connected properly.
- ECM GND should be connected in the correct position.

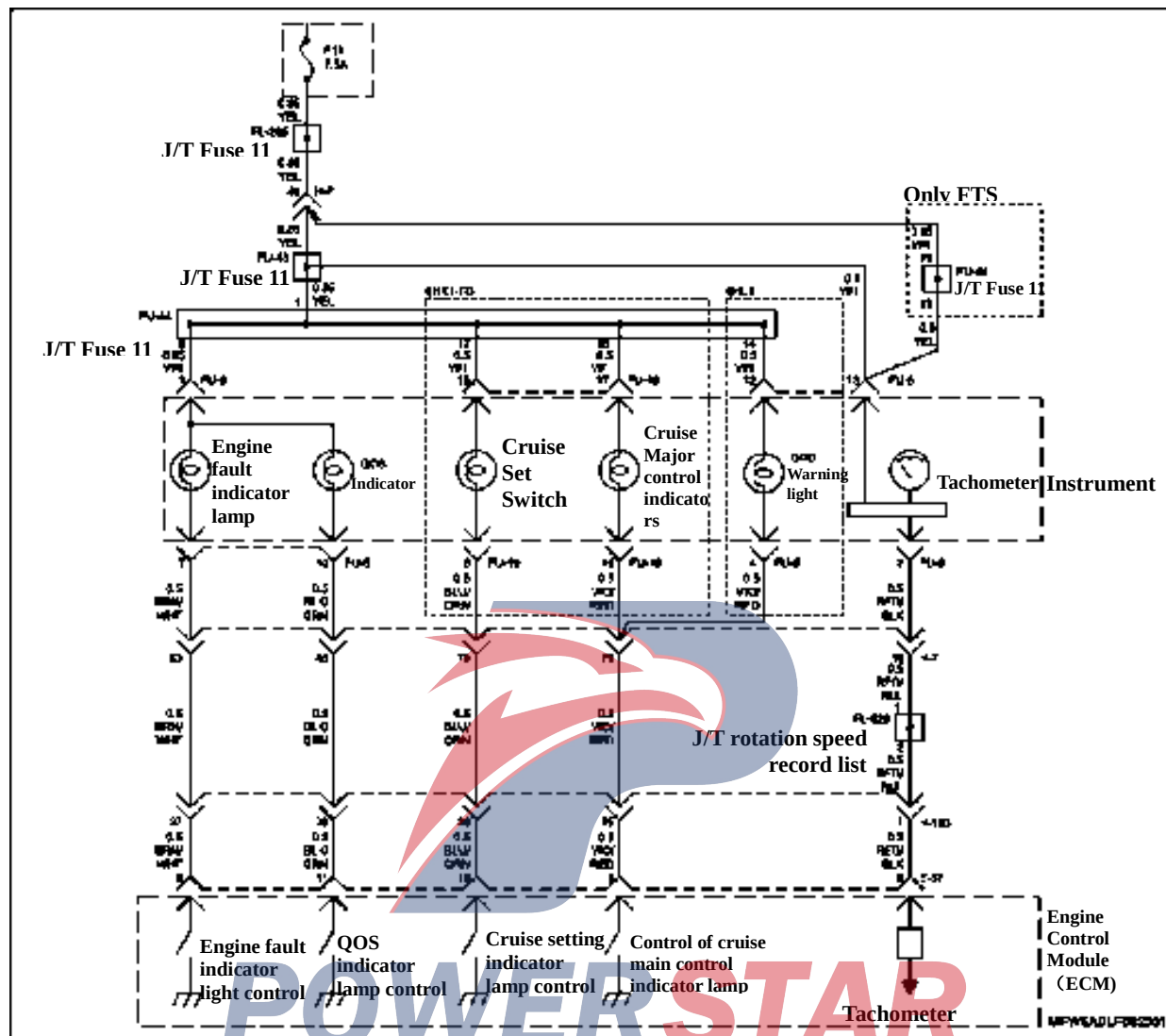
Step	Operation content	Datum value	YES	NO
1	1. Set the key switch to the [OFF] state 2. Check the engine failure indicator lamp. Is engine MIL on?	-	Go to step 2	Access to "confirmation of engine MIL circuit system"
2	1. Connect the diagnostic connector. 2. Check the engine failure indicator lamp. Is engine MIL on?	-	Go to step 3	Enter "Confirmation of Engine Fault Indicator lamp Flashing Circuit System"
3	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Set the scanner's power to [ON]. Is there a trend in scanning tools?	-	Go to step 4	Input scan tool Power circuit system
4	Use the scan tool to make it communicate with the ECM. Is the communication with ECM normal?	-	Go to step 5	Input scan tool Communication loop system
5	Set the key switch to the [START] position. Are the starters running?	-	Go to step 6	Go to "startup circuit system"
6	Set the key switch to the [START] position. Does the engine start?	-	Go to step 7	Access to "Starter system "
7	Check whether there is DTC detected with the scanning tool. Is there any DTC detected?	-	Enter the appropriate DTC diagnosis	Go to step 8

Step	Operation content	Datum value	YES	NO
8	1. Check for the following conditions: • The engine is off. • Engine jitter, unbalanced idle speed; • Insufficient engine power, poor injection, wheezing Much white smoke in emission gas • There are many white smoke emissions (vehicles with DPD); Much black smoke in emission gas • There are many black smoke emissions (vehicles with DPD); • Idle speed cannot be adjusted; • Idle speed cannot be reduced; • The DPD warning light flashes frequently. Is the fault detected?	-	Enter the diagnosis of the corresponding fault phenomenon	Normal system



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Engine fault indicator circuit system confirmation



Cycle Description

The engine MIL is powered through the key switch and illuminates according to the signal transmitted from ECM. ECM will inspect the lamp when the key switch is [ON], and the engine fault indicator light will be turned on when the DTC is detected. In addition, when the diagnostic connector is connected, the engine fault indicator will flash according to the detected DTC. ECM is equipped with a monitoring circuit to monitor the work of the lamp.

Diagnostic help

When there are doubts about intermittent problems, they can be taken into account from the following reasons:

- Harness connector bad connection;
 - Harness is not good;
 - Harness cover damage;
 - Wire breakage inside the harness cladding
- In order to find out these reasons, the following checks should be carried out.
- Connection between harness connector and ECM connector is bad;
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
 - Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

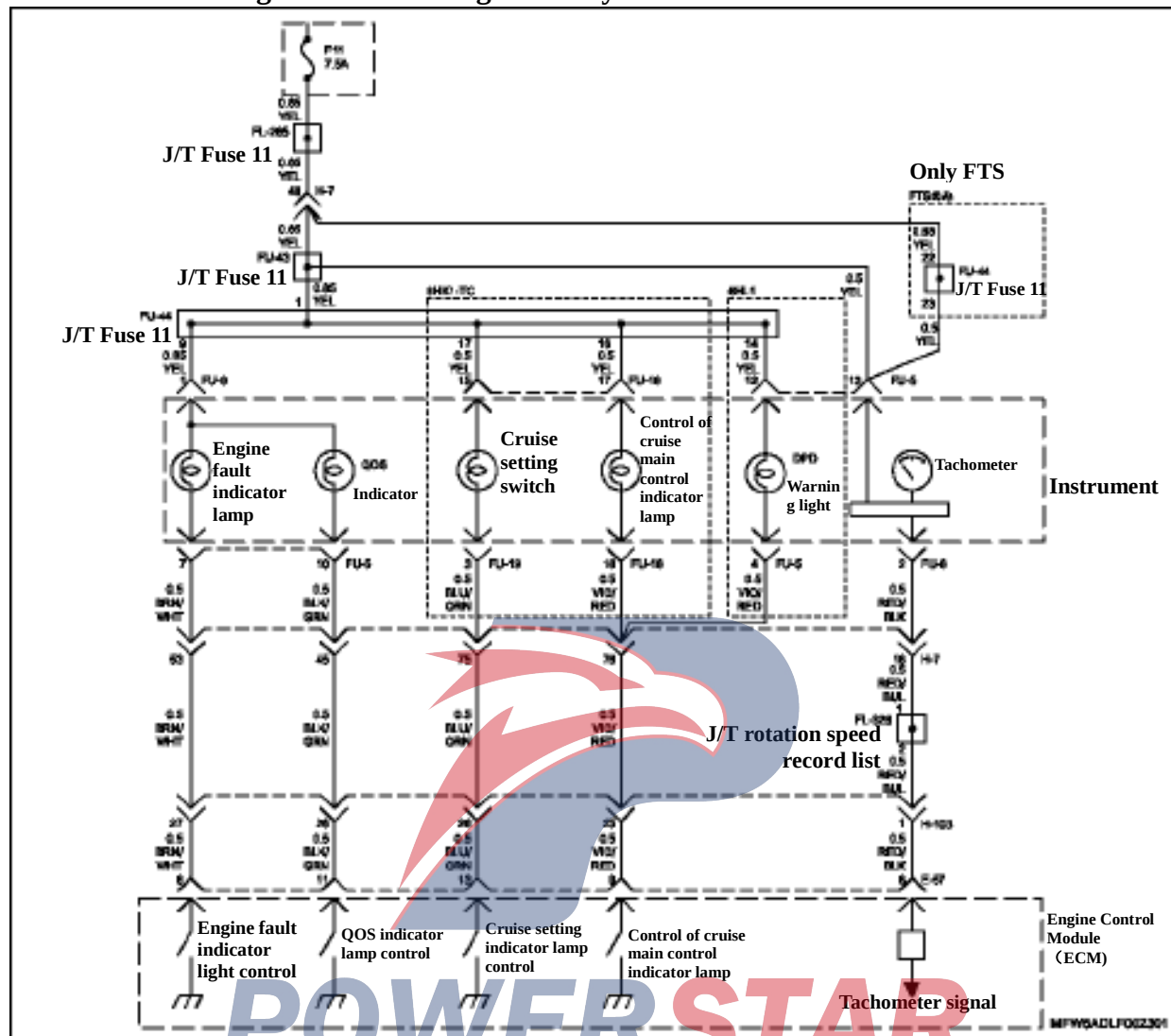
The numbers shown below indicate the step numbers in the table

2. When the output of the engine fault indicator is not consistent with the action of the engine fault indicator, the fault code of DTC P0650 will be detected.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Check whether there is DTC detected with the scanning tool. Has DTC P0650 been detected?	-	Enter into DTC P0650	Go to step 3
3	Confirm the lighting status of the engine fault indicator again. 1. Set the key switch to the [OFF] state 2. Check the engine failure indicator lamp. Is engine MIL on?	-	Enter diagnostic help	Go to step 4
4	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Confirm the repair conditions	-

POWERSTAR

Confirmation of engine MIL blinking circuit system



Cycle Description

The engine MIL is powered through the key switch and illuminates according to the signal transmitted from ECM. ECM will inspect the lamp when the key switch is [ON], and the engine fault indicator light will be turned on when the DTC is detected. In addition, after the diagnostic joint connected, engine MIL will blink according to the DTC. ECM is equipped with a monitoring circuit to monitor the work of the lamp.

Diagnostic help

When there are doubts about intermittent problems, they can be taken into account from the following reasons:

- Harness connector bad connection;
- Harness is not good;
- Harness cover damage;
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad;
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

2. When the output of the engine fault indicator is not consistent with the action of the engine fault indicator, the fault code of DTC P0650 will be detected.

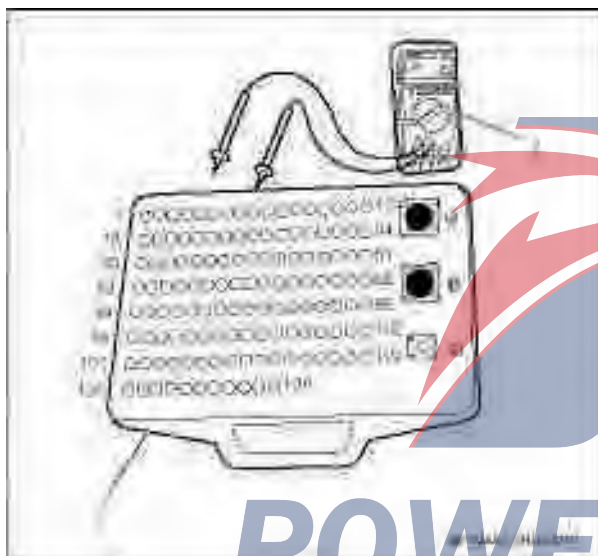
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System " Check
2	1. Set the key switch to the [OFF] state 2. Check whether there is DTC detected with the scanning tool. Has DTC N P0650 been detected?	-	Enter into DTC P0650	Go to step 3
3	1. Set the key switch to the OFF position. 2. Connect the DMM between the diagnostic connector (outer connector side) and GND. 3. Set the key switch to the [OFF] state Is the display value of the DMM above the reference value?	18V	Go to step 6	Go to step 4
4	Confirm the installation status of ECM terminal and middle joint. 1. Check the installation state of the connector for looseness and slack. 2. If fault detected, repair as needed. Whether the fault is detected or not?	-	Confirm the repair conditions	Go to step 5
5	1. Use the power distribution cabinet to check the circuit between ECM and diagnostic connector for the following states: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Confirm the repair conditions	Go to step 7
6	1. Check the circuit between the diagnostic connector and GND for the following states: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Confirm the repair conditions	Go to step 7
7	Confirm the lighting status of the engine fault indicator again. 1. Set the key switch to the [OFF] state 2. Connect the diagnostic connector. 3. Check the engine failure indicator lamp. Is engine MIL on?	-	Enter diagnostic help	Go to step 8

Step	Operation content	Datum value	YES	NO
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times; Refer to the description for programming in this chapter.	-	Confirm the repair conditions	-
	Is the disposal completed?			

About distribution cabinet

2. DMM

In a program that requires the use of a power distribution cabinet, check the following points.

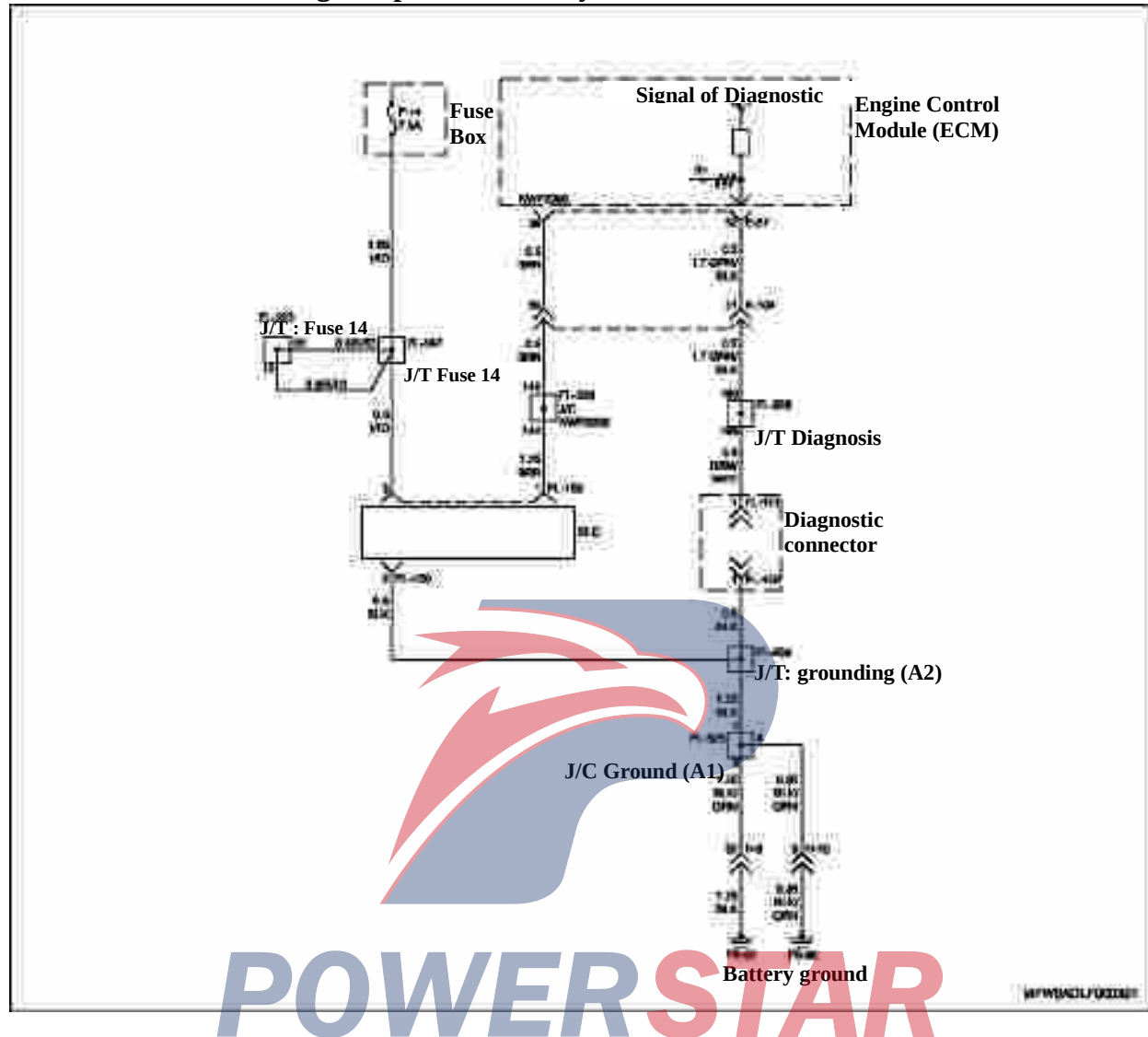


Name

1. Distribution cabinet

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Broken wire / high resistance	Measurement of resistance value	Key switch [OFF]	52 - Diagnostic Connector Terminal	Below 10MΩ (there is a guide)	10MΩ or more (no conduction)
	Short to GND	Measurement of resistance value	Key switch [OFF]	52 - GND	Above 10MΩ (No conduction)	Below 10MΩ (there is a guide)

Confirmation of scanning tool power circuit system



Cycle Description

ECM and other ECUs communicate with scanning tool through DLC. With the key switch on, the DLC will be powered on and supply the power voltage to the scanning tool.

Diagnostic help

When there are doubts about intermittent problems, they can be taken into account from the following reasons:

- Harness connector bad connection;
- Harness is not good;
- The harness coating is scratched and damaged;
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out:

- Connection between harness connector and ECM connector is bad.

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

● Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

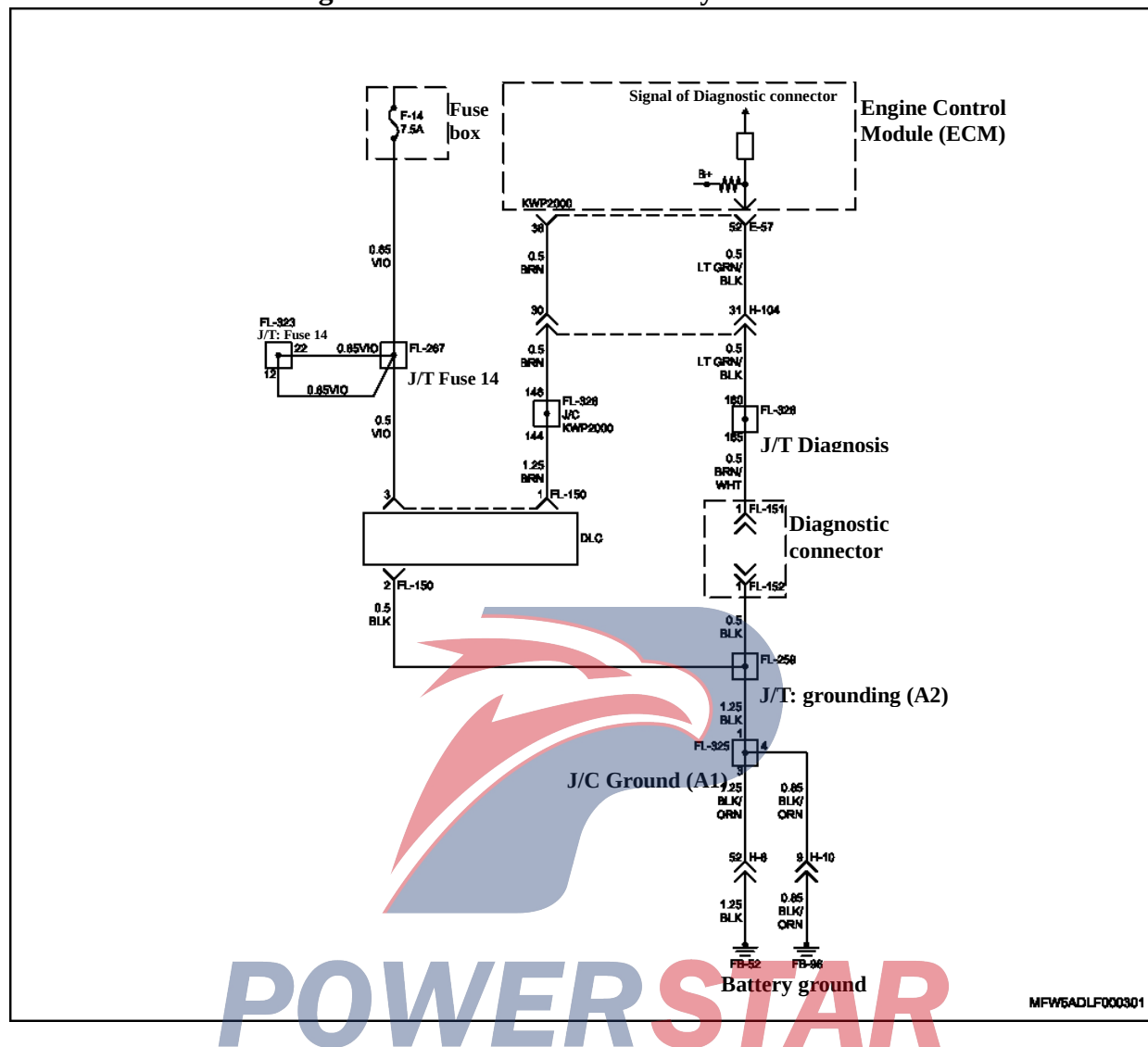
Test description

Important:

Before continuing with this diagnostic procedure, use another vehicle to confirm that the scanning tool is working properly.

Step	Operation content	Datum value	YES	NO
1	1. Set the key switch to the OFF position. 2. Check No.14 fuse in the glove box. Whether the fault is detected or not?	-	Go to step 2	Go to step 3
2	1. Repair the short circuit that caused the fuse to open and replace the No. 14 fuse. Is the disposal completed?	-	Go to step 7	-
3	1. Check if the DLC has the following status: • Terminal and connector separation; • The connection between the terminal and the line is poor. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 7	Go to step 4
4	1. Set the key switch to the OFF position. 2. Connect the DMM between the DLC power supply terminal and GND. 3. Set the key switch to the [OFF] state Is the display value of the DMM above the reference value?	18V	Go to step 6	Go to step 5
5	Repair the circuit between the fuse and the DLC. Is the disposal completed?	-	Go to step 7	Go to step 7
6	1. Check the loop between DLC and GND for the following condition: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 7	Enter diagnostic help
7	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Turn on the power of the scanning tool. Is the power passing through the scanning tool?	-	Confirm the repair conditions	Go to step 1

Confirmation of scanning tool communication circuit system



Cycle Description

The engine control module (ECM) performs Class 2 communication via the 3-pin data link connector (DLC) with the scan tool. In addition, the communication between the ECM and ECUs is made with CAN.

Diagnostic help

When there are doubts about intermittent problems, they can be taken into account from the following reasons:

- Harness connector bad connection;
- Harness is not good;
- Harness cover damage;
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out:

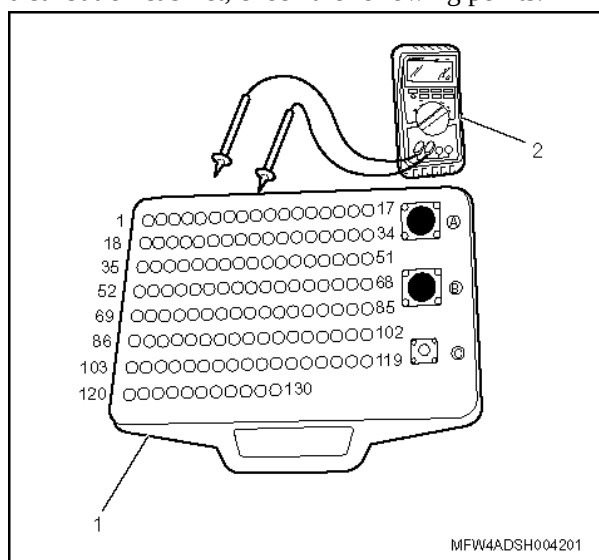
- Connection between harness connector and ECM connector is bad.

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.
- **Harness damage:**
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"
2	Confirm the installation status of ECM terminal and middle joint. 1. Check the installation state of the connector for looseness and slack. 2. If fault detected, repair as needed. Whether the fault is detected or not?	-	Go to step 7	Go to step 3
3	1. Check if the DLC has the following status: • Terminal and connector separation; • The connection between the terminal and the line is poor. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 7	Go to step 4
4	1. Use the power distribution cabinet to check the circuit between ECM and DLC for the following states: • Wire break • High resistance • Short connection with GND 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 7	Go to step 5
5	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Use the scan tool to communicate with the ECM. Is the communication with ECM normal?	-	Access to auxiliary diagnosis	Go to step 6
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value • Q adjustment value data • ID code of the ejector • Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 7	-
7	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Use the scan tool to communicate with the ECM. Is the communication with ECM normal?	-	Confirm the repair conditions	Go to step 2

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

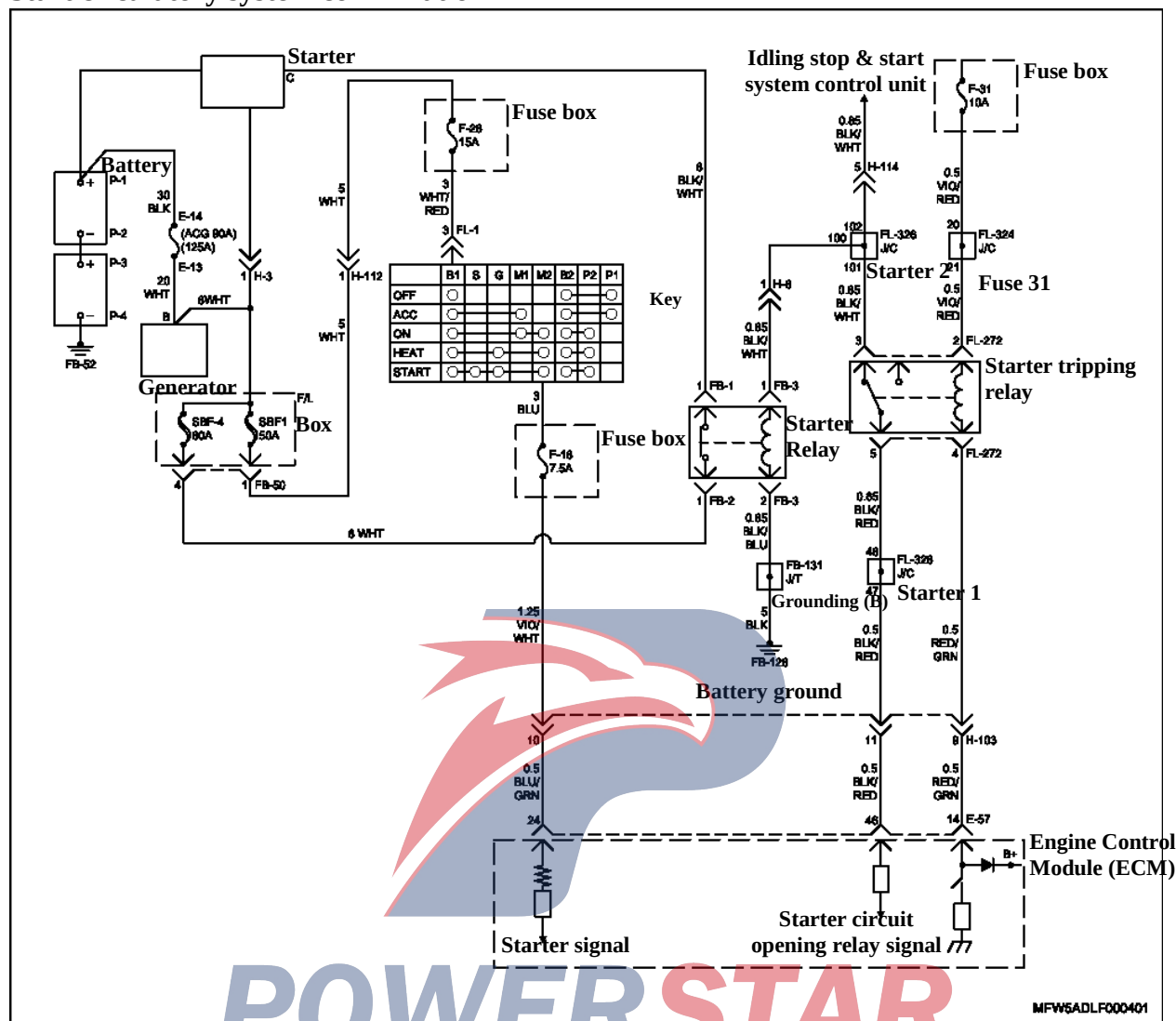


Name

1. Distribution cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Broken wire / high resistance	Measurement of resistance value	Key switch [OFF]	38 - DLC No.3 terminal	Below 10MΩ (there is a guide)	10MΩ or more (no conduction)
	Short to GND	Measurement of resistance value	Key switch [OFF]	38 - GND	Above 10MΩ (No conduction)	Below 10MΩ (there is a guide)

Start circulatory system confirmation



Cycle Description

When the key switch is located at the [START] position, the engine control module (ECM) will interrupt the output of the starter circuit opening relay. At this time the starter circuit opening relay and the starter relay will be in the [ON] state. After the starter relay becomes [ON], the starter will work and the engine will start.

Diagnostic help

When there are doubts about intermittent problems, they can be taken into account from the following reasons:

- Harness connector bad connection;
- Harness is not good;
- Harness cover damage;
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad;

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together
- Damage to the joint lock
- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test description

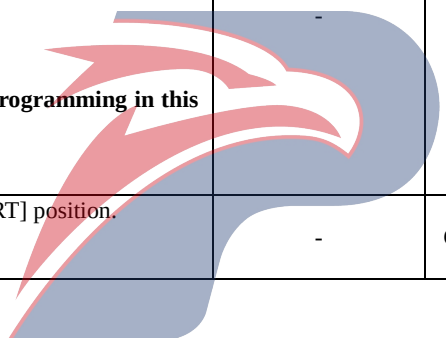
The following numbers indicate the step numbers in the diagnostics chart.

2. Sometimes the engine cannot start if these DTCs are detected.

3. If ECM does not identify the [START] position signal from the key switch, the starter tripping relay will become [OFF] and the starter relay will not become [ON].

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Check if DTC is detected using the scan tool? Is there any DTC P0340, P0341, P0615, P1345, P1625 detected?	-	Access to the detected DTC diagnostics	Go to step 3
3	1. Set the key switch to the [START] position 2. Use scanning tools to confirm the contents of the start switch (ST 1) in the data display. Is the display of the start switch (ST) [ON]?	-	Go to step 5	Go to step 4
4	1. Use the distribution cabinet to check whether there are the following states in the circuit between the key switch and the ECM: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 12	Go to step 11
5	1. Set the key switch to the OFF position. 2. Remove the starter disconnect relay. 3. Use the jumper wire to make short connections between the contact terminals and the coil side terminals of the starter circuit opening relay installation section. 4. Set the key switch to the [START] position. Does the starter work?	-	Go to step 6	Go to step 7
6	Replace the starter and disconnect the relay. Is the disposal completed?	-	Go to step 12	-
7	1. Set the key switch to the OFF position. 2. Remove the starter relay. 3. Use jumpers to short the contact side terminals of the starter relay mounting section. 4. Set the key switch to the [START] position. Does the starter work?	-	Go to step 8	Go to step 9
8	Replace the starter and disconnect the relay. Is the disposal completed?	-	Go to step 12	-

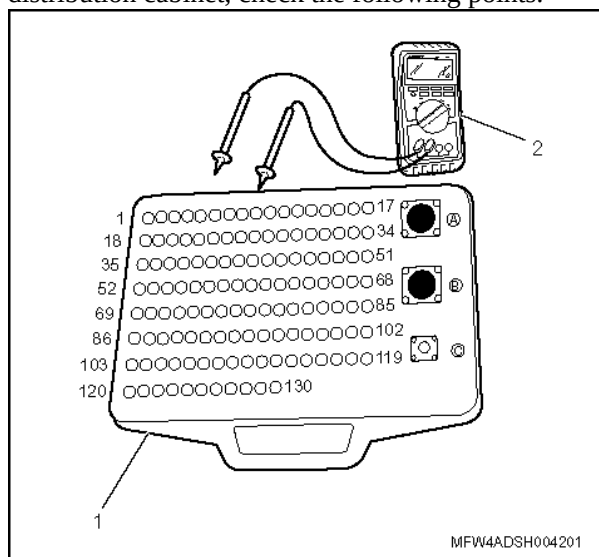
Step	Operation content	Datum value	YES	NO
9	1. Check the following circuit for wire break or high resistance: ● Key switch ~ Starter breaker relay; ● Starter breaker relay - Starter relay; ● starter relay grounded; ● Starter relay - Starter room. 2. If fault detected, repair as needed.	-		
	Whether the fault is detected or not?		Go to step 13	Go to step 10
10	Set the key switch to the [START] position. Does the starter work?	-	Enter diagnostic help	Go to step 11
11	1. Refer to the "Electrical System" in the "Engine" service manual for inspection. 2. When a fault is detected, make repairs as needed.	-		
	Whether the fault is detected or not?		Go to step 13	Go to step 12
12	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ● MAF correction value ● Q adjustment value data ● ID code of the ejector ● Engine starting times; Refer to the description for programming in this chapter.	-		
	Is the disposal completed?		Go to step 13	-
13	Set the key switch to the [START] position. Does the starter work?	-	Confirm the repair conditions	Go to step 2



POWERSTAR

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

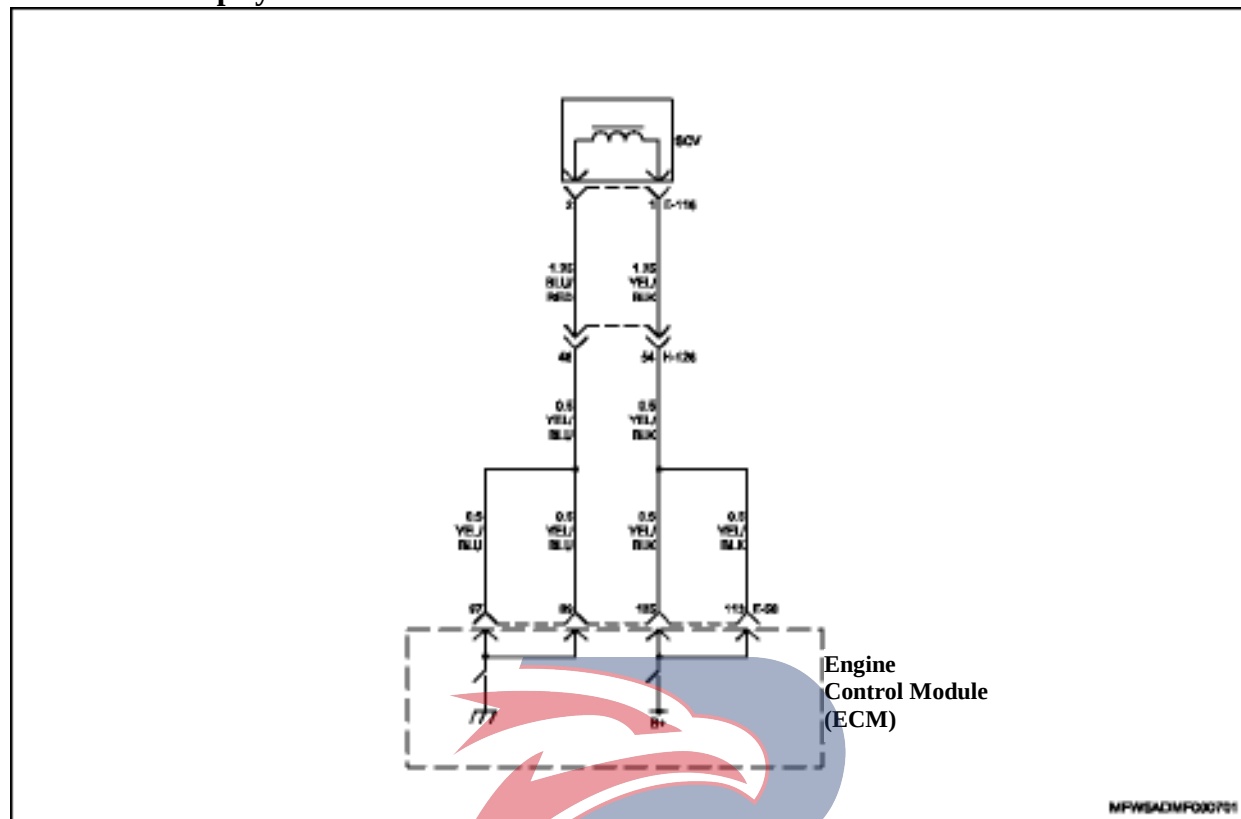
Distribution Cabinet

2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Broken wire / high resistance	Measurement of resistance value	Key switch [OFF]	46 - Key switch [START] terminal	Below 10MΩ (there is a guide)	10MΩ or more (no conduction)
	Short to GND	Measurement of resistance value	Key switch [OFF]	46 - GND	Above 10MΩ (No conduction)	Below 10MΩ (there is a guide)

POWERSTAR

Confirm startup system



About starting system confirmation

Confirming the boot system is one way to find a system that the engine cannot start. The following items are prerequisites for the diagnosis.

- The battery is fully charged and the battery cable is properly connected;
- The crankshaft rotation is normal;
- Adequate fuel;
- There is no air in the fuel;
- There is no abnormality in the air cleaner core and the fuel filter.

Important:

When the DTC related to fuel system is detected, diagnose DTC first.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Harness connector bad connection;

- Harness is not good;
 - Harness cover damage;
 - Wire breakage inside the harness cladding
- In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad;

- The terminals are removed from the wire harness
- The non-matched terminals are joined together
- Damage to the joint lock
- The connection between the terminal and the wire is poor.

Test instruction

The numbers shown below indicate the step numbers in the table

3. When the SCV-LOW circuit is shorted to GND, the SCV drive current cannot reach more than 900mA.

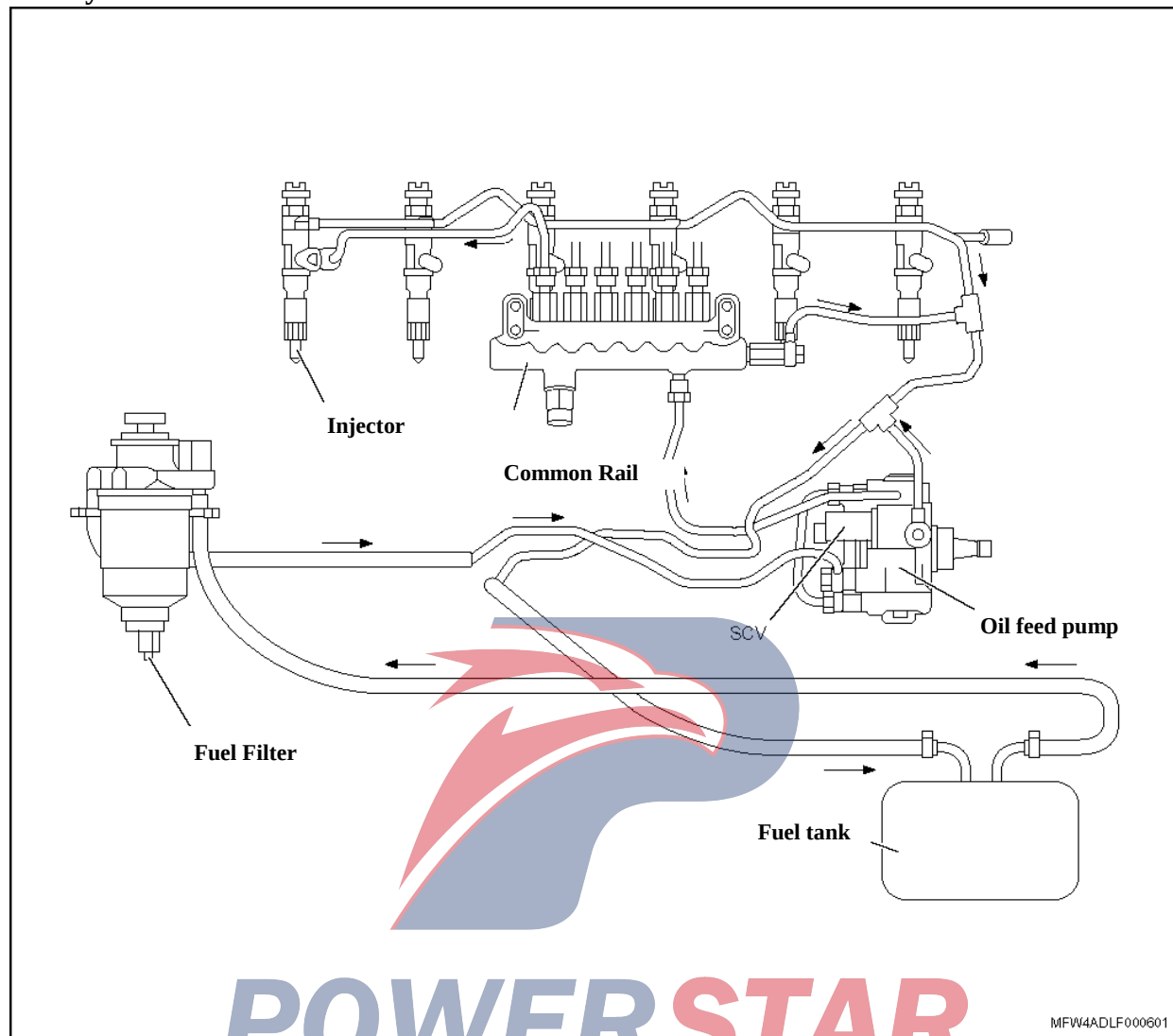
4. When the engine is able to start when the SCV joint is removed, the SCV system can be considered abnormal. In addition, when the engine cannot start, it can be considered an abnormal fuel system.

6. Confirm the signals and working conditions of sensors and ejectors.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system.	-	Go to step 2	Entry (OBD) System Check
	Is the disposal completed?			

Step	Operation content	Datum value	YES	NO
2	1. Set the key switch to the "ON" position. 2. Rotate the engine crankshaft for 15s. 3. Use a scanning tool to confirm DTC. Is DTC detected?	-	Access to the detected DTC diagnostics	Go to step 3
3	Confirm the contents of [SCV F/B] in data display with the scanning tool while rotating the engine crankshaft. Is the display value of [SCV F / B] higher than the reference value?	900mA	Go to step 4	Go to step 5
4	1. Remove the SCV connector. 2. Turn the crankshaft of the engine. Does the engine start?	-	Enter into DTC P0090	Go to step 6
5	Repair the SCV-LOW circuit between ECM and SCV. Is the disposal completed?	-	Go to step 8	-
6	1. Check whether the displayed values related to the following items are normal with the scanning tool. (Refer to [Tech2 Data Reference] for reference): ● Engine coolant temperature (ECT) sensor; ● Common rail pressure sensor ● MAF sensor (6HL1 only) ● EGR position sensor ● Injector 2. When a fault is detected, replace the circuit or sensor as needed. Whether the fault is detected or not?	-	Go to step 8	Go to step 7
7	1. Check whether there is an abnormality described below: ● Time deviation of engine mechanical components ● Flywheel installation position is not good ● The air intake system is seriously blocked ● Strictly punish the exhaust system 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 8	Enter "fuel system confirmation"
8	1. Clear the DTC. 2. Set the key switch to [OFF] state for more than 10 seconds. 3. Start the engine. Whether the fault is detected or not?	-	Go to step 9	Go to step 2
9	1. Check whether there is DTC detected with the scanning tool. Is there any DTC detected?	-	Access to the detected DTC diagnostics	Confirm the repair conditions

Fuel system confirmation



About fuel system

The fuel system consists of a fuel tank, a fuel filter, an oil supply pump, a common rail, and an injector. Each section is connected by a fuel pipe.

Diagnostic help

As a major cause of fuel system anomalies, consider the following:

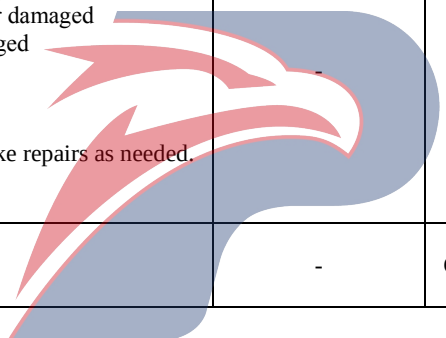
- Air entraining in the fuel system
- Block the fuel filter;
- Fuel pipe damage, clogging, bad connection;
- Abnormal fuel tank;
- Fuel feed pump abnormal
- Abnormal injector
- The pressure limiter is abnormal.

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Harness connector bad connection;
 - Harness is not good;
 - Harness cover damage;
 - Wire breakage inside the harness cladding
- In order to find out these reasons, the following checks should be carried out.
- Connection between harness connector and ECM connector is bad;
 - The terminals are removed from the wire harness
 - The non-matched terminals are joined together
 - Damage to the joint lock
 - The connection between the terminal and the wire is poor.

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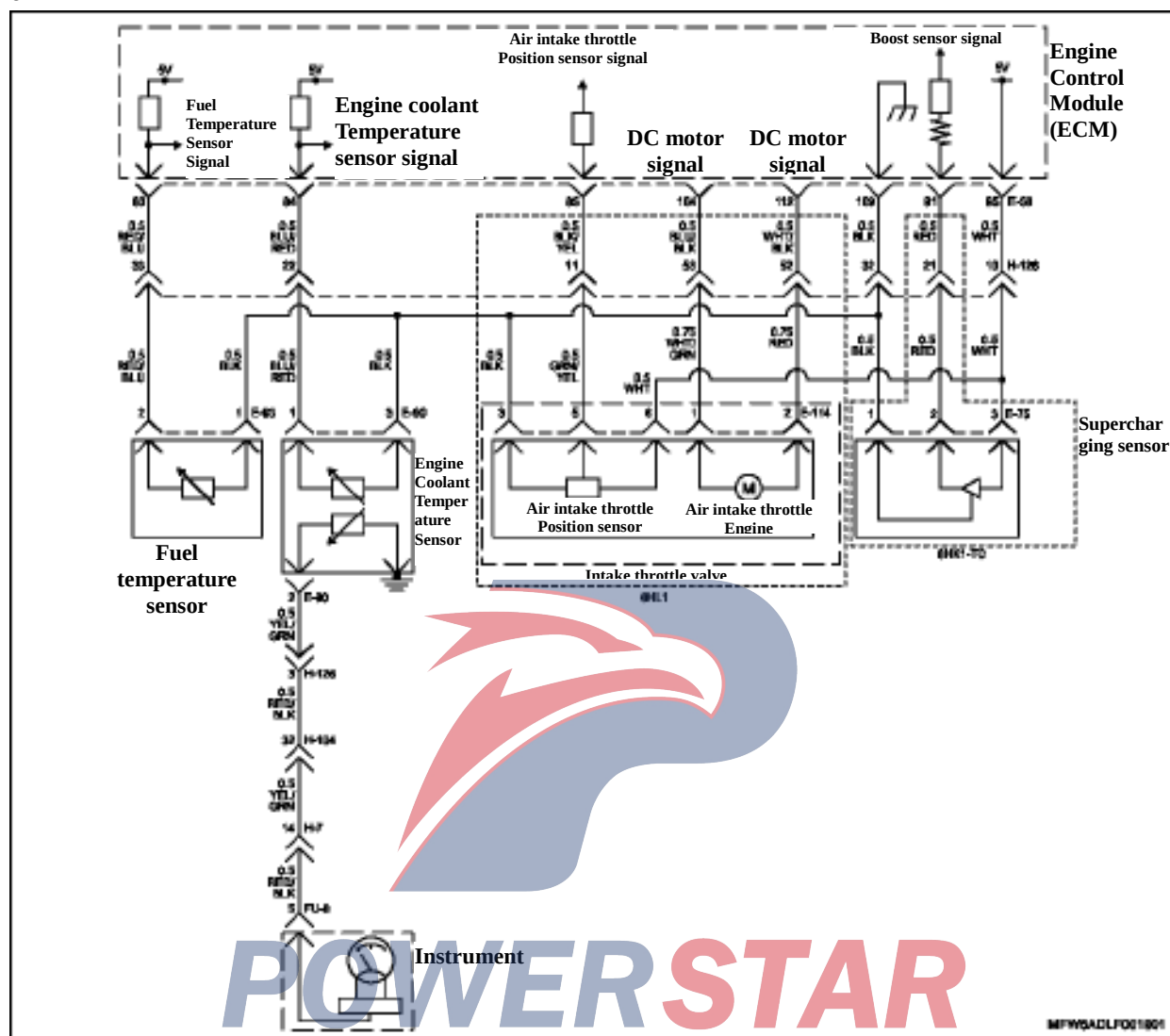
Step	Operation content	Datum value	YES	NO
1	Confirm the fuel quality. 1. Remove the fuel from the fuel tank. 2. Add appropriate fuel. 3. Deflate the gas in the fuel. 4. Start the engine. Does the engine start?	-	Go to step 6	Go to step 2
2	Check the fuel quantity. Is the fuel full?	-	Go to step 3	For fuel filling, go to Step 3
3	1. Check the fuel filter for serious dirt or clogging. 2. It should be cleaned or replaced immediately when detecting bad situations. Is the disposal completed?	-	Go to step 4	-
4	1. Check the fuel pipe for damage, clogging and poor connection. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	-	Go to step 5	-
5	1. Check the fuel tank for the following abnormal conditions: • Waste mixed with foreign objects • The supply pipe is crushed or damaged • Oil tank is damaged or damaged • fuel pipe assembly error • Fill plug • Water mixed in 2. When a fault is detected, make repairs as needed. Is the disposal completed?	-	Go to step 6	-
6	1. Drain the air in the fuel. Is the disposal completed?	-	Confirm the repair conditions	-



POWERSTAR

Confirmation of intake system

6HL1



About the intake system

The intake system consists of air filters, intake pipes, intake throttle valves and so on. The intake air is fed to the engine through the air filter and air intake duct. The amount of air inhaled is controlled by the intake throttle valve. The intake throttle valve is controlled by the DC motor for its open-close action. The air intake throttle valve is normally open and the valve is closed according to the ratio of the engine control module (ECM) output to the DC motor load.

Step	Operation content	Datum value	YES	NO
1	Check the air filter for severe dirt and clogging. ³ Whether the fault is detected or not?	-	Go to step 2	Go to step 3

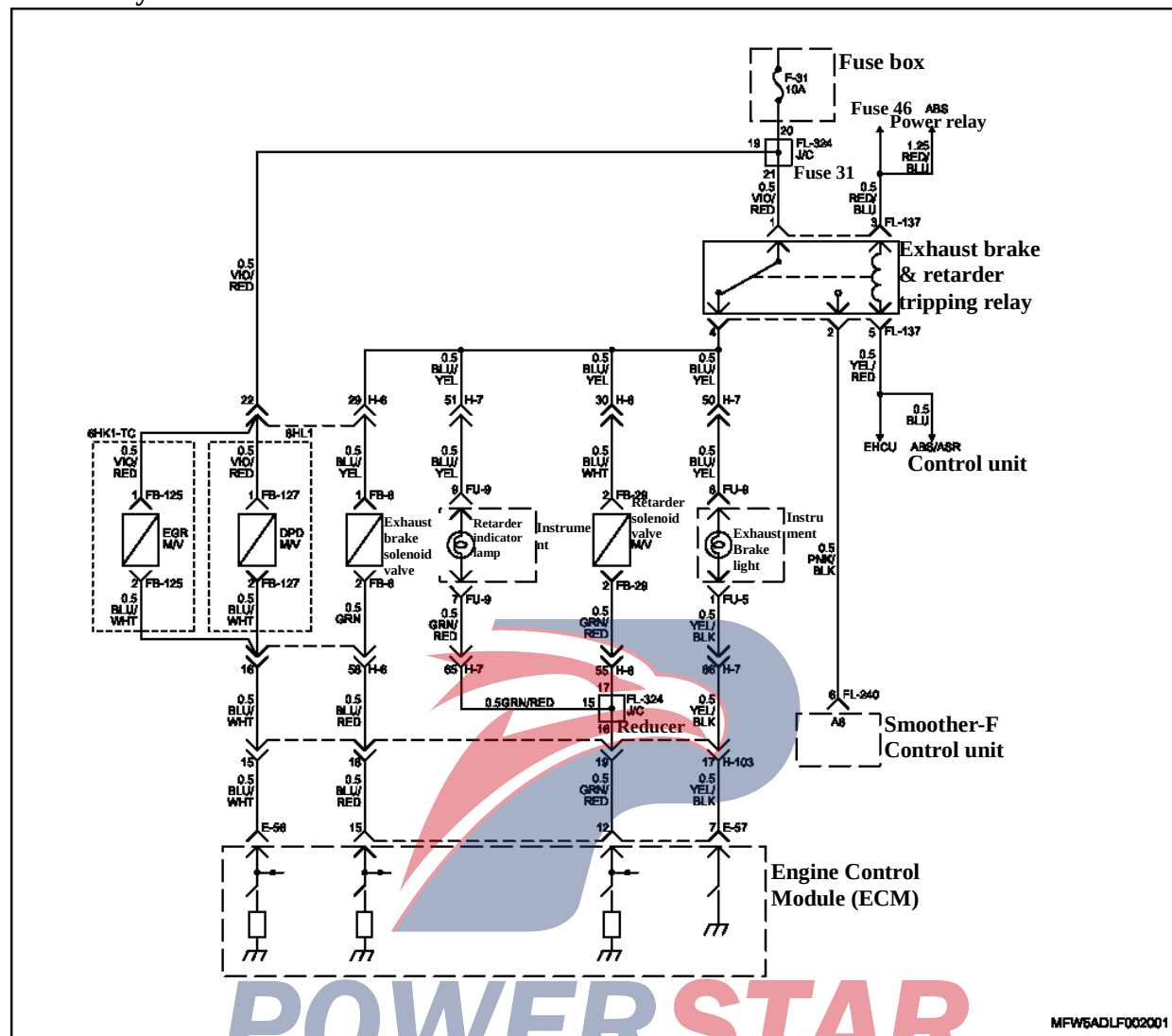
Step	Operation content	Datum value	YES	NO
2	Clean and replace the air filter. Is the disposal completed?	-	Go to step 3	-
3	Check the intake pipe for broken, damaged and gas leaks. Is the disposal completed?	-	Go to step 4	Go to step 5
4	Repair or replace intake piping. Is the disposal completed?	-	Go to step 5	-
5	Confirm DTC. 1. Set the key switch in the [OFF] position and start the engine after more than 10 seconds. 2. Check whether there is DTC detected with the scanning tool. Has DTC P1484, P1485, P1487 been detected?	-	Access to the detected DTC diagnostics	Go to step 6
6	1. Start the engine. 2. Confirm the displayed value of [air intake throttle duty ratio] in the data display with the scanning tool. Is the value of [Intake throttle valve duty] within the reference range?	$\pm 3\%$	Normal system	Go to step 7
7	Check the working condition of the intake throttle valve. 1. Start the engine. 2. Test the [intake throttle control] with the scanning tool. 3. Press the [UP] [DOWN] soft key. [Is the value of [intake throttle opening] within the reference value range?	$\pm 3\%$	Go to step 9	Go to step 8
8	Replace the intake throttle valve. Is the disposal completed?	-	Confirm the repair conditions	-
9	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times Refer to the description for programming in this chapter. Is the disposal completed?	-	Confirm the repair conditions	-

6HK1-TC**About the intake system**

The air intake system consists of the air filter, intake pipe and turbocharger etc. Inhaled air is supplied to the engine through an air filter and an air intake duct. In addition, the turbocharger will increase the intake air volume and enhance the combustion efficiency. Turbocharger uses the energy of the discharged gas to turn the turbine impeller and supercharge by rotating the compressor on the coaxial line. When the revolution of engine increases and that of the turbo reaches a certain number, the exhaust valve that is mounted on the turbocharger will open, and the exhaust gas supplied to the turbine is discharged to the exhaust pipe.

Step	Operation content	Datum value	YES	NO
1	Check the air filter for severe dirt and clogging. Whether the fault is detected or not?	-	Go to step 2	Go to step 3
2	Clean and replace the air filter. Is the disposal completed?	-	Go to step 3	-
3	Check the intake pipe for broken, damaged and gas leaks. Is the disposal completed?	-	Go to step 4	Go to step 5
4	Repair or replace the intake pipe. Is the disposal completed?	-	Go to step 5	-
5	Check the working status of the exhaust safety valve. 1. Remove the vacuum tube from the exhaust regulator. 2. Install the pressure gauge on the exhaust pressure relief valve. 3. Use a pressure gauge to apply negative pressure to the exhaust pressure reducing valve. 4. Confirm the negative pressure value when the control valve moves by 2mm with the pressure gauge. Is the value of the pressure gauge within the reference range?	184.7~ 200kPa (1385~ 1500mmHg)	Go to step 7	Go to step 6
6	Replace the boost converter ASM. Is the disposal completed?	-	Go to step 7	-
7	Repair the vehicle. Is the disposal completed?	-	Confirm the repair conditions	-

Exhaust system confirmation



About the exhaust system

Exhaust system consists of exhaust pipe, tail pipe, exhaust brake valve, air pipe and other components. The exhaust brake valve is [ON/OFF] air supplied by the solenoid valve. The exhaust valve is usually in the "on" state, but when the ECM enters the solenoid valve, the solenoid valve will be in the [ON] state, and the air will act on the exhaust valve and the exhaust valve will be closed. The engine control module (ECM) opens the exhaust brake solenoid only when the exhaust brake operating conditions are established.

Exhaust brake operating conditions

When the engine revolution exceeds 525rpm, and all the following conditions are established, the ECM will be at the [ON] state.

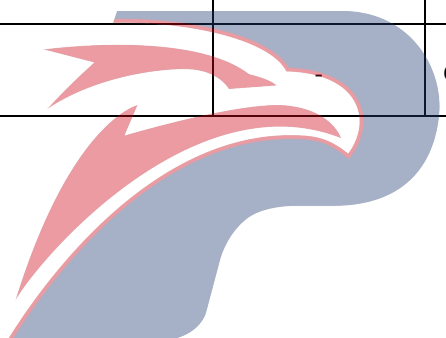
- Exhaust brake switches [ON].
- Accelerator pedal in idling position.
- Abnormal accelerator pedal position (APP) sensor, abnormal exhaust brake circuit, abnormal clutch switch, abnormal accelerator pedal position (APP) switch, abnormal A/D conversion.
- Clutch pedal not depressed
- System voltage is above 24V.

Step	Operation content	Datum value	YES	NO
1	Check the exhaust pipe and exhaust gas pipe for damage, breakage and exhaust leak.	-		
	Whether the fault is detected or not?		Go to step 2	Go to step 3

Step	Operation content	Datum value	YES	NO
2	Repair or replace exhaust pipe or tail pipe. Is the disposal completed?	-	Go to step 3	-
3	Check for broken, damaged or leaky signs. Whether the fault is detected or not?	-	Go to step 4	Go to step 5
4	Repair or replace the trachea. Is the disposal completed?	-	Go to step 5	-
5	Check the working conditions exhaust brake magnet valve. 1. Use the scan tool to test [exhaust brake control]. 2. Press the [ON] soft key. Is [Check Status] the [ON] status?	-	Go to step 6	Go to step 7
6	1. Press the [OFF] soft key. Is [Check Status] the [Off] status?	-	Go to step 18	Go to step 13
7	Use the scan tool to confirm whether DTC has been detected on the ABS side. Whether it is detected that the exhaust brake (and retarder) circuit breaker is abnormal?	-	Please refer to ABS Control of Brake Control in maintenance manual.	Go to step 8
8	Confirm the installation status of the joint, ECM joint and intermediate joint of the exhaust brake solenoid valve. 1. Check for loose and loose fittings. 2. If fault detected, repair as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 9
9	1. Set the key switch to the OFF position. 2. Remove the joint of the solenoid valve of the exhaust brake. 3. Measure the resistance value of the exhaust brake solenoid valve. Is the resistance value within the reference range?	20-60Ω	Go to step 10	Go to step 16
10	1. Check whether there are the following states in the circuit between the main relay and the exhaust brake and the retarder breaker relay. • Wire break • Short circuit to GND circuit • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 11

Step	Operation content	Datum value	YES	NO
11	1. Check the circuits between the exhaust brake & retarder tripping relay and exhaust brake magnet valve for the following status. • Wire break • Short circuit to GND circuit • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 12
12	1. Use the power distribution cabinet to check whether there are the following states in the circuit between the exhaust brake and the ECM. • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 17
13	Confirm the installation status of the joint, ECM joint and intermediate joint of the exhaust brake solenoid valve. 3. Check for loose and loose fittings. 2. If fault detected, repair as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 14
14	1. Set the key switch to the OFF position. 2. Remove the joint of the solenoid valve of the exhaust brake. 3. Measure the resistance value of the exhaust brake solenoid valve. Is the resistance value within the reference range?	20-60Ω	Go to step 15	Go to step 16
15	1. Check the circuit between the exhaust relay solenoid valve and the ECM for the following conditions. • Short circuit to GND circuit • Short circuit between power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 18	Go to step 17
16	Replace the solenoid valve of exhaust brake. Is the disposal completed?	-	Go to step 18	-

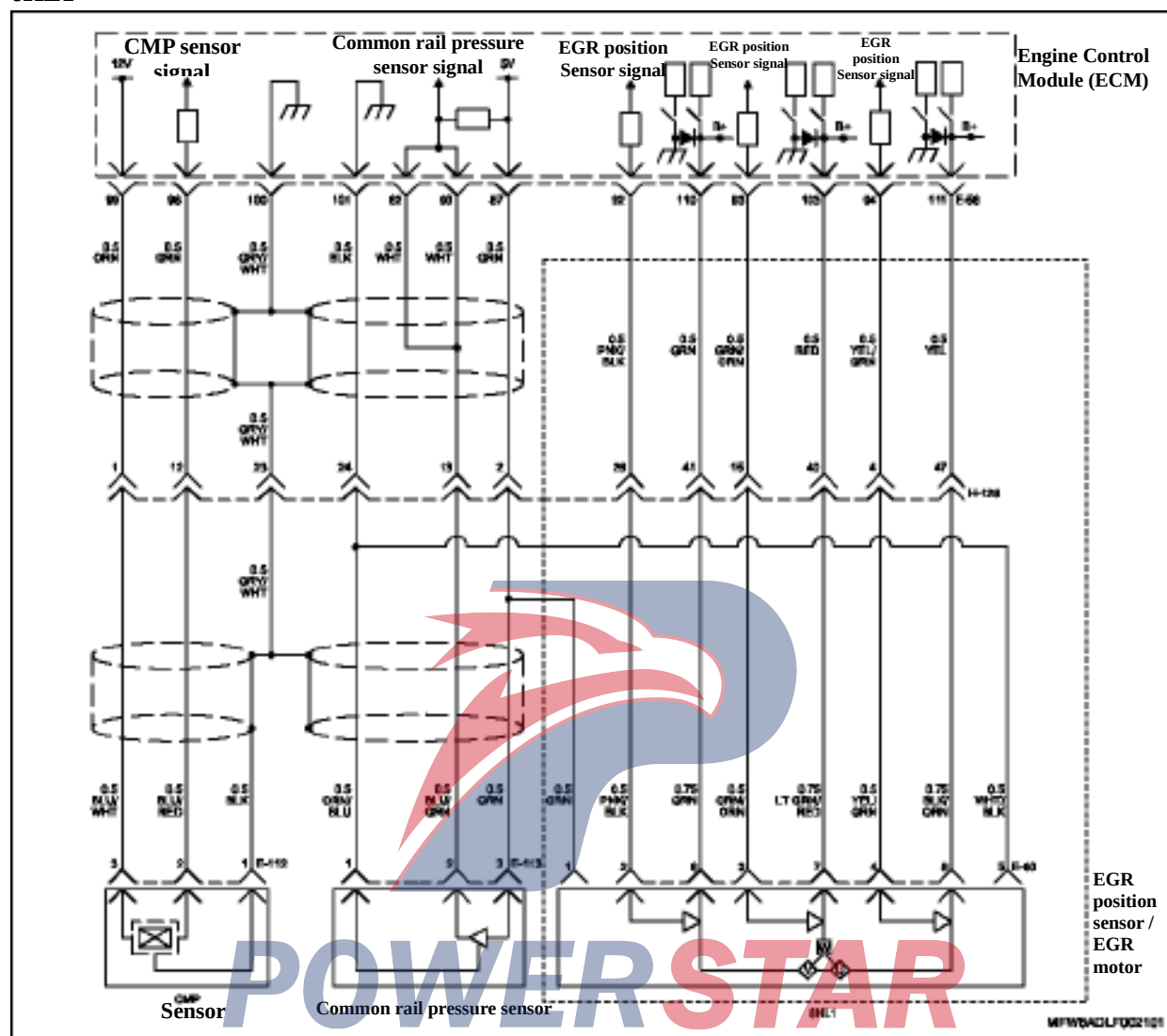
Step	Operation content	Datum value	YES	NO
17	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 18	-
18	Check the operation of the exhaust brake valve. 1. Remove the exhaust brake valve. 2. Supply the pneumatic cylinder with standard compression pressure and confirm that the valve is completely closed. Whether the exhaust brake valve fully closes;	588~ 686kPa (4410~ 5145mmHg)	Go to step 20	Go to step 19
19	Replace the exhaust brake valve. Is the disposal completed?	-	Go to step 20	-
20	Repair the vehicle. Is the disposal completed?	-	Confirm the repair conditions	-



POWERSTAR

Confirmation of EGR control system

6HL1



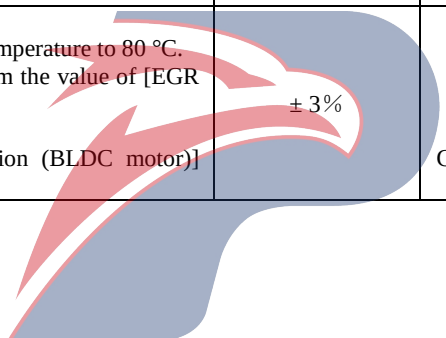
EGR

Cycle Description

ECM controls EGR valve according to the engine revolution, engine coolant temperature, intake air temperature, fuel injection volume and barometric pressure. The EGR valve is driven by EGR motor, and the opening of the EGR valve can be detected using EGR position sensor.

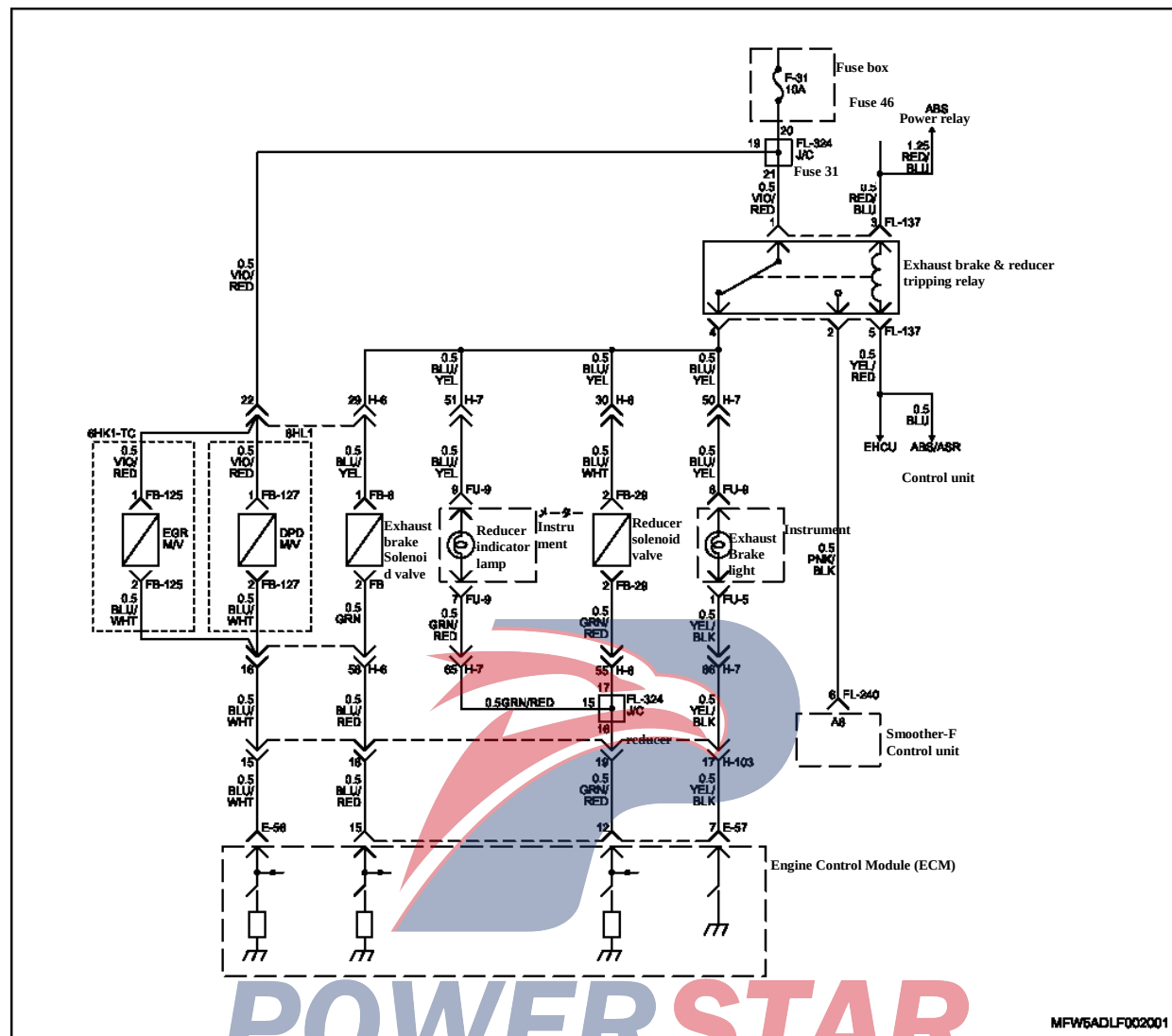
Step	Operation content	Datum value	YES	NO
1	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Use the scan tool to confirm whether DTC will be detected. Is there any DTC detected?	-	Access to the detected DTC diagnostics	Go to step 2

Step	Operation content	Datum value	YES	NO
2	1. Start the engine. 2. Raise the engine coolant temperature to 80 °C. 3. Use the scan tool to confirm the value of [EGR position (BLDC motor)]. Is the value of [EGR position (BLDC motor)] within the reference range?	± 3%	Normal system	Go to step 3
3	1. Use the scan tool to perform the [EGR control test]. 2. Press the [UP] [DOWN] soft key. 3. Check the value of [EGR position (BLDC motor)]. Is the value of [EGR position (BLDC motor)] within the reference range?	± 3%	Go to step 5	Go to step 4
4	Replace the EGR valve. Is the disposal completed?	-	Go to step 5	-
5	1. Check the EGR piping for damage or air leaks. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	-	Go to step 6	-
6	1. Start the engine. 2. Raise the engine coolant temperature to 80 °C. 3. Use the scan tool to confirm the value of [EGR position (BLDC motor)]. Is the value of [EGR position (BLDC motor)] within the reference range?	± 3%	Confirm the repair conditions	Go to step 2



POWERSTAR

6HK1-TC



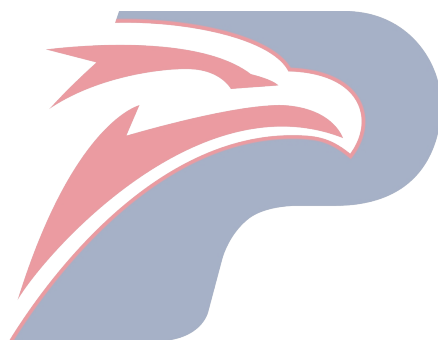
Cycle Description

ECM controls EGR valve according to the engine revolution, engine coolant temperature, intake air temperature, fuel injection volume and barometric pressure. EGR valve is driven by the magnet valve (M/V). ECM can detect the EGR valve operating state according to the signal transmitted from the EGR gas temperature sensor.

Step	Operation content	Datum value	YES	NO
1	1. Connect scanning tool 2. Set the key switch to the [OFF] state 3. Use the scan tool to confirm whether DTC will be detected. Is there any DTC detected?	-	Access to the detected DTC diagnostics	Go to step 2

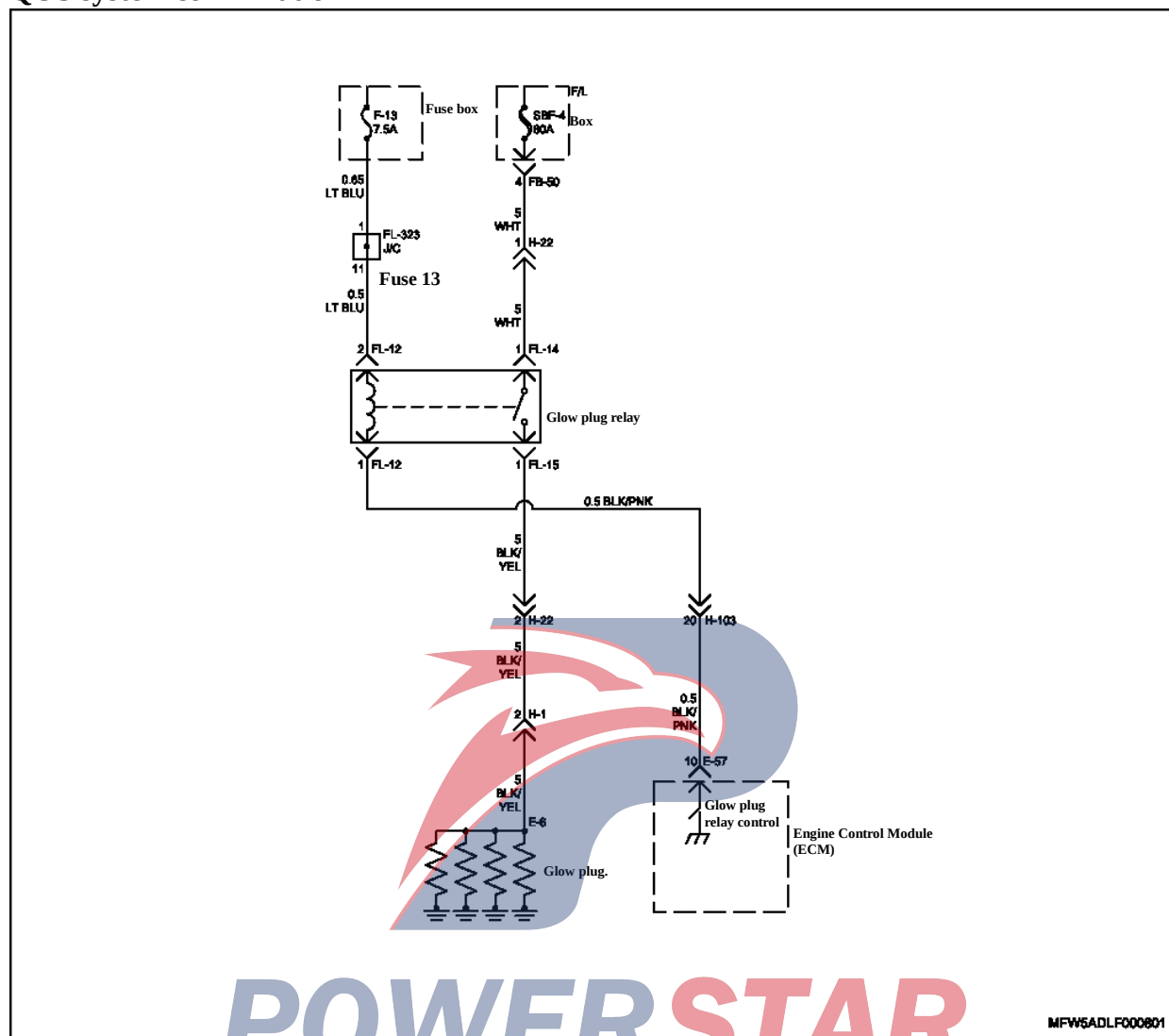
Step	Operation content	Datum value	YES	NO
2	1. Start the engine. 2. Raise the engine coolant temperature to 80 °C. 3. Use the scanning tool to confirm the [EGR valve control]] display. [EGR valve control] display [ON]?	-	Go to step 3	Go to step 4
3	1. Use the scan tool to confirm the display of the [EGR temperature sensor]. Is the value of [EGR temperature sensor] within the reference range?	4.7V	Normal system	Go to step 7
4	1. Use the scan tool to perform the [EGR control test]. 2. Press [ON] and [OFF] soft keys. 3. Confirm the display of [EGR valve control]]. With operation, does the display of [EGR valve control] change accordingly?	-	Go to step 5	Go to step 6
5	Replace the EGR solenoid valve. Is the disposal completed?	-	Go to step 12	-
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 12	-
7	Check the EGR valve. 1. Remove the EGR valve. 2. Confirm that the EGR valve is fully closed. Is the EGR valve closed?	-	Go to step 8	Go to step 9
8	Check the operation of the EGR valve. 1. Remove the EGR valve. 2. Deliver the standard compression pressure to the gas port and confirm that the valve should be open. Is the EGR valve open?	637.4~ 882.6kPa (4781~ 6620mmHg)	Go to step 10	Go to step 9
9	Replace the EGR valve. Is the disposal completed?	-	Go to step 12	-
10	1. Check the EGR piping for damage or air leaks. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 12	Go to step 11

Step	Operation content	Datum value	YES	NO
11	Replace the EGR gas temperature sensor. Is the disposal completed?	-	Go to step 12	-
12	1. Start the engine. 2. Raise the engine coolant temperature to 80 °C. 3. Use the scanning tool to confirm the [EGR valve control]] display. [EGR valve control] display [ON]?	-	Go to step 13	Go to step 4
13	1. Use the scan tool to confirm the display of the [EGR temperature sensor]. Is the value of [EGR temperature sensor] within the reference range?	4.7V	Confirm the repair conditions	Go to step 7



POWERSTAR

QOS system confirmation



Cycle Description

The QOS (Quick Start) system consists of an engine control module (ECM), glow plug relay, glow plug and QOS indicator. After the key switch is on when the temperature of coolant is low, the ECM will determine the preheating time according to the coolant temperature, and will make the preheating plug relay and the QOS indicator work. After a period of time, the ECM will cause the glow plug relay and the QOS indicator to be in the [OFF] state.

Test instruction

The numbers shown below indicate the step numbers in the table

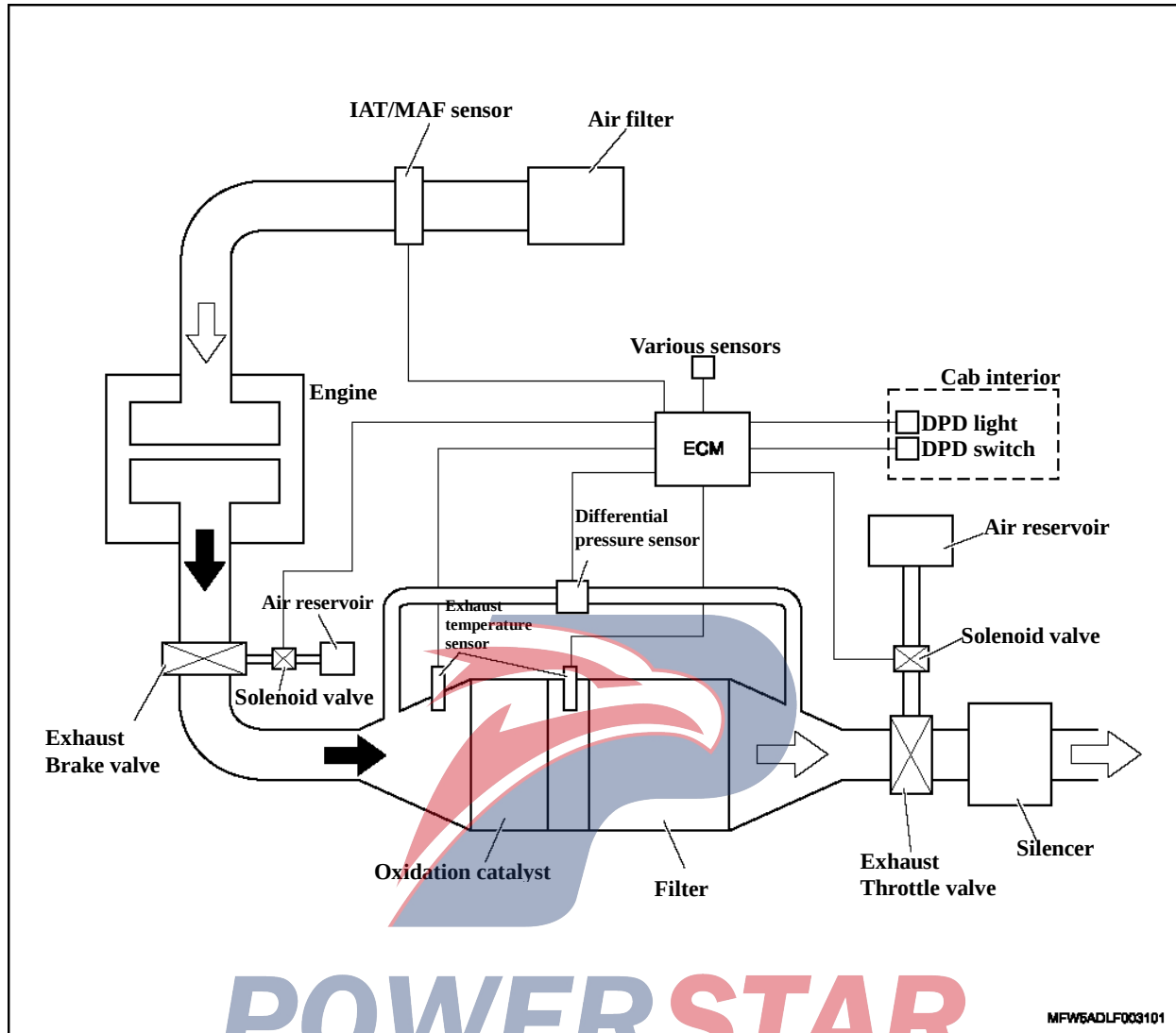
3. QOS indicator lamp also can illuminate with the key switch [ON] due to the lamp check function even if QOS not in operation.
4. The QOS indicator can be turned off within 5 seconds.
6. During the glow plug test, if the glow plug relay display does not change, it means ECM is abnormal.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Use scan tool to confirm if DTC is detected? Have you detected DTC related to engine coolant temperature sensor, QOS indicator and preheating plug relay?	-	Access to the detected DTC diagnostics	Go to step 3
3	1. Set the key switch to the OFF position. 2. Set the key switch to the [OFF] state 3. Confirm QOS indicator lamp. Is the QOS indicator lamp light on?	-	Go to step 4	Go to step 5
4	1. Set the key switch to the OFF position. 2. Set the key switch to the [OFF] state 3. Confirm QOS indicator lamp. Is the QOS indicator lamp off within 5 seconds?	-	Go to step 6	Go to step 5
5	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 6	-
6	1. Use scanning tool to test the glow plug relay. 2. Press [ON] and [OFF] soft keys. Is the glow plug relay displayed as [ON] and [OFF]?	-	Go to step 8	Go to step 7
7	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 6	-
8	1. Set the key switch to the OFF position. 2. Remove all glow plugs. 3. Measure the resistance between the terminals of each glow plug and the body. Is the measured resistance value a reference value?	约 4.3Ω	Go to step 9	Go to step 13

Step	Operation content	Datum value	YES	NO
9	1. Use the glow plug test of the scan tool to set the glow plug relay to the [ON] state. 2. Use DMM to confirm the voltage between each cylinder's hot wire harness and GND. Is the voltage of all cylinders higher than the reference?	18V	Go to step 10	Go to step 11
10	1. Use the glow plug test of the scan tool to set the glow plug relay to the [OFF] state. 2. Use DMM to confirm the voltage between each cylinder's hot wire harness and GND. Is the voltage of all cylinders higher than the reference?	0V	Normal system	Go to step 12
11	Repair short circuit between glow plug relay and glow plug and GND circuit. Is the disposal completed?	-	Go to step 14	-
12	Repair a short circuit between the glow plug relay and the glow plug power circuit or the ignition power circuit. Is the disposal completed?	-	Go to step 14	-
13	Replace the glow plug. Is the disposal completed?	-	Go to step 14	-
14	1. Use the glow plug test of the scan tool to set the glow plug relay to the [ON] state. 2. Use DMM to confirm the voltage between each cylinder's hot wire harness and GND. Is the voltage of all cylinders higher than the reference?	18V	Go to step 15	Go to step 9
15	1. Use the glow plug test of the scan tool to set the glow plug relay to the [OFF] state. 2. Use DMM to confirm the voltage between each cylinder's hot wire harness and GND. Is the voltage of all cylinders higher than the reference?	0V	Confirm the repair conditions	Go to step 10

DPD functional check



Cycle Description

The DPD functional test is a diagnostic method used to determine whether the DPD function is normal. Before performing this function check, carefully check the engine control module (ECM) and the engine grounding cable for dirt and air tightness. DPD functional check is to check the functions according to the problems due to the DPD insufficiency.

Diagnostic help

Bad harness connections and damage can sometimes cause intermittent malfunctions. Check for any of the following conditions. When a fault is detected, it should be repaired or replaced.

- Bad ECM wiring: Check the rear projection of the terminal block, poor contact, damage to the terminal, damage to the terminal or poor contact of the terminal, poor connection of the terminal and wiring.

- Harness damage: Check the insulation of the circuit for wear and if there is a broken wire inside the insulation.

Test instruction

The numbers shown below indicate the step numbers in the table

10. Confirm the values of [DPD PM Status] and [DPD TRIP Status] in the scan tool data list, and also check the status diagram to see the status. After confirming the status, take the corresponding measures.

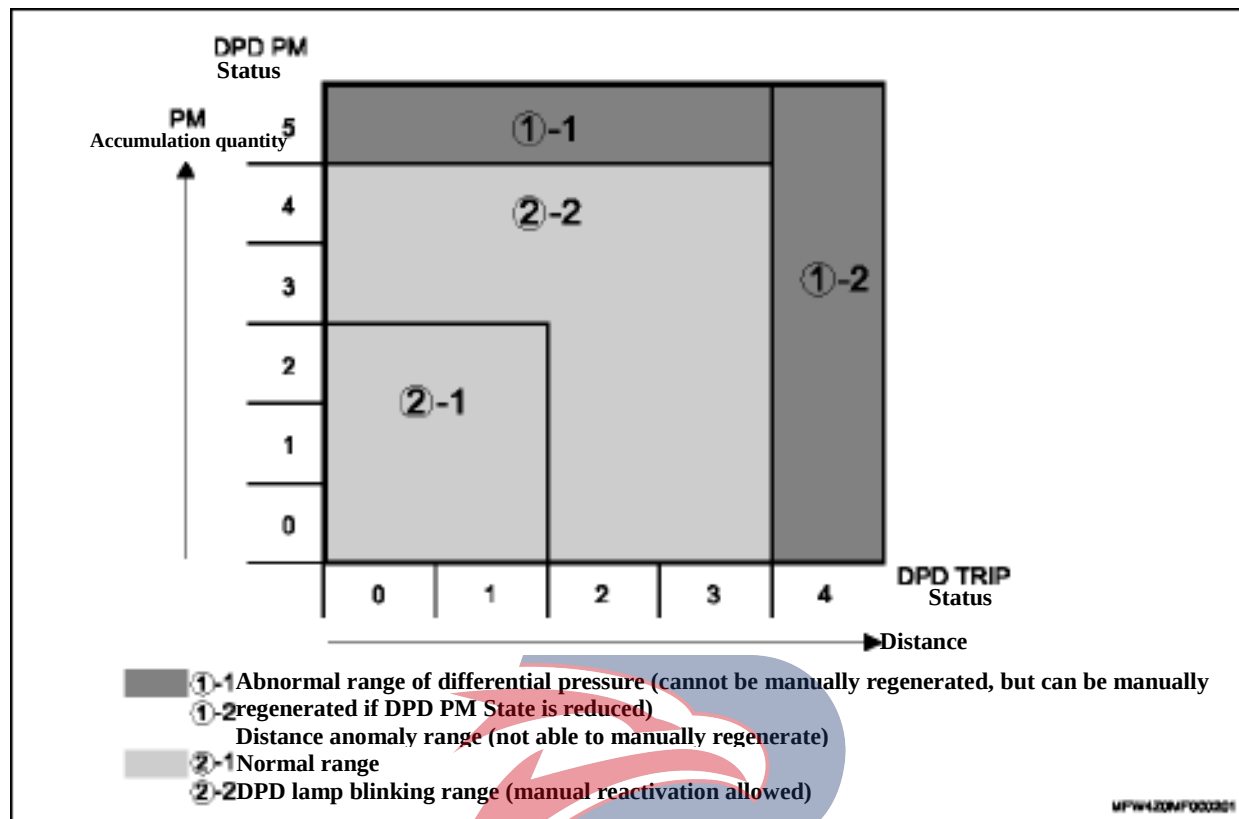
Step	Operation content	Datum value	YES	NO
1	Conduct an investigation. Is there a consultation?	-	Go to step 2	Access to [diagnosis]
2	Check whether DTC is detected. (MAF, injector, EGR, intake throttle related DTC) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	-	Access to [corresponding DTC inspection method]	Go to step 3
3	Confirm the working situation of the DPD warning light. 1. Set the key switch to the [OFF] state 2. Check that the DPD warning light is on. Is there an indicator lamp on?	-	Go to step 6	Go to step 4
4	Confirm DTC. Refer to the description on the DPD warning lamp related items and repair. Is the disposal completed?	-	Go to step 5	-
5	Check whether the DPD warning lamp flickers. 1. Only when the DPD alarm lights are flashing, please check the DPD switch, and then manually regenerate the DPD. 2. Confirm whether the engine fault indicator has lights when the DPD alarm lights flicker. Is the DPD warning lamp blinking and engine MIL on?	-	Access to [corresponding DTC inspection method]	Go to step 3
6	Check or confirm oil-related items. 1. Check the amount of oil, whether or not use the specified engine oil, and carry out proper maintenance at intervals specified for maintenance intervals. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	-	Go to step 7	-
7	Check the exhaust brake, exhaust throttle. 1. Check the operation (use scan tool and visual method check operation). 2. Check the air piping for leak. 3. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	-	Go to step 8	-
8	Check the exhaust pipe (between engine ~DPD) 1. Check the exhaust pipe for cavity (exhaust leak), damage, cave deformation etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	-	Go to step 9	-

Step	Operation content	Datum value	YES	NO
9	Check the DPD. 1. Check DPD host for air leak, looseness, abnormal noise, crack, pit deformation and abnormal installation state etc. 2. Check the differential pressure sensor hose and piping, joint and harness abnormal connection status and installation status. 3. Check whether the exhaust temperature sensor joint and harness connection status and installation status are normal. 4. If fault detected, repair or replace as needed. Is the disposal completed?	-	Go to step 10	-
10	Check the DPD status. 1. Use the scan tool to confirm the DPD state and take appropriate action according to its state. Is the disposal completed?	-	Go to step 11	-
11	Carry out DPD reactivation. 1. Connect scanning tool 2. Perform [DPD forced regeneration] in the scan tool. 3. Press DPD switch if DPD warning lamp blinks. Is the disposal completed?	-	Go to step 12	-
12	Differential memory storage. 1. Set the key switch from the [OFF] state to the [ON] state. (At this point the engine does not start) 2. DPD alarm light lights out after about 30 seconds. 3. After confirming that the DPD warning lamp is off, set the key switch to the [OFF] position and then leave it for 15 seconds. Is the disposal completed?	-	Go to step 13	-

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Step	Operation content	Datum value	YES	NO
13	Check for abnormal differential pressure, DPD abnormal (usually while driving) and dust accumulation in DPD filter. 1. Start the engine. 2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C. Mark: When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180°C, cool down the engine by idling (1500rpm). 3. Increase the engine speed to 2500rpm. 4. Check the differential pressure when the DPD filter front temperature reaches 250 °C with the scanning tool. Mark: When the differential pressure is below 1kPa, check the differential pressure sensor, differential pressure sensor and hose. If there are bad parts, they should be repaired or replaced. Is the pressure difference above the reference value?	3.1kPa		
14	1. Refer to Instructions on Dust Removal project to remove dust from the DPD filter. 2. Carry out the driving test. 3. Confirm DTC. Whether the fault is detected or not.	-	Go to step 14	Go to step 16
15	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value ●Q adjustment value data ●ID code of the ejector ●Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 16	-
16	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	-	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About DPD status



About the removal of dust

For the operation of dust removal device, refer to Operation Instructions for Dust Removal Device.

Dust removal step.

1. Remove the DPD assembly from the vehicle.

Please refer to Exhaust System of Engine in Maintenance Manual.

2. Disassemble the DPD assembly and remove the DPD filter.

Please refer to Exhaust System of Engine in Maintenance Manual.

3. Use a dust remover to clean the DPD filter.

a. Install the DPD filter on the dust removal unit.

B. Set the pressure of the tank of the dust removal device to 4 ± 0.5 air pressure.

C. Open the dust control valve for more than 3 seconds.

d. Repeat steps b and c above three times.

Mark:

Make sure that the filter is not deflected backwards and overflows the housing (case seal).

4. Confirm differential pressure sensor, differential pressure sensor piping and hoses for clogging. If clogged, repair or replace.

5. Assemble DPD assembly in the sequence reverse to the removal one.

Please refer to Exhaust System of Engine in Maintenance Manual.

Mark:

Because the washers and bolts before and after the DPD filter are used again, they will be replaced with new ones.

6. Install the DPD assembly on the car.

7. Start the engine to make sure that there is any gas leakage in all parts (especially the differential pressure sensor, the differential pressure sensor and hose).

8. Refer to step 13 in the diagnostics diagram to confirm the differential pressure and confirm that it is below the reference value.

Datum value: 3.1kPa

9. If the pressure difference is below the reference value, replace the DPD filter.

Important:

After replacing the DPD filter, be sure to clear the DPD data of the ECM and adjust the ECM and DPD to the same state.

Diagnostic code list

6HL1

6HL1 DTC list (1/8)

DTC	flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0087	227	Common rail pressure low pressure anomaly (no pressure feed from oil pump)	Common rail pressure is abnormally reduced	- The battery voltage is normal - DTCs P1630, P1635, P0195, P0192, P0193, P0090, P1261, P1262, P0201-P0204, P0611, and P0612 were not detected. - The engine is running - After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	- When the voltage of the common rail pressure sensor is one of the following states - Engine voltage is lower than 1V at 0-600rpm - The engine speed is lower than 1.2V at 900rpm	10 seconds or more	Lack of power due to lack of combustion Engine flameout	Lamp on
P0088	118	Common Rail Pressure High Pressure Abnormal(Phase 1)	Common rail pressure increased abnormally	- The battery voltage is normal - DTC P1630, P1635, P0192, P0193 were not detected. - After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	The common rail pressure sensor voltage is above 3.9V.	5 seconds or more	Insufficient power	Lamp on
		Common Rail Pressure High Pressure Abnormal(Phase 2)	Common rail pressure increased abnormally	- The battery voltage is normal - DTC P1630, P1635, P0192, P0193 were not detected. - After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	The common rail pressure sensor voltage is above 4.0V.	5 seconds or more	Insufficient power	Lamp on
P0089	151	Common rail pressure anomaly (no pressure feed from oil pump)	Supply surplus of common rail pressure (SCV performance degradation, fixation, supply piping anomaly to the engine side, etc.)	- The battery voltage is normal - DTC P1630, P1635, P0192, P0193, P0090, P1261, P1262, P0201-P0204, P0611, and P0612 were not detected. - The engine is running - After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	- SCV driving load ratio more than 40% 40Mpa when the engine speed is more than 900rpm	10 seconds or more	Insufficient power	Lamp on
P0090	247	Fuel pressure control valve circuit is abnormal	SCV electrical wiring break Short circuit aging	- The battery voltage is normal - DTC P1630 have not been detected. - SCV driver load rate is more than 10% and less than 90%	1) SCV feedback current above 2400mA 2) SCV feedback current below 50mV 3) SCV feedback current at idle speed is 200 ~ 1800mA When one of the above conditions 1), 2) and 3) is established	5 seconds or more	Insufficient power Poor ejection	Lamp on
P0102	91	Airflow (MAF) sensor input low	Sensor harness break, short circuit, damage	- The battery voltage is normal - DTC P1630 have not been detected. - The engine is running	MAF sensor voltage is below 0.1V	5 seconds or more		Lamp on
P0103	91	Airflow (MAF) sensor input high	Sensor harness short circuit	- The battery voltage is normal - DTC P1630 has not been detected. - The engine revolving speed is between 0rpm and 3000rpm.	MAF pressure sensor voltage less than 4.9V	5 seconds or more		Lamp on
P0107	71	Atmospheric pressure sensor circuit input is low	Sensor harness break, short circuit, damage	- The battery voltage is normal - No DTC detected P1630, P1635	The voltage of the atmospheric pressure sensor is below 0.5V.	10 seconds or more	Insufficient power	Lamp on
P0108	71	Atmospheric pressure sensor circuit input is high	Sensor harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1635	The voltage of the atmospheric pressure sensor is below 4.9V.	10 seconds or more	Insufficient power	Lamp on
P0112	22	Intake air temperature sensor circuit input low	Sensor harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1632	Intake air temperature sensor voltage is below 0.1V	10 seconds or more	Noise deterioration	Lamp on
P0113	22	Intake air temperature sensor circuit input high	Sensor harness break, short circuit, damage	- The battery voltage is normal - No DTC detected P1630, P1632 - After starting the engine for more than 3 minutes	Intake air temperature sensor voltage is above 4.8V	10 seconds or more	Noise deterioration	Lamp on
P0117	23	Engine water temperature sensor input low	Sensor harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1634	Engine coolant temperature sensor voltage is below 0.1V	10 seconds or more	Noise deterioration White smoke emitted at low temperature	Lamp on
P0118	23	Engine water temperature sensor input high	Sensor harness break, short circuit, damage	- The battery voltage is normal - No DTC detected P1630, P1634 - After starting the engine for more than 3 minutes	Engine coolant temperature sensor voltage is above 4.8V	10 seconds or more	Noise deterioration White smoke emitted at low temperature	Lamp on
P0182	211	Fuel temperature sensor loop input low	Sensor harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1634	Fuel temperature sensor voltage is less than 0.1	10 seconds or more		Lamp on
P0183	211	Fuel temperature sensor loop input high	Sensor harness break, short circuit, damage	- The battery voltage is normal - No DTC detected P1630, P1634 - After starting the engine for more than 3 minutes	Fuel temperature sensor voltage is more than 4.8	10 seconds or more		Lamp on
P0192	245	Common rail pressure sensor circuit input low	Sensor harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1635	The common rail pressure sensor voltage is below 0.7V.	5 seconds or more	Insufficient power	Lamp on

6HL1 DTC list (2/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0193	245	Circuit input of the common rail pressure sensor is high	Sensor harness break, short circuit, damage	1) The battery voltage is normal 2) No DTC detected P1630, P1635	The common rail pressure sensor voltage is above 4.5V.	5 seconds or more	Insufficient power	Lamp on
P0201	271	The first cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the first cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611 and P1261 were not detected. 4) DTC P0202 is not detected 5) DTC P0203 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of first cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0202	272	No.2 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the first cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611 and P1261 were not detected. 4) DTC P0201 is not detected 5) DTC P0203 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of 2nd cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0203	273	No.3 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 3rd cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611 and P1261 were not detected. 4) DTC P0201 is not detected 5) DTC P0202 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of 3rd cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0204	274	No.4 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 4th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0612 and P1262 were not detected. 4) DTC P0205 is not detected 5) DTC P0206 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of 4th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0205	275	No.5 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 5th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0612 and P1262 were not detected. 4) DTC P0204 is not detected 5) DTC P0206 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of 5th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on

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6HL1 DTC list (3/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0206	276	No.6 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 6th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0612 and P1262 were not detected. 4) DTC P0204 is not detected 5) DTC P0205 is not detected When all the contents and the above 1), 2), 3) and 4) or 5) are established. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Disconnection of drive circuit of 6th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0219	543	Abnormal engine speed	High-speed engine operation abnormal		2) The number of engine revolutions is more than 3900 rpm	5 seconds or more		No lights
P0335	15	Abnormal crankshaft position sensor circuit	Sensor • Broken wire	- Cam pulse normal - DTC P0335, P0336, P0340, P0341, P1345 were not detected. - The engine is running ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	No crankshaft pulse detected	-	Poor manipulation	Lamp on
P0336	15	Crankshaft position sensor anomaly	Excessive signal damage to teeth (short circuit to other lines, etc.)	- Cam pulse normal - DTC P0335, P0336, P0340, P0341, P1345 were not detected. - The engine is running ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	When excessive crankshaft pulses or crankshaft pulses are detected	-	Poor manipulation	Lamp on
P0340	14	Camshaft position sensor circuit abnormal	Sensor • Broken wire	- The crankshaft pulse is normal - DTC P0335, P0336, P0340, P0341, P1345 were not detected. - The engine is running ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	No cam pulse detected	-	Engine cannot start	Lamp on
P0341	14	Camshaft position sensor anomaly	Excessive signal damage to teeth (short circuit to other lines, etc.)	- The crankshaft pulse is normal - DTC P0335, P0336, P0340, P0341, P1345 were not detected. - The engine is running ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	When excessive camshaft pulses or insufficient camshaft pulses are detected	-	Engine cannot start	Lamp on
P0380	66	The glow plug relay circuit is abnormal	Relay • Harness Damage • Short Circuit • Damage	- The input power voltage of the key switch is above 16V and below 32V. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Detected Relay • Harness Damage • Short Circuit • Damaged	10 seconds or more	Cold start failure	Lamp on
P0381	67	Preheat indicator circuit is abnormal	Lamp harness disconnection short circuit, bulb puncture	- The power supply voltage is normal. ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	Light detected • Harness breakage • Short circuit • Light bulb breakage	10 seconds or more		Lamp on
P0470	47	DPD differential pressure sensor circuit input is low	Sensor harness break, short circuit, damage	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	DPD differential pressure sensor voltage less than 2.0V	10 seconds or more	Insufficient power DPD reactivation stop	Lamp on
P0471	47	Input of DPD differential pressure sensor circuit is high	Sensor harness short circuit	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 ※ After the DTC is detected, the key switch will not disappear within 15 seconds after it is turned off.	• ITP sensor voltage above 4.9V	10 seconds or more	Insufficient power DPD reactivation stop	Lamp on

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6HL1 DTC list (4/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0472	47	DPD differential pressure sensor 0 point storage abnormal	Differential pressure sensor aging	- With DPD vehicle - DPD differential pressure sensor 0 point storage [Storage order] 1. The key switch is set to the ON state from the OFF state (the engine does not start) 2. The DPD warning light is on and goes out after about 30 seconds 3. After confirming that the DPD warning light is off, close the key switch and hold it for more than 15 seconds. * After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	Stored value exceeds 4.5kPa	-	Insufficient power DPD reactivation stop	Lamp on
P0487	44	EGR valve BLDC position circuit is abnormal	Sensor harness break, short circuit, damage	- Battery voltage is normal 6HK1-TC is equipped with EGR valve. - DTC P1630, P1635, P0487 were not detected.	The signals input through the EGR position 1,2,3 are all turned on or off.	10 seconds or more	-	Lamp on
P0488	45	EGR valve abnormal	Drive motor side failure • Broken wire, partially stuck spool • Fixed	- Battery voltage is normal 6HK1-TC is equipped with EGR valve. - DTC P1630, P1635, P0487 were not detected. - Main relay input power supply voltage is greater than 20V, less than 32V * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	- EGR BLDC motor load ratio greater than 90% - EGR position (BLDC motor) greater than ±20%	20 seconds or more	-	Lamp on
P0500	25	Abnormal speed sensor circuit	Sensor harness break, short circuit, damage	- The battery voltage is normal - Do not step on the accelerator during your workout - Engine rotation is greater than 2000rpm * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The speed is 0km/h	5 seconds or more	Exhaust brake not work	Lamp on
P0501	25	Abnormal speed sensor	Noise and pulse mixing	- The battery voltage is normal 5 seconds or more after the key switch [ON] * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Noise detected • Pulse mixing	5 seconds or more	Exhaust brake not work	Lamp on
P0543	48	The input of exhaust temperature sensor 1 (in the front of filter) loop is low	Sensor harness short circuit	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Voltage of exhaust temperature sensor 1 is below 0.1V	5 seconds or more	DPD reactivation stop	Lamp on
P0544	48	The input of exhaust temperature sensor 1 (in the front of filter) loop is high	Sensor harness break, short circuit, damage	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 - After starting the engine for more than 10 minutes - The number of engine revolutions is 1700rpm or more and the state of operation * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Exhaust temperature sensor 1 voltage above 4.8V	160 seconds or more	DPD reactivation stop	Lamp on
P0546	49	Exhaust temperature sensor 2 (before oxidation catalyst) circuit input is low	Sensor harness short circuit	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Voltage of exhaust temperature sensor 2 is below 0.1V	5 seconds or more	DPD reactivation stop	Lamp on
P0547	49	The input of exhaust temperature sensor 2 (in the front of filter) loop is high	Sensor harness break, short circuit, damage	- The battery voltage is normal - With DPD vehicle - No DTC detected P1630, P1633 - After starting the engine for more than 10 minutes - The number of engine revolutions is 1700rpm or more and the state of operation * Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Exhaust temperature sensor 2 voltage above 4.8V	30 seconds or more	DPD reactivation stop	Lamp on

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6HL1 DTC list (5/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0571	26	Brake switch circuit is abnormal	One of the wiring switches of the brake light is broken and short circuited.	Battery voltage is normal	The state that the ON signal input times of brake switch 1 and brake switch 2 does not coincide more than 10 times.	-	Automatic cruise will not start Retarder does not move	Lamp on
P0601	53	Control module (ROM) abnormality	ROM abnormality	-	ROM abnormality detected	-	Unable to start Engine stop running	Lamp on
P0603	54	The control module (EEPROM) abnormal	EEPROM abnormal	-	EEPROM detected abnormal	-	Deterioration of engine vibration	Lamp on
P0606	51	Internal fault of the ECM (CPU anomalies)	CPU is anomaly	- After the ECM power is turned on for a while - Key switch input power supply voltage is greater than 20V, less than 16V	CPU abnormality was detected in the past	-	Insufficient power	Lamp on
	52	Internal fault of the ECM (auxiliary CPU anomalies)	Accessory □PU is anomaly	1 second after opening the button - Key switch input power supply voltage is greater than 20V, less than 16V	Detects normal secondary CPU	-	Insufficient power	Lamp on
P0611	34	EDU charging circuit 1 low voltage	EDU charging circuit □ 1 has abnormality (internal burn loss, broken line, etc.)	The battery voltage is normal	Abnormal EDU charging circuit detected 1	-	Insufficient power Deterioration of engine vibration	Lamp on
P0612	34	EDU charging circuit 2 low voltage	EDU charging circuit □ 2 has abnormality (internal burn loss, broken line, etc.)	The battery voltage is normal	Abnormal EDU charging circuit detected 2	-	Insufficient power Deterioration of engine vibration	Lamp on
P0615	19	Starter circuit breaker relay circuit abnormal	Relay • Harness Damage • Short Circuit • Damage	- The input power voltage of the key switch is above 16V and below 32V - Only vehicles equipped with MT or AT (cars with smoother-F cannot diagnose) - Once the DTC is detected, the code will not disappear □ within 15 seconds after the key switch is turned off.	Detected Relay • Harness Damage • Short Circuit • Damaged	15 seconds or more	-	Lamp on
P0650	77	Engine check control circuit is abnormal	Indicator light • Damaged harness • Short circuit • Damaged bulb	- Battery voltage is normal - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Indicator light • Damaged harness • Short circuit • Damaged bulb	5 seconds or more	-	No lights
P0652	75	Abnormal circuit of DPD alarm lamp	Indicator light • Damaged harness • Short circuit • Damaged bulb	- Battery voltage is normal - With DPD vehicle - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Indicator light • Damaged harness • Short circuit • Damaged bulb	5 seconds or more	DPD reactivation period uncertain	Lamp on
P1093	227	Fuel leaks(more)	Fuel leaks(more)	- Battery voltage is normal - DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 - The engine is running - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	- SCV driver load rate is less than 33% - The common rail pressure is above 30MPa when the engine revolution is below 1200rpm.	15 seconds or more	Insufficient power Engine flameout	Lamp on
P1094	226	Fuel leakage (less) or (fuel pump or injection system) control function	Fuel leaks(less) or other	- Battery voltage is normal - DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 - The engine is running - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	A small amount of fuel leak was detected during the 5-minute operation at idle speed	15 seconds or more	Insufficient power	Lamp on
		Fuel leakage (less) or (fuel pump or injection system) control function	Fuel leaks(less) or other	- Battery voltage is normal - DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 - Engine speed exceeds 1000rpm - Speed 50km/h or more - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Deceleration at a speed of 50km/h or more (a state in which the accelerator pedal is released), a small amount of fuel leak is more detected	15 seconds or more	Insufficient power	Lamp on
P1095	225	Pressure limiter start	Pressure limiter start	- Battery voltage is normal - DTC P1630, P1635, P0192, P0193 not detected - Common rail pressure sensor voltage is less than 0.7V and less than 4.5V - Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detecting the effect of pressure limiter	1 seconds or more	Insufficient power	Lamp on

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6HL1 DTC list (6/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P1173	542	Engine overheating	Overheat	- The battery voltage is normal - DTC P1630, P1634, P0117, P0118 not detected	Engine crankshaft temperature is higher than 110°C	15 seconds or more	Insufficient power	Lamp on
P1220	28	Abnormal circuit of PTO accelerator sensor	Harness short circuit	- The battery voltage is normal - No DTC detected P1630, P1631	PTO acceleration sensor voltage is higher than 4.8V.	5 seconds or more	PTO not act	Lamp on
P1225	31	UP/DOWN accelerator sensor circuit abnormal	Harness Damage • Short Circuit • Damage	The battery voltage is normal	Increasing signal of idle speed control and decreasing signal of idle speed control at the same time are at the ON state.	5 seconds or more	Cannot perform idle control operation	Lamp on
P1261	158	COMMON1 (Harness or Injector) Circuit Abnormal	Injector common rail 1 side electrical wiring break, short circuit, EDU output abnormal	- The battery voltage is greater than 18V - The number of engine revolutions is more than 40 rpm - DTC P0611 not detected - DTC P0201 not detected - DTC P0202 not detected - DTC P0203 not detected When one of the conditions 1), 2) and 3) and conditions 4), 5) and 6) above is satisfied. ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of abnormal drive circuits for all injectors of the first cylinder, second cylinder and third cylinder	10 seconds or more	Insufficient power Deterioration of engine vibration	Lamp on
P1262	159	COMMON2 (Harness or Injector) Circuit Abnormal	Injector common rail 2 side electrical wiring break, short circuit, EDU output abnormal	- The battery voltage is greater than 18V - The number of engine revolutions is more than 40 rpm - DTC P0612 not detected - DTC P0204 not detected - DTC P0205 not detected - DTC P0206 not detected When one of the conditions 1), 2) and 3) and conditions 4), 5) and 6) above is satisfied. ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of abnormal drive circuits for all injectors of the 4th cylinder, 5th cylinder and 6th cylinder	10 seconds or more	Insufficient power Deterioration of engine vibration	Lamp on
P1271	24	Accelerator Sensor 1-2 Correlation Abnormal	Sensors • Harness Short Circuit • Aging, Mechanical Misalignment • Damaged	- Battery voltage is normal - DTC P1630, P1631, P1632, P1277, P1278, P1282, P1283 are not detected ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 1 and acceleration sensor 2 are one of the following conditions - The value of [Acceleration Sensor 1 Voltage + Acceleration Sensor 2 Voltage] is higher than 6.4V - The value of [Acceleration Sensor 1 Voltage + Acceleration Sensor 2 Voltage] is lower than 3.6V	10 seconds or more	Unable to perform acceleration operation	Lamp on
P1277	24	Accelerator sensor 1 input low	Sensor harness break, short circuit, damage	- Battery voltage is normal - No DTC detected P1630, P1631 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 1 voltage is lower than 0.2V	5 seconds or more	Insufficient power	Lamp on
P1278	24	Accelerator sensor 1 input high	Sensor harness break, short circuit, damage	- Battery voltage is normal - No DTC detected P1630, P1631 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The acceleration sensor 1 has a voltage of 4.9 V or more	5 seconds or more	Insufficient power	Lamp on
P1282	24	Accelerator sensor 2 input low	Sensor harness break, short circuit, damage	- Battery voltage is normal - No DTC detected P1630, P1632 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 2 voltage is lower than 0.2V	5 seconds or more	Insufficient power	Lamp on
P1283	24	Accelerator sensor 2 input high	Sensor harness break, short circuit, damage	- Battery voltage is normal - No DTC detected P1630, P1632 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The acceleration sensor 2 has a voltage of 4.9 V or more	5 seconds or more	Insufficient power	Lamp on
P1345	16	Crankshaft - abnormal camshaft position relationship	Camshaft gear / crankshaft mounting angle misalignment and damage	- DTC P0335, P0336, P0340, P0341, P1345 were not detected. - Cam pulse normal - Crankshaft pulse normal ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of phase mismatch between cam pulse and crankshaft pulse		Engine cannot start	Lamp on

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6HL1 DTC list (7/8)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P1446	143	DPD exhaust temperature sensor 1 (filter front) high temperature abnormal	Abnormal high temperature exhaust temperature The possibility of filter soaking	- With DPD vehicle - DTCs P1630, P1633, P0543, P0544 were not detected. - After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	Exhaust temperature 1 is abnormally high	5 seconds or more	DPD reactivation stop Output low	Lamp on
P1449	145	DPD exhaust temperature sensor 2 (oxidation catalyst front) high temperature abnormal	Abnormal high temperature exhaust temperature The possibility of filter soaking	- With DPD vehicle - DTCs P1630, P1633, P0546, P0547 were not detected. - After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	Effluent gas temperature 2 in abnormal high temperature state	5 seconds or more	DPD reactivation stop Output low	Lamp on
P1453	139	DPD regeneration timed out too much	Regeneration time	- With DPD vehicle - Implement regeneration	Continuous generation without interruption	-	DPD reactivation stop Output low	Lamp on
P1454	131	Excessive DPD PM capture	DPD switch cannot be pressed	With DPD vehicle	PM catching volume is determined as the degree to disable the DPD reactivation	-	DPD reactivation stop Output low	Lamp on
P1471	149	DPD purification abnormal	Poor DPD regeneration	- With DPD vehicle - DTCs P1630, P1633, P0470, P0471, and P0472 were not detected.	When DPD regeneration continues to occur	-	DPD reactivation stop Output low	Lamp on
P1472	142	DPD pressure difference is too low	Filter, oxidation sensor dissolution damage and so on	- With DPD vehicle - When the accelerator opening is more than 50% (rotation speed is less than 1500rpm) or the accelerator opening is maintained at more than 75% (rotation speed is lower than 2500rpm) - DTCs P1630, P1633, P0470, P0471, and P0472 were not detected. - After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	When engine speed is above 1200rpm, DPD differential pressure is below - kPa	10 seconds or more	DPD reactivation stop Output low	Lamp on
P1473	141	DPD pressure difference is too high	Filter plugging, etc.	- With DPD vehicle - DTCs P1630, P1633, P0470, P0471, and P0472 were not detected. - The engine is running - After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	DPD pressure difference is higher than 60kPa	5 seconds or more	DPD reactivation stop Output low	Lamp on
P1484	43	Intake throttle position (ITP) sensor circuit input low	Sensor harness break, short circuit, damage	- The battery voltage is normal - With DPD vehicle - DTC P1630, P1634 not detected	Intake throttle position sensor voltage is below 0.1V	5 seconds or more	-	Lamp on
P1485	43	Intake throttle position (ITP) sensor circuit input high	Sensor harness break, short circuit, damage	- The battery voltage is normal - With DPD vehicle - DTC P1630, P1634 not detected	Intake throttle position sensor voltage is above 4.9V	5 seconds or more	-	Lamp on
P1487	61	Intake throttle anomaly	Drive motor side failure • Broken wire, partially stuck spool • Fixed	- The battery voltage is normal - DTC P1630, P1634, P1484, P1485 are not detected - After the DTC is detected, the code will not disappear if it is judged as abnormal in the next memory.	The intake throttle drive duty ratio above 47%, intake throttle opening above +20%	20 seconds or more	Hard to exercise	Lamp on
P1625	416	Main relay circuit constantly open	Harness Damage / GND (Short Circuit) • Relay Off Fixed	- The battery voltage is normal - DTC P1630 have not been detected. 5 seconds or more after the key switch [ON] - Main relay drive commands ON	Main relay input supply voltage is lower than 1V	5 seconds or more	Engine cannot start	Lamp on
		Main relay circuit is fixed and closed	Harness B - short circuit • relay-on fixed	-	ECM will not become OFF even if main relay drive commands OFF	10 seconds or more	Severe consumption battery	Lamp on
P1630	36	Analog/digital signal conversion abnormal	A/D conversion abnormal	-	Analog/digital conversion abnormality detected	-	Hard to exercise	Lamp on
P1631	55	5V 1# circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	- DTC P1630 have not been detected. - The input power voltage of the key switch is above 16V and below 32V.	Abnormal sensor or ECM power detected	5 seconds or more	Insufficient power	Lamp on
P1632	55	5V #2 circuit is abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	- DTC P1630 have not been detected. - The input power voltage of the key switch is above 16V and below 32V.	Abnormal sensor or ECM power detected	5 seconds or more	Insufficient power	Lamp on

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6HL1 DTC list (8/8)

DTC	flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P1633	55	5V # 3 circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	• DTC P1630 have not been detected. • The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P1634	55	5V # 4 circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	• DTC P1630 have not been detected. • The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P1635	55	5V # 5 circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	• DTC P1630 have not been detected. • The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P1636	55	12V circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken and the voltage-dropping resistor is broken.	• DTC P1630 have not been detected. • The input power voltage of the key switch is above 16V and below 32V.	Sensor detected, abnormal ECM power supply or buck resistance	5 seconds or more	Insufficient power	Lamp on
P1683	38	DPD exhaust throttle circuit input high	Harness Short Circuit • Damaged	• With DPD vehicle • The battery voltage is normal • DPD exhaust throttle control status is ON ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detected Harness Short Circuit • Damaged	5 seconds or more	Regeneration stops	Lamp on
P1684	38	DPD exhaust throttle circuit input low	Relay / Broken wire in valve harness • Short circuit • Damaged	• With DPD vehicle • The battery voltage is normal • DPD exhaust throttle control status is ON ※ Once the DTC is detected, the code will not disappear within 16 seconds after the key switch is turned off.	Detected Relay/valve • Harness Damage • Short Circuit • Damaged	5 seconds or more	Regeneration stops Hard to exercise	Lamp on
U2104	84	CAN communication abnormal	ECM side CAN output circuit abnormal		CAN communication failure detected	-	CAN abnormal control	Lamp on
U2106	85	Communication with TCM (transmission control module) is abnormal	CAN communication wiring, disconnection, short circuit or communication object unit malfunction		When communication with TCM is interrupted	-	Poor treatment of traditional Chinese medicine	Lamp on

6HK1-TC

6HK1-TC DTC list (1/7)

DTC	flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0087	227	Common rail pressure low pressure anomaly (no pressure feed from oil pump)	Common rail pressure abnormally low	- The battery voltage is normal ■ DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-P0204, P0611, P0612 ■ The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	- When the common rail pressure sensor voltage is in one of the following states ■ Engine speed is 0-600rpm, voltage is less than 1V ■ When the engine speed exceeds 900rpm, the voltage is lower than 1.2V	10 seconds or more	Insufficient power due to insufficient combustion Engine flameout	Lamp on
P0088	118	Common Rail Pressure High Pressure Abnormal(Phase 1)	Common rail pressure increased abnormally	- The battery voltage is normal ■ DTC P1630, P1635, P0192, P0193 not detected ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The common rail pressure sensor voltage is above 3.9V.	5 seconds or more	Insufficient power	Lamp on
		Common Rail Pressure High Pressure Abnormal(Phase 2)	Common rail pressure increased abnormally	- The battery voltage is normal ■ DTC P1630, P1635, P0192, P0193 not detected ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The common rail pressure sensor voltage is above 4.0V.	5 seconds or more	Insufficient power	Lamp on
P0089	151	Common rail pressure anomaly (no pressure feed from oil pump)	Common rail pressure excess supply (Deterioration of SCV performance, viscosity, anomalies in the supply pipe to the engine side, etc.)	- The battery voltage is normal ■ DTC P1630, P1635, P0192, P0193, P0090, P1261, P1262, P0201-P0204, P0611, and P0612 were not detected. ■ The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	■ SCV driving load ratio more than 40% ■ When the number of engine revolutions is more than 900 rpm, the pressure is 40MPa or more	10 seconds or more	Insufficient power	Lamp on
P0090	247	Fuel pressure control valve circuit abnormal	Electrical wiring of SCV broken	- The battery voltage is normal ■ DTC P1630 have not been detected. ■ SCV driver load rate is more than 10% and less than 90% ■ DTC P1630, P1635 not detected	1) SCV feedback current above 2400mA 2) SCV feedback current is less than 50mA 3) SCV feedback current upon idling not in 200 ~ 1800mA When one of the above conditions 1), 2) and 3) is established	5 seconds or more	Insufficient power Poor ejection	Lamp on
P0107	71	Atmospheric pressure sensor circuit input is low	Sensor harness break, short circuit, damage	- The battery voltage is normal ■ DTC P1630, P1635 not detected	Atmospheric sensor voltage lower than 0.5V	10 seconds or more	Insufficient power	Lamp on
P0108	71	Atmospheric pressure sensor circuit input is high	Sensor harness short circuit	- The battery voltage is normal ■ DTC P1630, P1635 not detected	Atmospheric sensor voltage higher than 4.9V	10 seconds or more	Insufficient power	Lamp on
P0112	22	Intake air temperature sensor circuit input is low	Sensor harness short circuit	- The battery voltage is normal ■ No DTC detected P1630, P1632	Intake air temperature sensor voltage is below 0.1V	10 seconds or more	Noise deterioration	Lamp on
P0113	22	Intake air temperature sensor circuit input is high	Sensor harness break, short circuit, damage	- The battery voltage is normal ■ No DTC detected P1630, P1632 ■ After starting the engine for more than 3 minutes	Intake air temperature sensor voltage is below 4.8V	10 seconds or more	Noise deterioration	Lamp on
P0117	23	Engine water temperature sensor input is low	Sensor harness short circuit	- The battery voltage is normal ■ DTC P1630, P1634 not detected	Engine coolant temperature sensor voltage is below 0.1V	10 seconds or more	Noise deterioration White smoke at low temperatures	Lamp on
P0118	23	Engine water temperature sensor input is high	Sensor harness break, short circuit, damage	- The battery voltage is normal ■ DTC P1630, P1634 not detected ■ After starting the engine for more than 3 minutes	Engine coolant temperature sensor voltage is above 4.8V	10 seconds or more	Noise deterioration White smoke at low temperatures	Lamp on
P0182	211	Fuel temperature sensor loop input low	Sensor harness short circuit	- The battery voltage is normal ■ DTC P1630, P1634 not detected	Fuel temperature sensor voltage is below 0.1V	10 seconds or more		Lamp on
P0183	211	Fuel temperature sensor loop input high	Sensor harness break, short circuit, damage	- The battery voltage is normal ■ DTC P1630, P1634 not detected ■ After starting the engine for more than 3 minutes	Fuel temperature sensor voltage is above 4.8V	10 seconds or more		Lamp on
P0192	245	Common rail pressure sensor circuit input is low	Sensor harness short circuit	- The battery voltage is normal ■ DTC P1630, P1635 not detected	The common rail pressure sensor voltage is below 0.7V.	5 seconds or more	Insufficient power	Lamp on
P0193	245	Circuit input of the common rail pressure sensor is high	Sensor harness break, short circuit, damage	- The battery voltage is normal ■ DTC P1630, P1635 not detected	The common rail pressure sensor voltage is above 4.5V.	5 seconds or more	Insufficient power	Lamp on

6HK1-TC DTC list (2/7)

DTC	flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0201	271	The first cylinder-injector circuit abnormal	Disconnection of the electrical wiring of the first cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611, P1261 not detected 4) DTC P0202 not detected 5) DTC P0203 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of first cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0202	272	No.2 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 2nd cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611, P1261 not detected 4) DTC P0201 not detected 5) DTC P0203 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of 2nd cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0203	273	No.3 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 3rd cylinder injector • Short circuit	1) The battery voltage is greater than 18V 2) The engine is running 3) DTC P0611, P1261 not detected 4) DTC P0201 not detected 5) DTC P0202 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of 3rd cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0204	274	No.4 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 4th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611, P1262 not detected 4) DTC P0205 not detected 5) DTC P0206 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of 4th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0205	275	No.5 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 5th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611, P1262 not detected 4) DTC P0204 not detected 5) DTC P0206 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of 5th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on

6HK1-TC DTC list (3/7)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0206	276	No.6 cylinder-injector circuit is abnormal	Disconnection of the electrical wiring of the 6th cylinder injector • Short circuit	1) The battery voltage is normal 2) The engine is running 3) DTC P0611, P1262 not detected 4) DTC P0204 not detected 5) DTC P0205 not detected All of the above 1), 2), 3) conditions and 4) or 5) when establishing ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Disconnection of drive circuit of 6th cylinder injector detected	5 seconds or more	Idling instability Insufficient power	Lamp on
P0219	543	Abnormal engine speed	The engine is running at an unusual height		2) The number of engine revolutions is more than 3900 rpm	5 seconds or more		No lights
P0234	42	High pressurization abnormality	Intake • Turbine Piping Abnormal • Exhaust Bypass Valve Abnormal	• The battery voltage is normal • DTC P1630, P1634, P0237, P0238 were not detected	Turbocharging is the following state • When the engine speed is 0-2200rpm, the pressure is greater than 130-240kPa • When the engine speed is 2600-3800rpm, the pressure is greater than 230-240kPa	5 seconds or more	Insufficient power	Lamp on
P0237	32	MAP sensor loop input low	Sensor harness break, short circuit, damage	• The battery voltage is normal • DTC P1630, P1634 not detected	The booster sensor voltage is below 0.1v.	5 seconds or more	Insufficient power Black smoke deteriorates	Lamp on
P0238	32	MAP sensor circuit input is high	Sensor harness short circuit	• The battery voltage is normal • DTC P1630, P1634 not detected	The booster sensor voltage is above 4.9v	5 seconds or more	Insufficient power Black smoke deteriorates	Lamp on
P0335	15	Abnormal crankshaft position sensor circuit	Sensor • Broken wire	• Cam pulse normal • DTC P0335, P0336, P0340, P0341, P1345 were not detected. • The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	No crankshaft pulse detected	-	Poor manipulation	Lamp on
P0336	15	Crankshaft position sensor anomaly	Excessive signal damage to teeth (short circuit to other lines, etc.)	• Cam pulse normal • DTC P0335, P0336, P0340, P0341, P1345 were not detected. • The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	When excessive crankshaft pulses or crankshaft pulses are detected	-	Poor manipulation	Lamp on
P0340	14	Camshaft position sensor circuit anomaly	Sensor • Broken wire	• Crankshaft pulse normal • DTC P0335, P0336, P0340, P0341, P1345 were not detected. • The engine is running ※ Once the DTC is detected, the code will not disappear within 16 seconds after the key switch is turned off.	No cam pulse detected	-	Engine cannot start	Lamp on
P0341	14	The camshaft position sensor is abnormal	Excessive signal damage to teeth (short circuit to other lines, etc.)	• Crankshaft pulse normal • DTC P0335, P0336, P0340, P0341, P1345 were not detected. • The engine is running ※ Once the DTC is detected, the code will not disappear within 16 seconds after the key switch is turned off.	When an extra cam pulse or an insufficient cam pulse is detected	-	Engine cannot start	Lamp on
P0380	66	The loop of preheating plug is abnormal	Relay • Harness Damage • Short Circuit • Damage	The input power voltage of the key switch is above 16V and below 32V. ※ Once the DTC is detected, the code will not disappear within 16 seconds after the key switch is turned off.	Detected Relay • Harness Damage • Short Circuit • Damaged	10 seconds or more	Cold start failure	Lamp on
P0381	67	Glow plug indicator circuit is abnormal	Indicator light • Damaged harness • Short circuit • Damaged bulb	The battery voltage is normal ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Indicator light • Damaged harness • Short circuit • Damaged bulb	10 seconds or more		Lamp on

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6HK1-TC DTC list (4/7)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P0401	44	EGR valve cannot be closed	The closed side of the EGR valve sticks Due to deformation and breakage of EGR without ventilation EGR temperature sensor intermediate value adhesion malfunction	• Vehicles with EGR M / V • DTC P1630, P1632, P1633, P2032, P2033, P2144, P2145, P0112, P0113, P0117, P0118 • Intake air temperature is higher than 5 °C • After starting the engine for more than 10 minutes • EGR valve control in ON status	EGR temperature sensor detected abnormal	-	-	Lamp on
P0402	44	EGR valve cannot be opened	The open side of the EGR valve sticks EGR temperature sensor intermediate value adhesion malfunction	• Vehicles with EGR M / V • DTC P1630, P1632, P1633, P2032, P2033, P2144, P2145, P0112, P0113, P0117, P0118 • Intake air temperature is higher than 5 °C • After starting the engine for more than 10 minutes • EGR valve control in ON status	EGR temperature sensor detected abnormal	-	-	Lamp on
P0500	25	Abnormal speed sensor circuit	Sensor harness break, short circuit, damage	• The battery voltage is normal • Exercising without stepping on accelerator pedal • Engine rotation is greater than 2000rpm ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The speed is 0km/h.	5 seconds or more	Exhaust brake valve not actuate	Lamp on
P0501	25	Abnormal speed sensor	Noise and pulse mixing	• The battery voltage is normal • Key switch is on for more than 5 seconds ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Noise detected • Pulse mixing	5 seconds or more	Exhaust brake valve not actuate	Lamp on
P0571	26	Anomaly of brake switch circuit	One of the wiring switches of the brake light is broken and short circuited.	• The battery voltage is normal	The state that the ON signal input times of brake switch 1 and brake switch 2 does not coincide more than 10 times.	-	Automatic cruise will not start Retarder does not move	Lamp on
P0601	53	Control module (ROM) abnormality	ROM abnormality	-	ROM abnormality detected	-	Unable to start Engine stop running	Lamp on
P0603	54	The control module (EEPROM) abnormal	EEPROM abnormal	-	EEPROM detected abnormal	-	Deterioration of engine vibration	Lamp on
P0606	51	Internal fault of the ECM (CPU anomalies)	CPU anomaly	• After the ECM power is turned on for a while • Key switch input power supply voltage is greater than 20V, less than 16V	CPU abnormality was detected in the past	-	Insufficient power	Lamp on
	52	Internal fault of the ECM (auxiliary CPU anomalies)	Accessory CPU anomaly	• 1 second or more after opening the button • Key switch input power supply voltage is greater than 20V, less than 16V	Detects normal secondary CPU	-	Insufficient power	Lamp on
P0611	34	EDU charging circuit 1 low voltage	EDU charging circuit 1 has abnormality (internal burn loss, broken line, etc.)	• The battery voltage is normal	Abnormal EDU charging circuit detected 1	-	Insufficient power Deterioration of engine vibration	Lamp on
P0612	34	EDU charging circuit 2 low voltage	EDU charging circuit 2 has abnormality (internal burn loss, broken line, etc.)	• The battery voltage is normal	Abnormal EDU charging circuit detected 2	-	Insufficient power Deterioration of engine vibration	Lamp on
P0615	19	Starter circuit breaker relay circuit abnormal	Relay • Harness Damage • Short Circuit • Damage	• The input power voltage of the key switch is above 16V and below 32V • Only vehicles equipped with M/T or A/T • (cars with smoother-F cannot diagnose) ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detected Relay • Harness Damage • Short Circuit • Damaged	15 seconds or more	-	Lamp on
P0650	77	Engine check control circuit is abnormal	Indicator light • Damaged harness • Short circuit • Damaged bulb	• The battery voltage is normal • Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Indicator light • Damaged harness • Short circuit • Damaged bulb	5 seconds or more	-	No lights
P1093	227	Fuel leaks(more)	Fuel leaks(more)	• The battery voltage is normal • DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 • The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	• SCV driver load rate is less than 33% • The common rail pressure is above 30MPa when the engine revolution is below 1200rpm.	15 seconds or more	Insufficient power Engine flamout	Lamp on

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6HK1-TC DTC list (5/7)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P1094	226	Fuel leakage (less) or (fuel pump or injection system) control function	Fuel leaks(less)other	Battery voltage is normal ■ DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 ■ The engine is running ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	A small amount of fuel leak was detected during the 5-minute operation at idle speed	15 seconds or more	Insufficient power	Lamp on
		Fuel leakage (less) or (fuel pump or injection system) control function	Fuel leaks(less)other	Battery voltage is normal ■ DTC P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-0206, P0611, P0612 ■ Engine speed exceeds 1000rpm ■ Speed 50km/h or more ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Deceleration at a speed of 50km/h or more (a state in which the accelerator pedal is released), a small amount of fuel leak is detected	15 seconds or more	Insufficient power	Lamp on
P1095	225	Pressure limiter start	Pressure limiter start	Battery voltage is normal ■ DTC P1630, P1635, P0192, P0193 not detected Common rail pressure sensor voltage is less than 0.7V and less than 4.5V ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detecting the effect of pressure limiter	1 seconds or more	Insufficient power	Lamp on
P1173	542	Engine overheating	overheat	Battery voltage is normal ■ DTC P1630, P1634, P0117, P0118 not detected	Engine crankshaft temperature is higher than 110°C	15 seconds or more	Insufficient power	Lamp on
P1220	28	Abnormal circuit of PTO accelerator sensor	Harness short circuit	Battery voltage is normal ■ No DTC detected P1630, P1631	PTO acceleration sensor voltage is higher than 4.8V.	5 seconds or more	PTO does not start	Lamp on
P1225	31	UP/DOWN accelerator sensor abnormal	Harness Damage • Short Circuit Damage	Battery voltage is normal	Increasing signal of idle speed control and decreasing signal of idle speed control at the same time are at the ON state.	5 seconds or more	Cannot perform idle control operation	Lamp on
P1261	158	COMMON1 (Harness or Harness) Circuit Abnormal	Injector common rail 1 side electrical wiring break, short circuit, EDU output abnormal	1) The battery voltage is greater than 18V 2) The number of engine revolutions is more than 40 rpm 3) DTC P0611 not detected 4) DTC P0201 not detected 5) DTC P0202 not detected 6) DTC P0203 not detected When one of the conditions 1), 2) and 3) and conditions 4), 5) and 6) above is satisfied ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of abnormal drive circuits for all injectors of the first cylinder, second cylinder and third cylinder	10 seconds or more	Insufficient power Deterioration of engine vibration	Lamp on
P1262	159	COMMON2 (Harness or Harness) Circuit Abnormal	Injector common rail 2 side electrical wiring break, short circuit, EDU output abnormal	1) The battery voltage is greater than 18V 2) The number of engine revolutions is more than 40 rpm 3) DTC P0612 not detected 4) DTC P0204 not detected 5) DTC P0205 not detected 6) DTC P0206 not detected When one of the conditions 1), 2) and 3) and conditions 4), 5) and 6) above is satisfied ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of abnormal drive circuits for all injectors of the 4th cylinder, 4th cylinder and 5th cylinder	10 seconds or more	Insufficient power Deterioration of engine vibration	Lamp on

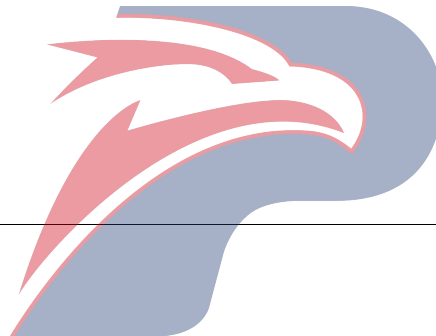
6HK1-TC DTC list (6/7)

DTC	Flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P1271	24	Accelerator Sensor 1-2 Correlation Abnormal	Sensors • Harness Short Circuit Aging, Mechanical Misalignment, Damaged	Battery voltage is normal ● DTC P1630, P1631, P1632, P1277, P1278, P1282, P1283 are not detected ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 1 and acceleration sensor 2 are one of the following conditions • The value of [Acceleration Sensor 1 Voltage + Acceleration Sensor 2 Voltage] is higher than 6.4V • The value of [Acceleration Sensor 1 Voltage + Acceleration Sensor 2 Voltage] is lower than 3.6V	10 seconds or more	Unable to perform acceleration operation	Lamp on
P1277	24	Accelerator sensor 1 input low	Sensor harness break, short circuit, damage	Battery voltage is normal ● No DTC detected P1630, P1631 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 1 voltage is lower than 0.2V	5 seconds or more	Insufficient power	Lamp on
P1278	24	Accelerator sensor 1 input high	Sensor harness break, short circuit, damage	Battery voltage is normal ● No DTC detected P1630, P1631 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The acceleration sensor 1 has a voltage of 4.9 V or more	5 seconds or more	Insufficient power	Lamp on
P1282	24	Accelerator sensor 2 input low	Sensor harness break, short circuit, damage	Battery voltage is normal ● No DTC detected P1630, P1632 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Acceleration sensor 2 voltage is lower than 0.2V	5 seconds or more	Insufficient power	Lamp on
P1283	24	Accelerator sensor 2 input high	Sensor harness break, short circuit, damage	Battery voltage is normal ● No DTC detected P1630, P1632 ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	The acceleration sensor 2 has a voltage of 4.9 V or more	5 seconds or more	Insufficient power	Lamp on
P1345	16	Crankshaft - abnormal camshaft position relationship	Camshaft gear / crankshaft gear mounting angle misalignment and damage	● DTC P0335, P0336, P0340, P0341, P1345 were not detected. ● Cam pulse normal ● Crankshaft pulse normal ※ Once the DTC is detected, the code will not disappear within 15 seconds after the key switch is turned off.	Detection of phase mismatch between cam pulse and crankshaft pulse	-	Engine cannot start	Lamp on
P1625	416	Main relay circuit constantly open	Harness Damage / GND Short Circuit • Relay Off Fixed	Battery voltage is normal ● DTC P1630 have not been detected. ● 5 seconds or more after the key switch [ON] ● Main relay drive commands ON	Main relay input supply voltage is lower than 1V	5 seconds or more	Engine cannot start	Lamp on
		Main relay circuit is fixed and closed	Harness B + short circuit • relay on - fixed		ECM will not become OFF even if main relay drive commands OFF	10 seconds or more	Severe battery consumption	Lamp on
P1630	36	Analog/digital signal conversion abnormal	A/D conversion abnormal		Analog/digital conversion abnormality detected		Hard to exercise	Lamp on
P1631	55	5V 1# circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	● DTC P1630 have not been detected. ● The input power voltage of the key switch is above 16V and below 32V.	Abnormal sensor or ECM power detected	5 seconds or more	Insufficient power	Lamp on
P1632	55	5V #2 circuit is abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	● DTC P1630 have not been detected. ● The input power voltage of the key switch is above 16V and below 32V.	Abnormal sensor or ECM power detected	5 seconds or more	Insufficient power	Lamp on
P1633	55	5V #3 circuit is abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	● DTC P1630 have not been detected. ● The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P1634	55	5V 4# circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	● DTC P1630 have not been detected. ● The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P1635	55	5V 5# circuit abnormal	The power supply wiring to sensor is short-circuit or the circuit of elements used in the internal power of ECM is damaged / the circuit is broken	● DTC P1630 have not been detected. ● The input power voltage of the key switch is above 16V and below 32V.	Abnormal power of sensor or ECM detected	5 seconds or more	Insufficient power	Lamp on
P2032	41	EGR temperature sensor loop input low	Sensor • Wire breakage • Short circuit • Damaged	Battery voltage is normal ● No DTC detected P1630, P1633	EGR temperature sensor voltage less than 0.1V	15 seconds or more	-	Lamp on

6HK1-TC DTC list (7/7)

DTC	flash Code	DTC name	Detection project	Preconditions for DTC setting	DTC Setting conditions	Decision time	Fault symptom	Engine fault indicator lamp
P2144	45	EGR control valve circuit input is low	Relay /Broken wire in valve harness • Short circuit • Damaged	• Key switch input power voltage more than 16V but less than 32V • EGR valve control in OFF status	Detected Relay/valve • Harness Damage • Short Circuit Damaged	10 seconds or more	-	Lamp on
P2145	45	EGR control valve circuit input is high	Short circuit in relay/valve harness • Damaged	• Key switch input power voltage more than 16V but less than 32V • EGR valve control in ON status	Detected Relay/valve • Harness Damage • Short Circuit Damaged	10 seconds or more	-	Lamp on

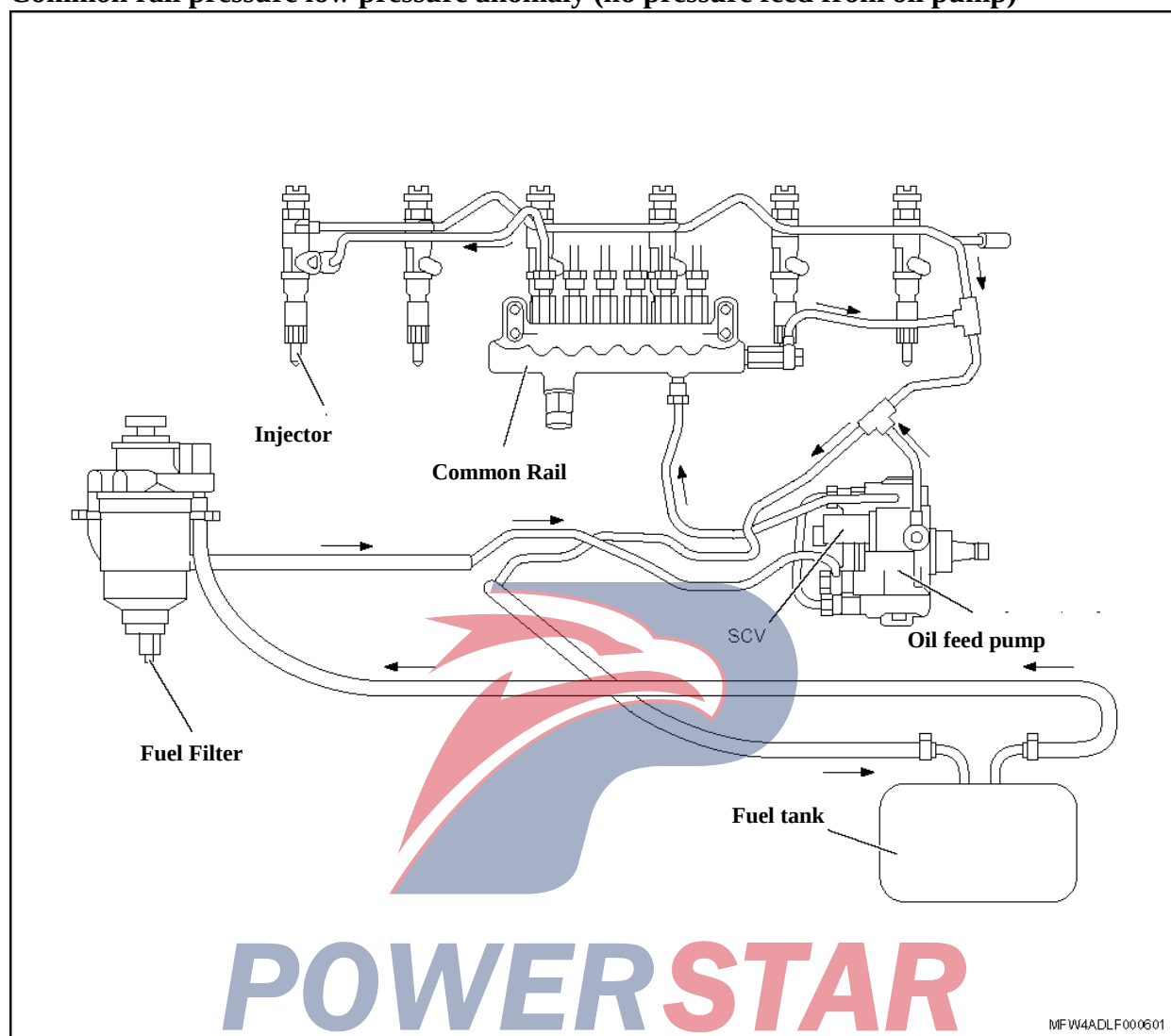
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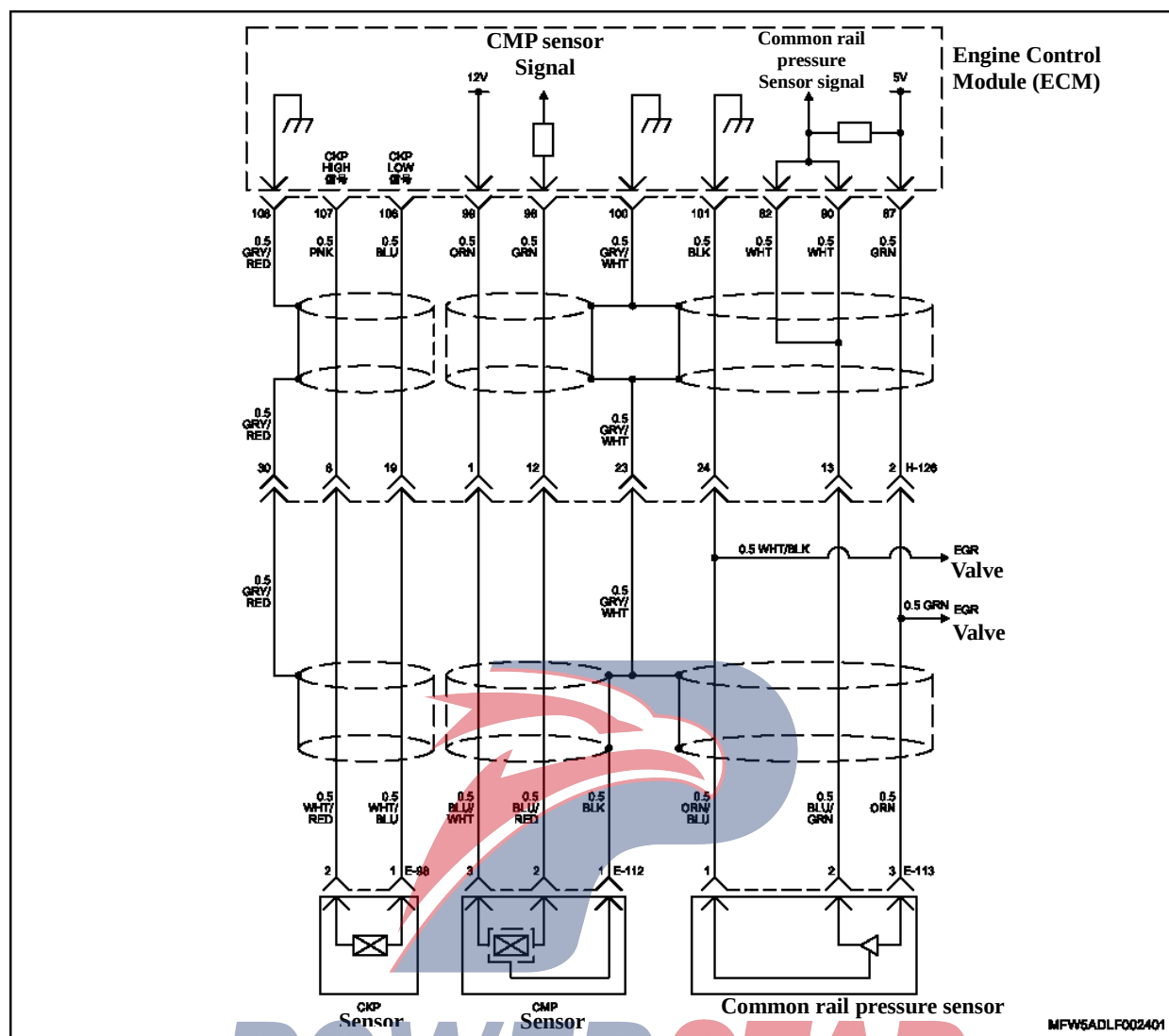


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DTC: P0087 (blinking code 227)

Common rail pressure low pressure anomaly (no pressure feed from oil pump)





Cycle Description

The engine control module (ECM) monitors the pressure in the common rail based on signals transmitted by the common rail pressure sensor. When the pressure is low, the ECM regulates the operating state of the supply pump SCV and raises the pressure in the common rail. When the fuel supplied to the fuel pump is insufficient and the fuel pump fails, the pressure in the common rail will not be increased.

Major adverse phenomena

- Insufficient combustion results in insufficient power.
- The engine is off.

Preconditions for DTC setting

- The battery voltage should be normal.
- DTCs P1630, P1635, P1095, P0192, P0193, P0090, P1261, P1262, P0201-P0204, P0611, P0612 were not detected.
- The engine should be running.

DTC setting conditions

- Common rail pressure is established when one of the following conditions is met:
 - Less than 1V when the engine speed is 0 to

600rpm.

- When the engine speed exceeds 900rpm - 1.2V or less.

Disposal when setting DTC

- The engine failure indicator lamp lights up.
- When DTC P0087 is detected; it will not turn off within 15 seconds after the key switch [OFF].

MIL/DTC clearing conditions

- Now the failure is out of the DTC's setting condition range (failure repaired), and at the same time it is the past failure (log).
- The engine failure indicator lamp turns into a past failure and will turn off when the [ON] key is turned on after the key switch [OFF].
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

Due to speculation the reason can be considered as follows:

- Internal failure of the injector
- Internal fault of the fuel pump
- (Fuel leaks, plugging, squeezing, etc.);
- Fuel filter abnormality (blockage)
- The fuel tank is abnormal (blocked).
- Limit voltage fault (action under specified pressure, aging of seal).
- Faults of the ECM.

- Common rail pressure sensor failure

Test instruction

The numbers shown below indicate the step numbers in the table

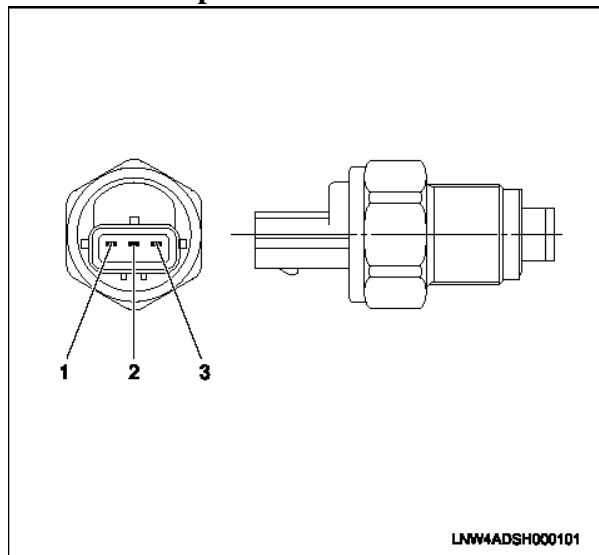
5. When it is confirmed that the oil level rises and light oil is detected, it is considered that DTC is detected due to fuel leakage from the injector portion.

9. When the connector of the common rail pressure sensor is removed, when the display value of the [common rail pressure sensor] is greater than the reference value, the circuit after the sensor including the ECM is normal.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	-	Go to step 2	Enter "OBD System Check"
2	Follow the instructions below to check the fuel system. 1. Check the high pressure piping and low pressure piping for connection looseness (fuel leak), deformation and clogging. 2. Check whether there is freezing and solidification in fuel piping, fuel filter and fuel tank. (cold) 3. Check the fuel supply tube inside the fuel tank for clogging. Is the fuel system normal?	-	Go to step 4	Go to step 3
3	Repair or replace defective parts found. Is the disposal completed?	-	Go to step 15	-
4	Use an oil level gauge to check the oil level rise and light oil taste. Have you noticed any undesirable phenomena?	-	Go to step 5	Go to step 6
5	Check each fuel injector's fuel line connection and replace the fuel injector where the fuel leak was found. Is the disposal completed?	-	Go to step 15	-
6	Check the syringe. 1. Use the scan tool to clear the DTC 2. Start the engine. 3. Select [actuator test] from the menu. 4. Select [Injection stop for each injector] in [Common Rail System]. 5. Press the [OFF] soft key to stop the injector of each cylinder and confirm that there is no change in the engine sound. Is there a cylinder with no change in engine sound?	-	Go to step 8	Go to step 7

Step	Operation content	Datum value	YES	NO
7	1. Replace the injector of the cylinder with no change in engine sound in step 6. Important: After replacing the ejector, be sure to register the ID code. Is the disposal completed?	-	Go to step 15	-
8	Check the common rail pressure. 1. Set the key switch to the [OFF] state 2. Use scanning tool to confirm the display contents of common rail pressure in data display. 3. Start the engine and run at idle state. 4. Use scanning tool to confirm the display contents of common rail pressure in data display. 5. Five times fast idle. 6. Confirm the contents displayed in the data display [common rail pressure] with the scanning tool. When the engine is stopped, is the value of each [common rail pressure] near idle or idle close to the reference?	When the engine stops: 30Mpa idle speed, idle speed:±5MPa	Go to step 13	Go to step 9
9	1. Set the key switch to the OFF position. 2. Dismantle the joint of the common rail pressure sensor 3. Set the key switch to the [OFF] state 4. Use scanning tool to confirm the display contents of common rail pressure in data display. Is the value of the common rail pressure sensor above the reference value?	4.8V	Go to step 12	Go to step 10
10	1. Use the distribution cabinet to check whether the common rail pressure sensor and ECM has short connection with other sensors. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	-	Go to step 15	Go to step 11

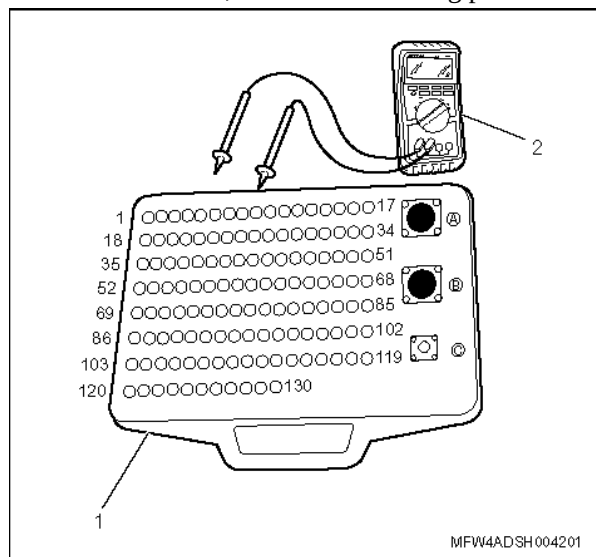
Step	Operation content	Datum value	YES	NO
11	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value •Q adjustment value data •ID code of the ejector •Engine starting times; Refer to the description for programming in this chapter. Is the disposal completed?	-	Go to step 15	-
12	Replace the common rail (common rail pressure sensor). Is the disposal completed?	-	Go to step 15	-
13	1. Use the scan tool to clear the DTC 2. Start the engine. 3. Select [actuator test] from the menu. 4. Select [Common Pressure Control] in [Common Rail System]. 5. The value of the [common rail pressure] is confirmed by the [UP] soft key. Is the value of the [common rail pressure] within the range of the reference value?	± 5MPa	Go to step 15	Go to step 14
14	Replace oil supply pump. Is the disposal completed?	-	Go to step 15	-
15	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Use a scanning tool to confirm DTC. Has DTC P0087 been detected?	-	Go to step 2	Go to step 16
16	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	-	Enter each DTC for diagnosis Diagnosis	Confirm the repair conditions

Common rail pressure sensor**Name**

1. Sensor 1 GND
2. Sensor signal
3. Sensor power supply

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

1. Distribution cabinet
2. DMM

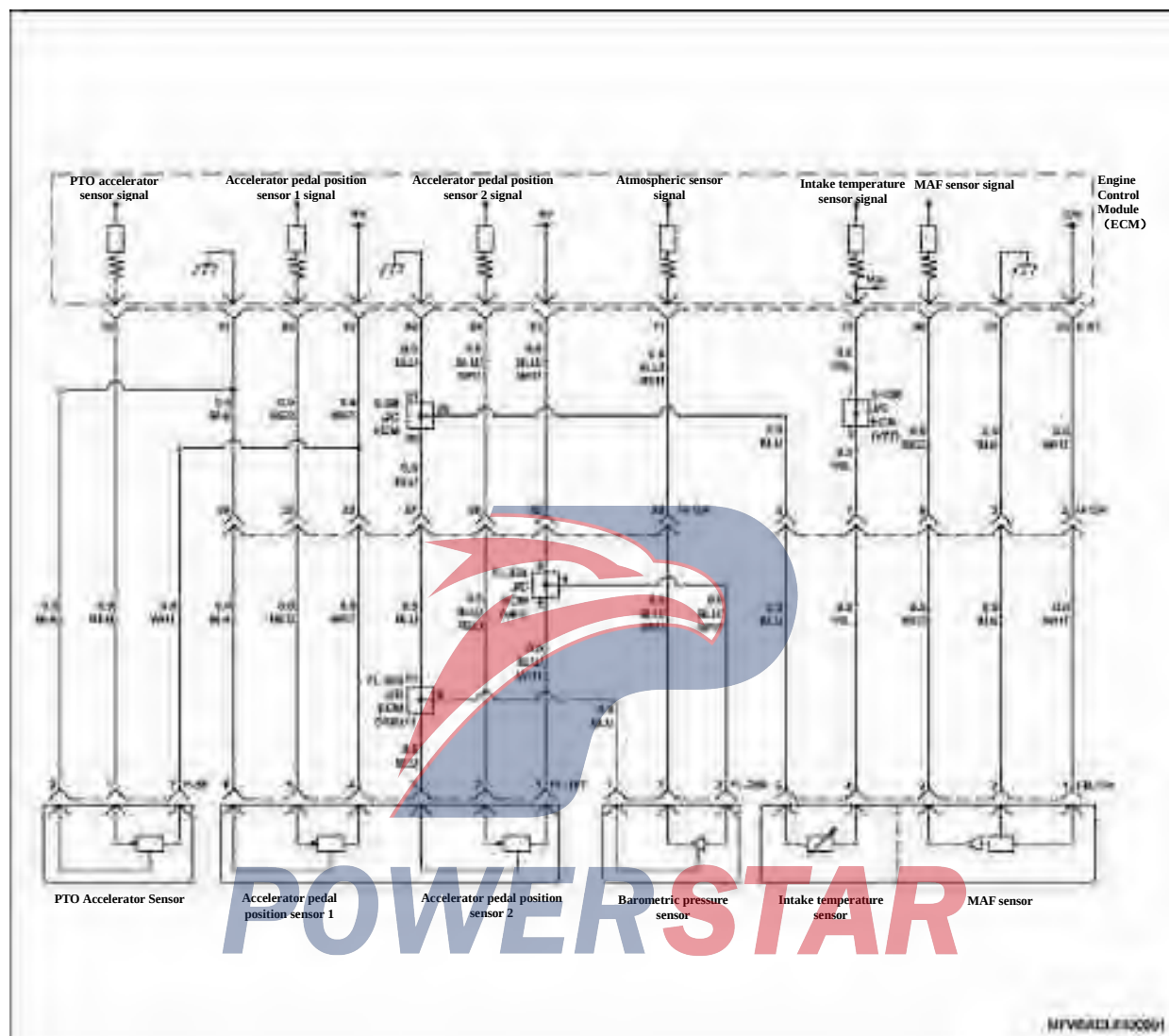
Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
10	Short circuit to other signal circuits	Voltage measurement	- Remove the sensor connector - Key switch [ON]	82 - GND 90 - GND	0V	Above 1V

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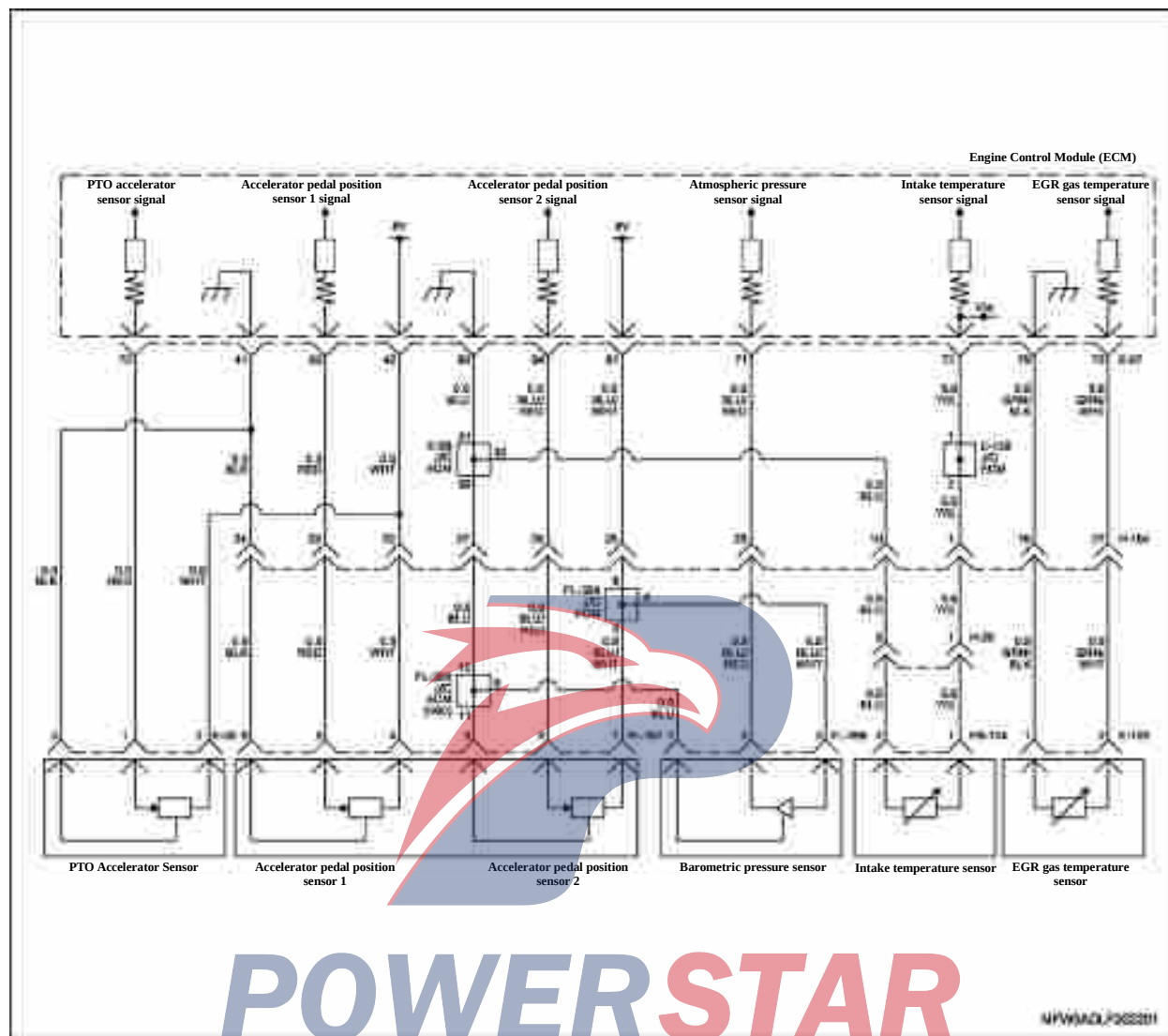
DTC: P1278 (Flash code 24)

Accelerator sensor 1 input high

6HL1



6HK1-TC



Cycle Description

The accelerator pedal position (APP) sensor is installed to the accelerator pedal. APP sensor is a potentiometric sensor and the output voltage varies from the accelerator pedal opening. There are 2 APP sensors which include the following circuits.

- 5V power circuit
- GND circuit
- APP sensor signal circuit

ECM supplies 5V power to APP sensor through 5V power circuit and then to ECM through the GND circuit. In addition, the APP sensor will yield the signal voltage to ECM through the APP sensor signal circuit. The APP sensor 1 output voltage is higher as the accelerator pedal opening increases. The APP sensor 2 output voltage is lower as the accelerator

pedal opening increases.

ECM will set DTC if the APP sensor 1 output signal voltage is lower than the normal range.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1631 have not been detected.

Setting conditions for DTC.

- Accelerator sensor 1 voltage above 4.9V

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1278 is detected, it will not go out within 15 seconds after the key switch is set to[OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by

DTC (fault repair).

- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

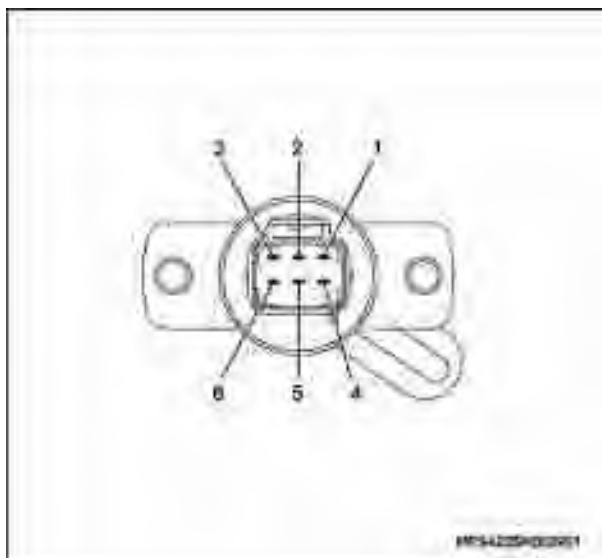
Test instruction

The numbers shown below indicate the step numbers in the table

2. In the case of current malfunction, the displayed value of [Accelerator Sensor 2] is above reference value. If below the reference value, please refer to the diagnosis assistance to explain why.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor 1] in the data display. [Is the displayed value of [accelerator sensor 1] below the reference value?	4.9V	Go to step 3	Enter diagnostic help
3	Confirm the installation status of APP sensor joint, ECM joint and middle joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Set the key switch to the [OFF] state 4. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor] in the data display. [Is the displayed value of [accelerator sensor 1] below the reference value?	0.1V	Go to step 6	Go to step 5
5	1. Check the signal circuit between the ECM and APP sensor 1 for short circuit to 5V power circuit and ignition power circuit with the distributing box. Important: If there is a short circuit in the power circuit of the APP sensor circuit, the sensor may be damaged. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 8

Step	Operation content	Datum value	YES	NO
6	1. 1. Check the GND circuit between the ECM and APP sensor 1 with the distributing box for the following state. • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 7
7	Replace the APP sensor. Is the disposal completed?	—	Go to step 9	—
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1278 been detected?	—	Go to step 2	Go to step 10
10	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About APP Sensor**Name**

1. Sensor 2 power supply
2. Signal of sensor 2
3. Sensor 2 GND
4. Sensor 1 power supply
5. Signal of sensor 1
6. Sensor 1 GND

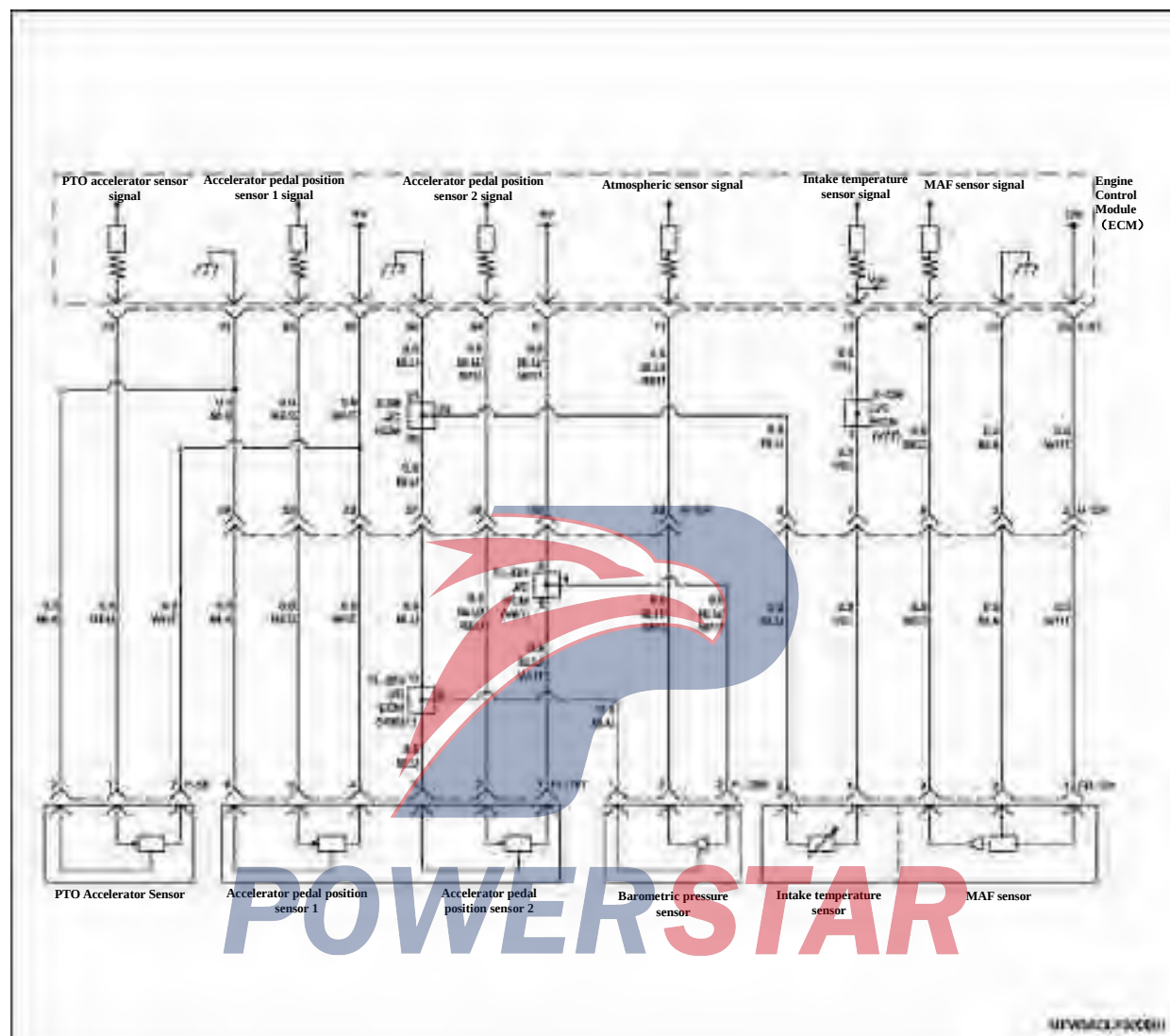
About distribution cabinet

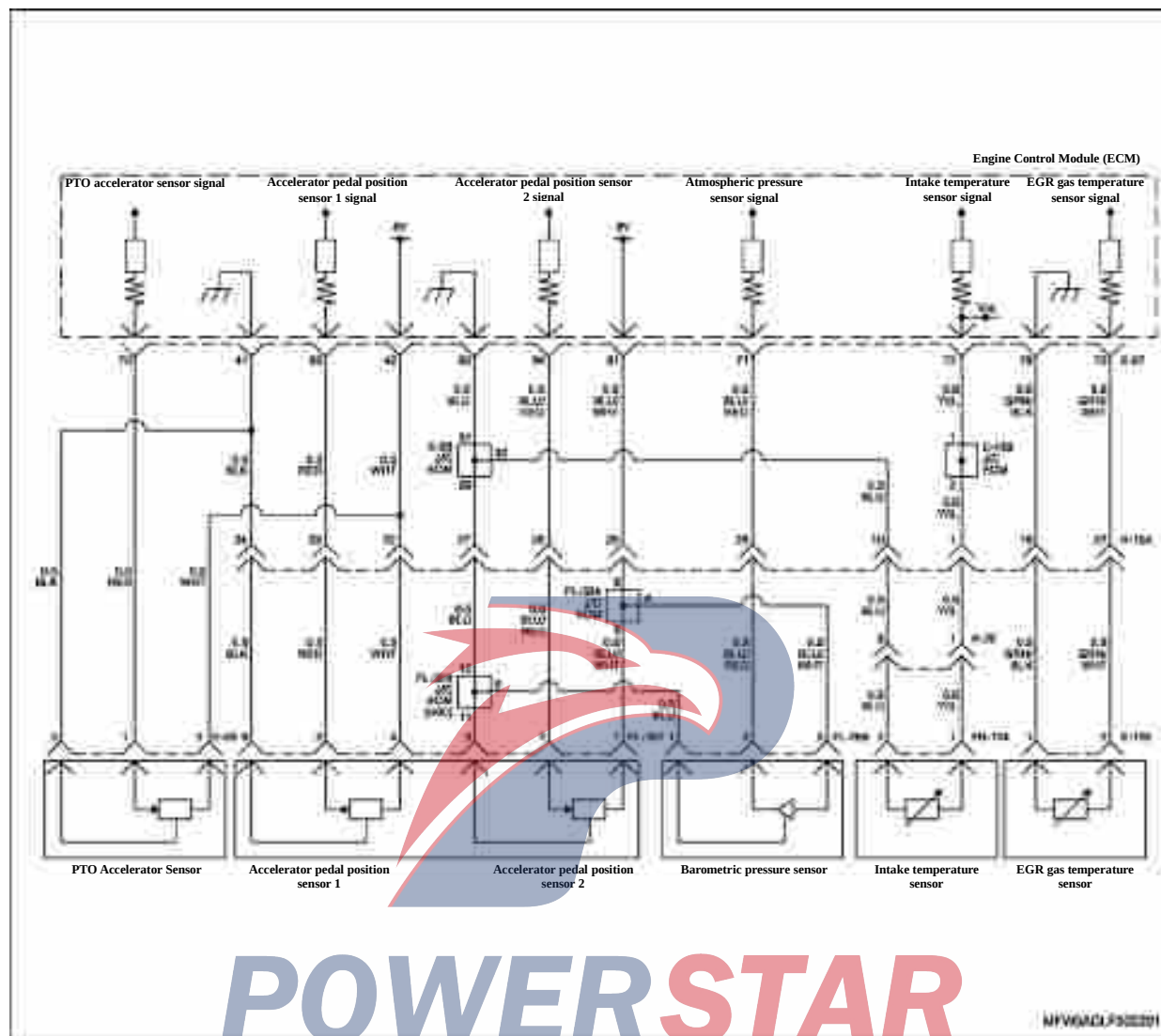
In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Short circuit to power supply circuit	Measurement of resistance value	- Remove the sensor connector - Key switch [ON]	63-GND	0V	Above 4.9V
6	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	41 Sensor Connector Sensor 1 GND Terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1282 (Flash code 24)**Accelerator sensor 2 input low****6HL1**

6HK1-TC**Cycle Description**

The accelerator pedal position (APP) sensor is installed to the accelerator pedal. APP sensor is a potentiometric sensor and the output voltage varies from the accelerator pedal opening. There are 2 APP sensors which include the following circuits.

- 5V power circuit
- GND circuit
- APP sensor signal circuit

ECM supplies 5V power to APP sensor through 5V power circuit and then to ECM through the GND circuit. In addition, the APP sensor will yield the signal voltage to ECM through the APP sensor signal circuit. The APP sensor 1 output voltage is higher as the accelerator pedal opening increases. The APP sensor 2 outputs a voltage is lower as the accelerator

pedal opening increases.

ECM will set DTC if the APP sensor 2 output signal voltage is lower than the normal range.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1632 have not been detected.

Setting conditions for DTC.

- Accelerator sensor 2 voltage below 0.2V

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1282 is detected, it will not go out within 15 seconds after the key switch is set to[OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by

DTC (fault repair).

- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

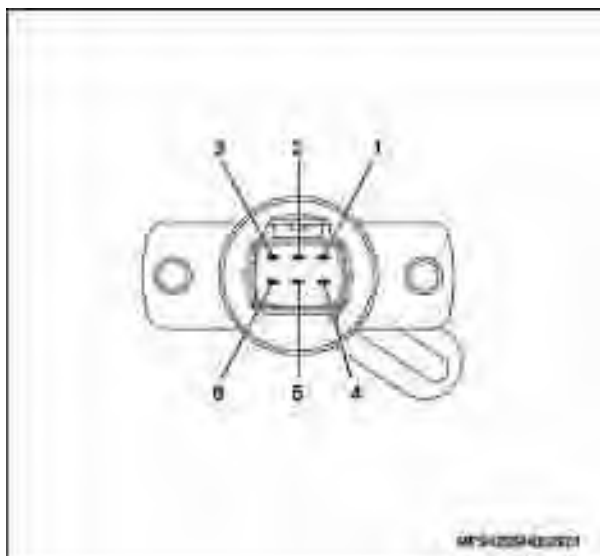
The numbers shown below indicate the step numbers in the table

2. In the case of current malfunction, the displayed value of [Accelerator Sensor 2] is above reference value. If below the reference value, please refer to the diagnosis assistance to explain why.

4. When there is short connection between the terminals of the APP sensor joint, [the accelerator sensor 2] shows that the value is above the reference value, indicating that the circuit after the sensor and the ECM is in the normal state.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	–	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor 2] in the data display. [Is the displayed value of [accelerator sensor 2] below the reference value?	0.2V	Go to step 3	Enter diagnostic help
3	Confirm the installation status of APP sensor joint, ECM joint and middle joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	–	Go to step 9	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Use a jumper to short-circuit the sensor 2 power terminal and the sensor 2 signal terminal of the APP sensor connector. 4. Set the key switch to the [OFF] state 5. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor] in the data display. Is the value of [accelerator sensor 2] above the reference value?	4.8V	Go to step 7	Go to step 5

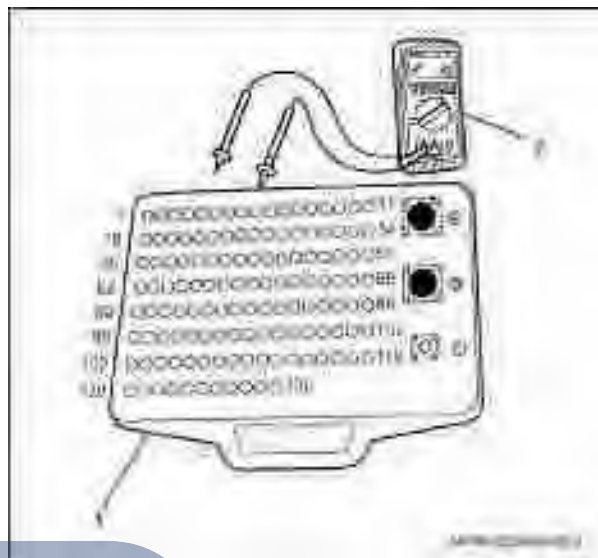
Step	Operation content	Datum value	YES	NO
5	1. Check the power circuit between the ECM and APP sensor 2 for the following state with the distributing box: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 6
6	1. Check the power circuit between the ECM and APP sensor 2 for the following state with the distributing box: • Short circuit to the APP sensor 1 GND circuit • Short connection with GND • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 8
7	Replace the APP sensor. Is the disposal completed?	—	Go to step 9	—
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1282 been detected?	—	Go to step 2	Go to step 10
10	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About APP Sensor**Name**

1. Sensor 2 power supply
2. Signal of sensor 2
3. Sensor 2 GND
4. Sensor 1 power supply
5. Signal of sensor 1
6. Sensor 1 GND

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

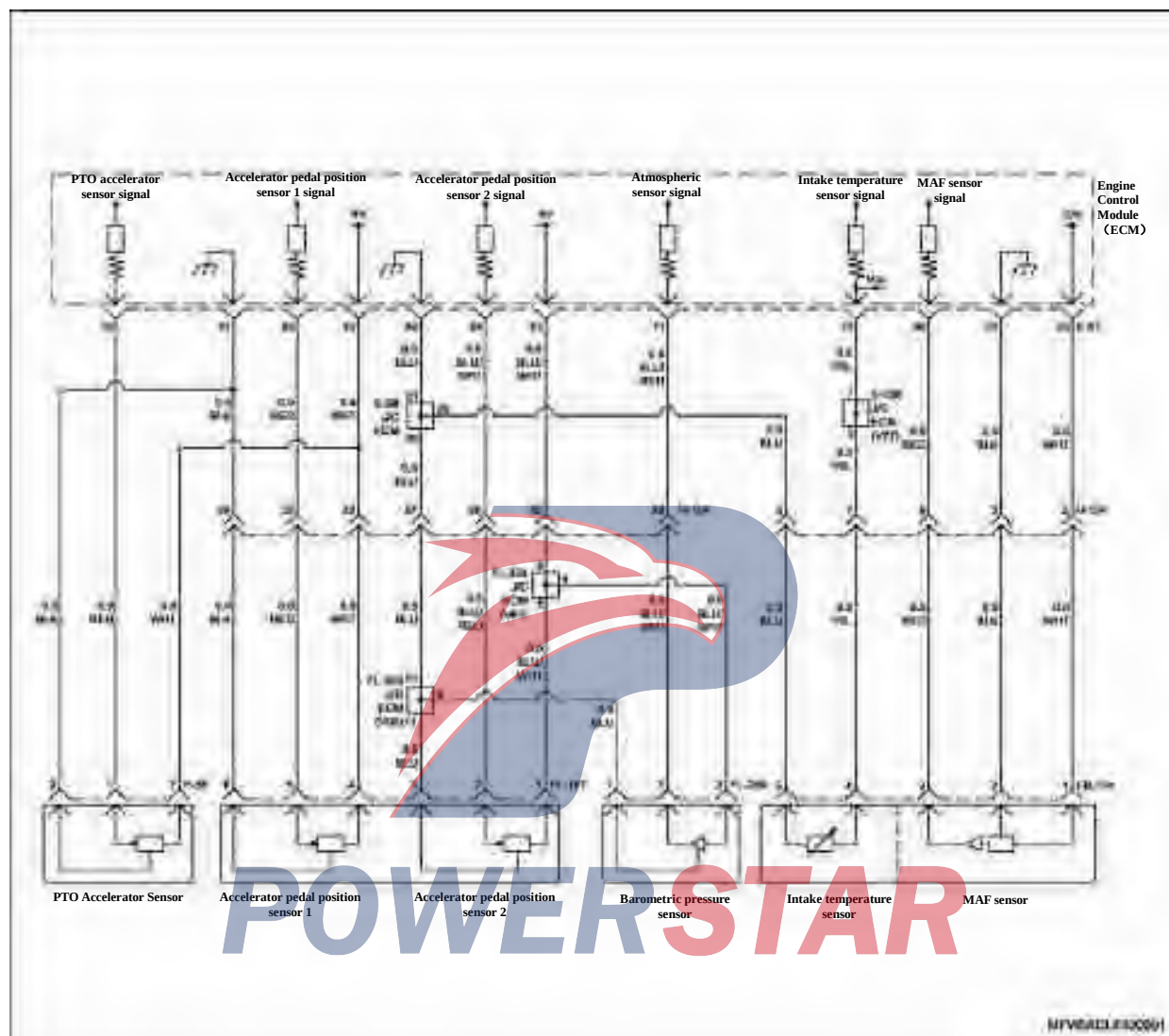
1. Distribution Cabinet
2. DMM

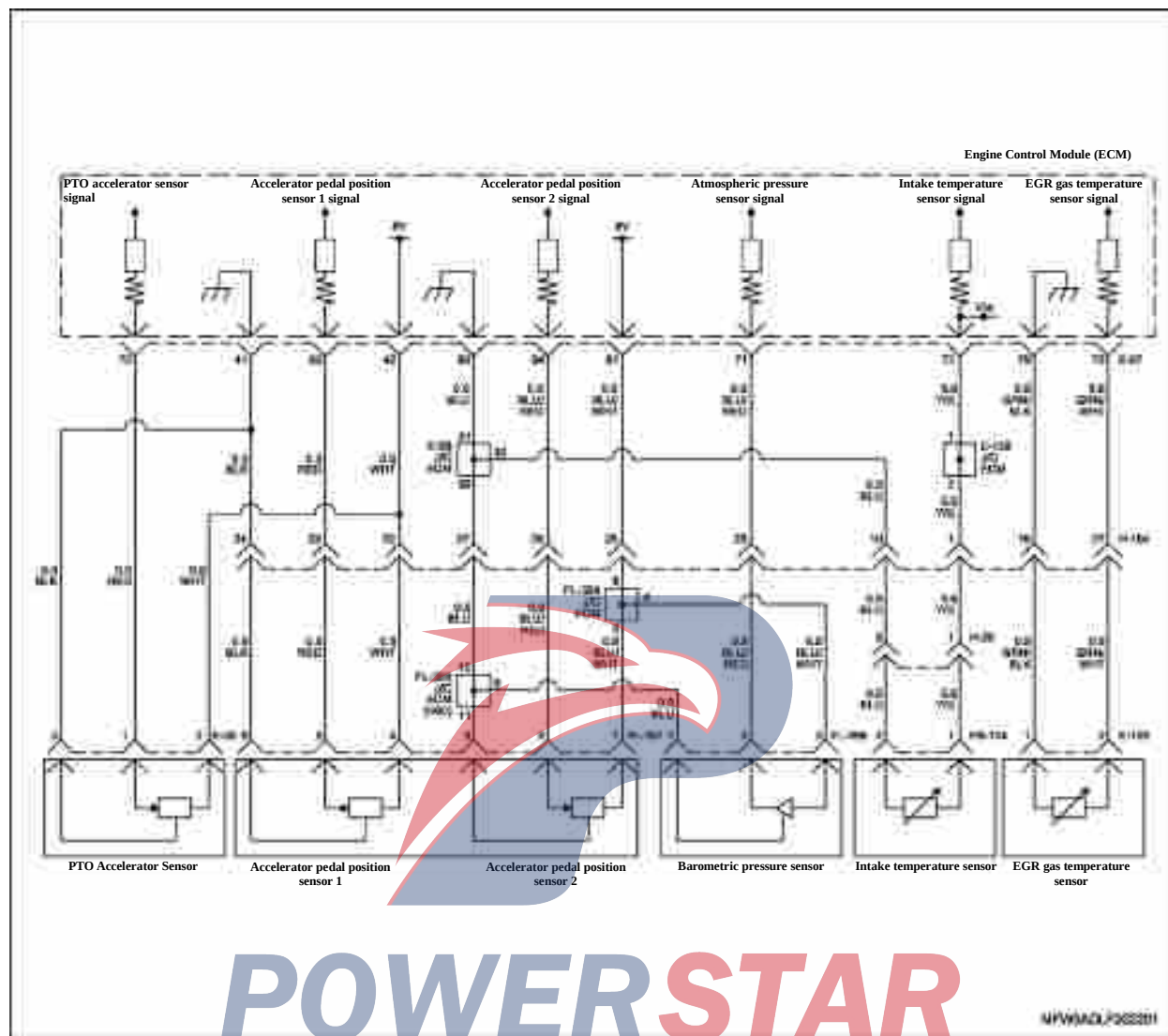
Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	61 Sensor Connector Sensor 2 Power Terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)
6	Short circuit to GND circuit/GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	64-61 64-GND	Above 10MΩ (no conduction)	Below 10MΩ (there is a guide)
	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	64- sensor joint sensor 1GND 2 signal terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1283 (Flash code 24)

Accelerator sensor 2 input high

6HL1



6HK1-TC**Cycle Description**

The accelerator pedal position (APP) sensor is installed to the accelerator pedal. APP sensor is a potentiometric sensor and the output voltage varies from the accelerator pedal opening. There are 2 APP sensors which include the following circuits.

- 5V power circuit
- GND circuit
- APP sensor signal circuit

ECM supplies 5V power to APP sensor through 5V power circuit and then to ECM through the GND circuit. In addition, the APP sensor will yield the signal voltage to ECM through the APP sensor signal circuit. The APP sensor 1 output voltage is higher as the accelerator pedal opening increases. The APP sensor 2 output voltage is lower as the accelerator

pedal opening increases.

ECM will set DTC if the APP sensor 2 output signal voltage is lower than the normal range.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1632 have not been detected.

Setting conditions for DTC.

- Accelerator sensor 2 voltage above 4.9V

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1283 is detected, it will not go out within 15 seconds after the key switch is set to[OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by

DTC (fault repair).

- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

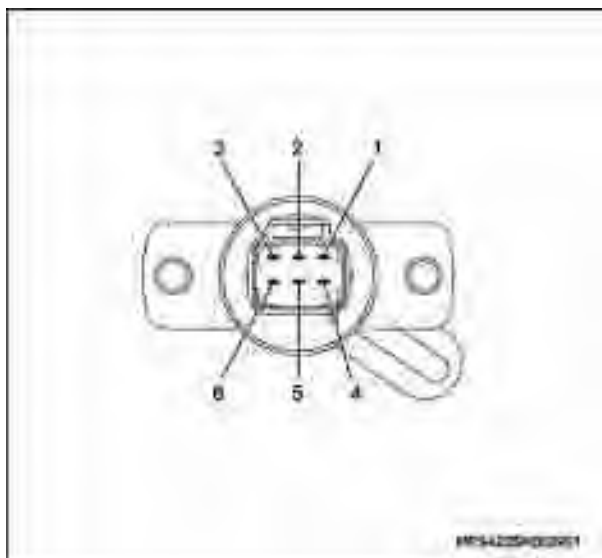
Test instruction

The numbers shown below indicate the step numbers in the table

2. In the case of current malfunction, the displayed value of [Accelerator Sensor 2] is above reference value. If below the reference value, please refer to the diagnosis assistance to explain why.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor 2] in the data display. [Is the displayed value of [accelerator sensor 2] below the reference value?	4.9V	Go to step 3	Enter diagnostic help
3	Confirm the installation status of APP sensor joint, ECM joint and middle joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Set the key switch to the [OFF] state 4. Use the scan tool to confirm the display value of the [PTO Accelerator Sensor] in the data display. [Is the displayed value of [accelerator sensor 2] below the reference value?	0.1V	Go to step 6	Go to step 8
5	1. Check the signal circuit between the ECM and APP sensor 2 for short circuit to 5V power circuit and ignition power circuit with the distributing box. Important: If there is a short circuit in the power circuit of the APP sensor circuit, the sensor may be damaged. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 8

Step	Operation content	Datum value	YES	NO
6	1. Check the GND circuit between the ECM and APP sensor 1 with the distributing box for the following state. • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 7
7	Replace the APP sensor. Is the disposal completed?	—	Go to step 9	—
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1283 been detected?	—	Go to step 2	Go to step 10
10	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About APP Sensor**Name**

1. Sensor 2 power supply
2. Signal of sensor 2
3. Sensor 2 GND
4. Sensor 1 power supply
5. Signal of sensor 1
6. Sensor 1 GND

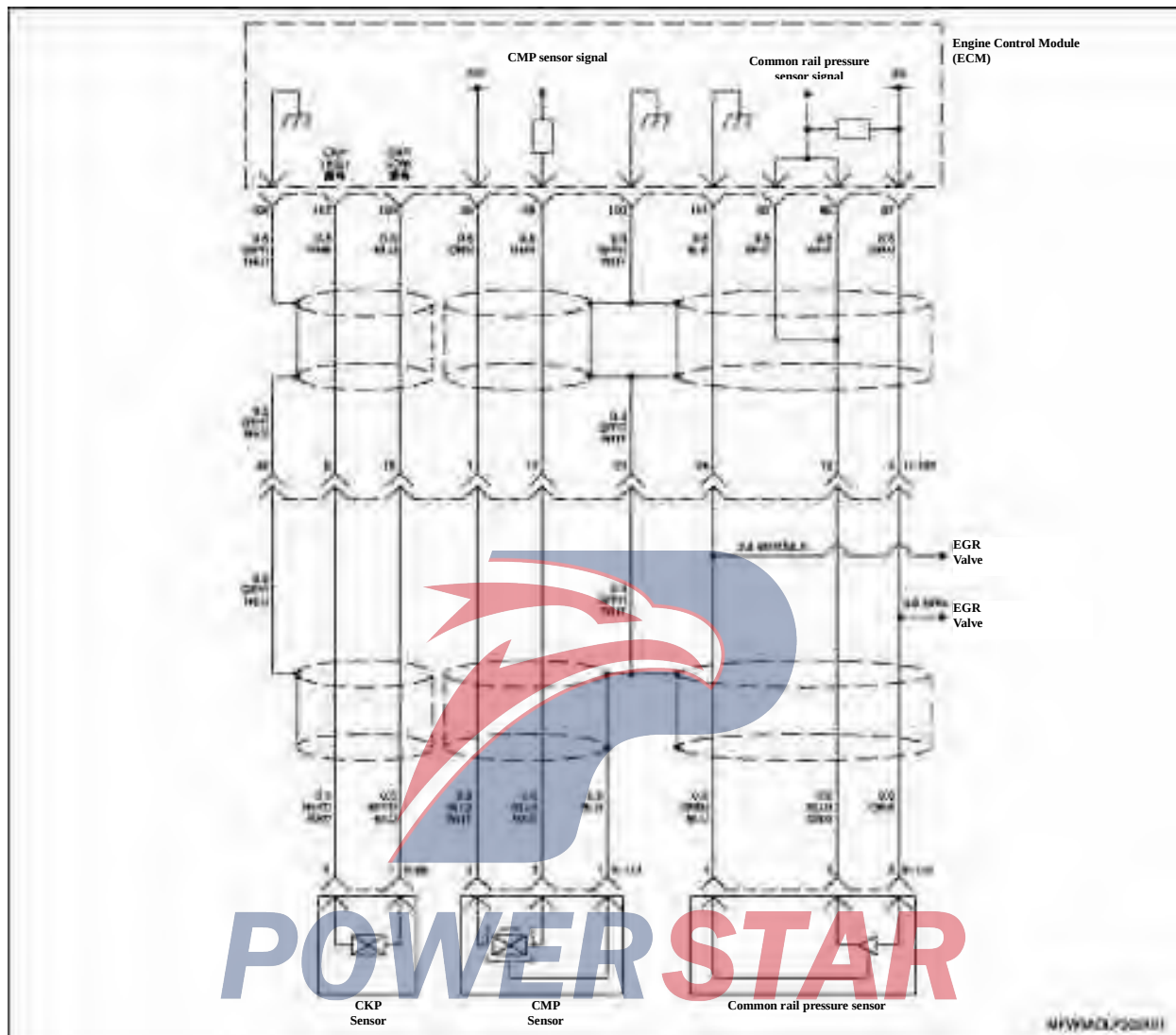
About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	64-GND	0V	Above 4.9V
6	Broken wire / high resistance	Voltage measurement	- Remove the sensor connector - Key switch [OFF]	60 Sensor Connector Sensor 2 GND Terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1345 (Flash code 16)**Crankshaft - abnormal camshaft position relationship****Cycle Description**

The crankshaft position (CKP) sensor generates pulse when passing through the flywheel teeth. Like the CKP sensor, the cam position (CMP) sensor generates pulse by the sensor hole on the camshaft gear for signal detection.

ECM will set DTC if CKP sensor pulse and CMP sensor pulse number are normal while the mutual pulses are out-of-sync.

Preconditions for DTC setting

- DTC P0335, P0336, P0340 and P0341 have not been detected.
- Cam pulse is normal
- Crankshaft pulse is normal.

Setting conditions for DTC.

- Cam pulse and crankshaft pulse phase detected inconsistent

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1345 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding
- Fuel bearing abundant air

In order to find out these reasons, the following checks should be carried out.

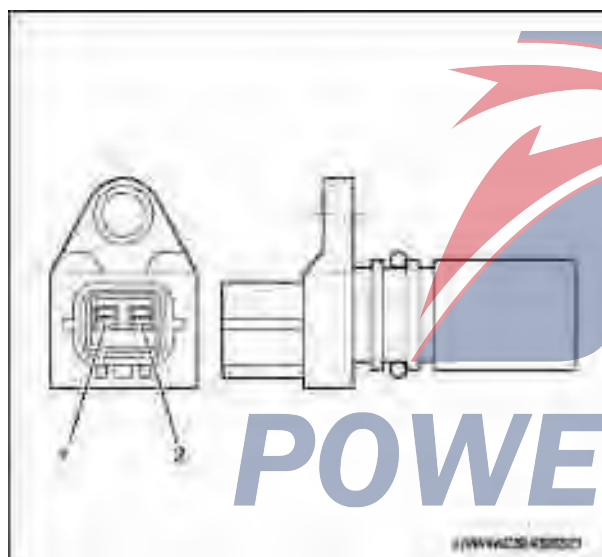
- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	–	Go to step 2	Enter "OBD System Check"
2	1. Confirm the installation status of the CMP sensor. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	–	Go to step 3	–
3	1. Confirm the installation status of the CKP sensor. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	–	Go to step 4	–
4	Confirm the DTC again. 1. Clear DTC with the scanning tool. 2. Set the key switch to the [OFF] state for 15 seconds or more. 3. Perform a test run under [Preconditions for DTC settings]. 4. Confirm DTC with the scanning tool. Has DTC P1345 been detected?	–	Go to step 5	Go to step 9
5	1. Check if the camshaft gear is properly installed. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	–	Go to step 6	–
6	1. Check whether the flywheel is properly installed. 2. When a fault is detected, make repairs as needed. Is the disposal completed?	–	Go to step 7	–
7	Confirm the DTC again. 1. Clear DTC with the scanning tool. 2. Set the key switch to the [OFF] state for 15 seconds or more. 3. Perform a test run under [Preconditions for DTC settings]. 4. Confirm DTC with the scanning tool. Has DTC P1345 been detected?	–	Go to step 8	Go to step 9

Step	Operation content	Datum value	YES	NO
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

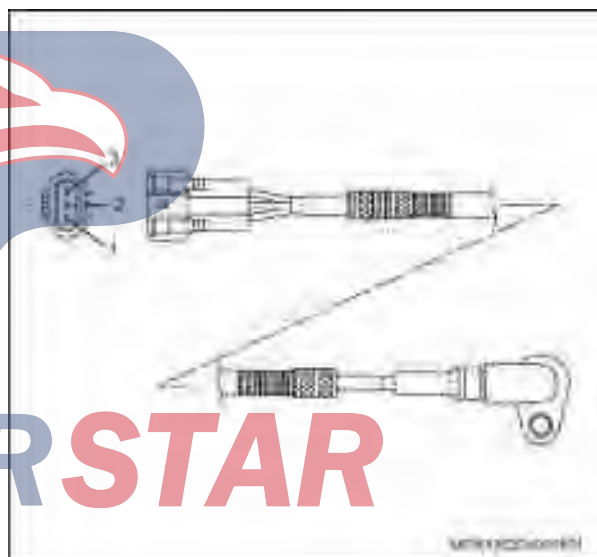
About CMP sensors



Name

1. HIGH signal
2. LOW signal

About distribution cabinet

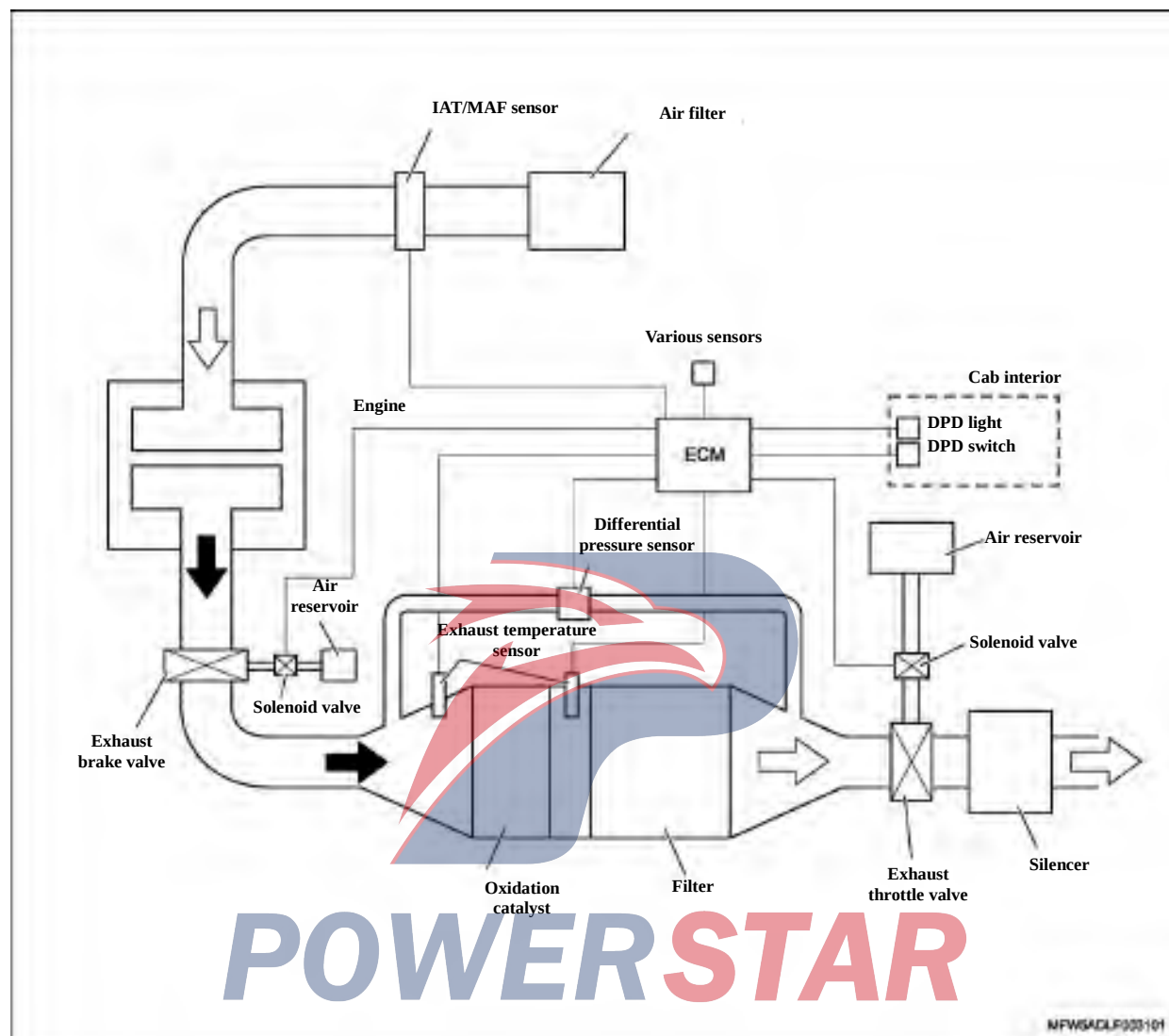


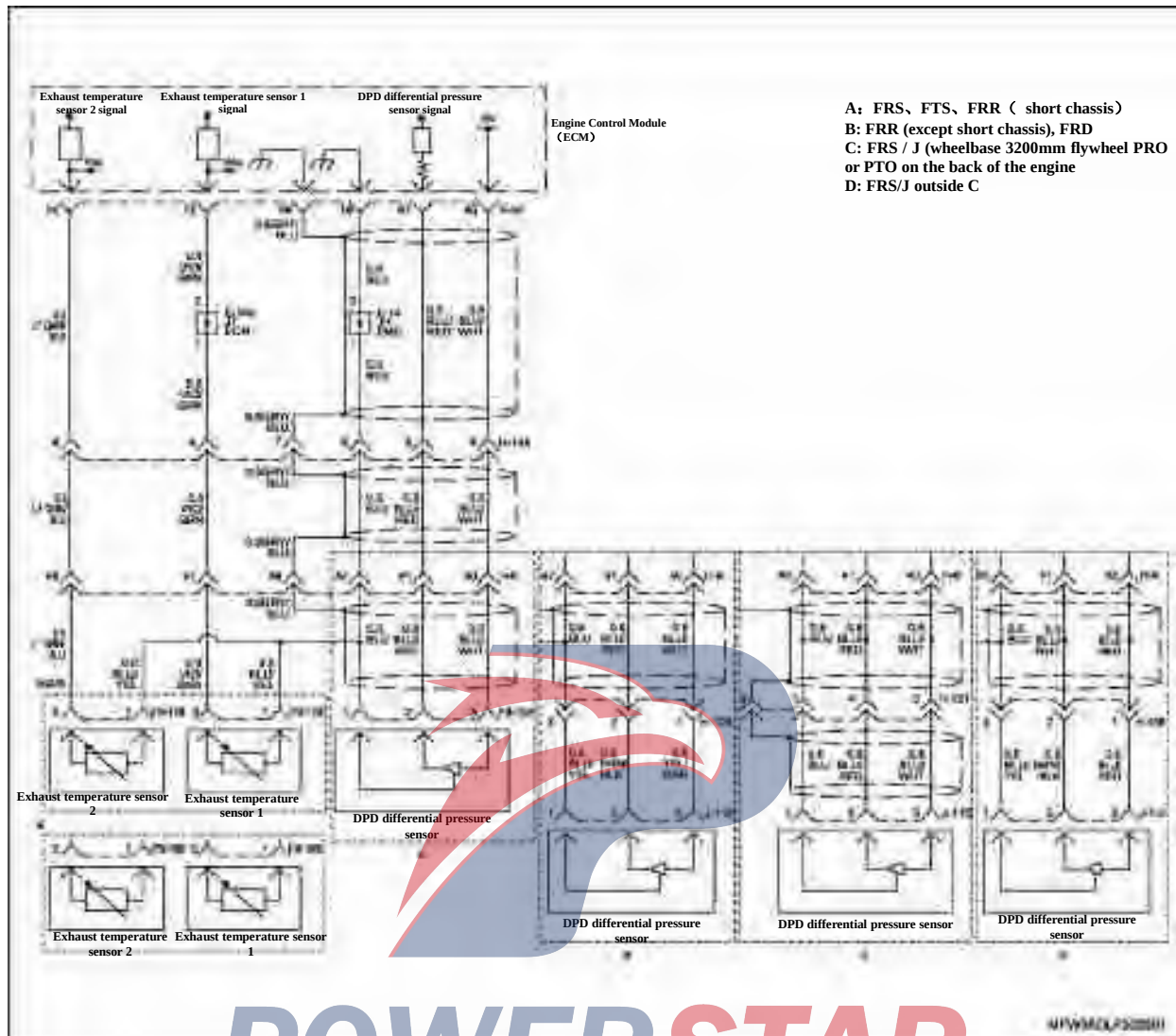
Name

1. Shield
2. -
3. +

DTC: P1446 (Flash code 143)

DPD Exhaust gas temperature 1 (before the filter) high temperature abnormal





Cycle Description

The exhaust temperature sensor 1 is set up to detect the temperature of the filter inlet in the DPD. The sensor is installed at the entrance of the DPD filter. The exhaust temperature sensor is a thermistor, and the internal resistance of the sensor changes due to temperature changes. When the temperature is high, the resistance value is low, and when the temperature is low, the resistance value is high. ECM is connected between the exhaust temperature sensor and signal circuit and GND circuit to apply 5V power voltage to the signal circuit. ECM can detect the exhaust temperature sensor temperature variation and the variation of voltage due to the resistance change (the voltage is low if the temperature is high and high if the temperature is low). ECM will set DTC if detecting the signal of exhaust temperature sensor 1 abnormal.

The main fault phenomenon

- DPD regeneration stop
- Output power insufficient

Preconditions for DTC setting

- DTC P1630, P1633, P0543 and P0544 have not been detected.
- Vehicles equipped with DPD

Setting conditions for DTC.

The exhaust temperature sensor 1 is at an abnormal high temperature.

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1446 is detected, it will not go out within 15 seconds after the key switch is set to[OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault

(resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data

display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

9. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status

10. After the DPD reactivation, check the function of DPD in step 12. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

12. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

When re-entry

4. When the differential pressure is equal to or higher than the reference value, use a dust removal device to clean the DPD filter.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Confirm whether multiple DTC is detected. (the temperature sensor series 5V series circuits etc.) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Confirm DTC, access to [corresponding DTC inspection method] to execute repair and then go to step 3	Go to step 3
3	Check the exhaust temperature sensor 1 . 1. Check harness, connector and mounting status of exhaust temperature sensor 1. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 4	—

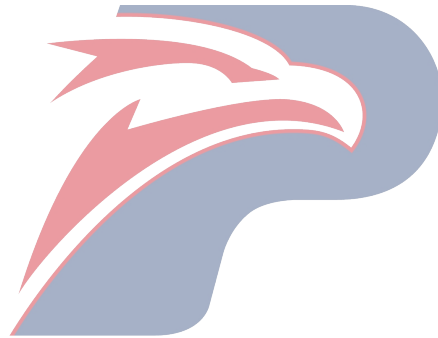
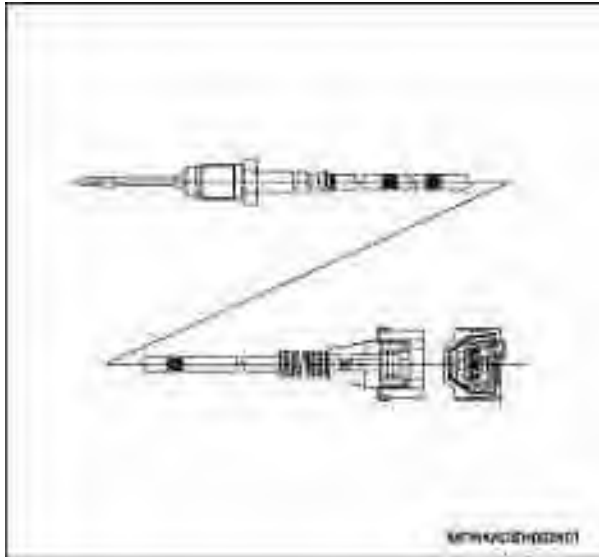
Step	Operation content	Datum value	YES	NO
4	Check the performance of exhaust temperature sensor 1. 1. Remove the wire harness from the ECM. 2. Install the power distribution cabinet. 3. Engage the gear to the neutral position. 4. Start the engine and fully open the accelerator for more than 2 minutes. 5. Stop the engine. 6. Use the scan tool to confirm the temperature. 7. Use the distribution cabinet to measure the resistance. (refer to sensor characteristic diagram and measure the resistance on 2 temperature spots) mark: In the process where the temperature of the DPD does not drop, the operation starts immediately after the engine is stopped. Does the measured value match the sensor's characteristic map?	—	Go to step 6	Go to step 5
5	Replace exhaust temperature sensor 1. Is the disposal completed?	—	Go to step 11	—
6	Check the DPD. 1. Check for air leak, looseness, abnormal noise, fracture, dent, abnormal installation status etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 11	Go to step 9
9	Confirm the status of DPD. 1. Install the DPD. 2. Refer to DPD status and check whether the DPD status is in the normal range with the scanning tool. Is the status within the normal range?	—	Go to step 12	Go to step 10
10	Refer to ECM for DPD reactivation and use the DPD reactivation ECM to reactivate DPD. Is regeneration completed?	—	Go to step 12	Go to step 11
11	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 12	—
12	Perform a functional check for DPD. Is the disposal completed?	—	Confirm the repair conditions	Access to [DPD function check

When coming back to the store

When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

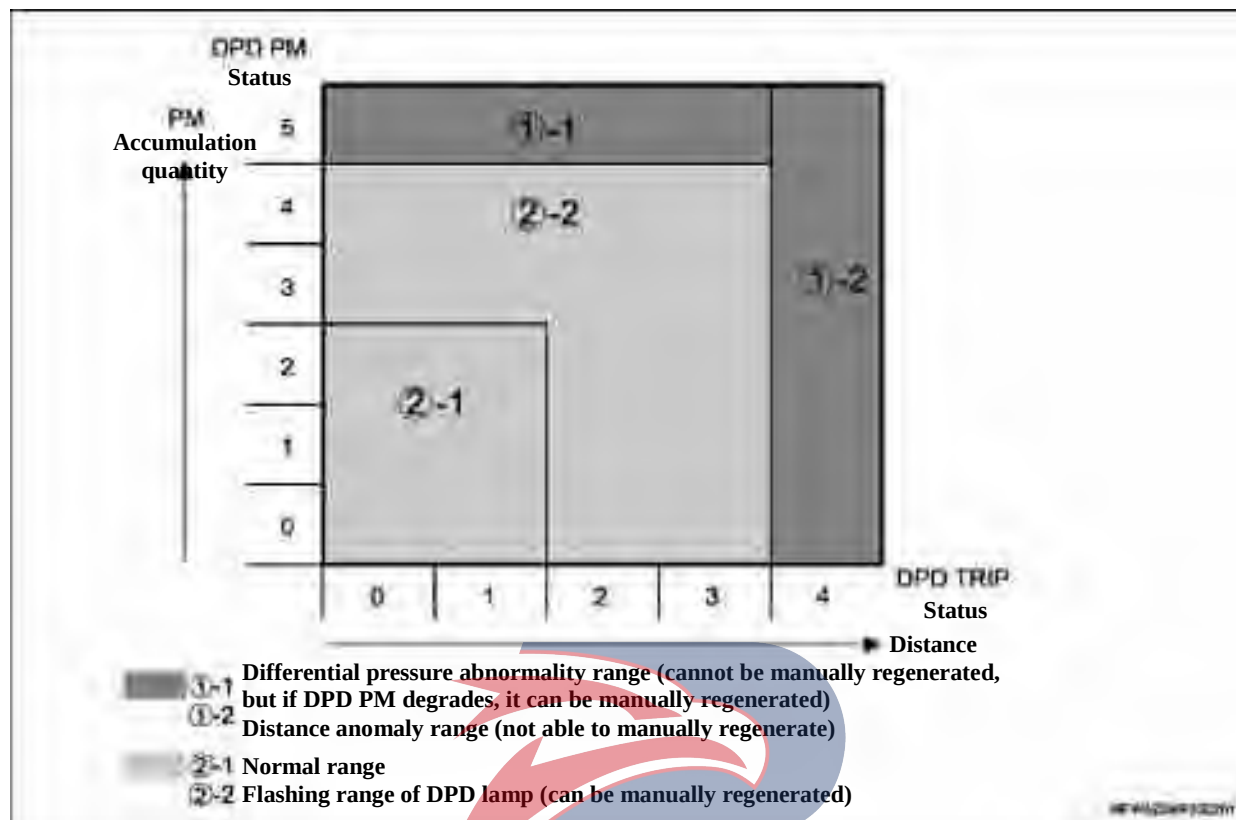
Step	Operation content	Datum value	YES	NO
1	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 2	—
2	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 3	Go to step 6
3	1. Check exhaust temperature sensor 1 for any abnormalities, such as damage. 2. Replace the exhaust temperature sensor 1 if any abnormality is detected. Is the disposal completed?	—	Go to step 4	—
4	1. Install the DPD. 2. Refer to ECM for DPD reactivation and install the DPD reactivation ECM to reactivate DPD. Is the disposal completed?	—	Go to step 5	—
5	Confirm the status of DPD. 1. Refer to DPD status and check whether the DPD status is in the normal range with the scanning tool. Is the status within the normal range?	—	Go to step 8	Go to step 6
6	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. 2. Run the test. 3. Confirm DTC. Is there any DTC detected?	—	Go to step 7	Go to step 8
7	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 8	—
8	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About exhaust temperature sensor

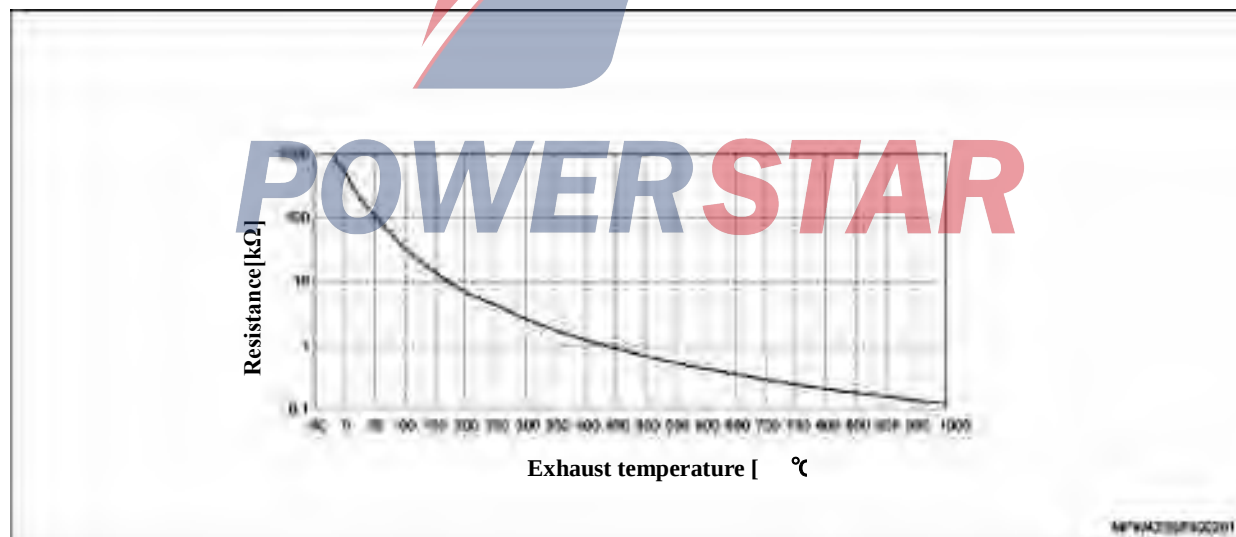


POWERSTAR

About DPD status



About characteristics of the exhaust temperature sensor (Reference)



About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front

temperature/DPD filter front temperature both reach 180°C, cool down (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.

4. Check the differential pressure when the DPD filter front temperature reaches 160°C with the scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.

7. Remove the ECM and install the DPD reactivation ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

mark:

When interrupting DPD regeneration, the DPD

regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECU have generated DTCs. Clear the DTC when generating the DTC.

Important:

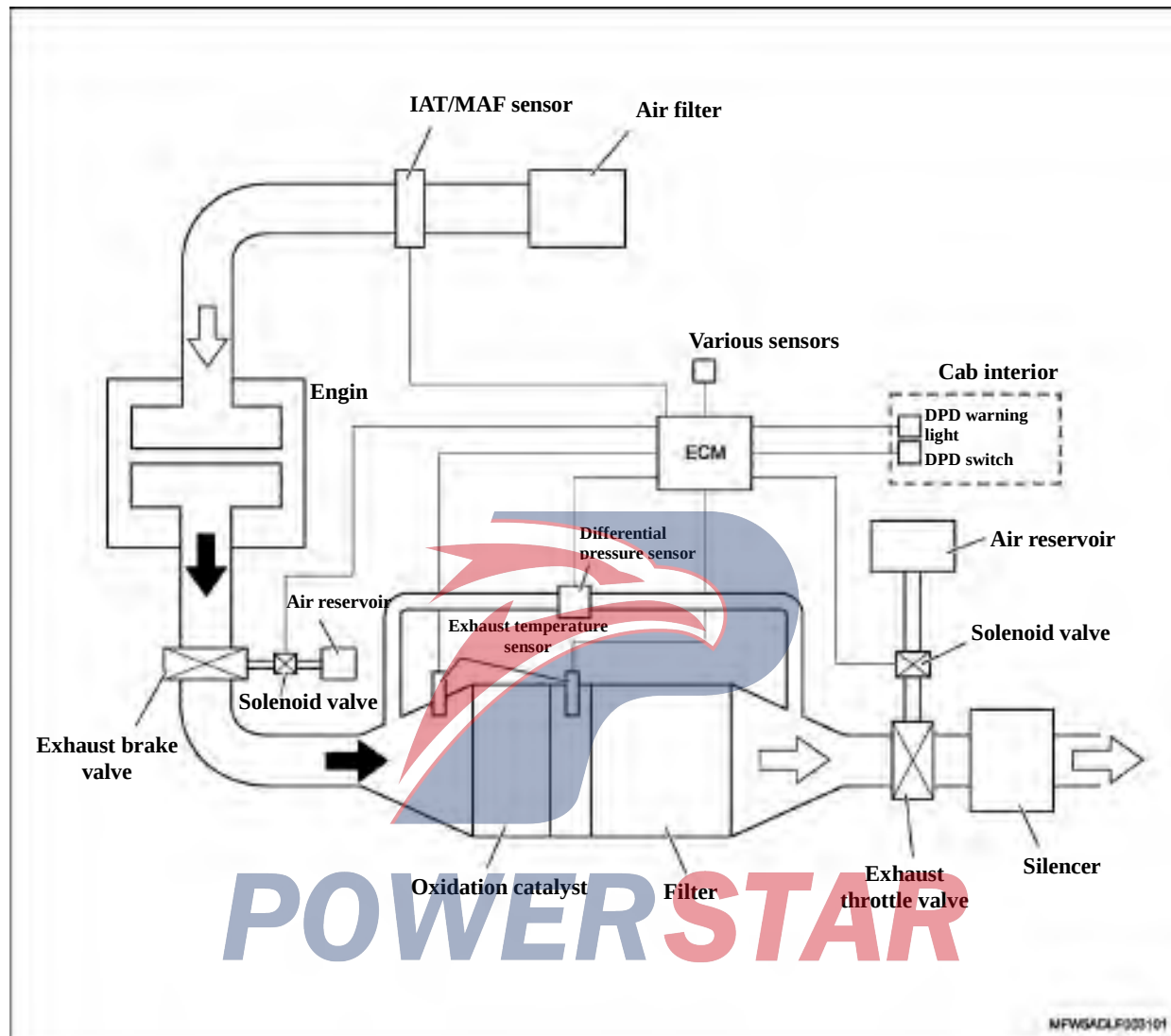
When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

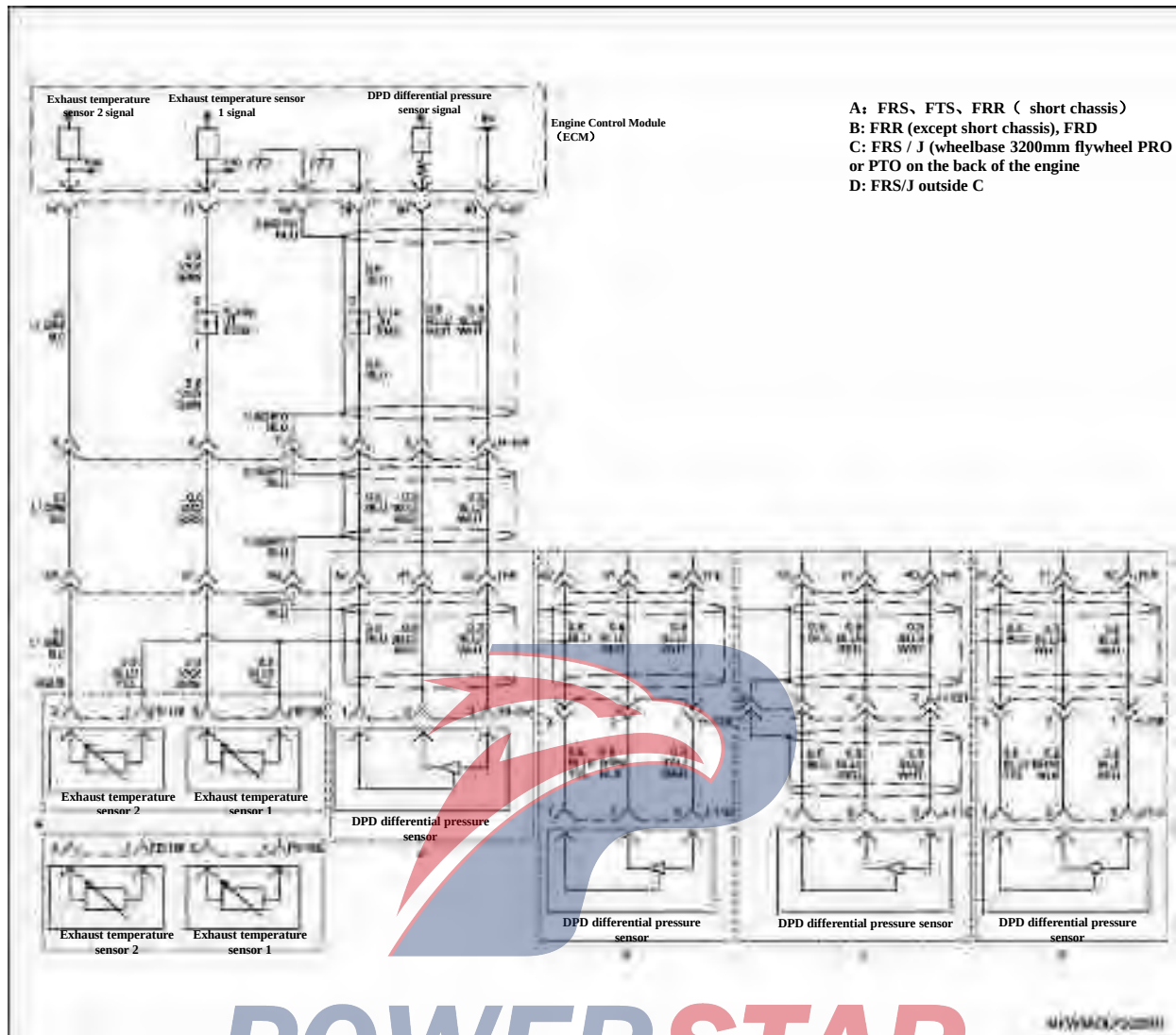
After the DPD is regenerated, check the pressure difference of the DPD. When the pressure difference is above the reference value, refer to the “Dust Removal” in “DPD Function Check” and clean the DPD filter with the dust removal device.

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DTC: P1449 (Flash code 145)

DPD Exhaust gas temperature 2 (before the oxidation catalyst) high temperature abnormal





Cycle Description

The exhaust temperature sensor 2 is set for detecting the temperature in the DPD. The sensor is installed at the inlet of the DPD catalyst. The exhaust temperature sensor is a thermistor, and the internal resistance of the sensor changes due to temperature changes. When the temperature is high, the resistance value is low, and when the temperature is low, the resistance value is high. ECM is connected between the exhaust temperature sensor and signal circuit and GND circuit to apply 5V power voltage to the signal circuit. ECM can detect the exhaust temperature sensor temperature variation and the variation of voltage due to the resistance change (the voltage is low if the temperature is high and high if the temperature is low). ECM will set DTC if detecting the signal of exhaust temperature sensor 2 abnormal.

The main fault phenomenon

- DPD regeneration stop
- Output power insufficient

Preconditions for DTC setting

- DTC P1630, P1633, P0546 and P0547 have not been detected.
- Vehicles equipped with DPD

Setting conditions for DTC.

The exhaust temperature sensor 2 is at an abnormal high temperature.

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1449 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault

(resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data

display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

9. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status

10. After the DPD reactivation, check the function of DPD in step 12. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

12. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

When coming back to the store

4. When the pressure difference is higher than the reference value, use a dust removal device to clean the DPD filter.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Confirm whether multiple DTC is detected. (the temperature sensor series 5V series circuits etc.) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Confirm DTC, access to [corresponding DTC inspection method] to execute repair and then go to step 3	Go to step 3
3	Check the exhaust temperature sensor 2 . 1. Check harness, connector and mounting status of exhaust temperature sensor 2. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 4	—

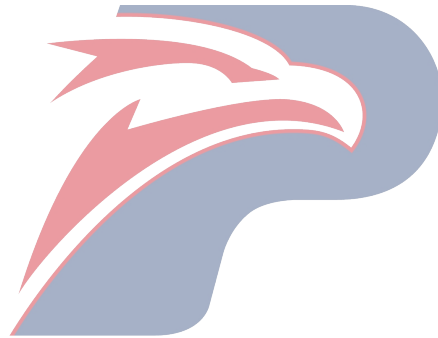
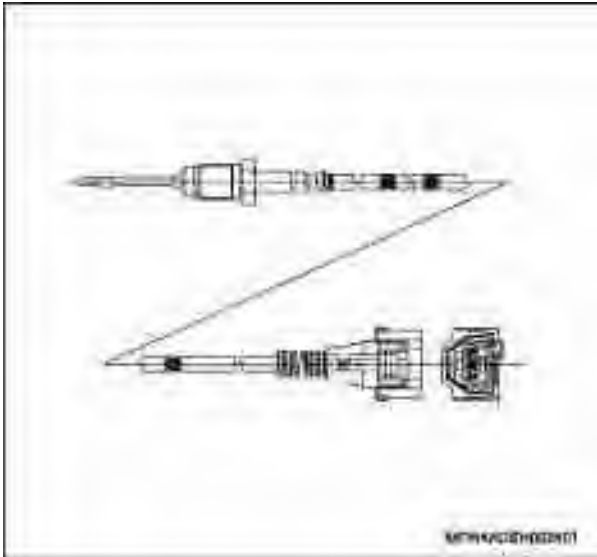
Step	Operation content	Datum value	YES	NO
4	Check the performance of exhaust temperature sensor 2. 1. Remove the wire harness from the ECM. 2. Install the power distribution cabinet. 3. Engage the gear to the neutral position. 4. Start the engine and fully open the accelerator for more than 2 minutes. 5. Stop the engine. 6. Use the scan tool to confirm the temperature. 7. Use the distribution cabinet to measure the resistance. (refer to sensor characteristic diagram and measure the resistance on 2 temperature spots) Mark: In the process where the temperature of the DPD does not drop, the operation starts immediately after the engine is stopped. Is the measurement reference value?	—	Go to step 6	Go to step 5
5	Replace exhaust temperature sensor 2. Is the disposal completed?	—	Go to step 11	—
6	Check the DPD filter. 1. Check for air leak, looseness, abnormal noise, fracture, dent, abnormal installation status etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 11	Go to step 9
9	Confirm the status of DPD. 1. Install the DPD. 2. Refer to DPD status and check whether the DPD status is in the normal range with the scanning tool. Is the status within the normal range?	—	Go to step 12	Go to step 10
10	Refer to ECM for DPD reactivation and use the DPD reactivation ECM to reactivate DPD. Is regeneration completed?	—	Go to step 12	Go to step 11
11	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 12	—
12	Perform a functional check for DPD. Is the disposal completed?	—	Confirm the repair conditions	Access to [DPD function check

When coming back to the store

When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

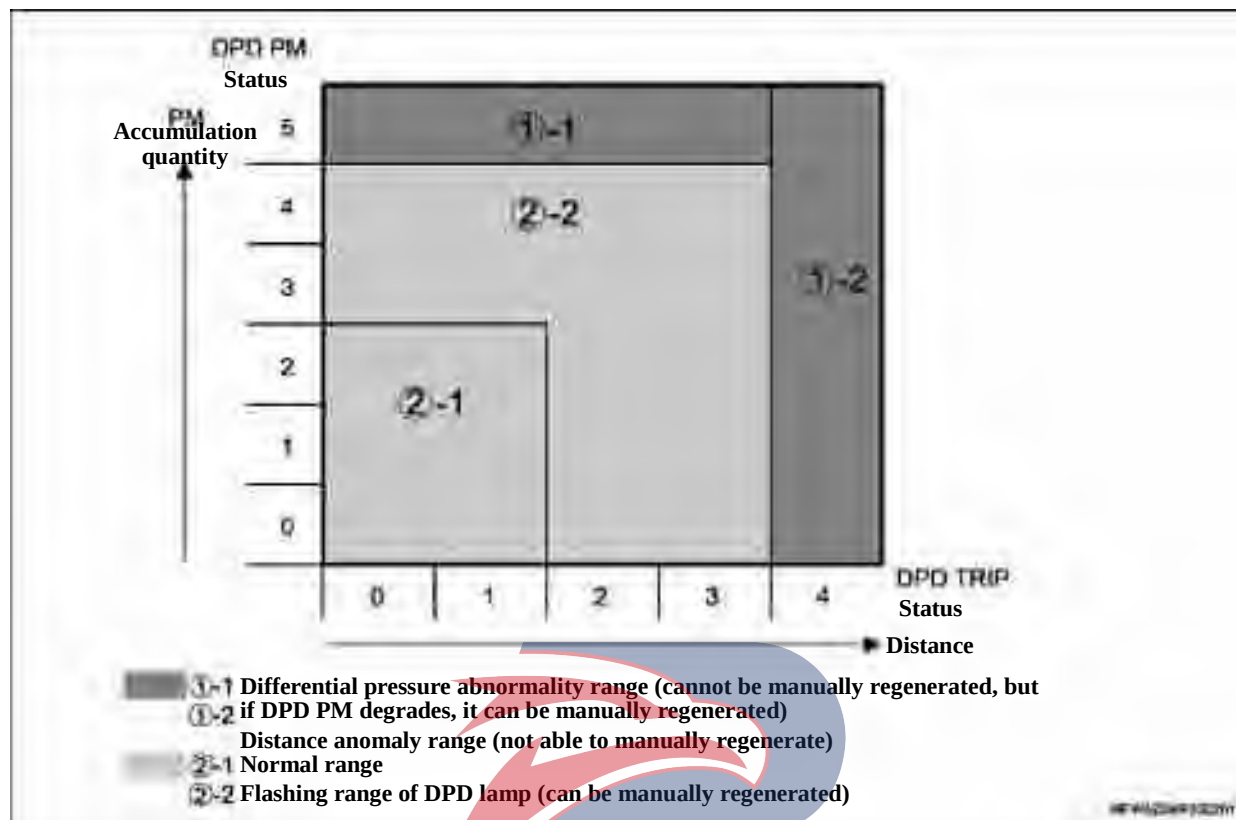
Step	Operation content	Datum value	YES	NO
1	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 2	—
2	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 3	Go to step 6
3	1. Check exhaust temperature sensor 2 for any abnormalities, such as damage. 2. Replace the exhaust temperature sensor 2 if any abnormality is detected. Is the disposal completed?	—	Go to step 4	—
4	1. Install the DPD. 2. Refer to ECM for DPD reactivation and install the DPD reactivation ECM to reactivate DPD. Is the disposal completed?	—	Go to step 5	—
5	Confirm the status of DPD. 1. Refer to DPD status and check whether the DPD status is in the normal range with the scanning tool. Is the status within the normal range?	—	Go to step 8	Go to step 6
6	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. 2. Run the test. 3. Confirm DTC. Is there any DTC detected?	—	Go to step 7	Go to step 8
7	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 8	—
8	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About exhaust temperature sensor

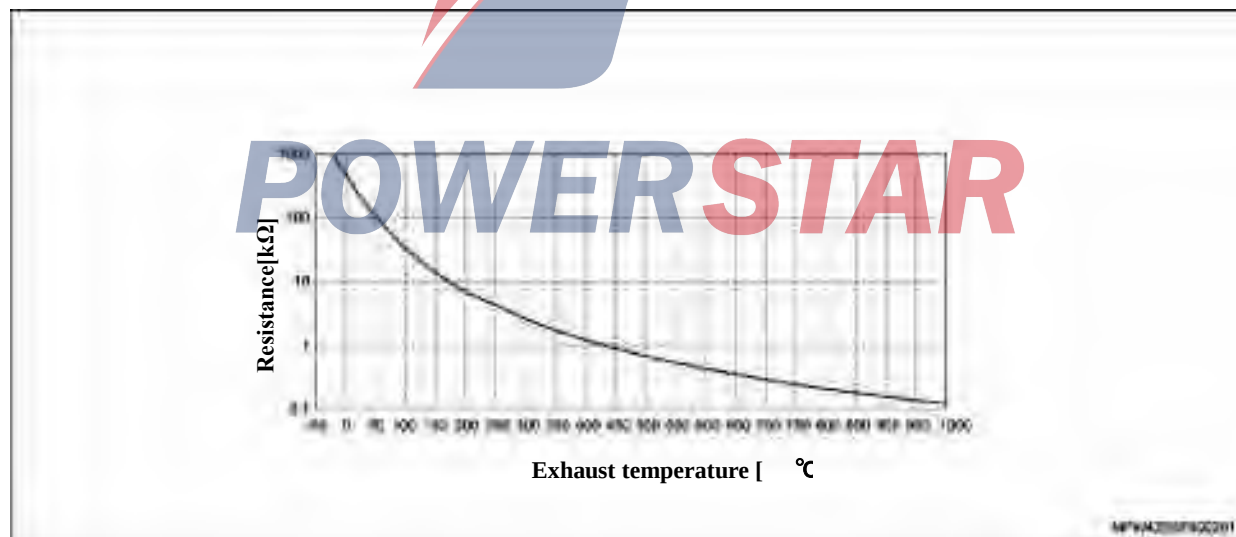


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About DPD status



About characteristics of the exhaust temperature sensor (Reference)



About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front

temperature/DPD filter front temperature both reach 180 °C, stop the engine by idling (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.

4. Check the differential pressure when the DPD filter front temperature reaches 160 °C with the scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.

7. Remove the ECM and install the DPD reactivation ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD

regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECUs have generated DTCs. Clear the DTC when generating the DTC.

Important:

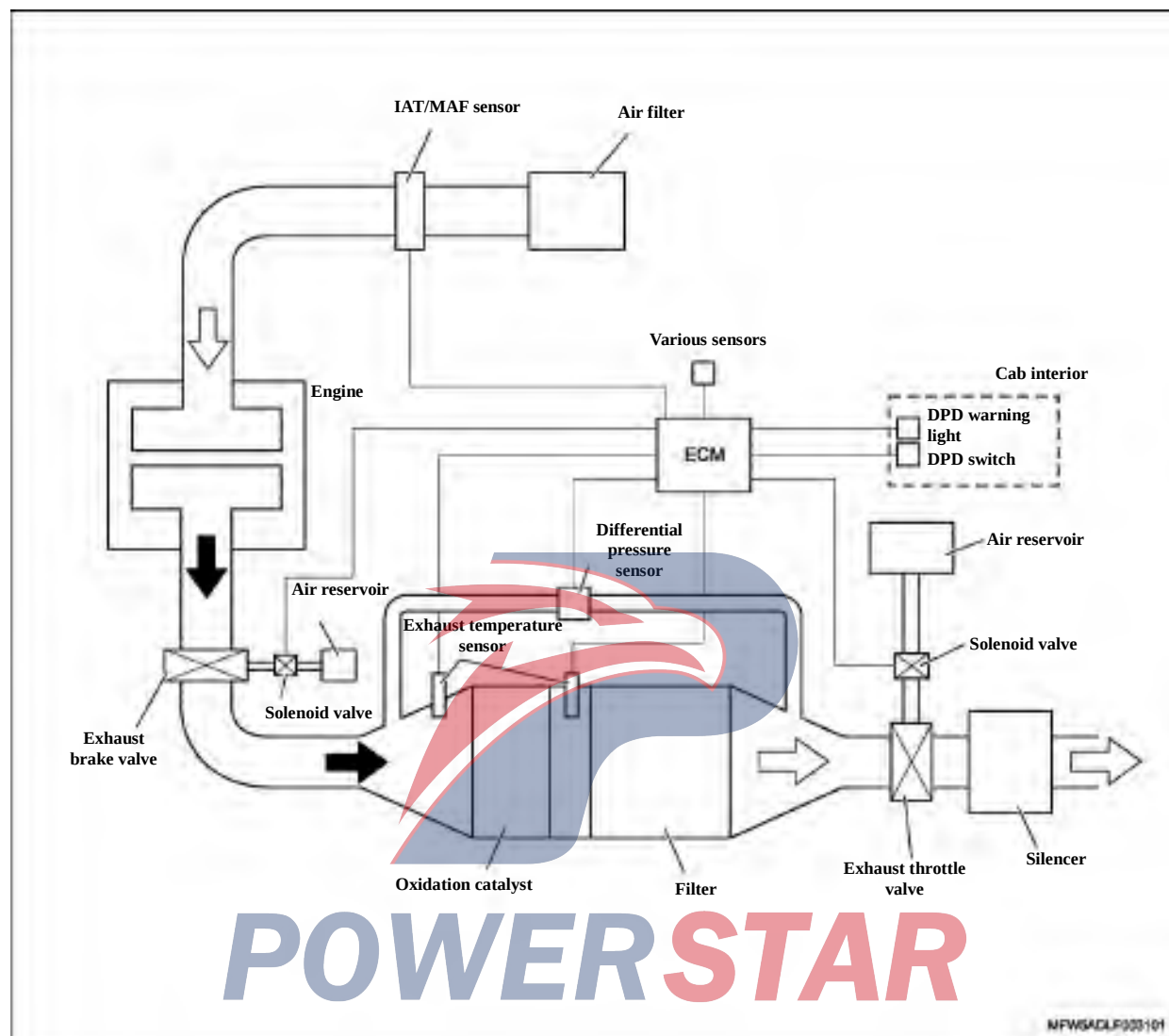
When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

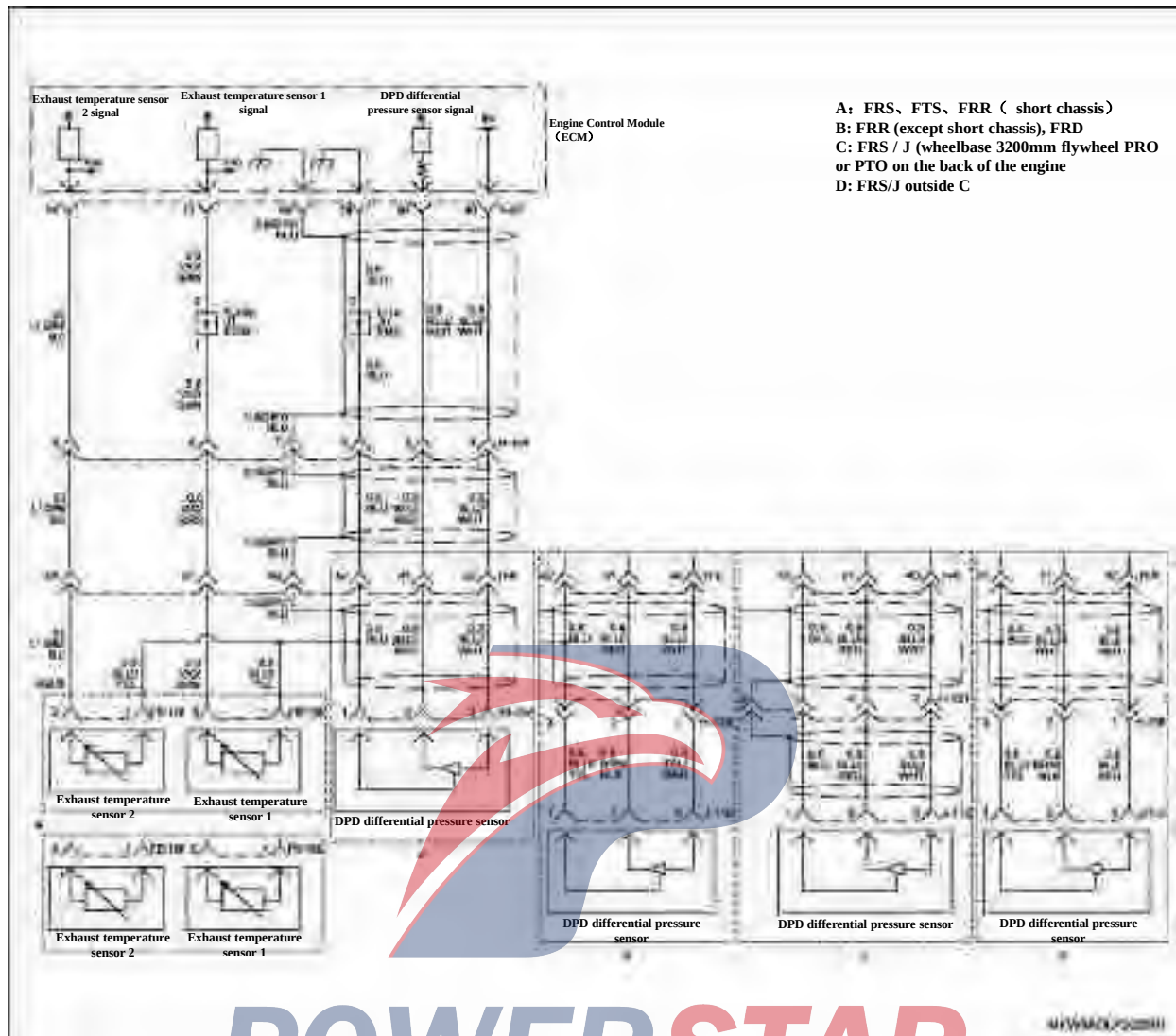
After the DPD is regenerated, check the pressure difference of the DPD. When the pressure difference is above the reference value, refer to the “Dust Removal” in “DPD Function Check” and clean the DPD filter with the dust removal device.

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DTC: P1453 (Flash code 139)

DPD regeneration timed out too much





Cycle Description

ECM can store the unfinished number of DPD reactivation. When the unfinished reactivation number is detected more than twice, it will determine DPD abnormal and DTC will be set.

The main fault phenomenon

- DPD regeneration stop
- Output power insufficient

Preconditions for DTC setting

- Regeneration implementation
- Vehicles equipped with DPD

Setting conditions for DTC.

The number of unfinished regenerations continues to occur.

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.

- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table.

3. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status

13. After the DPD reactivation, check the function of DPD in step 17. After confirming DPD differential pressure is above the reference value, clean DPD

filter with the dust removal device.

17. Implement functional diagnostics to verify that the DPD is working properly. In addition, it is confirmed that the data of the scanning tool shows the value of the internal [DPD TRIP state] and the [DPD PM state], and confirms which position is in the state with reference to the state diagram recorded. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

When coming back to the store

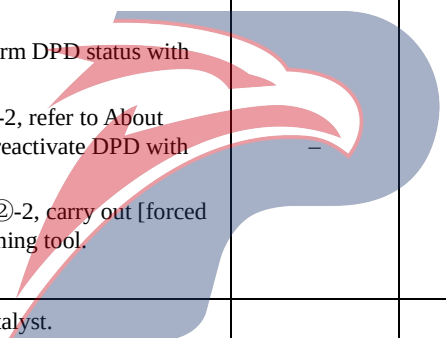
9. After the DPD reactivation, check the function of DPD in step 13. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

13. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Inquire the customer and confirm the DPD reactivation operation Is the disposal completed?	—	Go to step 3	
3	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 4	

Step	Operation content	Datum value	YES	NO
4	Explain the status to the customer. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD Is the disposal completed?	—	Go to step 5	—
5	The DTC is cleared and the DPD data is reset. 1. Clear DTC and reset DPD data with the scanning tool. Is the disposal completed?	—	Go to step 6	—
6	Check the DPD. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 16	Go to step 9
9	Check the exhaust brake and DPD exhaust throttle. 1. Install the DPD. 2. Use the scan tool to test [Exhaust Brake Control] and [Exhaust Throttle Control]. 3. Check vacuum piping for leaks, etc. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 10	—

Step	Operation content	Datum value	YES	NO
10	Reactivation DPD. 1. Refer to DPD status and confirm DPD status with the scanning tool. 2. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 17	Go to step 11
11	Check exhausts gas temperature sensors 1 and 2. 1. Ensure there is no air leak, body breakage and check the harness and joint installation status. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 12	—
12	Refer to Confirmation of fuel system to check the fuel system. Is the disposal completed?	—	Go to step 13	Access to [confirmation of fuel System]
13	Reactivation DPD. 1. Refer to DPD status and confirm DPD status with the scanning tool. 2. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 17	Go to step 14
14	1. Replace the oxidation type catalyst. Is the disposal completed?	—	Go to step 15	—



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Step	Operation content	Datum value	YES	NO
15	Confirm the pressure difference of the DPD. 1. Start the engine. 2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C. Mark: When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180 °C, increase the engine speed to 1500rpm). 3. Increase the revolution of engines to the maximum revolution without load. 4. Check the differential pressure when the DPD filter front temperature reaches 250 °C with the scan tool. Mark: If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault. Is the measurement above reference value?	3.1kPa	Go to step 16	Go to step 17
16	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 17	—
17	Check the function of DPD. Is the disposal completed?	—	Go to step 18	Access to [DPD function check]
18	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

When coming back to the store

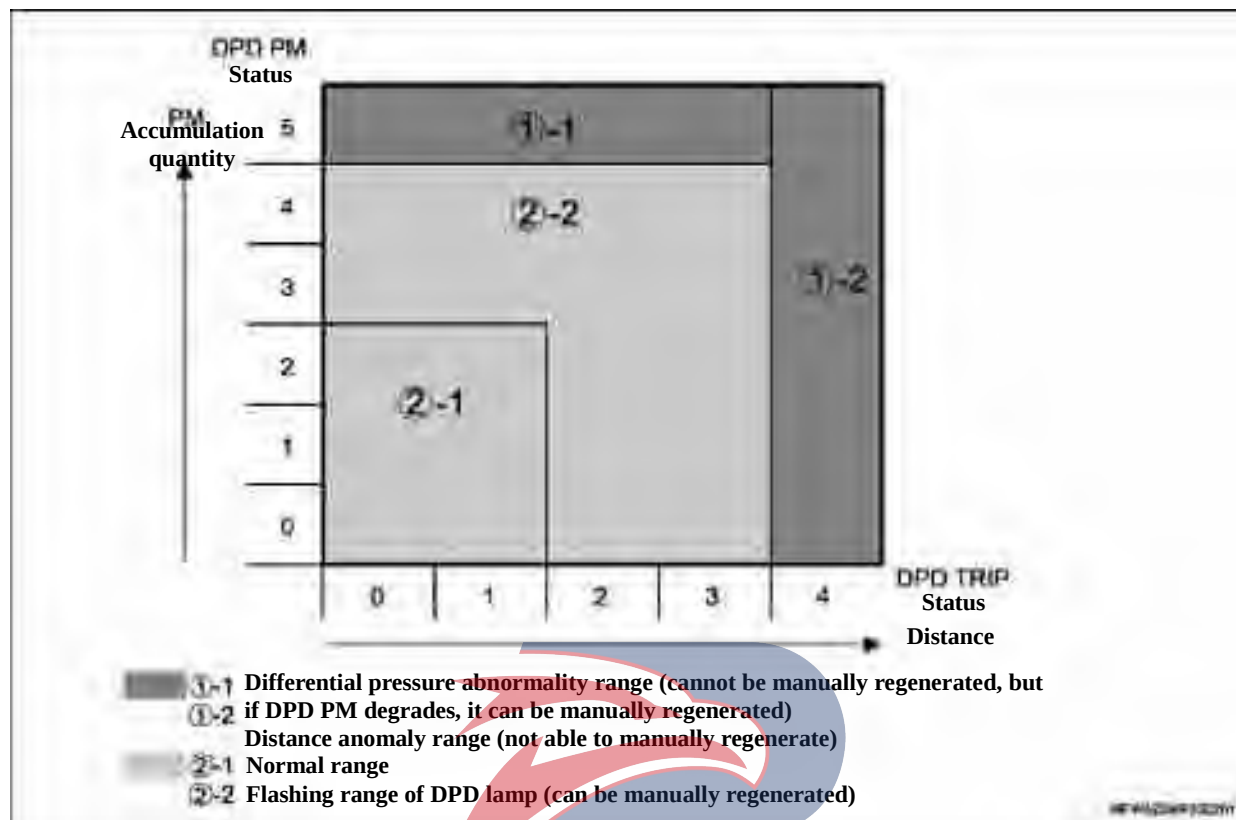
When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

Step	Operation content	Datum value	YES	NO
1	Inquire the customer and confirm the operation. Is the disposal completed?	—	Go to step 2	—
2	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 3	—

Step	Operation content	Datum value	YES	NO
3	Explain the status to the customer. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD Is the disposal completed?	—	Go to step 4	—
4	The DTC is cleared and the DPD data is reset. 1. Clear DTC and reset DPD data with the scanning tool. Is the disposal completed?	—	Go to step 5	—
5	Check the DPD filter. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 6	—
6	Refer to Confirmation of fuel system to check the fuel system. Is the disposal completed?	—	Go to step 7	Access to [confirmation of fuel System]
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 11	Go to step 9
9	Reactivation DPD. 1. Install the DPD. 2. Refer to DPD status and confirm DPD status with the scanning tool. 3. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 13	Go to step 10

Step	Operation content	Datum value	YES	NO
10	Check the DPD filter. 1. Check for signs of filter damage or damage. Is the filter abnormal?	—	Go to step 11	Go to step 12
11	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 13	—
12	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Important: When the ECM is changed and the DPD is not replaced, the forced regeneration of DPD can be implemented, so that ECM and DPD can be set to the same state. Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 13	—
13	Check the function of DPD. Is the disposal completed?	—	Go to step 14	Access to [DPD function check]
14	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About DPD status



About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180 (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.
4. Check the differential pressure when the DPD filter front temperature reaches 160 scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.

7. Remove the ECM and install the DPD reactivation ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECUs have generated DTCs. Clear the DTC when generating the DTC.

Important:

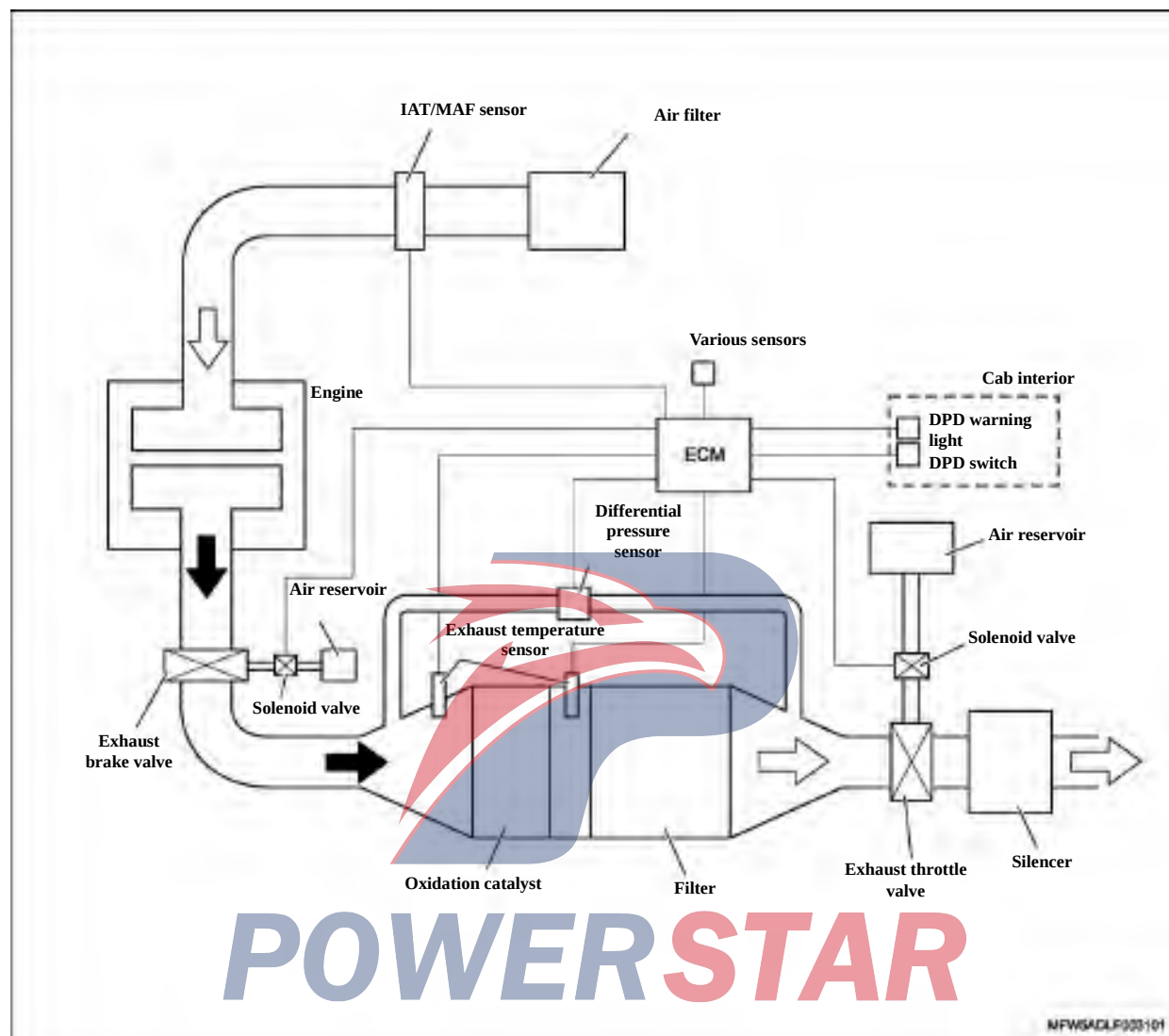
When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

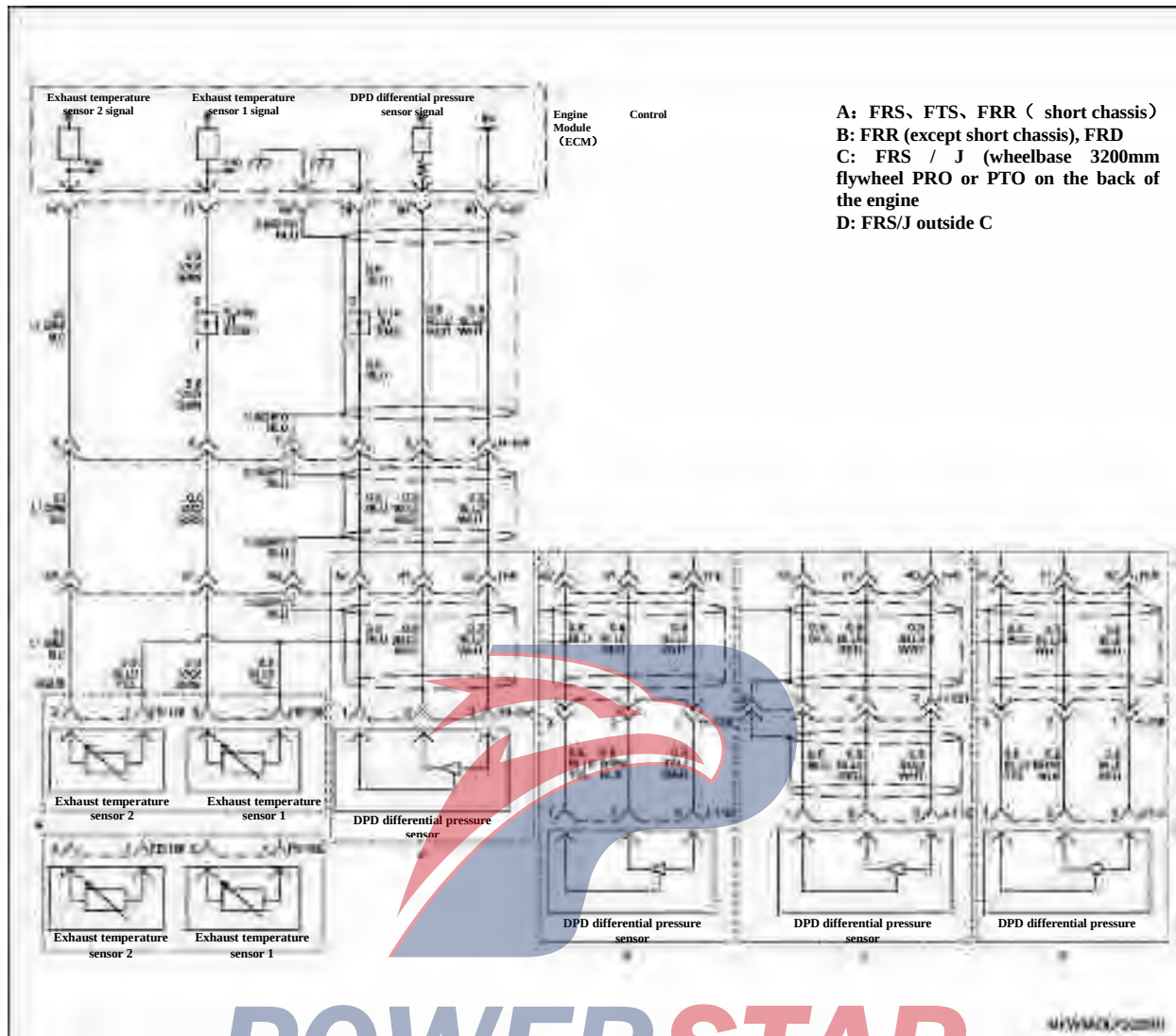
After the DPD is regenerated, check the pressure difference of the DPD. When the pressure difference is above the reference value, refer to the “Dust Removal” in “DPD Function Check” and clean the DPD filter with the dust removal device.



DTC: P1454 (Flash code 131)

Excessive DPD PM Capture





Cycle Description

The differential pressure sensor is set to detect the pressure difference in the DPD. The differential pressure sensor is installed to the DPD. ECM often uses the signal from differential pressure sensor to detect the differential pressure in DPD and detect the accumulation of PM. When determining PM accumulation disabling DPD reactivation, this DTC will be set.

The main fault phenomenon

- DPD regeneration stop
- Output drop

Preconditions for DTC setting

- Vehicles equipped with DPD

Setting conditions for DTC.

The cumulative amount of PM is judged to be at a level at which DPD cannot be regenerated.

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector

- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

3. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status

10. After the DPD reactivation, check the function of DPD in step 12. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

12. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

When coming back to the store

2. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status

9. After the DPD reactivation, check the function of DPD in step 12. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

12. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Inquire the customer and confirm the operation. Is the disposal completed?	—	Go to step 3	—

Step	Operation content	Datum value	YES	NO
3	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 4	—
4	Describe to the customer the status of the DPD and how to use the DPD vehicle and give instructions accordingly. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD 4. About the flashing of the warning light 5. Regeneration method of DPD Is the disposal completed?	—	Go to step 5	—
5	Perform DTC clear and DPD data clear. 1. Clear DTC and DPD data with the scanning tool. Is the disposal completed?	—	Go to step 6	—
6	When the following DTC are present, DTC should be processed first. 1. P1453 or P1471 2. P1471 or P1454 3) SCV feedback current is more than 2400mA Is the disposal completed?	—	Go to step 7	—
7	Check the DPD. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 8	—
8	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 9	—
9	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 11	Go to step 10

Step	Operation content	Datum value	YES	NO
10	Reactivation DPD. 1. Install the DPD. 2. Refer to DPD status and confirm DPD status with the scanning tool. 3. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 12	Go to step 11
11	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 12	—
12	Check the function of DPD. Is the disposal completed?	—	Go to step 13	Access to [DPD function check]
13	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

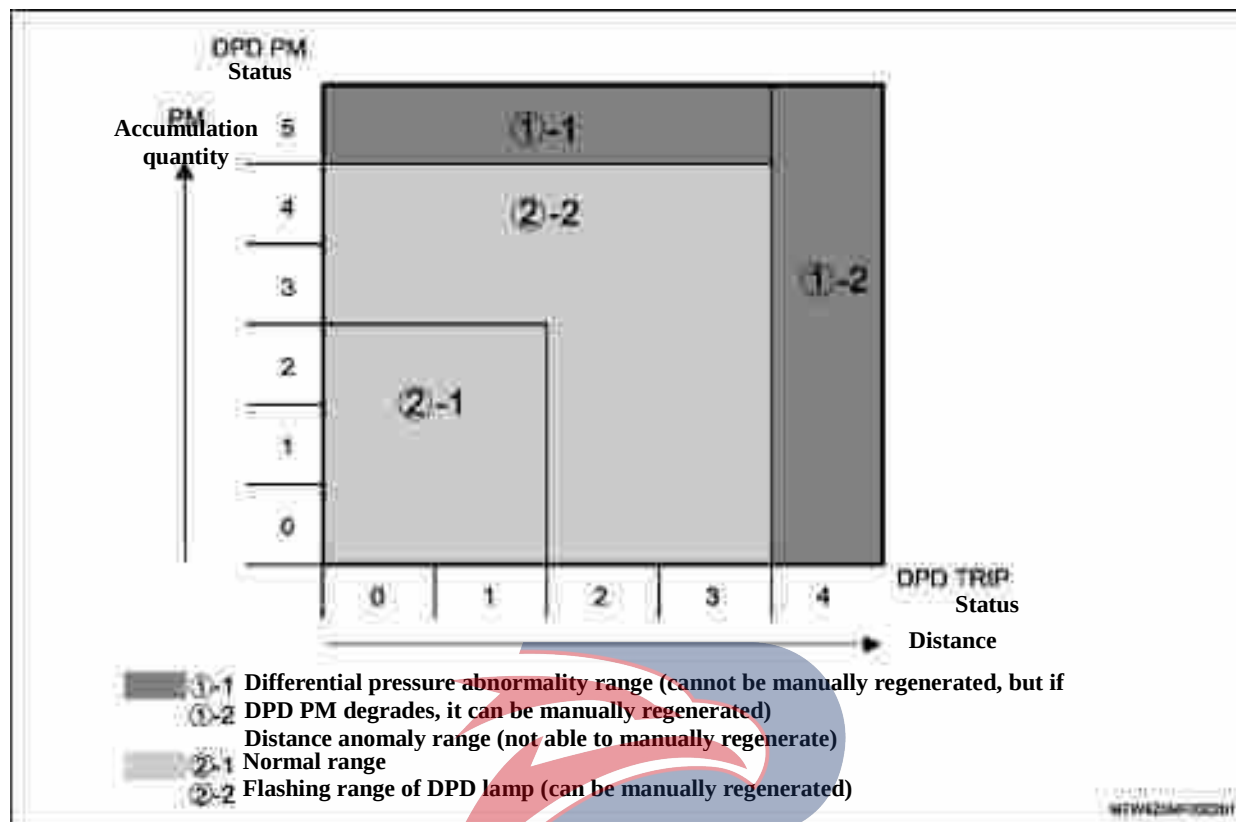
When coming back to the store

When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

Step	Operation content	Datum value	YES	NO
1	Inquire the customer and confirm the operation. Is the disposal completed?	—	Go to step 2	—
2	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 3	—
3	Describe to the customer the status of the DPD and how to use the DPD vehicle and give instructions accordingly. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD 4. About the flashing of the warning light 5. Regeneration method of DPD Is the disposal completed?	—	Go to step 4	—
4	Perform DTC clear and DPD data clear. 1. Clear DTC and DPD data with the scanning tool. Is the disposal completed?	—	Go to step 5	—

Step	Operation content	Datum value	YES	NO
5	Check the DPD filter. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 6	—
6	Refer to Confirmation of fuel system to check the fuel system. Is the disposal completed?	—	Go to step 7	Access to [confirmation of fuel System]
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 11	Go to step 9
9	Reactivation DPD. 1. Install the DPD. 2. Refer to DPD status and confirm DPD status with the scanning tool. 3. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 12	Go to step 10
10	Check the DPD filter. 1. Check for signs of filter damage or damage. Is the filter abnormal?	—	Go to step 11	Go to step 12
11	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 12	—
12	Check the function of DPD. Is the disposal completed?	—	Go to step 13	Access to [DPD function check]
13	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About DPD status



About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180 (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.

4. Check the differential pressure when the DPD filter front temperature reaches 160 scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.

7. Remove the ECM and install the DPD reactivation ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECUs have generated DTCs. Clear the DTC when generating the DTC.

Important:

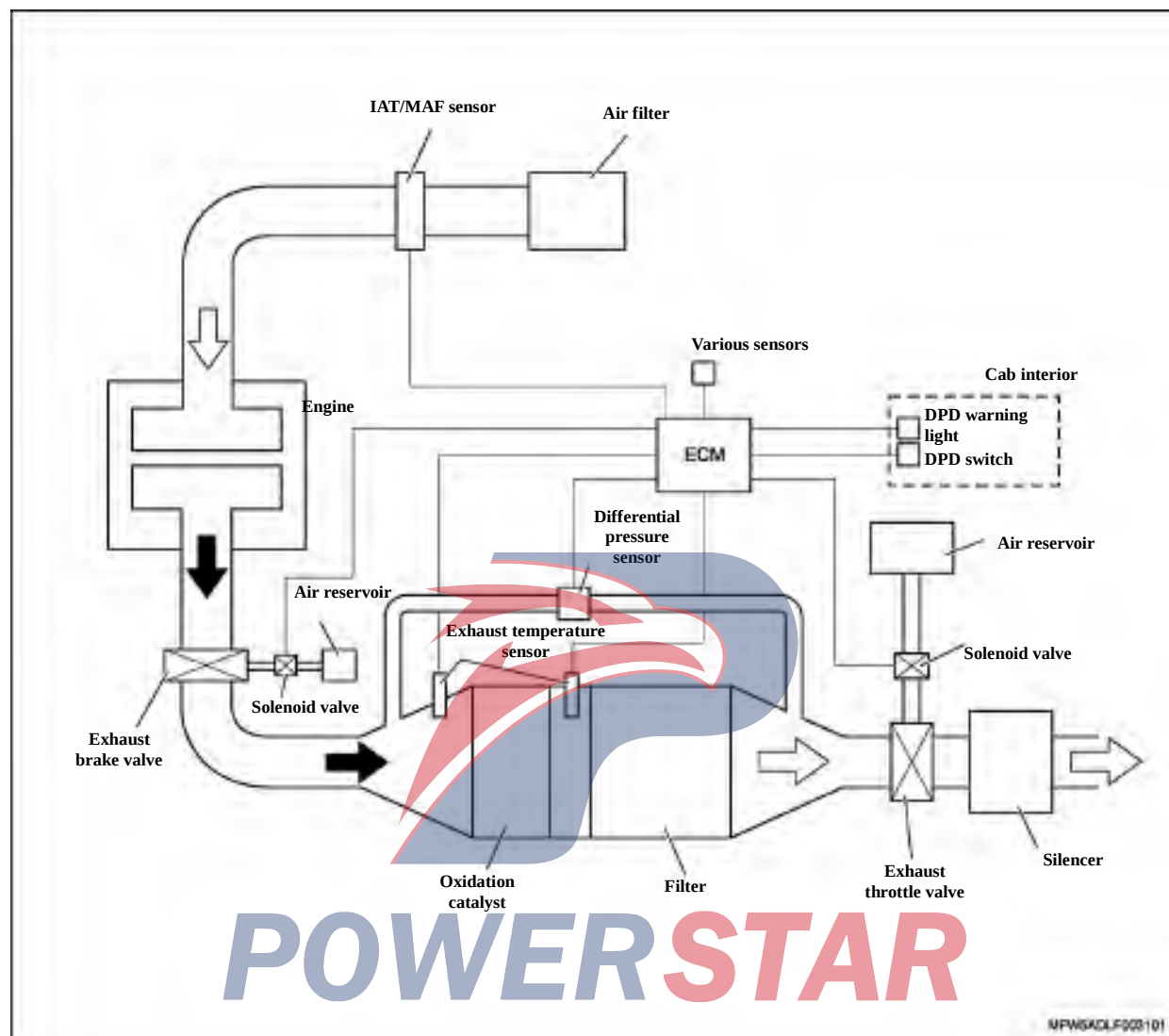
When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

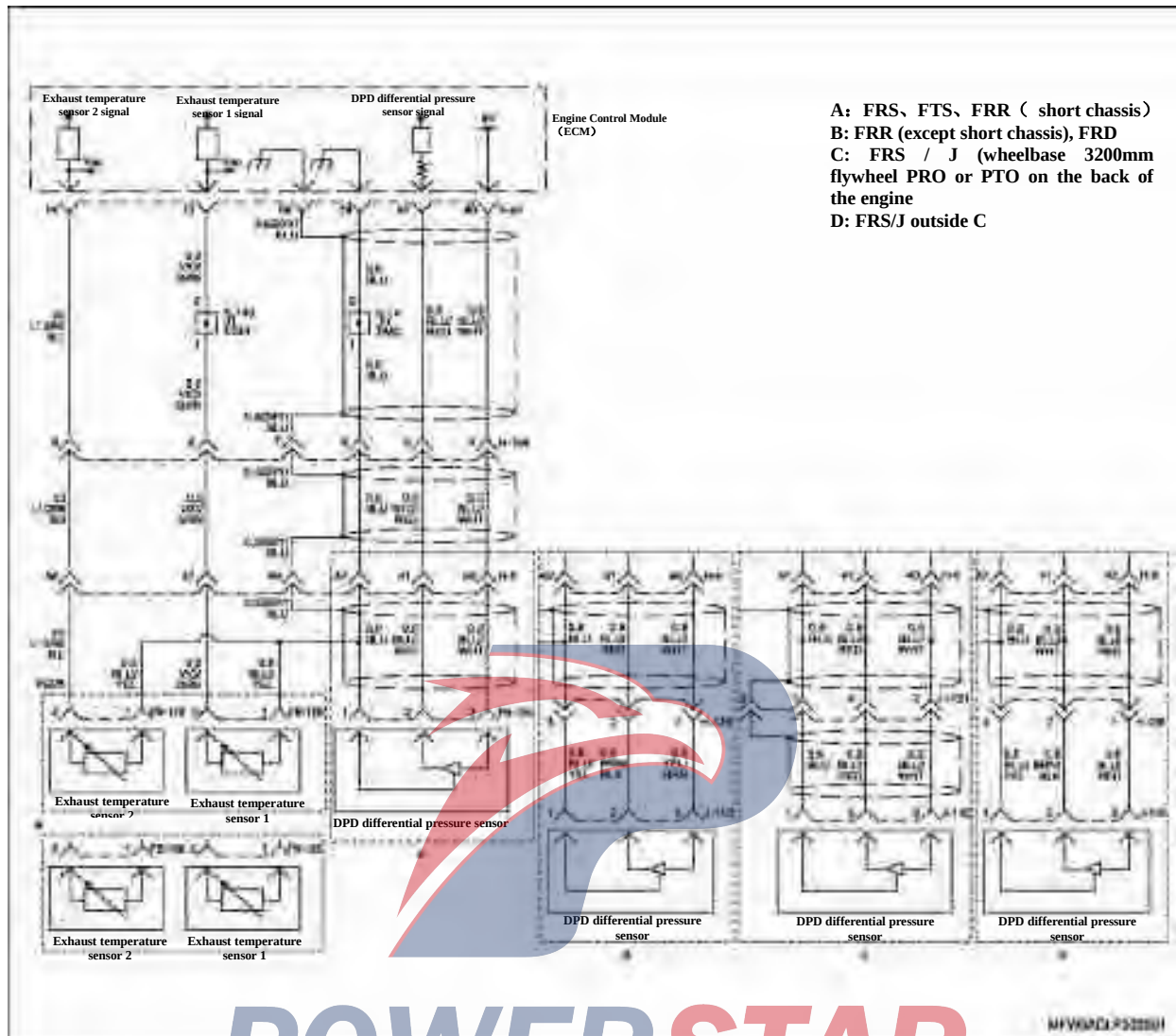
After the DPD is regenerated, check the pressure difference of the DPD. When the pressure difference is above the reference value, refer to the “Dust Removal” in “DPD Function Check” and clean the DPD filter with the dust removal device.



DTC: P1471 (Flash code 149)

DPD purification judgment abnormal





Cycle Description

The differential pressure sensor is set to detect the pressure difference in the DPD. ECM often uses the signal from differential pressure sensor to detect the differential pressure in DPD and detect the accumulation of PM. After the DPD regeneration is completed, the DTC is set when it is judged that the pressure before and after the DPD is not reduced.

The main fault phenomenon

- DPD regeneration stop
- Output drop

Preconditions for DTC setting

- DTC P1630, P1633, P0470, P0471 and P0472 have not been detected.
- Vehicles equipped with DPD

Setting conditions for DTC.

The regeneration of DPD continues to manifest itself in a bad state.

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table.

3. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing

according to the inspection method of each status

10. After the DPD reactivation, check the function of DPD in step 18. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

18. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

When coming back to the store

9. After the DPD reactivation, check the function of DPD in step 14. After confirming DPD differential pressure is above the reference value, clean DPD filter with the dust removal device.

14. Implement functional diagnostics to verify that the DPD is working properly. Check the values of [DPD TRIP Status] and [DPD PM Status] in the scan tool data display and confirm the status with reference to the status diagram. After confirming the status, perform the processing according to the inspection method of each status and confirm the pressure difference of the DPD. When the pressure difference is greater than or equal to the reference value, use a dust-removal device to clean the DPD filter.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	–	Go to step 2	Perform "OBD System Check" and go to step 2
2	Inquire the customer and confirm the operation. Is the disposal completed?	–	Go to step 3	–

Step	Operation content	Datum value	YES	NO
3	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 4	—
4	Explain the DPD status to the customer. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD Is the disposal completed?	—	Go to step 5	—
5	Perform DTC clear and DPD data clear. 1. Clear DTC and DPD data with the scanning tool. Is the disposal completed?	—	Go to step 6	—
6	Check the DPD filter. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. When detecting abnormal, clean or replace the oxidation type catalyst as required. Is the disposal completed?	—	Go to step 8	—
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 15	Go to step 9
9	Check the exhaust brake and DPD exhaust throttle. 1. Install the DPD. 2. Use the scan tool to test [Exhaust Brake Control] and [Exhaust Throttle Control]. 3. Check vacuum piping for leaks, etc. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 10	—

Step	Operation content	Datum value	YES	NO
10	Reactivation DPD. 1. Refer to DPD status and confirm DPD status with the scanning tool. 2. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 18	Go to step 11
11	Differential pressure sensor inspection and storage. 1. Check the hose and piping for breakage, falling off, air leak and main body breakage etc. 2. Set the key switch from the [OFF] position to the [ON] position. (The engine will not start at this time) 3. DPD alarm light lights out after about 30 seconds. 4. After confirming that the DPD warning lamp is off, set the key switch to [OFF] for more than 15 seconds. mark: After installation, please install the hose correctly and check for leaks. 5. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 12	—
12	Check the exhaust temperature sensor. 1. Ensure there is no air leak, body breakage and check the harness and joint installation status. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 13	—
13	Refer to Confirmation of fuel system to check the fuel system. Is the disposal completed?	—	Go to step 14	Access to [confirmation of fuel System]
14	Reactivation DPD. 1. Refer to DPD status and confirm DPD status with the scanning tool. 2. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 16	Go to step 15
15	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 16	—

Step	Operation content	Datum value	YES	NO
16	Confirm the pressure difference of the DPD. 1. Start the engine. 2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C. Mark: When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180°C, cool down the engine by idling (1500rpm). 3. Increase the revolution of engines to the maximum revolution without load. 4. Check the differential pressure when the DPD filter front temperature reaches 250 °C with scan tool. Mark: If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault. Is the measurement above reference value?	3.1kPa	Go to step 17	Go to step 18
17	Replace the oxidation catalyst. Is the disposal completed?	—	Go to step 18	—
18	Check the function of DPD. Is the disposal completed?	—	Go to step 19	Access to [DPD function check]
19	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

POWERSTAR

When coming back to the store

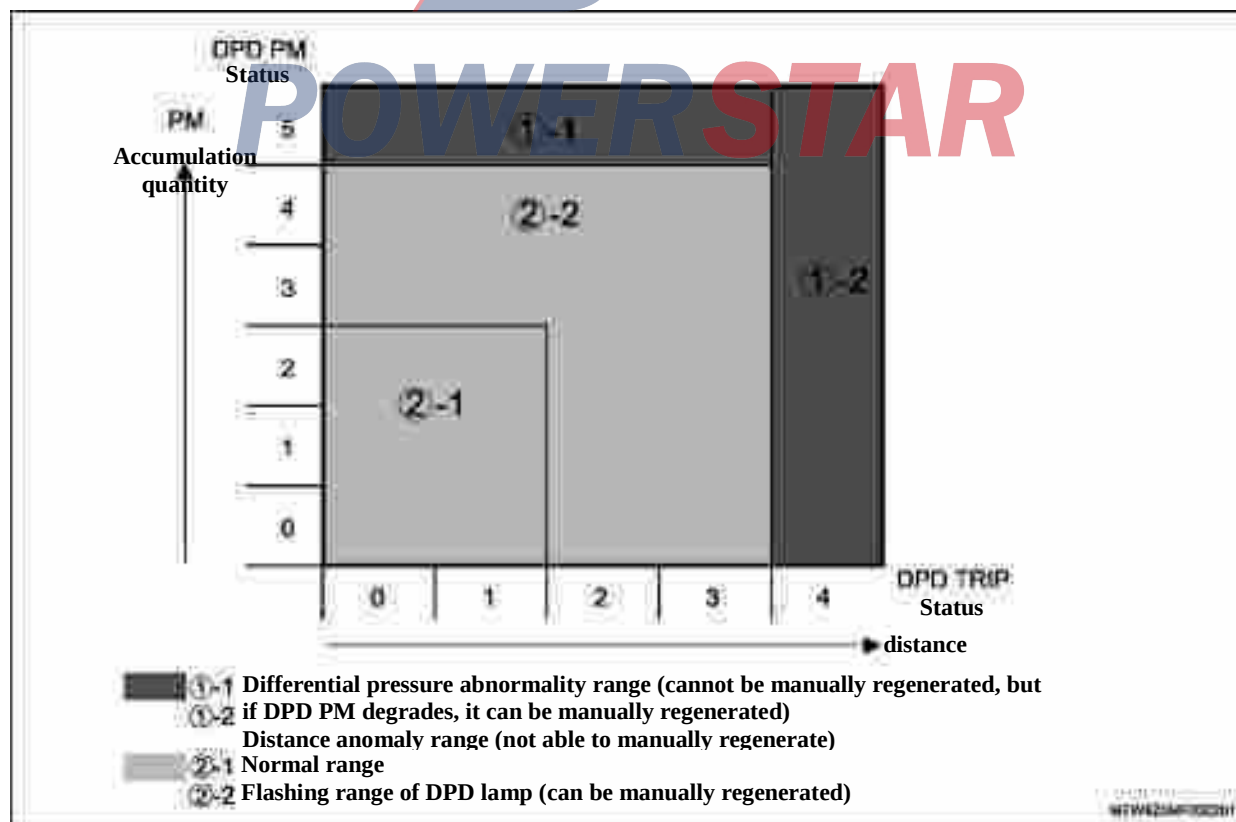
When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

Step	Operation content	Datum value	YES	NO
1	Inquire the customer and confirm the operation. Is the disposal completed?	—	Go to step 2	—
2	Confirm the status of DPD. 1. Refer to "About the Status of the DPD" and use the scan tool to confirm the status of the DPD. 2. Use the scan tool to confirm the display of [DPD differential pressure (high resolution)] in the data display. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 3	—
3	Explain the status to the customer. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD Is the disposal completed?	—	Go to step 4	—
4	Perform DTC clear and DPD data clear. 1. Clear DTC and DPD data with the scanning tool. Is the disposal completed?	—	Go to step 5	—
5	Check the DPD filter. 1. Check the DPD body for abnormalities such as air leaks, looseness, breakage, dents, and mounting conditions. 2. Confirm whether there is abnormality on the piping, connection, wiring harness, connection status and installation status of differential pressure sensor. 3. Check the exhaust temperature sensor's piping, fittings, wiring harness, connection status and installation status. 4. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 6	—
6	Refer to Confirmation of fuel system to check the fuel system. Is the disposal completed?	—	Go to step 7	Access to [confirmation of fuel System]
7	Oxidation catalyst inspection and cleaning. 1. Remove the DPD assembly and disassemble it. 2. Check the oxidized catalysts for dissolution or damage. 3. If abnormal detected, clear as needed. Is the disposal completed?	—	Go to step 8	—

Step	Operation content	Datum value	YES	NO
8	Check the DPD filter. 1. Check the DPD filter for dissolved damage, breakage and filter out-of-position. Is there an exception in the filter?	—	Go to step 12	Go to step 9
9	Reactivation DPD. 1. Install the DPD. 2. Refer to DPD status and confirm DPD status with the scanning tool. 3. If DPD status is ①-1 and ①-2, refer to About ECM for DPD reactivation and reactivate DPD with ECM. When DPD status is ②-1 and ②-2, carry out [forced DPD reactivation] with the scanning tool. Is regeneration completed?	—	Go to step 14	Go to step 10
10	Confirm the pressure difference of the DPD. 1. Start the engine. 2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C. mark: When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180°C, cool down to 150°C (1500rpm). 3. Increase the revolution of engines to the maximum revolution without load. 4. Check the differential pressure when the DPD filter front temperature reaches 250°C with the scanning tool. Mark: If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault. Is the measurement above reference value?	3.1kPa	Go to step 11	Go to step 13
11	Replace the oxidation catalyst. Is the disposal completed?	—	Go to step 14	—
12	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 14	—

Step	Operation content	Datum value	YES	NO
13	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Important: When the ECM is changed and the DPD is not replaced, the forced regeneration of DPD can be implemented, so that ECM and DPD can be set to the same state. Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 14	—
14	Check the function of DPD. Is the disposal completed?	—	Go to step 15	Access to [DPD function check]
15	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions

About DPD status



About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180 (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.
4. Check the differential pressure when the DPD filter front temperature reaches 160 °C with the scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.
7. Remove the ECM and install the DPD reactivation ECM.
8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be

cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECUs have generated DTCs. Clear the DTC when generating the DTC.

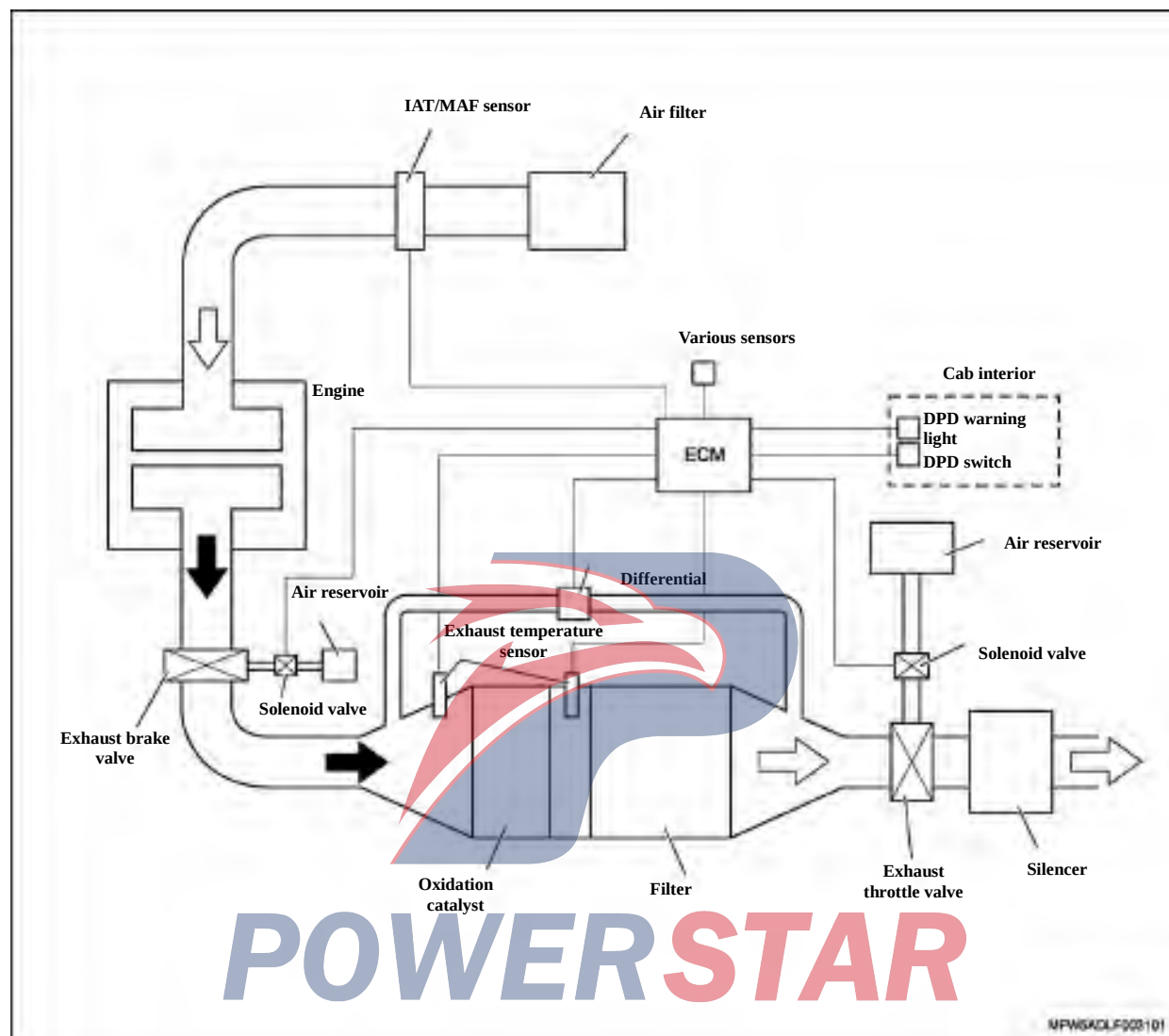
Important:

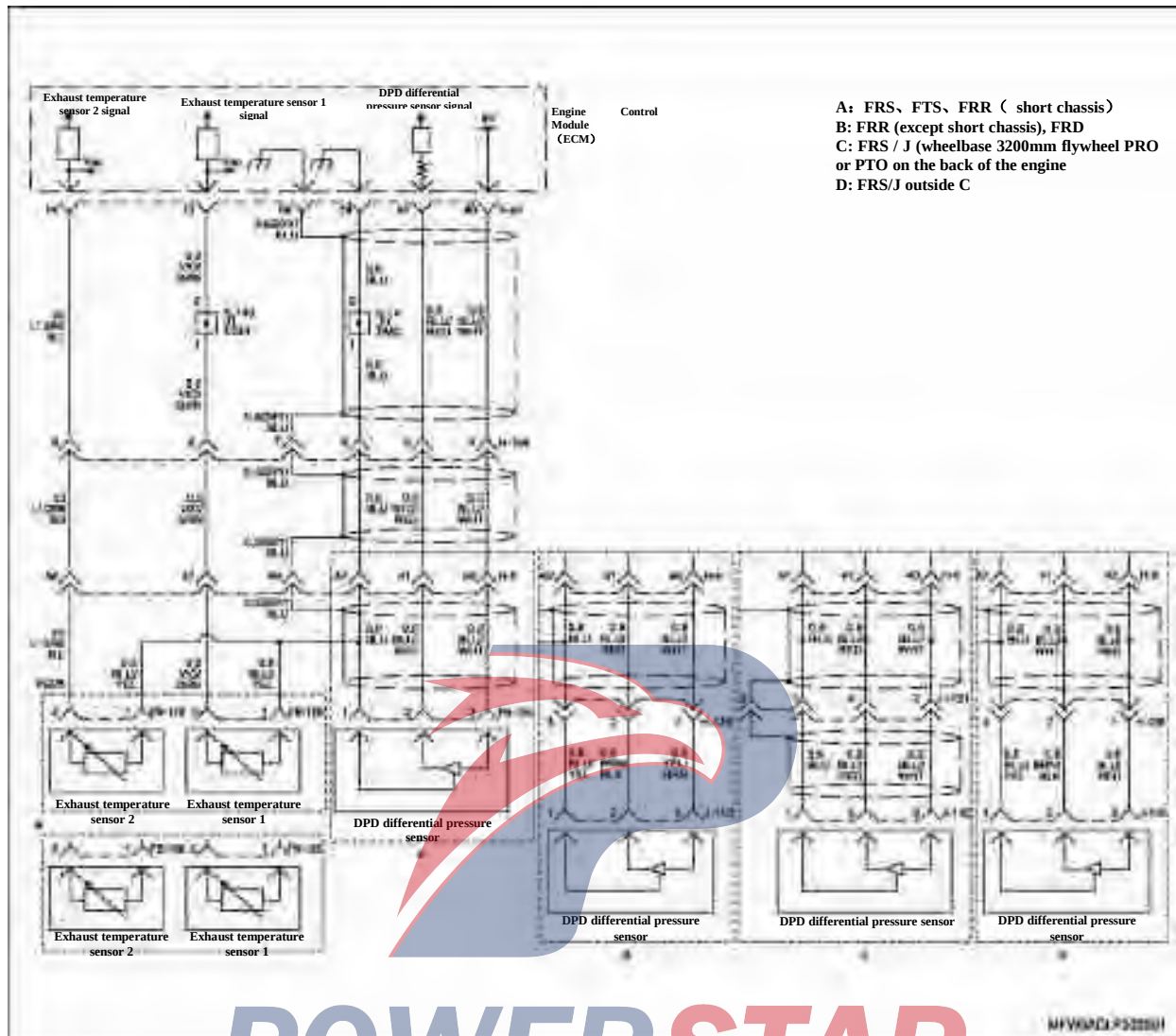
When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

After the DPD is regenerated, check the pressure difference of the DPD. When the pressure difference is above the reference value, refer to the "Dust Removal" in "DPD Function Check" and clean the DPD filter with the dust removal device.

DTC: P1472 (Flash code 142)

DPD pressure difference is too low





Cycle Description

The differential pressure sensor is set to detect the pressure difference in the DPD. The differential pressure sensor is installed to the DPD. ECM often uses the differential pressure sensors installed before and after DPD to detect the differential pressure in DPD and detect the accumulation of PM. DTC will be set when the DPD inside pressure is determined low.

The main fault phenomenon

- DPD regeneration stop
- Output drop

Preconditions for DTC setting

- Drive with the accelerator opening above 50% (the engine speed below 1500rpm) or the accelerator opening above 75% (the engine speed below 2500rpm).

- DTC P1630, P1633, P0470, P0471 and P0472 have not been detected.

- Vehicles equipped with DPD

Setting conditions for DTC.

- DPD differential pressure is less than - 2kPa when the engine revolution is above 1200rpm.

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1472 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch

in the [ON] position will be extinguished once again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

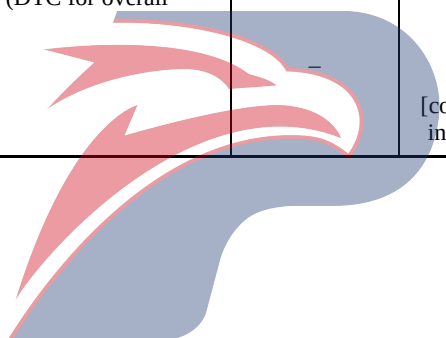
- Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

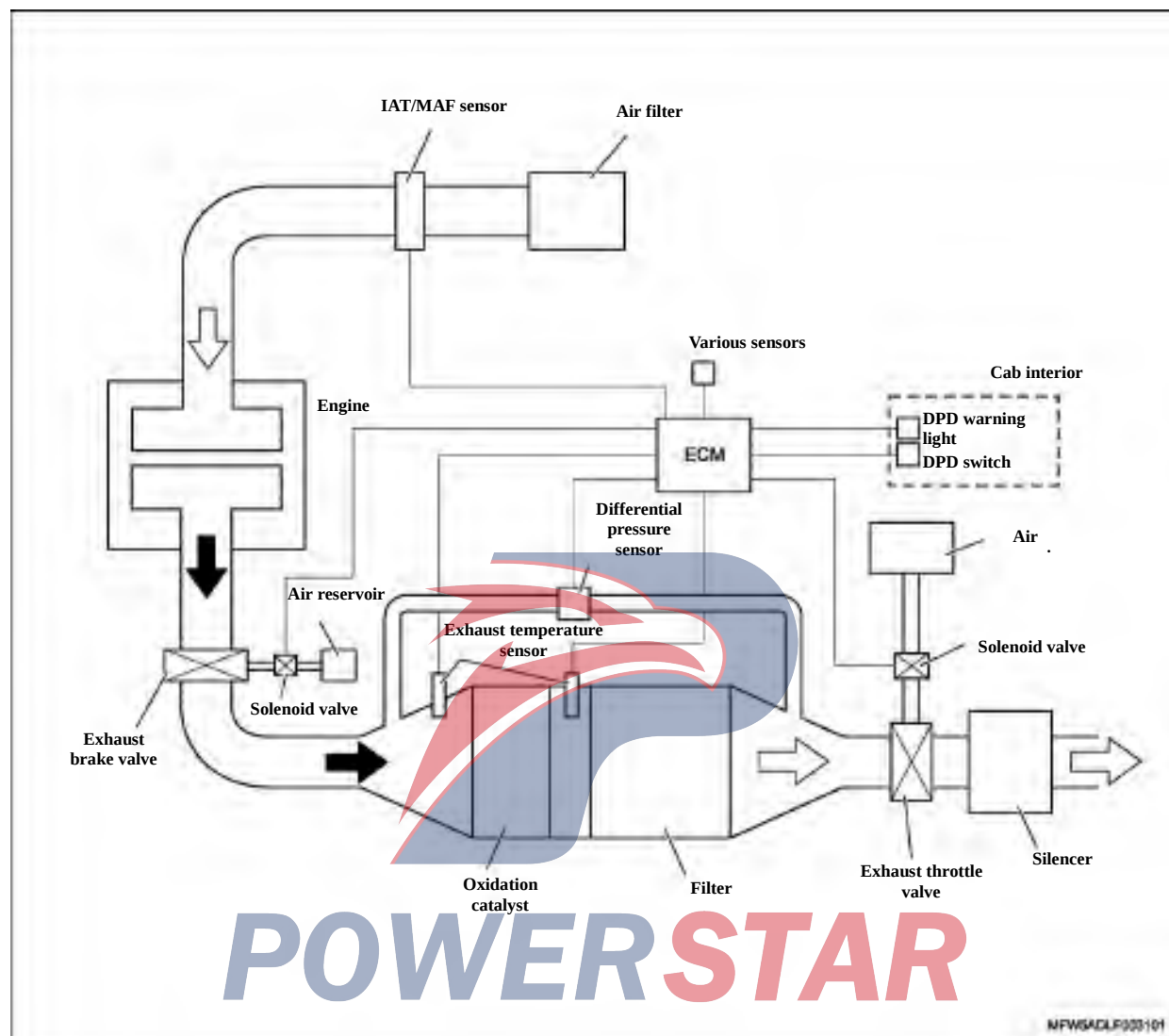
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Check the differential pressure sensor. 1. Check the hose and piping for breakage, falling off, air leak and main body breakage etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 3	—
3	Check the exhaust temperature sensor. 1. Check for air leak and main body breakage. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 4	—
4	Check the DPD filter. 1. Check for air leak, looseness, abnormal noise, fracture, dent, abnormal installation status etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 5	—
5	Check the differential pressure sensor. 1. Confirm the wiring harness, joint, and installation state. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 6	—
6	Check the exhaust pipe (between engine ~DPD) 1. Check the exhaust pipe for cavity (exhaust leak), scuffing, dent etc. 2. Check that the exhaust brake and exhaust throttle valve are stuck. 3. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—

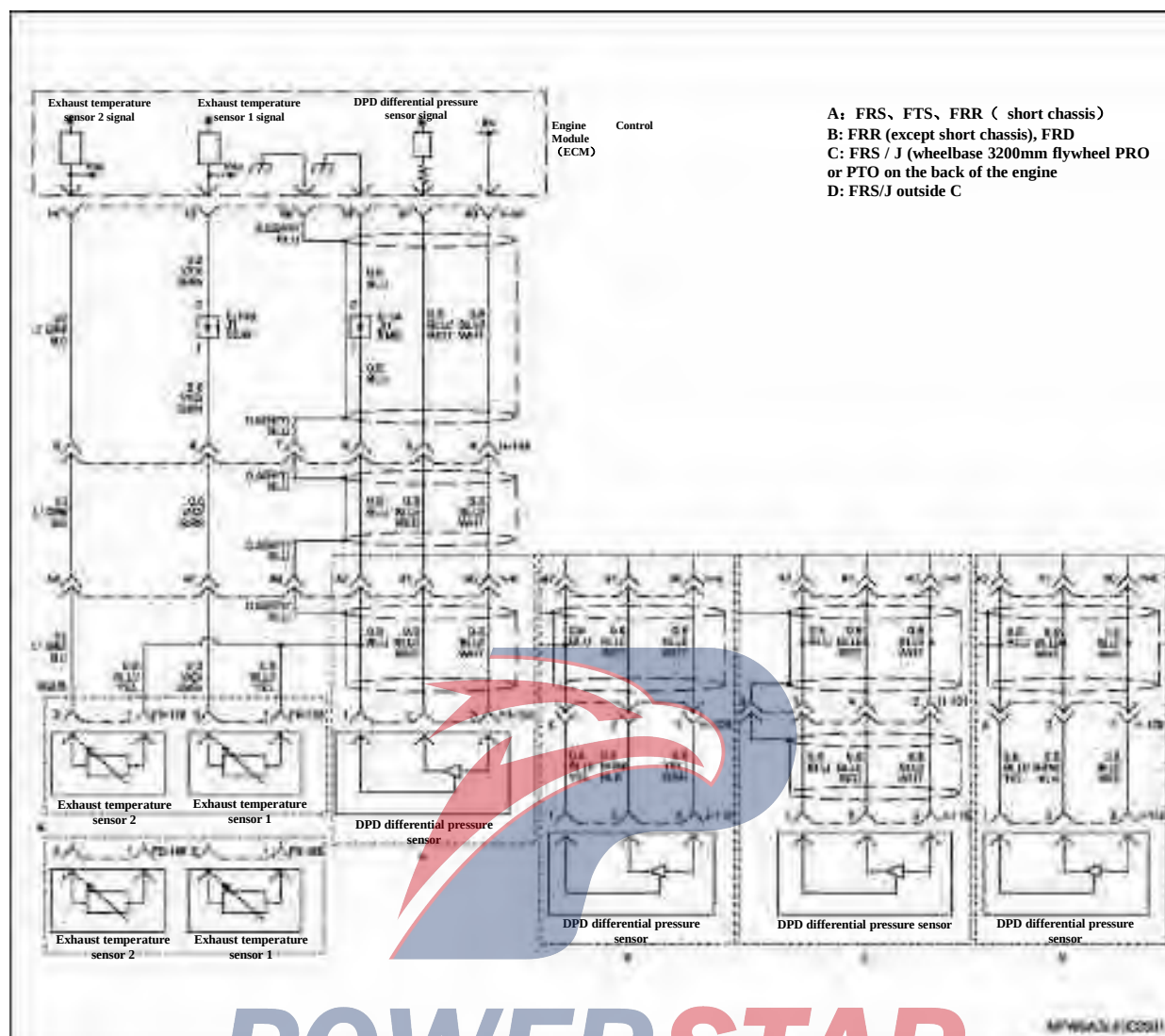
Step	Operation content	Datum value	YES	NO
7	Check the DPD main body. 1. Remove the DPD. 2. Check for stains, solvent loss, filter misalignment, leakage of the DPD body seal. 3. In the case of differential pressure abnormal, dissolved damage or filter out-of-position, replace the DPD. Is the disposal completed?	—	Go to step 8	—
8	Differential pressure sensor storage. 1. Set the key switch from the [OFF] position to the [ON] position. (The engine will not start at this time) 2. DPD alarm light lights out after about 30 seconds. 3. After confirming that the DPD warning lamp is off, set the key switch to [OFF] for more than 15 seconds. Is the disposal completed?	—	Go to step 9	—
9	Check for abnormal differential pressure, DPD abnormal (usually while driving) and dust accumulation in DPD filter. 1. Inquire and confirm the differential pressure. 2. Run the test. 3. Confirm DTC. Whether the fault is detected or not?	—	Go to step 10	Go to step 11
10	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 11	—
11	Perform a functional check for DPD. Is the disposal completed?	—	Go to step 12	—
12	Check for abnormal differential pressure, DPD abnormal (usually while driving) and dust accumulation in DPD filter. 1. Inquire and confirm the differential pressure. 2. Run the test. 3. Confirm DTC P1472. Important: This diagnosis is to confirm the phenomenon/DTC confirmation after returning the vehicle to the customer after inspection/maintenance is completed. In addition, since the dissolved damage and excessive PM catching may occur, check DPD inside again and replace the DPD as required. Is fault phenomenon detected?	—	Go to step 13	Go to step 14

Step	Operation content	Datum value	YES	NO
13	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Important: When the ECM is changed and the DPD is not replaced, the forced regeneration of DPD can be implemented, so that ECM and DPD can be set to the same state. Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 14	—
14	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions



POWERSTAR

DTC: P1473 (Flash code 141)**DPD pressure difference is too low**



Cycle Description

The differential pressure sensor is set to detect the pressure difference in the DPD. ECM often uses the differential pressure sensors installed before and after DPD to detect the differential pressure in DPD and detect the accumulation of PM. DTC will be set when the DPD inside pressure is determined low.

The main fault phenomenon

- DPD regeneration stop
- Output drop

Preconditions for DTC setting

- DTC P1630, P1633, P0470, P0471 and P0472 have not been detected.
- The engine is running.
- Vehicles equipped with DPD

Setting conditions for DTC.

When the pressure in the DPD is detected to be 60kPa or more

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC DTCP1473 is detected; it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator turns to a past fault, and when the key switch is set to [OFF], the key switch in the [ON] position will be extinguished once again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

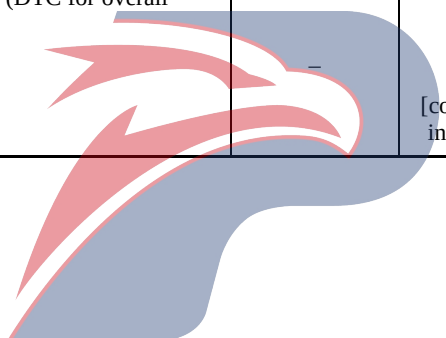
- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check" and go to step 2
2	Check the differential pressure sensor circuit. 1. Check whether there is short connection between the signal circuit of the differential pressure sensor and the power supply circuit. 2. Check for broken wire or high resistance in the GND circuit of the differential pressure sensor. 3. Check that the differential pressure sensor connector is loose or loose. 4. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 13	Go to step 3
3	Check the differential pressure sensor. 1. Check the hose and piping for breakage, falling off, damage, deformation, clogging, air leak and main body breakage etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 4	—
4	Check the exhaust temperature sensor. 1. Check for air leak and main body breakage. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 5	—
5	Check the DPD. 1. Check for air leak, looseness, abnormal noise, fracture, dent, abnormal installation status etc. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 6	—
6	Check the differential pressure sensor. 1. Check whether the harness, joint and installation are correct. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 7	—

Step	Operation content	Datum value	YES	NO
7	Check the DPD main body. 1. Remove the DPD. 2. Check for blot, dissolved damage, breakage and filter out-of-position. When the filter is misaligned, the DPD oxidation catalyst or DPD filter should be replaced. If there is blot, clean the DPD oxidation type catalyst (with gas). Refer to the exhaust system of the maintenance manual "engine" for the DPD cleaning (gas cleaning) method. 3. Install the DPD. 4. Refer to ECM for DPD reactivation and install the DPD reactivation ECM to reactivate DPD. Is the disposal completed?	—	Go to step 8	—
8	Differential pressure sensor storage. 1. Remove the DPD. 2. Set the key switch from the [OFF] position to the [ON] position. (The engine will not start at this time) 3. DPD alarm light lights out after about 30 seconds. 4. After confirming that the DPD warning lamp is off, set the key switch to [OFF] for more than 15 seconds. Is the disposal completed?	—	Go to step 9	—
9	Check for abnormal differential pressure, DPD abnormal (usually while driving) and dust accumulation in DPD filter. 1. Inquiries. 2. Run the test. 3. Confirm DTC. Is there any DTC detected?	—	Go to step 10	Go to step 11
10	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. Is the disposal completed?	—	Go to step 11	—
11	Perform a functional check for DPD. Is the disposal completed?	—	Go to step 12	Access to [DPD function check
12	Check for abnormal differential pressure, DPD abnormal (usually while driving) and dust accumulation in DPD filter. 1. Inquire and confirm the differential pressure. 2. Run the test. 3. Confirm DTC P1473. Important: This diagnosis is to confirm the phenomenon/DTC confirmation after returning the vehicle to the customer after inspection/maintenance is completed. In addition, since the dissolved damage and excessive PM catching may occur, check DPD inside again and replace the DPD as required. Is fault phenomenon detected?	—	Go to step 13	Go to step 14

Step	Operation content	Datum value	YES	NO
13	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Important: When the ECM is changed and the DPD is not replaced, the forced regeneration of DPD can be implemented, so that ECM and DPD can be set to the same state. Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 14	—
14	Check whether DTC is detected. (DTC for overall engine control) 1. Connect scanning tool 2. Check if DTC is detected. Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Confirm the repair conditions



POWERSTAR

About DPD regeneration using ECM**DPD regeneration method when using DPD to regenerate ECM**

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When DPD oxidation catalyst front temperature/DPD filter front temperature both reach 180 (1500rpm).

3. Increase the number of engine revolutions to 2500 rpm.
4. Check the differential pressure when the DPD filter front temperature reaches 160 °C with the scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.
7. Remove the ECM and install the DPD reactivation

ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Programmed the information downloaded from ECM to DPD reactivation ECM with the scanning tool.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

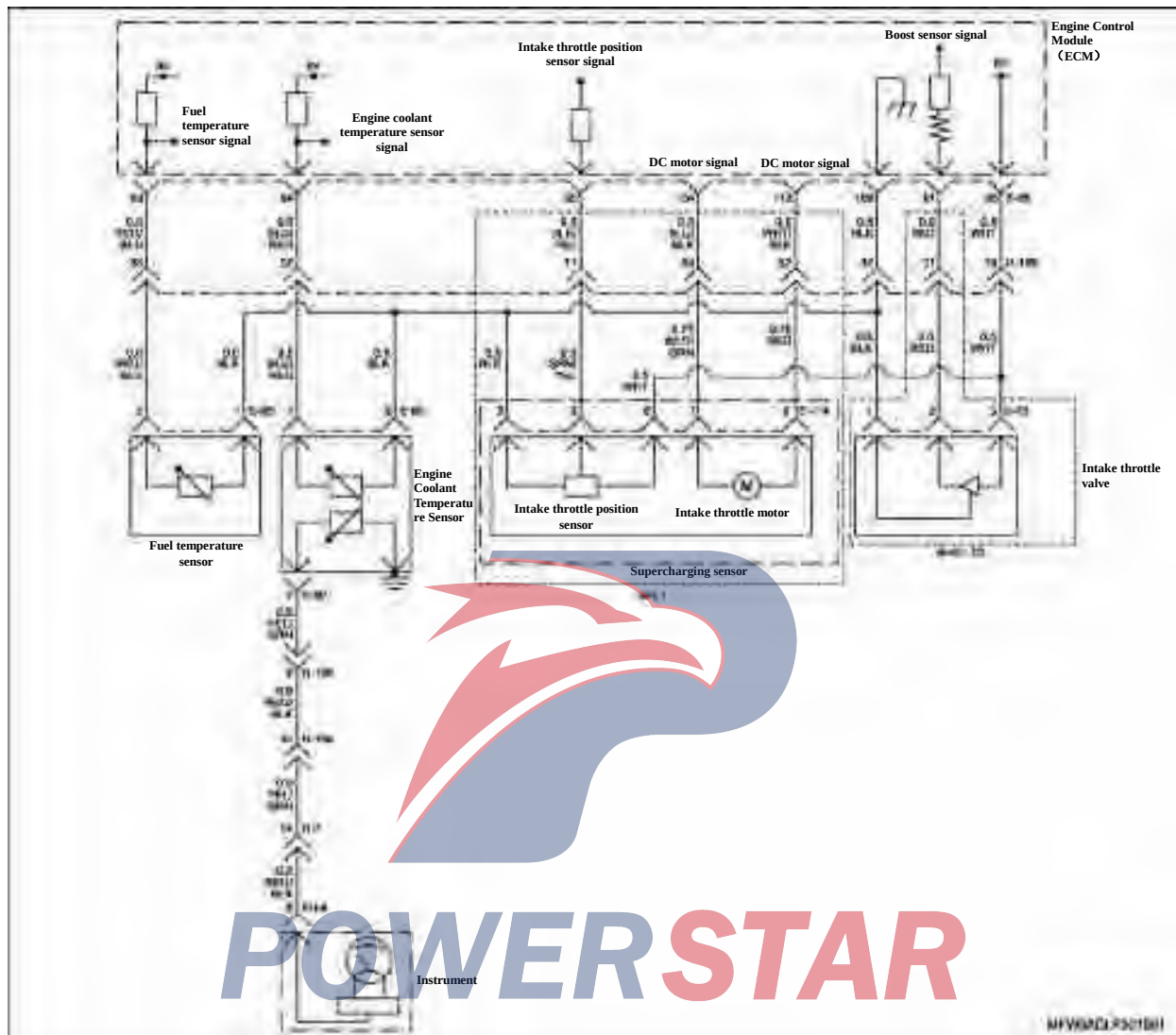
11. After the DPD light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECU have generated DTC. Clear the DTC when generating the DTC.

Important:

When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

DTC: P1484 (Flash code 43)**Intake throttle position (ITP) sensor circuit input low****Cycle Description**

The intake throttle position (ITP) sensor will supply the engine control module (ECM) with a voltage signal corresponding to the throttle opening. ECM can work out the throttle opening according to the voltage signal and use this information to control the air intake throttle.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1634 have not been detected.
- Vehicles equipped with DPD

Setting conditions for DTC.

- ITP sensor voltage below 0.1V

Disposal when setting DTC

- Engine fault indicator lights up.

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems

exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

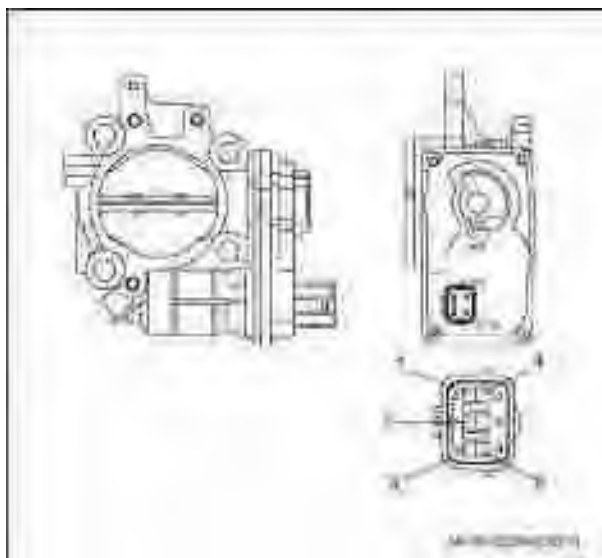
The numbers shown below indicate the step numbers in the table

2. In the case of current malfunction, the displayed value of [ITP sensor(Intake throttle position)] is below reference value. If above the reference value, please refer to the diagnosis assistance to explain why.

4. When a short circuit occurs between the terminals of the ITP sensor connector, the [ITP Sensor (intake throttle position)] is displayed above the reference value. This indicates that the sensor's subsequent circuit including the ECM is normal.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm the value of FT sensor with the scanning tool. 1. Set the key switch to the [OFF] state 2. Confirm the contents displayed in the data display [ITP sensor] with the scanning tool. [Is [ITP sensor] displayed value below the reference value?	0.1V	Go to step 3	Enter diagnostic help
3	Confirm the installation status of ITP sensor joint, ECM joint and middle joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Remove the ITP sensor connector 3. Use jumpers to short the sensor power and signal terminals of the ITP sensor connector. 4. Set the key switch to the [OFF] state 5. Confirm the contents displayed in the data display [ITP sensor] with the scanning tool. [Is ITP sensor value above the reference value?	4.8V	Go to step 7	Go to step 5
5	1. Check the power circuit between the ECM and ITP sensor for the following state with the distributing box: ● Wire break ● High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 6

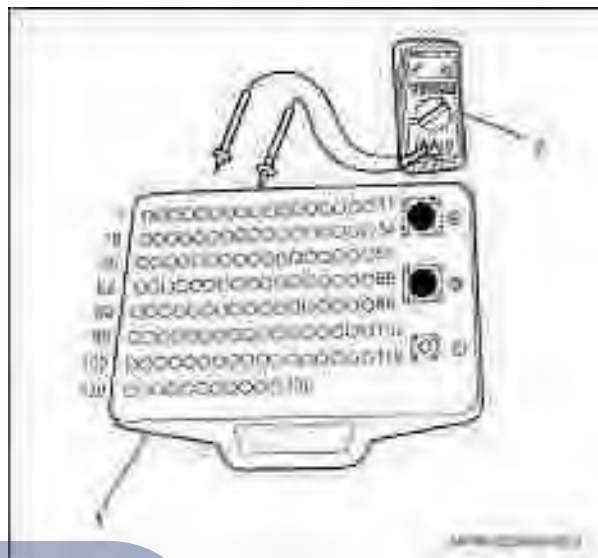
Step	Operation content	Datum value	YES	NO
6	1. Check the signal circuit between the ECM and ITP sensor for the following state with the distributing box: • There is a short circuit to the FT sensor GND circuit; • Short connection with GND • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 8
7	Replace the ITP sensor. Is the disposal completed?	—	Go to step 9	—
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1484 been detected?	—	Go to step 2	Go to step 10
10	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

Replace the ITP sensor.**Name**

1. Sensor 1 GND
2. Sensor output
3. Motor (-)
4. Sensor input
5. Motor (+)

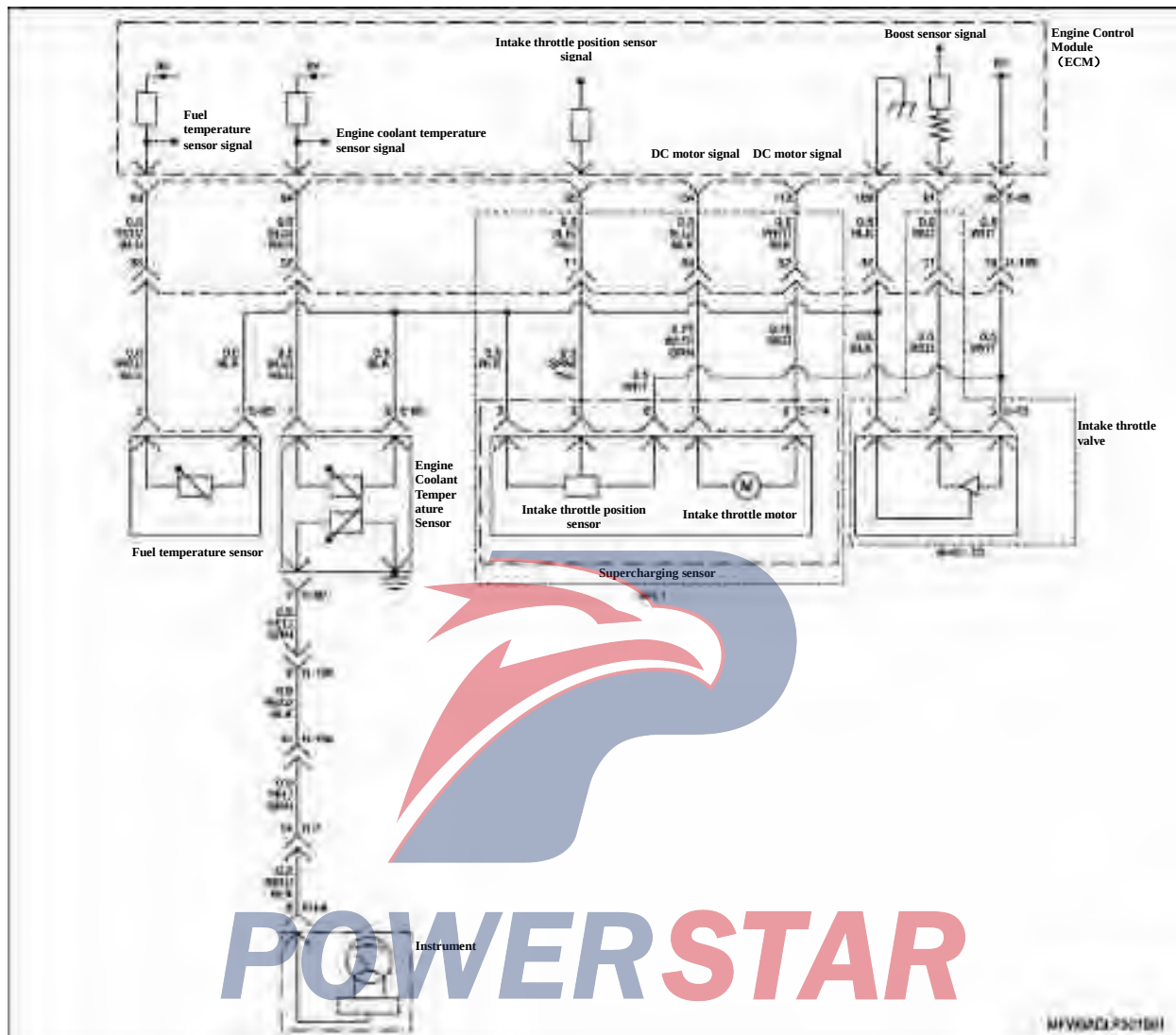
About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	95 - sensor connector power terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)
6	Short circuit to GND circuit/GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	85-109 85-GND	Above 10MΩ (no conduction)	Below 10MΩ (there is a guide)
	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	85-Sensor connector signal terminals	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1485 (Flash code 43)**Intake throttle position (ITP) sensor circuit input high****Cycle Description**

The intake throttle position (ITP) sensor will supply the engine control module (ECM) with a voltage signal corresponding to the throttle opening. ECM can work out the throttle opening according to the voltage signal and use this information to control the air intake throttle.

The main fault phenomenon

- Gas emission deteriorated

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1634 have not been detected.
- Vehicles equipped with DPD

Setting conditions for DTC.

- ITP sensor voltage above 4.9V

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

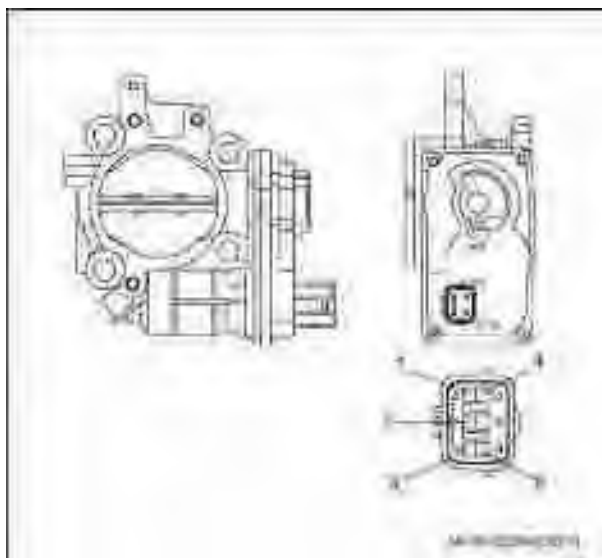
Test instruction

The numbers shown below indicate the step numbers in the table

2. In the case of current malfunction, the displayed value of [ITP sensor(Intake throttle position)] is above reference value. If below the reference value, please refer to the diagnosis assistance to explain why.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm the value of FT sensor with the scanning tool. 1. Set the key switch to the [OFF] state 2. Confirm the contents displayed in the data display [ITP sensor] with the scanning tool. [Is ITP sensor value above the reference value?]		Go to step 3	Enter diagnostic help
3	Confirm the installation status of ITP sensor joint, ECM joint and middle joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?		Go to step 9	Go to step 4
4	Confirm the value of FT sensor with the scanning tool. 1. Set the key switch to the [OFF] position. 2. Remove the ITP sensor connector 3. Set the key switch to the [OFF] state 4. Confirm the contents displayed in the data display [ITP sensor] with the scanning tool. [Is ITP sensor value below the reference value?]	0.1V	Go to step 6	Go to step 5
5	1. Use the distribution cabinet to check whether the signal circuit between the ECM and PTO accelerator sensors has short connection with the 5V power supply circuit and the ignition power supply circuit. Important: If there is a short circuit in the power circuit of the IEP sensor circuit, the sensor may be damaged. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 8

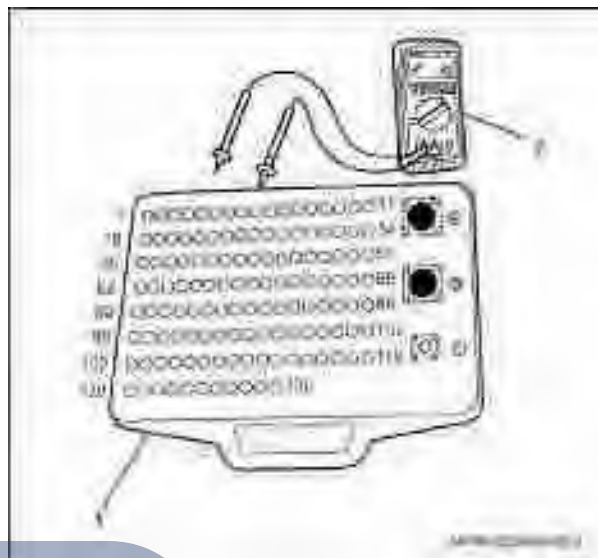
Step	Operation content	Datum value	YES	NO
6	1. Check the GND circuit between the ECM and ITP sensor for the following state with the distributing box: • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 9	Go to step 7
7	Replace the ITP sensor. Is the disposal completed?	—	Go to step 9	—
8	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
9	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1485 been detected?	—	Go to step 2	Go to step 10
10	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

Replace the ITP sensor.**Name**

1. Sensor 1 GND
2. Sensor output
3. Motor (-)
4. Sensor input
5. Motor (+)

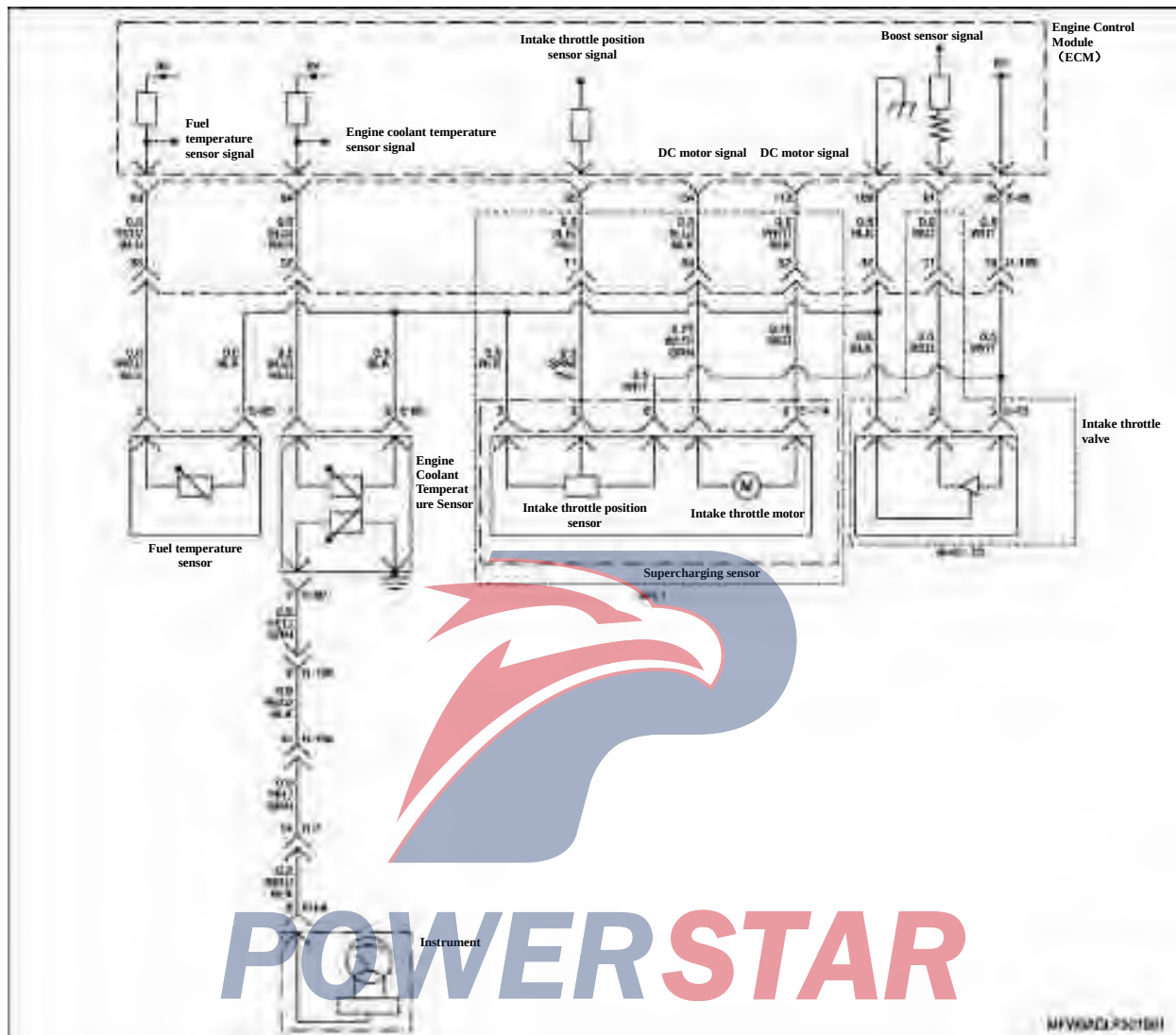
About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

**Name**

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	85-GND	0V	Above 4.8V
6	Broken wire / high resistance	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	109-Sensor connector GND terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1487 (Flash code 61)**Intake throttle anomaly****Cycle Description**

The intake throttle valve is controlled by a DC motor. The DC motor drives the drive voltage (load ratio) to change. The valve will be closed when the load ratio is increased, otherwise the valve will be open. The engine control module (ECM) calculates the target opening of the throttle valve according to the driving state of the vehicle and controls the DC motor.

The main fault phenomenon

- Driving difficult

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630, P1634, P1484 and P1485 have not been detected.

Setting conditions for DTC.

The intake throttle drive duty ratio above 47%, intake throttle opening above $\pm 20\%$

Disposal when setting DTC

- Engine fault indicator lights up
- When the DTC P1487 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state,

the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

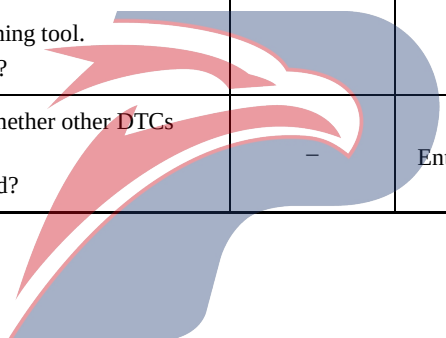
- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

- Harness damage:

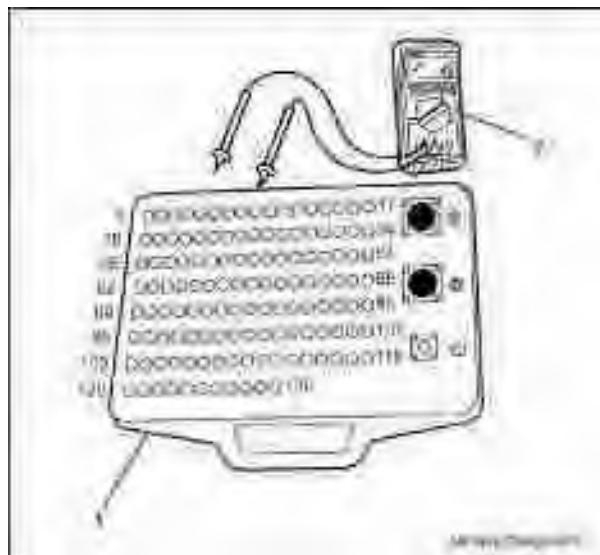
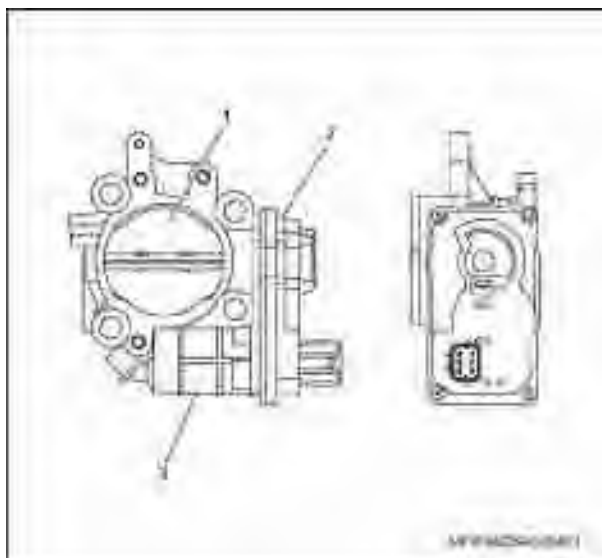
- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Use the power distribution cabinet to check the motor drive circuit between the ECM and the intake throttle motor for the following state. • Wire break • Short circuit to GND circuit • Short circuit between the power circuit and ignition power circuit • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 3
3	Confirm the DTC again. 1. Clear DTC with the scanning tool. 2. Set the key switch to the [OFF] state for 15 seconds or more. 3. Confirm DTC with the scanning tool. Has DTC P1487 been detected?	—	Go to step 4	Enter diagnostic help
4	1. Clear DTC with the scanning tool. 2. Start the engine. 3. Select [actuator test] from the menu. 4. Select the [intake throttle control] test in device control. 5. Press the [UP] soft key to confirm the display content of intake throttle opening. [Is the value of [intake throttle opening] within the reference value range?	±3%	Go to step 6	Go to step 5
5	Replace the intake throttle valve. Is the disposal completed?	—	Go to step 7	—

Step	Operation content	Datum value	YES	NO
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 7	—
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1487 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions



POWERSTAR

About intake throttle valve**Name****Name**

1. Air intake throttle valve
2. ITP sensor
3. DC motor

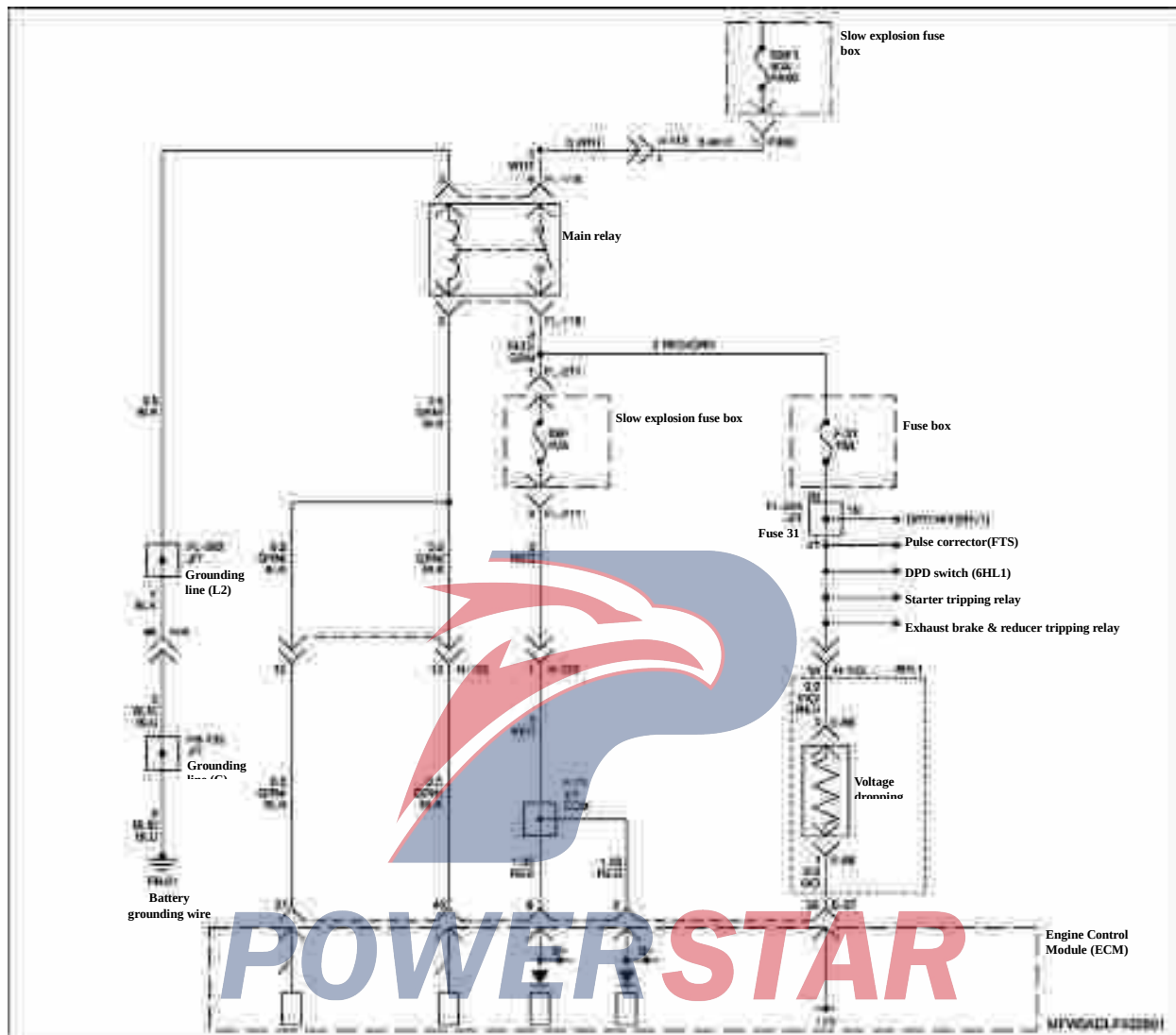
1. Distribution Cabinet

2. DMM

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
2	Short circuit to power supply circuit	Voltage measurement	- Remove the air intake throttle connector. - Key switch [ON]	112-GND 104-GND	0V	Above 18V
	Short to GND	Measurement of resistance value	- Remove the air intake throttle connector. - Key switch [OFF]	112-GND 104-GND	Above 10MΩ (no conduction)	Below 10MΩ (there is a guide)
	Broken wire / high resistance	Measurement of resistance value	- Remove the air intake throttle connector. - Key switch [OFF]	112-EGR valve joint motor (+) terminal 104-EGR valve joint motor (-) terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P1625 (Flash code 416)**Main relay circuit fixed open or fixed close****Cycle Description**

The main relay is installed near the blower motor in the dashboard. When the engine control module (ECM) receives the [ON] signal from the key switch input, it will put the main relay in the [ON] state, and when the key switch is in the [OFF] state, the main relay will be in the [OFF] state. The main relay is powered by a fuse. After the main relay enters the [ON] state, the ECM supplies the supply voltage.

ECM will set DTC when the output of the main relay is not consistent with the input from the main relay.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 have not been detected.
- 5 seconds or more after the key switch [ON]

- Main relay drive commands ON

Setting conditions for DTC.

For one of the following states:

- Main relay input power voltage below 1V.
- ECM will not become OFF even if main relay drive commands OFF.

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state,

the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Switch body abnormal
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire

is poor.

- Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

3. When the main relay is closed and cannot be opened again, the scan tool can communicate with the ECM even if the key switch is set to the [OFF] position.

4. Even if the main relay is removed and the scan tool is used to communicate with the ECM, if the DTC is detected, the ECM behind the relay or relay is in an abnormal state.

7. When a short circuit occurs between the terminals of the main relay mounting section, the case that the display of [Main Relay Voltage] is above the reference value indicates that the relay's subsequent circuit including the ECM is in normal.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Use the scan tool to confirm the display content of [Main Relay Voltage] in the data display. [Is the value of [main relay voltage] above the reference value?	24V	Go to step 3	Go to step 6
3	1. Set the key switch to the [OFF] position. 2. Confirm DTC with the scanning tool. Has DTC P1625 been detected?	—	Go to step 4	Enter diagnostic help
4	1. Remove the main relay. 2. Confirm DTC with the scanning tool. Has DTC P1625 been detected?	—	Go to step 5	Go to step 10
5	1. Check the connection point side circuit between the main relay and ECM for short circuit to power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 12
6	Confirm the installation status of main relay, ECM connector and intermediate joint. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 7

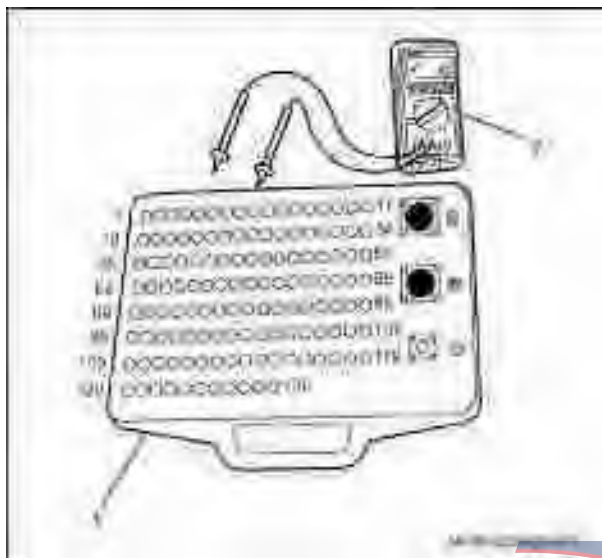
Step	Operation content	Datum value	YES	NO
7	1. Set the key switch to the [OFF] position. 2. Remove the main relay. 3. Use jumpers to short-circuit terminals 1-2 and 3-5 of the main relay mounting section. 4. Set the key switch to the [OFF] state 5. Use the scan tool to confirm the display content of [Main Relay Voltage] in the data display. [Is the value of [main relay voltage] above the reference value?]	24V	Go to step 10	Go to step 8
8	1. Check whether there are the following states in the circuit between slow-blow fuse and main relay, is there the following state? • Wire break • Short circuit to GND circuit • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 9
9	1. Use the distribution cabinet to check whether there are the following states in the circuit between the main relay and ECM. • Wire break • Short connection with GND circuits • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
10	Replace the main relay. Is the disposal completed?	—	Go to step 12	—
11	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 12	—
12	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1625 been detected?	—	Go to step 2	Go to step 13
13	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

Name

1. Distribution Cabinet
2. DMM



DTC: P1630 (Flash code 36)**Analog/digit signal conversion abnormal****Cycle Description**

The engine control module (ECM) can convert analog signals (voltage signals, etc.) input from various sensors into digital signals for processing, and can perform various controls and calculations.

ECM sets DTC when detecting abnormal analog / digital conversion.

Preconditions for DTC setting

- The analog/digital conversion is detected abnormal

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past

failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

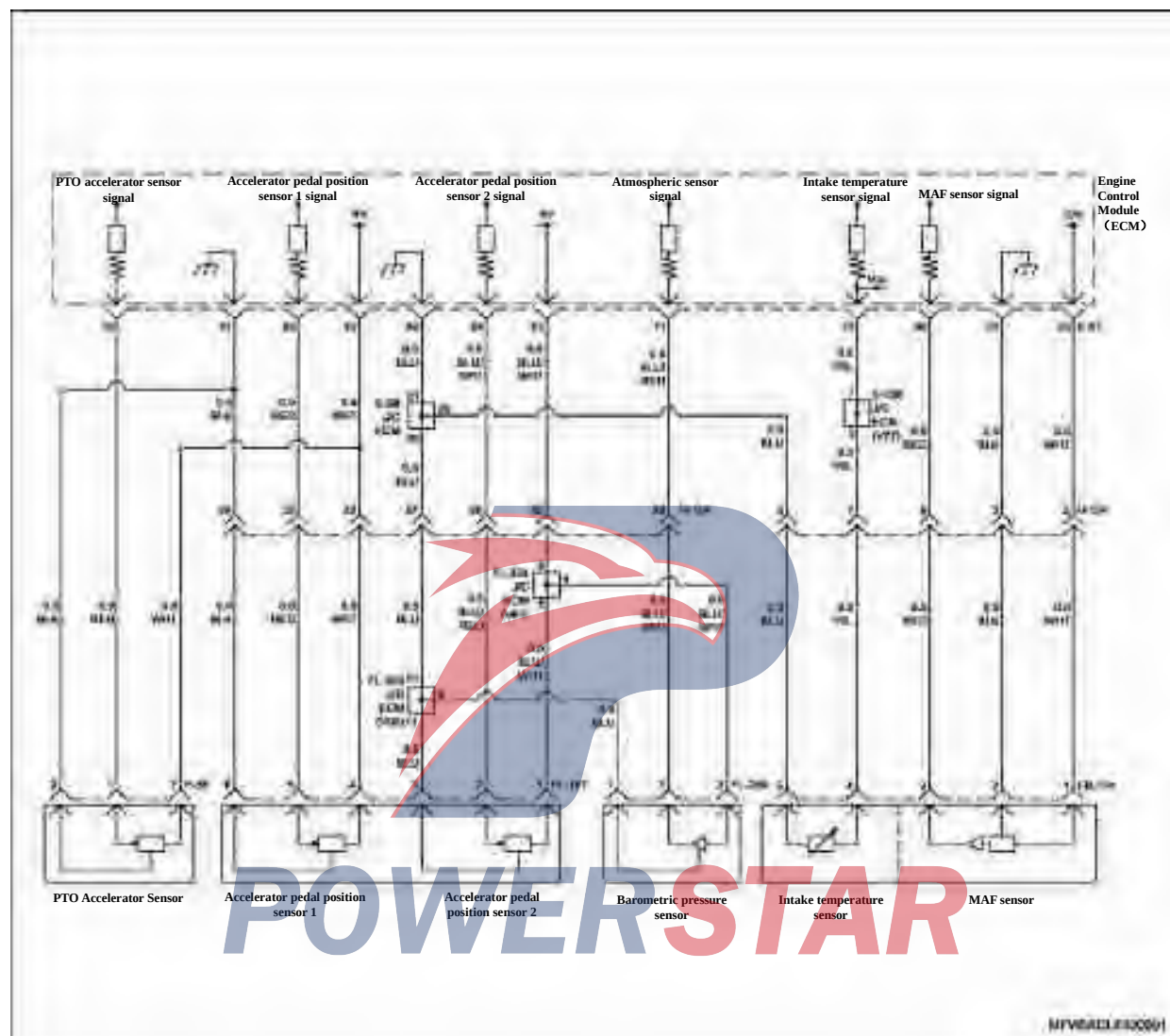
- Connection between harness connector and ECM connector is bad
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.

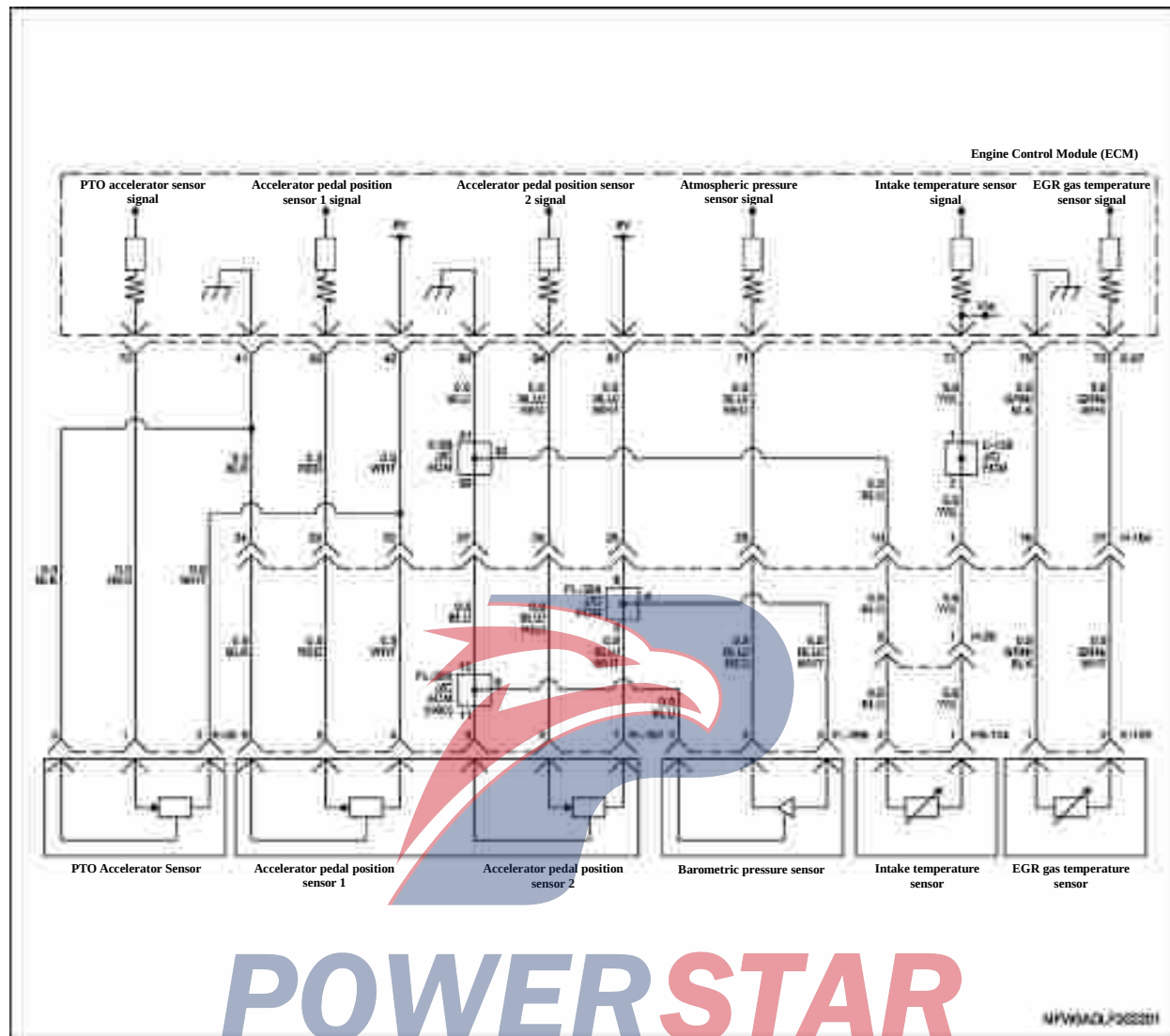
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm the DTC again. 1. Clear DTC with the scanning tool. 2. Set the key switch to the [OFF] state for 15 seconds or more. 3. Perform a test run under [Preconditions for DTC settings]. 4. Confirm DTC with the scanning tool. Has DTC P1630 been detected?	—	Go to step 3	Enter diagnostic help
3	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 4	—
4	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

DTC: P1631 (Flash code 55)

5V #1 circuit abnormal

6HL1



6HK1-TC**Cycle Description**

The engine control module (ECM) has 5 systems of 5V power circuits. The 5V power supply circuit 1 supplies power to an accelerator pedal position (APP) sensor 1 and a PTO accelerator sensor.

ECM will set DTC if the 12V power circuit voltage is lower or higher than the normal range.

Preconditions for DTC setting

- DTC P1630 have not been detected.
- Key switch input power voltage more than 16V but less than 32V

Setting conditions for DTC.

- Sensor or ECM power detected abnormal

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

-The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

2. Measure on the harness joint side.

3. Measure on the harness joint side.

6. Measure on the harness joint side.

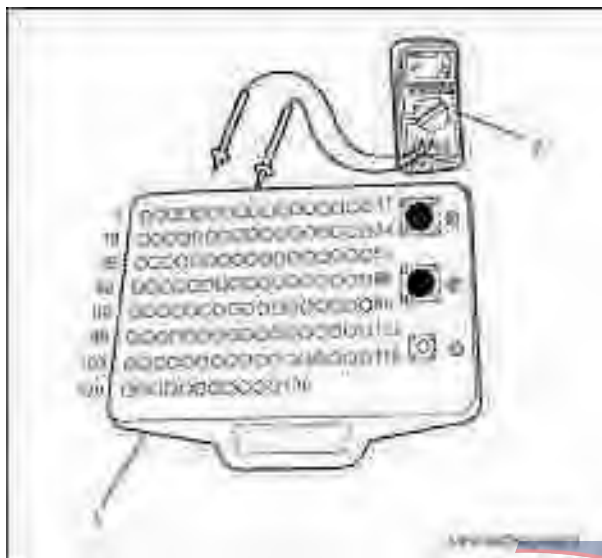
7. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	–	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Connect the DMM between the APP sensor 1 power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 4	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Connect the DMM between the APP sensor 1 power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 5	Go to step 6
4	1. Use the power distribution cabinet to check the power circuit between APP sensor 1 and ECM for short circuit between the battery power circuit and ignition power circuit. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	–	Go to step 11	Go to step 10
5	1. Check the power circuit between the APP sensor 1 and ECM for short circuit to GND with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	–	Go to step 11	Go to step 10
6	1. Set the key switch to the [OFF] position. 2. Remove the connector of the PTO accelerator sensor. 3. Connect the DMM between the PTO accelerator sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 8	Go to step 7

Step	Operation content	Datum value	YES	NO
7	1. Set the key switch to the [OFF] position. 2. Remove the connector of the PTO accelerator sensor. 3. Connect the DMM between the PTO accelerator sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 9	Enter diagnostic help
8	1. Check the power circuit between the PTO accelerator sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10
9	1. Check the power circuit between the PTO accelerator sensor and ECM for short circuit to GND circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10
10	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 11	—
11	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1631 been detected?	—	Go to step 2	Go to step 12
12	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

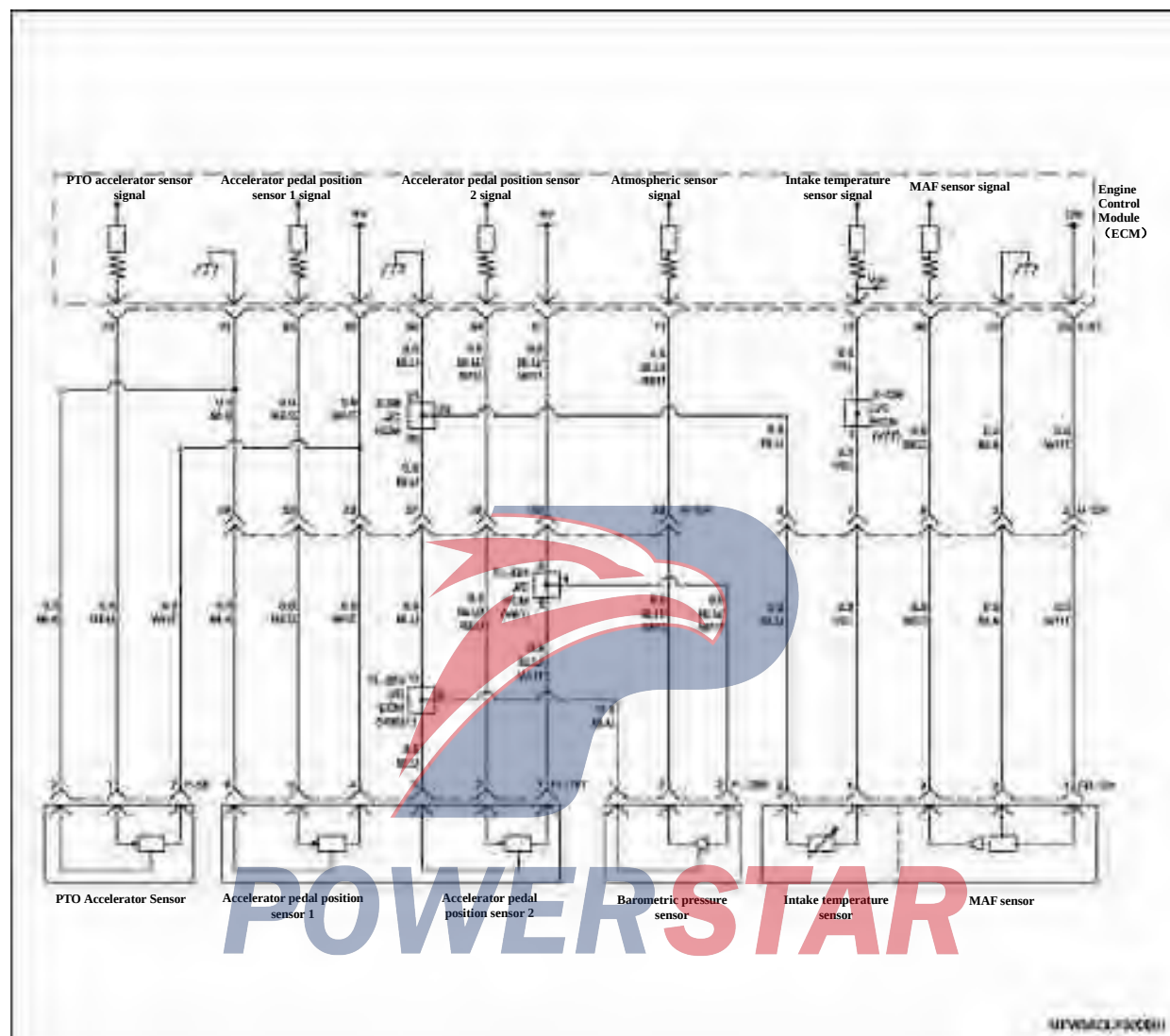
In a program that requires the use of a power distribution cabinet, check the following points.

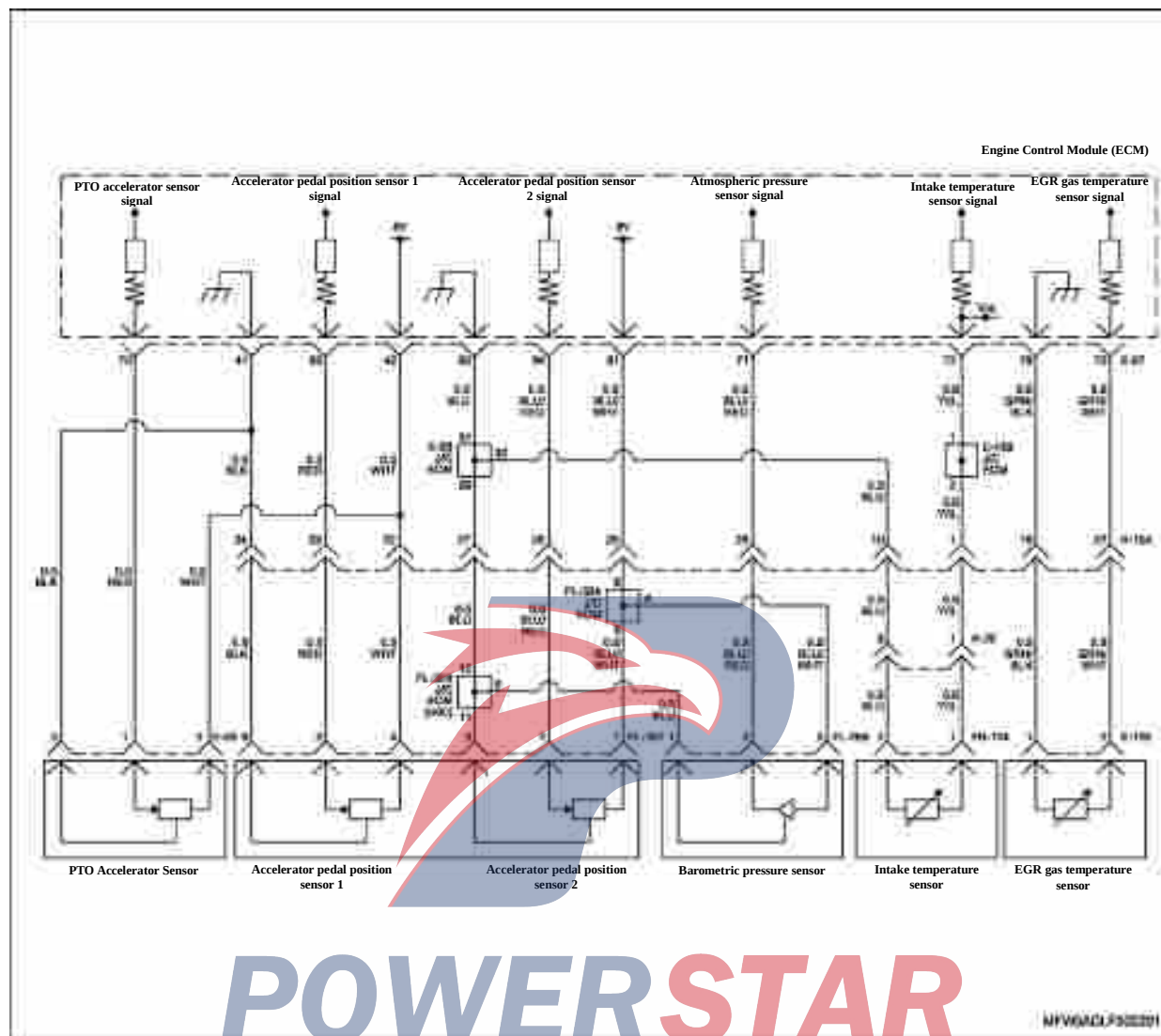


Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4 8	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	42-GND	0V	Above 18V
5 9	Short to GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	42-GND	Above 10MΩ (no conduction)	Below 10MΩ (there is a guide)

DTC: P1632 (Flash code 55)**5V #2 circuit abnormal****6HL1**

6HK1-TC**Cycle Description**

The engine control module (ECM) has 5 systems of 5V power circuits. 5V power circuit 2 can supply power to the accelerator pedal position (APP) sensor 2 and barometric pressure sensor.

ECM will set DTC if the 12V power circuit voltage is lower or higher than the normal range.

Preconditions for DTC setting

- DTC P1630 have not been detected.
- Key switch input power voltage more than 16V but less than 32V

Setting conditions for DTC.

- Sensor or ECM power detected abnormal

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

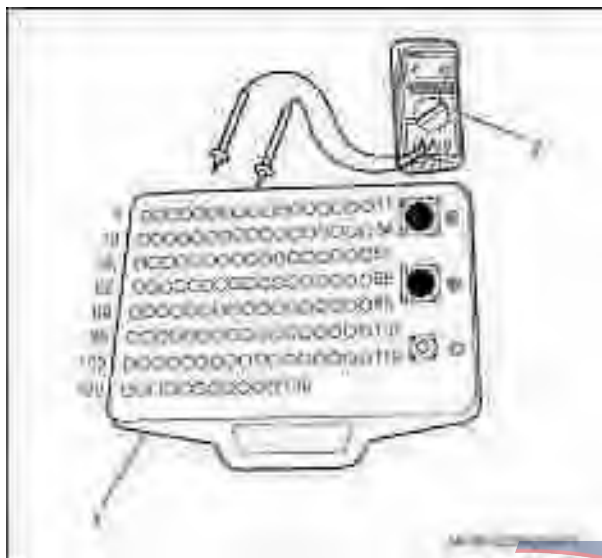
2. Measure on the harness joint side.
3. Measure on the harness joint side.
6. Measure on the harness joint side.
7. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Connect the DMM between the APP sensor 2 power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 4	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the APP sensor connector 3. Connect the DMM between the APP sensor 2 power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 5	Go to step 6
4	1. Use the power distribution cabinet to check the power circuit between APP sensor 2 and ECM for short circuit between the battery power circuit and ignition power circuit. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10
5	1. Check the power circuit between the APP sensor 2 and ECM for short circuit to GND with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10

Step	Operation content	Datum value	YES	NO
6	1. Set the key switch to the [OFF] position. 2. Remove the connector of the PTO accelerator sensor. 3. Connect the DMM between the barometric pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 8	Go to step 7
7	1. Set the key switch to the [OFF] position. 2. Remove the connector of the atmospheric pressure sensor. 3. Connect the DMM between the barometric pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 9	Enter diagnostic help
8	1. Check the power circuit between the barometric pressure sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10
9	1. Use a power distribution cabinet to check the short circuit between the atmospheric pressure sensor and the ECM and GND circuits. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 11	Go to step 10
10	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 11	—
11	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1632 been detected?	—	Go to step 2	Go to step 12
12	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.



Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4 8	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	61-GND	0V	Above 18V
5 9	Short to GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	61-GND	Above 10MΩ (no conduction)	Below 10MΩ (there is a guide)

- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

2. Measure on the harness joint side.

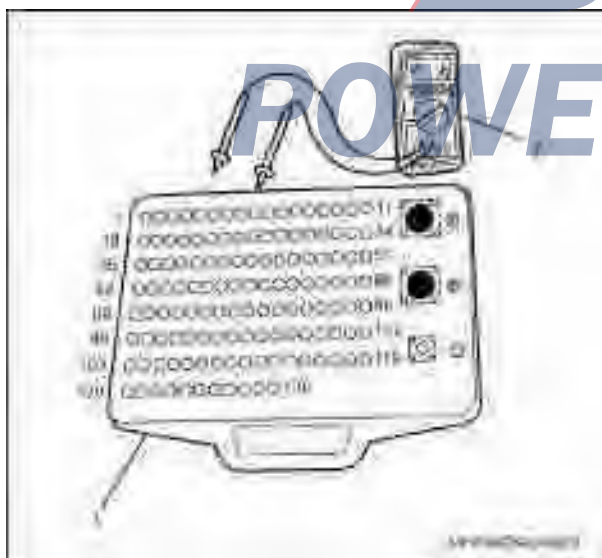
3. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] position. 2. Remove the DPD differential pressure sensor connector 3. Connect the DMM between the DPD differential pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 4	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the DPD sensor joint. 3. Connect the DMM between the DPD differential pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 5	Enter diagnostic help
4	1. Check the power circuit between the DPD differential pressure sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6
5	1. Use distribution cabinet to check whether the power supply circuit between the DPD differential pressure sensor and the ECM has short connection with GND. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6

Step	Operation content	Datum value	YES	NO
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 7	—
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1633 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

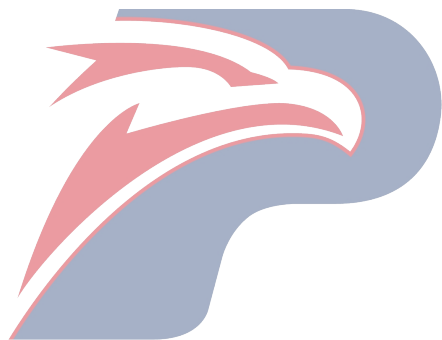
In a program that requires the use of a power distribution cabinet, check the following points.



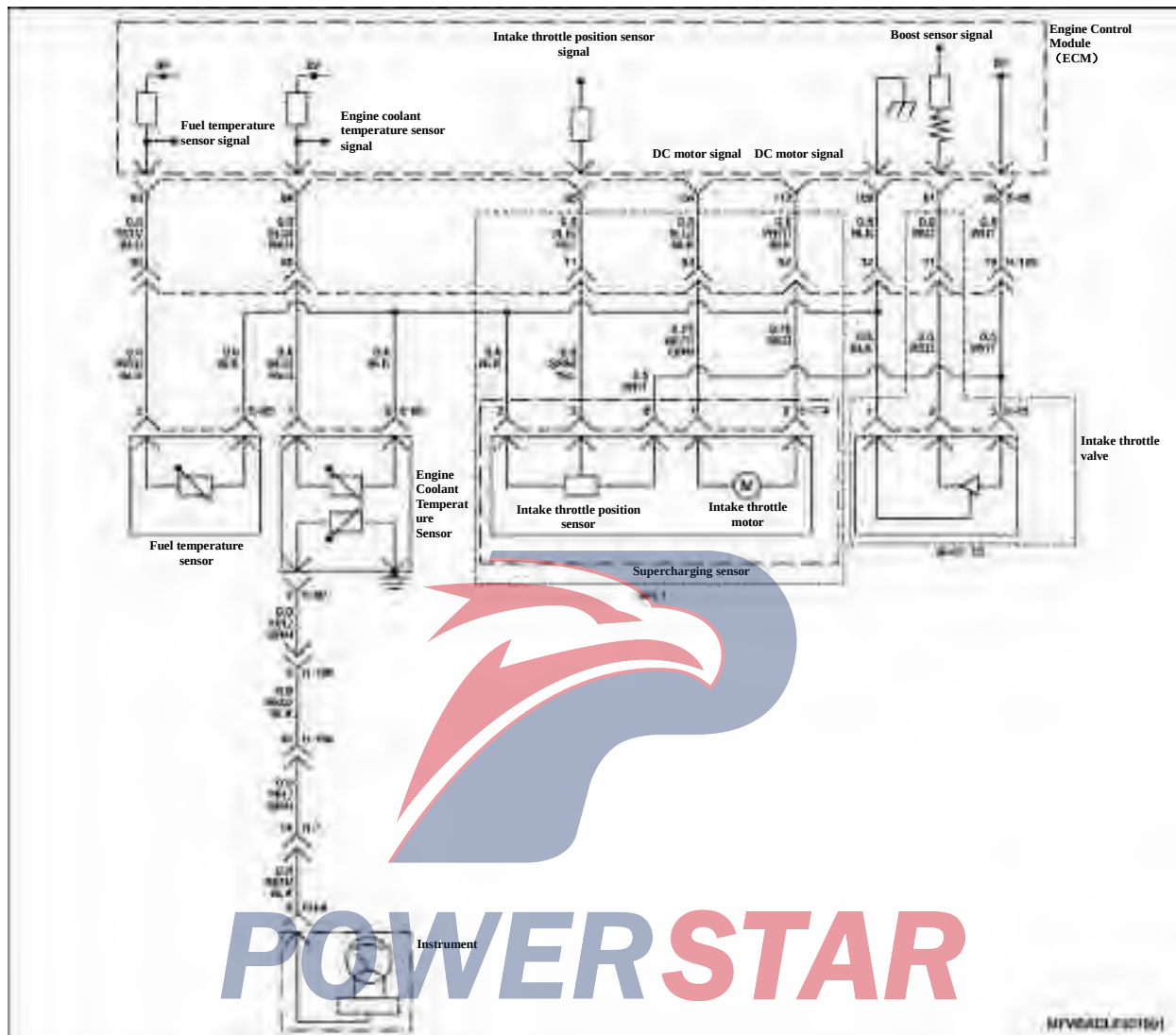
Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Short circuit to power supply circuit	Voltage measurement	● Remove the sensor connector ● Key switch [ON]	80-GND	0V	Above 18V
5	Short to GND	Measurement of resistance value	● Remove the sensor connector - Key switch [OFF]	80-GND	Above 10M Ω (no conduction)	Below 10M Ω (there is a guide)



POWERSTAR

DTC: P1634 (Flash code 55)**5V #4 circuit abnormal****Cycle Description**

The engine control module (ECM) has 5 systems of 5V power circuits. 5V power circuit will supply power to the intake throttle position (ITP) sensor (6HL1) or supercharging sensor (6HK1-TC).

ECM will set DTC if the 5V power circuit 4 voltage is lower or higher than the normal range.

Preconditions for DTC setting

- DTC P1630 have not been detected.
- Key switch input power voltage more than 16V but less than 32V

Setting conditions for DTC.

- Sensor or ECM power detected abnormal

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems

exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

– Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

2. Measure on the harness joint side.

3. Measure on the harness joint side.

7. Measure on the harness joint side.

8. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Is the engine equipped with 6HK1-TC?	—	Go to step 7 in the case of 6HK1-TC	Go to step 3 in the case of 6HL1
3	1. Set the key switch to the [OFF] position. 2. Remove the air intake throttle connector. 3. Connect the DMM between the ITF sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 5	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Remove the air intake throttle connector. 3. Connect the DMM between the ITF sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 6	Enter diagnostic help
5	1. Check the power circuit between the ITP sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
6	1. Check the power circuit between the ITP sensor and ECM for short circuit to GND with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11

Step	Operation content	Datum value	YES	NO
7	1. Set the key switch to the [OFF] position. 2. Remove the connector of the supercharging sensor. 3. Connect the DMM to the power supply terminal and GND of the booster sensor. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 9	Go to step 8
8	1. Set the key switch to the [OFF] position. 2. Remove the connector of the supercharging sensor. 3. Connect the DMM to the power supply terminal and GND of the booster sensor. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 10	Enter diagnostic help
9	1. Check the power circuit between the barometric pressure sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
10	1. Use a power distribution cabinet to check the short circuit between the boost sensor and the ECM and GND circuits. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
11	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 12	—
12	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1634 been detected?	—	Go to step 2	Go to step 13
13	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

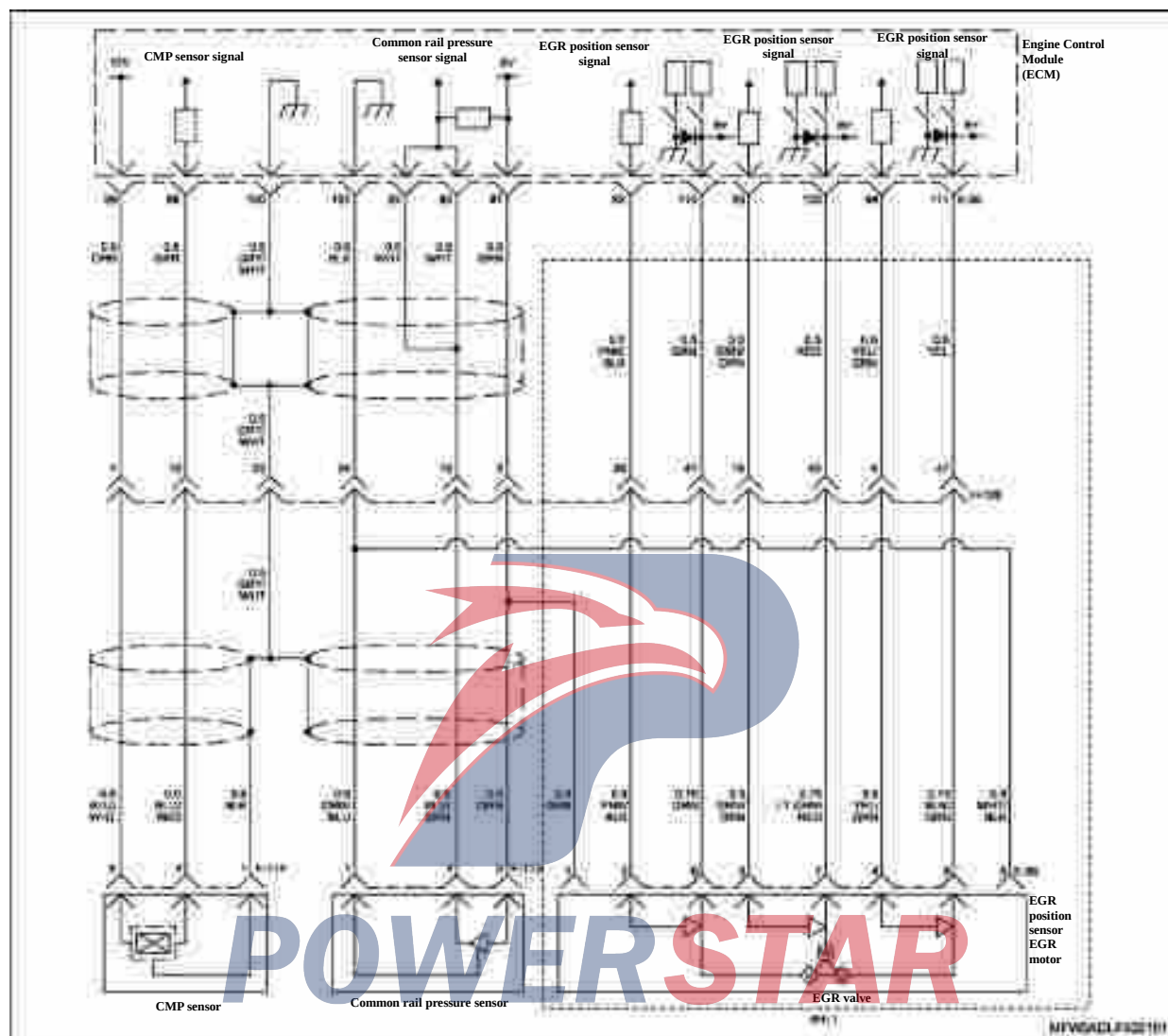
In a program that requires the use of a power distribution cabinet, check the following points.



Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5 9	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	95-GND	0V	Above 18V
6 10	Short to GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	95-GND	Above 10M Ω (no conduction)	Below 10M Ω (there is a guide)

DTC: P1635 (Flash code 55)**5V #5 circuit abnormal****Cycle Description**

The engine control module (ECM) has 5 systems of 5V power circuits. 5V power circuit 5 will supply power to the common rail pressure sensor and EGR position sensor (6HL1 only).

ECM will set DTC if the 5V power circuit 5 voltages are lower or higher than the normal range.

Preconditions for DTC setting

- DTC P1630 have not been detected.
- Key switch input power voltage more than 16V but less than 32V

Setting conditions for DTC.

- Sensor or ECM power detected abnormal

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems

exist, they can be taken into account from the following reasons.

- Switch body abnormal
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;

– The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

3. Measure on the harness joint side.

4. Measure on the harness joint side.

7. Measure on the harness joint side.

8. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Is the engine equipped with 6HK1-TC?	—	Go to step 7 in the case of 6HK1-TC	Go to step 3 in the case of 6HL1
3	1. Set the key switch to the [OFF] position. ● Remove the EGR valve connector 3. Connect the DMM between the EGR position sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 5	Go to step 4
4	1. Set the key switch to the [OFF] position. ● Remove the EGR valve connector 3. Connect the DMM between the EGR position sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 6	Go to step 7
5	1. Check the power circuit between the EGR position sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
6	1. Check the power circuit between the EGR position sensor and ECM for short circuit to GND with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11

Step	Operation content	Datum value	YES	NO
7	1. Set the key switch to the [OFF] position. 2. Dismantle the joint of the common rail pressure sensor. 3. Connect the DMM between the common rail pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	5.5V	Go to step 9	Go to step 8
8	1. Set the key switch to the [OFF] position. 2. Dismantle the joint of the common rail pressure sensor. 3. Connect the DMM between the common rail pressure sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	4.5V	Go to step 10	Enter diagnostic help
9	1. Check the power circuit between the common rail pressure sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
10	1. Use a power distribution cabinet to check the short circuit between the common rail sensor and the ECM and GND circuits. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 12	Go to step 11
11	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 12	—
12	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1635 been detected?	—	Go to step 2	Go to step 13
13	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?		Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.



Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5 9	Short circuit to power supply circuit	Voltage measurement	- Remove the sensor connector - Key switch [ON]	87-GND	0V	Above 18V
6 10	Short to GND	Measurement of resistance value	- Remove the sensor connector - Key switch [OFF]	87-GND	Above 10M Ω (no conduction)	Below 10M Ω (there is a guide)

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;
- The non-matched terminals are joined together;
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.
- Harness damage:
- Check the appearance of the wire harness;
- Move sensor-related connectors, harnesses, etc. to

confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

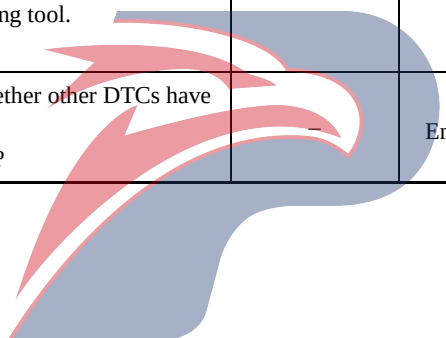
The numbers shown below indicate the step numbers in the table

2. Measure on the harness joint side.

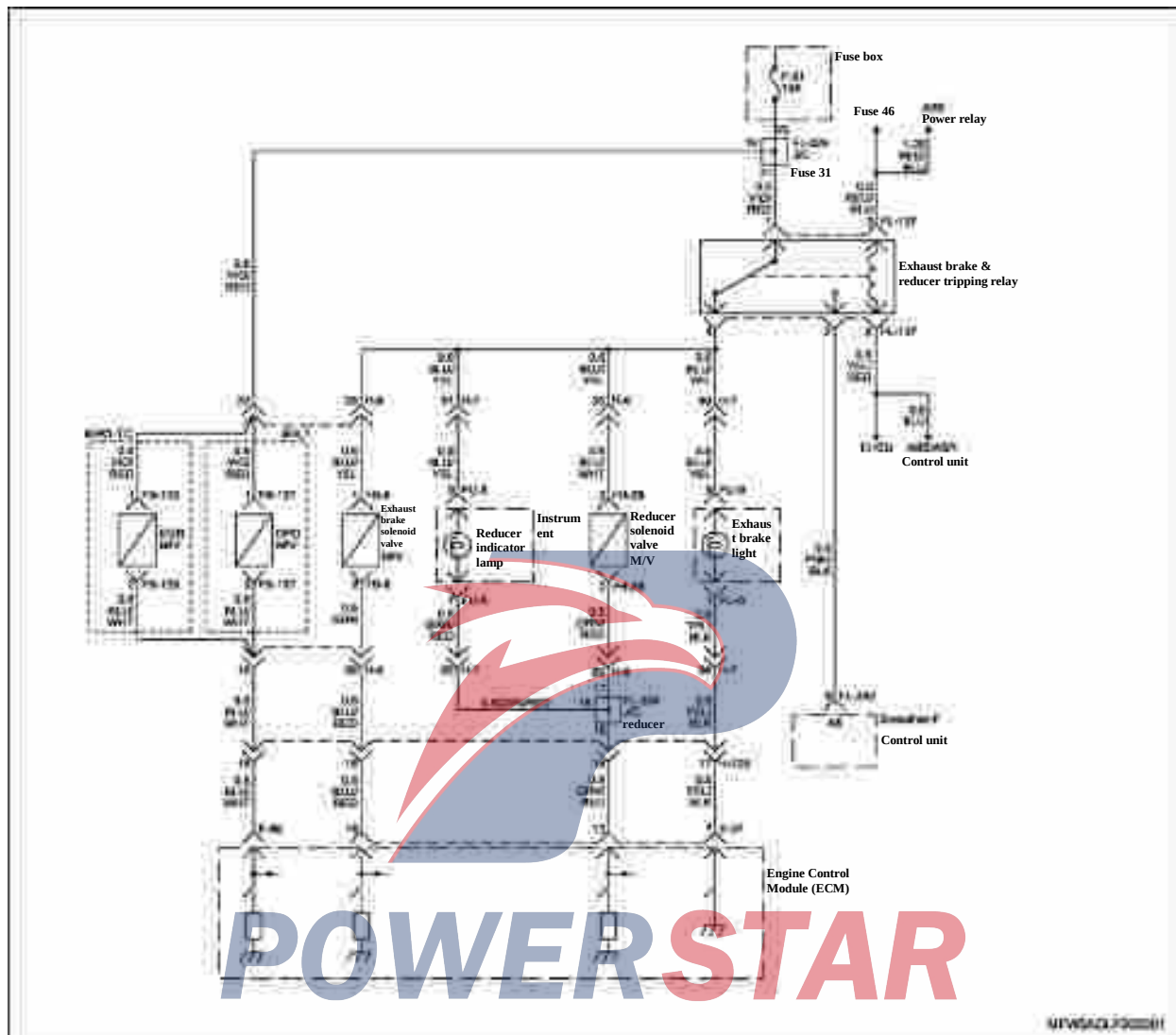
3. Measure on the harness joint side.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	–	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] position. 2. Remove the MAF sensor connector. 3. Connect the DMM between the MAF sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display of the digital multimeter higher than the reference value?	19V	Go to step 4	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the MAF sensor connector. 3. Connect the DMM between the MAF sensor power terminal and GND. 4. Set the key switch to the [OFF] state Is the display value of DMM below the reference value?	7V	Go to step 5	Enter diagnostic help
4	1. Check the power circuit between the MAF sensor and ECM for short circuit to the battery power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	–	Go to step 10	Go to step 6
5	1. Check the power circuit between the MAF sensor and ECM with the distributing box for the following state. ● Wire break ● High resistance ● Shorted to GND; 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	–	Go to step 10	Go to step 6
6	Measure the resistance of the step-down resistor. Is the resistance value a reference value?	50Ω	Go to step 7	Go to step 8
7	1. Check the step-down resistor circuit for the following states. ● Wire break ● High resistance ● Shorted to GND; 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	–	Go to step 10	Go to step 9
8	Replace the buck resistor. Is the disposal completed?	–	Go to step 10	–

Step	Operation content	Datum value	YES	NO
9	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 10	—
10	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1636 been detected?	—	Go to step 2	Go to step 11
11	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions



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DTC: P1683 (Flash code 38)**DPD exhaust throttle circuit input high****Cycle Description**

The [ON] and [OFF] of the exhaust throttle valve are controlled by switching of the DPD solenoid valve (M/V) control gas channel. When the DPD solenoid valve opens, the gas passage opens and the exhaust throttle valve operates. When the PDD solenoid valve is in the [OFF] state, the gas passage will be closed, and the atmosphere will flow in and the exhaust throttle will be released. Engine Control Module (ECM) the DPD solenoid valve is energized (ON) when the DPD regeneration switch is ON and all the vehicle conditions are met.

The main fault phenomenon

- DPD regeneration stop

Preconditions for DTC setting

Battery voltage is normal

- Exhaust throttle control state is ON
- Vehicles equipped with DPD

Setting conditions for DTC.

- Harness short circuit · breakage detected

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1683 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following

checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage:

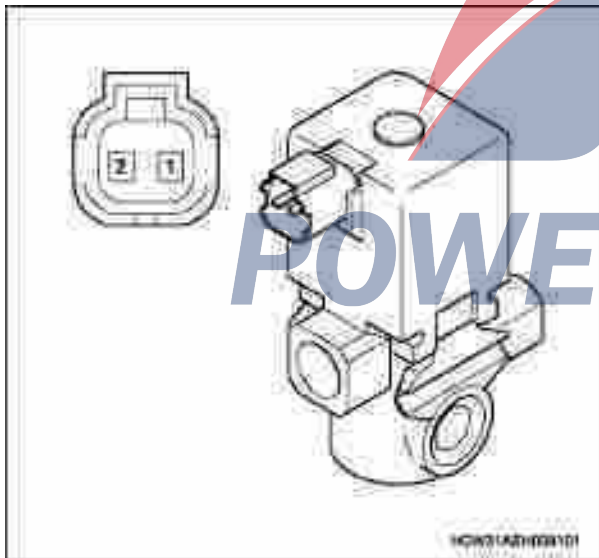
- Check the appearance of the wire harness;

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the installation state of the DPD solenoid valve connector, ECM connector and intermediate connector. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the DPD solenoid valve joint. 3. Measure the resistance value of the EGR solenoid valve. Is the resistance value within the reference range?	20~60Ω	Go to step 4	Go to step 5
4	1. Check the power circuit between the DPD solenoid valve and ECM with the distributing box for the following state. • Short circuit to GND circuit • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6
5	Replace the DPD solenoid valve. Is the disposal completed?	—	Go to step 7	—

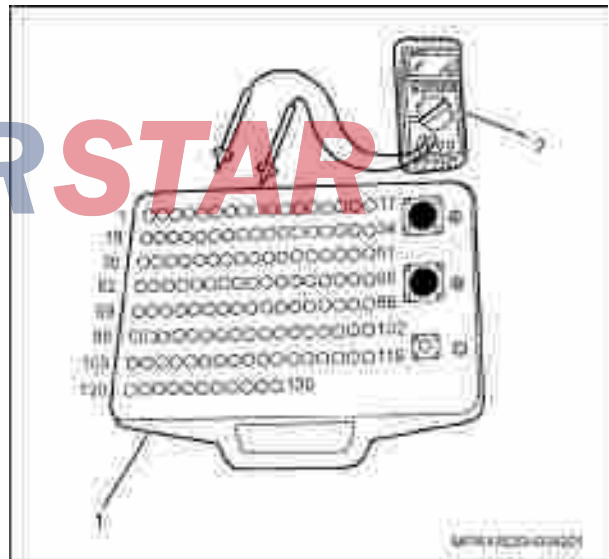
Step	Operation content	Datum value	YES	NO
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: ●MAF correction value; ●Q adjustment value data; ●ID code of the ejector; ● Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 7	—
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to [OFF], and conduct test operation for more than 15 seconds. 4. Confirm DTC with the scanning tool. Has DTC P1683 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About DPD M/V/ Solenoid valve.



About distribution cabinet

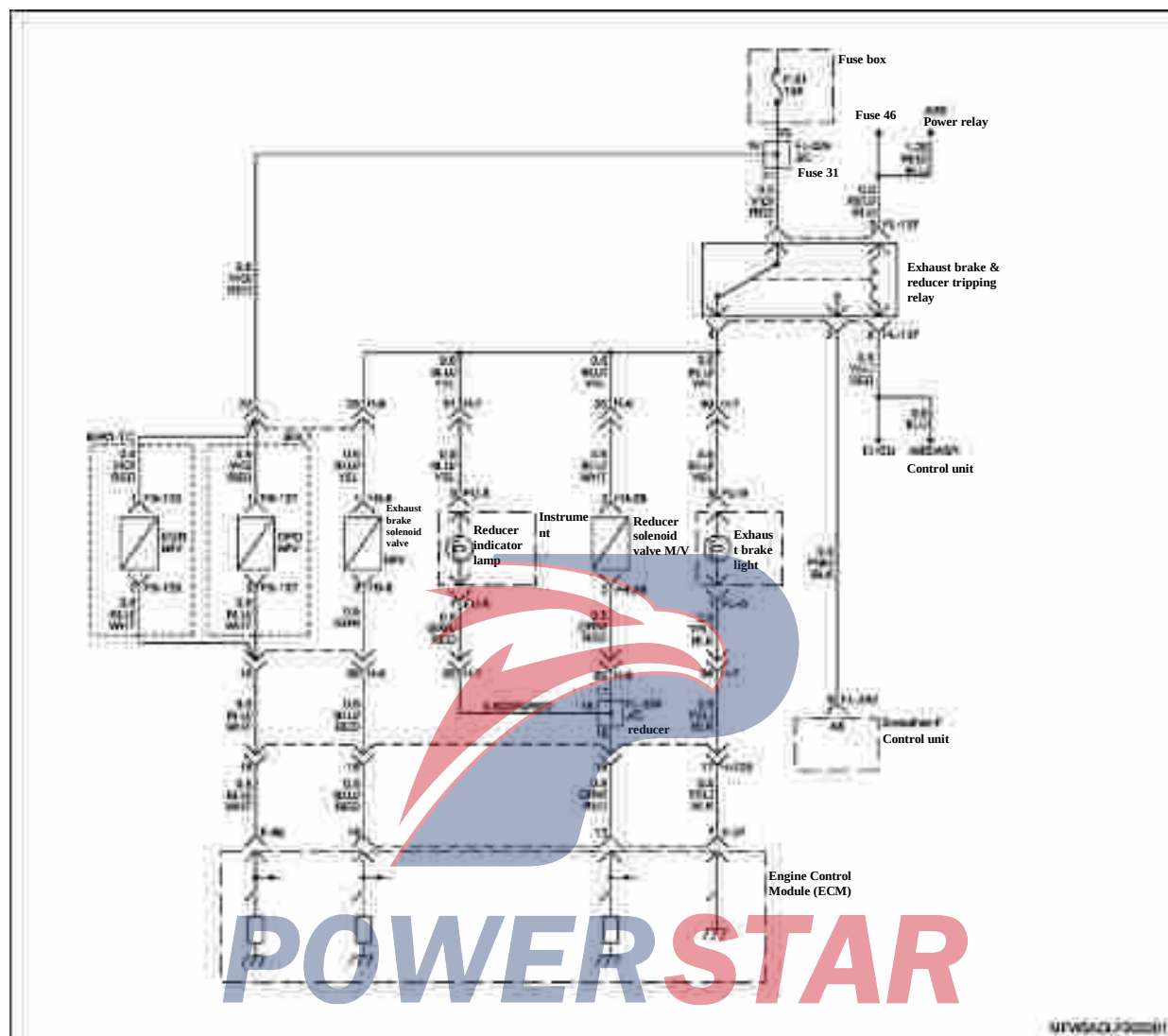
In a program that requires the use of a power distribution cabinet, check the following points.



Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Broken wire / high resistance	Measurement of resistance value	●Remove the solenoid valve joint Key switch [OFF]	102 - Solenoid valve joint No.2 terminal	Below 10MΩ(there is a guide)	Above 10MΩ (no conduction)

DTC: P1684 (Flash code 38)**DPD exhaust throttle circuit input low****Cycle Description**

The [ON] and [OFF] of the exhaust throttle valve are controlled by switching of the DPD solenoid valve (M/V) control gas channel. When the DPD solenoid valve opens, the gas passage opens and the exhaust throttle valve operates. When the PDD solenoid valve is in the [OFF] state, the gas passage will be closed, and the atmosphere will flow in and the exhaust throttle will be released. Engine Control Module (ECM) the DPD solenoid valve is energized (ON) when the DPD regeneration switch is ON and all the vehicle conditions are met.

The main fault phenomenon

- DPD regeneration stop
- Driving difficult

Preconditions for DTC setting

- The battery voltage is normal
- Exhaust throttle control state is OFF
- Vehicles equipped with DPD

Setting conditions for DTC.

- The relay/valve harness disconnection, short circuit or breakage detected

Measures upon DTC setting

- Engine fault indicator lights up
- When the DTC P1684 is detected, it will not go out within 15 seconds after the key switch is set to [OFF].

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by

DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

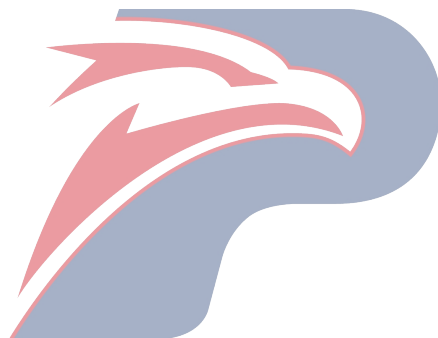
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
 - The terminals are removed from the wire harness;
 - The non-matched terminals are joined together;
 - Damage to the joint lock;
 - The connection between the terminal and the wire is poor.
- Harness damage:
 - Check the appearance of the wire harness;
 - Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

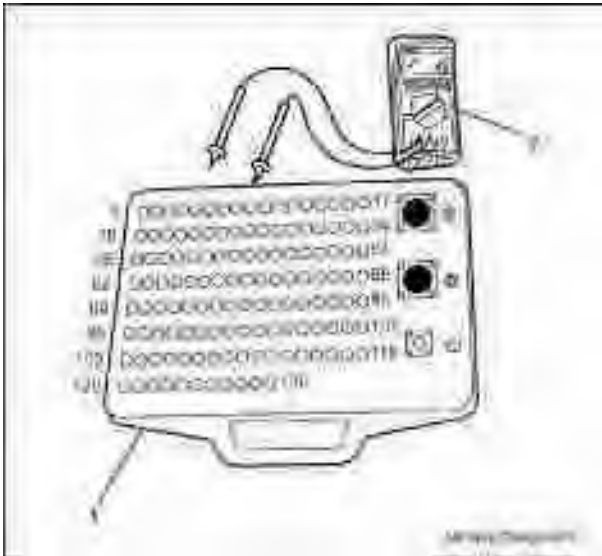
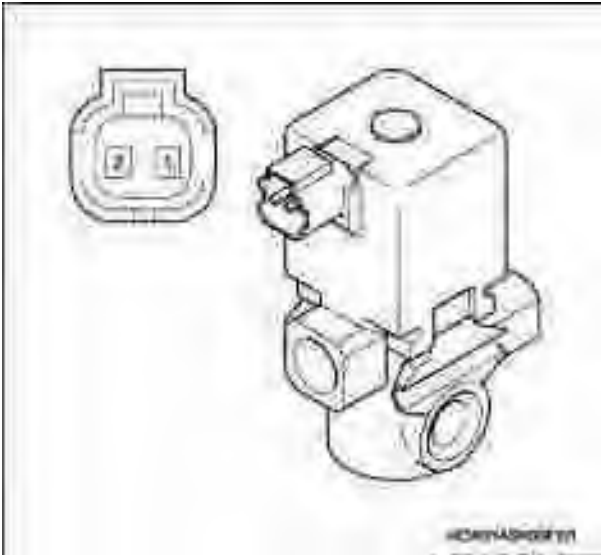
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the installation state of the DPD solenoid valve connector, ECM connector and intermediate connector. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the DPD solenoid valve joint. 3. Measure the resistance value of the EGR solenoid valve. Is the resistance value within the reference range?	20~60Ω	Go to step 4	Go to step 5
4	1. Check the power circuit between the DPD solenoid valve and ECM with the distributing box for the following state. • Short circuit to GND circuit • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6
5	Replace the DPD solenoid valve. Is the disposal completed?	—	Go to step 7	—
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 7	—

Step	Operation content	Datum value	YES	NO
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P1684 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions



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About DPD Solenoid Valves



Name

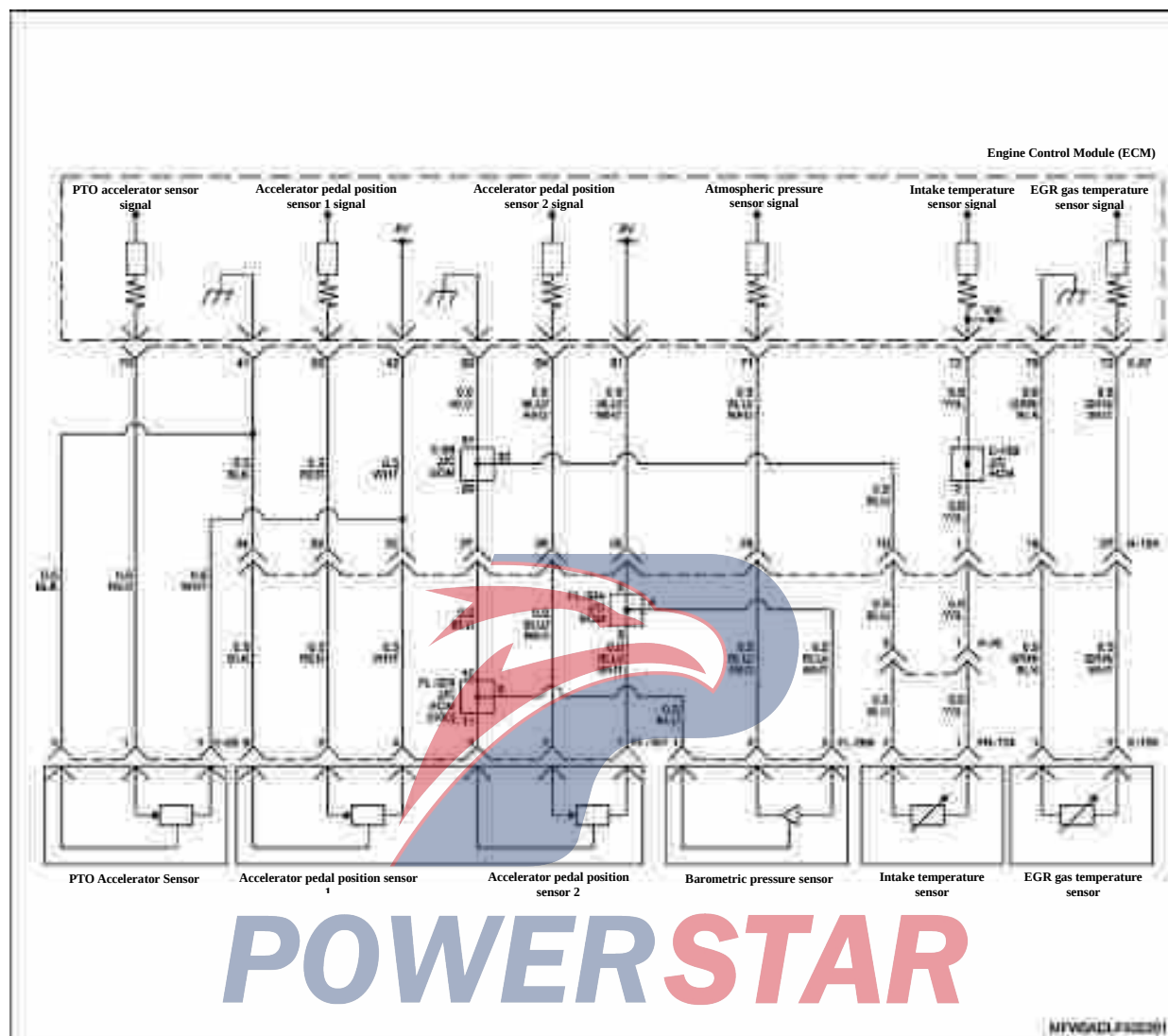
About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.

- 1. Distribution Cabinet
- 2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Short circuit to power supply circuit	Voltage measurement	●Remove the solenoid valve joint ● Key switch [ON]	102-GND	0V	Above 18V
	Short to GND	Measurement of resistance value	●Remove the solenoid valve joint Key switch [OFF]	102-GND	Above 10MΩ (no conduction)	Below 10MΩ(there is a guide)

POWERSTAR

DTC: P2032 (Flash code 41)**EGR temperature sensor loop input low****Cycle Description**

The EGR temperature sensor is installed on the upstream side of the EGR valve. The EGR gas temperature sensor is a thermistor-type sensor whose resistance value changes due to the EGR gas temperature, and the output voltage also changes correspondingly. As the EGR gas temperature rises, the resistance values decreases and the output voltage decreases. EGR gas temperature will increase as EGR valve opens and decreases as EGR valve closes. The engine control module (ECM) detects the gas temperature of EGR according to the signal transmitted by EGR gas temperature sensor and controls EGR.

ECM sets the DTC when the signal voltage out from

the EGR gas temperature sensor deviates from the normal range.

Preconditions for DTC setting

- The battery voltage is normal
- DTC P1630 and P1633 have not been detected.
- EGR valve of vehicles equipped with 6HK1-TC

Setting conditions for DTC.

- EGR gas temperature sensor voltage below 0.1V

Measures upon DTC setting

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage:

- Check the appearance of the wire harness;

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

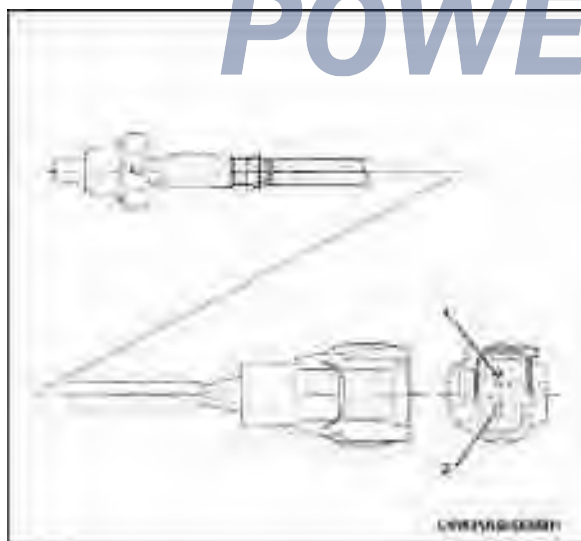
2. When the fault is present, the display value of [EGR Temperature Sensor] is below the reference value. When it is above the reference value, please refer to the diagnostic aid to clarify the reason.

4. When the EGR gas temperature sensor connector is removed and the [EGR temperature sensor] display is above the reference value, the sensor's subsequent circuit is in normal state, including ECM.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	1. Set the key switch to the [OFF] state 2. Confirm the contents displayed in the data display [EGR temperature sensor] with the scanning tool. [Is the value of [EGR temperature sensor] below the reference value?	0.1V	Go to step 3	Enter diagnostic help
3	Check the installation state of the EGR gas temperature sensor, ECM connector and intermediate connector. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 8	Go to step 4
4	1. Set the key switch to the [OFF] position. 2. Install the connector for the EGR gas temperature sensor. 3. Set the key switch to the [OFF] state 4. Confirm the contents displayed in the data display [EGR temperature sensor] with the scanning tool. [Is the value of [EGR temperature sensor] above the reference value?	4.8V	Go to step 6	Go to step 5
5	1. Check the power circuit between the ECM and EGR sensor for the following state with the distributing box: • Short circuit to the EGR gas temperature sensor GND circuit	—	Go to step 8	Go to step 7

Step	Operation content	Datum value	YES	NO
	• Shorted to GND; 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?			
6	Replace the EGR gas temperature sensor. Is the disposal completed?	—	Go to step 8	—
7	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector; • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 9	—
8	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to [OFF] and start the engine after 15s. 4. Confirm DTC with the scanning tool. Has DTC P2032 been detected?	—	Go to step 2	Go to step 9
9	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About EGR gas temperature sensor

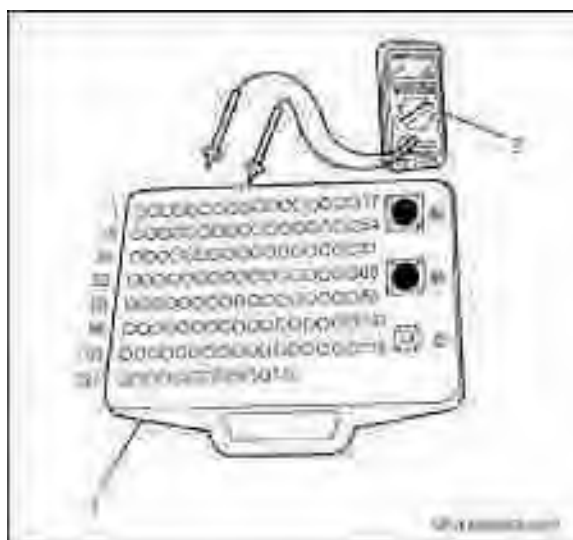


Name

1. GND
2. Signal

About distribution cabinet

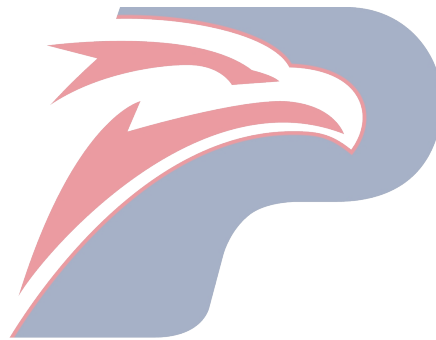
In a program that requires the use of a power distribution cabinet, check the following points.



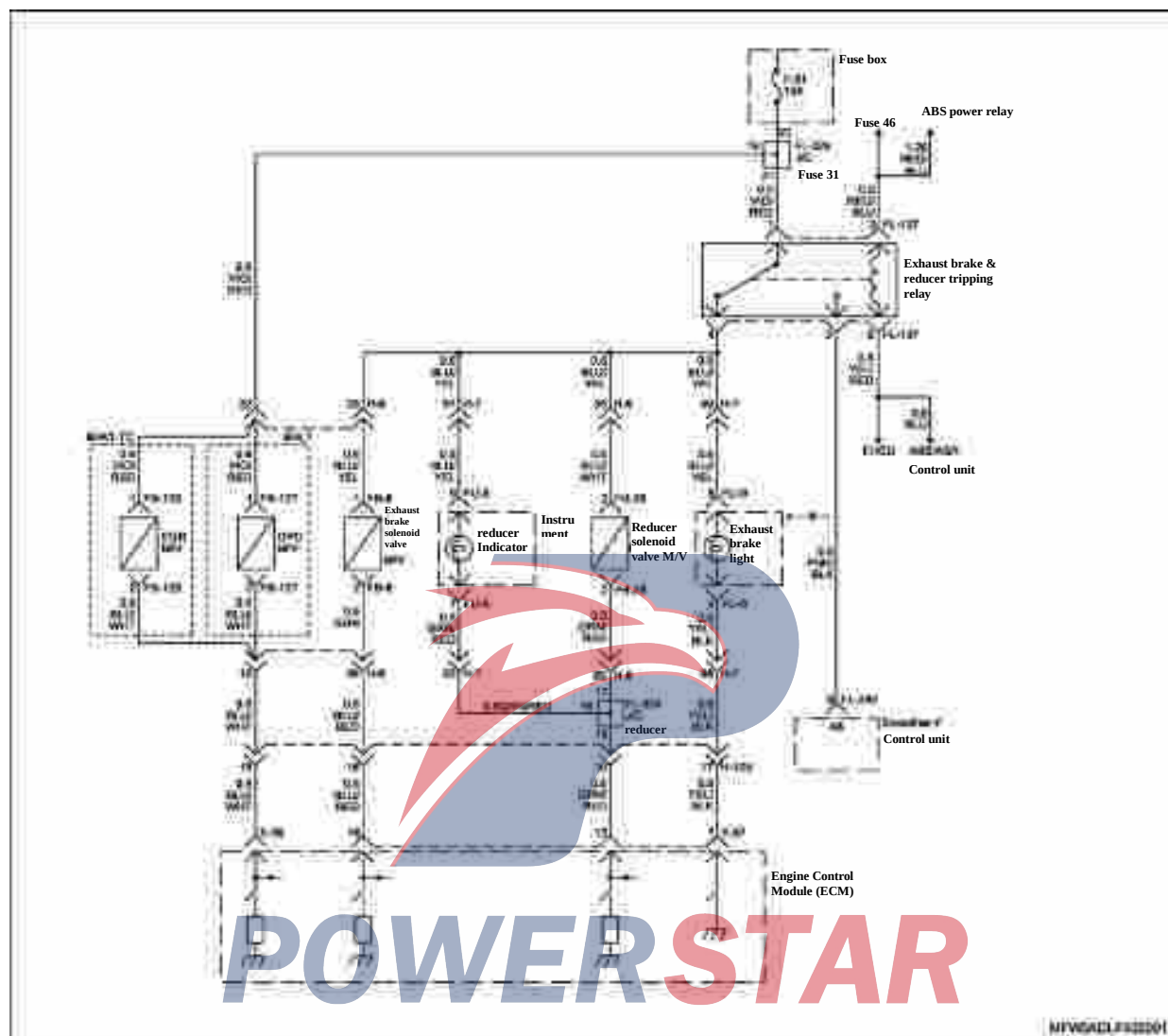
Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
5	Short circuit to GND circuit/GND	Measurement of resistance value	Remove the sensor connector Key switch [OFF]	73-79 73-GND	Above 10M Ω (no conduction)	Below 10M Ω (there is a guide)



POWERSTAR

DTC: P2144 (Flash code 45)**EGR control valve circuit input is low****Cycle Description**

The EGR solenoid valve (M/V) is installed on the left side of the cab. The EGR solenoid valve can control the gas supplied to the EGR valve. EGR magnet valve is powered by main relay. When it receives the driving signal from ECM, it will become [ON]. When the engine coolant temperature rises and EGR action conditions are established, ECM will change the EGR magnet valve to [ON]. ECM is equipped with a monitoring circuit, which will monitor the working condition of the solenoid valve. ECM will set DTC if ECM does not transmit drive signal to EGR magnet valve while the valve becomes [ON].

Preconditions for DTC setting

- EGR valve of vehicles equipped with 6HK1-TC
- Key switch input power voltage more than 16V but less than 32V
- EGR valve control is OFF

Setting conditions for DTC.

- Relay/valve harness disconnection short circuit · breakage detected

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness;

- The non-matched terminals are joined together;

- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage:

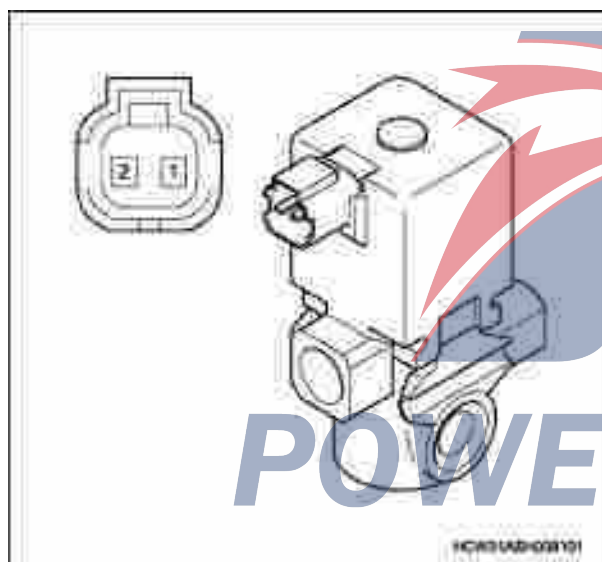
- Check the appearance of the wire harness;

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the installation state of the EGR solenoid valve connector, ECM connector and intermediate connector. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 3
3	1. Set the key switch to the [OFF] position. 2. Remove the connector of the EGR solenoid valve. 3. Measure the resistance value of the EGR solenoid valve. Is the resistance value within the reference range?	20~60Ω	Go to step 4	Go to step 5
4	1. Check the power circuit between the EGR solenoid valve and ECM with the distributing box for the following state. • Short connection with GND circuits • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6
5	Replace the EGR solenoid valve. Is the disposal completed?	—	Go to step 7	—
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: • MAF correction value; • Q adjustment value data; • ID code of the ejector;	—	Go to step 7	—

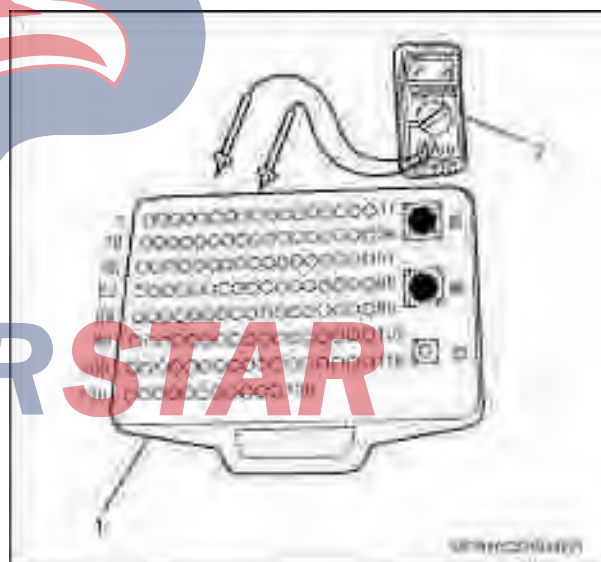
Step	Operation content	Datum value	YES	NO
	Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?			
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to [OFF] and carry out trial run after 15s. 4. Confirm DTC with the scanning tool. Has DTC P2144 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About EGR solenoid valve



About distribution cabinet

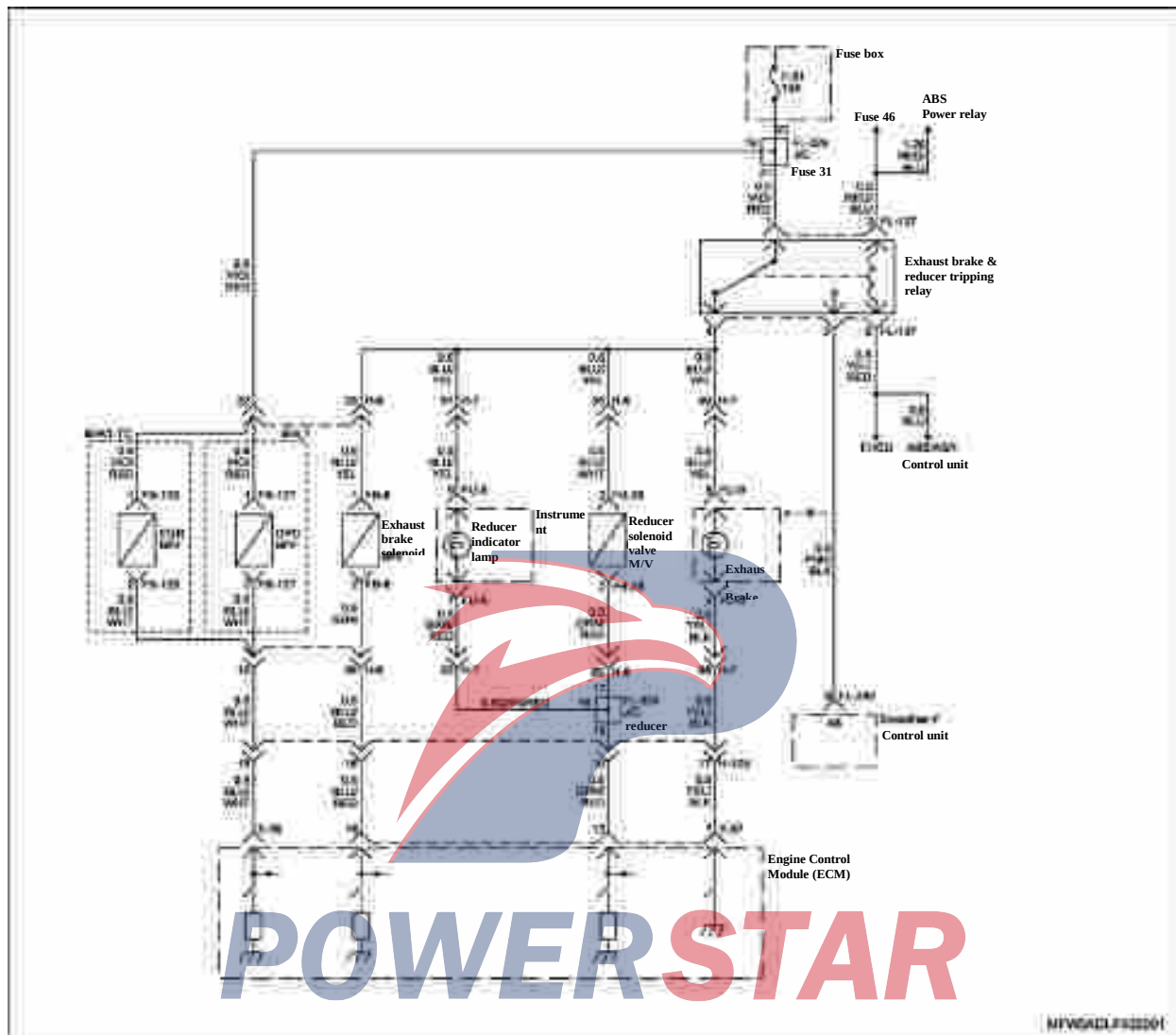
In a program that requires the use of a power distribution cabinet, check the following points.



Name

1. Distribution Cabinet
2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Broken wire / high resistance	Measurement of resistance value	- Remove the solenoid valve joint - Key switch [ON]	102 - Solenoid valve joint No.2 terminal	Below 10MΩ (there is a guide)	Above 10MΩ (no conduction)

DTC: P2145 (Flash code 45)**EGR control valve circuit input is high****Cycle Description**

The EGR solenoid valve (M/V) is installed on the left side of the cab. The EGR solenoid valve can control the gas supplied to the EGR valve. EGR magnet valve is powered by main relay. When it receives the driving signal from ECM, it will become [ON]. When the engine coolant temperature rises and EGR action conditions are established, ECM will change the EGR magnet valve to [ON]. ECM is equipped with a monitoring circuit, which will monitor the working condition of the solenoid valve. When the ECM does not output a drive signal to the EGR solenoid valve but the EGR solenoid valve goes to the [OFF] state, the DTC is set.

Preconditions for DTC setting

- EGR valve of vehicles equipped with 6HK1-TC
- Key switch input power voltage more than 16V but less than 32V
- EGR valve control in ON

DTC setting conditions

- Relay/valve harness disconnection short circuit breakage detected

Disposal when setting DTC

- Engine fault indicator lights up.

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness

- The non-matched terminals are joined together

- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage

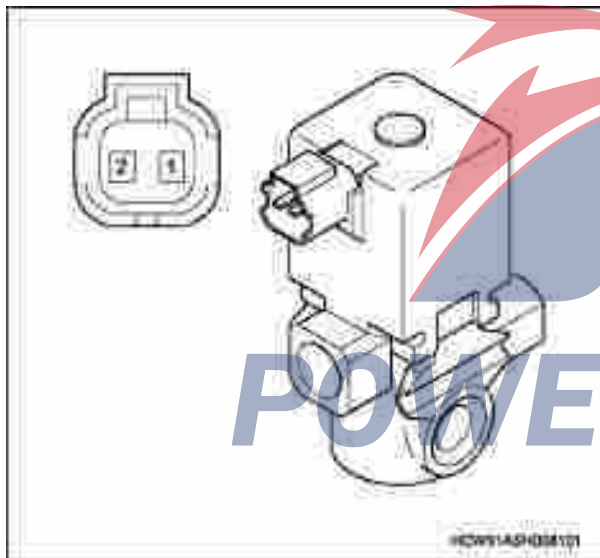
- Check the appearance of the wire harness

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the installation state of EGR solenoid valve connector, ECM connector and intermediate connector. 1. Check the installation status of the connector for looseness and slackness. 2. If fault detected, repair as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 3
3	1. Turn the key switch to the OFF position. 2. Remove the connector of the EGR solenoid valve. 3. Measure the resistance value of the EGR solenoid valve. Is the resistance value within the reference range?	20~60Ω	Go to step 4	Go to step 5
4	1. Use distribution cabinet to check whether there are the following states in the circuit between EGR solenoid valve and ECM • Short connection with GND circuits • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 7	Go to step 6
5	Replace the EGR solenoid valve. Is the disposal completed?	—	Go to step 7	—
6	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the ejector	—	Go to step 7	—

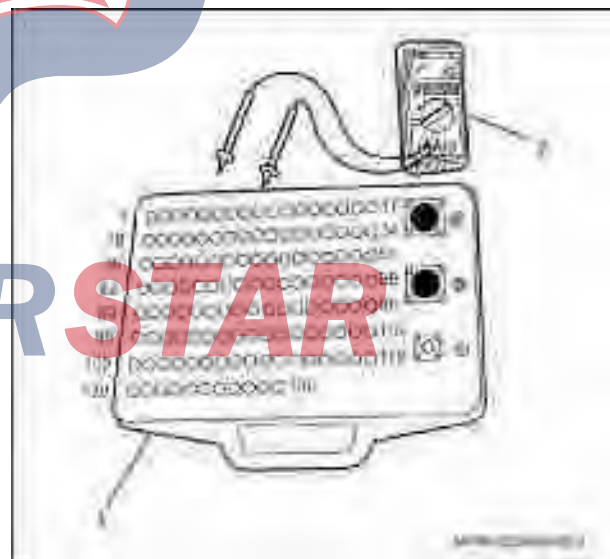
Step	Operation content	Datum value	YES	NO
	Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?			
7	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC P2145 been detected?	—	Go to step 2	Go to step 8
8	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About EGR solenoid valve



About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.



Name

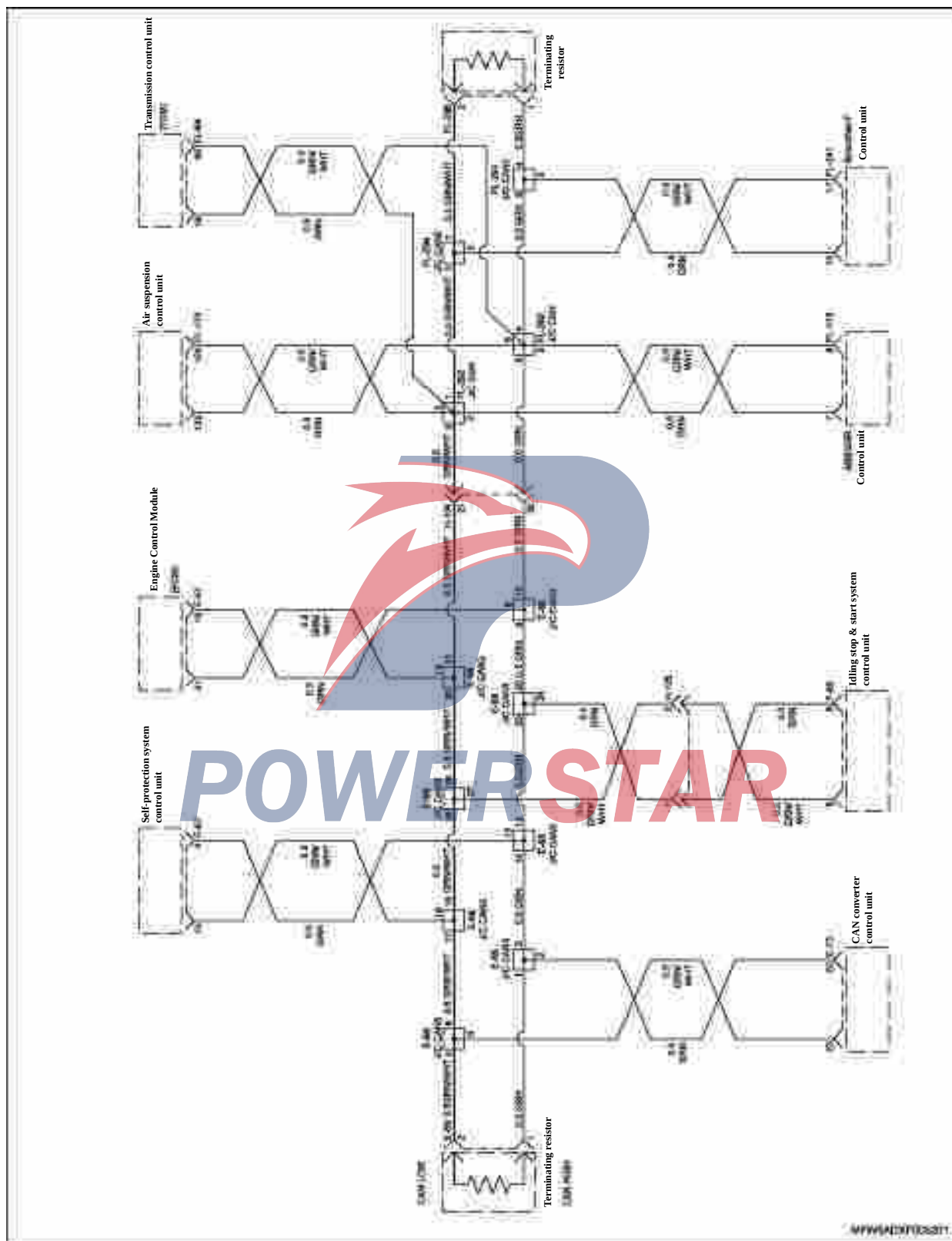
Distribution Cabinet

2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Short circuit to power supply circuit	Voltage measurement	- Remove the solenoid valve joint - Key switch [ON]	102-GND	0V	Above 18V
	Short to GND	Measurement of resistance value	- Remove the solenoid valve joint - Key switch [OFF]	102-GND	10MΩ or more (no conduction)	Below 10MΩ (there is a guide)

DTC: U2104 (Blinking code 84)

CAN communication abnormal



Cycle Description

The engine control module (ECM) communicates with other ECUs via the CAN communication circuit and the transmission control module (TCM), the Smoother-F control unit (ECU), and the ABS/ASR control unit. ECM data can be exported through CAN - HIGH and the data of other ECUs can be entered through CAN-LOW circuit. CAN always communicate at definite speed and the data output frequency and input frequency must be consistent.

If ECM detects CAN communication abnormal, this DTC will be set.

DTC setting conditions

- CAN communication failure detected

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).

- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.

- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the following reasons.

- Abnormality of main body of Switch

- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad
- The terminals are removed from the wire harness
- The non-matched terminals are joined together
- Damage to the joint lock;
- The connection between the terminal and the wire is poor.

- Harness damage

- Check the appearance of the wire harness

- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

The numbers shown below indicate the step numbers in the table

2. It can be considered intermittent if no DTC is detected. Locate the causes by referring to the diagnostic assistance description.

5. Measure on the harness joint side.

6. Make measurements on the side of the TCM harness connector.

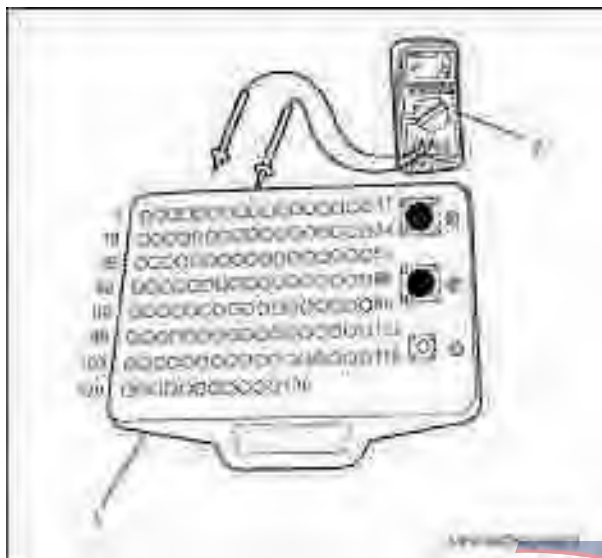
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check"
2	1. Set the key switch to the [ON] position for 15 seconds or more. 2. Connect the scan tool to the vehicle. 3. Start the engine. 4. Confirm DTC. Has DTC U2104 been detected?	—	Go to step 3	Enter diagnostic help
3	Has DTC U2106 been detected?	—	Enter into DTC U2106	Go to step 4

Step	Operation content	Datum value	YES	NO
4	1. Turn the key switch to the OFF position. 2. Remove all ECU related to CAN communication. 3. Connect the power distribution cabinet to the ECM. 4. Use the power distribution cabinet to measure the resistance between the CAN-HIGH side terminal and the CAN-LOW side terminal of the ECM. Is the resistance value within the reference range?	50~70Ω	Go to step 10	Go to step 5
5	1. Remove the joint from the resistor 1. 2. Use jumpers to short the terminals of the resistor 1 connector. 3. Use the distribution cabinet to measure the resistance between the CAN-HIGH side terminals and the CAN-LOW side terminals of ECM. Is the resistance value within the reference range?	Vehicle equipped with ABS: about 60Ω Vehicle with ABS: about 120Ω	Go to step 13	Go to step 6
6	1. Install the resistor 1 joint and remove the joint from the resistor 2. 2. Use jumpers to short the terminals of the resistor 2 connector. 3. Use the distribution cabinet to measure the resistance between the CAN-HIGH side terminals and the CAN-LOW side terminals of ECM. Is the resistance value within the reference range?	约 120Ω	Go to step 14	Go to step 7
7	1. Use the power distribution cabinet to check the CAN-HIGH circuit and CAN-LOW circuit between ECM and resistor 1 for disconnection or high resistance. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 8
8	1. Use the power distribution cabinet to check the CAN-HIGH circuit and CAN-LOW circuit between ECM and resistor 2 for disconnection or high resistance. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 15
9	1. Check the CAN-HIGH and CAN-LOW circuit for short circuit to GND with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 10
10	1. Check the CAN-HIGH and CAN-LOW circuit for short circuit to power circuit and ignition power circuit with the distributing box. 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 11
11	1. Check the CAN-LOW circuit and CAN-HIGH circuit between the ECM and ABS/ASR control unit with the distributing box for the following state. • Wire break • High resistance • Shorted to GND; 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 12

Step	Operation content	Datum value	YES	NO
12	1. Use the power distribution cabinet to check whether the CAN-LOW circuit and CAN-HIGH circuit between the ECM and the self-protection system control unit have the following states? • Wire break • High resistance • Shorted to GND; 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 16	Go to step 15
13	Replace resistor 1. Is the disposal completed?	—	Go to step 16	—
14	Replace resistor 2. Is the disposal completed?	—	Go to step 16	—
15	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the ejector • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 16	—
16	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC U2104 been detected?	—	Go to step 2	Go to step 17
17	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.



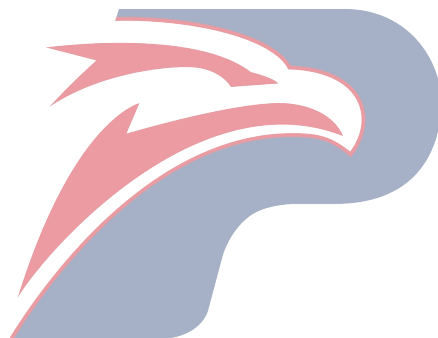
Name

Distribution Cabinet

2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
7	Broken wire / high resistance	Measurement of resistance value	- Remove all the relevant ECU joints and resistor 1 joint - Key switch [OFF]	37 - resistor 1 joint No.1 terminal 18 - resistor 1 joint No.2 terminal	Below 10MΩ (there is a guide)	10MΩ or more (no conduction)
8	Broken wire / high resistance	Measurement of resistance value	- Remove all the relevant ECU joints and resistor 2 joint - Key switch [OFF]	37 - resistor 2 joint No.1 terminal 18 - resistor 2 joint No.2 terminal	Below 10MΩ (there is a guide)	10MΩ or more (no conduction)
9	Short to GND	Measurement of resistance value	- Remove all the relevant ECU joints - Key switch [OFF]	18-GND 37-GND	10MΩ or more (no conduction)	Below 10MΩ (there is a guide)
10	Short circuit to power supply circuit	Voltage measurement	- Remove all the relevant ECU joints - Key switch [ON]	18-GND 37-GND	0V	Above 18V

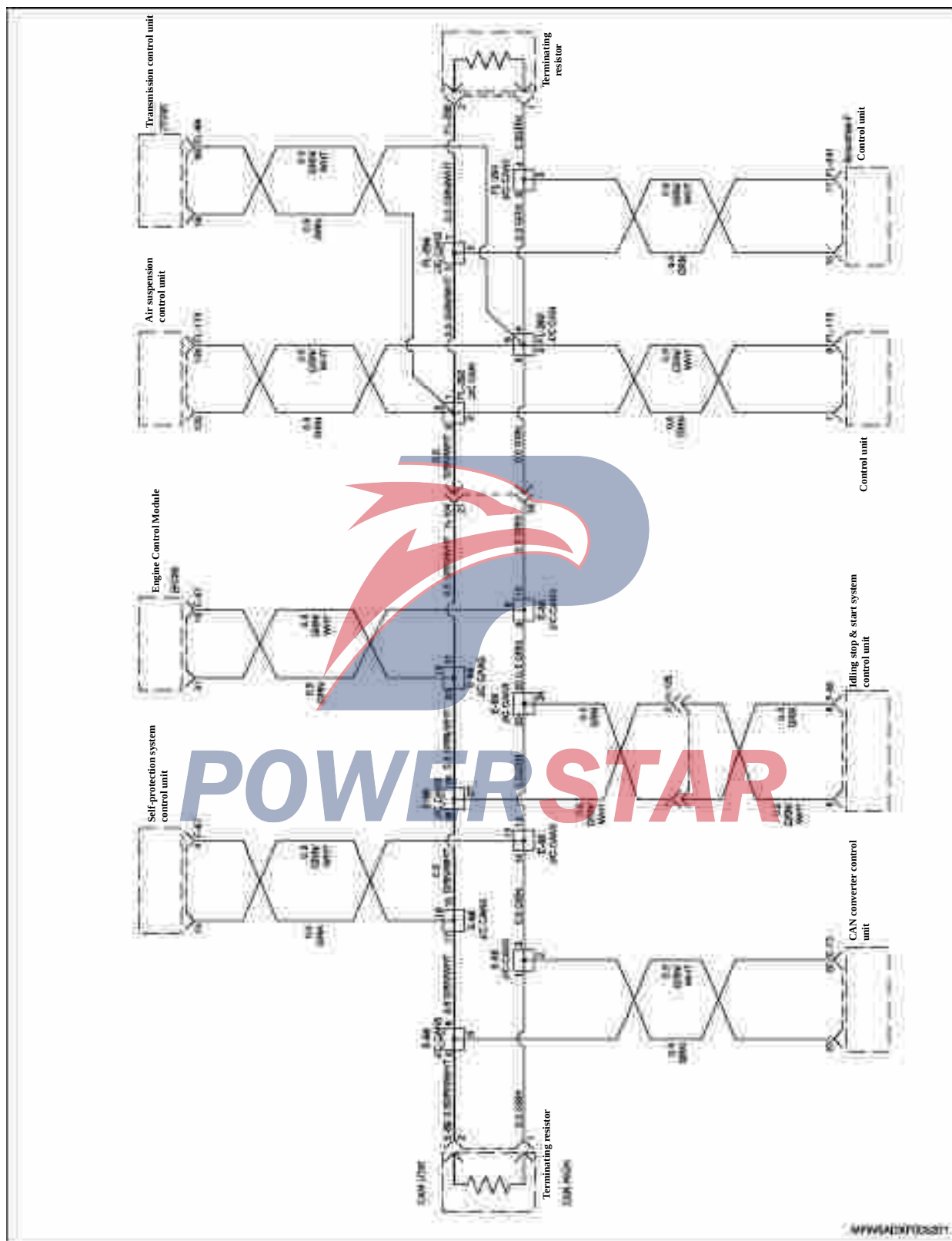
Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
11	Broken wire / high resistance	Measurement of resistance value	- Remove all the relevant ECU joints - Key switch [OFF]	37-ABS/ASR control unit connector No.14, 26 Terminal 18-ABS/ASR control unit connector No.15, 27 Terminal	Below 10MΩ(there is a guide)	10MΩ or more (no conduction)
12	Broken wire / high resistance	Measurement of resistance value	- Remove all the relevant ECU joints - Key switch [OFF]	37 - self-protection system ECU joint No, 4 Terminal 18 - self-protection system ECU joint No, 15 Terminal	Below 10MΩ(there is a guide)	10MΩ or more (no conduction)



POWERSTAR

DTC: U2106 (Blinking code 85)

Communication with TCM (transmission control module) is abnormal



Cycle Description

ECM communicates with transmission control unit (TCM), Smoother-F control unit (ECU) and ABS/ASR control unit through the CAN communication circuit. ECM data can be exported through CAN - HIGH and the data of other ECUs can be entered through CAN-LOW circuit. CAN always communicate at definite speed and the data output frequency and input frequency must be consistent.

The DTC is set when the ECM detects an interruption in CAN communication with the TCM or the Smoother-F ECUCAN.

DTC setting conditions

- Interruption of communication to TCM

Disposal when setting DTC

- Engine fault indicator lights up

MIL/DTC clearing conditions

- The current fault is changed to the past fault (resume) when it is within the set condition scope by DTC (fault repair).
- The engine fault indicator lamp turns to a past failure. After the key switch is set to the [OFF] state, the indicator lamp turns off when the key switch is set to the [ON] state again.
- The DTC can be cleared by operating the scan tool or accelerator pedal.

Diagnostic help

When there are suspicions that intermittent problems exist, they can be taken into account from the

following reasons.

- Abnormality of main body of Switch
- Maladjustment of the switch
- Poor connection of the harness connector
- Harness is not good.
- The cladding layer of the harness is broken.
- Wire breakage inside the harness cladding

In order to find out these reasons, the following checks should be carried out.

- Connection between harness connector and ECM connector is bad

- The terminals are removed from the wire harness
- The non-matched terminals are joined together
- Damage to the joint lock;

- The connection between the terminal and the wire is poor.

- Harness damage

- Check the appearance of the wire harness
- Move sensor-related connectors, harnesses, etc. to confirm the display of items related to the data display of the scan tool. If there is a change in the display, there is a faulty part.

Test instruction

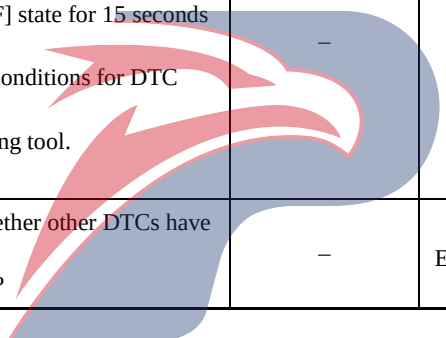
The numbers shown below indicate the step numbers in the table

2. It can be considered intermittent if no DTC is detected. Locate the causes by referring to the diagnostic assistance description.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Perform "OBD System Check"
2	1. Set the key switch to the [ON] position for 15 seconds or more. 2. Connect the scan tool to the vehicle. 3. Start the engine. 4. Confirm DTC. Has DTC U2106 been detected?	—	Go to step 3	Enter diagnostic help
3	Is the transmission automatic transmission (A/T)?	—	Go to step 4 in the case of A/T	Go to step 7 in the case of Smoother-F

Step	Operation content	Datum value	YES	NO
4	1. Check the CAN-LOW circuit and CAN-HIGH circuit between the ECM and TCM with the distributing box for the following state. • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 15	Go to step 5
5	1. Check the CAN-LOW circuit and CAN-HIGH circuit between the ECM and TCM with the distributing box for the following state. • Short circuit to GND circuit • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 15	Go to step 6
6	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC U2106 been detected?	—	Go to step 7	Enter diagnostic help
7	Do you detect DTC U2104 at the same time?	—	Go to step 14	Go to step 8
8	Replace the TCM. Is the disposal completed?	—	Go to step 15	—
9	1. Use the distribution cabinet to check whether there are the following states in the CAN-LOW circuit and CAN-HIGH circuit between ECM and Smoother-F ECU? • Wire break • High resistance 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 15	Go to step 10
10	1. Use the distribution cabinet to check whether there are the following states in the CAN-LOW circuit and CAN-HIGH circuit between ECM and Smoother-F ECU? • Shorted to GND; • Short circuit between the power circuit and ignition power circuit 2. When a fault is detected, make repairs as needed. Whether the fault is detected or not?	—	Go to step 15	Go to step 11
11	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC U2106 been detected?	—	Go to step 12	Enter diagnostic help

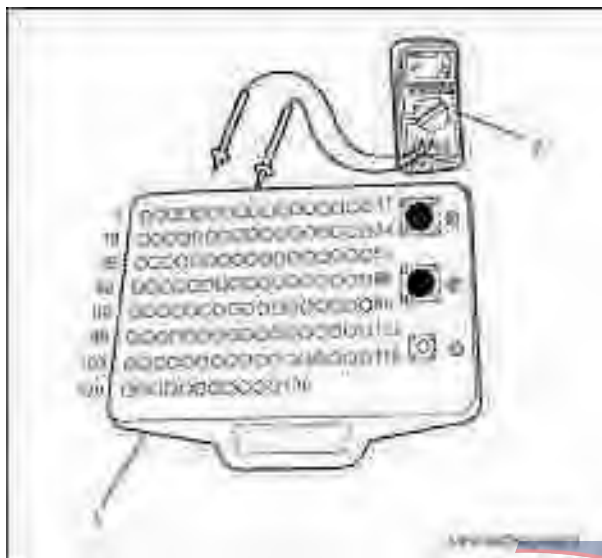
Step	Operation content	Datum value	YES	NO
12	Do you detect DTC U2104 at the same time?	—	Go to step 14	Go to step 13
13	Replace the Smoother-F ECU. Is the disposal completed?	—	Go to step 15	—
14	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the ejector • Starting times of the engine Refer to the description for programming in this chapter. Is the disposal completed?	—	Go to step 15	—
15	Confirm the DTC again. 1. Connect all the harnesses. 2. Clear DTC with the scanning tool. 3. Set the key switch to the [OFF] state for 15 seconds or more. 4. Perform a test run under [Preconditions for DTC settings]. 5. Confirm DTC with the scanning tool. Has DTC U2106 been detected?	—	Go to step 2	Go to step 16
16	Use the scan tool to confirm whether other DTCs have been detected? Is there any other DTC detected?	—	Enter each DTC for diagnosis	Confirm the repair conditions



POWERSTAR

About distribution cabinet

In a program that requires the use of a power distribution cabinet, check the following points.



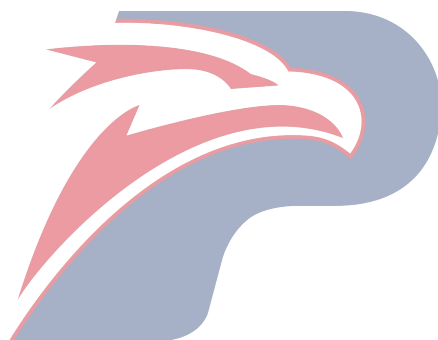
Name _____

Distribution Cabinet

2. DMM

Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
4	Broken wire / high resistance	Measurement of resistance value	Remove the joints of TCM • Key switch [OFF]	18- TCM joints No. 19 Terminal 37-TCM Joint No.46 Terminal	Below 10MΩ(there is a guide)	10MΩ or more (no conduction)
5	Short to GND	Measurement of resistance value	Remove the joints of TCM • Key switch [OFF]	18-GND 37-GND	10MΩ or more (no conduction)	Below 10MΩ(there is a guide)
	Short circuit to power supply circuit	Voltage measurement	Remove the joints of TCM • Key switch [ON]	18-GND 37-GND	0V	Above 18V
9	Broken wire / high resistance	Measurement of resistance value	• Remove the Smoother-F ECU joint • Key switch [OFF]	18-Smoother-F ECU joint No.B-15 terminal 37-Smoother-F ECU joint No.B-17 terminal	Below 10MΩ(there is a guide)	10MΩ or more (no conduction)

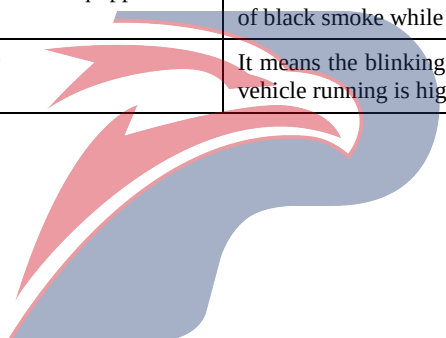
Step	Check item	Testing method	Measurement conditions	Measuring terminal No.	Normal value	Outliers
10	Short to GND	Measurement of resistance value	- Remove the Smoother-F ECU joint - Key switch [OFF]	18-GND 37-GND	10MΩ or more (no conduction)	Below 10MΩ(there is a guide)
	Short circuit to power supply circuit	Voltage measurement	- Remove the Smoother-F ECU joint - Key switch [ON]	18-GND 37-GND	0V	Above 18V



POWERSTAR

Failure symptoms list

Symptom	Definition
Engine flameout	This means that the crankshaft of the engine is rotating, but it cannot be started for a long time. Finally, it stops immediately after the start or after it begins.
Irregular engine vibration, idle speed regulation	This means that there is a deviation in engine idle speed or a change in engine idle speed. In severe cases, the engine or the vehicle sometimes vibrates. No matter what the situation is serious, it sometimes causes the engine to stall.
Engine output insufficient, injection failure, whoop	This means that the output of the engine is lower than the predicted energy. Even if you step on the accelerator pedal, the speed will not increase and the response will be slow.
More white smoke in the emitted gas	It means a great deal of white smoke is generated while vehicle running.
Much white smoke in emission gas (vehicles equipped with DPD)	It means the vehicle equipped with DPD generates a great deal of white smoke while running.
Much black smoke in emission gas	It means a great deal of black smoke is generated while vehicle running.
Much black smoke in emission gas (vehicles equipped with DPD)	It means the vehicle equipped with DPD generates a great deal of black smoke while running.
DPD alarm lights flicker more frequently	It means the blinking frequency of DPD warning lamp during vehicle running is high.



POWERSTAR

Engine flameout

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

In the step of confirming the symptoms, it is sometimes necessary to perform a visual inspection. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.
- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and

piping breakage, dent etc.

- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal

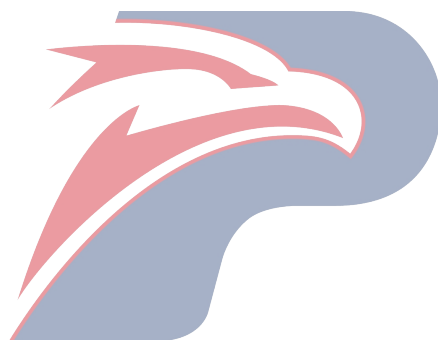
Diagnostic help

- Fuel system abnormal (fuel cutoff, fuel freezing, fuel piping gas entraining, filter abnormal { main body, screen binding), piping abnormal, fuel quality and fuel tank (waster or fuel induction abnormal)
- Abnormal intake system (filter blockage, abnormal intake piping, etc.)
- Supply pump abnormal (no fuel feed) ;
- Common rail abnormal (the flow regulator, pressure restrictor action/inside seal aging)
- Injector abnormal (no fuel injection)
- System function digression due to malfunction
- Engine main body abnormal (sintering, compression pressure insufficient, other mechanical faults)
- ACG malfunction
- Malfunction related to chassis (clutch, transmission, brake etc.)
- Impact from electrical equipment installed later (the radio unit, light fitting etc.)
- ECM failure (main body, power supply, GND, etc.)
- Check the connector for poor connection, abnormalities such as scratches and breakages in the harness, and whether or not the wires in the harness are scattered and shorted with other circuits. In addition, carry out the functional diagnosis to confirm the action and control of all parts and repair as needed.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the starting system. Is the disposal completed?	—	Go to step 3	Access to [confirmation of starting system]
3	Check whether DTC is detected. 1. Connect scanning tool. 2. Check if the following DTCs have been detected. DTC: P0335、P0336、P0340、P0341、P0601、P1261、P1262 Is there any DTC detected?	—	Access to [corresponding DTC inspection method]	Go to step 4

Step	Operation content	Datum value	YES	NO
4	Use the scan tool to check the signal status of the crankshaft position (CKP) sensor and camshaft position (CMP) sensor. 1. Connect scanning tool. 2. Turn the crankshaft of the engine. 3. Confirm the data list of scanning tool. Is there a show?	—	Go to step 5	Check and repair the CKP and CMP sensors
5	Start the engine and confirm that the engine is off again. Is the engine off?	—	Go to step 6	Confirm the repair conditions
6	Check ACG. Confirm the charging alarm lamp in the meter and whether there is a light in the operation of the engine. Is the charge warning light off?	—	Go to step 8	Go to step 7
7	Replace the ACG. In addition, check the charge warning lamp system and repair or replace the lamp as needed if abnormal. Is the disposal completed?	—	Go to step 8	—
8	Check the status of reinstall devices such as radios. Does the engine turn off even if you turn off the power to the loading device?	—	Go to step 9	Go to step 10
9	The installation state of the electrical equipment is set to a normal state, or the electrical equipment is removed. Is the disposal completed?	—	Go to step 10	—
10	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. •MAF correction value; •Q adjustment value data •ID code of the ejector • Starting times of the engine Refer to the description for programming in this chapter. 2. Start the engine. Is the engine off?	—	Go to step 11	Go to step 12
11	Restore the ECM to the state before the replacement. Is the disposal completed?	—	Go to step 12	—

Step	Operation content	Datum value	YES	NO
12	Check the mechanical parts of the engine and repair them if there is any abnormality. • Valve system • Injector • Compression pressure • Timing gear • Piston/crankshaft relevant items Is the disposal completed?	–	Go to step 13	–
13	Check the parts related to the chassis (such as clutch, transmission, etc.) and repair them if there is any abnormality. Is the disposal completed?	–	Go to step 14	–
14	Repair the vehicle and confirm maintenance. Is the disposal completed?	–	Confirm the repair conditions	–



POWERSTAR

Irregular engine vibration, idle speed regulation

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.
- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and

piping breakage, dent etc.

- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal

Diagnostic help

- Fuel system abnormal (fuel cutoff, fuel freezing, fuel piping gas entraining, filter abnormal { main body, screen binding), piping abnormal, fuel quality and fuel tank (waster or fuel induction abnormal)
- Abnormal intake system (filter blockage, abnormal intake piping, etc.)
- Supply pump(including fuel pump) abnormal (no fuel feed) ;
- Switch input circuit system abnormal (PTO switch, clutch switch, harness etc.)
- Sensor input circuit system abnormal (sensor, harness) (APP sensor, harness, ECT sensor)
- APP sensor, PTO accelerator sensor system abnormal (sensor, harness)
- Engine main body abnormal (sintering, compression pressure insufficient, other mechanical faults)
- Malfunction related to chassis (clutch, transmission, brake etc.)
- Impact from electrical equipment installed later (the radio unit, light fitting etc.)
- Check the connector for poor connection, abnormalities such as scratches and breakages in the harness, and whether or not the wires in the harness are scattered and shorted with other circuits. In addition, carry out the functional diagnosis to confirm the action and control of all parts and repair as needed.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm fuel. 1. Replace the fuel in the fuel tank and piping. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 21	Go to step 3
3	Check that the air in the exhaust gas is asymptomatic. Are the symptoms of the engine normal?	—	Go to step 21	Go to step 4

Step	Operation content	Datum value	YES	NO
4	Check the air intake system. Is the disposal completed?	—	Go to step 5	Access to [confirmation of air intake system]
5	Confirm the fuel system. Is the disposal completed?	—	Go to step 6	Access to[confirmation of fuel System]
6	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 21	Go to step 7
7	Check the state of the later mounted radio equipment and other electrical parts. Even if you turn off the power of the transformation, will the engine shut down?	—	Go to step 8	Go to step 9
8	The installation state of the electrical equipment is set to a normal state, or the electrical equipment is removed. Is the disposal completed?	—	Go to step 9	—
9	Check the installation status of the crank position (CKP) sensor and camshaft position (CMP) sensor. Check the CKP sensor and CMP sensor for looseness. Is the installation state normal?	—	Go to step 11	Go to step 10
10	Set the mounting state of the CKP sensor and CMP sensor to normal. Is the disposal completed?	—	Go to step 11	—
11	Use the scan tool to check the signal status of the crankshaft position (CKP) sensor and camshaft position (CMP) sensor. 1. Connect scanning tool. 2. Turn the crankshaft of the engine. 3. Confirm the data list of scanning tool. Is there a show?	—	Go to step 14	Go to step 12
12	Check the status of CKP sensor signal detection. 1. Start the engine. 2. Remove the harness from the CMP sensor. (Since DTC may be detected in this operation, ensure to clear DTC after repairing the vehicle) Is the idle state normal?	—	Go to step 20	Go to step 13
13	Check the status of CMP sensor signal detection. 1. Start the engine. 2. Remove the harness from the CKP sensor. (Since DTC may be detected in this operation, ensure to clear DTC after repairing the vehicle) Is the idle state normal?	—	Go to step 20	Go to step 14
14	Check APP sensor, PTO Accelerator sensor status. Refer to DTC P1271, P1277, P1278, P1282, P1283 to check. Is the disposal completed?	—	Go to step 15	—

Step	Operation content	Datum value	YES	NO
15	Check the common rail system. Refer to DTC P0201 ~ P0204 (injector), DTC P0087, P0088, P0089, P0192, P0193, P1094 (common rail, fuel feed pump) to check. When a fault is detected, it should be repaired or replaced. Is the disposal completed?	—	Go to step 16	—
16	Refer to DTC P0117 and P0118 for confirmation. Is the disposal completed?	—	Go to step 17	—
17	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 21	Go to step 18
18	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the injector • Starting times of the engine Refer to the description for programming in this chapter. 2. Start the engine. Is the engine off?	—	Go to step 21	Go to step 19
19	Restore the ECM to the state before the replacement. Is the disposal completed?	—	Go to step 20	—
20	Check the mechanical parts of the engine and repair them if there is any abnormality. • Valve system • Compression pressure • Flywheel • Camshaft gear • Timing gear • Piston/crankshaft relevant items Is the disposal completed?	—	Go to step 21	—
21	Repair the vehicle and confirm maintenance. Is the disposal completed?	—	Confirm the repair conditions	—

Engine output insufficient, injection failure, whoop

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.
- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.

- Whether the fuel system is free of oil leak and piping breakage, dent etc.
- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal

Diagnostic help

- Fuel system abnormal (fuel cutoff, fuel freezing, fuel piping gas entraining, filter abnormal { main body, screen binding), piping abnormal, fuel quality and fuel tank (waster or fuel induction abnormal)
- Abnormal intake system (filter blockage, abnormal intake piping, etc.)
- Exhaust system abnormal (exhaust brake, exhaust pipe abnormal, DPD, exhaust throttle etc.)
- Sensor input circuit system abnormal (Sensors, harnesses, etc.) (Accelerator pedal position (APP) sensors, harnesses, etc.)
- Switch input circuit system abnormal (PTO switch, clutch switch, harness etc.)
- Engine main body abnormal (compression pressure, valve play, turbocharger, fuel feed pump, injector, common rail, other mechanical faults)
- Malfunction related to chassis (clutch, transmission, brake etc.)
- Impact from electrical equipment installed later (the radio unit, light fitting etc.)
- Check the connector for poor connection, abnormalities such as scratches and breakages in the harness, and whether or not the wires in the harness are scattered and shorted with other circuits. In addition, carry out the functional diagnosis to confirm the action and control of all parts and repair as needed.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm fuel. 1. Replace the fuel in the fuel tank and piping. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 20	Go to step 3

Step	Operation content	Datum value	YES	NO
3	Evacuate the air in the fuel and reconfirm the symptoms. Are the symptoms of the engine normal?	—	Go to step 20	Go to step 4
4	Confirm the fuel system. Is the disposal completed?	—	Go to step 5	Access to [confirmation of fuel System]
5	Confirm the exhaust brake. (6HK1-TC) Check DPD, exhaust brake and exhaust throttle. (6HL1) Is the disposal completed?	—	Go to step 6	Access to [confirmation of exhaust system] (6HK1-TC) Access to [DPD functional check (6HL1)]
6	Check the air intake system. Is the disposal completed?	—	Go to step 7	Access to [confirmation of air intake system]
7	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 20	Go to step 8
8	Check the free stroke position of accelerator pedal. With the key switch in OFF state, fully depress the accelerator pedal to check whether the gap between the pedal and limiting bolt is normal. Is the clearance with the floor limit bolt normal?	Clearance 1-2mm	Go to step 10	Go to step 9
9	Adjust floor limit bolts. Is the disposal completed?	—	Go to step 10	—
10	Check the status of APP sensor. Refer to DTC P1271, P1277, P1278, P1282, P1283 to check. Is the disposal completed?	—	Go to step 11	—
11	Check the state of the later mounted radio equipment and other electrical parts. Does the engine turn off even if you turn off the power to the loading device?	—	Go to step 12	Go to step 13
12	The installation state of the electrical equipment is set to a normal state, or the electrical equipment is removed. Is the disposal completed?	—	Go to step 13	—
13	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 20	Go to step 14
14	Check the common rail system. Refer to DTC P0201 ~ P0206 (injector), DTC P0087, P0088, P0089, P0192, P0193, P1094 (common rail, fuel feed pump) to check. When a fault is detected, it should be repaired or replaced. Is the disposal completed?	—	Go to step 15	—
15	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 20	Go to step 16

Step	Operation content	Datum value	YES	NO
16	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the ejector • Starting times of the engine Refer to the description for programming in this chapter. 2. Start the engine. Is the engine off?	—	Go to step 18	Go to step 17
17	Restore the ECM to the state before the replacement. Is the disposal completed?	—	Go to step 18	—
18	Check the mechanical parts of the engine and repair them if there is any abnormality. • Valve system • Injector • Compression pressure • Timing gear • Piston/crankshaft relevant items Is the disposal completed?	—	Go to step 19	—
19	Check the parts related to the chassis (such as clutch, transmission, etc.) and repair them if there is any abnormality. Is the disposal completed?	—	Go to step 20	—
20	Repair the vehicle and confirm maintenance. Is the disposal completed?	—	Confirm the repair conditions	—

Much white smoke in emission gas

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the auxiliary power uses the power

accessories purchased from the market.

- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and piping breakage, dent etc.
- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal

Diagnostic help

- Fuel quality (use of non-qualified product etc.)
- Long time idling
- Engine coolant temperature (ECT) sensor, fuel temperature (FT) sensor, intake air temperature (IAT) sensor, supercharging sensor malfunction, barometric pressure sensor malfunction
- Engine main body malfunction (the compression pressure insufficient, piston related malfunction, too much/less oil)
- ECM malfunction (main body, power, GND etc.)

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm fuel. 1. Replace the fuel in the fuel tank and piping. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 12	Go to step 3
3	1. Use the scan tool to confirm the injector ID code. 2. Confirm injector ID code through the injector ID nameplate. Is the injector ID code of the two consistent?	—	Go to step 5	Go to step 4
4	1. Religion the injector ID code. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 12	Go to step 5
5	Confirm the fuel system. Is the disposal completed?	—	Go to step 6	Access to[confirmation of fuel System]

Step	Operation content	Datum value	YES	NO
6	Check each sensor, syringe, etc. Please refer to DTC P0117, P0118, P0182, P0183, P0112, P0113, P1261, and P1262 for inspection. If there is any abnormality, repair or replace it. Is the disposal completed?	—	Go to step 7	—
7	Check the supercharging sensor, barometric pressure sensor. Refer to DTC P0107, P0108, P0237, and P1283 to check. If any abnormality is found, it should be repaired or replaced. Is the disposal completed?	—	Go to step 8	—
8	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 12	Go to step 9
9	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. • MAF correction value; • Q adjustment value data • ID code of the ejector • Starting times of the engine Refer to the description for programming in this chapter. 2. Start the engine. Is the engine off?	—	Go to step 11	Go to step 10
10	Restore the ECM to the state before the replacement. Is the disposal completed?	—	Go to step 11	—
11	Check the mechanical parts of the engine and repair them if there is any abnormality. • Valve system • Injector • Compression pressure • Timing gear • Piston/crankshaft relevant items Is the disposal completed?	—	Go to step 12	—
12	Repair the vehicle and confirm maintenance. Is the disposal completed?	—	Confirm the repair conditions	—

Much white smoke in emission gas (vehicles equipped with DPD)

Prior check is conducted

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power

accessories purchased from the market.

- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and piping breakage, dent etc.
- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal
- Warm up sufficiently to distinguish between vapor and white smoke

Diagnostic help

- Fuel quality (use of non-qualified product etc.)
- Oil quality (use of non-qualified product etc.)
- There is no air entraining in the fuel system.
- Engine coolant temperature (ECT) sensor, fuel temperature (FT) sensor, intake air temperature (IAT) sensor failure
- Engine main body malfunction (the compression pressure insufficient, piston related malfunction, too much/less oil)
- ECM malfunction (main body, power, GND etc.)

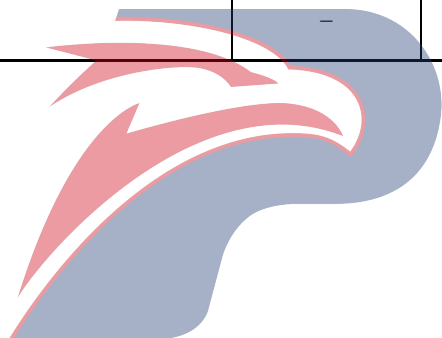
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm fuel. 1. Replace the fuel in the fuel tank and piping. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 12	Go to step 3
3	Confirm the fuel injection time. Check the time of the mechanical parts. Any abnormal timing in the mechanical part should be adjusted. Is the disposal completed?	—	Go to step 4	—
4	1. Make sure that white smoke is produced. • Generated during the DPD reactivation. • Generated during the time other than DPD reactivation. Is it confirmed?	—	Go to step 5	—

Step	Operation content	Datum value	YES	NO
5	Confirm the fuel system. Is the disposal completed?	—	Go to step 6	—
6	1. Check the ECT sensor, FT sensor, IAT sensor, atmospheric pressure sensor, exhaust temperature sensor 1 and 2. 2. Check the air cleaner, oil, valve clearance, compression pressure and other parts that are related to engine mechanics. 3. If found fault, it should be repaired or replaced. Is the disposal completed?	—	Go to step 7	—
7	The DPD was regenerated to confirm the degree of white smoke. 1. Use the scan tool to regenerate the DPD. Is the white smoke normal?	—	Go to step 12	Go to step 8
8	Check the exhaust throttle. 1. Use the scanning tool to confirm the working condition of exhaust throttle. 1) Remove the rear hose. 2). Plug the hole of the hose. 3). Start engine. 4) Measure the exhaust pressure (pressure difference). 2. Confirm whether the valve is abnormal. 3. If found fault, it should be repaired or replaced. Is the disposal completed?	—	Go to step 9	—
9	Confirm the degree of white smoke. White smoke is normal or not.	—	Go to step 11	Go to step 10
10	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. ●MAF correction value; ●Q adjustment value data ●ID code of the ejector ● Starting times of the engine Refer to the description for programming in this chapter. Important: If ECM is replaced while DPD filter is not, force the DPD to reactivate and set ECM and DPD to the same status. Is the disposal completed?	—	Go to step 12	—
11	Replace the DPD oxidation type catalyst. Is the disposal completed?	—	Go to step 12	—
12	Check the function of DPD. Is the disposal completed?	—	Confirm the repair conditions	—

When coming back to the store

When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm the fuel system. Is the disposal completed?	—	Go to step 3	—
3	Check the ECT sensor, FT sensor, IAT sensor, barometric pressure sensor. 2. If found fault, it should be repaired or replaced. Is the disposal completed?	—	Go to step 4	—
4	Confirm the degree of white smoke. White smoke is normal or not.	—	Go to step 6	Go to step 5
5	Replace the DPD oxidation type catalyst. Is the disposal completed?	—	Go to step 6	—
6	Check the function of DPD. Is the disposal completed?	—	Confirm the repair conditions	—



POWERSTAR

Much black smoke in emission gas

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

In the step of confirming the symptoms, it is sometimes necessary to perform a visual inspection. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.

- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and piping breakage, dent etc.
- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal

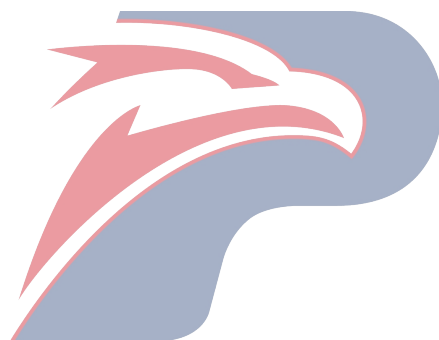
Diagnostic help

- Fuel quality (use of non-qualified product etc.)
- Abnormal intake system (filter blockage, abnormal intake piping, etc.)
- EGR control system abnormal
- Exhaust system abnormal (exhaust brake, exhaust pipe abnormal etc.)
- Engine coolant temperature (ECT) sensor abnormal
- Supercharged sensor failure (sensor, piping, etc.)
- Engine main body malfunction (the compression pressure insufficient, piston related malfunction, turbocharger, oil blow-by)

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Check the air intake system. Is the disposal completed?	—	Go to step 3	Access to [confirmation of air intake system]
3	Check EGR control system. Is the disposal completed?	—	Go to step 4	Access to [confirmation of EGR control system]
4	Confirm the fuel system. Is the disposal completed?	—	Go to step 5	Access to [confirmation of fuel System]
5	Confirm exhaust system. Is the disposal completed?	—	Go to step 6	Enter [confirmation of exhaust system]
6	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 18	Go to step 7

Step	Operation content	Datum value	YES	NO
7	Check the supercharging sensor, barometric pressure sensor and ECT sensor. Please refer to DTC P0117, P0118, P0237, P0238, P0107, P0108 for inspection. If there is any abnormality, repair or replace it. Is the disposal completed?	—	Go to step 8	—
8	Check the common rail system. Refer to DTC P0201 ~ P0206 (injector), DTC P0087, P0088, P0089, P0192, P0193, P1094 (common rail, fuel feed pump) to check. When a fault is detected, it should be repaired or replaced. Is the disposal completed?	—	Go to step 9	—
9	1. Use the scan tool to confirm the injector ID code. 2. Confirm injector ID code through the injector ID nameplate. Is the injector ID code of the two consistent?	—	Go to step 11	Go to step 10
10	1. Relogin the injector ID code. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 18	Go to step 12
11	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 18	Go to step 12
12	Check the mechanical parts of the engine and repair them if there is any abnormality. ● Valve series (valve play) Is the disposal completed?	—	Go to step 13	—
13	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 18	Go to step 14
14	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations. ● MAF correction value; ● Q adjustment value data ● ID code of the ejector ● Starting times of the engine Refer to the description for programming in this chapter. 2. Start the engine. Is the engine off?	—	Go to step 16	Go to step 15
15	Restore the ECM to the state before the replacement. Is the disposal completed?	—	Go to step 16	—
16	Confirm the symptoms of the engine again. Are the symptoms of the engine normal?	—	Go to step 18	Go to step 17

Step	Operation content	Datum value	YES	NO
17	Check the mechanical parts of the engine and repair them if there is any abnormality. • Valve system • Injector • Compression pressure • Timing gear • Piston/crankshaft relevant items Is the disposal completed?	—	Go to step 18	—
18	Repair the vehicle and confirm maintenance. Is the disposal completed?	—	Confirm the repair conditions	—



POWERSTAR

Much black smoke in emission gas (vehicles equipped with DPD)

Pre-inspection

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.

- Whether the ECM ground wire is polluted and installed to the correct position.

- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.

- Whether the fuel system is free of oil leak and piping breakage, dent etc.

- Whether the air intake system parts are abnormal

- Whether the air intake system parts are abnormal

Diagnostic help

- Fuel quality (use of non-qualified product etc.)

- Oil quality (use of non-qualified product etc.)

- There is no air entraining in the fuel system.

- Abnormal intake system (filter blockage, abnormal intake piping, etc.)

- Exhaust system abnormal (exhaust brake, exhaust pipe abnormal etc.)

- Engine coolant temperature (ECT) sensor abnormal

- Oil aging

- Engine main body malfunction (the compression pressure insufficient, piston related malfunction, oil blow-by)

- ECM malfunction (main body, power, GND etc.)

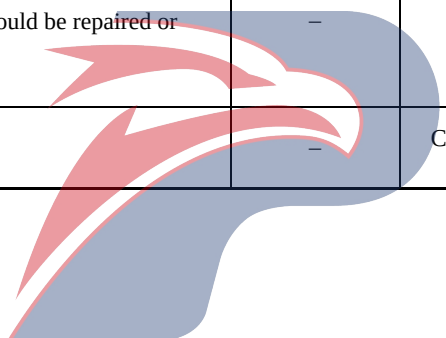
Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Confirm the black smoke degree. Is the disposal completed?	—	Go to step 3	—
3	Check the air intake system. Is the disposal completed?	—	Go to step 4	Access to [confirmation of air intake system]
4	Check the oil. 1. Change the oil and oil filter. 2. Start the engine to run the vehicle and confirm that there is no symptoms. Are the symptoms of the engine normal?	—	Go to step 11	Go to step 5
5	Confirm the fuel injection time. Check the machine's time. Adjust the mechanical timing if abnormal. Is the disposal completed?	—	Go to step 6	—

Step	Operation content	Datum value	YES	NO
6	Confirm the degree of black smoke. 1. Remove the DPD. 2. Install the exhaust pipe to substitute DPD for inspection. Important: Check the exhaust pipe for reference. mark: After the check is completed, the DPD is restored to its original state. 3. Confirm the degree of black smoke. Is the level of black smoke normal?	—	Go to step 10	Go to step 7
7	Confirm the fuel system. Is the disposal completed?	—	Go to step 8	—
8	Check engine related parts. Is the disposal completed?	—	Go to step 9	—
9	Confirm the degree of black smoke. Is the level of black smoke normal?	—	Go to step 10	Go to step 11
10	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. 2. Check the DPD oxidation type catalyst for dirt, damage and breakage. 3. The DPD oxidation catalyst should be replaced when a fault is detected. Is the disposal completed?	—	Go to step 12	—
11	1. Replace ECM. Important: After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; Starting times of the engine Refer to the description for programming in this chapter. Important: If ECM is replaced while DPD filter is not, force the DPD to reactivate and set ECM and DPD to the same status. Is the disposal completed?	—	Go to step 12	—
12	Check the function of DPD. Is the disposal completed?	—	Confirm the repair conditions	—

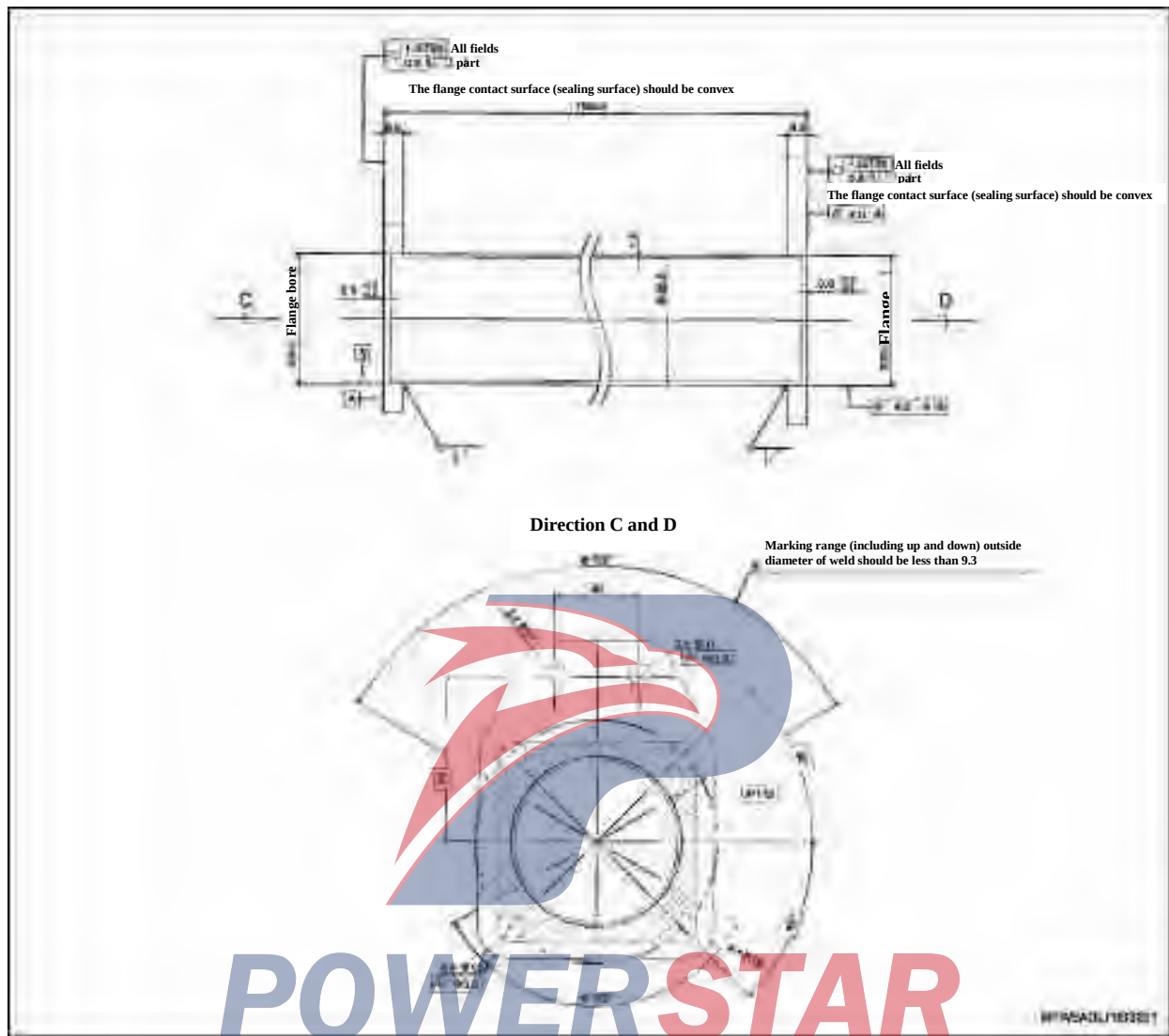
When coming back to the store

When the same vehicle is returned to the store, diagnosis should be based on the following diagnostic methods.

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check" Inspection"
2	1. Replace the DPD filter. Important: After replacing the DPD filter, be sure to clear the DPD data of the ECM so that the ECM and DPD are set to the same state. 2. Check the DPD oxidation catalyst. 3. The DPD oxidation catalyst should be replaced when a fault is detected. Is the disposal completed?	—	Go to step 3	—
3	1. Check the DPD differential pressure sensor's piping and hoses for clogging and deformation. 2. When a fault is detected, it should be repaired or replaced as needed. Is the disposal completed?	—	Go to step 4	—
4	Check the function of DPD. Is the disposal completed?	—	Confirm the repair conditions	—



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Exhaust pipe for DPD inspection (reference picture)

DPD lamp blinking frequency high

Prior check is conducted

Before using this item, implement [OBD system inspection] and confirm all the following items.

- Can ECM and engine MIL work normally?
- Confirm DTC.
- Is the scanning tool data in the normal working range?
- Confirm the vehicle status and search the symptom corresponding to the list. Then follow the steps shown in the symptom map.
- Ask the customer whether the specified oil and fuel are used.

Visual inspection

Sometimes careful visual inspections are required during the steps to confirm the symptoms. As a result, it is possible to repair the problems generated without further inspections, thereby saving valuable time.

The inspection includes the following items.

- Whether the wiring connection and tightening are correct without disconnection
- Whether the ECM power uses the power accessories purchased from the market.
- Whether the ECM ground wire is polluted and installed to the correct position.
- Whether the piping and hose related to the fuel, gas and oil are correctly connected free of fracture or distortion. Any leakage, blockage, etc. should be thoroughly checked.
- Whether the fuel system is free of oil leak and piping breakage, dent etc.
- Whether the air intake system parts are abnormal
- Whether the air intake system parts are abnormal
- Warm up sufficiently.

Diagnostic help

The twinkling frequency will increase in the following states

- Fuel quality (use of non-qualified product etc.)
- Engine oil quality (use of non-qualified product

etc.)

- Engine oil maintenance time interval unsuitable
- There is no air entraining in the fuel system.
- Air intake system abnormal (filter clogging, air intake piping abnormal, intake throttle abnormal, air intake duct deviation and looseness etc.)
- Exhaust system abnormal (exhaust brake, exhaust pipe abnormal etc.)
- Engine coolant temperature (ECT) sensor abnormal
- Engine oil aging
- Engine main body malfunction (the compression pressure insufficient, piston related malfunction, oil blow-by)
- ECM malfunction (main body, power, GND etc.)
- Idling time long
- Less load.
- PTO use time long
- Too much downtown driving
- MAF sensor abnormal
- DPD oxidation type catalyst, DPD filter performance aging
- Exhaust brake and exhaust throttle abnormal
- Too much dust.
- Differential pressure sensor is abnormal.
- Exhaust temperature sensor abnormal
- Barometric pressure sensor abnormal
- Intake air temperature (IAT) sensor abnormal
- Refitting vehicle (equipped with non-genuine parts)
- Reactivation overtime too much
- Reactivation purification abnormal frequency too much

Test instruction

The numbers shown below indicate the step numbers in the table

4. Refer to the diagnosis assistance and explain the stroke conditions for high warning lamp blinking frequency to the customer

Step	Operation content	Datum value	YES	NO
1	Check the OBD system. Is the disposal completed?	—	Go to step 2	Enter "OBD System Check"
2	Inquire the customer and confirm the DPD reactivation method Is the disposal completed?	—	Go to step 3	—
3	Confirm the status of DPD. 1. Use the scan tool to confirm the status and pressure difference of the DPD. Important: When the differential pressure is 8.7 kPa or more, replace the DPD filter. Is the disposal completed?	—	Go to step 4	—
4	Explain the status to the customer. Description item 1. With or without switches 2. Whether to ignore the blinking of the warning light. 3. PM deposit condition in DPD 4. About the flashing of the warning light 5. Regeneration method of DPD Is the disposal completed?	—	Go to step 5	—
5	Confirm the degree of black smoke. 1. Remove the DPD. 2. Install the exhaust pipe to substitute DPD for inspection. Important: Check the exhaust pipe for reference mark: After the check is completed, the DPD is restored to its original state. 3. Confirm the degree of black smoke. Is the level of black smoke normal?	—	Go to step 9	Go to step 6
6	Confirm the fuel system. Is the disposal completed?	—	Go to step 7	Access to [confirmation of fuel System]
7	Check the air intake system. Is the disposal completed?	—	Go to step 8	Access to [confirmation of air intake system]
8	Confirm the fuel injection time. Check the time of the mechanical parts. Adjust the mechanical part timing if abnormal. Is the disposal completed?	—	Go to step 9	—
9	1. Check whether the differential pressure sensor main body, piping and hose are normal. When a fault is detected, it should be repaired or replaced. Is the disposal completed?	—	Go to step 10	—
10	Refer to "Excessive number of DTC P1453 DPD regeneration timeouts" to check exhaust brake and exhaust throttle valve. Is the disposal completed?	—	Go to step 11	—
11	Reactivate DPD. 1. Use the scan tool to regenerate the DPD. Can DPD be reproduced normally?	—	Go to step 13	Go to step 12
12	1. Replace ECM. Important:	—	Go to step 13	—

Step	Operation content	Datum value	YES	NO
	After replacing the ECM, be sure to perform the following programming operations: •MAF correction value; •Q adjustment value data; •ID code of the ejector; Starting times of the engine Refer to the description for programming in this chapter. Important: If ECM is replaced while DPD filter is not, force the DPD to reactivate and set ECM and DPD to the same status. Is the disposal completed?			
13	Check the function of DPD. Is the disposal completed?	—	Confirm the repair conditions	—

About DPD regeneration using ECM

DPD regeneration method when using DPD to regenerate ECM

1. Start the engine.
2. Use a scan tool to confirm that the temperature inside the DPD (before the oxidation catalyst and before the filter) is less than 180°C.

Mark:

When the temperature before the DPD oxidation catalyst/temperature before the DPD filter is higher than 180°C, the engine should be cooled at idle (1500 rpm).

3. Increase the number of engine revolutions to 2500 rpm.
4. Check the differential pressure when the DPD filter front temperature reaches 160 °C with scanning tool.

Mark:

If the differential pressure is below 1kPa, check the differential pressure sensor and sensor piping and hose; repair or replace if there is fault.

5. Replace the DPD filter if the differential pressure is above the reference value.

If the differential pressure is below reference value, go to the steps after Step 6.

Reference value: 8.7kPa

6. Download MAF corrected value, injector ID code and Q adjustment data from ECM with the scanning tool.
7. Remove the ECM and install the DPD

reactivation ECM.

8. Check if the DTC is generated by the DPD regeneration ECM and various ECU. DTC should be cleared when DTC is generated.

9. Use the scan tool to program the information downloaded from the ECM into the DPD regeneration ECM.

10. After the DPD warning light flashes, press the DPD switch to start DPD regeneration.

Mark:

When interrupting DPD regeneration, the DPD regeneration will be interrupted when the DPD switch is pressed or the accelerator pedal is depressed during regeneration.

When restarting DPD regeneration, press the DPD switch to resume DPD regeneration.

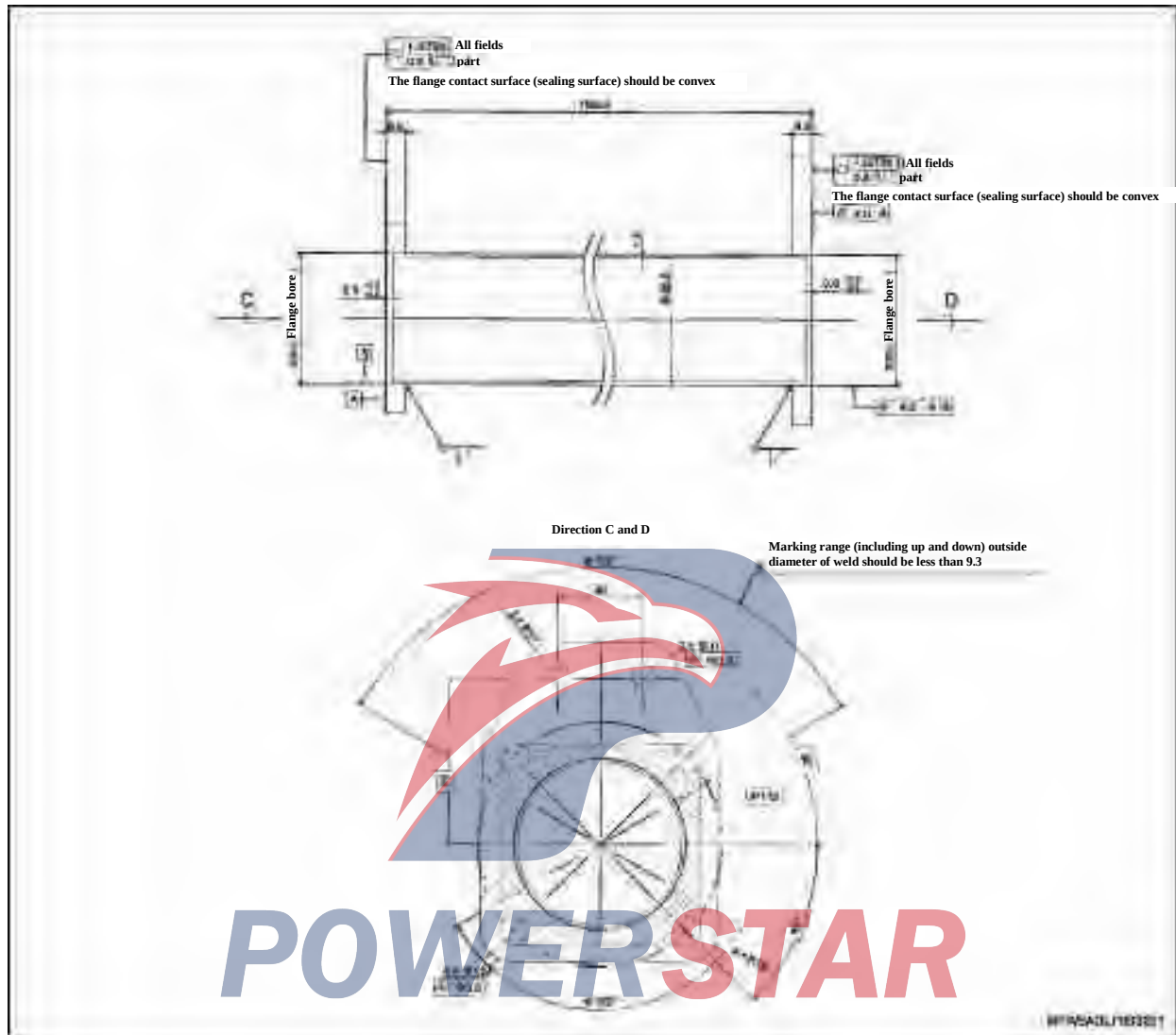
11. After the DPD warning light goes out (after the DPD regeneration is completed), check that DTC is generated in the DPD regeneration ECM. Clear the DTC when generating the DTC.

12. Remove the DPD regeneration ECM and install the ECM.

13. Check that the ECM and various ECUs have generated DTCs. Clear the DTC when generating the DTC.

Important:

When DPD regeneration ECM is used to carry out DPD regeneration, the fuel mixed with oil will be more than usual, so the oil should be changed after DPD regeneration.

Exhaust pipe for DPD inspection (reference picture)

Special tool

summary	Tool number Tool name
	5-8840-2691-0 Digital Multi-function Multimeter
	Tech2 Scan tool
	5-8840-0632-0 Terminal removal tool
	5-8840-0388-0 Weather Pack Waterproof seal terminal removal tool

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Programming

Function and action description

When changing ECM, ejector, MAF sensor, starter, DPD and so on, all kinds of data should be written to ECM.

Items to be confirmed in advance before programming

Confirm the following items as required upon programming:

- Engine type
- ID code of the ejector
- MAF correction value

- Engine starting number

About the project to be programmed

The following items need to be programmed when replacing:

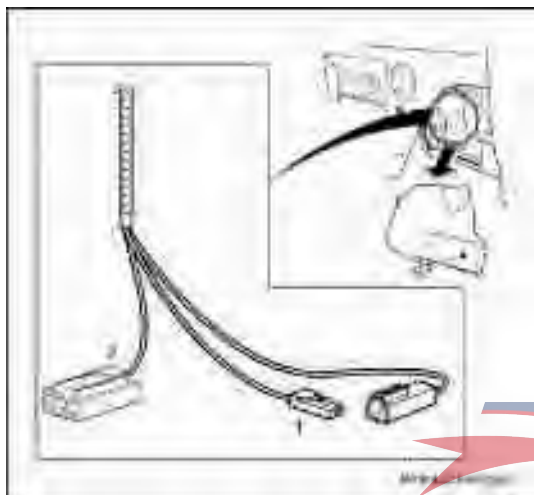
- ECM.
 - Engine (type, output partition modification)
 - Injector
2. Air Mass Flow (MAF) Sensor
- Starter
 - DPD

About each items

Item		How to use
F0: Q adjustment value correction	F0: Q adjustment value correction data uploading (Tech2)	Use it when replacing ECM, reading data from the ECM before replacing to Tech2.
	F1: Q adjustment value correction data downloading (ECM)	Use it when replacing ECM, logging in (writing in) data from the ECM before replacing to Tech2.
F1: Syringe ID code	F0: Syringe ID code	Ensure the injector ID code is programmed to each cylinder.
	F1: ID code registration	Use this when replacing ECM, unable to upload data from the ECM before replacement or logging in injector ID code when replacing the injector.
	F2: ID Code Upload (Tech2)	Use it when replacing ECM, reading data from the ECM before replacing to Tech2.
	F3: ID code downloading (ECM)	Use it when replacing ECM, logging in (writing in) data from the ECM before replacing to Tech2.
F2: MAF correction	F0: upload	Use it when replacing ECM, reading data from the ECM before replacing to Tech2.
	F1: Download	Use it when replacing ECM, logging in (writing in) data from the ECM before replacing to Tech2.
	F3: Aftermarket model	This function is used to write MAF correction data to the ECM when the ECM is replaced or data cannot be read from the pre-replacement ECM.
F3: engine starting number (6HL1) F2: engine starting number (6HK1-TC)		Use it when replacing ECM and writing in engine starting frequency.
F4: starting number after the starter replacement (6HL1) F3: starting number after the starter replacement (6HK1-TC)		When and after changing the starter, write the start times.
F5: DPD reactivation data clearing(6HL1)		Used when replacing DPD and clearing DPD regeneration data.

Programming preparation

1. Set the key switch to the OFF position.
2. Confirm that the vehicle has been repaired/replaced.
3. Confirm the specifications of the vehicle.
4. Connect the Tech2 accessory to the DLC.



Name

Diagnostic connector (red)

2. Data Link Connector (DLC)

5. Connect the Tech2 to the Tech2 accessory.
6. Turn the key switch to the ON position
7. Refer to "Using the Scan Tool for Fault Diagnosis" in this chapter to display the menu screen.
8. Use the ▲, ▼ (up/down) keys or F4 (function key) to select menu F4: Programming and press the [ENTER] key.

Engine
F0: Diagnostic Code (DTC) F: 1 data display F2: Snapshot Printing F3: Actuator Measurement F4: programming
LNW41ASH004001

Adjustment

Q Adjustment value correction data upload (Tech2)

1. Use the ▲, ▼ (up/down) keys or F0 (function key) to select the menu F0: Q Adjust the value correction, then press the [ENTER] key.
2. Select menu F1: Q adjustment value correction data upload (Tech2) with ▲, ▼ (UP/DOWN) key or F0 (function key) and then press [ENTER].

Programming
F0: Q adjustment value correction F1: Syringe ID code F2: MAF correction F3: Engine Start F4: number of starts after starter replacement F5: DPD reactivation data clearing
Q adjustment value correction
F0: Q adjustment value correction data uploading (Tech2) F1: Q adjustment value correction data downloading (ECM) F2: After-sale model
MFS4Z0MH002601

3. Press soft key [YES] to upload data. Press soft key [NO] to go to step 2 screen.

4. When the upload is complete, the "Upload completed" message will be displayed. Press soft key [ENTER] to go to step 2 screen.

Q Adjustment value correction data upload (Tech2)
Can data be uploaded?
YES NO

Q Adjustment value correction data upload (Tech2)
The upload has been completed!
Confirm

MFS4Z0MH002701

5. When the data uploading fails, there will show "data not uploaded!" This hint of information shows. At this point, press soft key [ENTER] to go to step 2 screen for uploading again.

Q Adjustment value correction data upload (Tech2)
Data could not be successfully uploaded!
Confirm

MFS4Z0MH002801

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Q Adjustment value correction data downloading (Tech2)

1. Use the ▲, ▼ (up/down) keys or F0 (function key) to select the menu F0: Q Adjust the value correction, then press the [ENTER] key.
2. Select menu F1: Q adjustment value correction data downloading (ECM) with ▲, ▼ (UP/DOWN) key or F1 (function key) and then press [ENTER].

Programming
F0: Q adjustment value correction
F1: Syringe ID code
F2: MAF correction
F3: Engine Start
F4: number of starts after starter replacement
F5: DPD reactivation data clearing
Q adjustment value correction
F0: Q adjustment value correction data uploading (Tech2)
F1: Q adjustment value correction data downloading (ECM)
F2: After-sale model
MFS4Z0MH002901

3. Press soft key [YES] to download data. Press soft key [NO] to go to step 2 screen.

4. After the download is complete, the "programming completed" message will be displayed. Press soft key [ENTER] to go to step 2 screen.

Q adjustment value correction data downloading (ECM)
Can data be downloaded?
YES NO
Q adjustment value correction data downloading (ECM)
Programming is finished!
Confirm
MFS4Z0MH003001

5. "Programming fails!" will occur when the data downloading fails this hint of information shows. At this point, press soft key [ENTER] to go to step 2 screen for downloading again.

Q adjustment value correction data downloading (ECM)
Programming fails!
Confirm
MFS4Z0MH003101

Injector ID code logging in

1. Select menu F1: injector ID code with ▲, ▼ (UP/DOWN) key or F1 (function key) and then press [ENTER].

Programming
F0: Q adjustment value correction
F1: Syringe ID code
F2: MAF correction
F3: Engine Start
F4: number of starts after starter replacement
F5: DPD reactivation data clearing
LNW41ASH004101

2. Select menu F1: ID code logging in with ▲, ▼ (UP/DOWN) key or F1 (function key) and then press [ENTER].

3. After the following screen is displayed, confirm that the engine is stopped, and then press the [OK] soft key.

Programming
F0: Syringe ID code
F1: ID code registration
F2: ID Code Upload (Tech2)
F3: ID code downloading (ECM)
F4: Bar Code Reader
Q adjustment value correction data downloading (ECM)
Please confirm the vehicle status! It is active only when the engine stops!
Confirm
MFS4Z0MH001401

4. Use the ▲, ▼ (up/down) keys to select the cylinder (1st to 6th cylinder) to register the ID code, and then press the [ENTER] key.

(Here describe the selection of No.1 cylinder.)

5. After the currently-logline-in ID code is displayed, press the Change key.

ID code logging in	
Please select the cylinder.	
No.1 cylinder (INJ.#1)	
No.2 cylinder (INJ.#4)	
No.3 cylinder (INJ.#2)	
No.4 cylinder (INJ.#3)	
ID code logging in	
Current first cylinder ID code	
CYL#1 ID code [MC-D0]	0000
CYL#1 ID code [D1-D2]	0000
CYL#1 ID code [D3-D4]	0000
CYL#1 ID code [D5-D6]	0000
CYL#1 ID code [D7-D8]	0000
CYL#1 ID code [D9-BC]	0000
Modification	
LNW41AMH000301	

6. After inputting to ID code input data [MC-D9], press [ENTER] key.

Since the model code (MC) and BCC code (BC) are automatically registered (written) in the Tech2 mode, no input is required.

Login ID code
When changing the value, use the up and down transposition arrow keys. Use the left / right transposition key to select the field.
ID code input data byte [MC-D9]
MCD0 D1D2 D3D4 D5D6 D7D8 D9 BC
0001-2345-6789-ABCD-EF01-23 *****
Keep pressing [ENTER]

7. After the following screen is displayed, confirm that the entered ID code is correct, and then press the [program] soft key.

When you want to terminate the program (login), press soft key [Interruption] or [EXIT] to go to step 2 screen

Login ID code	
When changing the value, use the up and down transposition arrow keys. Use the left / right transposition key to select the field.	
ID code input data byte [MC-D9]	
MCD0 D1D2 D3D4 D5D6 D7D8 D9 BC	
0001-2345-6789-ABCD-EF01-23**_***_**	
Program	Interruption
LNW41ASH004201	

8. When the input value is incorrect, there will be [data invalid - please retry!] This prompt message is displayed. Press [ENTER] again to enter it.

9. When the entered numerical value is correct, the status during data registration from Tech2 to ECM is displayed.

ID code logging in Invalid data – please operate again! ID code input data byte [MC-D9] MCD0 D1D2 D3D4 D5D6 D7D8 D9 BC 0001-2345-6789-ABCD-EF01-23**-*****-** Keep pressing ENTER
ID code logging in In programming! 0 100 MFS4Z0MH001901

10. When the entered ID code matches the registered ID code in ECM, [Complete programming!] This hint of information shows.

During ID code logging in, press soft key [YES] to return to the screen in Step 4 and then carry out the same logging in as before.

Press the [NO] soft key to return to step 2 of the screen.

11. If the entered ID code does not match the registered ID code in ECM, [Program failed!] This hint of information shows. Press soft key [EXIT] and go to step 2 screen to log in ID again.

ID code logging in Programming is finished! Is the ID code registration duplicate? YES NO
ID code logging in Programming fails! Confirm MFS4Z0MH002001

Injector ID code uploading (Tech2)

1. Select menu F2: ID code logging in with ▲, ▼ (UP/DOWN) key F2 (function key) Press ID code upload (Tech2) and then press [ENTER].

2. Data can be uploaded by pressing the soft key [YES]. Press soft key [NO] to go to step 2 screen.

ID code uploading (Tech2)
F0: Syringe ID code
F1: ID code registration
F2: ID Code Upload (Tech2)
F3: ID code downloading (ECM)
F4: Historical Information

ID code uploading (Tech2)
Can data be uploaded?
YES NO

MFS4Z0MH002101

3. After the upload is completed, the message "Data uploaded" will be displayed. Press soft key [ENTER] to go to step 1 screen.

4. When the data uploading fails, there will show "data not uploaded!" This hint of information shows. At this point, press soft key [ENTER] to go to step 1 screen for uploading again.

ID code uploading (Tech2)
Upload finish!
Confirm

ID code uploading (Tech2)
Data could not be successfully uploaded!
Confirm

MFS4Z0MH002201

ID code downloading (ECM)

1. Select menu F3: ID code logging in with ▲, ▼ (UP/DOWN) key F3 (function key) Press ID code downloading (ECM) and then press [ENTER].
2. After the following screen is displayed, confirm that the engine is stopped, then press the [OK] soft key.

ID code downloading (ECM)
F0: Syringe ID code
F1: ID code registration
F2: ID Code Upload (Tech2)
F3: ID code downloading (ECM)
F4: Historical Information

ID code downloading (ECM)
Please confirm the vehicle status! It is active only when the engine stops! confirm

MFS4Z0MH002301

3. Press soft key [YES] to download (log in) the uploaded ID codes (No.1 to No.6 cylinder) to ECM. Press soft key [NO] to go to step 1 screen.

ID code downloading (ECM)
Can data be downloaded? YES NO

LNW41ASH004301

4. Log in from No.1 cylinder to ECM in ID code.

ID code downloading (ECM)
No.1 cylinder (INJ.#1) In programming! 0 % 100

MFS4Z0SH001301

If the data downloading succeeds, after the logging in screen of No.6 cylinder displays, it will switch to the next screen.

If the data downloading is abnormal, repeat logging in of No.1 cylinder to No.6 cylinder for 3 times and then switch to the next screen.

5. After the data downloading completes, [programming finished!] will display this hint of information shows. Press soft key [ENTER] to go to step 1 screen.

ID code downloading (ECM)
Programming is finished! confirm

MFS4Z0SH001401

6. "Programming fails!" will occur when the data downloading fails this hint of information shows. At this point, press soft key [ENTER] to go to step 1 screen for downloading again.

ID code downloading (ECM) Programming fails! Confirm MFS4Z0SH001501
--

MAF correction data uploading (Tech2)

1. Select menu F2: MAF correction with ▲, ▼ (UP/DOWN) key or F2 (function key) and then press [ENTER].

2. Select the menu F0:MAF to correct the data upload (Tech2) by using the button ▲ and ▼ (up / down) key or F0 (function key), then press the [ENTER] key.

Programming F0: Q adjustment value correction F1: Syringe ID code F2: MAF correction F3: Engine Start F4: number of starts after starter replacement F5: DPD reactivation data clearing
MAF correction F0: MAF correction data uploading (Tech2) F1: MAF correction data downloading (ECM) F2: After-sale model
MFS4Z0MH003201

3. Press soft key [YES] to upload data. Press soft key [NO] to go to step 2 screen.

4. When the upload is complete, the "Upload completed" message will be displayed. Press soft key [ENTER] to go to step 6 screen.

MAF correction data uploading (Tech2) Can data be uploaded? Yes No
MAF correction data uploading (Tech2) The upload has been completed! Confirm MFS4Z0MH003301

5. When the data uploading fails, there will show "data not uploaded!" This hint of information shows. At this point, press soft key [ENTER] to go to step 2 screen for uploading again.

6. After uploading the data to the screen, press the [EXIT] key as prompted and set the key switch to the OFF position.

MAF correction data uploading (Tech2)

Data could not be successfully uploaded!

Confirm

MAF correction data uploading (Tech2)

Press EXIT and set ignition switch to OFF!

MFS4Z0MH003401

MAF correction

F0: MAF correction data uploading (Tech2)

F1: MAF correction data downloading (ECM)

F2: After-sale model

MAF correction data downloading (ECM)

Please confirm the vehicle status!
It is active only when the engine stops!

Confirm

MFS4Z0MH003501

MAF correction data downloading (ECM)

1. Select menu F1: MAF correction data downloading (ECM) with ▲, ▼ (UP/DOWN) key or F1 (function key) and then press [ENTER].
2. After the following screen appears, ensure the engine stalls and then press [ENTER].

3. When you press the [YES] soft key, downloading starts and the data registration status is displayed.

Press soft key [NO] to go to step 1 screen.

MAF correction data downloading (ECM)
Can data be downloaded?
YES NO

MAF correction data downloading (ECM)
0 % 100

MFS4Z0MH003701

4. After the download is complete, the "programming completed" message will be displayed. Press soft key [ENTER] to go to step 1 screen.

5. "Programming fails!" will occur when the data downloading fails this hint of information shows. At this point, press soft key [ENTER] to go to step1 screen for downloading again.

MAF correction data downloading (ECM)
Programming is finished!
confirm

MAF correction data downloading (ECM)
Programming fails!
confirm

MFS4Z0MH003801

After sale model (MAF correction data write)

- 1. Select menu F2: after-service mode with ▲, ▼ (UP/DOWN) key or F2 (function key) and then press [ENTER].
- 2. After the following screen appears, ensure the engine stalls and then press [ENTER].

MAF correction	
F0: MAF correction data uploading (Tech2)	
F1: MAF correction data downloading (ECM)	
F2: After-sale model	
Aftermarket Mode	
Please confirm the vehicle status!	
It is active only when the engine stops!	
Confirm	
MFS4Z0MH005201	

- 3. Enter the MAF correction value according to "About MAF corrections" and press the [ENTER] key.
- 4. After the following screen is displayed, confirm that the entered MAF correction value is correct, and then press the [Program] soft key.

When you want to terminate the program (login), press soft key [Interruption]or [EXIT] to go to step 1 screen.

Aftermarket Mode	

MAF correction value	***
If you continue, press the [ENTER] key.	
Aftermarket Mode	

MAF correction value	***
Program	Interruption
MFS4Z0MH005401	

5. When the input value is correct, after displaying the state of data logging from Tech2 to ECM, there will be a hint of "Programming Completed ". Press soft key [EXIT] to go to step 1 screen.

Aftermarket Mode

0 % 100

Aftermarket Mode

Programming is finished!

drop out

MFS4Z0MH005501

6. When the input MAF correction value and the registered MAF correction value in the ECM are not consistent, there will be "programming failure!" This hint of information shows. Press soft key [ENTER] and go to step 1 screen to write in MAF correction value again.

Aftermarket Mode

Programming fails!

confirm

MFS4Z0MH005601

About the MAF correction

Please search for the appropriate vehicle in the table and confirm the vehicle's corresponding MAF correction value.

No.	Chassis	GVW	T/M	SLD	ABS	ASR	MAF correction value
1	F** double cab	Below 8t	M/T	—	○	△	0.890
2	F** double cab	Below 8t	A/T	—	○	△	0.890
3	F** double cab	Above 8t	M/T	90km/h	○	△	0.890
4	F** double cab	Above 8t	A/T	90km/h	○	△	0.890
5	F*S-J low floor all drive double cab	Below 8t	M/T	—	○	△	0.890
6	F*S-J low floor all drive double cab	Above 8t	M/T	90km/h	○	△	0.890
7	F* S high floor all drive	Below 8t	M/T	—	○	△	0.900
8	F* S high floor all drive	Below 8t	M/T	75km/h	○	△	0.900
9	F* S high floor all drive	Above 8t	M/T	90km/h	○	△	0.900
10	F * S high-floor all-wheel drive snow forklift	Below 8t	M/T	—	○	△	0.900
11	F**-S no sleeper model	Below 8t	M/T	—	○	△	1.000
12	F**-S no sleeper model	Below 8t	Smoother-F	—	○	△	1.000

13	F**-S no sleeper model	Above 8t	M/T	90km/h	○	△	1.000
14	FRR STD & FSR	Below 8t	M/T	—	○	△	1.045
15	FRR STD & FSR	Below 8t	Smoother-F	—	○	△	1.045
16	FRR STD & FSR	Below 8t	A/T	—	○	△	1.045
17	FRR STD & FSR	Above 8t	M/T	90km/h	○	△	1.045
18	FRR STD & FSR	Above 8t	A/T	—	○	△	1.045
19	F*S-J low floor all drive	Below 8t	M/T	—	○	△	1.045
20	F*S-J low floor all drive	Above 8t	M/T	90km/h	○	△	1.045
21	F*S-J high floor all drive double cab	Below 8t	M/T	—	○	△	1.065
22	F*S-J high floor all drive double cab	Above 8t	M/T	90km/h	○	△	1.065
○: standard configuration △: optional configuration							

Starting times of the engine

1. Select menu F3: engine starting number with ▲, ▼ (UP/DOWN) key or F3 (function key) and then press [ENTER].
2. After the following screen appears, ensure the engine stalls and then press [ENTER].

Programming F0: Q adjustment value correction F1: Syringe ID code F2: MAF correction F3: Engine Start F4: number of starts after starter replacement F5: DPD reactivation data clearing
Engine starting times Please confirm the vehicle status! It is active only when the engine stops! confirm
MFS4Z0MH003901

3. Enter the number of engine starts (confirmed in the data display) and press the [ENTER] key.

4. After the following screen appears, check whether the entered engine start number is correct and then press [programming].

When you want to terminate the program (login), press the [Interrupt] or [EXIT] button to return to the screen in step 1.

Engine starting times ***** Counts
Engine starting times *Counting If you continue, press the [ENTER] key.
Engine starting times ***** Counts
Engine starting times *Counting
Program
Interrupt
MFS4Z0MH004101

5. When the input value is correct, after displaying the state of data logging from Tech2 to ECM, there will be a hint of "Programming Completed ". Press soft key [EXIT] to go to step 1 screen.

Engine starting times	
0	% 100

Engine starting times	
Programming is finished!	
Drop out	

MFS4Z0MH004201

6. If the entered engine start number is different from that in ECM, "programming fails!" will occur. This hint of information shows. Press soft key [ENTER] and go to step 1 screen to write in engine starting number again.

Engine starting times	
Programming fails !	
Confirm	

MFS4Z0MH004301

POWERSTAR

Number of starts after replacing the starter

- 1. Select menu F4: starting number after starter replacement with ▲, ▼ (UP/DOWN) key or F4 (function key) and then press [ENTER].
- 2. After the following screen appears, ensure the engine stalls and then press [ENTER].

Programming F0: Q adjustment value correction F1: Syringe ID code F2: MAF correction F3: Engine Start F4: Number of starts after starter replacement F5: DPD reactivation data clearing
Number of starts after replacing the starter Please confirm the vehicle status! It is active only when the engine stops! Confirm
MFS4Z0MH004701

- 3. Enter the starting number after the starter replacement (confirm it in data display) and then press [ENTER].

- 4. After the following screen appears, check whether the entered starting number after the starter replacement is correct and then press [programming].

When you want to terminate the program (login), press soft key [Interruption]or [EXIT] to go to step 1 screen.

Number of starts after replacing the starter ***** Counts Number of starts after replacing the starter *Counting If you continue, press the [ENTER] key.
Number of starts after replacing the starter ***** Counts Number of starts after replacing the starter *Counting Program Interruption
MFS4Z0MH004901

5. When the input value is correct, after displaying the state of data logging from Tech2 to ECM, there will be a hint of "Programming Completed ". Press soft key [EXIT] to go to step 1 screen.

Number of starts after engine replacement
0 % 100

Number of starts after replacing the starter
Programming is finished!
Drop out

MFS4Z0MH005001

6. If the entered starting number after the starter replacement is different from that in ECM, "programming fails!" will occur. This hint of information shows. Press soft key [ENTER] and go to step 1 screen to write in the engine starting number after the starter replacement again.

Number of starts after replacing the starter
Programming fails !
Confirm

MFS4Z0MH005101

POWERSTAR

DPD regeneration data removal

1. Select menu F5: DPD reactivation data clearing with ▲, ▼ (UP/DOWN) key or F5 (function key) and then press [ENTER].

2. Press soft key [YES] to clear the data.

Press soft key [EXIT] to go to step 1 screen.

Programming

F0: Q adjustment value correction

F1: Syringe ID code

F2: MAF correction

F3: Engine Start

F4: number of starts after starter replacement

F5: DPD reactivation data clearing

DPD regeneration data removal

Clear data?

YESNO

MFS4Z0MH004401

3. After pressing the soft key [YES] in step 2, the status of clear execution is displayed.

DPD regeneration data removal

Wait!

DPD regeneration data removal

0%100

MFS4Z0MH004501

4. After the clearing is completed, there will be “cleared successfully! ” This hint of information shows. Press soft key [EXIT] and set the key switch to [OFF].

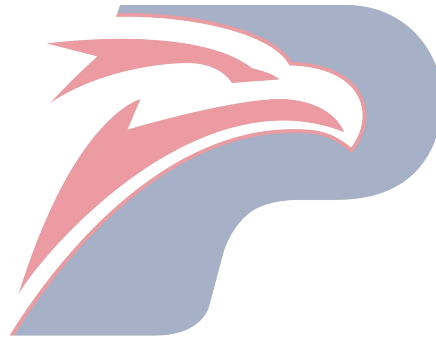
DPD regeneration data removal
You have cleared up! Set the key switch to the [OFF] state!
Drop out

MFS4Z0MH004601

POWERSTAR

Special tool

Summary	Tool number Tool name
	Tech2 Scan tool



POWERSTAR